1 Scope of the Volumes

1.1 Historical Aspects: 100 Years of Peptide Synthesis

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Peptides are ubiquitous in living systems. They are signal molecules which are involved in the control of critical biological functions such as cell division, mating, chemotaxis, pain perception, growth, and immunity. They are transport molecules which facilitate the passage of ions through cell membranes. They are protective agents that function as antibiotics which, through the wonder of evolution and discovery, protect both mold and man from their common enemy, the bacterium. They are important digestive intermediates and play a critical role in nutrition of both cells and intact organisms. At present, peptides are important commercial entities. The markets for small peptides that are either sweet or control blood pressure each surpass a billion US dollars, while peptide-like antibiotics are excellent antibacterial and antiviral agents.

Today the word "peptide" implies much more than simply a chain of amino acids. Even in nature, peptides are composed of more than the 20 amino acids that are the building blocks of proteins. Biologically active peptides contain D- as well as L-amino acid residues. Peptides are glycosylated, phosphorylated, myristoylated, farnesylated, palmitoylated, sulfated, and N-alkylated, and are even present in more complex conjugates.

The synthetic chemist has greatly extended the molecular diversity found in natural peptides by making unusual bond patterns that are generally classified under the rubric of peptidomimetics. These modifications confer unique properties on the resulting molecules, which often exhibit increased stability toward enzymatic degradation, and biological activities and specificities much higher than those of the natural products. The interplay of nature and the synthetic chemist has added greatly to the field of peptide research. This four-volume compendium celebrates a century of research, development, and applications of peptides, a critically important class of molecules.

When Emil Fischer[1] prepared the first unprotected synthetic peptide, glycylglycine, he gave birth to peptide chemistry. For several decades pioneers such as Curtius, Fischer, and Leuchs struggled valiantly to develop efficient approaches to build peptides from their component amino acids. In retrospect we marvel at the creativity of these giants. They could only utilize what today would be considered primitive tools. There was no high performance liquid chromatography, no mass spectroscopy, and no nuclear magnetic resonance spectroscopy; there were also no ground glass joints. Moreover, for many years these scientists were unaware of the structure of biologically active peptides, or of the mechanism of biosynthesis of peptides. Despite these handicaps the progenitors of our science made astounding discoveries. Indeed, viewed objectively, virtually all of the strategies that are used by modern peptide chemists were in place by the 1940s. What remained for the descendants of the early peptide chemists was to refine the approach of protection, peptide bond formation, and deprotection, and to make the entire process of peptide synthesis more efficient. The goal of these volumes is to gather and critically evaluate the detailed synthetic methods that have been developed in laboratories throughout the world so that the information will be available in a unified presentation.

This perspective section constitutes an overview of our field from its inception to the present. We hope to set the theme of these volumes in both a retrospective and prospective manner. We will attempt to highlight the evolution of synthetic peptide chemistry from the province of a few scientists to a field involving many chemists working both in industrial and academic settings. We will show how the early challenge to string together a few amino acids and to isolate pure compounds has grown so that the challenge at the beginning of the 21st century is to make peptides of ever increasing complexity. Ironically, whereas many of the

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early peptide chemists spent years to isolate a single molecule from a complex fermentation broth or cell extract, many peptide chemists today intentionally synthesize mixtures of millions of different peptides to obtain lead compounds. In writing this section it is our goal to point out the stages in the development of our field so that the reader will be informed of the origins, present activities, and future directions.

1880–1910: Amino Acids and Peptides

The field of peptide chemistry was born in the gastric-juice-derived digests of proteins. Early investigators, intrigued by the physical and chemical changes that accompanied digestion, isolated a host of natural products among which were a variety of amino acids and small- and medium-length peptides. Careful analytical investigations, the precursor of peptide synthesis, revealed the structures of the amino acids and led to a variety of theories on the molecular structure of proteins. It is intriguing that by the late 19th century, long before the covalent structure of proteins was established, organic chemists were struggling to recreate these molecules from the amino acid building blocks.

Of the early giants in our field, two individuals tower above their colleagues: Theodor Curtius and Emil Fischer. Working separately and often secretively and competitively, they established some of the basic principles that still guide the peptide chemist more than 100 years later. Curtius spent many years studying hippuric acid and attempting its synthesis. During his quest he developed the azide method[2,3] of peptide bond formation which, with modification nearly 50 years later,[4] remained one of the favored carboxy activation methods until the 1960s. Curtius was actually most interested in the chemistry of diazo compounds. Nevertheless, his studies from around 1880 to 1900 resulted in the synthesis of N-protected dipeptides of glycine and to the successful synthesis of benzoylpentaglycine ethyl ester.[2]

Emil Fischer must receive credit for the establishment of peptide and protein synthesis as a field of scientific endeavor. After his great success in carbohydrate chemistry, for which he was awarded the Nobel Prize in 1902, Fischer turned his attention to proteins. He also proposed the terms peptide, dipeptide, and so forth.[5] His initial approaches highlighted strategies that remain in place today. Fischer's laboratory was concerned with methods for separating amino acid derivatives, with the resolution of racemic mixtures of amino acids, and with the development of efficient chain-elongation reactions. He showed that the free bases of amino acid ethyl esters could be separated by fractional distillation. This resulted in the identification of two undiscovered components of protein hydrolysates, namely, proline and hydroxyproline. It is interesting to note that these early attempts to volatilize amino acids for analytical purposes eventually led to the gas chromatographic separation of amino acid derivatives and subsequently to efficient racemization assays.[6,7]

Fischer also demonstrated that amino acids could be resolved as diastereomeric salts and tenaciously pursued the synthesis of longer peptides, which he believed to be "proteinoids". His efforts culminated in the preparation of an octadecapeptide (Leu-Gly_3-Leu-Gly_3-Leu-Gly_9), reported in 1907.[8] Key aspects of this crowning achievement included the application of acid chlorides to peptide bond formation, and the use of α-halo fatty acids which eventually became the amino terminus of the peptide. Although the purity of the octadecapeptide could not be well established, and even though it consisted of only two types of amino acid, this synthesis was a seminal accomplishment and peptides of a given sequence with similar lengths were not synthesized again until the 1950s.

Fischer's laboratory was also involved in the use of gastric enzymes to prepare natural peptides for comparison with those synthesized by his collaborators, and to test the "protein-like nature" of his synthetic products. Given the fact that the dogma of the day held that proteins were compounds with molecular weights of several thousand, organic chemists thought that the total synthesis of a protein was within reach. In retrospect, even if the

assumption concerning molecular size had proven correct, a major component of peptide synthesis was lacking, namely, selectively removable protecting groups for the amino terminus.

Analysis of the work of Fischer and Curtius indicates that both of their laboratories experienced similar successes and problems. They employed esters to protect the carboxy terminus of the peptide and developed reasonably efficient methods for peptide bond formation (acid chlorides and acid azides). Indeed, after nearly a century of intense activity, there are few acylating reagents that are more reactive than Fischer's acid chlorides. Nevertheless, their progress suffered from the lack of selectively cleavable amino protecting groups and low yields for peptide bond formation because of the marked insolubility of intermediate length peptides. Although some progress was made in the 1930s and 1940s, these problems continued into the 1950s and 1960s when a burst of activity resulted in major advances in protecting groups, activation strategies, and eventually in the development of the solid-phase method of peptide synthesis.

1930s: Selective Removal of Amino Protecting Groups

Over billions of years of evolution, the ribosome has evolved to control peptide-chain elongation so that side-chain protection is unnecessary while stereochemical integrity is maintained and the protected carboxy terminus is not released until protein synthesis is complete. In addition, catalysis by the 16S ribosomal RNA is so efficient that assembly of the completed polypeptide occurs in a time frame of minutes. In contrast, synthetic peptide chemists must be concerned with coupling rates, with selective protection of amino acid N- and/or C-termini and side chains, and with maintenance of stereochemical purity during all stages of chain assembly. As mentioned above, the initial efforts to synthesize complicated peptides in an efficient manner were stymied, in particular, by the lack of a selectively removable protecting group on the amino terminus of the peptide chain. The earliest attempt to develop such protection for the α-amino group involved use of the ethoxycarbonyl group by Fischer. This was expected to decompose into carbon dioxide and the free amino terminus on treatment with base. In fact, a base-catalyzed rearrangement converted *N*-ethoxycarbonyl-protected peptides into derivatives of urea.[9]

Bergmann and Zervas successfully addressed the problems of chain elongation at the amino terminus through the development of the benzyloxycarbonyl protecting group,[10] which could be cleaved by hydrogenolysis. Both the peptide bond and the carboxy terminal alkyl esters are stable to hydrogenolysis allowing selective deprotection of the peptide amino terminus. This led to a new era in peptide chemistry where multifunctional amino acids could be incorporated into peptide chains. The adoption of the benzyloxycarbonyl protecting group was slow because of the small number of laboratories working on peptide synthesis prior to World War II. During the War, efforts halted in the field of synthetic peptide chemistry in Germany, while many laboratories in the United States and Great Britain were focused on the synthesis of penicillin,[11] which was subsequently shown to function as a dipeptidomimetic antibiotic. Knowledge concerning the structures of other biologically active peptides was growing slowly. There was, thus, little pressure to attempt the synthesis of different molecules. However, the benzyloxycarbonyl group was to provide stimulation to the field of peptide chemistry when it was reinvigorated in the late 1940s and 1950s. In particular, in 1952 it was shown that this group could be removed by acidolysis under conditions where the peptide linkage was essentially stable.[12] This extended the significance of the benzyloxycarbonyl group. More importantly, it awakened an interest in acid-sensitive protecting groups, which were to become the hallmark of peptide chemistry in the 1960s and 1970s.

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1940s: Rebirth of α-Amino Acid *N*-Carboxyanhydrides: Synthesis of Poly(α-amino acids)

Progress in peptide synthesis continued slowly in the late 1930s and 1940s. Toward the end of the 1940s there was increased interest in preparing polymers of α-amino acids. The impetus for these synthetic ventures arose from advances in the field of polymer chemistry and, in particular, in biophysical studies on fibrous proteins such as silk, keratin, and collagen. These studies created a need for model structures that could mimic natural biopolymers. The so-called Leuchs[13] anhydrides [α-amino acid *N*-carboxyanhydrides (NCAs), oxazolidine-2,4-diones] were the key synthetic reagent in these syntheses. These highly activated cyclic amino acid derivatives were first prepared in 1906 by rearrangement of ethoxycarbonylglycine chloride.[13] Subsequently, they were prepared by direct phosgenation procedures.[14,15] The NCA of glycine reacted spontaneously with even weak nucleophiles to generate carbon dioxide and a substitution product. Later, it was realized that, under appropriate conditions, high molecular weight poly(α-amino acids) could be prepared from the NCAs.

Efforts to prepare high molecular weight materials from amino acids were a continuing theme of many early synthetic peptide chemists who were seeking routes to prepare "synthetic proteins". Often the early studies with activated amino acids yielded oligomers and low molecular weight polymers as side products. In the 1920s and 1930s, efforts to prepare poly(α-amino acids) from amino acid esters were reported. Some success was achieved although these studies were primarily limited to polyglycines. These early achievements were quickly overshadowed by the development of methodologies to prepare and polymerize NCAs of the naturally occurring amino acids. By the end of the 1950s, the literature on poly(α-amino acids) had greatly expanded[16] and many efforts to discern mechanisms of the polymerization of the NCAs were reported.[17–19] The availability of these structures, both as homo- and copolymers, provided physical chemists with outstanding model compounds for proteins. These materials were used to develop spectroscopic methods, such as IR, UV, and CD, which could be applied to the determination of the secondary structure of proteins. In fact, Pauling was able to validate the α-helix structure using X-ray patterns of poly(γ-methyl glutamate).[20] Although the original goal of producing synthetic replacements for natural proteins such as silk has not been realized, a synthetic copolymer Copoxane (containing alanine, glutamic acid, lysine, and tyrosine) recently received FDA approval for the treatment of multiple sclerosis.

The original intention of Leuchs and Fischer was to use NCAs in peptide synthesis, but the high reactivity of these reagents and their tendency to oligomerize dampened the enthusiasm of most laboratories for these self-activated amino acids. Based on some successes in the early 1950s,[21,22] workers at Merck developed conditions where controlled-sequence peptides could be synthesized in aqueous media.[23] The crowning achievement involved the use of NCAs in the synthesis of ribonuclease A by the Merck group in 1969.

1950s: The Field of Peptide Synthesis Reborn

During the late 1930s and for most of the 1940s, little progress was reported in the actual synthesis of sequential peptides. In a major review, Fruton reported on the status of the field in 1949.[24] It was a striking conclusion that up to that date chemists had succeeded in preparing mostly dipeptides, a few tripeptides, and very few longer peptides. Of the 305 references cited by Fruton, nearly 50% contained the name Abderhalden, Curtius, Bergmann, or Fischer, demonstrating the dominance of the field by the German school in the period prior to World War II.

After World War II activity by peptide chemists increased dramatically. This explosive growth was accompanied by a significant broadening in the arena of peptide chemistry. Whereas efforts in this field had been to a large part the domain of German scientists until

World War II, afterward peptide research spread throughout the world. Driving this expansion were breakthroughs in biochemistry, molecular biology and protein chemistry. Researchers in the early 1950s were becoming aware of the biosynthesis of peptide chains and of the detailed primary and secondary structures of chains in biologically active peptides and proteins. In this period, Frederick Sanger[25] developed a method for determining the primary structure of peptides and reported the sequence of insulin for which he was subsequently awarded a Nobel Prize. This new knowledge created the impetus for the synthetic peptide chemist to develop methods to synthesize peptides efficiently for comparison with natural products and as models to understand biochemical and biophysical properties.

Many intersecting advances emerged in the 1950s. One important milestone was the development of convenient methods for peptide bond formation, including the use of mixed anhydrides,[26,27] active esters,[28–30] and carbodiimides.[31,32] In other laboratories, protecting groups were developed which could be cleaved under mildly acidic or basic conditions. The orthogonality of conditions for the removal of these protecting groups increased the proficiency of the peptide chemist to incorporate virtually any residue into a peptide chain. At the same time as the peptide chemist attacked problems of increasing complexity, the standards of the field became more exacting. In addition to synthesizing the peptide chain, issues of yield, purity, and optical integrity became paramount. This created increased demands for methods to distinguish closely related impurities and to measure the optical purity of each stereocenter in a peptide chain. Thus, new activation and coupling methods were required to meet standards for high-yield reactions and for the maintenance of stereochemical integrity of each asymmetric center (i.e., the absence of racemization).

The determination of the structure[33,34] of oxytocin and its synthesis[35] by du Vigneaud in 1953 were truly a milestone in peptide chemistry. Although the occurrence of hormones in the gastric fluids was suspected since the early 1900s, it was du Vigneaud's synthesis of oxytocin which provided the proof of structure of the hormone. For these achievements du Vigneaud received a Nobel Prize in 1955. A new era began in which biologically active peptides were synthesized for structure–activity studies. These target molecules increased the need for improved synthetic methods. The first synthesis of oxytocin was achieved with a yield of less than 1% and took many months to accomplish. The yield of the synthesis in solution was improved to nearly 40% by the use of active esters, and this nonapeptide can at present be assembled in high yield in a few hours using solid-phase approaches. This increase in yield and the ease of synthesis are indicative of the progress of synthetic peptide chemistry over the four decades after du Vigneaud's pioneering work. Just as oxytocin provided both interest and challenge to the synthetic chemist in the 1940s and 1950s, the discovery of numerous additional biologically active peptides compelled chemists to devise novel synthetic and purification methods.

The early studies on oxytocin, vasopressin, and gramicidin S created a need for the development of strategies for the cyclization of linear peptides. For peptides lacking a disulfide bond the approaches used employed activation of the carboxy terminus of a linear peptide in a manner similar to those utilized in standard peptide bond formation. Peptide chemists discovered that the critical aspect of peptide cyclization was working at high dilution so that polymerization reactions could be minimized.[36] Today, cyclic peptides remain important lead compounds for biochemical, biological, and biophysical investigations. A striking example is cyclosporin A that was first synthesized in 1984.[37] This 11-residue cyclopeptide has unique immunosuppressive activity and has revolutionized organ transplantation. The synthesis of cyclic peptides is thoroughly documented in Vol. E 22b, Section 6.8.

In many sections of this treatise, the unique characteristics and properties of the wide variety of protecting groups, coupling agents, and strategies are described. Since the early days of the benzyloxycarbonyl[10] and toluenesulfonyl[38,39] groups, literally hundreds of different protecting groups have been devised. As shown in Section 2, major advances were achieved using either electron-withdrawing or electron-releasing groups in the phenyl ring of

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the benzyloxycarbonyl group. This resulted in widely different acid stabilities for these protections, which have been utilized in the synthesis of numerous biologically active peptides.

Near the end of the 1950s, an important advance in amino protection emerged with the development of the *tert*-butoxycarbonyl group.[40,41] This group is quickly and completely removed from amino functions using dilute trifluoroacetic acid. Under these conditions the benzyloxycarbonyl group is quite stable. The fact that *tert*-butoxycarbonyl protection is completely stable to hydrogenolysis and base added greatly to its versatility. Peptide chemists enthusiastically adopted it and its success encouraged researchers to develop additional protecting groups with differential stabilities. The successes achieved for protection schemes are described in detail in Section 2.

As peptide chemists undertook the synthesis of peptides of increasing complexity and size the need for improved methods of amide bond formation became a central issue. In the 1950s, a significant expansion of the number of available methods for peptide bond formation was apparent in the literature. Many of these methods were rooted in discoveries in biochemistry that showed that phosphoanhydrides and thioesters could be used to efficiently drive reactions of carboxy groups.

The early studies in the use of phosphorus activation[42,43] did not meet the needs of the peptide community. This approach was never abandoned and resulted in significant contributions in the 1980s and 1990s. Thioesters found some utility and were the predecessors of active aryl esters, which were important reagents in the early syntheses of biologically active peptides. Peptide bond formation using isolated active esters required the synthesis and purification of a number of intermediates. A major breakthrough occurred with the introduction by Sheehan et al.[31,32] of *N,N'*-dicyclohexylcarbodiimide as a coupling agent which directly and efficiently joins the carboxy and amino components. This reagent greatly increased the speed of a synthesis. Later it was found to exhibit some deficiencies including racemization of the activated acyl amino acid and some tendency to undergo side reactions. Nevertheless, it set the stage for the development of improved coupling agents, and for numerous studies, where the undesired side reactions were eliminated or minimized; these are described in Section 3, which covers all aspects of peptide bond formation.

Beginning in the late 1950s work in peptide synthesis was facilitated by the availability of protected amino acids, coupling agents, and pure solvents from commercial vendors. These chemical companies utilized chemistry mostly developed in academic laboratories. This synergism continues to the present and will be ongoing in the future as new peptide chemistry develops, especially for the building blocks necessary for combinatorial chemistry.

1960s: Assembly Strategies, Mechanisms of Racemization, Solid-Phase Peptide Synthesis, and Hydrogen Fluoride Cleavage

As peptide chemistry entered the 1960s, it became important to consider the strategy chosen for chain assembly. It was realized that peptide chains could be built stepwise from the carboxy to the amino terminus, or from the amino to the carboxy terminus, or could be assembled by fragment condensation. With the availability of active esters, coupling agents, and selectively removable protecting groups, the choice of methodology affected the ultimate synthetic outcome. Problems were encountered such as the decrease in solubility found for many peptide fragments, which led to lower reactivity of the chain termini, loss of optical purity during fragment coupling, and various side reactions.

Since most biologically active peptides and proteins are composed of multiple stereogenic sites (usually L-amino acid residues) the chemist must be concerned with the optical purity of the target molecules synthesized. Peptide bond formation may be accompanied by racemization. For the purpose of this overview, we define racemization as a loss of optical

integrity at a stereogenic site, most often the activated acyl amino acid residue (see Vol. E 22b, Section 7.4 for additional definitions of racemization). Racemization at the α-carbon of an amino acid occurs by reversible removal of the proton from the α-carbon site. Although direct abstraction of this proton by base would suffice to racemize an amino acid residue, Bergmann and Zervas were the first to suggest that racemization during coupling was facilitated by the formation of oxazol-5(4*H*)-ones (formerly called azlactones or 5-oxazolinones).[44] Systematic studies on racemization in the late 1950s and the early 1960s strongly supported this hypothesis.[45–47]

Today it is widely accepted that the most prevalent route to racemization occurs through intermediate oxazol-5(4*H*)-ones. The finding that urethane protected amino acids manifest a markedly reduced tendency to racemize and that activated peptide free acids can racemize during activation or coupling (except under carefully defined conditions) strongly influenced the strategy that peptide chemists used for chain assembly. An early review on the synthesis and purification of oxazol-5(4*H*)-ones[48] and later reviews[49–51] of racemization present important insight into approaches used to determine racemization at the peptide carboxy terminus and the propensity of different amino acids to racemize. This topic is discussed in detail in Vol. E 22b, Section 7. Despite the general mechanistic considerations, many amino acids and peptide derivatives are unique in their tendencies to exhibit racemization during chain assembly.

Biology, through the process of evolution, has chosen to build peptide chains exclusively from the amino terminus to the carboxy terminus in a stepwise fashion. Peptide chemists do not, in general, follow this strategy because the carboxy group of a protected peptide would be activated at each coupling step and, therefore, highly susceptible to extensive racemization. Given the above, peptide chemists either assemble the chain by stepwise elongation from the carboxy teminal amino acid derivative or by the condensation of protected peptide segments. When carried out in solution both approaches require extensive purification of intermediates. This makes the synthesis of even medium length peptides labor intensive and the province of those highly skilled in the art. To deal with these difficulties, methods were developed which involved no isolation of intermediate length chains and gave efficient assembly of intermediate length peptides.[52,53] Despite these achievements, in situ syntheses by stepwise procedures have not been extensively utilized because they suffer from decreasing reactivity of the amino terminus of the peptide chain with increasing chain length. In most cases where stepwise synthesis has been used, the intermediate protected fragments were isolated and purified.[54]

Fragment coupling is the second strategy that has been widely used by peptide chemists for the assembly of peptide chains. The first synthesis of oxytocin involved a [3+4] coupling and a [2+7] coupling.[35] Fragment condensation allows the synthetic chemist to work with small peptides that have reasonable solubility in the solvents used for coupling. Since racemization often accompanies peptide bond formation, in the 1950s and 1960s investigators used strategies where either glycine or proline was located at the carboxy terminus of the peptide fragment to be coupled. In addition, the acyl azide method was used almost exclusively when peptides containing other residues at the carboxy terminus were assembled. This method was believed to be free of racemization; however, acyl azides react relatively slowly and can undergo rearrangements resulting in low yields of the product. Despite these shortcomings investigators achieved some remarkable successes including the total synthesis of gastrin (17 residues),[55] glucagon (29 residues),[56] insulin (51 residues in two chains),[57–59] and adrenocorticotrophic hormone (ACTH) (39 residues).[60,61] These syntheses employed a combination of stepwise construction of medium length fragments and fragment assembly to complete the polypeptide chains. In the case of insulin, two separate chains were oxidized to form the correct disulfide patterns.

It became obvious that the synthesis of peptides containing more than a few residues was a significant undertaking. Preparation of peptides with more than 20 residues required

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teams of investigators and many months, if not years, to complete. In addition, side reactions often plagued the final steps of a synthesis. These problems led to the development of better and more versatile protecting groups and more efficient coupling reagents; however, by the late 1950s and early 1960s several groups concluded that the efficient synthesis of large numbers of analogues and the assembly of small proteins could not be readily achieved by synthesis in solution.

Analysis of the difficulties encountered in solution indicated that the major problem involved the low reactivity of the amino terminus of long peptide fragments. This difficulty resulted in part from the low solubility of these fragments in coupling solvents that reduced the effective concentration of the chain ends. Therefore, the bimolecular amide bond formation proceeded slowly and in low yield. One approach to drive the reaction employed a large excess of one of the reagents; however, this resulted in major purification problems. Merrifield[62] and Letsinger[63] circumvented this problem by synthesizing peptides attached to styrene–divinylbenzene copolymers, which allowed for purification by filtration. This separated the resin-bound peptide from the reactants and soluble byproducts. Although the basic concept of both investigators was similar, the strategy of chain extension chosen by Letsinger featured growth from the N-terminus, exposing the growing peptide to reactions known to result in racemization. In contrast, Merrifield's approach involved growth from the carboxy terminal residue. This method has totally revolutionized synthetic peptide chemistry. For his discoveries, Merrifield was awarded a Nobel Prize in 1984.

The first synthesis reported by Merrifield produced the desired tetrapeptide (Leu-Ala-Gly-Val).[62] Amino acids, dipeptides, and tripeptides were all detected in the crude product released from the resin. Through continued improvements of the method, the high speed of the amino acid incorporation and automation, the solid-phase peptide synthetic methodology has become the method of choice for most laboratories synthesizing peptides. Shortly after the introduction of the solid-phase procedure it was used to synthesize insulin.[64] This impressive achievement awakened the biochemical community to the promise of synthetic chemistry and initiated a period in which large numbers of peptide analogues were prepared and analyzed.

In the solid-phase approach, the accumulation of impurities on the resin throughout the assembly of the peptide chain is a serious problem. These impurities arise as a result of incomplete coupling and deprotection reactions, and from various side reactions. Even small amounts of cleavage of a side-chain protecting group or release of the peptide from the resin during acidolysis of the α-amino protecting group cause significant side products. These problems were solved by the development of new side-chain protecting groups and resin attachments with increased acid stability. Many of these improvements were made by Merrifield and his colleagues[65] and specific details may be found in Section 4.3.

As the acid stability of the linkage and protecting groups increased, more powerful reagents were required to efficiently cleave the completed peptide chain from the resin. The hydrogen fluoride method introduced by Sakakibara[66] was a major advance overcoming this difficulty. This powerful acid cleaves nearly every known acid-labile protecting group and efficiently releases peptides even from the most acid-stable linkages to resins. It became the universally accepted method of cleavage by peptide chemists during the 1960s and 1970s; however, as is often the case, careful scrutiny showed that anhydrous hydrogen fluoride damaged certain peptides and resulted in a number of side reactions. Mechanistic studies by Merrifield demonstrated that additives could eliminate some of these difficulties.[67,68] To avoid some of these side reactions, other strong acids were also developed. In many cases, using the improved procedures, outstanding successes were realized for peptides containing up to 50 residues; however, there were target peptides that did not survive strong acid treatment. Therefore, it was necessary to develop synthetic routes that avoided exposure of the peptides to reagents such as anhydrous hydrogen fluoride.

At the end of the 1960s, the peptide community was elated by reports of the synthesis of ribonuclease A using both classical[69] and solid-phase[70] technologies. The classical approach was labor intensive. Both methods gave low overall yields of the desired protein; however, these achievements were milestones in that a protein with enzymatic activity had been synthesized. Emil Fischer's dream had been realized.

1970s: High Performance Liquid Chromatography and Peptidomimetics

Classical peptide chemists were steeped in the tradition of organic chemistry in which intermediates are carefully analyzed and purified to homogeneity. Since the assembly of a desired peptide by solid-phase peptide synthesis occurs without purification of intermediates it is likely that truncated peptides and other impurities accumulate on the resin; these are difficult to separate from the desired product. Prior to the 1970s, thin layer chromatography, paper chromatography, paper electrophoresis, and microchemical analysis were used to assess peptide homogeneity. None of these were sufficiently sensitive to distinguish the impurities that were expected from the solid-phase approach.

High performance liquid chromatography was an important breakthrough which helped the Merrifield method to gain widespread acceptance. This procedure was able to resolve relatively long peptides and was also amenable to a multimilligram-scale purification of crude product. It provided the peptide chemist of the 1970s with a powerful tool that in some ways was equivalent to the countercurrent distribution method developed in the 1940s, used extensively by du Vigneaud to purify oxytocin. Together with improved methods to sequence peptides and the development of Fourier-transform nuclear magnetic resonance spectroscopy, HPLC gave the synthetic peptide chemist confidence that the mixtures produced by a solid-phase strategy could be purified to homogeneity. Analytical techniques used by peptide chemists are described in Vol. E 22b, Section 7.

In the early 1970s, the discovery of families of biologically active peptides provided great impetus to the field of peptide synthesis. The search for the hypothalamic releasing factors led to an exciting and highly competitive hunt which culminated in Nobel Prize awards in 1977 to Guillemin[71] and Schally.[72] The biological significance of these factors, especially that of luteinizing hormone releasing hormone (LHRH) in female reproduction, induced many laboratories to generate hundreds of analogues in a search for practical biomedical applications. Also, the discovery of this new class of peptides taken together with increased studies of other peptide hormones, such as gastrin, secretin, ACTH, cholecystokinin (CCK), substance P, and insulin, to name just a few, initiated an extensive period of investigation in which biologically active peptides were lead compounds in medicinal chemistry. This driving force continues to the present and natural peptides from an ever-increasing set of tissues and cells provide the synthetic chemist with a challenge to understand the relationship between the primary, secondary and tertiary structures of these molecules and their biological properties. Researchers persist in their quest to seek ways to enhance and increase the potency and the selectivity of the bioactivities of the native peptides.

The synthetic methods available at the start of the 1970s were well suited to the synthesis of peptides containing from three to thirty residues. As biologically active peptides were targeted as lead compounds, it became clear that the serum stability, membrane permeability and bioavailability of the final molecule were of utmost importance for drug design. Natural peptides were extensively modified to take these factors into account. Moreover, during the 1950s and 1960s impressive advances in understanding the three-dimensional structure of peptides and in the development of spectroscopic procedures were achieved. In addition to the helical and sheet structures discovered by Pauling and Corey, peptides were shown to assume specific turn structures as well as a variety of other helices.

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Discussion of the field of conformational analysis and protein folding is beyond the scope of this section. Nevertheless, the concept that the most potent drugs mimic the three-dimensional ("biologically active") structure of the native peptide bound to its receptor became an important driving force in the field of peptide chemistry. The stabilization of this structure would result in increased potency and more efficacious pharmaceuticals.

The confluence of conformational concerns and the desire to increase the bioavailability of peptides resulted in the development of an ever-increasing variety of peptidomimetics. A peptidomimetic may be defined as a molecule in which a natural amino acid residue in a peptide is replaced by non-α-amino acid surrogates. In nature, peptidomimetics are found in the form of peptides containing D-residues, N-alkylated peptides, and hydroxy acids, which often function as antimicrobial agents and ionophores. Some of the earliest studies on such compounds involved efforts to understand the function of valinomycin. This cyclic depsipeptide contains 12 residues, is composed of D- and L-residues, and was first synthesized in 1963.[73] Cyclic peptides and peptidomimetics have become enormously important as target structures for drug discovery. In Vol. E 22b, Section 6.8, a discussion of the synthesis of cyclic peptides is presented.

The natural occurrence of compounds such as valinomycin provided peptide chemists with a starting point for stereochemical comparisons of homodetic peptides (peptides composed of only α-amino acids) and their topochemical analogues. From such analyses "retro-enantio" gramicidin S with high biological activity was designed and synthesized.[74] Numerous peptidomimetics and pseudopeptides began being synthesized in the 1970s. These included retro and retro-inverso peptides[75] and a plethora of backbone replacements.[76]

As the scope of peptide chemistry expanded new families of compounds were born. Novel and elegant methods to stabilize local conformational features such as β-turns and short α-helices were developed. These approaches have resulted in potent analogues of biologically active peptides including the enkephalins, Substance P, LHRH, and many others. The synthetic methods used to prepare peptidomimetics are thoroughly reviewed in Vol. E 22c, Sections 9–12, and Vol. E 22d, Sections 13–15.

1980s: 9-Fluorenylmethoxycarbonyl Chemistry, Synthesis of Proteins, and Recombinant DNA Procedures

Work in the 1980s reflected a change in thinking concerning the method used for protection of the α-amino group. Much discussion between peptide chemists considered the advisability of exposing a protein to strong acids such as hydrogen fluoride or trifluoromethanesulfonic acid. It became apparent that there was a need for a stable protecting group which could be efficiently removed by base. Indeed, in the early 1950s studies on trifluoroacetylamino acids demonstrated the utility of these derivatives in synthetic strategies; however, the tendency of acylated amino acids to racemize during carboxy activation eliminated the trifluoroacetyl moiety as a general protecting group for the α-amino group. In contrast to trifluoroacetyl, the 9-fluorenylmethoxycarbonyl[77] group led to minimal racemization of N^{α}-protected amino acids during activation. Although this group was originally reported in the early 1970s its use proliferated significantly in the 1980s. It is rapidly cleaved under anhydrous basic conditions and was readily incorporated into synthetic strategies. Together with *tert*-butyl side-chain-protection schemes, it has become a method of choice in automated solid-phase synthesizers. Thus, Carpino made a major contribution to the field through his development of *tert*-butoxycarbonyl and 9-fluorenylmethoxycarbonyl chemistries that now dominate much of synthetic peptide chemistry.

In addition to the impressive synthetic procedures elaborated for peptidomimetics, an important goal for the synthetic peptide chemist during the 1980s remained the efficient synthesis of longer and longer peptides. Several laboratories were committed to the synthesis

of proteins. Their approaches, however, were quite different. It is interesting that after the publication of the syntheses of ribonuclease A in 1969,[69,70] no new synthesis of polypeptides with 100 or more amino acids was reported in the 1970s. This sobering fact probably reflected the realization that the "enzyme with ribonuclease A activity" produced in solution and by the solid-state method was quite heterogeneous.

It is admirable that during the 1970s, Merrifield's group and others[78,79] expended great effort to understand and control the cause of side reactions during chain assembly using *tert*-butoxycarbonyl protection strategies. Critical discoveries included the fact that deletion sequences resulted from extraneous functional groups such as aldehydes on the resins used for synthesis, that truncated peptides were caused by trifluoracetylation of the amino terminus, and that chain loss was caused by instability of the peptide-resin linkage to acidolysis.[80–82] Recognition of these side reactions allowed Merrifield to solve many of these problems and nearly eliminate these sources of unwanted products. The expertise gained during these investigations resulted in additional attempts to synthesize proteins using solid-phase chemistry. Perhaps the most successful application of this approach was the synthesis of HIV protease before it was available by molecular biological expression.[83,84] Thus, the chemical synthesis validated the structure of this protein, facilitated studies of the mechanism of replication of the virus, and allowed screening for specific inhibitors.

In parallel with the attempts to improve the solid-phase method, several laboratories attempted to optimize the synthesis of proteins in solution. In 1981, Fujii and Yajima[85] reported the synthesis of fully active ribonuclease A using the azide strategy and purification featuring affinity chromatography. At about the same time Sakakibara developed a general strategy to synthesize long polypeptides and proteins.[86] Sakakibara has standardized the strategy used in the synthesis of proteins in solution. He considered such issues as maximal side-chain protection, orthogonality of the temporary and permanent protecting groups, solubility of the intermediate fragments in solvents suitable for peptide bond formation, racemization-free coupling procedures, minimization of degradation during final deprotection, and efficient approaches to disulfide formation where necessary. In reaching the optimum conditions, the Sakakibara group took advantage of many advances in protecting groups and coupling agents made during the 1960s, 1970s, and 1980s. Detailed reviews of these methodologies are presented in Sections 2 and 3. The fact that efficient coupling could be carried out in solvents such as dimethylsulfoxide or trifluoroethanol/dichloromethane was a most critical contribution. These powerful solvent mixtures allowed coupling to be carried out at high fragment concentration without significant side reactions. The problems associated with the correct formation of disulfide bonds in proteins and peptides were addressed by many researchers and are the subject of Vol. E 22b, Section 6.1. Between 1982 and 2000, Sakakibara et al. reported on the synthesis of about 10 peptides containing more than 45 residues, including the synthesis of a 121-residue heparin binding protein that contained five disulfide bonds.[87] Reported in 1998 was the total synthesis of the green fluorescent protein,[88] which contains 238 amino acid residues, a most remarkable achievement! These proteins are prepared by the condensation of fragments, permitting efficient purification and increasing the confidence in the homogeneity of the final product. At the conclusion of a century of peptide chemistry, it is clear that with high purity intermediate fragments, and the application of the best protection and coupling schemes, the chemical synthesis of many proteins can be undertaken. This field is thoroughly reviewed in Vol. E 22b, Section 5.

A discussion of peptide synthesis in the 1980s would not be complete without mentioning the impact of molecular biology on the field. By the beginning of the 1980s, reports of the efficient synthesis of peptides and proteins using recombinant DNA procedures were appearing. These reports increased in number, and commercial interest in molecules such as insulin, human growth hormone, interferon, and erythropoitin resulted in great advances in the biosynthesis and purification of these molecules through biotechnology. These biological alternatives compete with chemical methods and provide major challenges for the synthetic

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peptide chemist. It is clear, however, that peptide chemistry can use building blocks and functionalities to prepare proteins not readily available through biosynthetic routes. Thus, the combination of molecular biology and organic chemistry for peptide and protein design was envisioned as the 1980s came to a close.

1990s: Chemical Ligation and Combinatorial Chemistry

Despite the competition provided by recombinant DNA techniques, a number of laboratories recognized that synthetic peptide chemistry offered approaches to protein engineering that were beyond the realm of biology and the limitations of the genetic code. Protein analogues with unnatural residues would be of great interest to scientists studying protein folding and factors influencing protein stability, and could lead to novel biomaterials. Thus, there remained significant incentives to develop new approaches for the efficient synthesis of polypeptide chains.

As indicated above, the primary advantage of the solid-phase approach is speed and efficiency. There are synthetic difficulties inherent to this strategy that severely limit the attainment of homogeneity for peptides containing more than 60 or 70 residues. In contrast, fragment-coupling approaches for protein synthesis are slow and much less efficient, but result in intermediate and final products which can be confidently purified to homogeneity. Given these facts a number of groups began exploring convergent synthetic approaches to prepare long peptides. Perhaps the most exciting manifestation of the convergent strategy involves the use of solid-state methods to prepare medium length peptide chains containing from 10 to 60 residues, followed by chemical ligation strategies to couple these intermediates. This approach mimics protein splicing, but differs in the specifics of the chemistry and also in that no inteins are removed during the ligation sequence.

Chemical ligation techniques utilize efficient reactions between unprotected peptides to form stable peptide bonds between the α-carboxy group of one chain and the α-amino group of a second chain. The strategies appear to have evolved from mechanistic information concerning the activity of thiol and serine proteases and the principle of entropic activation, which uses proximity effects to increase the rate of formation of peptide bonds.[89,90] Impressive achievements have been reported using thioester activation of α-carboxy groups and transesterification with the thiol side chain of N-terminal cysteine and homocysteine residues in peptides.[91–94] Since free peptides are joined, the removal of side-chain protecting groups from the final polypeptide is avoided and the peptide bond formation can be carried out in aqueous media. This eliminates side reactions often associated with the acidolytic conditions required for final deprotection and allows relatively high concentrations of the condensing fragments to be obtained. In a recent study on parathyroid hormone analogues the ligation reaction occurred in less than four hours in high yield.[94] Somewhat earlier, this approach was used to assemble fully active phospholipase A_2 using a glycine-to-cysteine ligation from two fragments containing 58 and 66 residues.[95]

The chemical ligation strategy has been applied primarily to the formation of bonds involving cysteine and homocysteine; the latter residue can be converted into methionine by methylation. Moreover, the number of examples of this approach is rather limited and side reactions have been noted which will probably be unique for individual polypeptides. Nevertheless, this strategy would appear to have great potential in the chemical synthesis of long polypeptides. Section 4.1.5 extensively discusses the chemical ligation field.

It is paradoxical that during most of the 20th century peptide chemists exerted tremendous effort to isolate pure, biologically active peptides from complex fermentation broths or tissue homogenates, whereas in the 1990s many scientists developed methods to synthesize peptide libraries containing hundreds to billions of molecules. Molecular diversity was the key goal of these investigations. The entire area of multiple peptide synthesis emanated from the advances in solid-state peptide synthesis discussed above. Notable early

contributions reported the efficient synthesis of many hundreds of peptides;[96,97] however, the fact that the 20 naturally occurring amino acids could generate 1,280,000,000 (20^7) different heptapeptide sequences created a demand for alternative strategies. This led to the synthesis of soluble peptide libraries, peptide libraries attached to supports, and peptide libraries displayed on the surfaces of bacteriophage or other biological presenters. From a synthetic chemistry perspective one of the most novel achievements was the light-directed synthesis of combinatorial libraries,[98] which represented an interesting application of photochemically cleavable protecting groups.[99,100]

Initial studies on peptide libraries generated much interesting chemistry. Libraries were coded by a variety of chemical tags used for molecular biological, chromatographic and mass spectrometric decoding of the target sequences discovered during the high-throughput biological screens. As mentioned above, soluble libraries and libraries attached to solid supports have been prepared.[101,102] High-affinity ligands for antibodies and enzymes have been discovered in both matrix-bound and soluble libraries, and the original studies on peptide libraries have now been extended to include nonpeptide libraries of an ever-growing molecular diversity. Thus, the impact of combinatorial peptide libraries in drug discovery programs has been enormous. Combinatorial synthesis is presented in detail in Section 4.3.7.

A retrospective evaluation of the progress made in synthetic peptide chemistry during the 20th century is awe-inspiring. Building on the initial pioneering work of Fischer and Curtius, the peptide chemist of the year 2000 can now routinely synthesize peptides containing 10 to 15 residues in one day and can have pure compounds on a 100-mg scale after one purification by HPLC. Methods have been developed to synthesize proteins both in solution and on solid supports. Although neither approach is routine, each route augments molecular biological procedures by providing analogues with unnatural residues and peptide bond replacements. Excellent methods, analogous to post-translational modifications used in nature, have also been developed for making peptide derivatives. These include those containing tags which allow detection inside cells and efficient enzymatic assays, and alterations such as palmitoylation, myristoylation, farnesylation, geranylgeranylation, glycosylation, sulfation, and phosphorylation, which target these peptides to membranes and cell surfaces. Many of these topics are covered in specific sections of this compendium. Peptidomimetics based on conformational considerations have added new dimensions for the medicinal chemist. Through these advances, chemists have achieved increased serum stability, target specificity and receptor affinity in molecules they have designed. Moreover, the peptide chemist literally is providing tens of millions of compounds for screening purposes.

In reviewing accomplishments to date, it is clear that Emil Fischer's dream of synthesizing a small protein has been realized and surpassed. The peptide chemist has continually struggled to make longer peptide chains rapidly in high purity. In achieving these goals biology has often been imitated and in some cases even improved upon. Nevertheless, it is sobering to realize that despite these great achievements the lowly *Escherichia coli* remains the preeminent polypeptide synthesizer.

As we go forward, peptide chemists will continue to contend with many of the issues that have driven the field since the 1950s. We can speculate that conjugates of peptides with other biopolymers such as carbohydrates and nucleic acids will become important areas of focus. Peptide chemists will become increasingly involved in screening procedures to elucidate biological activities associated with the more sophisticated libraries being produced. Chemical ligation efforts will increase as convergent strategies combined with semisynthesis are expected to generate novel compounds of both fundamental and commercial interest. Engineering and de novo design of proteins will likely be a concept proffered with increasing frequency. The synthesis of proteins that function in membranes remains an important challenge. The solving of problems of insolubility and peptide aggregation will be high on the list of goals of many synthetic laboratories. Novel structures such as β-sheets, which can serve as interaction interfaces and that can inhibit pathological processes such as amyloid plaque

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formation, will be substantial synthetic targets, and this will result in increased activity in the development of templates and scaffolds for the synthesis of peptide chains. Peptide synthesis remains a vibrant, exciting, and challenging area of endeavor, which will undoubtedly yield remarkable contributions in the years ahead.

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1.2 General Concepts in Peptide Chemistry

L. Moroder

The ribosomal biosynthesis of polypeptide chains occurs under genetic control from the N- to the C-terminus with the 21 proteinogenic α-amino acids (including selenocysteine[1–3]) and with full retention of the L-configuration for all chiral amino acids. Co- and posttranslational modifications as well as processing of the polypeptide precursors to the functional bioactive forms occur in a highly specific manner by enzymatic processes, and the resulting molecules are more or less structured linear polyamide molecules containing as cyclic elements, apart from proline, hydroxyproline, pyroglutamic acid, and five-membered heterocycles such as oxazole and thiazole, only those deriving from side-chain disulfide and thioether bridges. In microorganisms a different pathway of peptide biosynthesis takes place under the control of large multifunctional enzyme complexes, termed nonribosomal peptide synthetases,[4,5] from an exceedingly diverse group of amino acids as precursors (to date more than 300 are known[6]) including nonproteinogenic N-methylated and D-amino acids. Moreover, linkage of the amino acid building blocks is not restricted to amide bonds, but in depsipeptides ester bonds from hydroxy acids are found. Thus, the resulting peptide backbone is more or less irregular, linear, or cyclic, and the occurrence of other cyclic structural motifs such as side-chain cross-linked rings or lactones and lactams leads to a great diversity of structures that often are further modified by acylations and glycosylations. These structurally diverse compounds are endowed with a broad spectrum of biological activities including antimicrobial, antiviral, antitumor, immunosuppressive, or enzyme-inhibiting properties.[7]

Both of the above natural biosynthetic pathways can be exploited for the efficient production of peptides and proteins. In fact, recent new developments in recombinant DNA technology allow for in vivo expression of those proteins that do not affect the physiology of the host cell[8–10] and in cell-free in vitro translation systems even of proteins that interfere with cell physiology.[11–14] Within certain limitations the genetic control can be bypassed and desired mutants can be obtained by replacing the natural proteinogenic amino acids with related analogues or synthetic amino acids carrying biophysical probes. For this purpose in vitro suppression methodologies based on the use of nonsense or missense codons[15,16] or frameshift mutations[17] were developed; alternatively, selective pressure can be applied in host cells[18] which forces bioincorporation of a variety of nonproteinogenic amino acids into recombinant proteins in an efficient manner when these resemble their natural counterparts in shape, size, and chemical properties.[19,20] Similarly, advances in understanding the module-based protein template of nonribosomal peptide synthetases enabled the development of techniques for the design of novel peptides[21,22] and for exploring the potential of these enzyme complexes in combinatorial synthesis for the generation of structural diversity.[23]

Despite the power and potential of biotechnological procedures, there are drawbacks and disadvantages such as the frequently encountered difficult isolation of desired products from the fermentation media. It is also difficult to produce the large numbers of analogues routinely required to delineate structure–activity relationships or mutants bearing biophysical probes to investigate molecular recognition processes. Thus, the chemical synthesis or semisynthesis of peptides remains a valid alternative, and for lower molecular weight peptides it is the method of choice. The immense pool of natural biologically active peptides derived from nucleic acid-dependent or nonribosomal protein-template-directed synthesis in microorganisms represents nature's lesson for chemists to design conformationally restricted molecules and peptidomimetics. Moreover, the evolution-dependent hypermutations as discovered, e.g. for toxins from cone snails,[24] are clearly relevant to the present-day efforts to develop methodologies of combinatorial peptide libraries.

In principle, the synthesis of peptides and proteins is based on a rather simple chemical reaction, i.e. polycondensation via amide bond formation of natural proteinogenic and

for references see p 34

nonproteinogenic or of nonnatural synthetic amino acids in a defined sequence order. The presence of reactive functionalities in the side chains of the amino acids, their sequence-dependent reactivities, and the sequence-dependent lability of amide bonds in many cases prevents the routine application of well-established synthetic procedures. This is further aggravated by the presence of heterodetic cyclic or branched structures such as *N*- and *O*-glycosyl, *N*- and *S*-fattyacyl, *O*-phosphate or *O*-sulfate moieties. Other difficulties are caused by posttranslational modifications as well as by artificial modifications specifically designed for production of biochemical or biophysical tools. But peptides and even proteins are needed as replicates of the natural bioactive compounds or as analogues and peptidomimetic surrogates for basic research; they are also required as drugs in human and veterinarian medicine as well as in agriculture. This fact has fostered tremendously the amelioration of existing and the development of new methodologies in the field since the first issue of a compendium on peptide chemistry in the *Houben–Weyl* series in 1974.[25]

To meet the demands in terms of speed and diversity of peptides, the development of new and more efficient synthetic procedures and of more sophisticated techniques in chromatography and analysis was required. This goal has been achieved by major efforts on various aspects of peptide synthesis on solid supports, the synthetic principle introduced by Merrifield.[26] The progress made in the field in terms of orthogonal protection schemes and coupling reagents for the peptide elongation steps fulfilled the requirements for efficient automated syntheses and created the basis for extension of these on-resin synthetic techniques to combinatorial chemistry of peptides and ultimately to other organic compounds.

In this context, a major impact has been the development of high-resolution reversed phase chromatographic (RP-HPLC) techniques for analytical and preparative purposes.[27] In fact, with the discovery of reversed stationary phases[28] and with applications to chromatographic resolution of amino acid and peptide derivatives,[29,30] HPLC very soon emerged as the most powerful technique for synthetic peptides.[31]

The cautiously optimistic prediction of E. Fischer about the synthetic accessibility of peptides[32] has been realized with the conceptual and methodological advances made in the last century. These have provided peptide chemists with many refined tools for essentially routine synthesis in solution and on solid supports of small- to medium-sized peptides. However, for peptides having a high incidence of more sensitive amino acids or containing particular side-chain modifications, it is still neither a routine nor a trivial matter. Similarly, only a few successful stepwise solid-phase syntheses of small proteins have been reported,[33] while with optimized segment condensation strategies in solution[34] and on resin[35–37] or by convergent technologies[35–38] total chemical syntheses of proteins can be attempted with at least a reasonable expectation of success. With the advent of chemoselective ligation methods of synthetic or bioexpressed protein fragments[39–46] new ingenious approaches have been devised for the synthesis of proteins of sizes that were inaccessible by the established procedures.

1.2.1 The Peptide Bond

Amide bonds between α-carboxy and α-amino groups of amino acids constitute the basic element of the polyamide structure of peptides, although even α,ω-groups are linked via amides in natural compounds, e.g. in glutathione or bacterial cell wall peptidoglycans. Additionally, amide bonds are used by nature to produce cyclic peptides of variable ring size as well as to generate rigid scaffolds such as β-lactams, hydantoins, benzodiazepines, and piperazine-2,5-diones. Conversely, ester bonds are found in addition to amides only in the less frequently occurring class of depsipeptides.

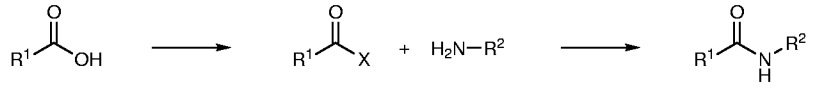
The amide bond with its hydrogen-donor and -acceptor properties is critically involved in the onset and stabilization of ordered structures of peptides and proteins as well as in the

biologically decisive molecular recognition processes where specific intermolecular complexes are tightened by networks of hydrogen bonds. Thus, amide bonds have been and still are one of the main targets in peptidomimetic research (see Vol. E 22c, Sections 10.4–10.12) and are used in the design of peptidomimetic structures derived from α-, β-, or ω-amino acids (see Vol. E 22c, Sections 10.12 and 10.13) as well as in peptoids (see Vol. E 22c, Section 10.1) and oligocarbamates (see Vol. E 22c, Section 10.14). Efficient methods for the formation of amide bonds are therefore critical to the synthesis of peptides and peptidomimetics.

1.2.1.1 Peptide Bond Formation

In addition to suitable protection of other coreactive functionalities, activation of the carboxy group prior to reaction with the amino group is required for a controlled formation of the amide bond between two α-amino acid or peptide components (Scheme 1). For this purpose activated species can be generated separately, isolated, and stored for subsequent use. Alternatively, the activation can be carried out in situ with specific coupling reagents. Independently of the mode of activation, an electron-withdrawing group X must be incorporated at the acyl carbon in a transient mode in the in situ activation or in the formation of reactive intermediates.

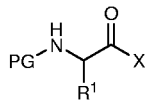
Scheme 1 Formation of the Amide Bond by Activation of the Carboxy Component

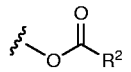
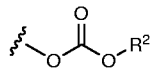
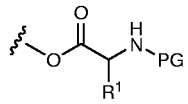


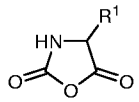
For efficient peptide bond formation acid halides **1**,[47] azides **2**,[48] and Leuchs' anhydrides[49] were employed as the first activated species in peptide synthesis. Since then, besides considerable improvements to the azide[50] (see Section 3.1) as well as the *N*-carboxyanhydride **7** procedure[51] (see Section 3.4.3), the methods have evolved over decades along a few basic principles as outlined in Scheme 2. The symmetrical **6**,[52] the mixed carboxylic acid **4**,[53–55] and the carbonic acid anhydrides **5**[56–58] were developed and remain useful despite the inherent drawbacks of loss of precious amino acid derivatives in the former and of potentially wrong nucleophilic attack in the latter case (see Sections 3.4.4 and 3.4.3, respectively).[59,60] Active esters **3** have been expanded from the initially discovered 4-nitrophenyl[61] and the *N*-hydroxysuccinimido esters[62,63] into a large variety of derivatives that find more or less routine application (for details see Section 3.2). However, to meet the demands of strong acylation potencies for efficient assembly of long polypeptide chains and particularly, for amide bond formation between sterically hindered amino acids in the synthesis of peptidomimetics, more recently increased attention has been focused on new developments related to acid chlorides,[64] fluorides,[65] and even bromides[66] (see Section 3.3). These were made possible by the use of N^{α}-9-fluorenylmethoxycarbonyl and the newly discovered N^{α}-arylsulfonyl amino acid derivatives, the latter being unable to racemize via the oxazol-5(4*H*)-one mechanism (*vide infra*).[67,68] Similarly efficient coupling reactions are achieved with the newly developed N^{α}-urethane-protected *N*-carboxyanhydrides **8** (UNCAs)[69] which are also devoid of racemization via the oxazol-5(4*H*)-one intermediate[70] and exhibit the additional advantage of having carbon dioxide as the sole byproduct (see Section 3.4.2).

for references see p 34

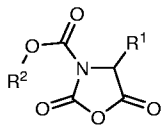
Scheme 2 Most Common Carboxy-Activated Amino Acid Derivatives Used in Peptide Synthesis[47,48,51–58,61–63,69]



Compound Number	X	Compound Name	Ref
1	F, Cl, Br	acid halides	[47]
2	N_3	acid azide	[48]
3	OR^2	active esters	[61–63]
4		mixed carboxylic anhydrides	[53–55]
5		mixed carbonic acid anhydrides	[56–58]
6		symmetrical anhydrides	[52]



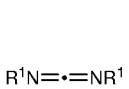
7 *N*-carboxyanhydrides



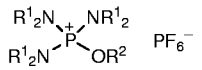
8 *N*-urethane-protected *N*-carboxyanhydrides

A similar evolution in the methodology is observable for in situ coupling reagents. Starting from carbodiimides **9** as the first reagent,[71] this procedure was optimized by addition of 1,2-dinucleophiles[72–74] (see Section 3.5). Conversely, exploiting the high reactivity of onium salts of polar aprotic solvents[75] and correspondingly, of hexamethylphosphoric triamide,[76] or of dimethylformamide,[77,78] and dimethylacetamide,[78] a large variety of phosphonium and uronium/guanidinium salts of the general structure shown in Scheme 3 were developed such as benzotriazol-1-yloxytris(dimethylamino)phosphonium hexafluorophosphate (BOP, **10**, R^1 = Me; R^2 = Bt)[79] and *N*-[(1*H*-benzotriazol-1-yl)(dimethylamino)methylene]-*N*-methylmethanaminium hexafluorophosphate *N*-oxide (HBTU, **11**)[80] (see Sections 3.7 and 3.8). The latter reagents which are nowadays widely used in the synthesis of polypeptide chains with natural, nonnatural, and sterically hindered amino acids,[37,81] satisfy even the severe demands of combinatorial chemistry in terms of efficient couplings in repetitive steps with a broad range of substrates.[82]

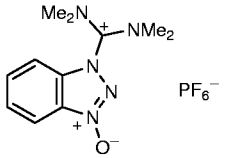
Scheme 3 Most Common Coupling Reagents Used in Peptide Synthesis[71,79,80]



9



10 R^2 = Bt, At

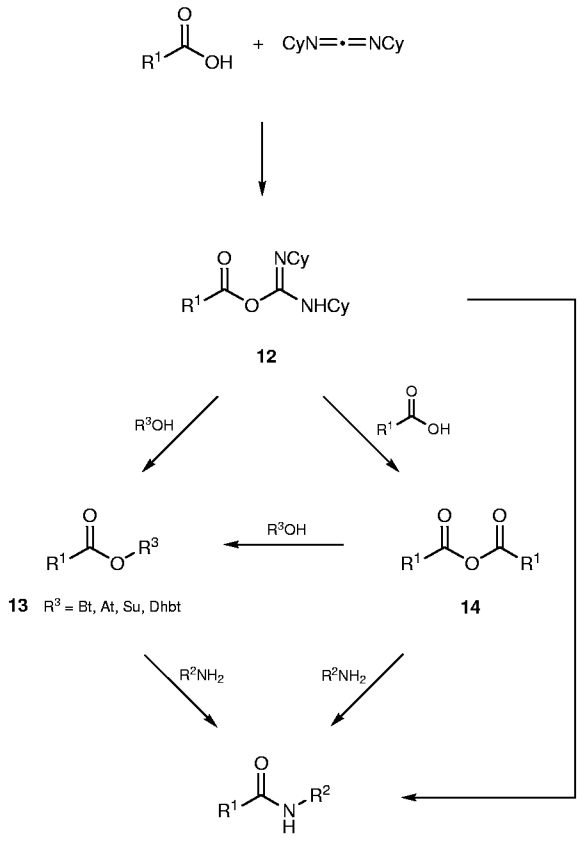


11

The main mechanistic pathways of amide bond formation by carbodiimides have been the object of intensive investigations.[83–87] There is general consensus that an *O*-acylisourea such as **12** is formed as the key intermediate which directly undergoes nucleophilic attack by an amino group to produce the amide or reacts with a second carboxy component to generate

the symmetrical anhydride **14** as the active species (Scheme 4); it may also rearrange to the inert *N*-acylurea as a side product.

Scheme 4 General Mechanism of Carbodiimide-Mediated Peptide Bond Formation in the Presence of 1,2-Dinucleophiles[83–87]

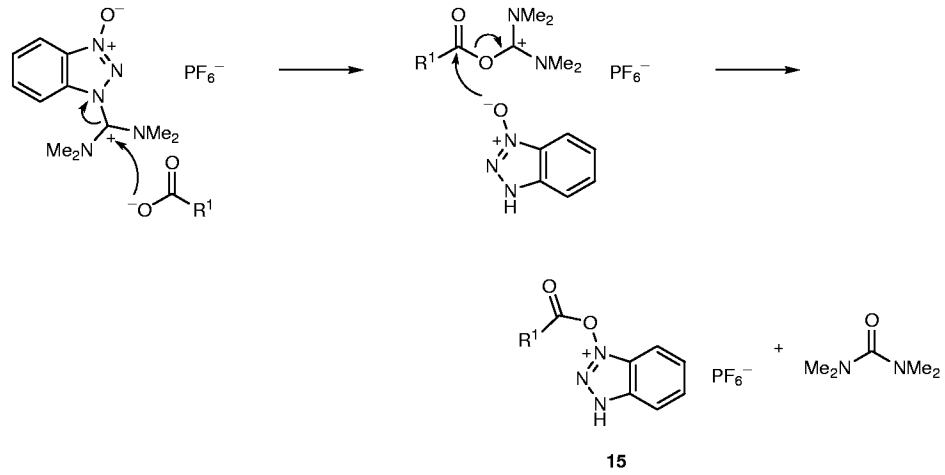


In the presence of weakly acidic additives, such as *N*-hydroxysuccinimide (HOSu),[72,73] 1,2,3-benzotriazol-1-ol (HOBt),[74] 3-hydroxy-1,2,3-benzotriazin-4(3*H*)-one (HODhbt),[88,89] or 7-aza-1,2,3-benzotriazol-1-ol (HOAt),[90] the O→N acyl migration as well as racemization/epimerization (*vide infra*) are largely prevented.[91] Concurrently, conversion of the highly reactive *O*-acylisourea derivatives and symmetrical anhydrides into the related less reactive active esters **13** derived from these additives takes place which leads to still sufficiently activated species to allow rapid in situ amide formation.

For the coupling reaction with phosphonium and uronium/guanidinium reagents an intermediate formation of the acyloxyonium salts (Scheme 5) is postulated which in the presence of the oxyanion of 1,2,3-benzotriazol-1-ol or 7-aza-1,2,3-benzotriazol-1-ol generate the related active esters **15** as the predominant transient species for aminolysis.[92–94]

for references see p 34

Scheme 5 General Mechanism of Onium Salt Mediated Peptide Bond Formation[92–94]

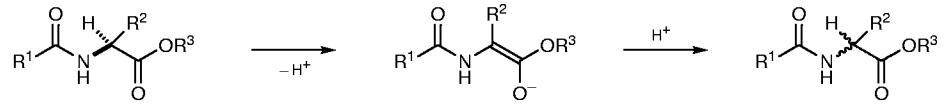


15

1.2.1.2 Racemization

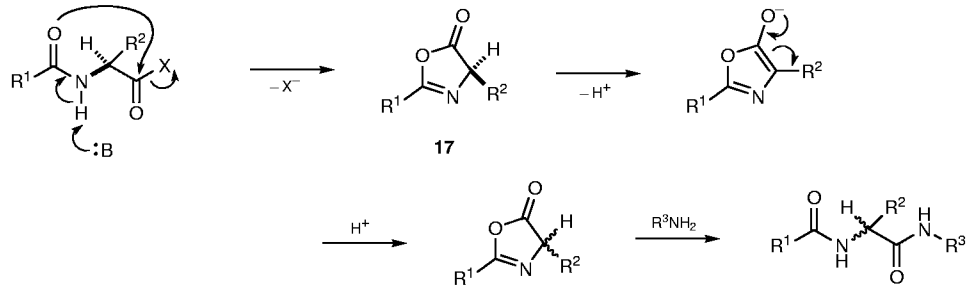
The intrinsic consequences of such strong activation of the carboxy group toward aminolysis are: (1) the increased acidity of the C^{α}-proton which favors enolization giving **16** and (2) the facile ring closure of the carboxy-activated amino acid or peptide component to oxazol-5(4*H*)-ones **17** by base catalysis. Both mechanisms lead to loss of stereochemical integrity, i.e. racemization or epimerization as illustrated in Schemes 6 and 7[95–98] (for the correct use of the terms racemization or epimerization see Vol. E 22b, Section 7.4).

Scheme 6 Enolization of Amino Acid and Peptide Derivatives by Base-Mediated C^{α}-Proton Abstraction



16

Scheme 7 Racemization/Epimerization of Amino Acid and Peptide Derivatives via Oxazol-5(4*H*)-one Formation

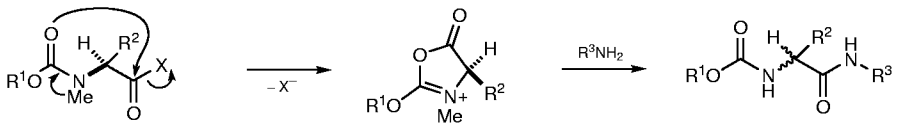


B = base
X = electron-withdrawing activating group

In the case of *N*-alkyl amino acids, the oxazolium ion **18** is formed even in the absence of base catalysis because of the electron-donating effect of the *N*-alkyl group,[97,99–101] thus increasing the risk of racemization (Scheme 8). Although proline can be considered as an N-alkylated amino acid, formation of the oxazolium salt is not observed, probably as a consequence of the steric constraints of the five-membered ring which disfavors cyclization.[95] As a result, in

the segment condensation approach proline (in addition to glycine) is a preferred choice for C-terminal amino acid (*vide infra*), although racemization has been reported in special cases.[102,103]

Scheme 8 Racemization of N-Alkylated Amino Acids by Oxazolium Ion Formation[97,99–101]

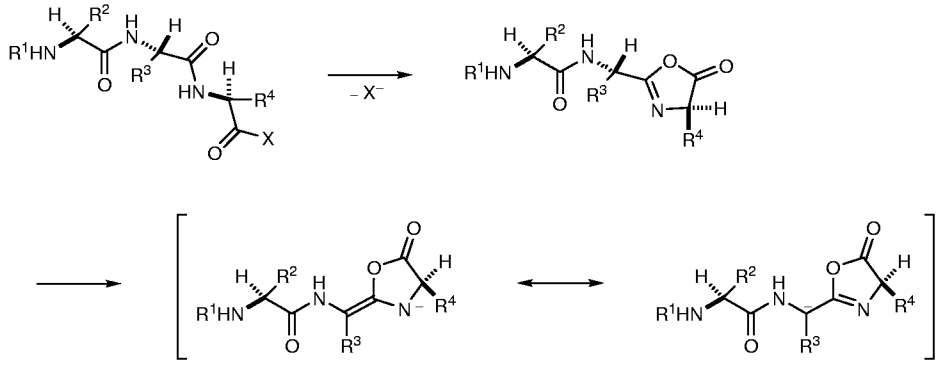


18

X = electron-withdrawing activating group

The formation of 2,4-dialkyloxazol-5(4*H*)-ones on C-terminal activation of peptides may even provoke epimerization of the penultimate amino acid residue (Scheme 9).[95,104–106] This is facilitated when the activated residue has no side chain (i.e., glycine).[106] Thus, even coupling at a C-terminal glycine residue may not be as free of epimerization as generally assumed. Oxazol-5(4*H*)-ones *per se* are activated species that provide an alternative method for amide bond formation for sterically hindered amino acids, although at low rates (see Vol. E 22c, Section 10.3.2).[92,107,108] Addition of Lewis acids such as zinc(II) chloride,[108,109] and copper(II) chloride,[110] or catalytic amounts of 10-camphorsulfonic acid,[108] or hydrogen cyanide[111] leads to rapid consumption of the oxazol-5(4*H*)-one species by aminolysis, which significantly reduces the extent of its racemization as well as of epimerization at the penultimate residue. Similarly, in the presence of 1,2,3-benzotriazol-1-ol or other dinucleophiles, which are released during formation of the oxazol-5(4*H*)-one from the active esters, or when these are added purposely in the free form[64,112] or as the related potassium salts,[113] the active esters are regenerated at high rates for subsequent aminolysis. Correspondingly, the extent of racemization strongly depends on the rate of amide bond formation, a fact that can be critical particularly in segment condensations in solution. In fact, such reactions can be sluggish as these often have to be performed at relatively high dilution because of poor solubility.

Scheme 9 Epimerization of the Penultimate Amino Acid Residue in Peptides via Oxazol-5(4*H*)-one Formation[95,104–106]



X = electron-withdrawing activating group

The loss of stereochemical integrity is one of the most troublesome side reactions in peptide synthesis,[95,114] because it is generally difficult to separate the resulting diastereomers. Thus, great attention has been paid to optimize the effectiveness of coupling reagents and additives as well as the nature of the solvents and bases (see Vol. E 22b, Section 7.4).

for references see p 34

Racemization of amino acid residues via the base-catalyzed mechanism of enolization (Scheme 6) does not occur under the normal conditions of peptide synthesis when the amino acid derivative is in the form of the free carboxylic acid. Racemization may occur when the carboxy group is protected and especially when the protection involves an electron-withdrawing group. In general, the extent of racemization by this mechanism is relatively low, although it depends on the nature of the amino acid, the type of N^{α}-protection applied, the amount and nature of the base, the solvent, and the reaction temperature (see Vol. E 22b, Section 7.4). Proteinogenic amino acids particularly susceptible to racemization include histidine and cysteine.[95,98] It has been a tenet for decades that, except for the *N*-(alkoxycarbonyl)-*N*-alkyl amino acids, the urethane-protected amino acids do not racemize during coupling, particularly in the absence of base, because of their inability to form 2-alkoxy-4-alkyloxazol-5(4*H*)-ones.[95] Thus, the observed sensitivity of urethane-protected histidine to racemization was rationalized on the basis of an imidazole-mediated intramolecular α-proton abstraction.[115] A recent discovery indicates that even *N*-alkoxycarbonyl amino acids form oxazol-5(4*H*)-ones[116] and that these are susceptible to base-catalyzed racemization.[117] This calls for attention to be paid, particularly with racemization-prone amino acids such as cysteine and serine even when N^{α}-protected as urethane derivatives (see Sections 2.6.6 and 3.8). In such cases the use of minimal amounts of base[118] or of sterically hindered bases,[119,120] or the use of base-free coupling methods[121,122] is recommended. In this context it is worthwhile to note that *N*-diacyl amino acids such as *N*-phthalimido derivatives, as well as the newly discovered *N*-arylsulfonyl amino acid which are unable to form oxazol-5(4*H*)-ones are interesting alternatives when highly activated species such as acid chlorides, fluorides, and bromides are used in difficult coupling reactions (see Sections 2.2.1.5.1 and 3.3.1).[66–68] In addition, N^{α}-urethane-protected *N*-carboxyanhydrides are well suited for such purposes because of their inability to racemize via the oxazol-5(4*H*)-one mechanism and their high reactivity even in the absence of base (see Section 3.4.4).

There is substantial evidence that the stronger the base, the greater the racemization/epimerization.[114] Therefore, onium salts as reagents in segment condensations, which require the base for carboxy anion formation, are less recommendable than carbodiimides in the presence of weakly acidic additives. In fact, the latter method currently represents the method of choice for the coupling of intermediate fragments (see Vol. E 22b, Section 5.2).

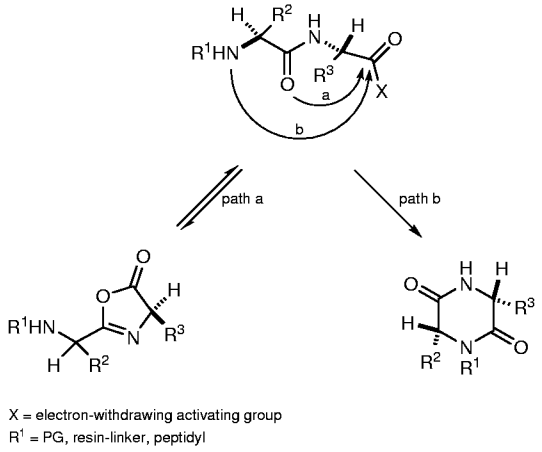
1.2.1.3 Difficult Sequences

The chemistry related to the repetitive N^{α}-deprotection and amino acid coupling steps required in peptide synthesis has been refined to a degree such that truncated sequences are generally reduced to a minimum, a fact particularly important in solid-phase synthesis where intermediate purification is excluded. However, the most serious potential problem facing both synthesis in solution and on solid support is the sequence-dependent poor solubility or solvation of the protected peptides, which prevents reactions from proceeding to completion. In solution and on resin, this phenomenon, described as difficult sequences,[123] derives from the intermolecular association of the protected peptide derivatives into β-sheet-like structures as well assessed by related conformational analysis.[124–126] In these aggregates the N-termini become inaccessible for further chemistry.[127,128] Such solubility problems may be circumvented with variable efficiencies by changes in the solvent composition, i.e. with 'magic mixtures',[129] or addition of chaotropic salts,[130,131] by operating at higher temperature,[132] or using solubilizing protecting groups[133] as well as by structural changes at the level of the amide bonds to allow for better solvation. For the latter purpose amide bond protection[134] or incorporation of 'pseudoprolines' as precursors of cysteine, serine, or threonine[135] are efficiently applied (see Section 2.3.2). In synthesis in solution such solubility problems may efficiently be bypassed by changes in the protection schemes that are known to affect the physicochemical properties of peptides, although most often a redesign of the intermediate segments represents the best remedy (see Vol. E 22b, Section 5.2).

1.2.2 C-to-N versus N-to-C Chain Elongation

The ribosomal biosynthesis of polypeptides occurs from the N- to C-terminus via aminolysis of the aminoacyl-tRNA. Similarly, the enzymatic biosynthesis proceeds from the N- to C-terminus via aminoacyl-enzyme intermediates. These routes exploit enzymatic control for full retention of the configuration of the amino acyl derivatives. In the biosyntheses an entropic activation, i.e. of the entropic proximity effect, is employed to facilitate the increased effectiveness of aminolysis with poorly activated esters. The chemical synthesis of amides is generally achieved by enthalpic activation of the carboxy group. Conversely, activation of the amino group is difficult and the related procedures such as the isocyanate,[136] phosphite amide,[137] and phosphorazo methods[138] are not efficient and are used only sporadically. In the N-to-C peptide synthesis in solution and on solid support the activation of the carboxy group of the growing peptide chain, i.e. of the peptidyl amino acid, with electron-withdrawing substituents, as required for efficient aminolysis, bears the intrinsic danger of (1) epimerization at the C-terminal amino acid via oxazol-5(4*H*)-one formation (path a, see Sections 1.2.1.2 and Vol. E22b, 7.4) and (2) capping of the peptide by piperazine-2,5-dione formation (path b, Scheme 10).

Scheme 10 Potential Cyclization Reactions in the N-to-C Peptide Synthesis in Solution and on Solid Supports: Oxazol-5(4*H*)-one (Path a) and Piperazine-2,5-dione Formation (Path b)



Although in one of the earliest approaches to solid-phase peptide synthesis, the N-terminal residue was attached to the solid support and used as the starting point for chain elongation in the N-to-C direction,[139] subsequent attempts to adopt these tactics in the azide procedure, which is less conducive to racemization, failed because of the low coupling yields and Curtius rearrangements.[140] Recent reinvestigation of the solid-phase N-to-C synthesis confirmed that piperazine-2,5-dione formation is not the main problem, but that racemization occurs at high degrees when more potent coupling reagents are used such as carbodiimide/1,2,3-benzotriazol-1-ol or uronium/guanidinium salts.[141] However, by the use of copper(II) 1,2,3-benzotriazol-1-olate [$Cu(OBt)_2$] as an additive to carbodiimides[142] racemization was significantly decreased.[143] Thus, the N-to-C direction of peptide synthesis is recommended only when required for particular purposes.

In view of these possible complications, the chemical synthesis of peptides is generally performed in the C-to-N direction by a stepwise elongation approach using preferentially N^{α}-alkoxycarbonyl-protected amino acid derivatives to produce the target polypeptide chain in solution and on solid supports directly. Specially designed intermediate segments, usually

for references see p 34

with glycine or proline as C-terminal residues, can be constructed for their subsequent assembly to form the final molecule in solution or on solid support (see Vol. E 22b, Sections 5.2 and 5.3). For this assembly of the segments again the C-to-N direction is generally the preferred mode, although even the N-to-C tactics are sometimes applied for particular purposes.

1.2.3 Stepwise Synthesis versus Segment Condensations

Since amide bond formation is a bimolecular reaction which involves a nucleophilic attack of the amino group on the activated carboxy group (Scheme 1), the kinetics are strongly affected by the type of carboxy activation, by steric effects exerted by the involved reaction partners, as well as by the solvents and catalysts employed (for comprehensive reviews, see refs[144–146]). Moreover, the concentration of the reactants plays a critical role and excess of one of the two components serves to drive the reaction to completion.

In peptide synthesis, the excess component is selected according to its solubility with respect to that of the product in order to facilitate purification of the peptide. For synthesis in solution, extraction procedures can be applied in the early steps thus allowing a choice of the excess component, i.e. amino or carboxy component, according to cost and availability. With increasing peptide chain length, application of the principle of excess reactant is favored by the differences in solubility between the acylating agent, i.e. the activated amino acid derivative, and the peptide amino component. Accordingly, the activated carboxy component is generally employed as the excess reactant. In this type of repetitive tactics of stepwise peptide-chain elongation, simple precipitations and washings with judiciously selected solvents suffice for quantitative removal of excess acylating agents together with the byproducts formed by the leaving groups and the salts when amines have to be employed in neutralization steps.

For synthesis in solution, this repetitive stepwise approach was successfully applied for the first time in the synthesis of oxytocin.[147] This route is generally adopted only for the preparation of relatively small peptides or of intermediate segments because solubility problems are quickly encountered with increased chain length. Conversely, in synthesis on solid supports the stepwise approach represents the general strategy since larger excesses of acylating components can be used to drive the reaction to completion without hampering the purification step since simple washings of the resin-bound peptide are sufficient for quantitative removal of excess reagents and resulting byproducts. Moreover, the solubility problem, although observable for particularly 'difficult sequences' even on solid support (Section 1.2.1.3), generally is not a limiting factor. However, this stepwise approach that renders the procedure conducive to automation, requires that coupling reactions are driven to completion and that the repetitive unmasking of the α-amino groups occurs quantitatively and with maximum selectivity. Otherwise, 'deletion sequences' and additional side products will accumulate and contaminate the final product to extents such that isolation of the desired material becomes an overwhelming task. Despite the application of the most advanced coupling and protection procedures, and even for synthesis on solid supports, limits in peptide size are reached that strongly depend upon the peculiar sequence characteristics of the target molecules.[27,123,148] Correspondingly, the most efficient strategies which emerged from recent optimization efforts for synthesis of larger peptides and proteins, are based on the preparation of intermediate segments in solution[34,149] or on solid support[123,150–152] and their subsequent assembly in solution[34,149,152] or on solid support (convergent solid-phase peptide synthesis)[38,153] as illustrated in Vol. E 22b, Sections 5.2 and 5.3.

1.2.4 Protection Schemes

In order to couple α-amino acids or peptides by chemical reactions in a selective manner all other functional groups not directly involved in amide bond formation that may compete with the amino group in its reaction with the activated carboxy group must be protected. Since the constituent natural and nonnatural amino acids of peptides and peptidomimetics may carry additional functional groups in their side chains, in general three types of protection are required. Taking into account that peptides are generally synthesized in the C-to-N direction, the α-amino group must be protected in a transient manner (also called temporary protection) in order to allow deprotection to occur selectively in the presence of all side-chain protections (semipermanent protection) that are generally removed in the last synthetic step. The C-terminal carboxy group can be part of this semipermanent protection, but its selective liberation is required in the presence of both the α-amino group and all the ω-functionalities, if protected segments are to be used to assemble the polypeptide chain. In solid-phase synthesis the C-terminal protecting group is generally the polymeric carrier to which the growing peptide chain is anchored via a linker.[26] In special cases, e.g. in the synthesis of cyclic peptides (see Vol. E 22b, Section 6.8), the polymeric carrier is attached to side-chain functionalities.[154,155] The linkers have to fulfill the stringent requirements of an ideal protecting group, i.e. facile introduction and, more importantly, quantitative cleavage under conditions that do not affect the structural and stereochemical integrity of the peptide. In this field great progress has been made in optimizing the resins and the linkers in terms of stability for the chemistry to be performed 'on resin' as well as of cleavage under conditions compatible with the synthetic tasks (see Section 4.3.1).

Because of the repetitive multistep nature of peptide synthesis, it is essential to maintain a maximum selectivity in the protection scheme, i.e. of the transient (temporary) versus semipermanent (permanent) protection. The term 'orthogonal' defines the ideal compatible protection strategy where each class of protecting groups is selectively and quantitatively removed in any order in the presence of all other classes of protecting groups.[156,157] This concept of orthogonality of protecting groups is attractive from a theoretical point of view, but it likely represents an extreme situation for the practice of synthetic strategies. It does not allow for the possibilities offered by the predetermined order of deprotections that take advantage of sensitivity differences, e.g. towards acidolysis. In fact, the protecting groups generally applied in peptide synthesis are not strictly orthogonal, but their utility is based on more or less differentiated rates of identical or similar cleavage reactions. Nonetheless, from the large variety of protecting groups developed over decades of work in the field, various efficient protection schemes have been derived which are currently successfully applied to the synthesis of increasingly larger peptides and even proteins (see Vol. E 22b, Sections 5.2 and 5.3).

The most stringent demand on the protection schemes can be seen in the selectivity necessary in the protection of the α- versus the ω-amino groups. With respect to basicity or nucleophilic character the two amines are too similar to permit selective acylation. Similarly, the thiol group of cysteine residues must be masked since the nucleophilic character of thiols renders them strong competitors to amino groups for acylating agents. A large variety of protection strategies have evolved which ranges from the minimal obligatory protection which can be applied in synthesis in solution particularly at the level of segment condensations, to the conditionally obligatory protection (see Table 1), and finally to the maximal (global) protection strategy that is required whenever the stepwise acylation reactions are to be forced to completion in solution and on solid supports to avoid, as much as possible, all side reactions arising from accessible side-chain functionalities. In practice, the option for suitable strategies to be employed can vary between the very minimum of obligatory protections to a maximum of protection, in special cases including even that of selected amide bonds (Table 1).

for references see p 34

Table 1 Categories of Side-Chain Functional Groups According to the Need and Frequency of Protection

Protection	Functional Group	Amino Acid
obligatory	amino	Lys, Orn
	thiol	Cys
conditionally obligatory[a]	carboxy	Asp, Glu
	guanidino	Arg
optional and frequently used	phenolic hydroxy	Tyr
	primary aliphatic hydroxy	Ser
	secondary aliphatic hydroxy	Thr
	imidazole	His
	indole	Trp
	primary carboxamide	Asn, Gln
seldomly used	aliphatic sulfide	Met
	secondary carboxamide	usually at the Gly NH

[a] The side-chain protection of Asp and Glu is essential in peptide-chain elongation steps with these residues or when present as C-termini in segments that are used as carboxy components. The only known way to introduce unprotected Asp and Glu residues in stepwise elongation steps is through their *N*-carboxyanhydrides.[51] Protonation of the guanidino group is sufficient in most cases of peptide synthesis in solution.

Since the basic choice of the selective α/ω-amino protection dictates the overall protection scheme, mainly two strategies have evolved for solid-phase peptide synthesis: (1) N^{α}-*tert*-butoxycarbonyl (Boc) in combination with N^{ω}-benzyloxycarbonyl- and benzyl/cyclohexyl-derived protecting groups (Boc/Bzl strategy) that relies on the markedly different sensitivity of the α- and ω-protection to acidolysis,[158] and (2) N^{ω}-*tert*-butoxycarbonyl (Boc) and related *tert*-butyl alcohol derived protecting groups in combination with N^{α}-9-fluorenylmethoxycarbonyl (Fmoc/tBu strategy) that allows for orthogonal N^{α}-deprotection with bases and N^{ω}-deprotection by acidolysis.[159,160] For synthesis in solution, in addition to the Boc/Bzl strategy that is frequently applied,[34] efficient alternative strategies have been proposed and successfully adopted.[161] The most popular is based on orthogonal N^{α}-protection with the benzyloxycarbonyl (Z) group (hydrogenolytic removal) versus the side-chain protection based on tertiary alcohol derived groups.[162] Moreover, in a manner similar to the Boc/Bzl strategy, a differentiated acid-sensitivity between the semipermanent ω-Boc/tBu and the transient N^{α}-protection is exploited with the 2-nitrophenylsulfanyl (Nps)[163] or 2-(biphenyl-4-yl)prop-2-yloxycarbonyl (Bpoc) group,[164,165] both in the presence of sulfur-containing amino acids, or with the 4-methoxybenzyloxycarbonyl group [Z(OMe)][166,167] (for more details, see Sections 2.1.2.2, 4.1, and 4.3).

1.2.4.1 Maximum versus Minimum Protection

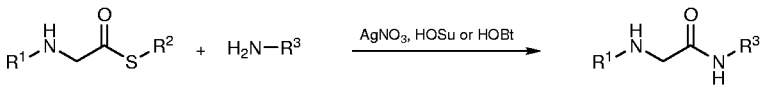
Both in synthesis in solution and on solid supports, nonquantitative yields in amino acid coupling steps and in N^{α}-deprotection reactions generate deletion peptides and truncated sequences that may be difficult to separate even with the most efficient high-resolution chromatographic methods. To overcome these well-recognized problems, increasingly more potent acylation methods have been developed to force coupling reactions to completion even for sterically hindered amino acid residues. As a consequence, the needs of global protection have been recognized.

In comparison to global protection, the minimal protection route offers, as the main advantage, increased solubility of intermediate segments in polar or even aqueous/organic media and, thus, enhanced opportunities for purification of intermediate segments even by ion-exchange chromatography. This aspect was crucial when high-resolution chromatographic methods were not available, but could still facilitate the synthesis of larger peptides and proteins (*vide infra*). The main restriction of this strategy lies in the severe limitation of the segment condensation procedures that must be essentially amine-specific. In this context

the azide method was most frequently used, since acid azides are fairly reactive towards aminolysis and are compatible with unprotected side chains including the free carboxy groups. Conversely, active esters might lead to possible transesterifications which in turn may form cyclic imides or branched peptides.[103,168] Thus, the minimal protection strategy with the acid azide procedure for assembly of the polypeptide chain has been the route of choice for various syntheses of larger peptides and proteins.[169] For a selective C-terminal carboxy activation also the conversion of inert 4-(methylsulfanyl)phenyl ester into an active 4-(methylsulfonyl)ester via oxidation with 3-chloroperoxybenzoic has been proposed. This procedure suffers from a lack of general applicability since it cannot be employed in the presence of oxidation-sensitive amino acid residues.[170]

The fast and nearly routine accessibility of medium-size segments by solid-phase synthesis has fostered more intensive research on the minimal protection strategy since it clearly represents a potentially very powerful synthetic approach for proteins. A significant advance in the acid azide procedure has been achieved by enhancing its relatively low acylating potency by in situ transformation of the acid azides into active esters with 7-aza-1,2,3-benzotriazol-1-ol (HOAt) or ethyl 1-hydroxy-1*H*-1,2,3-triazole-4-carboxylate (HOCt).[171] Moreover, a completely new concept was developed which is based on the selective conversion of C-terminal carbothioic *S*-acid groups[172,173] or more appropriately of almost inert carbothioate *S*-esters[174,175] into active esters in the presence of silver ions as illustrated in Scheme 11. Thereby, conditions were elaborated that facilitate the synthesis of glyco, phospho, and even cysteine peptides, despite the sensitivity of most thiol protecting groups to heavy metal ions.

Scheme 11 Silver Ion Mediated Activation of Carbothioic *S*-Acid or Carbothioate *S*-Ester Groups[172–175]



R^1 = N^{α},N^{ω}-protected peptide
R^2 = H, alkyl, $CH_2CH_2CONH_2$
R^3 = N^{ω}-protected peptide

1.2.4.2 Enzymatic versus Chemical Ligation without Protection of Functional Groups

The stereo- and regiospecificity of proteases and the possibility of reversing the proteolysis into peptide bond formation between essentially unprotected amino acids or peptides has attracted the attention of peptide chemists since the early days of this field of research.[176,177] Thus, a search was launched for enzymes suitable as biocatalysts for peptide bond formation in a manner independent of the nature of the individual amino acids, and for media in which proteolysis is optimally reversed into ligase activity. This led to a state of the art for this particular methodology that can be competitive with chemical methods for short peptides and in selected cases may even be superior in terms of economical industrial-scale preparations (for details see Section 4.2).[178,179] However, to overcome the intrinsic problem of proteolysis by the enzymes at undesired sites of the polypeptide chain, the minimum of protection is often required. Alternatively, to suppress proteolytic side effects which are crucial when assembling chemically synthesized peptide segments into proteins, specific chemical modifications and site-directed mutagenesis has been very successful in producing enzyme mutants that are effective ligases deficient of extensive protease and esterase activities.[180] An extension of this principle of enzymatic synthesis to peptide bond formation between segments containing noncoded amino acids, peptide mimetics, and peptide conjugates seems possible.[181] Similarly, the use of catalytic antibodies holds significant promise.[182]

In addition to the problem of partial proteolysis, the solubility of larger unprotected peptide segments may be crucial in the enzyme-assisted approaches, since solubilization with

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detergents or denaturants affects the catalytic activity of the enzymes. Therefore, more recently, attention has been paid to methods that allow for selective coupling of unprotected peptide segments via non-amide bonds (chemical ligation) or amide bonds (native chemical ligation) even in denaturing media. Moreover, selective conjugation of peptides to bioexpressed protein fragments is used when the linkage is not expected to affect the structural properties and thus, the biological functions of the conjugated domains (Vol. E 22b, Section 5.4). Similarly, conjugation procedures are used to graft peptides to carriers of different chemical nature and structure or for neo-glycosylation and -lipidation purposes.

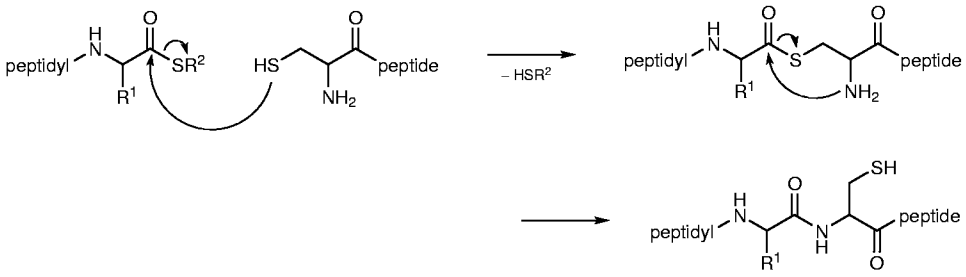
1.2.4.2.1 Chemical Ligation

Increasingly more detailed information has appeared from X-ray crystallographic and NMR structural analysis of proteins about the spatial array of domains and subdomains in their three-dimensional structures. This led to the finding that linkage of the domains often occurs via more or less flexible linker (hinge) peptides. These linker peptides offer the possibility of joining domains and subdomains by non-amides in a chemoselective manner without significantly affecting the structure and function of proteins. Chemical ligation is mainly based on (1) the selective thiol chemistry that can be exploited to couple two unprotected synthetic peptide segments via thioester[183] or sulfide bonds[184] (see Section 4.1.5), and (2) the imine chemistry to cross-link selectively peptide segments by hydrazone[185,186] or oxime formation[187,188] (see Vol. E 22b, Section 5.4). Therefore, it relies on the high and specific reactivity between two functional groups that are not present in the parent peptide segments.

1.2.4.2.2 Native Chemical Ligation

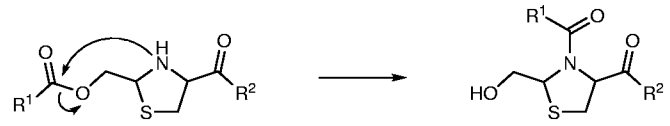
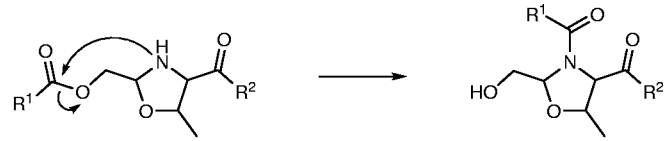
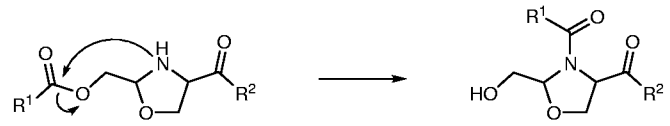
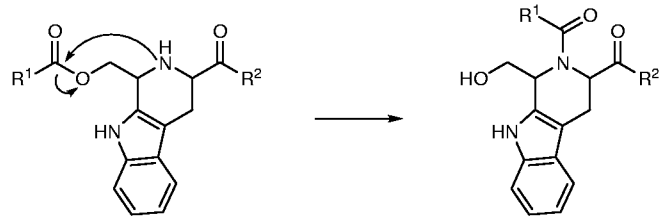
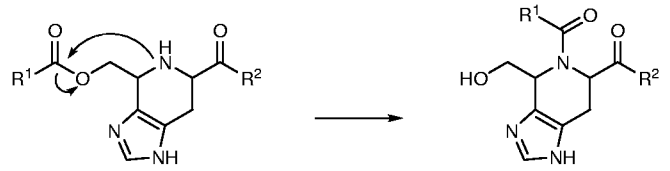
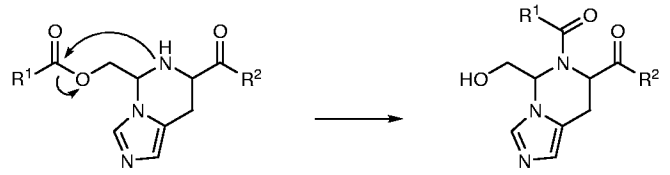
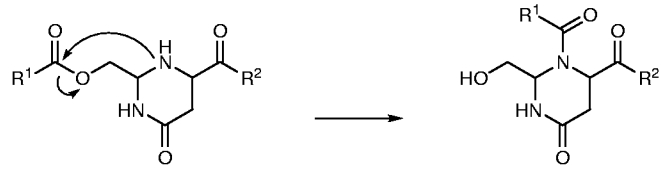
Previous studies on amide bond formation via conversion of the bimolecular coupling reaction (see Scheme 1) into an intramolecular reaction by grafting the carboxy and amino component on a template has clearly demonstrated the strong entropic effect, i.e. the high effective local concentration on the subsequent base-catalyzed intramolecular acyl transfer reaction.[189,190] This concept of a template-assisted coupling reaction of amino acids has been further developed to the 'prior thiol capture' procedure[191] from which the fast developing new methodology of native chemical ligation of unprotected synthetic or bioexpressed protein fragments into proteins has evolved.[33,41–43] It utilizes a capture step based on the selective reaction of a thioester of the carboxy component with an N-terminal cysteine of the amino component, followed by an entropy-driven intramolecular S→N acyl migration to form the amide bond,[39] as shown in Scheme 12 (for details see Section 4.1.5). Alternative thiol capture procedures have been proposed,[41,192–194] which all rely in the second step on the readily occurring S→N acyl migration. In addition, reaction of C-terminal thioesters with the imidazole group of N-terminal histidine residues and subsequent intramolecular nucleophilic attack of the amino group on the *N*-acylimidazolide leads to amide bond formation as well.[195]

Scheme 12 Native Chemical Ligation Based on S→N Acyl Shift[39]



For the initial step of chemical ligation even imine chemistry has been exploited with formation of thiazolidine[40,196] and oxazolidine intermediates,[197,198] which in turn undergo in efficient manner O→N acyl shifts to generate the *N*-acylthiazolidine or -oxazolidine compounds at the coupling site (Table 2). Since the original dipeptide sequence at the ligation site is not formed, but rather 'pseudoprolines' are introduced, this type of ligation has been termed orthogonal ligation,[41] although in principle some of these 'pseudoprolines' can be converted by acid treatment into the related amino acids.

Table 2 Heterocyclic Ligation Product Formed by Reacting Peptide Glycoaldehyde Esters with Various N-Terminal Residues of a Second Peptide and the Products Resulting Upon O→N Shift[41]

N-Terminus of C-Terminal Peptide	Ligation Intermediate and Ligation Product	Ref
Cys		[41]
Thr		[41]
Ser		[41]
Trp		[41]
His		[41]
His		[41]
Asn		[41]

for references see p 34

An alternative elegant native ligation approach has recently been proposed where the Staudinger reaction is exploited to couple the carboxy component as a thioester via trans-thioesterification to a suitable phosphine scaffold that reacts with azidoacyl peptides to produce the imino-λ^5-phosphine. This chemical ligation is followed by nucleophilic attack by the imino-λ^5-phosphine nitrogen on the thioester to form the amide bond[199,200] (Section 2.2.1.10).

Such S→N and O→N acyl shifts combined with aspartimide formation are used by nature in the intramolecular autocatalytic process of intein-mediated protein splicing, a newly discovered form of posttranslational processing (for recent reviews, see refs[201,202]). This process of protein splicing involves the excision of an intervening protein sequence (the intein) from the precursor protein and formation of a new peptide bond between the two flanking protein portions (the N- and C-exteins).[203,204] Due to the chemistry involved, it lends itself perfectly for the development of intein-mediated semisynthetic approaches[205–207] based on the production of C-terminal *S*-esters or N-terminal cysteine in bacterially expressed protein segments for use in chemical ligation with bioexpressed fragments or with synthetic segments that may be posttranslationally modified or may contain useful biophysical probes.[33,44,45]

1.2.4.2.3 Conjugation of Unprotected Peptide Segments

For the assembly of domains and subdomains into proteins it is generally necessary to retain the original peptidic or peptidomimetic structure for correct native folding. Conversely, for the attachment of synthetic peptides to natural proteins or synthetic carriers to produce efficient immunogens an amide bond is usually not essential. Similarly, to construct template-assembled synthetic proteins or to bundle synthetic peptides for the stabilization of tertiary structures, the nature and size of the linker may vary depending upon the required flexibility (see Vol. E 22d, Sections 13.1 and 14.1). However, maximum selectivity is required in the chemistry adopted for the conjugation of unprotected peptides. For this purpose mainly the thiol chemistry from the side-chain thiol group of cysteine residues is exploited. These residues can be introduced at defined positions of the protein by bioexpression of related mutants. Alternatively, thiol groups are artificially introduced on the carrier and for this purpose a large variety of bioconjugate techniques are available.[208] Conjugation is then accomplished via sulfide formation with synthetic peptides containing alkylating moieties such as alkyl halides or maleimido groups at defined sites. The latter reactive handle is stable to moderate acid treatment such as trifluoroacetic acid, thus allowing it to be incorporated during peptide synthesis.[209] It reacts with thiols in a very selective manner and at high rates,[209,210] thus facilitating specific conjugation of sensitive peptide derivatives.[211]

1.2.5 Cosynthetic versus Postsynthetic Modification of Peptides and Proteins

By analogy to the terms co- and posttranslational modifications of peptides and proteins to define these transformations in the in vivo biosynthesis, chemical manipulations at least theoretically can be carried out in a co- or postsynthetic manner. While nature exploits the sequence- and even conformation-dependent regioselectivity of enzymes to expand the molecular and functional diversity of peptides and proteins beyond the genetic code,[212] synthetic chemical reactions are insufficient for the required selectivity even with the most advanced conjugation techniques. Therefore, the tactics usually employed involves a co-synthetic approach, i.e. synthesis of polypeptide chains with amino acid derivatives or

building blocks that are already transformed in a native or artificial manner according to the known natural posttranslational modifications such as glycosylation, fattyacylation, phosphorylation, sulfation, etc., or constitute artificially modified derivatives as mimics of the natural ones.

Remarkable advances have been made in the synthesis of N- and O-glycosylated amino acids that, in addition to the *S*-fattyacyl-cysteine, are the most sensitive derivatives to the conditions of peptide synthesis. The fine tuning of both carbohydrate and peptide chemistry as well as the use of enzymatic methods are required for the synthesis of glycopeptides.[213–215] Similarly soft chemistry has to be applied in the case of S-acylated peptides, whereas fattyacylation at amino groups raises no additional difficulties.[216] The state of the art in the field of glyco- and lipopeptide synthesis is discussed in details in Vol. E 22b, Sections 6.3 and 6.4, respectively.

Through recent developments, an additional level of orthogonality has been achieved in protection schemes by the use of extremely acid-sensitive trityl-type or, more appropriately, of allyl-based protecting groups (see Sections 2.1.2 and 2.2.2). Thus, post-assembly tactics can be applied which anticipate selective deprotection on a resin or in solution of particular carboxy, hydroxy, or amino groups for native or artificial transformations. By this approach sensitive derivatives are exposed only to the acid treatment required for the final resin-cleavage/deprotection step. This post-assembly procedure is well suited to the phosphorylation of tyrosine, serine, or threonine as well as to the sulfation of tyrosine-peptides, although synthesis with the related protected or unprotected O-phosphorylated and O-sulfated amino acid residues is more or less routine.[217] Selective postsynthetic transformations are possible only with related enzymes, since the developed phosphorylating and sulfating agents suffer from lack of selectivity and from side reactions with other functionalities present in the peptides (for details see Vol. E 22b, Sections 6.5 and 6.6). An alternative approach to posttranslationally modified peptides is based on the incorporation of unique reactive handles at defined sequence positions of the peptides that are then allowed to react in a postsynthetic selective conjugation chemistry as reported, e.g. in refs[211,218–221].

The step from posttranslationally modified synthetic peptides to proteins, required for studying their central role in trafficking pathways and signaling networks, represents a challenge to future chemists. Only the newly developed peptide ligation procedures could possibly have a major impact.[33,222,223]

The question of the use of co- or postsynthetic transformation procedures is central to carry out the synthesis of multiple-cystine peptides with correct disulfide bridging. In vitro experiments on refolding of the ribonuclease A[224] were of seminal importance in establishing the principle of self-assembly of proteins. It is now recognized that efficient foldings in vitro and, particularly, in vivo are assisted by molecular chaperons and folding catalysts. These do not determine the final conformation of the polypeptide chain, but rather increase the efficiency of the folding processes by inhibiting off-pathway aggregation phenomena[225] or by rate-limiting catalysis of isomerization steps,[226] and most importantly by reshuffling of disulfides in the oxidative folding of cystine-containing proteins.[227] In vitro oxidative refolding of reduced or bioexpressed proteins is more or less successfully carried out by the addition of redox buffers.[228]

Folding and disulfide bond formation are thermodynamically coupled processes. Thus, formation of the native disulfide network in synthetic polypeptides critically depends on sequence information for correct folding. Bioactive cystine-peptides are generated in vivo by posttranslational processing of prefolded protein precursors. In the resulting metastable cystine-peptides the essential structural information required for correct oxidative refolding may have been lost. However, a larger number of disulfide bridges in relatively short polypeptide chains leads to compact globular structures that are reminiscent of protein domains or subdomains. In fact, in such cases according to general experience sufficient structural information is retained for a correct oxidative folding in acceptable yields under

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optimized experimental conditions (see Vol. E 22b, Section 6.1.5). Such postsynthetic tactics are highly attractive because of the simplicity of reactions compared to the synthetic strategies that have to be applied for regioselective disulfide bond formation. The advances made in the field of cysteine-thiol protection (see Section 2.6.6) allows for selective multiple intra- and intermolecular formation of disulfides with optimally tuned chemistries. These synthetic steps can be performed cosynthetically in successive steps during assembly of the peptide chain as was well documented in the classical synthesis of human insulin.[165] The preferred strategy, however, involves the post-assembly approach where cysteine residues are regioselectively paired via disulfide bonds on protected or deprotected peptides. In the latter case the cysteine residues must remain protected in a pairwise and selective manner. This requires sophisticated protection schemes that demand a high degree of orthogonality not only with regard to the remaining transient and semipermanent protections of other functionalities, but also among the S-protecting groups (see Vol. E 22b, Sections 6.1.1–6.1.4). It is for these reasons that at least in the chemical synthesis of multiple-cystine peptides the postsynthetic tactics of oxidative folding is the preferred route even if the sequence-dependent individual behavior of each peptide prevents general procedures to be applied. For each case optimization of experimental protocols is required.

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2 Protection of Functional Groups

L. Moroder

The protection of the amine group of the acylating component is essential for the condensation of α-amino acids via amide bonds in a defined sequential order. Depending upon the chemistry used for the formation of the amide bond, the carboxy group of the amino component may also require protection and so may other reactive side-chain functionalities. This principle of selective protection/deprotection was recognized by E. Fischer in 1903 as the prerequisite for successful peptide synthesis.[1] Since these early days of peptide chemistry, the development of protecting groups that satisfy as far as possible the demanding requirements of multistep synthesis has been the subject of intensive research. These requirements include facile and selective incorporation of the protecting group into the target functionalities, full resistance of the protecting group to the required transformation of the substrate, e.g. peptide bond formation, and mild and selective cleavage of the protecting group without affecting other functionalities. In peptide chemistry, protecting groups must meet additional criteria; for instance, the stereochemical integrity of the constituent amino acid residues must be completely maintained in the protection and deprotection steps as well as during the reactions involved in peptide bond formation. Moreover, protection of α-amino and/or α-carboxy groups, and to some extent even of side-chain functionalities, can affect amide bond formation by altering the steric environment and this affects the accessibility of the respective proximal carboxy and amino groups; hence, the steric bulk and thus the type of protecting group may also critically affect coupling yields.

Peptides are generally synthesized in the C→N direction; as a consequence, α-amino protection must be transient (temporary protection), i.e. deprotection has to occur selectively in the presence of all other side-chain and the C-terminal carboxy protecting groups (semipermanent protection) that are generally removed in the final synthetic step. With the increasing complexity of synthetic target molecules, the demands on the degree of selectivity of the transient versus semipermanent protection have increased greatly. The result is that this can only be accomplished with protection schemes based on completely different classes of derivatives that not only permit their selective preparation, but more importantly permit their selective and quantitative removal in any specific order and in the presence of all other classes of protecting groups. Such ideally compatible protection schemes have been termed orthogonal.[2] In reality, the combinations of protecting groups generally applied in peptide synthesis are not strictly orthogonal, but are rather based on strongly differentiated rates of identical or similar reactions. Nevertheless, great progress has been made in terms of optimized protection schemes, which presently allow the syntheses of complex peptides and even of small- to medium-sized proteins (see Section 4).

The intent of the present treatise is not to be an exhaustive compilation of the chemistry of all the protecting groups proposed and used in peptide chemistry. Rather, this section focuses mainly on the present state of the art in the synthesis of peptides and peptidomimetics. Correspondingly, only the most useful and versatile protecting groups for the different functionalities encountered in the synthesis of natural and nonnatural peptide-based compounds are discussed comparatively in terms of advantages and possible drawbacks for their basic applications. For more encyclopedic reviews of the available protecting groups the reader is directed to other treatises.[3–7]

for references see p 40

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2.1 Amino Group

J. Podlech

More than 250 amino protecting groups have been presented and used over the years, but only a relatively small number fulfill the majority of the requirements for an ideal protecting group in the synthesis of a complex peptide, particularly in terms of optimal compatibility with the protection of other functionalities. Correspondingly, only a few protection schemes have emerged as largely orthogonal and are routinely used for syntheses in the solution and solid-phase mode. For synthesis in solution, N^{α}-benzyloxycarbonyl (Z), N^{α}-2-nitrophenylsulfanyl (Nps), or N^{α}-2-(biphenyl-4-yl)prop-2-yloxycarbonyl (Bpoc) derivatives for temporary protection are suitably combined with the semipermanent *tert*-alcohol-based ω-protection, or N^{α}-*tert*-butoxycarbonyl (Boc) with benzyl (Bzl) derived side-chain protection. More recently, but with less impact than in SPPS, N^{α}-9-fluorenylmethoxycarbonyl (Fmoc) has been combined as temporary protection with *tert*-alkyl semipermanent protection. For peptide syntheses on resin the two main and routinely applied protection strategies are based on the Boc/Bzl and Fmoc/tBu chemistry. More demanding synthetic targets have been the object of intensive efforts such as the posttranslationally modified glyco-, lipo-, phospho-, and sulfated peptides or peptides bearing at preselected mostly side-chain sequence positions particular spectroscopic and biochemical probes or labels. As a consequence, a third generation of orthogonal protecting groups for syntheses in solution and on resin have had to be developed addressing particularly side-chain amino groups. Using the *N*-allyloxycarbonyl (Aloc) derivatives, which are selectively cleaved by palladium catalysis, the extremely acid-labile *N*-4-methyltrityl (Mtt) and 4-methoxytrityl (Mmt) derivatives, and also the promising 1-(4,4-dimethyl-2,6-dioxocyclohexylidene)ethyl (Dde) and 1-(4,4-dimethyl-2,6-dioxocyclohexylidene)-3-methylbutyl (known as ivDde) groups the synthetic challenges have been successfully mastered.

Since some of the amino protecting groups that were commonly used have now been replaced by more appropriate and efficient ones, the amino protecting groups which are presented in the following sections are only those that, based on a large amount of documented experience gained over the years, have found continuous application. For more detailed information on other amino protecting groups, the reader is referred to several comprehensive treatises.[1–14] In this context it is worthwhile to note that some of the protecting groups developed specifically for peptide chemistry have found wide application in other fields of organic synthesis and vice versa.

2.1.1 α-Amino Group

J. Podlech, M. Gurrath, G. Müller, and E. Lohof

All amino protecting groups are, fundamentally, suitable for masking the α-amino group of amino acids and related derivatives in peptide synthesis, unless selectivity is required versus the side-chain protection. In the latter case, transient α-amino protection has to be chosen which will permit selective and quantitative N^{α}-deprotection within the repetitive steps of peptide chain elongation. In reality, the choice of α-amino protecting groups is limited not only in terms of compatibility with other types of protection, but also by their possible sterical and electronic effects on the peptide coupling steps. In fact, steric bulkiness can drastically affect yields, but more importantly N^{α}-protecting groups can induce or reduce racemization via the oxazol-5(4*H*)-one or α-proton abstraction mechanisms (see Sections 1.2.1.2 and 7.4). In this context, similarly to what is observed in segment condensations, acyl

for references see p 149

derivatives such as N^{α}-benzoyl, N^{α}-formyl, or N^{α}-trifluoroacetyl favor racemization unless peptide bond formation is performed under enzyme catalysis.

In the following sections the protecting groups are classified according to their chemical structure as carbamoyl (urethane), acyl, sulfanyl, sulfonyl, alkyl, and silyl derivatives and not according to their cleavage conditions. Protecting groups that are photolytically or enzymatically cleaved are discussed in more detail in Sections 2.4 and 2.5, respectively.

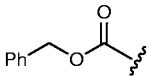
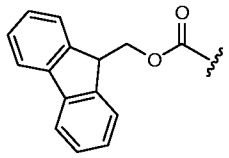
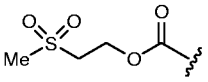
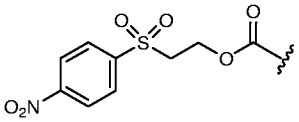
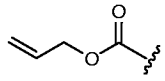
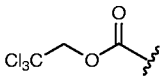
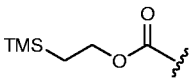
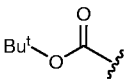
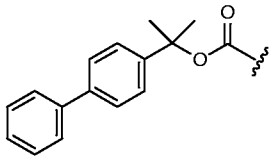
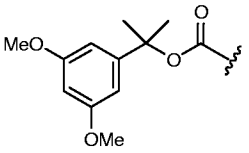
2.1.1.1 Urethane Protecting Groups

Due to the simple introduction of urethane protecting groups using either chloroformates, mixed carbonate esters, pyrocarbonates, or similar derivatives of carbonic acid, the high stability of urethane-protected amino acids together with their low tendency to racemize,[15–17] as well as their relative ease of cleavage, this type of amino protection has become the most widely applied not only in peptide chemistry, but also in organic synthesis. Most of the synthetic strategies for peptide synthesis in solution and in the solid-phase mode rely on the use of this urethane-type protection in different combinations (see Section 4). The most prominent representatives are the benzyloxycarbonyl (Z), *tert*-butoxycarbonyl (Boc), and 9-fluorenylmethoxycarbonyl (Fmoc) groups that by themselves represent three orthogonal dimensions for temporary and semipermanent amine protection strategies.[1–3,12,18,19]

The idea of using the carbamate of benzylic alcohol for the temporary protection of α-amino groups originates from the observation of Rosenmund and Zetzsche that O- and N-coupled benzyl groups are cleaved by catalytic hydrogenation.[20] This led to the pioneering discovery of the benzyloxycarbonyl group by Bergmann and Zervas which can be regarded as one of the dominant milestones in peptide chemistry.[21] Based on this first really useful urethane-type protecting group, a large series of related derivatives has been proposed which can roughly be divided into three classes according to their preferred cleavage conditions as shown in Table 1: (1) catalytic hydrogenolysis (Z); (2) acidolysis (Boc, Bpoc, Ddz); and (3) β-elimination (Fmoc, Msc, Nsc, Troc, Teoc). Due to the particular chemistry required for the cleavage of the Aloc group it represents a very useful special type of orthogonal protecting group.

Whilst the *N*-Z derivatives are cleaved by catalytic hydrogenation due to the low energy of the benzyl–oxygen bond, the easy acidolysis of urethanes derived from tertiary alcohols results from the high stability of the originated tertiary carbocations. A β-elimination from substituted ethanol-derived urethanes is either achieved by base-induced deprotonation at the β-carbon (Fmoc, Msc, or Nsc) or through formation of an anion by assisted cleavage (Troc, Teoc). Details of the cleavage mechanisms are given in the respective section.

Table 1 Urethane-Type Protecting Groups Most Frequently Used for N^{α}-Protection of Amino Acids

Protecting Group	Structure	Preferred Cleavage Conditions	Ref	First Cited in Ref
benzyloxycarbonyl (Z)		hydrogenation (H_2/Pd) strong acids: e.g. HBr/AcOH HF MsOH or TfOH	[21] [22] [23] [24]	[21]
9-fluorenylmethoxy-carbonyl (Fmoc)		basic, e.g. piperidine	[25]	[26]
2-(methylsulfonyl)-ethoxycarbonyl (Msc)		basic, e.g. 1 M NaOH	[27]	[27]
2-[(4-nitrophenyl)-sulfonyl]ethoxy-carbonyl (Nsc)		basic, e.g. 20% piperidine/DMF	[28]	[29]
allyloxycarbonyl (Aloc)		[Pd(0)] catalyst	[30]	[31]
2,2,2-trichloroethoxy-carbonyl (Troc)		Zn	[32]	[33]
2-(trimethylsilyl)-ethoxycarbonyl (Teoc)		F^-, e.g. TBAF strong acids: TFA	[34] [35]	[35]
tert-butoxycarbonyl (Boc)		H^+, e.g. HCl/dioxane TFA TosOH $BF_3 \cdot OEt_2$	 [36] [37] [38] [39,40]	[41,42]
2-(biphenyl-4-yl)prop-2-yloxycarbonyl (Bpoc)		H^+, e.g. 0.5% TFA/CH_2Cl_2	[43,44]	[45,46]
2-(3,5-dimethoxy-phenyl)prop-2-yloxy-carbonyl (Ddz)		H^+, e.g. dil TFA	[47]	[47]

2.1.1.1.1 Urethane Protecting Groups Derived from Primary Alcohols

The use of ethoxycarbonyl-protected amino acids as well as methoxycarbonyl derivatives for peptide synthesis was reported in 1903 by Fischer, although the protecting groups in related peptide derivatives could not be cleaved without affecting the peptide bonds,[48] since urethanes derived from aliphatic primary alcohols are about as stable as the amide bond. Thus, this type of carbamate can only be used for reversible protection of amino groups, at least in

for references see p 149

peptide chemistry, if cleavage occurs under conditions that are milder than acid or alkaline peptide and protein hydrolysis. The breakthrough was reached with the benzyloxycarbonyl group, which is readily removed by hydrogenolysis or by relatively strong acids.[21] Subsequent further modulation of the acid lability and/or photolability (see Section 2.4) has been achieved with various substituted benzyloxycarbonyl groups such as the 4-methoxybenzyloxycarbonyl [Z(OMe)] group which is significantly more acid labile.[49] The 4-acetoxybenzyloxycarbonyl group [Z(OAc)] was developed as an alkaline-labile protecting group;[50] however, it has not found due application as it has been replaced by other protecting groups which are cleaved under mild basic conditions, such as the 9-fluorenylmethoxycarbonyl (Fmoc)[26] or the sulfonylethoxycarbonyl groups, e.g. 2-(methylsulfonyl)ethoxycarbonyl (Msc)[27] and 2-[(4-nitrophenyl)sulfonyl]ethoxycarbonyl (Nsc).[29] Other urethane-type protecting groups that are cleaved under essentially neutral conditions are the allyloxycarbonyl (Aloc),[31] 2,2,2-trichloroethoxycarbonyl (Troc),[33] and 2-(trimethylsilyl)-ethoxycarbonyl (Teoc) groups.[35]

2.1.1.1.1.1 Benzyloxycarbonyl Group

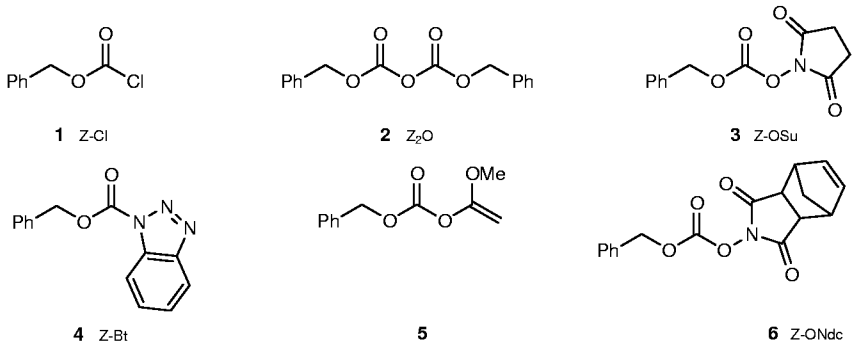
M. Gurrath, G. Müller, and J. Podlech

The introduction of the benzyloxycarbonyl group (Z) by Bergmann and Zervas[21] for amine protection of α-amino acids gave access to entirely new synthetic strategies in modern peptide chemistry. Over the decades following the pioneering study of Bergmann and Zervas, the Z group has emerged as one of the most widely used protecting groups in peptide synthesis in solution for temporary amine protection due to (1) the easy preparation of the related N^{α}-benzyloxycarbonyl amino acid derivatives; (2) the favorable stability of Z-protected amino acids and peptides; (3) the versatile cleavage conditions by hydrogenolysis[21] in intermediate chain-elongation steps or by strong acids[22,51] in the final deprotection; and finally (4) due to the suppression of racemization during peptide bond formation.[15,17,52]

2.1.1.1.1.1.1 Synthesis of N^{α}-Benzyloxycarbonyl Amino Acids

Over the past decades the originally proposed benzyl chloroformate (Z-Cl, **1**)[21] has been used as the main reagent for the preparation of N^{α}- or N^{ω}-benzyloxycarbonyl amino acids. Due to decomposition of benzyl chloroformate (**1**) on storage, the alternative reagents reported in Scheme 1 were proposed and have been used depending upon specific requirements.

Scheme 1 Reagents for the Introduction of the Z Group



2.1.1.1.1.1.1.1 Benzyl Chloroformate

Benzyl chloroformate (Z-Cl, **1**) is used mainly under Schotten–Baumann conditions in aqueous/organic solvent mixtures using sodium hydroxide, sodium hydrogen carbonate, or magnesium oxide as the base to acylate amino acids on a laboratory as well as on an industrial scale. The N^{α}-Z-protected amino acid derivatives are generally obtained as crystalline compounds or as oils that are easily converted into crystalline salts with bases such as cyclohexylamine (CHA) or dicyclohexylamine (DCHA). Alternatively, amino acids or peptides are acylated with benzyl chloroformate (**1**) in organic solvents using suitable organic bases for solubilization, whereas amino acid or peptide esters or amides are acylated with benzyl chloroformate (**1**) in inert organic solvents generally using TEA as the base. In the latter case, the addition of 1 equivalent of zinc to promote the acylation reaction is recommended.[53] The acylating chloroformate is readily accessible by reaction of benzyl alcohol with neat phosgene or with phosgene in inert solvents, e.g. toluene.[21,54,55]

A procedure for the preparation of Z-Lys-OH is given in Section 2.1.2.2.1.3.

Benzyl Chloroformate (Z-Cl, 1):[55]

CAUTION: *Phosgene is a colorless, poisonous gas or volatile liquid. This reagent should be handled with the utmost care.*

$COCl_2$ (1.26 L, 17.6 mol) was condensed at −20 °C and redistilled Bzl-OH (1.66 L, 16.0 mol) was added dropwise over 4 h at −10 °C. The reaction was stirred for several hours and was then allowed to reach rt. Compressed air, dried by passage through concd H_2SO_4 and then over flaked NaOH, was passed through the gas lead, with stirring for 24 h. The stirrer and the gas lead were removed and the flask evacuated for 5 min. A clear colorless liquid was obtained; yield: 2.6 kg (95%).

Z-Ala-OH; Typical Procedure:[56]

To an ice-cold (0–3 °C) soln of H-Ala-OH (267 g, 3 mol) in 2 M NaOH (1.5 mL, 3 mmol), Z-Cl (**1**; 445 mL, 3 mol) was added in portions together with 2 M NaOH (1.65 mL) within 3 h. After stirring for an additional 1 h, the alkaline soln was extracted with Et_2O, acidified to Congo red with 1 M H_2SO_4, and extracted with EtOAc. The organic layers were combined, washed with 1 M H_2SO_4 and H_2O, and dried (Na_2SO_4). The soln was concentrated and the residue recrystallized (Et_2O/petroleum ether); yield: 91.5%; mp 85–86 °C (lit.[1] mp 85–86 °C); $[\alpha]_D^{20}$ −14.9 (*c* 1, AcOH) {lit.[1] $[\alpha]_D^{22}$ −14.3 (*c* 1, AcOH)}.

Z-Arg-OH; Typical Procedure:[56]

To a soln of H-Arg-OH (784 g, 4.5 mol) in H_2O (6 L) containing $NaHCO_3$ (1.04 kg, 12.4 mol) was added Z-Cl (**1**; 725 mL, 4.95 mol) over 3 h at 0–3 °C. After stirring for an additional 1 h, the soln was adjusted with concd NH_3 to pH 8.5 and stirred at rt for 2 h. The precipitate was filtered off, washed with H_2O and recrystallized (boiling H_2O containing a few drops of NH_3); yield: 1.01 kg (73%), mp 174–177 °C (lit.[1] mp 175 °C); $[\alpha]_D^{20}$ −10.4 (*c* 2, 1 M HCl) {lit.[1] $[\alpha]_D^{20}$ −9.2 (*c* 5.5, 0.2 M HCl)}.

Z-Asn-OH; Typical Procedure:[56]

H-Asn-OH (1.32 kg, 10 mol) was dissolved at 60 °C in H_2O (7 L) containing $NaHCO_3$ (840 g, 10 mol). Upon addition of the residual $NaHCO_3$ (900 g, 11 mol) in H_2O the soln was cooled to rt, and Z-Cl (**1**; 1.55 L, 10.5 mol) was added over 3 h in portions. After stirring for an additional 1 h at pH 7–8, the mixture was extracted with Et_2O, and the aqueous phase was acidified with concd HCl to pH 2. The precipitate was filtered off, washed with cold H_2O and recrystallized (boiling MeOH); yield: 2.15 kg (81%); mp 165–166 °C (lit.[1] mp 164–165 °C); $[\alpha]_D^{20}$ −6.5 (*c* 1, 1 M $NaHCO_3$) {lit.[1] $[\alpha]_D^{22}$ −6.5 (*c* 1, 1 M $NaHCO_3$)}.

Z-Gln-Asn-OH; Typical Procedure:[57]

To a soln of H-Gln-Asn-OH (1.5 g, 5.76 mmol) in H_2O (30 mL) containing $NaHCO_3$ (1.0 g, 11.9 mmol), was added Z-Cl (**1**; 1.0 mL, 7.01 mmol) under vigorous stirring until a substantially clear soln was formed. More Z-Cl (**1**; 0.1 mL) and $NaHCO_3$ (0.1 g) were then added, the mixture shaken for 0.5 h, washed with Et_2O (3 ×), and acidified with concd HCl. The product separated as a gel which was collected, washed with H_2O, dried, and taken up in boiling H_2O (15 mL). The gel, which separated on cooling, was again collected, dissolved in the minimum amount of boiling H_2O and the soln was diluted with two volumes of EtOH. On cooling, the product crystallized; yield: 1.5 g (66%); mp 206 °C.

for references see p 149

2.1.1.1.1.1.1.2 Dibenzyl Dicarbonate

Due to the low stability of benzyl chloroformate (**1**) on storage with decomposition to benzyl chloride, numerous alternative Z donors have been proposed. Special emphasis was initially laid on the development of mixed carbonates with various phenols,[58,59] none of which proved to be an efficient substitute for benzyl chloroformate (**1**) because of their laborious and costly preparation and decreased acylating potential accompanied by the formation of side products. As a more efficient alternative Z-donor dibenzyl dicarbonate (Z_2O, **2**) was developed[60,61] in analogy to di-*tert*-butyl dicarbonate (Boc_2O).[62] This reagent is more stable on storage than benzyl chloroformate as it decomposes at 50 °C, and it leads to better yields of Z-protected amino acid derivatives than by using benzyl chloroformate (**1**). But formation of dipeptides (or even tripeptides) as contaminants, which were observed with benzyl chloroformate (**1**) as a result of mixed anhydride formation,[63] is probably not prevented with dibenzyl dicarbonate (**2**) if the results obtained with di-*tert*-butyl dicarbonate (see Section 2.1.1.1.3.1.1.6) are taken into account. The dicarbonate is synthesized by reacting benzyl chloroformate with the sodium salt of benzyl carbonate, the latter being prepared from benzyl alcohol with sodium hydride and carbon dioxide gas.[61] The Z-protected amino acid derivatives are obtained by reaction of amino acids with dibenzyl dicarbonate (**2**) under Schotten–Baumann conditions. Results from comparative studies indicate that product yields and purity of the dibenzyl dicarbonate (**2**) derived Z derivatives are comparable only with those obtained by using freshly prepared benzyl chloroformate of the best quality.[60,61]

Dibenzyl Dicarbonate (Z_2O, 2):[61]
Bzl-OH (21.6 g, 200 mmol) was treated with NaH (200 mmol) in refluxing THF (200 mL) for 2 h. The soln was cooled to rt and CO_2 was bubbled into the mixture under efficient stirring and external cooling. After 1 h, Z-Cl (**1**; 28.5 mL, 200 mmol) was added slowly. After 3 h stirring at rt, the precipitate was centrifuged and the supernatant concentrated. The resulting oil crystallized in the cold. The crystalline mass was triturated with cold hexane, filtered, and dried; yield: 45.8 g (79%); mp 28 °C.

Synthesis of a N^{α}-Z-Protected Amino Acid with Dibenzyl Dicarbonate (2); General Procedure:[60]
To a soln of amino acid (50 mmol) in 1 M NaOH (50 mL) and dioxane (50 mL) was added dropwise with stirring Z_2O (**2**; 50 mmol) in dioxane (50 mL). After 1 h at rt, the bulk of the dioxane was removed, and the resulting aqueous soln was acidified with 0.5 M H_2SO_4 to pH 2, and then extracted with EtOAc or MTBE (3 ×). The combined extracts were washed sulfate-free, dried (Na_2SO_4), and the solvent removed. The products were isolated by crystallization; yield: 83–95%.

2.1.1.1.1.1.1.3 Mixed Carbonates

Apart from benzyl chloroformate (**1**) and dibenzyl dicarbonate (**2**), attention has been paid to the mixed carbonates. Among the latter reagents benzyl succinimido carbonate (Z-OSu, **3**)[64,65] at present represents the most valuable alternative to benzyl chloroformate. It is commercially available and a stable reagent that results in satisfactory acylation yields, even of side-chain unprotected hydroxy-amino acids or esters.[66] It is particularly useful for the acylation of lipophilic amino acids using tetrabutylammonium hydroxide as a base.[67,68] However, attention has to be paid to the possible Lossen rearrangement of benzyl succinimido carbonate (Z-OSu, **3**) with formation of β-alanine derivatives as side products. This rearrangement can be measurably suppressed by lowering the pH or the temperature of the reaction medium.[69] The mixed carbonate benzyl succinimido carbonate (**3**) is synthesized by reaction of benzyl chloroformate with the DCHA salt of *N*-hydroxysuccinimide in organic solvents.

As an alternative to the mixed carbonates, benzyl 1,2,3-benzotriazole-1-carboxylate (Z-Bt, **4**) was introduced into peptide chemistry[60] which was originally prepared as an anti-inflammatory agent.[70,71] This acylating agent is readily accessible by reaction of 1,2,3-benzotriazole with benzyl chloroformate and it is stable on storage. The reactions of amino acid derivatives with benzyl succinimido carbonate (**3**) or benzyl 1,2,3-benzotriazole-1-carbox-

ylate (**4**) proceed under Schotten–Baumann conditions in aqueous organic mixtures. Due to the decreased acylation power of these acylating agents when compared to benzyl chloroformate or dibenzyl dicarbonate, longer reaction times are generally required, but the formation of dipeptides (or tripeptides) as contaminants is not observed. Moreover, these reagents are well suited for amine acylation of peptide intermediates in aqueous organic mixtures or inert organic solvents if an exchange of the protecting groups is required.

Z-Protected amino acid derivatives are further obtained by using benzyl 1-methoxyvinyl carbonate (**5**).[72–74] Due to the high reactivity of this acylating agent under mild conditions, together with the high yields of protected amino acid derivatives and the formation of methyl acetate as a byproduct, this amino-protecting reagent appears to be a versatile alternative to the reagents described above. The acylating agent is prepared by employing the reaction of bis[(methoxycarbonyl)methyl]mercury with phosgene in the presence of pyridine followed by in situ treatment with benzyl alcohol. The reaction proceeds via the enolate equivalent of the methyl acetate and yields the (1-methoxyvinyl)oxycarbonyl moiety transferring pyridinium complex. The resulting 1-methoxyvinyl carbonate is stable in common organic solvents and, upon distillation, can be stored at room temperature for weeks or in the refrigerator for several months under nitrogen or argon. The preparation of N^{α}-protected amino acids is achieved by reacting the amino acid in water/dioxane solution with equivalent amounts of the acylating agent for just a short period of time to give almost quantitatively N^{α}-Z-protected amino acid derivatives.

As an analogue of benzyl succinimido carbonate (**3**), benzyl 5-norbornene-2,3-dicarboximido carbonate (Z-ONdc, **6**) has been proposed[75–77] which is synthesized from the 5-norbornene-2,3-dicarboximido chloroformate. The corresponding carbonate is readily obtained in an elegant one-pot procedure without isolation of the chloroformate starting with Ndc-OH and phosgene. The product is reacted further with benzyl alcohol to produce the desired carbonate as a crystalline solid that can be stored for unlimited periods. This procedure is not restricted to the preparation of the mixed benzyl carbonate, but is also suitable for the preparation of the 5-norbornene-2,3-dicarboximido-activated carbonates of *tert*-butyl alcohol, 9-fluorenylmethanol, adamantan-1-ol, and 2-biphenyl-4-ylpropan-2-ol as Boc, Fmoc, Adoc, and Bpoc donors, respectively. For N^{α}-protection, the corresponding amino acid is reacted with the mixed carbonate in dioxane/water without formation of dipeptides and the *N*-hydroxy-5-norbornene-2,3-dicarboximide produced as a byproduct is easily removed due to its high solubility in water.[75–77]

Benzyl Succinimido Carbonate (Z-OSu, 3):[66]
To a stirred soln of Z-Cl (**1**; 54.4 mmol) in $CHCl_3$ (100 mL) was added HOSu · DCHA salt (obtained as a precipitate in acetone by mixing equimolar quantities of HOSu and DCHA) (54.4 mmol) in portions. After 12 h, the precipitate was filtered off and washed with $CHCl_3$. The filtrate and washings were combined and washed with one-third of the volume each of 10% citric acid, 10% $NaHCO_3$, and H_2O, and dried. The solvent was removed and the residue was recrystallized ($CHCl_3/Et_2O$); yield: 90%; mp 80–81 °C.

Benzyl 1,2,3-Benzotriazole-1-carboxylate (Z-Bt, 4):[60]
To a chilled soln of 1,2,3-benzotriazole (357 g, 3 mol) and freshly prepared Z-Cl (**1**; 512 g, 3 mol) in THF (4.5 L), was added TEA (417 mL, 3 mol) dropwise under stirring. The precipitate was filtered, and the filtrate concentrated to an oily residue. The product was crystallized twice (EtOAc); yield: 682 g (90%); mp 108–110 °C.

Benzyl 1-Methoxyvinyl Carbonate (5):[74]

CAUTION: *Phosgene is a colorless, poisonous gas or volatile liquid. This reagent should be handled with the utmost care.*

Through a stirred suspension of HgO (21.7 g, 100 mmol) and $Hg(OAc)_2$ (31.8 g, 100 mmol) in dry MeOH (217 mL) at rt was gently bubbled ketene (generated by the thermal decomposition of acetone). The reddish suspension turned into a white suspension. After stirring for 12 h, a gray clear soln was obtained

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which was concentrated to give a white solid. The solid was washed with Et_2O, dried, and recrystallized (EtOAc); yield of bis[(methoxycarbonyl)methyl]mercury: 18.7 g (54%); mp 98–99 °C.
To a well-stirred soln of 3.4 M $COCl_2$ in dry CH_2Cl_2 (33 mL, 115 mmol) was added dropwise a soln of pyridine (10.9 g, 138 mmol) in CH_2Cl_2 (100 mL) at −20 °C over 5 min under argon. White-blue crystals precipitated with the addition of pyridine, and stirring was continued for 15 min. Then a soln of bis-[(methoxycarbonyl)methyl]mercury (50 g, 140 mmol) in dry CH_2Cl_2 (115 mL) was added dropwise to the cooled stirred mixture, which then turned yellow. After 1 h, to the resulting orange mixture was added dropwise a soln of Bzl-OH (345 mmol) in dry CH_2Cl_2 (58 mL) over 5 min under argon. After 15 min at −20 °C and 30 min at 0 °C, pentane (300 mL) was added and the mixture was stirred for another 30 min and then at 25 °C for 30 min, while a red syrup was formed. Most of the excess $COCl_2$ was removed by a stream of argon and the soln was separated from the syrup. The syrup was extracted with Et_2O (2 × 50 mL) and the combined organic layers were washed with H_2O (3 × 100 mL) and brine (100 mL), dried ($MgSO_4$), and concentrated. By distillation (bp 105–107 °C/0.7 Torr) the desired carbonate was obtained; yield: 73%.

Benzyl 5-Norbornene-2,3-dicarboximido Carbonate (Z-ONdc, 6):[77]

CAUTION: *Phosgene is a colorless, poisonous gas or volatile liquid. This reagent should be handled with the utmost care.*

A soln of *N*-hydroxy-5-norbornene-2,3-dicarboximide (17.9 g, 100 mmol) and *N,N*-dimethylaniline (12 g, 95 mmol) in THF/benzene (1:3, 100 mL) was added dropwise to a soln of $COCl_2$ (9.9 g, 100 mmol) in benzene (50 mL) at 0–5 °C. The amine hydrochloride was filtered off, and the soln was concentrated. The residue was dissolved in CH_2Cl_2 (50 mL), and a trace of bis(5-norbornene-2,3-dicarboximido) carbonate was removed by filtration. The solvent was removed and the residue triturated with Et_2O (50 mL) to give 5-norbornene-2,3-dicarboximido chloroformate; yield: 20.5 g (85%); mp 98–100 °C.
To a soln of 5-norbornene-2,3-dicarboximido chloroformate (24.1 g, 100 mmol) in an inert solvent (toluene, benzene, THF, or a halogenated hydrocarbon; 150 mL) a soln of Bzl-OH (10.8 g, 100 mmol) and pyridine (8 mL, 100 mmol) in an inert solvent (40 mL) was added at 10–15 °C. The mixture was stirred at rt for 1 h and at 35 °C for 3 h. The pyridine hydrochloride was filtered off and the soln concentrated. Further purification was carried out either by recrystallization of the solid from 90% aq MeOH or by extraction of the impurities from a CH_2Cl_2 soln with 5% aq $NaHCO_3$ and then H_2O at 5 °C, followed by drying (Na_2SO_4) and evaporation to give the pure carbonate; no yield given; mp 121–123 °C.

Z-Ser-OH; Typical Procedure Using Z-OSu:[66]
To a stirred soln of serine (525 mg, 5 mmol) and $NaHCO_3$ (420 mg, 5 mmol) in a mixture of H_2O (7 mL) and acetone (7 mL) was added Z-OSu (1.24 g, 5 mmol). The mixture was stirred overnight, acetone was removed under reduced pressure, and the soln was washed with CH_2Cl_2 (2 × 3 mL). The aqueous layer was acidified to pH 2.5 with concd HCl and extracted with EtOAc (3 × 5 mL). The combined organic layers were washed with H_2O, dried, concentrated, and recrystallized; yield: 78%; mp 120 °C; $[\alpha]_D^{26}$ +5.7 (*c* 6, AcOH)

2.1.1.1.1.1.4 Via Isocyanato Intermediates

The synthesis of α-isocyanato carboxylic acid esters involves reaction of the amino acid esters with either gaseous phosgene[78,79] or a solution of phosgene in toluene in the presence of pyridine.[80] Alternatively, a phosgene-free synthesis of alkyl or aryl isocyanates by a DMAP-catalyzed reaction of the corresponding amines with di-*tert*-butyl dicarbonate (Boc_2O) (rt, 10 min)[81] has been used to convert amino acid or peptide derivatives in a racemization-free manner into their corresponding isocyanates that react in situ with alcohols such as benzyl alcohol to produce the related carbamates.[82] This approach may represent a promising new method for the synthesis of N^{α}-Z-protected amino acid or peptide derivatives in selected cases.

2.1.1.1.1.1.2 Peptide Synthesis with N^{α}-Benzyloxycarbonyl Amino Acid Derivatives

Due to the high stability of the benzyloxycarbonyl group under most conditions of peptide synthesis and its mild and selective hydrogenolytic cleavage, it is generally used for intermediate α-amino protection in chain-elongation steps in combination with the more acid-

labile *tert*-alkyl-derived side-chain protection (Z/tBu strategy) for the synthesis of peptides in solution (see Section 4.1). Although hydrogenolysis can be used for the deprotection of N^{α}-Z-protected peptides on resin by the use of catalytic transfer hydrogenation in the presence of palladium(II) acetate, the reduced palladium deposits on the resin, thus limiting its use for final deprotection or resin-cleavage steps.[83] Therefore, Z protection has only been applied in SPPS for side-chain amino groups as it is cleaved by strong acids, e.g. HBr/AcOH[84] or HF[51] (see Section 2.1.2.2.3.1.1).

Its use in solution synthesis is limited by the presence of sulfur-containing amino acid residues as these poison metal catalysts, albeit methods have been proposed that partially overcome this serious drawback (see Section 2.1.1.1.1.1.3.1). This limitation is also bypassed in cases where the target peptide molecule permits acidolysis with strong acids (HBr/AcOH or HF). As an additional limitation, the saponification of N^{α}-Z-protected peptide esters under drastic conditions, such as those required for longer peptide intermediates, was found to induce decomposition of the N^{α}-Z moiety with generation of N-terminal hydantoins.[85,86]

Acylation of amino acids or peptides with Z-protected amino acids can be performed with all known methods including the α-amino acid *N*-carboxyanhydrides (NCAs), since protocols for the preparation of the highly reactive Z-protected NCAs are now available (see Section 3.4.2).[87] Moreover, since formation of oxazol-5(4*H*)-ones is very slow with urethane-type protecting groups, racemization, at least by this mechanism, is largely suppressed (see Sections 1.2.1.2 and 7.4).[15,17,52]

2.1.1.1.1.1.3 Cleavage of the Benzyloxycarbonyl Group

Cleavage of the benzyloxycarbonyl group can be accomplished by various procedures, which in addition to the older procedure based on reductive cleavage by sodium in liquid ammonia[88–91] include catalytic hydrogenation over palladium catalysts and catalytic transfer hydrogenation employing a whole series of hydrogen donors that lead to carbon dioxide and toluene as especially innocuous byproducts. Conversely acidic deprotection requires the use of strong acids.[22,51]

2.1.1.1.1.1.3.1 Catalytic Hydrogenolysis

The catalytic hydrogenolysis of benzyloxycarbonyl groups over palladium catalysts[21] is a very mild and highly selective deprotection procedure for peptides with side-chain functionalities protected as acid-labile *tert*-butyl esters, *tert*-butyl ethers, or *N*-*tert*-butoxycarbonyl derivatives that all are completely resistant to hydrogenolysis whilst related *N*-9-fluorenylmethoxycarbonyl derivatives undergo partial deprotection under these conditions.[92] Hydrogenolysis is generally performed in alcohols (MeOH, iPrOH, BuOH, tBuOH), AcOH, DMF, NMP, DMA, or related aqueous mixtures using palladium black, palladium-on-carbon (Pd/C), or $Pd/BaSO_4$ as the catalyst and hydrogen at atmospheric pressure and ambient temperature. In situ reduction of palladium(II) chloride to palladium catalyst has also been proposed, since concomitantly HCl is generated that neutralizes the amino group.[93] Hydrogenolysis is faster in the presence of acids; thereby the use of AcOH as solvent or additive is advisable unless in subsequent synthetic steps the resulting amino component is acylated by in situ condensing agents, e.g. DCC, which may lead to acetyl derivatives as side products. Correspondingly, titration with mineral acids (aq HCl, HBr, or $HClO_4$) at apparent pH values of 4.5–5.5 is the most frequently used procedure, particularly in cases of larger peptide fragments and when concomitant deprotection of N^{ω}-nitro- or $N^{\omega,\omega'}$-dibenzyloxycarbonyl-arginine residues is the goal. Moreover, by titrating the liberated amine group with mineral acids side reactions are prevented such as the well-known piperazine-2,5-dione (DKP) formation, particularly, from aminoacyl-proline dipeptide esters and amides[94,95] and occasionally even from aminoacyl-prolyl-peptides with concomitant fragmentation of pep-

for references see p 149

tide chains,[96–99] in both cases catalyzed by either bases or weak acids. Similarly, ring closure of N-terminal glutamine residues to pyroglutamic acid is largely suppressed.[100] To circumvent the difficult and very slow hydrogenation of large peptide fragments in solvents such as NMP, HMPA, and TFE, has been successfully applied.[101] When using TFE attention has to be paid to the purity of the commercial solvent as trifluoroacetylation was reported to occur as a side reaction which probably derives from the trifluoroacetate present as an impurity.[102] In this context, it is worthwhile to note that the use of alcohols in catalytic hydrogenations may lead to related aldehydes or ketones which in turn are capable of producing stable imidazolid-4-one derivatives with the N-terminal amine group.[1] Moreover, when carrying out hydrogenations in an alcohol, oxygen has to be removed meticulously from the system to avoid as a serious side reaction N-alkylation via aldehyde and related imine derivatives which has been observed to occur readily, particularly with methanol as the solvent.[103–105] An additional inconvenience observed when air is not rigorously excluded, is the formation of palladium complexes with the peptides.[106]

The main drawback in the catalytic hydrogenolysis of Z-protected peptide derivatives is caused by the presence of sulfur-containing amino acid residues (Cys, Pen, or Met) due to catalyst poisoning. Attempts to overcome these restrictions by addition of either tertiary bases[107,108] or boron trifluoride–diethyl ether complex[109] to the hydrogenation mixture were of limited usefulness. More efficient appears to be the use of liquid ammonia (at −33 °C) as the solvent to prevent poisoning of the Pd/C catalyst.[110]

Catalytic hydrogenolysis of Z-protected peptides over Pd/C is accompanied by reductive cleavage of benzyl esters and ethers. However, addition of 0.1–0.5 equivalents of ammonia, pyridine, or ammonium acetate in methanol efficiently inhibits the hydrogenolysis of aliphatic benzyl ethers, e.g. in Z-Ser(Bzl)-OH or Z-Thr(Bzl)-OH,[111] whereas in the presence of 2,2′-bipyridine even phenolic benzyl protecting groups are retained.[112] Conversely, by deactivating the Pd/C catalyst via formation of an ethylenediamine complex [Pd/C(en)], the poisoned catalyst still catalyzes chemoselective hydrogenolysis of various functional groups such as nitro, benzyl ester, and azido groups without cleavage of *O*-benzyl ethers and of the Z protection;[113] for example, Z-Pro-OBzl is readily converted into Z-Pro-OH by this procedure.

It is also worthwhile to note that the relatively slow catalytic hydrogenolysis of N^{α}-Z-protected peptide derivatives permits some interesting chemical transformations to be performed in situ. For example, direct conversion of Z protection to Boc protection is possible when the hydrogenation is conducted in the presence of di-*tert*-butyl dicarbonate under neutral conditions.[114] Alternatively, the same transformation is achieved by the use of triethylsilane and di-*tert*-butyl dicarbonate in ethanol with catalytic amounts of palladium(II) acetate.[115] More efficiently this one-pot transformation is achieved by catalytic transfer hydrogenation in the presence of di-*tert*-butyl dicarbonate (cf. Section 2.1.1.1.3.1.1.6).[116] Similarly, peptide cyclization reactions have been performed in situ over Pd/C and the high yields of cyclic monomers are attributed to the high dilution effect as well as to catalysis of the charcoal surface.[117]

H-Pro-Leu-Gly-NH_2; Typical Procedure:[118]

Through a soln of Z-Pro-Leu-Gly-NH_2 (3.2 g, 7.6 mmol) in EtOH (30 mL) containing aq HCl (1 equiv) a current of hydrogen (1 atm) was bubbled in the presence of 5% Pd/C. On completion of the reaction, the catalyst was filtered off, the filtrate concentrated and the residue treated with a slight excess of NH_3 in $CHCl_3$. After 30 min, NH_4Cl was filtered off, the filtrate concentrated, and the residue recrystallized (H_2O); yield: 1.8 g (80%); mp 122–123 °C.

H-Arg(HBr)-Leu-Leu-Gly-Gln-Leu-Ser(tBu)-Ala-Lys(Adoc)-Lys(Adoc)-Tyr(tBu)-Leu-Gln-Ser(tBu)-Leu-Ile-NH_2·HBr; Typical Procedure:[101]

The peptide derivative Z-Arg(Z_2)-Leu-Leu-Gly-Gln-Leu-Ser(tBu)-Ala-Lys(Adoc)-Lys(Adoc)-Tyr-(tBu)-Leu-Gln-Ser(tBu)-Leu-Ile-NH_2 (4.3 g, 1.52 mmol) was dissolved in HMPA/NMP (1:3) and hydrogenated over 10% Pd/C at pH 5 (titration with 0.076 M HBr in MeOH). The catalyst was filtered off,

the filtrate was concentrated and the product precipitated with Et_2O. The product was collected and washed with Et_2O and H_2O; yield: 2.7 g (71%).

H-Thr(tBu)-Ser(tBu)-Asp(OtBu)-Phe-Ser(tBu)-Arg(HBr)-Leu-Leu-Gly-Gln-Leu-Ser(tBu)-Ala-Lys(Adoc)-Lys(Adoc)-Tyr(tBu)-Leu-Gln-Ser(tBu)-Leu-Ile-NH_2; Typical Procedure:[101]
The peptide derivative Z-Thr(tBu)-Ser(tBu)-Asp(OtBu)-Phe-Ser(tBu)-Arg(HBr)-Leu-Leu-Gly-Gln-Leu-Ser(tBu)-Ala-Lys(Adoc)-Lys(Adoc)-Tyr(tBu)-Leu-Gln-Ser(tBu)-Leu-Ile-NH_2 (1.0 g, 0.3 mmol) was hydrogenated over Pd/C in 90% aq TFE. The catalyst was filtered off, the filtrate was concentrated, and the residue was precipitated from MeOH with Et_2O; yield: 0.73 g (75%).

Conversion of Z-Pro-OMe into Boc-Pro-OMe Using $Et_3SiH/Pd(OAc)_2/(Boc)_2O$:[114]
To a dark suspension of Et_3SiH (0.304 mL, 1.90 mmol), $(Boc)_2O$ (0.261 mL, 1.14 mmol), $Pd(OAc)_2$ (22 mg, 0.10 mmol), and TEA (0.042 mL, 0.30 mmol) in EtOH (5 mL) at rt was added dropwise a soln of Z-Pro-OMe (249 mg, 0.95 mmol) in EtOH (3 mL) under N_2. After stirring for 15 h, the mixture was filtered and concentrated. The residue was chromatographed (silica gel, Et_2O/hexane, 1:4); yield: 204 mg (94%).

2.1.1.1.1.1.3.2 Catalytic Transfer Hydrogenolysis

For Z deprotection, the method of transfer hydrogenation using Pd/C or palladium black catalysts together with various hydrogen donors[119] has emerged as a more rapid reaction when compared to catalytic hydrogenation. As hydrogen donors cyclohexene,[120,121] cyclohexa-1,4-diene,[122] formic acid,[123,124] and ammonium formate[125] have been used in peptide chemistry. With cyclohexene as the hydrogen donor quantitative Z cleavage from related amino acid and peptide derivatives is achieved very rapidly (15 min at 65 °C).[120] Apart from Z groups C-terminal benzyl esters are removed simultaneously.[121] To avoid the higher temperatures at which some protecting groups are not sufficiently stable, the use of cyclohexa-1,4-diene was proposed as it is a more efficient hydrogen donor that can be used at room temperature. By this procedure Z groups, benzyl esters, and tyrosine benzyl ether are removed within 2 hours with 10% Pd/C as the catalyst.

Formic acid was found to act as an efficient hydrogen donor for the catalytic transfer hydrogenation of benzyl-type protecting groups with the main advantage over cyclohexene and cyclohexa-1,4-diene being its improved solvent characteristics.[123] Application of 4% formic acid in methanol results in quantitative cleavage of Z groups and benzyl esters and ethers in 5–10 minutes at room temperature; the acid-labile *tert*-alcohol-derived protecting groups are stable under these conditions. Apparently, even sulfur-containing derivatives impose no particular difficulties; attention has to be paid to possible formylation at high formic acid concentration.

Apart from the classical Pd/C or palladium black catalysts for transfer hydrogenation, a palladium–poly(ethyleneimine)-based catalyst was reported for the efficient removal of Z groups using formic acid as hydrogen donor.[124] The enhanced activity of the polymeric catalyst is attributed to the adsorption of formic acid to the poly(ethyleneimine) matrix and to the distribution of the palladium over a large surface area. Furthermore, this palladium catalyst does not appear to be pyrophoric and can be used repeatedly without apparent loss of activity. For hydrogenolysis of Z-protected dehydroamino acid derivatives, ammonium formate has been shown to be the hydrogen donor of choice for catalytic transfer hydrogenation, since the $C^{\alpha}=C^{\beta}$ bonds, normally susceptible to hydrogenation, remain unaffected under these reaction conditions.[126]

Hydrogenolysis of Z Amino Acids and Z Peptides with Cyclohexene as Hydrogen Donor; General Procedure:[120]
To a soln of Z amino acid or Z peptide in MeOH or EtOH were added cyclohexene (in excess over the molar proportion required) and 10% Pd/C catalyst [ratio (catalyst/substrate) 1:1 to 1:5 by weight]. The mixture was refluxed for 15 min, filtered from the catalyst, and concentrated; yield: 90–100%.

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Hydrogenolysis of Z Amino Acids and Z Peptides with Cyclohexa-1,4-diene as Hydrogen Donor; General Procedure:[122]
Through a soln of Z protected peptide (1.0 mmol) in EtOH (4 mL) a gentle stream of N_2 was passed and thorough agitation was provided by a vibro-mixer at 25 °C. An equal weight of 10% Pd/C (per protecting group) was added followed by the addition of cyclohexa-1,4-diene (0.94 mL, 10 mmol). After a minimum of 2 h, the mixture was filtered (Celite), washed with solvent (depending on the solubility of the product, a variety of solvents may be used, e.g. DMF, AcOH, H_2O) and concentrated. Products were generally obtained in 90–100% yields.

Hydrogenolysis of Z Amino Acids and Z Peptides with Formic Acid as Hydrogen Donor; General Procedure:[123]
A soln of Z protected peptide (200 mg) in 4.4% formic acid in MeOH (2–10 mL) was added to Pd black catalyst (200 mg) in 4.4% formic acid in MeOH (10 mL). The mixture was continuously stirred under N_2. After 10 min, products were isolated by filtering off the catalyst and washing with additional MeOH followed by H_2O. The combined filtrates and washes were concentrated and the products crystallized from appropriate solvents.

2.1.1.1.1.3.3 Acidolytic Cleavage of the Benzyloxycarbonyl Group

The benzyloxycarbonyl group is also readily cleaved under strong acidic conditions, e.g. by dry HBr in acetic acid with conversion into the corresponding amine hydrobromides, benzyl bromide, and CO_2.[22] Various alternative procedures have been proposed such as the use of TFA at higher temperatures,[127,128] or TFA/thioanisole where the push–pull mechanism remarkably increases the rate of acidolysis,[129] liquid HF,[51] HF/pyridine,[130] BBr_3/CH_2Cl_2,[131] or BBr_3/TFA for longer peptide fragments.[132] By these procedures also the benzyl-type side-chain protection is cleaved as well as Arg(Tos). Additionally, numerous sulfonic acids were shown to efficiently remove the Z group, such as methanesulfonic acid, trifluoromethanesulfonic acid, or fluorosulfonic acid mainly in CH_2Cl_2 or TFA.[24,133] Since the acidolytic removal of the benzyloxycarbonyl group proceeds under formation of the benzyl cation, unwanted side reactions such as C-benzylation of aromatic amino acids or S-benzylation of sulfur-containing amino acids occur. Therefore, addition of an efficient scavenger such as anisole or benzenethiol is strongly recommended. Deprotection conditions that favor a S_N2 mechanism, thus reducing the benzyl cation formation, are achieved by a mixture of HF/DMS/*p*-cresol.[134]

Deprotection of Z Amino Acids and Z Peptides with HBr/AcOH; General Procedure:[22]
A saturated soln (50 g) of dry HBr in glacial AcOH (36%) was added to the Z-protected amino acid or peptide derivative (10 g) in a reaction flask protected with a $CaCl_2$ tube. Immediate formation of CO_2 begins in some cases in markedly exothermic manner. The mixture was allowed to stand at rt with occasional shaking for 1 h, then dry Et_2O was added to precipitate the amine hydrobromide. The supernatant liquid was decanted and the solid triturated with Et_2O, filtered, and washed with Et_2O.

Deprotection of Z Amino Acids and Z Peptides with BBr_3 in CH_2Cl_2; General Procedure:[131]
To a cooled (−10 °C) soln of the protected amino acid or peptide derivative (2 mmol) in CH_2Cl_2 (50 mL) was added dropwise 1 M BBr_3 in CH_2Cl_2 (10 mL, 10 mmol) with stirring. After 1 h at −10 °C and 2 h at 25 °C, the reaction was terminated by careful dropwise addition of H_2O (50 mL). The layers were separated, the organic phase was washed with H_2O (3 × 25 mL), and the combined aqueous layers were concentrated. The residue was taken up in H_2O and purified by chromatography.

2.1.1.1.1.3.4 Cleavage of the Benzyloxycarbonyl Group by Other Procedures

Alkylsilanes have been shown to be versatile deprotection reagents for the benzyloxycarbonyl group. This is achieved by treatment with $Et_3SiH/PdCl_2$,[135] or by reaction with *tert*-butyldimethylsilane/$Pd(OAc)_2$ which allows the corresponding *tert*-butyldimethylsiloxycarbonyl-protected amino acids to be isolated.[115]

Apart from the above-mentioned deprotection procedures, photolytic deprotection[136] or electrochemical procedures[137,138] have been proposed.

2.1.1.1.1.2 Substituted Benzyloxycarbonyl Groups

J. Podlech

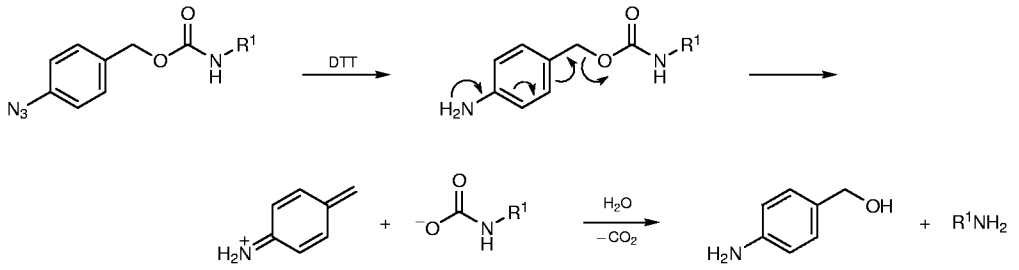
Various substituted benzyloxycarbonyl groups have been proposed for the protection of amino functionalities for various reasons.[139] Substitution of the benzyl part of the benzyloxycarbonyl group with electron-donating groups increases the acid lability, while substitution by electron-withdrawing groups would reduce it.[140] The latter is achieved with halo-[141–145] or nitro-substituted benzyloxycarbonyl groups;[146–151] these types of protecting groups are preferentially used for the side-chain protection of lysine and ornithine (see Section 2.1.2) in solution and solid-phase synthesis when Boc/Bzl chemistry is applied. Some substituents allow photolytic cleavage (see Section 2.4).

Further possible features addressed with substituents at the benzyl moiety are the improved tendency to crystallize, coloration (allowing for easier monitoring in chromatography), or the presence of heavy atoms suitable for X-ray structure analysis of peptides. Some of these protecting groups are introduced with crystalline reagents, which permit more precise dosage compared to the oily benzyl chloroformate (**1**) used for the preparation of *N*-Z derivatives. Moreover, linkage of the Z group to polystyrene resin leads to a resin-bound protecting group which was used by Letsinger for the solid-phase synthesis of peptides in the N→C direction.[152,153] This approach has enjoyed a recent revival, albeit with new polymer-type *N*-protecting groups such as resin-bound Boc[154] and Trt equivalents.[155,156]

For N^{α}-protection, among the substituted benzyloxycarbonyl derivatives only the 4-methoxybenzyloxycarbonyl group [Z(OMe)] has found general application.[49,157–166] It is cleaved by acidolysis under significantly milder conditions than the Z group (TFA, 0 °C;[49] 2–4 M $EtSO_3H$/AcOH or CH_2Cl_2, rt, 1 h;[162] TosOH, MeCN;[163] 5–10% TFA/CH_2Cl_2;[164] or 0.1% MsOH/20% *m*-cresol/CH_2Cl_2, rt, 30 min[166]). An additional interesting benzyloxycarbonyl derivative is the recently proposed 4-azidobenzyloxycarbonyl group, which relies on the safety-catch principle. In fact, it is stable towards acidic reaction conditions, but is cleaved instantaneously upon reduction with dithiothreitol (DTT) (Scheme 2).[167,168] So far, wider experimental application has not been reported.

For further variations of the Z group and related properties see Table 2 and previous reviews on this subject.[1,3] Most of these protecting groups are introduced by procedures similar to those described for the synthesis of *N*-Z derivatives, either with the corresponding chloroformates, azido formates or mixed carbonates. However, these reagents are usually not commercially available, except for 4,5-methoxy-2-nitrobenzyl chloroformate, 4-nitrobenzyl chloroformate, and the 4-methoxybenzyl carbazate [Z(OMe)-$NHNH_2$] as precursor of Z(OMe)-N_3.

Scheme 2 Cleavage of *N*-4-Azidobenzyloxycarbonyl Derivatives by Reduction with Dithiothreitol[167,168]



for references see p 149

Table 2 Substituted Benzyloxycarbonyl Groups

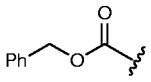
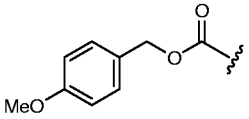
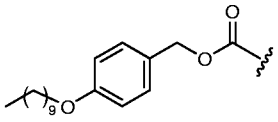
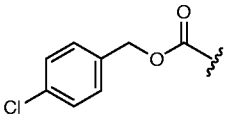
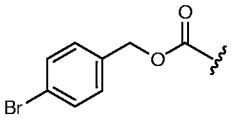
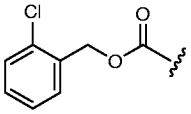
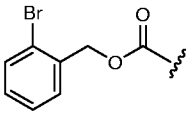
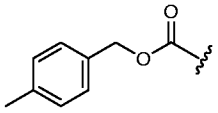
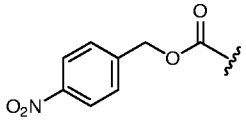
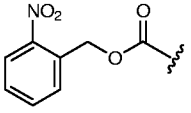
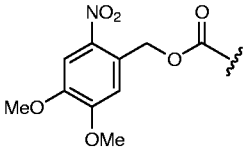
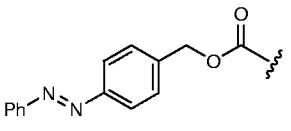
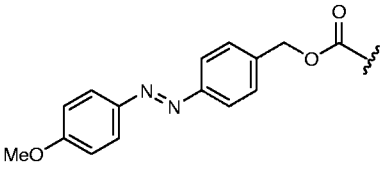
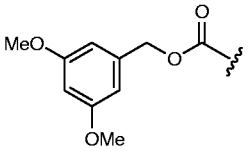
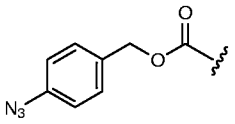
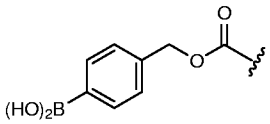
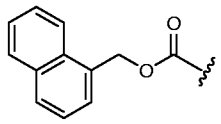
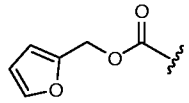
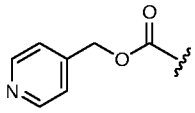
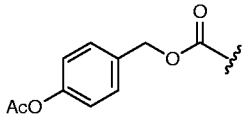
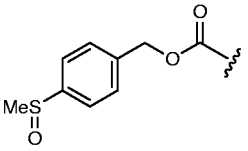
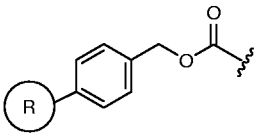
Protecting Group	Structure	Special Properties and Preferred Cleavage Conditions	Stability Toward Acidic Conditions, $t_{1/2}$[a] (min)	Ref
benzyloxycarbonyl (Z)			66	
4-methoxybenzyloxycarbonyl [Z(OMe)]		cleaved by weak acids such as HCl/AcOH, TosOH, or TFA	0.8	[49,157–166]
4-(decyloxy)-benzyloxycarbonyl		higher lipophilicity		[169]
4-chlorobenzyloxycarbonyl [Z(Cl)]		modified cleavage, heavy atom	120	[142]
4-bromobenzyloxycarbonyl [Z(Br)]		higher tendency to crystallize, modified cleavage, heavy atom		[141]
2-chlorobenzyloxycarbonyl [Z(2-Cl)]		higher stability		[144]
2-bromobenzyloxycarbonyl [Z(2-Br)]		higher stability		[145]
4-methylbenzyloxycarbonyl		higher tendency to crystallize, modified cleavage	7	[141]
4-nitrobenzyloxycarbonyl [Z(NO_2)]		higher tendency to crystallize, Z(NO_2)-Cl is crystalline, faster hydrogenolysis, higher stability towards acids, cathodic cleavage, cleaved with $Na_2S_2O_4$[170]	430	[146–148,151]
2-nitrobenzyloxycarbonyl Z(2-NO_2)		*hν*		[149,150]
4,5-dimethoxy-2-nitrobenzyloxycarbonyl (Nvoc)		*hν*		[149,150]

Table 2 (cont.)

Protecting Group	Structure	Special Properties and Preferred Cleavage Conditions	Stability Toward Acidic Conditions, $t_{1/2}$[a] (min)	Ref
4-(phenyldiazenyl)-benzyloxycarbonyl (Pz)		colored, faster hydrogenolysis	400	[171,172]
4-(4-methoxyphenyldiazenyl)-benzyloxycarbonyl (Mz)		colored		[171,172]
3,5-dimethoxybenzyloxycarbonyl		*hν*		[173]
4-azidobenzyloxycarbonyl		reduction with DTT leads to an amino derivative, which undergoes immediate cascade degradation to liberate the amino function (Scheme 2)		[167,168]
4-(dihydroxyboryl)-benzyloxycarbonyl		higher solubility, cleavage with H_2O_2		[174]
1-naphthylmethoxycarbonyl		higher tendency to crystallize, modified cleavage		[141]
furfuryloxycarbonyl		cleaved by 6.5% HBr/AcOH, HCl/AcOH, or TFA, similar to Z(OMe)		[175]
4-pyridylmethoxycarbonyl (iNoc)		acid stable, cleaved with Zn, 50% AcOH		[176]
4-(acetoxy)-benzyloxycarbonyl [Z(OAc)]		alkali labile		[50]
4-(methylsulfinyl)-benzyloxycarbonyl (Msz)		stable under basic and acidic conditions cleaved by reductive acidolysis (safety catch principle)		[177]

for references see p 149

Table 2 (cont.)

Protecting Group	Structure	Special Properties and Preferred Cleavage Conditions	Stability Toward Acidic Conditions, $t_{1/2}$[a] (min)	Ref
4-(polymer bound)-benzyloxycarbonyl[b]		polymer-bound protection group		[152]

[a] In 0.85 M HBr/AcOH, 25 °C.[2]
[b] R = Merrifield (polystyrene) resin.

4-Methoxybenzyl Chloroformate [Z(OMe)-Cl]; Typical Procedure:[18,158]

CAUTION: *Phosgene is a colorless, poisonous gas or volatile liquid. This reagent should be handled with the utmost care.*

To a soln of $COCl_2$ (100 g, 1 mol) in dry Et_2O (500 mL) was added dropwise within 30 min anisyl alcohol (69 g, 0.5 mol) with stirring at 0 °C. The soln was concentrated under reduced pressure at a bath temp of 0 °C or below to 200 mL. This soln was used directly for the preparation of *N*-Z(OMe)-protected amino acids.

Synthesis of N^{α}-Z(OMe)-Protected Amino Acids; General Procedure:[18,158]
To a stirred soln of the amino acid (100 mL) in 2 M NaOH (50 mL) and THF (50 mL) at 0 °C was added alternately (~5 portions each) an aliquot [corresponding to 125 mmol Z(OMe)-OH] of Z(OMe)-Cl in Et_2O as described above and 2 M NaOH (62 mL) within 30 min while the pH was maintained at 9–10, if necessary with addition of 2 M NaOH. Stirring was continued at rt for 1 h, and the pH was adjusted to 7 with NaOH or AcOH, and the THF was removed. The aqueous soln was extracted with Et_2O (100 mL), acidified with 2 M HCl, and extracted with EtOAc (200 mL). The organic layer was washed with H_2O, dried, concentrated, and recrystallized (EtOAc/hexane or EtOH/hexane); analytical data of the amino acid derivatives are reported in ref[158].

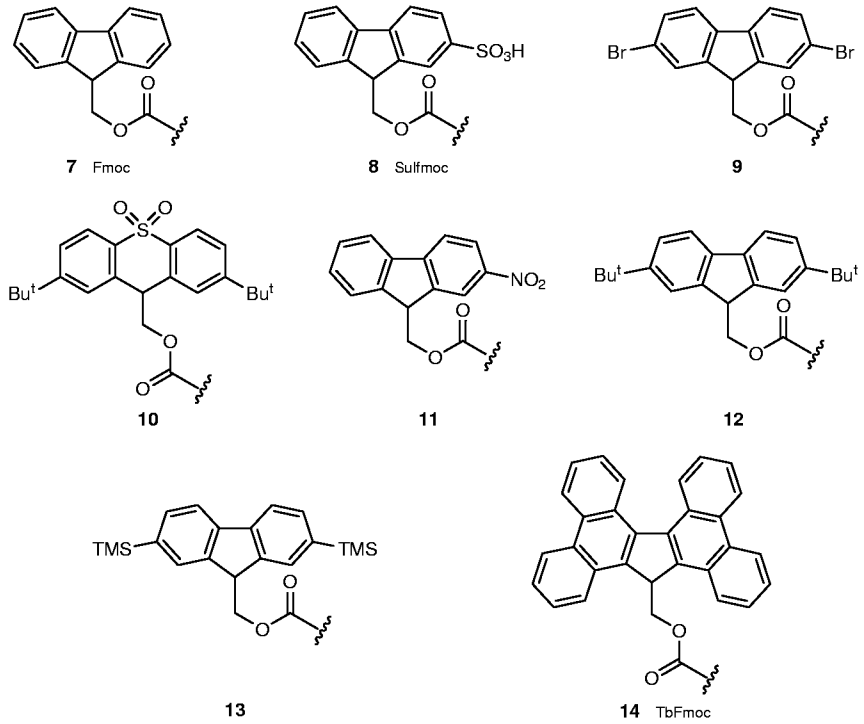
2.1.1.1.1.3 9-Fluorenylmethoxycarbonyl Group

J. Podlech, M. Gurrath, and G. Müller

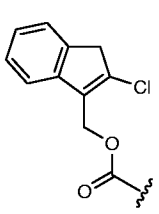
The 9-fluorenylmethoxycarbonyl group (Fmoc, **7**) has, since its first report by Carpino and Han[26] become one of the most frequently used α-amino protecting groups in solid-phase peptide synthesis. Although in the initial report liquid ammonia was used for cleavage of *N*-Fmoc derivatives, its usefulness was immediately recognized upon the observation that the Fmoc protection is readily removed even by amines such as piperidine.[178] The real breakthrough for this type of N^{α}-protection occurred with its application in solid-phase synthesis by Chang and Meienhofer[179] and Atherton et al.[180] in combination with the acid-labile, *tert*-butyl alcohol based side-chain protection, which resulted in the now well-established Fmoc/tBu chemistry (see Section 4.3.2.2). In contrast to the more traditional Boc/Bzl chemistry, no repetitive acidolytic steps are required for N-terminal deprotection in the stepwise chain elongation, which are known to cause a progressive loss of the side-chain protection, e.g. of N^{ω}-Z or -benzyl esters or ethers.[181] The relatively mild, versatile deprotection procedure with amines is operationally simple, directly liberating the free amino group and the by-product dibenzofulvene (DBF) which is readily removed as the corresponding amine adduct, at least from the solid support. Due to the base lability of the Fmoc group and its orthogonal nature relative to the acid-labile side-chain protection, this type of amino protection introduced an element of chemical versatility into solid-phase strategies that previously did not exist. Coupling of N^{α}-Fmoc-protected amino acids and cleavage of the Fmoc group can be monitored by UV or fluorescence spectroscopy. The chemistry and usefulness of this protecting group has been extensively reviewed.[92,182–187]

Major drawbacks of the N^{α}-Fmoc derivatives are their relatively low solubility in solvents generally used in peptide synthesis, which results in problems with automation, as well as their high lability even to traces of amines, particularly in the case of N-terminal proline residues.[185] In solution synthesis this high base lability raises problems especially with convergent strategies.[188,189] For this reason several variations on this protecting group have been proposed (Scheme 3), which either exhibit better solubility properties, or lead to an easier separation of the unreacted excess amino acid derivatives. In fact, the 2-sulfo-9-fluorenylmethoxycarbonyl group (Sulfmoc, **8**) leads to substrates which differ in their solubility properties from the uncoupled amino components and thus are easily separated by reverse-phase chromatography.[190] The 2,7-di-*tert*-butyl-9-fluorenylmethoxycarbonyl (**12**) derivatives are characterized by a higher solubility in common organic solvents,[191] and could well represent an interesting alternative.[192] A similar solubilizing effect is attributed to the newly proposed 2,7-bis(trimethylsilyl)-9-fluorenylmethoxycarbonyl group (**13**),[838] which compared to the 2,7-di-*tert*-butyl-9-fluorenylmethoxycarbonyl group (**12**) is cleaved more readily by amines and at rates comparable to the standard Fmoc group. The recently developed tetrabenzo[*a*,*c*,*g*,*i*]fluoren-17-ylmethoxycarbonyl group (**14**)[193,194] allows selective adsorption to porous graphitized carbon, thus leading to a facile separation of N^{α}-acetylated truncated peptides generated as side products in stepwise SPPS. The 2-chloro-1*H*-inden-3-ylmethoxycarbonyl (**15**) and 1*H*-cyclopenta[*b*]naphthylen-3-ylmethoxycarbonyl (**16**) show properties similar to those of the Fmoc derivatives.[195] The 2,7-dibromo-9-fluorenylmethoxycarbonyl (**9**)[196] and 2,7-di-*tert*-butyl-10,10-dioxothioxanthen-9-ylmethoxycarbonyl (**10**)[197] are cleaved by pyridine, against which the Fmoc group is rather stable. Conversely, 2-nitro-9-fluorenylmethoxycarbonyl (**11**) is cleaved by photolysis, but bears a stereogenic center which complicates analytical characterization of intermediates and products.[198]

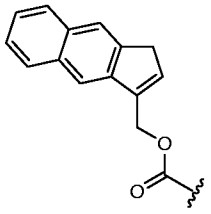
Scheme 3 Variations to the Fmoc Group



for references see p 149



15



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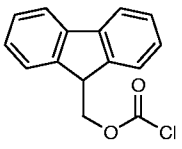
Despite the potential advantages of these Fmoc alternatives, none of them, so far, has received due attention and extensive experimental application in peptide synthesis.

Alternative base-labile protecting groups such as the 2-[4-nitrophenyl)sulfonyl]ethoxycarbonyl (Nsc) group have been developed to face the problems arising with the Fmoc group. These are discussed in Section 2.1.1.1.1.4.

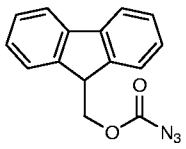
2.1.1.1.1.3.1 Synthesis of N^{α}-9-Fluorenylmethoxycarbonyl Amino Acids

N^{α}-Fmoc-protected amino acids are readily obtained under Schotten–Baumann conditions using 9-fluorenylmethyl chloroformate (Fmoc-Cl, **17**) (Scheme 4) as the acylating agent.[25] This reagent is stable when stored at low temperature under exclusion of moisture;[199] it is commercially available, but it is also readily prepared from the corresponding alcohol[196] and phosgene.[178,200] Acylation of amino acids is generally performed in dioxane/water with sodium carbonate as the base in 88–97% yield. The resulting Fmoc-protected amino acids are usually crystalline compounds, which are stable even as their respective acids when stored at low temperature with exclusion of moisture, and are commercially available. (For exhaustive compilations of Fmoc amino acid derivatives including related analytical data, see refs[182,186,201].) Recently, as an alternative to 9-fluorenylmethyl chloroformate (**17**) the related fluoride was suggested which is obtained as crystalline compound, stable on storage.[837]

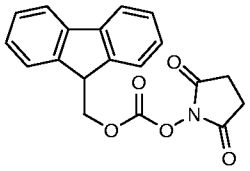
Scheme 4 Reagents for the Synthesis of N^{α}-Fmoc-Protected Amino Acids



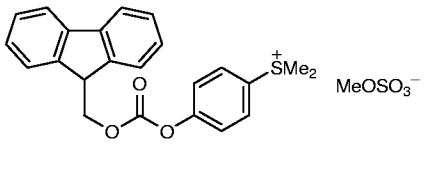
17 Fmoc-Cl



18 Fmoc-N_3



19 Fmoc-OSu



20

It was soon realized that major side products in this reaction are the corresponding Fmoc dipeptides (and even tripeptides) at extents of 1–5%, occasionally up to 14%. This side reaction occurs via a multistep mechanism involving, (1) Fmoc protection; (2) formation of a mixed anhydride with excess 9-fluorenylmethyl chloroformate (**17**); and (3) peptide coupling with unreacted amino acid.[202–205] The formation of Fmoc dipeptides is particularly favored with nonhindered amino acids, e.g. in the case of glycine 14.4% dipeptide and 0.2% tri-

peptide are formed, and for alanine 4.2% dipeptide is formed;[206] these side products are difficult to remove and correspondingly, yields of homogeneous material are significantly affected.

Several alternative procedures have been proposed to address this serious problem, a very simple one being changes in the reaction conditions. In fact, the amount of side product is substantially reduced when a slight excess of amino acid over acylating agent (1.25 equiv) and sodium carbonate (4 equiv) are used with a minimum amount of dioxane and with vigorous stirring at room temperature. This procedure, however, cannot be applied with hydrophobic amino acids such as leucine or valine.[202] Acylation of amino acids with 9-fluorenylmethyl azidoformate (Fmoc-N_3, **18**, prepared from Fmoc-Cl and NaN_3[178,203]) prevents formation of detectable amounts of dipeptidic material, but the reaction is more sluggish (5 d) and yields are somewhat lower.[178,203] Additionally a series of weaker acylating agents, mostly 9-fluorenylmethyl carbonates have been utilized,[77,182,202,207,208] among which only the corresponding succinimido carbonate (Fmoc-OSu, **19**) leads to acceptable yields, reaction times, and purities of the products as no detectable dipeptide formation has been observed with this reagent.[66,202,204,205]

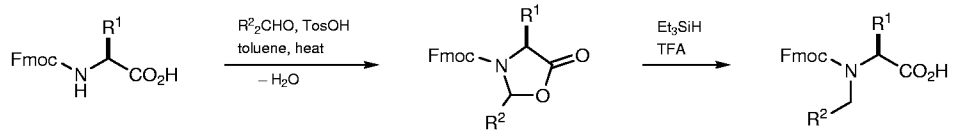
9-Fluorenylmethyl succinimido carbonate (**19**) is a commercially available or easy to prepare crystalline compound.[205] Acylation of amino acids with 9-fluorenylmethyl succinimido carbonate (**19**) is performed under standard conditions in dioxane/H_2O,[202] acetone/H_2O,[66] MeCN/H_2O,[205,209] or in DMF/H_2O[204] using sodium hydrogen carbonate,[66] sodium carbonate,[202,204] or triethylamine[205,209] as the base. An occasional precipitation of Fmoc-OSu or of the base during mixing of the aqueous and organic phases is avoided by using 1,2-dimethoxyethane as the solvent and sodium carbonate as the base.[183] A further extraction step during the standard workup has been reported to produce Fmoc-protected amino acids of enhanced purity.[209] The use of MeCN/H_2O as the solvent and triethylamine as base is reported to reduce the reaction time from 24 hours to 30 minutes without dipeptide formation or racemization and in 73–100% yield.[205]

Dipeptide formation is also prevented by an intermediate N,O-bis(trimethylsilylation) of the amino acid with chlorotrimethylsilane (TMSCl).[206] The silyl groups are cleaved during aqueous workup leading to Fmoc amino acids in 78–94% yields. Following the suggestion of Yadav et al.[53] that carbamates can be produced from amino acids under neutral conditions by reaction with chloroformates in the presence of zinc dust, N^{α}-Fmoc amino acids were obtained rapidly (15–20 min) in high yields without detectable dipeptide contaminants.[210]

Alternative Fmoc donors have been proposed for particular purposes; for example, [4-(9-fluorenylmethoxycarbonyloxy)phenyl]dimethylsulfonium methylsulfate (**20**) has been proposed by Okai et al. as a water-soluble reagent for the introduction of the Fmoc group in the presence of aqueous sodium carbonate.[211] Similarly, a polymer-bound 1,2,3-benzotriazol-1-ol can be converted by reaction with 9-fluorenylmethyl chloroformate (**17**) into a polymer-bound 1,2,3-benzotriazol-1-yl 9-fluorenylmethyl carbonate for the preparation of Fmoc-protected amines in organic and aqueous organic media with all the known advantages of insoluble reagents.[212]

For the synthesis of N^{α}-Fmoc-protected *N*-alkyl amino acids a multistep procedure has been proposed that relies on reduction of the corresponding oxazolidinones (Scheme 5). Overall yields are usually excellent despite the sterically hindered amino group.[213]

Scheme 5 Preparation of N^{α}-Fmoc-Protected *N*-Alkyl Amino Acids[213]



for references see p 149

Regarding the synthesis of N^{α}-Fmoc-protected amino acid *N*-carboxyanhydrides (Leuchs' anhydrides, amino acid-NCAs)[87] as highly activated derivatives, see Section 3.4.4.

Similarly, the exceptional acid stability of the N^{α}-Fmoc amino acid derivatives permits the preparation of the related acid chlorides[214–216] and fluorides,[217] which are also powerful acylating intermediates for peptide synthesis (see Section 3.3).

CAUTION: *9-Fluorenylmethyl chloroformate (Fmoc-Cl, **17**) can evolve CO_2 (pressure!) upon prolonged storage; therefore, bottles should be opened carefully. Fmoc-Cl is harmful if swallowed, inhaled, or absorbed through skin, is corrosive, and is a strong lachrymator. Inhalation may be fatal.*

9-Fluorenylmethyl Chloroformate (Fmoc-Cl, 17)[56,200]

CAUTION: *Phosgene is a colorless, poisonous gas or volatile liquid. This reagent should be handled with the utmost care.*

To an ice-cold soln of $COCl_2$ (110 g, 1.1 mol) in CH_2Cl_2 (600 mL) was added 9-fluorenylmethanol (196 g, 1 mol) with stirring and cooling. After 3 h, an almost clear soln was formed upon additional condensation of $COCl_2$. The mixture was filtered and the filtrate evaporated to an oil. The residue was crystallized twice from Et_2O. With large-scale preparations crystallization under stirring is recommended, to facilitate the filtration step; yield: 215 g (83%); mp 61–63 °C.

Synthesis of N^{α}-Fmoc Amino Acids with 9-Fluorenylmethyl Chloroformate (17); General Procedure:[25]
To an ice-cold soln (or suspension) of amino acid (50 mmol) in 10% aq Na_2CO_3 (100 mL) was added dioxane (50 mL) followed by the slow addition of a soln of Fmoc-Cl (**17**; 13 g, 50.2 mmol) in dioxane (75 mL). The mixture was stirred at 0 °C for 1 h and at 25 °C for 5–18 h. The mixture was poured into ice water (1.5 L) and extracted with Et_2O (2 × 400 mL). The aqueous layer was chilled in an ice bath and acidified with concd HCl to pH 2.0 (in the case of sensitive *tert*-butyl or trityl side-chain protections, acidification with $KHSO_4$ or citric acid is recommended). Crystalline precipitates were collected by filtration, washed thoroughly with 0.1 M HCl and H_2O, and either directly dried and recrystallized (from hot MeOH by the addition of Et_2O) or dissolved in EtOAc and washed again with 0.1 M HCl and H_2O, followed by drying, concentration, and crystallization (EtOAc/hexane). Most N^{α}-Fmoc amino acid derivatives precipitated as oils on acidification. These were taken up in Et_2O and washed thoroughly with 0.1 M HCl and H_2O. The organic phases were dried, and concentrated and the ensuing residues crystallized (CH_2Cl_2/hexane or EtOAc/petroleum ether); yield: 70–95%.

Fmoc-Ile-OH; Typical Procedure:[56]
To an ice-cold soln of H-Ile-OH (66 g, 0.5 mol) in 1 M Na_2CO_3 (1 L) and dioxane (300 mL) was added dropwise a soln of Fmoc-Cl (**17**; 130 g, 0.5 mol) in dioxane (300 mL) under stirring. After 1 h in ice bath and 8 h at rt, the mixture was diluted with H_2O and extracted twice with Et_2O. The aqueous layer was acidified with concd HCl and extracted with EtOAc. The organic layer was washed with H_2O, dried, and concentrated. The residue was crystallized (EtOAc/petroleum ether); yield: 160 g (90%); mp 145–147 °C; $[\alpha]_D^{20}$ −12.3 and $[\alpha]_{546}^{20}$ −15.2 (*c* 1, DMF); lit.[25] mp 145–147 °C; $[\alpha]_D^{20}$ −11.9 (*c* 1, DMF).

Synthesis of N^{α}-Fmoc Amino Acids with Fmoc-Cl (17) in the Presence of Zn Dust; General Procedure:[210]
After complete dissolution of the amino acid (10 mmol) in a minimum quantity of 1 M HCl in MeCN, activated zinc dust was added in small portions until the mixture attained neutral pH (alternatively the amino acid was dissolved by addition of the required quantity of a base such as TEA or NMM and the pH adjusted to neutral). A soln of Fmoc-Cl (**17**; 2.7 g, 10 mmol) in MeCN (10 mL) and zinc powder (650 mg, 10 mmol) were added in one portion and the mixture was stirred at rt for about 20 min. The mixture was filtered, washed with MeCN, and the organic layer was concentrated. The residue was dissolved in 2.5 M Na_2CO_3 (20 mL) and extracted with Et_2O (3 × 10 mL). The aqueous layer was acidified to pH 2.0 with 6 M HCl and the N^{α}-Fmoc amino acid was extracted with EtOAc (3 × 20 mL). The organic layers were washed with 0.1 M HCl and H_2O and dried. The solvent was removed and the residue crystallized from a suitable solvent; yield: 85–90%.

9-Fluorenylmethyl Azidoformate (Fmoc-N_3, 18):[178,203]
To an ice-cold stirred soln of NaN_3 (6.9 g, 106 mmol) in H_2O (27 mL) was added slowly a soln of Fmoc-Cl (**17**; 18 g, 70 mmol) in acetone (33 mL). The mixture was stirred for 2 h and the solid was filtered. The filtrate was washed with H_2O, dried, and recrystallized (acetone) leading to colorless needles; yield: 15.5 g (84%); mp 84 °C.

Synthesis of N^{α}-Fmoc Amino Acids with Fmoc-N_3 (18); General Procedure:[203]
To an ice-cold stirred soln of amino acid (25 mmol) in 10% Na_2CO_3 (66 mL) was added dropwise a soln of Fmoc-N_3 (**18**; 5.9 g, 22.24 mmol) in dioxane (65 mL). After stirring at 0 °C for 2 h and at rt for 5 d, the mixture was poured in H_2O (300 mL), extracted with Et_2O, and after cooling, acidified with concd HCl to pH 2.0. The precipitate was filtered, washed with H_2O (3 ×) and dried. Obtained by this procedure were, e.g., Fmoc-Gly-OH; yield: 5.2 g (78%); mp 176–178 °C; Fmoc-Ala-OH: yield: 5.2 g (75%); mp 146–147 °C; $[\alpha]_D^{20}$ −19.2 (*c* 1, DMF); no dipeptide could be detected.

9-Fluorenylmethyl Succinimido Carbonate (Fmoc-OSu, 19):[205]
Fmoc-Cl (**17**; 129 g, 0.5 mmol) was dissolved in dioxane (1.25 L) and HOSu (63.3 g, 0.55 mmol) was added. After cooling in an ice bath and with vigorous stirring, TEA (70 mL, 0.5 mol) was added slowly so as to maintain the temperature at 10–15 °C. After stirring at rt for 4 h the precipitated TEA · HCl was removed by filtration. The filtrate was concentrated to 250 mL and poured slowly into Et_2O (2.5 L) while stirring. The product that crystallized, was collected by filtration, washed with Et_2O, and dried; yield: 85–90%; mp ranging from 130–148 °C. The product can be used without further purification, however, if required, recrystallization ($CHCl_3$/Et_2O 4:1, 0.1 mol Fmoc-OSu in 75 mL) is possible; yield: 80%.

Synthesis of N^{α}-Fmoc Amino Acids with Fmoc-OSu (19); General Procedure:[209]
To a soln of amino acid (0.10 mol) in H_2O (100 mL) and TEA (1 equiv) Fmoc-OSu (**19**; 30 g, 90 mmol) in MeCN (100 mL) was added. The pH of the soln was brought to 8.5–9.0 with TEA and maintained until no further drop in the pH was detectable over 15 min. The mixture was filtered and concentrated and the residue slowly added to vigorously stirred 1.5 M HCl (0.5 L). If a solid formed, it was washed with 1.5 M HCl and H_2O and dissolved in EtOAc. If an oil precipitated, it was extracted with EtOAc (0.5 L). The respective organic layers were filtered through a porosity 4 sintered funnel to remove residual flocculate and washed with 1.5 M HCl (3 × 0.5 L), H_2O, and brine before drying. Crystallization of the product usually commenced during evaporation and the crystals were freed from EtOAc by trituration with petroleum ether (bp 40–60 °C) and dried; yield: 85–95%.

Fmoc-Ile-OH Using Fmoc-OSu (19); Typical Procedure:[218]
To a stirred soln of H-Ile-OH (66 g, 0.50 mol) and Na_2CO_3 (53 g; 0.50 mol) in H_2O/acetone (1.3 L, 1:1) was added Fmoc-OSu (**19**; 165 g, 0.49 mol) over a period of 1 h while the pH was kept at 9–10 by the addition of 1 M aq Na_2CO_3. After stirring overnight, EtOAc (2 L) was added and the mixture acidified with 6 M HCl. The organic layer was separated and washed with H_2O (4 ×, total of 1.5 L), dried, and concentrated to ~500 mL. On addition of petroleum ether, the product crystallized; yield: 162 g (93%); mp 143 °C; $[\alpha]_D$ −12.1 (*c* 1, DMF).

Synthesis of N^{α}-Fmoc Amino Acids with Fmoc-Cl (17) via *N,O*-Bis(trimethylsilyl)-Protected Amino Acids; General Procedure:[206]
To finely ground amino acid (37.5 mmol) suspended in CH_2Cl_2 (87.5 mL) was added with vigorous stirring TMSCl (9.52 mL, 75 mmol) in one portion. The mixture was refluxed for 1 h and cooled in an ice bath. Then DIPEA (11.3 mL, 65 mmol) and Fmoc-Cl (**17**; 6.47 g, 25 mmol) were added. The soln was stirred with cooling for 20 min and warmed to rt for 1.5 h. The mixture was concentrated and distributed between Et_2O (200 mL) and 2.5% $NaHCO_3$ (250 mL). The aqueous layer was extracted with Et_2O (2 × 50 mL) and the Et_2O layers backwashed with H_2O (2 × 25 mL). The combined aqueous layers were acidified to pH 2 with 1 M HCl and extracted with EtOAc (3 × 75 mL). The combined EtOAc layers were dried, concentrated, and the product was recrystallized to yield the dipeptide-free Fmoc-protected amino acids.

2.1.1.1.1.3.2 Peptide Synthesis with N^{α}-9-Fluorenylmethoxycarbonyl Amino Acids

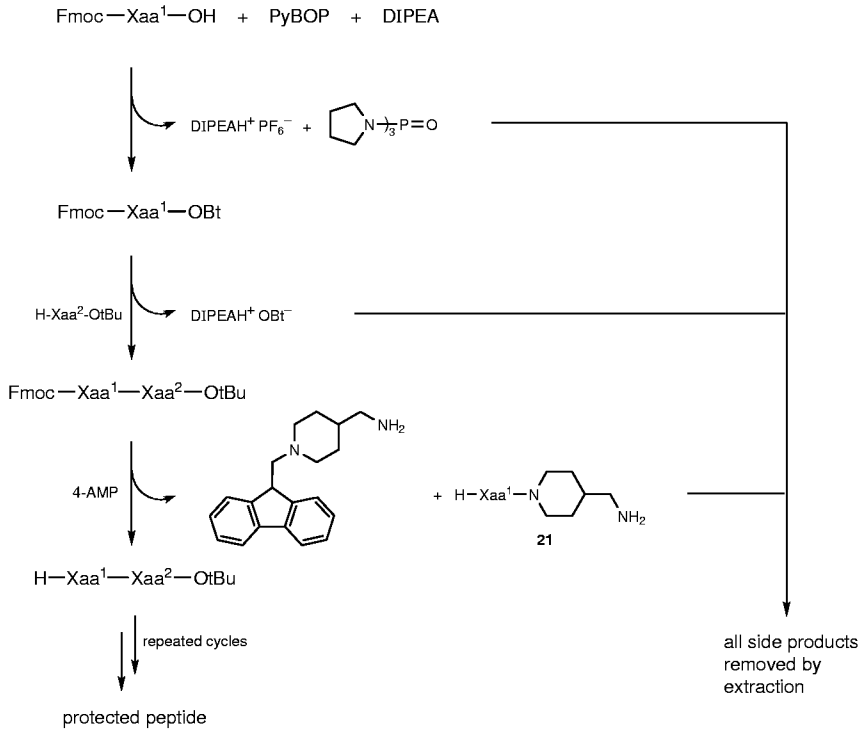
Due to the intrinsic advantages of the Fmoc group for temporary N^{α}-protection in combination with the semipermanent acid-labile *tert*-butyl alcohol based side-chain protection over the Boc/Bzl chemistry, this strategy can now probably be considered as the method of choice for the chemical synthesis of peptides. In fact, it has considerably enlarged the concept of solid-phase synthesis as discussed in more details in Section 4.3.2.2, since the chemistry involved in this strategy is well suited for the synthesis of most side-chain-modified peptides, including the phospho- and glycopeptides, as well as O-sulfated peptides. For experimental procedures regarding the use of N^{α}-Fmoc protection in the context of these particular peptide derivatives see Sections 6.3, 6.5, and 6.6.

for references see p 149

The coupling of N^{α}-protected amino acids can be performed with all the commonly utilized methods that are generally applied for the activation of the carboxy group, including as the most active species the related *N*-carboxyanhydrides,[219] acid chlorides,[214] and acid fluorides,[220] the latter also by in situ formation with TFFH.[221] The risk of racemization of even urethane-protected amino acids[52] is enhanced by the use of highly potent coupling reagents due to the increase in the acidity of the proton on the α-carbon and an enhanced tendency to oxazolone formation, both leading to racemization. In fact, significant racemization has been reported for Fmoc-Cys-OH, almost independently of the sulfur-protecting group,[222,225,226] and Fmoc-Ser(tBu)-OH[223,224] using uronium salts for activation in the presence of excess base. By coupling in the absence of base, e.g. via the pentafluorophenyl ester,[225,226] or even via TBTU/HOBt but with significantly decreased amounts of the base,[227] racemization is significantly reduced. Similarly beneficial is the use of hindered bases such as TMP.[222,223] Racemization of histidine which is largely promoted by its own ω-nitrogen, is prevented by the use of N^{ω}-*tert*-butoxymethyl protection.[228] Due to the difficult preparation of this derivative and complications deriving from the formaldehyde formed on removal, the N^{ω}-trityl derivative[229] is generally used, although racemization is expected to occur when couplings are sluggish because of steric hindrance or aggregation effects on resin.[230]

While in the solid-phase work with base-promoted Fmoc cleavage, the dibenzofulvene (DBF) and related amine adducts are removed by washing, in solution synthesis the lack of volatility and the reactivity of the DBF byproduct raise difficulties which make the Fmoc group less attractive for large-scale solution synthesis.[231] Piperidine in large excess as used in the solid-phase mode, serves as base to fragment the Fmoc group and as a scavenger to trap the liberated DBF via a Michael-type addition thereby outcompeting reaction with the product amine and preventing DBF polymerization.[92] As octanethiol was found to be an efficient DBF scavenger,[232] polymer-supported thiols were developed which allowed the reactive DBF to be trapped and to use catalytic amounts of DBU for efficient cleavage of Fmoc derivatives in solution.[233] Similarly by the use of 4-(aminomethyl)piperidine (4-AMP) for Fmoc deprotection the DBF–amine adduct is quantitatively extracted from the organic phase with phosphate buffer (pH 5.5), a fact which can be exploited in combination with Fmoc-protected amino acid chlorides for rapid continuous solution synthesis of peptides in a single repetitive operation as long as the growing peptide is soluble in a water-immiscible organic solvent.[214,215] As an alternative base for this rapid synthetic procedure tris(2-aminoethyl)amine (TAEA) has been proposed to facilitate separation of the DBF–amine adduct from the peptide.[234] This procedure has been successfully used for the synthesis in solution of numerous small peptides. It has also been used in a variation using PyBOP for the coupling steps (Scheme 6).[235]

Scheme 6 Rapid Route to Protected Fragments (Compound 21 Results from Quenching of Excess HOBt Ester)[235]



A similar one-pot deprotection–acylation procedure is realized using potassium fluoride for Fmoc cleavage, where intermediate isolation of the amine-free component is not required if coupling is performed with protected amino acid derivatives stable to fluoride anions.[236] Adopting the fluoride anion mediated Fmoc cleavage, a one-pot conversion from Fmoc-protected to Boc-protected amino acid and peptide is obtained with di-*tert*-butyl dicarbonate as acylating agent[237] (see Section 2.1.1.1.3.1.1.6); transacylation to the corresponding Z-protected derivatives was performed with *N*-(benzyloxycarbonyloxy)-5-norbornene-2,3-dicarboximide (Z-ONdc, **6**) and KF/TEA in DMF at room temperature.[238]

An additional difficulty in the solution synthesis of larger peptides arises from premature loss of the Fmoc group by the action of primary or secondary amines, which are the nucleophilic components of the coupling reactions, whereas cleavage rates by tertiary amines used as bases are slow and can be largely suppressed by acidic additives. Despite these possible difficulties successful syntheses have been reported.[239] The problem of premature N^{α}-deprotection is certainly faced to some extent even in the convergent strategies where N^{α}-protected peptide segments are coupled in solution or on resin to produce larger peptides and proteins.[19,189] Despite these inconveniences, only with the Fmoc-group protection schemes could be developed for the more demanding solution syntheses of glyco- and phosphopeptides or for the sulfated peptides.

Experimental procedures for synthesis of peptides on solid supports by the Fmoc/tBu chemistry are reported in Section 4.3.2.2.3, and for solution and solid-phase synthesis of glyco- and phosphopeptides or sulfated peptides in Sections 6.3, 6.5, and 6.6, respectively.

Method for Rapid Solution Synthesis of Peptides by the Fmoc/4-AMP Procedure; General Procedure:[215]

A 0.1 M soln of the C-terminal amino acid ester in $CHCl_3$ (5 mL) was combined with a soln of the appropriate Fmoc-Xaa-Cl (0.75 mmol) in $CHCl_3$ (5 mL) and 5% aq Na_2CO_3 (5 mL), and the two-phase mixture was stirred vigorously for 10 min (end point determined by TLC). The organic phase was

for references see p 149

separated and treated with 4-AMP (5 mL). After 30 min (end point determined by TLC), $CHCl_3$ (40 mL) was added, and the organic phase was washed with phosphate buffer, pH 5.5 [5 × 25 mL; prepared from $NaH_2PO_4 \cdot H_2O$ (90 g) and Na_2HPO_4 (32.7 g) in H_2O (500 mL)]. Alternatively, sat. NaCl (2 × 25 mL) was used to remove excess 4-AMP prior to extraction with the phosphate buffer. The organic phase was concentrated to a volume of 5 mL, and the resulting $CHCl_3$ soln was used analogously for treatment with the next N^{α}-Fmoc amino acid chloride; e.g. H-Phe-Phe-Gly-Leu-Nle-$NH_2 \cdot HCl$ was obtained in 79% yield,[215] and H-Tyr-Gly-Gly-Phe-Leu-OH in 56% overall yield by the PyBop coupling procedure.[235]

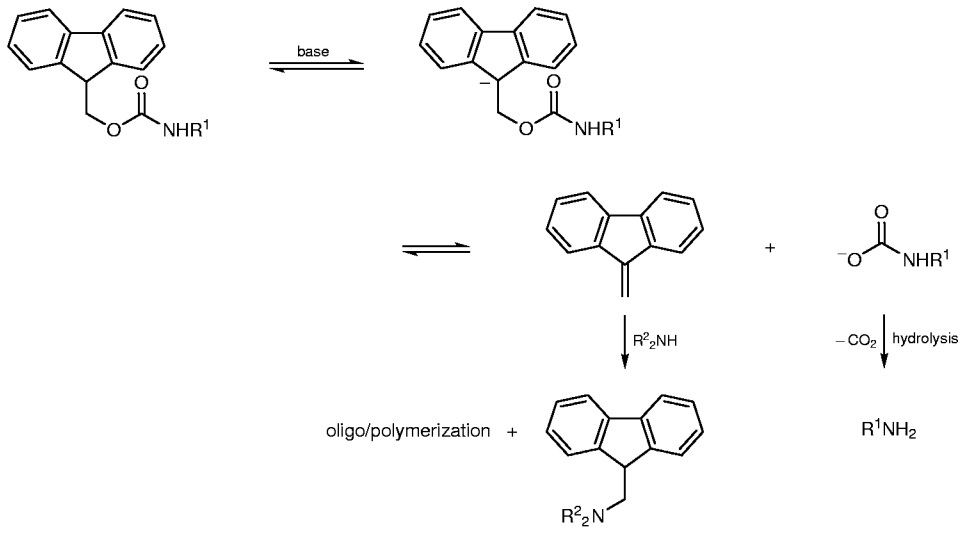
Conversion of Fmoc- into Z-Carbamates; General Procedure:[238]
To a stirred soln of Fmoc-protected compound (0.15 mmol) in DMF (1.4 mL) was added, under N_2, KF (1.05 mmol) followed by TEA (0.33 mmol). The soln was kept at rt and Z-ONdc (0.19 mmol) was added. After stirring for 7–12 h, the mixture was diluted with EtOAc, washed with H_2O, 5% HCl, 5% $NaHCO_3$, and brine, dried, and concentrated. The resulting crude material was purified by silica gel chromatography ($MeOH/CH_2Cl_2$). For example, the following compounds were obtained by this method: Z-Asp(OBzl)-Val-OMe; yield: 99%; Z-Ile-OMe; yield: 92%.

Z-Ala-Lys(Boc)-Gly-OMe; One-Pot Synthesis of Peptides with KF-Mediated Fmoc Cleavage:[236]
To a soln of Z-Ala-OH (511 mg, 2.29 mmol) in DMF (9 mL) was added DIPEA (617 mg, 4.77 mmol) and BOP-Cl (590 mg, 2.29 mmol) and after 3 min, the mixture was added to Fmoc-Lys(Boc)-Gly-OMe (1.03 g, 1.91 mmol) and KF (443 mg, 7.63 mmol) in DMF (10 mL). The mixture was stirred at rt until completion of the reaction. The crude product was purified by flash chromatography (EtOAc/hexane, 8:2); yield: 834 mg (84%); $[\alpha]_D^{26}$ −26.5 (*c* 0.5, MeOH).

2.1.1.1.1.3.3 Cleavage of the 9-Fluorenylmethoxycarbonyl Group

The versatility of the Fmoc group is given by its easy cleavage under weakly basic conditions (for a comprehensive review, see ref[184]), which makes this type of protection orthogonal to most of the other, generally acid-labile protecting groups. The use of anhydrous organic bases in deblocking reactions represents a novelty with regard to previous approaches and thus the key factor for the synthetic possibilities offered by this protecting group. The basic cleavage proceeds via the corresponding counter-base (E1cB mechanism), liberating the carbamate, which is further hydrolyzed to the free amine base. The protecting group is released as dibenzofulvene (DBF), which is trapped by the excess base used for cleavage (Scheme 7).[240–242] Further protecting groups that follow a similar mode of cleavage such as the Nsc group are reported in Section 2.1.1.1.1.4.

Scheme 7 E1cB Mechanism for the Cleavage of the Fmoc Group[240–242]



Cleavage of the *N*-Fmoc derivatives is fast with primary (cyclohexylamine, ethanolamine) and secondary amines such as piperidine or piperazine (Table 3), but substantially slower with tertiary amines (TEA, DIPEA).[25] The reaction is faster in polar solvents (DMF, NMP) than in apolar solvents (CH_2Cl_2).

Table 3 Reaction Conditions for Cleavage of the Fmoc Group[184][a]

Compound	Base	Solvent	Time (min)	Deprotection (%)	Ref
Fmoc-Gly-resin[b]	10% morpholine	CH_2Cl_2	240	18	[190]
Fmoc-Gly-resin[b]	10% morpholine	DMF	240	75	[190]
Fmoc-Gly-resin[b]	50% morpholine	CH_2Cl_2	240	100	[190]
Fmoc-Val-OH	50% morpholine	DMF	1	50	[243]
Fmoc-Ala-OtBu	50% morpholine	CH_2Cl_2	120	100	[25]
Fmoc-Gly-resin[b]	10% piperidine	CH_2Cl_2	240	100	[190]
Fmoc-Val-OH	20% piperidine	DMF	0.1	50	[243]
Fmoc-Gly-HMP-resin[b]	23% piperidine	NMP	0.25	50	[244]
Fmoc-Ala-OtBu	50% piperidine	CH_2Cl_2	<5	100	[25]
Fmoc-Val-OH	5% piperazine	DMF	0.33	50	[243]
Fmoc-Ala-OtBu	50% piperazine	CH_2Cl_2	60	100	[25]
Fmoc-Val-OH	50% DCHA	DMF	35	50	[243]
Fmoc-Ala-OtBu	50% DCHA	CH_2Cl_2	>1080	100	[25]
Fmoc-Val-OH	50% DIPEA	DMF	606	50	[243]
Fmoc-Ala-OtBu	50% DIPEA	CH_2Cl_2	>1080	100	[25]
Fmoc-Val-OH	10% DMAP	DMF	85	50	[243]
Fmoc-Ala-OtBu	50% DBU	CH_2Cl_2	<5	100	[25]
Fmoc-Ala-OtBu	50% pyrrolidine	CH_2Cl_2	<5	100	[25]
Fmoc-Ala-OtBu	50% DCHA	CH_2Cl_2	<5	100	[25]
Fmoc-Ala-OtBu	50% ethanolamine	CH_2Cl_2	<5	100	[25]
Fmoc-Ala-OtBu	50% DEA	CH_2Cl_2	180	100	[25]
Fmoc-Ala-OtBu	50% TEA	CH_2Cl_2	>1080	100	[25]
Fmoc-Ala-OtBu	50% NH_3	CH_2Cl_2	>1080	100	[25]
Fmoc-Ala-OtBu	50% Bu_3N	CH_2Cl_2	>1080	100	[25]
Fmoc-Ala-OtBu	1 mM DABCO	CH_2Cl_2	>1080	100	[25]
Fmoc-Ala-OtBu	1 mM $NH_2OH \cdot HCl$	CH_2Cl_2	>1080	100	[25]
Fmoc-Ala-OtBu	0.5 mM Proton-Sponge	CH_2Cl_2	>1080	100	[25]
Fmoc-Ala-OtBu	2 mM NaOH	30% MeOH/dioxane	<5	100	[25]

[a] Reprinted from (Fields, In *Methods in Molecular Biology*, Vol. 35), copyright (1994), p 17, with permission from Humana Press.
[b] Merrifield resin.

Among the amines proposed, piperidine is most frequently used in SPPS; it forms an adduct with dibenzofulvene which is readily removed by washing the resin. Typical conditions are 30% piperidine/DMF for 10 minutes,[245] 20% piperidine/DMF for 10 minutes,[180,246] 55% piperidine/DMF for 20 minutes,[247] 30% piperidine in toluene/DMF (1:1) for 11 minutes,[245,248–250] 23% piperidine/NMP for 10 minutes,[244] or 20% piperidine/NMP for 18 minutes.[251] Piperidine/CH_2Cl_2 should not be used, since a precipitate is formed, most probably deriving from decomposition of CH_2Cl_2, i.e. the piperidinium hydrochloride.[245] An inexpensive alternative to piperidine is diethylamine (DEA), with standard conditions being 60% DEA/DMF within 3 hours[252] or 10% DEA/DMA for 2 hours.[253]

The choice of deblocking amine strongly depends on the nature of the reaction system. For instance, the volatility of DEA or dimethylamine represents an advantage in peptide synthesis in solution.[239] The bulky character of diisopropylamine can be exploited for the selective removal of Fmoc protection in the presence of phthalimido residues which are sensitive to piperidine.[213] Consideration must also be given to possible subsequent reactions of the deprotected amine group with the initially formed byproduct dibenzofulvene. The factors governing the rates of adduct formation are still not completely understood, but both

for references see p 149

basicity and nucleophilicity as well as steric factors are clearly important. While piperazine and its *N*-methyl and *N*-phenyl derivatives, as well as piperidine and morpholine undergo facile adduct formation, the highly basic but sterically hindered 2,6-lutidine, 2,2,6,6-tetramethylpiperidine, and DBU do not. Among primary amines, propylamine and cyclohexylmethylamine react readily, whereas ethanolamine, ethylenediamine, and 2-methoxyethylamine require a large excess of amine or longer reaction times for extensive adduct formation.[92] DBU is a suitable substitute for piperidine, but it is only used in continuous-flow systems, since DBF does not form an adduct with DBU and thus must be washed rapidly from the peptide resin to avoid reattachment of DBF.[254] However, a solution of DBU/piperidine/DMF (1:1:48) is effective for batch syntheses, since the piperidine scavenges the DBF.

The byproduct DBF is not an easy compound to handle, as it polymerizes rapidly forming precipitates and gels. Thus adduct formation is a highly favorable side reaction; however, the process of adduct formation is an equilibrium reaction between DBF and the deblocking amine, where the product distribution at equilibrium depends on the solvent and identity of the amine.[92,255] This fact affects also the use of polymer-bound amines for Fmoc cleavage which would, upon adduct formation, facilitate separation of the DBF for synthesis in solution.[255,256] However, with nonquantitative adduct formation, this type of resin scavenger is more or less useless,[92] whereas, as discussed in Section 2.1.1.1.1.3.2, resin-bound thiols are much more efficient scavengers for DBF.

The Fmoc group is also efficiently cleaved by fluoride anions such as TBAF[257] or KF/18-crown-6.[258] Re-addition of DBF to the liberated amine is prevented by the addition of methanol (100 equiv) to 0.02 M TBAF · $3H_2O$/DMF to give the best reaction medium for the cleavage of Fmoc groups in SPPS; it is also reported to prevent piperazine-2,5-dione (DKP) formation (*vide infra*) and operating in a continuous flow mode, DBF does not accumulate in the system.[257] The use of thiols such as phenylmethanethiol or octane-1-thiol was found to be the most efficient method to trap the DBF.[232] With fluoride-mediated Fmoc cleavage, however, aspartimide formation, which is a side reaction in Fmoc/tBu chemistry, is not prevented. Moreover, the sensitivity of other protecting groups to fluoride anions such as silyl-based N- and O-protecting groups (see Section 2.1.1.1.1.7, 2.1.1.1.1.8, and 2.1.1.9) and ester-type linkages to the resin[259] as well as ω-*tert*-butyl esters[260,261] must be taken into account when selecting this deprotection procedure.

Although the C–O bond of Fmoc amino acid or Fmoc peptide derivatives should not be prone to hydrogenation, this urethane protecting group is cleaved hydrogenolytically with formation of 9-methylfluorene as the byproduct, but at rates significantly slower than the *N*-Z or *O*-benzyl derivatives.[92,262,263] Thus by appropriate monitoring of the reaction or by using a less active palladium catalyst[264] selectivity is retained to some extent, although this observation prevents the general use of the Fmoc group for orthogonal protection with respect to the Z group. Quantitative hydrogenolytic deprotection is achieved in 4–20 hours using 10% Pd/C in ethanol or MeOH/H_2O (3:1) with a few drops of acetic acid added.[262] The reaction is slower in DMF, but fairly rapid with palladium(II) acetate in this solvent.[263] Benzyl esters are selectively cleaved in the presence of Fmoc protection with poisoned catalysts; e.g. exclusive removal of the benzyl ester from Fmoc-Gly-OBzl is achieved with 10% Pd/$BaSO_4$ in methanol within ca. 1 hour,[265] or by addition of squaric acid derivatives.[266] In this context it is noteworthy that a one-pot conversion of benzyl into 9-fluorenylmethyl carbamates is possible by hydrogenolysis of the *N*-Z derivative over 10% Pd/C in the presence of 2,2′-bipyridine.[267]

While saponification of esters with sodium hydroxide in aqueous–organic media leads to relatively fast Fmoc cleavage, addition of calcium chloride allows for markedly selective ester hydrolysis.[268] Similarly, saponification of methyl esters is performed selectively in the presence of the *N*-Fmoc group with 0.2 M lithium hydroxide (2 equiv) in THF at 0 °C.[269]

Side Reactions

The facile base-(amine-) mediated cleavage of the Fmoc group, as expected, raises problems regarding the amine impurities in solvents such as DMF, DMA, or NMP,[263] which have to be freshly distilled prior to their use.[201] For precaution, particularly, when longer reaction times or storage are required, addition of HOBt is recommended.[270] Similarly, premature cleavage of the Fmoc group by the amino component as observed in solution synthesis[25,271] also occurs occasionally on solid phase.[185,272] The use of highly reactive coupling procedures is, therefore, recommended.[273]

An additional side reaction arising from the basic Fmoc cleavage is the rapid piperazine-2,5-dione (DKB) formation, especially when Pro, Gly, D-amino acids, or N-methylated amino acids are present in the peptide strand. In SPPS, piperazine-2,5-dione formation is suppressed with higher concentrations of piperidine and thus shorter reaction times, and by the use of highly activated amino acid derivatives for the peptide coupling steps;[274] more efficiently this side reaction, at least in C-terminal dipeptides, is prevented by using sterically hindered supports such as the 2-chlorotrityl resin.[275] In the most severe cases, e.g. with Pro-Pro sequences, intramolecular aminolysis can not be prevented, and the only recommendable remedy is the use of suitable Fmoc-protected peptide blocks.[276–281,842,843] This, however, implies a risk of epimerization at the C-terminal amino acid on coupling of the peptide block, and precautions have to be taken to address this problem. Fmoc cleavage with TBAF quenched with methanol has also been recommended (*vide supra.*)[257]

Cyclization of aspartic acid and asparagine to form aspartimides, and to a lesser extent of glutamic acid and glutamine to form glutarimide is an acid- and base-catalyzed common side reaction in peptide synthesis (see also Section 2.2.2). In SPPS it is particularly troublesome when Asp-Gly, Asp-Ala, and Asp-Ser sequences are present,[282] but also with Asp-Asn.[283,284] Piperidine-catalyzed aspartimide formation can be very rapid,[285,286] and in this context DBU is even worse than piperidine.[287] The formation of aspartimide is reduced by the addition of HOBt or 2,4-dinitrophenol, but more efficiently it is reduced by protecting the aspartyl amide bond with the 2-hydroxy-4-methoxybenzyl (Hmb) group (see Section 2.3.2).[288]

C-Terminal cysteine residues linked via ester bonds to the solid support suffer extensive epimerization in the repetitive piperidine-mediated Fmoc cleavage steps of SPPS.[289–294] Moreover, β-elimination occurs to some extent from the S-protected cysteine residues with formation of dehydroalanine that reacts with the excess piperidine to form β-(piperidino)alanine as an adduct.[254,294–297] These cysteine-related side reactions are thoroughly discussed in Section 2.6.6.

Protocols used for standard Fmoc cleavage in SPPS and related spectroscopic monitoring techniques are reported in Section 4.3.2.2, while protocols used for Fmoc removal in solution- and solid-phase synthesis of glyco- and phosphopeptides, and O-sulfated peptides are reported in Sections 6.3, 6.5, and 6.6, respectively.

Ala-Gly-Cys(tBu)-Lys-Asp-Phe-Phe-Trp-Lys-Thr-Phe-Thr-Ser-Cys(tBu)-OH, Fmoc Cleavage with Piperidine on Resin; Typical Procedure:[247]

Fmoc-Ala-Gly-Cys(tBu)-Lys-Asp-Phe-Phe-Trp-Lys-Thr-Phe-Thr-Ser-Cys(tBu)-$OCH_2C_6H_4$-$OCH_2C_6H_4$-resin (3.45 g, 0.846 mmol) was N^{α}-deprotected by following steps (35 mL each): (a) CH_2Cl_2, 2×2 min; (b) DMF, 2×2 min; (c) 55% piperidine/DMF, 20 min; (d) DMF, 2×2 min; (e) H_2O/dioxane (1:2), 2×5 min; (f) DMF, 3×2 min; (g) CH_2Cl_2, 3×2 min; (h) DMF, 2 min. The resin was then treated with a mixture of 55% TFA in CH_2Cl_2 (35 mL), anisole (7 mL), and 1,2-ethanedithiol (0.35 mL), and stirred under N_2 for 2.5 h. The mixture was filtered and the resin washed with CH_2Cl_2 (65 mL). The combined filtrate and washings were precipitated with anhyd Et_2O and the product was reprecipitated (MeOH/Et_2O); yield: 995 mg (67%); mp 240 °C (dec).

for references see p 149

H-Ser(Ac_3DGal)-Asp-Arg-Asp-Tyr(SO_3H)-Met-Gly-Trp-Met-Asp-Phe-NH_2, Fmoc Cleavage with Morpholine; Typical Procedure:[298]
To a soln of Fmoc-Ser(Ac_3DGal)-Asp-Arg-Asp-Tyr(SO_3H)-Met-Gly-Trp-Met-Asp-Phe-NH_2 (260 mg, 0.132 mmol) in DMF (10 mL) under argon was added freshly distilled morpholine (10 drops). After 10 h at rt, the soln was concentrated and the residue was precipitated from DMF with EtOAc; yield: 0.22 g (98%).

H-Trp-Leu-Asp(OtBu)-Phe-NH_2, Fmoc Cleavage with Dimethylamine in Solution; Typical Procedure:[299]
A soln of Fmoc-Trp-Leu-Asp(OtBu)-Phe-NH_2 (0.7 g, 0.82 mmol) in 10% Me_2NH in DMF (5 mL) was stirred for 5 min, and the solvent removed under reduced pressure. The residue was triturated with petroleum ether and filtered off; yield: 0.47 g (90%); mp 181–184 °C.

H-Asp(OtBu)-pNA by Fmoc Cleavage with DBU in Solution; Typical Procedure:[300]
A soln of Fmoc-Asp(OtBu)-pNA (2.13 g, 4.0 mmol) and DBU (609 mg, 4.0 mmol) in dry DMF (40 mL) was stirred at rt for 1 h. The mixture was diluted with Et_2O (200 mL) and extracted with 1 M HCl (3 × 20 mL). The aqueous extracts were washed with Et_2O and neutralized with K_2CO_3 (8.3 g, 60 mmol). The aqueous soln was extracted with Et_2O (9 × 50 mL), and the combined extracts were concentrated and taken up in EtOAc/hexane (1:1, 50 mL). The soln was filtered through a pad of silica gel to remove traces of DBU. The soln was evaporated to a solid; yield: 955 mg (77%); the product can be recrystallized (EtOAc/cyclohexane).

H-Lys(Boc)-AMC by Fmoc Cleavage with DBU/Octane-1-thiol; Typical Procedure:[233]
To Fmoc-Lys(Boc)-AMC (15.9 g, 25.5 mmol) in dry THF (250 mL) was added octane-1-thiol (44.2 mL, 255 mmol) followed by dropwise addition of catalytic amount of DBU (114 µL, 0.77 mmol, 3 mol%) over 15 min. The reaction was kept under N_2 and monitored by TLC or HPLC. After 4 h at rt, the solvent was removed, the residue triturated with Et_2O and filtered off; yield: 10.3 g (100%).

Fmoc Cleavage with TBAF on Resin, Typical Procedure:[257]
Starting from Fmoc-Val-$OCH_2C_6H_4OCH_2$-resin the peptide chain was prolonged to Fmoc-Leu-Ala-Gly-Val using a standard protocol except for the intermediate deprotection steps which were achieved with 0.1 M TBAF·$3H_2O$/DMF (2 × 2 min). The resin was treated with 90% TFA/H_2O to afford Fmoc-Leu-Ala-Gly-Val-OH; yield: 73%; mp 197–200 °C (dec); $[\alpha]^{20}_D$ –25.5 (*c* 1, EtOH).

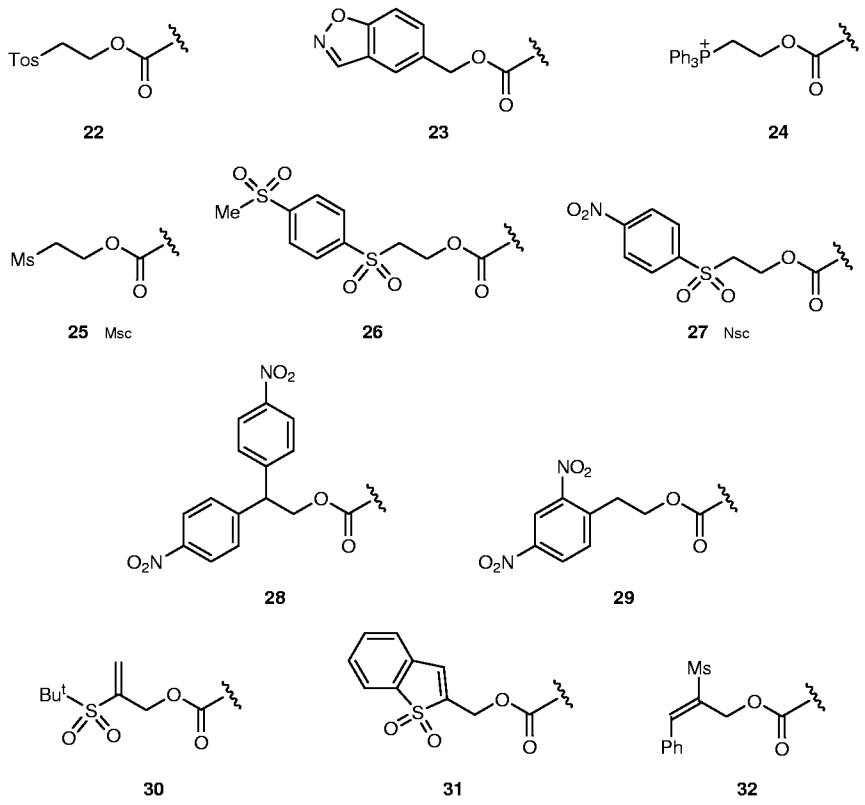
2.1.1.1.1.4 Substituted Sulfonylethoxycarbonyl Groups

J. Podlech

β-Elimination was first exploited by Kader and Stirling[301] in the design of base-labile urethane protecting groups with the 2-tosylethoxycarbonyl residue (**22**) (Scheme 8). In addition to the classical Fmoc group (see Section 2.1.1.1.1.7), various urethane-type protecting groups have been developed which undergo β-elimination under alkaline conditions[185] such as benzisoxazol-5-ylmethoxycarbonyl (**23**),[302] 2-(triphenylphosphonio)ethoxycarbonyl (**24**),[303] 2-(methylsulfonyl)ethoxycarbonyl (Msc, **25**),[27] 2-[4-(methylsulfonyl)phenylsulfonyl]ethoxycarbonyl (**26**),[29] 2-[4-(nitrophenyl)sulfonyl]ethoxycarbonyl (Nsc, **27**),[29,304] 2,2-bis(4-nitrophenyl)ethoxycarbonyl (**28**),[305] and 2-(2,4-dinitrophenyl)ethoxycarbonyl (**29**)[306] (Scheme 8). Additional *para*-substituents of the phenylsulfonyl moiety were examined, e.g. bromo and trifluoromethyl.[307]

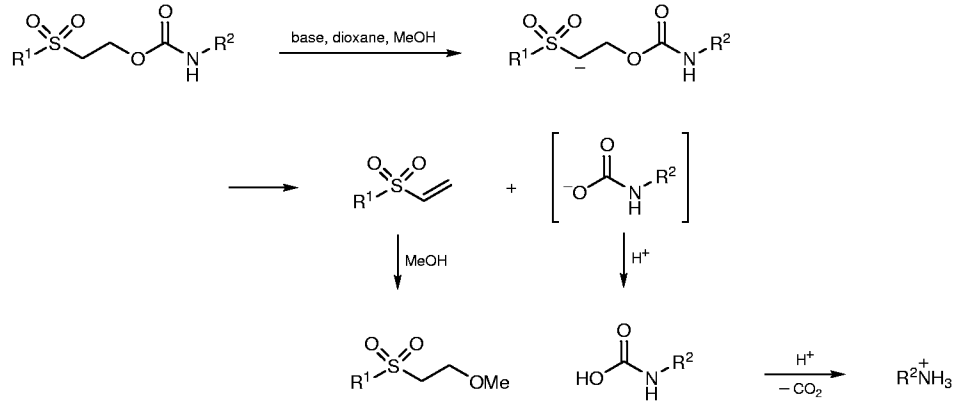
A similar principle has been utilized with protecting groups, which are cleaved by nucleophilic attack to a Michael system forming an enolate that subsequently eliminates the carbamate anion. Protecting groups following this principle are 2-(*tert*-butylsulfonyl)allyloxycarbonyl (**30**),[308] 1,1-dioxobenzo[*b*]thien-2-ylmethoxycarbonyl (**31**),[309] and 2-(methylsulfonyl)-3-phenylallyloxycarbonyl (**32**)[310] (Scheme 8).

Scheme 8 Base-Labile Urethane Protecting Groups



All the sulfonylethyl carbamates are stable, even towards strong acids, but easily cleaved with bases, and thus allow for orthogonal amino-protection in respect to acid-labile groups in a manner similar to the more familiar Fmoc group. Their deprotection occurs via an E1cB mechanism, where the fragmentation product vinyl sulfone is immediately attacked by a nucleophile, either the base or the solvent (Scheme 9). The intermediate anion of the carbamoic acid is stable and carbon dioxide only evolves on acidification of the solution.[27] The sulfone moiety in the protecting group results in increased solubility of the amino acid or peptide derivatives in polar solvents, occasionally even in water. Generally, the amino acid derivatives are crystalline and easy to purify.

Scheme 9 Basic Cleavage of Sulfonylethyl Carbamates[27]



for references see p 149

2.1.1.1.1.4.1 2-(Methylsulfonyl)ethoxycarbonyl Group

The N^{α}-2-(methylsulfonyl)ethoxycarbonyl (Msc, **25**) protection is highly stable against acidic conditions (even towards 12 M HCl, 40 °C, 1 h),[27] and in contrast to the Fmoc group which is known to be partially cleaved also by hydrogenolysis,[92,262,263] the Msc residue is fully resistant to catalytic hydrogenation without poisoning the catalyst.[311] In solution synthesis the Msc group represents a valid alternative to the N^{α}-2-nitrophenylsulfanyl (Nps) or N^{α}-2-biphenyl-4-ylprop-2-yloxycarbonyl protection in the presence of sulfur-containing amino acid residues that prevent application of the classical Z/tBu strategy. The N^{α}-Msc-protected amino acids are readily synthesized using 2-(methylsulfonyl)ethyl 4-nitrophenyl carbonate (Msc-ONp) in 69–100% yield.[27] Alternatively, the commercially available 2-(methylsulfonyl)ethyl succinimido carbonate (Msc-OSu) can be used for the introduction of the Msc group.[312] Optimized conditions for Msc cleavage consist of the mixture DMF (or dioxane)/MeOH/4 M NaOH, (14:5:1) where deprotection occurs in 10–20 seconds.[313] Alternatively, deprotection is achieved with 0.05 M $Ba(OH)_2$ within 10 minutes at room temperature; under these conditions even sulfate esters are not cleaved[314] (see Section 6.6). The Msc group has been used in synthesis in solution in general, but more extensively for particular synthetic purposes such as in the semisynthesis of insulin analogues[312] and of cytochrome c.[315] Generally, Msc protection was found to favorably affect the solubility of peptides in polar solvents.

Aspartic acid β-esters are incompatible with this group since cyclization to aspartimide occurs during the base-mediated deprotection. Even aspartic acid β-*tert*-butyl ester residues undergo this reaction with the related α→β transpeptidation (see Section 2.2.2).[27,314,316] Similarly, with C-terminal asparagine *tert*-butyl ester rapid succinimide formation was observed.[316]

The Msc group is well-suited for side-chain protection of basic amino acids (see Section 2.1.2.2.3.1.3), but has also been proposed as suitable for the protection of the guanido group of arginine either as the mono- or bis-Msc derivative of Boc-Arg-OH.[317]

2-(Methylsulfonyl)ethyl 4-Nitrophenyl Carbonate (Msc-ONp):[27]
To a cooled (0 °C) soln of 2-(methylsulfonyl)ethanol (80.5 g, 648 mmol) in abs pyridine (200 mL) was added with stirring 4-nitrophenyl chloroformate (119 g, 590 mmol). The mixture was left for 5 h at rt, concentrated, and poured into 1.5 M HCl (1.5 L). The product crystallized and was collected by filtration, washed with iPrOH (4 × 50 mL) and iPr_2O (3 × 50 mL); yield: 128 g (75%); mp 102 °C.

N^{α}- and N^{ω}-Msc-Protected Amino Acids; General Procedure:[27]
The amino acid (20 mmol) was suspended or dissolved in MeCN/H_2O (25 mL, 4:1), and Msc-ONp (5.78 g, 20 mmol) was added. The reaction was started by addition of TEA (2.8 mL, 20 mmol) and the mixture was stirred for 15 min (Boc-Lys-OH, Z-Lys-OH), 1 h [H-Ala-OH, H-Asp-OH, H-Glu-OH, H-Glu(OtBu)-OH, H-Gly-OH, H-Pro-OH, H-Ser-OH, H-Thr-OH, H-Tyr(tBu)-OH, H-Trp-OH], or >3 h (H-Arg-OH, H-Ile-OH, H-Leu-OH, H-Met-OH, H-Phe-OH, H-Val-OH), respectively. The mixture was filtered, concentrated, and dissolved in H_2O. (A) When the N^{α}-Msc-protected amino acid was considerably soluble in cold H_2O, the residue was passed through strongly acidic ion-exchange resin (H^+), the resin was washed with H_2O, and the eluent was concentrated to give the pure amino acid derivative; yield: 66–96%. (B) Msc Derivatives showing a limited solubility in cold H_2O [Msc-Glu(OtBu)-OH, Msc-Leu-OH, Msc-Phe-OH, Msc-Trp-OH, Msc-Tyr(tBu)-OH] were liberated by addition of 2 M $KHSO_4$ (pH 5). Either the product crystallizes, or the oil was extracted into EtOAc after addition of NaCl. Removal of the solvent yielded the N^{α}-Msc amino acid derivatives; yield: 75–100%.

H-[4,5,6,7-F_4]Trp-Met-Asp-Phe-NH_2; Typical Procedure for Msc Cleavage:[314]
To a suspension of Msc-[4,5,6,7-F_4]Trp-Met-Asp-Phe-NH_2 (819 mg, 1 mmol) in dioxane/MeOH (20 mL, 14:5) was injected 1 M NaOH (4 mL, 4 mmol) under vigorous stirring. After about 1 min, the clear soln was acidified to pH 4, the resulting suspension was concentrated and after addition of H_2O (20 mL) the pH was adjusted to 7. The precipitate was collected by filtration, washed with H_2O, and dried; yield: 562 mg (84%); mp 227–229 °C.

2.1.1.1.1.4.2 2-(Methylsulfanyl)ethoxycarbonyl Group

Based on the properties of the 2-(methylsulfonyl)ethoxycarbonyl (Msc) group, with the 2-(methylsulfanyl)ethoxycarbonyl group the safety-catch principle was realized.[318] This group is stable to acids and bases under moderate conditions, but can be converted into the sulfone by oxidation (ammonium molybdate/H_2O_2), i.e. to the Msc group and then be removed as described in the preceding section. 2-(Methylsulfanyl)ethoxycarbonyl-protected amino acids are easily prepared from the amino acid benzyltrimethylammonium salts with either 2-(methylsulfanyl)ethyl phenyl carbonate or with 2-(methylsulfanyl)ethyl 4-nitrophenyl carbonate. The former method gives higher yields (73–90%), most probably due to the easier removal of phenol as byproduct than 4-nitrophenol. To date this promising approach has not received due attention.

2.1.1.1.1.4.3 2-[(4-Nitrophenyl)sulfonyl]ethoxycarbonyl Group

The 2-[(4-nitrophenyl)sulfonyl]ethoxycarbonyl group (Nsc, **27**) has been developed as an alternative to the Fmoc group for solid-phase peptide synthesis,[304,319–321] to counter certain shortcomings of the Fmoc/tBu strategy such as the insufficient stability of N^{α}-Fmoc-protected amino acids and peptides in weakly basic conditions[263,272] and unfavorable hydrophobic interactions and steric hindrance of the bulky Fmoc group.[92,322]

N^{α}-Nsc amino acids are crystalline compounds, a physical property strongly preferred for automated SPPS. In terms of coupling efficiency, N^{α}-Nsc-protected amino acids are comparable to the related Fmoc derivatives.[272,323] Racemization as observed by coupling of N^{α}-Fmoc-protected His,[230] Cys,[225,226] and Ser[223] with standard procedures (TBTU or HBTU/DIPEA 1:2) was significantly reduced in the case of N^{α}-Nsc amino acids, a fact that is possibly correlated with the lower acidity of the β-proton of the protecting group that results in the lower acidity of the α-proton of the amino acid.[272] The lower acidity of this β-proton is also responsible for the observed higher stability of the Nsc derivatives in DMF[323] and the more basic conditions required for its removal in a short period of time (Table 4). The 3–8-fold lower rate of Nsc cleavage if compared to the Fmoc group under standard 20% piperidine in DMF[28] is compensated by addition of 1% DBU to 20% piperidine in DMF or DMF/dioxane (1:1) as shown in Table 4.[319] The deprotection can be monitored in real-time by measuring the UV absorption of the solution resulting from the cleavage (no dilution is necessary before measurements are taken, since the intensity of the vinyl sulfone is much lower than that of dibenzofulvene).[323] Unlike dibenzofulvene, the primary product of Nsc cleavage, 4-nitrophenyl vinyl sulfone, does not tend to polymerize and is trapped very effectively by piperidine.[319] The recurring problem of Asp-X rearrangement to succinimide which in the Fmoc/tBu SPPS leads during the repeated piperidine-mediated cleavage of the N^{α}-Fmoc group to α- and β-piperidides of aspartic acid,[283,284] is apparently reduced significantly,[323] although the use of DBU, even at a concentration of 1%, can cause aspartimide formation in sensitive sequences.[324] Possibly the use of DMF/dioxane (1:1) is favorable in this context.

for references see p 149

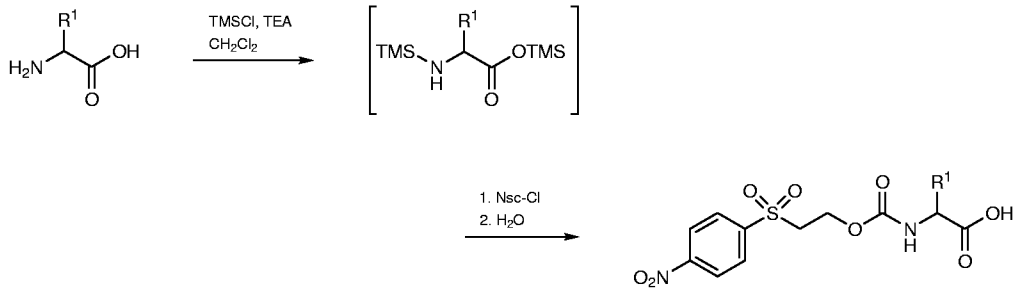
Table 4 Comparison Between Fmoc and Nsc Protected Peptides[323]

Conditions	Fmoc	Nsc	Ref
20% piperidine/DMF	10–15 s[a]	90–110 s[a]	[323]
1% DBU/20% piperidine in DMF/dioxane (1:1)	–[b]	12–15 s[a]	[323]
decomposition in DMF solution			[323]
1 week	10%	<1%	[323]
3 weeks	40%	2%	[323]
alkene/amine adduct formation	fast, reversible	very fast, irreversible	[323]
polymerization during deprotection	yes	no	[323]
available UV-monitoring range	302 nm	380 nm	[323]

[a] Cleavage rate ($t_{1/2}$).
[b] Not reported.

The N^{α}-Nsc amino acids are easily synthesized and commercially available (Hyundai Pharm./Bucheon, Korea), crystalline, and stable compounds (Scheme 10). They are best prepared by acylation of the trimethylsilylated amino acids with 2-[4-(nitrophenyl)sulfonyl]ethyl chloroformate (Nsc-Cl)[29] in nonaqueous solution (78–90% yield).[304]

Scheme 10 Synthesis of N^{α}-Nsc Amino Acids[29,304]



2-[(4-Nitrophenyl)sulfonyl]ethyl Chloroformate (Nsc-Cl):[29]

CAUTION: *Phosgene is a colorless, poisonous gas or volatile liquid. This reagent should be handled with the utmost care.*

To a soln of 1-chloro-4-nitrobenzene (40.4 g, 256 mmol) in 2-sulfanylethanol (46 mL, 654 mmol), was added 4 M NaOH in H_2O (72 mL, 288 mmol). The two-phase system was vigorously stirred at 70 °C for 3 h. During this period, EtOH (15 mL) was added along the inner walls of the vessel to return creeping crystalline 1-chloro-4-nitrobenzene. The cooled soln was extracted with CH_2Cl_2 (200 mL) and the organic layer was washed with 10% Na_2CO_3 (3 ×) and with aq $KHSO_4$. The organic layer was washed with H_2O until neutral, then dried, concentrated, and crystallized (iPr_2O, 150 mL) to give 2-(4-nitrophenylsulfanyl)ethanol; yield: 43.7 g (86%); mp 60 °C.

A fraction of this 2-(4-nitrophenylsulfanyl)ethanol (5.0 g, 25 mmol) was dissolved in EtOH (25 mL) and 2% aq $Na_2WO_4 \cdot 2H_2O$ soln (3 mL) was added. 35% H_2O_2 (2 mL) was added slowly to the warmed soln (40 °C), the temperature was raised to 80 °C and another portion of 35% H_2O_2 (2 mL) was added. The mixture was refluxed for 1 h to complete the oxidation and decompose traces of H_2O_2. 2-(4-Nitrophenylsulfonyl)ethanol crystallized upon cooling (0 °C); yield: 5.0 g (86%); mp 128 °C.

$COCl_2$ (57 mL) was condensed in a 100-mL three-necked flask and, with stirring, a soln of 2-(4-nitrophenylsulfonyl)ethanol (10.0 g, 43 mmol) in THF (75 mL) was added slowly at an internal temperature of −78 °C. The cooling bath was removed and the reaction flask was closed with a $CaCl_2$ tube and left in the hood at rt overnight. The following day, residual $COCl_2$ and HCl were removed by purging with N_2 for 1 h after which time the solvent was removed; yield: 12.6 g (99%); mp 119–122 °C.

N^{α}-Nsc Amino Acids; General Procedure:[304]

To a soln of the amino acid (20 mmol) in CH_2Cl_2 (40 mL) was added TEA (40 mmol) and TMSCl (40 mmol) dropwise with stirring. The mixture was heated to reflux for 1.5 h, cooled (0 °C), and Nsc-Cl (17 mmol) was added. After stirring at 0 °C for 30 min and at rt for 2 h, the soln was concentrated, and the residue partitioned between Et_2O and 5% $NaHCO_3$. The aqueous layer was acidified to pH 2–2.5 with 1 M H_2SO_4, extracted with EtOAc, dried (Na_2SO_4), and concentrated. The residues were recrystallized (generally EtOAc/petroleum ether). For example: Nsc-Ser(tBu)-OH; yield: 82%; mp 109–111 °C; $[\alpha]_D^{25}$ +3.3 (*c* 1, DMF); Nsc-Asp(OtBu)-OH; yield: 83%; mp 73–75 °C; $[\alpha]_D^{25}$ −12.7 (*c* 1, DMF).

H-Ala-Ser-Ser-Thr-Ile-Ile-Lys-Glu-Gly-Ile-Asp-Lys-OH, SPPS with N^{α}-Nsc Amino Acids; Typical Procedure:[304]

Synthesis was performed on aminomethylated 1% divinylbenzene–styrene copolymer (1.01 mequiv $NH_2 \cdot g^{-1}$) modified with [4-(hydroxymethyl)phenoxy]propanoic acid. The resin (150 mg) was placed into a polypropylene syringe reactor with inserted polypropylene porous disc and esterified with the C-terminal amino acid. [Nsc-Lys(Boc)-OH (0.6 mmol), DCC (0.6 mmol), DMAP (0.06 mmol), CH_2Cl_2/NMP (2:1, 3 mL), rt, 12 h] After careful washings ($CHCl_3$, DMF), the synthesis was performed as follows: (1) prewash: 33% piperidine/DMF, 4 mL, 0.5 min; (2) deblocking: 33% piperidine/DMF, 4 mL, 15 min; (3) wash: DMF (6 × 4 mL, 0.5 min); (4) coupling: N^{α}-Nsc amino acid [Asp(OtBu), Ile, Gly, Glu(OtBu), Lys(Boc), Ile, Thr(tBu), Ser(tBu), Ala in the required order, 0.5 mmol], BOP (0.5 mmol), HOBt (0.5 mmol), NMM (0.75 mmol), DMF (2 mL), 60 min (for Nsc-Ile-OH, 90 min); (5) wash: DMF (5 × 4 mL, 0.5 min). After completion of peptide assembly, the peptidyl-resin was thoroughly washed with $CHCl_3$, dried, and treated with 50% TFA in CH_2Cl_2 (rt, 60 min). The resin was filtered off and washed with 50% TFA in CH_2Cl_2. The combined washings were then diluted with cold, dry Et_2O. The precipitate was filtered, washed with Et_2O and dried (170 mg of crude product) which was further purified by gel-permeation chromatography (6% aq AcOH) and lyophilized; yield: 104 mg (41%).

2.1.1.1.1.5 Allyloxycarbonyl Group

A. Loffet, F. Guibé, E. Lohof, and J. Podlech

The protection of amino groups as allyloxycarbonyl (Aloc) derivatives has been known for many years.[31] Unfortunately, apart from cleavage by strong acids such as HBr in acetic acid, no specific methods were available for the deprotection of N^{α}-Aloc derivatives; even using hydrogenolysis only partial removal occurs due to conversion of the Aloc residue to the highly stable propoxycarbonyl derivative.[31] Correspondingly, this urethane-type protecting group did not offer any advantage over others such as the well-established benzyloxycarbonyl group[139] until the development of the Tsuji–Trost reaction[325,326] which offered new possibilities for the use of allyl-based protecting groups in the synthesis of peptides both in solution and on solid supports (for reviews, see refs[30,327]). In fact, allylic protecting groups are selectively cleaved through catalytic η^3-allylpalladium chemistry, i.e. under mild and highly selective conditions that make the allyl-type protecting groups orthogonal to a variety of other types of protection such as the widely used acid-labile *tert*-butyl alcohol derived protection and the base-labile N^{α}-Fmoc derivatives. An obvious limitation of allyl protection is the incompatibility with reagents that affect the C=C bond such as oxidizing agents and some reducing conditions like those used for the hydrogenolysis of benzyl-type protecting groups.

2.1.1.1.1.5.1 Synthesis of N^{α}-Allyloxycarbonyl Amino Acids

The protection of α-amino groups is usually effected with allyl chloroformate under Schotten–Baumann conditions,[31] but diallyl dicarbonate ($Aloc_2O$) can also be used.[328] As with other chloroformates, a severe side reaction is the formation of N^{α}-Aloc-protected dipeptides. Since most N^{α}-Aloc amino acids are not crystalline compounds, they are preferentially isolated as the corresponding DCHA salts (for N^{α}-Aloc amino acids and their analytical characterization, see ref[329]).

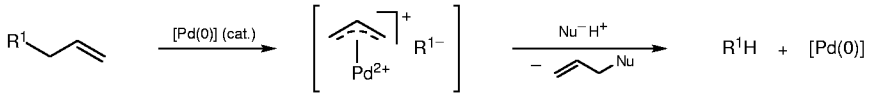
for references see p 149

Synthesis of N^{α}-Aloc Amino Acids; General Procedure:[31]
To a stirred soln of the amino acid (0.1 mol) in 4 M NaOH (25 mL) was added at 0 °C Aloc-Cl (10.6 mL, 0.1 mol) and 4 M NaOH (25 mL) alternately in 8 portions within ~30 min to keep the soln alkaline. After an additional 15 min at rt, the mixture was extracted with Et_2O, and the aqueous layer was acidified with concd HCl. After cooling for several hours, the derivatives, which crystallized, were collected, dried, and recrystallized. If the product was not crystalline, the cold soln was extracted with Et_2O, and the organic layers were concentrated to an oil.
The DCHA salts were obtained by adding the amine (1 equiv) to a soln of the oily Aloc derivatives in EtOH or EtOAc and precipitation with Et_2O or Et_2O/petroleum ether; for yields and analytical characterization, see ref[329].

2.1.1.1.1.5.2 Cleavage of the Allyloxycarbonyl Group

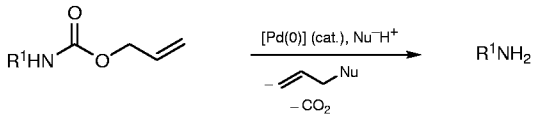
The deprotection of allyl-protected functional groups is achieved through a palladium-catalyzed transfer of the allylic group to a nucleophilic species as represented in Scheme 11 (for a review, see ref[30]). The reaction proceeds through the intermediate formation of an electrophilic π-allyl complex, which then undergoes attack by the nucleophile. The cleavage between the allyl group and the function, therefore, occurs during the first step of the process. The second step allows the regeneration of the catalytic species.

Scheme 11 Palladium-Catalyzed Transfer of an Allyl Group to a Nucleophile[30]



Many complexes of palladium in the zerovalent state, differing by the nature of the ligand on the metal, can be used for allylic cleavage but in the majority of the cases it is the commercially available tetrakis(triphenylphosphine)palladium(0) that is used. Tetrakis(triphenylphosphine)palladium(0) partially dissociates in solution into triphenylphosphine and $Pd(PPh_3)_2$, which is believed to be the true catalytic species. Typically, 2 to 5 mol% of catalyst are used in deprotection reactions in solution but greater amounts are often required in solid-phase chemistry. The reaction is preferentially run under an inert atmosphere (argon, N_2). A necessary condition for successful removal of allyl groups is that the protected function possesses sufficient leaving group ability to allow the formation of the π-allyl complex. As far as peptide synthesis is concerned, allyl esters of carboxylic, phosphonic, and phosphoric acids, allyl phenoxides, and allyloxycarbonyl derivatives of amines are the main protected functions, which fulfill this condition. On the contrary allyl ethers or sulfides and allylamines in their nonprotonated form are not cleaved by the palladium(0) catalyst. In most cases allylic carbamates undergo decarboxylation concomitantly with allylic cleavage (Scheme 12) thus leading directly to the amine.

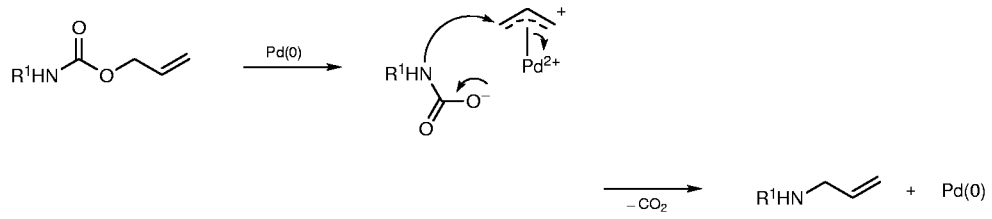
Scheme 12 Allyl Cleavage with Concomitant Decarboxylation of Allyl Carbamates[30]



The deprotected amine may then compete with the nucleophilic scavenger in the trapping of the η^3-allylpalladium complex. To avoid the formation of allylamines it is therefore better to use deprotection systems that lead to the amine in a protonated or another masked nonnucleophilic form or to use an allyl group scavenger in large excess. In the presence of the palladium catalyst alone allyl carbamates may also form allylamines[330,331] through unimo-

lecular decarboxylative rearrangement of the corresponding carbamato π-allyl complex[331] as represented in Scheme 13. Fortunately, this process, which is quite rapid in the case of carbamates derived from secondary amines, is relatively slow with Aloc derivatives of primary amines in general and N^{α}-Aloc derivatives of amino acids in particular.[30] It is, nevertheless, highly advisable not to expose N^{α}-Aloc derivatives to the catalyst without adding nucleophilic scavengers.

Scheme 13 Formation of Allylamines from Allyl Carbamates[30,330]

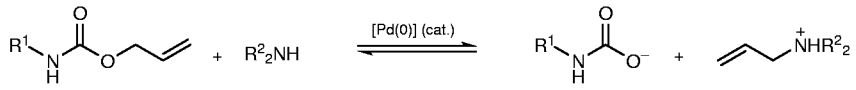


A large variety of nucleophiles may be used as allyl group scavengers in the palladium-catalyzed deprotection of allylic derivatives, including oxygen, nitrogen, carbon, or sulfur nucleophiles, and hydride donors.[30]

2.1.1.1.1.5.2.1 Deprotection with Secondary Amines

Following a first report by Kunz and Waldmann[332] secondary amines such as diethylamine, piperidine, and in particular the less basic morpholine[333] have been extensively used for the deprotection of allyl esters. As the allylammonium species formed in this process (Scheme 14) may also behave as an allyl donor towards palladium zerovalent complexes,[30] the use of an excess of scavenger is highly recommended in order to totally displace the equilibrium towards deprotection. Secondary amines can also be used in the deprotection of allyl carbamates. Here again an excess of scavenger is necessary to avoid kinetically competitive allylation of the liberated amine.

Scheme 14 Formation of an Allylammonium Species in the Deprotection of Allylic Groups with Secondary Amines[30]

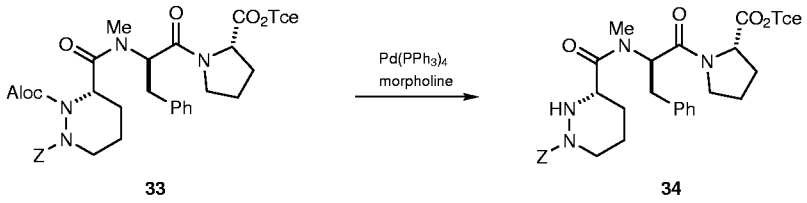


Most secondary amines, including morpholine, are not compatible with the presence of *N*-Fmoc derivatives. A method to ensure Fmoc compatibility is to use the weakly basic *N*-methylaniline as a scavenger.[334,335] Another possible solution is to use 4-methylmorpholine in multicomponent systems, i.e. DMSO/THF/aq 0.1 M HCl[336–338] or $CHCl_3$/AcOH.[339–341] In these cases the ultimate allyl group scavenger is probably not the tertiary amine but more likely an oxygen nucleophile (H_2O in the first case, the acetate anion in the second one). *N*-Hydroxysuccinimide, an oxygen nucleophile, is also Fmoc compatible.[342–344]

For experimental protocols regarding cleavage of *N*-Aloc groups on resin, see Section 4.3.2.3.3.

for references see p 149

Scheme 15 Cleavage of the Aloc Group with Pd(0) in the Presence of Morpholine[333,345]



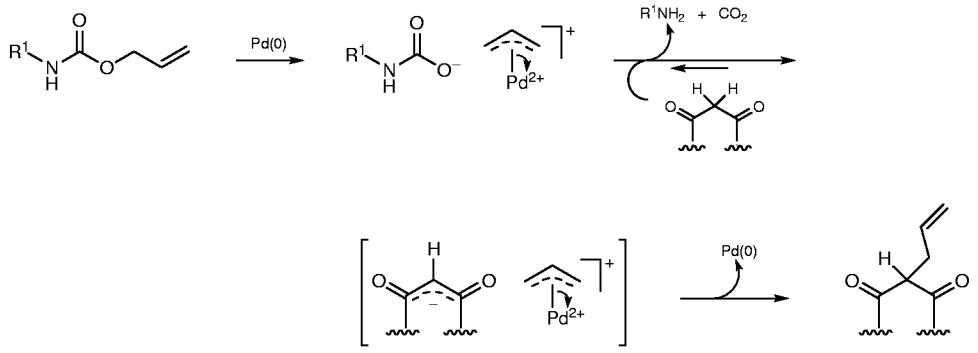
***N*-[(3*S*)-1-(Benzyloxycarbonyl)hexahydropyridazin-3-ylcarbonyl]-*N*-methyl-(*R*)-phenylalaninyl-(*S*)-proline 2,2,2-Trichloroethyl Ester (34) (Scheme 15):**[333,345]

The allyloxycarbonyl-substituted starting material **33** (1.7 g, 2.3 mmol) in abs THF (25 mL) was treated with morpholine (2 mL, 23 mmol) and $Pd(PPh_3)_4$. The mixture was stirred at rt for 2 h, concentrated under reduced pressure, triturated in EtOAc, washed with 10% citric acid, sat. $NaHCO_3$, and brine, dried ($MgSO_4$), concentrated under reduced pressure, and filtered through a short pad of silica gel (EtOAc) to give the product; yield: 1.4 g (94%); $[\alpha]_D^{25}$ +25.5 (*c* 0.91, $CHCl_3$).

2.1.1.1.1.5.2.2 Deprotection with Carbon Nucleophiles

5,5-Dimethylcyclohexane-1,3-dione (dimedone) and *N*,*N*-dimethylbarbituric acid are relatively acidic β-dicarbonyl compounds (pK_a 5.3 and 4.7, respectively). The deprotection reaction involves (Scheme 16) a prototropic equilibrium between the β-dicarbonyl compounds and the deprotected carbamate species (shifted further to right by the formation of carbon dioxide and the respective amine).

Scheme 16 Deprotection of Allylic Groups with Dimedone or *N*,*N*-Dimethylbarbituric Acid[30]



The enolate formed in this process then condenses irreversibly with the η^3-allylpalladium species. At the outset, dimedone and *N*,*N*-dimethylbarbituric acid had been more specifically devised[346,347] for the deprotection of allyl carbamates in solution or in solid-phase[348] chemistry in which case the enolate species also forms by proton transfer from the β-dicarbonyl allyl scavenger to the liberated amine; but since then it has also been used for the deprotection of allyl carboxylic esters (see Section 2.2.1.1.3).[349–351]

Due to the relative rapidity of the reactions and to the protonation of the free amine by the excess scavenger during the deprotection of allyl carbamates, the risk of formation of allylamine seems very limited and has not been reported. In the reactions with dimedone, a possible complication is the formation of substantial amounts of hydrolytically stable oxoenamine by condensation with the liberated amine. For this reason, the use of *N*,*N*-dimethylbarbituric acid is probably a safer solution.

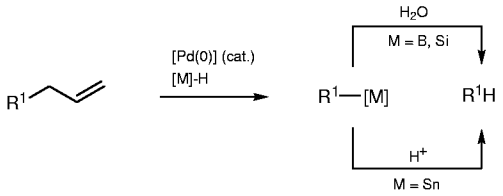
H-Met-Phe-OtBu; Typical Procedure:[346]
To a soln of Aloc-Met-Phe-OtBu (0.57 g, 1.30 mmol) in THF (10 mL) under argon was added $Pd(PPh_3)_4$ (0.15 g, 0.13 mmol) and 5,5-dimethylcyclohexane-1,3-dione (1.4 g, 10 mmol). The mixture was stirred for 30 min, the solvent distilled off under reduced pressure, the residue taken up in Et_2O (100 mL), filtered, and extracted with 0.5 M HCl (4 × 50 mL). The combined aqueous phases are adjusted to pH 10 with Na_2CO_3 and extracted with Et_2O (4 × 100 mL). The organic layers were combined, dried, and concentrated. The product was obtained as oil; yield: 0.41 g (88%); $[\alpha]_D^{22}$ +29.4 (*c* 1, $CHCl_3$).

2.1.1.1.1.5.2.3 Deprotection with Tin or Silicon Hydrides or Amine–Borane Complexes

Various hydride donors have been used as allyl group scavengers in palladium-catalyzed deallylation reactions leading to propene as the byproduct. These include formic acid[352] and several semimetal hydrides, including tributyltin hydride (Bu_3SnH),[329,353–355] phenylsilane ($PhSiH_3$),[356] lithium or sodium borohydrides,[357,358] and amine–borane complexes such as borane–ammonia complex ($BH_3 \cdot NH_3$) and borane–dimethylamine complex ($BH_3 \cdot NHMe_2$).[359] They are efficient not only for the deprotection of allyl carbamates and esters but also for the deprotection of allyl phenoxides, in contrast to the other allyl group scavengers mentioned (*vide supra*). With borohydride and tributyltin hydrides the air-stable palladium divalent complex dichlorobis(triphenylphosphine)palladium(II) may be used instead of tetrakis(triphenylphosphine)palladium(0) as it is reduced in situ to $Pd(PPh_3)_2$. Tributyltin hydride,[329] phenylsilane[360] and the amine–borane complexes[359] have been used in solid-phase peptide synthesis for Aloc cleavage as these procedures lead to complete deprotection in less than 10 minutes at room temperature. Strong activation by palladium of the otherwise poor nucleophile tributyltin hydride is likely to take place through the formation of an intermediate palladium hydrido species. Tributyltin hydride is not indefinitely stable in the presence of palladium zerovalent complexes but decomposes to hexabutyldistannane ($Bu_3SnSnBu_3$) and H_2. This process is sufficiently slow so as not to interfere with the deprotection reaction in the majority of cases. Phenylsilane gives slower deprotection reactions compared to tributyltin hydride, but is apparently indefinitely stable in the presence of the palladium catalyst.[356] When hydride donors and amine–borane complexes are used as allyl group scavengers the primary products of deprotection are not the free functional groups but the corresponding metallic salts (Scheme 17). For example, allyl carbamates do not decarboxylate but are converted into tin, silicon, or boron carbamates, which can be used as such in peptide coupling reactions. Otherwise isolation of the free functional groups from their metallic salts is readily achieved by protonolysis (tin salts) or by hydrolysis (silicon and boron salts). The hydrostannolytic deprotection of allyl carbamates is best carried out in the presence of a weak acid species[329] such as acetic acid under conditions in which the free amine is obtained directly without formation of allylamine as a potential side product.

It should be noted that tin salts or complexes are toxic materials.

Scheme 17 The Use of Hydride Donors To Form Metallic Salts[30]



Experimental procedures for the cleavage of N^{α}-Aloc derivatives on resin by palladium(0)-catalyzed hydrostannolysis are reported in Section 4.3.2.3.2.

for references see p 149

$Bu_3SnOC(=O)$-Phe-OBzl by Pd(0)-Catalyzed Hydrostannolytic Deprotection of Aloc-Phe-OBzl; Typical Procedure:[329]

To Aloc-Phe-OBzl (260 mg, 0.77 mmol) and $PdCl_2(PPh_3)_2$ (11 mg, 0.016 mmol) in CH_2Cl_2 (8 mL) and H_2O (80 µL) was added Bu_3SnH (220 µL, ca. 0.85 mmol). After 10 min, the solvent was removed and the crude product purified on silica gel by elution with cyclohexane/Et_2O (8:2) followed by EtOAc; the EtOAc fraction was concentrated to oil to give N^{α}-[(tributylstannyloxy)carbonyl]phenylalanine benzyl ester; yield: 0.33 g (74%). Alternatively, the residue is taken up in Et_2O and a sat. soln of TosOH (2.1 equiv) in Et_2O was added, the soln concentrated and the H-Phe-OBzl·TosOH filtered off.

Pd(0)-Catalyzed Aloc Cleavage with $PhSiH_3$; General Procedure:[360]

To a peptide-resin (200 mg) washed with CH_2Cl_2 (5 × 30 s) under argon was added a soln of $PhSiH_3$ (24 equiv) in CH_2Cl_2 (1 mL) and the resin was manually stirred. A soln of $Pd(PPh_3)_4$ (0.10 equiv) in CH_2Cl_2 (3 mL) was added, argon was passed through the resin and the reaction was mechanically stirred for 10 min. The peptide-resin was washed with CH_2Cl_2 (8 × 30 s) and the process was repeated once.

Pd(0)-Catalyzed Aloc Cleavage with $BH_3·NH_3$ or $BH_3·NHMe_2$; General Procedure:[359]

The deprotection was performed as with $PhSiH_3$ in CH_2Cl_2 at rt by adding first the $BH_3·NH_3$ or $BH_3·NHMe_2$ (6 equiv) and then 10 mol% (based on the substitution level of the resin) of $Pd(PPh_3)_4$. After 10 min, the Pd soln was drained and the resin washed with CH_2Cl_2 (3 × 30 s) and the deprotection procedure was repeated once more. Before the next coupling, the resin was washed successively with CH_2Cl_2 (8 × 30 s), 0.2% TFA/CH_2Cl_2 (2 × 60 s), CH_2Cl_2 (5 × 30 s), 5% DIPEA/CH_2Cl_2 (3 × 60 s), and CH_2Cl_2 (5 × 30 s).

2.1.1.1.5.3 Synthesis of Peptides with N^{α}-Allyloxycarbonyl Amino Acids

N^{α}-Protection as allyl carbamate is used generally only for specific synthetic purposes where acid- or base-mediated deprotection procedures must be avoided or reduced to a minimal extent as in the case in the synthesis of glyco- and phosphopeptides (see Sections 6.3 and 6.5). Otherwise the orthogonality of this type of amino protection has been mainly exploited in side-chain protection to allow for selective chemistry such as selective acylations, e.g. cyclizations, to be performed on protected peptides in solution or in the resin-bound state (see Section 4.3.2.3.4). An additional interesting application of the Aloc group is the one-pot deprotection and acylation with carboxy components resistant to the palladium(0)-catalyzed allyl cleavage by hydrostannolysis as reported in Section 4.3.2.3.4.[334–339]

N^{α}-Aloc-protected amino acids are stable to all the common conditions of peptide-bond formation and as urethane-type protected derivatives can be activated and coupled with minimal risk of racemization.[15,52] For the coupling of N^{α}-Aloc-protected peptide segments as used in the synthesis of glycopeptides, methods to minimize the risk of epimerization must be used.

2.1.1.1.6 2,2,2-Trichloroethoxycarbonyl Group

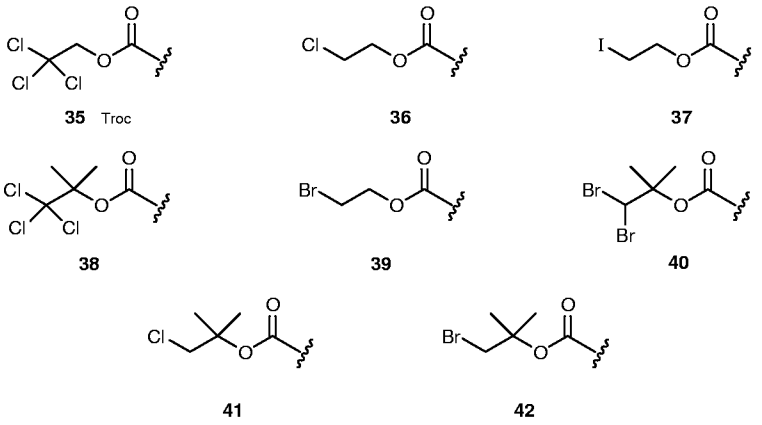
J. Podlech

The 2,2,2-trichloroethoxycarbonyl group (Troc, **35**) (Scheme 18) is orthogonal to most other commonly used protecting groups for the amino function since it is cleaved by Grob fragmentation. It can be removed in the presence of Z, Boc, Aloc, and Fmoc groups. It easily survives the conditions for acidic Boc (stable even at pH 1[12,361]) and basic Fmoc cleavage, but it is unstable towards hydrogenolysis as used for removal of the Z group.[362] It was first used by Woodward et al. in the synthesis of cephalosporin C,[33] and since then it has been adapted for use in peptide synthesis.[66,204,363,364]

Several additional 2-haloethoxycarbonyl groups have been suggested as alternatives for the protection of the amino group. Their cleavage conditions differ in some cases; whilst the 2-chloroethoxycarbonyl (**36**),[365–367] the 2-iodoethoxycarbonyl (**37**),[365,366] and the 2,2,2-trichloro-1,1-dimethylethoxycarbonyl group (**38**)[368–370] are cleaved with various reductive reagents similar to those used for the *N*-Troc derivatives, the 2-bromoethoxycarbonyl group

(**39**) is cleaved with silver salts,[1] and the 2,2-dibromo-1,1-dimethylethoxycarbonyl group (**40**) by solvolysis with alcohols (MeOH or EtOH).[371] 2-Chloro-1,1-dimethylethoxycarbonyl (**41**) and 2-bromo-1,1-dimethylethoxycarbonyl (**42**) are either cleaved under acidic reaction conditions or by solvolysis with alcohols.[372]

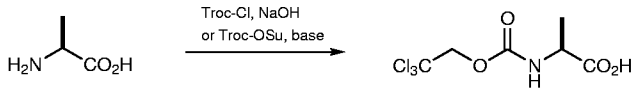
Scheme 18 2-Haloethoxycarbonyl Groups



2.1.1.1.1.6.1 Synthesis of N^{α}-2,2,2-Trichloroethoxycarbonyl Amino Acids

2,2,2-Trichloroethyl chloroformate was initially used as a reagent for the introduction of the Troc group (Scheme 19).[33,364] However, there are several drawbacks to this reagent; rather strong base (NaOH) must be used which leads to the formation of bis(2,2,2-trichloroethyl) carbonate and this is a severe side reaction.[364,373] Using milder bases such as sodium carbonate or sodium hydrogen carbonate gives significantly lower yields of the Troc-protected product.[364] Moreover, the preparation of the N^{α}-Troc derivatives of serine or tryptophan failed with this reagent. The best results are obtained by alternating the addition of 2,2,2-trichloroethyl chloroformate (Troc-Cl) and sodium hydroxide and by using a large excess of the chloroformate; under these conditions the products were obtained in 60–95% yield.[364] Even better results are achieved with the succinimido 2,2,2-trichloroethyl carbonate (Troc-OSu) which results in the desired N-acylation of either amino acid salts or esters within minutes.[66,204] The N^{α}-Troc-protected amino acids are accessible by this method in high yields and purities, without formation of bis(2,2,2-trichloroethyl) carbonate as a side product (Scheme 19). Both Troc-Cl and Troc-OSu are commercially available, and the latter reagent is readily prepared from Troc-Cl and *N*-hydroxysuccinimide under Schotten–Baumann conditions (94%).[204] The protection of amines via silylation using *N,O*-bis(trimethylsilyl)acetamide (BTMSA) followed by acylation with Troc-Cl has been proposed, but this procedure has not as yet been used for the synthesis of N^{α}-Troc amino acid derivatives.[374]

Scheme 19 Synthesis of N^{α}-Troc Amino Acids[364]



Troc-Met-OH; Typical Procedure for Synthesis of N^{α}-Troc Amino Acids with Troc-Cl:[364]
To a stirred soln of H-Met-OH (7.00 g, 46.9 mmol) in 1 M NaOH (55 mL) in an ice bath was added Troc-Cl (20.0 g, 94 mmol) and 1 M NaOH (12.5 mL) alternately in small portions over 7 h under N_2. The mixture was stirred overnight, extracted with EtOAc (4 × 75 mL) to remove the bis(2,2,2-trichloroethyl) carbonate, the alkaline soln was chilled and acidified with HCl, which precipitated an oil. The oily

for references see p 149

suspension was extracted with EtOAc (3 × 75 mL) and the organic layers were dried, concentrated, and the resulting oil was recrystallized (Et_2O/hexane 1:3); yield: 11.1 g (73%); mp 80–81 °C; $[\alpha]_D^{25}$ −12.3 (*c* 3.5–4%, EtOH).

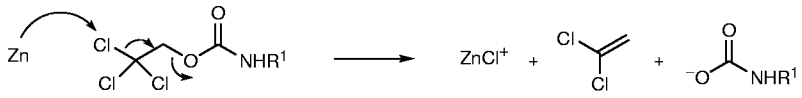
N^{α}-Troc Amino Acids via Troc-OSu; General Procedure:[204]
To a soln of the amino acid (10 mmol) in 1 M NaOH (10 mL) and 1 M $NaHCO_3$ (20 mL) was added Troc-OSu (2.9 g, 10 mmol) in dioxane (20 mL) in one portion at 0 °C under efficient stirring and stirring was continued for 10 min. The dioxane was removed and the residue extracted with EtOAc (2 × 15 mL). The alkaline soln was cooled and acidified with concd HCl to pH 2. The oily suspension was extracted with EtOAc (6 × 25 mL), and the organic layers were washed with H_2O (2 × 15 mL), dried, and concentrated. The residue was recrystallized (EtOAc/petroleum ether). For example: Troc-Trp-OH; yield: 86%; mp 104–106 °C; $[\alpha]_D^{20}$ −32.4 (*c* 5, DMF); Troc-Phe-OH; yield: 94%; mp 126–127 °C; $[\alpha]_D^{20}$ −29.9 (*c* 3, DMF).

Troc-Ser-OMe; Typical Procedure:[204]
To a soln of H-Ser-OMe (1.3 g, 11 mmol) in 2 M $NaHCO_3$ (20 mL, 40 mmol) was added in one portion at rt Troc-OSu (2.9 g, 10 mmol) in dioxane (20 mL) and stirring was continued for 10 min. The mixture was diluted with H_2O (200 mL) and extracted with EtOAc (~200 mL). The organic layer was washed with H_2O (3 × 15 mL), 1 M HCl (2 × 15 mL), H_2O (3 × 15 mL), dried, and concentrated to an oil; yield: 2.5 g (77%); $[\alpha]_D^{20}$ −13.8 (*c* 2, MeOH).

2.1.1.1.1.6.2 Cleavage of the 2,2,2-Trichloroethoxycarbonyl Group

A variety of reagents have been proposed for the efficient cleavage of *N*-Troc derivatives which do not affect other protecting groups such as Z, Boc, and Fmoc. Most commonly used is the Grob fragmentation with zinc (Scheme 20),[33] zinc in buffered media (pH 4.2–7.2),[32] Zn–Pb couple,[375] cadmium in 50% AcOH/DMF,[362] or the 10% Cd–Pb couple.[376] Further single-electron reductions for this purpose are based on lithium cobalt phthalocyanine (prepared from cobalt phthalocyanine and the ketyl radical from lithium and benzophenone),[377] sodium thiophene-2-tellurolate (prepared from di-2-thienyl ditelluride and sodium borohydride),[370] or electrolytic reduction at mercury electrodes.[366,378,379] Side reactions, which have been reported to occur when zinc is used as reducing metal are avoided with cadmium.[362]

Scheme 20 Cleavage of the Troc Group[33]



The Troc group is not stable towards hydrogenation over Pd/C catalyst, but is stable to hydrogenations with Ru/C or Pt/C as catalysts.[362]

Zinc Dust Reduction of Troc-Phe-Met-OEt; Typical Procedure:[364]
To a soln of Troc-Phe-Met-OEt (4.40 g, 8.8 mmol) in THF (120 mL) under argon was added with vigorous stirring activated Zn dust (14 g) followed immediately by 1 M KH_2PO_4 (24 mL). After 7.5 h stirring, the slurry was filtered, and the Zn was washed with THF (100 mL) and EtOH (100 mL). The filtrate was diluted with H_2O (20 mL), acidified with 1 M HCl (10 mL) and concentrated to ~50 mL. The acidic suspension was extracted with Et_2O (3 × 75 mL) to remove starting material and the aqueous layer was made alkaline with aq $NaHCO_3$. The soln was again extracted with EtOAc (3 × 75 mL), and the combined extracts were treated with 1 M HCl (8 mL) and concentrated to 50 mL. H_2O was removed by 3 repeated additions of 75 mL portions of EtOH and concentrated until the product was obtained as a foam which crystallized on addition of Et_2O; H-Phe-Met-OEt was isolated as the hydrochloride; yield: 2.60 g (82%); mp 86–87 °C; $[\alpha]_D^{25}$ −16.8 (*c* 3.4, DMF).

2.1.1.1.1.6.3 Synthesis of Peptides with N^{α}-2,2,2-Trichloroethoxycarbonyl Protection

Similarly to the Aloc protection, the Troc group is mainly used for specific synthetic purposes where acidic or basic deprotection steps are avoided as encountered in the synthesis of glycopeptides[346,380] (see Section 6.3).

It has also been used to create a third dimension in protection schemes such as for peptide hydrazide protection in the synthesis of ribonuclease A by Yajima and Fujii.[381] Due to its full stability to HF treatments it is used for the SPPS synthesis of minimally protected peptide segments for assembly of protein sequences in a manner similar to the 4-pyridylmethoxycarbonyl group (iNoc, see Section 2.1.1.1.1.2)[176] which is also stable to HF, but cleaved with Zn dust in acetic acid.[382]

2.1.1.1.1.7 2-(Trimethylsilyl)ethoxycarbonyl and 2-Phenyl-2-(trimethylsilyl)ethoxycarbonyl Groups

J. Podlech

Besides the Fmoc (Section 2.1.1.1.1.3) and the substituted sulfonylethyl groups (Section 2.1.1.1.1.4) that are cleaved by β-elimination under basic conditions, the N^{α}-2-(trimethylsilyl)ethoxycarbonyl (Teoc, **43**)[35] and 2-phenyl-2-(trimethylsilyl)ethoxycarbonyl (**44**)[383] derivatives (Scheme 21) are readily removed by the action of the fluoride anion, the elimination products of the Grob-type reaction being TMSF, ethene or styrene, and a carbamate which releases carbon dioxide upon aqueous workup. The nature of the cleavage byproducts allows for easy isolation of the free amino compound.

Scheme 21 Silylethoxycarbonyl Protecting Groups



43 Teoc **44**

N^{α}-2-(Trimethylsilyl)ethoxycarbonyl and N^{α}-2-phenyl-2-(trimethylsilyl)ethoxycarbonyl amino acid derivatives are prepared by the Schotten–Baumann procedure using the corresponding chloroformate (Teoc-Cl),[35] the mixed carbonates Teoc-ONp,[261] Teoc-OBt or Teoc-OSu[384] or the imidazolide Teoc-Im[384] and 4-nitrophenyl 2-phenyl-2-(trimethylsilyl)-ethyl carbonate, respectively.[383] Teoc-ONp is commercially available; Teoc-OBt and Teoc-OSu have been reported to give higher yields, though dipeptide formation has been observed as a side reaction (~5%) when Teoc-OBt was used for the protection of amino acids.[384] Both type of N^{α}-protected amino acids or peptides are stable toward hydrogenolysis and common basic conditions, e.g. NMM or piperidine. The 2-phenyl-2-(trimethylsilyl)ethoxycarbonyl group is stable to weak acids, e.g. acetic acid, but is cleaved rapidly by TFA. Both groups are stable to conditions used for the cleavage of allyl-based protecting groups and for coupling in peptide synthesis, but are cleaved with tetraalkylammonium fluorides, with Lewis acids or with strong acids. Cleavage of the N^{α}-2-phenyl-2-(trimethylsilyl)ethoxycarbonyl derivatives occurs in quantitative manner using 2 equivalents of tetrabutylammonium fluoride in 10 minutes, while under these conditions only 30–40% of Teoc is removed. Attention must be paid when using fluoride sources as aspartic acid ω-*tert*-butyl ester is cleaved quantitatively with formation of aspartimide and the ω-benzyl ester is cleaved partially.[260,261] In this context, the use of a milder fluoride source such as tris(dimethylamino)sulfonium difluorotrimethylsilicate could be useful.[385]

for references see p 149

2-(Trimethylsilyl)ethyl Chloroformate (Teoc-Cl):[384]

CAUTION: *Phosgene is a colorless, poisonous gas or volatile liquid. This reagent should be handled with the utmost care.*

To a soln of 2-(trimethylsilyl)ethanol (2.0 g, 17 mmol) in toluene (10 mL) was added at −10 °C powdered anhyd K_2CO_3 (2 g, 14.5 mmol). While vigorously stirring, a 12.5% soln of $COCl_2$ in toluene (20 mL) was added within 30 min, and the mixture was further stirred at −15 °C for 30 min and at rt for 1 h. N_2 was bubbled through the soln in a closed hood to remove the excess $COCl_2$ and the last traces were removed under vacuum. The solids were removed by filtration through a pad of anhyd $MgSO_4$ and washed with Et_2O. The filtrate was concentrated (below 50 °C) and the residue used without further purification.

1,2,3-Benzotriazol-1-yl 2-(Trimethylsilyl)ethyl Carbonate (Teoc-OBt):[384]
Teoc-Cl (3.07 g, 17 mmol) was dissolved in MeCN (50 mL), the soln was cooled (0 °C) and HOBt·H_2O (3.0 g, 22 mmol) was added with vigorous stirring followed by a soln of TEA (2.0 g, 20 mmol) in MeCN (5 mL). The mixture was stirred at low temperature for 15 min and then at rt overnight, poured into H_2O (200 mL), and extracted with Et_2O (3 × 75 mL). The organic extracts were combined, washed with H_2O (2 × 75 mL), 1 M HCl (75 mL), and H_2O (2 × 75 mL), dried, and concentrated. The residue was taken up in boiling hexane (250–350 mL), and the soln allowed to cool. Crystallization was completed by storage at −15 °C; yield: 3.19 g [68% based on 2-(trimethylsilyl)ethanol]; mp 104–106 °C.

4-Nitrophenyl 2-Phenyl-2-(trimethylsilyl)ethyl Carbonate:[383]
To a stirred soln of 4-nitrophenyl chloroformate (343 mg, 1.7 mmol) in CH_2Cl_2 (3 mL) was added under N_2 over 5 min a soln of 2-phenyl-2-(trimethylsilyl)ethanol (300 mg, 1.5 mmol) and pyridine (0.137 mL, 1.7 mmol) in CH_2Cl_2 (1 mL). The soln was further stirred for 4 h at rt and then diluted with CH_2Cl_2 (25 mL), washed with H_2O (2 × 10 mL) and with brine, dried ($MgSO_4$), and concentrated to yield crude product, which was used without further purification.

N^{α}-Teoc Amino Acids with Teoc-ONp; General Procedure:[261]
Method A: A soln of amino acid methyl or ethyl ester hydrochloride (10 mmol), TEA (1.01 g, 10 mmol), and Teoc-ONp (2.83 g, 10 mmol) in THF (50 mL) was stirred at rt for 24 h. The mixture was concentrated, and the residue partitioned between EtOAc and H_2O. The organic layer was washed with 2% aq $KHSO_4$, 5% aq Na_2CO_3, and H_2O, dried, and evaporated. The N^{α}-Teoc amino ester was dissolved in dioxane (50 mL) and treated with 1 M NaOH (1 equiv). After completion of the reaction, the bulk of dioxane was removed, H_2O was added and the mixture acidified to pH 2–3 with 1 M HCl. The product was extracted with EtOAc or Et_2O and the combined extracts were washed with H_2O, dried, and concentrated. The product was crystallized from a suitable solvent or isolated as its DCHA salt. For example: Teoc-Ala-OH·DCHA (crystallized from iPr_2O); yield: 66%; mp 123–125 °C; $[\alpha]_D^{20}$ −11.7, $[\alpha]_{546}^{20}$ −13.9 (*c* 1, 80% AcOH).

Method B: The amino acid (10 mmol) was dissolved in 1 M NaOH (10 mL, 10 mmol) and reacted with Teoc-ONp (2.83 g, 10 mmol) in tBuOH or dioxane (20 mL). The mixture was stirred overnight at rt and the solvent was removed. The residue was diluted with H_2O, layered with Et_2O or EtOAc, and acidified with 2% aq $KHSO_4$ to pH 2–3. The aqueous layer was extracted with Et_2O or EtOAc and the combined organic layers were washed with H_2O, dried, and concentrated. The product was crystallized from a suitable solvent or isolated as its DCHA salt. For example: Teoc-Ala-OH·DCHA; yield: 70%; Teoc-Trp-OH (crystallized from Et_2O/petroleum); yield: 81%; mp 100–105 °C; $[\alpha]_D^{20}$ −24.7, $[\alpha]_{546}^{20}$ −29.4 (*c* 1, DMF); Teoc-Pro-OH (crystallized from Et_2O); yield: 89%; mp 145–148 °C; $[\alpha]_D^{20}$ −28.3, $[\alpha]_{546}^{20}$ −33.6 (*c* 1, MeOH).

N^{α}-Teoc Amino Acids with Teoc-OBt; General Procedure:[384]
To a stirred soln of the amino acid (1 mmol) in H_2O (1 mL) was added a soln of TEA (152 mg, 1.5 mmol) in dioxane (1 mL) followed by solid Teoc-OBt (0.31 g, ~1.1 mmol). The mixture was stirred at rt overnight, then diluted with H_2O (5 mL), acidified with sat. $KHSO_4$, and extracted with Et_2O (3 × 15 mL). The combined organic extracts were washed with H_2O (4 × 20 mL), dried, and concentrated. The crude products were purified by crystallization or as CHA or DCHA salts. For example: Teoc-Ala-OH·CHA; yield: 86%; Teoc-Ser-OH·CHA (crystallization from CH_2Cl_2/Et_2O, 1:1 and hexane); yield: 88%; mp 120–123 °C; $[\alpha]_D^{25}$ +12.8 (*c* 1.9, MeOH); Teoc-Val-OH·CHA (crystallization from $iPrO_2$/hexane); yield: 95%; mp 127–133 C; $[\alpha]_D^{25}$ −2.0 (*c* 3.7, MeOH).

N^{α}-2-Phenyl-2-(trimethylsilyl)ethoxycarbonyl Amino Acids with 4-Nitrophenyl 2-Phenyl-2-(trimethylsilyl)ethyl Carbonate; General Procedure:[383]
To a soln of crude 4-nitrophenyl 2-phenyl-2-(trimethylsilyl)ethyl carbonate (as synthesized from 1.5 mmol starting material, *vide supra*) dissolved in DMF (3 mL) was added a soln of the hydrochloride or 4-tolu-

enesulfonate of the amino acid derivative (1.7 mmol) and DIPEA (0.582 mL, 3.4 mmol) in DMF (3 mL). The mixture was stirred for 48 h under argon, the solvent was removed under reduced pressure and the residue was dissolved in EtOAc (20 mL), washed with $NaHCO_3$ soln (3 × 15 mL) and with brine (15 mL). The organic layer was dried ($MgSO_4$) and the solvent removed under reduced pressure. The resulting residue was purified by silica gel chromatography (hexane/EtOAc) yielding the protected amino acid derivative.

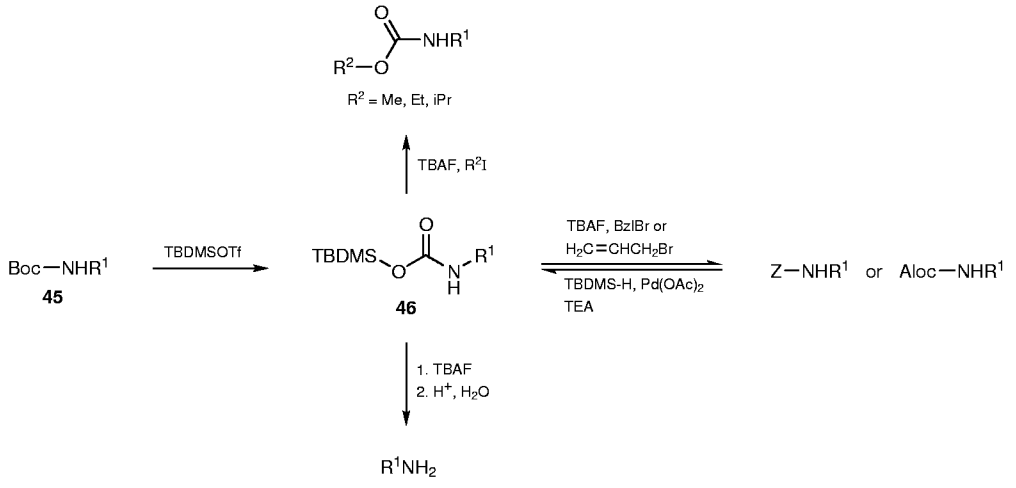
Deprotection of N^{α}-2-Phenyl-2-(trimethylsilyl)ethoxycarbonyl Amino Acids; General Procedure:[383]
The N^{α}-2-phenyl-2-(trimethylsilyl)ethoxycarbonyl-protected amino acid derivative (0.1 mmol) was dissolved in CH_2Cl_2 (5 mL) and TBAF·$3H_2O$ (63 mg, 0.2 mmol) was added. After stirring for 15 min at rt (end of reaction checked by TLC), the mixture was diluted with EtOAc (15 mL) and H_2O (10 mL), the organic layer was washed with half-sat. $NaHCO_3$ soln (2 × 15 mL) and concentrated. The residue was purified by silica gel chromatography (CH_2Cl_2/MeOH/AcOH/H_2O) to yield the deprotected amino acid derivative.

2.1.1.1.1.8 *tert*-Butyldimethylsiloxycarbonyl Group

J. Podlech

Though *N*-trialkylsiloxycarbonyl derivatives (silyl carbamates) should be well-suited to peptide chemistry, their use is very limited. The *N*-*tert*-butyldimethylsiloxycarbonyl group can be considered as a masked carbamoic acid, which in the unprotected form is a very unstable compound (Scheme 22). The related amino acid derivatives are obtained from N^{α}-Boc amino acids **45** with the commercially available TBDMSOTf. The *tert*-butyl group is cleaved by the reagent and the intermediately formed carbamoic acid reacts in situ with the reagent to produce the N^{α}-*tert*-butydimethylsiloxycarbonyl-protected amino acid **46**. Similarly, N^{α}-Z- or N^{α}-Aloc-protected amino acids or peptides can also be used; the respective carbamoic acids are formed by reduction with palladium(II) acetate in the presence of tributylsilane and TEA; *tert*-butyldimethylsilane acts as hydride donor and silylating reagent.[386] The opposite transformation of N^{α}-*tert*-butyldimethylsiloxycarbonyl amino acids into any other urethane-protected derivative is readily achieved by liberating the carbamate with TBAF and trapping the anion with the alkyl halides thus leading to methoxycarbonyl, ethoxycarbonyl, isopropoxycarbonyl, Z, and Aloc derivatives, but not to Boc-protected amino acids or peptides. Conversely, cleavage of the N^{α}-*tert*-butyldimethylsiloxycarbonyl protection with TBAF and subsequent aqueous workup yields the corresponding unprotected amino compound.

Scheme 22 Synthesis and Cleavage of N^{α}-*tert*-Butyldimethylsiloxycarbonyl-Protected Amino Acids and Peptides



for references see p 149

An alternative silyl carbamate proposed recently is the triisopropylsiloxycarbonyl group, which is prepared from the carbamic acids by reaction with TIPSOTf. It is stable to hydrogenation, TFA, and bases, but cleaved with TBAF in THF at 0 °C to room temperature in 15–30 minutes.[387]

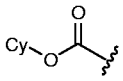
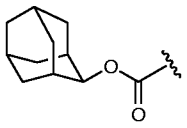
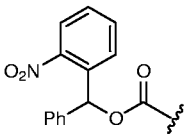
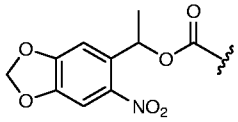
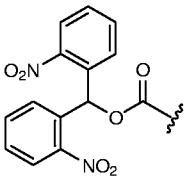
N^{α}-*tert*-Butyldimethylsiloxycarbonyl-Alg-Thr(OTBDMS)-OMe; Typical Procedure:[386]
To a stirred soln of Boc-Alg-Thr-OMe (24.0 mg, 0.07 mmol) and 2,6-lutidine (0.024 mL, 0.21 mmol) in abs CH_2Cl_2 (0.5 mL) at rt was added TBDMSOTf (0.041 mL, 0.18 mmol) in CH_2Cl_2. The mixture was stirred for 15 min, quenched with sat. NH_4Cl, and extracted with Et_2O several times. The combined organic phases were washed with H_2O and brine, dried, and concentrated to give the product as an oil; yield: 43.3 mg (Note: this is a >100% yield, it is likely that the product was impure).

Cleavage of the N^{α}-*tert*-Butyldimethylsiloxycarbonyl Group, H-Phe-OMe; Typical Procedure:[386]
To a stirred soln of N^{α}-*tert*-butyldimethylsiloxycarbonyl-Phe-OMe (187 mg, 0.5 mmol) at rt was added 1 M TBAF in THF (0.500 mL, 0.5 mmol). The mixture was stirred for 1 h, quenched with sat. NH_4Cl, and extracted with $CHCl_3$ several times. The combined organic phases were washed with brine, dried, and concentrated under reduced pressure to yield the product as an oil; yield: 84 mg (93%).

2.1.1.1.2 Urethane Protecting Groups Derived from Secondary Alcohols

Urethane protecting groups derived from secondary alcohols have not found widespread application for the protection of amino groups in peptide chemistry due to the lack of particular complementary properties in terms of stability/cleavage with respect to other classes of carbamates as outlined in Table 5. The *N*-cyclopentyloxycarbonyl (Poc), *N*-cyclohexyloxycarbonyl (Hoc) (predominantly used for side-chain protection of tryptophan[388]), and *N*-2-adamantyloxycarbonyl (2-Adoc) derivatives exhibit acid stability similar to that of the Z derivatives, but in contrast to the latter derivatives they are stable toward hydrogenolysis. Although this property should allow their use in combination with Z protection for particular syntheses in solution, this feature has not yet been exploited. Similarly, the higher solubility in organic solvents imparted, for example by the 2-Adoc group, particularly if used for side-chain protection, might be advantageous in convergent peptide synthesis.[389,390] Furthermore, photosensitive protecting groups have been derived from secondary alcohols such as 2-nitrobenzhydryloxycarbonyl, 1-(6-nitro-1,3-benzodioxazol-5-yl)ethoxycarbonyl (Menpoc) and bis(2-nitrophenyl)methoxycarbonyl {dinitrobenzhydryloxycarbonyl, Z[2,4-$(NO_2)_2$]} which are discussed in detail in Section 2.4.

Table 5 Urethane Protecting Groups Derived from Secondary Alcohols

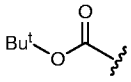
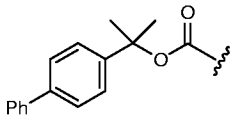
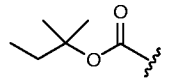
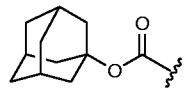
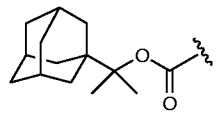
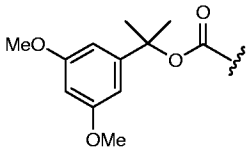
Protecting Group	Structure	Preferred Cleavage Conditions	First Cited in Ref
cyclohexyloxycarbonyl (Hoc)		HBr/AcOH HF	[41]
2-adamantyloxycarbonyl (2-Adoc)		TfOH or HF at 0 °C	[390]
α-2-nitrophenylbenzyloxy-carbonyl		*hν*	[391]
1-(6-nitro-1,3-benzodioxol-5-yl)ethoxycarbonyl (Menpoc)		*hν*	[392]
bis(2-nitrophenyl)methoxy-carbonyl {Z[2,4-$(NO_2)_2$]}		*hν*	[149]

2.1.1.1.3 Urethane Protecting Groups Derived from Tertiary Alcohols

Several urethane protecting groups have been derived from tertiary alcohols due to their facile cleavage under relatively mild acidic conditions. A selection of this type of protecting groups is summarized in Table 6, among which the *tert*-butoxycarbonyl (Boc), the 2-(biphenyl-4-yl)prop-2-yloxycarbonyl (Bpoc), and the 2-(3,5-dimethoxyphenyl)prop-2-yloxycarbonyl group (Ddz) have found wide application for the synthesis of peptides both in solution and on solid supports. Although other *tert*-alkyl carbamates that have been proposed do not exhibit particularly advantageous properties, at least for N^{α}-protection, they are occasionally applied for side-chain amino protection such as, for example, the 2-(1-adamantyl)prop-2-yloxycarbonyl[393,394] and the 1-adamantyloxycarbonyl group (1-Adoc), the latter was shown to enhance the solubility of protected peptide intermediates,[395] a feature that might be particularly useful for peptide synthesis in solution or by convergent strategies. 1-(3,5-Di-*tert*-butylphenyl)-1-methylethoxycarbonyl (*t*-Bumeoc) is an extremely acid-labile protecting group, though it has not found further utilization.[396] A comparison of different acid-labile protecting groups is shown by the following relative cleavage rates (80% AcOH, rt): Boc (1), 2-(1-adamantyl)prop-2-yloxycarbonyl (2400), Bpoc (2800), *t*-Bumeoc (4000).[396]

for references see p 149

Table 6 Urethane Protecting Groups Derived from Tertiary Alcohols

Protecting Group	Structure	Preferred Cleavage Conditions	Ref	First Cited in Ref
tert-butoxycarbonyl (Boc)		HCl/dioxane	[36]	[41,42]
		TFA	[37]	
		TosOH	[38]	
		$BF_3 \cdot OEt_2$	[39,40]	
2-(biphenyl-4-yl)prop-2-yloxycarbonyl (Bpoc)		0.5% TFA/CH_2Cl_2	[43,44]	[45,46]
tert-pentyloxycarbonyl		TFA	[23]	[23]
		2 M HCl/AcOH	[397]	
1-adamantyloxycarbonyl (1-Adoc)		TFA	[398]	[398]
2-(1-adamantyl)prop-2-yloxycarbonyl		TFA, 0 °C	[393,394]	[393,394]
2-(3,5-dimethoxyphenyl)prop-2-yloxycarbonyl (Ddz)		dil TFA	[47]	[47]

2.1.1.1.3.1 *tert*-Butoxycarbonyl Group

J. Podlech

The *tert*-butoxycarbonyl (Boc, **47**) group represents together with the benzyloxycarbonyl and the 9-fluorenylmethoxycarbonyl group the basic protecting groups of amino functions. It is readily introduced, very stable even under strongly nucleophilic or basic conditions and is cleaved by relatively mild acids. Consequently, it is one of the most commonly used amino protecting groups, and not only in peptide chemistry. The reader is directed to a number of excellent reviews on *N*-Boc protection.[1,12,13,399,400]

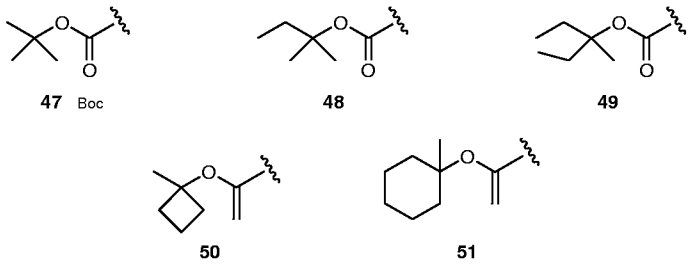
The outstanding features of the Boc group were recognized very rapidly after its first use,[41,42] but it was the fact that it is complementary to the benzyloxycarbonyl group in terms of stability and cleavage that was exploited by Schwyzer et al.[401–404] to develop the Z/tBu and by Sakakibara et al.[23,405] and Bodanszky et al.[406,407] to elaborate the Boc/Bzl strategy, respectively, for peptide synthesis in solution (see Section 4.1). The experience gained with these protection schemes was then transferred to the synthesis on solid support where the selectivity of the Boc/Bzl strategy was the essential prerequisite for successful repetitive synthetic steps (Section 4.3.2.1).[84,408,409] More recently, in analogy to the Z/tBu strategy the Boc group has been used in the Fmoc/tBu chemistry on solid phase for semipermanent N^{ω}-protection (Section 4.3.2.2).[179,180]

2.1.1.1.3.1.1 Synthesis of N^{α}-*tert*-Butoxycarbonyl Amino Acids

tert-Butyl chloroformate is a potential reagent[410] for the introduction of the Boc group, however it is only fairly stable above −10 °C and thus it has been used infrequently;[411–413] the first N^{α}-Boc amino acids were obtained from *N*-carbonyl amino acid esters (isocyanates) via addition of *tert*-butyl alcohol[41] or with *tert*-butyl 4-nitrophenyl carbonate.[42] In the subsequent years a number of reagents such as *tert*-butyl azidoformate[414–416] and fluoroformate[413,417,418] as well as various mixed carbonate esters were developed that allow preparation of N^{α}-Boc amino acids or peptides in high yields. Some of these reagents are disadvantageous, since byproducts are formed which are either highly toxic or hard to remove. The real breakthrough was achieved with di-*tert*-butyl dicarbonate (Boc_2O).[62,419–421] This stable and easily accessible compound leads under mild reaction conditions to carbamate formation in high yields. At present on laboratory scale Boc_2O is used for Boc protection of amino acids and related derivatives or peptides almost exclusively, on an industrial scale however *tert*-butyl fluoroformate still finds application.[422] Similarly, also the isocyanate method has enjoyed a revival for particular synthetic purposes.[423,424] Since the reactivity of tertiary alcohols is lower than that of primary and secondary alcohols due to steric hindrance,[424] various catalysts have been developed amongst which the combination of SmI_2 and HMPA appears to be particularly efficient.[425]

During the time between recognition of the special features of the Boc group and development of a practical method for its introduction, Sakakibara et al. proposed the *tert*-pentyloxycarbonyl group (**48**) as an alternative (Scheme 23). It is readily introduced with *tert*-pentyl chloroformate and cleaved with acidic reaction conditions.[23,397,412,426,427] Further variations of the Boc group are 3-methylpent-3-yloxycarbonyl (**49**),[23] 1-methylcyclobutyloxycarbonyl (**50**), and 1-methylcyclohexyloxycarbonyl (**51**).[428]

Scheme 23 *tert*-Alkoxycarbonyl Protection Groups



2.1.1.1.3.1.1.1 *tert*-Butyl Chloro- and Fluoroformate

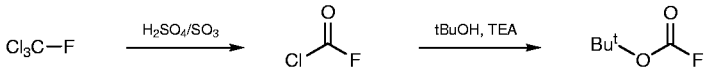
Despite the instability of *tert*-butyl chloroformate[410,413] it can be used for the incorporation of the Boc group if the reagent is freshly prepared and used immediately.[429] More convenient is *tert*-butyl fluoroformate as it is significantly more stable,[413,417,418] and N^{α}-Boc amino acids are obtained in high yields.[418,422,430] The fluoroformate is reported to be even more reactive than *tert*-butyl azidoformate (*vide infra*); thus, under selected conditions the nucleophilic side-chain functionalities of Ser, Thr, Tyr, His, and Cys, if unprotected, are Boc derivatized too. To increase the nucleophilicity of amino acids, generally Schotten–Baumann conditions are used which provokes a concurrent faster hydrolysis of the reagent. Therefore, acylation with *tert*-butyl fluoroformate is conveniently carried out under pH control.[418,430]

***tert*-Butyl Fluoroformate from 1-Chloroethyl Chloroformate:**[422]
Pyridine (27.0 g, 0.34 mol) was added dropwise over 20 min into a stirred, cooled (0 °C) soln of 1-chloroethyl chloroformate (44.0 g, 0.31 mol) and tBuOH (21.5 g, 0.29 mol) in CH_2Cl_2 (200 mL). After 15 h at rt, the mixture was diluted with CH_2Cl_2 (100 mL), washed with 0.5 M HCl (100 mL) and H_2O

for references see p 149

(3 × 100 mL), dried, concentrated, and distilled to give pure 1-chloroethyl *tert*-butyl carbonate; yield: 47.6 g (91%); bp 58–60 °C/10 Torr. A fraction of the material (20.4 g, 0.11 mol), dried KF (15.0 g, 0.26 mol), and 18-crown-6 (1.90 g, 7.2 mmol) was heated (oil bath at 70 °C) at 37 Torr. Volatile products were condensed in a −78 °C trap as formed. After 30 h, a mixture of acetaldehyde and *tert*-butyl fluoroformate had been collected, which was heated to 50 °C to remove the aldehyde and then distilled to isolate pure product; yield: 11.4 g (86%); bp 40–42 °C/175 Torr.

Scheme 24 Synthesis of *tert*-Butyl Fluoroformate from Trichlorofluoromethane[431]



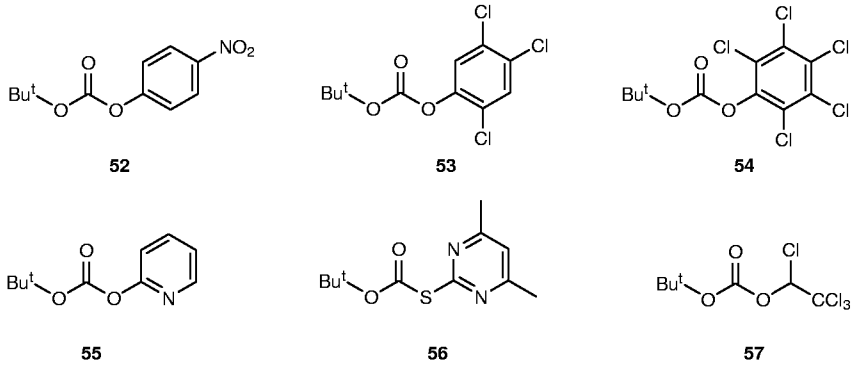
***tert*-Butyl Fluoroformate from Trichlorofluoromethane (Scheme 24):**[431]
To 65% oleum (600 g) in a flask equipped with a dropping funnel, a condenser, and a stirrer was added dropwise over 4–5 h CCl_3F (275 g, 2.0 mol). The evolved gas was passed from the top of the condenser through an absorption bottle filled with H_2SO_4 and via a dry ice condenser in a second flask filled with tBuOH (126 g, 1.70 mol), TEA (250 mL), and dry Et_2O (750 mL), which was kept at −40 °C with stirring. TEA·HCl precipitated and when the evolution of gas had ceased, the temperature was raised to 20 °C. Stirring was continued for 1 h and the mixture was degassed at 200 Torr, filtered, and the precipitate was washed with dry Et_2O. The filtrate contained Boc-F and can be used directly or after concentration; yield: 139–144 g (90–93%).

Boc-Asp-OH; Typical Procedure for N-Acylation with *tert*-Butyl Fluoroformate:[430]
To an ice-cold suspension of H-Asp-OH·H_2O (3.0 g, 20 mmol) in dioxane/H_2O (15 mL, 2:1) was added an ether soln of Boc-F (68%, 4.50 mL, 25 mmol) and the mixture was titrated with 4 M NaOH at pH 8.5. When no further base was needed, the suspension was filtered and the filtrate washed repeatedly with Et_2O (30 mL each), cooled, and acidified with 6 M HCl to pH 6.5 and with citric acid to pH 3.5. After standing for several hours at 4 °C, the precipitate was filtered off, washed with small amounts of H_2O and dried. The residue was recrystallized (MeOH); yield: 3.95 g (85%); mp 176–177 °C; $[\alpha]_{578}^{22}$ −8.25 (*c* 1, DMF), $[\alpha]_{578}^{22}$ +11.8 (*c* 1, 2 M NaOH).

2.1.1.1.3.1.1.2 Mixed *tert*-Butyl Carbonates

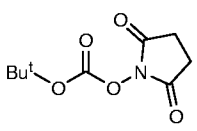
Historically, the first practicable reagent for the synthesis of N^{α}-Boc amino acids, *tert*-butyl 4-nitrophenyl carbonate (**52**), was developed by Anderson and McGregor[42] (Scheme 25). Due to the low reactivity of this mixed carbonate, drastic reaction conditions are required and yields remain unsatisfactory because of the difficult and laborious separation of the by-product 4-nitrophenol. Alternative mixed carbonates have been proposed such as *tert*-butyl 2,4,5-trichlorophenyl carbonate (**53**),[432] *tert*-butyl pentachlorophenyl carbonate (**54**),[433] and *tert*-butyl 2-pyridyl carbonate (**55**)[434,435] as well as *O*-*tert*-butyl *S*-4,6-dimethylpyrimidin-2-yl thiocarbonate (**56**).[436] With the latter two reagents the byproducts, pyridin-2-ol and 4,6-dimethylpyrimidine-2-thiol, respectively, are acid soluble and, thus, readily removed.

Scheme 25 Mixed Carbonates and Thiocarbonates Used for the Synthesis of N^{α}-Boc Amino Acids

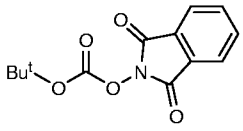


52 **53** **54**

55 **56** **57**



58



59

More recently, *tert*-butyl 1,2,2,2-tetrachloroethyl carbonate (**57**) has been proposed as an efficient reagent which leads to yields comparable to those obtained with *tert*-butyl azidoformate or Boc_2O.[437] Mixed carbonate esters of *N*-hydroxy heterocyclic compounds have also been used as reagents for Boc protection of amino groups.[438,439] In fact, *tert*-butyl succinimido carbonate (**58**) and *tert*-butyl phthalimido carbonate (**59**) (Scheme 25) are efficient Boc-donating reagents,[64] whereby the respective byproducts are *N*-hydroxysuccinimide, which is water soluble and thus readily removed during workup of the reaction mixture, and *N*-hydroxyphthalimide, which is precipitated during the reaction as its sodium salt.

***tert*-Butyl 2-Pyridyl Carbonate** (55):[435]

CAUTION: *Phosgene is a colorless, poisonous gas or volatile liquid. This reagent should be handled with the utmost care.*

To 2.5 M $COCl_2$ in toluene (20 mL, 50 mmol) with CH_2Cl_2 (20 mL) at −10 °C was slowly added a soln of pyridin-2-ol (951 mg, 10 mmol) and pyridine (870 mg, 11 mmol) in CH_2Cl_2 (40 mL). After stirring for 15 min, excess $COCl_2$ and solvents were removed under reduced pressure and then the residue in dissolved in CH_2Cl_2 (30 mL). A soln of tBuOH (815 mg, 11 mmol) and pyridine (830 mg, 10.5 mmol) in CH_2Cl_2 (20 mL) was added at rt. The mixture was stirred for 5 h, diluted with CH_2Cl_2, washed with sat. $NaHCO_3$ (30 mL), and brine (30 mL), dried, concentrated, and passed through a short column of silica gel (CH_2Cl_2) affording the product; yield: 1.37 g (70%), which could be further purified by recrystallization from hexane; mp 48–49 °C.

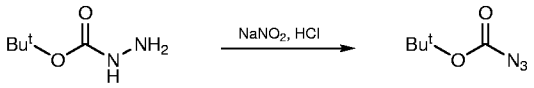
Boc-Pro-OH; Typical Procedure for Acylation with *tert*-Butyl 2-Pyridyl Carbonate (55):[435]
To a soln of H-Pro-OH (230 mg, 2.0 mmol) in DMF/H_2O (2:1, 12 mL) was added TEA (303 mg, 3.0 mmol). After 10 min, *tert*-butyl 2-pyridyl carbonate (**55**; 392 mg, 2.0 mmol) was added and the resulting mixture was stirred at rt for 6 h. The pH was adjusted to 3.5 with 0.025 M oxalic acid and the soln was extracted with CH_2Cl_2 (3×30 mL). The organic layers were washed with H_2O (20 mL) and brine (20 mL), dried, and concentrated; yield: 424 mg (98%). The product was a syrup that could be crystallized (EtOAc/petroleum ether); mp 133–135 °C; $[\alpha]_D^{20}$ −60.7 (*c* 0.9, AcOH).

2.1.1.1.3.1.1.3 *tert*-Butyl Azidoformate

tert-Butyl azidoformate (Boc-N_3, **60**) was proposed by Carpino[414,415,440] for the synthesis of carbamates and was then applied by Schwyzer et al.[416] for the preparation of N^{α}-Boc amino acids. Although it is not a highly reactive Boc donor, under optimized and constant pH values for each amino acid satisfactory yields are obtained.[440–442] The best solvent systems are dioxane or acetone/H_2O, and KOH or NaOH, carbonates or magnesium oxide are favored as the base.[416] The major drawback of this method is the formation of HN_3 as byproduct which is a strong ichthyotoxin. In addition, Boc-N_3 is explosive when heated, e.g. during distillation or as the distillate,[443–446] and therefore it is advisable to prepare the azide from the commercially available Boc-hydrazide in situ and to use the reaction mixture directly for the acylation of amino acids and related derivatives (Scheme 26). Under optimized conditions, i.e. in DMF and TEA as the base, even Boc-Asp(OBzl)-OH and Boc-Glu(OBzl)-OH are obtained in satisfactory yields.[447]

for references see p 149

Scheme 26 Conversion of Boc-$NHNH_2$ into Boc-N_3 for the In Situ Acylation of Amino Acids[56,440]



60

Boc-Ile-OH; Typical Procedure for Acylation with Boc-N_3 (60):[442]

To an ice-cold soln of Boc-$NHNH_2$ (477 g, 3.6 mol) in dioxane (2 L) and H_2O (500 mL) was added slowly 5 M HCl (720 mL, 3.6 mol) followed by a soln of $NaNO_2$ (255 g, 3.6 mol) in H_2O (750 mL). After additional stirring for 1–3 h at rt the azide soln was ready for use. H-Ile-OH (393 g, 3 mol) was suspended in the azide soln and dioxane (500 mL) was added and the soln titrated at pH 10 with 4 M NaOH (1.5 L) over a period of 12 h. The mixture was neutralized to pH 7 with 1 M H_2SO_4 and concentrated. It was then acidified to pH 3–3.5 and extracted with EtOAc. The organic phase was washed with H_2O, dried (Na_2SO_4), and concentrated to an oily residue which solidified under petroleum ether; yield: 642 g (93%); $[\alpha]_{578}^{20}$ +3.8 (*c* 1.0, 80% AcOH).

***tert*-Butyl Azidoformate (Boc-N_3, 60):**[56,440]

CAUTION: *As with all azides of low molecular weight,* tert*-butyl azidoformate is sensitive to both heat (>80 °C) and shock.*[443–446] *It is usually purified by distillation under reduced pressure and detonation under these conditions has been reported (distillate and distillation residue are prone to explode). All operations involving this reagent should be carried out with great caution in a fume hood and behind a safety shield! Hydrogen azide is an ichthyotoxic compound.*

To an ice-cold soln of Boc-$NHNH_2$ (476 g, 3.6 mol) in H_2O (500 mL) and 5 M HCl (720 mL, 3.6 mol) was added slowly a soln of $NaNO_2$ (248 g, 3.6 mol) in H_2O (750 mL). After additional stirring for 1 h, the mixture was diluted with H_2O and extracted with Et_2O. The organic layer was washed with H_2O, dried and concentrated. The residue was distilled to yield **60**; yield: 400 g (78%); bp 70–73 °C/70 Torr; bp 36.5–37.5 °C/10 Torr.

Boc-Asp(OBzl)-OH:[56]

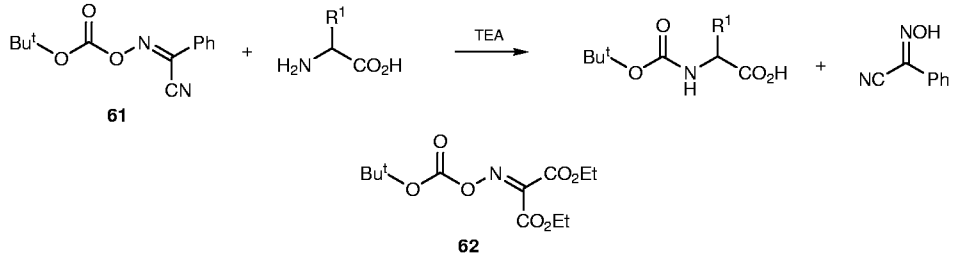
To a soln of H-Asp(OBzl)-OH (937 g, 4.2 mol) and *N,N,N′,N′*-tetramethylguanidine (484 g, 4.2 mol) in DMF (5 L) Boc-N_3 (**60**; 673 g, 4.7 mol) was added, followed by TEA (588 mL, 4.2 mol) in four portions over 36 h. The mixture was concentrated, diluted with H_2O, acidified to pH 3–3.5 and extracted with EtOAc; the organic layers were washed with H_2O, dried, and concentrated. The product was recrystallized (EtOAc/Et_2O/petroleum ether); yield: 940 g (70%); mp 101–102 °C; $[\alpha]_D^{20}$ +8.6 (*c* 1.0, AcOH); $[\alpha]_D^{20}$ −19.5 (*c* 2, DMF).

2.1.1.1.3.1.1.4 2-(*tert*-Butoxycarbonyloxyimino)-2-phenylacetonitrile

2-(*tert*-Butoxycarbonyloxyimino)-2-phenylacetonitrile (Boc-ON, **61**) reacts rapidly with amino acids to provide Boc-protected derivatives in high yields (Scheme 27).[448–450] It is a stable, crystalline and commercially available reagent, which is soluble in a variety of solvents (*inter alia* EtOAc, Et_2O, benzene, $CHCl_3$, dioxane, acetone, or alcohols; insoluble in H_2O). The oxime byproduct and excess reagent are easily removed by extraction with Et_2O, EtOAc, or benzene, though it is sometimes advantageous to remove most of the solvent by distillation before extraction to prevent co-extraction of product. Boc-ON (**61**) is preferentially used as a 10% excess in either 50% aqueous dioxane or 50% aqueous acetone and with 1.5 equivalents TEA as the base. The reaction is usually complete after 4–5 hours at 20 °C or after 1 hour at 45 °C. Under these reaction conditions even α-methylated amino acids are acylated in satisfactory yields.

Diethyl 2-(*tert*-butoxycarbonyloxyimino)malonate (**62**) (Scheme 27) is a potential alternative to Boc-ON (**61**) that is a noncrystalline compound and is, in fact, more reactive than Boc-ON, but removal of the oxime byproduct formed when it is used as a *tert*-butoxycarbonylating agent requires a critical pH adjustment.[449]

Scheme 27 2-(*tert*-Butoxycarbonyloxyimino)-2-phenylacetonitrile (**61**) and Diethyl 2-(*tert*-Butoxycarbonyloxyimino)malonate (**62**) as N-Acylating Agents

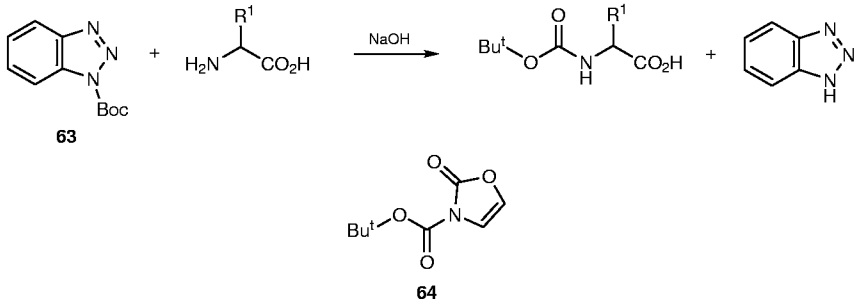


Boc-Trp-OH; Typical Procedure for Acylation with Boc-ON (61):[449]
To a soln of H-Trp-OH (2.05 g, 10.0 mmol) and TEA (2.10 mL, 15.0 mmol) in H_2O (6 mL) was added dioxane (6 mL) and Boc-ON (**61**; 2.71 g, 11.0 mmol) at rt. The mixture became homogenous within 1 h and was stirred for additional 2 h. After addition of H_2O (15 mL) and EtOAc (20 mL), the aqueous layer was separated, washed with EtOAc (20 mL), acidified with 5% citric acid, and extracted with EtOAc. The soln was taken to dryness; yield: 3.0 g (99%); mp 137–138 °C (dec).

2.1.1.1.3.1.1.5 *N-tert*-Butoxycarbonyl Heterocycles

Among the *N*-Boc-protected heterocycles the most familiar *N*-Boc-imidazole does not act as a transacylating reagent.[451] With 1-(*tert*-butoxycarbonyl)-1,2,3-benzotriazole (Boc-Bt, **63**) (Scheme 28) the good leaving group character of the 1,2,3-benzotriazole is exploited and *N*-Boc amino acids are obtained in satisfactory yields although the reaction is slow.[452] Conversely, 3-(*tert*-butoxycarbonyl)oxazol-2(3*H*)-one (**64**) is a less reactive reagent which even in the presence of DMAP as catalyst requires several hours for high yields of acylation products.[453]

Scheme 28 *N*-Boc-Protected Heterocycles as N-Acylating Agents[452,453]



1-(*tert*-Butoxycarbonyl)-1,2,3-benzotriazole (Boc-Bt, 63):[452]

CAUTION: *Phosgene is a colorless, poisonous gas or volatile liquid. This reagent should be handled with the utmost care.*

To a soln of 20% $COCl_2$ in toluene (100 mL) was added dropwise over 1 h a soln of 1,2,3-benzotriazole (5.95 g, 50 mmol) in THF (50 mL) under N_2 at −20 °C. The mixture was allowed to warm to rt and stirred for another 2 h. The solvents and excess $COCl_2$ were removed at reduced pressure to give 1,2,3-benzotriazol-1-ylcarbonyl chloride in almost quantitative yield (based on ^1H NMR), which was dissolved in THF (40 mL) and the resulting soln was added dropwise to a soln of tBuOH (7.41 g, 100 mmol) and pyridine (3.96 g, 50 mmol) in THF (30 mL) under N_2 at −20 °C. The mixture was allowed to warm to rt and was stirred overnight. The precipitate was filtered off, and the filtrate was washed with sat. $NaHCO_3$ (3 × 50 mL), dried, and concentrated to yield the product as an oily substance; yield: 7.54 g (68%).

for references see p 149

Boc-Phg-OH; Typical Procedure for Acylation with Boc-Bt (63):[452]
To a suspension of H-Phg-OH (756 mg, 5.0 mmol) in dioxane/H_2O (1:1, 20 mL) was added NaOH (0.20 g, 5.0 mmol). The mixture was stirred until all solids had dissolved. It was then cooled to 0 °C using an ice bath. A soln of Boc-Bt (**63**; 1.10 g; 5 mmol) in dioxane (10 mL) was added dropwise and the mixture was stirred at rt for 24 h. The dioxane was removed, the residue was diluted with H_2O (30 mL) and extracted with EtOAc (3 × 30 mL). The aqueous layer was acidified with 1 M H_2SO_4 (pH 2) and extracted with Et_2O (3 × 50 mL). The combined organic layers were washed with H_2O (3 × 50 mL), dried, and concentrated; yield: 1.11 g (88%); mp 110–112 °C; $[\alpha]_D^{20}$ +0.2 (*c* 1.5, EtOH).

2.1.1.1.3.1.1.6 Di-*tert*-Butyl Dicarbonate

Since the initial reports on the use of di-*tert*-butyl dicarbonate (Boc_2O, **65**) for Boc protection of amino groups (Scheme 29)[62,400,419–421] this reagent has virtually replaced all others. It is a stable, crystalline (mp 22–24 °C), commercially available compound that results in the rapid protection of amine functionalities even at temperatures below 0 °C (sometimes within 15 min) in high yield.[446] The byproducts are carbon dioxide and *tert*-butyl alcohol, which do not require laborious workup procedures, and thus it has found widespread application in various transformations. On comparison of several Boc-donating reagents such as the mixed carbonates with Boc_2O (**65**) for the *N*-Boc protection of insulin, the dicarbonate proved to be the most efficient in terms of reaction rates and purity of the product, the tris-acylated insulin.[454] Correspondingly, it is also used for efficient postsynthetic derivatization of peptides and for interchange of amino-protecting groups.

Boc_2O (**65**) is prepared from potassium *tert*-butoxide and carbon dioxide to form the *tert*-butyl potassium carbonate, which is converted with phosgene into the di-*tert*-butyl tricarbonate. The tricarbonate is smoothly converted by basic catalysis with 1,4-diazabicyclo[2.2.2]octane (DABCO) into the dicarbonate.[419,455]

N^{α}-*tert*-Butoxycarbonyl Amino Acids

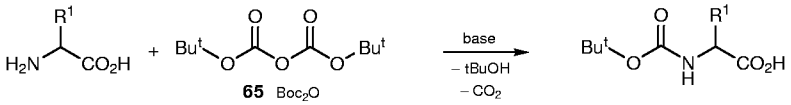
Boc_2O (**65**) is soluble in almost all commonly used organic solvents, preferentially it is used in solvent systems such as dioxane/H_2O, which guarantee the solubility of all components. Performing amine acylation under pH control may be advantageous especially when base- or water-sensitive compounds are transformed, but generally reagent and base are added without special precautions. However, like the chloro- and fluoroformates, Boc_2O can produce at high organic solvent concentration di- and tripeptides as contaminants via in situ mixed anhydride formation (see Section 3.9).[63]

Due to the easy removal of the byproducts, Boc_2O (**65**) can also be used for the acylation of amino acids (Scheme 29) or peptide esters in organic solvents.[456] Amino acid esters or their respective hydrochlorides react sluggishly when the reaction is performed in alcohols with sodium hydrogen carbonate as a solid, poorly soluble base. Sonication increases reaction rates leading to Boc-protected amino acid esters in good to quantitative yields within few hours instead of more than 24 hours.[457]

Amino acids are acylated in organic solvents with Boc_2O (**65**) using organic bases such as TEA, *N*,*N*,*N*′,*N*′-tetramethylguanidine, benzyltrimethylammonium hydroxide (Triton B) or tetramethylammonium hydroxide, although the products must be carefully analyzed for peptides as contaminants. Under these conditions Boc-protected, ^{17}O-labeled amino acids, are obtained on reaction in the absence of water as assessed by determining the ^{17}O enrichment by NMR spectroscopy (14–49%).[458] Similarly, slowly reacting sterically hindered amino acids result in the significant hydrolysis of Boc_2O (**65**) in aqueous solvents; e.g. Boc-Aib-OH is obtained only in 50% yield in aqueous solvents. Performing the acylation in organic solvents, the Boc derivatives are obtained in high yields. α-Substituted prolines have been N^{α}-Boc-protected under such conditions in >90% yield and Boc-(Me)Aib-OH in 95% yield.[459] The latter conditions have been reported to suppress a peptide formation.

Conversion of Boc_2O (**65**) with hydroxylamine into the related Boc-ONH_2 leads to a reagent of 1.5–2.5 enhanced reactivity. This reagent can be prepared in situ and used for acylating sterically hindered amino groups.[460]

Scheme 29 Acylation of Amino Acids with Boc_2O



Boc-Phe-OH; Typical Procedure for Acylation with Boc_2O (65):[446]

CAUTION: *Boc_2O is a crystalline compound, mp 22–24 °C. The liquid is flammable and must be stored in a refrigerator in the absence of moisture. Do not heat above 80 °C.*

To a soln of H-Phe-OH (165 g, 1.0 mol) in H_2O (1.1 L) and tBuOH (750 mL) containing NaOH (44.0 g, 1.10 mol) was added dropwise Boc_2O (**65**; 223 g, 1.00 mol, liquidified in a 35 °C water bath) over 1 h. After stirring overnight, the mixture was extracted with pentane (2 × 250 mL) and the organic layers were extracted with sat. $NaHCO_3$ (3 × 100 mL). The combined aqueous layers were acidified to pH 1–1.5 by careful addition of $KHSO_4$ (224 g, 1.65 mol) in H_2O (1.5 L), which was accompanied by evolution of CO_2. The mixture was extracted with Et_2O (4 × 400 mL). The organic layers were washed with H_2O (2 × 200 mL), dried, and concentrated (below 30 °C). After addition of hexane (150 mL), the mixture was allowed to stand overnight and within 1 d the following portions of hexane were added with stirring to the partially crystallized product: 2 × 50 mL, 4 × 100 mL, 1 × 200 mL. After standing in the cold overnight, the precipitate was filtered and washed with cold pentane. The mother liquor was concentrated and treated as above; yield: 207–230 g (78–87%); mp 86–88 °C; $[\alpha]_D^{20}$ +25.5 (*c* 1, EtOH). Further examples with detailed information are summarized in Table 7.[446]

Boc-Cys-OMe; Typical Procedure for Acylation of Amino Acid Esters with Boc_2O (65):[456]
To a well-stirred slurry of H-Cys-OMe·HCl (1.72 g, 10.0 mmol) in CH_2Cl_2 (20 mL) was added TEA (1.01 g, 10.0 mmol) followed after 10 min by Boc_2O (**65**; 2.18 g, 10.0 mmol). The mixture was stirred for 16 h, washed with H_2O, the organic layer was dried, and concentrated to give the product as a colorless oil; yield: 2.29 g (97%); $[\alpha]_D^{21}$ +28.5 (*c* 318 mM, $CHCl_3$).

***N*$^{\alpha}$-Boc-Protected ^{17}O-Amino Acids; General Procedure:**[458]
To a 10% soln of TEA in MeOH or DMF (30 mL) was added the ^{17}O-amino acid or its hydrochloride (for Gly, Phe, Pro in both solvents: 20 mmol; Phe, Tyr in DMF only: 5 mmol), followed by Boc_2O (**65**; 2 mol equiv) under vigorous stirring. The mixture was heated to 40–50 °C for 5 min (Pro and Phe in MeOH only) or 10 min (Phe and Tyr in DMF) or was heated at 60 °C for 30 min (Gly in DMF) or was refluxed for 30 min (Gly in MeOH). After the amino acid had dissolved, stirring was continued for 30 min, the solvent removed, and the residue stirred for 10 min with ice-cold dil HCl (pH 2.15, 5–10 mL) and extracted immediately with EtOAc (5 × 40 mL). The organic extract was dried and concentrated, and the residues were crystallized (Et_2O). For example: Boc-Phe-OH (in DMF); yield: 97%; mp 85 °C; 49% [^{17}O]. Boc-Pro-OH (in DMF); yield: 99%; mp 136 °C; 14% [^{17}O].

for references see p 149

Table 7 Preparation and Properties of N^{α}-Boc-Protected Amino Acids[44e][a]

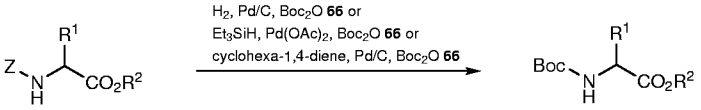
Boc-Xaa-OH	Solvent	Base	Time[b] (h)	Yield (%)	mp (°C)	$[\alpha]_D^{20}$	Remarks
Boc-Ala-OH	tBuOH/H_2O	NaOH	16	93	82–83	−25.5 (*c* 2, AcOH)	pH 8[d]
Boc-Arg-OH	dioxane/H_2O		15	88	159–160 (dec)	−6.8 (*c* 1, AcOH)	extraction with BuOH
Boc-Arg(NO_2)-OH[c]	dioxane/H_2O	NaOH	15	82	107	−22.0 (*c* 2, pyridine)	pH 8.5[d]
Boc-Asn-OH	DMF/H_2O	NaOH	18	80	176 (dec)	−7.2 (*c* 2, DMF)	45–50 °C, 5 h
Boc-Asp(OBzl)-OH	tBuOH/H_2O	NaOH	16	85	101–102	−19.7 (*c* 2, DMF)	pH 8[d]
Boc-Cys(Bzl)-OH	dioxane/H_2O	NaOH	15	65	86–87	−43.4 (*c* 1, AcOH)	
(Boc-Cys-OH)$_2$	MeOH/H_2O	NaOH	16	85	143–145 (dec)	−115.6 (*c* 2, AcOH)	
Boc-Gln-OH	MeCN/H_2O	NaOH	18	76	125 (dec)	−3.4 (*c* 2, EtOH)	pH 8[d]
Boc-Glu(OBzl)-OH[e]	dioxane/H_2O	NaOH	15	86	142–143	+13.2 (*c* 1, MeOH)	pH 8.5–9[d]
Boc-Gly-OH	tBuOH/H_2O	NaOH	16	96	87–88	–	
Boc-His(Boc)-OH	tBuOH/H_2O	$KHCO_3$	18	75	170 (dec)	+19.5 (*c* 2, $CHCl_3$)	
Boc-Ile-OH	tBuOH/H_2O	NaOH	16	78	69–71	+2.8 (*c* 2, AcOH)	
Boc-Leu-OH[f]	tBuOH/H_2O	NaOH	18	96	85–87	−24.7 (*c* 2, AcOH)	
Boc-Lys(Z)-OH[c]	tBuOH/H_2O	NaOH	18	96	oil	–	
Boc-Met-OH[e]	tBuOH/H_2O	NaOH	18	85	139–140	+18.2 (*c* 2, EtOH)	
Boc-Pro-OH	tBuOH/H_2O	NaOH	12	95	134–135	−60.6 (*c* 2, AcOH)	
Boc-Ser-OH	tBuOH/H_2O	NaOH	16	66–82	86–88	−3.6 (*c* 2, AcOH)	pH 8.5–9[d]
Boc-Ser(Bzl)-OH	dioxane/H_2O	NaOH	16	90	62–63	+19.2 (*c* 2, 80% EtOH)	pH 8.5–9[d]
Boc-Thr-OH	tBuOH/H_2O	NaOH	16	85	71–73	−8.2 (*c* 1, AcOH)	
Boc-Trp-OH[g]	tBuOH/H_2O	NaOH	16	96	137–138 (dec)	−18.2 (*c* 1, DMF)	
Boc-Trp(For)-OH[e]	DMF	Et_3N	48	61	158–159[h]	+36.0 (*c* 2, EtOH)	
Boc-Tyr-OH	tBuOH/H_2O	NaOH[i]	24	75	137[e]	+2.6 (*c* 1, AcOH)	
Boc-Tyr(Bzl)-OH	dioxane/H_2O	NaOH	18	70	110–111	+27.6 (*c* 1, EtOH)	pH 10.4[d]
Boc-Val-OH	tBuOH/H_2O	NaOH	16	85	76–78	−7.5 (*c* 1, AcOH)	

[a] Reprinted from (Keller; Keller; van Look; Wersin, *Organic Syntheses*), copyright (1990), p 70, with permission from John Wiley & Sons, Inc.
[b] Reactions were performed at rt after the exothermic starting period had subsided. Progress of the reaction was monitored by TLC. Reaction times were not optimized.
[c] Crystallized with ~15% EtOAc.
[d] pH control is necessary.
[e] DCHA salt.
[f] Monohydrate.
[g] Crystallized with ~4% EtOAc.
[h] Decomposes below 130 °C if heated rapidly.
[i] 2 equivalents of base were employed.
[j] Resolidifies at 138 °C and does not melt below 300 °C.

Transprotection to *tert*-Butoxycarbonyl Derivatives

For synthetic purposes often an exchange of protecting groups is desirable and this requires two reaction steps, i.e. deprotection and reprotection. Such transformation of a Z-protected derivative into a Boc-protected derivative is readily performed under hydrogenation conditions in the presence of Boc_2O (**65**). Generally such a reaction is performed using Pd/C as catalyst for hydrogenation;[114,461] however, when benzyl ethers or esters are present in the substrate hydrogen donors such as triethylsilane and palladium(II) acetate are used[114] or transfer hydrogenation with cyclohexa-1,4-diene in the presence of 10% Pd/C is applied (Scheme 30).[116]

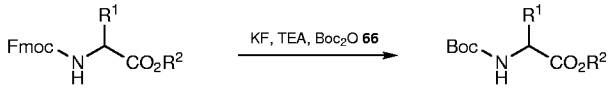
Scheme 30 Conversion of Z into Boc Carbamates[116]



α-Azidocarboxylic acids are also transformed to N^{α}-Boc-protected amino acids in a one-pot reaction using either H_2, Pd/C[462] or Et_3SiH, $Pd(OH)_2/C$[463] in the presence of Boc_2O. Yields are good to excellent with both protocols (cf. Section 2.1.1.10). This transformation is useful for the preparation of unnatural amino acids, that are, for example, accessible via the diastereoselective bromination of carboxylic acid derivatives and subsequent substitution with azide.[464]

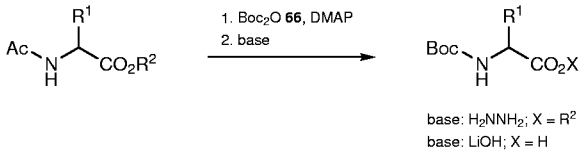
An exchange of the Fmoc with the Boc group is achieved in a one-pot reaction with potassium fluoride in the presence of TEA/Boc_2O (Scheme 31, cf. Section 2.1.1.1.1.3.2).[237]

Scheme 31 Conversion of Fmoc into Boc Carbamates[237]



Similarly, conversion of N^{α}-acetyl amino acid esters into the related N^{α}-Boc-protected derivatives is performed with Boc_2O/DMAP which in the first step leads to acylation of the acetyl amide group. This is then in turn hydrolyzed with hydrazine or lithium hydroxide to produce the N^{α}-Boc-protected amino acid ester or free carboxy derivative, respectively (Scheme 32). No racemization was detected under these reaction conditions.[465]

Scheme 32 Conversion of N^{α}-Ac- into N^{α}-Boc-Protected Amino Esters[465]



Boc-Ser(tBu)-OtBu; Typical Procedure for the Conversion of Z into Boc Carbamates by Hydrogenation over Pd/C:[461]

A cooled (0 °C) soln of Z-Ser-OH (5.00 g, 21 mmol) in dry CH_2Cl_2 (100 mL) was saturated with isobutene, after which H_2SO_4 (1 mL) was added and the mixture sealed at 0 °C and stirred at rt for 4 d. The reaction vessel was then carefully depressurized, and the mixture was washed with sat. $NaHCO_3$ (2 × 50 mL), dried, and concentrated. The crude Z-Ser(tBu)-OtBu (6.73 g) was dissolved in dry MeOH (50 mL) and to this soln were added Boc_2O (**65**; 5.00 g, 23 mmol) and 5% Pd/C (600 mg). The mixture was stirred under H_2 overnight, after which it was filtered through Celite, concentrated and chromatographed to give the product as an oil; yield: 5.32 g (80%); $[\alpha]_D^{25}$ +5 (*c* 1.4, $CHCl_3$).

for references see p 149

Boc-Thr-OMe; Typical Procedure for Conversion of Z into Boc Carbamates by Transfer Hydrogenation:[116]
To a soln of Z-Thr-OMe (200 mg, 0.75 mmol) and Boc_2O (**65**; 245 mg, 1.12 mmol) in abs EtOH (5 mL) was added under N_2 10% Pd/C (200 mg, 1:1 catalyst/substrate by weight) followed by cyclohexa-1,4-diene (0.74 mL, 7.5 mmol). After stirring for 80 min at rt, the mixture was filtered and the filtrate was concentrated. The residue was chromatographed (silica gel, EtOAc/hexane 1:4); yield: 166 mg (95%).

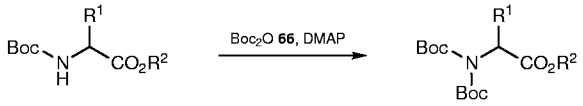
Boc-Thr(tBu)-Ile-OMe; Typical Procedure for Conversion of Fmoc into Boc Carbamates:[237]
To a stirred soln of Fmoc-Thr(tBu)-Ile-OMe (73 mg, 0.14 mmol) in DMF (1.2 mL) was added, under N_2, KF (56 mg, 0.98 mmol) and TEA (42 µL, 30 mg, 0.30 mmol). The soln was kept at rt and Boc_2O (**65**; 39 mg, 0.18 mmol) was added. After stirring for 10 h, the reaction was diluted with EtOAc and washed with H_2O, 5% HCl, 5% $NaHCO_3$, and brine. The organic layer was dried and concentrated. The resulting crude product was purified by chromatography; yield: 48 mg (85%).

Boc-Val-OMe; Typical Procedure for Conversion of N^{α}-Ac- into N^{α}-Boc-Protected Amino Acids:[465]
To a soln of Ac-Val-OMe (200 mg, 1.15 mmol) and DMAP (28 mg, 0.23 mmol) in THF (4 mL). Boc_2O (**65**; 0.53 mL, 2.30 mmol) was added, and the mixture was heated to reflux for 4 h. After the soln was cooled to rt, MeOH (4 mL) and hydrazine (0.14 mL, 4.6 mmol) were added and the mixture was stirred at rt for 4 h. The mixture was poured into CH_2Cl_2, washed with 1 M HCl, $CuSO_4$, and $NaHCO_3$, dried, and concentrated. The crude product was chromatographed (silica gel); yield: 240 mg (92%); $[\alpha]_D^{22}$ +12.9 (*c* 2.43, $CHCl_3$).

N^{α},N^{α}-Di-*tert*-Butoxycarbonyl Amino Acids

For particular applications the Boc carbamate N–H may be still too acidic and double protection is required. The second Boc group can be introduced by reaction of a N^{α}-Boc amino acid derivative with Boc_2O (**65**) in the presence of the acylation catalyst DMAP (Scheme 33). By this procedure N^{α},N^{α}-bis-Boc-protected amino esters are accessible in excellent yields within about 5 hours at room temperature.[466,467] The N^{α},N^{α}-bis-Boc amino acid esters are stable and easy to handle compounds. In addition to treatment with hydrazine or lithium hydroxide (*vide supra*) for the selective cleavage of one Boc group from a bis-acylated amino acid derivative, this is also achieved with magnesium perchlorate; e.g. $(Boc)_2$-Phe-OMe is converted into Boc-Phe-OMe at room temperature over 16 hours in 97% yield when 20% magnesium perchlorate is added to a solution of the amino acid in acetonitrile.[468] A similar reaction sequence can be used for the preparation of *N*-Boc-protected 1,2-amino alcohols. Reaction of the amino alcohol with phosgene leads to a cyclic carbamate, which can be *N*-Boc-protected with Boc_2O/DMAP. Subsequent opening of the cyclic carbamate with cesium carbonate gives the *N*-Boc-protected 1,2-amino alcohols.[469]

Scheme 33 N^{α},N^{α}-Bis-Boc-Protection of Amino Esters[466,467]



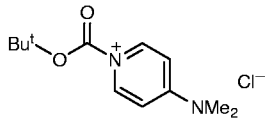
$(Boc)_2$-Gly-OBzl; Typical Procedure for the Synthesis of N^{α},N^{α}-Bis-Boc Amino Acid Esters:[467]
To a soln of Boc-Gly-OBzl (265 mg, 1.0 mmol) in abs MeCN (2–3 mL) was added with stirring DMAP (12 mg, 0.10 mmol) and Boc_2O (**65**; 240 mg, 1.10 mmol). After stirring at rt for 5 h, the solvent was removed and the oily residue was partitioned between Et_2O (50 mL) and 1 M $KHSO_4$ (25 mL). The organic layer was washed with 1 M $KHSO_4$, 1 M $NaHCO_3$, and brine, and dried. After evaporation of the solvent, the residue was essentially pure. It was recrystallized (petroleum ether, addition of activated charcoal); yield: 347 mg (95%); mp 30–31 °C.

2.1.1.1.3.1.1.7 Water-Soluble *tert*-Butoxycarbonylating Reagents

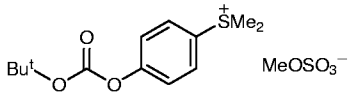
Water-soluble Boc donors are interesting N-acylating agents for postsynthetic protection of amino groups. Since acylpyridinium ions are very reactive species,[470] the unstable *tert*-butyl chloroformate was converted into the 1-(*tert*-butoxycarbonyl)-4-(dimethylamino)pyridinium chloride (**66**), which is a stable compound that reacts with amines in water (Scheme 34).[471]

Similarly, reaction of Boc_2O (**65**) with (4-hydroxyphenyl)dimethylsulfonium methylsulfate generates [4-(*tert*-butoxycarbonyloxy)phenyl]dimethylsulfonium methylsulfonate (**67**), which is stable on storage, soluble in organic solvents such as DMF, alcohols, and acetonitrile, but also very soluble in water (>30 wt%).[211] It has proved to be well-suited for the N-acylation of peptides in aqueous solution.

Scheme 34 Water-Soluble Boc Donors



66



67

2.1.1.1.3.1.2 Peptide Synthesis with N^{α}-*tert*-Butoxycarbonyl Amino Acids

N^{α}-Boc-protected amino acids are generally crystalline compounds, but if oily they are converted for storage into solids as DCHA or CHA salts; they are stable on storage and resistant to the majority of the reaction conditions used for peptide bond formation. The N^{α}-Boc protection is used in the standard Boc/Bzl chemistry in solution (see Sections 4.1 and 5.2) and on solid support (see Sections 4.3.2.1 and 5.3); for coupling of N^{α}-Boc-protected amino acids all condensation procedures, i.e. active esters, mixed anhydrides, and uronium and phosphonium salts as well as the acyl fluoride, can be applied except the acyl chloride method, since preparation of the acyl chlorides is accompanied by partial cleavage of the Boc group. Moderate acidic conditions as required during aqueous workup procedures do not affect the Boc group (stable >pH 4, essentially stable at pH >2). For such purposes the use of citric or oxalic acid or aqueous potassium hydrogensulfate is recommended rather than dilute solutions of HCl or H_2SO_4. The Boc group is resistant against hydrogenolysis, Na/NH_3 treatments and against nucleophilic or basic reagents (even at pH >12 cleavage of the carbamate is slow). Even with butyllithium deprotonation of the carbamate N–H, rather than addition to the carbamate carbonyl with subsequent cleavage of the Boc group is observed.

As with most other urethane-protected amino acids, racemization is largely suppressed (see Sections 1.2.1.2 and 7.4). Although in the activation process oxazol-5(4*H*)-one is formed, it has a low tendency to deprotonate and thus to racemize.[472]

2.1.1.1.3.1.3 Cleavage of the *tert*-Butoxycarbonyl Group

One of the major advantages of the Boc protection is the easy cleavage of this group under mild acidic conditions. Numerous reagents and cleavage cocktails have been proposed for this purpose and these consist almost exclusively of Brønsted or Lewis acids. That Boc-protected amino acids and peptides undergo thermal cleavage should be a warning, since the temperature of 180 °C which is required for fast and complete deprotection, might not be compatible with sensitive substrates.[473]

Reagents and conditions most frequently used for the deprotection of Boc amino acid and peptide derivatives are listed in Table 8.

for references see p 149

Table 8 Reagents and Conditions Most Frequently Used for the Deprotection of *N*-Boc Amino Acid and Peptide Derivatives

Conditions	Ref	
	In Solution	SPPS
HCl/AcOH	[41,42]	
HCl/dioxane	[36]	
HCl/EtOAc	[474]	
HF		[475]
HBr/AcOH	[41]	
AcCl/MeOH	[476]	
10% H_2SO_4/dioxane		[477]
HCO_2H	[478,479]	
70% TFA/H_2O	[40]	
50% TFA/CH_2Cl_2	[37]	[480]
TFA/AcOH	[481]	
TFA, phenol, *p*-cresol	[481]	
TosOH·H_2O	[38,482,483]	[484]
2-sulfanylethanesulfonic acid ion-exchange resin	[485] [486,487]	
BF_3·OEt_2/AcOH	[39,40]	
$AlCl_3$	[488]	
$SiCl_4$, phenol		[489]
TMSCl, phenol		[490]
TMSI	[491]	
TMSOTf, TFA	[492]	[492]
TMSOTf, 2,6-lutidine		[493]
silica gel at low pressure	[494]	

In the simplest cases where the Boc group is removed in the absence of additional acid-labile or partially acid-labile protecting groups inorganic acids such as HCl or HBr in organic solvents, e.g. AcOH, dioxane, EtOAc, or Et_2O, are used to facilitate isolation of the product. Similarly, neat TFA or 4-toluenesulfonic acid or TFA in CH_2Cl_2 and 4-toluenesulfonic acid in MeCN[495] are used. A simultaneous deprotection and purification of Boc-protected amino acids or esters has been reported on Amberlyst 15 ion-exchange resin. In this case the protecting group is cleaved within 12 hours and the deprotected amino acid is liberated from the resin by addition of ammonia in methanol in ~90% yield.[487] When repetitive Boc cleavage is required for the synthesis of peptides and proteins in solution and on solid support according to the Boc/Bzl strategy, conditions have been developed under which the semipermanent protection of the side-chain functionalities are resistant, and these are mainly 25–50% TFA in CH_2Cl_2 (for experimental procedures, see Sections 4.3.2.1 and 5.2.1). When the Boc protection is cleaved together with other *tert*-butyl alcohol derived esters and ethers from fully protected peptides in solution or deprotection/resin cleavage is the goal, optimized cocktails containing scavengers have been developed to facilitate cleavage and to prevent side reactions at sensitive amino acid side chains (Sections 4.1 and 4.3.5).

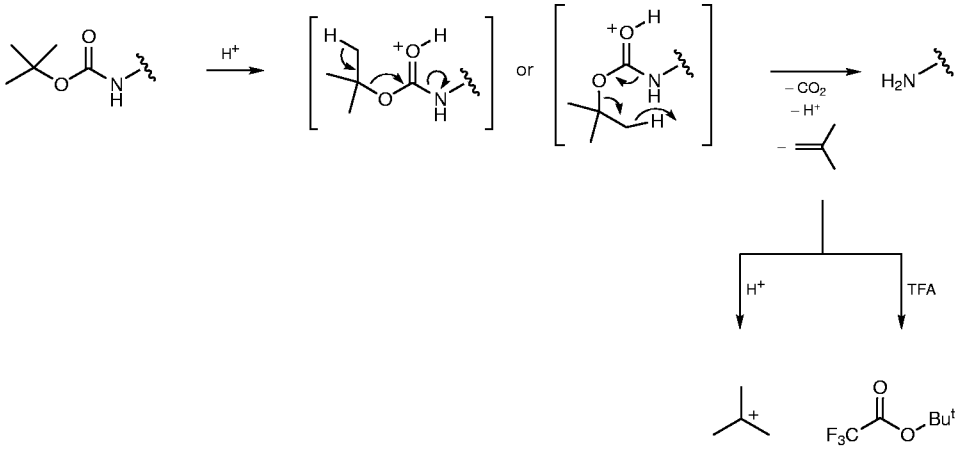
More difficulties are encountered in selective Boc cleavage in the presence of other groups of similar acid sensitivity, although some procedures are available in this context; the use of 1 M TMSOTf/1.5 M 2,6-lutidine/CH_2Cl_2 results in a clean, selective cleavage of the Boc group, without removal of peptides from Rink-amide resin.[493] Cleavage with TFA has been reported to not affect bulky *O*-silyl groups such as TBDPS[496] or TBDMS.[497] The selective cleavage of the Boc group in the presence of *tert*-butyl esters with the acidic ion-exchange resin Zeo-Carb 225/H^+ has also been reported. Similarly, selective Boc cleavage in the presence of ω-benzyl esters, which is not possible even with diluted TFA, occurs with TFA/phenol/cresol,[481] BF_3·OEt_2[40] or TMSCl/phenol,[490] respectively. Selective cleavage of the Boc group in the presence of *tert*-butyl esters is achieved with 1 M HCl/EtOAc[461] or by

using concentrated H_2SO_4 (1.5–3.0 equiv) in *tert*-butyl acetate or methanesulfonic acid (1.5–3.0 equiv) in *tert*-butyl acetate/CH_2Cl_2 (4:1) where acid-catalyzed re-esterification is taking place to compensate partial ester cleavage.[498] The selective removal of the Boc group in the presence of *tert*-butyl or 4-methoxybenzyl esters has been reported with TosOH.[482] A possible decomposition of tryptophan residues during acidolytic cleavage of the Boc group is substantially suppressed with 2 M TosOH/dioxane containing 2% anisole.[483]

The oxidative cleavage of *N*-Boc groups with ammonium cerium(IV) nitrate (CAN) in acetonitrile has been reported,[499] but the fact that, again, acidic or Lewis acidic rather than oxidative reaction conditions might affect the reaction (CAN is, at least in aqueous solution, strongly acidic) should be kept in mind. Boc-protected amino acids have been reported to be stable against other oxidative reagents.[473]

The acidic cleavage of the Boc group is most probably proceeded by the protonation of the carbamate carbonyl with subsequent elimination of isobutene, either by an open or, more likely, by a cyclic transition state. Protonation leads to the liberated amine or the ammonium salt, respectively (Scheme 35). The byproduct isobutene is either protonated to give the stable carbenium ion, or generates the *tert*-butyl trifluoroacetate,[500] both of which act as *tert*-butylating agents, especially, when nucleophilic amino acid side chains are present in the molecule (*vide infra*).

Scheme 35 General Mechanism for the Acidic Cleavage of the Boc Group



The *tert*-butylations of methionine,[475,501,502] histidine, and, in particular, tryptophan[500,503–509] have been reported. These side reactions, the extent of which is strongly dependent upon the topochemical environment of the amino acid residues, can be reduced or largely prevented by addition of suitable scavengers such as water, ethers, sulfides, or thiols, as well as indoles.[500] Such side reactions are also prevented using particular acids; for example, $AlCl_3$/anisole was used for the synthesis of cephalosporin derivatives without alkylation of the sulfur,[510] with TMSI the intermediate cation is immediately trapped by iodide,[491] and 2-sulfanylethanesulfonic acid has been proposed as an acidic reagent that bears a scavenger within its molecule.[485]

H-Ala-Pro-OH·HCl; Typical Procedure for Cleavage with HCl/Dioxane:[511]
Boc-Ala-Pro-OH (43 g, 151 mmol) was dissolved in dioxane (200 mL) and reacted at rt with 4 M HCl in dioxane (200 mL). After 4 h the dioxane was removed and the oily residue triturated with Et_2O. The product was filtered, washed extensively with Et_2O and reprecipitated from iPrOH with Et_2O; yield: 32.1 g (96%); mp 173–175 °C; $[\alpha]_D^{20}$ –112 (*c* 1, H_2O); $[\alpha]_{546}^{20}$ –134 (*c* 1, H_2O).

for references see p 149

H-Pro-Pro-Cys(Acm)-Pro-Ala-Pro-OH·HCl; Typical Procedure for Cleavage with HCl/Dioxane:[511]
Boc-Pro-Pro-Cys(Acm)-Pro-Ala-Pro-OH (47.2 g, 62.7 mmol) was treated with 2 M HCl in dioxane (350 mL), kept at rt for 2 h and concentrated. The oily residue was dissolved in iPrOH and on addition of Et_2O, the precipitate was filtered and extensively washed with EtOAc; yield: 42.4 g (98%); $[\alpha]_D^{20}$ −211 (*c* 1, H_2O); $[\alpha]_{546}^{20}$ −251 (*c* 1, H_2O).

H-Met-Met-OH; Typical Procedure for Cleavage with TFA:[1]
Boc-Met-Met-OH (19 g, 50 mmol) was dissolved in ice-cold TFA (100 mL), kept for 1 h at rt, and concentrated. The residue was dissolved in a small amount of H_2O, neutralized with concd NH_3 soln and treated with hot EtOH. After cooling for several h, the precipitate was collected and recrystallized (EtOH/H_2O with a few drops of thioglycol); yield: 13 g (93%); $[\alpha]_D^{20}$ +26.1 (*c* 1.9, H_2O).

H-Met-Met-OMe·HCl; Typical Procedure for Cleavage with HCl:[1]
Boc-Met-Met-OMe (60 g, 152 mmol) was treated with 3 M HCl in EtOAc (250 mL). The precipitation was completed by addition of abs Et_2O and cooling. The filtered precipitate was dried under reduced pressure; yield: 49.3 g (98%).

H-Cys(Trt)-Gly-OH; Typical Procedure for Cleavage with $BF_3 \cdot OEt_2$:[39]
A soln of Boc-Cys(Trt)-Gly-OH·DCHA (7.02 g, 10.0 mmol) in AcOH (50 mL) was treated with $BF_3 \cdot OEt_2$ (4 mL) for 30 min and then poured onto a soln of NaOAc (18 g) in ice water (100 mL). The gelatinous precipitate was collected, washed with H_2O (6 ×) and Et_2O (5 ×) and dried; yield: 4.05 g (96%); mp 114–116 °C; $[\alpha]_D^{25}$ −6.03 (*c* 1, HMPA).

H-Leu-Leu-Gln-Gly-Leu-Val-NH_2·TFA; Typical Procedure for Cleavage with TFA/PhOH/*p*-cresol:[481]
To a suspension of Boc-Leu-Leu-Gln-Gly-Leu-Val-NH_2 (0.37 g, 0.50 mmol) in PhOH/*p*-cresol (3 mL, 1:1) was added TFA (2 mL) and the resulting soln kept at rt for 1 h. The TFA was removed under reduced pressure and Et_2O (25 mL) was added to the residue. The precipitate was collected by centrifugation, washed with Et_2O (8 × 20 mL), dried in air and then under reduced pressure over P_4O_{10} to yield the TFA salt which has no defined melting point; yield: 380 mg (100%).

H-Ala-OtBu·TosOH; Typical Procedure for Cleavage with TosOH·H_2O:[482]
A soln of Boc-Ala-OtBu (2.45 g, 10.0 mmol) in Et_2O (50 mL) was added dropwise at −10 °C to a soln of TosOH·H_2O (1.90 g, 10.0 mmol) in EtOH (10 mL) over 30 min. The mixture was warmed to rt and stirred for 3 h. The solvents were removed and the residue recrystallized (EtOAc) to provide the product; yield: 2.91 g (91%); mp 222 °C; $[\alpha]_D^{22}$ +1.7 (*c* 1.6, MeOH).

2.1.1.1.3.2 1-Adamantyloxycarbonyl Group

E. Lohof, H.-J. Musiol, and J. Podlech

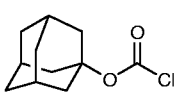
An alternative to the *N*-Boc protection the bulky 1-adamantyloxycarbonyl (Adoc) group has been proposed by Haas et al.[398] This urethane-type protection is removed under identical mild acidic conditions to the Boc group, but the Adoc group has as an advantage higher stability due to steric hindrance and a significant solubilizing effect derived from the bulky hydrophobic group. It has therefore been advantageously used in solution synthesis, particularly for the side-chain protection of amino groups[395,602] as well as for N^{α},N^{im}-bis-protection of N-terminal histidine residues[398,512,513] or N^{α},N^{ω},$N^{\omega'}$-protection of N-terminal arginine residues[515] for peptide segment condensations.

2.1.1.1.3.2.1 Synthesis of N^{α}-1-Adamantyloxycarbonyl Amino Acids

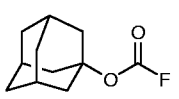
The reagents for the synthesis of N^{α}-1-adamantyloxycarbonyl amino acid derivatives are similar to those used for other urethane-type protecting groups (Scheme 36). Although the related 1-adamantyl chloroformate (Adoc-Cl, **68**) is obtained as crystalline solid, the acylation of amino acids under Schotten–Baumann conditions leads to significant decomposition of the reagent and thus its use does not result in satisfactory yields in all cases. Better results are obtained with 1-adamantyl fluoroformate (Adoc-F, **69**) (yields of N^{α}-Adoc amino acids: 85–95%)[514] which is commercially available or readily prepared from adamantan-1-ol and

fluorophosgene.[417,431] Alternatively, Adoc-Cl (**68**) is converted into the mixed carbonate 1-adamantyl succinimido carbonate (Adoc-OSu, **70**).[515] Acylation of amino acids with Adoc-F (**69**) is conveniently performed under Schotten–Baumann conditions whereas amino acid and peptide derivatives are N-acylated in organic solvents using *N,N,N′,N′*-tetramethylguanidine or benzyltrimethylammonium hydroxide as a base.

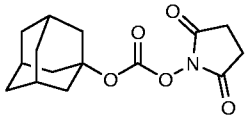
Scheme 36 Reagents Used for the Introduction of the 1-Adamantyloxycarbonyl Group



68 Adoc-Cl



69 Adoc-F



70 Adoc-OSu

1-Adamantyl Chloroformate (Adoc-Cl, 68):[398]

CAUTION: *Phosgene is a colorless, poisonous gas or volatile liquid. This reagent should be handled with the utmost care.*

To a cooled soln of $COCl_2$ (30 g, 0.30 mol) in benzene (100 mL) was added dropwise a soln of adamantan-1-ol (8 g, 53 mmol) and pyridine (7 g, 88 mmol) in benzene (200 mL) at 4 °C over 1 h; when a precipitate formed additional benzene was added. After 1 h at rt the mixture was filtered into ice-cold H_2O. The organic layer was dried (Na_2SO_4) and concentrated. The product, which was obtained in an essentially quantitative yield, crystallized (petroleum ether at −20 °C), no yield was given for the crystallized product; mp 46–47 °C.

1-Adamantyl Fluoroformate (Adoc-F, 69):[56]
To 65% oleum (3 L) was added at 35 °C slowly CCl_3F (550 mL) over 5 h. The resulting fluorophosgene was washed via two condensers (−10 °C) in two washing flasks containing H_2SO_4 and then condensed over a low temperature condenser (−60 °C) into an ice-cold soln of adamantan-1-ol (300 g, 2 mol) and TEA (280 mL, 2 mol) in CH_2Cl_2 (3.5 L). The mixture was allowed to reach rt and stirred for additional 60 min. The excess fluorophosgene was extracted under reduced pressure over NaOH for 12 h, and then the suspension was concentrated. The residue was taken up in hexane and after 24 h the precipitate (TEA·HCl) was filtered off, the filtrate was decolorized on charcoal and concentrated. The oily residue solidified on standing in the cold; yield: 394 g (99%); mp 29 °C (lit. mp 31–32 °C).

1-Adamantyl Succinimido Carbonate (Adoc-OSu, 70):[515]
To a stirred and ice-cold soln of Adoc-Cl (**68**; 21.5 g, 100 mmol) and HOSu (15 g, 130 mmol) in dioxane (100 mL) was added pyridine (8.2 mL, 101 mmol) in dioxane (65 mL). After 20 h at rt, the mixture was filtered and the filtrate taken to dryness. The residue was recrystallized (MeOH); the mother liquors were concentrated and the residue crystallized from MeOH/H_2O. The combined crystalline fractions were washed with Et_2O; yield: 19.6 g (67%); mp 140–141 °C.

Synthesis of N^{α}-Adoc Amino Acids; General Procedure:[514]
To the ice-cold soln of amino acid (10 mmol) in 1 M NaOH (10 mL) was added $NaHCO_3$ (0.92 g, 11 mmol) followed by Adoc-F (**69**; 2.18 g, 11 mmol) in dioxane (10 mL). The mixture was stirred for several hours at rt. After completion of the reaction, the bulk of dioxane was removed and the aqueous soln was extracted with Et_2O (3 ×). The aqueous layer was acidified with aq $KHSO_4$ and extracted with EtOAc (2 ×). The organic layers were washed with H_2O, dried ($NaSO_4$), and concentrated. The residue was recrystallized from suitable solvents, e.g. EtOAc/petroleum ether.

Adoc-His(Adoc)-OH:[56]
To a soln of H-His-OH (3.1 g, 20 mmol) in 2 M NaOH (10 mL) and dioxane (100 mL) was added Adoc-F (**69**; 8.7 g, 44 mmol) in dioxane and $NaHCO_3$ (3.7 g, 44 mmol) in H_2O (25 mL). After 3 h, the dioxane was removed, the mixture diluted with H_2O, acidified with 5% $KHSO_4$ and extracted with Et_2O (3 ×). The organic layers were combined, washed with H_2O, dried (Na_2SO_4), and concentrated. The product was precipitated with petroleum ether; yield: 9.3 g (91%); mp 144–145 °C (lit.[398] mp 140–145 °C); $[\alpha]_D^{20}$ +11.8 (*c* 2, MeOH).

for references see p 149

2.1.1.1.3.2.2 Cleavage of the 1-Adamantyloxycarbonyl Group

Cleavage of *N*-Adoc derivatives is performed concurrently with the cleavage of *tert*-butyl alcohol derived protecting groups, i.e. under the relatively mild TFA conditions in the presence of suitable scavengers (see Section 2.1.1.1.3.1.3).

Cleavage of *N*-Adoc Peptide Derivatives with TFA; Typical Procedure:[101]
The Glu^{24}-PHI heptacosapeptide derivative Adoc-His(Adoc)-Ala-Asp(OtBu)-Gly-Val-Phe-Thr(tBu)-Ser(tBu)-Asp(OtBu)-Phe-Ser(tBu)-Arg(HBr)-Leu-Leu-Gly-Gln-Leu-Ser(tBu)-Ala-Lys(Adoc)-Lys-(Adoc)-Tyr(tBu)-Leu-Glu(OtBu)-Ser(tBu)-Leu-Ile-$NH_2 \cdot 2H_2O$ (0.3 g, 0.069 mmol) was dissolved in ice-cold TFA (25 mL) containing anisole (2.5 mL) and ethane-1,2-dithiol (1 mL). The soln was kept at rt for 2 h and then concentrated. On addition of Et_2O, the crude product was precipitated with Et_2O and was purified by CM-Biogel A ion-exchange chromatography followed by partition chromatography on Sephadex G-25F; yield: 88 mg (30%).

2.1.1.1.3.3 2-(Biphenyl-4-yl)prop-2-yloxycarbonyl Group

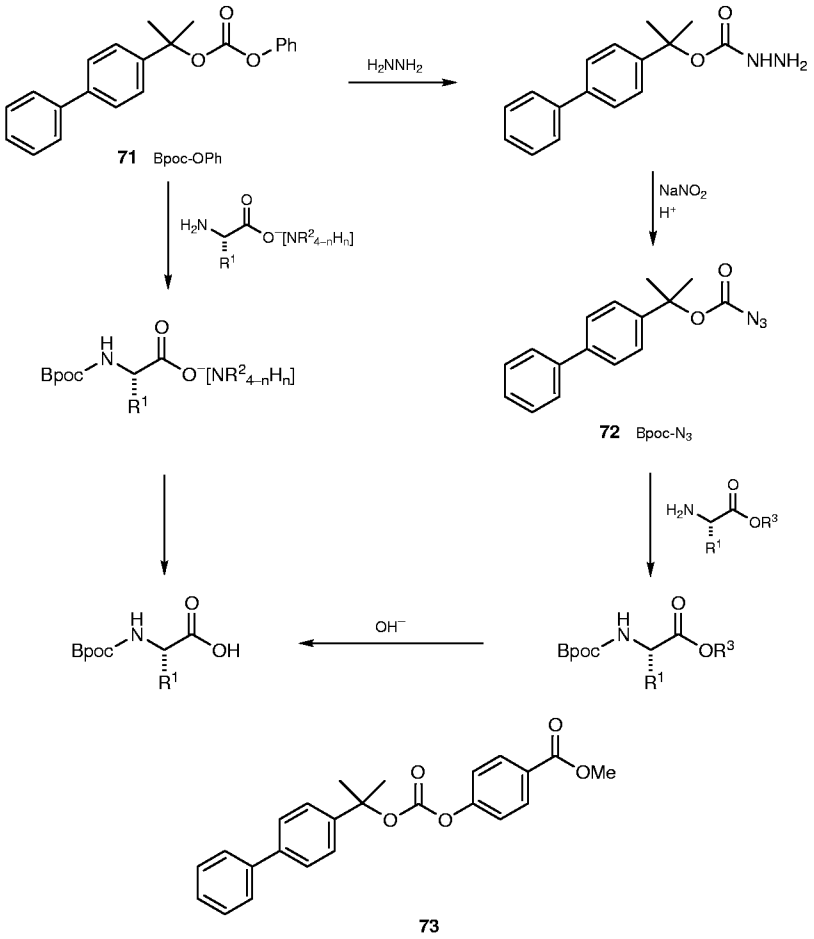
M. Gurrath, G. Müller, and J. Podlech

The 2-(biphenyl-4-yl)prop-2-yloxycarbonyl (Bpoc) group has been proposed by Sieber and Iselin[45,46] as an α-amino protecting group that is cleaved under weakly acidic conditions. It has been successfully used in precedent-setting peptide syntheses of historical note, in particular in the first total synthesis of human insulin.[531] Despite additional applications in solution synthesis[516,517] or in SPPS[518,519] this type of amino protection has not gained widespread use. It was used for a specific purpose by Kemp et al. in combination with *tert*-butyl alcohol derived protecting groups for solid-phase synthesis on phenyl ester resin where the Fmoc/tBu chemistry is not compatible with the base-labile linker group.[520–524] The main advantage of this amino protecting group is its high acid lability which ensures rapid and complete deprotection at low acid concentrations.[43–45] For comprehensive reviews, see refs[1,2,186].

2.1.1.1.3.3.1 Synthesis of N^{α}-2-(Biphenyl-4-yl)prop-2-yloxycarbonyl Amino Acids

For the preparation of N^{α}-Bpoc-protected amino acids generally 2-(biphenyl-4-yl)prop-2-yl azidoformate (Bpoc-N_3, **72**) and the mixed carbonate 2-(biphenyl-4-yl)prop-2-yl phenyl carbonate (Bpoc-OPh, **71**) are used (Scheme 37),[45,46] although the related 2-(biphenyl-4-yl)prop-2-yl fluoroformate (Bpoc-F) has been proposed as the more reactive reagent.[525] 2-(Biphenyl-4-yl)prop-2-yl phenyl carbonate (Bpoc-OPh, **71**) can be used directly as an acylating agent or is further converted into the corresponding hydrazide as a shelf-stable precursor of Bpoc-N_3 (**72**).[44–46,526] Both Bpoc-OPh (**71**) and Bpoc-N_3 (**72**) are thermally unstable and decompose to 2-(biphenyl-4-yl)propene or 2-(biphenyl-4-yl)-2-phenoxypropane, respectively.[45,526,527] The stability of 2-(biphenyl-4-yl)prop-2-yl phenyl carbonate (**71**) is increased with the introduction of *para* substituents in the phenyl moiety, i.e. with 4-methoxy- and, particularly, with 4-methoxycarbonyl derivatives such as 2-(biphenyl-4-yl)-prop-2-yl 4-(methoxycarbonyl)phenyl carbonate (**73**).[525] According to the original work of Sieber and Iselin,[45,46] the Bpoc group can be introduced by reaction of Bpoc-N_3 (**72**) with amino acid esters followed by the ester saponification or by acylation of amino acid salts with Bpoc-N_3 (**72**) or Bpoc-OPh (**71**) in dry solvents. These procedures were later optimized[44] and of the tetraalkylammonium salts the *N,N,N′,N′*-tetramethylguanidinium salt was found to give the best results[526] (for a compilation of N^{α}-Bpoc amino acid derivatives, see ref[186]). Due to the sensitivity of the N^{α}-Bpoc group to the free α-carboxy group, the related amino acid derivatives are converted into and stored as CHA or DCHA salts.

Scheme 37 Reagents for the Synthesis of N^{α}-Bpoc Amino Acid Derivatives



2-(Biphenyl-4-yl)prop-2-yl Phenyl Carbonate (Bpoc-OPh, 71):[526]
To a cold (−10 °C) stirred soln of 2-(biphenyl-4-yl)propan-2-ol (107.3 g, 0.5 mmol) and pyridine (49.6 mL, 0.61 mol) in CH_2Cl_2 (500 mL) was added over 1 h phenyl chloroformate (87.7 g, 0.56 mol) in CH_2Cl_2 (150 mL). The mixture was stirred overnight at 4 °C, filtered, and poured over ice water (600 mL). The layers were separated and the organic phase was washed with H_2O (3 × 1 L) and brine (500 mL), dried (Na_2SO_4), and concentrated to solid residue; yield: 133 g (80%), mp 111–115 °C; the product is moisture sensitive and must be stored in the cold.

Bpoc-Ala-OH·DCHA; Typical Procedure:[526]
To 2.1 M benzyltrimethylammonium hydroxide in MeOH (20 mL, 42 mmol) was added H-Ala-OH (4.0 g, 45 mmol) and the resulting suspension was concentrated. The oily residue was re-concentrated from DMF (3 × 15 mL) and the syrup was dissolved in DMF (10 mL) and treated with Bpoc-OPh (**71**; 10 g, 30 mmol) for 3 h at 50 °C. The solvent was removed and the residue was taken up in H_2O (50 mL). Upon addition of small amount of Na_2SO_4 to prevent emulsion, the aqueous layer was extracted with Et_2O (2 × 50 mL; all extracts are back-extracted). The aqueous layer and aqueous washes were combined, overlayered with Et_2O (100 mL), chilled by the addition of ice, and acidified with 0.5 M citric acid/sodium citrate buffer (pH 3.5). The aqueous phase was extracted twice with Et_2O, and the pooled extracts were washed with citrate buffer, H_2O, and brine, and then dried (Na_2SO_4). Upon addition of DCHA (6.5 g, 35.8 mmol) the precipitate was collected and recrystallized twice (EtOAc or from boiling Et_2O with fast filtration through Celite); yield: 12.9 g (85%); mp 167–168 °C (lit.[528] mp 163–165 °C).

for references see p 149

Synthesis of N^{α}-Bpoc Amino Acids with Bpoc-N_3 (72); Typical Procedures:[526]

2-(Biphenyl-4-yl)prop-2-yl Azidoformate (Bpoc-N_3, 72):[526]
To a soln of Bpoc-$NHNH_2$[526] (10 g, 37 mmol) in MeCN (100 mL) was added in one portion at −25 °C a mixture of 6 M HCl (19 mL) and MeCN (36 mL), followed by dropwise addition of 5 M $NaNO_2$ (8.3 mL, 42 mmol) over 5 min under vigorous stirring and maintaining the temperature at −15 to −20 °C. After another 10 min stirring at −14 °C, the soln was cooled to −20 °C and neutralized to pH 7 under vigorous stirring by rapid addition of 1 M Na_2CO_3. The mixture was then poured into cold H_2O (400 mL) and the resulting yellow solid was collected by filtration and dissolved in Et_2O. The soln was dried ($MgSO_4$) and concentrated. The azide can be stored under N_2 at freezer temperature for up to three months. Decomposition of a stored sample can be tested by inspection of the ^1H NMR spectrum for olefinic resonances at δ 5.1 and 5.5.

Bpoc-Cys(Acm)-OH · CHA:[526]
To a soln of H-Cys(Acm)-OH (1.1 g, 5.3 mmol) in freshly distilled DMF (20 mL) Bpoc-N_3 (**72**; 2.4 g, 8.5 mmol) was added, and the mixture was stirred at 25 °C for 5 min under N_2 before addition of *N,N,N′,N′*-tetramethylguanidine (1.4 mL, 11 mmol). After 2 h stirring, the mixture was poured onto an aq $NaHCO_3$/ice slurry, and the resulting aqueous phase was extracted with Et_2O (2 × 100 mL), chilled by the addition of ice and acidified with citrate buffer to pH 3.5. The aqueous phase was extracted with Et_2O (4 × 75 mL) and the organic phases were washed with brine, dried, and treated with CHA (0.6 mL, 5.3 mmol); yield: 2.1 g (76%); mp 78 °C.

Bpoc-Asn(Trt)-OH:[529]
To a soln of H-Asn(Trt)-OH (600 mg, 1.6 mmol) and Bpoc-N_3 (**72**; 674 mg, 2.39 mmol) in DMF (10 mL) was added *N,N,N′,N′*-tetramethylguanidine (0.4 mL, 3.3 mmol). After stirring under N_2 for 4 h, the reaction was poured into cold sat. $NaHCO_3$ and extracted with Et_2O (2 × 20 mL). The aqueous layer was cooled to 4 °C by the addition of ice, overlayered with EtOAc (25 mL), and acidified to pH 3.5 by addition of 1 M citric acid. The aqueous layer was extracted twice more with EtOAc, and the EtOAc layers were combined and washed with H_2O (2 ×), brine (1 ×), dried ($MgSO_4$), and concentrated to a lumpy solid which was triturated with Et_2O (2 × 15 mL); yield: 700 mg (71%); mp 134–135 °C.

2.1.1.1.3.3.2 Peptide Synthesis with N^{α}-2-(Biphenyl-4-yl)prop-2-yloxycarbonyl Amino Acids

A major disadvantage in the use of Bpoc protection in peptide synthesis is the autocatalyzed decomposition of N^{α}-Bpoc amino acids when stored as free acids.[43] In contrast, the corresponding CHA or DCHA salts, which are generally obtained in a crystalline form, are stable when stored even upon repeated warming and recooling cycles. The N^{α}-Bpoc amino acids must be recovered from the salts prior to their use in peptide synthesis, a fact which *per se* poses no complications, since the Bpoc group is sufficiently stable at pH 3.5 to allow essentially complete recovery of the free acid. For synthesis in solution N^{α}-Bpoc amino acids are used together with the respective *N*-hydroxysuccinimide, 4-nitrophenyl, 2,3,5-trichlorophenyl, or pentachlorophenyl esters.[516,517,528,530,531] For solid-phase synthesis the pentafluorophenyl esters have been proposed.[532] These active esters are readily accessible by standard procedures and are generally obtained as crystalline compounds more stable on storage than the corresponding free acids. The N^{α}-Bpoc amino acid pentafluorophenyl esters have been successfully used, for example, in the solid-phase synthesis of brakykinin, substance P, and a fragment of the insulin B chain.[532]

The initially recommended conditions for Bpoc cleavage (Table 9) have been refined for use in solid-phase synthesis with 0.5% TFA in CH_2Cl_2.[43,44] Although such conditions are compatible with most of the Boc, tBu, and even Trt side-chain protections, repetitive treatment of growing peptide chains under these acidic conditions leads to partial deprotection of Lys(Boc), Tyr(tBu), and His(Trt) derivatives whilst the residues Asn(Trt) and Gln(Trt) are sufficiently stable.[529] The 2-(biphenyl-4-yl)propene formed in the acidolytic deprotection step as well as its dimer represent a source for the formation of related cations which may react with nucleophilic side-chain functionalities as present in tryptophan and methionine residues, thus requiring efficient cation scavengers, e.g. indoles or Trp derivatives.

Magnesium perchlorate in acetonitrile as a very mild reagent for the cleavage of the Bpoc group has also been reported. With this, neither resins like Siebe- and Rink-amide and SARSIN resins nor extremely acid labile thioxo peptides are cleaved.[839]

Table 9 Relative Rates of Acidic Cleavage of N^{α}-Bpoc and N^{α}-Boc Amino Acid Derivatives at 22–25 °C[46]

Acidic Conditions	Cleavage Time	N^{α}-Bpoc/N^{α}-Boc	N^{α}-Bpoc/N^{ε}-Boc	Ref
$AcOH/H_2O$ (8:2)	3.5–7 h	2000–4000:1	4300–8700:1	[46]
$AcOH/83\%\ HCO_2H/H_2O$ (7:1:2)	100 min	1800–2700:1	4900–6800:1	[46]
$CH_2Cl_2/75\%\ ClCH_2CO_2H$ (50:37)	10 min	6200–9000:1		[46]

Solution Synthesis and Deprotection of N^{α}-Bpoc Peptides:

Bpoc-Ser(tBu)-Thr(tBu)-Cys(Trt)-Met-Leu-Gly-OMe:[533]
To a soln of Bpoc-Ser(tBu)-Thr(tBu)-$NHNH_2$ (4.4 g, 7.7 mmol) in DMF (40 mL) was added at −10 °C 2 M HCl in EtOAc (9.6 mL, 19.2 mmol) and *tert*-butyl nitrite (0.91 mL, 8 mmol) followed after 15 min, by a precooled soln of H-Cys(Trt)-Met-Leu-Gly-OMe·AcOH (5.18 g, 7 mmol) and TEA (2.7 mL, 19.2 mmol) in DMF (35 mL). After 1 h at −10 °C, the TEA·HCl salt was filtered off, and the soln was concentrated. The solid was triturated with H_2O and suspended in benzene/petroleum ether (1:2). The insoluble product was recrystallized (MeOH); yield: 7.0 g (82%); mp 175–177 °C; $[\alpha]_D^{20}$ −33 (*c* 1.1, MeOH).

H-Ser(tBu)-Thr(tBu)-Cys(Trt)-Met-Leu-Gly-OMe·AcOH:[533]
A soln of Bpoc-Ser(tBu)-Thr(tBu)-Cys(Trt)-Met-Leu-Gly-OMe (8.46 g, 6.95 mmol) in 80% AcOH (100 mL) was kept at 45 °C for 90 min and then concentrated. The residue was precipitated from MeOH (15 mL) and EtOAc (65 mL) with petroleum ether; yield: 7.2 g (quant).

Solid-Phase Synthesis of Substance P by N^{α}-Bpoc Protection:[532]
Starting with an MBHA resin, loaded with Met (0.57 mmol·g^{-1}, 100 mg), ten cycles of (i) coupling were conducted with 0.1 M Bpoc-Xaa-OPfp (0.228 mmol), DIPEA (0.456 mmol), and HOBt (0.114 mmol) in DMF, 2 h; (ii) deprotection of the Bpoc group with 0.5% TFA in CH_2Cl_2, (1 × 3 min and 1 × 15 min); (iii) washing with CH_2Cl_2 (4 × 8 mL); and (iv) washing with DMF (2 × 8 mL). For side-chain protection Gln(Trt), Lys(Tfa), Arg(Pmc) were used. Upon completion of the synthesis, the Trt and Pmc groups were removed with 95% TFA/1% H_2O/4% ethane-1,2-dithiol, 30 min. The deprotected, resin-bound peptide was cleaved with 1 M TfOH/1 M thioanisole in TFA (4 °C, 4 h), precipitated with Et_2O and triturated with Et_2O (6 ×). The crude product was treated with 20% piperidine in H_2O for 1 h, concentrated, triturated with Et_2O, and purified on C18 Vydac preparative HPLC column, and lyophilized; yield: 41 mg (53%).

2.1.1.1.3.4 2-(3,5-Dimethoxyphenyl)prop-2-yloxycarbonyl Group

J. Podlech

The 2-(3,5-dimethoxyphenyl)prop-2-yloxycarbonyl group (Ddz) (Scheme 38) was developed by Birr et al. as an alternative to the Bpoc group which for most synthetic purposes is too acid labile, since N^{α}-Bpoc-protected amino acids are occasionally cleaved by their intrinsic acidity.[47] Kinetic measurements revealed that the relative cleavage rates of Boc, Ddz, and Bpoc are 1:1400:3000 (for various acid deprotection procedures, see Table 10).[47]

Magnesium perchlorate in acetonitrile has also been reported to be a very mild reagent for the cleavage of the Bpoc- and the Ddz group (*vide supra*).[839]

for references see p 149

Scheme 38 Introduction and Cleavage of the Ddz Group[47]

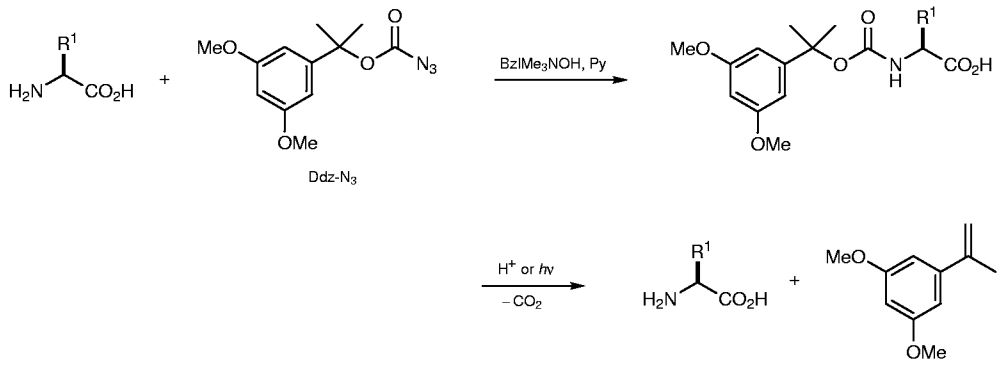


Table 10 Acid Lability of the Ddz Group[47]

Conditions	50% Cleavage (min)	Complete Cleavage (min)	Ref
80% AcOH/H_2O, 20 °C	60	180	[47]
50% AcOH/H_2O, 20 °C	30	80	[47]
AcOH, 50 °C	30	90	[47]
50% AcOH/H_2O, 50 °C	8	20	[47]
TFA, 20 °C	immediate	immediate	[47]
10% TFA/H_2O, 20 °C	immediate	immediate	[47]
5% TFA/H_2O, 20 °C	3	8	[47]
1% TFA/H_2O, 20 °C	25	60	[47]

A further feature of this protecting group is its photolytic cleavage (see Section 2.4). Absorption maxima are at 276 and 282 nm, but deprotection is usually performed by simple irradiation through a quartz apparatus. Only traces of side products can be detected under these reaction conditions.[47,136,534]

2.1.1.1.3.4.1 Synthesis of N^{α}-2-(3,5-Dimethoxyphenyl)prop-2-yloxycarbonyl Amino Acids

The synthesis of N^{α}-Ddz-protected amino acids is generally achieved with the 2-(3,5-dimethoxyphenyl)prop-2-yl azidoformate (Ddz-N_3) which is obtained from 3,5-dimethoxybenzoic acid by a multistep procedure.[47] Protection is performed either under Schotten–Baumann conditions, or preferentially, by acylation of the amino acid benzyltrimethylammonium salts in pyridine.[47] N^{α}-Acylation of the amino acid methyl esters followed by saponification has been reported to produce the desired derivatives in higher overall yields.[535]

The 2-(3,5-dimethoxyphenyl)prop-2-yloxycarbonyl group has been used in continuous solid-phase synthesis of a number of biologically active peptide fragments.[536,537]

Synthesis of N^{α}-Ddz Amino Acids; General Procedure:[47]
The amino acid (230 mmol) was dissolved at 40–60 °C in a 40% soln of benzyltrimethylammonium hydroxide in MeOH (100 mL, ca. 240 mmol) and MeOH (200 mL). The mixture was concentrated twice and each time dissolved in benzene (2 × 500 mL). After concentration, the residue was dissolved in pyridine (400 mL) at 40 °C with exclusion of moisture and with stirring. Ddz-N_3 (81 g, 305 mmol) in pyridine (200 mL) was added dropwise, and the mixture was stirred for 48 h at 40 °C. The mixture was concentrated, 1 M NaOH (500 mL) was added and the soln was extracted with Et_2O (4 × 300 mL). The aqueous phase was chilled with ice, acidified with citric acid to pH 4 (evolution of HN_3, work under a fumehood), extracted with CH_2Cl_2 (4 × 300 mL), dried, and concentrated. The residue was dissolved in boiling Et_2O and DCHA (1.1 equiv) was added. The precipitate can be recrystallized. For example: Ddz-Phe-OH · DCHA (recrystallized from MeOH/H_2O); yield: 97%; mp 182–184 °C; Ddz-Pro-OH · DCHA (recrystallized from EtOAc/petroleum ether); yield: 98%; mp 114–116 °C.

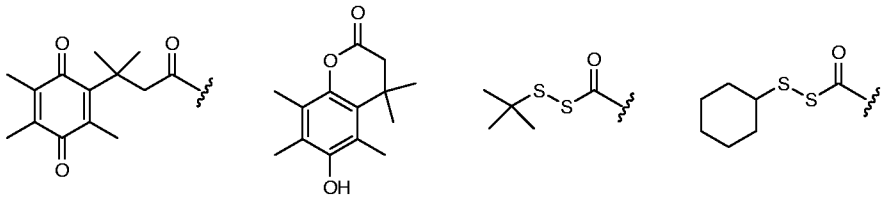
N^{α}-Ddz Amino Acid Esters; General Procedure:[535]
To a soln of the amino ester hydrochloride (0.1 mol) in DMF (45 mL) was added under stirring TEA (14 mL, 0.1 mol) and Ddz-N_3 (31.8 g, 0.12 mol). After 5 and 30 min, further portions of TEA were added (14 mL and 2.8 mL, respectively) and stirring was continued at rt for 24 h. The mixture was partitioned between EtOAc (100 mL) and H_2O (100 mL), the organic layer was extracted with citrate buffer (pH 3, 3 × 50 mL), $NaHCO_3$ (100 mL), and H_2O (50 mL), dried, and concentrated. For example: Ddz-Ile-OMe; yield: 90%; mp 91–92 °C; $[\alpha]_D^{22}$ −15.7 (*c* 1, MeOH); Ddz-Tyr(tBu)-OMe; yield: 99%; oil.

H-Leu-Glu(OtBu)-Asn-Tyr(tBu)-Cys(Trt)-Asn-OtBu·TFA; Typical Procedure for Cleavage with TFA:[535]
Ddz-Leu-Glu(OtBu)-Asn-Tyr(tBu)-Cys(Trt)-Asn-OtBu (27.8 g, 20 mmol) was slowly added at rt to a mixture of TFA (17.5 mL, 0.22 mol), H_2O (3.5 mL), and CH_2Cl_2 (330 mL). Stirring was continued for 2 h, the soln was chilled to 0 °C and neutralized with pyridine (17.8 mL, 0.22 mol). After evaporation, the residue was diluted in EtOAc (250 mL) and the soln was extracted with H_2O (3 × 25 mL). The organic layer was (without drying) concentrated (0.1 Torr), triturated with Et_2O, and dried. Then, 6-g portions were dissolved in boiling EtOAc and centrifuged. The precipitate was triturated with Et_2O (2 ×), centrifuged and dried; yield: 22.3 g (86%); mp 178–182 °C; $[\alpha]_D^{22}$ +3.5 to +4.0 (*c* 1, 90% AcOH).

2.1.1.2 Monoacyl Protecting Groups

Monoacyl-type protecting groups are not recommendable for N^{α}-protection, since racemization of the related amino acid derivatives during carboxy activation for their coupling represents a serious problem. Consequently, *N*-acyl derivatives such as formyl or trifluoroacetyl have been used mainly for side-chain protection. Nevertheless, some new N^{α}-derivatives, which address this problem, have been proposed. With the quinone-derived acyl-type protecting group 3-methyl-3-(2,4,4-trimethyl-3,6-dioxocyclohexa-1,4-dienyl)butyryl (**74**) (Scheme 39) racemization is almost totally suppressed, and it is readily cleaved with sodium dithionite with formation of the chromanone byproduct **75**.[538] The N^{α}-(alkyldisulfanyl)carbonyl derivatives **76** and **77** were found to be stable under the usual conditions of peptide synthesis, but are cleaved by reduction by trialkylphoshanes or thiols.[539]

Scheme 39 Acyl Protection Groups



74 **75** **76** **77**

2.1.1.3 Diacyl Protecting Groups

While the N^{α}-phthaloyl (Phth)[540,541] and N^{α}-dithiasuccinoyl (Dts)[542] derivatives of amino acids are used as reversible protecting groups, the aliphatic *N*-maleoyl group[543,544] is mainly exploited as a reactive handle for postsynthetic modifications of peptides with thiol-functionalized labels and probes, e.g. carbohydrates,[545] lipids,[546] and chromophores.[547] These N^{α}-diacyl derivatives are highly stable to even strong acids, but are cleaved under basic conditions.

for references see p 149

2.1.1.3.1 Phthaloyl Protection

M. Gurrath, G. Müller, and J. Podlech

The N^{α}-phthaloyl group (Phth), well known for the preparation of primary amines in the Gabriel synthesis,[548] was used more extensively as a temporary backbone amine protecting group in the early period of amino acid and peptide chemistry.[540,541,549–551] The resulting phthalimides ensure exhaustive substitution of the primary amine, i.e. removal of both acidic hydrogens, thus, moderating the nucleophilic character.

Due to the synthetic vagaries in the introduction as well as the removal of the phthaloyl group, superior alternatives now exist for many of its original applications and, therefore, in recent times it has received only scant attention.[2] Nevertheless, due also to the development of new phthaloyl derivatives, i.e. of tetrachlorophthaloyl[552,553] and 4-nitrophthaloyl,[554] that have improved the cleavage conditions, this type of N-protection has gained considerable interest in the synthesis of aminoglycosides,[552,555–559] β-lactam antibiotics,[560] phosphorus-containing amino acid analogues,[561,562] and amino acid–oligonucleotide analogues.[563] So far, the advantages of the tetrachloro- and 4-nitrophthaloyl groups in terms of cleavage conditions have not been investigated in peptide synthesis.

In stereoselective syntheses, the phthaloyl group is extensively used to control regioselectivity and to achieve side-chain functionalization of amino acid derivatives, e.g. in free radical reactions. Further, the manipulation of the phthaloyl group provides self-reproduction of chirality during the synthesis of optically active amino acid derivatives. The steric bulk together with the electronic characteristics give neighboring group effects that can be exploited in reactions of N^{α}-Phth amino acid derivatives.[564–566]

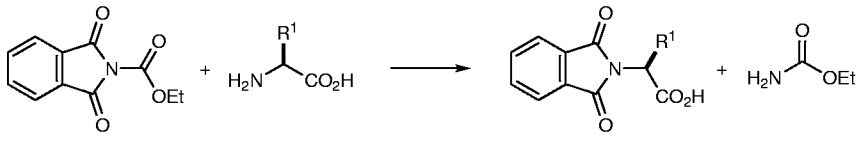
2.1.1.3.1.1 Synthesis of N^{α}-Phthaloyl Amino Acids

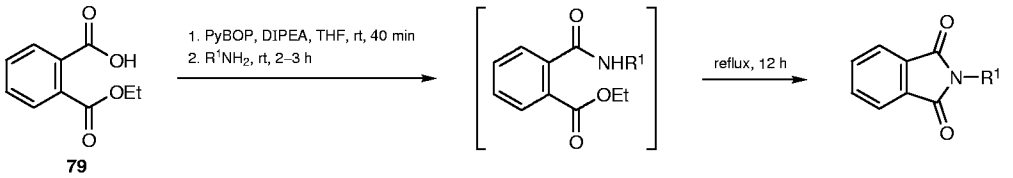
Several methods have been developed for phthaloylation of primary amines. Originally, the phthaloyl group was introduced by treatment of amino acids with phthalic anhydride at elevated temperatures of around 180 °C.[540,541,549–551] The harsh reaction conditions of this fusion method caused extensive racemization, thus severely limiting the utilization of phthaloyl protection in routine peptide chemistry. An improvement in the phthaloylation was achieved by conducting the reaction of phthalic anhydride with amino acids or amino acid esters in the presence of boiling solvents such as benzene, pyridine, dioxane, or acetic acid.[567]

With the development of *N*-(ethoxycarbonyl)phthalimide (**78**) an efficient and mild phthaloylation reagent was obtained which upon reaction with amino acids liberates ethyl carbamate as shown in Scheme 40.[568]

Recently, monoethyl phthalate (**79**) has been described as a reagent for the mild and selective conversion of primary amines into their corresponding phthalimides. The procedure is based on the PyBOP-promoted reaction of the amine with monoethyl phthalate (**79**) yielding an intermediate phthalamic ester that cyclizes to give the desired phthalimide (Scheme 40).[569]

Scheme 40 Phthaloylation of Amino Acids with *N*-(Ethoxycarbonyl)phthalimide or Monoethyl Phthalate[568,569]

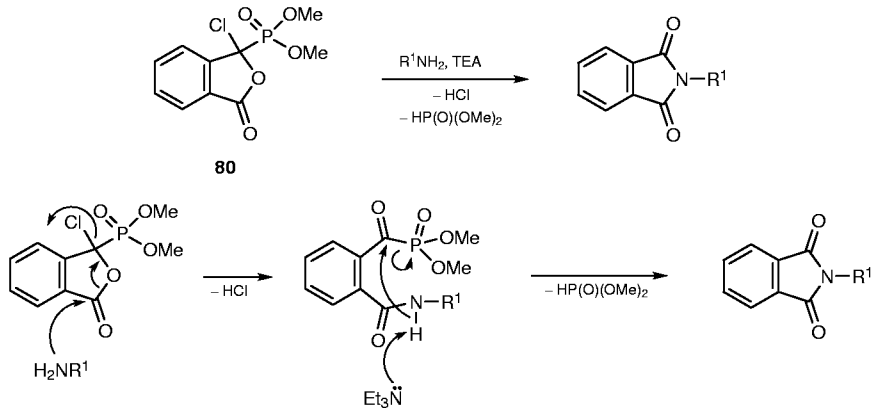




79

As a further alternative, 3-chloro-3-(dimethoxyphosphoryl)isobenzofuran-1(*3H*)-one (**80**) was proposed as a reagent for the rapid and convenient phthaloylation of amino acids in high yields (Scheme 41).[570] Reaction of this novel reagent with amino acids in the presence of tertiary amines yields dimethyl phosphite as the byproduct.

Scheme 41 Phthaloylation of Amino Acids with 3-Chloro-3-(dimethoxyphosphoryl)isobenzofuran-1(*3H*)-one[570]



80

Phth>Glu-OH; Typical Procedure:[568]
To a chilled suspension (0 °C) of H-Glu-OH (2.4 g, 16.3 mmol) and Na_2CO_3 (4.2 g, 40 mmol) in H_2O (20 mL) finely powdered *N*-(ethoxycarbonyl)phthalimide[568] (**78**; 5 g, 22.8 mmol) was added. After stirring at 0 °C for 5 min and at rt for 30 min, the mixture was acidified with 6 M HCl to pH 2.5 and the resulting oil crystallized upon cooling. The product was collected, washed with cold H_2O, and dried; yield: 3.62 g (80%); mp 160 °C.

Phth>Phe-OMe:[569]
To a suspension of PyBOP (0.24 g, 0.46 mmol) in THF (1 mL) were added a soln of monoethyl phthalate (**79**; 0.089 g, 0.46 mmol) in THF (1 mL) and DIPEA (0.105 mL, 0.63 mmol) and the resulting mixture was stirred at rt for 40 min. The soln was then added via cannula to H-Phe-OMe (0.075 g, 0.42 mmol) in THF (0.5 mL). The mixture was stirred at rt for 2–3 h and then heated to reflux overnight. After cooling to rt, the mixture was poured into sat. aq $NaHCO_3$ (10 mL) and extracted with EtOAc (2 × 20 mL). The combined organic layers were dried (Na_2SO_4) and concentrated. The crude product was purified by column chromatography (silica gel, hexane/EtOAc 9:1); yield: 0.117 g (91%); mp 89–91 °C.

N^{α}-Phth Amino Acids; General Procedure:[570]
A soln of 3-chloro-3-(dimethoxyphosphoryl)isobenzofuran-1(3*H*)-one[570] (**80**; 1.38 g, 5.3 mmol), amino acid derivative (5.25 mmol), and DIPEA (3.5 mL, 20 mmol) in MeCN/H_2O (1:1, 25 mL) was stirred at rt for 10 min. ^{31}P NMR showed complete conversion of 3-chloro-3-(dimethoxyphosphoryl)isobenzofuran-1(*3H*)-one into dimethyl phosphite. The mixture was concentrated to a small volume, acidified with 2 M HCl, and cooled to 0 °C. The white precipitate was filtered, washed with ice cold 2 M HCl and dried (P_2O_5). For example: Phth>Phe-OH; yield: 80%; mp 175–177 °C; Phth>Glu-OH; yield: 77%; mp 158–160 °C.

for references see p 149

2.1.1.3.1.2 Cleavage of the Phthaloyl Group

The removal of the phthaloyl group is routinely carried out by hydrazinolysis, generally in an alcohol.[571] A significantly milder deprotection procedure is based on an efficient two-stage, one-pot operation using $NaBH_4$/iPrOH for reductive phthalimide ring opening, followed by an acid-initiated lactonization, thus liberating the free amine.[572] Alternatively, the intermediate phthalyl amides are hydrolyzed enzymatically by phthalyl amidase.[573–576]

H-Phe-Gly-OH·H_2O; Typical Procedure for Cleavage of the Phthaloyl Group with Hydrazine Hydrate:[551]

A soln of Phth>Phe-Gly-OH (3.52 g, 10 mmol) and $H_2NNH_2 \cdot H_2O$ (3.0 mL, 60 mmol) in EtOH (100 mL) was heated under reflux for 1 h. H_2O (50 mL) was added and the solvent was evaporated. The residue was taken up in H_2O (50 mL) and the soln acidified to pH 5 with glacial AcOH. After 1 h, the phthalhydrazide was removed by filtration and washed with H_2O. The combined filtrates were concentrated. To the resulting syrup was added dropwise with stirring acetone (15 mL). After a short period, a further quantity of acetone (35 mL) was added to the mass of fine, colorless crystals. The product was collected by filtration, washed with acetone, and dried; yield: 2.37 g (99%); mp 259.5–260.5 °C.

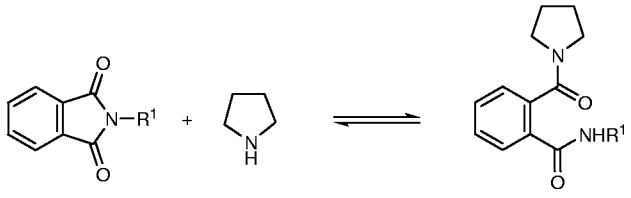
H-Aib-OH·NH_3; Typical Procedure for Reductive Cleavage of Phthalimides by $NaBH_4$:[572]

To a stirred soln of Phth>Aib-OH (0.2 g, 0.86 mmol) in iPrOH (7.7 mL) and H_2O (1.3 mL) was added $NaBH_4$ (163 mg, 4.3 mmol). After stirring for 24 h, TLC indicated complete consumption of starting material. Glacial AcOH (0.9 mL) was added carefully and when the foaming subsided, the flask was stoppered and heated to 80 °C for 2 h. The crude mixture was then eluted onto a Dowex 50 (H^+) column (2.7 × 10 cm), washed with H_2O (150 mL), and then eluted with 1 M NH_4OH (200 mL). Ninhydrin-positive fractions were collected and pooled for freeze drying and thus afforded the product; yield: 0.1 g (97%).

2.1.1.3.1.3 Peptide Synthesis with N^{α}-Phthaloyl Amino Acids

Within N^{α}-Phth-protected amino acid derivatives the C–N bond is extremely susceptible to basic and nucleophilic conditions, thus, leading to unwanted base-catalyzed side reactions of the Phth derivatives in the course of synthetic steps. The applicability of the phthalimido chemistry has been extended by converting the phthalimido moiety into the corresponding 2-(pyrrolidinocarbonyl)benzamide derivative **81**, thereby eliminating the base- and nucleophile-sensitivity of the parent phthaloyl derivative (Scheme 42).[560]

Scheme 42 Conversion of Phthalimido into 2-(Pyrrolidinocarbonyl)benzamide Derivatives[560]



81

After converting N^{α}-Phth-protected amines into the corresponding 2-(pyrrolidinocarbonyl)benzamide derivative, Dieckmann reactions, ester hydrolysis, and transesterfication can be carried out. This strategy of pyrrolidine-based deactivation of phthaloyl-protected amines followed by reclosure of the phthaloyl group has successfully been applied in the synthesis of β-lactam antibiotics.[560]

In peptide chemistry, in addition to the initially used acid chloride method[551] N^{α}-Phth amino acids can be coupled by the entire repertoire of coupling methods, i.e. mixed anhydrides, symmetrical anhydride, active ester, and the azide method.[1] Due to the acid stability of N^{α}-phthaloyl derivatives, the acid hydrolyses of esters are readily performed.[551]

2.1.1.3.2 Dithiasuccinoyl Protection

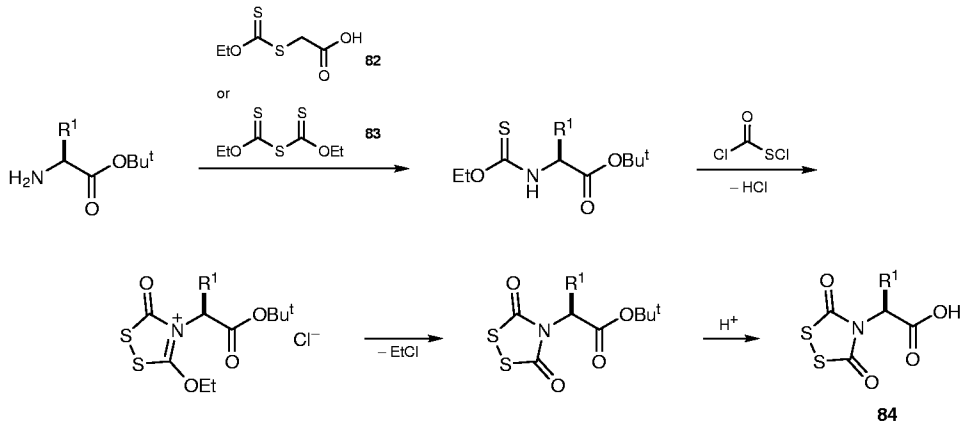
E. Lohof, H.-J. Musiol, J. Podlech

The dithiasuccinoyl (Dts) amino protecting group was proposed by Barany and Merrifield for developing orthogonal protection schemes in peptide synthesis.[542] The N^{α}-Dts derivatives are stable toward acid treatments as required for removal of both the *tert*-butyl alcohol and benzyl-based protecting groups, and are also resistant to photolysis as used to cleave the acid-stable 2-nitrobenzyl and 1-methyl-2-oxo-2-phenylethyl (α-methylphenacyl) esters. However, the cumbersome preparation of N^{α}-Dts amino acids has limited their use to specific synthetic targets. The Dts group has also been successfully applied in the synthesis of peptide nucleic acids.[577]

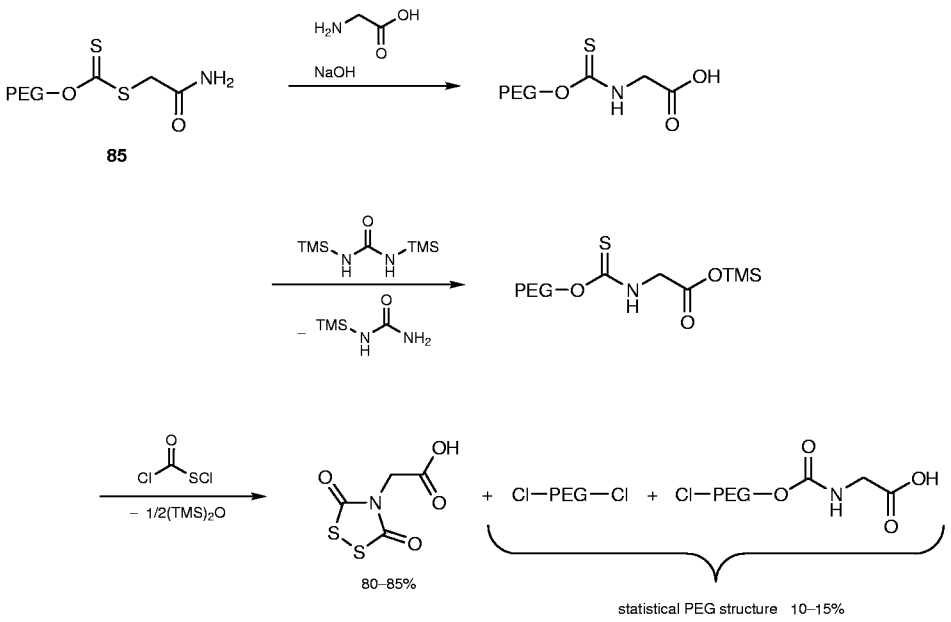
2.1.1.3.2.1 Synthesis of N^{α}-Dithiasuccinoyl Amino Acids

The N^{α}-Dts amino acids **84** are prepared from amino acid esters with *O*-ethyl *S*-carboxymethyl dithiocarbonate (**82**)[578] or alternatively with *O*,*O*-diethyl trithiodicarbonate (**83**)[579] according to Scheme 43.[542] The intermediate ethoxythiocarbonyl derivatives are treated with chlorocarbonylsulfenyl chloride (prepared by partial hydrolysis of trichlorosulfenyl chloride[542]) in dry solutions (optionally in the presence of a tertiary base) for the ring closure which is followed by the loss of chloroethane to produce the N^{α}-Dts derivatives **84**. Although the conversion of the *O*-alkyl thiocarbamates upon reaction with chlorocarbonylsulfenyl chloride to produce N^{α}-Dts amino acids proceeds in good yields, a number of low-level byproducts have been observed amongst which are the most troublesome *O*-alkyl carbamates. Therefore, the polymeric PEG-O(C=S)SCH_2CONH_2 (PEG-xanthate) (**85**) was developed[580] which allows removal of any urethane-blocked amino acids formed upon production of the Dts heterocycle by aqueous workup (Scheme 43). N^{α}-Dts amino acids are generated from the esters by acid treatments or directly when silylation of the *O*-PEG thiocarbamates is used before treatment with chlorocarbonylsulfenyl chloride (for a list of N^{α}-Dts amino acids and related analytical data, see ref[580]).

Scheme 43 Preparation of N^{α}-Dts Amino Acids[542]



for references see p 149

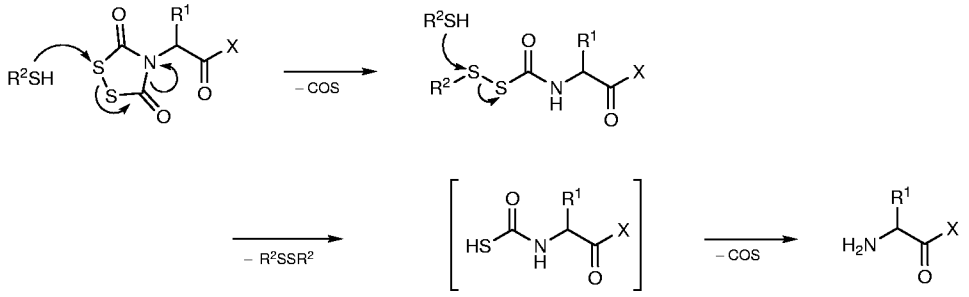


N^{α}-Dts Amino Acids; General One-Pot Procedure Using PEG-Xanthate:[580]
To a soln or suspension of the amino acid or side-chain protected amino acid (25 mmol), 1 M NaOH (25 mL), and THF (25 mL) was added PEG-O(C=S)SCH_2CONH_2 (**85**; PEG-xanthate) (31.3 g, 27.5 mmol). Over the next 30 min, 2 M NaOH (~13 mL) was added to adjust and maintain the pH at 9.5. After 15 min of additional stirring, the soln was acidified with 6 M HCl (~13 mL) to pH 3, and then concentrated. The residue was dissolved in CH_2Cl_2 (150 mL), dried ($MgSO_4$), and concentrated. The resultant PEG-thiocarbamate was dried overnight under reduced pressure over P_2O_5, at which point it had solidified (the polymer has to be dry). Using a mixture of dry MeCN/CH_2Cl_2 (1:1, 100 mL), the polymer was first dissolved in a minimal volume, then solid 1,3-bis(trimethylsilyl)urea (5.5 g, 27 mmol) was added, and finally rinsing into the viscous soln was carried out until no suspended particles remained. This soln was maintained at rt for 45 min, and then chilled to 5 °C with an ice bath. Chlorocarbonylsulfenyl chloride (3.2 mL, 38 mmol) was added all at once to the chilled soln, which spontaneously exothermed and was kept at rt for 10 min. Solvents were removed by rotary evaporation, and the residue was taken up with a mixture of Et_2O (80 mL) and 20% aq NaCl (50 mL). The aqueous phase was extracted with Et_2O (2 × 80 mL), EtOAc (2 × 80 mL), and Et_2O (80 mL). The combined organic phases were finally washed with H_2O (50 mL), dried ($MgSO_4$), and concentrated to provide the N^{α}-Dts amino acid; N^{α}-Dts amino acids solidified after standing for several days under petroleum ether or an equivalent hydrocarbon at –20 °C; yield: 60–80%; the N^{α}-Dts amino acids are stable on storage at rt or 4 °C for extra safety.

2.1.1.3.2.2 Cleavage of the Dithiasuccinoyl Group

The Dts group is rapidly and quantitatively removed both in solution and on resin under mild conditions with nucleophiles such as thiols, or under mildly reducing conditions with borohydrides and trialkylphosphines (Scheme 44). Different thiols can be employed such as pyridine-2-thiol, benzenethiol, dithiothreitol, or *N*-(methylsulfanyl)acetamide in various solvents,[581] but the 2-sulfanylethanol/DIPEA mixture is used most frequently.[582] Model experiments established that quantitative thiolytic removal (>99.8%) of the Dts group occurs with 2-sulfanylethanol/DIPEA (0.5 M:0.5 M) in CH_2Cl_2 (2 × 2 min), or *N*-(methylsulfanyl)acetamide/NMM (0.5 M:0.5 M) in CH_2Cl_2 (2 × 2 min), or *N*-(methylsulfanyl)acetamide/1,2,3-benzotriazol-1-ol (0.5 M:0.1 M) in DMF (2 × 2 min).[270] Similar conditions were used for cleavage of the Dts group from glycopeptides.[583,584]

Scheme 44 Reaction Mechanism of N^{α}-Dts Cleavage



The Dts-aminoacyl resins were exposed for 24 hours to 2-sulfanylethanol/DIPEA (0.5 M:0.5 M) in CH_2Cl_2 with a 65% of the peptide chain retained. A shorter reaction period of 15 minutes led to a 92% of integrity of the peptide chain, while an even shorter period of 2 minutes with the above mentioned thiol mixture gave 97%.[582]

Although the Dts protection has been used in the SPPS of some model peptides,[270,582,585] formation of terminated peptides has been observed which limits its use for such purpose.[185] The Dts group has been more efficiently employed in the synthesis of glycopeptides as a N-protecting group on the carbohydrate itself, for example, the nitrogen group in glucosamine, to circumvent the side reactions in the synthesis with *N*-acetylglucosamine.[583,584,586,587]

H-Tyr(tBu)-Gly-Gly-Phe-Leu-OH[582]

Synthesis in Solution: Purified Dts>Tyr(tBu)-Gly-Gly-Phe-Leu-OH (4.5 μmol) was fully dissolved in a mixture (1 mL) of 2-sulfanylethanol/DIPEA (0.5 M:0.5 M) in CH_2Cl_2, which was maintained at 25 °C for 5 min following which solvents and most of the reagents were removed by rotary evaporation at 25 °C, at 1 Torr, 15 min (coevaporated 3 × with CH_2Cl_2). The resulting film was then immediately passed through a MPLC column (convex gradient formed from 175 mL each of 1:19 and 3:2 mixtures of MeCN and 0.1% TFA in H_2O to provide pure peptide); FAB-MS *m/z*: 634 $[M+Na]^+$.

Solid-Phase Synthesis: Dts>Tyr(tBu)-Gly-Gly-Phe-Leu-ONb-resin (30 mg, 11 μmol) was treated with 2-sulfanylethanol/DIPEA (0.5 M:0.5 M) in CH_2Cl_2 (2 × 2 min), washed with CH_2Cl_2 (5 × 1 min), and photolyzed to provide the title peptide (5.9 μmol, 54%), 96% pure by analytical HPLC.

2.1.1.4 Sulfenamide-Based Protecting Groups

M. Gurrath, G. Müller, H.-J. Musiol, and J. Podlech

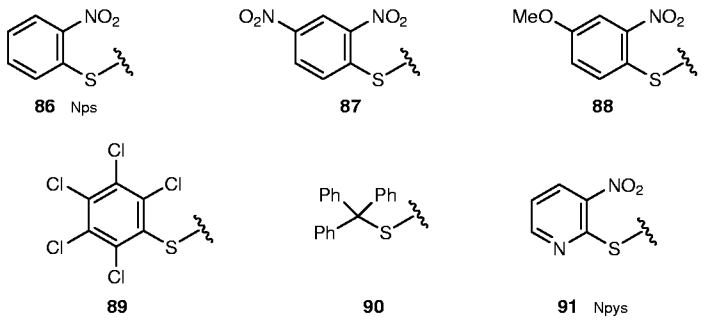
Among the various sulfanyl derivatives proposed, e.g. 2,4-dinitrophenylsulfanyl (**87**),[588] 2-nitro-4-methoxyphenylsulfanyl (**88**),[589] pentachlorophenylsulfanyl (**89**),[590] and tritylsulfanyl (**90**),[591] the 2-nitrophenylsulfanyl (Nps, **86**)[592] group has emerged as the most suitable in terms of stability and cleavage conditions (Scheme 45). Additionally the 3-nitro-2-pyridylsulfanyl (Npys, **91**) has been proposed as an amino, hydroxy, and, in particular, as a thiol protecting group.[593] While the Npys group (**86**) has found significant application for activated thiol protection in the formation of intra- and intermolecular disulfide bridges (see Sections 6.1.3 and 6.1.4), it is not commonly used for amino protection. The N^{α}-Npys protection is stable to trifluoroacetic and formic acid, but is cleaved by triphenylphosphine and pyridine-2-thiol.[593] Based on the rapid cleavage of Npys with thiols, a general thiolytic deprotection scheme has been proposed,[594] which is based on semipermanent *O*-, *S*-, and *N*-Npys protection combined with the benzenethiolate-sensitive phenacyl group for carboxy group protection[595] and dinitrophenyl protection for the imidazole function.[596] Conversely, as an alternative to the highly acid-labile urethane-type protecting groups or to trityl derivatives for temporary N^{α}-protection in chain elongation steps, the N^{α}-Nps group in combination with *tert*-butyl alcohol derived side-chain protection represents a well-estab-

for references see p 149

lished strategy for the synthesis of sulfur-containing peptides in solution as documented by the large-scale synthesis of various bioactive peptides, e.g. glucagon,[597] gastrin and CCK peptides,[598–601] or somatostatins.[395,602] In the synthesis of larger size peptides, the Nps group offers the advantage of rapid and clean deprotection steps with 1 to 2 equivalents of HCl in protic and aprotic solvents such as MeOH, TFE, DMF, etc., compared to the generally low rates of hydrogenolysis of the benzyloxycarbonyl group of the standard Z/tBu chemistry in solution, when for solubility reasons solvents such as DMF, DMA, NMP, or HMPA are required.

Apart from the syntheses of peptides, the Nps group has found useful application in the preparation of aminopenicillins,[603] of O-sulfated β-lactam hydroxamic acids,[604] and of the β-lactone antibiotic (+)-obafluorin.[605]

Scheme 45 Sulfenamide-Based Protecting Groups

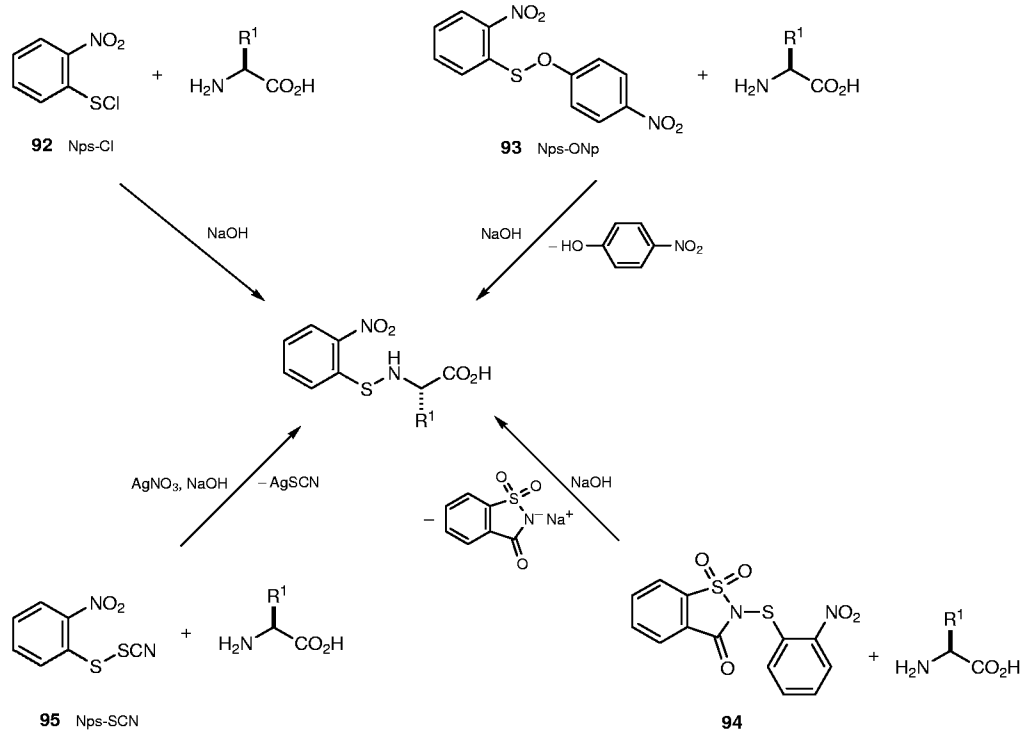


The Nps group (**86**)[606] was first introduced by Goerdeler and Holst for N^{α}-protection of amino acids and related esters.[592] Its application in peptide synthesis was pioneered by Zervas and co-workers[591,607] in a comparative study where coupling efficiencies of N^{α}-trityl, N^{α}-tritylsulfanyl, and N^{α}-2-nitrophenylsulfanyl amino acids were analyzed using the 4-nitrophenyl ester, mixed anhydride, carbodiimide, and diphenylphosphoryl method. The problems associated with trityl- and tritylsulfanyl-protected amino acids were mainly attributed to steric hindrance[591] that was not observed with the Nps derivatives. Moreover, under the reaction conditions required for cleavage of the Nps group the semipermanent benzyl and *tert*-butyl alcohol based side-chain protection were unaffected.[607] Due to the facile synthetic accessibility of the N^{α}-Nps amino acid derivatives on a large preparative scale and their remarkable stability on storage as DCHA salts as well as because of the highly selective cleavage of the sulfenamide bond under mild conditions, the Nps group has achieved considerable popularity in peptide synthesis in solution.

2.1.1.4.1 Synthesis of N^{α}-2-Nitrophenylsulfanyl Amino Acids

For the synthesis of N^{α}-Nps amino acid derivatives various Nps-donating reagents have been proposed (Scheme 46): (1) 2-nitrophenylsulfenyl chloride (Nps-Cl, **92**),[591,592,607] (2) 2-nitrophenylsulfenyl thiocyanate (Nps-SCN, **95**),[608] (3) 4-nitrophenyl 2-nitrophenylsulfenate (Nps-ONp, **93**),[608] and (4) 2-(2-nitrophenylsulfanyl)-1,2-benzoisothiazole-3(2*H*)-one 1,1-dioxide (Nps-saccharin, **94**).[609]

Scheme 46 Synthesis of N^{α}-2-Nitrophenylsulfanyl Amino Acids[591,592,607–609]



A major drawback of Nps-Cl (**92**) is its instability on storage with partial conversion into the unreactive disulfide,[608] although this means that it must be recrystallized prior to its use, Nps-Cl (**92**) remains the predominant Nps-donating reagent used due to its low cost. Among the alternative Nps donors, most are even more unstable than Nps-Cl (**92**).[610] Only the 2-nitrophenylsulfenyl thiocyanate (**95**) is a stable compound that can be stored indefinitely without special precautions; it is easily accessible by reaction of Nps-Cl (**92**) with sodium thiocyanate.[608,611] Using this reagent for the preparation of N^{α}-Nps amino acids, the thiocyanate released into the reaction must be captured with silver nitrate to avoid cleavage of the newly formed sulfenamide bond.[608] Alternatively, the use of 4-nitrophenyl 2-nitrophenylsulfenate (**93**) has been proposed and this is obtained by reacting Nps-Cl (**92**) with sodium 4-nitrophenolate.[608] Finally, based on the observation that *N*-acyl derivatives of saccharin act as acylating reagents,[612,613] Nps-saccharin (**94**), a stable Nps donor, was readily prepared by reaction of Nps-Cl (**92**) with the sodium salt of saccharin.[609]

N^{α}-Nps amino acids are generally prepared under Schotten–Baumann conditions, whereas amino-unprotected peptide fragments are converted into the related Nps derivatives in organic solvents using TEA as a base.[614]

Synthesis of N^{α}-Nps Amino Acids with Nps-Cl (92); General Procedure:[591,607]

To a soln of the amino acid (20 mmol) in 2 M NaOH (10 mL) and dioxane (20 mL) freshly prepared[615] or recrystallized (from petroleum ether) Nps-Cl (**92**; 22 mmol) was added in 10 equal portions over 15 min, whilst 2 M NaOH (12 mL) was added dropwise with vigorous stirring. The mixture was diluted with H_2O (200 mL), filtered from precipitated disulfide, overlayered with EtOAc and acidified with 0.5 M H_2SO_4 (or aq $KHSO_4$). Upon extraction with EtOAc or Et_2O, or with their mixture (1:1), the combined extracts were repeatedly washed with H_2O and then dried (Na_2SO_4). The soln was concentrated and upon addition of DCHA (4 mL) the crystalline products were collected and washed with Et_2O/petroleum ether; yield: 40–90%.

Note: The use of 5% aq $KHSO_4$ for acidification is recommended instead of dil H_2SO_4.

for references see p 149

Synthesis of N^{α}-Nps Amino Acids with Nps-SCN (95); General Procedure:[608]
To a soln of amino acid (1 mmol) in H_2O (2 mL) adjusted to pH 8–9 by addition of 1 M NaOH, Nps-SCN (**95**; 1 mmol) was added in portions over 15 min while 1 M $AgNO_3$ (0.5 mL, 0.5 equiv) was added dropwise under stirring and maintaining the pH at 8 to 9 with 1 M NaOH. Then additional 1 M $AgNO_3$ (0.5 mL) was added dropwise over 15 min at pH 8–9. The AgSCN was filtered off and washed with H_2O. The combined filtrates were cooled to 0 °C and acidified with 1 M H_2SO_4. The N^{α}-Nps amino acid derivatives were extracted with EtOAc or Et_2O. The extracts were washed with H_2O and dried (Na_2SO_4). Upon addition of DCHA (0.22 mL) the salts were collected by filtration; yield: 75–90%.

Synthesis of N^{α}-Nps Amino Acids with Nps-Saccharin (94); General Procedure:[609]
To a chilled soln of the amino acid (10 mmol) in 1 M NaOH (10 mL) and dioxane (10 mL), Nps-saccharin (**94**; 3.36 g, 10 mmol) in THF (40 mL) was added dropwise under vigorous stirring. The mixture was kept at rt for 12 h, then the organic solvents were removed, the aqueous soln diluted with H_2O (50 mL), acidified with 2% aq $KHSO_4$ to pH 3, and extracted with EtOAc (3 × 50 mL). The extracts were washed with H_2O and dried (Na_2SO_4). Upon addition of DCHA (2.2 mL) the N^{α}-Nps amino acid salts were collected; yield: 75–87%.

Nps-[Glu(OtBu)]$_5$-OH with Nps-Cl (92):[614]
To a soln of H-[Glu(OtBu)]$_5$-OH (7.6 g, 8.1 mmol) in DMF (200 mL) and TEA (1.13 mL, 8.10 mmol), were added Nps-Cl (**92**; 2.26 g, 11.9 mmol) and TEA (1.68 mL, 12.0 mmol) at rt. After 12 h, the solvent was removed and the residue partitioned between EtOAc and 5% aq $KHSO_4$. The organic layer was washed with H_2O, dried (Na_2SO_4), concentrated, and the residue reprecipitated from EtOAc/Et_2O; yield: 6.8 g (77%); mp 160–162 °C; $[\alpha]_D^{20}$ −16.8 (*c* 1, DMF).

2.1.1.4.2 Peptide Synthesis with N^{α}-2-Nitrophenylsulfanyl Amino Acids

For the purpose of coupling N^{α}-Nps amino acids with amino acid or peptide derivatives, generally the free acid is liberated from the DCHA salt by acidification with a 5% aqueous solution of $KHSO_4$ in the presence of EtOAc,[616] thus avoiding the numerous disadvantages associated with the use of citric acid for the acidification of acid-labile protected amino acid salts. In particular cases, the hydrochloride of the amino component may serve to liberate the carboxy group by precipitation of the DCHA · HCl salt in organic solvents. N^{α}-Nps-protected amino acid derivatives are coupled via the DCC, mixed anhydride, or active ester procedures, in the latter case using preferentially the *N*-hydroxysuccinimide esters which are readily prepared by standard DCC/HOSu protocols (see Section 3.2)[395,617,618] and are stable on storage. Alternatively, the 4-nitrophenyl[591] or pentachlorophenyl esters[617,619] are applied. Coupling of N^{α}-Nps-protected peptide fragments is generally carried out by the DCC/HOSu method. More acidic 1,2-dinucleophilic additives to DCC such as HOBt, HOAt, and HODhbt result in Nps cleavage.[620,621] Saponification of Nps-protected peptide fragment alkyl esters can be performed with precautions.[622]

N-Carboxy amino acid anhydrides (NCAs) react almost quantitatively with Nps-Cl (**92**) to form the related N^{α}-Nps-protected NCAs[623,624] which have been used for the synthesis of peptides[624] and even of depsipeptides, in the latter case taking advantage of the rapid formation of the ester bond between a Nps-NCA and a hydroxy acid without further protection.[615,625] Since the development of urethane-protected NCAs, the related Nps derivatives are no longer used (see Section 3.4.4).

For N^{α}-protection in SPPS the Nps group has not found general application, despite some early attempts.[590] It could, however, represent an interesting approach for the linkage of the first amino acid to the solid support,[626] since esterification of hydroxymethylated resin with Nps-Phe-OH by DCC/DMAP occurs without racemization.[627] Interesting use of this protection has been reported in convergent SPPS and solution synthesis.[628]

Nps-Val-Phe-OMe by the DCC Method:[591]
To a soln of H-Phe-OMe · HCl (2.15 g, 10 mmol) in $CHCl_3$ (30 mL) was added TEA (1.4 mL) and Nps-Val-OH (2.7 g, 10 mmol) followed by DCC (2.2 g, 10.7 mmol) at 0 °C. After 12 h at rt, 50% AcOH was added dropwise and the DCU filtered off. The filtrate was washed with H_2O, dil H_2SO_4, aq $KHCO_3$, and

H_2O, dried (Na_2SO_4), and the solvent removed. The product was recrystallized (MeOH or EtOAc/petroleum ether); yield: 3 g (70%); mp 123–124 °C.

Nps-Phe-gTrp(For)-mLys(Boc)-OEt by the Mixed Anhydride Method:[622]
To a soln of Nps-Phe-OH (1.06 g, 3.32 mmol) in THF at −15 °C was added NMM (0.372 mL, 3.38 mmol) followed by isobutyl chloroformate (0.445 mL, 3.43 mmol). After 2 min, a precooled soln of H-gTrp(For)-mLys(Boc)-OEt·TFA (2.0 g, 3.32 mmol) in THF and NMM (0.372 mL, 3.38 mmol) was added. After 15 min at −15 °C and 90 min at 0 °C, the mixture was warmed to rt and after additional stirring for 30 min, the solvent was removed and the residue dissolved in EtOAc. The organic layer was washed with aq $NaHSO_4$, brine, dried (Na_2SO_4), and concentrated. The product was precipitated with iPr_2O; yield: 2.17 g (83%); mp 132–136 °C; $[\alpha]_D^{20}$ +21.5 (*c* 1, DMF).

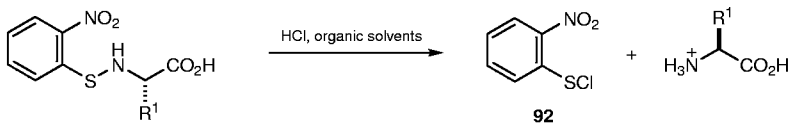
Nps-Lys(Adoc)-Asn-Phe-Phe-Trp-OH by the Active Ester Method:[395]
To an ice-cold soln of H-Asn-Phe-Phe-Trp-OH·H_2O·0.5MeCN (15.4 g, 23.7 mmol) in DMF (150 mL) was added TEA (3.3 mL, 23.7 mmol) followed by Nps-Lys(Adoc)-OSu (14.3 g, 24.8 mmol) and pyridine (2 mL, 24.8 mmol). After 2 h in an ice bath and 12 h at rt, the product was precipitated with aq $KHSO_4$ (6.8 g). The solid was filtered off, washed with H_2O, and reprecipitated from DMF/Et_2O; yield: 24 g (95%); mp 190–195 °C; $[\alpha]_D^{20}$ −20 (*c* 1, DMF).

Nps-Lys(Adoc)-Asn-Phe-Phe-Trp-Lys(Boc)-Thr(tBu)-Phe-Thr(tBu)-Ser(tBu)-Cys(StBu)-OtBu by DCC/HOSu Fragment Condensation:[395]
To a soln of Nps-Lys(Adoc)-Asn-Phe-Phe-Trp-OH (23.5 g, 21.9 mmol), H-Lys(Boc)-Thr(tBu)-Phe-Thr(tBu)-Ser(tBu)-Cys(StBu)-OtBu (21.9 g, 19.9 mmol) and HOSu (2.75 g, 23.9 mmol) in DMF (450 mL) was added DCC (4.52 g, 21.9 mmol) at −10 °C. After 12 h at 4 °C and 48 h at rt, DCU was filtered off and the soln concentrated and poured into H_2O-sat. EtOAc. The precipitate was collected and reprecipitated twice from DMF/MeCN; yield: 42 g (98%); mp 218–219 °C; $[\alpha]_D^{20}$ −17.6 (*c* 1, DMF).

2.1.1.4.3 Cleavage of the 2-Nitrophenylsulfanyl Group

Cleavage of the sulfenamide bond occurs in a two-step mechanism based on protonation of the amide nitrogen followed by nucleophilic attack on the sulfur atom. While the latter is under the pronounced electron-withdrawing influence of the 2-nitrophenyl group and thus very sensitive to nucleophiles, its large electron shell might enhance the basic character of the nitrogen, thus allowing the use of acetic acid as acid catalyst (Scheme 47). As listed in Table 11, the Nps group is cleaved in quantitative manner by 1 to 1.1 equivalents of HCl or HBr in alcohols such as MeOH, since the intermediate Nps-Cl (**92**) is converted into the sulfenic acid ester liberating HCl for the protonation of the amino group.[629] Conversely, in aprotic solvents, such as EtOAc, acetone, or DMF, 2 equivalents of the hydrogen halide are required to protect the amino group as the hydrohalide from reaction with the intermediate Nps-Cl (**92**).[591] To bypass the difficult solubility problem of larger protected peptide fragments TFE has been used successfully.[101,601] As alternative to hydrogen halides, perchloric acid and 4-toluenesulfonic acid[629] have been proposed as well as dibenzenesulfimide.[630] To lower the proton activity of the hydrogen halides in alcohols and thus to prevent partial cleavage of the acid-labile protecting groups, a large excess of 2-methylindole is added which concomitantly reacts with the intermediately formed Nps-Cl (**92**) thereby releasing 1 equivalent of hydrogen halide for protonation of the amino group.[395]

Scheme 47 Cleavage of the Sulfenamide Bond by Nucleophiles[629]



for references see p 149

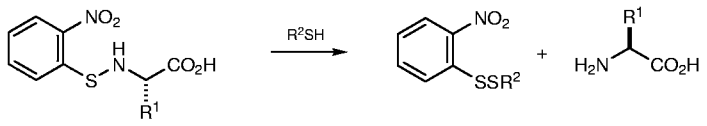
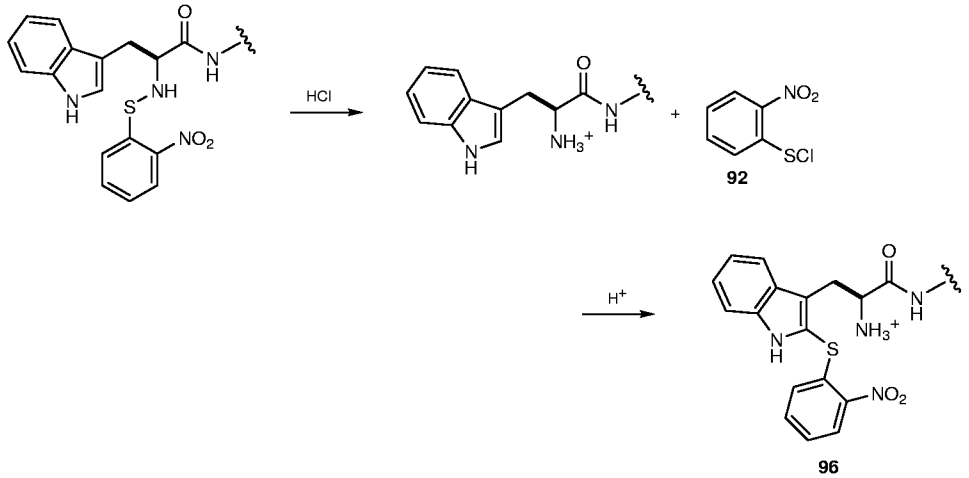


Table 11 Most Common Nps-Cleavage Methods and Their Compatibility with Sequence Composition and Protection Schemes in Solution Synthesis

Cleavage Method	Trp	Boc, OtBu, tBu	*S*-Acm, *S*-Trt, *S*-tBu	*S*-StBu	Ref
HCl or HBr (1–1.1 equiv), alcohols, 0–20 °C, 30 min	2-indole sulfenylation	stable	cleavage	fairly stable[631]	[630]
HCl or HBr (2 equiv), EtOAc, acetone, DMF, 0–20 °C, 30 min	2-indole sulfenylation	stable	cleavage	fairly stable[631]	[591]
HCl or HBr (1–2 equiv)/2-methylindole (10–20 equiv), DMF, MeOH, TFE, 0–20 °C, 30 min	no side reaction	fully stable	stable	fully stable	[395,632]
Ph_3P or Bu_3P (5 equiv), MeOH, TFE, rt, 1 h	no side reaction	fully stable	stable	unstable	[633]
thiols, MeOH, rt, 30–60 min	no side reaction	fully stable	stable	unstable	[590]

In the absence of efficient Nps-Cl (**92**) capture, this intermediate reacts with the indole side chain of tryptophan residues to produce the related 2-Nps derivatives **96** (Scheme 48). Although reaction of sulfenyl halides with indoles was exploited by Wieland et al.[634] for the synthesis of phalloidin, this side reaction leads to irreversible modification of Trp-containing peptides.[635–638] This side reaction does not occur via an intramolecular electrophilic substitution as postulated previously,[635] but by a direct attack of the Nps-Cl (**92**); in fact, it is efficiently suppressed by the addition of a large excess of an indole derivative as a scavenger.[636,638] These scavengers serve also to decrease the proton activity of the acids (*vide supra*). Due to the unpleasant odor of 2-methylindole the less volatile 1-acetyltryptophan butyl ester has been proposed as scavenger.[639]

Scheme 48 Sulfenylation of Tryptophan by Nps-Cl as an Intermediate of HCl-Mediated Nps Cleavage[636,642]



A further side reaction involving Nps-Cl (**92**) as an intermediate in hydrogen halide mediated sulfenamide cleavage, is its fast reaction with the majority of the thiol protecting groups, e.g. *S*-Acm, *S*-Trt, and *S*-tBu[618,640,641] to produce the unsymmetrical disulfides (see Section 2.6.6). Again operating in the presence of a large excess of 2-methylindole this side reaction is completely suppressed and the potential attack on the unsymmetrical disulfide of the *S-tert*-(butylsulfanyl)cysteine residues is also suppressed.[395,632]

In view of the side reactions with tryptophan residues, cleavage of the Nps group by desulfuration with Raney nickel has been proposed.[643] Since hydrogenolytically labile protecting groups are affected and the method cannot be applied in the presence of Cys or Met residues, it is of limited use.

The findings by Kessler and Iselin[590] that the Nps group can be removed with nucleophiles such as thioacetamide, thiourea, or thiols, thus entirely avoiding acidic conditions, opened a new avenue in deprotection procedures for this group. A large series of inorganic and organic reagents have been proposed such as benzenethiol or thioglycolic acid,[636,638] 2-nitrobenzenethiol,[644] thiosulfate,[603] thiocyanate,[645,646] pyridine-2-thiol,[647] 3-nitro-4-sulfanylbenzoic acid,[626] and even polymeric reagents based on pyridine-2-thiol and 2-nitrobenzenethiol.[648] As an elegant alternative for the cleavage of the sulfenamide bond the use of triphenylphosphine was proposed by Mukaiyama et al.[633] More recently, a smooth Nps cleavage by the use of HOBt was obtained in the presence of aniline or *N*-methylaniline which act as a Nps acceptor releasing HOBt for salt formation with the amine component.[620]

H-Lys(Adoc)-Asn-Phe-Phe-Trp-Lys(Boc)-Thr(tBu)-Phe-Thr(tBu)-Ser(tBu)-Cys(StBu)-OtBu; Nps Cleavage by HCl/2-Methylindole in DMF:[395]
To an ice-cold soln of Nps-Lys(Adoc)-Asn-Phe-Phe-Trp-Lys(Boc)-Thr(tBu)-Phe-Thr(tBu)-Ser(tBu)-Cys(StBu)-OtBu (18.5 g, 8.59 mmol) and 2-methylindole (22.5 g, 172 mmol) in DMF (200 mL) was added 0.26 M HCl in MeOH (72.7 mL, 18.9 mmol) dropwise under stirring over 30 min. The mixture was kept at rt for 2 h and then poured into iPr_2O/petroleum ether. The precipitate was collected, dissolved in DMF and treated with Na_2CO_3 (1 g) in H_2O (50 mL). The product was precipitated with H_2O, filtered off, washed with H_2O, and reprecipitated from DMF/MeCN; yield: 16.2 g (94%); mp 220–221 °C; $[\alpha]_D^{20}$ −17.7 (*c* 1, DMF).

H-Gly-Gln-Leu-Ser(tBu)-Ala-Lys(Adoc)-Lys(Adoc)-Tyr(tBu)-Leu-Glu(OtBu)-Ser(tBu)-Leu-Ile-NH_2; Nps Cleavage with HCl/2-Methylindole in TFE:[101]
To a chilled soln of Nps-Gly-Gln-Leu-Ser(tBu)-Ala-Lys(Adoc)-Lys(Adoc)-Tyr(tBu)-Leu-Glu(OtBu)-Ser(tBu)-Leu-Ile-NH_2 (6.9 g, 3.16 mmol) and 2-methylindole (8.3 g, 63.2 mmol) in TFE (350 mL) was added dropwise over 30 min 0.1 M HCl in MeOH (35 mL). The soln was kept at rt for 60 min and concentrated. Upon addition of Et_2O, the precipitate was collected, redissolved in TFE and treated with 10% aq Na_2CO_3 (3.5 mL). The product was precipitated with H_2O; yield: 6.0 g (92%); $[\alpha]_D^{20}$ −13.7 (*c* 1, TFE).

H-Val-Cys(Trt)-Gly-Glu(OtBu)-OH·HCl; Nps Cleavage with HCl/2-Methylindole in DMF:[631]
To a chilled soln of Nps-Val-Cys(Trt)-Gly-Glu(OtBu)-OH (22.9 g, 26.7 mmol) and 2-methylindole (35 g, 267 mmol) in DMF (200 mL) 0.55 M HCl in dioxane (97.1 mL, 53.4 mmol) was added dropwise in 30 min. After 1 h stirring the soln was concentrated and the product precipitated with iPr_2O. The oil was reprecipitated twice from EtOAc and Et_2O to obtain a solid; yield: 18.0 g (91%).

H-Ala-Lys(Adoc)-Lys(Adoc)-Tyr(tBu)-Leu-Glu(OtBu)-Ser(tBu)-Leu-Ile-NH_2; Nps Cleavage with Tributylphosphine:[101]
To a soln of Nps-Ala-Lys(Adoc)-Lys(Adoc)-Tyr(tBu)-Leu-Glu(OtBu)-Ser(tBu)-Leu-Ile-NH_2 (1.6 g; 0.92 mmol) in TFE (100 mL) was added Bu_3P (1.1 mL, 4.45 mmol) and H_2O (5 mL). The soln was stirred at rt for 1 h and then concentrated. The product was precipitated with Et_2O/petroleum ether, and reprecipitated from TFE/H_2O; yield: 1.33 (89%); mp 261–263 °C; $[\alpha]_D^{20}$ −17.0 (*c* 1, TFE).

H-Thr(tBu)-Ser(tBu)-Cys(StBu)-OtBu·TosOH; Nps Cleavage with Pyridine-2-thiol:[395]
To a soln of Nps-Thr(tBu)-Ser(tBu)-Cys(StBu)-OtBu (0.72 g, 1.0 mmol) in MeOH (20 mL) was added pyridine-2-thiol (0.33 g, 2.7 mmol) and AcOH (0.18 mL). After 15 min at rt, the soln was concentrated, and upon addition of iPr_2O/petroleum ether, the precipitate was collected, dissolved in EtOAc, and

for references see p 149

treated at 0 °C with TosOH (1 equiv) in Et_2O. The precipitate was filtered off; yield: 0.7 g (95%); mp 160–162 °C; $[\alpha]_D^{20}$ −32.8 (*c* 1, MeOH).

H-Glu(OtBu)-Leu-Cys(Acm)-His-NHNH-Boc·AcOH; Nps Cleavage with Thioacetamide:[618]
To a soln of Nps-Glu(OtBu)-Leu-Cys(Acm)-His-NHNH-Boc (2.0 g, 2.23 mmol) in MeOH (50 mL) and AcOH (5 mL) was added thioacetamide (0.67 g, 8.9 mmol). After 40 min at rt, the precipitate was filtered off and the filtrate evaporated. The residue was distributed between EtOAc and H_2O. The aqueous layer was removed and the product precipitated from MeOH/Et_2O; yield: 1.27 g (71%); $[\alpha]_D^{25}$ −45.2 (*c* 1, MeOH).

H-Lys(Boc)-O-Resin; On-Resin Nps Cleavage with Thiocyanate:[646]
To a stirred suspension of Nps-Lys(Boc)-O-resin (50 mg) CH_2Cl_2/MeOH/AcOH (15:2:2, 19 mL) NH_4SCN (0.50 mmol) was added at rt, followed by dicyclohexylammonium 2-(2-methylindol-1-yl)acetate (0.57 mmol) after 5 min. After a total of 18 min, the resin was filtered off and washed with solvent to give a colorless product.

2.1.1.5 Sulfonyl Protecting Groups

J. Podlech

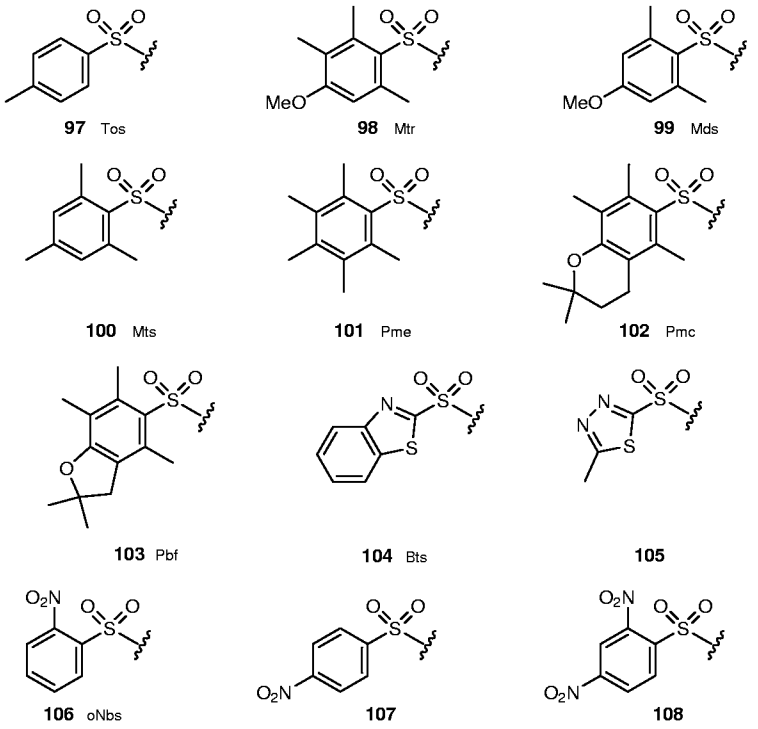
In the early phases of peptide chemistry arylsulfonyl protecting groups were generally used for the semipermanent side-chain protection of amino and particularly of imidazole and guanidino functions (see Section 2.6.1). For use in peptide synthesis the original tosyl group (Tos, **97**) has been replaced more recently by other sulfonamide-based protecting groups (Scheme 49) due to side reactions occurring in coupling of N^{α}-tosyl amino acids which lead to tosylamide and aldehydes via intermediate tosylimine formation followed by decomposition.[649] Moreover, harsh deprotection conditions are required, i.e. sodium in liquid ammonia,[650] since the tosylamide bond is stable to HBr/AcOH (except at elevated temperatures), alkali, and liquid HF. For deprotection procedures that have been applied in organic syntheses, see ref,[12] and for more information on the N^{α}-tosyl protection in peptide chemistry the reader is directed to earlier exhaustive reviews.[1,651]

Although the tosylguanidine bond is cleaved by HF, for the protection of the guanidino function the Tos group has been replaced by more efficiently cleaved sulfonyl derivatives such as 4-methoxy-2,3,6-trimethylphenylsulfonyl (Mtr, **98**),[652] 4-methoxy-2,6-dimethylphenylsulfonyl (Mds, **99**),[653] mesitylsulfonyl (Mts, **100**),[654] 2,3,4,5,6-pentamethylphenylsulfonyl (Pme, **101**),[652] 2,2,5,7,8-pentamethylchroman-6-ylsulfonyl (Pmc, **102**),[655] and 2,2,4,6,7-pentamethyl-2,3-dihydrobenzofuran-5-ylsulfonyl (Pbf, **103**),[656] as described in Section 2.6.1.4.

2.1.1.5.1 Arylsulfonyl Groups

Since the discovery by Carpino et al.[657] that N^{α}-Pbf-protected amino acids can be activated even as acid chlorides, such derivatives have been found to be well suited for the acylation of sterically demanding amino components (for details and experimental procedures, see Section 3.3.1). Oxazolone formation is prevented by this type of N^{α}-protection, racemization is not observed during activation and coupling.[658] In line with these findings, further sulfonamide derivatives have been proposed such as 1,3-benzothiazol-2-ylsulfonyl (Bts, **104**),[659,660] 5-methyl-1,3,4-thiadiazol-2-ylsulfonyl (**105**),[659] 2-nitrophenylsulfonyl (oNbs, **106**),[658,661–664] 4-nitrophenylsulfonyl (**107**),[664] and 2,4-dinitrophenylsulfonyl group (**108**) (Scheme 49).[665]

Scheme 49 Sulfonyl Protecting Groups Used in Peptide Chemistry



Amongst the previously proposed *N*-sulfonyl derivatives, the Mtr group has most frequently been used for the side-chain protection of lysine,[666,667] although it has occasionally been used for N^{α}-protection of amino acids.[666] It is cleaved by strong acids such as 0.3 M methanesulfonic acid in TFA/thioanisole (9:1) at room temperature within 1 hour.[666]

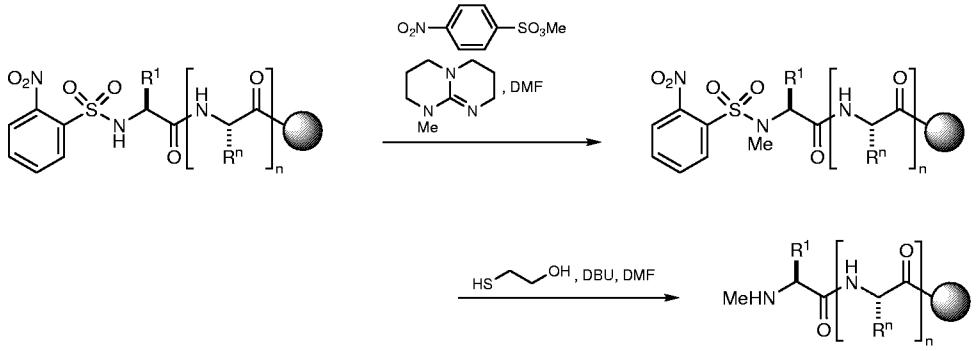
N^{α}-Pbf amino acid derivatives are prepared in analogy to the N^{ω}-arginine derivative[668] by reaction of amino acid or amino acid derivatives under Schotten–Baumann conditions with 2,2,4,6,7-pentamethyl-2,3-dihydrobenzofuran-5-sulfonyl chloride (Pbf-Cl), which is commercially available or is readily synthesized in two steps from 2,3,5-trimethylphenol and isobutyraldehyde using H_2SO_4 as a catalyst in toluene with subsequent chlorosulfonation. Conversion of the N^{α}-Pbf amino acids into the corresponding acyl chlorides is performed as for the related N^{α}-oNbs derivatives[658] by simply heating with thionyl chloride and evaporation of excess reagent. Cleavage of the N^{α}-Pbf-protected compounds is achieved with 10% DMS/TFA.[657] The 1,3-benzothiazol-2-ylsulfonyl (Bts, **104**) and 5-methyl-1,3,4-thiadiazol-2-ylsulfonyl (**105**) groups show similar properties. The related N^{α}-protected amino acids are prepared with 1,3-benzothiazole-2-sulfonyl chloride or 5-methyl-1,3,4-thiadiazole-2-sulfonyl chloride, respectively, which are prepared by chlorination of 1,3-benzothiazole-2-thiol and 5-methyl-1,3,4-thiadiazole-2-thiol, respectively. Cleavage of the 1,3-benzothiazol-2-ylsulfonyl (Bts, **104**) group is achieved with zinc in 10% AcOH/EtOH, with H_3PO_2,[659] or with a thiol in the presence of a base.[669] The 5-methyl-1,3,4-thiadiazol-2-ylsulfonyl (**105**) group is removed with either zinc, 10% AcOH/EtOH, with zinc, concentrated HCl/THF (1:15), with Al/Hg in Et_2O/H_2O, or with 2 M HCl/MeOH (2:1).[659]

The *N*-sulfonyl protection in general,[2] and more recently the 2-nitrophenylsulfonyl (oNps, **106**), the 1,3-benzothiazol-2-ylsulfonyl (Bts, **104**), and the 5-methyl-1,3,4-thiadiazol-2-ylsulfonyl (**105**) groups have been used for terminal N-alkylation (Scheme 50), e.g. for the synthesis of polyamines.[665,670,671] In fact, as the sulfonamide N–H bond is very acidic, deprotonation with 7-methyl-1,5,7-triazabicyclo[4.4.0]dec-5-ene and alkylation is possible

for references see p 149

without loss of the protecting group.[658,660–664] This N-alkylation has been advantageously exploited for the solid-phase synthesis of *N*-alkyl amino acids.[662,672] In analogy to other arylsulfonyl groups, the *N*-2-nitrophenylsulfonyl derivatives are prepared by the conventional Schotten–Baumann procedure using 2-nitrobenzenesulfonyl chloride (commercially available). Cleavage is either achieved with benzenethiol and potassium *tert*-butoxide in DMF or with DBU/2-sulfanylethanol in DMF[662,664] resulting in the formation of a colored byproduct which is of further advantage in automated syntheses, as it can be followed photometrically.[662] Cleavage with nucleophiles proceeds by initial formation of a Meisenheimer complex with subsequent loss of sulfur dioxide and liberation of the amine.[673] This addition/elimination mechanism proceeds less regioselectively, for the corresponding *para*-derivative, 4-nitrophenylsulfonyl (**107**), making this protecting group less favorable.[674]

Scheme 50 Preparation of N-Alkylated Amino Acid Derivatives[662]



1,3-Benzothiazole-2-sulfonyl Chloride (Bts-Cl):[659]
Cl_2 gas was bubbled at 0 °C into 33% AcOH in H_2O (60 mL). After a yellow precipitate formed, a suspension of 1,3-benzothiazole-2-thiol (10.0 g, 60 mmol) in 33% AcOH in H_2O (60 mL) was added in ca. 20 portions over 2 h still with Cl_2 bubbling through the mixture (internal temperature was maintained at 0 °C). A large excess of Cl_2 is essential. The yellow suspension was stirred at 0 °C for further 15 min. The mixture was filtered through a cooled filter, the solid was washed with ice water (100 mL) and dissolved in CH_2Cl_2 (100 mL), and the clear soln was washed with $NaHCO_3$ (100 mL), and cold brine. The organic phase was dried (powdered molecular sieves 4 Å) at 0 °C over 1 h, and filtered. The soln was concentrated and dissolved in Et_2O (20 mL), cooled to −78 °C, filtered and washed with Et_2O (10 mL); yield: 9.6–10.0 g (69–72%); white needles; mp 105–108 °C (dec). Freshly prepared 1,3-benzothiazole-2-sulfonyl chloride is stable at rt for several hours, a smell of SO_2 indicates decomposition after several days.

5-Methyl-1,3,4-thiadiazole-2-sulfonyl Chloride:[659]
5-Methyl-1,3,4-thiadiazole-2-sulfonyl chloride was prepared as described for 1,3-benzothiazole-2-sulfonyl chloride except that 5-methyl-1,3,4-thiadiazole-2-thiol was added as a solid over 30 min instead of as a suspension in aq AcOH over 2 h. Moreover, Et_2O was used instead of CH_2Cl_2 for extraction, and Na_2SO_4 instead of molecular sieves; yield: 80%; mp 44–48 °C (Et_2O).

Bts-Phg-OH; Typical Procedure for Preparation of N^{α}-1,3-Benzothiazol-2-ylsulfonyl and N^{α}-5-Methyl-1,3,4-thiadiazol-2-ylsulfonyl Amino Acids:[659]
To a soln of H-Phg-OH (483 mg, 3.20 mmol) in 0.25 M NaOH (11 mL, 2.8 mmol) was added at 10 °C 1,3-benzothiazole-2-sulfonyl chloride (1.10 g, 4.72 mmol) and the suspension was stirred at 10 °C for 10 h. The pH was maintained at 10–10.5 by addition of 1.3 M NaOH (total 5 mL, 6.5 mmol). The soln was extracted with Et_2O (10 mL), and the aqueous phase was acidified to pH 1 with concd HCl, and extracted with EtOAc (3×10 mL). The organic phases were dried and concentrated to yield 1,3-benzothiazol-2-ylsulfonyl-Phg-OH as white crystals; yield: 1.09 g (97%); purity >98%. Pure material was obtained by recrystallization (EtOAc/$CHCl_3$); mp 166 °C (dec); $[\alpha]_D^{25}$ −139 (*c* 2, EtOH).
Further examples: 1,3-benzothiazol-2-ylsulfonyl-Val-OH; yield: 94%; mp 101–103 °C; $[\alpha]_D^{25}$ +29.8 (*c* 2.3, EtOH); 5-methyl-1,3,4-thiadiazol-2-ylsulfonyl-Pro-OH; yield: 94%; mp 109–110 °C; $[\alpha]_D^{25}$ −112 (*c* 2.3, EtOH).

N^{α}-oNbs Amino Acids; General Procedure:[658,675]
To a soln of the amino acid (30 mmol) in 1 M NaOH (30 mL, 30 mmol) and dioxane (30 mL) were added at 0 °C simultaneously 1 M NaOH (50 mL, 50 mmol) and 2-nitrobenzenesulfonyl chloride (9.97 g, 45 mmol) in dioxane (50 mL) (in about 10 portions with 15–20 min stirring between each addition). Stirring was continued at rt for 1–2 h, the soln was made basic (litmus paper), filtered, and the filtrate washed with Et_2O (2 × 40 mL). The aqueous soln was acidified with concd HCl (Congo red), saturated with NaCl, and extracted with EtOAc (3 × 50 mL). The organic layers were washed with H_2O (30 mL), petroleum ether (100 mL) was added, and the soln was extracted with $NaHCO_3$ (3 × 50 mL). The extracts were concentrated to remove EtOAc, and slowly acidified with concd HCl (Congo red) with vigorous stirring. The separating product was recrystallized or chromatographed. For example: 2-nitrophenylsulfonyl-Leu-OH (recrystallization from CH_2Cl_2/hexane); mp 145 °C; $[\alpha]_D^{25}$ −67.8 (*c* 1, DMF); 2-nitrophenylsulfonyl-Phe-OH (recrystallization from CH_2Cl_2/hexane); mp 139 °C; $[\alpha]_D^{25}$ −28.4 (*c* 1, DMF).

H-(Me)Aib-(Me)Aib-OMe·TFA; Typical Procedure for Deprotection of the N^{α}-Pbf Group with DMS/TFA:[657]
A soln of Pbf-(Me)Aib-(Me)Aib-OMe (1 mg, 0.002 mmol) in 10% DMS in TFA (0.1 mL) was kept at rt for 1 h. The product was precipitated by addition of Et_2O; yield: 0.4 mg (86%); mp 162–164 °C.

H-Phg-OH; Typical Procedure for Deprotection of the N^{α}-Bts Group with Zn/AcOH:[659]
To a soln of Bts-Phg-OH (0.18 g, 0.52 mmol) in EtOH (5 mL) was added Zn powder (0.56 g, 8.5 mmol) and AcOH (0.70 mL, 12.2 mmol). The mixture was stirred at rt for 14 h (as monitored by TLC). The reaction was quenched with cold sat. oxalic acid soln (10 mL), the mixture filtered through Celite, and the Celite cake washed with cold sat. oxalic acid (10 mL). The filtrate was washed with Et_2O (2 × 15 mL), and the aqueous layer desalted on an ion-exchange column (Dowex 50 × 8) to give the product: yield: 78 mg (100%).

H-Phg-OH; Typical Procedure for Deprotection of the N^{α}-5-Methyl-1,3,4-thiadiazol-2-sulfonyl Group with Al/Hg:[659]
To a soln of 5-methyl-1,3,4-thiadiazol-2-ylsulfonyl-Phg-OH (0.10 g, 0.32 mmol) in THF (3 mL) was added Al (foil, polished with sandpaper and cut to small pieces, 86 mg, 3.2 mmol), followed by a few crystals of $HgCl_2$ and H_2O (0.12 mL, 6.7 mmol). The mixture was stirred at rt for 3.5 h (as monitored by TLC). The reaction was quenched with cold sat. oxalic acid (10 mL), the mixture filtered through Celite, and the Celite cake washed with cold sat. oxalic acid (10 mL). The filtrate was washed with Et_2O (2 × 15 mL), and the aqueous layer desalted on an ion-exchange column (Dowex 50 × 8) to give the product; yield: 48 mg (100%).

Boc-Ala-(Et)Leu-OMe, Site-Specific N-Alkylation of Peptides on Solid Phase Using 2-Nitrophenylsulfonyl Protection; Typical Procedure:[664]
Fmoc was removed from Tentagel-OH-bound Fmoc-Leu-OH by treatment with 20% piperidine/NMP for 0.5 h, followed by washing with DMF. The free amine was treated with 2-nitrobenzenesulfonyl chloride (5 equiv), DIPEA (5 equiv) in DMF for 0.5 h, followed by washing with DMF and CH_2Cl_2. The sulfonamide was treated with Ph_3P (5 equiv), EtOH (5 equiv), and DEAD (5 equiv) in CH_2Cl_2 for 0.5 h, followed by an extensive washing with DMF. The 2-nitrophenylsulfonyl group was removed by tBuOK (3 equiv), PhSH (5 equiv) in DMF or DBU (3 equiv), 2-sulfanylethanol (5 equiv) in DMF for 0.5 h, followed by washing with 25% AcOH/DMF, DMF, 5% DIPEA/DMF, and DMF, respectively. The resulting secondary amine was treated with Boc-Ala-OH (3 equiv), PyBroP (3 equiv), HOAt (3 equiv), and DIPEA (9 equiv) in DMF for 16 h, followed by washing with DMF and CH_2Cl_2. The peptide was cleaved from the resin by treatment with catalytic amount of NaCN in MeOH for 16 h and purified by chromatography (silica gel).

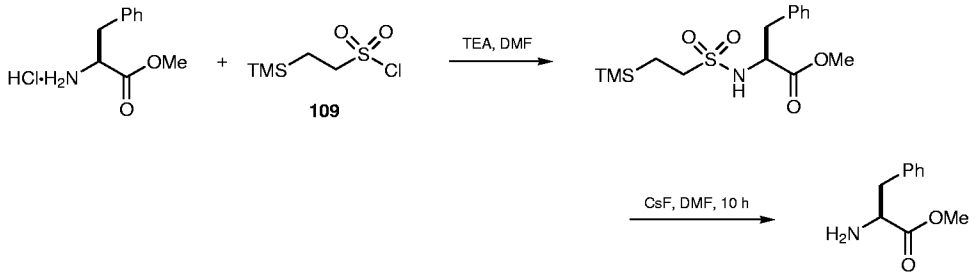
2.1.1.5.2 2-(Trimethylsilyl)ethylsulfonyl

The N^{α}-2-(trimethylsilyl)ethylsulfonyl (SES) amino acid derivatives[676] have a similar stability as the arylsulfonyl amides described in Section 2.1.1.5.1, but are easily cleaved with cesium fluoride to yield the deprotected amino compound and the low-boiling side products TMSF, ethene, and SO_2, which are removed by workup or evaporation (Scheme 51). The SES group is remarkably stable, even against boiling TFA, 6 M HCl in refluxing THF, or against boron trifluoride–diethyl ether complex, and its greatest advantage when other silyl protecting groups are present in the molecule is that it is not cleaved with mild fluoride compounds like lithium tetrafluoroborate in refluxing MeCN, 1 M TBAF in refluxing THF,

for references see p 149

or with 40% HF in EtOH. Conversely, cleavage is achieved with 2–3 equivalents CsF in DMF at 90 °C over 9–40 hours, or with 10 equivalents of TBAF in THF (25 °C, 48 h)[676] and in cases where these procedures are unsatisfactory, with anhydrous HF (neat, anisole, 0 °C, 1.5 h).[677] Due to these favorable features, this protecting group has been applied for the synthesis of depsipeptides[677,678] as well as for the synthesis of polyamines.[679]

Scheme 51 Synthesis and Cleavage of N^{α}-SES Amino Acids and Peptides[676]



2-(Trimethylsilyl)ethanesulfonyl Chloride (SES-Cl, 109):[676]
To trimethyl(vinyl)silane (26.0 g, 259 mmol) and *tert*-butyl peroxybenzoate (1.02 g, 5 mmol) in MeOH (100 mL) was slowly added $NaHSO_3$ (52.6 g, 505 mmol) in H_2O (100 mL) at rt. The mixture was refluxed for 48 h, carefully concentrated (**CAUTION:** *peroxides!*) and H_2O was removed by azeotropic distillation with MeOH (3 ×) by rotary evaporation. The residue was extracted with MeOH (3 × with a total of 250 mL), and the resulting soln was taken to dryness and dried at 150 °C under reduced pressure giving sodium 2-(trimethylsilyl)ethanesulfonate; yield: 37.0 g (70%).
To a fraction of sodium 2-(trimethylsilyl)ethanesulfonate (ca. 90% purity; 16.8 g, 80 mmol) in CCl_4 (50 mL) was added PCl_5 (51.3 g, 247 mmol) over 15 min in portions and the mixture was stirred at rt for 90 min, carefully poured into ice water (30 mL), and extracted with CH_2Cl_2. The extract was washed with $NaHCO_3$ (10 mL), dried, concentrated, and distilled; yield: 10 g (60%); bp 60 °C/0.1 Torr.

N^{α}-SES Amino Acids; General Procedure:[676]
A primary or secondary amine (1.93 mmol) and TEA (1.5 mL) in dry DMF (2 mL) at 0 °C was treated dropwise with SES-Cl (**109**; 0.58 g, 2.89 mmol) in DMF (1.5 mL) over 15 min. The mixture was stirred at 0 °C for 1.5 h, poured into H_2O and extracted with Et_2O. The organic layer was washed with brine, dried, and concentrated. Purification on silica gel yielded the N^{α}-SES-protected amino acid, e.g. SES-Phe-OMe; yield: 93%.

H-Phe-OMe; Typical Procedure for the Cleavage of SES:[676]
A soln of SES-Phe-OMe (0.61 mmol) and CsF (0.30 g, 1.96 mmol) in dry DMF (1 mL) was heated at 95 °C (TLC control, 9–40 h). Then MeOH (0.5 mL) was added, and the mixture concentrated under reduced pressure. The residue was diluted with Et_2O (5 mL), filtered and the solvent removed. The crude product was purified on silica gel; yield: 87% within 10 h.

2.1.1.6 *N*-Alkyl Groups

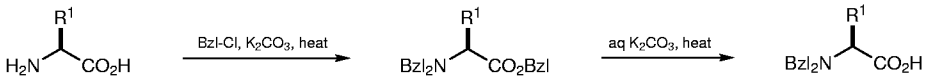
J. Podlech

The use of *N*-alkyl derivatives for the protection of α-amino functionalities in amino acids is usually not advantageous, since the *conditio sine qua non* of suitable protecting groups, i.e. easy removal, is not given. However, N^{α}-benzyl, N^{α}-benzhydryl, and N^{α}-triarylmethyl derivatives have gained considerable importance as these derivatives are cleaved by hydrogenolysis and the latter two also by acidolysis. A fine tuning between stability and cleavage is achieved by substitution of the aryl groups with additional electron-donating substituents. As an essential feature of amino protection, the reduction of the amine nucleophilicity is usually found with these derivatives. Although the protected amino group is protonated by strong acids, it no longer acts as a nucleophile (which would compete, for example, during peptide coupling), partly for electronic and partly for steric reasons.

2.1.1.6.1 *N*-Benzyl Group

While the *N*-benzyl protection of amino acids has found almost no application in peptide chemistry, the *N*,*N*′-dibenzyl protection can be of considerable interest in specific cases. In fact, Reetz et al.[680–682] used such derivatives for the preparation of *N*,*N*′-dibenzylamino aldehydes by reduction of the corresponding amino acid esters to amino alcohols followed by oxidation. These aldehydes have been reacted, for example, with Grignard reagents to yield amino alcohols with excellent *anti* selectivity. Enantiopure N^{α},$N^{\alpha'}$-dibenzyl amino acids are accessible by reacting the amino acid with a benzyl halide under basic conditions (Scheme 52). The resulting N^{α},$N^{\alpha'}$,*O*-tribenzyl-protected amino acids are subsequently saponified.[683–685] When the respective amino aldehydes are required, saponification is not necessary as the dibenzyl-protected amino esters are directly reduced with diisobutylaluminium hydride.[682] Hydrogenolytic cleavage of the benzyl groups is achieved with either Pd/C or $Pd(OH)_2/C$.[680]

Scheme 52 Synthesis of N^{α},$N^{\alpha'}$-Dibenzyl-Protected Amino Acids[684]

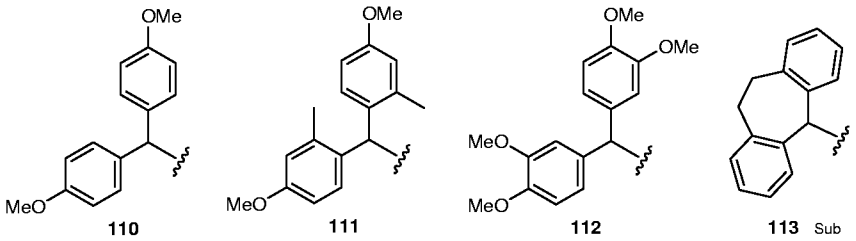


Bzl_2>Leu-OH; Typical Procedure:[684]
To a refluxing soln of H-Leu-OH (6.55 g, 50.0 mmol) in EtOH/H_2O (2:1, 75 mL) and 3.5 M K_2CO_3 (30 mL) was added dropwise Bzl-Cl (25 mL, 0.22 mol). After refluxing for 1 h, the EtOH was removed, the soln was acidified with AcOH and extracted with $CHCl_3$. The solvent and benzyl alcohol were removed by distillation at 120 °C. Saponification of the resulting benzyl ester was performed in 20% K_2CO_3 in propane-1,2-diol (40 mL) on refluxing for 5 min. Then H_2O (30 mL) was added and the soln acidified with AcOH, extracted with $CHCl_3$, and recrystallized (AcOH/H_2O 1:1, 56 mL); yield: 13.6 g (86%); mp 104–106 °C; $[\alpha]_D$ −68 (*c* 2%, MeOH).

2.1.1.6.2 *N*-Benzhydryl Groups

N-Benzhydryl-substituted amino acids are considerably more stable than the *N*-trityl derivatives (*vide infra*) to acidic reaction conditions; nevertheless, they are deprotected by rather mild procedures. Several variations have been presented such as 4,4′-dimethoxybenzhydryl (**110**), 4,4′-methoxy-2,2′-dimethylbenzhydryl (**111**), and 3,3′,4,4′-tetramethoxybenzhydryl (**112**) derivatives (Scheme 53),[686] but none of these protecting groups have found wider application.

Scheme 53 *N*-Benzhydryl Protecting Groups



A further variation has been proposed by Pless with the 10,11-dihydro-5*H*-dibenzo[*a*,*d*]cyclohepten-5-yl derivative (dibenzosuberyl, Sub, **113**), in which two phenyl groups are linked via an ethylene bridge.[687] The Sub group represents an alternative to the more commonly used trityl group, since it is more stable under acidic conditions (stable even against 5 M HCl/dioxane), but cleaved by various hydrogenolytic or acidolytic procedures, e.g. 10% TFA/

for references see p 149

CH_2Cl_2, 22 °C, 0.5 h; 60% AcOH/H_2O, reflux, 1 h; or H_2, Pd/C, MeOH, 22 °C, 1 h. As the N^{α}-Sub-protected amino acids are not sterically hindered in the region of the carboxy group, peptide coupling (e.g., of Sub-Phe-OH) is possible with the DCC or the mixed anhydride method. The amino group, on the other side, is effectively blocked; although protonation is still possible with strong acids, N-acylation has not been observed on Sub-protected amino functions. The Sub group is easily introduced using commercially available 5-chloro-10,11-dihydro-5*H*-dibenzo[*a*,*d*]cycloheptene (Sub-Cl). Nevertheless, Schotten–Baumann conditions are not recommended, since the chloride undergoes fast hydrolysis under basic conditions; phase-transfer catalysis has therefore been applied. Alternatively, the N-protection of the corresponding amino acid esters with subsequent saponification leads to the enantiopure Sub-protected amino acids, although yields are not always satisfactory.

Sub-Phe-OH·HCl from H-Phe-OMe·HCl; Typical Procedure:[687]
To a soln of H-Phe-OMe·HCl (4.3 g, 20 mmol) and Sub-Cl (5.5 g, 24 mmol) in CH_2Cl_2 (160 mL) was added at rt TEA (8.4 mL, 60 mmol). After several h, the mixture was washed with dil HCl, dried, and the solvent removed. To the residue was added HCl/Et_2O and petroleum ether and the precipitate was filtered to give Sub-Phe-OMe·HCl; yield: 7.0 g (86%); mp 143–146 °C (dec); $[\alpha]_D^{20}$ +28.5 (*c* 5%, MeOH).
To Sub-Phe-OMe·HCl (4.1 g, 10 mmol) in MeOH (100 mL) was added 2 M NaOH (25 mL) and the soln was heated to reflux. MeOH was removed, the residue was treated with dil HCl and extracted with CH_2Cl_2. The organic layer was dried, filtered, concentrated, and treated with HCl/Et_2O; yield: 3.5 g, (89%); mp 108–110 °C (dec); $[\alpha]_D^{20}$ +33.4 (*c* 5, DMF).

Sub-Phe-OH·HCl from H-Phe-OH; Typical Procedure:[687]
To a soln of H-Phe-OH (1.7 g, 10 mmol) and TBAB (6.4 g, 20 mmol) in 1 M NaOH (35 mL), 10% Na_2CO_3 (35 mL), and CH_2Cl_2 (100 mL) was added Sub-Cl (4.6 g, 20 mmol) under vigorous stirring. Stirring was continued for 16 h at rt and the organic layer was washed with dil H_2SO_4. The organic layer was treated with 2 M HCl in Et_2O (50 mL) for 1 h and the solvent removed. The residue was recrystallized (Et_2O/petroleum ether 1:1); yield: 2.1 g (53%); mp 105–110 °C (dec); $[\alpha]_D^{20}$ +33.4 (*c* 5, DMF).

2.1.1.6.3 *N*-Trityl Groups

Among the trityl protecting groups (Scheme 54), the trityl group (Trt, **114**) has found most application, although more frequently for the side-chain protection of cysteine (see Section 2.6.6.2.1.3.3), histidine (see Section 2.6.2.1), asparagine and glutamine (see Section 2.3.3.3), serine and threonine (see Section 2.6.4.1.1.3), tyrosine (see Section 2.6.4.2.1.3), and ω-amino functions (see Section 2.1.2.2.5.1) than for the α-amino protection. In fact, N^{α}-Fmoc- and N^{ω}-Trt-protected amino acids are useful intermediates in SPPS,[226,688–690] since these trityl protections are completely stable toward the basic conditions of N^{α}-Fmoc cleavage as well as toward the generally utilized coupling reagents in peptide-chain-elongation steps, but are readily removed by acidolysis at the desired synthetic step.

A number of variations to the trityl group have been proposed, which differ in their electron-donating substituents. Acidic cleavage is facilitated with these substitutions (see Table 12), since the intermediately formed trityl cations are better stabilized. Among these substituted trityl groups the 4-methyltrityl (Mtt, **115**),[691] 4-methoxytrityl (Mmt, **116**),[692–694] and bis(4-methoxyphenyl)phenylmethyl (DMTr, **117**)[692–694] have received attention, particularly for side-chain protection as their selective acidolytic cleavage with respect to the more stable *tert*-butyl alcohol-derived protecting groups allows regioselective chemistry to be performed on the peptides protected at all the residual functionalities.

Scheme 54 *N*-Trityl Protecting Groups

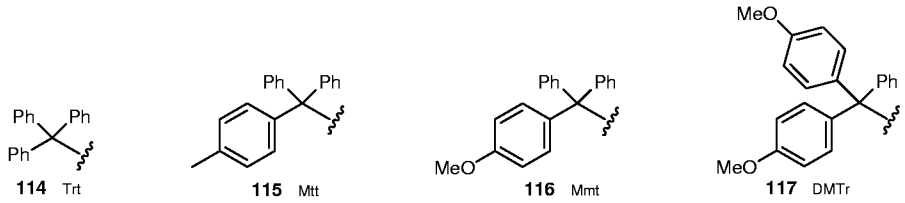


Table 12 Acidolytic Cleavage of *N*-Trityl Amino Acid and Peptide Derivatives[90,185,691,693–707]

N-Trityl Derivatives	Cleavage Conditions	Ref
trityl (Trt)	1 M HCl/EtOH	[695]
	1% TFA/CH_2Cl_2 or HOBt/TFE	[696]
	pyridinium hydrochloride	[697]
	hot 50% AcOH, 1–2 min	[695]
	80% AcOH, rt, 1 h	[698]
	HOBt, phenol, cresol	[699]
	CCl_3CO_2H, iPrOH	[700]
	0.1 M HOAt, 0.12 M TMSCl, TFE or 0.02 M TMSCl, TFE or 0.2% TFA/1% H_2O, CH_2Cl_2	[701]
	3% CCl_3CO_2H/CH_2Cl_2	[185]
	H_2, Pd/C	[695]
	Na/NH_3	[90]
4-methyltrityl (Mtt)	1% TFA/CH_2Cl_2, rt, 30 min	[691,702]
	AcOH/TFA/CH_2Cl_2 (1:2:7), rt, 1 h	[691]
	1% TFA/5% TIS in CH_2Cl_2	[703]
4-methoxytrityl (Mmt) and bis(4-methoxyphenyl)-phenylmethyl (DMTr)	$CHCl_2CO_2H$ (10 equiv), anisole (100 equiv), CH_2Cl_2	[693]
	AcOH, TFE, CH_2Cl_2	[694]
9-phenyl-9-fluorenyl (PhFl)	TFA, aq MeCN, 0 °C, 1 h	[704,705]
	H_2, Pd/C	[704,705]
	Li/NH_3, THF	[706]
	I_2, MeOH (*N*-PhFl derivatives of secondary amines)	[707]

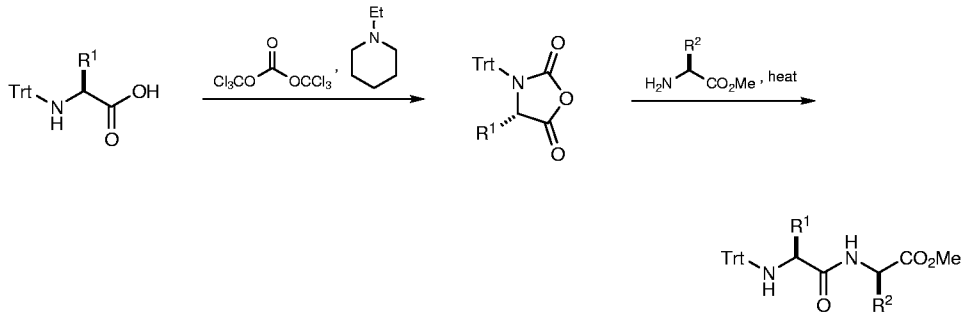
2.1.1.6.3.1 *N*-Trityl Group

The trityl group was proposed by Helferich et al.[708] as a N-protecting group that is stable to basic conditions, but is very acid sensitive. Synthesis of N^{α}-Trt amino acids by the Schotten–Baumann procedure is not recommended, since hydrolysis of the trityl halides is faster than the amine alkylation, thus generally resulting in low yields. Better preparations are achieved by an intermediate esterification of the amino acid. The resulting *N,O*-ditrityl amino acids are saponified by methanolysis and isolated as diethylammonium salts [Gly, Ala, Val, Leu, Ile, Phe, Lys(Boc), Lys(Fmoc), yields ranging from 80–86%]. Moreover, trityl bromide (Trt-Br) has proved to be superior to trityl chloride (Trt-Cl) in this procedure.[709] Comparable or even better results have been obtained by intermediate trimethylsilyl protection of the carboxy group.[710] Serine or threonine are hydroxy- and carboxy-protected either with excess TMSCl or with bifunctional silyl chlorides such as dichlorodimethylsilane or dichlorodiphenylsilane. These intermediates are then N-tritylated under very mild conditions and the silyl entities are subsequently removed by the aqueous workup. Alternatively, N-tritylation is performed on amino acid methyl esters by reaction with Trt-Cl followed by saponification of the methyl ester with $LiOH/H_2O$/MeOH at 25–40 °C. The latter procedure has been reported to produce better yields (>75% compared to 40–60% by the intermediate silylation method) and chromatography is not required to isolate homogeneous derivatives.[185,711] No racemization was observed with this procedure.

for references see p 149

A major disadvantage of the N^{α}-trityl group is its bulkiness, which sterically disfavors peptide couplings at the C-terminus or even saponification of the corresponding methyl esters. Nevertheless, couplings with DCC/HOBt,[700,710] with the corresponding 1,2,3-benzotriazol-1-yl esters,[712] HATU/DIPEA,[185] or PyAOP[701] proceed in satisfactory manner and without racemization. Saponification of sterically nonhindered N^{α}-trityl-protected amino acids is achieved with ethanolic KOH;[695] for sterically more demanding amino acid derivatives 20% ethanolic KOH at elevated temperature[695,713] or LiOH in aqueous methanol (*vide supra*) also at higher temperature are used.[711] As a way to bypass the reduced reactivity of N^{α}-trityl amino acids, the use of the corresponding NCAs has been proposed (see Section 3.4.4).[714] Recently, N^{α}-trityl-protected NCAs (Scheme 55) have been shown to be highly stable crystalline compounds that can be used directly without additives for peptide bond formation without racemization in THF at 40 °C or at reflux for 6–20 hours (80–90% yield).[715]

Scheme 55 Synthesis and Use of N^{α}-Trt-Protected NCAs[715]



The trityl group is essentially stable against basic conditions, although cleavage under strongly basic conditions has been reported.[708] Conversely, clean and rapid cleavage of *N*-trityl derivatives is usually observed by acidolysis or even hydrogenolysis (Table 12). Selective cleavage of the trityl group in the presence of the highly acid-labile Bpoc group is also possible by pH-controlled titration with HCl in aqueous TFE.[531,716] The trityl cation formed during acidic cleavage is poorly electrophilic compared to the *tert*-butyl cation formed during Boc cleavage; therefore, side-product formation due to the electrophilic attack of this moiety at nucleophilic centers (e.g., Trp side chain) plays a minor role.

The N^{α}-trityl protection has been successfully applied for the synthesis in solution of suitable peptide segments particularly when containing Cys(Trt) residues, e.g. in the synthesis of insulin.[531] Conversely, in SPPS this type of N^{α}-protection has only been used sporadically,[712] although occasionally it may be advantageous. Basic cleavage of N^{α}-terminal protecting groups from resin-bound dipeptides is known to generate piperazine-2,5-diones (DKP) as noxious side products when the proper sequence favors such a cyclization reaction (see Section 2.1.1.1.1.3.3). In such cases N^{α}-trityl protection represents a useful alternative since the Trt group is readily cleaved on resin with 0.1 M HOAt/0.12 M TMSCl in TFE, 0.02 M TMSCl in TFE or 0.2% TFA/1% H_2O in CH_2Cl_2 without piperazine-2,5-dione formation.[701,711] This side reaction is additionally suppressed when the subsequent coupling is performed with PyAOP/DIPEA in DMF. These cleavage cocktails are furthermore compatible with the 3-[4-(hydroxymethyl)phenoxy]propanoic acid linker. Similarly, cleavage with 0.2% TFA/1% H_2O in CH_2Cl_2 or 3% trichloroacetic acid in CH_2Cl_2 is compatible with the highly labile Riniker handle.[185]

During cleavage of N^{α}-trityl derivatives protonation of the amino group is sufficient to prevent its realkylation by the trityl cations; alternatively, addition of reducing silanes as

routinely used for *S*-trityl cleavage[717] (see Section 2.6.6.2.1.3.3) should definitely prevent alkylations at sensitive functionalities.

Synthesis of N^{α}-Trt Amino Acids with Trt-Br via Intermediate Trityl Esters, General Procedure:[709]
A suspension of the amino acid (10 mmol) in a soln of Trt-Br (7.11 g, 22 mmol) in $CHCl_3$/DMF (2:1, 50 mL) was vigorously stirred at rt until a clear soln was obtained (<1 h). Then, TEA (5.6 mL, 40 mmol) in $CHCl_3$/DMF (10 mL, 2:1) was added over 20 min, stirring was continued for 30 min, MeOH (50 mL) was added and the mixture was heated to 50 °C for 1–2.5 h. The solvents were removed, Et_2O (100 mL) was added, and the soln was extracted with 10% citric acid (3 × 50 mL) and H_2O (3 × 50 mL), and dried. Then DEA (1.0 mL, 10 mmol) in Et_2O (10 mL) was added dropwise to the soln. The precipitate was filtered off, washed several times with cold Et_2O, and dried. For example: Trt-Leu-OH·DEA; yield: 85%; $[\alpha]_D$ +2.2 (*c* 5, MeOH); Trt-Lys(Fmoc)-OH·DEA; yield: 80%; $[\alpha]_D$ +10.2 (*c* 0.5, MeOH).

Trt-Leu-OH·DEA; Typical Procedure by Intermediate Protection as the TMS Ester:[710]
To a stirred suspension of H-Leu-OH (1.31 g, 10 mmol) in $CHCl_3$/MeCN (5:1, 18 mL) was added TMSCl (1.27 mL, 10 mmol) at rt. The mixture was heated under reflux for 2 h and then cooled to rt. Addition of TEA (2.79 mL, 20 mmol) at a rate sufficient to maintain gentle reflux was followed by addition of Trt-Cl (2.79 g, 10 mmol) dissolved in $CHCl_3$ (10 mL). The mixture was stirred for 1 h and MeOH (50 mL) was added. The mixture was concentrated and the residue was partitioned between Et_2O (50 mL) and cooled 5% citric acid (50 mL). The organic layer was washed with 1 M NaOH (2 × 20 mL) and H_2O (2 × 10 mL). The combined aqueous layers were washed with Et_2O (20 mL), cooled to 0 °C, neutralized with glacial AcOH, and extracted with Et_2O (2 × 30 mL). The organic layers were washed with H_2O (2 ×), dried, and the solvent removed to yield spectroscopically pure Trt-Leu-OH; yield: 3.47 g (93%), which upon dissolving in Et_2O (20 mL) and addition of DEA (1 mL, 10 mmol) afforded the corresponding diethylammonium salt; yield: 83%; mp 152–154 °C; $[\alpha]_D^{25}$ +3.5 (*c* 5, MeOH).

Trt-Thr-OH·DEA:[710]
A mixture of H-Thr-OH (3.57 g, 30 mmol) and Ph_2SiCl_2 (6.23 mL, 30 mmol) in CH_2Cl_2 (40 mL) was refluxed for 2 h and then allowed to reach rt. The TEA (12.6 mL, 90 mmol) was added dropwise, followed by a soln of Trt-Cl (8.36 g, 30 mmol) in CH_2Cl_2 (20 mL), and the resulting suspension was stirred for 5 h at rt. Excess of MeOH and TEA (4.18 mL, 30 mmol) was added, the mixture was concentrated and the residue was partitioned between EtOAc (50 mL) and cooled 5% citric acid (50 mL). The organic layer was washed with 1 M NaOH (2 × 20 mL) and H_2O (2 × 10 mL). The combined aqueous layers were washed with EtOAc (20 mL), cooled to 0 °C, neutralized with glacial AcOH, and extracted with EtOAc (2 × 30 mL). The organic layers were washed with H_2O (2 ×), dried, and concentrated. The residue was dissolved in EtOAc (20 mL) and DEA (3 mL, 30 mmol) was added, and the precipitate recrystallized (acetone/petroleum ether); yield: 10.8 g (83%); mp 165 °C; $[\alpha]_D^{25}$ −6.7 (*c* 2, MeOH).

Synthesis of N^{α}-Trt Amino Acid NCAs; General Procedure:[715]
To a soln of bistrichloromethyl carbonate (triphosgene) (1.51 g, 5.0 mmol) in EtOAc (280 mL) was added the N^{α}-Trt amino acid (14 mmol), then a soln of 1-ethylpiperidine (1.74 g, 15.4 mmol) in EtOAc (20 mL) was added dropwise within 40 min, and the mixture was stirred for another 2 h. The mixture was filtered, the solvent removed and chromatographed on a short column (silica gel, EtOAc). Concentration and crystallization of the residue gave the N^{α}-Trt-NCAs. For example: *N*-Trt-Ala-NCA (crystallized from EtOAc/petroleum ether); yield: 71%; mp 205–206 °C; $[\alpha]_D^{21}$ +34.3 (*c* 0.5, EtOAc).
Experimental protocols for selective detritylation of peptides containing Cys(Trt) are reported in Section 2.6.6.2.1.3.3.

Bpoc-Tyr(tBu)-Gln-Leu-Glu(OtBu)-Asn-Tyr(tBu)-Cys-Asn-OtBu/H-Leu-Val-Cys-Gly-Glu(OtBu)-Arg(HCl)-Gly-Phe-Phe-Tyr(tBu)Thr(tBu)-Pro-Lys(Boc)-Thr(tBu)-OtBu; Insulin A-[14–21]/B-[17–30]: Detritylation by Titration with HCl in TFE:[531]
The fully protected insulin segment N^{α}-Bpoc-A-[14–21]/N^{α}-Trt-B-[17–30] (4.17 g, 1.1 mmol) was dissolved in 90% aq TFE (208 mL). The soln (apparent pH 5.9) was titrated at pH 3.5 with 0.15 M HCl; after a consumption of 1.08 mmol HCl within 5 min the pH was raised with pyridine (0.8 mL) to 5.5, and the soln was concentrated and lyophilized.

2.1.1.6.3.2 Substituted *N*-Trityl Groups

The 4-methyltrityl group (Mtt, **115**) has been used mainly for ω-protection of lysine, ornithine, or histidine residues.[689–691,718] It is cleaved under very mild acidic conditions (Table 12) where the *tert*-butyl alcohol-based protecting groups are unaffected.[691] Similarly, the

for references see p 149

N-4-methoxytrityl (Mmt, **116**),[692–694] and *N*-bis(4-methoxyphenyl)phenylmethyl (DMTr, **117**) derivatives are stable against base, but are cleaved smoothly by very mild acid treatment. These *N*-alkyl-type protecting groups are introduced with the commercially available 4-methyltrityl chloride (Mtt-Cl), 4-methoxytrityl chloride (Mmt-Cl), and bis(4-methoxyphenyl)phenylmethyl chloride (DMTr-Cl) generally using TEA as the base.[692] The N^{α}-DMTr amino acids slowly decompose in protic solvents, whereas the N^{α}-Mmt-protected compounds are somewhat more stable, especially as their DIPEA salts.[694] Correspondingly, this type of α-amino protection has found only limited application. For example, N^{α}-Mmt protection was used for the preparation of a dipeptidyl-RNA, which is labile to base, hydrogenolysis, and acids. Even trityl protection was unsuitable in this case, since the conditions required for its removal were found to decompose the compound. In this case, the Mmt group was cleaved with 5% trichloroacetic acid within 5 minutes at 4 °C, leaving the synthetic target intact.[719]

Mmt-Gly-OH; Typical Procedure:[719]
To a soln of H-Gly-OH (750 mg, 10 mmol) in H_2O (4 mL), DEA (3 mL), and iPrOH (8 mL) was added with stirring in 20 portions Mmt-Cl (3.7 g, 12 mmol) over a period of 90 min. H_2O (50 mL) was added, the soln filtered, and the precipitate washed with H_2O. The filtrate was cooled (0 °C) and acidified with a cold soln of AcOH (0.6 mL) in H_2O (25 mL). The precipitate was filtered, washed with cold H_2O, and dried; yield: 3 g (87%); mp 76–77 °C.

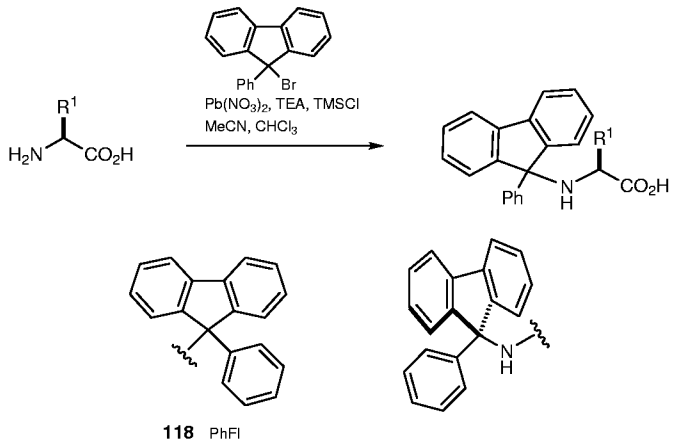
Mmt-βAla-OH·DIPEA or DMTr-βAla-OH·DIPEA; General Procedure:[694]
To a suspension of H-βAla-OH (10.0 mmol) in abs DMF/pyridine (1:1, 20 mL) was added with stirring either Mmt-Cl or DMTr-Cl (15 mmol) and the mixture was heated under reflux for 60–90 min. After cooling, MeOH (3 mL) was added and the mixture was concentrated. The residue was diluted in EtOAc (300 mL) and poured into sat. $NaHCO_3$ (100 mL). The organic layer was washed with brine (2 × 100 mL) and the resulting aqueous layers were re-extracted twice with EtOAc. The combined organic layers were dried, concentrated, and chromatographed (silica gel, 30 g, toluene/EtOAc/MeOH 10:10:3, containing 1% DIPEA) to yield a colorless foam of either Mmt-βAla-OH·DIPEA (79%) or DMTr-βAla-OH·DIPEA (16%).

2.1.1.6.4 *N*-9-Phenyl-9-fluorenyl Group

The 9-phenyl-9-fluorenyl (PhFl, **118**, Scheme 56) amino-protecting group has been proposed by Christie and Rapoport[704] as it is significantly more stable to solvolysis in acid conditions than the trityl group because of the antiaromatic character of the fluorenyl cation. In fact, solvolysis of Trt-Cl is 6000 times faster than PhFl-Cl.[720] The N^{α}-PhFl amino acid derivatives are prepared with PhFl-Br (commercially available or prepared from fluorenone and phenyllithium and subsequent substitution with HBr) and lead nitrate, which is required to trap the bromide (Scheme 56).[704,721] A lead-free alternative using nitromethane as solvent has also been reported.[722]

Cleavage of the *N*-PhFl derivatives occurs either with TFA in aqueous organic solvents or by hydrogenolysis over Pd/C,[704,705] preferentially in the presence of AcOH.[723] When hydrogenation is performed in the presence of Boc_2O the corresponding Boc-protected derivatives are obtained directly in excellent yields.[724] *N*-PhFl derivatives are also cleaved with Li/NH_3 in THF[706] or in the case of *N*-PhFl derivatives of secondary amines by iodine in MeOH.[707]

Scheme 56 Synthesis of N^{α}-9-Phenyl-9-fluorenyl Amino Acids, and the 9-Phenyl-9-fluorenyl (PhFl) Group and Its Preferred Conformation[704,721]



118 PhFl

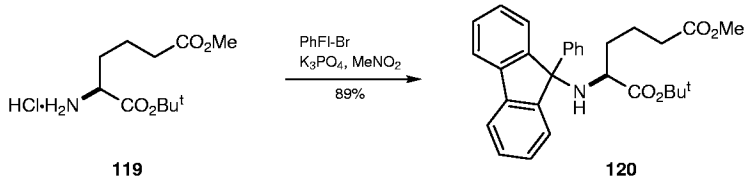
The PhFl group attached to the α-amino group prevents racemization of the amino acid derivatives by sterically shielding the α-carbon proton from abstraction by a strong base and by positioning the α-proton and the carbonyl in a planar geometry in which deprotonation is stereoelectronically disfavored.[721,725] These properties permit the synthesis of a wide variety of peptidomimetic structures from proteinogenic amino acids with the chirality of the α-carbon determined by that of the initial amino acid. In this context, this type of amino protection has attracted considerable attention.[721,723,726,727] Since the conformation of the PhFl group (Scheme 56) differs from that of the trityl group, which is more like a propeller, it has been reported to occasionally affect the stereochemistry of reactions in its vicinity.[704]

In analogy to the N^{α}-Trt amino acids, the N^{α}-PhFl amino acids are readily converted into the NCAs for synthesis of small peptides.[715]

PhFl-Ala-OH; Synthesis with PhFl-Br and $Pb(NO_3)_2$:[721]

A mixture of H-Ala-OH (13.5 g, 152 mmol), $CHCl_3$ (375 mL), MeCN (75 mL) and TMSCl (19.04 mL, 150 mmol) was refluxed for 2 h under inert gas. The mixture was cooled to rt and TEA (46 mL, 330 mmol) was added slowly to maintain a gentle reflux and the mixture was stirred for 15 min after which $Pb(NO_3)_2$ (33.1 g, 100 mmol) was added. Then a soln of PhFl-Br (57.8 g, 180 mL) in $CHCl_3$ (180 mL) was added. The mixture was stirred under inert gas vigorously for 48 h. After ~20 h, the mixture becomes orange and darkens over the time. MeOH (15.2 mL, 375 mmol) was added and the mixture stirred for an additional 30 min. The solid was filtered off and the filtrate concentrated to a residue, which was partitioned between Et_2O (750 mL) and 5% aq citric acid. The aqueous layer was extracted with Et_2O (4 × 250 mL) and the organic layers were re-extracted with 1 M NaOH (300 mL). The aqueous layer was extracted with Et_2O (300 mL), cooled to 0 °C and the pH adjusted to 7 with AcOH and extracted with 25% iPrOH in $CHCl_3$ (3 × 300 mL). The combined organic soln were washed with brine, dried, and concentrated to a light yellow foam; yield: 39.2 g (80%).

Scheme 57 Synthesis of N^{α}-PhFl Amino Acids without Lead Reagents[722]

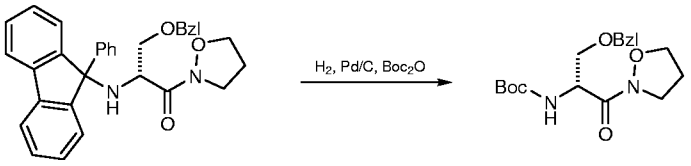


119 **120**

for references see p 149

1-*tert*-Butyl 6-Methyl (*S*)-2-[(9-Phenyl-9-fluorenyl)amino]hexanedioate (120); Typical Procedure for the Synthesis of N^{α}-PhFl Amino Acids without Lead Reagents (Scheme 57):[722]
A soln of 1-*tert*-butyl 6-methyl (*S*)-2-aminohexanedioate hydrochloride (**119**; 0.60 g, 2.2 mmol), PhFl-Br (0.84 g, 2.6 mmol), and K_3PO_4 (0.93 g, 4.4 mmol) in $MeNO_2$ (4 mL) was stirred at rt for 72 h, after which MeOH (2 mL) was added and stirring continued for 1 h. Filtration, removal of the solvent, and chromatography of the residue (10% EtOAc/hexane) gave a clear oil; yield: 0.92 g (89%); $[\alpha]_D^{25}$ −180 (*c* 4.4, $CHCl_3$).

Scheme 58 Hydrogenolytical Cleavage of the PhFl Group[724]



121 122

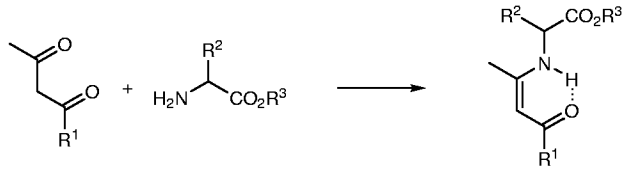
Conversion of the N^{α}-PhFl into the N^{α}-Boc Derivative in 2-[*O*-Benzyl-*N*-(9-phenyl-9-fluorenyl)-D-serinyl]isoxazolidine (122) under Hydrogenolytic Conditions; Typical Procedure (Scheme 58):[724]
To a degassed soln of the N^{α}-PhFl-protected amino acid (**121**; 1.5 g, 3.05 mmol) and Boc_2O (2.26 g, 10.4 mmol) in MeOH (25 mL) was added 10% Pd/C (0.38 g). The resulting slurry was degassed once more and then stirred under a balloon of H_2 over 3 h. The slurry was degassed under N_2, filtered through Celite, and the Celite pad was washed with MeOH (280 mL). Removal of the solvent and chromatography (EtOAc/hexane 1:1) yielded the amino acid derivative **122**: 1.05 g (98%); mp 60–61 °C; $[\alpha]_D^{20}$ +1.3 (*c* 2.1, $CHCl_3$).

2.1.1.7 *N*-Vinyl Groups

J. Podlech

1,3-Diketones or β-oxocarboxylic acid esters and amides react with amino acid salts to yield the corresponding enamines **123** (Scheme 59).[728,729] These derivatives are stable to basic reaction conditions and can be used for peptide couplings. Cleavage of the *N*-vinyl derivatives is readily achieved with mild acids. The enamines are usually stabilized by an intramolecular hydrogen bond, as has been confirmed by NMR and IR spectroscopy.[730]

Scheme 59 Vinyl-Protecting Groups[728,729]



123

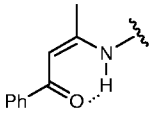
R^1 = alkyl, aryl, alkoxy, alkylamine
R^2 = amino acid side chains
R^3 = Me, Et, K, DCHA

2.1.1.7.1 *N*-2-Benzoyl-1-methylvinyl, *N*-2-Acetyl-1-methylvinyl, and *N*-2-(Trifluoroacetyl)vinyl Groups

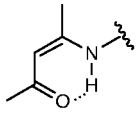
By reacting amino acids with acetylacetone or benzoylacetone in methanolic KOH the corresponding *N*-enamine derivatives potassium salts (Scheme 60) are obtained generally as crystalline and stable compounds in 65–95% yield.[728] Acetylacetone reacts at room temperature, other substrates need a short warming, but even in these cases, racemization has not been observed.[731] Acidification to liberate the carboxylic acid is possible with due

precaution, but generally this is not required for the coupling step. Most commonly used are the 2-benzoyl-1-methylvinyl (1-methyl-3-oxo-3-phenylprop-1-enyl, Mbv **124**) and 2-acetyl-1-methylvinyl (1-methyl-3-oxobut-1-enyl, **125**) derivatives, which are advantageously used for the synthesis of short peptides, particularly when containing methionine residues.[600,614,732]

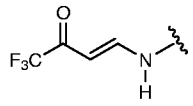
Scheme 60 Enamine-Type Protecting Groups



124 Mbv



125



126

The N^{α}-Mbv amino acid derivatives are used as potassium[729] or DCHA salts,[733,734] the latter being less hygroscopic and more stable, crystalline compounds soluble in organic solvents. *N*-Enamine derivatives have also been used for intermediate protection in the synthesis of esters, whereby best results were obtained with the methyl or ethyl acetoacetate derived enamines.[735]

Cleavage of the *N*-enamine derivatives occurs with 2 M HCl or AcOH,[728] with 0.4 M HCl in THF or preferably with 0.1 M TosOH·H_2O in THF.[733] Cleavage of the enamine group can be conducted under conditions where even the acid-sensitive *S*-Trt group remains intact.[736]

More recently, the 2-(trifluoroacetyl)vinyl group (4,4,4-trifluoro-3-oxobut-1-enyl, **126**) was proposed for the N-protection of amino acids.[737] This enamine derivative is readily prepared with 4-ethoxy-1,1,1-trifluorobut-3-en-2-one and NaOH at room temperature in 1–3 hours. Acidic workup leads to the protected amino acids in yields ranging from 70–89% without racemization. Cleavage is achieved with 3 M HCl in dioxane at room temperature in 10 hours.

The main advantage of the enamine-type protection with benzoylacetone is the easy recovery of latter reagent, which can be recycled, a fact that is exploited in the large industrial-scale production of di- and tripeptides.

Mbv-D-Phg-OK; Typical Procedure:[729]
To a refluxing soln of benzoylacetone (3.84 g, 23.7 mmol) in MeOH (25 mL) was added dropwise H-D-Phg-OH (3.26 g, 21.6 mmol) in 1.08 M methanolic KOH over a period of 4 h. The mixture was filtered and the solvent was removed, leading to an orange residue, which crystallized by trituration with EtOAc; yield: 7.05 g (98%). The product can be recrystallized (EtOAc/EtOH); mp 231–234 °C.

Mbv-Met-OH:[1]
To a soln of Mbv-Met-OK[732] (16.6 g, 50 mmol) in ice-cold H_2O (200 mL) was added dropwise precooled AcOH (48 mL) under stirring. The semisolid mass was triturated with H_2O and the precipitate was filtered off, washed with ice-cold H_2O, dried, and recrystallized (EtOH/petroleum ether); yield: 12.8 g (87%); mp 136–137 °C.

Mbv-Met-OH·DCHA:[56,734]
To a suspension of H-Met-OH (22.4 g, 0.15 mmol) in H_2O (15 mL) was added DCHA (30 g, 0.165 mmol) in MeOH (140 mL), followed by a hot soln of benzoylacetone (26.76 g, 0.165 mmol) in EtOH (140 mL). The mixture was refluxed for 3 h, and the solvent removed. The residue was dissolved in $CHCl_3$ (150 mL), insoluble material was filtered off and washed with $CHCl_3$ (40 mL). After addition of Et_2O (300 mL) the product crystallized; yield: 68.2 g (96%); mp 173–175 °C; $[\alpha]_D^{20}$ +21.5 (*c* 1, EtOH).

Mbv-Met-Asp(OtBu)-Phe-NH_2:

From Mbv-Met-OH:[614]
To Mbv-Met-OH (13 g, 44.3 mmol) and TEA (6.16 mL, 44.3 mmol) in THF (200 mL), was added at −15 °C with vigorous stirring ethyl chloroformate (4.2 mL, 44.2 mmol). After 5 min at −10 °C a precooled soln of H-Asp(OtBu)-Phe-NH_2 (13.4 g, 39.9 mmol) in THF (200 mL) was added. After 2 h in an

for references see p 149

ice bath and overnight at rt, the mixture was concentrated and the residue partitioned between H_2O and EtOAc. The organic layer was washed with aq $KHSO_4$ and H_2O, dried, and the solvent removed. The residue was crystallized (MeOH/H_2O); yield: 19 g (78%); mp 192–193 °C; $[\alpha]_D^{20}$ +57.2 (*c* 1, MeOH).

From Mbv-Met-OH·DCHA:[600]
To a cooled (−10 °C) soln of Mbv-Met-OH·DCHA (24.9 g, 52.5 mmol) and H-Asp(OtBu)-Phe-NH_2 (16.7 g, 49.8 mmol) in DMF (250 mL) was added 3.73 M HCl in dioxane (14.1 mL) followed by HOSu (6.9 g, 60 mmol) and DCC (11.35 g, 55 mmol). After 1 h in ice bath and 12 h at rt, DCU was filtered off, and the filtrate was worked up as described above; yield: 26 g (86%).

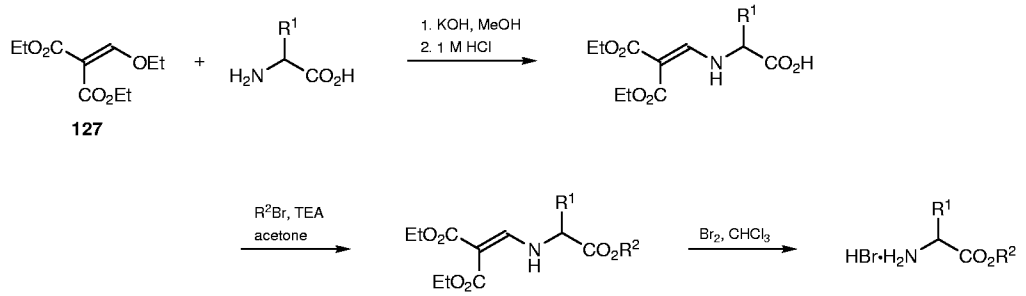
H-Met-Asp(OtBu)-Phe-NH_2·TosOH; Typical Procedure for the Cleavage of the 2-Mbv Group:[600]
To a cold soln of Mbv-Met-Asp(OtBu)-Phe-NH_2 (23.2 g, 38 mmol) in CH_2Cl_2, Tos-OH·H_2O (7.2 g, 38 mmol) was added. After 12 h stirring, the product was filtered off; yield: 24.2 g (~100%); mp 216–218 °C; $[\alpha]_D^{20}$ −3.0 (*c* 1, MeOH).

Amino Acid 4-Nitrobenzyl Esters via N^{α}-Mbv Protection; General Procedure:[735]
To a soln of the amino acid (10 mmol) in 0.5 M KOH in MeOH (20 mL, 10 mmol) was added ethyl acetoacetate (1.39 mL, 11.0 mmol), and the mixture was refluxed for 10 min. The soln was concentrated. To the solid was added dry DMF (10 mL) and Bzl-Cl (10 mmol), and the mixture was stirred at rt for 16–24 h. The resulting suspension was diluted with 1 M $NaHCO_3$ (50 mL) and EtOAc (70 mL), and the organic phase was washed with 1 M $NaHCO_3$ (50 mL) and H_2O (2 × 50 mL), and dried. Removal of the solvent yielded the crude protected ester, to which 1 M methanolic HCl (20 mL) was added to dissolve the solid. After 10 min at rt, the soln was concentrated and the residue was triturated with Et_2O. For example: H-Ser-ONbz·HCl (recrystallization from EtOAc/MeOH); yield: 68%; mp 190.5–192.5 °C (dec); $[\alpha]_D^{20}$ −15.5 (*c* 1, DMF).

2.1.1.7.2 *N*-2,2-Bis(alkoxycarbonyl)vinyl, *N*-2-Acyl-2-(alkoxycarbonyl)vinyl, and *N*-2,2-Bis(acyl)vinyl Groups

N^{α}-2,2-Bis(ethoxycarbonyl)vinyl-protected amino acids are prepared by reaction of commercially available diethyl 2-(ethoxymethylene)malonate (**127**) with the respective amino acid in methanolic KOH. This rapid reaction is complete within 5 minutes and leads to the potassium salts. Subsequent acidification with 1 M HCl yields the amino acid derivative in 75–90% yield.[738] This intermediate enamine-type N-protection is of particular interest in chemistry to be performed on the carboxy groups of the amino acids such as esterification with alkyl bromides in the presence of a base. Since cleavage of the enamine entity is achieved by treatment with bromine in chloroform at room temperature, it cannot be used for amino acids sensitive to halogenation such as tyrosine, tryptophan, and methionine (Scheme 61). Based on the experience gained with the enamine-type protection the *N*-2-(4,4-dimethyl-2,6-dioxocyclohexylidene)ethyl (Dde) and *N*-2-(4,4-dimethyl-2,6-dioxocyclohexylidene)isovaleryl derivatives were developed as specific side-chain protecting groups (see Section 2.1.2.2.5.2).

Scheme 61 Diethyl Alkylidene Malonates for N^{α}-Protection of Amino Acids



N^{α}-2,2-Bis(ethoxycarbonyl)vinyl Amino Acids; General Procedure:[738]
To a soln of the amino acid (10 mmol) in MeOH (15 mL) containing KOH (670 mg, 12.0 mmol) was added diethyl 2-(ethoxymethylene)malonate (2.16 g, 10.0 mmol). The mixture was left for 5 min at rt, concentrated, and the residue was dissolved in H_2O (10 mL). The soln was brought to pH 3–4 with 1 M HCl, and the separated oil was extracted with EtOAc (3 × 10 mL). The extract was dried and concentrated. For reaction with H-Asp-OH and H-Glu-OH, 24 mmol KOH, and for H-Lys-OH·HCl 24 mmol KOH and 20 mmol diethyl 2-(ethoxymethylene)malonate are required.

N^{α}-2,2-Bis(ethoxycarbonyl)vinyl Amino Acid Esters; General Procedure:[738]
A soln of N^{α}-2,2-bis(ethoxycarbonyl)vinyl amino acid (0.4 mmol) in acetone (5 mL) containing TEA (0.26 mL) was treated with the alkyl bromide (0.4 mmol). The mixture was left for 30 min at rt. The TEA·HBr was filtered off, the filtrate was concentrated, and the residue recrystallized (EtOH); yield: 79–80%.

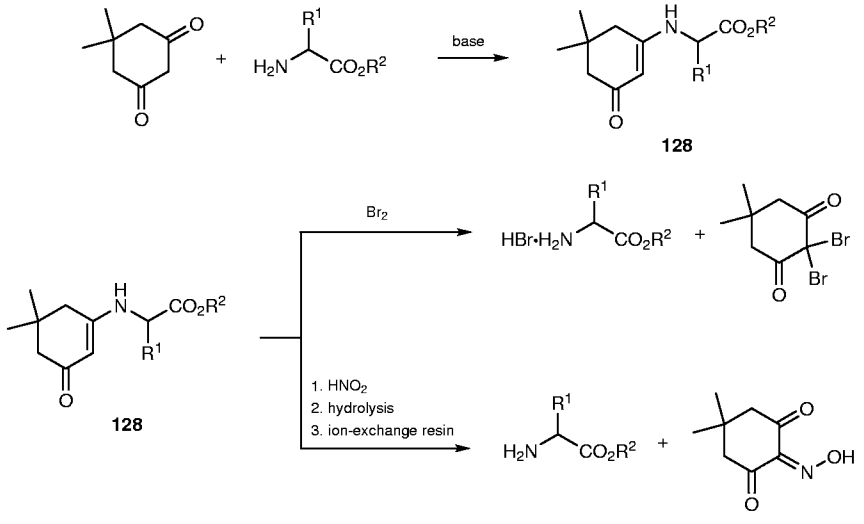
Removal of the 2,2-Bis(ethoxycarbonyl)vinyl Group; General Procedure:[738]
A soln of N^{α}-2,2-bis(ethoxycarbonyl)vinyl amino acid ester (0.3 mmol) in $CHCl_3$ (5 mL) was treated dropwise with a soln of Br_2 in $CHCl_3$ until no further decoloration was observed. The pure hydrobromides crystallized on cooling and could be recrystallized (EtOH); yield: 46–85%.

2.1.1.7.3 Other Enamine-type N-Protections

Condensation of 5,5-dimethylhexan-1,3-dione (dimedone) with amino acid esters leads to N^{α}-5,5-dimethyl-3-oxocyclohex-1-enyl derivatives **128** as particularly stable enamines that are converted into the free acids by saponification, for coupling by carbodiimides, carbonyldiimidazole, or Woodward's reagent K, or are transformed into the hydrazides for coupling via the azide method (Scheme 62).[739–743]

Condensation can be performed at room temperature, leading to optically pure, crystalline peptides which are stable toward strong acids and bases, hydrogenolysis, and Birch conditions (Na/NH_3). Deprotection occurs with bromine, leading to the corresponding amino acid or peptides as hydrobromides. The side product, 2,2-dibrominated dimedone is insoluble and easily separated.[740] Alternatively, cleavage is achieved with nitric acid, which generates an oximo-substituted dimedone imine of the amino acid; the imine is subsequently removed by hydrogenolysis. The cleavage can also be performed in a one-pot reaction and the free amino acid or peptide is isolated by ion-exchange chromatography (>90%). No racemization has been observed, even with racemization-prone amino acids.[742]

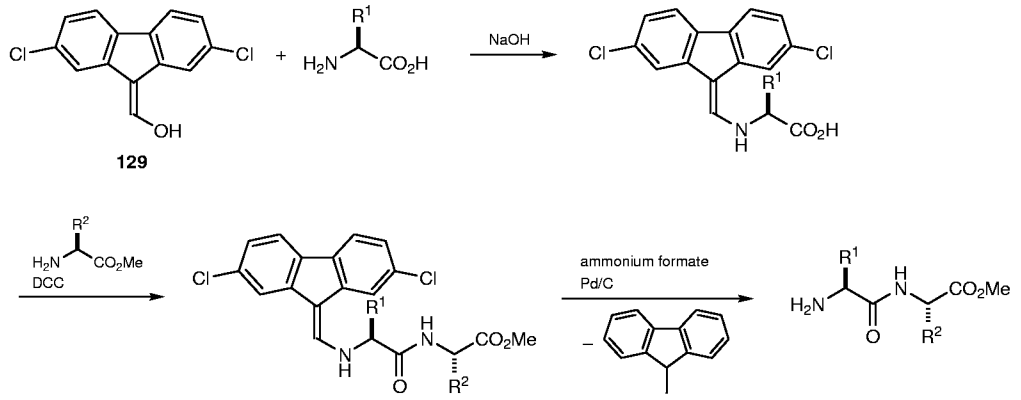
Scheme 62 Dimedone as a N-Protecting Group for Amino Acids[739–743]



for references see p 149

Reaction of 2,7-dichloro-9-(hydroxymethylene)fluorene (**129**) with amino acids and amino acid esters yields the corresponding enamines (Scheme 63), which are sufficiently stable to allow normal peptide coupling reactions, e.g. via DCC.[744] The protecting group is removed by catalytic transfer hydrogenolysis, but also by mild acid hydrolysis. Moreover, this type of protecting group is less prone to cause racemization than a urethane protecting group.

Scheme 63 Synthesis of N^{α}-2,7-Chloro-9-fluorenylidenemethyl Amino Acids and Cleavage of the Protecting Group[744]

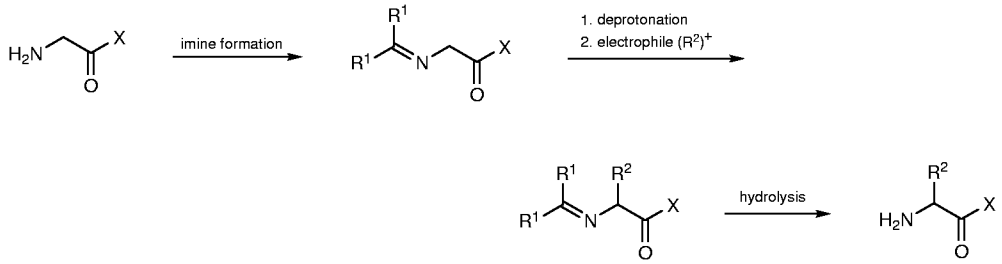


2.1.1.8 *N*-Alkylidene Groups

J. Podlech

The α-amino group of amino acids can be protected as the imine derivatives, i.e. aldimine or ketimine, however this protecting group is too labile for efficient peptide coupling due in particular to facile transimination occurring with the amino component.[1] Although with 2-hydroxy-substituted benzaldimines and naphthaldimines,[745–748] or with 5-chloro-2-hydroxybenzaldimines[746] and 4-chlorobenzaldimines[749,750] the Schiff bases are more stable, this type of protection is not used for peptide synthesis, but has found an interesting application in the selective N^{α}-protection of diamino acids (see Section 2.1.2.2.1.3) and, particularly, in the synthesis of unnatural amino acids (for additional imine derivatives of amino acids see ref[12]). For the latter purpose, the typical reaction sequence includes imine formation of the amino acid derivative, usually of a glycine ester, deprotonation, addition of an electrophile, and hydrolysis of the imine (Scheme 64). The alkylation step can be performed either diastereoselectively using components that bear centers of chirality within the imine or the ester function, or enantioselectively in the presence of chiral additives. A plethora of methods has been proposed and used in this context (for reviews on the subject, see refs[464,751,752]).

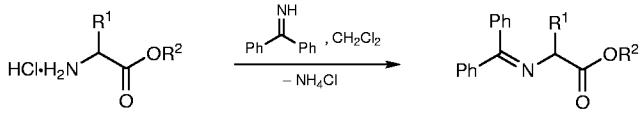
Scheme 64 Synthesis of Unnatural Amino Acids Using *N*-Alkylidene Protection[464,751,752]



2.1.1.8.1 *N*-Benzylidene and *N*-Diphenylmethylene Groups

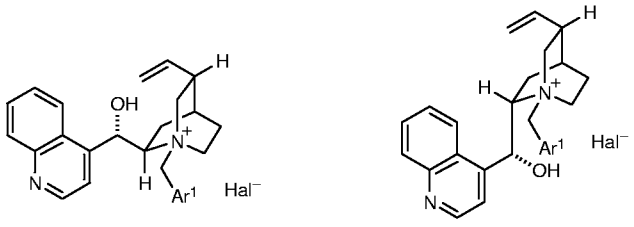
While aromatic aldimines of amino acid esters are readily accessible from benzaldehyde or related derivatives and amino acid esters with concomitant removal of water,[745,753–756] the corresponding ketimines require harsher reaction conditions, e.g. the benzophenone imine of glycine methyl ester is formed in boiling xylene in the presence of boron trifluoride–diethyl ether complex as a Lewis acid in 82% yield.[757] These or similar reaction conditions lead also to piperazine-2,5-dione (DKP) formation; therefore milder reaction conditions have been developed such as the transimination with benzophenone imine in dichloromethane, which occurs at room temperature overnight in 80–95% yield (Scheme 65).[749,758] The N^{α}-diphenylmethylene-protected glycine ethyl or *tert*-butyl esters are commercially available.

Scheme 65 Preparation of N^{α}-Diphenylmethylene-Protected Amino Esters with Benzophenone Imine[758]



N-Alkylidene derivatives of glycine esters are the preferred starting compounds for the synthesis of unnatural amino acids via the deprotonation/alkylation procedure. Enolate formation is achieved with a strong base (LDA, BuLi, or phosphazene bases such as 2-(*tert*-butylimino)-2-(ethylamino)-1,3-dimethyl-1,3,2-diazaphosphinane) or by phase-transfer catalysis (tetraalkylammonium salts with NaOH or KOH),[757,759,760] the latter method allows much milder reaction conditions. When the phase-transfer catalyst is a chiral, enantiopure derivative of a *cinchona* alkaloid, the subsequent alkylation can be performed enantioselectively with up to 70% ee (Scheme 66).[759,760] The use of glycine equivalents bearing a chiral auxiliary can result in significantly enhanced stereoselectivity (*vide infra*).

Scheme 66 Phase-Transfer Catalysts Derived from *cinchona* Alkaloids



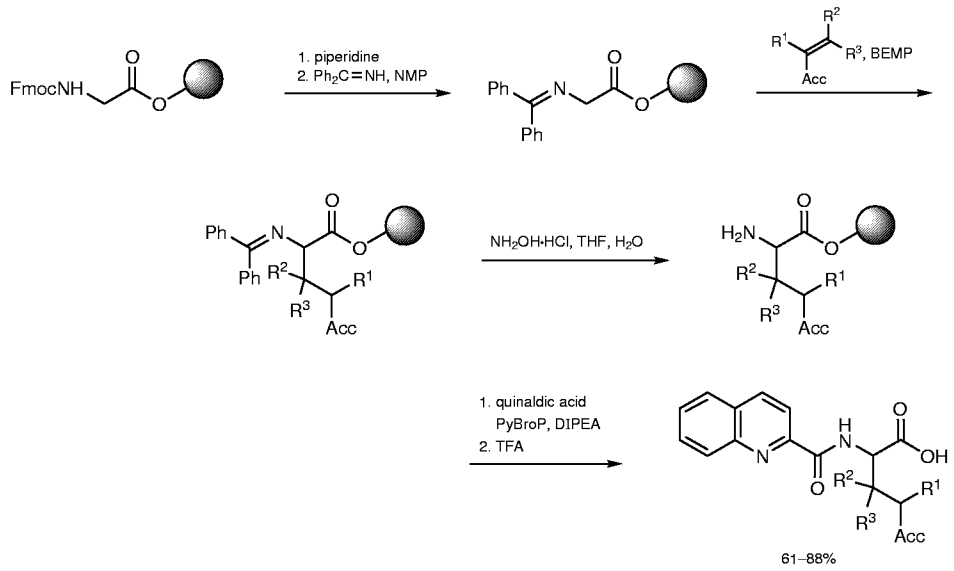
(+)-*N*-benzylcinchonium halide (+)-*N*-benzylcinchonidinium halide

for references see p 149

Cleavage of *N*-benzylidene and *N*-diphenylmethylene derivatives is performed either by acidolysis (1 M HCl) or hydrogenolysis (H_2, Pd/C).

The N^{α}-diphenylmethylene protection has also been used in the solid-phase mode for the synthesis of either unnatural amino acids[761] or peptides (Scheme 67).[762] Thereby, both Merrifield or Wang resins were used and the best base proved to be the organic soluble, nonionic phosphazene bases of Schwesinger, e.g. 2-*tert*-(butylimino)-2-(ethylamino)-1,3-dimethyl-1,3,2-diazaphosphinane (BEMP). As electrophiles alkyl halides,[762] aldehydes, and Michael acceptors[761] have been used.

Scheme 67 Synthesis of Unnatural Peptides[761]

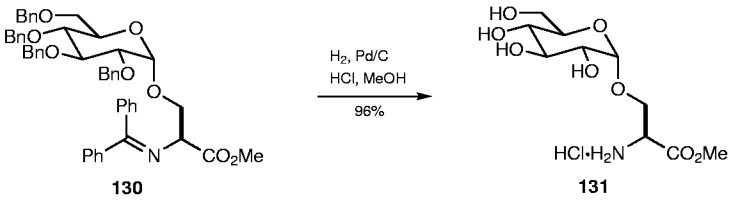


Acc = acceptor, e.g. CO_2R^4, CN

N^{α}-Diphenylmethylene Amino Acid or Dipeptide Esters; General Procedure:[758]
The amino acid or dipeptide methyl (ethyl) ester hydrochloride (5.0 g) and diphenylketimine (1 equiv) were stirred in CH_2Cl_2 (100 mL) at rt for 18–24 h under exclusion of moisture. The mixture was filtered to remove NH_4Cl, and the filtrate was washed with H_2O (4 × 100 mL), brine (50 mL), and dried. The solvent was removed and the resulting solid was recrystallized from an appropriate solvent. For example: Ph_2CH>Gly-Gly-OEt (recrystallized from EtOH); yield: 91%; mp 123–124 °C; Ph_2CH>Gly-D-Phe-OMe (recrystallized from EtOH); yield: 92%; mp 90–92 °C.

N^{α}-4-Chlorobenzylidene Amino Acid Esters; General Procedure:[758]
4-Chlorobenzaldehyde (2.22 g, 15.8 mmol), amino acid ester hydrochloride (1 equiv), and anhyd $MgSO_4$ (1.1 equiv) were mixed in CH_2Cl_2 (50 mL). The soln was cooled to 0 °C with an ice bath and TEA (2 equiv) was added with stirring. The soln was stirred at rt for 18 h under exclusion of moisture, filtered, and concentrated. The residue was taken up in Et_2O, washed quickly with cold 0.1% $NaHCO_3$ (2 × 75 mL), 2 M NaOH containing 2 g · 100 mL^{-1} $NH_2OH \cdot HCl$ (2 × 75 mL), 0.1% $NaHCO_3$ (2 × 75 mL), and brine (2 × 75 mL). The organic phase was dried, concentrated, and used without further purification. For example: 4-ClC_6H_4CH>Phe-Leu-OMe (oil); yield: 90%.

Scheme 68 Cleavage of Diphenylmethylene Derivatives by Hydrogenolysis[763]



130 131

Methyl O^3-(α-D-Glucopyanosyl)serinate Hydrochloride (131) from Methyl *N*-(Diphenylmethylene)-O^3-(2,3,4,6-tetra-*O*-benzyl-α-D-glucopyranosyl)serinate (130); Typical Procedure for the Cleavage of N^α-Diphenylmethylene Derivatives by Hydrogenolysis (Scheme 68):[763]

Methyl *N*-(diphenylmethylene)-O^3-(2,3,4,6-tetra-*O*-benzyl-α-D-glucopyranosyl)serinate (**130**; 25 mg, 31 μmol) was dissolved in MeOH (5 mL), and 36% HCl (2.6 μL) and 5% Pd/C (25 mg) were added. The mixture was stirred under H_2 (balloon) for 2 h, diluted with CH_2Cl_2 (20 mL), filtered, and concentrated to give an amorphous solid; yield: 9.5 mg (96%); $[\alpha]_D^{20}$ +38 (*c* 0.4, MeOH).

H-*rac*-Phe-OEt·HCl; Typical Procedure for the Cleavage of N^α-Diphenylmethylene Derivatives by Acidolysis:[749]

To a stirred soln of Ph_2CH>*rac*-Phe-OEt (1.79 g, 5.00 mmol) in Et_2O (10 mL) was added dropwise 1 M HCl (6.00 mmol, 1.2 equiv) at 0 °C over 30 min, and the mixture was stirred overnight at rt. The layers were separated, and the aqueous phase was concentrated; yield: 1.15 g (100%); mp 122–124 °C.

Synthesis of Unnatural Peptides on Resin; Typical Procedure:[761]

Fmoc-Gly-Wang resin (100 mg, 0.054 mmol, 0.54 mmol·g^{-1}) was suspended in 30% piperidine/NMP (1.5 mL) and mixture was shaken for 30 min. The resin was filtered and washed with NMP (4 × 1.5 mL). Benzophenone imine (0.54 mmol, 10 equiv) in NMP (1.5 mL) was added followed by glacial AcOH (0.54 mmol, 10 equiv) and the suspension was shaken overnight. The resin was filtered and washed with NMP (4 × 1.5 mL).

The resin, BEMP (3 equiv), a Michael acceptor (5 equiv), and NMP (1.5 mL) were shaken for 16 h and the resin was filtered and washed with NMP (4 × 1.5 mL).

The resin was shaken for 5 h in 1 M aq NH_2OH·HCl/THF (3:7, v/v). The resin was filtered and washed with THF (4 × 1.5 mL), NMP (4 × 1.5 mL), and mixed with 1 M DIPEA in NMP (1.5 mL) for 30 min. The resin was filtered and washed with NMP (4 × 1.5 mL).

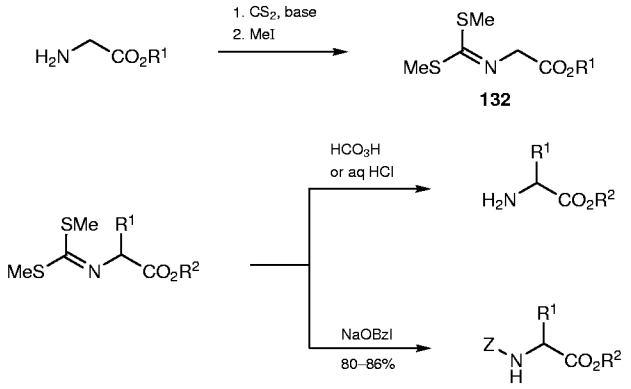
The resin, quinaldic acid (10 equiv), PyBroP (10 equiv), and DIPEA (20 equiv) were shaken overnight. The resin was filtered, washed with NMP, THF, MeOH, and CH_2Cl_2 (4 × 1.5 mL each) and suspended in 95% aq TFA for 3 h. The filtrate was concentrated; yield: 61–88%; purity 74–97% (HPLC).

2.1.1.8.2 *N*-Bis(methylsulfanyl)methylene Group

N^α-Bis(methylsulfanyl)methylene amino acid esters such as **132** are prepared from the amino acid esters hydrochlorides upon deprotonation with a base by reaction with carbon disulfide and iodomethane (Scheme 69).[764] The corresponding glycine methyl ester is commercially available.

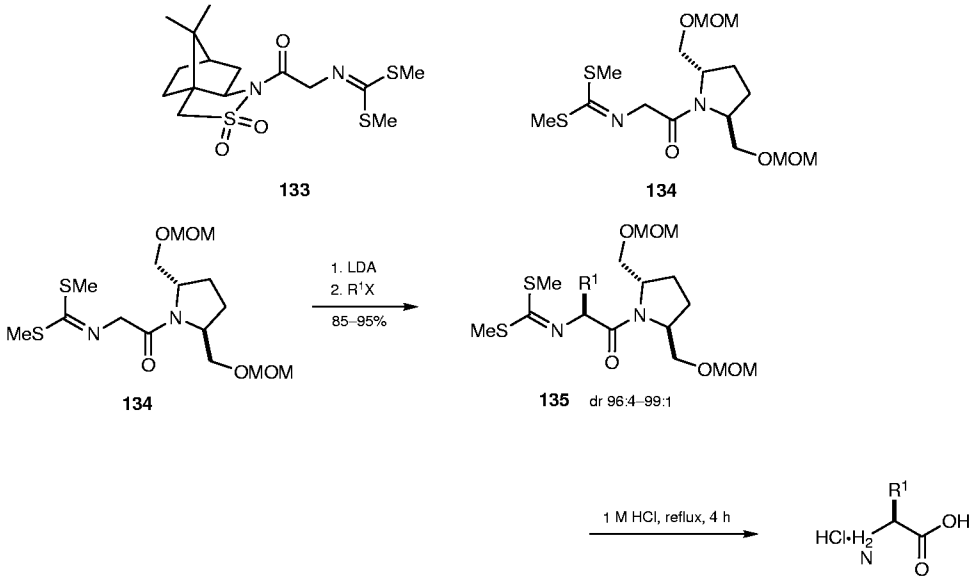
for references see p 149

Scheme 69 Preparation and Cleavage of N^{α}-Bis(methylsulfanyl)methylene Amino Acid Esters[764,766]



Reaction of these N^{α}-protected amino acid derivatives with electrophiles have been shown to produce the desired compounds with superior stereoselectivity and in higher yields compared to the alkylations of N^{α}-benzylidene-protected derivatives.[765] Best results were obtained with the Oppolzer's sultam-derived glycine equivalents **133** (Scheme 70). Diastereomeric ratios were usually better than 98:2, and yields were between 85–96%. Liberation of the unnatural amino acids is quantitative and the products are virtually enantiomerically pure (>99.8% ee).[766] Good results have been further achieved with chiral glycine amides derived from 2,5-bis(methoxymethoxymethyl)pyrrolidines **134** (Scheme 70); yields and selectivities were essentially within the same range.[765]

Scheme 70 Chiral Auxiliaries Used in the Derivatization of Glycine Equivalents and Diastereoselective Alkylation of Chiral N^{α}-Alkylidene Amino Acid Derivatives[765]



Cleavage of the bis(methylsulfanyl)methylene group is achieved either with performic acid (generated in situ from formic acid and hydrogen peroxide)[764] or with 0.5 M aqueous HCl in THF (Scheme 69).[766] A conversion of this protection group into the Z group is possible in a one-pot reaction with the sodium salt of benzyl alcohol in anhyd THF at 30 °C for 6 hours (80–86%).[767]

$(MeS)_2CH$>Gly-OEt; Typical Procedure:[764]

CAUTION: *Carbon disulfide is a colorless, poisonous liquid that is stored in brown bottles, CS_2 forms explosive mixtures with air and is teratogenic. Use in a fume hood.*

To a soln of H-Gly-OEt·HCl (70 g, 0.5 mol) in $CHCl_3$ (500 mL) and CS_2 (40 g) was added dropwise at 20–40 °C TEA (106 g, 1.05 mmol) and stirring was continued for 1 h. MeI (78.1 g, 0.55 mol) was added under gentle reflux of the soln and stirring was continued for 1 h. The soln was extracted with H_2O (2 × 250 mL), the solvent was removed and the residue was dissolved in Et_2O (300 mL), extracted with H_2O (2 × 100 mL), dried, and concentrated. The residue was dissolved in abs acetone (300 mL), MeI (85.2 g, 0.6 mol) and anhyd K_2CO_3 (100 g) was added, and the mixture was heated to reflux for 2 h. The cooled soln was filtered, the filtrate was concentrated, dissolved in Et_2O (300 mL), washed with H_2O (2 × 50 mL), dried, and distilled; yield: 78 g (77%); bp 88–90 °C/0.05 Torr.

Ethyl 2-Amino-2-(4-bromophenyl)butanoate; Typical Procedure for the Cleavage of the Bis(methylsulfanyl)methylene Group:[764]

To a soln of ethyl 2-[[bis(methylsulfanyl)methylene]amino]-2-(4-bromophenyl)butanoate (4.05 g, 10 mmol) in formic acid (98–100%, 15 mL) was added with stirring at 0 °C 30% H_2O_2 (7.4 g, 65 mmol) and TosOH (0.05 equiv). The mixture was kept at 10 °C (ice bath) for 4 h and at rt for 20 h. The solvent was removed at 50 °C by rotatory evaporation, formic acid (50%, 10 mL) was added and again removed. The cooled residue was covered with a layer of Et_2O (30 mL), cold 2 M NaOH (10 mL, 20 mmol) was added and the mixture was stirred until complete dissolution. This was repeated with a second portion of 2 M NaOH (5 mL). The organic layer was washed with a small amount of H_2O, dried, and concentrated to give the product as a colorless oil; yield: 2.7 g (94%).

Conversion of N^{α}-Bis(methylsulfanyl)methylene Amino Acid Derivatives into the Z-Protected Derivatives; General Procedure:[767]

To NaOBzl (260 mg, 2 mmol) in dry THF (10 mL) was added the N^{α}-bis(methylsulfanyl)methylene amino acid derivative (2 mmol) and the mixture was stirred at 30 °C for 6 h. After complete consumption of the starting material (as monitored by TLC), H_2O (0.1 mL) was added to the mixture, which was then stirred overnight at 30 °C. The THF was removed and the aqueous mixture was extracted with EtOAc (25 mL). The organic layer was washed with 1% HCl (2 × 20 mL) and brine (20 mL), dried, concentrated, and chromatographed (silica gel, acetone/petroleum ether 1:9).

1-{*N*-[Bis(methylsulfanyl)methylene]-2-prop-2-ynylglycyl}-2,5-bis(methoxymethoxymethyl)pyrrolidine (135, $R^1 = CH_2C{\equiv}CH$); Typical Procedure for the Diastereoselective Alkylation of Chiral Alkylidene Protected Amino Acid Derivatives:[765]

To 0.631 M LDA in THF (1.05 equiv) was added a soln of 1-{*N*-[bis(methylsulfanyl)methylene]glycyl}-2,5-bis(methoxymethoxymethyl)pyrrolidine (**134**; 48.4 mg, 0.127 mmol) in THF (0.6 mL) at −78 °C and the temperature was raised to −20 °C. After 5 min, the mixture was again cooled to −78 °C, 3-bromoprop-1-yne (1.2 equiv) was added, stirring was continued at −20 °C for 24 h, and 5% H_3PO_4 (150 μL) was added at this temperature. After warming to rt and extraction with CH_2Cl_2, the organic layer was dried, concentrated, and chromatographed (silica gel, hexane/EtOAc) to give the product as an oil; yield: 50.6 mg (95%); dr 99:1; $[\alpha]_D^{26}$ −71.2 (*c* 3.6, MeOH).

2.1.1.9 Silyl Groups

J. Podlech

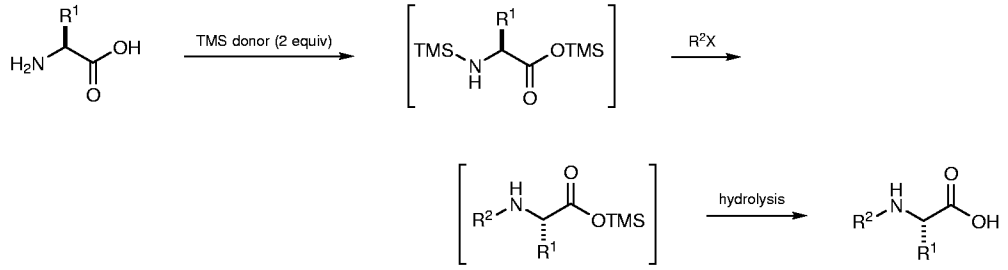
The simple cleavage of virtually all *N*- and *O*-silicon-containing protecting groups with the fluoride anion confers on them a high degree of orthogonality toward various other important protecting groups (see also Sections 2.2.1.7, 2.6.4.1.3, and 2.6.4.2.3 for *O*-silyl derivatives). Although some of these silyl-based protections are not very stable and can only be used under restricted reaction conditions, their particular features may be advantageously exploited for specific synthetic tasks. For amine protection *N*-silyl derivatives are known as well as silyl-containing carbamates and *N*-sulfonyl derivatives such as *N*-2-(trimethylsilyl)-ethoxycarbonyl (TMSEC; see Section 2.1.1.1.1.7), *N*-2-(trimethylsilyl)ethylsulfonyl derivatives (SES, see Section 2.1.1.5.2) and N-*tert*-butyldimethylsilyloxy derivation (Section 2.1.1.1.1.8).

for references see p 149

2.1.1.9.1 *N*-Silyl Groups

N^{α}-Trimethylsilyl (TMS) and N^{α}-*tert*-butyldimethylsilyl (TBDMS) derivatives of amino acids and peptides have not been extensively used for amine protection in peptide chemistry due to their high reactivity to moisture. But as *N*-silyl, *O*-silyl, or *N,O*-bis-silyl amino acid derivatives are easily prepared and highly soluble in organic solvents, they represent interesting intermediates in the synthesis of other suitably protected amino acid derivatives as well as of small peptides since they react rapidly with acylating agents, e.g. chloro- or fluoroformates, as well as with N-protected amino acid mixed anhydrides, halides or active esters (Scheme 71); thereby, racemization is largely suppressed and the silyl entities as transient protections are removed in the workup steps by simple hydrolysis.[768,769] Selective formation of silyl esters of amino acids by the use of 1 equivalent of silylating reagent has been reported;[770] however, quantitative N,O-bis-silylation by excess reagent is preferable, since these derivatives can be directly employed in the coupling reactions (*vide supra*) without intermediate isolation.

Scheme 71 Acylation of Transiently N,O-Silylated Amino Acids or Peptides[768,769]



R^2 = alkyloxycarbonyl, N-protected amino acyl

Silylation with silyl chlorides or silyl triflates is performed in the presence of auxiliary bases, e.g. pyridine, TEA, imidazole, etc., in solvents such as CH_2Cl_2, $CHCl_3$, DMF, or THF.[768,771–773] As milder reagents *N*-methyl-*O*-(trimethylsilyl)acetamide and preferably *N,O*-bis(trimethylsilyl)acetamide (BTMSA, **136**) (Scheme 72) are used for such purposes.[768] Since the acetamide that results from this reaction is difficult to remove from the reaction medium, cyanotrimethylsilane[774] or 1-(trimethylsilyl)imidazole (**137**).[775] may be used as alternatives. For quantitative silylation of amino acids containing additional side-chain functionalities, excess silylating agent is required.

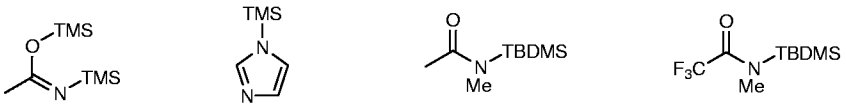
The stability of *O*- but also *N*-silyl derivatives strongly depends on the steric bulk of the silicon substituents. Therefore, the *N*-*tert*-butyldimethylsilyl derivatives are more stable and are accessible in virtually quantitative yields either with *N*-(*tert*-butyldimethylsilyl)-*N*-methylacetamide (**138**) (Scheme 72) or with the corresponding trifluoroacetamide (**139**).[776]

N,O-Bis-silylation of amino acids is used for reactions with acylating agents for solubilizing effects in organic media, e.g. in the synthesis of N^{α},$N^{\omega,\omega'}$-protected arginine derivatives with Z-Cl[777] or with benzyl pentachlorophenyl carbonate[778] (see Section 2.6.1) or when water-sensitive acylating agents are used, e.g. (alkyldisulfanyl)carbonyl chloride.[539] Similarly, N,O-bis-silylation of amino acids can be used advantageously to avoid dipeptide formation in N-acylation with chloroformates[206] and dicarbonates.[779] This type of protection is also used to prevent racemization and increase the yield in peptide couplings[780,781] or to enhance acylation rates of poorly nucleophilic amino components on solid support.[227]

For specific purposes such as the synthesis of unnatural amino acids the carbamate NH of urethane-protected amino acid esters can be further protected with a silyl group, when TMSOTf in the presence of triethylamine is used for the silylation.[782] The resulting N^{α}-Boc-N^{α}-TMS-protected amino acid esters are stable against basic reaction conditions, but are

deprotected with acids. Since the TMS group is removed more easily by controlled conditions, the N^{α}-Boc-protected derivative is isolated.

Scheme 72 Examples of Some Silylating Reagents



136 BTMSA **137** **138** **139**

Protocols for synthesis of Z-Arg(Z_2)-OH or N^{α}-Fmoc amino acids via intermediate N,O-silylation are reported in Sections 2.6.1 and 2.1.1.1.1.3, respectively.

Z-Val-Alg-OH; Typical Procedure for Synthesis via Intermediate N,O-Silylation:[783]
A suspension of allylglycine (Alg; 230 mg, 2.0 mmol) and 1-(trimethylsilyl)imidazole (**137**; 560 mg, 4.0 mmol) in dry DMF (5 mL) was stirred at rt for 30 min. A soln of Z-valine thiopyridine ester (550 mg, 1.6 mmol) in dry DMF (3 mL) was added, and after stirring for 2 h, 1 M HCl (5 mL) was added. Stirring was continued at rt for 30 min, H_2O was added and the mixture was extracted with EtOAc (3 ×). The combined organic layers were washed with H_2O, dried, and concentrated. The crystalline residue was recrystallized ($CHCl_3$/hexane); yield: 467 mg (84%) as white crystals.

Boc-Gly-Glu(tBu)-Phe-Ser(tBu)-Val-Cys(StBu)-Asp(OtBu)-Ser(tBu)-OH; Typical Procedure for the Silylation of the Amino Component in Segment Condensation:[781]
To a suspension of H-Cys(StBu)-Asp(OtBu)-Ser(tBu)-OH · HCl (1.2 g, 2.0 mmol) in THF were added NMM (0.22 mL, 2.0 mmol) and BTMSA (0.98 mL, 4.0 mmol). After 4 h at 70 °C, a clear soln was formed which was cooled to −10 °C and combined with the mixed anhydride prepared from Boc-Gly-Glu(OtBu)-Phe-Ser(tBu)-Val-OH (0.75 g, 1.0 mmol), NMM (0.11 mL, 1.0 mmol) and isobutyl chloroformate (0.13 mL, 1.0 mmol) at −10 °C in THF/DMF (3:1, 20 mL) in 2 min. The mixture was allowed to reach rt in 6 h, and then concentrated. On addition of H_2O, the precipitate was collected, washed with H_2O, and reprecipitated from DMF/Et_2O; yield: 0.85 g (68%); mp 263–265 °C; $[\alpha]_D^{20}$ −48.7 (*c* 0.3, DMF).

H-Phe-[4-(4-aminophenylazo)benzoyl]-Ala-Cys(StBu)-Ala-Thr(tBu)-Cys(StBu)-Asp(OtBu)-Gly-OH; Typical Procedure for SPPS with Silylation on Resin:[227]
To 4-(4-aminophenylazo)benzoyl-Ala-Cys(StBu)-Ala-Thr(tBu)-Cys(StBu)-Asp(OtBu)-Gly-chlorotrityl-resin (loading: 0.56 mmol · g^{-1}) BTMSA (2 equiv) in NMP was added. After 45 min, coupling with Fmoc-Phe-F (2 × 3 equiv for 2 × 45 min) was performed. The resin was washed with NMP and Fmoc cleavage was performed in standard manner with 20% piperidine/NMP. Cleavage from the resin with CH_2Cl_2/AcOH/TFE/Et_3SiH (8:1:1:0.15) (2 × 30 min at rt) was followed by washing with CH_2Cl_2. The solvent was removed and the residue precipitated from TFE with Et_2O; yield: 54%; ESI-MS *m/z*: 1298.4 $[M+H]^+$.

2.1.1.9.2 *N,N*-Bis-Silyl Groups

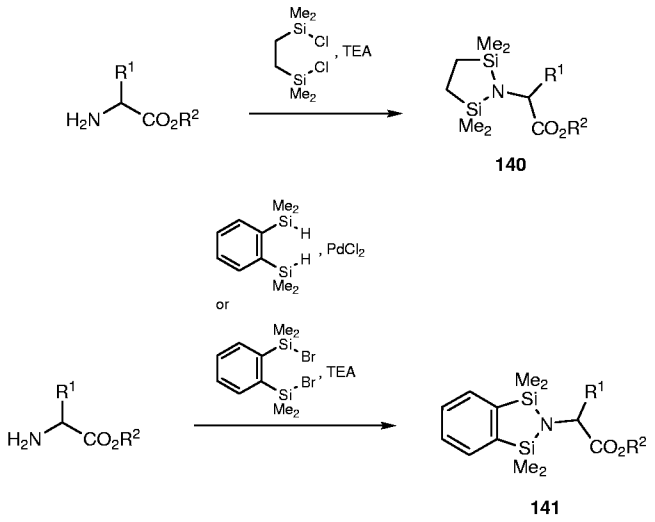
Occasionally double N,N-protection of amino esters is essential and for this purpose *N,N*-bis-silyl-protection is sometimes more advantageous than *N,N'*-bis-Boc protection, since the former is less prone to racemization. Such N^{α},$N^{\alpha'}$-bis(trimethylsilyl) derivatives are accessible only for sterically less demanding amino acids; e.g. H-Gly-OMe is converted into the *N,N',O*-tris(trimethylsilyl) derivative by reaction with TMSOTf with concomitant cleavage of the methyl ester in 91% yield; similarly, H-Ala-OMe is transformed into the N,N',O-tris-protected derivative only in 61% yield with 27% N,O-derivative as contaminant. For entropic reasons, cyclic protection using bis-silyl compounds is much more effective (Scheme 73). In fact, the 2,2,5,5-tetramethyl-1-aza-2,5-disilacyclopentane adducts (STABASE adduct) **140** are readily obtained with commercially available 1,2-bis(chlorodimethylsilyl)ethane and triethylamine. Even sterically hindered amino acid derivatives such as valine or leucine esters are protected in 42–92% yield. These amino acid derivatives are stable against butyllithium at −25 °C, lithium diisopropylamide, saturated ammonium chloride, water, methanol, pyridinium dichromate, and even against potassium fluoride dihydrate in THF/

for references see p 149

H_2O. On the other hand, cleavage occurs with 0.1 M hydrochloric acid, 1 M potassium hydroxide, 75% acetic acid, 0.1 M acetic acid, sodium borohydride in ethanol, or pyridinium chlorochromate.[784]

The 1,1,3,3-tetramethyl-1,3-disilaisoindoline adducts **141** (Scheme 73) are superior, since the rigidity of the protecting reagent results in easier derivatization and thus in higher yields.[785,786] Introduction of the protection group is either achieved with 1,2-bis(dimethylsilyl)benzene/$PdCl_2$ or with 1,2-bis(bromodimethylsilyl)benzene/TEA. Cleavage is achieved under weakly acidic conditions, with TMSI/TEA, or with cesium fluoride. Saponification of the ester is possible, but peptide couplings have not as yet been performed.

Scheme 73 STABASE and 1,1,3,3-Tetramethyl-1,3-disilaisoindoline Adducts of Amino Acid Esters[784–788]



1,2-Bis(dimethylsilyl)benzene:[785]
A soln of 1,2-dibromobenzene (23.6 g, 0.10 mol) in anhyd THF (100 mL) was added over 1 h to a stirred mixture of Mg (5.3 g, 0.218 mol) and chlorodimethylsilane (23.5 mL, 19.9 g, 0.210 mol) in anhyd THF (15 mL) and the mixture was refluxed for 1 h and cooled to rt. A precipitate was formed, which was collected and triturated with a large amount of hexane, filtered, concentrated, and distilled; yield: 14.6 g (75%); bp 75–80 °C/8 Torr.

1,2-Bis(bromodimethylsilyl)benzene:[785]
A soln of 1,2-bis(dimethylsilyl)benzene (15.0 g, 77 mmol) in CCl_4 (90 mL) was added over 45 min to a cooled (0 °C) soln of Br_2 (9 mL, 28.0 g) in CCl_4 (90 mL) under N_2. N_2 was bubbled through the soln for an additional 30 min before removing the solvent at rt. The residue was distilled; yield: 21.7 g (80%); bp 160–170 °C/15 Torr.

STABASE and 1,1,3,3-Tetramethyl-1,3-disilaisoindoline Adducts of Amino Acid Esters; General Procedure:[785]
1,2-Bis(chlorodimethylsilyl)ethane or 1,2-bis(bromodimethylsilyl)benzene (10.5 mmol) and TEA (6 equiv, 7 mL) were added to a stirred suspension of the amino acid ester hydrochloride (8 mmol) in an appropriate solvent [40 mL; CH_2Cl_2 (STABASE: Gly, Ala), benzene (STABASE: Leu, Val; 1,1,3,3-tetramethyl-1,3-disilaisoindoline: Ala, Leu, Val, Phe)] under N_2. The mixture was stirred [rt, 2 h (STABASE: Gly), rt, 24 h (STABASE: Ala), 80 °C, 2 h (STABASE: Leu, Val; 1,1,3,3-tetramethyl-1,3-disilaisoindoline: Ala, Leu, Val, Phe)] and diluted with Et_2O (100 mL) and the precipitate was removed by filtration. The filtrate was concentrated, affording an oil, which was purified by distillation or chromatography. For example STABASE of Ala-OMe; yield: 80% (after chromatography); 1,1,3,3-tetramethyl-1,3-disilaisoindoline of Ala-OMe; yield: 90%; bp 120–125 °C/10–15 Torr; 1,1,3,3-tetramethyl-1,3-disilaisoindoline of Val-OMe; yield: 62%; bp 120–122 °C/10–15 Torr.

2.1.1.10 Precursors of Amino Groups

E. Lohof, H.-J. Musiol, and J. Podlech

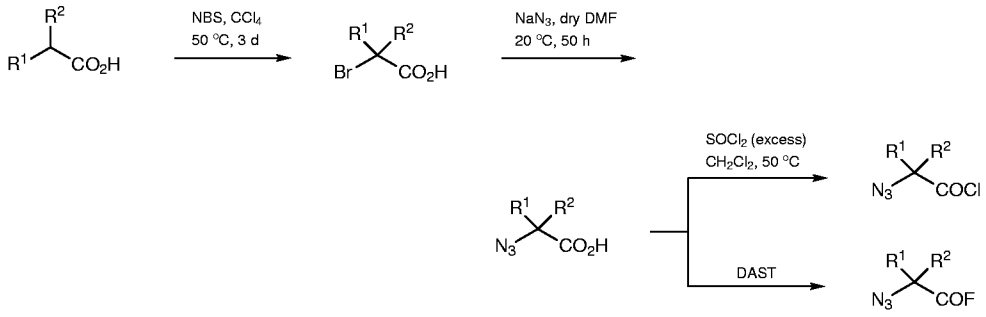
Due to the lack of efficient protecting groups and the difficult accessibility of natural amino acids, the use of amino acid precursors for the synthesis of peptides was thoroughly investigated during the earliest periods of peptide chemistry. For this purpose α-halo acids,[787] α-azido acids,[788–790] and α-oxo acids[791,792] were used (for a comprehensive review of the older literature on this subject, see ref[793]).

With a surge of interest focused on peptidomimetic structures, new methods are needed to prepare new oligomeric backbones such as carbamates, peptoids, ureas, sulfonamides, azatides, as well as a range of nonnatural amino acids that are to be efficiently incorporated into peptidic structures. In this context, the α-halo acids as precursors of amino acids are currently used, in particular, for the synthesis of peptoids (see Section 10.1.2).[794] Since several enantiocontrolled routes to α-azido acid derivatives are available[795–804] and their conversion into amino acids occurs using mild procedures,[805] α-azido acids have also experienced a revival. In fact, α-azido acids have the advantage that they have a small (in terms of size) and sterically nondemanding 'pseudo-protection' of the amino group that additionally permits activation of the carboxy group via acyl halides by preventing oxazolone formation that is known to occur during slow acylation reactions; these are all advantages that help circumvent the problems encountered in difficult coupling reactions.[806]

2.1.1.10.1 Synthesis of α-Azido Acids

Racemic or achiral α-azido acids are synthesized by direct azide substitution on commercially available α-bromo carboxylic acids or by radical bromination of carboxylic acids followed by azide substitution. In general, azido acids are stored in the dark to avoid photolytic degradation with loss of nitrogen; temperatures above 50 °C should be avoided. Radical α-bromination of α-branched carboxylic acids as required for the synthesis of α,α-dialkyl or α,α-diaryl amino acids is performed with *N*-bromosuccinimide. This is followed by nucleophilic substitution with sodium azide or other azide donors, e.g. tetrabutylammonium azide, to produce achiral or racemic α-azido-α,α-dialkyl or α-azido-α,α-diaryl carboxylic acids (Scheme 74).[464] Synthesis of more sterically hindered α,α-disubstituted azido acids leads to hydroxy compounds when prolonged reaction times are required and not sufficient care is taken to operate under dry conditions and an inert atmosphere.[803,807]

Scheme 74 Synthesis of Activated Azido Acids[464]

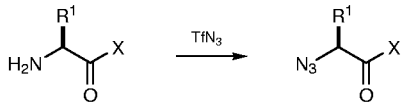


Optically active α-azido acids as versatile α-amino acid synthons are obtained by diastereoselective halogenation of chiral auxiliary-based enols or enolates[797,799] followed by stereospecific azide displacement by *N,N,N′,N′*-tetramethylguanidinium azide or alternatively

for references see p 149

by direct reaction of the chiral auxiliary-based enolates with 2,4,6-triisopropylbenzenesulfonyl azide.[799] Conversely, α-azido acids related to natural amino acids are readily synthesized in good yields and with retention of the chirality at the α-carbon by direct diazo transfer from trifluoromethanesulfonyl azide (triflyl azide, TfN_3)[808,809] to the amino acids in a two-phase reaction, where yields are only slightly improved by the use of a phase-transfer catalyst (Scheme 75).[804] If amino acid esters are used for the reaction this is best performed in organic solvents using methyltrioctylammonium chloride as the base catalyst, while the two-phase reaction in the presence of the phase-transfer agent results in substantial racemization. Addition of copper sulfate as a catalyst has been recommended.[803] Due to the basic conditions of the diazo-transfer reaction, even side-chain protected amino acids containing *tert*-butyl alcohol-derived acid-labile groups are smoothly converted into the related α-azido acids. The trifluoromethanesulfonamide byproduct can be removed by complexation and precipitation with mercury(II) acetate,[810] except in the case of tryptophan and asparagine, or by chromatographic methods.

Scheme 75 Synthesis of Azido Acids by the Diazo-Transfer Procedure[804]



X = OH, OR^2, peptide

CAUTION: *Explosions may be expected to result in situations involving neat triflyl azide,[808] and, although problems have been rarely experienced while handling CH_2Cl_2 solutions of the azide, as fundamental safety precautions the solutions should not be allowed to evaporate under any circumstances and they should not be stored for any length of time. When acidifying solutions containing NaN_3, toxic HN_3 will be evolved and, therefore, it is vital that care is taken to ensure good ventilation.*

2-Azido-2-methylpropanoic Acid; α-Azido Acids from α-Bromo Acids and NaN_3:[803]
2-Bromo-2-methylpropanoic acid (4.96 g, 29.7 mmol) and NaN_3 (2.88 g, 44.3 mmol) were mixed in dry DMF (50 mL) and stirred under argon for 2 d. The mixture was concentrated, redissolved in H_2O (30 mL) and acidified to pH 2 with 3 M HCl. The aqueous phase was extracted with $CHCl_3$ (3 × 30 mL) and the organic phase was dried, concentrated, and purified by vacuum liquid chromatography (VLC[811]) (AcOH/EtOAc/heptane 5:20:75); yield: 2.68 g (70%).

***rac*-2-Azido-2-methylbutanoic Acid; α-Azido Acids from α-Branched Carboxylic Acids:**[803]
rac-2-Methylbutanoic acid (10.4 g, 0.102 mol) and NBS (19.0 g, 0.107 mol) were refluxed in CCl_4 (100 mL) under argon. The red color from bromine disappeared overnight and the solid succinimide was filtered off. The filtrate was concentrated affording a yellow oil, which was distilled under reduced pressure and the α-bromo acid was collected at 53 °C/0.2 Torr; yield: 14.4 g (78%).
rac-2-Bromo-2-methylbutanoic acid (13.7 g, 75.7 mmol) and NaN_3 (7.38 g, 114 mmol) in dry DMF (35 mL) were stirred under argon for 3 d. The soln was concentrated, redissolved in H_2O (40 mL) and acidified to pH 2.0 with 3 M HCl. It was extracted with $CHCl_3$ (3 × 50 mL), the combined organic phases were dried, concentrated, and purified by vacuum liquid chromatography (AcOH/EtOAc/heptane 5:10:85); yield: 9.86 g (91%).

Triflyl Azide:[808]
To a magnetically stirred soln of NaN_3 (8 g, 0.123 mol) in H_2O (20 mL) over CH_2Cl_2 (25 mL) at 0 °C was added Tf_2O (7 g, 25 mmol). The low temperature and stirring were continued for 2 h. The organic layer was washed once with 1 M NaOH (1 equiv) to remove all traces of acid. The yield was taken to be 50% for the purposes of determining the amounts of triflyl azide used in subsequent reactions. (Base titration of TfOH formed during the reaction indicated that about 50% of the anhydride had hydrolyzed during the reaction.)

α-Azido Acids with Triflyl Azide; General Procedure:[804]
To a soln of amino acid (0.012 mol) in sufficient 1 M NaOH to adjust the pH to 9.5, triflyl azide in CH_2Cl_2 (ca. 1 M, 1 equiv) was added and the mixture was stirred at rt, with occasional addition of 1 M NaOH to maintain pH 9–9.5. After overnight stirring, the phases were separated, and the organic phase

was discarded. The aqueous phase was treated with 1 M aq $Hg(OAc)_2$ (1 equiv), and the resulting white precipitate (a mercuric salt of triflylamide) was filtered off and washed with H_2O (50 mL). The pH of the combined filtrates was adjusted to 3.5 with 1 M H_2SO_4 and the mixture was extracted with Et_2O (4 × 25 mL). The combined organic extracts were dried (Na_2SO_4), filtered, and concentrated; when oils were obtained the α-azido acids were isolated as DCHA salts from the soln in Et_2O; yield: 20–60%.

Ethyl 2-Azidopropanoate; Typical Procedure:[804]
To H-Ala-OEt (5.73 g, 49 mmol) in CH_2Cl_2 (10 mL), triflyl azide in CH_2Cl_2 (ca. 1 M, 1 equiv) was added along with a catalytic amount (ca. 0.1 mmol) of methyltrioctylammonium chloride, and the soln was stirred at rt for 10–15 h. The mixture was washed with 0.25 M citric acid (2 × 30 mL), the CH_2Cl_2 layer was dried (Na_2SO_4) and concentrated and the azido ester distilled at reduced pressure; yield: 4.32 g (62%).

(*S*)-2-Azido-4-(methylsulfanyl)butanoic Acid:[803]
A soln of H-Met-OH (0.886 g, 5.94 mmol) and $CuSO_4 \cdot 5H_2O$ (32.4 mg, 0.13 mmol) in H_2O (18 mL) was adjusted to pH 9–10 with K_2CO_3. MeOH (36 mL) and crude triflyl azide (11.9 mmol) in CH_2Cl_2 (30 mL) were added, the pH adjusted again to 9–10 with K_2CO_3 and the reaction was stirred vigorously at rt for 20 h. The two phases were separated by addition of CH_2Cl_2, the organic phase was washed with H_2O (2 × 40 mL) and the combined aqueous phases were acidified with 3 M HCl to pH 2. The aqueous phase was extracted with CH_2Cl_2 (4 × 50 mL). The combined extracts were dried, concentrated to a small volume, and purified by vacuum liquid chromatography (AcOH/EtOAc/heptane 5:20:75); yield: 0.96 g (92%); $[\alpha]_D^{25}$ −99 (*c* 1, $CHCl_3$).

2.1.1.10.2 Synthesis of Peptides with α-Azido Acids

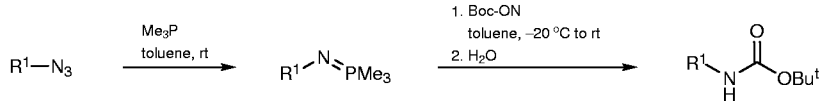
The azido acids are coupled by the common procedures used in solution or solid-phase peptide synthesis, but can also be converted into the highly activated acid chlorides or fluorides in situ and then used in the synthesis of sterically hindered peptides containing, for example, up to four successive diphenylglycine or Aib residues.[806] Acid chlorides are prepared in dichloromethane with thionyl chloride over 1–2.5 hours when acid-sensitive protecting groups are absent (Scheme 74),[802,803] and the related acid fluorides are prepared using DAST (see Section 3.3).[812] Coupling of the chlorides and fluorides proceeds without racemization. Similarly, in situ activation with TBTU/4-ethylmorpholine is used for the synthesis of peptides containing less sterically demanding acylation reactions.[802]

Conversion of the azide into amino derivatives occurs by hydrogenolysis over palladium catalysts or by the Staudinger reaction, i.e. treatment of the aliphatic azides with phosphines to give phosphazenes, followed by hydrolysis to the amine (for reviews on the Staudinger reaction and phosphazene chemistry, see refs[813,814]). Correspondingly, there are numerous procedures for the reduction of azides to amines,[815] and it remains a matter of convenience and/or practicality that the best method should be used for the synthetic strategy employed. The classical, heterogeneous reduction of azides with Pd/C is performed in solution. Thereby, one-pot procedures can be applied for in situ transformation of the α-azido acids or α-azidoacyl peptides into protected α-amino derivatives, e.g. N^{α}-Boc-protected compounds when hydrogenation is carried out in the presence of Boc_2O over Pd/C[463,816] (see Section 2.1.1.1.3.1) or over Rh/C catalyst.[817] Reduction can also be performed with Fe/NH_4Cl under sonication in the presence of the dicarbonate or even chloroformates to produce all possible carbamates.[818] These reductive procedures may not always proceed in a satisfactory manner for sterically hindered derivatives. Similarly, the Staudinger reduction of azides with trialkyl- or triarylphosphines permits one-pot conversion of azides into carbamates. Among the numerous procedures utilized for this purpose, the reduction with trimethylphosphine and subsequent reaction with 2-(*tert*-butoxycarbonyloxyimino)-2-phenylacetonitrile (Boc-ON) (Scheme 76)[819] appears to be superior to other variants such as the use of Bu_3P/Boc_2O,[820] at least in terms of yields. By this procedure the azido compound can also be directly transformed in other carbamates by reaction with chloroformates to produce, for example, *N*-Z-, *N*-Aloc-, and *N*-Fmoc-protected amino acids or aminoacyl peptides.[821] The phosphazenes

for references see p 149

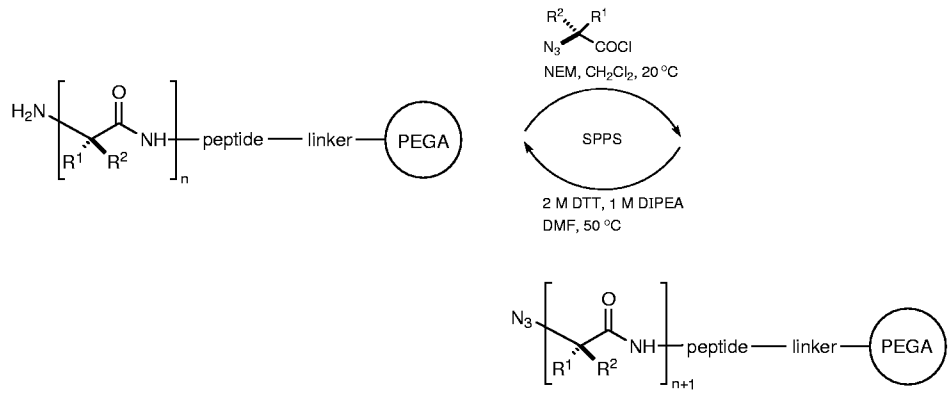
that are produced in the Staudinger reduction as intermediates, are readily acylated with suitably activated acids in solution[822,840] or on solid support.[823]

Scheme 76 Preparation of In Situ Boc-Protected Amino Acids[819]



Since azides are smoothly reduced on solid supports with $SnCl_2$/benzenethiol/TEA (5 mmol/25 mmol/25 mmol for 1 mmol resin-bound azide; rt, 2 h),[824] this alternative procedure has been exploited for the synthesis of oligoureas on resin using α-azido acids.[825] Similarly, azides are efficiently reduced to amines with dithiothreitol (DTT),[826–828] a procedure that was successfully transferred to the reduction of α-azidoacyl peptides on resin for the SPPS with α-azido acids.[802] In the case of sterically hindered azides, addition of small amounts of 2-sulfanylethanol to the DTT/DIPEA mixture or the use of DBU as a base enhances the reaction rates (Scheme 77).[829] Other reductions of azides include the use of H_2S/pyridine/water[830,831] or $NaBH_4/CuSO_4$.[832]

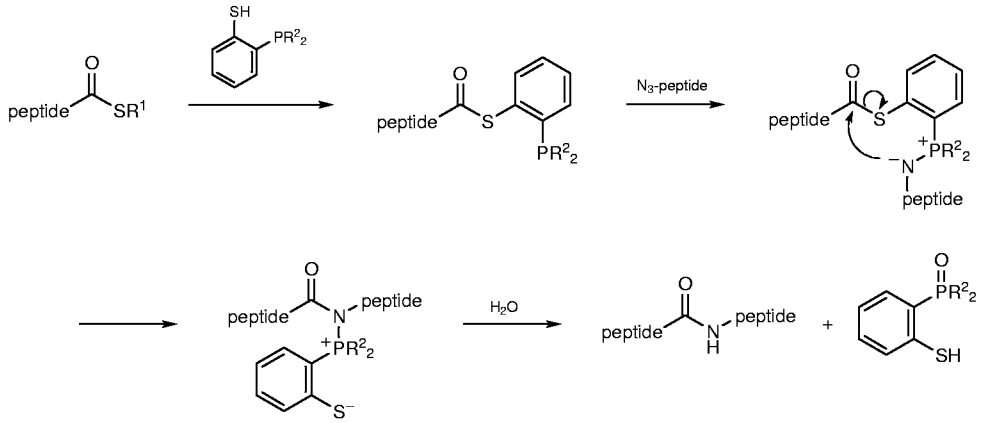
Scheme 77 Solid-Phase Synthesis of Peptides Using α-Azido Acids[802]



An alternative interesting approach, which is based on the Mukaiyama reduction/oxidation procedure for peptide bond formation[633,833] uses tributylphosphine in the presence of diphenyl diselenide and the N-protected amino acid for in situ condensation with the α-azido compound. The phenyl carboselenolate is formed as an activated carboxylic acid and the resulting benzeneselenol reduces the azide to the amine for reaction with the active ester.[834]

A promising new chemoselective ligation procedure has also been developed that utilizes α-azidoacyl peptides that are acylated by peptide thioesters under the Staudinger reduction conditions using 2-phosphanylbenzenethiol for transesterification of the thioester, trapping of the azido component as the phosphazene followed by intramolecular O→N acyl shift (Scheme 78).[835,836,841]

Scheme 78 Staudinger Ligation Starting with Peptide-Derived Thioesters and Azides[836]



Transformation of α-Azido Compounds into Boc-Protected Amino Compounds; General Procedure:[819]
To a soln of azido compound (1 mmol) in toluene (3 mL) under argon at rt, was added 1 M Me_3P in toluene (1.05 mL, 1.05 mmol). After 1 h, the mixture was cooled to −20 °C and Boc-ON (259 mg, 1.05 mmol) in toluene (1 mL) was added under exclusion of moisture. After stirring at rt for 5 h, CH_2Cl_2 or Et_2O (50–100 mL) was added and the soln was extracted with H_2O (3 ×) and with sat. aq NaCl; the organic layers were combined and dried, concentrated, and filtered through a small pad of alumina (CH_2Cl_2/MeOH 99:1). Alternatively, washing of the above-mentioned CH_2Cl_2 or Et_2O soln with 2 M NaOH, drying of the organic layer, removal of the solvent, and purification of the crude product through a short column of silica gel (CH_2Cl_2/MeOH) gave identical results; yield: 90–100%.

Transformation of Azido Compounds into Other Carbamates; General Procedure:[821]
A 1 M soln of Me_3P in THF (1 mL) was added at rt to the azide (1 mmol) in THF (3 mL); when the phosphazene formation (1–2 h, TLC) was complete Z-Cl (or other chloroformates) (1.1–1.2 mmol) was added. A precipitate was formed; stirring was continued for 20–60 min and the reaction was quenched by addition of phosphate buffer (pH 7). The carbamates were extracted with CH_2Cl_2, the organic layers washed, dried, and evaporated; yield: 80–100%.

Bz-Leu-Gly-OEt via Diphenyl Diselenide and Tributylphosphine:[834]
Bu_3P (0.77 mL, 3.15 mmol) was added to a stirred soln of Bz-Leu-OH (744 mg, 3.2 mmol), PhSeSePh (940 mg, 3.0 mmol) in CH_2Cl_2/DMF (9 mL) at rt under argon. After 1–1.5 h at rt, the mixture was cooled in a ice bath and ethyl azidoacetate (390 mg, 3.0 mmol) was added slowly. A rapid evolution of N_2 was observed. The mixture was allowed to reach rt and stirred overnight. The mixture was diluted with EtOAc (50 mL) and washed with $NaHCO_3$ soln and brine, dried (Na_2SO_4), and concentrated. The residue was triturated with petroleum ether/Et_2O (5:1); yield: 93%.

Reduction of α-Azidoacyl Peptides on Resin:[803]
Resin-bound α-azidoacylpeptide was reduced with 1 M DTT and 0.5 M DIPEA in DMF at 50 °C for 2 h. The resin was washed and further coupling steps were performed by standard procedures.

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2.1.2 ω-Amino Group

J. Podlech, H.-J. Musiol, E. Lohof, and L. Moroder

The chemistry elaborated for selective N^{α},N^{ω}-protection of lysine as the only proteinogenic diamino acid and of ornithine, frequently found in microbial peptides, can largely be extended to lower homologues such as diaminobutanoic acid and diaminopropanoic acid and also to higher homologues. The ω-amino group of these diamino acids is more nucleophilic than the α-amino group thus it competes in all the various types of reaction that are involved in amide bond formation; correspondingly, protection of the ω-amino group is mandatory in peptide synthesis if branching of the growing peptide chain is to be avoided. In principle, all the groups proposed and used for α-amino protection are also suitable for ω-protection. Additionally, since no stereogenic center is adjacent to the ω-amino function, it is possible that *N*-acyl-type protection can be used, while it is difficult to control the racemization that occurs for N^{α}-acyl derivatives during the time that it takes to activate the carboxy group and form the amide bond. Depending upon the sequence position of the diamino acids and upon the synthetic targets the N^{α},N^{ω}-protection can be symmetric (i.e., identical) or asymmetric (i.e., reciprocally selective) if the peptide-chain extension is to occur selectively at the α- or ω-amino group.

Within this section only protecting groups that still find considerable use for N^{ω}-protection are included. Further information is available on protecting groups that have been superseded by other more efficient protecting groups, such as the various sulfonyl (e.g., Tos) or acyl groups (e.g., acetyl, phthaloyl), and on the analytical data of a large variety of N^{ω}-mono- and N^{α},N^{ω}-diprotected diamino acids.[1,2] Obviously, the stability properties and deprotection procedures of the N^{ω}-protected derivatives are identical to those of the related N^{α}-amino acid derivatives. For related information on N^{α}-protection, see Section 2.1.1; protecting groups that are used almost exclusively for N^{ω}-protection are described in more detail in this section.

2.1.2.1 Symmetric N^{α},N^{ω}-Protection of Diamino Acids

Independent of the general protection scheme used for synthesis in solution or on solid support, N-terminal lysine or ornithine residues are generally N^{α},N^{ω}-derivatized with identical protecting groups that are compatible with the overall semipermanent protection applied, unless selective N^{α},N^{ω}-protection is necessary for a specific purpose, e.g. to exploit particular physical properties of the protecting groups for affinity chromatography.[3–6] The synthesis of symmetrically protected lysine or ornithine such as the N^{α},N^{ω}-bis(benzyloxycarbonyl), N^{α},N^{ω}-bis(*tert*-butoxycarbonyl) or N^{α},N^{ω}-bis(9-fluorenylmethoxycarbonyl) derivatives is straightforward, since identical procedures are applied to those described for the preparation N^{α}-protected amino acids. Recently, N^{α},N^{ω}-symmetrically protected diamino acids have found extensive use in the synthesis of polyfunctionalized dendrimers for the multiple presentation of peptides to molecular recognition, e.g. as antigens,[7–9] or biochemical and biophysical probes for the purpose of signal amplification.[10] For such syntheses, generally Boc-Lys(Boc)-OH[7] or Fmoc-Lys(Fmoc)-OH[11] is used to build up the inner levels of the dendrimers, whereas the outer level may also be charged with Fmoc-Lys(Boc)-OH to allow, at least to some degree, selective chemistry to be applied for grafting different molecular probes to the surface of the dendrimers.[8] Lysine residues are preferred over shorter homologues to reduce the density on the higher levels of the dendrimers. For details on the synthesis of dendrimers, see Vol. E 22d, Section 14.1.

Fmoc-Lys(Fmoc)-OH:[11]

To an ice-cold soln of H-Lys-OH·HCl (0.91 g, 5 mmol) in H_2O (40 mL) containing $NaHCO_3$ (1.68 g, 20 mmol) and Na_2CO_3 (1.06 g, 10 mmol) was added dropwise Fmoc-OSu (3.37 g, 10 mmol) in dioxane

(40 mL) over 1 h. After 4 h stirring at rt, the soln was poured into ice water (150 mL) containing concd HCl (4 mL). The precipitate was collected, washed with H_2O (50 mL) and dissolved in CH_2Cl_2 (100 mL). The organic layer was washed with 0.1 M $KHSO_4$, dried ($MgSO_4$), and taken to dryness. The solid was recrystallized (MeOH); yield: 1.7 g (57%); mp 118–120 °C.

2.1.2.2 Asymmetric N^{α},N^{ω}-Protection of Diamino Acids

In peptides and proteins derived from ribosomal synthesis the amino acid residues are exclusively linked by α-amide bonds, a fact that has dictated the search and development of selective protection schemes based on side-chain blocking groups that are resistant to all the repetitive steps involved in peptide-chain assembly, and this contrasts with the temporary N^{α}-protection, which must be selectively removed after each coupling step. The main protection strategies that have been developed and have now matured over the last few decades for the synthesis of peptides and even proteins in solution and on solid support are summarized in Table 1. Although strategy 1 (Z/tBu, Table 1) can be considered as strictly orthogonal in terms of the chemistry applied in the deprotection steps,[12,13] its disadvantages are that it is limited to synthesis in solution, it does not tolerate the presence of sulfur-containing residues (catalyst poisoning), and larger peptides are poorly soluble in solvents that allow for efficient hydrogenation. These latter two drawbacks are efficiently bypassed by strategy 2, which in terms of chemistry cannot be considered as strictly orthogonal since this is only based on the highly differentiated rates of acidolysis of the N^{α}- versus N^{ω}-deprotection.[14,15] The strategies 3, 4, and 5 are based on a similar concept. While the Bpoc/tBu strategy 3[16,17] and the Z(OMe)/Bzl strategy 4[18–20] have found elegant application mainly in solution syntheses, the Boc/Bzl strategy 5, in addition to representing in its original form the first protection scheme applied to solid-phase peptide synthesis,[21,22] has developed into a state-of-the-art form that permits efficient synthesis of both large peptides and even proteins in solution[23] (see Vol. E 22b, Section 5.2) and on solid supports[24] (see Vol. E 22b, Section 5.3). The orthogonal Msc/tBu protection scheme proposed by Tesser et al.[25,26] has the disadvantage that the relatively strong alkaline Tesser base (dioxane/MeOH/4 M NaOH, 7.5:2.25:0.25) is required for efficient N^{α}-deprotection. Thus, upon the introduction of the Fmoc group,[27] this protection strategy was completely replaced by the Fmoc/tBu chemistry (strategy 6)[28,29] as an orthogonal system that has proved to be optimal for synthesis in the solid-phase mode, and at present it represents a valid alternative to the Boc/Bzl chemistry (strategy 5).

With the recently developed convergent strategies for the synthesis of large peptides and proteins an additional level of selectivity is required for the resin linker.[30,31] Similarly, with the increasing complexity of synthetic targets, even for N^{ω}-protection additional levels of orthogonality are needed to permit chemical manipulations of assembled peptide chains on resin or in solution, upon suitable cleavage from the resin. While for synthesis in solution more versatile combinations of protecting groups can be selected, in SPPS the choice from the pool of available protecting groups is more limited. In the context of the Boc/Bzl chemistry (strategy 5) an additional level of selectivity can be obtained for the N^{ω}-protection, e.g. with N^{ω}-trifluoroacetyl groups[32,33] that are removed by separate treatment with piperidine solutions,[34,35] or with N^{ω}-Fmoc groups that are cleaved both on resin or in solution by the well-established piperidine procedure.[36,37] These amino protections are fully stable to the intermediate acidolytic N^{α}-deprotection steps with trifluoroacetic acid as well as to the final hydrogen fluoride treatment. Compatible with both the Boc/Bzl and the Fmoc/tBu chemistry are the N^{ω}-Aloc derivatives which are cleaved by suitable nucleophiles in the presence of a palladium catalyst[38] and the N^{ω}-Npys group that is removed with phosphines or thiols.[39–41] The interesting stability/deprotection properties of N^{ω}-Troc derivatives have not yet been fully exploited. In fact, this amine protecting group is completely compatible with the Fmoc/tBu strategy as well as with the Aloc protection, but is removed by the Grob

fragmentation with zinc.[42] More attention has been paid in the context of the Fmoc/tBu chemistry to N^{ω}-protections that are highly acid-sensitive such as the 2-(1-adamantyl)prop-2-yloxycarbonyl group,[43,44] the trityl derivatives Mtt[45] and Mmt,[46] or the newly developed 1-(4,4-dimethyl-2,6-dioxocyclohexylidene)ethyl (Dde) or 1-(4,4-dimethyl-2,6-dioxocyclohexylidene)-3-methylbutyl groups.[47,48]

Table 1 Main Protection Schemes Presently Used for Synthesis of Peptides and Proteins in Solution and on Solid Support or by Convergent Strategies[12–22,28,29]

Strategy	N^{α}-Protection	N^{α}-Deprotection	Side-Chain Protection[a]	Final Deprotection[b]	Mode of Synthesis	Ref
1	Z	H_2, Pd	Boc (Adoc), tBu	TFA (scavengers)	solution synthesis	[12,13]
2	Nps	HCl (1–2 equiv), 2-methylindole	Boc (Adoc), tBu	TFA (scavengers)	solution synthesis	[14,15]
3	Bpoc	80% aq AcOH or 1% TFA/CH_2Cl_2	Boc, tBu	TFA (scavengers)	solution and solid-phase synthesis	[16,17]
4	Z(OMe)	TFA, anisole	Bzl	1 M TfOH/TFA (scavengers)	mainly solution synthesis	[18–20]
5	Boc	TFA or 50% TFA/CH_2Cl_2	Z(2-Cl), Bzl, Cy	HF (scavengers)	solution and solid-phase synthesis	[21,22]
6	Fmoc (Msc, Nsc)	piperidine in solvents	Boc (Adoc), tBu	TFA (scavengers)	mainly solid-phase synthesis	[28,29]

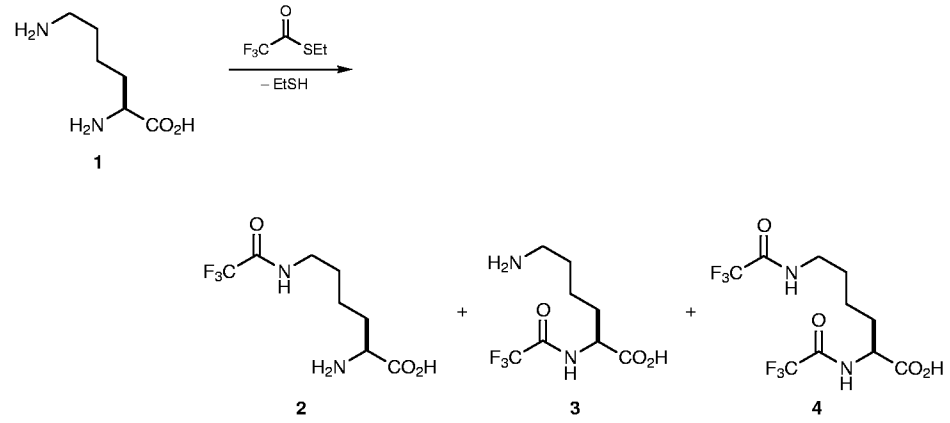
[a] Only the most indicative side-chain protections are listed; for additional details regarding optimal protection of side-chain functionalities and deprotection cocktails, see Section 4.

[b] The protection schemes have been derived over the years from experiences of many laboratories and are still continuously being optimized; for references to the original literature, see Section 4 and for examples of successful syntheses according to these strategies, see Vol. E22b, Section 5.

2.1.2.2.1 General Synthetic Methods for Asymmetric N^{α},N^{ω}-Protection of Diamino Acids

It is well established that the nucleophilicity of the α- and ω-amino groups are sufficiently different to permit selective N^{ω}-modifications of lysine residues in peptides if moderately reactive reagents are used. In fact, lysine residues are converted by reaction with *O*-methylisourea at alkaline pH into homoarginine[49–51] and ornithine to arginine residues[52] in high yields without affecting the α-amino group, while with 3,5-dimethylpyrazole-1-carboximidamidium nitrate less discrimination between α- and ω-amino groups is reported.[53] A similar preferred reaction of the ω-amino groups of lysine (**1**) and ornithine was obtained with *S*-ethyl trifluorothioacetate giving N^{ε}-(trifluoroacetyl)lysine (**2**) (Scheme 1) (for an experimental procedure, see Section 2.1.2.2.2).[54] Although in this particular case the small amounts of N^{α}-acyl and N^{α},N^{ε}-diacyl derivatives **3** and **4**, respectively, can be removed by crystallization, generally this approach is not suitable for large-scale preparation of the desired asymmetrically protected N^{α},N^{ω}-derivatives. Nevertheless, the resulting H-Lys(Tfa)-OH (**2**) can serve as a useful intermediate for further transformations.

Scheme 1 Selective N^{ω}-Protection of a Diamino Acid with a Weak Acylating Agent[54]



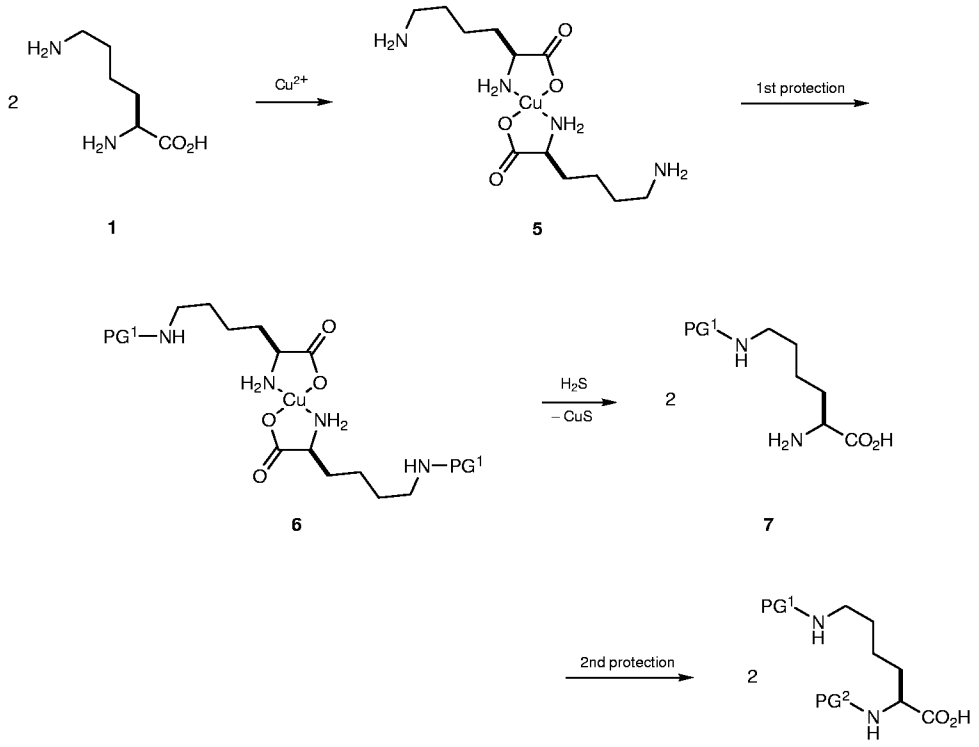
2.1.2.2.1.1 N^{ω}-Derivatization of Copper(II) Complexes of Diamino Acids

The most successful approach to the direct protection of the ω-amino group is the conversion of lysine (**1**) and ornithine into the corresponding copper(II) complexes, e.g. **5**, which trap the α-amino and α-carboxy group allowing selective acylations at the ω-amino group to be performed in weak alkaline media giving e.g. **6** (Scheme 2).[55] The copper complexes are prepared with basic copper carbonate, of the approximate formula $CuCO_3 \cdot Cu(OH)_2 \cdot H_2O$,[56] or more appropriately with $CuSO_4 \cdot 5H_2O$;[1,57] the latter procedure does not require heating and filtration. From the resulting derivatives the N^{ω}-protected diamino acids must be liberated by destruction of the Cu^{2+} complex; this is readily achieved with gaseous hydrogen sulfide,[58–62] or preferentially with hydrogen sulfide sources such as thioacetamide.[63,64] Alternatively, to avoid the unpleasant smell of the highly toxic hydrogen sulfide and the difficult separation of copper sulfide, the use of EDTA is recommended although decomposition of the copper complexes requires long reaction times.[62,65,66] The latter method is especially mild and allows the synthesis of large amounts of ω-protected amino acids. The workup can be performed under acidic or basic conditions.[62] The copper complex is also decomposed with the chelating ion-exchange resin Na^+ Chelex 100;[67] this resin is rather expensive and, therefore, an optimized procedure has been proposed for the decomposition of, for example, Cu[Lys(Boc)]$_2$ (**6**, PG^1 = Boc) with quinolin-8-ol which proceeds readily generating H-Lys(Boc)-OH (**7**, PG^1 = Boc) in almost quantitative yields.[57] Additionally, the use of potassium cyanide[68] and in the case of acid-resistant N^{ω}-derivatives treatment with acids (2 M HCl)[69,70] have been reported. More recently, thiazolidine-2-thione has been proposed as a highly efficient reagent for the decomposition of the complexes and trapping of the copper(II) ions.[71]

The copper(II) complex method has been used for the selective N^{ω}-derivatization of lysine and ornithine as well as of 2,4-diaminobutanoic acid, but not of 2,3-diaminopropanoic acid where the preference for complexation of the α- versus ω-amino group is questionable. The additional N^{α}-protection of the N^{ω}-protected amino acid is then performed according to standard procedures under Schotten–Baumann conditions.

for references see p 188

Scheme 2 Selective N^{ω}-Protection by Acylation of the Copper(II) Complexes of Diamino Acids[55–57]



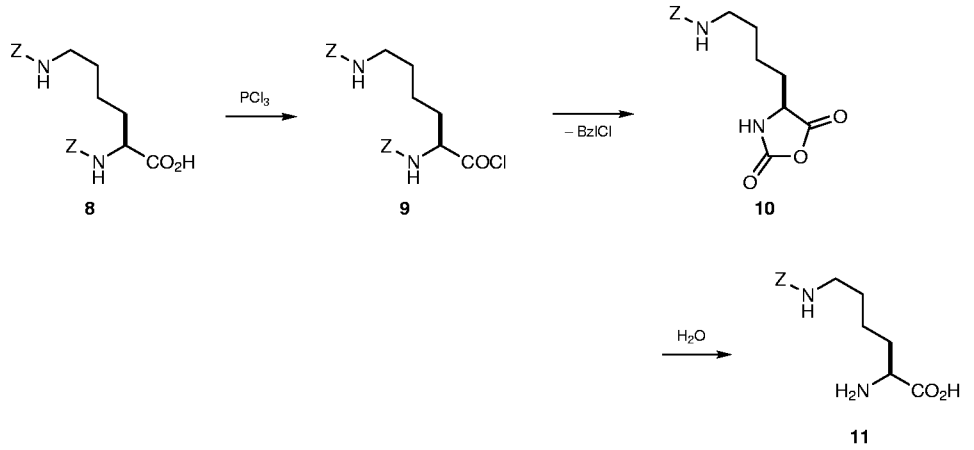
Lysine–Copper(II) Complex 5; Typical Procedure:[56]
A soln of H-Lys-OH·HCl (200 g, 1.09 mol) in H_2O (2 L) was heated to 85–90 °C and treated, over 10 min, with basic copper carbonate [180 g, approximate formula $CuCO_3 \cdot Cu(OH)_2 \cdot H_2O$]. The mixture was stirred and refluxed for 20 min and filtered through Celite. The filter pad was washed with hot H_2O (4 ×, last filtrate colorless). The combined filtrates were concentrated to dryness; yield: quant; generally, upon concentration of the copper(II) complex soln it was used directly for N^{ω}-acylations.

Lysine–Copper(II) Complex 5; Typical Procedure:[57]
To a stirred soln of H-Lys-OH·HCl (365 g, 2 mol) in aq 2 M $NaHCO_3$ (2 L) a soln of $CuSO_4 \cdot 5H_2O$ (250 g, 1 mol) in H_2O (2 L) was added and the resulting blue colored soln was used directly for N^{ω}-acylations.

2.1.2.2.1.2 Selective N^{α}-Deprotection of N^{α},N^{ω}-Bis(benzyloxycarbonyl) Diamino Acids

An alternative approach to the preparation of N^{ω}-protected lysine, e.g. N^{ε}-(benzyloxycarbonyl)lysine (**11**), and ornithine derivatives is based on selective N^{α}-deprotection of N^{α},N^{ε}-bis(benzyloxycarbonyl)lysine (**8**) (or the respective ornithine derivative) which proceeds via the corresponding acid chloride **9** and *N*-carboxyanhydride **10** (Scheme 3).[72]

Scheme 3 Synthesis of N$^{\omega}$-Protected Diamino Acids via Selective N$^{\alpha}$-Deprotection[72]



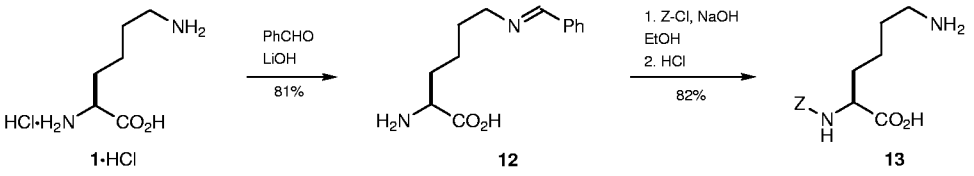
H-Lys(Z)-OBzl·HCl; Typical Procedure:[1]
To a suspension of Z-Lys(Z)-OH (**8**; 207 g, 0.5 mol) in Et_2O (300 mL) was added dropwise $SOCl_2$ (92 mL, 1.26 mol) at −10 °C within 30 min. The mixture was stirred for 30 min at rt, and refluxed for 3 h. The starting crystallization was completed in the cold. The product was collected, dried (P_2O_5), and the Lys(Z)-NCA was taken up in BzlOH (500 mL) containing HCl (19 g). After 16 h at rt, the product was precipitated with Et_2O; yield: 154 g (76%); mp 139–140 °C; $[\alpha]_D^{20}$ −4.9 (*c* 0.5, 0.1 M HCl).

2.1.2.2.1.3 Selective N$^{\alpha}$-Protection of Diamino Acids

An additional alternative is the selective preparation of N$^{\alpha}$-protected diamino acids and their use for the synthesis of the asymmetrically N$^{\alpha}$,N$^{\omega}$-protected derivatives. One approach is based on the use of the α-protected Tfa-Lys-OH (**3**) and Tfa-Orn-OH derivatives, which are accessible in good yields by direct reaction of lysine and ornithine with trifluoroacetic anhydride in trifluoroacetic acid.[73]

A more practical approach is based on *N*$^{\omega}$-benzylidene protection of diamino acids by their reaction with benzaldehyde and lithium hydroxide.[67,74–77] This transient protection is very convenient for the selective introduction of N$^{\alpha}$-protecting groups, e.g. the Z group, with subsequent hydrolysis of the imine derivative, e.g. **12**, to release the desired product, i.e. **13** (Scheme 4).[67,76,77] Furthermore, if the α-amino function remains protected, e.g. as the Z derivative, the *N*$^{\omega}$-benzylidene group can then be reduced with sodium borohydride, in the case of Z-Lys(=CHPh)-OH leading to the *N*$^{\varepsilon}$-monobenzyllysine derivative Z-Lys(Bzl)-OH which can be methylated to produce ε-methyllysine Z-Lys(Me)-OH.[77]

Scheme 4 Synthesis of Z-Lys-OH via *N*$^{\varepsilon}$-Benzylidenelysine[67,77]



H-Lys(=CHPh)-OH (12); Typical Procedure:[67]
To a soln of H-Lys-OH·HCl (**1**·HCl; 1.31 kg, 7.2 mol) in 2 M LiOH (3.6 L) was added dropwise at 0 °C under N_2 freshly distilled benzaldehyde (764 mL, 7.48 mol) over a period of 4 h (internal temperature ≤4 °C). Product precipitation made stirring difficult. The mixture was kept at 3–5 °C overnight, diluted with cold EtOH (2 L) and slowly filtered over coarse sintered glass in a cold room. The solid was washed with cold EtOH (300 mL), triturated by stirring at 3–5 °C with cold EtOH (1 L) and again filtered. The

for references see p 188

filter cake was washed with cold EtOH (2 × 500 mL), and dried (20 °C/0.5 Torr); yield: 1.37 kg (81%); mp 203–206 °C (dec).

Z-Lys-OH (13); Typical Procedure:[67]
A stirred mixture of 1 M NaOH (1.5 L) and EtOH (1.5. L) was cooled in a −30 °C bath to an internal temperature of −20 °C. To this soln was added, in one portion, finely ground N^{ε}-benzylidenelysine (**12**; 352 g, 1.50 mol). Immediately the addition, from a jacketed addition funnel kept at −20 to −10 °C, of Z-Cl (440 mL, 3.00 mol) was begun. Simultaneously, from a separate funnel, a precooled (−20 °C) mixture of 1 M NaOH (3.75 L) and EtOH (2.98 L) was added. The addition rates were kept as rapid as possible and proportionally equal so that they were complete at the same time. The reaction was exothermic, the temperature rising rapidly to −15 °C, then more slowly to −10 °C by the time the additions were complete at 17–18 min. During this time, the bath temperature rose to −23 °C. The reaction was stirred with cooling until the internal temperature began to drop (3 min) and then without cooling until the temperature reached −5.5 °C (35 min). At this point, precooled (3 °C) concd HCl (374 mL, 4.5 mol) was added. The mixture was warmed to 50 °C on a steam cone and then distilled at 15–20 Torr with a 40 °C water bath until 4.5 L of distillate had been collected. The residue was washed with Et_2O (3 × 760 mL, discarded) and brought from pH 1 to pH 6.2 by the addition of 2 M NaOH (ca. 1.2 L). (Any of the less soluble H-Lys(Z)-OH formed will precipitate at this point and can be removed by filtration.) The soln was kept at 3–5 °C overnight, filtered through medium sintered glass to remove a small amount of fine brown debris and concentrated on a rotary evaporator (40 °C/1 Torr) to a volume of 1.5 L. The resultant slurry was stored at 3 °C overnight and filtered on coarse sintered glass. The solid was washed with cold H_2O (3 × 150 mL) and dried ($CaSO_4$ at 25–30 °C/0.5 Torr) to give a white powder; yield: 345 g (82%); mp 228–229 °C.

2.1.2.2.2 Acyl Protecting Groups

N-Acyl derivatives are not advisable for the protection of α-amino groups due to the high risk of racemization (see Section 2.1.1.2). Since no stereogenic center is adjacent to the ω-amino groups of the diamino acids generally encountered in peptide synthesis, related N^{ω}-acyl derivatives have found extensive use, particularly, in synthetic strategies based on minimum protection and thus on segment condensations by the azide procedure as originally proposed by Hofmann et al.[78,79] These strategies foresee selective transient protection of the α-amino group versus the C-terminal hydrazide and the ω-amino functions, whereby deprotection of the hydrazides must occur concurrently with removal of all the protection groups of side-chain functionalities not reactive with azides, i.e. hydroxy and carboxy groups, but again in selective mode versus the N^{ω}-protection (Table 2). This type of three-dimensional orthogonality, which is used in the solution synthesis of larger peptides, is mainly based on combinations such as those reported in Table 2. By this procedure minimally protected intermediate segments are obtained which generally are sufficiently water-soluble to enable the unprotected carboxy groups to be exploited for ion-exchange chromatography.

Table 2 Protection Schemes for Synthesis of Peptides with N′-Protected Peptide Hydrazides[78–92] [a]

N′-Hydrazide Protection	N^{α}-Amino Protection	ω-Amino, ω-Hydroxy and ω-Carboxy Protection	Ref
Z	Boc, Bpoc	For, Phth, tBu	[78–81]
Boc	Z	For, Bzl, tBu	[82–87]
Trt	Bpoc, Z	Tfa, tBu	[88]
Boc	Z	Tfa, tBu	[89,90]
Boc	Nps	Tfa, tBu	[89,91]
Z	Boc	Tfa	[92]

[a] For a more complete compilation of protection schemes applied in this strategy, see Section 3.1.1.4.

The N^{ω}-acyl groups used in these protection schemes are the formyl and trifluoroacetyl groups. With the advent of alternative alkali-labile amine protecting groups that are cleaved via β-elimination under milder basic conditions, e.g. 9-fluorenylmethoxycarbonyl (see Section 2.1.1.1.1.3) and substituted sulfonylethyloxycarbonyl groups (see Section 2.1.1.1.1.4), the

trifluoroacetyl group has partly lost its significance. Nevertheless, for particular synthetic tasks it may well represent an interesting alternative.[33,93,94] Similarly, the small size and low hydrophobicity of the formyl group (it is sufficiently resistant to the basic and anhydrous acid conditions required for the cleavage of both the Fmoc and Boc groups) may facilitate selective postsynthetic manipulations.[95]

An additional interesting acyl-type N^{ω}-protection is the phenylacetyl group that is enzymatically removed with penicillin G acylase (see Section 2.5).

2.1.2.2.2.1 Formyl Group

Formylation of amino groups occurs with boiling formic acid,[96] with a mixture of formic acid and acetic anhydride at 0–20 °C,[97–99] or more efficiently with alkyl esters[100] or active esters of formic acid such as the 4-nitrophenyl,[101,102] 2,4,5-trichlorophenyl,[102] pentafluorophenyl[103] or cyanomethyl esters,[104] as well as with 1,2,3-benzotriazole-1-carbaldehyde.[105] Direct N^{ω}-formylation of ornithine and lysine is possible using commercially available 4-nitrophenyl formate (For-ONp);[99] alternatively, the classical approach via the copper(II) complexes is used.[98]

The formyl group is routinely used for the protection of the tryptophan indole function in Boc/Bzl chemistry in solution and on solid support (see Section 2.6.3), and cleavage of the N^{in}-formyl derivative with HF/ethane-1,2-dithiol[106] or TFA/2-sulfanylethanol[107] as well as by exposure to alkaline conditions (pH 8–9)[108] is well established. Less detailed information is available on the stability and cleavage of *N*-formyl derivatives of primary amines. These are stable to neat trifluoroacetic acid, but are partly cleaved by alkali at higher temperatures or longer reaction times.[109] Although N^{ω}-formyl protection has been extensively used in combination with the Z group, its absolute stability toward catalytic hydrogenation is doubtful, since cleavage has been reported in particular cases.[110]

Cleavage of the formyl group is readily achieved using aqueous alcoholic 0.5–1 M HCl.[98] Under milder acidic conditions such as 50% thioglycolic acid, which is used to concomitantly reduce methionine sulfoxides (37 °C, 44 h), only ~50% deformylation results.[111] Alternatively, oxidative cleavage with hydrogen peroxide has been utilized;[112] however, it cannot be applied in the presence of oxidation-prone amino acids such as Met and Trp. Deformylation is also achieved by hydrazinolysis via formyl transfer to hydrazine.[113–115] This latter procedure was the most efficient in a detailed comparative analysis, although a side product of unknown nature and involving tyrosine residues was detected.[95]

H-Orn(For)-OH:[101]
Ice-cold aq H-Orn-OH·HCl (10.1 g, 60 mmol) was neutralized with 1 M NaOH (60 mL, 1 equiv) and commercial For-ONp (15 g, 90 mmol) in THF was added. After 6 h at rt, the mixture was extracted with Et_2O and concentrated. Unreacted H-Orn-OH was removed with Amberlite IRC-50, and the aqueous soln was concentrated. On addition of MeOH, the product crystallized in the cold. It was recrystallized (H_2O/MeOH); yield: 5 g (52%); mp 217 °C (dec); $[\alpha]_D^{15}$ +17.1 (*c* 2.17, sat. aq $NaHCO_3$).

H-Lys(For)-OH:[100]
To an ice-cold soln of lysine–Cu(II) complex **5** (5.2 g, 14.9 mmol; see Section 2.1.2.2.1.1) in 1 M NaOH (60 mL) and MeOH (60 mL) was added ethyl formate (24 mL, 298 mmol), and the mixture was stirred vigorously for 2 h with ice-cooling. The pH was maintained between 8 and 9 by addition of 5 M NaOH. The organic solvent was removed, and the precipitated copper complex was filtered off and washed with several portions of H_2O. The complex was suspended in H_2O, H_2S was passed through the solution and the copper sulfide was removed by filtration with the aid of Norite and Filter-cel. The filtrate was concentrated to a small volume, and EtOH was added to bring about crystallization. The crystals were collected, dissolved in a small volume of H_2O and recrystallized by addition of EtOH; yield: 1.72 g (33%); mp 214–215 °C (dec); $[\alpha]_D^{26}$ +4.1 (*c* 0.91, H_2O); $[\alpha]_D^{26}$ +15.5 (*c* 1.4, sat. aq $NaHCO_3$).

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2.1.2.2.2.2 Trifluoroacetyl Group

Generally, *N*-trifluoroacetyl amino acids[32] are prepared by reaction with trifluoroacetic anhydride in trifluoroacetic acid as a solvent.[73] Other reagents such as methyl and phenyl esters of trifluoroacetic acid and *S*-ethyl trifluorothioacetate are also well suited to the trifluoroacetylation of peptides.[54,73,116] More recently 1-(trifluoroacetyl)-1,2,3-benzotriazole[117] and *N*-(trifluoroacetyl)succinimide[118] have been reported as useful reagents that can be stored. Lysine and ornithine are directly acylated at the side-chain amino group with *S*-ethyl trifluorothioacetate,[54] and the resulting ω-protected diamino acids are then converted into the desired N^{α}-protected forms. By the use of the 1,2,3-benzotriazole or succinimide derivatives the foul-smelling byproduct ethanethiol can be avoided.

The N^{ω}-trifluoroacetyl protection is stable to acidolysis and is, therefore, compatible with Boc protection; it is cleaved by mild alkali treatment[32] or by exposure to aqueous piperidine.[34,35,119] Alternatively, the trifluoroacetyl protection is removed by reduction with sodium borohydride,[120] i.e. under conditions where protecting groups such as Z, Boc, *tert*-butyl, and benzyl ethers and esters are stable. Cleavage of the N^{ω}-trifluoroacetyl groups under basic conditions bears the known danger of aspartimide formation and of cyclization of N-terminal glutamine residues to pyroglutamyl derivatives.

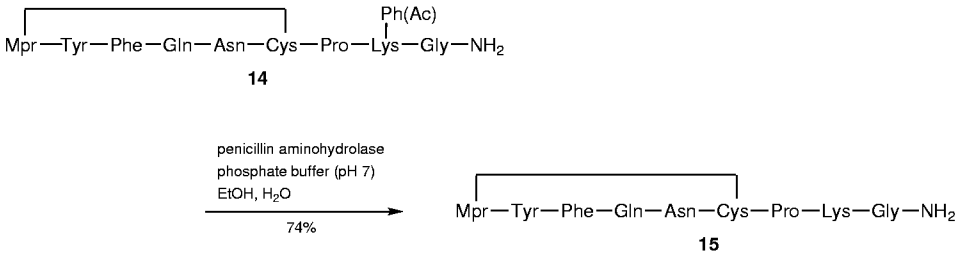
H-Lys(Tfa)-OH:[121]
S-Ethyl trifluorothioacetate (10 mL, 78 mmol) was added to a soln of H-Lys-OH·HCl (9.15 g, 50 mmol) in 1 M NaOH (50 mL) and the mixture was stirred for 12 h. The mixture was cooled and the precipitate was collected and recrystallized (aq EtOH); yield: 7.15 g (59%); mp 260–262 °C (dec); $[\alpha]_D^{20}$ +17.5 (*c* 2.1, AcOH).

Cleavage of N^{ω}-Trifluoroacetyl Groups with Aqueous Piperidine; Typical Procedure:[119]
A soln of H-Gly-Thr-Lys(Tfa)-Met-Ala-Phe-Gly-Gly-Leu-Lys(Tfa)-Lys(Tfa)-Glu-Lys(Tfa)-Asp-Arg-Asn-Asp-Leu-Ile-Thr-Tyr-Leu-Lys(Tfa)-Lys(Tfa)-Ala-Thr-Glu-OH (0.1 g, 0.026 mmol) in aq 1 M piperidine (10 mL) was stirred in an ice bath for 4 h and then lyophilized. The residue was dissolved in H_2O (100 mL) and chromatographed on CM-23 cellulose, followed by gel permeation on Sephadex G-25 and ion-exchange chromatography on IRA-400; the product was collected as lyophilizate; yield: 0.046 g (58%).

2.1.2.2.2.3 Phenylacetyl Group

The mild and highly selective enzymatic cleavage of N^{ω}-phenylacetyl amino acid or peptide derivatives under almost neutral conditions (pH 7–8, rt) with penicillin G acylase (see Section 2.5) makes the use of this type of protecting group particularly attractive when additional levels of orthogonality in side-chain protection are required. This N^{ω}-Ph(Ac) protection was successfully utilized in the synthesis of 1-deamino-Lys^8-vasopressin (**15**) where following disulfide bond formation the Ph(Ac) group could be removed enzymatically from **14** without side reactions (Scheme 5).[122] Similarly, this group could be removed enzymatically in quantitative manner from N^{α},$N^{\alpha'}$,N^{ω}-tris[Ph(Ac)]insulin without the danger of disulfide scrambling.[123] A concurrent enzymatic cleavage of Cys(Phacm) and Lys[(Ph(Ac)] groups from a synthetic (1–32) salmon calcitonin with penicillin acylase proceeded successfully.[6]

Scheme 5 Synthesis of 1-Deamino-Lys^8-Vasopressin[122]



2.1.2.2.3 Urethane Protecting Groups

For the protection of α-amino functions, urethane-type protecting groups are the most convenient even for N$^{\omega}$-protection, since they are readily prepared and stable on storage. Among the large number of these type of protecting groups, a relatively limited number has emerged as useful for ω-amino protection in the various strategies discussed in Section 2.1.2.2 (Table 1).

2.1.2.2.3.1 Urethane-type Protecting Groups Derived from Primary Alcohols

In terms of stability in the repetitive N$^{\alpha}$-deprotection steps required for the synthesis of larger peptides and proteins, only the Z group and particularly its halogenated derivatives are compatible with the Boc/Bzl strategy for N$^{\omega}$-protection, while among the other groups discussed in Section 2.1.1.1.1, the Fmoc, Teoc, Msc, and particularly Aloc groups have found application in terms of an additional level of selectivity as required in special cases for postsynthetic site-specific chemical manipulations of the peptide chain on resin or in solution.

2.1.2.2.3.1.1 Benzyloxycarbonyl Group and Related Derivatives

In solution synthesis, the Boc/Z combination allows the selective hydrogenolytic cleavage of the Z group to be exploited; it can also be removed together with the Boc group by HBr in acetic acid or in TFA,[124] or even with TFA in the presence of thioanisole.[125] Since a quantitative cleavage is obtained with liquid HF,[126] N^{ω}-Z protection was the first choice in the original Boc/Bzl strategy of solid-phase synthesis. However, it was soon realized that when applied to the synthesis of larger peptides, the repetitive N$^{\alpha}$-deprotection with TFA or 50% TFA in dichloromethane results in partial N^{ω}-Z cleavage and, thus, to branching in the growing peptide chain.[127,128]

Although for synthesis in solution the Z group is sufficiently stable toward TFA/AcOH (7:3), room temperature, 25 minutes[124] or 2–4 M ethanesulfonic acid in AcOH or dichloromethane,[129] it is partially cleaved by neat TFA or HCl solutions.[130] Correspondingly, more acid-stabile Z derivatives containing electron-withdrawing substituents such as halo or nitro groups have been developed and carefully analyzed in their kinetics of cleavage under the standard acidic conditions used for N^{α}-Boc deprotection.[130,131] The results indicate that the Z(NO$_2$) group is the most stable, but at the same time its quantitative removal in the final deprotection step with acids is difficult,[131] although it is cleaved faster than the Z group by hydrogenolysis.[132] Correspondingly, for the Boc/Bzl strategy mainly the halobenzyloxycarbonyl derivatives have been considered and their cleavage rates determined with 50% TFA in dichloromethane as the standard procedure[133] for SPPS by the Boc/Bzl chemistry. The rank order of stability under these acidic conditions is Z < Z(4-Cl) ≪ Z(2-Cl) < Z(2,4-Cl$_2$) < Z(3,4-Cl$_2$) < Z(3-Cl) < Z(2,6-Cl$_2$) (Table 3).[128] All these chloro derivatives except Z(2,6-Cl$_2$) are quantitatively removed by treatment with liquid HF for 60 minutes at

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0 °C.[127,128,131] At present, the Z(2-Cl) group is predominately used for N^{ω}-protection of lysine residues in the context of the Boc/Bzl chemistry in solution and on solid supports; it is up to 60 times more stable than the related N^{ω}-Z derivatives. Similarly, the N^{ω}-Z(2-Br) derivatives show approximately a 60-fold enhanced acid stability versus the Z protection.[66]

The benzyloxycarbonyl group and the related substituted derivatives are introduced selectively into the ω-position of diamino acids via the carboxyanhydrides starting with N^{α},N^{ω}-disubstituted derivatives (see Section 2.1.2.2.1.2),[72] and H-Lys(Z)-OH has also been prepared directly by reaction of lysine with Z-OPh[134] or with benzyl 1-piperidyl carbonate, although yields by the latter procedures are rather poor (40 and 58%, respectively).[135] Thus, the routine approach is based on reaction of the copper(II) complexes with the related chloroformates.

Besides the halo-subsituted Z derivatives, the Msz group, a group that is based on the safety-catch principle, is of potential interest for particular synthetic tasks as it is stable toward acidic and basic reaction conditions, but it is readily cleaved in a one-pot procedure with silicon tetrachloride/TFA in the presence of scavengers such as anisole, phenols, sulfides, selenides, and thiols.[136]

Among the other benzyloxycarbonyl derivatives the Z(OMe) group shows a slightly higher acid-lability than the Boc group and may find occasional use for N^{ω}-protection instead of the Boc group when *tert*-alcohol-derived groups are selected for semipermanent side-chain protection.

Table 3 Deprotection of Chlorinated Derivatives of N^{ω}-(Benzyloxycarbonyl)lysine in 50% TFA[128]

Lysine Derivative	k^a (10^{-8} sec^{-1})	k_{rel}	Loss per Cycle[b] (%)	Ref
Lys(Z)	396	100	0.47	[128]
Lys[Z(4-Cl)]	138	35	0.17	[128]
Lys[Z(2-Cl)]	6.3	1.6	0.008	[128]
Lys[Z(2,4-Cl_2)]	4.9	1.2	0.006	[128]
Lys[Z(3,4-Cl_2)]	2	0.6	0.003	[128]
Lys[Z(3-Cl)]	0.49	0.12	0.0006	[128]
Lys[Z(2,6-Cl_2)]	0.4	0.1	0.0005	[128]

[a] Apparent first-order rate constant for deprotection with 50% TFA/CH_2Cl_2 at 20 °C.
[b] Based on treatment with 50% TFA for 20 min/cycle.

H-Lys(Z)-OH via the Copper(II) Complex; Typical Procedure:[1]
To a soln of H-Lys-OH·HCl (365 g, 2.0 mol) and NaOH (160 g, 4.0 mol) in H_2O (1.6 L) was added at 30 °C $CuSO_4 \cdot 5H_2O$ (250 g, 1.0 mol) in H_2O (800 mL). The soln was cooled to 0 °C and then $NaHCO_3$ (200 g, 2.4 mol) was added followed by Z-Cl (380 mL, 2.7 mol) in portions over 45–60 min while maintaining the mixture at 0 °C. This was stirred for an additional 3 h at 0 °C and overnight at rt. The precipitate was filtered off and washed with H_2O (4 L) and acetone (2 L). The copper complex was suspended in H_2O (4 L) and treated with portions of concd HCl (720 mL). Through the resulting dark green soln H_2S(g) was bubbled for 60–90 min. The suspension was shortly degassed under reduced pressure and then filtered, and the filter cake was washed extensively with H_2O (8 L). The filtrates were cooled to 5–10 °C and neutralized with ammonia to pH 7. After 1 h at 5–10 °C, the precipitate was filtered off, washed with H_2O (4 L) and MeOH (2 L), and dried; yield: 480–505 g (85–90%); mp 256–258 °C (dec); $[\alpha]_D^{20}$ +15 (*c* 2, 3 M HCl).

H-Lys(Z)-OH by Copper(II) Complex Decomposition with Thioacetamide:[63]
To a suspension of pulverized Cu[Lys(Z)]$_2$ (15.0 g, 24 mmol) in boiling H_2O (350 mL) was added AcOH (20 mL) followed by thioacetamide (7.5 g, 0.10 mol). After refluxing for 5 min, the CuS precipitate was filtered off on a hot funnel. The product crystallized immediately from the filtrate. After several hours at 4 °C, the crystals were collected, washed with a little cold H_2O, and dried; yield: 10.2 g (73% calcd from H-Lys-OH·HCl); mp 252–254 °C; $[\alpha]_D^{20}$ +6.4 (*c* 6.6, 2 M NaOH).

H-Orn(Z)-OH by Copper(II) Complex Decomposition with Thioacetamide:[64]
To a suspension of Cu[Orn(Z)]$_2$ (5.4 g, 9 mmol) and thioacetamide (1 g, 13.3 mmol) in H_2O (50 mL) was added 2 M NaOH to adjust the pH to 8, and the mixture was stirred at rt for 26 h. The soln was acidified with 2 M HCl to pH 1.6 and the precipitated CuS was removed by filtration and washed with dil HCl and then H_2O. The filtrate was adjusted to pH 8 with 2 M NaOH and the precipitate was collected, washed with H_2O and recrystallized (HCl/NaOH); yield: 4.2 g (88%); mp 235 °C; $[\alpha]_D^{20}$ +18 (*c* 0.9, H_2O/acetone 1:1 and 2 equiv HCl).

H-Lys(Z)-OH via the *N*-Carboxyanhydride; Typical Procedure:[72]
To a cooled (0 °C) soln of Z-Lys(Z)-OH (45 g, 109 mmol) in dry Et_2O (175 mL) was added powdered PCl_5 (26 g, 125 mmol). After shaking for 30 min, the mixture was filtered, concentrated with exclusion of moisture, and extracted with EtOAc (2 ×). The organic layers were concentrated, and the residue recrystallized (EtOAc/petroleum ether) to yield the anhydride; yield: 28.1 g (85%); mp 100 °C (dec).
To Lys(Z)-NCA (15.7 g, 51 mmol) in acetone (75 mL) was added 5 M HCl (20 mL). After standing overnight at rt, the mixture was concentrated, 1 M HCl (75 mL) was added, and the solvent was again removed. The residue was dissolved in hot H_2O and the soln was neutralized with ammonia; the precipitate was filtered off; yield: 14.3 g (~100%); mp 255 °C (dec); $[\alpha]_D^{25}$ +14.0 (*c* 1, H_2O with 2 molar equiv HCl).

H-Lys[Z(2-Br)]-OH, Typical Procedure:[66]
To a boiling soln of H-Lys-OH·HCl (22.3 g, 122 mmol) in H_2O (400 mL) was added basic copper carbonate (16.4 g, 74 mmol). After 10 min the soln was cooled and filtered. The filtrate was diluted with H_2O (1.2 L) and DMF (4.9 L). Z(2-Br)-ONp (50 g, 142 mmol) and $NaHCO_3$ (22.3 g, 265 mmol) were added, and the mixture was stirred for 2 d. The blue solid was collected, washed with H_2O and EtOH, and dissolved in boiling H_2O (6 L) in the presence of EDTA disodium salt (~50 g). The soln was cooled and stored at 4 °C overnight. The colorless product was collected and washed with H_2O and EtOH; yield: 42 g (96%); mp 220–223 °C; $[\alpha]_D^{24}$ +9.6 (*c* 1.09, 80% AcOH).

H-Lys[Z(2-Cl)]-OH:[128]
The title compound was prepared as described for the Z(2-Br) derivative using Z(2-Cl)-Cl to acylate Cu[Lys]$_2$ and by decomposing the copper(II) complex with EDTA;[65] the product was recrystallized (90% aq EtOH); mp 219–222 °C.

2.1.2.2.3.1.2 9-Fluorenylmethoxycarbonyl Group

The N^{α}-Fmoc protection is the essential feature of the Fmoc/tBu strategy of SPPS, but the stability/deprotection characteristics of this group (see Section 2.1.1.1.1.3) can be conveniently exploited in the context of the Boc/Bzl protection scheme for site-directed chemical manipulations at ω-amino functions on resin and in solution.[36,137–139]

For the synthesis of N^{ω}-Fmoc-protected lysine and ornithine the route via the related copper(II) complexes has been used exclusively and the acylation performed with Fmoc-N_3,[37] Fmoc-OSu,[71] and, most conveniently, with Fmoc-Cl.[140] The resulting ω-protected diamino acids are readily converted into the N^{α}-Boc derivatives.

H-Lys(Fmoc)-OH; Typical Procedure:[140]
To a soln of H-Lys-OH·HCl (1.1 g, 6.0 mmol) in H_2O (10 mL), NaOH (0.24 g, 6.0 mmol) was added followed by $CuSO_4 \cdot 5H_2O$ (0.8 g, 3.2 mmol) in H_2O (10 mL). The soln was cooled in an ice bath and $NaHCO_3$ (0.6 g, 7.0 mmol) was added followed by a soln of Fmoc-Cl (1.6 g, 6.2 mmol) in dioxane (5 mL) over 5 min. The mixture was stirred overnight at rt, and then the blue precipitate was collected and washed with cold H_2O (10 mL), EtOH (10 mL), and EtOAc (10 mL) to give Cu[Lys(Fmoc)]$_2$; yield: 2.0 g (85%); mp 211–213 °C.
To the finely powdered copper(II) complex (1.0 g, 1.3 mmol) in H_2O (60 mL) was added EDTA disodium salt dihydrate (0.6 g, 1.6 mmol). The soln was stirred at 80 °C for 2 h. The white crystalline product was filtered off and washed with cold H_2O (20 mL) and EtOH (20 mL); yield: 0.9 g (99%); mp 210–211 °C.

Boc-Lys(Fmoc)-OH:[37]
Following the typical procedure in Section 2.1.1.1.3.1.1.4, H-Lys(Fmoc)-OH was acylated with Boc-ON; yield: 70%; mp 88–91 °C; $[\alpha]_D$ −1.4 (*c* 1, MeOH).

for references see p 188

2.1.2.2.3.1.3 2-(Methylsulfonyl)ethoxycarbonyl Group

The Msc group is fully stable to acid treatments such as TFA (rt, 2 d), liquid HF (0 °C, 30 min), and even 12 M HCl (40 °C, 1 h) as well as to hydrogenolysis as required for the cleavage of *N*-Z derivatives, but it is readily cleaved under alkaline condition, e.g. by 0.025–0.05 M $Ba(OH)_2$ or by the Tesser base (dioxane/MeOH/4 M NaOH 7.5:2.25:0.25) yielding as byproduct a volatile and highly soluble ether in the presence of methanol.[25] Due to these characteristics the Msc group is well suited for N^{ω}-protection in combination with either the Z or the Boc group for N^{α}-protection. The Z/Msc combination was successfully applied to the solution synthesis of the cytochrome c fragment 66–104[141] and the variation Boc/Msc to the synthesis of α-melanotropin[142] and to a large-scale synthesis of somatostatin.[143] The high polarity of the Msc group results in the enhanced solubility of protected peptides in polar solvents which facilitates, at least in solution synthesis, purification and processing of such intermediates.

The N^{ε}-Msc derivatives of lysine are prepared by reaction of the related copper(II) complex with Msc-Cl, followed by decomposition of the $Cu[Lys(Msc)]_2$ complex with H_2S and conversion into Z-Lys(Msc)-OH and Boc-Lys(Msc)-OH by standard procedures.[25] The latter derivative was also obtained directly by reaction of Boc-Lys-OH with Msc-ONp.[142] The commercially available Msc-OSu should serve with equal efficiency for such purpose.

Boc-Lys(Msc)-OH:[142]
A suspension of Boc-Lys-OH (4.93 g, 20 mmol) and Msc-ONp (5.8 g, 20 mmol) in DMF (40 mL) was stirred overnight at rt. The resulting soln was concentrated and the residue was dissolved in H_2O (80 mL) by addition of sat. aq $NaHCO_3$. The soln was overlayered with Et_2O and CO_2 was bubbled through the aqueous layer until full decoloration. The aqueous layer was separated, cooled with ice, acidified with 1 M H_2SO_4 to pH 2, and extracted with EtOAc (3 × 150 mL). The organic layers were washed with brine, dried (Na_2SO_4), and concentrated to give an oil that crystallized slowly under Et_2O; yield: 7.4 g (93%); mp 83–86 °C.

[Cys(Acm)^3,Cys(Acm)^{14}]-Somatostatin; Msc Cleavage with $Ba(OH)_2$:[143]
To a cold (<5 °C) soln of H-Ala-Gly-Cys(Acm)-Lys(Msc)-Asn-Phe-Phe-Trp-Lys(Msc)-Thr-Phe-Thr-Ser-Cys(Acm)-2-[4-(phenylazo)phenylsulfonyl]ethyl ester trifluoroacetate (57.2 g, 23 mmol) in DMF (1.15 L) and MeOH (115 mL) was added slowly ice-cold 0.2 M $Ba(OH)_2$ (1150 mL, 0.23 mol) over 40 min maintaining <15 °C. The mixture was allowed to reach 18 °C and a current of CO_2 was bubbled through the soln until it was slightly acidic (pH 5–6). The $BaCO_3$ was removed by centrifugation and the supernatant was concentrated at 35–40 °C to about 600 mL. This soln was poured into EtOAc (4 L) under vigorous stirring. After 1 h, the precipitate was filtered off and washed with EtOAc, and dried; yield: 42 g of crude product. This was purified by partition chromatography on Sephadex G25F or by countercurrent distribution using Partridge ($BuOH/H_2O$/AcOH) systems; yield: 20.2 g (48%).

2.1.2.2.3.1.4 Allyloxycarbonyl Group

The allyl-type protecting groups are characterized by full stability towards moderate to strong acids and toward basic conditions as used in the Boc/Bzl and Fmoc/tBu strategies in solution and on solid supports, but are readily cleaved by palladium-catalyzed isomerization (see Section 2.1.1.1.1.5).[38,144] Correspondingly, these carboxy, hydroxy, and amino protecting groups offer a new level of full orthogonality in side-chain protection for regioselective deprotection of selected functionalities on resin or in solution for additional postsynthetic site-directed chemical reactions (see Section 4.3.2.3).

The N^{ω}-Aloc derivatives of lysine or ornithine are readily prepared via the copper(II) complexes,[145] and the desired N^{α}-protection is carried out by standard procedures under conventional Schotten–Baumann conditions.[144]

H-Lys(Aloc)-OH·HCl; Typical Procedure:[145]
A soln of H-Lys-OH·HCl (50.0 g, 0.28 mol) and basic copper carbonate (63.5 g, 0.29 mol) in H_2O (500 mL) was refluxed for 30 min. Solids formed during reflux were removed by hot filtration. The

filtrate was cooled to 0 °C and adjusted to pH 9 by the addition of solid $Na_2CO_3 \cdot H_2O$ (~5 g). Aloc-Cl (42 mL, 0.41 mol) was added dropwise over 1 h with stirring at 0 °C. The pH was maintained during the reaction at pH 9 by the addition of solid $Na_2CO_3 \cdot H_2O$ (a total of 70 g). The mixture was allowed to warm to rt, and after 12 h, there was no starting material (as monitored by TLC). The blue solid formed was collected and suspended in H_2O (1 L) and thioacetamide (42.1 g, 0.56 mol) was added. The alkaline suspension was stirred at 50 °C for 3 h, during which time the solid slowly dissolved. The soln was then acidified to pH 2 with 2 M HCl, and was further refluxed for 5 min. The CuS was removed by filtration and the filtrate was concentrated in vacuo to ca. 300 mL, at which point the product precipitated as a white solid; yield: 74.7 g (~100%); mp 228–231 °C; $[\alpha]_D^{25}$ +9.2 (*c* 1, 8% aq K_2CO_3).

Fmoc-Lys(Aloc)-OH; Typical Procedure:[144]
A suspension of H-Lys(Aloc)-OH (6.92 g, 0.03 mol) in H_2O (150 mL) and dioxane (10 mL) was adjusted with 2 M NaOH to a pH of 11.5, yielding a clear soln. To this Fmoc-F (0.027 mol) in DME was added dropwise over 30 min, maintaining the pH at 11.5 by addition of 2 M NaOH. The mixture was further stirred for 1 h and washed with Et_2O (2 ×). Standard workup afforded a crude product, which was recrystallized (Et_2O/hexane); yield: 11.6 g (85%); mp 90–93 °C; $[\alpha]_D$ −11.9 (*c* 1, DMF).

Deprotection of the Allyl and Aloc Groups; General Procedure:[144]
To a soln of the allyl- or Aloc-protected amino acid (1 mmol) in CH_2Cl_2 (5 mL) were added $PdCl_2(PPh_3)_2$ (0.04 mmol) and AcOH (2.5 mmol). The mixture was stirred and Bu_3SnH (2 mmol) was added rapidly. Stirring was continued for 10 min at rt until, in most cases, a color change from yellow to purple showed the end of the reaction.

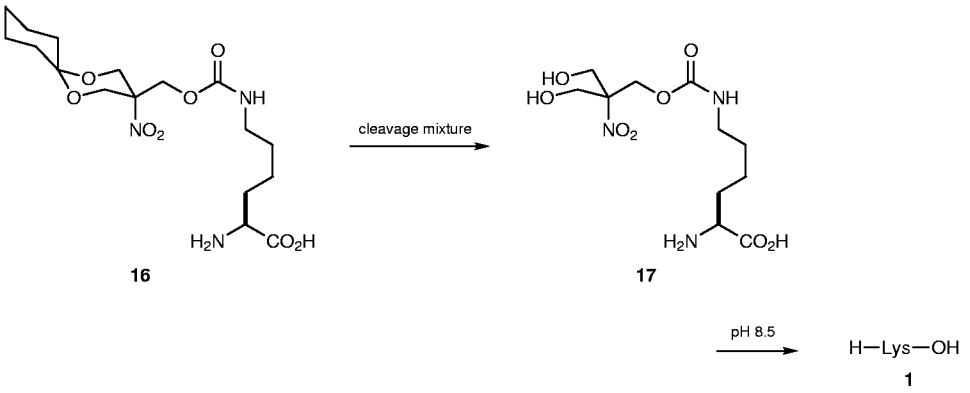
2.1.2.2.3.1.5 Other Urethane-type N^{ω}-Protecting Groups Derived from Primary Alcohols

Due to the relatively high stability of the *N*-2,2,2-trichloroethoxycarbonyl (Troc) group to acids and its selective removal by zinc reduction even in the presence of *N*-benzyloxycarbonyl derivatives, it can offer interesting possibilities in the syntheses of particular peptidic compounds; for example, it has been used for N^{ω}-protection of lysine in the synthesis of MSH.[146] The N^{ω}-Troc-protected lysine is readily prepared by reaction of the copper(II) complex with 2,2,2-trichloroethyl chloroformate (Troc-Cl)[147] or by reaction of Z-Lys-OH (obtained via the benzylidene route, see Section 2.1.2.2.1.3) with 2,2,2-trichloroethyl chloroformate.[148]

An additional promising side-chain amino protection is the (4-nitro-2,6-dioxaspiro-[5.5]undecan-4-yl)methoxycarbonyl group that is based on the safety-catch principle (Scheme 6).[6] This group is fully compatible with the Fmoc/tBu strategy. The N^{ω}-Boc groups at the desired amino functions are removed by trifluoroacetic acid treatment, while the N^{ω}-(4-nitro-2,6-dioxaspiro[5.5]undecan-4-yl)methoxycarbonyl derivatives are hydrolyzed to the 3-hydroxy-2-(hydroxymethyl)-2-nitropropoxycarbonyl groups, e.g. **16** is hydrolyzed to **17**, which still protect the selected side-chain amino functions, but at the same time enhance the solubility of the partially protected peptides. After transformations at the desired amino functions, the N^{ω}-3-hydroxy-2-(hydroxymethyl)-2-nitropropoxycarbonyl groups are removed in aqueous or aqueous organic media by the action of piperidine.[149]

for references see p 188

Scheme 6 Conversion of N^{ε}-(4-Nitro-2,6-dioxaspiro[5.5]undecan-4-yl)methoxycarbonyl-Protected Lysine into the N^{ε}-[3-Hydroxy-2-(hydroxymethyl)-2-nitropropoxycarbonyl] Derivative by Acid Cocktails Used in the Cleavage of *tert*-Butyl Alcohol Derived Protecting Groups, Followed by Cleavage of the N^{ε}-[3-Hydroxy-2-(hydroxymethyl)-2-nitropropoxycarbonyl] Derivative Under Basic Conditions[149]



16 17 1

2.1.2.2.3.2 Urethane-type Protecting Groups Derived from Secondary Alcohols

For α-amino protection, the urethane-type protecting groups derived from secondary alcohols do not offer improved properties compared to those of more classical groups derived from primary alcohols. The main advantage possibly derived from the use of this type of urethane-derived N^{ω}-derivative is the increased solubility, e.g. such as that reported for the 2-adamantyloxycarbonyl derivatives (see Section 2.1.1.1.2).[150,151]

2.1.2.2.3.2.1 Cyclohexyloxycarbonyl Group

The cyclohexyloxycarbonyl group (Hoc) was initially proposed by McKay et al., but it was not immediately used for peptide synthesis due to its high acid-stability.[152,153] Nevertheless, this property makes it well suited for semipermanent side-chain protection for lysine and ornithine in the context of the Boc/Bzl strategy. In fact, the Hoc group is stable toward TFA or HCl in organic solvents at room temperature,[154] and toward 1 M TMSOTf/thioanisole/TFA,[155] but is cleaved with anhydrous HF.[154,155]

The related N^{ω}-derivatives are readily accessible by reaction of the copper(II) complexes of diamino acids with the relatively stable chloroformate (Hoc-Cl),[155] and subsequent liberation of the ω-protected amino acid. These partially protected amino acids are then N^{α}-protected, e.g. as Boc derivatives, under conventional Schotten–Baumann conditions.[154]

A further urethane protecting group derived from a secondary alcohol is the 2-adamantyloxycarbonyl group.[150,151,156] It is more stable toward acidic reaction conditions than the corresponding 1-adamantyloxycarbonyl group (1-Adoc, Section 2.1.2.2.3.3.2), but it is removed with trifluoromethanesulfonic acid or anhydrous HF in a few minutes at 0 °C.[150]

Hoc-Cl:[155]

CAUTION: *Phosgene ($COCl_2$) is a colorless, poisonous gas or volatile liquid. This reagent should be handled with the utmost care.*

A 1-L round-bottomed flask equipped with a gas inlet tube, pressure-equalizing dropping funnel and a dry ice/acetone cooled, jacketed Vigreux condenser was placed in an ice bath. The exit tube of the condenser was connected to a $CaCl_2$ drying tube and suitable traps for $COCl_2$ and HCl. $COCl_2$ (99 g, 1 mol) was washed with linseed oil, and concd H_2SO_4, and was condensed into the reaction vessel. While stirring, cyclohexanol (95 mL, 0.9 mol) was added from the dropping funnel at such a rate that the mixture was kept in gentle reflux for ca. 2 h. Stirring was continued for 1 h and then the condenser was

removed and dry N_2 was bubbled through the mixture for 6 h. The escaping fumes were treated properly. The resulting oil was used without further purification and stored at 4 °C. The Hoc-Cl content of the product was determined by pipetting a 0.1 mL aliquot into 10% NH_3 (10 mL). Cyclohexyl urethane crystals were collected, dried and analyzed; yield: 135 g (92%); purity 99%; bp 38–44 °C/2 Torr.

H-Lys(Hoc)-OH; Typical Procedure:[154]
A soln of H-Lys-OH·HCl (18.3 g, 100 mmol) and NaOH (8 g, 200 mmol) in H_2O (100 mL) was treated with a soln of $CuSO_4 \cdot 5H_2O$ (12.5 g, 50 mmol) in H_2O (40 mL). At 0 °C with stirring were added within 30 min $NaHCO_3$ (10 g, 119 mmol) and Hoc-Cl (16 g, 100 mmol). Stirring was continued at 0 °C for 2 h and at rt for 20 h. The precipitate was filtered, washed with H_2O, EtOH, and acetone, and dried. The residue was dissolved in 1 M HCl (200 mL per 25 g complex) and saturated with H_2S. The CuS was removed and the filtrate was adjusted to pH 6 with ammonia. The precipitate was collected, washed with H_2O, EtOH, and acetone, and recrystallized (H_2O).

Boc-Lys(Hoc)-OH; Typical Procedure:[154]
The above synthesized H-Lys(Hoc)-OH (31.2 g, 0.11 mol) was suspended in dioxane (70 mL) and H_2O (70 mL) and treated with Boc-N_3 (15.6 g, 0.11 mol). The pH was maintained at 10.2 by addition of 4 M NaOH. After stirring for 6 h, the dioxane was removed, and the residue was acidified with 20% H_3PO_4, and extracted with EtOAc. The organic layers were washed with H_2O, dried, and the solvent was removed to give a slowly crystallizing residue; the product was recrystallized (Et_2O/petroleum ether); yield: 31 g (76%); mp 57–60 °C; $[\alpha]_D^{25}$ −7.2 (*c* 1, DMF).

2.1.2.2.3.3 Urethane-type Protecting Groups Derived from Tertiary Alcohols

In addition to N^{ω}-Boc protection, which is integral part of all semipermanent protection schemes (see Section 2.1.2.2, Table 1) based on cleavage with the moderate acid trifluoroacetic acid, only the Adoc group has found wide application; in particular, in solution synthesis it is used to alleviate solubility problems related to the critically sized 8- to 15-membered intermediate segments. In view of their built-in additional level of selectivity at side-chain amino groups, the highly acid-labile Bpoc group and the 2-(1-adamantyl)prop-2-yloxycarbonyl group could offer interesting possibilities.

2.1.2.2.3.3.1 *tert*-Butoxycarbonyl Group

The N^{ω}-Boc protection, proposed by Anderson and McGregor,[157] has been successfully used to elaborate the classical Z/tBu protection scheme, which permitted the first syntheses of larger biologically active peptides, e.g. of β-corticotropin[158] and glucagon,[14] by the strategy of maximum protection[12,13,159] or minimum protection.[160] In fact, the N^{ω}-Boc derivatives are completely stable towards the hydrogenolysis that is used for the repetitive cleavage of N^{α}-Z protection. To avoid the problem of catalyst poisoning by sulfur-containing amino acid residues, the addition of bases such as cyclohexylamine or triethylamine has been proposed,[161] as has catalytic hydrogenation in liquid ammonia[162] as demonstrated by the synthesis of oxytocin[163] and somatostatin.[164] Hydrogenation over a palladium catalyst in the presence of 5 equivalents of boron trifluoride–diethyl ether complex has also been proposed, but under these conditions the stability of N^{ω}-Boc is questionable.[115]

Alternatively, N^{α}-protecting groups are used which are significantly more acid labile than N^{ω}-Boc derivatives; for this purpose the Nps and Bpoc groups have been used extensively. The N^{α}-Nps peptides are deprotected in aprotic solvents such as DMF, DMA, or NMP with 2 equivalents of HCl in the presence or absence of 2-methylindole (used as scavenger to prevent sulfenylation of tryptophan residues, as discussed in Section 2.1.1.4) without partial cleavage of the N^{ω}-Boc group.[14] Conversely, in protic solvents such as methanol or trifluoroethanol deprotection must be carried out with 1 equivalent of HCl in the presence of 2-methylindole to lower the proton activity of HCl and to capture the intermediate Nps-Cl and, thus, to liberate the required equivalent of HCl for protonation of the α-amino group. In the absence of 2-methylindole partial cleavage of N^{ω}-Boc can occur.[15]

for references see p 188

The selectivity of the N^{α}-Bpoc protection versus N^{ω}-Boc derives from its facile cleavage with 80% acetic acid or with trifluoroethanol at higher temperature.[165] As alternative to N^{α}-Bpoc protection the related trityl derivatives can be used, which are also cleaved by aqueous acetic acid.[166]

The N^{ω}-Boc-protected diamino acids are prepared via the copper(II) complexes of lysine or ornithine,[67] or by using suitable N^{α}-protected derivatives obtained by the benzylidene route under conventional Schotten–Baumann conditions.[167] Since previous reports differ in yields and are not completely reproducible, the procedures have been optimized.[67] Thereby, the chelating resin Chelex 100 can efficiently be used for decomposition of the Cu[Lys(Boc)]$_2$ complex. However, as this resin is expensive, an alternative procedure has been proposed for large-scale synthesis that is based on the use of the inexpensive quinolin-8-ol to quantitatively remove the copper(II) ions.[57] The resulting H-Lys(Boc)-OH intermediate is then converted into the desired N^{α}-protected form by the classical procedures reported in Section 2.1.1. For cleavage of the N^{ω}-Boc protection by various acidic reaction conditions, see Section 2.1.1.1.3.1.3.

H-Lys(Boc)-OH; Typical Procedure:[67]
A soln of H-Lys-OH·HCl (200 g, 1.09 mol) in H_2O (2 L) was heated to 85–90 °C and treated, over 10 min, with basic copper carbonate [180 g, approximate formula $CuCO_3 \cdot Cu(OH)_2 \cdot H_2O$]. The mixture was stirred and refluxed for 20 min and filtered through Celite. The filter pad was washed with hot H_2O (4 ×, last filtrate colorless). The combined filtrates were cooled to 20 °C, treated with $NaHCO_3$ (91.5 g, 1.09 mol) and diluted to 4.0 L with H_2O. To this soln was immediately added another portion of $NaHCO_3$ (91.5 g, 1.09 mol). The soln was stirred and a soln of Boc_2O (310 g, 1.42 mol) in dioxane (2.6 L) was added over 1 h (a precipitate formed after 0.5 h). After stirring overnight, the suspension was filtered, and the filter cake was washed with H_2O (6 × 500 mL), slurried in H_2O (5 L), refiltered, and washed with H_2O (8 × 500 mL) and MeOH (8 × 500 mL). A mixture of the resulting damp copper complex and MeOH (3 L) was stirred at 20 °C overnight and diluted with deionized H_2O (4 L). Na^+ Chelex 100 resin (6 L, Bio-Rad) was washed with 1 M AcOH (6 × 2 L) and deionized H_2O (15 × 2 L), and immediately added to the copper complex. The total was stirred at 20 °C for 3 h and filtered. The resin was washed with 50% MeOH (5 × 800 mL) and deionized H_2O (1 L). The combined filtrates were filtered through Celite and stripped of solvent at high vacuum (25–30 °C bath); yield: 205 g (76%); mp 256–258 °C (dec); $[\alpha]_D^{25}$ +6.8 (*c* 1, 2 M NH_3). The resin could be regenerated from its copper form.

H-Lys(Boc)-OH; Typical Procedure:[57]
To an intensely stirred suspension of Cu[Lys(Boc)]$_2$ (277 g, 0.5 mol) in H_2O (10 L), quinolin-8-ol (189 g, 1.3 mol) was added. After 5 h, copper(II) quinolin-8-ate was filtered off and washed with H_2O (0.5 L). The precipitate was suspended in H_2O (5 L) and left standing for several h. It was filtered off and washed with H_2O (2 × 0.5 L). All combined filtrates and washings were extracted with EtOAc (2 × 3 L). The aqueous phase was taken to dryness; yield: 237 g (96%).

Z-Lys(Boc)-OH from Z-Lys-OH; Typical Procedure:[167]
To a soln of Z-Lys-OH (280 g, 1 mol; prepared by benzyloxycarbonylation of N^{ε}-benzylidenelysine[76]) in H_2O (1.4 L) was added at rt dioxane (0.8 L), Boc-N_3 (160 g, 1.1 mol), and MgO (60 g, 1.5 mol). The mixture was stirred at 40 °C for 8 h and at rt overnight, filtered and concentrated at reduced pressure (<60 °C) to about one-third of its volume. To the residue was added H_2O (2 L) and EtOAc (3 L) and, with shaking, concd aq citric acid, until the aqueous phase had pH 3. The organic layer was separated, washed with H_2O (200 mL), and dried ($MgSO_4$), and the solvent was removed; yield: 330 g (86%).

Z-Lys(Boc)-OH·DCHA from Cu[Lys(Boc)]$_2$; Typical Procedure:[57]
To an intensively stirred suspension of Cu[Lys(Boc)]$_2$ (55.4 g, 100 mmol) in acetone (0.2 L) and 10% aq Na_2CO_3 (0.4 L) was added quinolin-8-ol (29.9 g, 206 mmol) and the mixture left for 1 h (mixture 1). To a soln of HOSu (25.3 g, 220 mmol) in H_2O (125 mL) placed in a separate vessel, Na_2CO_3 (11.7 g, 110 mmol) was introduced, followed by acetone (0.1 L). Z-Cl (30 mL, 200 mmol; 96% purity) was added dropwise and the whole mixture was left standing for 0.5 h with occasional stirring. This mixture was poured into mixture 1 and, after 1 h, the precipitate was filtered off and washed with H_2O (4 × 0.1 L). The filtrate was acidified with 1 M HCl under stirring to pH 2 and extracted with EtOAc (0.8 L). The organic phase was washed with H_2O, dried, and treated with DCHA (40 mL, 200 mmol). After several hours, the crystalline solid was filtered off and washed with a mixture of EtOAc/hexane (1:1); yield: 103 g (91%); mp 152–154 °C.

Fmoc-Lys(Boc)-OH; Typical Procedure:[57]
Mixture 1 was obtained as described above. To a soln of HOSu (24.3 g, 211 mmol) in H_2O (120 mL) placed in a separate vessel, Na_2CO_3 (11.2 g, 106 mmol) was introduced, followed by acetone (0.1 L). Fmoc-Cl (49.6 g, 192 mmol) was added and the whole was left for 0.5 h with occasional stirring. This mixture was diluted with acetone (340 mL) and then poured into the stirred mixture 1. After 1 h, the precipitate was filtered off and washed with H_2O (4 × 0.1 L). The filtrate was acidified with 1 M HCl under stirring to pH 2. The resulting precipitate was filtered off, washed with H_2O, dried, and crystallized (EtOAc/hexane); yield: 80.5 g (86%); mp 129–130 °C.

2.1.2.2.3.3.2 Other Urethane-type N^{ω}-Protecting Groups Derived from Tertiary Alcohols

As an alternative to the Boc group, N^{ω}-Adoc protection has found extensive application, particularly in solution synthesis, due to its favorable effects on the solubility of intermediate fragments in organic solvents as documented in the synthesis of a 129-membered peptide chain related to an analogue of hen egg lysozyme,[168–170] but also of shorter peptide fragments.[15] Moreover, N^{ω}-Adoc protection is more acid-stable than the respective Boc protection[168] and, thus, also better suited for a combination with the N^{α}-Bpoc protection.[94] The Adoc group is introduced into ω-amino functions by identical procedures as the Boc group and it is cleaved under identical acidic conditions. For more detailed information see Section 2.1.1.1.3.2.

Among other urethane-type N^{ω}-derivatives derived from tertiary alcohols, the high acid lability of the Bpoc group and, in particular, of the 2-(1-adamantyl)prop-2-yloxycarbonyl group[43] can be utilized for the selective side-chain protection of a diamino acid residue among others present in the peptide chain when regioselective transformations are the goal. This protecting group is quantitatively removed on resin by treatment with 1% trifluoroacetic acid in dichloromethane for 10 minutes. This approach was successfully used for side-chain to side-chain cyclization of peptides on resin[44] in combination with the 2-phenylprop-2-yl ester,[171,172] which is concurrently removed by the 1% trifluoroacetic acid solution. Thus, such a combination of side-chain amino and carboxy protection offers a valid alternative to the Aloc/allyl ester approach.

H-Orn(Adoc)-OH; Typical Procedure:[169]
H-Orn-OH·HCl (84.3 g, 0.5 mol) was dissolved in 1 M NaOH (1 L) and a soln of $CuSO_4 \cdot 5H_2O$ (62.5 g, 0.25 mol) in H_2O (750 mL) was added. After stirring for 15 min, the soln was cooled to 0 °C and 0.55 M Adoc-Cl in dioxane (1 L, 0.55 mol) was added. Solid Na_2CO_3 (159 g, 1.5 mol) was added as required to maintain the pH at ca. 10. The soln was maintained at 0 °C for 2 h, then allowed to warm to rt overnight. The resulting blue suspension was diluted with H_2O (2 L) and filtered. The solid was washed with H_2O until the washings were colorless, then washed with ice-cold EtOH and Et_2O to remove unreacted Adoc-Cl. The finely powdered copper complex (94.0 g, 138 mmol) was suspended in a mixture of H_2O (3 L), AcOH (60 mL), and MeOH (1 L), and H_2S was passed through the suspension for 4 h. Air was passed through the suspension to remove excess H_2S, which was then warmed to nearly boiling and filtered using a double-fluted filter paper. The volume of the filtrate was then concentrated under reduced pressure until the crystalline product was obtained; this was then washed with cold H_2O; yield: 59.7 g (39%); mp 214–216 °C; $[\alpha]_D^{24}$ −7.8 (*c* 1, MeOH).

H-Lys(Adoc)-OH; Typical Procedure:[169]
H-Lys-OH·HCl (91.3 g, 0.50 mol) was reacted with Adoc-Cl (0.55 mol) as described for H-Orn-OH to yield the copper complex (100 g, 141 mmol). The finely powdered complex was added to a boiling soln of EDTA disodium salt (142 g) in H_2O (3.6 L) and stirred until dissolved. The soln was allowed to cool to rt and the product crystallized. After cooling to 0 °C for 2 h to complete crystallization the crystals were collected and washed with ice water yielding H-Lys(Adoc)-OH; yield: 65.0 g (40%); recrystallization may be carried out (MeOH/Et_2O); mp 227–229 °C; $[\alpha]_D^{24}$ −1.3 (*c* 1, MeOH).

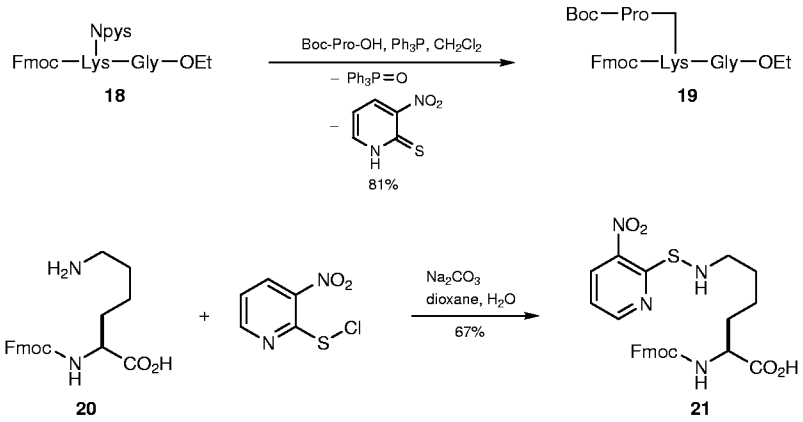
for references see p 188

2.1.2.2.4 Sulfenamide-Based N^{ω}-Protecting Groups

While the 2-nitrophenylsulfanyl group (Nps) is only suitable for N^{α}-protection due to its high acid lability (see Section 2.1.1.4), the 3-nitro-2-pyridylsulfanyl (Npys) group, which is used for amino and hydroxy protection[39,173] as well as for protection/activation of cysteine thiol functions (see Section 2.6.6),[174] offers interesting applications even for ω-amino protection. The 3-nitro-2-pyridylsulfanyl group is stable towards trifluoroacetic acid and formic acid, but cleaved by triphenylphosphine, dilute HCl (e.g., 0.1–0.2 M HCl in dioxane),[39] 2-sulfanylpyridine 1-oxide,[39,40,175] pyridine-2-thiol or 1-methyl-1*H*-imidazole-2-thiol.[41] Occasionally it has proved advantageous to add HOSu, HOBt, or 2-sulfanylpyridine 1-oxide, to facilitate cleavage.[39] Quantitative removal of the N^{ω}-Npys groups on resin has proved to be more difficult than in solution, and as well documented by a comparative analysis, 2-sulfanylpyridine 1-oxide in large excess is the reagent of choice.[40] These stability and cleavage properties of the Npys group allow for interesting combinations with the Z, Boc, Bpoc, Fmoc, and Aloc groups, particularly in cases where several amino functions must be protected in a selective manner, e.g. in the synthesis of template-assembled synthetic peptides.[40] As a special feature of this protecting group, it can be exploited for peptide bond formation via the Mukaiyama reaction[176] by in situ deprotection with phosphines in the presence of the carboxy component, e.g. formation of the ε-Pro peptide **19** from **18** (Scheme 7).[39]

The Npys group is readily introduced under Schotten–Baumann conditions using Npys-Cl, e.g. protection of Fmoc-Lys-OH (**20**) with Npys-Cl to give Fmoc-Lys(Npys)-OH (**21**) (Scheme 7).[39,177] This reagent is prepared by chlorination of bis(3-nitro-2-pyridyl) disulfide;[173,175] it is also commercially available, but it is worthy to note that it is extremely sensitive to moisture.

Scheme 7 Introduction of the Npys Group; Cleavage and In Situ Coupling of a Further Amino Acid[39]



3-Nitropyridine-2-sulfenyl Chloride (Npys-Cl):[173]
To a suspension of bis(3-nitro-2-pyridyl) disulfide (3.1 g, 10 mmol) in CH_2Cl_2 (100 mL), Cl_2(g) was introduced with vigorous stirring at 0 °C for 30 min and at rt for 1 h. The mixture became almost clear and was filtered to remove a trace of tar in the mixture; the solvent was then removed and Npys-Cl was obtained as yellow needles from the residue; yield: quant; mp 217–222 °C (dec); other authors have reported a lower mp such as 91–94 °C;[40] 96–98 °C;[175] 95 °C.[178]

Fmoc-Lys(Npys)-OH (21); Typical Procedure:[39,177]
To a soln of Fmoc-Lys-OH (**20**; 7.3 g, 20 mmol) in H_2O (10 mL) and dioxane (25 mL) was added Na_2CO_3 (2.12 g, 20 mmol) followed by Npys-Cl (4.2 g, 22 mmol) in four equal portions over a period of 10 min, while Na_2CO_3 (2.12 g, 20 mmol) in H_2O (12 mL) was added dropwise with vigorous stirring at 0 °C. After 30 min, the soln was diluted with H_2O (200 mL), filtered, and extracted with Et_2O (100 mL). The aqueous layer was acidified with 0.5 M H_2SO_4 and extracted with EtOAc (2 × 200 mL). The organic layer

was washed with H_2O, dried, and the solvent removed under reduced pressure; yield: 7.0 g (67%); mp 112–115 °C (dec); $[\alpha]_D^{22}$ +32.1 (*c* 1, MeOH).

H-Lys-Tyr-OH; Cleavage of the Npys Group:[41]
Thiolysis of Npys-Lys(Npys)-Tyr(Npys)-OH (77 mg, 0.1 mmol) was carried out with pyridine-2-thiol (56 mg, 0.5 mmol) in DMF (0.5 mL) and was completed, as examined by TLC, in less than 30 min. The product was isolated by trituration with dry Et_2O; yield: 25 mg (82%); purity >95%; $[\alpha]_D^{25}$ −40.4 (*c* 0.5, H_2O).

Fmoc-Lys(Boc-Pro)-Gly-OEt (19); Typical Procedure:[39]
To a stirred soln of Fmoc-Lys(Npys)-Gly-OEt (**18**; 152 mg, 0.25 mmol) and Boc-Pro-OH (59.3 mg, 0.275 mmol) in CH_2Cl_2 (50 mL), Ph_3P (131 mg, 0.5 mmol) was added and stirring was continued. After 1 h, Ph_3P (6.5 mg, 0.025 mmol) and 2,2′-dipyridyl disulfide (5.5 mg, 0.025 mmol) were added to complete the reaction since a small amount of Fmoc-Lys-Gly-OEt, hydrolyzed product of the intermediate of oxidation–reduction condensation, was observed along with the main product on TLC. The mixture was stirred overnight. The solvent was removed, and the residue was dissolved in EtOAc, followed by successive washing with 5% citric acid, 10% $NaHCO_3$, and H_2O. After removal of solvent, the residue was crystallized (EtOAc/petroleum ether); yield: 131 mg (81%); mp 140–149 °C (dec); $[\alpha]_D^{25}$ −26.6 (*c* 1, DMF).

2.1.2.2.5 N^{ω}-Alkyl Protecting Groups

Whilst the N^{ω}-benzylidene diamino acids are employed only for the production of asymmetrically protected derivatives (see Section 2.1.2.2.1.3), of the *N*-alkyl-type protecting groups the N^{ω}-triarylmethyl derivatives are used as an extension of the Fmoc/tBu strategy,[179] since this type of amino protection is significantly more acid-sensitive than the Boc group (see Section 2.1.1.6.3). Similarly, the 1-(4,4-dimethyl-2,6-dioxocyclohexylidene)ethyl (Dde)[47] and 1-(4,4-dimethyl-2,6-dioxocyclohexylidene)-3-methylbutyl group (known as ivDde)[48] are valuable tools for a third level of selective protection of ω-amino functions in the Fmoc/tBu approach as well as for synthesis of polyamines. Both classes of protecting groups permit the construction of branched, cyclic, and side-chain-modified peptides.

2.1.2.2.5.1 N^{ω}-Triarylmethyl Derivatives

The N^{ω}-Trt groups are somewhat more stable than the N^{α}-Trt derivatives. In fact, cleavage of Trt-Lys-OH or Trt-Orn-OH with 1% trifluoroacetic acid in dichloromethane occurs within 3 minutes at 0 °C, while for the corresponding N^{ω}-derivatives 2 hours at room temperature are required for deprotection with 5% trifluoroacetic acid in dichloromethane.[180] This differentiated acid lability is exploited for the synthesis of N^{ω}-triarylmethyl-protected lysine and ornithine derivatives that are readily obtained by selective N^{α}-deprotection of α,ω-bis-protected diamino acids via treatment with mild acids.[45,180] The subsequent N^{α}-protection, e.g. with the Fmoc group, is then carried out by standard procedures under Schotten–Baumann conditions.[181]

Although N^{ω}-trityl derivatives of diamino acids are significantly more acid-labile than the related Boc derivatives, as shown by a comparative study on a model peptide using 65% trifluoroacetic acid in dichloromethane,[179] the N^{ω}-Mtt derivatives are better suited for the selective protection of ω-amino groups versus other N^{ω}-Boc derivatives. In fact, this protection is smoothly removed by mild acid treatments (Table 4) without affecting *tert*-butyl alcohol derived protecting groups. The deprotection of Lys(Mtt)-containing peptides on resin is easily monitored by following the UV absorbance, but as other trityl groups are also removed, monitoring of MttOH versus TrtOH by RP-HPLC is more advantageous.[182] Cleavage of the Mtt groups with AcOH/TFE/CH_2Cl_2 (1:2:7) fails when hydrophilic resins such as TentaGel or cellulose are used.[183] In these cases, N^{ω}-Mmt protection is more advisable as this type of protection is cleaved on resin with AcOH/TFE/CH_2Cl_2 (1:2:7).[183] The N^{ω}-DMTr derivatives are even more sensitive toward acidic reaction conditions, though a systematic investigation has not been reported.[183]

for references see p 188

Table 4 Acidolytic Cleavage of N^{ω}-Protected Diamino Acid Residues[45,179,183–187]

Protecting Group	Cleavage Conditions	Ref
Trt	5% TFA/CH_2Cl_2, rt, 30 min	[179]
Mtt	>5% TFA/CH_2Cl_2, rt, 1 min	[45]
	1% TFA/CH_2Cl_2, rt, 30 min	[45,184]
	1% TFA and 5% TIS/CH_2Cl_2	[185]
	0.5% TFA/CH_2Cl_2, rt	[186]
	AcOH/TFE/CH_2Cl_2 (1:2:7), rt, 1 h	[45]
Mmt	AcOH/TFE/CH_2Cl_2 (1:2:7), rt, 1 h	[183]
	0.15–0.2 M Cl_2CHCO_2H/anisole (100 equiv), CH_2Cl_2, rt, 1 h	[187]

The Trt group, as well as the related Mtt and Mmt derivatives, are sometimes preferable over Boc protection, since the low electrophilicity of the extremely bulky triarylmethyl cations that are formed during cleavage does not lead to any side reaction at the level of sensitive amino acid residues such as Trp, Met, and Tyr.[179,184] As an example, the peptide H-Thr-Thr-Trp-Ser-Met-Ser-Trp-Tyr-OH was obtained with 94% homogeneity by using the Fmoc/Trt strategy as compared to 44% of the peptide prepared by using the standard Fmoc/tBu protection.[184]

Trt-Lys(Trt)-OH·DEA and Trt-Orn(Trt)-OH·DEA:[180]
To a suspension of H-Orn-OH or H-Lys-OH (3.30 g or 3.65 g, 25 mmol) in $CHCl_3$/MeCN (3.5:1, 100 mL) was added TMSCl (3.17 mL, 25 mmol), and the mixture was refluxed for 2 h. Then TEA (10.5 mL, 75 mmol) was added and stirring was continued at rt for 30 min. After addition of Trt-Cl (13.6 g, 50 mmol) and further stirring at rt for 2 h, the mixture was concentrated to a slurry, and partitioned between 10% citric acid and $CHCl_3$. The organic layer was washed with brine (2 ×), and H_2O, dried, and the solvent was removed. The residue was dissolved in acetone, DEA (2.5 mL, 25 mmol) was added and the mixture was left at rt for 1 d. The precipitate was collected, washed with Et_2O, and recrystallized (acetone); yields: Trt-Lys(Trt)-OH·DEA (58%); mp 145–147 °C (softening), 158–159 °C (melting); $[\alpha]_D^{20}$ +14.2 (*c* 1, MeOH); Trt-Orn(Trt)-OH·DEA (64%); mp 153–155 °C; $[\alpha]_D^{20}$ +16.7 (*c* 1, MeOH).

H-Lys(Trt)-OH·TFA and H-Orn(Trt)-OH·TFA by Selective Removal of the N^{α}-Trt Group:[180]
To the dityrityl-protected diamino acid (2–5 mmol) was added at 0 °C 1% TFA in CH_2Cl_2 (20–50 mL), and the soln was stirred for 3 min and concentrated (0 °C). The residue was triturated with Et_2O/petroleum ether (1:1), and the precipitated product was collected, washed with Et_2O and dried; yields: H-Lys(Trt)-OH·TFA (77%); mp 85–89 °C (dec); H-Orn(Trt)-OH·TFA (82%); mp 85–88 °C; $[\alpha]_D^{20}$ +7.1 (*c* 2, $CHCl_3$).

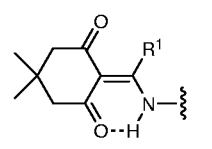
H-Lys(Mtt)-OH; Typical Procedure:[45]
To a suspension of H-Lys-OH·HCl (2.5 g, 13.7 mmol) in CH_2Cl_2 (25 mL) was added TMSCl (2.6 mL, 20.5 mmol) and the resulting mixture was refluxed for 30 min. After cooling to rt, TEA (8.5 mL, 61.5 mmol) and Mtt-Cl (8.0 g, 27.4 mmol) were added, stirring was continued at rt for 15 h and H_2O was added. The organic phase was separated, dried, and concentrated. To the oily residue ice-cold 1% TFA in CH_2Cl_2 (120 mL) was added and the mixture was stirred at 0 °C for 30 min and at rt for 1 h, and then TEA (2.3 mL, 16.4 mmol) was added. After concentrating the mixture, the residue was partitioned between Et_2O and 5% citric acid, and the organic phase was washed once more with 5% citric acid. The aqueous soln was washed with Et_2O and the pH was adjusted to 6.5 with 4 M NaOH. The resulting precipitate was filtered, washed with H_2O, dried on air, washed with Et_2O, and dried; yield: 3.5 g (64%); mp 170–172 °C; $[\alpha]_D^{22}$ −2.1 (*c* 1, MeOH).

2.1.2.2.5.2 N^{ω}-1-(4,4-Dimethyl-2,6-dioxocyclohexylidene)ethyl and N^{ω}-1-(4,4-Dimethyl-2,6-dioxocyclohexylidene)-3-methylbutyl Derivatives

The N^{ω}-1-(4,4-dimethyl-2,6-dioxocyclohexylidene)ethyl (Dde **22**, R^1 = Me)[47] and N^{ω}-1-(4,4-dimethyl-2,6-dioxocyclohexylidene)-3-methylbutyl (**22**, R^1 = iBu)[48,199] derivatives of diamino acids are stable towards trifluoroacetic acid, piperidine/DMF (20% piperidine in NMP for 117 min cleaves the Dde group to an extent of only 2.1%[188]) and DBU, but are easily cleaved with 2% hydrazine in DMF. The preferential protection of primary versus secondary amines is explained by the formation of hydrogen bonds in the product.[189] Further deriv-

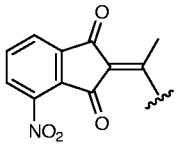
atives of this type of protecting group have been developed (Scheme 8), but they have not found widespread application in peptide chemistry, e.g. the N^{ω}-1-(4-nitro-1,3-dioxo-1,3-dihydro-2*H*-inden-2-ylidene)ethyl group **23**,[190] which is prepared from 2-acetyl-4-nitro-1*H*-indene-1,3(2*H*)-dione.[191] Preliminary experiments confirmed an even faster cleavage with 2% hydrazine in DMF.[190,192]

Scheme 8 Dimedone-[48] and Nitroindanedione-Derived Protecting Groups[190]



R^1 = Me (Dde), Et, Pr, iBu, Bzl

22

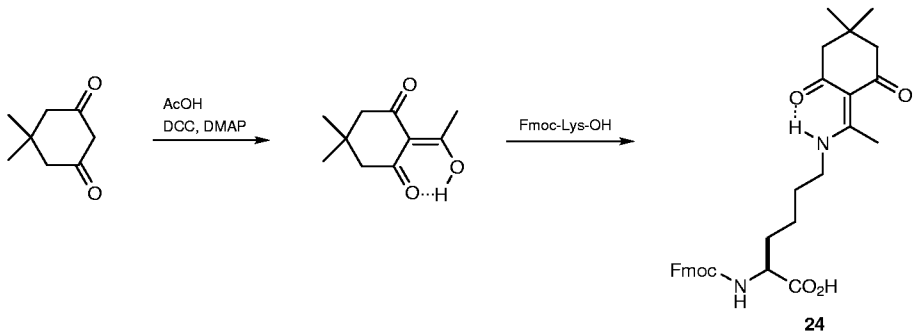


23

The N^{ω}-Dde protection has some limitations that derive from migration of the Dde group during piperidine-mediated N^{ε}-Fmoc deprotection to the unprotected Lys residue[193,194] and from the β- to the α-amino group of diaminopropanoic acid.[195] The latter side reaction is observed even with DBU as the cleavage base.[195] However, it can be prevented by cleaving the Fmoc group with DBU using short reaction times (2%, 3 × 3 min).[193,194] Nevertheless, the use of DBU is restricted to solid-phase applications, since in solution removal of the byproduct dibenzofulvene is difficult when this base is used. Therefore, the subsequent reaction of the dibenzofulvene with the N-terminal amino function would lead to a chain termination. During the synthesis of longer peptides by the Fmoc/tBu chemistry partial loss of the Dde group has been noted.[48] This loss of the Dde protection is negligible when using dilute piperidine solutions (0.5 M in DMF), but it reaches about 5% when 50% piperidine is used for 2 hours. As much as 70% loss has been observed with 0.5 M pentylamine for 2 hours.[194] To address this problem, the 1-(4,4-dimethyl-2,6-dioxocyclohexylidene)-3-methylbutyl group has been developed, which is fully stable towards 20% pyridine in DMF, and intramolecular N→N′ migration has not been observed.[48,199]

The Dde-protected amino acids are readily synthesized by acylation of dimedone (5,5-dimethylcyclohexane-1,3-dione) with acetic acid (DCC/DMAP activation),[47,196] followed by condensation with the primary amine, e.g. Fmoc-Lys-OH, in refluxing ethanol for 60 hours, affording the orthogonally protected amino acid **24** (Scheme 9).

Scheme 9 Synthesis of Dde-Protected Amino Acids[47]

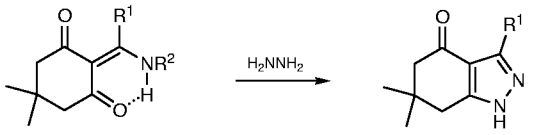


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for references see p 188

Cleavage of the N^{ω}-Dde and N^{ω}-1-(4,4-dimethyl-2,6-dioxocyclohexylidene)-3-methylbutyl groups is performed with 2% anhydrous hydrazine or hydrazine hydrate in DMF at room temperature (on resin, 3×2 min). This leads to a 1,5,6,7-tetrahydro-4*H*-indazol-4-one **25** as the byproduct (Scheme 10), which can be monitored by UV spectroscopy at 220 or 290 nm.[48] Since hydrazine will also remove Fmoc groups, the assembly of the peptide by Fmoc/tBu chemistry must be completed prior to the cleavage of the Dde or 1-(4,4-dimethyl-2,6-dioxocyclohexylidene)-3-methylbutyl groups.

Scheme 10 Formation of 1,5,6,7-Tetrahydro-4*H*-indazol-4-ones



25

It must be remembered that hydrazine treatment of peptides containing allyl-based protecting groups also results in reduction of the allyl moiety;[197] this is successfully prevented by adding a 200-fold excess of allyl alcohol as a scavenger.[198] Moreover, hydrazine treatment of peptides may lead to aspartimide formation in the case of cyclization-prone Asn-Xaa and Asp-Xaa sequences (see Section 2.2.2).

Fmoc-Lys[1-(4,4-dimethyl-2,6-dioxocyclohexylidene)-3-methylbutyl]-OH; Typical Procedure:[48]
To a stirred suspension of Fmoc-Lys-OH (0.72 g, 1.95 mmol) and 2-(1-hydroxy-3-methylbutylidene)-5,5-dimethylcyclohexane-1,3-dione (0.874 g, 3.9 mmol) in EtOH (15 mL) at rt TFA (15 µL, 0.195 mmol) was added. The mixture was heated under reflux for 60 h and the reaction was monitored by analytical TLC. The solvent was removed, and the orange residue was redissolved in EtOAc (50 mL). The organic phase was washed with aq 1 M $KHSO_4$ (2×40 mL), dried, and concentrated to a yellow oil. The oil was triturated with hexane to remove any unreacted starting material and then it was recrystallized (EtOAc/hexane) to afford the desired product as a colorless crystalline solid; yield: 0.72 g (64%); mp 69–71 °C.

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2.2 Carboxy Group

2.2.1 α-Carboxy Group

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α-Carboxy protection is recommended during peptide synthesis to avoid any reaction of the carboxylate anion with the acylating agent, to promote the solubility of reactants under coupling conditions and to facilitate purification of the products.[1] Competition of the α- as well as of ω-carboxylate anions, if present, with the amino group for the acylating agent has been observed in several cases.[2–5] The transient formation of a reactive intermediate might then lead to a loss of the chiral purity of the C-terminal amino acid residue, if they are used without carboxy protection.[6] Nevertheless, peptide syntheses are often performed without protection of the carboxy group(s) of the amino component.[7] Sodium salts of amino acids and peptides in water or aqueous dioxane, salts with tertiary bases (e.g., TEA, DIPEA, NMM, or 1,1,3,3-tetramethylguanidine), with quaternary amines such as benzyltrimethylammonium hydroxide, or quaternary phase transfer reagents (e.g., benzyltrimethylammonium chloride or tetrabutylammonium hydrogen sulfate) in organic solvents or aqueous organic mixtures are efficiently acylated by active esters, mixed anhydrides, or azides.[8–10] The use of either tetrabutylammonium salts[11,12] or DMF containing strong acids (TFA, HBF_4, or TosOH) and an excess of tertiary base with $pK \leq 6$, preferably pyridine,[13] has also been proposed as a method to dissolve amino acids in organic solvents for their acylation. Amino acids and peptides are acylated efficiently even in the zwitterionic form as a suspension in organic solvents such as DMF,[14,15] whereby sonication of the suspension was found to facilitate the reaction.[16]

The most important considerations in choosing semipermanent protecting groups for the α-carboxy group are their stability during synthesis and the conditions required for their removal after the peptide chain lengthening process to ensure chemical as well as chiral integrity of the product. For semipermanent carboxy protection methyl-, ethyl-, benzyl-, *tert*-butyl-, silyl-, or aryl-type ester groups are commonly used. Protected hydrazides offer an alternative efficient α-carboxy protection, which additionally allow conversion into azides or diimides[17] for further coupling steps at the carboxy terminus (see Section 3.1.1.4). Together with aryl esters, e.g. 4-(methylsulfanyl)phenyl and O-protected 2-hydroxyphenyl esters, the protected hydrazides represent an interesting approach for dual-function protecting groups. Both aryl esters are readily converted into reactive acylating agents, the former by oxidation to the 4-(methylsulfonyl)phenyl ester and the latter by removal of the hydroxy protection to form the catechol monoester. The dual-function protecting groups are of obvious interest for the preparation of activated intermediates for the synthesis of sequential polypeptides, cyclic peptides, and peptide intermediates for segment condensations.[18]

Several active esters (e.g., ONp, OPcp, and OTcp) generally used as acylating agents in peptide synthesis, are occasionally used for α-carboxy protection with the objective of synthesizing protected peptide active esters.[19–21] This backing-off[20] approach (see Section 3.2.2.2) is limited to syntheses in which highly reactive acylating amino acid species such as mixed anhydrides are used in the coupling step for N-acylation of amino acid or peptide active esters. The success of this method strongly depends upon the relative rates of formation of the peptide in comparison to intra- or intermolecular condensations of the active ester component.[18] Pentafluorophenyl esters have also been used for the temporary protection/activation of the α-carboxy group in the preparation of side-chain glycosylated N^{α}-Fmoc amino acid synthons for solid-phase synthesis of glycopeptides (see Vol. E 22b, Section 6.3).[22]

Due to the absence of any sufficiently selective chemical deprotection procedures, amides generally represent irreversible carboxy protection. An enzymatic method for the hydrolysis of α-amides using highly selective peptide amidases has been proposed.[23]

for references see p 228

2.2.1.1 Alkyl Esters

2.2.1.1.1 Methyl and Substituted Methyl Esters

The methyl and substituted methyl esters most frequently used in peptide chemistry and the corresponding deprotection procedures are summarized in Table 1.

Table 1 Methyl and Substituted Methyl Esters[24,32–36,38,39,72–74,77,80,81,83,85,87–89,92–96,98–100,102,106,109–114,118,146]

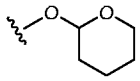
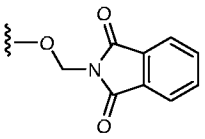
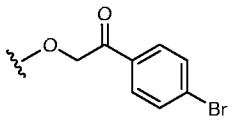
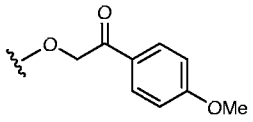
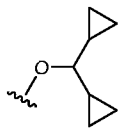
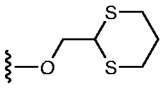
Esters	Structure	Deprotection Procedures[a]	Ref
methyl (Me) ester	OMe	alkali, aq organic mixtures	[24,33,34]
		Bu_4NOH, aq organic solvents, ≤0 °C	[32]
		BBTO, toluene, 70–100 °C, 24–36 h	[35,36]
		TMSI	[38,39]
methoxymethyl (Mom) ester	OCH_2OMe	$MgBr_2$ (2 equiv), Et_2O, rt, 1–24 h	[77]
(2-methoxyethoxy)methyl (Mem) ester	$OCH_2OCH_2CH_2OMe$	3 M HCl, THF, 40 °C, 12 h	[72]
		$MgBr_2$ (2 equiv), Et_2O, rt, 1–24 h	[77]
benzyloxymethyl (Bom) ester	OCH_2OBzl	concd HCl/H_2O/THF (8:6:36), rt, 2 h	[73]
		H_2 (1 atm), Pd/C, EtOH, rt	[73]
tetrahydropyran-2-yl ester		AcOH/THF/H_2O (4:2:1), 45 °C, 3.5 h	[74]
(methylsulfanyl)methyl (Mtm) ester	OCH_2SMe	TFA, rt, 15 min	[80]
		1. $H_2O_2/(NH_4)_6Mo_7O_{24}$ 2. NaOH, pH 11	[81]
		$MgBr_2$ (2 equiv), Et_2O, rt, 1–24 h	[77]
		1. MeI, acetone, reflux, 24 h 2. 1 M NaOH	[81]
2-(trimethylsilyl)ethoxy-methyl (SEM) ester	$OCH_2OCH_2CH_2TMS$	48% HF/MeCN (15:85), −10 °C, 5 h	[83]
		$MgBr_2$ (2 equiv), Et_2O, rt, 1–24 h	[77]
phthalimidomethyl ester		2 M HBr/AcOH, rt, 10–15 min	[85]
		Zn, AcOH, rt, 12 h	[87]
		DEA, MeOH, H_2O, rt, 24 h	[85]
		$H_2NNH_2 \cdot H_2O$, MeOH, rt, 3 h	[85]
		NaOH (2 equiv), MeOH/H_2O (1:1), rt, 45 min	[85]
phenacyl (Pac) ester	OCH_2COPh	Zn, AcOH, rt, 1 h	[92,106,109,110]
		NaSPh, DMF, rt, 30 min	[88,89,94,95]
		H_2, Pd/C, aq MeOH, rt, 1h	[89,93]
		TMTOH or BBTO, $(CH_2Cl)_2$, reflux, 15–25 h	[96]
		$H_2NNH_2 \cdot H_2O$, DMF/MeOH (4:1), rt, 20 h	[98]
		TBAF, DMF, rt, 30 min	[99,100]
4-bromophenacyl ester		NaSPh, DMF, rt, 30 min	[111]
4-methoxyphenacyl ester		*hν* (313 nm), dioxane or EtOH, rt, 5–17 h	[112]

Table 1 (cont.)

Esters	Structure	Deprotection Procedures[a]	Ref
1-methyl-2-oxo-2-phenyl-ethyl (α-methylphenacyl) ester	OCH(COPh)Me	*hν* (313 nm), dioxane or EtOH, rt, 6 h	[112]
		$H_2NNH_2 \cdot H_2O$, DMF, rt, 20 h	[113]
		KCN, 18-crown-6, NMP, rt, 8–24 h	[102]
acetonyl ester	OCH_2COMe	TBAF, THF, rt, 5–45 min	[114]
dicyclopropylmethyl (Dicycpropme) ester		1% TFA, CH_2Cl_2, 15 min	[118]
1,3-dithian-2-ylmethyl ester		1. $H_2O_2/(NH_4)_6Mo_7O_{24}$ 2. NaOH, pH 8	[146]

[a] v/v unless otherwise stated.

2.2.1.1.1.1 Methyl Ester

Methyl esters are excellent permanent protecting groups, which additionally allow conversion of these amino acid or peptide derivatives into amides or hydrazides by simple aminolysis or hydrazinolysis, respectively. Their use for the semipermanent protection of the carboxy group is, however, rather limited as their removal, especially with growing peptide chain length, is fraught with difficulties. The simplest and most frequently used method for the hydrolysis of methyl esters is saponification with aqueous alkali, mostly in the presence of varying amounts of organic solvents such as dioxane, MeOH, EtOH, acetone, DMF, etc.[24] The saponification of methyl esters of larger peptides is difficult and requires harsher conditions such as excess alkali and long reaction times, which increase the possibility of many side reactions occurring. Even at low temperature and in the absence of excess alkali, side reactions might accompany saponification. Noteworthy are hydantoin formation at Z-[25–27] and Boc-protected N-termini,[28] decomposition of serine-containing peptides,[29,30] and racemization of the C-terminal amino acid residues,[31] especially when represented by S-protected cysteine.[26] To overcome the need for harsh conditions for the saponification of peptide methyl esters, tetrabutylammonium hydroxide has been suggested as an alternative that apparently drastically reduces the possibility of racemization;[32] tetrabutylammonium hydroxide is generally used in aqueous organic solvents, e.g. aqueous THF, DMF, acetonitrile, and acetone, at temperatures $\leq 0\,^{\circ}C$.

Preservation of the N^{α}-Fmoc protecting group under alkaline conditions is possible either by using 2 equivalents of lithium hydroxide (0.2 M in THF) at $0\,^{\circ}C$[33] or by addition of 0.8 mmol calcium chloride to a solution of 1.2 equivalents of sodium hydroxide in iPrOH/H_2O (7:3).[34] The latter mixture enabled efficient preparation of Fmoc-protected peptide segments by alkaline hydrolysis; for example, Fmoc-Gly-Phe-Pro-OH was obtained after 7 hours in 85% yield from Fmoc-Gly-Phe-Pro-OMe.

An efficient non-acidolytic and non-nucleophilic method for the selective cleavage of N^{α}-protected amino acid and short N^{α}-protected peptide methyl esters is the use of bis-(tributyltin) oxide (BBTO).[35,36] BBTO also cleaves phenacyl, benzyl, and several other primary alkyl esters.

Unfortunately, transesterification with 2-(dimethylamino)ethanol, followed by hydrolysis, a method which serves well in the case of benzyl esters (see Section 2.2.1.2.1), can be used for methyl esters only if thallium salts are used as a catalyst.[37] The cleavage of methyl esters with trimethylsilyl iodide which takes place under mild and neutral conditions and

for references see p 228

leaves amines, amides, and sulfides unaffected, has been proposed as a method of methyl ester cleavage in peptide synthesis, but so far has not found wide application.[38,39] Urethane-type protecting groups are concomitantly cleaved under these conditions.

Amino acid methyl esters are fairly simple to prepare. The classical method uses a suspension of amino acid hydrochloride in MeOH and gaseous HCl as catalyst;[40–42] dicarboxylic acids yield the dimethyl esters.[43] Hydrolytic and alcoholytic cleavage of side-chain amide groups of glutamine and asparagine as well as peptide bond cleavage prevents the use of this method for the esterification of peptide fragments.

Highly recommended for the efficient and racemization-free synthesis of amino acid methyl esters is esterification using thionyl chloride and MeOH as the solvent in the cold.[44,45] Chlorosulfinic acid methyl ester (MeOSOCl) is thought to be the reactive intermediate resulting from the reaction of thionyl chloride with the alcohol;[46] HCl and SO_2 as byproducts are easily removed from the equilibrium. The ideal ratio of reaction components is 10.4 equivalents amino acid/11.5 equivalents $SOCl_2$/78.1 equivalents MeOH.[47] During the addition of thionyl chloride to MeOH low temperatures (−10 °C) are recommended. In general, the reaction mixture is heated after suspension of the amino acid for 2–4 hours to 40 °C to assure complete esterification. For sensitive amino acids such as methionine, low temperatures (0 °C) and longer reaction times (2 d) are recommended. Alternatively, unprotected amino acids are reacted with 2,2-dimethoxypropane with HCl as catalyst.[48]

Diazomethane is highly recommended as a mild and efficient reagent for the preparation of methyl esters of N^{α}-protected amino acids or peptides.[49] Diazo(trimethylsilyl)-methane is reported to be a safer alternative to the volatile, toxic, and explosive diazomethane in organic syntheses.[50,51]

α-Methyl esters of amino dicarboxylic acids are accessible via reaction of their internal anhydrides with MeOH. Amino acids such as glucosaminic acid are esterified in MeOH in the presence of 2 equivalents of trimethylsilyl chloride.[52,53] The intermediate trimethylsilyl esters are then converted in situ into the corresponding methyl esters. A similarly efficient one-pot esterification of N-protected amino acids with various alcohols employs isopropenyl chloroformate whereby the reactive species is thought to be the mixed anhydride.[54] The protocol involves addition of the chloroformate (1.1 equiv) to the N-protected amino acid in the presence of TEA (1 equiv) and DMAP (0.1 equiv), followed by the alcohol (1 equiv) in CH_2Cl_2 at 0 °C.

A series of Z-protected amino acids was successfully methylated by refluxing a methanolic solution of the amino acid in the presence of 5 equivalents of boron trifluoride–diethyl ether complex.[55] Alkylation of cesium salts of protected amino acids and even protected peptides using iodomethane at room temperature generated the corresponding esters in efficient yields under neutral conditions.[56] No racemization was observed during the very rapid process, which required only a few minutes for completion. Esterification of amino acid residues with the corresponding dicarbonates is catalyzed by DMAP. In fact, the reaction of dimethyl dicarbonate with Boc-Phe-OH in THF in the presence of 0.1 equivalents of DMAP proceeds smoothly within 15 minutes generating the methyl ester in 99% yield.[57]

Commonly used peptide coupling agents may also be used for the conversion of activated carboxylic acids into esters. Treatment of N-protected amino acids or peptides with ethyl 2-ethoxy-1,2-dihydroquinoline-1-carboxylate (EEDQ) results in mixed anhydride intermediates that produce methyl esters in the presence of excess MeOH (solvent) at room temperature.[58] The methyl esters are conveniently isolated by aqueous acid washing of the crude residue to remove the quinoline.

Another efficient method for the methyl ester protection of amino acids under mild conditions is the application of lithium hydroxide monohydrate/dimethyl sulfate in THF where the simultaneous electrophilic activation of the carboxy function and nucleophilic activation of dimethyl sulfate by virtue of coordination with Li^+ is exploited.[59]

Easy access to peptide methyl esters is offered by the transesterification of peptides linked to benzyl alcohol type resins such as the HMBA resin.[60] Tertiary amines generally serve as catalysts,[61] although even better results are obtained with potassium cyanide[62,63] or lithium bromide/DBU[64] as base catalysts. Note that Asp or Glu side-chain esters are likewise transformed under these conditions.

H-His-OMe · 2HCl; Typical Procedure for Esterification with MeOH/HCl:[65–67]
HCl(g) was bubbled through a suspension of H-His-OH · H_2O · HCl (985 g, 4.7 mol) in abs MeOH (1 L) at rt until saturation. After stirring overnight, the solvent was removed under reduced pressure (temperature $\leq$50 °C). The residue was immediately redissolved in warm MeOH, precipitated with Et_2O and recrystallized (MeOH/Et_2O); yield: 1073 g (93%); mp 202–203 °C (lit.[66] mp 205–206 °C); $[\alpha]_D^{20}$ +9.2 (*c* 2, H_2O).

H-Ser-OMe · HCl; Typical Procedure for Esterification with $SOCl_2$:[30]
Abs MeOH (100 mL) was cooled to −10 °C before freshly distilled $SOCl_2$ (26 mL) was added over 10 min. H-Ser-OH (10.5 g, 0.1 mol) was suspended in the mixture and stirred at rt until complete dissolution. After 24 h, the solvent was removed and the residue recrystallized (MeOH/Et_2O); yield: 15.5 g (99%); mp 168 °C; $[\alpha]_D^{23}$ −2.5 (*c* 1.8, DMF).

Amino acid methyl esters are obtained in about 90% yield applying this method.

H-Met-OMe · HCl; Typical Procedure for Esterification with 2,2-Dimethoxypropane/HCl:[48,68]
To a suspension of H-Met-OH (1.49 g, 10 mmol) in 2,2-dimethoxypropane (100–150 mL) was added concd HCl (10 mL) and the mixture was kept at rt overnight. After removal of the solvent (<60 °C), the residue was crystallized twice (dry MeOH/Et_2O, 250 mL); yield: 1.9 g (95%); mp 147–150 °C; $[\alpha]_D^{19.5}$ +25.2 (*c* 5.1, H_2O).

In the case of H-Lys-OH · HCl, the mixture was supplemented with MeOH (30–40 mL) to enhance solubility and the mixture was heated to reflux for 2 h prior to storage at rt overnight. The same modification, but with a reflux period of 5 h, was needed for the preparation of H-Glu(OMe)-OMe · HCl. Yields of amino acid methyl esters range from 80–95%.

Z-Gln-OMe; Typical Procedure for Esterification with Diazomethane:[69]
To an ice-cold stirred soln of Z-Gln-OH (21.5 g, 76 mmol) in abs EtOH (120 mL) freshly generated diazomethane in Et_2O was added carefully until persistence of the yellow color. After additional stirring for 30 min at low temperature, the excess diazomethane was destroyed with a few drops of AcOH. The product crystallized on storage at 0 °C; yield: 16.3 g (72%); mp 140–141 °C; $[\alpha]_D^{23}$ −19.4 (*c* 1, MeOH).

Z-Asp-OMe · DCHA; α-Methylation via Internal Anhydride:[70]
A mixture of freshly distilled Ac_2O (700 mL) and Z-Asp-OH (1.005 kg, 3.75 mol) was stirred at 45 °C for 4 h and the solvent removed. The solid residue was triturated with abs MeOH (4 L) and heated under reflux for another 4 h. After removal of the solvent, the oil was dissolved in Et_2O and extracted with several portions of aq $NaHCO_3$, and each fraction was separately acidified and stored in the cold. Fractions in which the Z-Asp-OMe separated in crystalline form were combined and dissolved in Et_2O. Upon addition of DCHA (1.05 equiv), the DCHA salt was filtered off and recrystallized (EtOH/petroleum ether); yield: 961 g (52%); mp 159–160 °C; $[\alpha]_D^{20}$ +14.9 (*c* 1, EtOH).

2.2.1.1.1.2 Substituted Methyl Esters

In a formal sense, alkyl ester type carboxy protecting groups, e.g. benzyl, allyl, 4-picolyl, and (9,10-dioxo-2-anthryl)methyl esters, can be viewed at as modified methyl esters. However, because of their specific properties and cleavage conditions these esters will be discussed separately. In this section a series of acetal, phthalimidomethyl, various phenacyl, acetonyl, 9-anthrylmethyl, and cyclopropylmethyl esters will be presented.

Acetal esters such as methoxymethyl (Mom),[71] (2-methoxyethoxy)methyl (Mem),[72] benzyloxymethyl (Bom),[73] tetrahydropyranyl,[74] 2-(trimethylsilyl)ethoxymethyl (SEM)[75] and (methylsulfanyl)methyl (Mtm)[76] are sensitive to mild aqueous acids in THF as well as to mild Lewis acids such as magnesium bromide in diethyl ether (2 equiv, rt, 1–24 h).[77] In addition, Bom esters may be cleaved by catalytic hydrogenation.[73] In the absence of acid, acetal esters display reactivities typical for unhindered alkyl esters. Surprisingly, these esters

for references see p 228

are seldom used in peptide synthesis although their preparation is facile and their removal achieved under mild conditions.

2.2.1.1.1.2.1 (Methylsulfanyl)methyl Ester

The (methylsulfanyl)methyl (Mtm) ester was introduced as a carboxy protecting group into peptide chemistry by Dossena et al.[78] A new convenient method for the preparation of these esters in which the use of chloroalkyl or hydroxyalkyl sulfide precursors is avoided, is based on the reaction of N-protected amino acids with mixtures of DMSO and *tert*-butyl bromide in the presence of sodium hydrogen carbonate for a few hours at room temperature.[76,79] Good yields were obtained with a series of Z-, Boc-, Nps-, or Trt-protected amino acids; Z-Glu-OH and Z-Asp-OH gave the diesters in excellent yields. Deprotection may be accomplished by acid hydrolysis using either a saturated solution of HCl in anhydrous diethyl ether (≈2 M) for a few minutes or TFA at room temperature (15 min).[80] The Mtm group can also be selectively removed from Z- and Boc-protected amino acids or peptides by oxidation with hydrogen peroxide/ammonium molybdate to yield the corresponding (methylsulfonyl)methyl esters which are readily hydrolyzed with sodium hydroxide at pH 11. An alternative deprotection method involves conversion of the sulfides into the corresponding sulfonium salts with iodomethane.[81] Mild alkaline hydrolysis of the intermediate sulfonium salts generates the free acids. Among the various N^{α}-protecting groups, only Nps and Trt have proved to be selectively removable with total retention of the Mtm ester. No racemization was observed during methylsulfanylmethylation and the deprotection procedure. It is noteworthy that the strongly enhanced lability of the intermediate (methylsulfonyl)methyl ester toward nucleophiles may also be utilized for carboxy activation in aminolytic processes.[82]

2.2.1.1.1.2.2 2-(Trimethylsilyl)ethoxymethyl Ester

As has been demonstrated, 2-(trimethylsilyl)ethoxymethyl (SEM) esters are selectively removed from amino acids and peptide derivatives in the presence of the most common N- and O-protecting groups applied in peptide chemistry including the urethane-type Boc, Z, Fmoc, and Troc as well as Bzl, tBu, and TBDMS ethers.[75] The SEM ester is removed by acidolysis or with a fluoride ion source, e.g. TBAF in THF or HMPA or with aqueous HF in MeCN (−10 °C).[83] Deprotection with magnesium bromide in Et_2O represents an even milder alternative that allows increased selectivity toward fluoride-labile silyl ethers or Fmoc groups. The SEM esters are prepared in 60–80% yield by stirring a mixture of 0.25 M N-protected amino acids in DMF with 0.8 equivalents of SEM-Cl and 1.1 equivalents of lithium carbonate at room temperature for 16 hours.

2.2.1.1.1.2.3 Phthalimidomethyl Ester

Phthalimidomethyl esters of Z-protected amino acids are easily accessible by their reaction with *N*-(chloromethyl)- or *N*-(bromomethyl)phthalimide followed by catalytic hydrogenolysis of the Z group.[84–86] N-Protected peptides can be esterified by the same procedure. Esterification via cesium salts[56] is equally suitable for the preparation of this type of ester.

These esters undergo acidolysis within 10–15 minutes by treatment with 2 M HBr in acetic acid at room temperature. They are stable toward standard catalytic hydrogenation, but are cleaved by reduction with Zn/AcOH (12 h, rt)[87] or by nucleophiles; sodium benzenethiolate readily cleaves the phthalimidomethyl esters (DMF, rt, 30 min).[88] Removal of the phthalimidomethyl group is also accomplished using 2 equivalents of sodium hydroxide in MeOH/H_2O (1:1) or 2–4 equivalents of diethylamine in MeOH at room temperature for

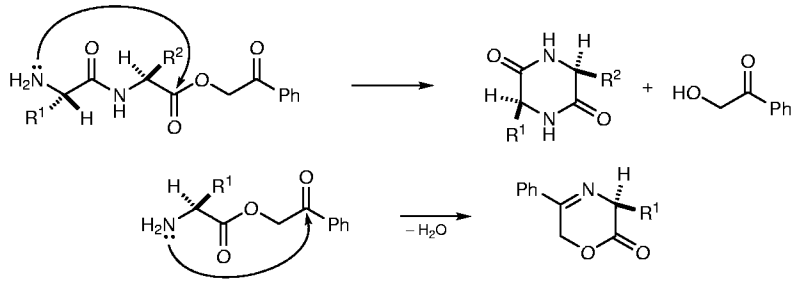
24 hours or alternatively, by refluxing for 2 hours. Reaction of phthalimidomethyl ester with excess hydrazine hydrate for 3 hours at room temperature results in the free carboxylic acid and phthalhydrazide.

2.2.1.1.1.2.4 Phenacyl Ester

Phenacyl (Pac) esters are smoothly prepared by reaction of phenacyl bromide with a carboxylate salt.[89,90] They are stable toward acid hydrolysis, e.g. concentrated HCl,[89] HBr/AcOH,[89] TFA/CH_2Cl_2 (1:1),[91] or HF (0 °C, 1 h).[91] Reduction with Zn/AcOH at room temperature removes the phenacyl group quantitatively,[92] while catalytic hydrogenation suffers from additional reduction of the oxo group to form the stable phenethyl ester.[89,93] Deprotection can be achieved under mild conditions very rapidly by treatment with sodium benzenethiolate in DMF at room temperature.[88,89,94,95] The nucleophile cleaves the alkyl–oxygen bond, leaving other esters like the benzyl ester unaffected. Trimethyltin hydroxide (TMTOH) and the sterically more demanding bis(tributyltin) oxide (BBTO) are reported to cleave Boc amino acids and peptides from phenacyl-functionalized polystyrene resin cleanly and efficiently in refluxing 1,2-dichloroethane (83 °C).[96,36] Trimethyltin hydroxide also cleaves Boc-protected amino acids and peptides from Merrifield, PAM, and Wang resins, but leaves ω-cyclohexyl esters of Asp and Glu almost completely intact, thus offering a potential orthogonal deprotection protocol.[97] Alternatively, hydrazinolysis[98] or treatment with tetrabutylammonium fluoride[99,100] are recommended for the cleavage of this type of peptidyl resin. Addition of HCN converts the Pac esters into the activated 2-cyano-2-phenylvinyl ester via cyanohydrin intermediates.[101] The active vinyl-type esters readily undergo aminolysis.

Due to their increased reactivity toward nucleophiles, dipeptide α-phenacyl esters are prone to piperazine-2,5-dione (DKP) formation.[102] Coupling of amino acid phenacyl esters with an activated carboxy component for dipeptide synthesis often results in substantial cyclization of the amino acid phenacyl esters via Schiff base formation (Scheme 1).[102–104]

Scheme 1 Piperazine-2,5-dione and Schiff Base Formation in Phenacyl Esters[102–104]



Numerous biologically active peptides and proteins have been successfully synthesized by Sakakibara et al.[105–107] adopting the maximum protection strategy based on N^{α}-Boc peptide phenacyl esters as intermediates for solution-phase fragment condensations (see Section 5.2). The suitably protected peptide phenacyl esters are stable under coupling conditions (WSCI, HOBt, TFE, $CHCl_3$), mostly offer good solubility properties in organic solvents,[108] and are generally crystalline compounds and thus easy to purify. Moreover, either the Boc or Pac group can be removed independently using TFA to yield an amino component or Zn/AcOH to produce the carboxy component for subsequent segment condensations. For the synthesis of the linear 238 residue *Aequorea* green fluorescent protein precursor molecule, the Pac group at the C-terminus of each large protected Boc-peptide segment was removed by heating with Zn powder in a mixture of 20% AcOH in HFIP at 40 °C for 2 hours.[109]

for references see p 228

Recently, similar conditions were applied to the deprotection of a Boc-peptide phenacyl ester fragment (1–41) during successful combined solid-phase and solution synthesis of muscarinic toxin 1 (MTX1).[110]

Z-Glu-OPac · DCHA; Typical Procedure for Esterification with Phenacyl Bromide:[93]
Phenacyl bromide (2 g, 10 mmol) was added to a soln of Z-Glu-OH (2.8 g, 10 mmol) and TEA (1.4 mL, 10 mmol) in DMF (7 mL). After 48 h, the mixture was diluted with EtOAc (100 mL) and washed several times with H_2O. The organic layer was dried (Na_2SO_4) and concentrated to 50 mL. Upon addition of DCHA (3 mL) the product precipitated in the cold; yield: 3.1 g (54%); mp 149–151 °C; $[\alpha]_D^{14}$ +16.5 (*c* 3, MeOH).

Z-Val-Val-OH; Typical Procedure for Thiolysis of Phenacyl Esters:[93]
NaSPh (390 mg, 3 mmol) was added to Z-Val-Val-OPac (710 mg, 1.5 mmol) in DMF (4 mL). After 30 min, H_2O (15 mL) was added followed by subsequent extractions with Et_2O. Upon acidification of the aqueous phase with 2.5 M H_2SO_4, Z-Val-Val-OH precipitated in the cold; yield: 380 mg (72%); mp 137–138 °C; $[\alpha]_D^{15}$ −30.8 (*c* 3.2, aq 1 M KOH).

Boc-MTX1(1–41)-OH; Typical Procedure for Cleavage of Peptide Phenacyl Esters via Reduction with Zn/AcOH:[110]
Fully protected Boc-MTX1(1–41)-OPac (3.6 g, 0.495 mmol) was dissolved in HFIP/AcOH (1:4, 50 mL). Zn dust (1.62 g, 24.7 mmol) was added to the soln and the whole mixture was stirred vigorously under argon at 45 °C for 2 h. Excess Zn was removed by filtration and the solvent removed under reduced pressure. The residue was treated with 1 M HCl, washed with H_2O and MeCN, and dried. NMP was added and the crude product was washed with MeCN and Et_2O. Boc-MTX1(1–41)-OH was dried in vacuo; yield: 3.41 g (97%).

2.2.1.1.1.2.5 Ring-Substituted Phenacyl Esters

Ledger and Stewart[111] proposed the readily crystallizable 4-bromophenacyl ester derivatives for carboxy protection in peptide synthesis. Preparation and cleavage, e.g. by thiolysis, are identical to those of the phenacyl esters. The electron-withdrawing substituent in the *para* position enhances the activated ester character and, therefore, the 4-bromophenacyl esters may not be sufficiently inert toward nucleophiles to provide unequivocal protection of the carboxy group. The 1-methyl-2-oxo-2-phenylethyl (α-methylphenacyl) and 4-methoxyphenacyl esters undergo photolysis by irradiation (313 nm, dioxane or EtOH, rt, 6 h, and 5–17 h, respectively; see Section 2.4).[112] Correspondingly, α-methylphenacyl ester linkages are used in SPPS due to their cleavage by photolysis and hydrazinylosis[113] as well as by reaction with potassium cyanide in DMF in the presence of dicyclohexyl-18-crown-6.[102]

2.2.1.1.1.2.6 Acetonyl Ester

Acetonyl esters as carboxy protecting groups were described by Kundu.[114] For their preparation Z- or Boc-protected amino acids are esterified with hydroxyacetone (acetol) using the DCC/DMAP procedure. The esters are stable under acidic conditions (HCl, dioxane or TFA, CH_2Cl_2) and catalytic hydrogenolysis, and are hydrolyzed by tetrabutylammonium fluoride in THF at room temperature, generally within 5–45 minutes, in a selective manner versus Bzl and tBu protecting groups.

2.2.1.1.1.2.7 Dicyclopropylmethyl Ester

A new carboxy protecting group that takes advantage of the facile solvolysis of cyclopropyl methyl substrates was proposed by Carpino et al.[118] The dicyclopropylmethyl (Dicycpropme) ester can be used for carboxylic acid protection where selective removal is required in the presence of tBu or Trt side-chain protection. The ester is stable toward conditions of

N^{α}-Fmoc deprotection and chain elongation. The final deblocking step involves treatment of the Dicycpropme ester with 1% TFA in CH_2Cl_2 (15 min).

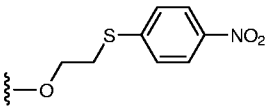
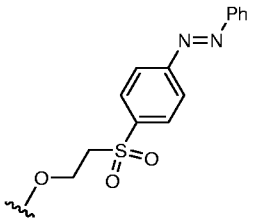
2.2.1.1.1.2.8 1,3-Dithian-2-ylmethyl Ester

1,3-Dithian-2-ylmethyl esters[146] are prepared by transesterification of amino acid methyl esters in the presence of aluminum isopropoxide in good yield. At room temperature, these esters are stable toward TFA as well as HCl in either CH_2Cl_2 or Et_2O and also against mild basic conditions, e.g. TEA. Thus, 1,3-dithian-2-ylmethyl esters are compatible with base-catalyzed coupling reactions and selective acidolytic deprotection of N-terminal Boc groups. Cleavage is carried out in a two-step procedure based on oxidation of the dithiane group with hydrogen peroxide/ammonium molybdate that results in the disulfone cleavable in a β-elimination process (E1cB) using mild basic aqueous solutions (pH 8, rt). 1,3-Dithian-2-ylmethyl esters like the other sulfanylethyl esters are not recommended for tryptophan- and sulfur-containing peptides.

2.2.1.1.2 Ethyl and Substituted Ethyl Esters

The ethyl and substituted ethyl esters most frequently used in peptide chemistry and the corresponding deprotection procedures are listed in Table 2.

Table 2 Ethyl and Substituted Ethyl Esters[120–122,134–144,151,152,156–158,161–166,172–174]

Esters	Structure	Deprotection Procedures[a]	Ref
ethyl (Et) ester	OEt	alkali, aq organic mixtures (see Me esters, Table 1)	
		trypsin, chymotrypsin	[120,121]
		carboxypeptidase Y	[122]
2-(methylsulfanyl)ethyl ester	OCH_2CH_2SMe	1. $H_2O_2/(NH_4)_6Mo_7O_{24}$, acetone, rt, 2 h 2. NaOH, pH 10–11, rt, 12–24 h	[135]
		1. MeI 2. pH 10–10.5, rt, 10 min	[134]
2-(4-nitrophenylsulfanyl)-ethyl ester		1. $H_2O_2/(NH_4)_6Mo_7O_{24}$, aq acetone 2. NaOH, pH 10.5	[136]
2-tosylethyl (Tse) ester	OCH_2CH_2Tos	NaOH, pH 11.5, dioxane, H_2O, rt, 3–40 min	[137,138]
		Na_2CO_3, dioxane, H_2O, rt, 2 h	[138]
		KCN, dioxane, H_2O, rt, 2 h	[138]
		DBN or DBU, benzene, rt	[139,140]
2-(4-tolylsulfanyl)ethyl ester	OCH_2CH_2S-4-Tol	1. $H_2O_2/(NH_4)_6Mo_7O_{24}$ 2. NaOH	[141]
2-(4-phenylazophe-nylsulfonyl)ethyl ester		NaOH, pH 10–11, dioxane, H_2O	[142]
2-cyanoethyl (Cne) ester	OCH_2CH_2CN	10% aq K_2CO_3	[143]
2-(diphenylphos-phanyl)ethyl ester	$OCH_2CH_2PPh_2$	1. MeI 2. KF, MeCN	[144]
		1. MeI 2. K_2CO_3, H_2O, MeCN	[144]

for references see p 228

Table 2 (cont.)

Esters	Structure	Deprotection Procedures[a]	Ref
2,2,2-trichloroethyl (Tce) ester	OCH_2CCl_3	Zn, AcOH/H_2O (9:1), 0 °C, 1.5–2 h	[151,152]
		electrolysis (−1.85 V)	[156]
		Se, $NaBH_4$, DMF, 40–50 °C	[157]
		TBAF	[158]
2-haloethyl esters	OCH_2CH_2X	$NaS(CH_2)_2SNa$, MeCN, reflux, 2 h	[162]
		NaSeH, EtOH, rt, 1 h, reflux, 6 min	[163]
		$(NaS)_2CS$, MeCN, reflux, 1.5 h	[164]
		Co(I) phthalocyanine, MeOH, 0–20 °C, 40 min–60 h	[165]
		1. NaI, DMF	[161]
		2. Zn powder	
		1. TEA, acetone, rt	[161]
		2. NaOH, pH 10–11, rt	
choline ester	$OCH_2CH_2NMe_3^+X^-$ (X = OH, Br)	NaOH, pH 10–11, rt	[161]
		choline esterases or proteases	[166]
2-(trimethylsilyl)ethyl (TMSE) ester	OCH_2CH_2TMS	[TEA]F, DMF or DMSO, rt	[172]
		TBAF, DMF or DMSO, rt	[172]
		TBACl, $KF \cdot H_2O$, DMF, rt	[173]
2-phenyl-2-(trimethylsilyl)ethyl ester	$OCH_2CH(TMS)Ph$	TBAF, DMF or DMSO, rt	[174]

[a] v/v unless otherwise stated.

2.2.1.1.2.1 Ethyl Ester

There are no significant differences between ethyl and methyl esters concerning synthesis or cleavage. Related protocols of the methyl esters (see Section 2.2.1.1.1) are, therefore, applied to the ethyl esters. The usefulness of ethyl esters is somewhat limited by the difficulties encountered in their saponification. Hydrolysis with alkali is feasible, but ethyl esters are less sensitive to nucleophilic attack than methyl esters. Aminolysis and hydrazinolysis as well as cleavage of the alkyl–oxygen bond with lithium iodide in pyridine[119] proceed several times slower in the case of ethyl esters. Mild enzyme-catalyzed hydrolysis by trypsin and chymotrypsin,[120,121] or by carboxypeptidase Y[122] remains an attractive alternative.

Analogously to methyl esters, esterification is performed by introducing gaseous HCl into an ethanolic suspension of the amino acid,[123] or by treating this suspension with either thionyl chloride in the cold,[124] or TosOH under reflux.[125] The use of ethyl 4-toluenesulfonate in refluxing EtOH[126] or the reaction of N-protected amino acids with 1 equivalent of triethyl orthoacetate, tetraethyl orthocarbonate, or triethyl orthoformate in DMF (90–120 °C, 1 h) is recommended.[127,128] Esterification of N-protected amino acids and also N-protected peptides in aqueous media may be accomplished using triethyloxonium tetrafluoroborate (Et_3OBF_4, Meerwein's reagent);[129] side-chain functionalities such as hydroxy, phenoxy, amide, and guanidine remain unchanged under these conditions while thiol, imidazole, and sulfide functions are alkylated.

H-Tyr-OEt; Typical Procedure for Esterification with EtOH/HCl:[41,130]
Dry HCl(g) was bubbled through a suspension of H-Tyr-OH (18.1 g, 0.10 mol) in abs EtOH (125 mL) without cooling until a clear soln was formed. More abs EtOH (250 mL) was added and the mixture was refluxed for 6 h, then cooled to rt and the solvent removed. The residue was dissolved in H_2O (100 mL) and neutralized with 5 M NaOH (~20 mL) in the cold. The precipitate was recrystallized (EtOAc); yield: 17 g (81%); mp 108–109 °C; $[\alpha]_D^{20}$ +20.4 (*c* 4.9, EtOH).

H-Met-OEt · TosOH; Typical Procedure for Esterification with EtOH/TosOH:[130]
To a suspension of H-Met-OH (1.49 g, 10 mmol) in abs EtOH (50 mL) was added TosOH · H_2O (3.8 g, 20 mmol) and the resulting soln was refluxed for 24 h. The alcohol was removed and the residue triturated with dry Et_2O (100 mL). The crystalline product was washed with Et_2O (100 mL) and dried; yield: 3.44 g (98%); mp 124–126 °C; $[\alpha]_D^{22}$ +13 (*c* 3.3, 95% EtOH).

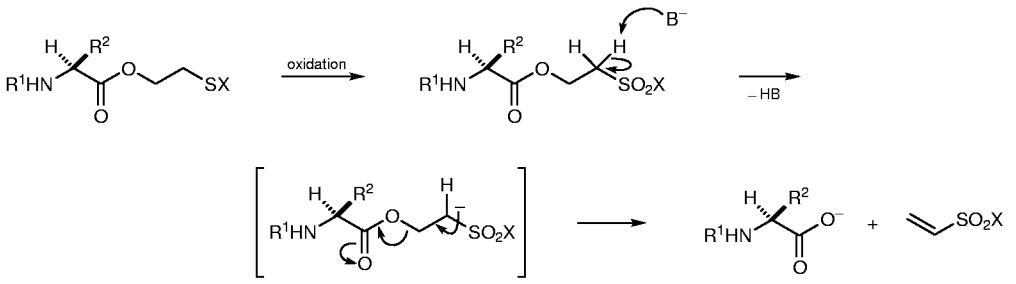
H-D-Glu(OEt)-OEt · HCl; Typical Procedure for Esterification with $SOCl_2$:[131]
At −40 °C H-D-Glu-OH (31.6 g, 0.22 mol) and then $SOCl_2$ (15 mL) were added to abs EtOH (200 mL) under stirring. The mixture was left at rt for 30 min, refluxed for 2 h and the solvent removed. The residual syrup was submitted to the same procedure with a smaller amount of $SOCl_2$ (10 mL). After careful elimination of excess $SOCl_2$, the residue was dissolved in EtOH (100 mL) and precipitated with dry Et_2O (500 mL); yield: 43.8 g (83%); mp 112–114 °C; $[\alpha]_D^{22}$ −22 (*c* 2, H_2O).

Z-Xaa-OEt; General Procedure for Esterification with Meerwein's Reagent:[129]
To a soln of Z-Xaa-OH (1 mmol) in H_2O (10 mL) containing $NaHCO_3$ (21 mmol), Et_3OBF_4 (3.8 g, 20 mmol) in MeCN was added over a period of 10 min under stirring. After 20 min, the mixture was extracted with EtOAc and the organic layer was washed with aq $KHCO_3$ and H_2O before drying and removing the solvent; yields of amino acid ethyl esters range from 82–98%.

2.2.1.1.2.2 Substituted Ethyl Esters

A number of β-substituted ethyl esters have been proposed for carboxy protection in peptide synthesis, which are cleaved by fragmentation via an E1cB base-catalyzed β-elimination (Scheme 2). Appropriate electron-withdrawing substituents such as sulfones facilitate proton abstraction from the β-carbon under mild alkaline conditions. An elimination step to stabilize the carbanion occurs which provides the release of the free carboxy group (Scheme 2).

Scheme 2 Cleavage via the E1cB Elimination Process



X = Me, Ph, 4-Tol, 4-$O_2NC_6H_4$
B = base

The remarkable lability of such esters toward alkali has resulted in the development of several carboxy protecting groups based on the generation of either ethyl sulfone[132] or ethyl sulfonium salt[133] intermediates. 2-(Methylsulfanyl)ethyl esters, after conversion into the corresponding sulfonium iodides by alkylation with iodomethane, are cleaved at pH 10–10.5 within 10 minutes at room temperature.[134] Obviously, alkylation of some side-chain functionalities in the presence of excess iodomethane reduces the general applicability of this procedure. Oxidation with hydrogen peroxide/ammonium molybdate in 4% acetone/water yields the corresponding 2-(methylsulfonyl)ethyl esters which require 12–24 hours for hydrolysis under the same conditions.[135] N^{α}-Z-Protection is not suitable in combination with the 2-(methylsulfanyl)ethyl ester group owing to formation of sulfonium salts with benzyl bromide, an intermediate of standard acidolytic deprotection with HBr/AcOH. 2-(Methylsulfanyl)ethyl esters do not interfere with standard coupling procedures such as DCC, azides, or active esters and are stable against the tertiary amines that are often used as bases. The alternative 2-(4-nitrophenylsulfanyl)ethyl esters are readily prepared as crystalline com-

for references see p 228

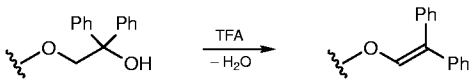
pounds by direct esterification of amino acids with the corresponding 2-(4-nitrophenylsulfanyl)ethanol.[136] The 2-tosylethyl (Tse) esters are obtained by reaction of N-protected amino acids with 2-tosylethanol using DCC in pyridine and are saponified in 3–40 minutes at pH 11.5.[137,138] Alternatively, sodium carbonate in aqueous dioxane (rt, 2 h), potassium cyanide in aqueous dioxane (rt, 2 h), as well as DBN or preferentially DBU in benzene at room temperature[139,140] are proposed as reagents for efficient hydrolysis. Hydrazinolysis of the Tse esters yields the corresponding hydrazides. Application of the corresponding safety-catch-type 2-(4-tolylsulfanyl)ethyl ester reduces the problems resulting from the base-sensitivity of the sulfone in multistep synthesis.[141] Cleavage under mild basic conditions is accomplished by oxidation of the 2-(4-tolylsulfanyl)ethyl group with hydrogen peroxide/ammonium molybdate. Easy visual detection of peptide intermediates is accomplished by introducing a chromophore into the masking group, e.g. in 2-(4-phenylazophenylsulfonyl)ethyl esters.[142]

β-Elimination is not only restricted to sulfonyl- or sulfonium-containing groups. The 2-cyanoethyl (Cne) ester has been proposed by Misra et al.[143] and the corresponding derivatives are prepared in good yields by DCC/DMAP using suitably protected amino acids. The 2-cyanoethyl ester is stable under acidic (Boc cleavage) and hydrogenolytic (Z cleavage) conditions. Hydrolysis of this ester occurs readily by exposure to 10% aqueous potassium carbonate.

The use of 2-(diphenylphosphanyl)ethyl esters for carboxy protection of amino acids or peptides has been described by Chantreux et al.[144] This group is introduced by esterification using 2-(diphenylphosphanyl)ethanol in the presence of DCC and DMAP. 2-(Diphenylphosphanyl)ethyl esters, generally obtained as oils, are stable under the standard conditions of Boc and Z cleavage. Mild deprotection is achieved upon quaternization using iodomethane, followed by β-elimination induced by fluoride ion sources (KF, MeCN) or 10% aqueous potassium carbonate.

The 2-hydroxy-2,2-diphenylethyl ester represents a dual-function group which ensures suitable carboxy protection during chain elongation in peptide synthesis while being readily converted into the active 2,2-diphenylvinyl ester using TFA (Scheme 3).[145]

Scheme 3 Conversion of the 2-Hydroxy-2,2-diphenylethyl Ester into the Corresponding Active Ester Species[145]

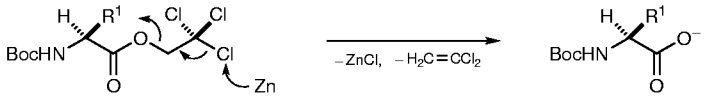


2.2.1.1.2.2.1 2,2,2-Trichloroethyl Ester

The 2,2,2-trichloroethyl (Tce) ester group,[149–151] which was introduced to peptide chemistry by Marinier et al.,[152] has recently found application in the synthesis of peptide antibiotics and β-oligopeptides.[153,154] Preparation of these esters is based on the treatment of N-protected amino acids with phosphoryl chloride or phosphorus pentachloride followed by in situ esterification of the resulting acid chlorides with excess 2,2,2-trichloroethanol in the presence of TEA. Alternatively, treatment of N-protected amino acids and the alcohol with DCC in pyridine[149] or in the presence of TosOH[155] produces the esters. The Z group is selectively removed by acidolysis (HBr/AcOH) in the presence of Tce esters. Removal of the carboxy protecting group is effected in a β-elimination process (Scheme 4) by dissolving the compound in 90% AcOH and treating it with an excess of Zn powder for 1.5–2 hours, while stirring and cooling in an ice bath. Other reductive methods include electrolysis (−1.85 V)[156] or treatment with catalytic amounts of selenium in the presence of sodium borohydride in DMF at 40–50 °C.[157] The Tce esters are also cleaved with tetrabutylammonium fluoride.[158]

In the presence of phosphorus(III) reagents such as triphenylphosphine or hexamethylphosphorous triamide the trihaloethyl esters are reduced in situ yielding activated acyloxyphosphonium intermediates that undergo trapping by amine and alcohol nucleophiles.[159]

Scheme 4 Elimination of Tce Induced by Zn/AcOH



Boc-β-HSer(Bzl)-OTce; Typical Procedure for Esterification with 2,2,2-Trichloroethanol:[154]
To a soln of Boc-β-HSer(Bzl)-OH (4.34 g, 14 mmol) in CH_2Cl_2 (25 mL) and were added 2,2,2-trichloroethanol (1.56 mL, 16 mmol), DCC (3.3 g, 16 mmol), and DMAP (171 mg, 1.4 mmol) at 0 °C. The mixture was allowed to warm to rt and stirring was continued for 16 h. The solvent was removed under reduced pressure, the residue redissolved in EtOAc, filtered through Celite, and the solvent removed from the filtrate. The crude product was purified by flash chromatography (EtOAc/petroleum ether, 1:5); yield: 5.8 g (94%); mp 40.5–42.5 °C, $[\alpha]_D^{20}$ +2.3 (*c* 0.84, $CHCl_3$).

Boc-D-Asp[β-HGlu(OBzl)-β-HSer(Bzl)-OH]-OBzl; Typical Procedure for Cleavage of the Trichloroethyl Ester:[154]
Zn powder (4 g, 80 equiv) was activated using 6 M aq HCl, filtered off and washed successively with H_2O and Et_2O. The resulting solid was added to a soln (0.03 M) of Boc-D-Asp[β-HGlu(OBzl)-β-HSer(Bzl)-Tce]-OBzl (500 mg, 0.57 mmol) in 90% AcOH/H_2O (24 mL). After 16 h at rt, the suspension was filtered and the solvent removed. The residue was dissolved in EtOAc and washed successively with 1 M $NaHSO_4$ and brine. The organic layer was dried ($MgSO_4$), and the solvent removed under reduced pressure.

2.2.1.1.2.2.2 2-Haloethyl Esters

2-Haloethyl esters, introduced quite early in peptide chemistry,[160] were first described by Kunz and Buchholz as α-carboxy protecting groups several decades later.[161] 2-Chloroethyl as well as 2-bromoethyl esters are efficiently prepared from N-protected amino acids by treatment with thionyl chloride in the presence of excess 2-haloethanol at −10 °C. Conversely, esterification via DCC results in substantial racemization. Both 2-haloethyl esters are stable toward mild alkaline (2 M NaOH, pH 10) as well as acidic conditions used for Z and Boc cleavage (40% HBr/AcOH). The more labile 2-bromoethyl ester necessitates low temperatures (−15 °C) during acidolytic removal of the Z group to avoid decomposition. 2-Haloethyl esters are cleaved by strong nucleophiles, e.g. by disodium ethane-1,2-diyldidithioate in refluxing acetonitrile (2 h),[162] sodium hydroselenide in EtOH (rt, 1 h; reflux, 6 min),[163] trithiocarbonates in acetonitrile (1.5 h),[164] or Co(I) phthalocyanine in MeOH (0–20 °C, 40 min to 60 h).[165] Amino acid 2-chloroethyl and 2-bromoethyl esters are easily converted into the corresponding 2-iodoethyl esters by sodium iodide in DMF, and subsequently decomposed in a one-pot reaction by reduction with zinc powder.[161] A very interesting alternative is the conversion of 2-haloethyl esters into choline esters by treatment with trimethylamine in acetone at room temperature.[161] These choline esters show increased stability toward acidolysis which is advantageous in terms of selective Z or Boc cleavage at room temperature. Their base lability is also increased, allowing deblocking of the carboxy group at pH 10–11 at room temperature without any racemization. Moreover, choline esters represent a hydrophilic moiety that increases the solubility of protected amino acids and peptides in aqueous solutions. This solubilizing effect promotes enzymatic cleavage of protected amino acid and peptide choline esters under mild conditions by different choline esterases or by proteases such as trypsin, chymotrypsin, papain, subtilopeptidase A, nagarse, or proteinase K (see Section 2.5.3.1).[166]

for references see p 228

2.2.1.1.2.2.3 2-(Trimethylsilyl)ethyl Esters

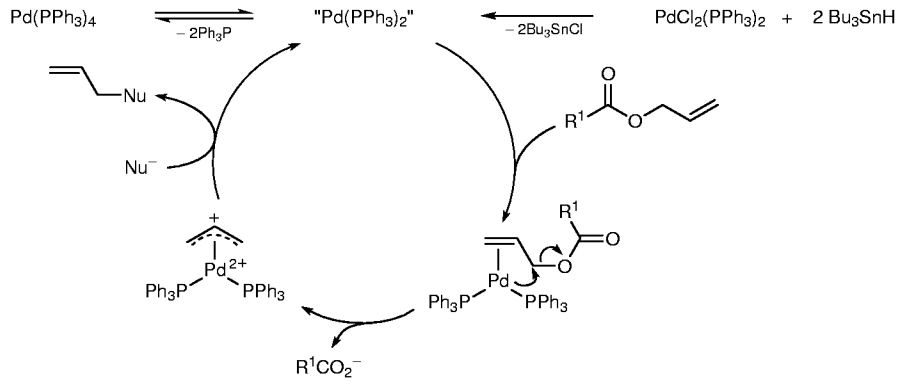
2-(Trimethylsilyl)ethyl (TMSE) esters as carboxy protecting groups were developed independently by Sieber and Gerlach.[167,168] They are prepared either from carboxylic acids, 2-(trimethylsilyl)ethanol, and DCC in pyridine/acetonitrile (0 °C, 5–15 h) or from acid chlorides and 2-(trimethylsilyl)ethanol in pyridine (rt, 3 h). Esterification is also accomplished using DCC/DMAP[169] or 2-chloro-1-methylpyridinium chloride (Mukaiyama's reagent).[170] Decomposition of 2-(trimethylsilyl)ethanol via Peterson olefination in the presence of acids or bases represents a problem when acylation rates are low. Dimethyl[2-(trimethylsilyl)-ethoxy]aluminum is successfully used to solve this problem.[171] The TMSE group is removed by treatment with fluoride ion sources at room temperature.[172] It is cleaved in 5–60 minutes using excess tetraethylammonium fluoride or tetrabutylammonium fluoride in DMF or DMSO at room temperature by a fragmentation reaction that results in ethene, trimethylsilyl fluoride and the tetraalkylammonium salt of the carboxylic acid which is then recovered on acidification. Fragmentations encountered with acidification of the resulting peptide tetrabutylammonium salts have been suppressed by the use of a mixture of tetrabutylammonium chloride and potassium fluoride monohydrate in DMF.[173] Generally, when treated with fluoride ions the N^{α}-Boc groups tend to form hydantoins and Asp(OtBu) residues tend to cyclize to aspartimide, while benzyl esters are partially cleaved and disulfides undergo disproportionation. To bypass these noxious side reactions, the 2-phenyl-2-(trimethylsilyl)ethyl group has been recently developed which is cleaved by treatment with tetrabutylammonium fluoride more rapidly than the TMSE ester.[174]

2.2.1.1.3 Allyl Ester

Allyl (Al) esters are suitable for the temporary protection of carboxy groups in peptide and glycopeptide synthesis[247–249] and also in the synthesis of cyclic peptides.[250] In the latter case these esters allow, for example, on-resin head-to-tail cyclizations of peptides that are attached via the side chains of aspartic acid and glutamic acid residues to the solid support.[251] The application of allylic protecting groups in peptide synthesis has been comprehensively reviewed by Guibé.[252]

Allyl esters are stable under mild basic and acidic conditions, thus providing full orthogonality to the tBuOH and Fmoc-OH derived protecting groups. Amino acid or peptide allyl esters are readily cleaved with high selectivity by an allyl-transfer reaction in the presence of the catalyst tetrakis(triphenylphosphine)palladium(0) and a nucleophilic allyl acceptor (Scheme 5). Various nucleophiles have been employed, e.g. secondary amines (morpholine, piperidine, NMA, NMM, pyrrolidine),[253–257] highly acidic β-dicarbonyl compounds [*N*,*N*′-dimethylbarbituric acid, 5,5-dimethylcyclohexane-1,3-dione (dimedone)],[249,253] oxygen nucleophiles (HOBt),[258] or hydride donors (Bu_3SnH).[259,260] Tributyltin hydride may also be used in combination with the catalyst dichlorobis(triphenylphosphane)palladium(II). Allyl ester cleavage by isomerization to form the corresponding labile propenyl esters is accomplished using chlorotris(triphenylphosphane)rhodium(I) in aqueous EtOH at 70 °C.[247]

Scheme 5 Deprotection of Allyl Esters by Allyl Transfer in the Presence of "Pd(0)" Catalyst



The allyl protecting groups were recently found to undergo partial hydrogenation of the alkene group during hydrazinolysis (2% hydrazine in DMF) of the Dde group (see Section 2.1.2.2.5.2). This was attributed to the presence of small amounts of diazine in the hydrazine.[261] Addition of allyl alcohol as a scavenger prevents this side reaction, providing complete orthogonality between Dde and allyl protection.

Allyl esters are synthesized by the azeotropic esterification of amino acids with allyl alcohol in the presence of TosOH[247] or by reaction of N^{α}-protected amino acid cesium salts with allyl bromide.[56,250] They are also available from N^{α}-protected amino acids by treatment with allyl alcohol/DCC in THF[251] or with diallyl dicarbonate in THF with DMAP as catalyst.[57]

H-Ala-OAl · TosOH; Typical Procedure for Esterification with Allyl Alcohol/TosOH:[247]
A mixture of H-Ala-OH (2.2 g, 25 mmol), allyl alcohol (15.0 g, 0.25 mol), and TosOH · H_2O (5.7 g, 30 mmol) in benzene (200 mL) was refluxed until the calculated amount of H_2O was trapped in a Dean–Stark receiver (4–24 h). Most of the benzene was removed to precipitate H-Ala-OAl · TosOH by addition of Et_2O; yield: 7.38 g (98%); mp 84–85 °C; $[\alpha]_D^{22}$ −1.6 (*c* 1, MeOH).

Yields of amino acid allyl ester tosylates range from 80–99%.

Z-Thr-OAl; Typical Procedure for Esterification with Allyl Bromide/Cs_2CO_3:[250]
To a stirred soln of Z-Thr-OH (12.7 g, 50 mmol) and allyl bromide (4.5 mL, 52 mmol) in DMF (50 mL) was added Cs_2CO_3 (8.5 g, 26 mmol). The reaction was slightly exothermic. After 2 h stirring, the precipitate was filtered off and the solvent removed. The oily residue was dissolved with Et_2O (200 mL) and washed with H_2O (3 × 30 mL). The organic phase was dried ($MgSO_4$), filtered, and the solvent removed to give the product as an oil; yield: 14.4 g (98%).

Fmoc-Asp-OAl and Fmoc-Glu-OAl:[251]
Fmoc-Asp(OtBu)-OH or Fmoc-Glu(OtBu)-OH was refluxed in neat allyl bromide (5 mL · g^{-1} Fmoc derivative) in the presence of DIPEA (2 equiv with respect to the Fmoc derivative) for 30 min. The mixture was diluted with EtOAc, extracted successively with 0.1 M HCl, carbonate buffer (pH 9.5),[262] and brine. The soln was dried ($MgSO_4$) and the solvent removed. The solid was dissolved in TFA/CH_2Cl_2 (1:1), and after 30 min, the solvent was removed to provide a solid which was recrystallized (Et_2O/pentane); yield of Fmoc-Asp-OAl: 84%; mp 92–93 °C; yield of Fmoc-Glu-OAl: 66%; mp 120–121 °C.

Allyl Ester Cleavage; Using Chlorotris(triphenylphoshine)rhodium(I); General Procedure:[253]
The allyl ester protected peptide (0.47 mmol) and $RhCl(PPh_3)_3$ (40 mg, 43 μmol) were dissolved at 70 °C in a mixture of EtOH (4.5 mL) and H_2O (0.5 mL). The course of the reaction was monitored by TLC. After stirring at 70 °C for 30 min, the solvent was distilled off and the residue purified chromatographically.

Allyl Ester Cleavage; Using Tetrakis(triphenylphosphine)palladium(0); General Procedure:[253]
The allyl ester protected peptide (0.47 mmol) was dissolved under argon in THF (10 mL) at rt. In the dark, $Pd(PPh_3)_4$ (50 mg, 43 μmol) and morpholine (0.12 g, 1.4 mmol) or 5,5-dimethylcyclohexane-1,3-

for references see p 228

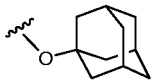
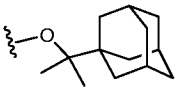
dione (0.2 g, 1.4 mmol) were added. The reaction was monitored by TLC. After stirring for 10 min, the solvent was removed and the remaining residue purified chromatographically.

Allyl Ester Cleavage; Using Dichlorobis(triphenylphosphine)palladium(II); General Procedure:[260]
To a stirred soln of the N^{α}-protected amino acid allyl ester (1 mmol) in CH_2Cl_2 (5 mL), $PdCl_2(PPh_3)_2$ (31.6 mg, 45 μmol) and AcOH (2.5 mmol) under argon was added Bu_3SnH (582 mg, 2 mmol). Stirring was continued at rt for about 10 min. In most cases, the end of the reaction was shown by a color change of the soln from yellow to purple.

2.2.1.1.4 *tert*-Alkyl Esters

The *tert*-alkyl esters developed and most frequently used in peptide chemistry are listed in Table 3 together with the most common cleavage conditions.

Table 3 *tert*-Alkyl Esters[179–186,193,242–246]

Esters	Structure	Deprotection Procedures[a]	Ref
tert-butyl (tBu) ester	OtBu	TFA, CH_2Cl_2, rt, 1 h	[184]
		2 M HBr/AcOH	[180]
		TosOH, benzene, reflux	[179]
		HCl/EtOAc, HCl/CH_2Cl_2, or HCl/AcOH	[181–183]
		$BF_3 \cdot OEt_2$/AcOH	[185,186]
		$ZnBr_2$, CH_2Cl_2	[193]
trityl (triphenylmethyl, Trt) ester	$OCPh_3$	AcOH, rt, 5 min	[242]
		2 M HCl, rt, 1 min	[242]
1-adamantyl (1-Ada) ester		TFA, rt, 1 h	[243]
		TFA/CH_2Cl_2 (1:2), rt, 15 h	[243]
2-phenylprop-2-yl ester	$OCMe_2Ph$	2–4% TFA, CH_2Cl_2	[244,245]
2-(1-adamantyl)prop-2-yl ester		3% TFA, CH_2Cl_2	[246]

[a] v/v unless otherwise stated.

2.2.1.1.4.1 *tert*-Butyl Ester

Acid-labile protecting groups are among the most useful in peptide synthesis, since the treatment of protected peptides with mild acids generally causes fewer side reactions. The need for these groups, selectively cleaved by less strong acids, led to the development of the *tert*-butyl ester.[178,179] Amino acid *tert*-butyl esters can be distilled and stored for months, and dipeptide *tert*-butyl esters do not cyclize readily to piperazine-2,5-diones (DKP). The stability of the intermediate *tert*-butyl carbocation allows acidolysis with 2 M HBr in AcOH,[180] TosOH in refluxing benzene,[179] HCl in ethyl acetate,[181] CH_2Cl_2,[182] or AcOH,[183] TFA (neat or in CH_2Cl_2),[184] and boron trifluoride–diethyl ether complex in AcOH.[185,186] While *tert*-butyl groups are removed with moderately strong acids, they are sufficiently resistant to weak acids to allow washing of organic solutions of protected peptides with dilute aqueous acids, conversion of hydrazides into azides with nitrites in the presence of moderate excesses of HCl,[187] or addition of 1 equivalent of HCl to the hydrogenation mixture for efficient Z cleavage.[188] The *tert*-butyl esters also remain intact under the mild acidic conditions required for cleavage of the Bpoc, Trt, and Nps amino protecting groups.[189–191] A general protocol for the selective cleavage of N^{α}-Boc groups in the presence of *tert*-butyl esters has not yet been established.[192] However, under appropriate conditions (CH_2Cl_2), the *tert*-butyl

ester is cleaved selectively in the presence of *N*-Boc protection with zinc(II) bromide;[193] in other solvents the selectivity of this procedure is lost.

The *tert*-butyl cation released in the hydrolysis of the corresponding ester represents a powerful electrophile, prone to react, for example, with the Met,[194,195] Tyr,[196] or Trp[197–200] side-chain functionalities in the peptide.[201–203] Therefore, nucleophilic scavengers have to be added to the reaction mixture[204] (see also Section 4.3.5). Electrophilic substitution of the phenolic Tyr side chain is efficiently suppressed by cocktails containing phenol and *p*-cresol.[205] N^{in}-*tert*-Butylation may also be followed by alkyl-group migration to the ring positions 2, 5, and 7, resulting in trisubstituted indoles.[206] Addition of scavengers, such as thiols, dialkyl sulfides, indole, or skatole, to the deprotection mixture or protection of the indole group is an essential requirement.[207] It should be mentioned that other acidolytic deprotection protocols have been reported in natural product syntheses such as formic acid in refluxing benzene[208] or a mixture of AcOH/iPrOH/H_2O (4:4:1) under reflux.[209]

Compared to primary alkyl esters, *tert*-butyl esters offer a degree of steric shielding which makes them resistant to attack by a wide range of nucleophiles. The α-*tert*-butyl ester group is stable against alkaline hydrolysis, hydrazinolysis, aminolysis, and hydrogenolysis. Selective alkaline hydrolysis of *n*-alkyl esters in the presence of α- or of the even more labile ω-*tert*-butyl esters is possible if large excesses of base are avoided.[210–212] However, the saponification of methyl esters of long-chain peptides requires high base concentrations (*vide supra*) which may lead to concomitant hydrolysis of *tert*-butyl esters.[213] A general procedure for efficient alkaline cleavage of *tert*-butyl esters involving potassium hydroxide and 18-crown-6 in toluene (100 °C, 5 h) has been proposed by Pedersen.[214]

tert-Butyl esters have been proposed as substrates for enzymatic hydrolysis.[215] Alkaline serine protease (thermitase) from *Thermactinomyces vulgaris* is capable of hydrolyzing sterically hindered esters, which are unaffected by other hydrolases. The high esterase/protease activity of the enzyme is exploited for the deprotection of peptide as well as glycopeptide *tert*-butyl esters.

The classical method for *tert*-butyl esterification involves mineral acid catalyzed addition of the amino acid to isobutene. Both, N-protected, e.g. Z-Xaa-OH, and unprotected amino acids form *tert*-butyl esters with isobutene in the presence of catalytic amounts of sulfuric acid or TosOH.[178,179,216–220] Another efficient method is the transesterification of an acetic acid *tert*-butyl ester catalyzed by perchloric acid.[182,221–224] Amide *tert*-butylation was recognized as a side reaction in the presence of perchloric acid, but could be completely suppressed by using sulfuric acid.[225,226] Both methods, acid-catalyzed addition to isobutene and the transesterification of acetic acid *tert*-butyl esters, result in simultaneous *tert*-butylation of hydroxy and sulfanyl groups.[227,228] A synthetic route to Z-Thr-OtBu, Z-Ser-OtBu, and Z-Hyp-OtBu without concomitant O-alkylation involves the hydroxy protection by the acetoacetyl group, which is readily cleaved by treatment with 2 equivalents of hydrazine in EtOH for 30 minutes.[229]

The reaction of Z and Phth amino acids with phosphoryl chloride and tBuOH in the presence of excess pyridine[230] or of N^{α}-protected amino acid silver salts with *tert*-butyl iodide[179] gives the corresponding *tert*-butyl esters, whilst the cesium salts, although very successful in the preparation of different alkyl and benzyl esters, give poor results in terms of *tert*-butylation.[56] *O-tert*-Butyl 2,2,2-trichloroacetimidate (TBTA), prepared from potassium *tert*-butoxide and trichloroacetonitrile in 70% yield, readily esterifies carboxylic acids in the presence of catalytic amounts of boron trifluoride–diethyl ether complex at room temperature over 16–21 hours.[231,232] This method also converts hydroxyamino acids into the corresponding ethers. *tert*-Butylation with TBTA was also used for C-terminal esterification of protected peptide segments for subsequent solution-phase fragment condensations.[233,234] Originally prepared by SPPS according to the Fmoc/tBu strategy and detached from a highly acid-labile HMPB linker, the C-terminally unprotected peptide segments are successfully esterified by a mixture of $CHCl_3$/TFE/TBTA (7:2:1) at room temperature for 1 hour. This

for references see p 228

mixture efficiently replaces boron trifluoride, a fact that is advantageous when other acid-labile groups are present in the molecule. This combined solid-phase and solution strategy was applied in the syntheses of calcitonin (1–33) and human neuropeptide Y.[234]

A very convenient method is based on the reaction of N^{α}-protected amino acids with tBuOH in the presence of anhydrous magnesium sulfate and catalytic sulfuric acid as additional reagents.[235] An additional approach to the preparation of *tert*-butyl esters involves the treatment of the carboxylic acid with either 2-*tert*-butyl-1,3-dicyclohexylisourea[236] or 2-*tert*-butyl-1,3-diisopropylisourea[237] under essentially neutral conditions. The presence of alcohols is tolerated, but not those of thiols or unhindered amines. Another mild method for forming *tert*-butyl esters is based on esterification via mixed anhydride intermediates, prepared by reaction of the carboxylic acid with 2,4,6-trichlorobenzoyl chloride and TEA in THF. The mixed anhydrides, tBuOH, and catalytic amounts of DMAP yield the corresponding esters.[238] Intermediate mixed anhydrides, resulting from reaction of carboxylic acids with EEDQ yielded the *tert*-butyl esters upon refluxing in tBuOH for 5 hours.[58]

The reaction of 2 equivalents of *tert*-butyl dicarbonate with N^{α}-Z-protected amino acids in tBuOH and catalytic DMAP (0.1 equiv) proceeds smoothly over 30 minutes to give the corresponding esters in high yields (see Section 3.9.3.2).[57] Preparation of *tert*-butyl esters of N-protected amino acids via *tert*-butyl mixed carbonic anhydride generated in situ by reaction with *tert*-butyl fluorocarbonate has been reported by Loffet et al.[239] Finally, N^{α}-Z-protected amino acids are esterified in good yields with *tert*-butyl bromide.[240]

H-Pro-OtBu; Typical Procedure for Esterification of N^{α}-Protected Amino Acids with Isobutene:[179,241]

Z-Pro-OtBu:
To a soln of Z-Pro-OH (24.9 g, 100 mmol) in CH_2Cl_2 (200 mL) was added concd H_2SO_4 (1 mL) and the mixture was saturated with isobutene, resulting in an increase of volume to about 300 mL. The flask was stoppered and kept at rt for 3 d. After this time, aq Na_2CO_3 soln (10:150, w/v) was added slowly and the organic layer was separated and washed with H_2O (3 × 10 mL). The solvent was removed under reduced pressure allowing the bath temperature to rise to 60 °C. The oily residue of Z-Pro-OtBu slowly crystallized (Et_2O/petroleum ether); yield: 28.8 g (94%); mp 44–45 °C; $[\alpha]_D^{25}$ −52.5 (*c* 2.2, EtOH).

H-Pro-OtBu:
To yield H-Pro-OtBu, Z-Pro-OtBu was hydrogenated in abs EtOH (250 mL) over 10% Pd/C. The catalyst was filtered off and the solvent removed. The residue was redissolved in Et_2O (150 mL), extracted several times with 10% aq Na_2CO_3 and finally dried ($MgSO_4$); yield: 12.5 g (73%); bp 57 °C/1.5 Torr; $[\alpha]_D^{25}$ −41.5 (*c* 1.8, EtOH).

H-Tyr-OtBu; Typical Procedure for the Esterification of Unprotected Amino Acids with Isobutene:[218,241]
Liq isobutene (30 mL) was slowly added to H-Tyr-OH (3.62 g, 20 mmol) in dioxane (30 mL) containing TosOH · H_2O (7.6 g, 40 mmol) in a 500-mL pressure bottle. The flask was stoppered properly and shaken at rt for about 20 h. The soln was poured into an ice-cold mixture of EtOAc and 0.25 M NaOH (120 mL/120 mL). The pH was adjusted to 9.1 to extract the *tert*-butyl ester into EtOAc (2 × 200 mL). The organic solvent was removed under reduced pressure; yield: 2.15 g (45%); mp 140–143 °C; $[\alpha]_D^{25}$ +24.4 (*c* 2, EtOH).

Transesterification with *tert*-Butyl Acetate; General Procedure:[223,241]
The amino acid (10 mmol), *tert*-butyl acetate (150 mL) and $HClO_4$ in H_2O (60%, 11 mol equiv) were mixed and stirred for 15 min until dissolution. After 4 d at rt, the mixture was cooled in an ice bath and extracted with 0.5 M HCl (4 × 25 mL). The aqueous extracts were immediately neutralized with solid $NaHCO_3$, combined, and extracted with Et_2O (4 × 100 mL). The pooled Et_2O extracts were dried ($MgSO_4$) and the solvent was removed under reduced pressure. Upon addition of dry HCl in Et_2O, the amino acid *tert*-butyl esters were isolated as hydrochlorides; yield: 40–70%; H-Pro-OtBu · HCl was obtained in modest yields.

Z-Xaa-OtBu; General Procedure for the Esterification of Z Amino Acids with Boc-F:[239]
Boc-F (1.5 equiv) was added to Z-Xaa-OH in the presence of TEA (1.5 equiv) and DMAP (0.2 equiv) in a CH_2Cl_2/tBuOH (1:1) soln (0.2 mL per mmol) at rt. After 4 h, the ester was easily purified by the usual workup procedures; yield: 82–96%.

Z-Xaa-OtBu; General Procedure for the Esterification of Z Amino Acids with *tert*-Butyl Bromide:[240]
To a soln of Z-Xaa-OH (10 mmol) and TEBAC (2.27 g, 10 mmol) in DMA (75 mL) was added dried K_2CO_3 (36 g, 0.260 mol), followed by tBuBr (55.3 mL, 0.480 mol). The mixture was stirred at 55 °C for 24 h and, after cooling, cold H_2O (1 L) was added, and the resulting solid precipitate was filtered and washed with H_2O. In the case of oily compounds, the precipitate was extracted into EtOAc (250 mL). The organic layer was separated, washed with H_2O (2 × 100 mL), dried, and the solvent removed; the products were recrystallized (Et_2O/hexane); yield: 80–100%.

Esterification of Protected Peptides with TBTA; General Procedure:[233,234]
Solns of the protected peptides (5–10%) in $CHCl_3$/TFE/TBTA (7:2:1) were kept at rt until TLC probes indicated completion of the esterification (usually 1 h). The products were then precipitated by addition of iPr_2O or MTBE, if necessary in a 1:1 mixture with hexane.

2.2.1.1.4.2 Other *tert*-Alkyl Esters

In order to enhance the acid lability of *tert*-butyl esters, several derivatives have been proposed based on the substitution of one or more methyl groups by residues which increase the stability of the intermediate tertiary carbocation in acidolytic deprotection. Phenyl, cyclopropyl, and adamantyl substituents have been used for such purposes.

Trityl (triphenylmethyl, Trt) esters are extremely labile toward acidic conditions and are selectively cleaved in the presence of *tert*-butyl esters by treatment with AcOH at room temperature for a few minutes.[89,242] The hydrochloride salts of amino acid trityl esters are hydrolyzed in aqueous and methanolic solutions. Amino acid trityl esters are obtained in satisfactory yields by treatment of the corresponding N^{α}-protected amino acid cesium salts with trityl chloride.[56]

The 1-adamantyl (1-Ada) esters exhibit increased acid stability if compared to the *tert*-butyl esters.[243] These amino acid esters are prepared by reaction of the amino acid tosylates with adamantan-1-ol and dimethyl sulfite in boiling toluene. Removal of the adamantyl ester is effected by neat TFA within 1 hour or TFA/CH_2Cl_2 (1:2) within 15 hours. N-Terminal Boc groups are selectively cleaved with 4 M HCl in anhydrous dioxane (rt, 25 min) in the presence of the 1-adamantyl ester.

2.2.1.1.4.2.1 2-Phenylprop-2-yl Ester

The 2-phenylprop-2-yl (1-methyl-1-phenylethyl) ester, although closely resembling the *tert*-butyl group, is more sensitive toward acids due to its phenyl group.[244,245] It can also be considered as a substituted benzyl ester. 2-Phenylprop-2-yl esters of Fmoc-, Z-, or Boc-protected amino acids are prepared via the 2-phenylprop-2-yl trichloroacetimidates in the absence of any catalyst.[231,232] These esters are cleaved with 2–4% TFA in CH_2Cl_2, leaving *N*-Boc and *tert*-butyl esters and ethers intact.

2.2.1.1.4.2.2 2-(1-Adamantyl)prop-2-yl Ester

The 2-(1-adamantyl)prop-2-yl [1-(1-adamantyl)-1-methylethyl] ester is resistant to catalytic hydrogenation and is removable by mild acid treatment.[246] These esters are prepared by reaction of Z-protected amino acids with 2-(1-adamantyl)prop-2-ene in the presence of sulfuric acid. The 2-(1-adamantyl)prop-2-yl ester is stable under conditions required for cleavage of the N^{α}-Fmoc and N^{α}-Nps groups and are hydrolyzed by treatment with 3% TFA in CH_2Cl_2 which leaves *tert*-butyl esters and even Boc groups largely intact.

for references see p 228

2.2.1.2 Aralkyl Esters

The benzyl, substituted benzyl, and other aralkyl esters most frequently used in peptide chemistry and the corresponding deprotection procedures are summarized in Table 4.

Table 4 Benzyl, Substituted Benzyl, and Other Aralkyl Esters

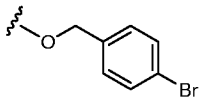
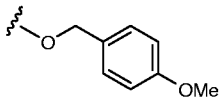
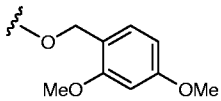
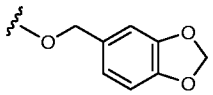
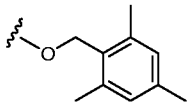
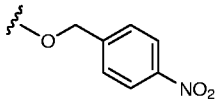
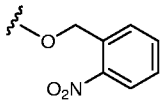
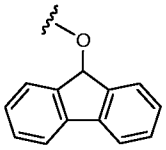
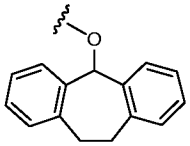
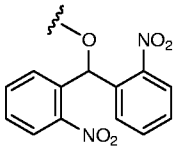
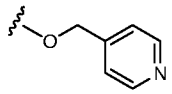
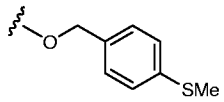
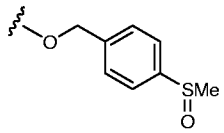
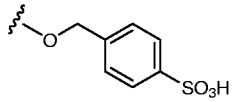
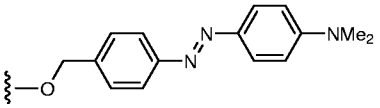
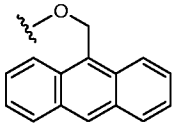
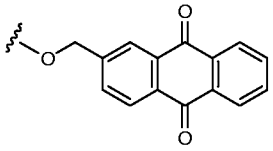
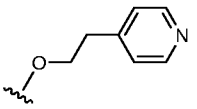
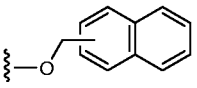
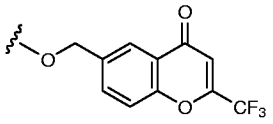
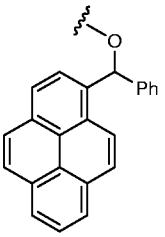
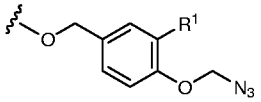
Esters	Structure	Deprotection Procedures[a]	Ref
benzyl (Bzl) ester	OCH_2Ph	H_2, Pd/C, EtOH, rt, 45 min–24 h	[267,268]
		Pd/C, cyclohexene, rt	[270,271]
		Pd/C, cyclohexa-1,4-diene, rt	[272]
		Pd/C, H_2NNH_2, rt	[273]
		Pd/C, HCO_2H, rt	[274,275]
		Pd/C, ammonium formate, rt	[276,277]
		Na/liq NH_3, alkali, aq organic mixtures	[278–281]
		1. TBDMSH, $Pd(OAc)_2$ 2. AcOH, H_2O	[282]
		alkali in aqueous organic mixtures (see Me esters, Table 1)	[283]
		1. TMSI 2. hydrolysis	[38,39]
		1. 2-(dimethylamino)ethanol 2. hydrolysis	[37]
		HF, 0 °C	[475]
4-bromobenzyl ester		HF, 0 °C, 10 min	[299]
4-methoxybenzyl (Mob) ester		TFA, 0 °C, 3–5 min	[329]
		HCO_2H, rt, 1 h	[330]
		HCl, $MeNO_2$, 30 °C, 30 min	[330]
2,4-dimethoxybenzyl (2,4-Dmb) ester		1% TFA, CH_2Cl_2, 6 × 5 min	[302]
3,4-(methylenedioxy)-benzyl ester		2 M HBr/AcOH, rt, 5 min	[303]
		TFA, rt, 5 min	[303]
2,4,6-trimethylbenzyl ester		2 M HBr/AcOH, rt, 10 min	[304,308]
		TFA, rt, 60 min	[307]
4-nitrobenzyl ester (ONbz)		H_2, Pd/C	[334]
		Zn, HCl	[335,336]
		Zn, AcOH, 0 °C, 1.5 h	[338,339]
		$Na_2S_2O_4$, pH 8–9	[340]
		$SnCl_2$, AcOH, DMF	[341]
		Na_2S, H_2O, acetone, 0 °C, 30 min	[342]
		In, NH_4Cl, H_2O, EtOH	[337]
		TBAF · H_2O, THF	[158,343]
2-nitrobenzyl ester (O2Nbz)		*hν* (350 nm)	[315–319]

Table 4 (cont.)

Esters	Structure	Deprotection Procedures[a]	Ref
benzhydryl (diphenylmethyl, Dpm) ester	$OCHPh_2$	TFA, rt, 30 min	[89,353]
		sat. HCl/AcOH, rt	[353]
		AcOH, 70 °C	[353,354]
		HCO_2H, 40 °C	[359]
		$BF_3 \cdot OEt_2$/AcOH, $CHCl_3$	[355]
		H_2, Pd/C, MeOH, THF, 3 h	[89,362–364]
		NaOH, dioxane, rt, 1 h	[353]
9-fluorenyl ester		piperidine, CH_2Cl_2, rt, 2 h	[175,176]
		50% TFA, CH_2Cl_2, rt, 30 min	[310]
		H_2, Pd/C, MeOH	[310]
10,11-dihydro-5*H*-dibenzo[*a*,*d*]cyclohepten-5-yl (5-dibenzosuberyl) ester		H_2, Pd/C	[311]
		aq HCl, THF, rt, 30 min	[311]
		TFA	[311]
2,2′-dinitrobenzhydryl [bis(2-nitrophenyl)methyl] ester		*hv* (320 nm), dioxane, 1–24 h	[320]
		all methods for Dpm esters (*vide supra*)	
4-picolyl (4-pyridylmethyl, Pic) ester		H_2, Pd/C, H_2O, EtOH, 10 h	[348]
		Na/liq NH_3, 1.5 h	[348]
		1 M NaOH, dioxane, rt, 1 h	[348]
4-(methylsulfanyl)benzyl ester		50% TFA, $CHCl_3$, DMS, 1 h	[313]
4-(methylsulfinyl)benzyl (MsOb) ester		1. TMSCl, Ph_3P	[313]
		2. TFA, DMS	
		1. TMSCl, DMS	[313]
		2. TFA, DMS	
		1. anhyd HCl, dioxane	[313]
		2. TFA, DMS	
4-sulfobenzyl ester		H_2, Pd/C	[312]
		1 M NaOH, dioxane/H_2O (3:7)	[312]
4-[4-(dimethylamino)phenylazo]benzyl ester		H_2, Pd/C, MeOH, 1 h	[314]
9-anthrylmethyl ester		NaSMe, THF, HMPA, −20 °C, 0.5–60 min	[117]
		2 M HBr/AcOH, rt, 10–30 min	[115]
		0.1 M NaOH, dioxane, rt, 15 min	[115]
		catalytic hydrogenolysis	[116]

for references see p 228

Table 4 (cont.)

Esters	Structure	Deprotection Procedures[a]	Ref
(9,10-dioxo-9,10-dihydro-2-anthryl)methyl ester		$Na_2S_2O_4$, H_2O, dioxane, 8 h 9-hydroxyanthrone, TEA, DMF, 5 h 9,10-dihydroxyanthracene-polystyrene resin, 1.5 h *hν* (350 nm), iPrOH, NMM, 4 h	[370] [370] [370] [370]
2-(4-pyridyl)ethyl ester		1. MeI 2. morpholine, CH_2Cl_2	[147,148]
1-naphthylmethyl or 2-naphthylmethyl ester		H_2, Pd/C, EtOH, 2 h $Pd(dba)_2$/dppe, ammonium formate, DMSO, 50 °C, 12 h	[328] [329]
[4-oxo-2-(trifluoromethyl)-4*H*-benzopyran-6-yl]-methyl ester		$PrNH_2$ dil H_2NNH_2, EtOH	[324] [324]
phenyl(pyren-1-yl)methyl ester		*hν* (340 nm), MeOH, MeCN	[321]
4-(azidomethoxy)benzyl ester	 R¹ = H, Cl, OMe	$SnCl_2$, MeOH, 4–12 h	[325]

[a] v/v unless otherwise stated.

2.2.1.2.1 Benzyl Ester

Benzyl (Bzl) ester protection was introduced into peptide chemistry by Bergmann and Zervas.[263] Cleavage, both by catalytic hydrogenation and by acidolysis, has contributed to the popularity of benzyl and substituted benzyl esters as protecting groups for carboxy functionalities. Before the introduction of the N^{α}-Boc protecting group, benzyl esters were often used in combination with the benzyloxycarbonyl group as both groups are removed simultaneously in the final deprotection step by hydrogenolysis. The Z group is rapidly cleaved by HBr/AcOH while benzyl esters are to some extent stable toward this reagent. The main drawback of the benzyl ester protection is the insufficient stability under conditions of repetitive N^{α}-Z as well as N^{α}-Boc cleavage in the synthesis of larger peptides.[264] Although the benzyl ester undergoes partial deprotection even under these relatively mild conditions, its complete acidolysis requires drastic conditions (sat. HBr/AcOH, rt, 12 h; 50–60 °C, 1–2 h) which may also affect peptide bonds in larger peptides.[265,266] Like the ω-benzyl esters (see Section 2.2.2.2), the α-benzyl esters are quantitatively hydrolyzed with HF[475] and this procedure is routinely applied in the final deprotection step if peptides are synthesized by Boc/Bzl chemistry in solution.[476] An extension of the benzyl ester to the

anchorage of C-terminal amino acids to solid supports led to benzyl ester type resin linkage[477] (see Section 4.3.1.2) where again HF-mediated cleavage/detachment from the resin[478] proved to be the method of choice.[479]

The simultaneous removal of benzyl esters and benzyloxycarbonyl groups from peptides is accomplished by catalytic hydrogenation with H_2, Pd/C in the usual organic solvents (e.g., EtOH, dioxane, DMF) at ambient temperatures (25–45 °C).[267,268] By employing catalytic transfer hydrogenation[269] with hydrogen donors such as cyclohexene,[270,271] cyclohexadiene,[272] hydrazine,[273] formic acid,[274,275] or ammonium formate[276,277] efficient deprotection is frequently accomplished within a few minutes at room temperature, even in the presence of sulfur-containing residues.

Reduction with sodium/liquid ammonia was introduced 70 years ago for the removal of benzyl-type and tosyl protecting groups and still remains a well-established protocol in peptide synthesis.[278–281] Benzyl esters are also cleaved with *tert*-butyldimethylsilane in the presence of palladium(II) acetate yielding the corresponding TBDMS esters which are readily hydrolyzed in the presence of AcOH/H_2O (see Section 2.2.1.3).[282]

Saponification of benzyl esters is achieved under similar conditions as for methyl and ethyl esters (see Sections 2.2.1.1.1.1 and 2.2.1.1.2.1).[283] Hydrazinolysis of peptide benzyl esters affords the corresponding hydrazides which may be readily transformed into active azide species.[264] Peptide derivatives with various C-terminal functionalities are obtained from resin-bound peptide benzyl esters by aminolysis, hydrazinolysis, saponification, and transesterification (see Section 4.3.1.2).[60–64,284] Transesterification of benzyl esters with 2-(dimethylamino)ethanol or with trimethylsilyl iodide, followed by hydrolysis of the 2-(dimethylamino)ethyl and trimethylsilyl ester intermediates with water yields the free carboxy termini.[37–39]

Amino acid benzyl esters are readily prepared by azeotropic esterification with benzyl alcohol in the presence of an acid catalyst such as HCl,[285] TosOH,[286] or benzenesulfonic acid.[287–289] Azeotropic distillation with HCl as catalyst affords the dibenzyl ester hydrochlorides of aspartic and glutamic acid (see Section 2.2.2.2) which are easily transformed into the α-benzyl monoesters by selective acidolysis of the side-chain protection with HI.[290] Milder procedures for esterification with benzyl alcohol involve the use of tribenzyl borate,[291] *N*-carboxyanhydrides (NCAs),[292] polyphosphoric acid,[293] or dimethylformamide–dineopentyl acetal.[294] Highly recommended for facile and efficient benzylation of N^{α}-protected amino acid is the treatment of the cesium salt of the corresponding amino acid with benzyl bromide.[56] Similarly, esterification of N-protected amino acids with benzyl alcohol is accomplished using DCC/DMAP[295,296] or 2-chloro-1-methylpyridinium chloride (Mukaiyama's reagent).[170] Alternatively, Yamaguchi esterification via 2,4,6-trichlorobenzoyl mixed anhydride,[238] isopropenyl chloroformate in the presence of TEA and DMAP,[54] or commonly used peptide coupling reagents such as EEDQ,[58] which generate in situ activated carboxylic acid species, can be applied. Furthermore, successful application of 2-benzylisoureas[236,237] and dibenzyl dicarbonate/DMAP in THF[57] have been reported.

H-Gly-OBzl · TosOH; Typical Procedure for the Esterification with Bzl-OH/TosOH:[286]
A mixture of H-Gly-OH (18.8 g, 0.25 mol) and TosOH · H_2O (48.5 g, 0.25 mol) in freshly distilled Bzl-OH (100 mL) and benzene (50 mL) was heated to reflux and the H_2O formed in the reaction was trapped in a Dean–Stark receiver. When no more H_2O appeared in the distillate, the mixture was cooled to rt, diluted with Et_2O (500 mL) and cooled in an ice-water bath for 2 h. The crystalline product was washed with Et_2O (200 mL) and dried. The salt was recrystallized (MeOH/Et_2O); yield: 70.8 g (84%); mp 132–134 °C.

Boc-Asn-OBzl; Typical Procedure for the Esterification of N-Protected Amino Acid Cesium Salts with Bzl-Br:[56]
A soln of Boc-Asn-OH (23.2 g, 100 mmol) in MeOH (400 mL) and H_2O (40 mL) was neutralized with 20% aq Cs_2CO_3 before removing the solvent. DMF (250 mL) was added and the solvent removed at 45 °C and the procedure repeated. Then the solid was reacted in DMF (250 mL) with Bzl-Br (18.8 g, 13.1 mL, 110 mmol). After 6 h at rt, the solvent was removed and the residue triturated with H_2O (1 L).

for references see p 228

The solid was dissolved in EtOAc (300 mL), the organic layer was washed with H_2O (150 mL), dried (Na_2SO_4), and the solvent removed. The crude ester was recrystallized (EtOAc/hexane); yield: 29 g (90%); mp 120–122 °C; $[\alpha]_D^{25}$ –17.3 (*c* 1, DMF).

H-Glu-OBzl:[290]
To a soln of H-Glu(OBzl)-OBzl · 2HCl (20 g, 50 mmol) in AcOH (200 mL) 57% HI (20 mL, 24 mmol) was added. After 6 h at 50 °C, the solvent was removed and the residual oil was thoroughly dried by repetitive addition and removal of benzene (4 × 100 mL). The remaining dark syrup was diluted with precooled (–10 °C) 95% EtOH (110 mL) containing Bu_3N (13 mL, 54 mmol), the mixture was neutralized with Bu_3N. Upon precipitation of H-Glu-OBzl in the cold, the crystals were collected, washed with abs EtOH and Et_2O, dried, dissolved in 3 M HCl (21 mL, 63 mequiv) and treated with activated charcoal (about 0.5 g). After filtration, the mixture was diluted with an equal volume of EtOH and neutralized with Bu_3N. The precipitate was collected and washed with cold abs EtOH; yield: 8 g (67%); mp 147–148 °C; $[\alpha]_D^{22}$ +12.0 (*c* 3, 0.1 M HCl).

2.2.1.2.2 Substituted Benzyl Esters

To avoid any loss of benzyl ester protection during acidolytic removal of the benzyloxycarbonyl and *tert*-butoxycarbonyl groups, electron-withdrawing substituents were used to destabilize the intermediate benzyl cation and thus to increase the acid stability. In addition to the very useful 4-nitrobenzyl esters (*vide infra*), the picolyl ester (see Section 2.2.1.2.2.3) as well as halo-[297–299] or cyano-[300] substituted benzyl esters have been reported, the latter being rarely used for α-carboxy protection. Conversely, an increase in sensitivity toward acids can be achieved by introduction of electron-releasing substituents, such as methoxy or methyl groups. Addition of scavengers to quench intermediate carbocations and to prevent electrophilic substitutions at sensitive amino acid side chains is beneficial in the deprotection of such esters.

An alternative to the 4-methoxybenzyl ester (*vide infra*) are the 2,4-dimethoxybenzyl (2,4-Dmb) esters, proposed by Stelakatos et al.[301] for side-chain protection of aspartic and glutamic acid, which have also been successfully used for α-carbon-protection in syntheses of cyclic peptides.[302] In fact, the 2,4-Dmb group is removed by treating the peptidyl resin with 1% TFA in CH_2Cl_2. These mild acidic conditions leave the *tert*-butyl-based and Mtr side-chain protection employed according to Fmoc/tBu strategy intact. However, 1% TFA will partly cleave the trityl protecting group from Cys and His residues within 20 minutes.

3,4-(Methylenedioxy)benzyl esters (piperonyl esters) are cleaved more readily than the 4-methoxybenzyl ester by acid hydrolysis with 2 M HBr/AcOH or TFA at room temperature within 5 minutes.[303] 3,4-(Methylenedioxy)benzyl esters are readily prepared either by transesterification of amino acid 4-nitrophenyl esters with 3,4-(methylenedioxy)benzyl alcohol in imidazole/dioxane (rt, 12 h) or from Nps-protected amino acids with 3,4-(methylenedioxy)benzyl chloride/TEA in DMF at room temperature.

The 2,4,6-trimethylbenzyl ester has been proposed by Stewart,[304] since it is slightly more stable toward acidolysis than the *tert*-butyl ester, thus providing a greater margin of safety when it is used in combination with highly acid-labile protecting groups such as N^{α}-Nps and N^{α}-Bpoc as well as N^{α}-Trt protection.[305] Unfortunately, selective removal of the Boc group in the presence of 2,4,6-trimethylbenzyl esters fails.[306] The 2,4,6-trimethylbenzyl esters of N-protected amino acids are prepared by reaction with (chloromethyl)mesitylene in DMF at room temperature in the presence of TEA. The esters are hydrolyzed with 2 M HBr/AcOH (rt, 10 min), or with neat TFA within 1 hour, in a selective manner versus the benzyloxycarbonyl group.[307,308] 4-Methylbenzyl and pentamethylbenzyl esters have also been proposed,[309] the latter showing similar properties to the 2,4,6-trimethylbenzyl esters.

The 4-sulfobenzyl ester has been proposed as an anionic protecting group by Bindewald et al.[312] The incorporation of a charged group in totally protected peptides increases the solubility in polar solvents. For the synthesis of 4-sulfobenzyl esters, N-protected amino acid cesium or DCHA salts are reacted with sodium 4-(bromomethyl)benzenesulfonate in DMF

at room temperature. The esters are stable under conditions necessary to remove the Boc group. After acidolytic N^{α}-deprotection the amino acid 4-sulfobenzyl ester forms a zwitterionic structure. These esters are hydrolyzed under mild basic conditions or by hydrogenolysis.

The 4-(methylsulfinyl)benzyl (MsOb) ester has been recommended as a carboxy protecting group that is stable under various basic (TEA) and acidic (TFA) conditions present during peptide synthesis.[313] These esters are exceptionally stable to TFA, but are readily deoxygenated by excesses of TMSCl/Ph_3P, TMSCl/DMS, or anhyd HCl/dioxane to the corresponding 4-(methylsulfanyl)benzyl esters which undergo rapid TFA solvolysis preferentially performed in the presence of DMS. Additionally, MsOb esters are cleaved slowly by catalytic hydrogenation. Peptide or amino acid MsOb or 4-(methylsulfanyl)benzyl esters are prepared from the corresponding alcohols or alkyl halides by standard protocols. Additionally, MsOb esters may be obtained from 4-(methylsulfanyl)benzyl esters by oxidation with 3-chloroperoxybenzoic acid.

An interesting modification of the benzyl ester group is the introduction of chromophores that allow visual monitoring of the isolation and purification steps. Esters of 4-(dimethylamino)-4′-(hydroxymethyl)azobenzene additionally provide a basic substituent that enables attachment to ion-exchange resins and thus facilitates removal of excess starting materials, e.g. acylating agents, and possible byproducts.[314]

[4-Oxo-2-(trifluoromethyl)-4*H*-benzopyran-6-yl]methyl esters were proposed by Kemp and Hanson as a benzyl-like protecting group.[324] After reaction of Boc amino acid cesium salts with the corresponding 6-(bromomethyl)-2-(trifluoromethyl)-4*H*-benzopyran-4-one esters are obtained in good yields. These esters can be used in Boc chemistry, since the N^{α}-Boc group can be cleaved in the presence of [4-oxo-2-(trifluoromethyl)-4*H*-benzopyran-6-yl]methyl esters. They are stable to secondary amines, but are cleaved by unhindered primary amines such as propylamine or dilute ethanolic hydrazine.

The 2- and 4-(azidomethoxy)benzyl esters have been described by Loubinoux and Geradin.[325] Both groups offer good stability toward nucleophiles, basic and less harsh acid treatments. Therefore, removal of Boc protecting groups in the presence of 2- or 4-(azidomethoxy)benzyl esters is feasible using either gaseous HCl in diethyl ether for 20 minutes or TosOH (1 equiv) in AcOH (or TFE) for 1 hour. The latter procedure is compatible with sulfur-containing amino acid residues.[326] Cleavage of (azidomethoxy)benzyl esters is accomplished at room temperature by mild reduction using tin(II) chloride in MeOH (4–12 h) which releases the free carboxy group and quinomethane. The quinomethane is quenched by addition of 0.8 mmol diethylamine in CH_2Cl_2 (rt, 4 h) and the resulting [(diethylamino)-methyl]phenol is removed by acid washings (1 M HCl).

2.2.1.2.2.1 4-Methoxybenzyl Ester

The acid lability of 4-methoxybenzyl (Mob) esters is comparable with that of *tert*-butyl or 2,4,6-trimethylbenzyl esters. Cleavage is accomplished by TFA at 0 °C , formic acid at room temperature, or by HCl/$MeNO_2$ (30 °C).[329,330] The Mob esters are compatible with highly acid-labile N^{α}-protecting groups such as Bpoc and Nps, but are not selectively removed in the presence of *tert*-butyl or benzhydryl esters. Similarly to the benzyl esters, they are cleaved by catalytic hydrogenolysis as well as by saponification.

The 4-methoxybenzyl esters are prepared by transesterification of N-protected amino acid nitrophenyl esters with 4-methoxybenzyl alcohol in the presence of several equivalents of imidazole, followed by subsequent acidolytic cleavage of the N-protecting group.[331] Alternatively, reaction of N-protected amino acids silver(I) salts with 4-methoxybenzyl halides (TEA/$CHCl_3$, rt, 24 h, 60% yield) or treatment of N-protected amino acids with 4-meth-

for references see p 228

oxybenzyl chloride/TEA in ethyl acetate or DMF yield the 4-methoxybenzyl esters.[332] Diesters of aminodicarboxylic acid are not available under these conditions.

Z-Gly-OMob; Typical Procedure for the Esterification with 4-Methoxybenzyl Chloride/TEA:[332]
Z-Gly-OH (2.09 g, 10 mmol), 4-methoxybenzyl chloride (1.59 g, 10 mmol), and TEA (2.1 mL, 15 mmol) were refluxed in EtOAc (10 mL) for 2 h. Upon addition of EtOAc (40 mL) and H_2O (50 mL), the organic layer was washed with 0.5 M HCl, 5% aq Na_2CO_3, and H_2O (50 mL of each), dried, and the solvent removed. The residue was recrystallized (Et_2O/hexane); yield: 3.1 g (94%); mp 61–62.5 °C.

2.2.1.2.2.2 4-Nitrobenzyl Ester

The 4-nitrobenzyl esters (ONbz) were developed due to the insufficient stability of the benzyl ester under acidic conditions, e.g. 2 M HBr/AcOH, room temperature, 30–60 minutes.[333,334] This type of ester has increased stablity to acidolysis, and it is easily removed by catalytic hydrogenolysis or saponification. Deprotection with Zn/HCl,[335,336] with indium in aqueous ethanolic ammonium chloride[337] or selective cleavage in the presence of the benzyl ester by reduction with powdered zinc in 90% AcOH (0 °C, 1.5 h)[338,339] have been reported. Similarly, 4 equivalents of sodium hydrosulfite, a reducing agent for the nitro group at pH 8–9, readily cleaves 4-nitrobenzyl esters.[340] A mixture of $SnCl_2$/DMF/AcOH is also used to reduce the nitro group, ensuring efficient deprotection of polymer-bound peptides.[341] Alternative deprotection methods are treatment with sodium sulfide in aqueous acetone (0 °C, 30 min)[342] or with 2–9 equivalents of tetrabutylammonium fluoride trihydrate in THF.[158,343]

The preparation of 4-nitrobenzyl esters is accomplished following the same methods as used for benzylation. Azeotropic distillation of the amino acid, 4-nitrobenzyl alcohol [or alternatively bis(4-nitrobenzyl) sulfite],[344] and TosOH in benzene, toluene, carbon tetrachloride, or chloroform[345] yields the corresponding esters. Intramolecular anhydrides of N-protected aminodicarboxylic acid offer access to 4-nitrobenzyl esters by trituration with the 4-nitrobenzyl alcohol and benzenesulfonic acid.[219] N-Protected amino acids may be esterified with 4-nitrobenzyl chloride either in the presence of TEA or via cesium salts.[56,334] Esterification methods via active esters, mixed, or symmetrical anhydrides in the presence of, for example, imidazole, TEA, and/or DMAP are also applicable for N-protected amino acids.

Intermediate 4-nitrobenzyl ester protection has been used in the efficient syntheses of N^{α}-Z-protected aminodicarboxylic acid ω-*tert*-butyl esters[220] via *tert*-butylation of Z-Asp-ONbz followed by selective saponification of Z-Asp(OtBu)-ONbz.[346]

H-Ala-ONbz · TosOH; Typical Procedure for the Esterification with Nbz-OH/TosOH:[347]
A suspension of H-Ala-OH (8.9 g, 0.1 mol), 4-nitrobenzyl alcohol (76.5 g, 0.5 mol), and TosOH · H_2O (57 g, 0.3 mol) in anhyd $CHCl_3$ (300 mL) was heated to reflux to form a clear soln. The solvent was distilled off and replaced by fresh anhyd $CHCl_3$ until no H-Ala-OH was identified by TLC (6 h). The solvent was removed, the residue triturated with Et_2O, and recrystallized (iPrOH, 250 mL); yield: 36.4 g (92%); mp 155–157.5 °C.

2.2.1.2.2.3 4-Picolyl Ester

4-Picolyl (4-pyridylmethyl, Pic) esters provide an attractive approach to carboxy protection due to their basic character. They form a "handle" on peptide intermediates that allows selective adsorption on ion-exchange resins.[348–351] Washing of these resins with THF or DMF allows excess reactants, e.g. acylating agents, and nonbasic contaminants to be removed before the peptide is eluted by tertiary amines in a suitable solvent. In the presence of acid-labile protecting groups preliminary saturation of the resin with 3-bromopyridine is required to avoid any decomposition. 4-Picolyl esters are resistant to 2 M HBr/AcOH, but are removed by catalytic hydrogenation (H_2, Pd/C, aq EtOH, 10 h), Na/liquid NH_3 (1.5 h), elec-

trolytic reduction,[352] and saponification (1 M NaOH, dioxane, rt, 1 h). They are readily transformed into hydrazides or amides.

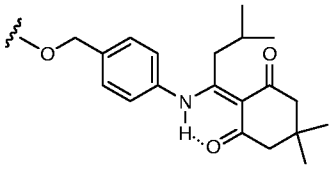
The 4-picolyl esters are prepared from N-protected amino acids by treatment with 4-(chloromethyl)pyridine/TEA (or tetramethylguanidine) in DMF (90–100 °C, 2 h) or by using DCC and pyridine-4-methanol.

Z-Val-OPic; Typical Procedure for the Esterification with Pyridine-4-methanol/DCC:[348]
To a soln of Z-Val-OH (2.76 g, 11 mmol) and pyridine-4-methanol (1.09 g, 10 mmol) in CH_2Cl_2 (25 mL) was added DCC (2.27 g, 11 mmol), and the mixture was stirred overnight at rt. The DCU was removed by filtration, and the filtrate washed with 1 M $NaHCO_3$ (2 × 30 mL) and H_2O (10 mL). After removal of the solvent, the residue was extracted with 1 M aq citric acid (3 × 75 mL). The combined extracts were neutralized with $NaHCO_3$ and extracted into EtOAc. The organic layer was dried ($MgSO_4$) and the solvent removed. The residue was recrystallized (Et_2O/petroleum ether); yield: 2 g (58%); mp 63–65 °C; $[\alpha]_D^{20}$ −11.6 (*c* 1, DMF).

2.2.1.2.2.4 4-{[1-(4,4-Dimethyl-2,6-dioxocyclohexylidene)-3-methylbutyl]amino}benzyl Ester

The 4-{[1-(4,4-dimethyl-2,6-dioxocyclohexylidene)-3-methylbutyl]amino}benzyl (Dmab) ester[371] (Scheme 6) was introduced as a further development of the 4-{[1-(4,4-dimethyl-2,6-dioxocyclohexylidene)ethyl]amino}benzyl group.[372] It retains all the desired chemical properties and is completely stable under the Fmoc deprotection conditions, i.e. 20% piperidine/DMF. The application of the Dmab ester as an orthogonal protecting group is demonstrated by the synthesis of a model cyclic peptide by on-resin head-to-tail cyclization of the corresponding linear peptide assembled using Fmoc-Glu-ODmab and standard Fmoc/tBu procedures on 2-chlorotrityl polystyrene resin.[371] In accordance with the safety-catch principle cleavage is accomplished by trituration with hydrazine hydrate/DMF (2:98) which initially removes the Dde-derived N-protecting group of the labile 4-aminobenzyl ester.

Scheme 6 The 4-{[1-(4,4-Dimethyl-2,6-dioxocyclohexylidene)-3-methylbutyl]amino}benzyl (Dmab) Ester



2.2.1.2.3 Other Aralkyl Esters

2.2.1.2.3.1 Benzhydryl Ester

Benzhydryl (diphenylmethyl, Dpm) esters are about as labile as *tert*-butyl esters, thus they are quantitatively cleaved under mild acidic conditions, e.g. TFA (rt, 30 min), saturated HCl solution in AcOH at room temperature, AcOH at 70 °C, or fivefold excess of boron trifluoride–diethyl ether complex/AcOH in chloroform (rt, 1 h).[89,353–356] To obtain good yields it is recommended to admix scavengers such as thioanisole or anisole for acidolysis with TFA because of the release of highly electrophilic benzhydryl carbocations into the reaction medium.[357,358] The use of strong acids such as neat formic acid at 40 °C[359] or HF/$MeNO_2$/AcOH (12:2:1)[360] and Lewis acids (e.g., $AlCl_3$)[361] have been described in organic syntheses. Dpm esters are particularly prone to cleavage by catalytic hydrogenation (e.g., with H_2, Pd black, MeOH, THF, 3 h).[89,362,363] Simultaneous deprotection together with the Z group by hydrogenolysis over Pd/C was described in the clavalamine synthesis.[364] In contrast to the *tert*-butyl esters and in spite of obvious steric hindrance the Dpm esters are saponified

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rapidly [NaOH (2 equiv), dioxane, rt, 1 h] and undergo fast hydrazinolysis [hydrazine hydrate (2 equiv), MeOH, reflux].[353]

Amino acid 4-toluenesulfonate or 2-naphthalenesulfonate salts are esterified in high yield with diazodiphenylmethane in DMF (50 °C, 10 min).[362–365] Diazodiphenylmethane is prepared by oxidation of benzophenone hydrazone, e.g. with yellow mercury(II) oxide in the presence of base, and can be stored in the dark.[366,367] In situ oxidation of the benzophenone hydrazone is performed with peracetic acid/I_2 (trace) in the presence of the N-protected amino acids or peptides.[368,369] This method is not compatible with sulfur-containing amino acid residues. Alternatively, Dpm esters of protected amino acids and peptides are obtained using benzhydryl chloride or diphenylmethanol by standard esterification procedures.[89,353]

9-Fluorenyl esters have been recommended as a useful alternative to the benzhydryl (diphenylmethyl) esters in the protection of amino acids.[310] They are readily prepared from 9-diazofluorene and the amino acid following a similar procedure to that for the benzhydryl esters. Cleavage is accomplished by mild acidolysis or hydrogenolysis.

The 10,11-dihydro-5*H*-dibenzo[*a*,*d*]cyclohepten-5-yl (5-dibenzosuberyl) group was proposed by Pless[311] as a system with broad applications in amino, hydroxy, sulfanyl, and also carboxy protection. The ester is prepared from N-protected amino acids with 5-chloro-10,11-dihydro-5*H*-dibenzo[*a*,*d*]cycloheptene/TEA in dioxane (reflux, 4 h). It is cleaved like *tert*-butyl esters by acid hydrolysis with aqueous HCl/THF (rt, 30 min) or with neat TFA; it is also removed by catalytic hydrogenolysis.

H-Xaa-ODpm·TosOH; General Procedure for the Esterification of Amino Acid TosOH Salts with Diazodiphenylmethane:[363]

To a soln of H-Xaa-OH·TosOH (10–20 mmol) in DMF (5 mL) was added at 50 °C a soln of diazodiphenylmethane (15 mmol) in DMF (10 mL) while stirring. After 10 min, the solvent was removed and the residue recrystallized (MeCN); yields range from 75–90%. For Arg and His the di-2-naphthalenesulfonate salts were used; yield: 78 and 53%, respectively.

2.2.1.2.3.2 Photocleavable Aralkyl Esters

In contrast to the 4-nitrobenzyl esters (*vide supra*) the 2-nitrobenzyl derivatives are photolytically cleaved in H_2O/dioxane at pH 7.[315–317] Excellent yields were obtained in the syntheses of protected peptide acids related to LHRH by Rich and Gurwara using a 2-nitrobenzyl ester linkage to polystyrene;[318,319] cleavage was achieved by irradiation at 350 nm for 18–24 hours in MeOH. 2,2′-Dinitrobenzhydryl [bis(2-nitrophenyl)methyl] esters are formed and cleaved by the same methods as those used for benzhydryl esters. Additionally, quantitative photolysis is obtained by irradiation ($h\nu$ 320 nm), e.g. in dioxane or THF over 1–24 hours.[320] Photolabile phenyl(pyren-1-yl)methyl esters were introduced into peptide chemistry by Iwamura et al. who favored them over pyren-1-ylmethyl esters for the stability and easier handling of the corresponding diazopyrene precursor.[321–323] 1-[Diazo(phenyl)methyl]pyrene is stable at room temperature for months and reacts with 4-toluenesulfonic acid salts of amino acids in DMF at 50 °C to generate the highly fluorescent phenyl(pyren-1-yl)methyl esters in moderate yields. Upon irradiation in MeOH/MeCN (0.05 mmol ester in 5 mL) at 340 nm the deprotected amino acids are recovered. Phenyl(pyren-1-yl)methyl esters are stable to HCl in acetonitrile for about 4 hours, but are susceptible to catalytic hydrogenation (for photocleavable esters see also Section 2.4).

2.2.1.2.3.3 9-Anthrylmethyl Ester

9-Anthrylmethyl esters are prepared from N-protected amino acids by reaction with 9-(chloromethyl)anthracene in acetonitrile and in the presence of TEA as base.[115] They are readily cleaved by acidic (2 M HBr/AcOH, rt, 10–30 min) and alkaline (0.1 M NaOH/dioxane, rt, 15 min) hydrolysis as well as by catalytic hydrogenolysis.[116] These esters are very reactive toward thiolate anions (NaSMe, THF, HMPA, −20 °C) with quantitative release of the carboxylate within 30 seconds to 1 hour.[117]

2.2.1.2.3.4 (9,10-Dioxo-9,10-dihydro-2-anthryl)methyl Ester

(9,10-Dioxo-9,10-dihydro-2-anthryl)methyl esters are readily available by reaction of carboxylic acids with 2-(hydroxymethyl)anthraquinone using DCC/HOBt for activation. They are cleaved in high yields by a variety of selective, mild reducing agents as well as by photolysis and catalytic hydrogenation.[370] The following cleavage conditions are recommended: 5 equivalents of sodium hydrosulfite in H_2O/dioxane (8 h, 100%), 4 equivalents of 9-hydroxyanthrone in TEA (2 equiv)/DMF (5 h, 99%), excess polystyrene resin functionalized with 9,10-dihydroxyanthracene (1.5 h, 100%), photolysis ($h\nu$ 350 nm) in iPrOH in the presence of 4 equivalents of 4-methylmorpholine (4 h, 99%). The (9,10-dioxo-9,10-dihydro-2-anthryl)methyl group is fully stable to TFA (rt, 1 h), HCl in CH_2Cl_2 (rt, 1 h), or TEA (2 equiv) in H_2O/dioxane (24 h) and thus compatible with several different protection schemes in peptide synthesis.

2.2.1.2.3.5 2-(4-Pyridyl)ethyl Ester

2-(4-Pyridyl)ethyl esters for protection of the carboxy group were introduced by Kunz and Kneip[147] and by Kessler et al.[148] Amino acid 2-(4-pyridyl)ethyl esters, generally isolated as oils, are prepared from 2-(4-pyridyl)ethanol, Z- or Boc-protected amino acids, and DCC/DMAP. These esters show good solubility in protic solvents and are cleaved in a two-step process: methylation with iodomethane yields the pyridinium salts which are decomposed under mild basic conditions, e.g. with morpholine in CH_2Cl_2. Hydrogenolysis of the Z group in the presence of 2-(4-pyridyl)ethyl esters is rather slow, while selective acidolysis of the Z and Boc groups proceeds efficiently. No racemization was observed during esterification and deprotection. The use of 2-(4-pyridyl)ethyl esters is not recommended when the peptide includes sulfur-containing amino acids.

2.2.1.2.3.6 9-Fluorenylmethyl Ester

The 9-fluorenylmethyl (Fm) ester was first introduced by Bodanszky as an extension of the Fmoc protecting group.[175] The amino acid esters are obtained by esterification of N-protected amino acids with 9-fluorenylmethanol using DCC/DMAP[176] or by transesterification of active esters. The Fm esters are also synthesized by reacting 9-fluorenylmethyl chloroformate with N-protected amino acids in the presence of DIPEA as base and DMAP as catalyst.[177] The 9-fluorenylmethyl esters are stable under the conditions of peptide synthesis and no transesterification was observed on heating in alcohols. The cleavage is similar to the deprotection procedures used for Fmoc groups, by treatment with diethylamine or piperidine in CH_2Cl_2 at room temperature for 2 hours. No racemization has been observed during formation and cleavage of N-protected amino acid or peptide esters.

for references see p 228

2.2.1.2.3.7 Naphthylmethyl Esters

1-Naphthylmethyl and 2-naphthylmethyl esters have been proposed as protecting groups as they are selectively cleaved in the presence of a benzyl ester by homogeneous palladium-catalyzed hydrogenolysis using ammonium formate.[327,328]

2.2.1.3 Silyl Esters

Silyl esters are stable to nonaqueous reaction conditions, but are often too labile to mild acid or base or even neutral aqueous media to survive many simple manipulations. Thus, they have not found wide application in peptide synthesis. Due to easy formation and cleavage they may play an important role as intermediates in the synthesis of amino acid derivatives and for temporary carboxy protection in the preparation of small peptide fragments. The TMS group has been used for the solubilization of H-Arg-OH for the synthesis of Z-Arg(Z_2)-OH[373] and in the synthesis of N^{α}-Nps- and N^{α}-Tfa-protected amino acids.[374] Amino acid trimethylsilyl esters as well as the related N^{α}-TMS derivatives react rapidly with acylating agents and are used for the preparation of peptides with amino acid active esters, e.g. *N*-hydroxysuccinimide-, 4-nitrophenyl-, or 2,4,5-trichlorophenyl esters, or mixed anhydrides.[375–380]

Trimethylsilyl esters are intermediate products in the cleavage of *tert*-butyl, benzyl, or methyl esters using trimethylsilyl iodide[38,39] and also in the preparation of methyl esters using trimethylsilyl chloride.[52,53] An interesting feature of the silyl esters is the protection of acids against reduction with hydroborating agents.[381]

The stability of silyl esters is strongly dependent on steric effects and parallels the steric bulk of the substituents on the silicon atom.[382] A TMS ester is cleaved by, for example, refluxing in EtOH,[383,384] while silyl esters with more bulky substituents are cleaved by mild acidic or basic hydrolysis. The *tert*-butyldimethylsilyl (TBDMS) ester is cleaved at room temperature either by AcOH/H_2O/THF (3:1:1), by K_2CO_3/MeOH/H_2O, or by tetrabutylammonium fluoride in DMF.[385,386] The di-*tert*-butylmethylsilyl group affords the most stable of the silyl esters allowing selective removal of a tetrahydropyranyl group with pyridine/TsOH in warm EtOH.[387]

Silyl esters are generally prepared from carboxylic acids and the corresponding silyl chlorides or silyl triflates in the presence of base, e.g. pyridine, TEA, imidazole, in CH_2Cl_2, $CHCl_3$, DMF, or THF.[375,385,388,389] *N*-Methyl-*O*-(trimethylsilyl)acetamide or *N*,*O*-bis(trimethylsilyl)acetamide in dioxane or DMF have successfully been used for the synthesis N,O-bis-trimethylsilylated amino acids.[375] As selective formation of the silyl ester by using 1 equivalent of silylamide is not possible, sufficient reagent is used to assure formation of the N,O-bis-silylated amino acid which is then directly submitted to the coupling reaction (*vide supra*) without isolation. Since acetamide results from the reaction and is hard to remove from peptide intermediates trimethylsilyl cyanide has been proposed as an alternative reagent.[390] Silyl esters are generally accessible with full retention of the chiral integrity of the amino acids.[375]

2.2.1.4 Stannyl Esters

Triethylstannyl and tributylstannyl esters are prepared for temporary protection of the carboxy function of amino acids.[391,392] Stannyl esters of N-protected amino acids are stable to anhydrous amines as well as to water and alcohols. Aqueous amines convert them into the ammonium salts. Stannyl esters of N-unprotected amino acids are cleaved in quantitative

yield by water or alcohols. Tributylstannyl esters of protected peptides are intermediates in palladium-catalyzed allyl ester cleavage in the presence of tributyltin hydride.[259]

2.2.1.5 Aryl Esters

Phenyl esters show notable reactivity in aminolysis compared to alkyl esters.[393] In spite of their weak active ester character, acylation of amino acid phenyl esters with, for example, mixed anhydrides or 2-acylisoureas, generates peptides in good yields even in demanding syntheses.[394–397] Amino acid phenyl esters are readily prepared from the corresponding N-protected derivatives.[398] The esters offer complete stability toward acidolysis (HBr/AcOH) and hydrogenolysis (H_2, Pd/C). Rapid and racemization-free cleavage of the phenyl esters is accomplished within 15 minutes by alkaline hydrolysis under moderate basic conditions (pH 10.5, aq DMF) in the presence of 1 equivalent of hydrogen peroxide.[395] When sensitive amino acid residues are present, such as Trp, Met, and Cys, addition of a large excess of DMS to the cleavage mixture is recommended to prevent unwanted oxidations. The saponification of phenyl esters in the absence of peroxide as well as acidolysis with mineral acids in dioxane results in substantial racemization.

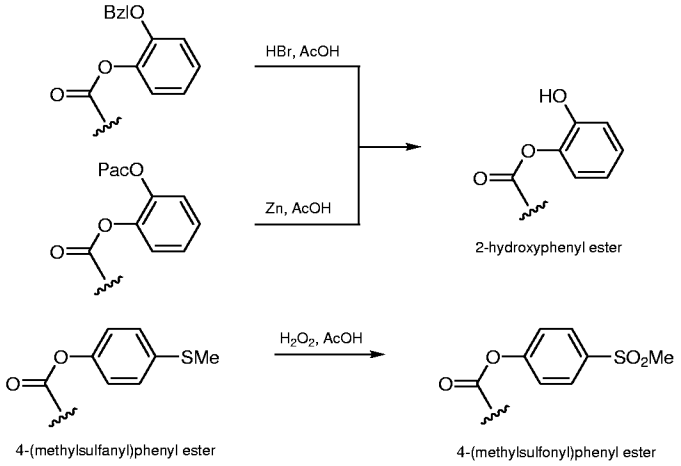
4-Nitrophenyl, pentachlorophenyl, and 2,4,5-trichlorophenyl esters have been used for carboxy protection in the so-called backing-off[20] approach, illustrated in the preparation of several activated peptide fragments for convergent peptide syntheses.[19,20,399–401]

Electron-releasing substituents in the aromatic nucleus reduce the electrophilic character of the aryl ester carbonyl carbon, thus increasing the stability of the esters toward aminolysis. This feature has been used in the design of several dual-function ester groups for peptide synthesis.[18] 4-(Methylsulfanyl)phenyl esters (Scheme 7) provide satisfactory carboxy protection toward aminolysis and, simultaneously allow facile racemization-free conversion into the active ester.[402–405] The conversion of the protecting group into the activated 4-(methylsulfonyl)phenyl ester is achieved by oxidation with H_2O_2/AcOH, 3-chloroperoxybenzoic acid, or magnesium monoperoxyphthalate (phase-transfer catalysis).[406,407] 4-(Methylsulfanyl)phenyl esters are readily prepared by esterification of N-protected amino acids using DCC. Removal of the N-terminal protecting group from activated polymer-bound peptide 4-(methylsulfonyl)phenyl ester results in self-aminolysis and thus release of protected cyclic peptides from the resin.[408] The introduction of a C-terminal amino acid 4-(methylsulfanyl)phenyl ester residue via aminolysis of a peptide attached to an oxime-type solid support was reported as a novel route for the preparation of a peptide active ester.[407]

The use of N-protected amino acid 2-hydroxyphenyl esters in peptide synthesis was suggested owing to good activation for aminolysis and high resistance to racemization.[409,410] By protecting the phenolic group of the catechol monoesters as benzyl or phenacyl derivatives, ester-type protecting groups with dual-functionality are obtained (Scheme 7). These esters offer appropriate carboxy protection while the active 2-hydroxyphenyl ester is easily recovered by removal of the phenol protection by catalytic hydrogenation or HBr/AcOH for benzyl, and Zn/AcOH for phenacyl, respectively.[410–413]

for references see p 228

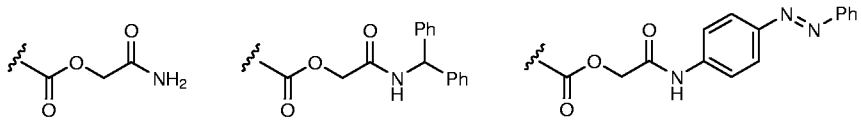
Scheme 7 Conversion of 2-(Benzyloxy)phenyl, 2-(Phenacyloxy)phenyl, and 4-(Methylsulfanyl)phenyl Esters into the Corresponding Active Ester Species



2.2.1.6 Aminocarbonylmethyl Esters

The aminocarbonylmethyl (carbamoylmethyl, Cam) ester group (Scheme 8) can be used in combination with Boc, Z, and Fmoc as N^{α}-protecting groups.[414,415] Preparation of such an ester is achieved by the reaction of a N-protected amino acid cesium salt with an α-haloacetamide. Aminocarbonylmethyl esters are not affected under the usual coupling conditions and are stable toward TFA as well as catalytic hydrogenation. They are cleaved by alkaline hydrolysis using 0.5 M sodium hydroxide or sodium carbonate in DMF/H_2O. The considerable water solubility of aminocarbonylmethyl esters sometimes results in reduced yields after the workup of reaction mixtures. Therefore, the less water soluble [(benzhydrylamino)carbonyl]methyl ester has been proposed as an alternative carboxy protecting group which offers almost the same stability as the aminocarbonylmethyl ester.[416] Additionally, the [(benzhydrylamino)carbonyl]methyl ester is readily converted into the corresponding amide or hydrazide derivative. As a colored carboxy protecting group the [(4-phenylazophenylamino)carbonyl]methyl ester was introduced to peptide chemistry.[417]

Scheme 8 Aminocarbonylmethyl Esters



Cam ester [(benzhydrylamino)carbonyl]methyl ester [(4-phenylazophenylamino)carbonyl]methyl ester

2.2.1.7 N′-Protected Amino Acid Hydrazides

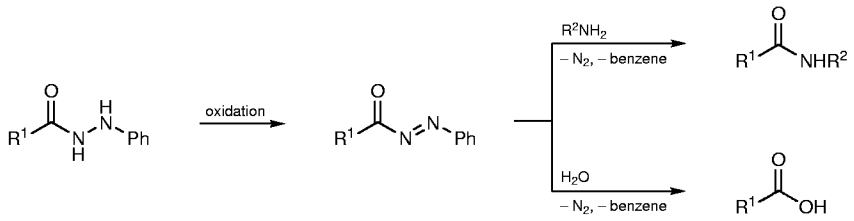
N′-Substituted hydrazides are effective semipermanent α-carboxy protecting groups which have simultaneously the possibility of conversion into azides or diimides for further coupling steps involving the carboxy terminus (see Section 3.1.1.4).[18,418] The hydrazide requires the use of an additional amine protecting group.[419,420] Removal of the *N′*-substituent after chain-elongation releases the peptide hydrazide thus bypassing the frequently troublesome hydrazinolysis of the corresponding peptide esters. In principle, almost every amino protecting group can be used for this purpose provided that it is fully compatible with the overall protection scheme. Several syntheses of large peptides illustrated the general applicability of

N'-benzyloxycarbonyl[420–422] and *N'*-*tert*-butoxycarbonyl[184,422–425] hydrazides. Other suggested protection schemes include masking of hydrazides by trityl,[426,427] trifluoroacetyl,[428] formyl,[429] trichloroethoxycarbonyl,[430] and 4-picolyloxycarbonyl[431] moieties.

N'-Protected amino acid or peptide hydrazides are prepared from the corresponding N^{α}-protected amino acids or peptides and monosubstituted hydrazine by any of the conventional coupling methods (see Section 3.1.1.4).

Amino acid *N'*-phenylhydrazides ensure efficient carboxy protection, but upon chain elongation the phenyl substituent cannot be removed to release a peptide hydrazide as the precursor for azide formation. Nevertheless, the hydrazide can be oxidized to the corresponding peptide *N'*-phenyldiimide which, as an acylating agent, offers diverse reactivity in the presence of nucleophiles for aminolysis, alcoholysis, or hydrolysis (Scheme 9).[145,432–434] Activating oxidation is accomplished using NBS in CH_2Cl_2,[145,432] I_2,[145,432] electrolysis (Pt anode),[435] tetrabutylammonium perchlorate,[435] enzymes (oxidoreductase-peroxidase, lactase),[436] or O_2/Cu(II)–pyridine complex.[436] Amino acid *N'*-phenylhydrazides are synthesized from the N-protected amino acids and phenylhydrazine using papain as a catalyst or by the azide method.[433]

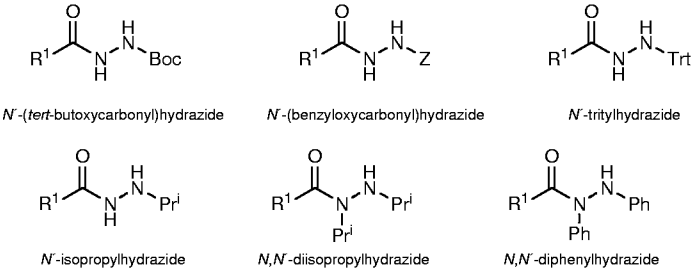
Scheme 9 Conversion of Phenylhydrazide into the Activated *N'*-Phenyldiimide[145,432–434]



Further substituted hydrazides proposed for carboxy protection in peptide synthesis are *N'*-isopropylhydrazides[435] and the disubstituted *N,N'*-diphenylhydrazide[145] and *N,N'*-diisopropylhydrazide[437,438] derivatives.

In Scheme 10, several N'-substituted hydrazides employed for α-carboxy protection are shown.

Scheme 10 Several N'- and N,N'-Substituted Hydrazides for α-Carboxy Protection



N'-(*tert*-butoxycarbonyl)hydrazide | *N'*-(benzyloxycarbonyl)hydrazide | *N'*-tritylhydrazide

N'-isopropylhydrazide | *N,N'*-diisopropylhydrazide | *N,N'*-diphenylhydrazide

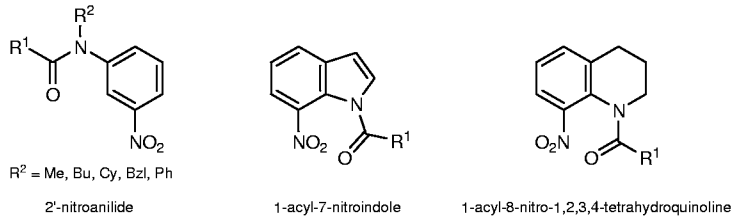
2.2.1.8 Amide

Unsubstituted amides generally represent an irreversible carboxy protection because of the lack of selective chemical deprotection procedures in the presence of peptide bonds. The use of highly selective peptide amidases has been described as an efficient alternative[23] (see Section 2.5). Amides complement ester protecting groups in the cleavage conditions necessary to release the free carboxylic acid. They are stable to alkaline hydrolysis while being readily cleaved in the presence of nitrous acid.[439] Mild cleavage is accomplished using

for references see p 228

nitrogen dioxide in carbon tetrachloride[440] or *N*-(chloromethylene)cyclohexylamine *N*-oxide/silver tetrafluoroborate followed by acidolysis.[441] N,N-Disubstituted amides are removed by tBuOK/H_2O in Et_2O.[442] 2-Nitroanilides, 1-acyl-7-nitroindoles, and 1-acyl-8-nitro-1,2,3,4-tetrahydroquinolines shown in Scheme 11 are susceptible to photolytic cleavage (*hν*, 350 nm, 5–10 h).[443–446] Amines with low nucleophilicity such as 4-nitroaniline require efficient activating agents, e.g. isobutyl chloroformate or pivaloyl chloride, for the synthesis of the corresponding amides.[447,448]

Scheme 11 Photolytically Cleavable Amides for α-Carboxy Protection[443–446]



N-Acyl-4-polystyrenesulfonamides, readily available from amino acid active esters and resin-bound sulfonamides were successfully introduced into SPPS by Kenner.[449] *N*-Acyl-sulfonamides are stable to alkaline hydrolysis and conditions present during acidolysis of Z or Boc groups. According to the safety-catch principle, they are rapidly cleaved under mild basic conditions (0.5 M NaOH) after conversion into the *N*-methyl derivatives by reaction with diazomethane.

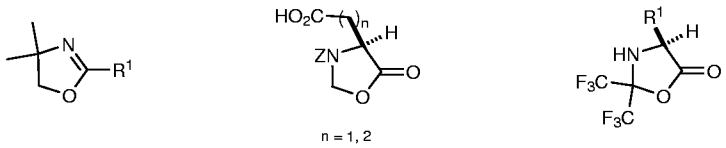
2.2.1.9 Dihydrooxazole, Oxazolidin-5-one, Oxazole, and 4-Methyl-2,6,7-trioxabicyclo[2.2.2]octane Ester

2-Substituted dihydrooxazoles (Scheme 12) require harsh conditions for hydrolysis, e.g. strong mineral acids (3 M HCl, EtOH, reflux) or alkaline hydrolysis (aq NaOH) of the corresponding iminium salts, obtained by N-methylation with iodomethane.[450] They offer excellent resistance to attack by strong nucleophiles such as lithium aluminum hydride, Grignard or organolithium reagents incompatible with the presence of an ester group. 2-Substituted 4,4-dimethyl-4,5-dihydrooxazoles are prepared by treatment of the carboxylic acid with 2-amino-2-methylpropan-1-ol/thionyl chloride or with dimethylaziridine in the presence of DCC.[451,452]

4-Alkyloxazolidin-5-ones (Scheme 12) are prepared for intermediate α-carboxy protection to allow selective manipulation of the ω-carboxy groups of aspartic and glutamic acid (see Section 2.2.2.4).[453] They are synthesized as the *N*-benzyloxycarbonyl derivatives from the corresponding N^{α}-protected amino dicarboxylic acids by treatment with HCO_2H/Ac_2O/AcOH/$SOCl_2$ or by azeotropic distillation with HCHO/TosOH in benzene. The heterocyclic ring is removed either by alkaline hydrolysis or by catalytic hydrogenation to yield the free acid. Ammonolysis, hydrazinolysis, or even slow aminolysis yield the corresponding amides, hydrazides, and peptides.

Hexafluoroacetone reacts with α-amino acids to give 2,2-bis(trifluoromethyl)oxazolidin-5-ones (Scheme 12). This heterocyclization process results in a simultaneous protection of the α-amino and carboxy group. Furthermore, the carboxylic group is obtained as an active ester that can be functionalized in various ways on reaction with nucleophiles. Since additional groups present in the amino acid side chain such as carboxy or hydroxy groups remain unaffected, this method can be applied successfully to regioselective functional group transformations in multifunctional α-amino acids.[454]

Scheme 12 4,4-Dimethyl-4,5-dihydrooxazoles and Oxazolidin-5-ones Used in α-Carboxy Protection

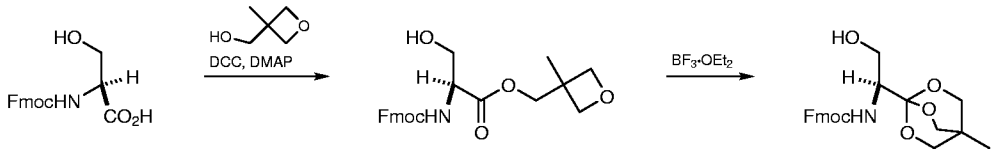


4,4-dimethyl-4,5-dihydrooxazole | 3-(benzyloxycarbonyl)oxazolidin-5-one | 2,2-bis(trifluoromethyl)oxazolidin-5-one

4,5-Diphenyloxazole as a carboxy protecting group shows resistance to the conditions of alkaline hydrolysis as well as of acidolytic removal of Z and Boc groups.[455] Furthermore, the strong ultraviolet activity of the oxazole moiety facilitates the identification and purification of intermediates in peptide synthesis. Formation of 4,5-diphenyloxazoles takes place readily by the standard Davidson procedure employing the N-protected amino acid, benzoin, and DCC to form the oxo ester, which is then converted into the corresponding oxazole with ammonium acetate in glacial AcOH (yield: 80–85%). Rapid and nearly quantitative transformation of the oxazole into a triamide species is accomplished by dye-sensitized photooxygenation in dry CH_2Cl_2 (see Section 2.4.2.3.2). The resulting triamide is highly activated and hydrolysis to recover the free carboxylic acid is facile. Aminolysis of N-protected amino acid dibenzoyl triamides with a second amino acid in DMF yields dipeptides. Application of the photooxidative process may be limited by residues such as His, Trp, Cys, and Met, which are sensitive to singlet oxygen.

The ortho ester 4-methyl-2,6,7-trioxabicyclo[2.2.2]octane (OBO) protecting group (Scheme 13) was developed by Corey[456] in order to shield the carbonyl group of carboxylic acids from attack by strong nucleophiles. Syntheses of a wide range of nonproteinaceous and unnatural α-amino acids were accomplished taking advantage of the stability of α-carboxy OBO esters toward metal hydrides, Grignard and organolithium reagents, Reformatsky and Wittig reactions, alkoxides, diazomethane, and hydrogen azide under Mitsunobu conditions.[457–460] Preparation of the N^{α}-protected amino acid OBO ester is achieved by boron trifluoride catalyzed rearrangement of the corresponding 3-(hydroxymethyl)-3-methyloxetane ester which is available from reaction of the N-protected amino acid and 3-(hydroxymethyl)-3-methyloxetane (20 equiv) in the presence of DCC (1.2 equiv)/DMAP (0.05 equiv).[457] Alkylation of a N-protected amino acid cesium salt by the corresponding 3-methyl-3-(tosyloxymethyl)oxetane in the presence of sodium iodide (0.1 equiv) in DMF was found to be much more convenient.[458] The OBO ester is removed either by a two-step procedure starting with mild acid hydrolysis by TFA/H_2O/CH_2Cl_2 (0.35:0.25:15) followed by saponification of the intermediate 2,2-bis(hydroxymethyl)propyl ester with cesium carbonate (5.1 equiv) in MeOH/H_2O (3:5) or by treatment with trimethylsilyl iodide (14 equiv, 75 °C, 24 h) followed by hydrolysis of the resulting trimethylsilyl ester with 0.5 M NaOH.[457]

Scheme 13 Synthesis of the N^{α}-Protected Amino Acid OBO Ester[456]



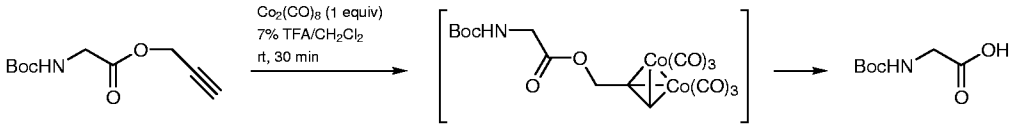
2.2.1.10 Propargyl Ester

The use of the propargyl ester function for α-carboxy protection has been described.[461] These esters are obtained quantitatively from N-protected amino acids and propargyl alcohol in the presence of DCC/DMAP. Propargyl esters are stable to neat TFA but are

for references see p 228

readily cleaved by acidolysis after formation of an alkyne–cobalt complex that stabilizes the cationic charge at the α-position (Scheme 14). Thus, near quantitative deprotection is achieved applying $Co_2(CO)_8$ (1 equiv) and TFA (7%) in CH_2Cl_2 (rt, 30 min).

Scheme 14 Removal of Propargyl Ester Protection via an Alkyne–Cobalt Complex



2.2.1.11 Enzymatically Cleavable α-Carboxy Protection

The use of enzymes for cleaving protecting groups in peptide synthesis has been reviewed[462–464] and is described in Section 2.5.3.

Thermolysine,[465] a protease from *Bacillus thermoproteolyticus*, effects selective peptide bond hydrolysis on the amino side of hydrophobic amino acid residues (Leu, Ile, Val, Phe), thus enabling removal of a "supporting" peptide ester sequence from the protected target fragment.

Enzyme-catalyzed hydrolysis, exploiting the esterase activity of proteases such as trypsin and chymotrypsin[120,121] or carboxypeptidase Y[122] has opened alternative routes to the deprotection of several peptide methyl, ethyl, and *tert*-butyl esters. In fact, methyl, ethyl, and benzyl esters are successfully hydrolyzed from protected peptides using the alkaline protease from *Bacillus subtilis DY*[466] or alcalase from *Bacillus licheniformis* which accepts highly hydrophobic and in aqueous solutions insoluble Fmoc amino acids as substrates.[467,468] Thermitase from *Thermoactinomyces vulgaris* shows broad amino acid side-chain tolerance and cleaves methyl, ethyl, benzyl, and *tert*-butyl esters.[469]

Ester hydrolysis catalyzed by proteases may be accompanied by undesired cleavage of peptide bonds; the use of lipases was proposed to overcome this evident problem. The selective removal of heptyl (Hep) esters is achieved by lipase N from *Rhizopus niveus* in high yields without damaging urethane protecting groups or peptide bonds.[470,471] In order to improve the enzymes access to the substrate in lipase-catalyzed ester hydrolysis markedly more hydrophilic 2-morpholinoethyl esters, available from the corresponding 2-bromoethyl esters and morpholine (see Section 2.2.1.1.2.2.2), were proposed.[472] Polar esters derived from ethylene glycol proved to be more favorable concerning substrate acceptance of the lipase, yields of hydrolysis, physicochemical properties, and solubility. 2-(2-Methoxyethoxy)ethyl esters[473] and the highly recommended 2-methoxyethyl esters[474] have successfully been applied to glycopeptide synthesis.

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2.2.2 ω-Carboxy Group

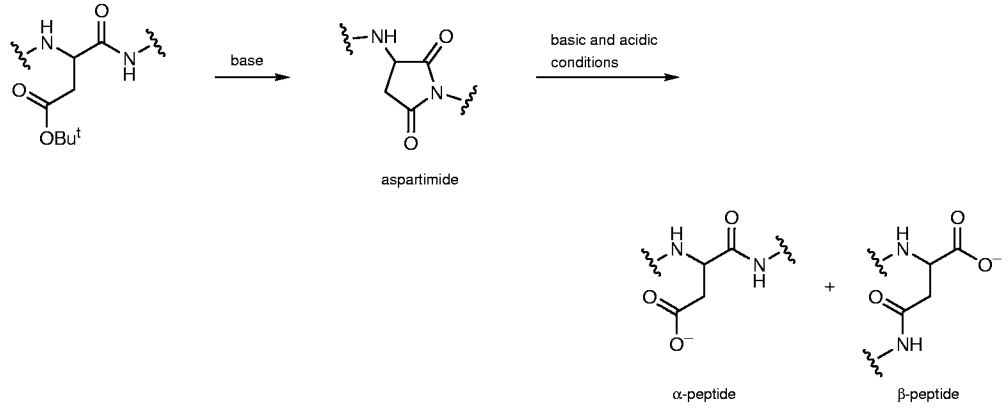
R. Thürmer, H.-J. Kohlbau, and W. Voelter

The synthesis of aspartic and glutamic acid-containing peptides without protection of the side-chain carboxy groups is feasible, but often unsatisfactory. Their protection, generally as esters, represents the better approach to the synthesis of complex peptides, as concerns about counterions associated with the carboxylate group, and more importantly about reactions with acylating agents, are prevented. The type of protection of the side-chain carboxy groups has to differ in its reactivity from that of the α-carboxy group whenever additional synthetic steps have to be performed on either the ω- or α-carboxy group in the peptide. Accordingly, in such cases, protection of dicarboxylic amino acids requires specific introduction of two selectively removable ester groups.

In solid-phase peptide synthesis (SPPS), either the acid-labile Boc or the base-labile Fmoc group is commonly used for N^{α}-protection. In the first case, the side-chain protecting groups are cleaved from the final peptide simultaneously with detachment from the resin, e.g. with HF or TfOH, and benzyl esters fulfill these requirements in most cases in a satisfactory manner. In the Fmoc strategy, final cleavage of the peptide from the resin is commonly achieved with TFA, which also cleaves the *tert*-butyl ester groups commonly used in this synthetic strategy. In general, protecting groups initially proposed for α-carboxy groups are equally well suited for the protection of side-chain carboxy groups.[1,2] Their regioselective introduction often involves simple kinetically controlled esterification of the free amino acids, taking advantage of the higher reactivity of the side-chain carboxy groups.

Aspartimide formation is a major side reaction during the synthesis of peptides containing aspartyl sequences such as Asp-Gly, Asp-Ser, and Asp-His. Aspartimide is formed either by an acid- or a base-catalyzed mechanism,[3–7] and the reaction rates depend upon the acid, the base, or the sequence of the peptide and/or upon the protecting group used. Opening of the aspartimide ring leads to β-aspartyl peptides as the major component (Scheme 1). Since the chromatographic separation of either the aspartimide derivative or the β-aspartyl peptide from the desired peptide is generally difficult, the acid-catalyzed aspartimide formation represents a serious problem, particularly in Boc/Bzl chemistry where treatment with strong acids like HF for the final resin cleavage and deprotection step results in aspartimide formation to variable extents. Conversely, base-catalyzed aspartimide formation is strongly affected by the type of base used in the synthetic procedures. Correspondingly, with Fmoc/tBu chemistry[8] in several sequence-dependent cases, high levels of aspartimide have been reported upon standard piperidine treatment.[9,10]

Scheme 1 Reaction Mechanism of Base-Catalyzed Aspartimide Formation



In order to suppress this most troublesome side reaction in peptide synthesis, different side-chain protecting groups have been proposed and among these some cycloalkyl esters, especially the cyclohexyl group (see Section 2.2.2.6), have been found to be very efficient. The fluorenylmethyl group (see Section 2.2.2.8) is another candidate for prevention of aspartimide formation. Similarly efficient is the acid- and base-stable allyl group (see Section 2.2.2.7) which, however, is mainly exploited for introduction of an additional dimension into the orthogonal protection scheme, thus allowing selective chemistry to be performed at the carboxy group like glycosylation or cyclization reactions.

2.2.2.1 Methyl and Ethyl Esters

Methyl and ethyl esters of aspartic and glutamic acids are prepared by esterification of the free amino acids with methanol and ethanol, respectively, in the presence of hydrogen chloride, thionyl chloride, acetyl chloride, or perchloric acid,[11] whereby the related α-esters are produced as contaminants as well. Tetrafluoroboric acid–diethyl ether complex[12] is also used for the synthesis of ω-methyl, -ethyl, and other esters. Transesterification with methyl or ethyl acetate in the presence of perchloric acid yields almost exclusively the ω-esters, since the positive charge on the amino group interferes with protonation of the α-carboxy group.[13] Selective hydrolysis of dimethyl esters catalyzed by copper(II) ions is an alternative approach to the synthesis of ω-esters.[14] The dimethyl esters are prepared by the HCl/MeOH method,[15] or more conveniently using $SOCl_2$/MeOH.[16] Cleavage of methyl and ethyl esters is usually effected by alkaline hydrolysis. As, under strong basic conditions, numerous side reactions may occur in peptides, methyl and ethyl esters cannot be recommended as ω-protecting groups for the synthesis of larger peptides, although selective deprotection of a variety of esters can now be achieved with bis(tributyltin) oxide (BBTO).[17,18]

H-Asp(OMe)-OH·HCl:[16,19,20]
Method A: Using MeOH/HCl:[19] To dry MeOH (70 mL) containing HCl (3.8 g, 0.1 mol) was added H-Asp-OH·HCl (11.0 g, 65 mmol). After 10 min, Et_2O (300 mL) was added and the precipitate recrystallized (EtOH/Et_2O); yield: 8.0 g (67%); mp 190 °C.

Method B: Using MeOH/$SOCl_2$:[20] To MeOH (200 mL) at −10 °C was added $SOCl_2$ (31 mL, 0.43 mol) and H-Asp-OH (40 g, 0.3 mol). A clear soln was obtained by slowly warming to rt. After 25 min at rt, abs Et_2O (600 mL) was added and the crystalline product filtered and recrystallized twice (MeOH/Et_2O containing a small amount of HCl); yield: 33.6 g (61%); mp 191–193 °C; $[\alpha]_D^{25}$ +21.4 (*c* 1, EtOH/H_2O 1:3).

A similar procedure was reported by Yoshioka et al.;[21] mp 194–195 °C; $[\alpha]_D^{20}$ +16.4 (*c* 1, MeOH).

Method C: Using AcCl/MeOH:[16] To a stirred suspension of H-Asp-OH (40 g, 0.3 mol) in dry MeOH (100 mL) was added AcCl (33.4 g, 0.42 mol) in MeOH (100 mL, both compounds were premixed at 5 °C and stirred in an ice bath for 30 min) at 8–10 °C. After 3 h, the ice bath was removed and stirring continued for 15 h; then the soln was poured into Et_2O (600 mL) and cooled for 2 h. The precipitate was collected; yield: 53 g (77%); mp 191–193 °C.

Method D: Via partial saponification of the diester:[16] The diester H-Asp(OMe)-OMe·HCl (3.95 g, 20 mmol) and $CuCO_3 \cdot Cu(OH)_2$ (22.1 g, 100 mmol) were stirred in a mixture of EtOH (150 mL) and H_2O (525 mL) at 70 °C for 2 h. After cooling to rt, excess H_2S was passed through the mixture and this was filtered through Celite. The procedure was repeated once more and the filtrate removed; yield: 3.6 g (98%); mp 190–191 °C.

H-Asp(OMe)-OH:[16]
H-Asp(OMe)-OH·HCl (53 g, 0.29 mol) and propene oxide (75 mL) were refluxed in dry EtOH (225 mL) for 6 h. After cooling the mixture, the precipitate was filtered off and the mother liquors were removed; yield: 38.8 g (91%); mp 194–195 °C; ^{13}C NMR.[12]

H-Glu(OMe)-OH·HCl:[19,22]
To a soln of HCl (15.0 g, 0.41 mol) in redistilled (but not necessarily dry) MeOH (200 mL) was added H-Glu-OH (56.0 g, 0.38 mol) (the monohydrate salt is not adequate for this reaction) under vigorous

stirring. Upon dissolution of the solid, the mixture was poured into Et_2O (1.5 L) and the product allowed to crystallize at 4 °C for 1 h. The product was filtered off and recrystallized (EtOH); yield: 71.5 g (95%).

H-Asp(OMe)-OMe·HCl:[16]
To a suspension of H-Asp-OH (13.3 g, 0.1 mol) in MeOH (75 mL) was added $SOCl_2$ (16.7 g, 0.14 mol) dropwise at 0 °C. The bath was removed and the soln was stirred at rt for 45 h and then concentrated. The residual oil was triturated with Et_2O and the resulting crystalline solid was washed with cold Et_2O; yield: 19.4 g (98%); mp 114–115 °C.

H-Glu(OEt)-OH:[19,23]
To rigorously anhyd EtOH (330 mL) containing HCl (24.0 g, 0.66 mol), H-Glu-OH (80.0 g, 0.54 mol) was added under stirring. Upon dissolution of the solid (20 min), two volumes of Et_2O were added and after 1 h the soln was filtered from H-Glu-OH·HCl. More Et_2O (2.5 L) was added and the crystalline precipitate was recrystallized (EtOH/Et_2O), giving H-Glu(OEt)-OH·HCl; yield: 55 g (48%); mp 134 °C.

To an ice-cold soln of the hydrochloride (5.0 g) in MeOH (30 mL), 10 M NH_3 (2.3 mL) was added and the soln left overnight at 4 °C, whereby the product crystallized; yield: 3.0 g (72.5%); mp 187 °C,[19] 192 °C,[23] 195–198 °C;[12] $[\alpha]_D^{20}$ +14 (*c* 1, H_2O).[23]

H-Asp(OEt)-OH:[12]
Finely ground H-Asp-OH (4 g, 30 mmol) and anhyd Na_2SO_4 (4 g) were suspended in EtOH (50 mL) and $HBF_4 \cdot OEt_2$ (54%, 8.2 mL, 60 mmol) was added through a syringe. After stirring at 60 °C for 15 h, Na_2SO_4 was removed by suction, washed with EtOH, and the filtrate was neutralized with TEA (9.2 mL, 66 mmol). EtOH was removed and the residue was triturated with EtOAc (200 mL) and recrystallized (H_2O/acetone); yield: 3.0 g (62%); mp 206–208 °C; $[\alpha]_D^{20}$ +24.0 (*c* 2, 1 M HCl); ^{13}C NMR data of ω-ethyl esters of Asp and Glu are available.[12]

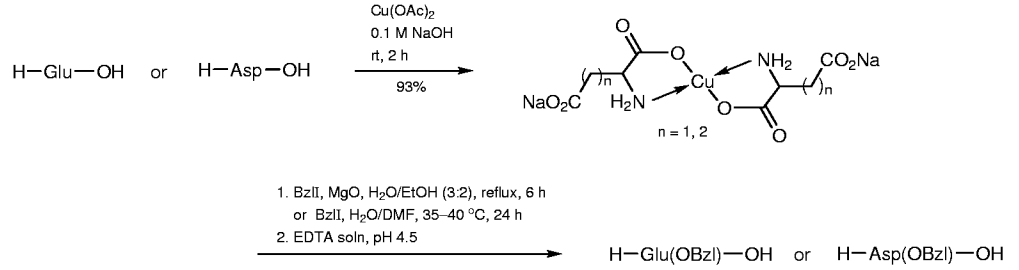
2.2.2.2 Benzyl Esters

Benzyl esters are relatively stable to mild acidic and basic conditions as used in peptide synthesis, but are readily hydrolyzed by exposure to strong acids, e.g. HF, TfOH, or by hydrogenolysis. The main drawback of the benzyl esters, however, is their insufficient stability under the conditions commonly used for repetitive N^{α}-Boc cleavage, e.g. TFA/CH_2Cl_2 (1:1), in SPPS applying the Boc/Bzl strategy. This lability causes difficulties in the synthesis of larger peptides, giving rise to cumulative loss of side-chain protection and hence branching of the peptide chain. For the synthesis of ω-benzyl esters of aspartic and glutamic acids the free amino acids are partially esterified with benzyl alcohol in the presence of hydrogen iodide, hydrogen bromide, 4-toluenesulfonic acid, concentrated sulfuric acid,[24,25] or tetrafluoroboric acid diethyl ether complex.[12] Using higher reaction temperatures and hydrogen chloride as a catalyst, the dibenzyl esters of aspartic and glutamic acids are readily obtained.[26] β-Benzyl aspartic acid is also prepared by partial saponification of its dibenzyl ester,[27] preferably with lithium hydroxide in acetone,[28] while the controlled cleavage of the diester with hydrogen iodide affords the α-benzyl ester.[28] The diesters (benzyl, 4-chlorobenzyl, 4-nitrobenzyl, methyl, ethyl)[14] can also be hydrolyzed by aqueous copper(II) sulfate; the resulting monoester copper complexes are decomposed to the ω-benzyl esters with ethylenediaminetetraacetic acid (EDTA).[29]

An alternative approach to the selective ω-esterification is offered by the copper(II) complexes of Asp and Glu. In these complexes, only the α-carboxy groups are engaged as ligands while the side-chain carboxy groups form alkali salts and thus are available for esterification with alkyl halides (Scheme 2).[29] Upon first unsatisfactory attempts to use copper salts for the selective esterification of glutamic acid,[30] this method of simultaneous protection of the α-amino and α-carboxy groups by the formation of stable chelate complexes has been optimized for the synthesis of H-Glu(OBzl)-OH.[29] This method is also applied for the preparation of a series of substituted benzyl esters (2-, 3-, or 4-nitrobenzyl, 2-cyanobenzyl, 2,4,6-trimethylbenzyl, 4-methoxybenzyl, 4-iodobenzyl, 4-phenylazobenzyl, etc.) of aspartic and glutamic acids using different benzyl halide derivatives.[29] The alkylation of alkali salts of amino acid–copper(II) complexes has been improved by the use of *N,N,N′,N′*-tetramethylguanidinium instead of lithium or sodium salts of amino acid–copper(II) com-

plexes.[31] Thereby the amount of water in the reaction mixture can be 10-fold reduced and good yields (44–86% over the three steps) are obtained for a large number of different ω-esters.

Scheme 2 Regioselective Synthesis of ω-Benzyl Esters via Copper Complexes[29]



Boroxazolidinones are zwitterionic boron–amino acid complexes that can be used for the simultaneous protection of the α-amino and α-carboxy groups (Scheme 3).[32,33] Treatment of aspartic and glutamic acids with triethylborane in THF results in the respective boroxazolidinones in almost quantitative yield. These boroxazolidinones are then selectively esterified at the ω-carboxy group by their conversion into the dicyclohexylamine (DCHA) salts followed by reaction with benzyl bromide to produce the ω-benzyl derivatives. After reconversion of the heterocycles into the amino acids with hydrogen bromide or hydrogen chloride in ethyl acetate, followed by neutralization, the ω-benzyl esters of aspartic and glutamic acids are obtained.

A summary of the most common aspartic and glutamic acid ω-benzyl ester derivatives is given in Table 1.

Scheme 3 Synthesis of ω-Benzyl Esters via Boroxazolidinones[32,33]

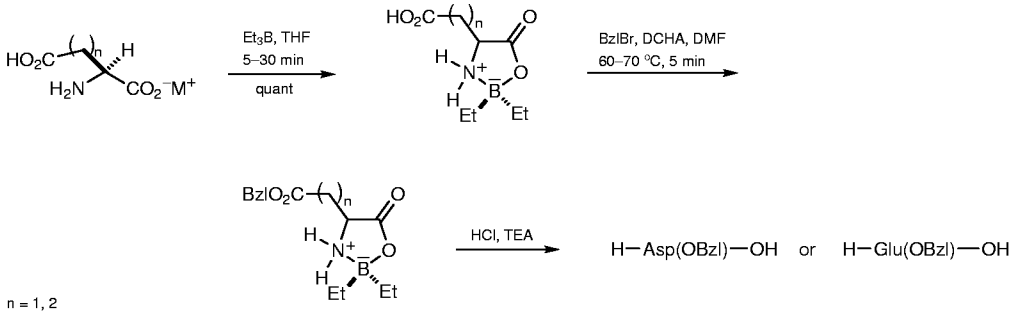


Table 1 ω-Benzyl Esters of Aspartic and Glutamic Acid Derivatives[8,12,25,34,35,38,39]

Product	mp (°C)	$[\alpha]_D^{20}$	Ref
H-Asp(OBzl)-OH	217–219	+27.9 (*c* 2.1, 1 M HCl)	[12]
Boc-Asp(OBzl)-OH	99–103	+8.5 (*c* 1, AcOH)	[25]
Fmoc-Asp(OBzl)-OH	121–123	+1.1 (*c* 1, DMF)	[8]
Z-Asp(OBzl)-OH	107–108	+11.9 (*c* 10, AcOH)	[34]
H-Glu(OBzl)-OH	188–189	+19.6 (*c* 6, AcOH)	[35]
Fmoc-Glu(OBzl)-OH	136–138	+6.2 (*c* 0.9, MeOH)	[8,39]
Boc-Glu(OBzl)-OH	55–59	−5.2 (*c* 1, AcOH)	[25]
Z-Glu(OBzl)-OH	75–76	−22.7 (*c* 1.41, 1 M $KHCO_3$)	[38]

for references see p 257

H-Asp(OBzl)-OH:[34]
Freshly distilled benzyl alcohol (100 mL, 0.1 mol) was added to a mixture of dry Et_2O (100 mL) and concd H_2SO_4 (10 mL). The Et_2O was removed and finely powdered H-Asp-OH (13.4 g, 0.1 mol) was added in small portions under stirring. The resulting soln was kept at rt for about a day, diluted with EtOH (95%, 200 mL), and neutralized by dropwise addition of pyridine (50 mL) under vigorous stirring. The mixture was stored overnight in the cold, then the crystalline product was collected and washed (Et_2O). The ester was recrystallized from hot water containing a few drops of pyridine; yield: 8.9–9.9 g (40–45%); mp 218–220 °C; $[\alpha]_D^{20}$ +28.1 (*c* 1, 1 M HCl); ^{13}C NMR.[12]

H-Glu(OBzl)-OH:
Method A: Via H_2SO_4 catalysis:[35] To a mixture of Et_2O (500 mL) and concd H_2SO_4 (50 mL), aldehyde-free benzyl alcohol (500 mL) was carefully added. The Et_2O was removed and H-Glu-OH (73.5 g, 0.5 mol) was added in small portions. After 20 h at rt the soln was diluted with EtOH (96%, 1 L) and neutralized with pyridine (250 mL) under vigorous stirring. After 10 h at 0 °C the product was filtered off and recrystallized from boiling H_2O (950 mL) containing pyridine (10 mL); yield: 90 g (76%); mp 189 °C; $[\alpha]_D^{20}$ +19.6 (*c* 6, AcOH).

Method B: Via $HBF_4 \cdot OEt_2$ catalysis:[12] To a suspension of finely powdered H-Glu-OH (2.0 g, 13.6 mmol) and anhyd Na_2SO_4 (2.0 g) in benzyl alcohol (25 mL, 242 mmol), $HBF_4 \cdot OEt_2$ (54%, 3.7 mL, 27.2 mmol) was added by syringe. The suspension was stirred at rt for 15 h. The mixture was diluted with anhyd THF (75 mL) and filtered through activated charcoal. The filtrate was neutralized with TEA (4.1 mL, 29.6 mmol) and concentrated under reduced pressure (14 Torr, with the bath temperature not exceeding 50–60 °C) until a slurry was formed. The viscous residue was triturated with EtOAc (100 mL) and the resulting solid was filtered off and washed with additional solvent; yield: 3.02 g (94%); mp 188–189 °C; $[\alpha]_D^{20}$ +29.5 (*c* 2, 1 M HCl); ^{13}C NMR.

Method C: Via partial saponification of the diester:[14] To a suspension of powdered TosOH (100 g, 0.525 mol) and H-Glu-OH (75 g, 0.5 mol) in benzyl alcohol (500 mL), benzene (2 L) was added and the suspension placed in a three-necked flask. H_2O was removed azeotropically. After refluxing for 4 d the mixture was cooled, the crystalline product H-Glu(OBzl)-OBzl · TosOH was collected and washed with Et_2O to give the tosylate salt; yield: 196 g (78.5%); mp 144–145 °C,[36] 123–125 °C;[37] $[\alpha]_D^{20}$ +7.3 (*c* 1, MeOH).

To a soln of H-Glu(OBzl)-OBzl · TosOH (10 g, 20 mmol) in EtOH (140 mL), an aq soln of $CuSO_4 \cdot 5H_2O$ (20 g, 80 mmol) in H_2O (350 mL) was added. The pH was adjusted to 8.0 with 1 M NaOH and the soln maintained at that pH and 32 °C for 1 h. The pH was then lowered to 3.0 with 3 M HCl and the precipitated copper complex of H-Glu(OBzl)-OH was filtered off and washed (H_2O, EtOH, and Et_2O). EDTA sodium salt (7.8 g, 21 mmol) in H_2O (100 mL) was added, the soln boiled and filtered, and upon cooling the precipitate was filtered and washed (H_2O, EtOH, and Et_2O); yield: 3.5 g (74%); mp 169–170 °C,[14] 174 °C;[38] $[\alpha]_D^{25}$ +19.3 (*c* 5.5, AcOH),[14] +21.8 (*c* 1.9, AcOH).[38]

From this compound, Fmoc-Glu(OBzl)-OH can easily be obtained.[39]

H-Glu(OBzl)-OBzl · HCl:[26]
A suspension of H-Glu-OH (10 g, 68 mmol) in benzyl alcohol (150 mL) was warmed to 55 °C and stirred while dry HCl was passed through for 1 h and the temperature permitted to rise. The mixture was transferred to a still, and benzene (75 mL) was added which was distilled off with most of the H_2O at a bath temperature of about 40 °C. The mixture was left under reduced pressure (ca. 10 Torr) for 1 h at a bath temperature of 85 °C. Then, dry HCl was again passed in for 1 h as described above. Unreacted H-Glu-OH · HCl (about 2 g) was filtered off, benzene added, and the process described above was repeated. Dry HCl was passed in for a third time; after removal of about one half of the benzyl alcohol under reduced pressure (steam bath), the diester hydrochloride was precipitated with Et_2O (5–7 volumes) and recrystallized (MeOH/Et_2O); yield: 15 g (61%); mp 100–102 °C; $[\alpha]_D^{22}$ +9.4 (1.5% in 0.1 M HCl).

Sodium Copper L-Glutamate:[29]
The copper complex was prepared essentially as described by Abderhalden and Kautzsch,[40] except that $Cu(OAc)_2$ was used instead of the oxide. $Cu(OAc)_2$ (0.522 g, 1.2 mmol) was shaken with a soln of H-Glu-OH (294 mg, 2 mmol) in 0.1 M NaOH (40 mL) for 2 h at rt. The soln was filtered and the filtrate concentrated (ca. 10 mL). The product was precipitated with EtOH; yield: 0.744 g (93%).

H-Glu(OBzl)-OH via the Copper Complex:[29]
Method A: A soln of sodium copper glutamate (0.8 g, 2 mmol) in H_2O (4 mL) was added to DMF (4 mL) containing benzyl iodide (0.872 g, 4 mmol). The resulting suspension was stirred at 35–40 °C for 24 h and then diluted with acetone (10 mL). The crude γ-benzyl glutamate copper complex was washed thor-

oughly with H_2O and acetone. The copper complex (446 mg) was dissolved in boiling EDTA soln (20 mL) at pH 4.5 and the soln filtered. The product crystallized on cooling. The maximum overall yield obtained by this method was 27%.

Method B: A mixture of benzyl iodide (0.872 g, 4 mmol), sodium copper glutamate (0.8 g, 2 mmol), and MgO (0.08 g, 2 mmol) in EtOH (4 mL) and H_2O (6 mL) was refluxed for 6 h, and filtered. The crude copper complex was treated as in method A and the γ-benzyl ester was obtained in a maximum yield of 18%.

Synthesis of Boroxazolidinones; General Procedure:[32]
To a suspension of finely powdered H-Asp-OH or H-Glu-OH (10 mmol) in THF (5–10 mL), a 1 M soln of Et_3B (12 mmol) in THF (12 mL) was added and the mixture stirred until the amino acid had dissolved. The reaction time strongly depended on the effectiveness of the grinding; with finely powdered amino acids the reaction was completed within 5–30 min. The soln was then filtered, if necessary, and the solvent removed. The residue was treated either with toluene or cyclohexane, the product was collected by filtration and washed with cyclohexane or iPr_2O; yield: quant; Asp derivative: $[\alpha]_D^{20}$ +79.2 (*c* 1, DMF); Glu derivative: $[\alpha]_D^{20}$ −17.3 (*c* 1, DMF).

H-Asp(OBzl)-OH via the Boroxazolidinone Derivative; Typical Procedure:[32]
To the Asp boroxazolidinone derivative (2.0 g, 10 mmol) in DMF (10 mL), DCHA (10 mL, 50 mmol) was added. The mixture was heated to 60–70 °C and then benzyl bromide (1.71 g, 10 mmol) was added. After 5 min the mixture was cooled, the DCHA · HBr salt was filtered off, and the filtrate concentrated. The residue was dissolved in EtOAc, washed (H_2O), and dried ($MgSO_4$). The soln was concentrated and gaseous HCl was passed over the soln, affording β-benzyl aspartate as the hydrochloride. This salt was dissolved in H_2O and the product precipitated by neutralization with TEA; yield: 1.45 g (62%); mp 218–219 °C; $[\alpha]_D^{20}$ +7.73 (*c* 1, AcOH).

2.2.2.3 Substituted Benzyl Esters

The introduction of substituents in the phenyl ring of the benzyl ester changes in a decisive manner its stability toward acids. The 4-methoxy group increases the acid lability, whereas the related chloro- and nitrobenzyl esters are more stable than the benzyl ester itself. A proper choice of the substituents therefore allows variation of the stability against acid cleavage, and differently substituted benzyl esters are presently available for their appropriate use, depending upon the synthetic strategies. A variety of other substituted benzyl esters, e.g. 2,4,6-trimethylbenzyl, 1-naphthylmethyl, or biphenyl-4-ylmethyl, have been additionally prepared.[41] The substituted benzyl esters are obtained following procedures similar to those of the benzyl esters (see Section 2.2.2.2).

2.2.2.3.1 4-Methoxybenzyl Esters

Weygand and Hunger introduced the 4-methoxybenzyl ester and showed that this group can be cleaved with TFA at 0 °C.[42] For the preparation of these esters, various methods have been proposed such as the imidazole-catalyzed reaction with 4-methoxybenzyl alcohol and subsequent acid cleavage of the N^{α}-protecting group.[43] The use of 4-methoxybenzyl chloride and TEA in EtOAc or DMF as solvent is recommended;[42] the latter is especially suitable for alkylations of this type.

2.2.2.3.2 4-Chlorobenzyl Esters

4-Chlorobenzyl esters have been proposed by Prestidge et al.[44] These esters are readily prepared by the copper-catalyzed hydrolysis method.[14] The 4-chlorobenzyl ester is twice as stable as the benzyl ester toward hydrolysis with TFA/CH_2Cl_2 (1:1) at 45 °C, but is quantitatively cleaved by liquid HF at 0 °C within 30 min. The 4-bromobenzyl esters have also been proposed,[45] although only sporadic application is reported [H-Asp(O-4-BrBzl)-OH: mp 220–223 °C; $[\alpha]_D^{20}$ +14.1 (*c* 2, 80% AcOH); Boc-Asp(O-4-BrBzl)-OH: mp 128.5–130 °C; $[\alpha]_D^{20}$ −15.5 (*c* 2.08, DMF)].

for references see p 257

2.2.2.3.3 4-Nitrobenzyl Esters

4-Nitrobenzyl (Nbz) esters were introduced by Schwarz and Arakawa[46] and by Schwyzer and Sieber;[47] these are 50 times more stable than the benzyl ester against TFA/CH_2Cl_2 (1:1), completely stable to HF, and can be removed by hydrogenolysis. Schwyzer and Sieber prepared the 4-nitrobenzyl esters with 4-nitrobenzyl bromide and TEA. Also the preparation of the 2- and 3-nitrobenzyl esters was reported.[48] Nitrobenzyl chloride as a reagent was used by Tesser and Nivard,[49] whereas Goodman et al.[50] obtained the γ- and β-4-nitrobenzyl esters of Glu and Asp in 20 and 30% yield, respectively, by nitration of the corresponding benzyl esters with nitronium fluoroborate. Alternatively, direct nitration of the benzyl group is possible using nitric acid.[25] Several 4-nitrobenzyl esters of *N*-acylglutamic acid derivatives were prepared by selective hydrolysis of α,γ-diesters.[48,51] The Fmoc derivatives are prepared from the respective amino acid by ω-esterification with 4-nitrobenzyl alcohol and sulfuric acid as a catalyst followed by reaction with Fmoc-OSu under standard conditions.[52] The nitrobenzyl esters are suggested in peptide synthesis to prevent aspartimide formation.[52–54] Their cleavage can be accomplished by several methods such as tetrabutylammonium fluoride, sodium sulfide, zinc, tin(II) chloride, or sodium dithionite reduction, as well as by catalytic hydrogenolysis and Birch reduction.[52,55]

2.2.2.3.4 4-Picolyl Esters

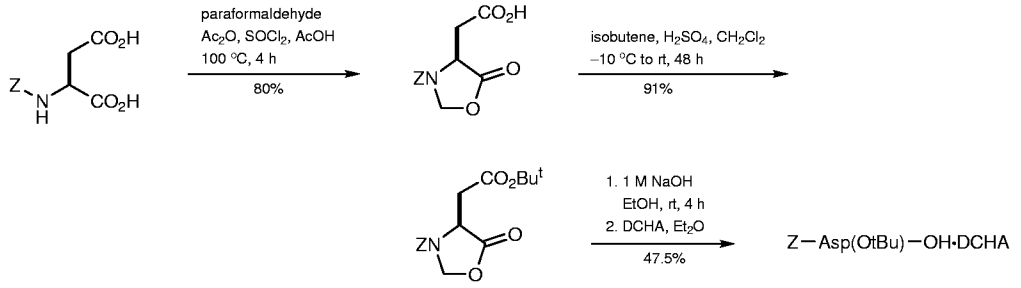
The 4-picolyl (Pic) esters were introduced by Young and et al.[56] and are also used for the side-chain protection of Asp and Glu.[57] Synthesis of Boc-Asp(OPic)-OH and Boc-Glu(OPic)-OH is achieved by reaction of the Boc amino acids with 4-pyridylmethanol/DCC in dichloromethane and subsequent partial saponification of the diesters with lithium hydroxide to give the β- and γ-monoesters, respectively.

2.2.2.4 *tert*-Butyl Esters

The *tert*-butyl ester is one of the most important carboxy protecting groups and numerous different methods are described for its preparation. The reaction of carboxylate silver salts with *tert*-butyl chloride is possible,[58] but esterification is mainly achieved with isobutene/sulfuric acid,[59–63] acetic acid *tert*-butyl ester/perchloric acid,[13,61,64–67] or *tert*-butyl alcohol/phosphoryl chloride.[68] During the acid-catalyzed esterification with *tert*-butyl acetate, amide alkylation is observed as a side reaction[69] when perchloric acid is used as a catalyst, whilst with sulfuric acid this side reaction is not detected.[68,70] The *tert*-butyl ester group is easily cleaved by acids like TFA, HCl/AcOH, and $BF_3 \cdot OEt_2$/AcOH, but is stable against basic hydrolysis and amminolysis as well as hydrogenolysis.

4-Substituted oxazolidin-5-one derivatives, which are prepared from N^{α}-protected α-amino dicarboxylic acids and paraformaldehyde, are employed for dual protection of the α-amino and α-carboxy groups in the synthesis of β-aspartyl and γ-glutamyl esters (Scheme 4).[71] For this purpose the oxazolidinone derivatives are synthesized by treatment of the Z amino acids with paraformaldehyde in a mixture of acetic anhydride, acetic acid, and traces of thionyl chloride or by azeotropic distillation of the Z amino acids with paraformaldehyde and 4-toluenesulfonic acid in benzene. The resulting heterocyclic compounds are readily converted into the *tert*-butyl esters with isobutene under acid catalysis. Esterification is achieved with *tert*-butyl bromide[72] or with Boc-F.[73] Finally, the oxazolidinone ring is opened by alkaline hydrolysis or catalytic hydrogenolysis to yield the *tert*-butyl esters.

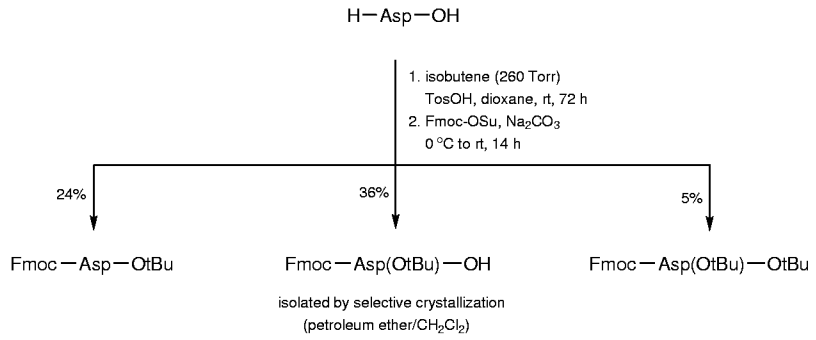
Scheme 4 Regioselective *tert*-Butylation via Oxazolidinones[71]



Upon first attempts to use boroxazolidinones (BON) for the synthesis of H-Asp(OtBu)-OH or H-Glu(OtBu)-OH, Wang et al.[74] designed a new boroxazolidinone compound that would be less affected by acids in replacing the alkyl substituents of the boron atom with electron-withdrawing groups. By reaction of the amino acids with equivalent amounts of base, organic and inorganic salts are prepared, dried, and then treated with more than 2 equivalents of boron trifluoride–diethyl ether complex. The resulting *B,B*-difluoroboroxazolidinones (DFBON) are converted into the *tert*-butyl esters using isobutene and boron trifluoride/phosphoric acid as catalysts. After alkaline hydrolysis, H-Asp(OtBu)-OH and H-Glu(OtBu)-OH are obtained.

A simple one-pot procedure for the synthesis of Fmoc-protected aspartic and glutamic acid *tert*-butyl esters is shown in Scheme 5.[75] *tert*-Butylation of unprotected Asp and Glu can be effected in dioxane with excess isobutene using 4-toluenesulfonic acid as catalyst. Subsequent Fmoc derivatization results in a mixture of the mono-*tert*-butyl esters (ω/α 65/35 for Glu and 50/40 for Asp) with an overall yield of 55–60% contaminated with small amounts of di-*tert*-butyl esters. This procedure has the advantage that the Fmoc group is introduced without prior isolation or purification of the aspartic and glutamic *tert*-butyl esters. Whilst the di-esters are readily removed by extraction with organic solvents, the Fmoc-Asp/Glu(OtBu)-OH derivatives are isolated by crystallization from dichloromethane/petroleum ether.

Scheme 5 One-Step Procedure for Regioselective *tert*-Butylation[75]



Also, the syntheses of Fmoc-Asp(OtBu)-OH[76] and Fmoc-Asp(OtBu)-OBzl[77] are reported in the literature and a summary of the most common aspartic and glutamic acid *tert*-butyl ester derivatives is given in Table 2.

for references see p 257

Table 2 ω-*tert*-Butyl Esters of Aspartic and Glutamic Acid Derivatives[39,58,60,63,67,72,75–79]

Product	mp (°C)	$[\alpha]_D^{20}$	Ref
Z-Asp(OtBu)-OBzl	45–47.5	–	[60]
H-Asp(OtBu)-OH	189–190	+8.5 (*c* 1.02, 90% AcOH)	[60]
Fmoc-Asp(OtBu)-OH	148–149	−20.3 (*c* 1, DMF)	[76]
Boc-Asp(OtBu)-OH	63–64	−21.7 (*c* 1, DMF)	[63]
Fmoc-Asp(OtBu)-OBzl	107–108	−19.4 (*c* 1, DMF)	[77]
Z-Asp(OtBu)-OH	75–78	+0.6 (*c* 1, EtOH)	[72]
Z-Glu(OtBu)-OH	79–80	−8.9 (*c* 1, EtOH)	[72]
H-Glu(OtBu)-OH	182	+9.8 (*c* 2, H_2O)	[78]
	168–169	+8.1 (*c* 2, H_2O)	[67]
H-Glu(OtBu)-OH·HCl	180	+22.6 (*c* 5, H_2O)	[79]
H-Glu(OtBu)-OtBu	bp 110/0.05 Torr	+16.6 (*c* 5.4, EtOH)	[58]
Fmoc-Glu(OtBu)-OH	76–77	+0.8 (*c* 1, EtOAc)	[76]
	78–80	−15.7 (*c* 1, DMF)	[75]
Fmoc-D-Glu(OtBu)-OH	–	+8.4 (*c* 1, MeOH)	[39]
Nps-Glu(OtBu)-OH·DCHA	179–180	−25.0 (*c* 4, MeOH)	[79]

Z-Asp(OtBu)-OBzl:[60]

To a cooled soln of Z-Asp-OBzl (4.92 g, 13.76 mmol) in CH_2Cl_2 (28 mL), isobutene (15 mL) and 17.75 M H_2SO_4 (0.14 mL, 2.47 mmol) were added and the mixture left at rt for 65 h. The mixture was cooled to 0 °C and carefully neutralized with cold H_2O (25 mL) containing 1 M Na_2CO_3 (4 mL). The excess isobutene was removed and the mixture extracted under cooling with CH_2Cl_2. Problems with emulsions were avoided by the addition of EtOAc and sat. Na_2SO_4 soln. The organic layers were washed with H_2O until neutral, dried ($MgSO_4$), filtered, and concentrated. The resulting oil was dissolved in EtOAc, washed with cold aq $NaHCO_3$ and H_2O, dried ($MgSO_4$), and recrystallized (EtOAc/petroleum ether); yield: 5.5 g (98%); mp 45–47.5 °C.

H-Glu(OtBu)-OH:[26,78–80]

To a cooled soln of K_2CO_3 (16.8 g, 120 mmol) in H_2O (100 mL), H-Glu-OBzl (23.7 g, 0.1 mol) was added. Upon dissolution of the solid, benzyl chloroformate (20.5 g, 17.1 mL, 0.12 mol) was added in four portions over a period of 30 min. The pH of the mixture was maintained around 8 by addition of 10% aq K_2CO_3. After additional 10 min stirring, the mixture was extracted with Et_2O (2 × 150 mL). The aqueous phase was acidified to Congo red with 6 M HCl and the oily precipitate solidified on standing. The solid was collected, washed with H_2O, and dried. Recrystallization (EtOH/H_2O or CCl_4) afforded pure Z-Glu-OBzl; yield: 26–31 g (70–83%); mp 95–96 °C; $[\alpha]_D^{20}$ −0.4 (*c* 1.7, AcOH).

A soln of Z-Glu-OBzl (18.6 g, 50 mmol) in dry dioxane (55 mL) was placed in a thick-walled glass vessel, cooled to −10 °C, and mixed with liq isobutene (210 mL). After addition of concd H_2SO_4 (1 mL) the flask was stoppered, wrapped in a towel, and shaken at rt for about 20 h. Then H_2O (250 mL) was added and the excess isobutene was removed in vacuo. The residue was extracted with Et_2O (2 × 150 mL) and the Et_2O extracts were combined, extracted with cold sat. aq $NaHCO_3$ (4 × 100 mL), washed with ice water until the washes were neutral, dried ($MgSO_4$), and the solvent was removed to dryness. The residue crystallized on standing; yield: 13.5 g (63%); mp 40–44 °C.

The derivative Z-Glu(OtBu)-OBzl (4.3 g, 10 mmol) was hydrogenated in MeOH (50 mL) and H_2O (25 mL) over 10% Pd/C (0.9 g). After completion of the reaction, the catalyst was filtered off and the soln concentrated. The residue was triturated with acetone and, upon storage in the cold, the solid was collected and washed with acetone; yield: 1.8 g (89%); mp 182 °C; $[\alpha]_D^{20}$ +9.8 (*c* 2, H_2O).

Z-Asp(OtBu)OH·DCHA via 3-Benzyloxycarbonyl-5-oxooxazolidine-4-acetic Acid:[71]

A mixture of Z-Asp-OH (5.35 g, 20 mmol), paraformaldehyde (1.8 g, 60 mmol), Ac_2O (4.0 g, 40 mmol), and $SOCl_2$ (0.3 mL, 4.1 mmol) in AcOH (80 mL) was heated at 100 °C for 4 h. Removal of the AcOH gave an oily residue which was dissolved in EtOAc and extracted with 5% aq $NaHCO_3$. The aqueous layer was acidified with 6 M HCl on ice cooling and extracted with EtOAc. The extract was washed with H_2O, dried ($MgSO_4$), and concentrated to a pale yellow syrup. Purification by column chromatography using $CHCl_3$/EtOAc (3:1) as eluent afforded pure 3-benzyloxycarbonyl-5-oxooxazolidine-4-acetic acid; yield: 4.4 g (80%).

3-Benzyloxycarbonyl-5-oxooxazolidine-4-acetic acid (4.4 g, 15.8 mmol) was dissolved in dry CH_2Cl_2 (30 mL) containing concd H_2SO_4 (0.2 mL). Gaseous isobutene (ca. 15 g) was bubbled into the stirred soln at −10 °C. The soln was kept in a stoppered flask at rt for 48 h. After addition of EtOAc (100 mL), the soln was washed with 5% aq $NaHCO_3$ and H_2O and then dried ($MgSO_4$). Solvents were removed to give

tert-butyl 3-benzyloxycarbonyl-5-oxooxazolidine-4-acetate as a pale yellow syrup, which was used for further reactions without purification; yield: 4.9 g (91.3%).

A solution of *tert*-butyl 3-benzyloxycarbonyl-5-oxooxazolidine-4-acetate (4.8 g, 14.4 mmol) in EtOH (20 mL) was treated with 1 M NaOH (7.1 mL) at rt for 4 h. The mixture was neutralized with 1 M HCl and the EtOH was removed. The product in EtOAc was extracted with 5% aq $NaHCO_3$. The extract was washed once with EtOAc and acidified with 6 M HCl. The acidified soln was extracted with EtOAc. The organic layer was washed with H_2O and dried ($MgSO_4$). Removal of the solvent gave a syrup (2.8 g). Upon adding DCHA (1.7 g, 9.4 mmol) to a soln of the syrup in Et_2O, the crystalline DCHA salt was obtained which was recrystallized (H_2O); yield: 3.4 g (47.5%); mp 123–124 °C; $[\alpha]_D^{20}$ +6.5 (*c* 1.7, 90% AcOH).

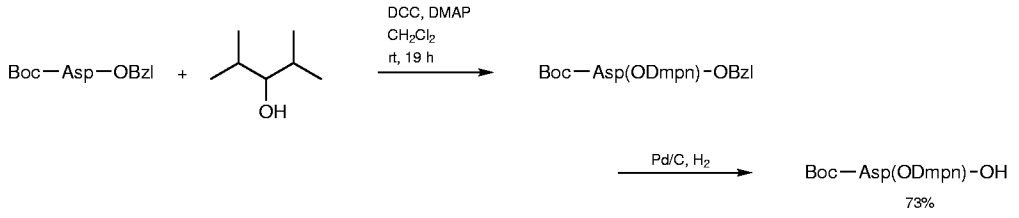
Fmoc-Asp(OtBu)-OH:[75]

A mixture of H-Asp-OH (3.9 g, 29 mmol) and TosOH (11.1 g, 58 mmol) in dioxane (140 mL) was stirred under isobutene gas (260 Torr) for 3 days at rt. Following the addition of 10% aq Na_2CO_3 (175 mL), Fmoc-OSu (9.9 g, 29 mmol) in dioxane (50 mL) was added dropwise at 0 °C. After stirring overnight at rt, the mixture was poured into ice water (300 mL) and extracted with Et_2O (3 × 300 mL). The aqueous phase was chilled (0 °C), acidified to pH 5.5 with 1 M HCl, and extracted with EtOAc (3 × 300 mL). The combined EtOAc fractions were washed with brine, dried, and the solvent was removed. This mixture of α- and β-*tert*-butyl esters was dissolved in 1:1 CH_2Cl_2/petroleum ether (30–60 °C) and chilled overnight. The resulting crystalline product was collected and washed with petroleum ether; yield: 5.53 g (35%); mp 149–150 °C; $[\alpha]_D^{20}$ −21.3 (*c* 1, DMF).

2.2.2.5 2,4-Dimethylpent-3-yl and 3-Methylpent-3-yl Esters

In order to prevent base-catalyzed aspartimide formation, a protecting group which provides steric hindrance in the form of both bulkiness and conformational flexibility (compared to, for example, the *tert*-butyl group) is required. For this purpose, the 2,4-dimethylpent-3-yl (Dmpn) ester of aspartic acid was proposed (Scheme 6),[81,82] which was shown to be very efficient in suppressing base-catalyzed aspartimide formation and which exhibits an acid lability similar to other protecting groups used in Boc chemistry. By comparing the Dmpn group with other esters like β-pent-3-yl (3-Pn), β-2,6-dimethylhept-4-yl (Dmh), β-hept-4-yl (4-Hep), β-cyclohexyl (Cy) (see Section 2.2.2.6.2), and β-adamant-2-yl (2-Ada, see Section 2.2.2.6.5), this ester led to the best results in terms of preventing aspartimide formation. The Dmpn group is more acid labile than the cyclohexyl group, more stable than the benzyl group, and is cleaved by a mixture of TfmSA/thioanisole/EDT/TFA (1:1:0.5:10). The protection against base-catalyzed aspartimide formation provided by the Dmpn group is sufficient for most of the orthogonal protection strategies that require base treatment to remove particular protecting groups or elevated temperatures to overcome difficult couplings.

Scheme 6 Regioselective Dmpn Protection by Indirect Means[81,82]



For the Fmoc strategy, the properties of the β-3-methylpent-3-yl (Mpe) group was investigated.[83] Fmoc-Asp(OMpe)-OH is synthesized from Fmoc-Asp-OBzl by its conversion into the acid chloride with oxalyl chloride. This was dissolved in a small volume of THF, diluted with a 20-fold volume excess of the sterically hindered 3-methylpentan-3-ol, and heated under reduced pressure to 50 °C for 1 hour. Catalytic hydrogenation gives Fmoc-Asp(OMpe)-OH in only 20–25% yield (probably due to partial loss of the Fmoc group during hydrogenation). As observed in some model peptide syntheses, Mpe protection is

for references see p 257

superior to that of the *tert*-butyl and 1-adamantyl protecting groups (see Sections 2.2.2.4 and 2.2.2.6.5) in terms of base-catalyzed aspartimide formation. Though the new protecting group is structurally rather similar to the commonly used *tert*-butyl ester, the additional methyl groups have a substantial effect on the rate of aspartimide formation. The flexibility and sterical hindrance of the alkyl chains is increased, thereby interfering with the attack of the amide bond nitrogen on the β-carboxy group and/or the abstraction of the NH proton at the amide bond. Some related useful aspartic and glutamic acid derivatives are listed in Table 3.

Table 3 ω-Alkyl Esters of Aspartic Acid Derivatives[82,83]

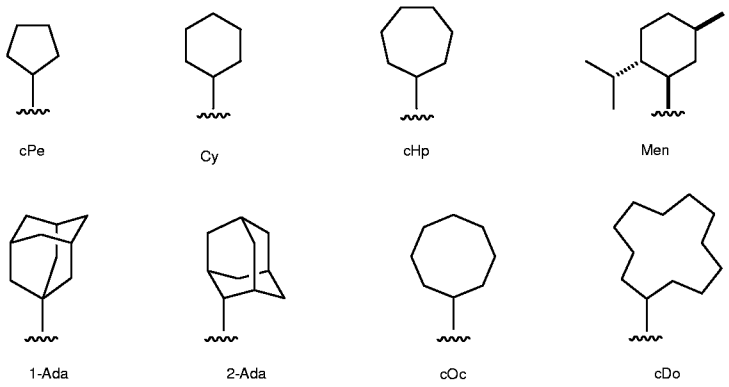
Product	mp (°C)	$[\alpha]_D^{20}$	Ref
Boc-Asp(ODmpn)-OH	102–104	+33 (*c* 1, $CHCl_3$)	[82]
Boc-Asp(O3-Pn)-OH	76–77	+34 (*c* 1, $CHCl_3$)	[82]
Boc-Asp(ODmh)-OH	96–97	+31 (*c* 1, $CHCl_3$)	[82]
Boc-Asp(O4-Hep)-OH	78–79	+34 (*c* 1, $CHCl_3$)	[82]
Fmoc-Asp(OMpe)-OH	–	+31 (*c* 1, $CHCl_3$)	[83]

Boc-Asp(ODmpn)-OH:[82]
To a cooled soln of Boc-Asp-OBzl (11.3 g, 35 mmol) in CH_2Cl_2 (50 mL), 2,4-dimethylpentan-3-ol (14.7 mL, 105 mmol) and DCC (3.61 g, 17.5 mmol) in CH_2Cl_2 (25 mL) were added, followed by DMAP (214 mg, 1.75 mmol). The mixture was stirred at rt for 3 h. Additional DCC (3.61 g, 17.5 mmol) was added and the mixture stirred for 16 h. The precipitated DCU was removed by filtration. The filtrate was concentrated, redissolved in EtOAc/petroleum ether (20:80) and filtered through a short column of silica gel in order to remove residual DCU. The eluate was washed three times each with 0.1 M HCl, 0.1 M $NaHCO_3$, and sat. NaCl soln, and dried ($MgSO_4$). The solvent was removed and the resulting oil was hydrogenated in EtOH (150 mL) over 5% Pd/C (0.75 g) for 4 h. The catalyst was removed, the solvent removed, and the resulting oil dissolved in EtOAc and extracted into a 0.1 M Na_2CO_3 soln adjusted to pH 10.5 by addition of HCl. The aq soln was washed once with Et_2O, acidified with citric acid, extracted into Et_2O, dried ($MgSO_4$), and concentrated. The oil was dissolved in Et_2O and concentrated. This procedure was repeated twice to remove residual EtOH. The resulting oil was dissolved in warm hexane and, upon cooling, crystallized; yield: 8.5 g (73%); mp 102–104 °C; $[\alpha]_D^{20}$ +33 (*c* 1, $CHCl_3$).

2.2.2.6 Cycloalkyl Esters

Cycloalkyl esters for the side-chain protection of aspartic acid in SPPS have been developed to increase resistance to aspartimide formation. Based on mechanistic studies of this side reaction, these protection groups should fulfill the following criteria: provide steric hindrance to intramolecular aminolytic attack of the ester by the amide nitrogen in acidic and basic media, provide increased stability toward repetitive TFA treatments but quantitative cleavage by HF, as well as stabilization of the carbenium ion produced by cleavage of the protecting group to prevent recapture by the peptide. The secondary cycloalkyl esters are more acid stable and more sterically hindered if compared to the primary benzyl esters. In Scheme 7, different cycloalkyl esters are shown.

Scheme 7 Cycloalkyl Esters for Carboxy Side-Chain Protection



2.2.2.6.1 Cyclopentyl Esters

The cyclopentyl (cPe) ester group was introduced by Blake[84] for the protection of carboxy groups. The ester bond is stable against TFA and readily cleaved with HF at 0 °C. This ester is 14 times more stable than the corresponding benzyl ester when exposed to 55% TFA in dichloromethane. However, there is little difference in the extent of HF-catalyzed succinimide formation between peptidyl-resins containing benzyl- or cyclopentyl-protected aspartyl residues, although the latter protection reduces base-catalyzed succinimide formation.

Boc-Asp(OcPe)-OH:[81]
To Boc-Asp-ONbz (10.6 g, 29 mmol) in EtOH (45 mL), 2.4 M aq $CsHCO_3$ was added until pH 7.0. The soln was concentrated to a residue that was further dried by resuspension in benzene and reconcentration. The benzene treatment was repeated and the residue was dried over P_2O_5. The cesium salt was dissolved in DMF (40 mL) and then cyclopentyl bromide (4.2 mL, 5.84 g, 39 mmol) was added. After 17 h at 50–55 °C, the DMF was concentrated and the residue distributed between EtOAc (100 mL) and H_2O (50 mL). The organic layer was further washed with 5% $NaHCO_3$ (2 × 50 mL), H_2O (50 mL), and dried. The residue (11.6 g) was hydrogenated in MeOH (90 mL) over Pd/C for 4 h. The mixture was filtered and the filtrate concentrated to a residue which was distributed between 5% $NaHCO_3$ (100 mL) and Et_2O (50 mL). Solid NaCl was added to the aqueous layer until saturation and the aqueous mixture was extracted with EtOAc (2 × 100 mL). The combined EtOAc extracts were washed with brine (20 mL), cold 20% citric acid (35 mL), and H_2O (3 × 40 mL). After drying, the soln was concentrated and the residue crystallized (EtOAc/petroleum ether) and recrystallized (Et_2O/petroleum ether); yield: 3.4 g (38%); mp 87–89 °C; $[\alpha]_D^{20}$ +27 (*c* 1, MeOH + 1.2 equiv DIPEA).

2.2.2.6.2 Cyclohexyl Esters

The cyclohexyl (Cy) ester is a protecting group proposed for the prevention of the aspartimide formation and is widely used in SPPS.[85] The Cy ester group is by one order of magnitude more stable than the benzyl ester group in 50% TFA/CH_2Cl_2. It is cleaved by HF or MsOH within 1 h at 0 °C, but stable against mild saponification, hydrazinolysis, and hydrogenation. These esters are prepared from aspartic and glutamic acids by ω-esterification with cyclohexanol and sulfuric acid[86] or tetrafluoroboric acid diethyl etherate[12] as a catalyst.

H-Asp(OCy)-OH:[86]
To cooled Et_2O (250 mL) containing H_2SO_4 (25 mL, 0.5 mmol), cyclohexanol (250 mL, 2.5 mol) and H-Asp-OH (33 g, 0.25 mol) were added under stirring. The resulting suspension was heated in a rotary evaporator at 70 °C for 2 h under reduced pressure. The bulk of the solvent was removed during this procedure. The resulting oil was partitioned between EtOAc (250 mL) and 5% aq $KHCO_3$ (300 mL). The pH was adjusted to 7.0 with 4 M NaOH and the aqueous layer was concentrated until precipitation occurred. The resulting suspension was chilled overnight and filtered; yield: 32 g (60%); mp 220–223 °C,[86] 208–209 °C;[12] $[\alpha]_D^{20}$ +20.1 (*c* 2, H_2O),[86] +23.3 (*c* 2, 1 M HCl).[12]

for references see p 257

(Without rigorous control of temperature and reaction time, a mixture of the ω-monoester, the diester, and the dicarboxylic acid is formed. Moreover, the presence of the residual unreacted cyclohexyl alcohol hinders the isolation of the desired compound. Therefore, complete removal of the alcohol is required.)

H-Glu(OCy)-OH:
Method A: Via H_2SO_4 catalysis:[86] This compound was prepared according to the procedure described for the synthesis of H-Asp(OCy)-OH; yield: 28 g (49%); mp 195–198 °C.

Method B: Via $HBF_4 \cdot OEt_2$ catalysis:[12] Finely powdered H-Glu-OH (4.4 g, 30 mmol) and anhyd Na_2SO_4 (4 g, 28 mmol) were suspended in cyclohexanol (50 mL, 470 mmol) and $HBF_4 \cdot OEt_2$ (54%, 8.2 mL, 60 mmol) was added through a syringe. After stirring at rt for 15 h the mixture was diluted with THF (100 mL), the water scavenger removed by suction, washed with cyclohexanol, and the filtrate neutralized with TEA (9.2 mL, 66 mmol). The alcohol was removed, the residue triturated with EtOAc (200 mL), and the remaining solid recrystallized (H_2O); yield: 3.1 g (45%); mp 175–176 °C; $[\alpha]_D^{20}$ +29.8 (*c* 2, 1 M HCl); ^{13}C NMR.

2.2.2.6.3 Cycloheptyl, Cyclooctyl, and Cyclododecyl Esters

To examine the effect of the steric hindrance as a function of the ring size of cyclic aliphatic esters, the cycloheptyl (cHp) and cyclooctyl (cOc) aspartic acid derivatives were prepared[87] according to the procedure described for the synthesis of the cyclohexyl derivatives (see Section 2.2.2.6.2).[85] Both groups are stable under the conditions of Boc deprotection and are cleaved quantitatively with 1 M TfOH/thioanisole in TFA in an ice bath or alternatively with HF within 1 h at 0 °C. By comparing the effect of different side-chain protecting groups on the succinimide formation in a model tripeptide, only slightly differences were observed for cPe, Cy, cHp, and cOc. Similar properties were reported for the β-cyclododecyl (cDo) ester of aspartic acid,[88] which is prepared according to the procedure reported by Tam et al.[85] Some of the cycloalkyl ester derivatives of aspartic acid are listed in Table 4.

Table 4 ω-Cycloalkyl Esters of Aspartic Acid Derivatives[82,87–89]

Product	mp (°C)	$[\alpha]_D^{20}$	Ref
Boc-Asp(OcHp)-OH	96–98	−3.4 (MeOH)	[87]
Boc-Asp(OcOc)-OH	84–86	−2.0 (MeOH)	[87]
Boc-Asp(OcDo)-OH	99–100	−15.6 (*c* 1, DMF)	[88]
Boc-Asp(OMen)-OBzl	90–91	−48.4 (*c* 1, MeOH)	[89]
Boc-Asp(OMen)-OH	135–137	−17.7 (*c* 1, MeOH)	[89]
	141–142	−11 (*c* 1, $CHCl_3$)	[82]

2.2.2.6.4 Menthyl Esters

A comparative study of the β-menthyl (Men) ester of aspartic acid with other cycloalkyl esters showed that this ester is less susceptible to base-catalyzed aspartimide formation than the cPe, Cy, cHp, or cOc derivatives.[89] Boc-Asp(OMen)-OH is prepared by esterification of Boc-Asp-OBzl with menth-1-ol by the DCC/DMAP procedure (see Section 2.2.2.5), followed by hydrogenolysis. The menthyl ester is stable to TFA treatment and is cleaved by HF or 1 M TfOH/thioanisole in TFA within 60 min in an ice bath. For some syntheses, dependent on the peptide sequence, an excess (1.4 equiv) of Boc-Asp(OMen)-OSu has to be used, presumably owing to the bulkiness of the protected side chain. For the cleavage of the Men group, in some cases diphenyl sulfide (20 equiv) has to be added as a scavenger to the 1 M TfOH/thioanisole/TFA mixture to facilitate its removal. However, acid-catalyzed aspartimide formation cannot be fully prevented.

2.2.2.6.5 Adamantyl Esters

The search for a ω-carboxy protection resistant to base-catalyzed aspartimide formation led to the β-1-adamantyl (1-Ada) and β-2-adamantyl (2-Ada) esters of aspartic acid[90–92] and to the γ-2-adamantyl ester of glutamic acid[93] and their evaluation in peptide synthesis (Table 5). The 1-adamantyl ester is easily hydrolyzed by TFA at room temperature, as well as 7 M HCl/dioxane, whilst the 2-adamantyl ester is stable against TFA but quantitatively cleaved by MsOH within 5 minutes at room temperature or HF at 0 °C. 2-Ada ester is more stable against bases, such as 1 M sodium carbonate or 55% piperidine. These results indicate that the 1- and 2-adamantyl esters could serve as a good alternative to *tert*-butyl protection in the Fmoc strategy and to benzyl protection in Boc/Bzl chemistry, respectively. The adamantyl group is bulky and rigid enough to minimize the acid- and base-catalyzed aspartimide formation. In fact, in peptide model syntheses containing Asp-Ser or Asp-Gly sequences, only small amounts of aspartimide formation are observed. Acid-catalyzed aspartimide formation is more suppressed when compared to benzyl and cyclohexyl protection. In a study using Fmoc chemistry, the 1-adamantyl ester is less effective in suppressing piperidine-catalyzed aspartimide formation compared to the *tert*-butyl protection.[10] Esterification of Z-Asp-OBzl with adamantan-1-ol is performed by the DCC/DMAP method and hydrogenation of Z-Asp[O(1-Ada)]-OBzl delivers H-Asp[O(1-Ada)]-OH. The respective Fmoc derivative is obtained with the standard procedure using Fmoc-OSu.[91] Boc-Asp[O(2-Ada)]-OH is synthesized in a similar way. The 1-adamantyl diesters of Asp and Glu were also prepared and isolated as their hemioxalate salts.[94]

Table 5 ω-1-Adamantyl and ω-2-Adamantyl Esters of Aspartic and Glutamic Acid Derivatives[91,93,94]

Product	mp (°C)	$[\alpha]_D^{20}$	Ref
Boc-Asp[O(2-Ada)]-OH	111–114	+21.8 (*c* 1.2, $CHCl_3$)	[91]
H-Asp[O(1-Ada)]-OH	240 (dec)	−15 (*c* 0.6, MeOH)	[91]
H-Asp[O(2-Ada)]-OH	217–221	−11.5 (*c* 0.5, MeOH)	[91]
Fmoc-Asp[O(1-Ada)]-OH	–	+0.2 (*c* 1, MeOH)	[91]
		+36.6 (*c* 1, $CHCl_3$)	[91]
H-Glu[O(2-Ada)]-OH	183–186	+4.2 (*c* 1, MeOH)	[93]
Boc-Glu[O(2-Ada)]-OH·DCHA	144–148	+12.3 (*c* 1, MeOH)	[93]
Fmoc-Glu[O(2-Ada)]-OH	–	+3.2 (*c* 0.5, MeOH)	[93]
H-Asp[O(1-Ada)]-O(1-Ada)	105–107	+6.3 (*c* 0.5, 96% EtOH)	[94]
H-Glu[O(1-Ada)]-O(1-Ada)	89–91	+11.7 (*c* 1.5, $CHCl_3$)	[94]

H-Asp[O(1-Ada)]-OH:[91]

Z-Asp-OBzl (2.0 g, 5.6 mmol), adamantan-1-ol (0.94 g, 6.2 mmol), and DMAP (68 mg, 0.56 mmol) were dissolved in CH_2Cl_2 (50 mL) and cooled with an ice/salt bath. DCC (1.27 g, 6.2 mmol) was added and the mixture was stirred at 4 °C overnight. DCU and solvent were removed and the residue extracted with EtOAc, washed with H_2O, dried (Na_2SO_4), and concentrated. Light petroleum ether was added, and the crystals were collected by filtration and recrystallized (EtOH) to give Z-Asp[O(1-Ada)]-OBzl; yield: 2.0 g (73%); mp 94–95 °C; $[\alpha]_D^{23}$ +8.6 (*c* 1, $CHCl_3$).

Z-Asp[O(1-Ada)]-OBzl (1.2 g, 2.44 mmol) in MeOH (40 mL) was hydrogenated over Pd/C for 4 h. The catalyst and solvent were removed, Et_2O was added, and the crystals were collected by filtration; yield: 0.6 g (89%); mp 240 °C (dec); $[\alpha]_D^{23}$ −15.0 (*c* 0.6, MeOH).

Boc-Asp(O2-Ada)-OH:[91]

Boc-Asp-OBzl (5.0 g, 14.5 mmol), adamantan-2-ol (2.34 g, 15.4 mmol), and DMAP (171 mg, 1.4 mmol) were dissolved in CH_2Cl_2 (50 mL) and cooled with an ice/salt bath. DCC (3.18 g, 15.4 mmol) was added, and the mixture was stirred at rt overnight. DCU and solvent were removed, and the residue was dissolved in EtOH (2 mL). The crystals Boc-Asp[O(2-Ada)]-OBzl that formed were collected by filtration; yield: 5.0 g (78%); mp 74–75 °C; $[\alpha]_D^{23}$ +8.3 (*c* 1, $CHCl_3$).

Boc-Asp[O(2-Ada)]-OBzl (1.34 g, 2.93 mmol) in MeOH (30 mL) was hydrogenated over Pd/C for 4 h. After removal of the catalyst and solvent, EtOH and H_2O were added and the crystals collected by filtration; yield: 0.88 g (82%); mp 111–114 °C; $[\alpha]_D^{23}$ +21.8 (*c* 1.2, $CHCl_3$).

for references see p 257

2.2.2.7 Allyl Esters

For temporary protection of the carboxy group of amino acids, allyl esters were proposed by Kunz et al. for the synthesis of glycopeptides.[95–97] The application of the allyl group in peptide synthesis is comprehensively reviewed by Guibé.[98] Allyl esters are stable under standard basic or acidic conditions used in peptide chemistry. Therefore, the side-chain protection of aspartic and glutamic acids with the allyl ester provides a new orthogonal dimension in protecting schemes based on the Boc or Fmoc groups.[99,100] This protecting group is especially attractive for the synthesis of glycopeptides with specific labeling, as well as for the synthesis of cyclic peptides which involve the regioselective side-chain-to-side-chain or side-chain-to-backbone amide bond formation. The allyl group is relatively small and neither steric hindrance nor decreased solubility of the building blocks are observed during coupling reactions. The aspartic acid β-allyl ester can be prepared from the free amino acid, allyl alcohol, and hydrogen chloride or chlorotrimethylsilane as catalyst, but can also be obtained by enzymatic hydrolysis of the α,β-diallyl ester (Scheme 8). In Table 6 the most common allyl ester derivatives of Asp and Glu are listed. Allyl esters are readily and selectively cleaved with tetrakis(triphenylphosphine)palladium(0) in the presence of morpholine or pyrrolidine,[99] dichlorobis(triphenylphosphine)palladium(II)/acetic acid/tributylstannane,[100] or with chlorotris(triphenylphosphine)rhodium(I).[99] During cleavage of the allyl ester group, rearrangements to cyclic imides are not observed. The influence of nucleophilic amines during palladium-catalyzed removal of the allyl protection group has been analyzed in detail,[101] as well as the solvent effect during cleavage from an allyl anchoring group.[102]

Scheme 8 Regioselective Synthesis of β-Allyl Aspartic Acid Derivatives[105]

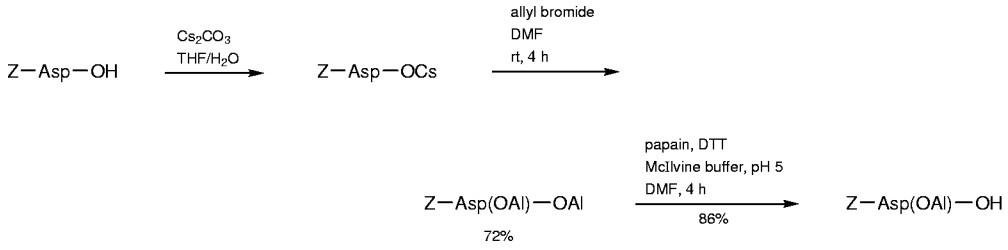


Table 6 ω-Allyl Esters of Aspartic and Glutamic Acid Derivatives[100,103–104]

Product	mp (°C)	$[\alpha]_D^{20}$	Ref
H-Asp(OAl)-OH·HCl	183–186	+19.0 (*c* 1, MeOH)	[103,104]
Z-Asp(OAl)-OH	oil	+15.3 (*c* 1.1, $CHCl_3$)	[103]
Z-Asp(OAl)-OAl	43–44	−17.1 (*c* 1, MeOH)	[105]
Z-Asp(OAl)-OtBu	oil	+18.2 (*c* 1.2, $CHCl_3$)	[103]
Boc-Asp(OAl)-OH	51–53	−1.0 (*c* 1, MeOH)	[100]
Boc-Asp(OAl)-OH·DCHA	110–111	+15.5 (*c* 1, MeOH)	[104]
Fmoc-Asp(OAl)-OH	114–115	−26.7 (*c* 1, DMF)	[100]
H-Glu(OAl)-OH	187–188	+23.3 (*c* 2, AcOH)	[100]
H-Glu(OAl)-OH·HCl	130–132	+22.5 (*c* 1, MeOH)	[104]
Boc-Glu(OAl)-OH	50–51	−10.0 (*c* 1, MeOH)	[100]
Boc-Glu(OAl)-OH·DCHA	134–136	+10.9 (*c* 1, MeOH)	[104]
Fmoc-Glu(OAl)-OH	102–105	−17.1 (*c* 1, DMF)	[100]

H-Asp(OAl)-OH·HCl:

Method A: Via HCl catalysis:[103] To a soln of anhyd HCl (13.5 g, 0.37 mol) in allyl alcohol (125 mL, 1.84 mol), H-Asp-OH (16 g, 0.12 mol) was added and the resulting suspension was stirred at rt for 18 h. The mixture was poured into Et_2O (500 mL) and the precipitate filtered and dried; yield: 23 g (91%); mp 185–186 °C.

Method B: Via TMSCl catalysis:[104] To a stirred suspension of H-Asp-OH (4.0 g, 30 mmol) in dry allyl alcohol (150 mL, 2.21 mol) under N_2, TMSCl (8.15 g, 9.5 mL, 75 mmol) was added dropwise, and the resulting soln was stirred at rt for 18 h. Upon addition of Et_2O (1 L) the precipitate was collected, washed with Et_2O, and dried; yield: 5.82 g (93%).

Z-Asp(OAl)-OAl:[105]
To a soln of Z-Asp-OH (3.5 g, 13.1 mmol) in peroxide-free THF (15 mL), H_2O was added dropwise until the soln became turbid. Neutralization was accomplished by addition of Cs_2CO_3 (4.3 g, 26.3 mmol) in H_2O (5 mL) at rt. After a few minutes, the soln was concentrated to dryness, the salt suspended in DMF (15 mL), and treated with allyl bromide (3.63 g, 2.54 mL, 30 mmol). The mixture was stirred for 4 h at rt. Removal of the solvent and excess reagent afforded an oil which was redissolved in EtOAc and washed with 5% aq $NaHCO_3$ and H_2O. The organic layer was dried ($MgSO_4$) and concentrated. The pale yellow oil was recrystallized (MeOH/H_2O); yield: 3.2 g (72%); mp 43–44 °C; $[\alpha]_D^{20}$ –17.1 (*c* 1, MeOH).

Z-Asp(OAl)-OH:[105]
To a soln of papain (5.3 mg) and DTT (40 mg, 0.26 mmol) in McIlvine buffer (pH 5, 30 mL), Z-Asp-(OAl)-OAl (1 g, 3 mmol) in DMF (10 mL) was added dropwise over a period of 4 h. The reaction was allowed to continue under stirring at 25 °C until a clear soln was obtained. The pH was lowered to 2 with dil HCl and lyophilized. The residue was suspended in EtOAc and washed (H_2O). The organic layer was dried and the solvent removed; yield: 0.78 g (86%); $[\alpha]_D^{20}$ –4.4 (*c* 1, MeOH).

2.2.2.8 9-Fluorenylmethyl Esters

The 9-fluorenylmethyl (Fm) ester for the protection of the carboxy group of amino acids was first proposed by Bednarek and Bodanszky.[106] Its main utility in peptide synthesis is the temporary side-chain protection of aspartic and glutamic acids, since these derivatives are stable to TFA, HBr in AcOH, and TFA/thioanisole, but are readily cleaved with secondary amines, e.g. diethylamine and piperidine in DMF, as well as by catalytic hydrogenation. Different strategies were reported for the preparation of side-chain-protected Asp and Glu derivatives. One approach is based on esterification of Boc-Asp/Glu-OBzl derivatives with 9-fluorenylmethanol and DCC/DMAP in CH_2Cl_2 in 82 (Asp) and 91% (Glu) yield, respectively.[107] Hydrogenation of the benzyl group over 10% palladium/carbon in methanol or ethanol for 1.5–3.5 hours generates Boc-Asp/Glu(OFm)-OH, leaving the Fm esters almost intact. Alternatively, the H-Asp/Glu(OFm)-OH derivatives are obtained via boroxazolidinones by reaction with 9-fluorenylmethanol in the presence of DCC/DMAP, followed by hydrolysis with gaseous HCl in 83 and 84% yield, respectively.[108] It is worth mentioning that the preparation of boroxazolidinones is simple but time consuming (two days for Asp and several days for Glu). Conversely, 4-substituted 5-oxooxazolidines can be synthesized quantitatively in 2 hours[71] and are then reacted with 9-fluorenylmethanol in the presence of DCC/DMAP (84–87% yield).[109] These intermediates are converted by acid treatment into the hydrochlorides of the esters in 72–78% yield. A one-step procedure starting directly from aspartic or glutamic acid was also reported (Scheme 9).[110] ω-Fluorenylmethyl esters of aspartic and glutamic acid derivatives are listed in Table 7.

for references see p 257

Scheme 9 Regioselective Introduction of the Fluorenylmethyl Group[110]

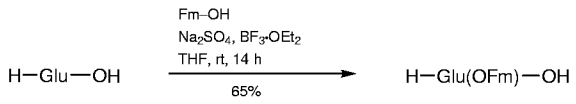


Table 7 ω-Fluorenylmethyl Esters of Aspartic and Glutamic Acid Derivatives

Product	mp (°C)	$[\alpha]_D^{20}$	Ref
Boc-Asp(OFm)-OBzl	74–76	−11.7 (*c* 1, MeOH)	[107]
Boc-Glu(OFm)-OBzl	102–103	−15.2 (*c* 1, MeOH)	[107]
Boc-Asp(OFm)-OH	141–142	+5.5 (*c* 1, MeOH)	[107]
	133–135	+30.4 (*c* 1, $CHCl_3$)	[110]
Boc-Glu(OFm)-OH	130–131	+9.3 (*c* 1, MeOH)	[107]
	129–130	+13.4 (*c* 1, $CHCl_3$)	[110]

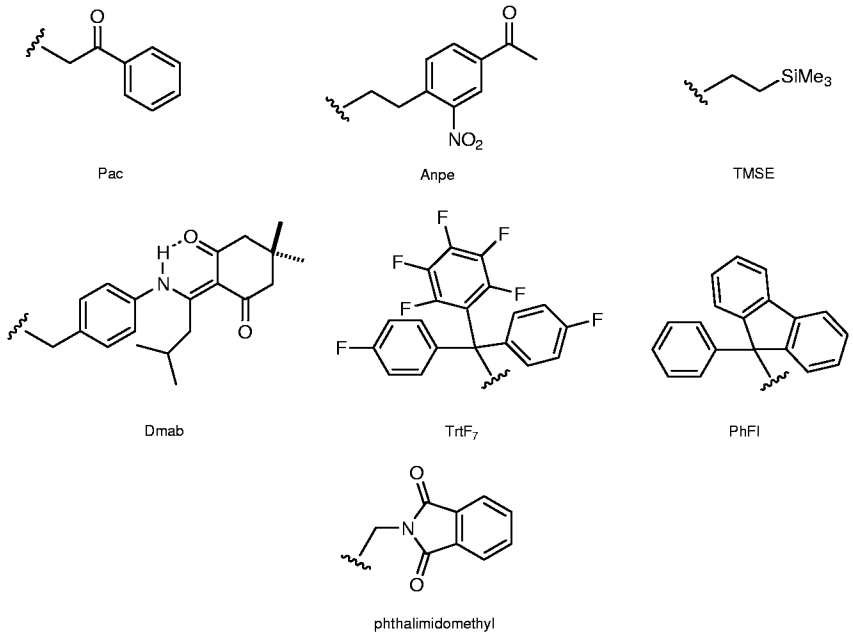
H-Glu(OFm)-OH:[110]
To a suspension of H-Glu-OH (5.0 g, 34 mmol), 9-fluorenylmethanol (33.4 g, 170 mmol), and anhyd Na_2SO_4 (5.0 g) in dry THF (30 mL) was added fresh $BF_3 \cdot OEt_2$ (11.6 mL, 85 mmol, 54%). The mixture was stirred at rt for 14 h, diluted with THF (100 mL) and filtered through Celite. The filtrate was neutralized by the addition of DIPEA (11 g, 14.8 mL, 85 mmol) and triturated with EtOAc (200 mL). The product was isolated by filtration and dissolved in 1 M HCl (200 mL). The pH was adjusted to 6.5 with $NaHCO_3$ and the precipitate filtered; yield: 7.2 g (65%); mp 156–158 °C; $[\alpha]_D^{20}$ +8.0 (*c* 0.5, DMSO).

H-Asp(OFm)-OH:[110]
For aspartic acid, the protocol was similar to the described synthesis of H-Glu(OFm)-OH, except that the mixture was heated at 60 °C for 12 h. The amorphous solid obtained upon trituration was converted into a fine powder by sonication in EtOAc, then filtered; yield: 35–40%; mp 187–189 °C; $[\alpha]_D^{20}$ −7.3 (*c* 0.5, DMSO).

2.2.2.9 Other Esters

Other ester groups used for the side-chain protection of aspartic and glutamic acids are shown in Scheme 10 and are described in more detailed in the following sections.

Scheme 10 Other Esters for Carboxy Side-Chain Protection



2.2.2.9.1 Phenacyl Esters

Phenacyl (Pac) esters for protection of side-chain carboxy groups were proposed by Zervas et al.[111] and later applied by Yang and Merrifield to avoid aspartimide formation during peptide syntheses.[53,112,113] The Pac ester is stable under the conditions of coupling reactions and is cleaved by zinc reduction in acetic acid without affecting other permanent protecting groups. The phenacyl esters are prepared from the Boc amino acids or peptides using phenacyl bromide. In general, Pac ester derivatives are often crystalline and easily purified by recrystallization from ordinary organic solvents.[114] Phenacyl esters are cleaved to the parent acids by zinc/90% acetic acid (3–6 h), hydrogen and palladium/carbon, sodium benzenethiolate, phenylselane, potassium cyanide/18-crown-6, hydrazine, tetrabutylammonium fluoride trihydrate, copper(II)/oxygen/dimethylformamide/water,[115] and bis(tributyltin) oxide (BBTO).[17] Since formation of succinimide derivatives was observed to significant extents when the Pac ester was used,[116,117] this type of side-chain protection of aspartyl residues is not recommended.

2.2.2.9.2 2-(4-Acetyl-2-nitrophenyl)ethyl Esters

As an alternative base-labile side-chain protection, the 2-(4-acetyl-2-nitrophenyl)ethyl (Anpe) group was proposed.[118,119] Its lability to bases is similar to that of the 9-fluorenylmethyl ester, but the Anpe derivatives have improved solubility. Two alternative methods are used for the synthesis of the Boc-Asp(OAnpe)-OH and Boc-Glu(OAnpe)-OH derivatives: (1) bis-esterification of the Boc-protected amino acid with Anpe-OH and selective hydrolysis of the α-carboxylic ester with lithium hydroxide and (2) simultaneous protection of both the α-amino and α-carboxy groups by reaction of the amino acid with triethylborane and selective esterification of the ω-carboxy group, followed by hydrolysis of the oxazaborole complex with hydrogen chloride and α-amino protection with the Boc group. The Anpe group is stable under conditions used for Boc chemistry and can be cleaved with 0.01 M TBAF in THF (5 min) or 0.25 M LiOH in methanol/dioxane/water (45 min). Displacement of the β-Anpe group by benzyl cations generated during HF treatment of protected peptides is not observed, whilst this does occur in the case of the fluorenylmethyl esters.[119] Two other Boc-protected aspartic acid β-esters, i.e. the 2-(4-nitrophenyl)ethyl (Npe)[120] ester (cleavable within 1.5–2 h with 0.1 M TBAF) and the 2-cyanoethyl (Cne) ester (cleavable within 45 min with 0.1 M TBAF) have been proposed.

2.2.2.9.3 Trimethylsilylethyl Esters

The 2-(trimethylsilyl)ethyl (TMSE) ester, originally reported by Sieber[121,122] as an α-carboxy protecting group, is applied as orthogonal side-chain protection of aspartic acid in solid-phase peptide chemistry for peptide cyclization on a TFA-labile resin.[123] TMSE esters are cleaved under acid and basic conditions, but are stable toward hydrogenolysis. Fmoc-Asp-(OTMSE)-OH is prepared in four steps from Boc-Asp-OBzl by esterification with trimethylsilylethanol/DCC in 50% pyridine/CH_2Cl_2 followed by catalytic hydrogenolysis and Boc deprotection with 50% TFA/CH_2Cl_2 and finally standard introduction of the Fmoc group.

2.2.2.9.4 Safety-Catch Esters

Safety-catch protecting groups are stable under a wide range of reaction conditions used during peptide synthesis, but are easily cleaved under mild conditions upon their activation. Their use in peptide chemistry is reviewed by Patek.[124] 4-{[1-(4,4-Dimethyl-2,6-dioxocyclo-

for references see p 257

hexylidene)-3-methylbutyl]amino]}benzyl (Dmab) esters are quasi-orthogonal to the Fmoc/tBu strategy, since they are stable to 20% piperidine in DMF and TFA, but are quantitatively cleaved with 2% hydrazine in DMF within minutes.[125,126] Fmoc-Asp/Glu(ODmab)-OH are commercially available and are recommended for the synthesis of cyclic peptides.

2.2.2.9.5 Modified Trityl Esters

Trityl esters of aspartic acid and glutamic acids are too labile to be used in peptide synthesis, so modified trityl esters with increased stability are proposed as protecting groups for aspartic and glutamic acids.[127] For this purpose, seven fluorine substituents were introduced into the phenyl rings as electron-withdrawing substituents and the bis(4-fluorophenyl)-(2,3,4,5,6-pentafluorophenyl)methyl ($TrtF_7$) ester should exhibit increased stability. The phenylfluorenyl group can be regarded as a trityl-type protection group with a 6000-fold higher stability compared to the trityl group. Esterification of Fmoc-Glu-OAl and the Glu-oxazolidinone derivative with heptafluorotrityl chloride or phenylfluorenyl bromide in CH_2Cl_2 with DIPEA at room temperature produces the ω-$TrtF_7$ and ω-9-phenylfluoren-9-yl (PhFl) esters, respectively. These esters are stable to AcOH/EtOAc (1:1), but are cleaved with 1% TFA and thus with a high degree of selectivity versus the *tert*-butyl group. During Fmoc cleavage with morpholine a side reaction, resulting in the formation of a pyroglutamic acid derivative, has been observed.

2.2.2.9.6 Phthalimidomethyl Esters

The phthalimidomethyl ester protection[128] has been used for the synthesis of poly-L-glutamic acid.[129] This protecting group is cleaved by piperidine or sodium hydroxide, but it is also sensitive to acids and catalytic hydrogenation. The ω-ester is prepared by reacting Z-Glu-OBzl with *N*-chloromethylphthalimide in the presence of DCHA. Owing to its characteristic absorption at 299 nm, the reaction can be monitored spectrometrically.

2.2.2.10 Enzymatic Methods

Enzyme-cleavable protecting groups in peptide synthesis have been comprehensively reviewed (see also Section 2.5).[130–132] Enzymatic transformations are useful for the synthesis of selectively functionalized aspartic and glutamic acid derivatives. For instance, α-chymotrypsin cleaves stereospecifically the α-ester group from *N*-acetylaspartic and -glutamic acids.[133,134] The alkaline protease alcalase hydrolyzes not only selectively the α-methyl ester group from H-Asp/Glu(OMe)-OMe,[135] but also the α-benzyl ester of H-Asp(OBzl)-OBzl and H-Glu(OBzl)-OBzl in 82 and 85% yields, respectively, on a decagram scale.[136] Similarly, aspartyl and glutamyl peptides can be deprotected selectively at the C-terminus by this enzyme; however, in these cases an undesired attack on the peptide bond may occur.[137] Either the α- or β-ester of L-aspartic acid may be preferentially hydrolyzed by chymotrypsin by changing the alcohol moiety of the esters.[138] In the cases of dicyclopentyl- and dicyclohexyl-L-aspartate, subtilisin Carlsberg still preferentially cleaves the α-ester; conversely, chymotrypsin attacks preferentially the β-esters. A lipase from *Candida cylindracea* is able to differentiate between the two carboxylic groups of glutamic acid. From H-Glu(OcPe)-OcPe it preferably (ratio 20:1) removes the γ-ester in 90% yield.[139] Pronase specifically hydrolyses the α-benzyl ester group from the dibenzyl esters of Asp and Glu in 73 and 72% yields, respectively.[140] From a variety of enzymes which catalyze the selective hydrolysis of the α-ester of H-Asp(OMe)-OMe, pig liver esterase (PLE) gives the best results.[141] Ethyl, allyl, and *tert*-butyl diesters of aspartic acid are readily and specifically hydrolyzed to the respective β-esters by PLE and only for the dibenzyl ester are longer reaction times required.

In addition, Z-Asp(OAl)-OAl is converted into Z-Asp(OAl)-OH in quantitative yield by papain.[105] Some Z-protected Glu and Asp diesters are regioselectively hydrolyzed by several microbial proteases, yielding the ω-protected derivatives.[142]

H-Asp(OBzl)-OH:[136]
To H-Asp(OBzl)-OBzl (62.6 g, 0.2 mol) in a mixture of acetone (150 mL) and H_2O (450 mL) at 37 °C was added alcalase 2.5 L (4 mL) and the pH maintained at 8.2 for 4 h. After 30% of the reactant had been hydrolyzed, the product started to precipitate. At the end of the reaction, the mixture was adjusted to pH 6.2 and cooled to complete precipitation. The product was collected by filtration and washed with cold H_2O (50 mL) and acetone (2 × 50 mL); yield: 18.3 g (82%); mp 222–223 °C; $[\alpha]_D^{25}$ +29.0 (*c* 1, AcOH).

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2.3 Amide Group

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Peptides may contain three different kinds of amide group, i.e. C-terminal α-amides (as present in numerous, naturally occurring bioactive peptides), the side-chain β- and γ-amide groups of asparagine and glutamine, respectively, and the primary and secondary amides of the peptide backbone. Although amides are stable under normal conditions of peptide chemistry, they may give rise to side reactions, as well as to solubility problems, making their protection advisable in selected cases.

2.3.1 C-Terminal α-Amide Protection

α-Carboxamides, as relatively stable groups, are not dehydrated by the usual peptide-coupling reagents. On the contrary, the α-carboxamide group itself may be considered as a protecting group for C-terminal carboxy groups, since it can be cleaved enzymatically in a highly selective manner to yield the free carboxy group (see Section 2.5.3.2).[1] In the synthesis of peptide amides on solid support, the C-terminal α-amide group is protected as a precursor by the handles linked to the resin and are thus generated in the final resin-cleavage step. Handles for this purpose have been derived from known amide protecting groups and optimized for suitable stability and cleavage properties (see Section 4.3.1.2). In contrast, for the synthesis of peptides in solution, protection of α-carboxamides has only been used to enhance the solubility of synthetic intermediates, to decrease the rates of piperazine-2,5-dione (DKP) formation, or in the case of C-terminal isoasparagine, and to a lesser extent isoglutamine, to prevent rearrangement to cyclic imides in the presence of base (e.g., 0.1 M NaOH or 0.1 M TEA).[2,3] In fact, the imides hydrolyze with concomitant deamidation.

2.3.1.1 Di- and Trimethoxybenzyl Protection

The unsubstituted benzyl group is inappropriate for amide protection as it is not cleaved under the relatively mild standard conditions of peptide chemistry. Acid hydrolysis, however, is markedly enhanced by incorporation of electron-withdrawing groups into the aromatic moiety, as in the case of the di- or trimethoxybenzyl groups.[4] The related 2,4-Dmb- and Tmb-protected amino acid amides are readily obtained by coupling N^{α}-protected amino acids with 2,4-dimethoxy- or 2,4,6-trimethoxybenzylamine, respectively, e.g. with DCC.[4] It is, however, recommended that the *N*-hydroxysuccinimide ester is generated first, as formation of insoluble salts between amino acids and amines is prevented and yields are increased. While these amide protecting groups are very stable toward hydrogenolysis, they can be removed under acidolytic conditions (Table 1). The addition of efficient scavengers is required not only to avoid alkylations at susceptible amino acid side chains, but also to drive the hydrolysis to completion.[3,4]

Table 1 Removal of Amide Protecting Groups under Acidolytic Conditions[4–10]

2,4-Dmb	Tmb	Mbh
TFA, anisole, rt, 20 h[6]	TFA, phenylmethanethiol, rt, 4 h[4]	TFA/anisole (10:1), rt, 3–16 h[5,9]
TFA, phenylmethanethiol, rt, 3 h[4]	TFA, phenylmethanethiol, rt, 20 h[8]	TFA, 1 M TMSBr, methyl phenyl sulfide in the presence of ethane-1,2-dithiol and *m*-cresol, 0 °C, 2 h[10]
HF, rt, 3 h[7]	TFA/anisole (10:1), rt, 1 h[9]	

N^{α}-Protected Amino Acid 2,4-Dmb- or Tmb Amide; General Procedure:[4]
To a cooled soln of the N^{α}-protected amino acid (50 mmol) and HOSu (6.9 g, 60 mmol) in $CHCl_3$ or DMF was added DCC (11.3 g, 55 mmol), followed after 5–10 min at rt by 2,4-Dmb-NH_2 or Tmb-NH_2 (50 mmol). After 12 h, the DCU was filtered off and the filtrate (in the case of DMF, diluted with $CHCl_3$ or EtOAc) was washed with 1 M HCl, sat. $NaHCO_3$, and H_2O, dried (Na_2SO_4), and concentrated. The residue was crystallized (EtOAc/petroleum ether); yield: 70–95%.

2.3.1.2 4,4′-Dimethoxybenzhydryl Protection

In contrast to 2,4-Dmb- and Tmb amide protection (Section 2.3.1.1), 4,4′-dimethoxybenzhydryl (Mbh) amide derivatives are obtained by reaction of N^{α}-protected amino acid amides with 4,4′-dimethoxybenzhydrol under acid catalysis.[5] The Mbh protecting group is stable toward hydrogenolysis, but cleavage can be achieved with TFA in the presence of scavengers for cation capture (see Table 1).

Z-Gly-NH(Mbh):[5]
To a soln of Z-Gly-NH_2 (2.1 g, 10 mmol) and Mbh-OH (2.4 g, 10 mmol) in AcOH (20 mL) was added one drop of concd H_2SO_4. After 20 h at rt the soln was poured into H_2O (75 mL). The resulting precipitate was dissolved in EtOAc and the organic layer washed with H_2O, dried (Na_2SO_4), and concentrated. The residue was recrystallized (THF/petroleum ether); yield: 3.6 g (83%); mp 148–150 °C.

2.3.1.3 Safety-Catch-type Benzhydryl Protection

The concept of converting a very stable protecting group into a labile group is known as the safety-catch principle. To apply this concept to amide protection, extremely acid-stable, *N*-benzhydrylamide derivatives are obtained by coupling N^{α}-protected amino acids with 4-(methylsulfinyl)benzhydrylamine or 4,4′-bis(methylsulfinyl)benzhydrylamine.[11] Upon reduction of the sulfinyl groups, the resulting methylsulfanyl derivatives are readily cleaved by acidolysis. Reduction and hydrolysis can be combined into one step by treatment with silicon tetrachloride or halotrimethylsilane and TFA in the presence of scavengers.[11] Although very promising, this type of amide protection has, so far, found only limited application.

2.3.1.4 Benzyloxy Protection

An interesting new α-amide protecting group, the *N*-benzyloxyamide derivative, has been proposed which is stable toward acids (e.g., TFA, formic acid in CH_2Cl_2, HCl in organic solvents) and bases (e.g., piperidine in CH_2Cl_2, NaOH in aq MeOH).[12,13] Conversely, deprotection is achieved by hydrogenolysis over a palladium catalyst followed by reduction of the resulting hydroxamic acid with titanium(III) chloride. The amino acid amide derivatives are readily obtained by DCC-mediated coupling of N^{α}-protected amino acids with *O*-benzylhydroxylamine.[14]

2.3.2 Peptide Backbone Protection

The reasons for applying peptide backbone protection are manifold. Unhindered backbone amides, as present in Gly peptides, are acylated by highly activated carboxy components, and aspartic acid or asparagine residues bear the inherent risk of forming aspartimides, in a sequence-dependent manner, with subsequent transpeptidation to β-aspartyl peptides. Since both of these side reactions are fully suppressed by protecting the related peptide amide groups, it is advisable to use such protection, particularly if larger peptide sequences are the

synthetic target.[15,16,97] Likewise opportune would be backbone protection in fragment-assembly strategies where the major drawback is represented by the facile epimerization of the C-terminal amino acid residue via the oxazol-5(4*H*)-one intermediate. Again, proper protection of the C-terminal backbone amide would eliminate this side reaction. The main advantage of backbone protection, however, is the strong solubilizing effect on synthetic intermediates.

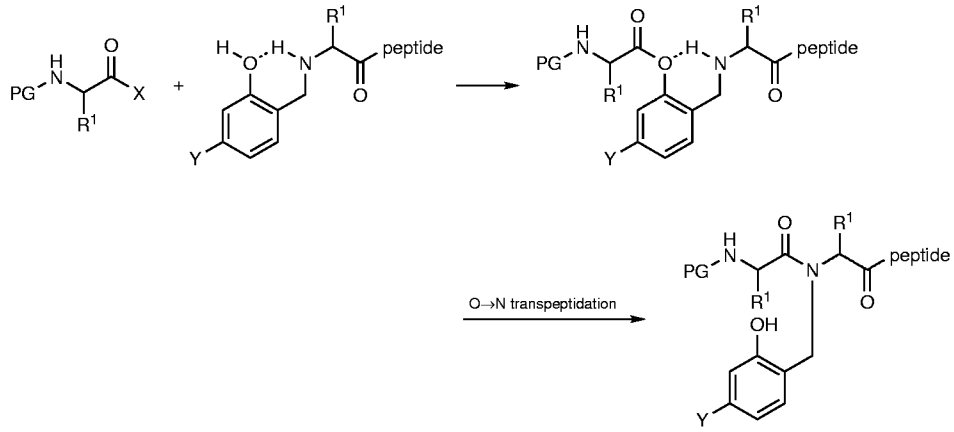
Successful assembly of peptides is often hampered by inherent problems such as poor solvation of the growing peptide chain during synthesis on solid supports, as well as limited solubility of fully protected peptide fragments in the solution approach, often leading to poor coupling yields. These undesirable properties originate from intermolecular hydrophobic aggregation of the protected peptide chains and/or from the formation of hydrogen-bond-stabilized secondary structures, most notably of β-sheets.[16,17] Particular sequences that favor such phenomena have come to be known as "difficult sequences".[16] Many methods have been proposed to bypass this solvation problem, including addition of chaotropic salts,[18–21] special solvent systems[22] or "magic mixtures",[23,24] elevated temperature,[25] solubilizing protecting groups,[26] and disruption of the symmetry of side-chain protecting groups;[27] however, backbone protection is presently believed to be the best approach, since it addresses the problem at its root, i.e. formation of hydrogen bonds. To prevent this type of interaction and, correspondingly, aggregation phenomena, substitution of every backbone amide is probably superfluous, but in the course of peptide-chain elongation such derivatization must be implemented before aggregation occurs. Empirically, it has been found that backbone protection at every sixth residue is sufficient. For shorter sequences, or intermediate fragments of approximately ten residues, placing the backbone protection around the center of the molecule is apparently sufficient, although some exceptions are known.[22]

There are two main types of reversible backbone protection. These are based on (i) the replacement of the amide hydrogen with a protecting group, and (ii) bridging the amide nitrogen to a side-chain group of the amino acid residue. In the first approach, amino acids are N^{α}-substituted with a group that guarantees reversible backbone protection, and these derivatives are additionally protected by a second group at the amino group to prevent aminoacylation during their incorporation into the peptide chain. As noted previously (Section 2.3.1.1), introduction of methoxy groups into the benzyl moiety is known to enhance acidolytic cleavage; however, steric hindrance for aminoacylation of the related derivatives is simultaneously drastically increased. A suitable compromise is found with N^{α}-(2,4-Dmb), which was proposed by Weygand et al.[6] as the first backbone protecting group. Alkylation of the amino groups is achieved by reaction of amino acids with 2,4-dimethoxybenzaldehyde to yield the azomethines which in turn are reduced by sodium borohydride[6] or by catalytic hydrogenation[28] to produce the desired *N*-aryl derivatives. Appropriate protection of the secondary amine, as well as the use of these amino acid derivatives in peptide synthesis, occurs under standard conditions whereby cleavage of the 2,4-Dmb group is achieved by acidolysis in the presence of scavengers (see Table 1). The aim of solubilizing otherwise insoluble peptides has been fully achieved. Thus, $(Gly)_5$, which is completely insoluble in standard organic solvents in the underivatized form, becomes fully soluble in ethyl acetate.[6] This solubilizing effect has been confirmed, but the limitations of this protecting group have also been recognized.[28] Additionally, Tmb protection has been applied to glycine residues in the synthesis of difficult transmembrane sequences and again a strong solubilizing effect was observed.[29,30]

2.3.2.1 2-Hydroxybenzyl Protection

Although N^{α}-benzylated amino acids or peptides are known to be sterically hindered with respect to aminoacylation, acylation of N^{α}-(2-hydroxybenzyl) (Hbz) amino acid derivatives proceeds surprisingly well.[31] As shown in Scheme 1, the initial step in aminoacylation of N^{α}-Hbz-derivatized amino acid residues (Y = H) is esterification of the hydroxy group, which is followed by intramolecular, base-catalyzed O→N acyl migration to produce the peptide bond.[31]

Scheme 1 Aminoacylation of N^{α}-(2-Hydroxybenzyl) and N^{α}-(2-Hydroxy-4-methoxybenzyl) Amino Acid Derivatives[31,32]



X = leaving group; Y = H, OMe

N^{α}-2-Hydroxybenzylation of amino acids can be performed using a reductive amination procedure[6] with salicylaldehyde and sodium borohydride.[31,32] For the use of these N-alkylated amino acids as carboxy components, an additional protection of the N^{α}-amino group and of the Hbz moiety is required so that these two protecting groups can be simultaneously removed for the subsequent aminoacylation step. To meet this goal, conversion into the *N,O*-bis-Fmoc derivatives by reaction with Fmoc-Cl has been proposed,[31,32] as the Fmoc group can be readily removed from the amino and hydroxy groups by exposure to nucleophiles, e.g. piperidine. Acylation of peptide chains with N-terminal Hbz-amino acyl residues is mainly performed using symmetrical anhydrides[31,33] and urethane-protected *N*-carboxyanhydrides (UNCAs).[31] The use of fluorides should be similarly efficient, whereas reaction with amino acid active esters, e.g. pentafluorophenyl esters, in the presence of HOBt,[33] proceeds at a low rate.

The Hbz group is stable to TFA treatment, but is hydrolyzed with TfOH[31] and, as expected, even with HF, but preferably in the presence of scavengers. Due to this stability toward TFA, Hbz protection is more compatible with Boc/Bzl chemistry than with the Fmoc/tBu strategy.[31]

H-(Hbz)Ala-OH:[32]
To a soln of H-Ala-OH (0.9 g, 10 mmol) in H_2O (10 mL) containing KOH (0.56 g, 10 mmol) was added salicylaldehyde (1.22 g, 10 mmol) in EtOH (10 mL), and the mixture was stirred for 30 min at rt. A soln of $NaBH_4$ (0.42 g, 10.1 mmol) in H_2O (5 mL containing 5 drops of 1 M NaOH) was then added dropwise within 15 min and after an additional 15 min stirring, the mixture was kept at 60 °C for 10 min and then cooled in an ice bath. After acidification with concd HCl, the product precipitated on standing overnight at 4 °C. The precipitate was collected, and washed with ice-cold water and 50% aq MeOH; yield: 74%; mp >300 °C; $[\alpha]_D^{20}$ −16.9 (*c* 1, 1 M NaOH).

The related Fmoc derivatives were prepared by standard procedures with Fmoc-Cl, e.g. Fmoc-(Fmoc-Hbz)Ala-OH;[32] mp 165–172 °C; $[\alpha]_D^{20}$ −2.3 (*c* 1, DMF).

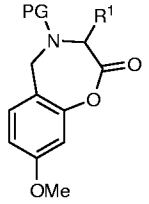
for references see p 273

2.3.2.2 2-Hydroxy-4-methoxybenzyl Protection

For a more appropriate adaptation of the benzyl-type backbone protection to the Fmoc/tBu strategy, the 2-hydroxy-4-methoxybenzyl (Hmb) group was proposed because incorporation of the 4-methoxy moiety leads to significantly enhanced acid lability.[33] The Hmb amino acids are prepared analogously to the Hbz compounds (Section 2.3.2.1),[32] and the N^{α}-Fmoc-protected N^{α}-Hmb amino acids can be obtained by reaction of the Hmb amino acids with Fmoc-OSu[34] or, upon silylation, with Fmoc-Cl[35] in the presence of a base.

Studies of the mechanism of acylation of amino groups with N^{α}-protected Hmb amino acid derivatives revealed a new type of reactive intermediate, which is formed independently of the coupling procedure applied. This intermediate was identified as an 8-methoxy-4,5-dihydro-1,4-benzoxazepin-2(3*H*)-one derivative (Scheme 2), which is believed to be the active component that undergoes aminolysis to produce the desired peptide bond.[35] The acylating potency of this intermediate is, however, inferior to that of normal active esters and coupling yields are reported to be low.[36] However, aminoacylation with N^{α}-Fmoc-mono-protected Hmb amino acids proceeds efficiently if the combination TBTU/HOBt/DIPEA is used as the acylating agent.[34] To prevent formation of the intermediate lactone, the phenolic group has to be protected. Due to acid lability, the Hmb group is mainly used in combination with Fmoc/tBu chemistry and thus the *N,O*-bis-Fmoc derivatives of Hmb amino acids are the intermediates of choice. These are preferably prepared by reaction with Fmoc-Cl under standard conditions,[32] as the related azide and *N*-hydroxysuccinimide ester are not sufficiently reactive for this purpose.[35] The purity of the derivatives is strongly dependent upon temperature and pH, which must be strictly controlled.[35]

Scheme 2 8-Methoxy-4,5-dihydro-1,4-benzoxazepin-2(3*H*)-one Derivative: Intermediate in the Activation of N^{α}-Protected N^{α}-(2-Hydroxy-4-methoxybenzyl) Amino Acids[35]



Coupling of *N,O*-bis-Fmoc-protected Hmb amino acids to amino groups is preferably carried out with the corresponding pentafluorophenyl esters[32,33] or with carbodiimides.[33] The yields strongly depend on the nature of the side chains and this can be explained in terms of steric hindrance, especially with β-branched amino acids. This observation suggests that only amino acids with unbranched side chains are suitable as holders of backbone protection,[35] and among these glycine and phenylalanine are particularly recommended.[37] The acylation of N-terminal Hmb amino acid residues is equally as difficult and thus the sterically most unhindered sites are generally preferred for this backbone substitution. For this acylation, the use of UNCAs in a large excess appears to be the method of choice.[38] Similarly efficient is the use of acyl fluorides, demonstrated in the case where Fmoc-Val-OH was coupled to N-terminal H-(Hmb)Ile or H-(Hmb)Val residues.[39] In the course of N-terminal peptide extension, the phenolic group of Hmb is acylated first as it is less hindered than the secondary amino group. Again, O→N migration in a second step leads to peptide bond formation, as shown in Scheme 1 (Y = OMe). In some cases, the phenolic group remains partially esterified by the Fmoc amino acid derivatives; quantitative cleavage of these esters, however, is assured in the subsequent Fmoc-removal steps with piperidine.[38]

Cleavage of the Hmb groups can be performed with TFA, although care has to be taken to avoid alkylation of tryptophan residues. An efficient cocktail for acidolytic removal of all

acid-labile protecting groups is a mixture of TFA/phenol/ethane-1,2-dithiol/triethylsilane (92:3:3:2) where treatment for two hours is recommended;[38] in this context, triisopropylsilane (TIS) is an equally efficient scavenger.[32]

With the removal of the Hmb groups, their solubilizing effect is lost. Therefore, an alternative procedure has been proposed for peptides that are insufficiently soluble to allow for chromatographic purification steps. This alternative is based on the observation that base-catalyzed acetylation of the Hmb phenolic group confers full stability to TFA treatment for more than 16 hours.[40] Thus, the N^{α}-deprotected peptide is capped at the N-terminal amino group as the Boc derivative and acetylation is performed prior to the resin-cleavage step.[38] After release from the resin as the multiple (2-acetoxy-4-methoxybenzyl)-peptide derivative and its subsequent chromatographic purification, the acetyl groups are removed by exposure to hydrazine. Finally, the Hmb groups are cleaved by acidolysis using the efficient cocktail (*vide supra*). A similar approach is based on the use of N^{α}-(2-acetoxy-4-methoxybenzyl) N^{α}-Fmoc amino acids where selective cleavage of the N^{α}-Fmoc group is achieved by treatment with a 1% solution of 1,8-diazabicyclo[5.4.0]undec-7-ene in DMF.[41,42] With this approach, however, the anchimeric assistance of the hydroxy group in the acylation step is lost.

The efficiency of postsynthetic acetylation of the Hmb groups is well demonstrated by the successful synthesis of the extremely insoluble β-amyloid-(1–43) peptide.[38] Additional examples of the advantageous use of this backbone protection are the recent syntheses of *N*-glycopeptides[43,44] and phosphopeptides.[45] There are, however, also cases where even with multiple backbone protection only minor improvements in solubility were obtained,[37] and polar aprotic solvents with strong hydrogen-bond acceptor properties, such as DMSO, proved to be more efficient.[22]

To facilitate aminoacylation of N^{α}-(2-hydroxybenzyl)amino acids or -peptide derivatives via O→N acyl migration various related nitro derivatives were proposed,[98,99] among which the N^{α}-(2-hydroxy-6-nitrobenzyl) and N^{α}-(2-hydroxy-5-nitrobenzyl) proved significantly more reactive in this context than the Hmb amino acid residues. Incorporation of these protecting groups is performed by reductive alkylation as for the Hmb derivatives and removal of the N^{α}-protecting group is readily achieved by mild photolysis at 366 nm for 3 hours.

Hmb Amino Acids; General Procedure:[35]
A soln of the amino acid (11.4 mmol) in H_2O (20 mL) was adjusted to pH 9.0 with 2 M NaOH. Finely powdered 2-hydroxy-4-methoxybenzaldehyde (1.7 g, 11.4 mmol) was added and the resulting suspension was left for 2 h, under stirring at rt, while the pH was kept at 9–9.5 with 2 M NaOH. Then the mixture was cooled to 5 °C, and 0.76 M aq $NaBH_4$ (11.4 mmol) and 2 M NaOH (3 drops) were added dropwise within 20 min. After 1 h at 5–10 °C, the mixture was neutralized to pH 6–7 with 2 M HCl under vigorous stirring while the temperature was kept below 20 °C. The product precipitated in the cold and was collected by filtration; yield: 60–80%.

Fmoc-(Hmb)amino Acids; General Procedures:
With Fmoc-Cl after Silylation:[35] To an argon-sat. suspension of the Hmb amino acid (8.9 mmol) in anhyd CH_2Cl_2 (30 mL) were added DIPEA (9.3 mL, 54.4 mmol) and TMSCl (6.7 mL, 53.0 mmol). After stirring for 3 h at rt the soln was cooled to −65 °C, a soln of 0.89 M Fmoc-Cl (1 equiv) in anhyd CH_2Cl_2 was added dropwise within 10 min, and then the mixture was allowed to reach rt. The reaction was quenched with 2 M HCl (40 mL), and the organic layer was washed with 2 M HCl, dried ($MgSO_4$), and evaporated. The resulting orange-colored, oily crude product was chromatographed [silica gel, $CHCl_3$ then $CHCl_3$/MeOH (9:1)]. The product was obtained as a pale yellow solid; yield: 75–85%.

With Fmoc-OSu:[34] To a suspension of the Hmb amino acid (5 mmol) in MeCN/H_2O (3:1) was added TEA (2.09 mL, 15 mmol) followed by Fmoc-OSu (3.37 g, 10 mmol). After 90 min the soln was concentrated and then distributed between 5% citric acid and EtOAc (200 mL). The EtOAc layer was washed with sat. NaCl, dried (Na_2SO_4), and concentrated to give an oily residue which was chromatographed (silica gel, CH_2Cl_2 then 5% MeOH in $CHCl_3$); yield: 70–80%.

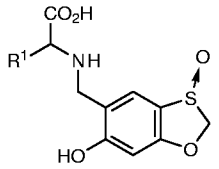
for references see p 273

Fmoc-(Fmoc-Hmb)amino Acids; General Procedure:[32]
To an ice-cold soln of Hmb amino acid (5 mmol) in H_2O (50 mL) containing $Na_2CO_3 \cdot 10\ H_2O$ (4.58 g, 16 mmol) and dioxane (30–50 mL) was added Fmoc-Cl (2.64 g, 10.2 mmol) in dioxane (10 mL) over 1h. The soln was stirred for an additional 1–2 h and a precipitate formed. The mixture was extracted with Et_2O (3 × 100 mL) and the combined ethereal solutions were washed with 0.5 M HCl (3 × 100 mL) and dried (Na_2SO_4). After removal of the solvent, the oily residue was chromatographed (silica gel, packed with $CHCl_3$). The sample was loaded in Et_2O and eluted with $CHCl_3$/MeOH mixtures (4–8% MeOH). The product was recrystallized (EtOAc/hexane); yield: 10–40%.

2.3.2.3 N^{α}-[(6-Hydroxy-3-oxido-1,3-benzoxathiol-5-yl)methyl] Protection

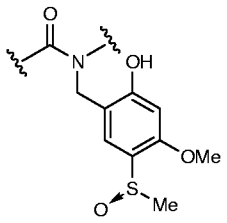
Recently, an interesting new backbone protecting group, the N^{α}-[(6-hydroxy-3-oxido-1,3-benzoxathiol-5-yl)methyl] group (Scheme 3), was proposed[36] to realize the safety-catch principle (see Section 2.3.1.3). This group was developed to improve the acylating potency of 8-methoxy-4,5-dihydro-1,4-benzoxazepin-2(3*H*)-one derivatives (Scheme 2) that normally seem to be formed on activation of N^{α}-Fmoc-monoprotected Hmb amino acids[35] (Section 2.3.2.2). Thus, an electron-withdrawing group, namely a sulfoxide in the position *para* to the hydroxy group, was introduced into the Hmb moiety (Scheme 3). The advantage of this group over the Hmb group is an increased coupling rate and complete suppression of epimerization on carboxy activation of peptide fragments. As an *S*-oxide derivative it is fully stable to TFA, but after reduction with ammonium iodide/dimethyl sulfide it can be cleaved with TFA, thus representing an attractive alternative to the 2-acetoxy-4-methoxybenzyl approach (Section 2.3.2.2).[36]

Scheme 3 N^{α}-[(6-Hydroxy-3-oxido-1,3-benzoxathiol-5-yl)methyl] Amino Acids[36]



A further development of this safety-catch principle led to N^{α}-[2-hydroxybenzyl-4-methoxy-5-(methylsulfinyl)] (SiMB) derivatives (Scheme 4) where the acylation potency of the intermediate 8-methoxy-4,5-dihydro-1,4-benzoxazepin-2(3*H*)-one derivatives (Scheme 2) is significantly enhanced and similarly the rate of intramolecular O→N acyl migration.[100] The related Fmoc-protected amino acid derivatives are readily prepared and cleavage is achieved by reductive acidolysis ($SiCl_4$/TFA/anisole/ethandithiol 5:90:2.5:2.5, 2 h at room temperature).

Scheme 4 N^{α}-[2-Hydroxybenzyl-4-methoxy-5-(methylsulfinyl)] Derivative[100]

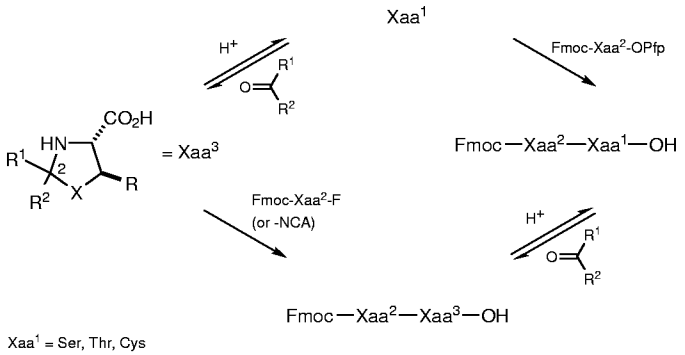


SiMB

2.3.2.4 "Pseudoproline" Protection

Studies on host–guest peptides have shown that incorporation of a proline residue into peptides which have a high tendency to fold into ordered secondary structures disrupts the onset of helical as well as β-sheet conformations and increases solubility and coupling rates.[46,47] Based on this observation, serine- and threonine-derived oxazolidine, and cysteine-derived thiazolidine derivatives (pseudoprolines) were proposed as valuable tools for combining protection of their side-chain functions with the simultaneous solubilization of the peptide chain.[48,49] Due to the induction of "kink" conformations in the peptide backbone, originating from the preference of these pseudoproline residues to adopt the *cis*-imide bond configuration,[50] insertion of such derivatives into peptides prevents self-association and β-structure formation.

Scheme 5 Synthesis of "Pseudoproline" Dipeptides as Building Blocks for Peptide Synthesis Utilizing Fmoc/tBu Chemistry[52–54]



The pseudoprolines are readily accessible by cyclization of serine, threonine, and cysteine with aldehydes and ketones. As reported[51] for the thiazolidine derivatives, ring-opening of the pseudoprolines regenerates the amino acids; therefore, the oxazolidine and thiazolidine derivatives act as reversible protection in peptide synthesis. Due to steric hindrance of the C2 substituents, aminoacylation of N-terminal pseudoprolines is a slow reaction and thus the use of dipeptides as building blocks is preferred over stepwise synthesis. As shown in Scheme 5, the dipeptide building blocks are readily prepared by acylation of the oxazolidine prepared in situ, or of the isolated thiazolidine derivatives, with N^{α}-protected amino acid fluorides or UNCAs.[48,52] In the case of oxazolidine derivatives, the dipeptide synthons are more conveniently prepared from suitably protected Thr/Ser dipeptides.[52,53] Alternatively, the Thr/Ser-derived oxazolidine derivatives are converted in situ into N^{α}-protected pseudoprolines.[54] Variation of the C2 substituents directly affects ring stability, a fact which allows for tailoring of the chemical stability according to the strategy used in the peptide synthesis. Table 2 indicates the effect of the substituents on acidolysis.[52] In the case of the thiazolidine derivatives, the high stability of Thz toward strong acids allows for its use even with Boc/Bzl chemistry, as ring-opening is accomplished with Hg^{2+} ions.[55] Similarly, Thz(2,2-Me_2) is readily opened with Hg^{2+} ions,[51,56] but also with acids.[52] Of the derivatives listed in Table 2, Thz(2-Dmp), Oxa(2,2-Me_2), and Oxa(2,2,5-Me_3) are fully compatible with standard Fmoc/tBu protection, whereas Thz(2,2-Me_2), Oxa, and Oxa(5-Me) are stable under these conditions and require stronger acidic conditions for their reversal into the amino acid residues. All pseudoprolines are fully stable under conditions normally used for N^{α}-Fmoc deprotection. Coupling of the dipeptide building blocks to the growing peptide chains occurs without racemization, as formation of oxazol-5(4*H*)-one intermediates is prevented, in

for references see p 273

analogy with proline. The strong effect of pseudoprolines on solubility has been well-documented in the syntheses of peptides with known difficult sequences.[52,57,101]

Table 2 Acid Hydrolysis of Pseudoprolines[52]

Pseudoproline[a]	Cleavage Conditions	Ref
Thz(2,2-Me_2)	TFA within hours	[52]
Thz(2-Dmp)	TFA within minutes	[52]
Thz	stable to strong acids	[52]
Oxa(2,2-Me_2)	TFA within minutes	[52]
Oxa	TfOH within hours	[52]
Oxa(2,2,5-Me_3)	TFA within minutes	[52]
Oxa(5-Me)	TfOH within hours	[52]

[a] Mutter and co-workers[48] have defined oxa- and thiaproline derivatives of serine, threonine, and cysteine with Ser(ΨPro), Thr(ΨPro), and Cys(ΨPro), respectively, where the abbreviation "ΨPro" indicates the relationship to proline (with heteroatomic ring substitution in position 4). Pseudoprolines with substitution in position 2 of the proline ring are named Ser/Thr/Cys($\Psi^{R',R''}$Pro).

It is noteworthy that unsymmetrical substitution at the C2 atom generates a mixture of epimers. Although separation of the diastereomers per se is possible, it is not required for the purpose of solubilization of protected peptides, since the final acidolytic deprotection results in the loss of this chiral center.

1,3-Thiazolidine-4-carboxylic Acid (H-Thz-OH):[58]
To a soln of H-Cys-OH·HCl·H_2O (3.5 g, 0.02 mol) in H_2O (10 mL) was added aq CH_2O (40%, 4 mL) and the mixture was left overnight at rt. Addition of abs EtOH (10 mL) and Py (6 mL) gave crystals which were collected, washed with EtOH and Et_2O, and air-dried; yield: 2.45 g (92%); mp 196–197 °C.

1,3-(2,2-Dimethyl)thiazolidine-4-carboxylic Acid [H-Thz(2,2-Me_2)-OH]:[58]
A soln of H-Cys-OH·HCl·H_2O (3.5 g, 0.02 mol) and 2,2-dimethoxypropane (50 mL) was refluxed for 1 h in acetone (250 mL). The resulting white platelets were collected; yield: 3.83 g (96.5%); mp 161–163 °C.

Oxazolidine-4-carboxylic Acid (H-Oxa-OH) and Oxazolidine-5-methyl-4-carboxylic Acid [H-Oxa(5-Me)-OH]:[52,54,59]
A 37% aq soln of CH_2O (19.3 mL, 260 mmol) was added to H-Ser-OH or H-Thr-OH (103 mmol) in aq 2.5 M Na_2CO_3 (32 mL) at rt. The clear soln was left standing overnight at 4 °C, after which the pH was 7.8. This soln was used directly for aminoacylation with Fmoc-protected amino acid fluorides or UNCAs to produce the dipeptide synthons,[48,52] or with Z-Cl and $(Boc)_2O$ to produce the related N^{α}-protected derivatives.[54]

Fmoc-Xaa-Oxa(2,2-Me_2)-OH and Fmoc-Xaa-Oxa(2,2,5-Me_3)-OH; General Procedure:[52]
To a suspension of Fmoc-Xaa-Ser-OH or Fmoc-Xaa-Thr-OH (5.0 mmol) in dry THF (100 mL) were added pyridinium toluene-4-sulfonate (250 mg, 1.0 mmol) and 2,2-dimethoxypropane (3.0 mL, 25.0 mmol). The mixture was refluxed for several hours under an argon atmosphere, the condensate being passed over molecular sieves (4 Å). After cooling, TEA (0.21 mL, 1.5 mmol) was added and the yellow soln concentrated. The residue was partitioned between EtOAc (150 mL) and H_2O, and the organic phase washed with H_2O, dried ($MgSO_4$), and concentrated. The residue was purified by flash chromatography (silica gel) or crystallized (EtOAc/hexane) to give the dipeptide derivative as a white powder; yield: 70–90%.

Fmoc-Xaa-Thz-OH, Fmoc-Xaa-Thz(2,2-Me_2)-OH and Fmoc-Xaa-Thz(2-Dmp)-OH; General Procedure:[52]
To a soln of H-Thz-OH,[57] or H-Thz(2,2-Me_2)-OH,[57] H-Thz(2-Dmp)-OH[57] (10 mmol) in DMF (100 mL) and DIPEA (3.25 mL, 19 mmol) was added dropwise a soln of the Fmoc-protected amino acid fluoride or Fmoc-N^{α}-carboxy amino acid anhydride in dry DMF (100 mL), and the mixture was stirred for 2 h at rt. The soln was concentrated and the residue purified by flash chromatography (silica gel, petroleum ether/EtOAc/AcOH); yield: 60–95%.

2.3.3 ω-Amide Protection

The side-chain carboxamide group of asparagine and glutamine can give rise to various side reactions in peptide-coupling steps, among which dehydration to nitriles is known to occur to a higher extent with asparagine than glutamine. Dehydration strongly depends upon the method applied for carboxy activation,[60,61] and does not solely occur with carbodiimides, as well-assessed on model compounds (Table 3).[62]

Table 3 Dehydration of Unprotected Asparagine During Coupling of Boc-Asn-OH to H-X-Peptide[62]

Xaa	Formation of Boc-Ala(CN)-Xaa-peptide (%)				Ref
	HATU	HBTU/HOBt	DCC/HOBt	DCC/HODhbt	
Gly	5	4	0.7	<0.1	[62]
Pro	65	58	11	2	[62]

Once asparagine and glutamine residues are incorporated into the peptide chain, they are no longer susceptible to dehydration.[63] Base-catalyzed cyclization to imides, besides leading to more facile racemization, is followed by transamidation. Aspartimide formation on treatment with inorganic[2,5,64] or organic bases,[65] and, occasionally, also in the presence of acetate ions,[65] is much more likely than the corresponding cyclization to glutarimide.[66] In contrast, N-terminally positioned glutamine residues show a high tendency to undergo cyclization to 5-oxopyrrolidine-2-carboxylic acid (pyroglutamic acid). This side reaction is often observed when N^{α}-protecting groups of glutamyl peptides are cleaved by acidolysis[67,68] or by catalytic hydrogenolysis in acetic acid as the solvent.[69] This cyclization occurs easily even during acylation of N-terminal glutamine residues, and is dependent upon the rate of peptide-bond formation.[70] Moreover, peptides with N^{ω}-unmasked carboxamides exhibit low solubility and a high tendency to undergo self-aggregation. Surprisingly, this property is more pronounced for glutamine than asparagine;[71] this may be due to the tendency of the γ-carboxamide group of glutamine to undergo intermolecular hydrogen bonding, in contrast to the β-carboxamide group of asparagine which preferentially forms hydrogen bonds to its own peptide backbone.[72] Thus, asparagine may favor dissociation rather than association. In view of the disadvantages of using unprotected glutamine and asparagine residues in peptide synthesis, great attention has been devoted to developing useful protecting groups, a goal that has become more urgent with the advent of solid-phase peptide synthesis.

2.3.3.1 Xanthenyl Protection

Although 9*H*-xanthen-9-yl (Xan) protection was proposed in the early 1960s,[73,74] it has not found widespread application because of its instability under the standard acid conditions required for Boc removal.[75,76] Recently, however, it has been rediscovered. Taking into account that asparagine and glutamine residues, once incorporated into peptide chains, are not susceptible to dehydration,[63] the Xan group can be used for temporary protection in the Boc/Bzl strategy.[77–79] Moreover, as the Xan group is cleanly removed with TFA/H_2O/ethane-1,2-dithiol (18:1:1) or with TFA/methyl phenyl sulfide/ethane-1,2-dithiol/anisole (90:5:3:2),[80] and in particular cases even better than the trityl (Trt) group[81] (see Section 2.3.3.3), it is also well-compatible with Fmoc/tBu chemistry.[80] Incorporation of a methoxy group into position 2 or 3 of 9*H*-xanthen-9-yl does not improve acid lability.[80]

Analogues of the Xan group, 9-phenylxanthen-9-yl (Pix),[82] 10,11-dihydro-5*H*-dibenzo[*a*,*d*]cyclohepten-5-yl (Sub), and 5*H*-dibenzo[*a*,*d*]cyclohepten-5-yl (Dbs), which as the carboxamide derivatives are cleaved under even milder acid conditions, have been recently

for references see p 273

proposed.[83] The Pix group is completely stable to piperidine treatment, as required for Fmoc cleavage, but cleanly removed upon exposure to 50% TFA/CH_2Cl_2 in less than 5 minutes; cleavage of the Trt group under these conditions is negligible. The Dbs and Sub derivatives are also acid-sensitive and are rapidly removed by the various common, acid-cleavage mixtures containing ethane-1,2-dithiol, methyl phenyl sulfide, or anisole as cation scavengers. Thus, even these new asparagine and glutamine derivatives are fully compatible with Fmoc/tBu chemistry.

H-Gln(Xan)-OH:[74]
9*H*-Xanthen-9-ol (9.8 g, 50 mmol) and Z-Gln-OH (13.9 g, 50 mmol) were dissolved in AcOH (200 mL) and the soln was allowed to stand for 2 d at rt. The precipitate was filtered off, dried (KOH), and recrystallized (THF) to give Z-Gln(Xan)-OH; yield: 18.5 g (80%); mp 182–183 °C.
Z-Gln(Xan)-OH was transformed to H-Gln(Xan)-OH by hydrogenation over Pd/C in EtOH at 75–80 °C. The precipitate was dissolved in H_2O by the addition of TEA and filtered to remove the catalyst. The filtrate was acidified to pH 4.5–5 with 2 M HCl, and the precipitate was collected and washed with MeOH and Et_2O; yield: 60%.

Fmoc-Asn(Xan)-OH and Fmoc-Gln(Xan)-OH:[80]
TFA (1.75 mL, 22.7 mmol) was added in small portions to a stirred soln of Fmoc-Asn-OH (1.77 g, 5.0 mmol) and 9*H*-xanthen-9-ol[78] (1.24 g, 7.5 mmol) in DMF (35 mL). The mixture was stirred at rt for 4 h, then poured into ice water (400 mL). The precipitate was collected and washed with H_2O, ice-cold MeOH, and Et_2O; yield: 2.65 g (99%); mp 206–208 °C.
Prepared in a similar manner was Fmoc-Gln(Xan)-OH; yield: 94%; mp 211–212 °C.

Z-Asn(Pix)-OH and Z-Gln(Pix)-OH:[82]
To Z-Xaa-OH (5 mmol) and 9-phenylxanthen-9-ol (2.74 g, 10 mmol) in AcOH (15 mL) were added $(Ac)_2O$ (0.95 mL) and H_2SO_4 (0.05 mL), and the mixture was kept at 50 °C for 12 h. After cooling, the mixture was poured into H_2O. The precipitate was collected by filtration, then recrystallized (CH_2Cl_2/Et_2O/petroleum ether); yield: 61% for Z-Asn(Pix)-OH and 78% for Z-Gln(Pix)-OH. After hydrogenation, conversion into the Fmoc derivatives was performed using Fmoc-OSu under standard conditions.

Similar methods were used for the preparation of Fmoc-Asn(Sub)-OH, Fmoc-Asn(Dbs)-OH, Fmoc-Gln(Sub)-OH, and Fmoc-Gln(Dbs)-OH.[83]

2.3.3.2 4,4′-Dimethoxybenzhydryl Protection

Similar to the observations in the case of protection of α-amide functions with the 2,4-Dmb[8] or Tmb[8] groups (Section 2.3.1.1), the alkylating properties of ω-amides upon acidolytic removal of the protecting group represent a serious drawback; moreover, alkali-induced side reactions are not suppressed, at least in the case of asparagine residues.[7] Therefore, other protecting groups are preferred, such as the 4,4′-dimethoxybenzhydryl (Mbh) group, despite its inherent disadvantage in that quantitative acidolysis is difficult (Table 1). The ω-amide derivatives are obtained by reaction of the N^{α}-protected asparagine or glutamine, preferably the benzyloxycarbonyl derivatives, with 4,4′-dimethoxybenzhydrol under acid catalysis, e.g. sulfuric acid, toluenesulfonic acid, or the boron trifluoride–diethyl ether complex.[5] If attached to glutamine, the N^{ω}-Mbh group is stable toward catalytic hydrogenation, alkaline conditions, and organic bases. In contrast, in the case of treatment of asparagine with aqueous alkali, formation of side products is not prevented.[5] The bulkiness of the protecting group exerts some steric hindrance[84] and, therefore, strong acylating procedures are required for incorporation of these amino acid derivatives into growing peptide chains.

As shown in Table 1, cleavage of the Mbh group with TFA is slow[9] and, furthermore, is sequence-dependent.[5] Scavengers are required to prevent alkylation at susceptible functions and incomplete cleavage can be well-detected spectroscopically.[85] Moreover, the impact of the Mbh group on solubility is not as strong as expected in some cases.[86]

Z-Xaa(Mbh)-OH:[5]
To a soln of the Z-Xaa-OH (100 mmol) and Mbh-OH (24.0 g, 100 mmol) in AcOH (250 mL) was added concd H_2SO_4 (0.5 mL) at rt. The mixture was left to stand at rt overnight and then poured into H_2O (750 mL). The crystals were collected by filtration and dissolved in EtOAc; this soln was washed with H_2O, dried (Na_2SO_4), and concentrated. The residue was triturated with Et_2O and recrystallized (THF/petroleum ether); yield: 45.8 g (90%) for Z-Gln(Mbh)-OH and 98% for Z-Asn(Mbh)-OH.
After hydrogenation of the Z-Xaa(Mbh)-OH in AcOH/MeOH (1:1), the mixture was concentrated and the residue recrystallized (H_2O); yield: 76% for H-Gln(Mbh)-OH; mp 205–206 °C, and 94% for H-Asn(Mbh)-OH; mp 215–217 °C.[5]

Fmoc-Gln(Mbh)-OH:[84]
H-Gln(Mbh)-OH was reacted with Fmoc-Cl (1.2 equiv) under standard conditions and the mixture was worked up by washing the product, upon acidification and extraction into EtOAc, with H_2O. The soln was concentrated and the residue recrystallized (EtOAc); yield: 48.5%; mp 158–164 °C.

2.3.3.3 Trityl Protection

Tritylation of the ω-carboxamide group is known to significantly enhance the solubility of both the amino acid derivatives and of growing peptide chains with overall beneficial effects in the synthesis of peptides.[87,88] While condensation of the ω-carboxy group with tritylamine using DCC/HOBt fails, direct acid-catalyzed tritylation of ω-carboxamides at 50 to 60 °C, in the presence of acetic anhydride to trap the water formed during the reaction, allows for the preparation of both the asparagine and glutamine derivatives in satisfactory yields.[87] This type of ω-protection is stable to mineral acids in aqueous solution, as well as to nucleophiles and bases; it is completely stable to 20% piperidine in DMF, strong aqueous alkali, catalytic hydrogenolysis, 1 M HCl in 50% *tert*-butyl alcohol, and acetic acid/water (4:1), but is quantitatively removed with 95% aqueous TFA within 10 minutes,[87] except when the α-amino group is free, e.g. with H-Asn(Trt)- in the N-terminal position of a peptide, where cleavage requires at least four hours.[88] This longer cleavage time is not observed with Pix[82] or Xan protection[81] (see Section 2.3.3.1). Alkylation, for example of tryptophan residues, is observed,[87] therefore the use of efficient scavengers is required.

H-Asn(Trt)-OH:[87]
A suspension of H-Asn-OH (13.2 g, 100 mmol), Trt-OH (52 g, 200 mmol), Ac_2O (18.9 mL, 200 mmol), and concd H_2SO_4 (6.1 mL, 115 mmol) in AcOH (300 mL) was stirred at 60 °C for 75 min. The soln was then poured into cold H_2O (600 mL) and adjusted to pH 6 by the addition of 10 M NaOH. After 1 h at 0 °C, the crystals were filtered off, washed with H_2O and toluene, and dried; yield: 28.7 g (75%, as the semihydrate).

H-Gln(Trt)-OH:[87]
A suspension of Z-Gln-OH (27.9 g, 100 mmol), Trt-OH (52 g, 200 mmol), Ac_2O (18.9 mL, 200 mmol), and concd H_2SO_4 (0.5 mL, 9 mmol) in AcOH (300 mL) was stirred for 1 h at 50 °C. The soln was then slowly added to cold H_2O (3 L), and the precipitate filtered off and dissolved in EtOAc (800 mL). The organic layer was washed with H_2O and taken to dryness. The residue was crystallized (EtOAc/hexane) to give Z-Gln(Trt)-OH; yield: 35 g (67%).

Z-Gln(Trt)-OH (24 g, 46 mmol) was hydrogenated over Pd/C in 90% aq MeOH (240 mL) containing 1 M HCl (46 mL). After 90 min, the starting material was completely dissolved, and after hydrogenolysis for an additional 60 min, the catalyst was filtered off, and the soln concentrated to 200 mL and neutralized with TEA (6.4 mL). After stirring for 1 h at 0 °C, the crystals were collected by filtration, then washed with H_2O and dried; yield: 17 g (93%, as the semihydrate).

2.3.3.4 Other Alkyl-type Protections

The experience with the Trt group (Section 2.3.3.3) made apparent the need for an ω-amide protecting group that could be cleaved more rapidly, especially for N-terminally positioned asparagine residues. Introduction of a methyl group to the 4-position of one of the benzene rings stabilizes the Trt cation through hyperconjugative effects and facilitates acidolytic

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cleavage.[89] The 4-methyltrityl (Mtt) group can be introduced by procedures similar to those used for the Trt group (Section 2.3.3.3).[89,90] Cleavage rates of the Mtt group with 50% TFA/CH_2Cl_2 are two to three times more rapid than that of the related N^{ω}-Trt derivatives. With 95% TFA the asparagine derivative is cleaved within 20 minutes; the glutamine derivative is cleaved even more rapidly.[90]

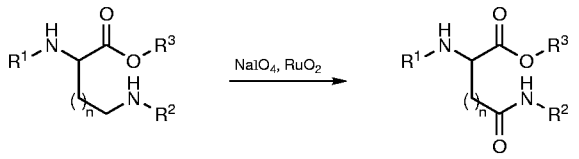
Recently, the cyclopropyldimethylcarbinyl group has been proposed as a new, acid-labile carboxamide protecting group.[91,92] This group is fully compatible with Fmoc/tBu chemistry and is removed significantly faster than the Trt group from asparagine, even if positioned in the N-terminal position.

2.3.3.5 Newer Protecting Groups

As in the case of C-terminal α-carboxamides, the benzyloxy group could become an interesting protecting group for ω-carboxamide groups.[12] The related amino acid derivatives are readily obtained from suitably protected aspartic and glutamic acid derivatives by benzyloxyamidation of the β- and γ-carboxy groups via a DCC-mediated reaction with *O*-benzylhydroxylamine. As noted in Section 2.3.1.4, the benzyloxy group is stable in both acidic and alkaline media,[12,13] and deprotection is accomplished by hydrogenolysis over a palladium catalyst followed by reduction with titanium(III) chloride.[14] Although this group may not replace the standard amide protecting groups, it could represent an interesting alternative in particular synthetic tasks.

An alternative, new approach for the synthesis of ω-carboxamide-containing amino acid derivatives is based on the oxidation of suitably protected α,α-diamino acids (2,4-diaminobutyric acid, ornithine, lysine) with permanganate[93] or ruthenium(IV) oxide,[94] as shown in Scheme 6. The ω-Boc-carboxamide group is stable toward catalytic hydrogenation, acetic acid, and 10% TEA/CH_2Cl_2, is unstable in the presence of 1 M NaOH/MeOH, and is cleaved with methanolic hydrazine hydrate, 25% TFA/CH_2Cl_2, 25% HBr/AcOH, or HF.[95]

Scheme 6 Selective Oxidation of Protected α,ω-Diamino Acids to Urethanyl-Protected N^{ω}-Carboxamide Amino Acids[94]



R^1 = Boc; R^2 = Boc, Z; R^3 = Me, tBu, Nbz

Peptide syntheses with the N^{α},N^{ω}-bis-protected amino acids are performed using standard methods, e.g. via the mixed anhydride or carbodiimide[95] procedure, although both the N^{ω}-Boc and N^{ω}-Z derivatives do not fully prevent dehydration to the nitrile derivatives.[96]

Boc-Gln(Boc)-ONbz:[94]
To a soln of Boc-Orn(Boc)-ONbz (0.935 g, 2 mmol) in EtOAc (40 mL) were added $RuO_2 \cdot xH_2O$ (120 mg) and 10% aq $NaIO_4$ (120 mL) under vigorous stirring at rt over 24 h. The crude Boc-Gln(Boc)-ONbz was purified by chromatography (silica gel); yield: 73%

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2.4 Photocleavable Protecting Groups

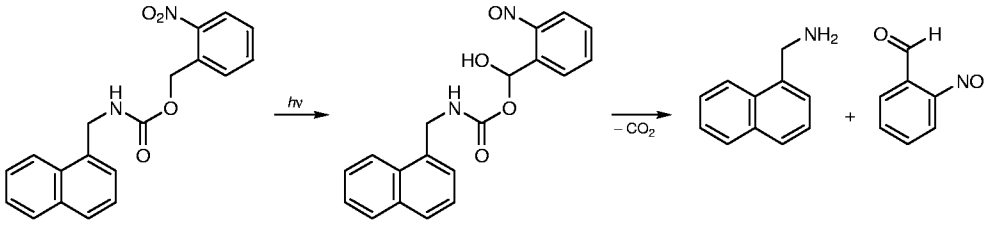
M. A. Clark and T. T. Romoff

The ideal set of protecting groups for peptide synthesis has as its most important aspect the orthogonality of the various groups; in the best of all worlds each protecting group could be attached and cleaved independently of all the others.[1] Since this ideal state has not yet been achieved, significant development of new protecting groups continues to be an active area of research. Traditional approaches to orthogonality have been based on the relative lability of protecting groups according to their tolerance to acid or base. More recent advances in protecting group chemistry are moieties that can be cleaved under neutral conditions, such as the Pd(0)-sensitive allyl-based groups described in Sections 2.1 and 2.2. The ultimate neutral condition is to use no chemical reagent at all, and protecting groups have indeed been developed that are cleaved by light of appropriate intensity and wavelength. This section will review a number of photocleavable protecting groups used in peptide synthesis.

The Photochemistry of Deprotection

The principle of photocleavage is absorption of light of a specific energy that initiates electronic transitions, ultimately resulting in the breakage of a chemical bond. The most common pathway for photocleavage of chemical bonds begins with an $n \rightarrow \pi^*$ or a $\pi \rightarrow \pi^*$ transition.[2] Any chemical bond can be cleaved given sufficient energy. As with all protecting group chemistries, selectivity is required so that only the particular bond of interest is cleaved while leaving all other bonds intact. In the case of a photocleavable bond, the selectivity is achieved by arraying appropriate functional groups in the vicinity of the bond. The ideal absorption energy lies in the ultraviolet region in the vicinity of 320 nm: higher energies will result in widespread bond cleavage; lower energies will render the molecule sensitive to visible light. The most common photocleavable groups are 2-nitrobenzyl-derived carbamates, esters, and ethers. Upon irradiation, the excited nitro group abstracts a hydrogen atom from the benzylic position and the intermediate rapidly rearranges to a nitroso hemiacetal or hemiaminal.[3,4] A typical route for photocleavage of an amino protecting group is shown in Scheme 1.[5] Photochemical removal of a protecting group can also proceed by light-induced bond homolysis, excitation and solvolysis, electron transfer, photocyclization, or photooxygenation.

Scheme 1 Typical Route for Photocleavage of 2-Nitrobenzyl Protecting Groups[5]



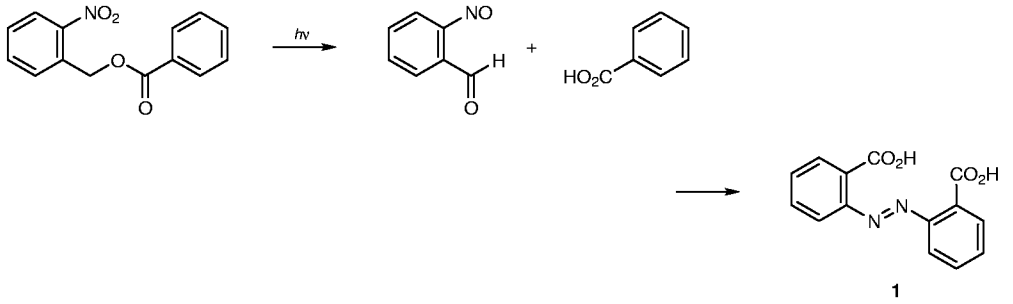
2.4.1 Amino Protecting Groups

Amino protecting groups fall into four broad categories: nitrobenzyl (e.g., nitrobenzyloxycarbonyl and 4,5-dimethoxy-2-nitrobenzyloxycarbonyl), phenacyl (e.g., 4-methoxyphenacyloxycarbonyl), benzyloxycarbonyl (e.g., α,α-dimethyl-3,5-dimethoxybenzyloxycarbonyl), and arylsulfonamides (e.g., tosyl). All but the sulfonamides mask the amino group as a carbamate. Removal of the blocking moiety releases an unstable carbamic acid, which spontaneously decarboxylates to give the unprotected amine.

2.4.1.1 Nitrobenzyl-Based Groups

One of the first uses of a 2-nitrobenzyl-based photocleavable protecting group was simply an 2-nitrobenzyl ester, reported in 1966 by Barltrop et al.[6] 2-Nitrobenzyl benzoate was irradiated with ultraviolet light from a medium pressure mercury arc source, but the yield of benzoic acid was only 17%. The low yield was largely a result of the further reaction of the byproduct nitroso compound, which is converted into the light-filtering azo compound **1** (Scheme 2).

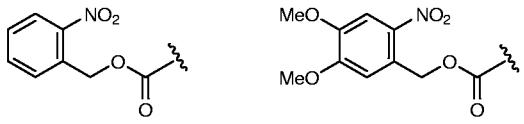
Scheme 2 Formation of **1** from Irradiation of 2-Nitrobenzyl Benzoate[6]



1

Much of the further work in this area was devoted to working around this problem by structural modification of the photocleavable group. Amine protection was developed using 2-nitrobenzyl-derived carbamates as the blocking moiety. An additional problem with amine deprotection is the fact that the released amine can react with the aldehyde photoproduct. This problem was addressed by Patchornik et al.[7] They introduced the 2-nitrobenzyloxycarbonyl Z(2-NO_2) and 4,5-dimethoxy-2-nitrobenzyloxycarbonyl (Nvoc) groups for protection of the α-amino group (Scheme 3). The authors found that the presence of aldehyde trapping agents in the photodeprotection greatly increased the yields of deprotected products.

Scheme 3 The Nboc and Nvoc Photocleavable Protecting Groups[7]



Z(2-NO_2) Nvoc

2.4.1.1.1 2-Nitrobenzyloxycarbonyl Group

N-Z(2-NO_2)-protected amino acids and peptides are normally prepared by reacting the free amine with 2-nitrobenzyl chloroformate, which is in turn prepared from 2-nitrobenzyl alcohol by reaction with phosgene or a phosgene equivalent.[8] 2-Nitrobenzyloxy-4′-nitrophenylcarbonate has also been used to install the Z(2-NO_2) group. The Z(2-NO_2) group is stable to strong acids (100% TFA, HCl/dioxane); thus, Boc and other acid-labile protecting groups can be selectively cleaved in its presence. Like all benzyloxycarbonyl-based groups, the Z(2-NO_2) derivatives are unstable to catalytic hydrogenation conditions. The deprotection of Z(2-NO_2) requires the use of additives such as sulfuric acid or semicarbazide hydrochloride.[7] These minimize side reactions with the nitrosoaldehyde byproduct and increase the yields of deprotected amino acids or peptides.

2-Nitrobenzyloxycarbonyl Chloride:[6]

CAUTION: *Phosgene is a colorless, poisonous gas or volatile liquid. This reagent should be handled with the utmost care.*

Phosgene (200 g) was bubbled through a soln of 2-nitrobenzyl alcohol (50 g, 330 mmol) in dioxane (400 mL) at such a rate that the temperature of the mixture was kept below 40 °C. The mixture was left at rt overnight, the excess phosgene was then removed by bubbling nitrogen through the soln, and the solution was evaporated at a bath temperature <40 °C. The residual oil (70 g, quant yield) constituted pure product. Little decomposition was observed when a soln was kept at 0 °C for 1 month.

2-Nitrobenzyl 4-Nitrophenyl Carbonate:[9]
An ice-cold soln of 4-nitrophenyl chloroformate (9.0 g, 45 mmol) in acetone (25 mL) was added dropwise to a soln of nitrobenzyl alcohol (7.8 g, 50 mmol) and pyridine (3.6 g, 45 mmol) in acetone (25 mL) with stirring. The product began to precipitate immediately after the last portion was added. The mixture was then poured into H_2O (300 mL) and stirred for 15 min with cooling in ice water. The product was then filtered, washed with H_2O, and dried; yield: 13.8 g (96%). Recrystallization (MeOH/$CHCl_3$) afforded the product as light yellow crystals, mp 132 °C.

Z(2-NO_2)-Ala-OH:[9]
A mixture of H-Ala-OH (0.53 g, 6 mmol), 2-nitrobenzyl 4-nitrophenyl carbonate (2.3 g, 7.2 mmol), 2 M NaOH soln (6.0 mL), and THF (20 mL) was stirred for 24 h at rt. On removal of the THF, sodium 4-nitrophenolate precipitated. This was filtered off and washed with $NaHCO_3$ soln. The filtrate was acidified with 2 M HCl to pH 5–6, and the excess reagent and 4-nitrophenol were removed by extraction with Et_2O. The remaining aq phase was acidified to pH 1 and extracted thoroughly with Et_2O or EtOAc. Removal of the solvent afforded Z(2-NO_2)-Ala-OH as light yellow crystals; yield: 1.2 g (75%); mp 131 °C.

Photolysis of an *N*-Z(2-NO_2) Amino Acid or Peptide; General Procedure:[9]
A soln of Z(2-NO_2) amino acid or peptide (0.01–0.05 M) in EtOH, dioxane, or $CHCl_3$ was irradiated in a water-cooled Pyrex immersion-type photochemical reactor with a 125-W Hg quartz lamp. In the case of photosensitive amino acids such as phenylalanine or tryptophan, light of wavelength less than 320 nm was avoided using a $CuSO_4$ filter soln between the light source and the substrate soln. The irradiation periods for different amino acid and peptide derivatives varied from 0.5 to 3.5 h. The reaction was followed by TLC. In the presence of H_2SO_4 (5 equiv) or semicarbazide hydrochloride (10 equiv), the deprotection occurred with substantially higher yields. After irradiation, the solvent was removed from the photolysate under reduced pressure. The residue was treated with dil HCl and the mixture was then extracted with Et_2O to remove the protecting group photoproducts. The aqueous layer was neutralized carefully; the separated N-deprotected product was filtered and dried or extracted with suitable solvents. Yields range from 77% for H-Lys(Boc)-OH to 100% for H-Ala-OH.

2.4.1.1.2 4,5-Dimethoxy-2-nitrobenzyloxycarbonyl Group

N-Nvoc-protected amino acids and peptides are prepared by procedures analogous to those used for Z(2-NO_2) derivatives. The removal is also similar to Z(2-NO_2). However, Nvoc deprotections often proceed in higher yield than when Z(2-NO_2) is used. This is likely due to the lower electrophilicity of the aldehyde byproduct. In some cases, additives are not re-

quired to obtain a high yield of deprotected product. Nvoc also resembles Z(2-NO_2) in terms of its stability to acid and base. The Nvoc group has been used for side-chain amine protection as well as α-amine blocking.[10]

The Nvoc group has also been used in nontraditional peptide synthesis. Fodor et al. used Nvoc-protected amino acids for the light-directed synthesis of peptide arrays on a solid surface.[11] The Nvoc group has also been used for amine protection in the coupling of amino acids to transfer RNAs[12] and as the phototrigger in a peptide-based fluorescent protecting group for oligonucleotides.[13]

The procedures for the installation and removal of Nvoc groups are largely identical to those for the Z(2-NO_2) group (Section 2.4.1.1.1).

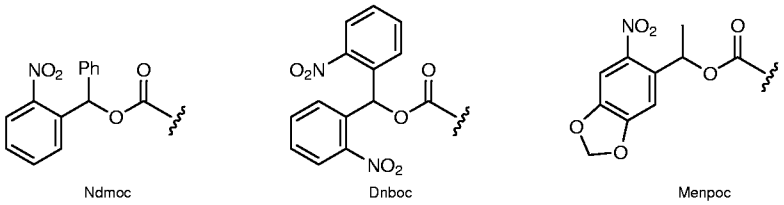
Fmoc-Lys(Nvoc)-OH:[10]
To Fmoc-Lys-OH (3.385 g, 10 mmol) in H_2O/dioxane/MeCN (1:1:1, 60 mL) in a foil covered flask was added $NaHCO_3$ (1.060 g, 10 mmol). Additional amounts of H_2O (10 mL) and MeCN (5 mL) were added so that all solids dissolved. 4,5-Dimethoxy-2-nitrobenzyl chloroformate (2.756 g, 10 mmol) in dioxane (20 mL) was added dropwise to the mixture (20 min.), and stirred overnight at rt (18 h). The solvent was removed, the residue was partitioned between 1 M $NaHSO_4$ (50 mL) and CH_2Cl_2 (125 mL), and the aq phase washed with additional CH_2Cl_2 (125 mL). The combined organic layers were dried (Na_2SO_4) and the solvent removed under reduced pressure. The resulting orange oil was purified by flash chromatography (silica gel, 5% MeOH/$CHCl_3$/1% AcOH), affording the product as a yellow foam; yield: 4.540 g, (75%) which was stable for >6 months at −10 °C in a foil-covered flask.

2.4.1.1.3 Other Nitrobenzyl-Based Groups

Most modifications of the Z(2-NO_2) and Nvoc protecting groups have involved the addition of substituents on the benzylic carbon. Such modifications improve the efficiency of the photocleavage by presumably accelerating the hydrogen abstraction by the photoexcited nitro group. Additionally, the ketone that is formed after cleavage is less reactive than the aldehyde that results from Z(2-NO_2) and Nvoc deprotection. Thus, additives are not always needed to obtain good yields in the photocleavage step.

Barltrop et al.[6] studied the photodeprotection of an N^{α}-2-nitrophenyl(phenyl)methoxycarbonyl (Ndmoc)-protected dipeptide and found that deblocking proceeded in 84% yield. Similarly, Patchornik et al.[7] prepared the N^{α}-bis(2-nitrophenyl)methoxycarbonyl (Dnboc) derivative of alanine and found that it could be deblocked in 95% yield without the addition of aldehyde trapping agents. Z(2-NO_2) and Nvoc gave 35 and 80% yields, respectively, under similar conditions. Dnboc also has an advantage over other α-substituted 2-nitrobenzyl groups in that it is nonchiral and thus diastereomers are not formed upon N-protection of amino acids. The 1-(6-nitro-1,3-benzodioxol-5-yl)ethoxycarbonyl (Menpoc) group has been studied for amine protection.[14] This group is photoremoved from amino acids four times faster than Nvoc (Scheme 4).

Scheme 4 The Ndmoc, Dnboc, and Menpoc Photocleavable Protecting Groups[6,14]

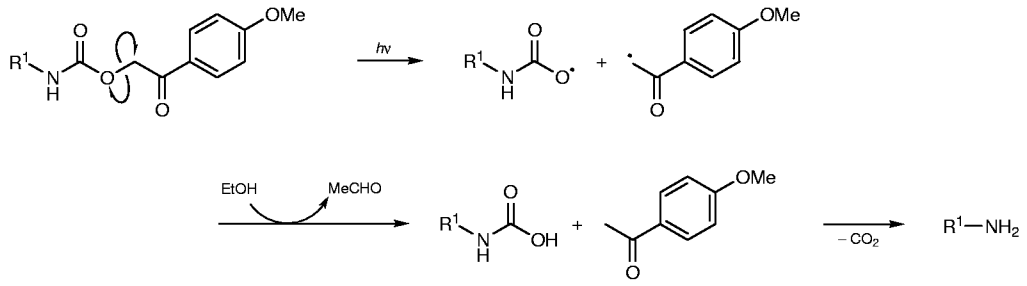


for references see p 293

2.4.1.2 Phenacyl-Based Groups

An alternative to nitrobenzyl groups for photocleavable deprotection are the phenacyl (2-oxo-2-phenylethyl) groups, first reported in 1973 by Sheehan and Umezawa.[15] In contrast to the 2-nitrobenzyl protecting groups, the choice of solvent for photocleavage of phenacyl groups is crucial because the solvent contributes to the reaction as a hydrogen donor. The cleavage of the phenacyl group is thought to take place via a radical mechanism (Scheme 5). Ethanol is the preferred solvent provided the phenacyl derivatives are soluble; if not, dioxane can be used, or even benzene provided the hydrogen radical donor cumene is present.

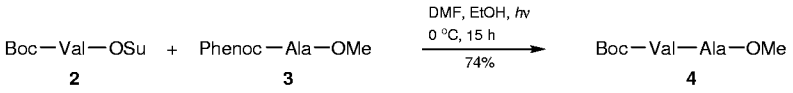
Scheme 5 Mechanism of Phenacyl Carbamate Photocleavage



2.4.1.2.1 4-Methoxyphenacyloxycarbonyl Group

The 4-methoxyphenacyloxycarbonyl (Phenoc) group was introduced by Church et al. for the protection of the α-amino group of amino acids.[16] It has been shown to be stable to both acid (50% TFA in CH_2Cl_2) and base (20% piperidine in DMF). The authors demonstrated photodeprotection under neutral conditions with yields ranging from 50 to 90%. Also demonstrated was an inventive use of the phenacyl group for the synthesis of a dipeptide. Using the Phenoc group for α-amino protection, an in situ deprotection and coupling was carried out (Scheme 6). The activated Boc-Val-OSu (**2**) and Phenoc-Ala-OMe (**3**) were mixed in DMF/EtOH at 0 °C and irradiated with a Rayonet RPR 208 lamp at 350 nm for 15 hours. The released free amine reacted with the activated amino acid ester, and the yield of isolated dipeptide **4** was 74%. This experiment is an excellent illustration of the orthogonality of photocleavable protecting groups.

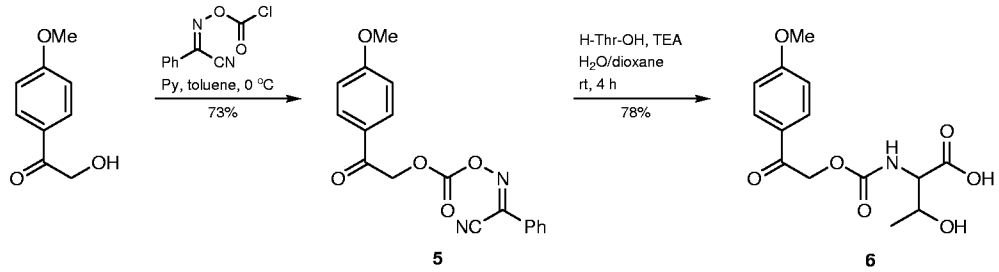
Scheme 6 In Situ Deprotection and Coupling Using 4-Methoxyphenacyloxycarbonyl Protection[16]



2 **3** **4**

Phenoc-protected amines were prepared in two ways. The first involved the reaction of an amino acid with the oxime carbonate **5** to yield **6** as shown in Scheme 7. The second method consisted of reaction of the α-hydroxy ketone with an isocyanate. Preparation of the phenacyl chloroformate was attempted, resulting in a cyclic carbonate which was inert to amines.

Scheme 7 Preparation of Phenoc-Protected Threonine[16]



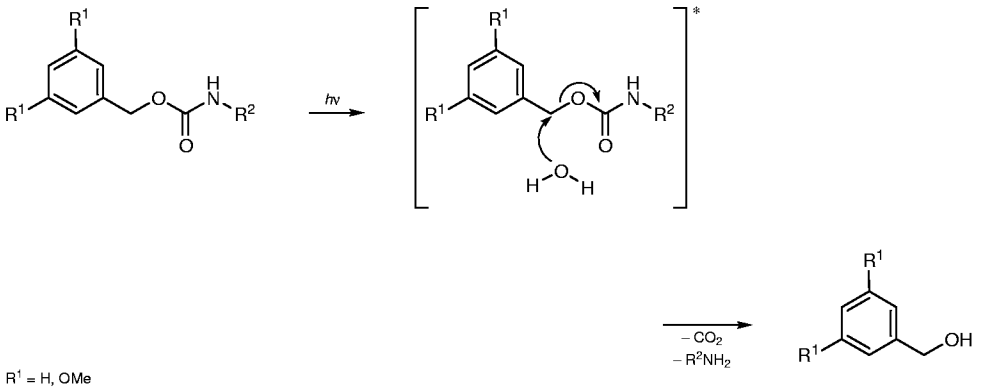
Phenoc-Thr-OH (6):[16]
A mixture of H-Thr-OH (2.00 g, 16.8 mmol) and the carbonate **5** (6.25 g, 18 mmol) was stirred in 30% aq dioxane (50 mL) at rt in the presence of TEA (3.50 mL, 25 mmol). After 4 h, the solvent was removed, the residue was diluted with H_2O (60 mL) and the mixture was extracted with EtOAc (2×20 mL). The aq layer was acidified with 5% citric acid and extracted twice with EtOAc. The combined extracts were worked up in the usual manner to afford a crude solid which was recrystallized (EtOAc/hexane) to give the product **6** as white crystals; yield: 4.08 g (78%); mp 128–130 °C.

Photoremoval of Phenoc Group; General Procedure:[16]
A soln of Phenoc-Thr-OH (3 mmol) in EtOH (100 mL) was irradiated under argon at 0 °C using a Rayonet RPR 208 photochemical reactor fitted with 300 nm lamps. TLC analysis of the mixture after 4 h showed a small amount of starting material. After 14 h, the soln was concentrated and solid residue triturated in Et_2O; yield of H-Thr-OH: 75%; mp 255–258 °C.

2.4.1.3 Benzyloxycarbonyl-Based Groups

Benzyloxycarbonyl protecting groups are designed for photolytic removal via solvolysis. Unlike nitrobenzyl-based groups, no highly reactive nitroso or aldehyde groups are formed, nor are free radicals. Instead, photoexcitation results in a highly activated species which is subject to solvolysis or even spontaneous heterolysis (Scheme 8). Thus, benzyloxycarbonyl deprotections are usually run in polar, protic solvents, except in the case of the highly active 2-(3,5-dimethoxyphenyl)prop-2-yloxycarbonyl group, which can be removed by photolysis in dry THF.

Scheme 8 Photoinduced Solvolysis of Benzyloxycarbonyl-Based Groups[3]



for references see p 293

2.4.1.3.1 Benzyloxycarbonyl Group

Barltrop and Schofield first reported the photolysis of an N^{α}-benzyloxycarbonyl-protected amino acid in 1962.[17,18] Irradiation of Z-Gly-OH at 254 nm in basic solution gave glycine in 75% yield, as well as the side-products phenethylamine and *N*-benzylglycine. Due to the sluggishness of its photocleavage, the benzyloxycarbonyl group can be preserved during the photosolvolysis of more active groups (see Section 2.4.1.3.2).

2.4.1.3.2 3,5-Dimethoxybenzyloxycarbonyl Group

Zimmerman and Sandel found that methoxy substituents accelerated the photo-solvolysis of benzyl acetates.[19,20] Based on this finding, Chamberlin designed the 3,5-dimethoxy-benzyloxycarbonyl group as a more active alternative to the benzyloxycarbonyl group.[21] Thus, irradiation of the glycine derivative in dioxane/water (1:1) gave glycine in 85% yield. Similar conditions caused only 10% removal of a benzyloxycarbonyl group. This difference in reactivity allowed the removal of a 3,5-dimethoxybenzyloxycarbonyl group from the α-amine of Lys in 75% yield without cleavage of a benzyloxycarbonyl group on the ε-amino group.

3,5-Dimethoxybenzyl 4-Nitrophenyl Carbonate:[21]
To an ice-cold soln of 3,5-dimethoxybenzyl alcohol (28.0 g, 0.167 mol) and pyridine (5.90 g, 0.150 mol) in acetone (100 mL) was added with stirring 4-nitrophenyl chloroformate (30.4 g, 0.150 mol). Immediately after the last portion had been added, the product began to precipitate. The mixture was added to H_2O (700 mL); the product was collected by filtration, washed several times with H_2O, and recrystallized (MeOH, 2500 mL); yield: 30.7 g (61%), mp 114–115 °C.

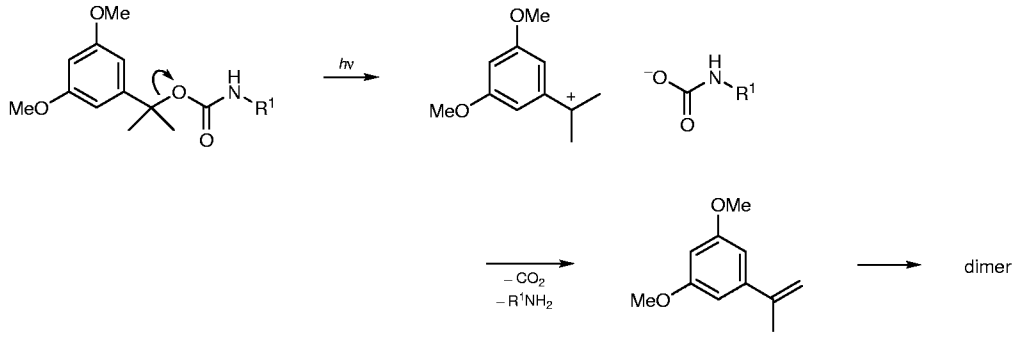
3,5-Dimethoxybenzyloxycarbonyl-D-Phg-OH:[21]
A mixture of H-D-Phg-OH (3.66 g, 0.024 mol), 3,5-dimethoxybenzyl 4-nitrophenyl carbonate (10.0 g, 0.030 mol), 2 M NaOH (24.0 mL), and THF (48 mL) was stirred at rt for 21 h. The THF was removed under reduced pressure, the precipitated solid was collected by filtration and extracted several times with 1 M $NaHCO_3$ soln. The pH of the combined filtrates was adjusted to 5.8 with 1 M HCl, and the resulting mixture was extracted three times with Et_2O to remove byproducts. The aq phase was then over-layered with Et_2O and the pH adjusted to 2.0. The Et_2O layer was separated and the aq layer was extracted twice more with Et_2O. The Et_2O extracts were combined, washed twice with H_2O and once with brine, and dried (Na_2SO_4). The solvent was removed under reduced pressure and the residue crystallized (MeOH/H_2O) to afford the product; yield: 6.40 g (77%); mp 130–132.5°C. Further recrystallization (MeOH/H_2O) raised the mp to 132.5–133.5 °C; $[\alpha]_D^{25}$ −100 (*c* 1.05, $CHCl_3$).

Irradiation of 3,5-Dimethoxybenzyloxycarbonyl-D-Phg-Gly-OH:[21]
A soln of 3,5-dimethoxybenzyloxycarbonyl-D-Phg-Gly-OH (1.00 g, 0.00249 mol) in dioxane/H_2O (1:1, 1 L) was irradiated with a Hanovia 654A-36 high-pressure Hg lamp housed in a water-cooled quartz immersion well fitted with a Vycor glass filter sleeve. After 1.5 h, the solvent was removed from the soln to dryness under reduced pressure and the residue treated with a minimal volume of acetone. Filtration afforded H-D-Phg-Gly-OH; yield: 0.335 g (65%); mp 226–228 °C; $[\alpha]_D^{25}$ −112 (*c*, 0.484 in 1 M HCl).

2.4.1.3.3 2-(3,5-Dimethoxyphenyl)prop-2-yloxycarbonyl Group

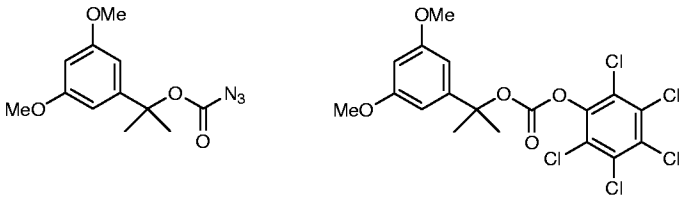
The 2-(3,5-dimethoxyphenyl)prop-2-yloxycarbonyl (Ddz) group was developed by Birr et al. as a more active analogue of the 3,5-dimethoxybenzyloxycarbonyl group.[22] In addition to being photolabile, the Ddz group can be removed by acidolysis, under much milder conditions than a Boc group (see Section 2.1.1.1.3.4.1). In practice, acid-removal of the Ddz group is much more common than photoremoval. One advantage of the Ddz group is its ability to heterolyse upon irradiation. Thus a polar, protic solvent is unnecessary, and photocleavage can be performed in THF. The styrene byproduct is dimerized almost quantitatively via a photochemical [2+2] cycloaddition (Scheme 9).

Scheme 9 Light-Induced Cleavage of the Ddz Group[22]



N^{α}-Ddz-protected amino acids are usually prepared by reacting the free amine with Ddz-N_3 (**7**). Sometimes the pentachlorophenyl mixed carbonate **8** is used (Scheme 10). N^{α}-Ddz amino acids are also available commercially. The Ddz group has been extensively used in solid-phase and automated peptide synthesis, however, usually being removed by acidolysis.

Scheme 10 Reagents for Preparation of N^{α}-Ddz-Protected Amino Acids



7 **8**

Synthesis of N^{α}-Ddz Amino Acids; General Procedure:[22]
The amino acid (230 mmol) was dissolved in trimethylbenzylammonium hydroxide soln (Triton B, 40% soln in MeOH 100 mL, 240 mmol) and MeOH (200 ml) at 40–60° (H_2O added if necessary). The soln was evaporated under reduced pressure and the residue dissolved twice in benzene (500 mL) and evaporated under reduced pressure again. With exclusion of moisture, the residue was dissolved in anhyd pyridine (500 mL, DMF added if necessary). To this stirred soln was added Ddz-N_3 (81 g, 305 mmol) dissolved in pyridine (200 mL) dropwise over 30 min. After stirring 48 h at 40 °C, the solvent was removed under reduced pressure. The residue was dissolved in 1 M NaOH (500 mL) and extracted with Et_2O (4×300 mL). Ice (200 g) was added to the alkaline aq layer, which was then acidified with 50% citric acid to pH 4 (ammonia evolution!) and extracted with CH_2Cl_2 (4×300 mL). The combined organic layers were washed three times with H_2O, dried (Na_2SO_4), and the solvent removed under reduced pressure. The residue was dissolved in a small amount of dry boiling Et_2O and DCHA (1.1 equiv) were added to the soln, from which the product crystallized as the DCHA salt on cooling.

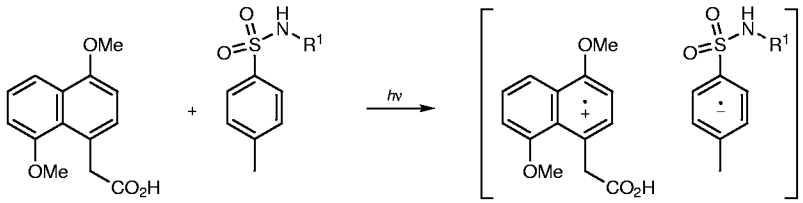
2.4.1.4 Photocleavage via Donor-Acceptor Ion Pairs

In general, photolysis of a protecting group is possible because the group is appropriately substituted with a functionality that renders a bond labile to ultraviolet energy. It is possible, however, to photosensitize certain ordinary protecting groups if specific types of molecules are present in the reaction mixture. The necessary ingredients for such a procedure include both a strong electron donor and a strong electron acceptor. Hamada et al. reported the application of photohydrolysis of the electron-withdrawing tosyl group to the deprotection of lysine peptides.[23] Classical deprotection of the tosyl group is carried out with sodium in liquid ammonia, which is often described as being a solution of solvated electrons. The authors showed that the solvated electrons could be generated under mild conditions by addition of a variety of electron-rich aromatic compounds. This photosensitizer, upon irra-

for references see p 293

diation, transfers an electron to the tosyl group, initially forming a radical ion pair (Scheme 11). The radical anion partner is identical with the known species generated in the initial step of sodium/liquid ammonia reduction. A reducing agent is required to complete the photochemically initiated deprotection.

Scheme 11 Formation of a Radical Ion Pair between a Photosensitizer and a Tosyl Group



2.4.1.4.1 Tosyl Group

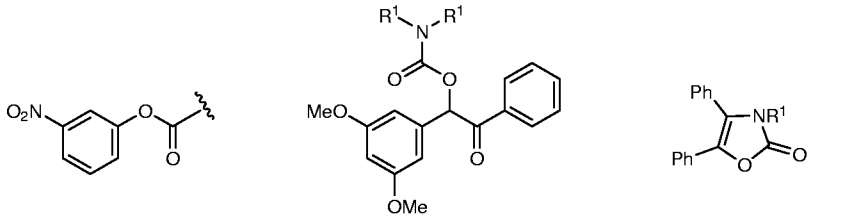
The tosyl group has been used for protection of side-chain amino as well as of α-amino groups (see Section 2.1.1.5). The resulting tosylamides are very stable functionalities, resistant to acid, base, and nucleophiles. They are usually prepared by reacting an amine with tosyl chloride in the presence of base. Traditional removal requires use of sodium in liquid ammonia, which can lead to unwanted side reactions. The photochemical electron transfer reaction discussed above allows for the mild removal of the tosyl protection. No cleavage of peptide bonds or Boc groups has been observed under these conditions.

Photolysis of Boc-Gly-Pro-Lys(Tos)-OMe:[23]
The tripeptide derivative Boc-Gly-Pro-Lys(Tos)-OMe (0.23 g, 0.5 mmol), 1,5-dimethoxynaphthalene (94 mg, 0.5 mmol), and ammonia-borane (155 mg, 5 mmol) in 83% aq EtOH (120 mL) were irradiated with a 300-W Hg lamp (Pyrex filter) for 2 h. After the solvent was removed under reduced pressure the residue was chromatographed on a silica gel column ($MeOH/H_2O/AcOH$, 40:20:1) to give Boc-Gly-Pro-Lys-OMe, which was converted into the benzyloxycarbonyl derivative in the usual way to yield Boc-Gly-Pro-Lys(Z)-OMe, colorless oil; yield 219 mg (80%).

2.4.1.5 Miscellaneous Amino Groups

Several photolabile amino protecting groups have been studied which do not fit into the four categories above (Scheme 12). These groups are less widely used and may not be as applicable as the groups described above. They are summarized below.

Scheme 12 Miscellaneous Amino Protecting Groups[24,25,28]



3-nitrophenyloxycarbonyl 3',5'-dimethoxybenzoin carbamate 4,5-diphenyl-1,3-oxazol-2(3*H*)-one

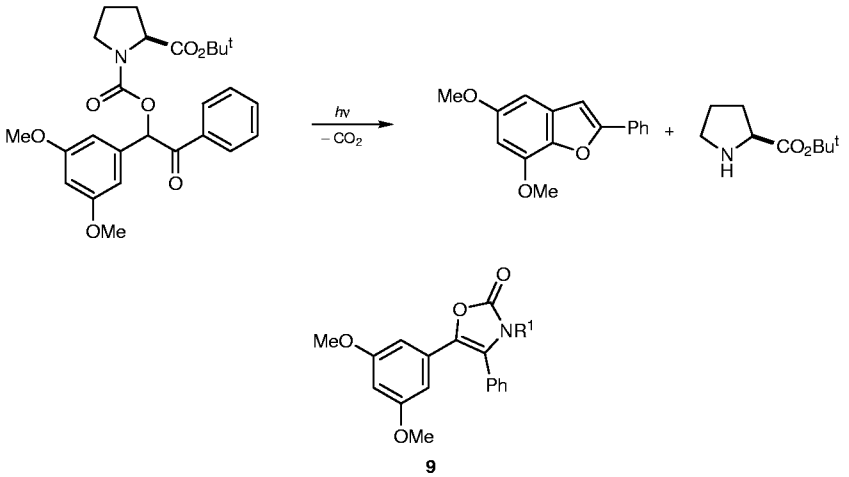
2.4.1.5.1 3-Nitrophenyloxycarbonyl Group

This group was developed by Wieland et al., based on a previous observation that 3-nitrophenyl phosphates were photolabile.[24] Like the benzyloxycarbonyl-based groups, the 3-nitrophenyloxycarbonyl group is cleaved by a photoinduced solvolysis, liberating CO_2 in the process. Photodeprotection of 3-nitrophenyloxycarbonyl amino acids has been found to be efficient. However, peptides suffer from some degree of hydantoin formation during the photolysis which lowers the yield.

2.4.1.5.2 Dimethoxybenzoin Carbamate

This group was developed for peptide chemistry by Pirrung and Huang,[25] and additionally studied by other authors.[26,27] It is based on the observation by Sheehan that benzoin esters cyclize on irradiation to produce a phenylbenzofuran and a carboxylic acid. The benzoin carbamates were designed for amine protection. While dimethoxybenzoincarbamate-protected secondary amines can be photodeprotected in yields ranging from 60–100%, primary amine derivatives are inert. This is due to the formation of a cyclic carbamate **9** that does not fragment on photolysis. Thus, this type of amine protection is applicable only to proline, an example of which is given by Pirrung and Huang (Scheme 13).[25]

Scheme 13 Photocyclization of Dimethoxybenzoin Carbamate Derivatives and Structure of the Inert Cyclic Carbamate 9[25]



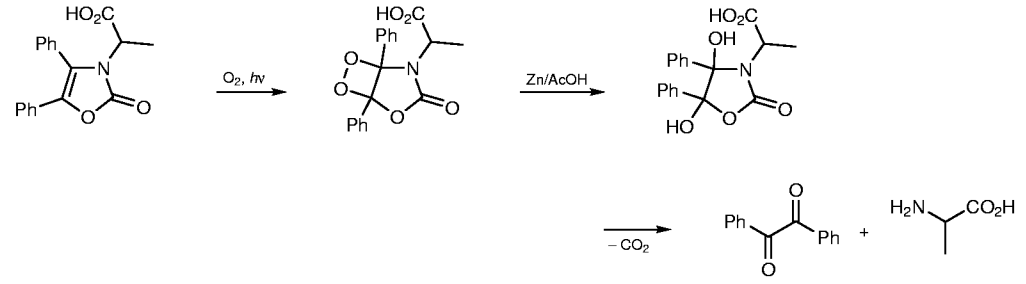
9

2.4.1.5.3 4,5-Diphenyl-1,3-oxazol-2(3*H*)-one

The 4,5-diphenyl-1,3-oxazol-2(3*H*)-one protection group was developed by Sheehan and Guziec.[28] The group is installed by reaction of the amino acid with either a benzoin urethane or a 4,5-diphenyl cyclic unsaturated carbonate. This type of N-protected amino acid is stable to a variety of vigorously basic and acidic conditions. The removal of the 4,5-diphenyl-1,3-oxazol-2(3*H*)-one group is affected by photolysis in the presence of air. This causes a photooxygenation of the oxazolinone, which, following in situ reduction, fragments to afford the unprotected amino acid. Alanine as model compound was thus recovered in nearly quantitative yield (Scheme 14).[29]

for references see p 293

Scheme 14 Mechanism of Photooxygenation and Cleavage of the 4,5-Diphenyl-1,3-oxazol-2(3*H*)-one Group



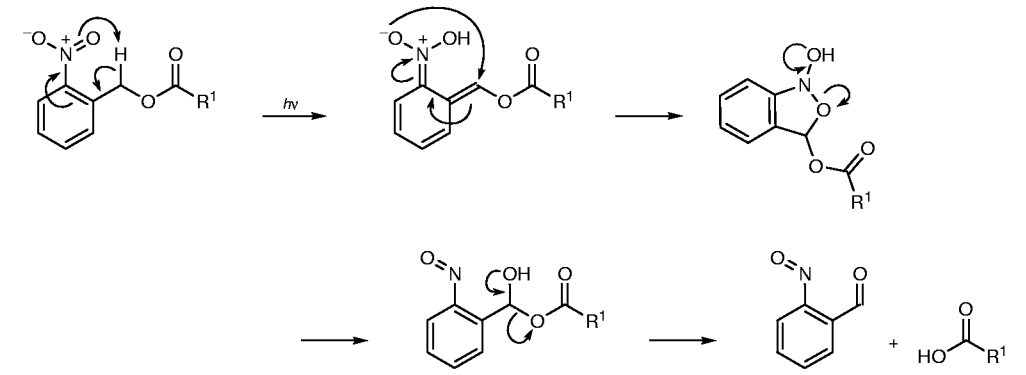
2.4.2 Carboxy Protecting Groups

In peptide chemistry, two groups are generally utilized as photocleavable carboxy protecting groups. The first is the 2-nitrobenzyl ester and its derivatives. The other is the phenacyl ester. These groups are closely related to their carbamates, which were discussed above in the context of amine protection. The mechanism of photocleavage is the same, except that instead of cleavage liberating an unstable carbamic acid, a carboxyl group is directly generated. Unlike the carbamates, the ester moiety is unstable to strongly nucleophilic conditions.

2.4.2.1 2-Nitrobenzyl-Based Groups

The mechanism for the photoremoval of a 2-nitrobenzyl ester [Bzl(2-NO_2)] is essentially identical to that discussed above for the Nboc group. Accordingly, the issue of light filtering by the azo-byproduct and the presence of reactive nitrosoaldehyde species is a concern. Despite these complications, photodeprotection of 2-nitrobenzyl esters often occurs with high yield (Scheme 15). 2-Nitrobenzyl esters have been used in side-chain as well as α-carboxy protection.

Scheme 15 Mechanism of Photocleavage of 2-Nitrobenzyl Esters



2.4.2.1.1 2-Nitrobenzyl Ester

The 2-nitrobenzyl ester [Bzl(2-NO_2)] has seen little use as a traditional peptide protecting group since its introduction by Barltrop et al.[6] However, it has been used as both a photolabile linker in solid-phase peptide synthesis[30] and as a caging group for biological studies.[31] In one such caging study, Pirrung and Nunn found that irradiation of a 2-nitrobenzyl ester in a phosphate buffer (pH 7) proceeded cleanly with a half-life of 6.9 minutes.[32] No aldehyde-trapping reagents were necessary.

H-Ser-O2Nbz·TosOH:[32]
A soln of H-Ser-OH (500 mg, 4.76 mmol), 2-nitrobenzyl alcohol (5.14 g, 23.8 mmol), and TosOH·H_2O (4.5 g, 23.6 mmol) in $CHCl_3$ (60 mL) was refluxed through a Soxhlet extractor containing molecular sieves for 18 h. The solvent was removed under reduced pressure and the title product precipitated with Et_2O; yield: 1.21 g (62%); $[\alpha]_D^{20}$ –7.3 (*c* 1.05, H_2O).

2.4.2.1.2 4,5-Dimethoxy-2-nitrobenzyl Ester

Like the 2-nitrobenzyl ester, the 4,5-dimethoxy-2-nitrobenzyl ester has attracted attention as a photolabile caging group. It has also been used for side-chain carboxy protection in the synthesis of misacylated transfer RNAs.[33] In this case, both the α- and side-chain carboxyls were first protected; a selective saponification of the α-ester afforded the side-chain protected material.

Bis(4,5-dimethoxyl-2-nitrobenzyl) *N*-(4-Pentenoyl)-L-aspartate:[33]
To a soln of dimethyl *N*-(4-pentenoyl)-L-aspartate (203 mg, 0.83 mmol) in THF (2 mL) was added 0.1 M NaOH (25 mL). After stirring at rt for 3 h, the mixture was acidified to pH 2 with 10% aq $NaHSO_4$ and extracted with EtOAc (3×30 mL). The combined organic extracts were dried (Na_2SO_4) and the solvent removed under reduced pressure. The residue was dissolved in DMF (2 mL) and CsF (429 mg, 2.82 mmol) was added, followed by 4,5-dimethoxy-2-nitrobenzyl bromide (276 mg, 1.0 mmol). The mixture was stirred at rt for 6 h, diluted with EtOAc (30 mL), and washed with H_2O (2×10 mL). The organic layer was dried (Na_2SO_4) and concentrated. The crude product was purified by column chromatography over silica gel; yield: 200 mg (40%); mp 167.5 – 168.5 °C.

2.4.2.1.3 Other 2-Nitrobenzyl-Based Groups

A variety of 2-nitrobenzyl esters containing α-substituents have been studied. Patchornik et al. showed that bis(2-nitrophenyl)methyl esters were effective protecting groups for the carboxy groups of amino acids.[7] Irradiation led to quantitative recovery of products without the use of additives. Barltrop et al. examined the 2-nitrophenyl(phenyl)methyl ester of *N*-phthalimidoglycine and found that irradiation led to deprotection in 75% yield.[6] 2-Nitrobenzyl esters containing carboxy substitution at the α-position have been used as caging groups for bioactive peptides.[34]

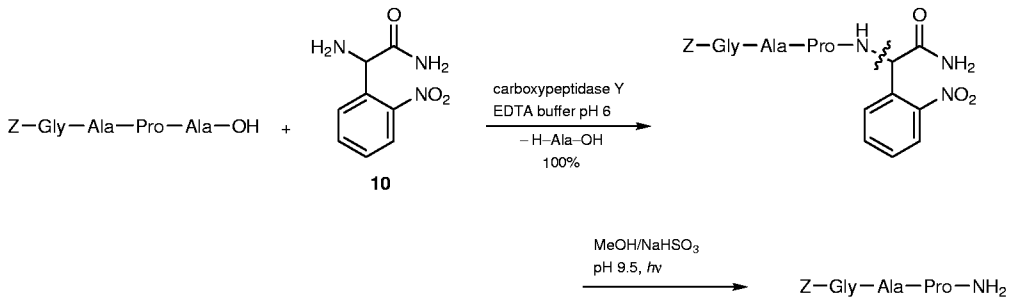
A novel use of the 2-nitrobenzyl group was reported by Henriksen et al.[35] The authors wished to prepare C-terminal peptide amides using an in vitro synthesis. Classical enzymatic methods using carboxypeptidases to add the C-terminal residue sometimes fail when amino acid amides are used as the substrate. The approach by the authors was to incorporate 2-nitrophenylglycine amide (**10**) as the terminal amino acid. Upon photolysis, the bond between the α-nitrogen and the α-carbon is cleaved, generating the peptide amide. The 2-nitrophenylglycine amide group was cleverly introduced in a transpeptidation reaction catalyzed by carboxypeptidase Y (Scheme 16).

Alanine was initially incorporated in the C-terminal position as a dummy amino acid to act as a substrate for the enzyme. The transacylation was carried out in EDTA buffer at pH 6.0 and the yield was essentially quantitative. Photolysis was carried out in methanol solution containing aqueous sodium bisulfite at pH 9.5 at 20 cm from a mercury SP 200 lamp filtered

for references see p 293

through Pyrex. The yields were in excess of 85%. This procedure was carried out on peptides as large as the calcitonin fragment 9–32. Trypsin can also be utilized as the catalyst.

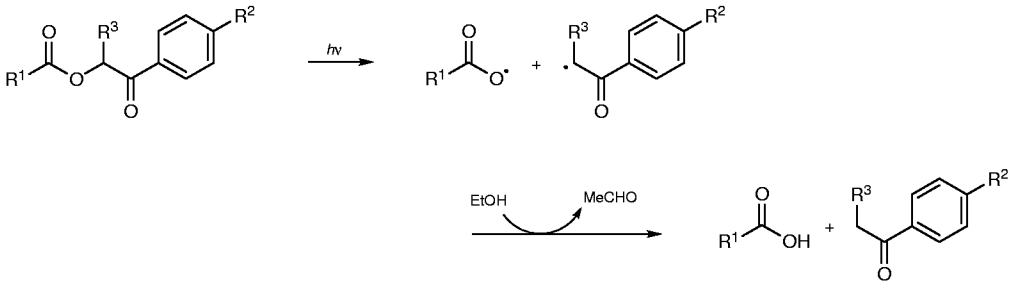
Scheme 16 Synthesis of Z-Gly-Ala-Pro-NH_2 via Enzymatic 2-Nitrophenylglycine Incorporation and Subsequent Photocleavage[35]



2.4.2.2 Phenacyl-Based Esters

The general mechanism of the light-induced fragmentation of phenacyl groups has been discussed in Section 2.4.1.2. In the case of phenacyl esters, irradiation promotes bond homolysis, which directly produces the deprotected carboxy group as a radical. H-atom abstraction from the solvent affords the product (Scheme 17). To date, phenacyl esters have been used only for α-carboxy protection.

Scheme 17 Mechanism of Phenacyl Ester Photocleavage



4-methoxyphenacyl: R^2 = OMe, R^3 = H
α-methylphenacyl: R^2 = H, R^3 = Me

2.4.2.2.1 4-Methoxyphenacyl Ester

4-Methoxyphenacyl esters were first described by Sheehan and Umezawa in 1973.[15] They studied the deprotection of a series of amino acids and found that the best yields were obtained when the photolysis was carried out in ethanol. The 4-methoxyphenacyl esters were formed by the reaction of the free acid with 4-methoxyphenacyl bromide in DMF containing triethylamine. Photolyses were carried out with a 450-W Hanovia mercury immersion lamp fitted with a Pyrex filter. The 4-methoxyphenacyl group was used in a synthesis of thymopentin analogues.[36] The authors used zinc dust as the standard deprotection procedure for cleavage of phenacyl esters (see Section 2.2).

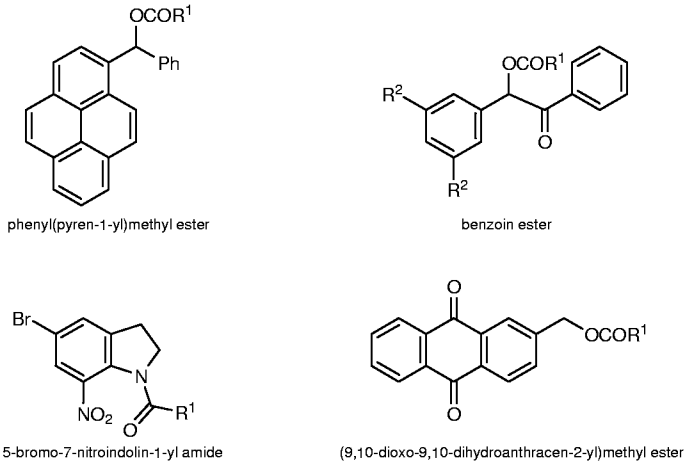
2.4.2.2.2 α-Methylphenacyl Ester

This group was also developed by Sheehan and Umezawa as an alternative to the 4-methoxyphenacyl group, and both have comparable activity.[15] The deprotection of dipeptide α-methylphenacyl esters proceeded in higher yield than with the corresponding 4-methoxyphenacyl esters. In the case of the α-methylphenacyl ester, dioxane was shown to be the preferred solvent for photolysis. The procedures for preparation and photolysis of the α-methylphenacyl ester are otherwise identical to those for the 4-methoxyphenacyl ester.

2.4.2.3 Miscellaneous Carboxy Groups

Other carboxy protecting groups which have been reported but have not gained widespread use are summarized in Scheme 18.

Scheme 18 Miscellaneous Carboxy Protecting Groups[37,38,41,42]



2.4.2.3.1 Phenyl(pyren-1-yl)methyl Esters

Diazophenyl(pyren-1-yl)methane was developed by Iwamura et al. for the esterification of carboxy groups in peptide synthesis.[37] The reagent is a stable red solid that reacts with amino acid 4-toluenesulfonate salts to form the phenyl(pyren-1-yl)methyl esters. The mechanism of ester formation is analogous to that of diazomethane. The resulting esters are fluorescent, unstable to catalytic hydrogenation, and removable by irradiation with UV light. The authors reported deprotection yields ranging from 68 to 83% for related protected dipeptides.

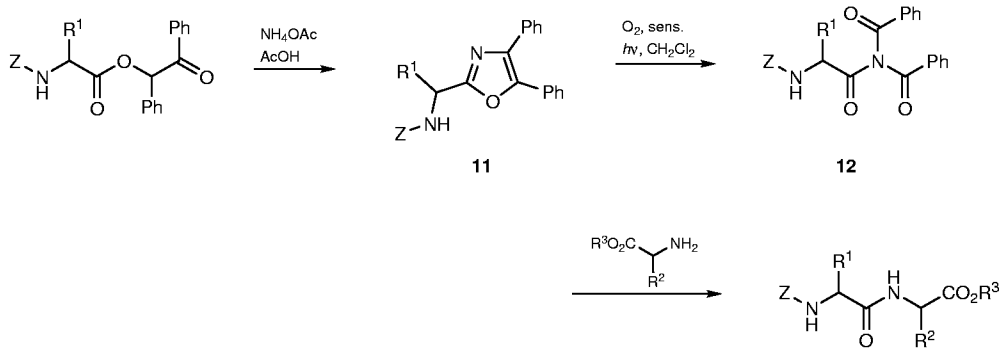
Amino Acid Phenyl(pyren-1-yl)methyl Esters; General Procedure:[37]
To an amino acid 4-toluenesulfonate salt (1.0–1.5 mmol) in DMF (5 mL) at 50 °C, a DMF soln of diazophenyl(pyren-1-yl)methane (1.5 equiv) was added dropwise with stirring. After the addition, stirring was continued at 50 °C until the red color almost disappeared and then the solvent was removed under reduced pressure. Addition of MeCN to the residue gave a precipitate of crude ester which was collected and purified by washing with MeCN and H_2O; yields: 18–88%.

for references see p 293

2.4.2.3.2 Benzoin Esters

Benzoin esters were first studied as carboxy protecting groups by Sheehan et al.[38] The mechanism of photocleavage is identical to that discussed in the section describing benzoin carbamates for amine protection. The authors compared several substituted benzoin esters and found that 3′,5′-dimethoxybenzoin esters of Phth-Gly-OH gave the best yields. Unsubstituted benzoin esters have also been used as caging groups.[39] A novel use of a benzoin ester was reported by Wasserman and Lu.[40] A protected amino acid benzoin ester was reacted with ammonium acetate to form the diphenyloxazole **11**. This was photooxygenated to yield, after rearrangement, the triamide **12**, which reacted with another amino acid to create a dipeptide. This deprotection/activation procedure was utilized to synthesize a variety of dipeptides (Scheme 19).

Scheme 19 Deprotection/Coupling of a Benzoin Ester[40]



3′,5′-Dimethoxybenzoin Phthaloyl Glycinate:[38]

CAUTION: *Benzene is carcinogenic!*

To a soln of 3′,5′-dimethoxybenzoin (1.5 g, 5.52 mmol) and pyridine (0.638 g, 8.08 mmol) in CH_2Cl_2 (25 mL) at 0 °C was added over a period of 15 min with shaking a soln of Phth-Gly-Cl (1.8 g, 8.07 mmol) in CH_2Cl_2 (25 mL). The soln was allowed to warm to rt, and stand overnight. The mixture was then diluted with CH_2Cl_2, washed with H_2O (25 mL), dil HCl (2×25 mL), H_2O (3×15 mL), and dried. The solvent was removed under reduced pressure and the residue recrystallized (benzene/MeOH) to afford the title compound; yield: 2.21 g (87%); mp 181–182 °C.

Photolysis of 3′,5′-Dimethoxybenzoin Phthaloyl Glycinate:[38]

CAUTION: *Benzene is carcinogenic!*

The title ester (0.487 g, 1.05 mmol) was irradiated in benzene soln for about 45 min. Removal of the solvent afforded a crystalline residue which after several washings with benzene gave Phth-Gly-OH as the insoluble crystalline material; yield: 0.187 g (87%). Recrystallization (MeOH/benzene) yielded pure material: mp and mixed mp 195–198 °C.

2.4.2.3.3 5-Bromo-7-nitro-1-indolinyl Group

The 5-bromo-7-nitro-1-indolinyl group was applied to peptide chemistry by Pass et al. as both a protecting and activating group for the carboxy moiety.[41] These amides are stable to nucleophiles under normal conditions. However, irradiation at 420 nm activates the group towards nucleophiles such as amines, and allows for photoinduced peptide coupling reactions. The 5-bromo-7-nitro-1-indolinyl amides are prepared by reacting the substituted indoline with an Fmoc-protected amino acid in the presence of thionyl chloride.

2.4.2.3.4 (9,10-Dioxo-9,10-dihydroanthracen-2-yl)methyl Ester

Kemp and Reczek developed the (9,10-dioxo-9,10-dihydroanthracen-2-yl)methyl ester as a photocleavable protecting group.[42] The ester was prepared by conventional esterification techniques using DCC and HOBt. This ester is stable to 100% TFA, HCl in CH_2Cl_2, and triethylamine in dioxane/water. It reacts slowly with HBr in acetic acid. The ester group was designed for mild removal under reductive conditions. The dihydroanthroquinone intermediate rapidly fragments by a push–pull mechanism. It was found that the ester could also be removed under photochemical conditions (350 nm in propan-2-ol containing 4 equiv *N*-methylmorpholine, 99%). However, the removal method of choice was determined to be nonphotochemical, using a specially prepared reducing resin.

2.4.3 Side-Chain Protecting Groups

Amino acids contain a variety of reactive functional groups on their side chains that can interfere with peptide synthesis. Therefore, the protection of side chains is a crucial element to a successful synthetic strategy. The amino acids are discussed individually below.

2.4.3.1 Lysine

The protection of the side-chain amino group of lysine can be accomplished with any of the groups discussed in the section pertaining to amino protection. A procedure for the ε-Nvoc protection of Boc-Lys is given in Section 2.4.1.1.2.[10]

2.4.3.2 Arginine

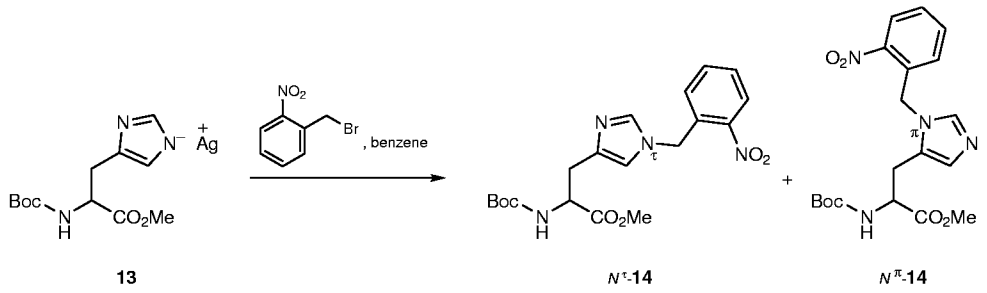
There are no examples in the literature of light-induced deprotection of the guanidine moiety of arginine. The tosyl group and other aryl sulfonamides are commonly used. It is conceivable that the photochemical donor–acceptor protocol developed by Hamada et al.[23] (*vide supra*) could be utilized to affect photodeprotection of arginine arylsulfonamides.

2.4.3.3 Histidine

The imidazole on the histidine side chain may be protected as a 2-nitrobenzyl derivative which leads to a mixture of N^{π}- and N^{τ}-derivatives. Generally, the τ-derivative dominates, and can often be isolated pure by recrystallization. The group is installed by reaction of the silver salt of Boc-His-OMe (**13**) with 2-nitrobenzyl bromide in refluxing benzene (Scheme 20).[43] The product **14** was immediately saponified and, after acidification, extracted into ethyl acetate to give the acid in yield 60–70%. Photolysis of this material was affected by irradiating a 0.001 M solution in dioxane for 1 hour with Pyrex-filtered UV light from a mercury vapor lamp to give Boc-His-OH in quantitative yield.

for references see p 293

Scheme 20 Formation of Side-Chain Protected Histidine as a Mixture of N^{π}- and N^{τ}-Derivatives[43]



2.4.3.4 Tyrosine

For the photocleavable protection of the hydroxy group of tyrosine the 2-nitrobenzyl ether has been investigated.[44] It is completely analogous to the 2-nitrobenzyl carbamates and esters discussed previously, and the mechanism of its cleavage is identical. The 2-nitrobenzyl ether is stable to both base (1 M NaOH) and acid (45% HBr in acetic acid). As with the previously discussed groups, aldehyde trapping agents increase the yields of deprotection. The 2-nitrobenzyl group has also been used as a photolabile caging agent for tyrosine-containing peptides.[45]

Z-Tyr(2Nbz)-OMe:[44]
A soln of Z-Tyr-OMe (2.5 g, 7.6 mmol) in dry MeOH (35 mL) containing NaOMe (9 mmol) was stirred for 15 min at rt, then the MeOH was removed under reduced pressure. The oily residue was dissolved in dry DMF (5 mL) and the solvent was removed under reduced pressure again. The product was dissolved in dry DMF (15 mL) and a soln of 2-nitrobenzyl chloride (1.7 g, 9.9 mmol) in dry DMF (10 mL) was added. The yellowish soln immediately turned violet and NaCl precipitated. The mixture was stirred for 15 min at rt, then the DMF removed and the residual oil dissolved in EtOAc (30 mL). The soln was washed with H_2O, dried (Na_2SO_4), and the solvent removed under reduced pressure. In order to remove unreacted 2-nitrobenzyl chloride, the oily product was extracted three times with boiling petroleum ether under reflux. The insoluble residue was recrystallized (Et_2O) to afford the title compound; yield: 2.82 g (80%); mp 96–98 °C; $[\alpha]_D^{23}$ +0.5 (*c* 1.05, DMF).

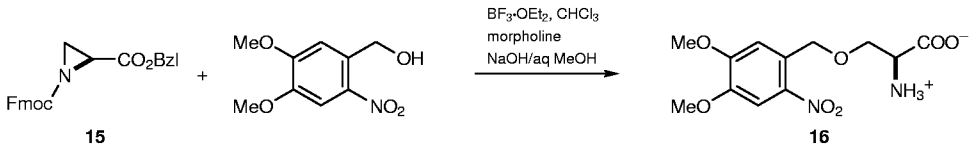
2.4.3.5 Aspartic Acid and Glutamic Acid

Any of the carboxy protecting groups discussed above are applicable to the side-chain protection of aspartic and glutamic acids. A procedure for the preparation of a di(4,5-dimethoxy-2-nitrobenzyl) aspartate was given in Section 2.4.2.1.2. Selective saponification with LiOH leads to the side-chain protected product.[33]

2.4.3.6 Serine and Threonine

The 4,5-dimethoxy-2-nitrobenzyl ether derivative of serine has been prepared by Pirrung and Nunn and its photolysis studied.[32] The side-chain protected serine **16** was prepared by Lewis acid catalyzed opening of benzyl (*S*)-*N*-Fmoc-aziridine-2-carboxylate (**15**) with 4,5-dimethoxy-2-nitrobenzyl alcohol. Removal of the Fmoc group and saponification afford the side-chain protected serine **16** (Scheme 21). The authors found that the photolysis reaction proceeded most efficiently in a bisulfite buffer, presumably due to trapping of the 4,5-dimethoxybenzaldehyde as its bisulfite adduct. Combs et al. have patented a method for incorporating 2-nitrobenzyl ethers of Ser or Thr into proteins via unnaturally acylated tRNAs.[46] The protecting groups are removed by the normal photolytic procedure. No other reports of photolabile protecting groups for the threonine side chain have been reported.

Scheme 21 Synthesis of Benzyl *N*-Fmoc-*O*-(4,5-Dimethoxy-2-nitrobenzyl)-L-serinate (**16**)[32]



Benzyl *N*-Fmoc-*O*-(4,5-Dimethoxy-2-nitrobenzyl)-L-serinate (16):[32]
To a soln of benzyl (*S*)-*N*-Fmoc-aziridine-2-carboxylate (**15**; 899 mg, 2.25 mmol) and 4,5-dimethoxy-2-nitrobenzyl alcohol (1.44 g, 6.75 mmol) in dry, EtOH-free $CHCl_3$ (40 mL) at 5 °C was added a soln of BF_3·OEt_2 (25 μL) in $CHCl_3$ (1 mL). The soln was stirred for 3 h and then quenched with aq $NaHCO_3$. The layers were separated, and the aq layer extracted twice with CH_2Cl_2. The combined organic layers were washed with brine, dried (Na_2SO_4), and filtered. Silica gel (5 g) was added and the solvents were removed under reduced pressure. The mixture was loaded directly onto the top of a silica gel column and eluted (hexane/EtOAc 8:2) to give the title product as a pale yellow oil; yield: 703 mg (51%); $[\alpha]_D^{20}$ −8.2 (*c* 0.73, $CDCl_3$).

2.4.3.7 Cysteine

Photoremovable 2-nitrobenzyl-based groups have been used for both the protection and caging of Cys and Cys-containing peptides. Hazum et al. reported the use of the *S*-2-nitrobenzyl derivative of cysteine.[47] The protecting group was stable to most peptide synthesis conditions and removed by the usual photolytic procedure. Additives were needed to obtain high yields (*vide supra*). A cysteine-containing tetrapeptide was prepared using this method. Marriott and Heidecker have reported the use of the 4,5-dimethoxy-2-nitrobenzyl group as a cysteine caging group in heavy meromyosin.[48] The caged peptide was synthesized by exposing heavy meromyosin to 4,5-dimethoxy-2-nitrobenzyl bromide in a pH 7.4 buffer.

H-Cys(2Nbz)-OH:[47]
To a soln of H-Cys-OH·HCl (1.57 g, 10 mmol) and Bzl(2-NO_2)-Cl chloride (1.88 g, 11 mmol) in H_2O/MeOH (1:1, 50 mL,) cooled to 4 °C, TEA (3 mL, 21.6 mmol) was added dropwise with stirring over 5–10 min. The mixture was vigorously stirred at 4 °C for 2 h and the precipitated product was then filtered, washed successively with H_2O and EtOAc and dried; yield: 2.05 g (80%), mp 188–192°C (dec.).

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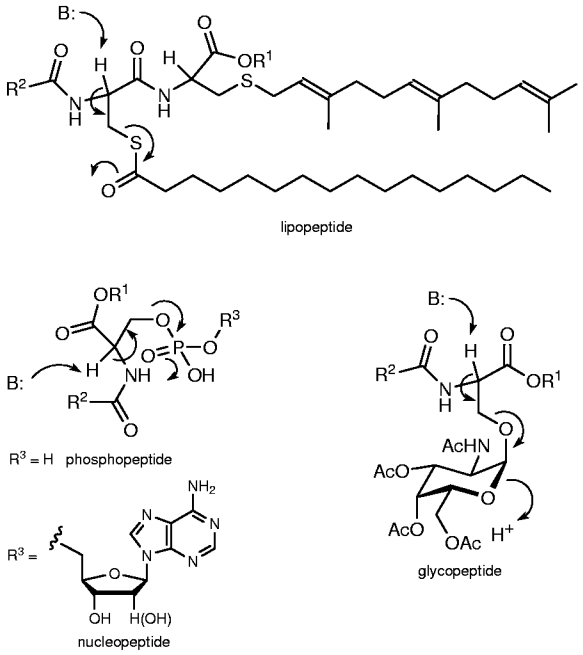
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2.5 Enzymatic Cleavable Protecting Groups

H. WALDMANN and T. PATHAK

The selective protection and deprotection of the α-amino group, the carboxy group, and the various side-chain functional groups of polyfunctional amino acids constitute some of the most fundamental synthetic problems in peptide chemistry. Although there are numerous classical chemical methods available for the manipulation of protecting groups under mild conditions,[1–3] the synthesis of complex polyfunctional molecules presents difficulties using chemical methods alone. For example, in the construction of biologically important glyco-, phospho-, nucleo-, and lipopeptides, the hydroxy groups of the carbohydrates, the amino groups of the amino acids and the nucleotides, the carboxy group and the side-chain functional groups of the amino acids and the internucleotide phosphates all have to be masked with blocking functions that are orthogonally stable to each other, i.e. functions that can be removed selectively leaving the other protecting groups intact. This synthetic challenge is further enhanced by the pronounced chemical lability of the peptide conjugates (Scheme 1).

Scheme 1 Problems Associated with the Chemical Synthesis of Peptide Conjugates



Even under weakly basic conditions (pH >9) the entire fatty acid, carbohydrate, nucleotide, and phosphate parts can be cleaved from these compounds. While under acidic conditions an attack on the *N*- or *O*-glycosidic bonds in nucleopeptides and glycopeptides may occur, resulting in anomerization and even a complete deglycosidation.

Since the mid-1970s, a systematic search for enzyme-labile blocking groups has been carried out. In addition to their stereo-discriminating properties, enzymes offer the opportunity to carry out highly chemo- and regioselective transformations. Enzymes often operate at neutral, weakly acidic, or weakly basic pH values and in many cases combine a high selectivity for the reactions they catalyze and the structures they recognize with a broad substrate tolerance. The use of enzyme-labile protecting groups in combination with classical chemical methods has substantially enriched the arsenal of protecting-group chemistry.[3–7]

for references see p 313

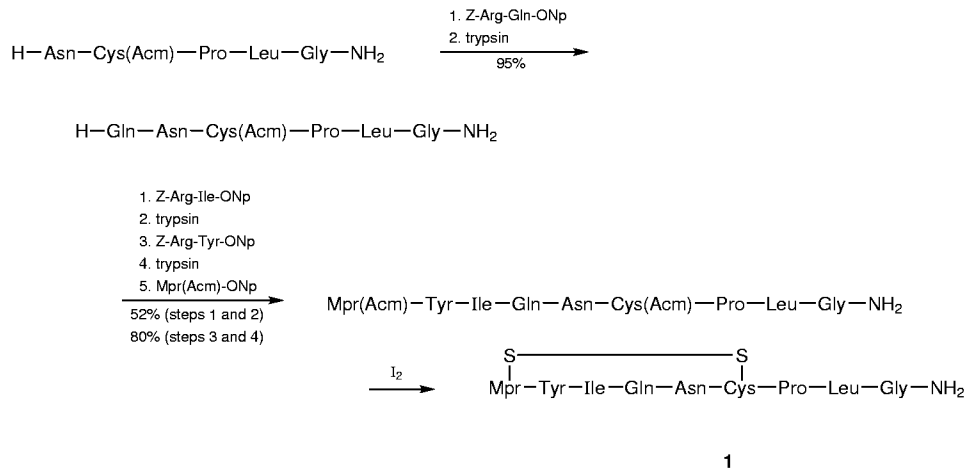
For peptide synthesis, enzyme-labile protecting groups are particularly useful for the construction and manipulation of larger peptide units, i.e. for transformations which, for solubility reasons, have to be carried out in aqueous systems. There are also applications for this technique in the reprocessing of peptides obtained by recombinant DNA technology.

2.5.1 Amino Group Protection

Initial attempts to introduce an enzyme-labile amino protecting group involved the use of chymotrypsin for the removal of *N*-benzoylphenylalanine (Bz-Phe) from the tripeptide Bz-Phe-Leu-Leu-OH[8] to generate H-Leu-Leu-OH. However, chymotrypsin is an endopeptidase of rather broad substrate tolerance, catalyzing the hydrolysis of peptide bonds on the carboxy group of hydrophobic and aromatic amino acid residues. Since such amino acids appear widely in peptides and no method is available to protect them against attack by the enzyme during the attempted deprotection, the use of chymotrypsin is problematic; its uses are limited to special cases.[9]

A protease with a much narrower specificity is trypsin, which catalyzes the hydrolysis of peptide bonds at the carboxy group of lysine and arginine. These amino acids carry polar, chemically reactive side-chain functional groups, which can be masked by various techniques.[10] The narrow specificity of trypsin together with the possibility of hiding the critical amino acids which function as primary points of tryptic cleavage has allowed the development of a general method for the protection of the N-terminus of peptides.[11,12–14] In several studies the application of trypsin-labile protecting groups, along with suitable protecting groups for the side chains of arginine and lysine are described.[12–18] Thus, for instance Z-Arg-OH served as an enzymatically removable blocking group in a stepwise synthesis of deamino-oxytocin (**1**) (Scheme 2).[13,14] Unfortunately, the preparation of the protected arginine 4-nitrophenyl esters is difficult, thus preventing this method from becoming generally useful for the stepwise assembly of larger peptides.

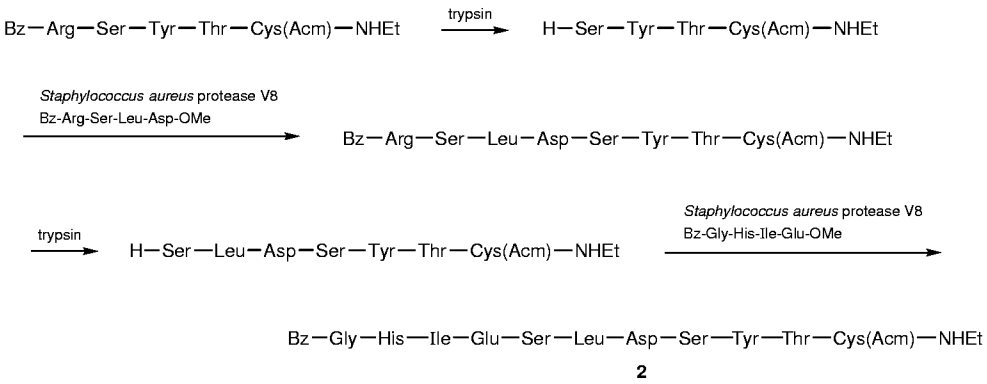
Scheme 2 The Trypsin-Assisted Synthesis of Deamino-oxytocin[13,14]



1

However, the trypsin-labile blocking groups have proved to be very useful for the construction of oligo- and polypeptides via condensation of preformed peptide fragments. An illustrative example is the chemoenzymatic construction of the 21–31 fragment of murine epidermal growth factor **2** (Scheme 3). In the course of this synthesis deblocking by trypsin is applied twice.[9]

Scheme 3 The Chemoenzymatic Synthesis of the 21–31 Fragment of Murine Epidermal Growth Factor[9]



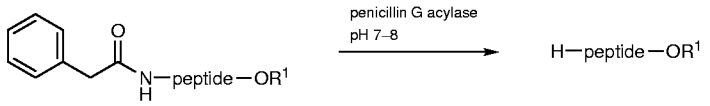
The enzyme first liberated the N-terminus of a tetrapeptide and subsequently of a heptapeptide. In a synthesis[19] of human β-lipotropin, an Ac-Arg residue is introduced by the solid-phase technique at the N-terminus of the 29 C-terminal amino acids of the desired polypeptide. After cleavage from the resin and protection of the side-chain functional groups with citraconic acid, the arginine moiety is removed with trypsin, leaving the peptide chain intact. Finally, coupling of the citraconylated 61–89 fragment, thus obtained, to the partially protected 1–60 segment and subsequent deprotection delivered β-lipotropin, i.e. a polypeptide consisting of 89 amino acids. Further examples are found in syntheses of oxypressin,[11] Met-enkephalin[20] and Glu^4-oxytocin.[11] In addition to chymotrypsin and trypsin, the collagenase from *Clostridium histolyticum* has been proposed as a catalyst for the removal of N-terminally attached dummy amino acids from peptides.[21] The enzyme recognizes the tetrapeptide Pro-X-Gly-Pro and splits the X-Gly bond. The use of this biocatalyst permitted the construction of des-pyroglutamyl-[15-Leu] human little gastrin I by selective hydrolysis of the dipeptide Pz-Pro-Leu [Pz = 4-(phenyldiazenyl)benzyloxycarbonyl] from the N-terminus of the octadecapeptide Pz-Pro-Leu-Gly-Pro-Trp-Leu-(Glu)$_5$-Ala-Tyr-Gly-Trp-Leu-Asp-Phe-NH_2.

2.5.1.1 Phenylacetic Acid Amide Group

Despite the impressive syntheses that have been realized using proteases, the use of these enzymes is always accompanied by the danger of a competitive (and sometimes unexpected) cleavage of the peptide backbone at an undesired site. This disadvantage can be overcome if a biocatalyst devoid of peptidase activity is used for the liberation of the N-terminal amino group. This idea has been realized by applying penicillin G acylase (PGA) from *Escherichia coli*.[22–36] This enzyme attacks phenylacetic acid amides and esters but does not hydrolyze peptide bonds. The acylase accepts a broad range of protected dipeptides as substrates and selectively liberates the N-terminal amino group under almost neutral conditions (pH 7–8, rt), leaving the amide bonds as well as the C-terminal methyl, allyl, benzyl, and *tert*-butyl esters unaffected (Scheme 4).[23–30,33] Considerable amounts of organic cosolvents are tolerated in this transformation.

for references see p 313

Scheme 4 The Penicillin G Acylase Assisted Selective Removal of a Phenylacetyl Group[23–30,33]

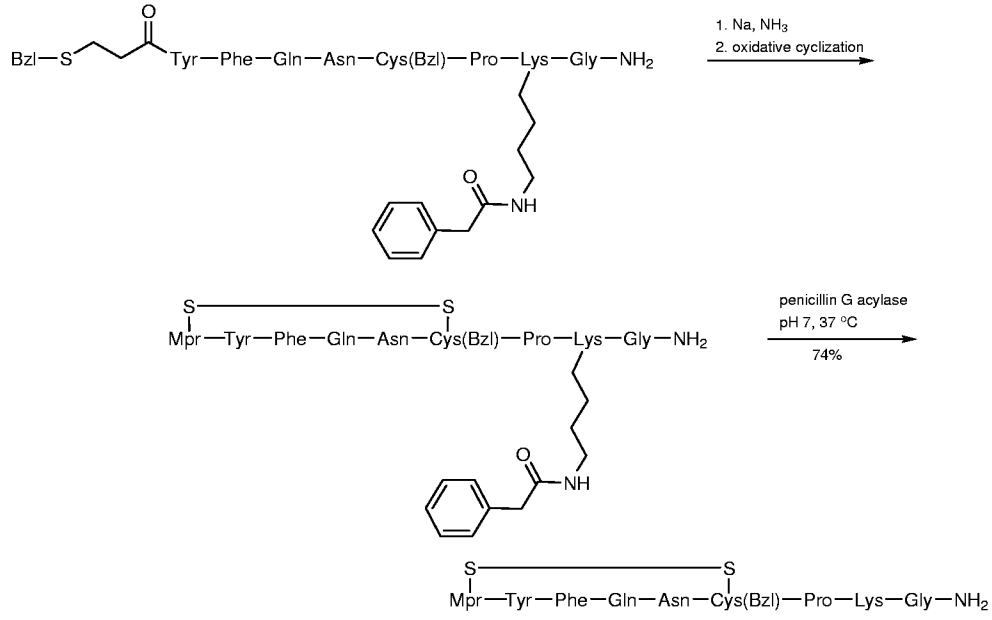


Peptide	R^1	Yield (%)
Gly-Phe	tBu	65
Thr-Ala	tBu	75
Ala-Ala	tBu	90
Ser-Leu	tBu	95
Asp(OH)-Phe	Me	87

The phenylacetyl group is easily introduced into amino acids by chemical[37] or enzymatic[33,38] methods. It is stable during the alkaline saponification of methyl esters, the hydrogenolysis of benzyl esters, the acidolysis of *tert*-butyl esters, and the Pd(0)- or Rh(I)-catalyzed removal of allyl esters. If the construction of phenylacetyl dipeptides is carried out by chemical activation of the phenylacetyl amino acids, the application of the non-urethane blocking group results in approximately 6% racemization.[24,25] However, this disadvantage can be overcome by forming the peptide bonds enzymatically, e.g. with trypsin,[39] chymotrypsin,[39] or carboxypeptidase Y (CpY).[34,39] For these condensation reactions and the subsequent enzymatic removal of the Ph(Ac) group, a continuous process has been developed which has the potential to be used for large-scale reactions.[34]

The applicability of PGA-catalyzed deprotection for the construction of larger peptides has been demonstrated in the complete deprotection of porcine insulin carrying three phenylacetyl groups[22] at the N-terminus of the glycine of the A chain, the N-terminal phenylalanine of the B chain, and the side chain of the lysine in position 29 of the B chain, respectively. The enzymatic hydrolysis proceeds to completeness and the peptide backbone is not attacked. A further interesting example is the biocatalyzed synthesis of Leu-enkephalin *tert*-butyl ester[33] in which all critical steps are performed by enzymes: (1) phenylacetyl groups are introduced for N-terminal protection of the amino acid esters by using PGA, (2) the elongation of the peptide chain is carried out with papain or α-chymotrypsin, and (3) the deprotection of the N-terminal amino group is achieved again by means of PGA. These examples and also the application of this technique for the synthesis of aspartame[23,35,36] and glutathione derivatives[30] demonstrate that PGA can advantageously be used for the N-terminal unmasking of peptides. If trypsin is used for the construction of the peptide bonds or N-terminal deprotection during chemical synthesis of peptides, the side-chain amino group of lysine generally has to be protected to prevent side reactions.[10] This is achieved using the PGA-labile phenylacetyl group.[40] Thus, the first application of the phenylacetyl group in peptide chemistry was reported in the synthesis of 1-deamino-Lys^8-vasopressin (**3**), where the lysine side chain was masked as the phenylacetamide (Scheme 5).

Scheme 5 The Penicillin G Acylase Mediated Deprotection of a Phenylacetyl Group in the Synthesis of 1-Deamino-Lys^8-vasopressin[40]

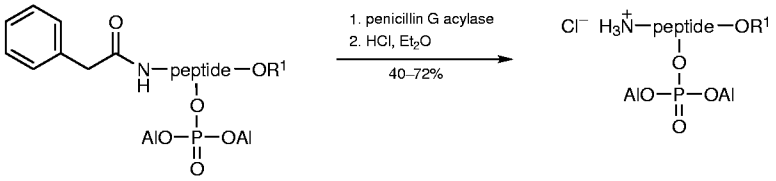


3

After peptide chain extension and disulfide bond formation by oxidative cyclization, the phenylacetyl group could be removed enzymatically in 74% yield. The technique has been applied further in the synthesis of even larger peptides, for example in the deprotection of tris(phenylacetamido)porcine insulin.[22] Since PGA is commercially available and devoid of any peptidase activity, this method appears to be generally useful for the construction of lysine-containing oligopeptides.

Enzymatic protecting-group techniques have recently been used[41,42] in the synthesis of phosphopeptides, in which the phenylacetamide group is removed from the respective phenylacetyl amides by means of PGA. In the course of the ensuing enzymatic transformations only the phenylacetamide-blocking group is attacked. The peptide bonds, the C-terminal ester, and the phosphates remain intact (Scheme 6). More importantly, the reaction conditions are so mild that no trace of β-elimination is observed.[41,42]

Scheme 6 Enzymatic Selective Removal of the N-Terminal Phenylacetyl Group in the Synthesis of Phosphopeptides[41,42]

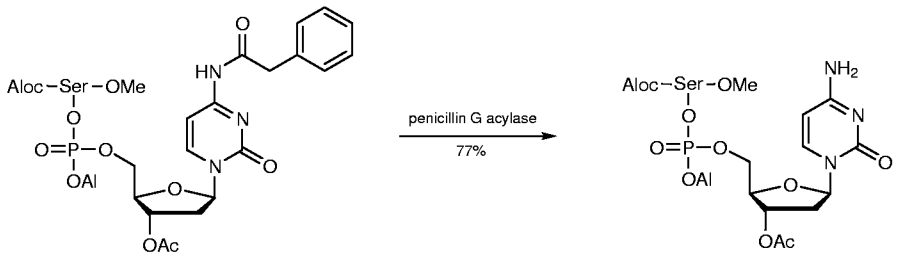


peptide-OR^1 = Ala-Ser(P)-OtBu, Ala-Thr(P)-OtBu, Leu-Ala-Thr(P)-OAl, Leu-Ala-Ser(P)-OAl, Ala-Ala-Ser(P)-OtBu

PGA-mediated selective deprotection of the phenylacetyl-protected nucleobase has been implemented successfully in the synthesis of nucleopeptides (Scheme 7).[43]

for references see p 313

Scheme 7 Penicillin G Acylase Mediated Deprotection of a Nucleobase[43]



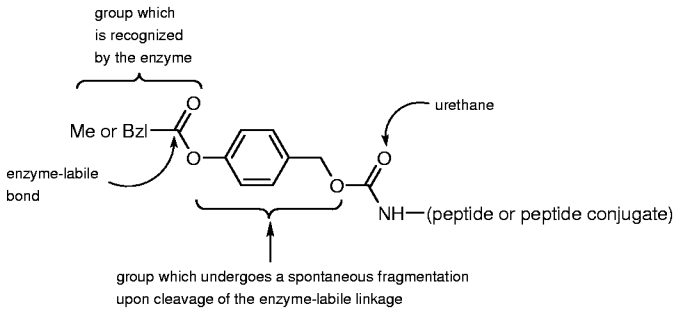
Enzymatic Deprotection of Phenylacetyl-Protected Phosphopeptides; General Procedure:[42]
To a soln of the fully protected phosphopeptide (0.1–0.3 mmol) in MeOH (30 mL), H_2O (70 mL) was added slowly. The pH value was adjusted to pH 6.5 by addition of 5% AcOH or 0.02 M NaOH. After stirring the soln for 1 h, PGA (1200 units, immobilized on Eupergit C) was added. The course of the reaction was followed by TLC (EtOAc/hexane/MeOH) or by HPLC (RP18, MeOH/H_2O 7:3). The reactions went to completion in 6 to 36 h. The enzyme was filtered and the residue washed successively with H_2O and MeOH, and the MeOH was removed from the aq methanolic filtrate under reduced pressure. The pH value was adjusted between 1–2 with dil HCl, and the soln was extracted with Et_2O (3 × 50 mL). The pH was readjusted to 9 by addition of Na_2CO_3 and the aq soln was immediately extracted with CH_2Cl_2 (5 × 80 mL). The organic layer was dried ($MgSO_4$) and sat. HCl in Et_2O was added. The solvent was removed under reduced pressure and the N-terminally deprotected phosphopeptides were isolated as the hydrochloride salts.

2.5.1.2 4-Acetoxybenzyloxycarbonyl Group

Protein conjugates (Scheme 1) are the focus of intense biochemical research because of their role in cellular signal transduction cascades. Problems associated with the chemical synthesis of these biologically important complex macromolecules have been highlighted (Scheme 1) and discussed in detail.[44–46] Hydrolytic enzymes are particularly useful for the chemoenzymatic synthesis of such peptide conjugates. However, there is a risk of racemization with enzyme-labile N-protecting acyl groups during the chemical synthesis of peptides and sensitive peptide conjugates. The use of enzyme-labile urethane blocking functions, e.g. the 4-acetoxybenzyloxycarbonyl [Z(OAc)] group,[47,48] overcomes this problem.

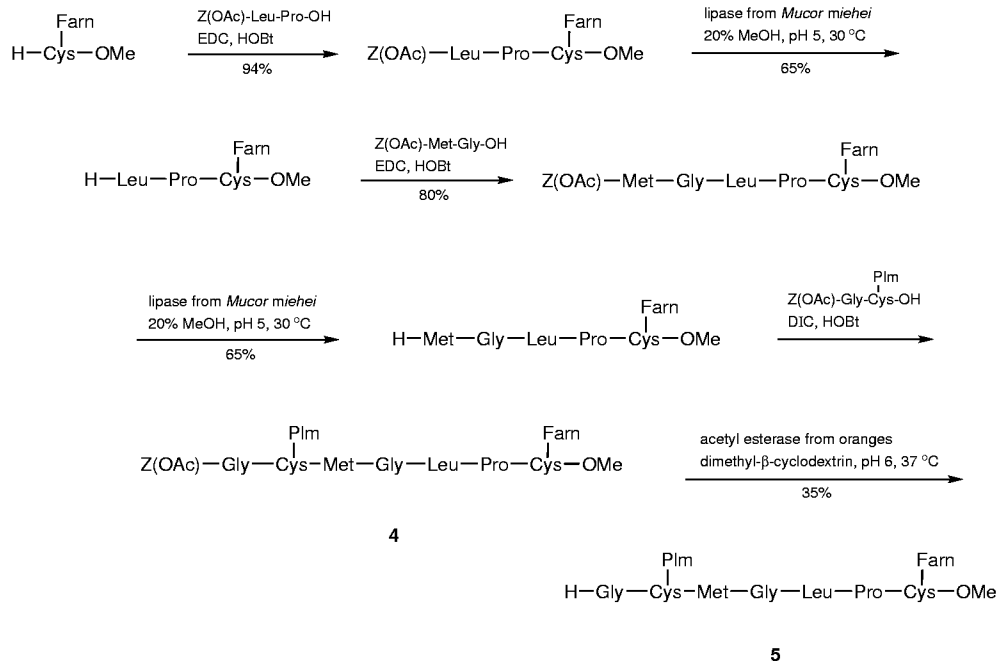
The urethane-based protecting group Z(OAc)[47,48] contains a functional group (here: an acetyl) that is recognized by the biocatalyst (here: esterase or lipase) and that is bound by an enzyme-labile linkage (here: an ester) to a functional group that undergoes a spontaneous fragmentation upon cleavage of the enzyme-sensitive bond (here: a 4-hydroxybenzyl urethane). This results in the liberation of a carbamic acid derivative, which decarboxylates to give the desired peptide or peptide conjugate (Scheme 8).

Scheme 8 Principle behind the Development of the Urethane-Based 4-Acetoxybenzyloxycarbonyl and 4-Phenylacetoxybenzyloxycarbonyl Protecting Groups



The S-palmitoylated and S-farnesylated C-terminal heptapeptide of human N-*Ras* protein has been synthesized using this protecting group (Scheme 9).[48]

Scheme 9 Application of the 4-Acetoxybenzyloxycarbonyl Protecting Group in the Synthesis of an S-Palmitoylated and S-Farnesylated C-Terminal Heptapeptide of Human N-*Ras* Protein[48]



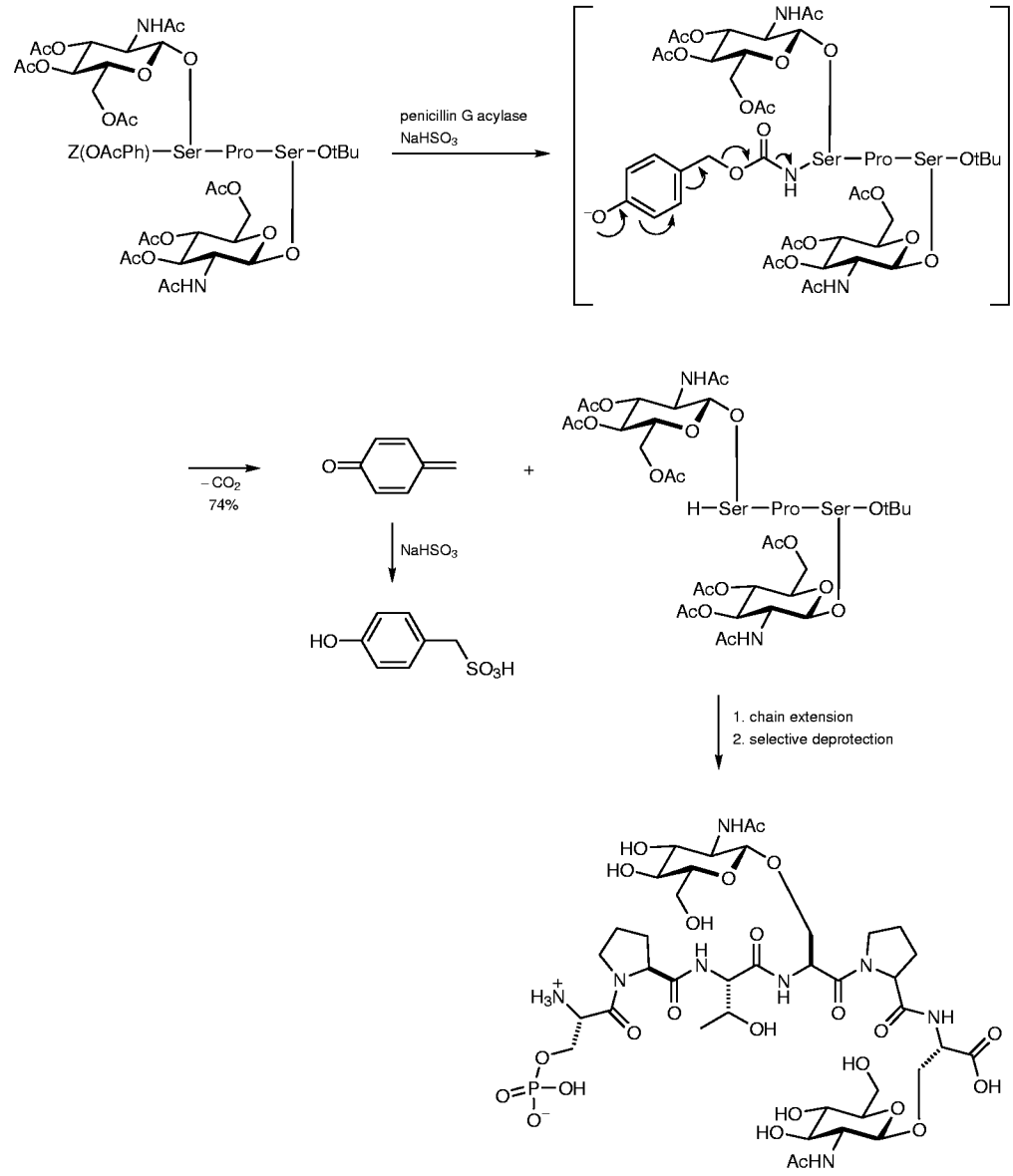
Removal of the Z(OAc) Group from Z(OAc)-Gly-Cys(Plm)-Met-Gly-Leu-Pro-Cys(Farn)-OMe (4):[48] Z(OAc)-Gly-Cys(Plm)-Met-Gly-Leu-Pro-Cys(Farn)-OMe (**4**; 8 mg, 6 μmol) and dimethyl-β-cyclodextrin (720 mg, 600 μmol) were dissolved in MeOH (5 mL). The soln was poured into 20 mM phosphate buffer (100 mL, pH 6). Acetyl esterase (8 units) was added and the mixture was stirred at 37 °C at pH 6 for 24 h. The soln was extracted with $CHCl_3$ (4×50 mL) and the organic layer was dried ($MgSO_4$). $CHCl_3$ was removed under reduced pressure and the residue was purified by flash chromatography (silica gel, EtOAc/MeOH 9:1) to generate H-Gly-Cys(Plm)-Met-Gly-Leu-Pro-Cys(Farn)-OMe (**5**); yield: 2.5 mg (35%).

2.5.1.3 4-Phenylacetoxybenzyloxycarbonyl Group

The idea behind the urethane-based protecting group, 4-phenylacetoxybenzyloxycarbonyl Z(OAcPh), is similar to that described for the Z(OAc) group (Scheme 8). However, unlike the Z(OAc) group, the phenylacetyl part of the Z(OAcPh) group is saponified by treatment with PGA. The usefulness of this protecting group was demonstrated in the synthesis of a phosphoglycohexapeptide (Scheme 10).[49,50]

for references see p 313

Scheme 10 Application of the 4-Phenylacetoxybenzyloxycarbonyl Protecting Group in the Synthesis of a Phosphoglycohexapeptide[49,50]



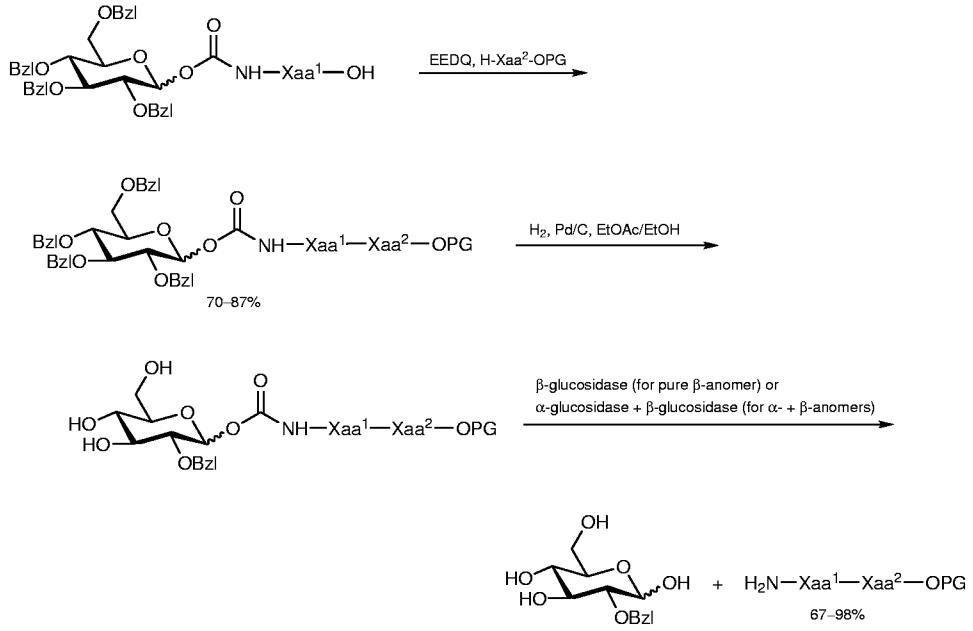
Removal of the Z(OAcPh) Group from Peptides and Glycopeptides; General Procedure:[50]
A soln of the respective Z(OAcPh)-protected peptide (0.05 mmol) in a mixture of MeOH (33 mL), 40% $NaHSO_3$ soln (13 mL) and 0.07 M Na_2HPO_4 buffer (68 mL) was treated with PGA (300 units) at pH 7.5 for 24 h at rt. The immobilized enzyme was removed by filtration and MeOH was removed under reduced pressure. The aq residue was extracted with $CHCl_3$ (3 ×) and the combined organic layers were dried ($MgSO_4$) and the solvent removed. The crude residue was purified by flash chromatography (silica gel, $CHCl_3$/MeOH and 1% TEA).

2.5.1.4 2,3,4,6-Tetra-*O*-benzylglucosyloxycarbonyl Group

Another urethane-protecting group has been derived from carbohydrates.[51] 2,3,4,6-Tetra-*O*-benzylglucosyloxycarbonyl protected amino acids are coupled with different amino acid *tert*-

butyl or heptyl (Hep) esters to deliver the fully protected dipeptides. For the deprotection, the benzyl ethers are first removed quantitatively by hydrogenolysis to generate glycosyloxycarbonyl-protected dipeptides. These glycosyloxycarbonyl-protected dipeptides are treated either with a mixture of α-glucosidase and β-glucosidase (in case of a mixture of α- and β-anomers of the glycosyloxycarbonyl-protected dipeptides) or with β-glucosidase (in case of pure β-anomer of the glycosyloxycarbonyl-protected dipeptides) to generate N-terminally deprotected dipeptides (Scheme 11).

Scheme 11 Use of Glucosidase-Labile Glycosyloxycarbonyl Protecting Group in the Synthesis of Dipeptides[51]



Enzymatic Removal of Glycosyloxycarbonyl Group from Dipeptides; General Procedure:[51]
Method A: A soln of the *N*-(β-D-glucopyranosyloxycarbonyl)dipeptide ester (0.2 mmol) in MeOH (10 mL) was added to a soln of β-glucosidase (40 units) in 0.25 M acetate buffer (pH 5.0, 190 mL). The reaction mixture was shaken for 12–36 h at 37 °C, the pH was adjusted to 10–11 and the soln was extracted with EtOH (4×100 mL). The combined organic phases were dried ($MgSO_4$), filtered, and the solvent removed to give the N-terminal deprotected dipeptide esters as colorless oils.

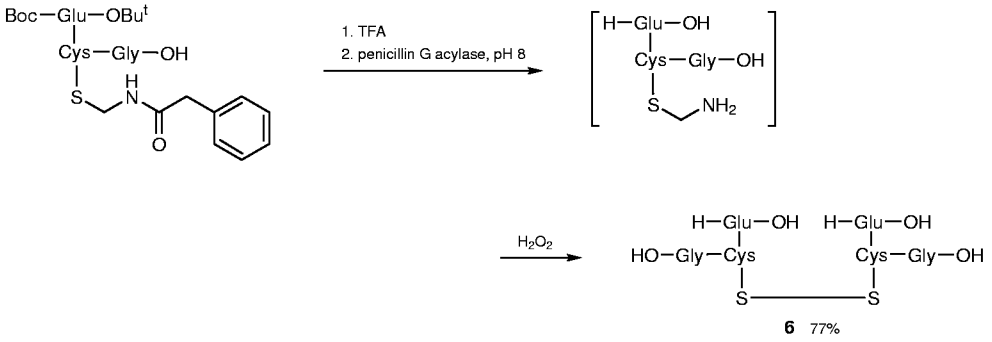
Method B: A soln of the *N*-(α/β-D-glucopyranosyloxycarbonyl)dipeptide ester (0.2 mmol) in MeOH (10 mL) was added to a soln of α-glucosidase (16 units) and β-glucosidase (40 units) in 0.07 M phosphate buffer (pH 5.5, 190 mL). The mixture was shaken for 12–36 h at 37 °C and then lyophilized. The residue was dissolved in sat. aq $NaHCO_3$ (20 mL) and the soln was extracted with CH_2Cl_2 (3×40 mL). The combined organic phases were dried ($MgSO_4$), filtered, and the solvent removed to give the N-terminal deprotected dipeptide esters as colorless oils.

2.5.2 Thiol Group Protection

The liberation of the β-sulfanyl group of cysteine is achieved by means of PGA-mediated hydrolysis of phenylacetamides.[28–30] For this purpose, the thiol group is masked with the phenylacetamidomethyl (Phacm) blocking function. After penicillin acylase catalyzed hydrolysis of the amide incorporated in the acylated thioacetal, the labile *S*-aminomethyl compound is formed and immediately liberates the desired thiol. This method has been used in a synthesis of glutathione which is isolated as its disulfide **6** (Scheme 12).

for references see p 313

Scheme 12 The Use of Penicillin G Acylase Mediated Deprotection of a Phenylacetyl Group in the Synthesis of Glutathione[28–30]



In a related glutathione synthesis, the method was used for the simultaneous liberation of the thiol and the N-terminal amino group of glutamic acid.[29,30] The *S*-phenylacetamidomethyl group was removed selectively by means of penicillin G acylase in the presence of the *S*-acetamidomethyl group (see Section 2.6.6.2.4.3).[30,52,96] The importance of such a strategy has also been highlighted in the regioselective disulfide pairing required for the synthesis of multiple-cystine peptides.[52] For the synthesis of Cys(Phacm) derivatives and additional examples of enzymatic cleavage see Section 2.6.6.2.4.3.

Enzymatic Removal of Phacm from Boc-Cys(Phacm)-OH:[96]
Solns (1 mM) of Boc-Cys(Phacm)-OH were incubated with different amounts of penicillin amidohydrolase (0.13, 1, 4 and 5 ue/μmol phenylacetamidomethyl) either in soln or immobilized on dry fibre or on acrylic beads (Eupergit C) in 0.02 M Na_2HPO_4-NaH_2PO_4 buffer at pH 7.8 at 35 °C for 24 h. Other aq buffers such as 0.02 M TEAOAc and 0.05 M NH_4OAc were also studied as well as the addition of β-ME (2%), NMP (25%), MeOH (30%), and DMF (25%) to Na_2HPO_4-NaH_2PO_4 buffer. The reaction was monitored by HPLC. Deprotections carried out in NH_4OAc buffer gave slightly better results in terms of both overall yield and purity of the final product. Deprotection was more effective in aqueous media. In the presence of organic solvents, a reaction time of 48 h was required for complete deprotection and oxidation.

2.5.3 Carboxy Group Protection

Initial attempts to achieve an enzyme-catalyzed deprotection of the carboxy group of peptides centred around the use of the endopeptidases chymotrypsin,[53–55] trypsin,[54,56,58] and thermolysin.[57] Thermolysin is a protease obtained from *Bacillus thermoproteolyticus* that hydrolyzes peptide bonds on the amino side of the hydrophobic amino acid residues (e.g., leucine, isoleucine, valine, phenylalanine). It cleaved the "supporting" tripeptide ester H-Leu-Gly-Gly-OEt from a protected undecapeptide (pH 7, rt). The octapeptide, thus obtained, is composed exclusively of hydrophilic amino acids. Due to the broad substrate specificity of thermolysin and the resulting possibility of unspecific peptide hydrolysis, this method is of limited application.

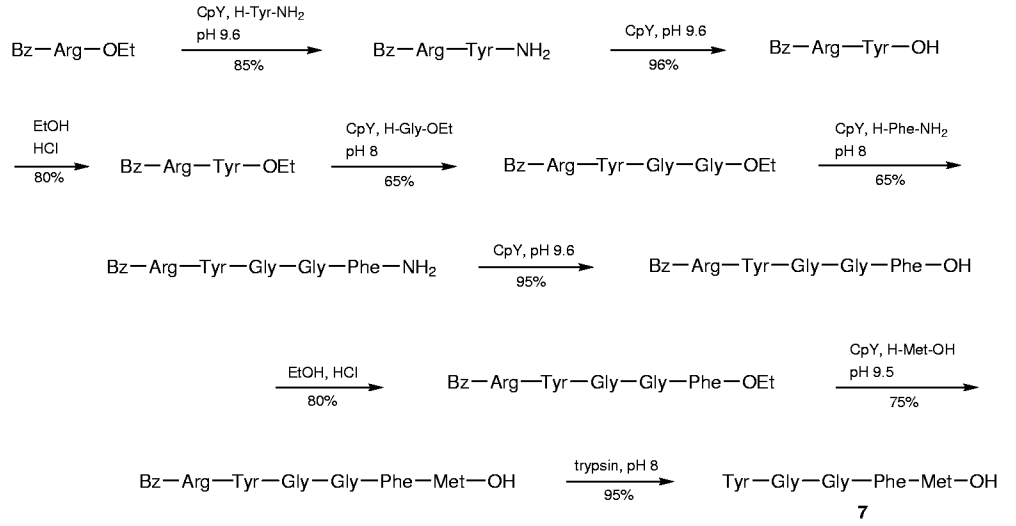
The exploitation of the esterase activities of chymotrypsin and trypsin opened routes to the hydrolysis of several peptide methyl and ethyl esters at pH 6.4–8 and room temperature.[53,54] The transformation is not only successful with peptides carrying the respective enzyme-specific amino acids at the C-terminus, but in several examples different amino acids are tolerated at this position. However, a severe drawback of this approach is that numerous peptides are poor substrates or not accepted at all. Moreover, a competitive cleavage of the peptide bond occurs if the peptides contain trypsin- or chymotrypsin-labile sequences. Therefore, these proteases are not generally useful for a safe C-terminal deprotection.

2.5.3.1 Alkyl Esters Group

2.5.3.1.1 Cleavage with Proteases and Esterases

The disadvantages displayed by the endopeptidases can be overcome by using CpY from baker's yeast.[20,58,59] This serine-exopeptidase also has esterase activity and is characterized by quite different pH optima for the peptidase and the esterase activity (pH >8.5). Even in the presence of various organic cosolvents the enzyme selectively removes the carboxy protecting groups from a variety of differently protected di- and oligopeptide methyl and ethyl esters[20,59] without attacking the peptide bonds. An additional attractive feature is that its esterase activity is restricted to α-esters. Therefore, β- and γ-esters of aspartic and glutamic acid, respectively, are not cleaved. CpY is advantageously used for the stepwise elongation of the peptide chain in aqueous solution employing a solubilizing polymeric support as the N-terminal blocking group.[58] The peptide chain is elongated in the C-terminal direction by successive couplings on amino acid ethyl esters with CpY at pH 8.5 and room temperature. Optically pure H-Leu-Phe-Leu-OEt is obtained in 64% overall yield. In another synthesis, which did not use a polymeric N-protecting group, Met-enkephalin **7** is built up by employing CpY for C-terminal deprotection as well as for the formation of the peptide bonds (Scheme 13).[20]

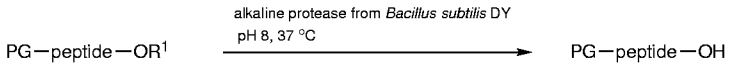
Scheme 13 Synthesis of Met-enkephalin by Employing Carboxypeptidase Y for C-Terminal Deprotection as well as for the Formation of Peptide Bonds[20]



An enzyme which generally displays a high esterase/protease ratio, such as the alkaline protease from *Bacillus subtilis DY*, selectively removes methyl, ethyl, and benzyl esters from a variety of Trt-, Z-, and Boc-protected di- and tripeptides and a pentapeptide at pH 8 and 37 °C (Scheme 14).[59]

for references see p 313

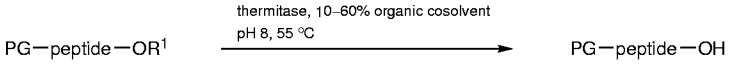
Scheme 14 Alkaline Protease Assisted Selective Removal of C-Terminal Protecting Groups[59]



Protecting Group	Peptide	R^1	Yield (%)
Z	Tyr(tBu)-Glu-Leu	Me	93
Boc	Leu-Glu-Val	Bzl	85
Trt	Ala-Glu-Asp-Leu-Glu	Bzl	80

Thermitase, a thermostable extracellular serine protease from the thermophilic microorganism *Thermoactinomyces vulgaris*, with an esterase/protease activity ratio >1000:1 shows a broad amino acid side-chain tolerance and cleaves methyl, ethyl, benzyl, ethoxybenzyl, and *tert*-butyl esters from a variety of Nps-, Boc-, Bpoc-, and Z-protected di- and oligopeptides in high yields at pH 8 and 33–55 °C (Scheme 15).[28,29,60–62] In addition, it is specific for the α-carboxy groups of Asp and Glu.

Scheme 15 Deblocking of C-Terminal Methyl Esters by Thermitase in the Presence of Organic Solvents[28,29,60–62]

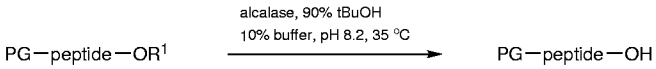


Protecting Group	Peptide	R^1	Yield (%)
Z	Leu-Val-Glu(tBu)-Ala	Me	92
Boc	Pro-Gly	Me	73
Bpoc	Tyr(tBu)-Glu-Leu	Me	55
Nps	Ser(Bzl)-His(Dnp)-Leu-Val-Glu(tBu)-Ala	Me	90

To enhance the solubility of the substrates, up to 50 vol % of organic cosolvents such as DMF and DMSO may be added which also serve to reduce the remaining peptidase activity to a negligible amount.[29,62]

Alcalase, a serine endopeptidase from *Bacillus licheniformis* and whose major component is subtilisin A (subtilisin Carlsberg)[63–65] has been employed to selectively saponify peptide methyl and benzyl esters (Scheme 16) in a solvent system consisting of 90% *tert*-butyl alcohol and 10% buffer (pH 8.2) leaving the N-terminal Fmoc group intact. A selective classical alkaline saponification of methyl esters would have been impossible due to the base sensitivity of the Fmoc group.

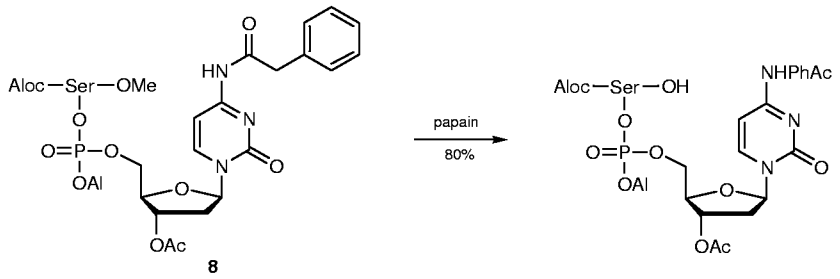
Scheme 16 Alcalase-Mediated Selective Saponification of Methyl Esters in the Presence of the Base-Labile Fmoc Group[63–65]



Protecting Group	Peptide	R^1	Yield (%)
Fmoc	Ala-Val-Ile	Me	85
Fmoc	Asn-Phe	Bzl	90
Boc	Met-Leu-Phe	Me	80
Z	Met-Asp(OMe)-Phe	Me	90

Proteases, such as thermitase,[29,66] papain,[67–69] and subtilisin[70] have been employed for the deprotection of *tert*-butyl and methyl esters from various glycopeptides. More interestingly, the C-terminus of a nucleopeptide methyl ester **8** was deprotected selectively and in very high yield by papain.[43] Neither the acetate, the N-terminal urethane, the allyl phosphate, nor the phenylacetamide protection of the nucleobase were attacked (Scheme 17).

Scheme 17 Selective Deprotection of the C-Terminus of a Nucleopeptide Methyl Ester by Papain[43]

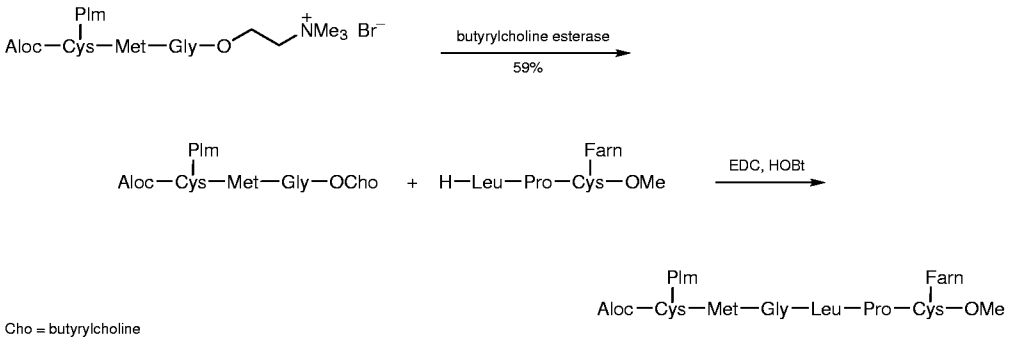


8

Proteases have been used for the unmasking of the side-chain carboxy groups of glutamic and aspartic acid. Z-Asp(ArgOMe)-NH_2 and Z-Glu(ArgOMe)-NH_2 could be converted into Z-Asp(OH)-NH_2 and Z-Glu(OH)-NH_2 in two steps. Trypsin hydrolyzes the arginine methyl esters and porcine pancreatic carboxypeptidase B splits off the arginines.[71] The second step is slow and requires a high concentration of the enzyme which may result in competitive side reactions. Therefore, this method is of limited use. However, alcalase selectively hydrolyzes the α-benzyl esters of H-Asp(Bzl)-OBzl and H-Glu(Bzl)-OBzl in high yields on a decagram scale.[72] Similarly, peptidyl-glutamic and -aspartic acid derivatives can be deprotected selectively at the C-terminus by this enzyme; however, in these cases an undesired attack on the peptide bond may occur.[73] Z-Asp(OAl)-OAl is converted into Z-Asp(OAl)-OH in quantitative yield by papain.[74]

An application of the concept of combining enzyme-labile protecting groups with chemically labile ones is illustrated by the chemoenzymatic synthesis of the S-palmitoylated and S-farnesylated C-terminal lipohexapeptide of the human N-*Ras* protein (Scheme 18).[75]

Scheme 18 An Alternative Approach to the Chemoenzymatic Synthesis of a S-Palmitoylated and S-Farnesylated C-Terminal Lipohexapeptide[75]

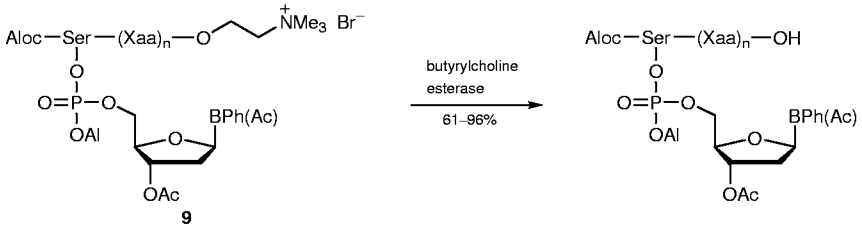


This type of modified lipopeptide cannot be deblocked under basic conditions because the labile palmitic acid thioester group would be preferentially hydrolyzed. The C-terminus of the peptide chain was selectively deprotected by removing the choline ester with choline esterase without affecting the palmitic acid thioester bond. The observed chemoselectivity here is exactly opposite to that found in nonenzymatic conversions. Some of the synthesized N-terminally deprotected lipopeptides, have been labeled with biotin[76] and are expected to serve as anchors for a protein moiety in an artificial membrane. Butyrylcholine esterase mediated cleavage of the choline ester has been utilized[77] as the key step in the synthesis of S-palmitoylated peptides such as Myr-Gly-Cys(Plm)-Thr-Leu-Ser-Ala-OH, which represents the characteristic N-terminus of the α-subunit of human $G_{\alpha o}$ protein.

for references see p 313

The C-terminus of a nucleopeptide choline ester **9** was deprotected[43] smoothly by butyrylcholine esterase (Scheme 19). Neither the acetate, the N-terminal urethane, the allyl phosphate, the phenylacetamide and the peptide bonds, nor the acid- or base-labile purine nucleosides and serine phosphates are attacked.[43] The methods have been exploited further for the synthesis of several biologically relevant nucleopeptides.[78]

Scheme 19 Selective Deprotection of the C-Terminus of a Nucleopeptide Choline Ester by Butyrylcholine Esterase[43]



B = nucleotide base

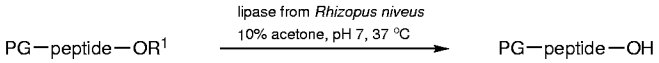
Saponification of Fmoc-Ala-Val-OMe:[63]
Alcalase (4 mL, 10 AU) was added to a soln of Fmoc-Ala-Val-OMe (4.42 g, 10 mmol) dissolved in a mixture of tBuOH (200 mL) and 0.5 M phosphate buffer (20 mL, pH 8.2) at 35 °C. The soln was stirred until the substrate completely disappeared (about 30 min, monitored by HPLC). The mixture was diluted with EtOAc, washed with aq 0.1 M HCl (6×30 mL) and H_2O (3×50 mL). The resulting soln was concentrated and Et_2O was added to give Fmoc-Ala-Val-OH (3.68 g, 86%).

Enzymatic Conversion of Z-Asp(OAl)-OAl into Z-Asp(OAl)-OH:[74]
To a soln of Z-Asp(OAl)-OAl (0.10 g, 0.3 mmol) in either DMF or MeCN (4.2 mL), a buffer soln (5.8 mL) containing papain (4.5 mg) and DTT (30 mg, 0.19 mmol) was added. The mixture was kept at 25 °C and stirred for 2.5 h or 24 h depending on the reaction rate. Monitoring was performed by removal of 100 µL samples, which were diluted 1/11 with MeCN/H_2O (5:5), frozen, and subsequently analyzed by HPLC. Chemically synthesized intermediates Z-Asp(OAl)-OAl, Z-Asp-OAl, and Z-Asp(OAl)-OH were used for the calibration purposes. Reactions performed in MeCN were always slower and nearly quantitative yields were always achieved except for carbonate buffer/MeCN containing mixtures.

2.5.3.1.2 Cleavage with Lipases

The problem of undesired hydrolysis of peptide bonds occurring during deprotection can be avoided completely by choosing enzymes devoid of any protease activity. For example by using Hep[26,27,79–82] and 2-bromoethyl[79,26,27,80,83] esters as carboxy protecting groups for peptide synthesis and lipases for their removal (Scheme 20).

Scheme 20 Lipase-Assisted Selective Saponification of C-Terminal (Hep), 2-Bromoethyl, and 2-(*N*-Morpholino)ethyl Esters[26,27,79–83]



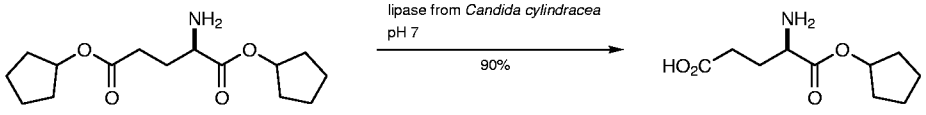
Protecting Group	Peptide	R^1	Yield (%)
Boc	Ser-Thr	Hep	95
Z	Thr-Ala	Hep	85
Aloc	Met-Gly	Hep	90
Z	Ser-Phe	EtBr[a]	84
Boc	Val-Ala	EtBr[a]	95
Boc	Val-Phe	MoEt[b]	91

[a] 2-bromoethyl.
[b] 2-(*N*-morpholino)ethyl.

The Hep esters, which can be synthesized by azeotropic esterification, are chemically stable during the removal of the N-terminal Z, Boc, and the Aloc group from the corresponding dipeptides. The lipase, which is best suited for the hydrolysis of the heptyl ester group in terms of substrate tolerance and reaction rate, is isolated from the fungus *Rhizopus niveus*. The enzyme accepts a variety of Boc-, Z-, and Aloc-protected dipeptide heptyl esters as substrates and hydrolyzes the ester groups in high yields at pH 7 and 37 °C without damaging the urethane protecting groups and amide bonds (Scheme 20).[81,82] 2-Bromoethyl ester-protected dipeptides also behave in the same way. However, heptyl esters of hydrophobic peptides are very often not hydrolyzed by lipases. This problem is circumvented by using more hydrophilic 2-(*N*-morpholino)ethyl esters of protected peptides which are cleaved in good yields by means of lipase from *R. niveus*.[84] It should be emphasized that lipases are not specific for L-amino acids, but also tolerate the presence of the D-enantiomer.[85] Moreover, in contrast to proteases and esterases, they operate at the interface between water and organic solvents.[86]

A lipase from *Candida cylindracea* is able to differentiate between the two carboxylic acid groups of glutamic acid. Using dicyclopentyl ester glutamate **10**, it preferably unmasks the γ-ester in 90% yield in a ratio 20:1 (Scheme 21).[87]

Scheme 21 Enzymatic Regioselective Deprotection of the γ-Ester of Glutamic Acid[87]

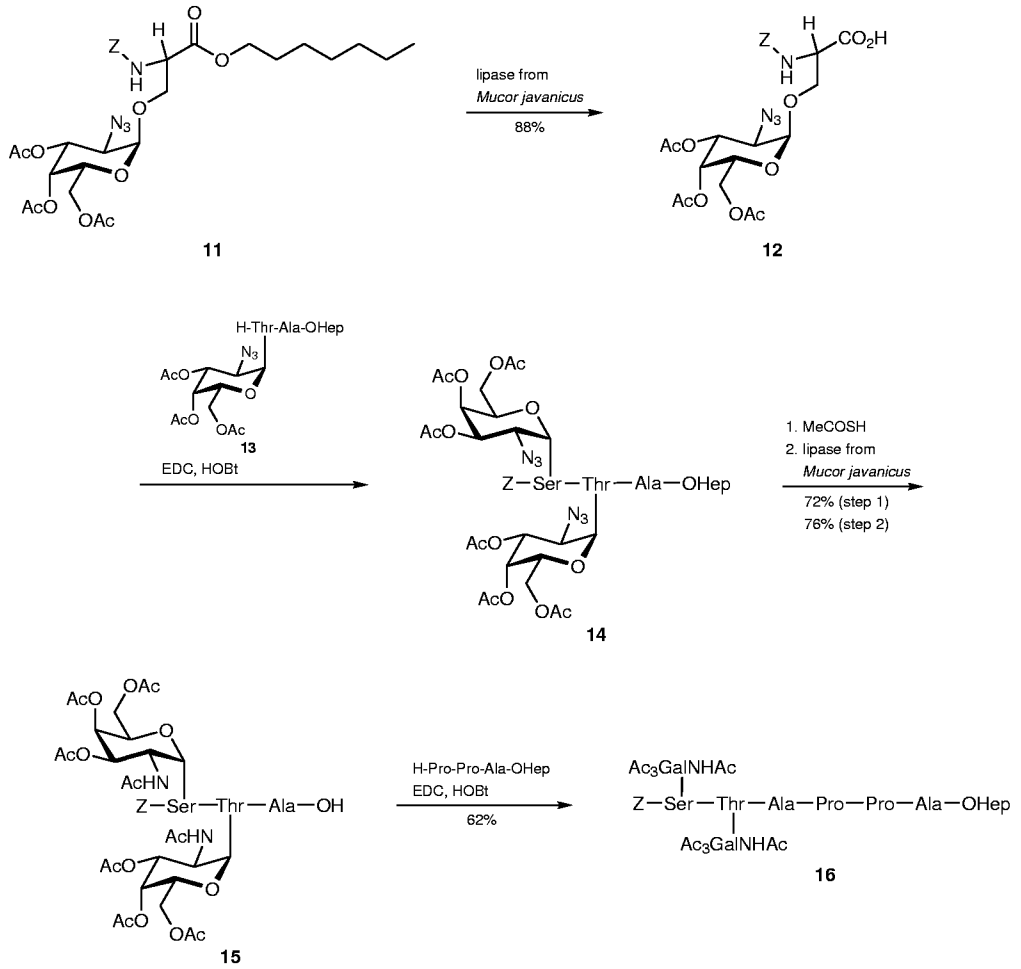


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The full capacity of the lipase-mediated technique for C-terminal deprotection is demonstrated in the synthesis of complex *O*-glycopeptides, which are sensitive to both acids and bases.[88,89] The key starting material, serine glycoside heptyl ester **11**, is selectively deprotected at the C-terminus by lipase obtained from the fungus *Mucor javanicus* (Scheme 22). The free carboxylic acid derivative **12** is coupled with an N-terminally protected glycodipeptide **13** to generate the tripeptide **14**.

for references see p 313

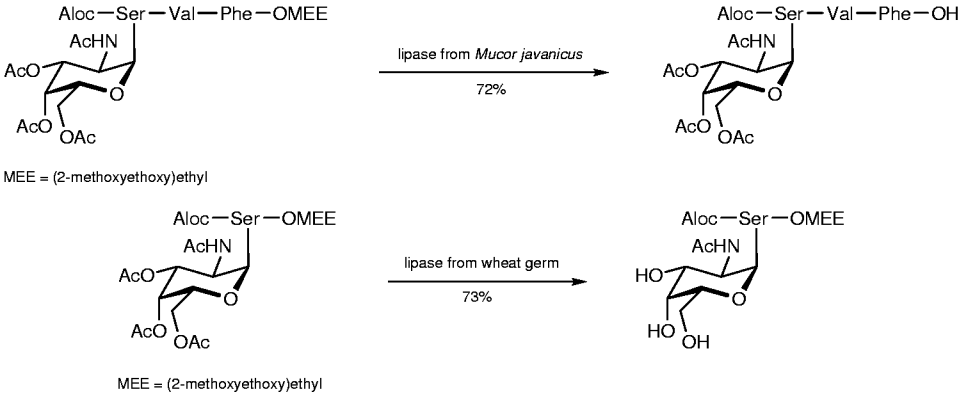
Scheme 22 Selective Deprotection of the C-Terminal Heptyl Esters of Glycopeptides by Lipase[88,89]



After conversion of the azido groups into acetamides, lipase-mediated deprotection yields the C-terminally deprotected glycotripeptide **15**. This compound was finally condensed with a tripeptide to generate the target diglycohexapeptide **16**. In the course of these enzymatic transformations, the N-terminal urethanes, the peptide bonds, the acid- and base-labile glycosidic linkages, and the base-sensitive acetyl protecting groups are not attacked.

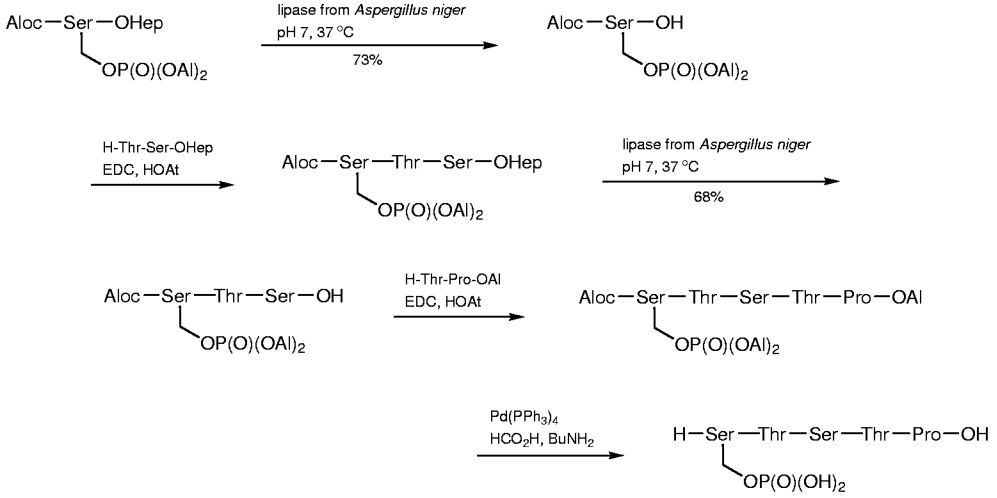
The lipase is also capable of removing the 2-(*N*-morpholino)ethyl ester protecting group from *O*-glycopeptides.[86] A report has appeared[90] on the selective C-terminal deprotection of *O*-glycopeptide (2-methoxyethoxy)ethyl esters that are more hydrophilic than the corresponding heptyl esters. This reaction takes place under mild conditions (pH 6.6, 37 °C) by enzymatic hydrolysis using papain or lipase from *Mucor javanicus*, and has been used to prepare building blocks for extending the chain of glycopeptides. The selective removal of acetyl protecting groups from the glycoside part of the glycopeptides is accomplished by an alternative enzymatic hydrolysis with lipase from wheat germ (Scheme 23).

Scheme 23 Lability and Stability of the C-Terminal (2-Methoxyethoxy)ethyl Ester of Glycopeptides to Lipases from *Mucor javanicus* and Wheat Germ[90]



Lipase (*Aspergillus niger*)-mediated saponification of heptyl esters is employed for the C-terminal deprotection of several phosphopeptides. For example this method has been used in the synthesis (Scheme 24) of a consensus sequence of the *Raf*-1 kinase.[42,91]

Scheme 24 Enzymatic Selective Saponification of the C-Terminal Heptyl Esters in the Synthesis of Phosphopeptides[42,91]

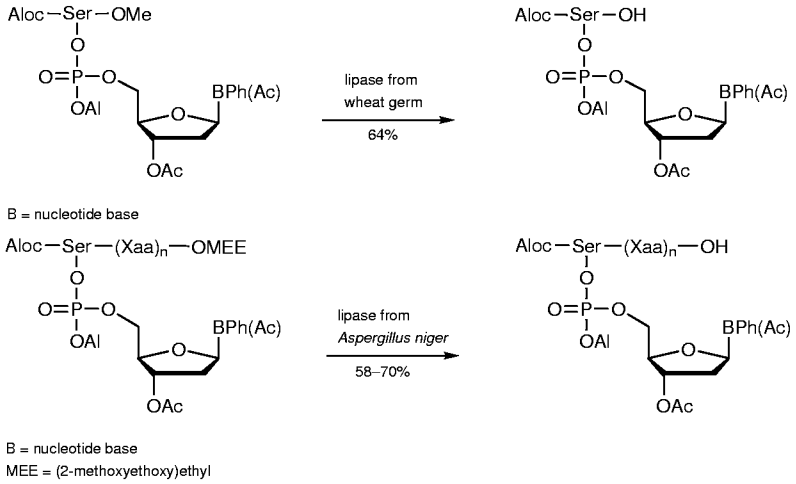


Recently, a comparison of the lipase-catalyzed cleavage of polar esters of peptides and *N*-glycopeptides has been reported.[92] This study established the superiority of 2-methoxyethyl esters over other esters derived from ethylene glycol. In comparison to these esters, peptide 2-methoxyethyl esters of various structures are accepted as substrates by lipases. In addition, these 2-methoxyethyl esters show enhanced solubility in water-containing solutions. Most 2-methoxyethyl esters crystallize easily. The selective removal of 2-methoxyethyl esters by lipases has been achieved under mild conditions (pH 7.0 and 37 °C) leaving all other linkages including peptide bonds and other ester-type protecting groups unaffected.

Lipase from wheat germ[43] and lipase from *Aspergillus niger*[93] selectively unmasked the methyl and (2-methoxyethoxy)ethyl protected C-terminals of nucleopeptides (Scheme 25).

for references see p 313

Scheme 25 Lipase-Mediated Selective Deprotection of the C-Terminal Esters of Nucleopeptides[43,93]



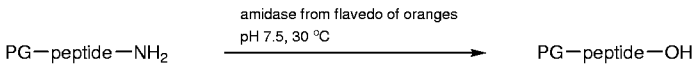
Enzymatic Deprotection of Phosphopeptide Heptyl Esters; General Procedure:[42]
To a soln of lipase from *Aspergillus niger* (0.1 g-1.2 g, 0.4–4.8 units) in 0.2 M phosphate buffer (20–100 mL, pH 7.0) was added a soln of the respective fully protected phosphopeptide (85–950 μmol) in acetone (2–10 mL) at 37 °C and, if required, deoxytaurocholic acid sodium salt (0.1 equiv.). The mixture was shaken at 37 °C for 12–48 h, sat. with NaCl, and the pH was adjusted to 4.5 with concd HCl. The soln was extracted with EtOAc (5×100 mL), the combined organic layers were dried ($MgSO_4$), and the solvent removed. If necessary, the crude product was purified by flash chromatography (EtOAc/hexane 10:3 containing 1–2% AcOH).

Lipase-Catalyzed Hydrolysis of N-Protected Dipeptide 2-Methoxyethyl Esters; General Procedure:[92]
N-Protected dipeptide 2-methoxyethyl esters (3 mmol) were dissolved in acetone (5–15 mL). 0.2 M Na_2PO_4 buffer (50 mL, pH 7) was added, and the mixture was warmed to 37 °C. After addition of lipase from *Aspergillus niger* (300 mg), the mixture was shaken for 15 h to 2 d at 37 °C, then sat. with NaCl, and the dipeptide extracted with EtOAc. If necessary, AcOH was added before extraction. The organic layer was dried ($MgSO_4$) and the solvent removed. The residue was purified by flash chromatography (EtOAc/AcOH 100:1). The partially deprotected dipeptides were obtained in 79–94% yields.

2.5.3.2 Amide Protection

The additional opportunity to hydrolyze selectively C-terminal peptide amides is of particular interest. For example, a peptide amidase from the flavedo of oranges has broad substrate specificity and accepts Boc-, Trt-, Z-, and Bz-protected and N-terminal unprotected peptide amides (Scheme 26).[94,95] The C-terminal amides are saponified by this biocatalyst in high yields at pH 7.5 and 30 °C without affecting the N-terminal blocking groups or the peptide bonds.

Scheme 26 The Amidase-Mediated Selective Hydrolysis of C-Terminal Amides[94,95]



Protecting Group	Peptide	Conv. (%)
Bz	Tyr-Ser	100
Boc	Leu-Val	20
Trt	Gly-Leu-Val	80
Z	Gly-Gly-Leu	100

Selective Enzymatic Deamidation of Peptide Amides; General Procedure:[94]
The enzyme preparation was obtained from the flavedo of oranges, that is, from the outmost colored layer of the peel, by a two-step purification involving fractionating salt precipitation and ion-exchange chromatography. The peptide amidase had an activity of 2 to 100 $U \cdot mg^{-1}$, depending on the substrate, where an activity of 1 U corresponds to the formation of 1 μmol of product per minute at 30 °C and

pH 7.5. Peptide amide (10–50 mmol) in 50 mM Tris/HCl buffer at pH 7.5 was treated with 20–40 $\mu g \cdot mL^{-1}$ of the peptide amidase. The determination and quantification of the peptide was accomplished by HPLC on an RP-18 column, which was eluted isocratically with various compositions of $Bu_4N(HSO_4)$/MeCN at the rate of $1.0\,mL \cdot min^{-1}$.

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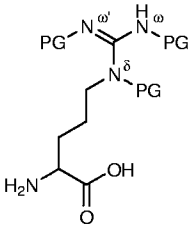
2.6. Other Side-Chain Protections

2.6.1 Guanidino Group

P. Thamm, W. Kolbeck, H.-J. Musiol, and L. Moroder

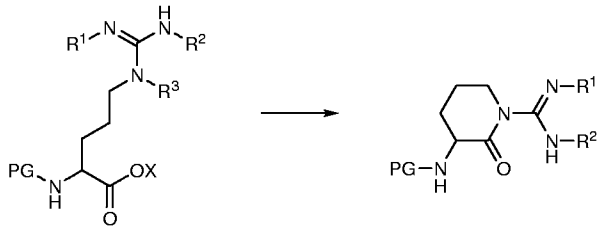
The trifunctional guanidino group of arginine displays a strong nucleophilic character and, thus, if unprotected or improperly protected, side reactions occur which mainly consist of intramolecular cyclization to δ-lactam derivatives, acylation with subsequent decomposition to ornithine, and intramolecular cyclization to 4-carboxy-2-imino-1,3-diazacycloheptane derivatives. The origin and mechanism of these side reactions have been repeatedly reviewed in detail.[1–4] Depending upon the synthetic strategy, the demands for efficient protection vary, with the most stringent requirements in stepwise syntheses, particularly when performed on solid supports where strong acylating agents are required to fulfill the need for quantitative aminoacylation in the chain-elongation steps. Many diverse protecting groups based on nitro, urethane, arylsulfonyl, or aryl derivatives have been proposed, but so far none satisfies all the characteristics of an ideal protecting group, i.e. full protection against undesired side reactions, and clean and smooth removal under mild conditions.

Scheme 1 $N^{\delta,\omega,\omega'}$-Protected Arginine Guanidino Group



An ideal protection would envisage $N^{\delta,\omega,\omega'}$-per-derivatization of the guanidino group (Scheme 1). This has been realized with the tris(trityl) derivative[5] which, however, has not found application in peptide synthesis. In contrast, $N^{\delta,\omega}$-protection should at least prevent δ-lactam formation (Scheme 2), but $N^{\omega'}$-acylation with the related cascade into side products (Scheme 3) may still occur.[6–8] In reality, most of the proposed protections are simple N^{ω}- or $N^{\omega'}$-derivatives which act via their electron-withdrawing properties and bulkiness with a range of efficiencies to prevent both δ-lactam formation and aminoacylation.

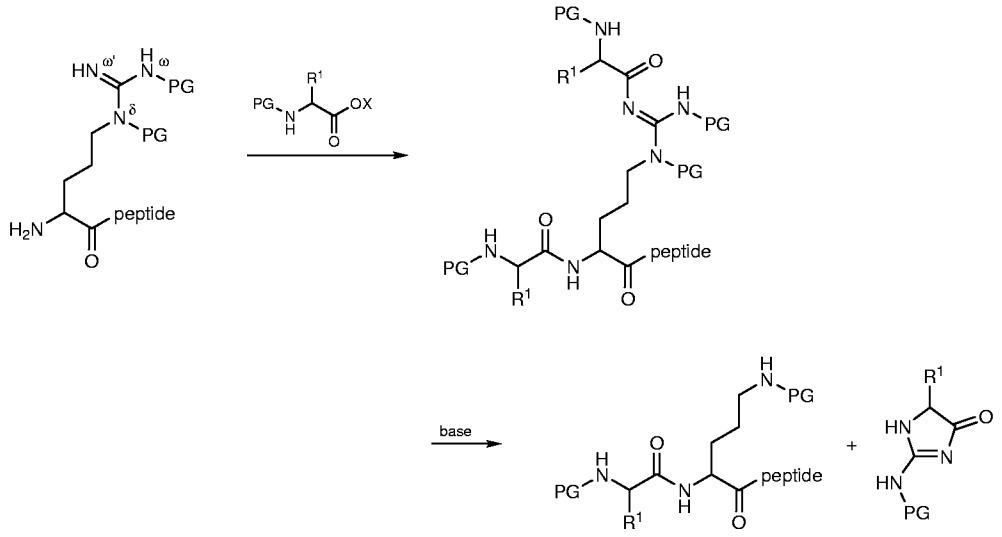
Scheme 2 δ-Lactam Formation with Carboxyl-Activated Arginine Derivatives when $R^1 = R^2 = R^3 = H$, or $R^3 = H$; $R^2 = PG$ and $R^1 = H$ or PG[164,165]



X = activating group

for references see p 330

Scheme 3 N$^{\omega'}$-Acylation of Partially Protected Arginine Residues, Followed by Base-Catalyzed Decomposition[8]



2.6.1.1 Protection by N$^{\omega}$-Protonation

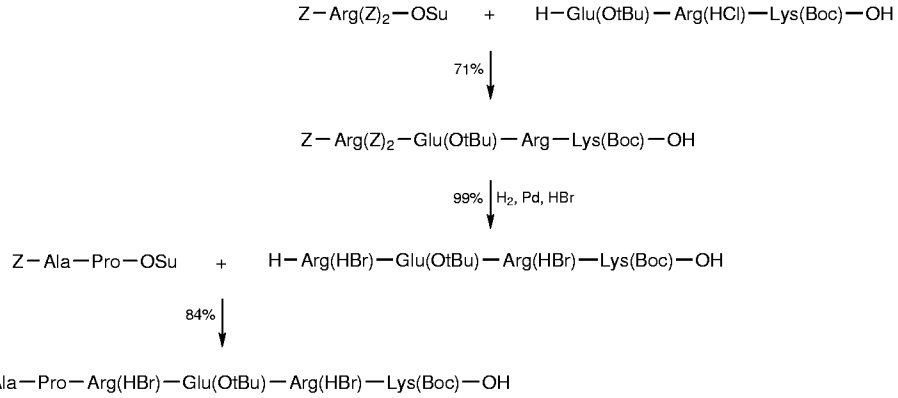
In contrast to the amino group, protection of the very basic guanidino group ($pK_a = 12.5$) is not absolutely necessary, since under most conditions prevailing in peptide synthesis the guanidino group is protonated and, thus, acylation is prevented to a large extent.[9,10] In fact, N$^{\alpha}$-acylation of N$^{\omega}$-protonated arginine, e.g. in the zwitterionic form, with active esters of protected amino acids or peptides proceeds smoothly and allows facile synthetic access to peptides with C-terminal arginine.[11,12] Similarly, DCC-mediated condensation of suitably protected peptide fragments containing protonated arginine residues in *endo* or N-terminal positions has been repeatedly used in the past, whereas a protonated arginine residue in the C-terminal position has generally been avoided because of possible δ-lactam formation. However, fragment condensations with C-terminal protonated arginine residues,[12,13] as well as aminoacylation reactions with N$^{\alpha}$-protected N$^{\omega}$-protonated arginine residues, have been applied to both synthesis in solution[14,15] and on solid supports.[16–18]

Regarding the acids used for protonation, the choice of the counteranion apparently plays a crucial role. For unknown reasons, bromide represents the anion of choice, in terms of exchange rates, compared to chloride.[14,19] Nevertheless, a sufficient bromide content has to be secured in the course of syntheses based on repetitive base treatments. Excess HBr added to ensure complete protonation is preferentially removed with NMM[14,19] rather than TEA.[9] If deprotonation occurs, the guanidino group is readily acylated, even with moderately active esters,[20] resulting in a progressive conversion of the arginine residues into ornithine.[16] Among other counterions proposed, perchlorate, which is presented as the pyridinium salt, exerts advantageous solubilizing effects,[21] and even HOBt can be used as an acidic additive.[7,22]

Besides the use of Z-Arg(HX)-OH (X = Cl, Br),[15] Boc-Arg(HCl)-OH,[17] and Z-Arg(HClO$_4$)-OH[21] for incorporation of the arginine residue, Z-Arg(NO$_2$)-OH (see Section 2.6.1.2) and Z-Arg(Z)$_2$-OH (see Section 2.6.1.3.3) are useful for this purpose. After N$^{\alpha}$-deprotection or, in the case of the two latter arginine derivatives, N$^{\alpha,\omega}$-deprotection, subsequent chain-elongation steps are performed with the protonated arginine residues, as shown in Scheme 4 for the synthesis of a subfragment of somatostatin-28.[23] This protonated arginine approach is well-suited for peptide synthesis in solution and has been successfully

applied in numerous large-scale syntheses of peptides.[15,23,24] Successful syntheses on solid support have also been reported,[16,18,22] and this approach may even be useful for large-scale solid-phase synthesis, as demonstrated by Arzeno et al.[17]

Scheme 4 Synthesis of the Peptide Fragment 9–14 of Somatostatin-28 Using the Protonated Arginine Approach[23]



Z-Arg-OH:[25]
To an ice-cold soln of H-Arg-OH · HCl (63 g, 0.3 mol) in 1 M NaOH (300 mL) was added Z-Cl (55 mL, 0.39 mol) together with 2 M NaOH (165 mL) in small portions to keep the pH at 9–10. The mixture was stirred for an additional 2 h, during which the pH dropped to 7.0–7.5. The precipitate was filtered off, washed with ice-cold H_2O and recrystallized (boiling H_2O, 400 mL). The product was washed with acetone and Et_2O; yield: 82.7 g (89.5%); mp 184 °C; $[\alpha]_D^{23}$ –9.3 (*c* 2, 1 M HCl).
Conversion into the HCl or HBr salt was performed by dissolving Z-Arg-OH in MeOH containing an equivalent amount of HCl or HBr, followed by precipitation with Et_2O.

Boc-Arg(HCl)-OH:[26]
To a suspension of H-Arg-OH (1.75 g, 10 mmol) in H_2O was added a soln of Boc_2O (2.55 g, 11.7 mmol) in tBuOH (7 mL). The mixture was stirred for 2 h at rt and then taken to dryness. The residue was triturated with Et_2O, dissolved in EtOH (10 mL) and, upon cooling to 0–5 °C, 12 M HCl (1 mL) in acetone (9 mL) was added. The product was precipitated with Et_2O and washed with Et_2O/acetone (1:1); yield: 3.0–3.1 g (91–95%); mp 110–115 °C.

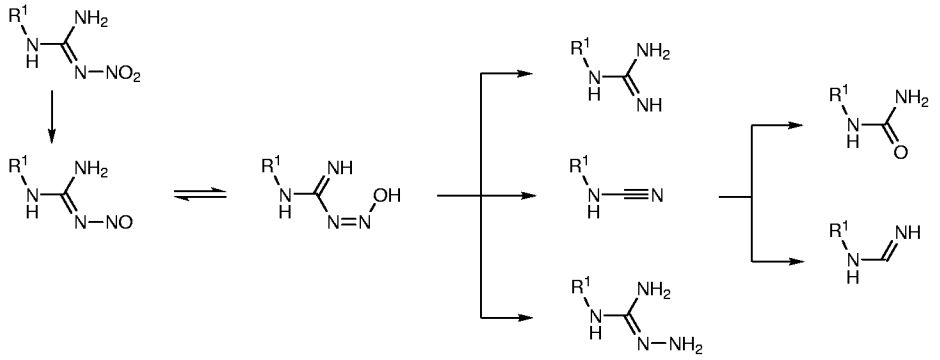
2.6.1.2 $N^{\omega'}$-Nitro Group

With the strongly electron-withdrawing nitro group, introduced by Bergmann et al.[27] for the side-chain protection of arginine, the nucleophilic character of the guanidino group is significantly reduced, and thus N^{ω}-aminoacylation as well as δ-lactam formation is largely prevented, although not in all cases.[28,29] The nitroarginine, which has been structurally defined by ^1H and ^{15}N NMR spectroscopy as the $N^{\omega'}$-derivative,[30] is stable upon treatment with mild acid, such as TFA, but is quantitatively removed with HF;[31,32] with MsOH and TfOH[33,34] complete deprotection is difficult to achieve. Although nitro protection would be compatible with Boc/Bzl chemistry, the difficulties encountered in the removal of this protecting group makes it more useful as a temporary protection. In fact, it is generally removed by hydrogenolysis after incorporation of the nitroarginine residue,[27] and further synthetic steps are then performed with the protonated guanidino group. In the Z/tBu synthetic strategy in solution, N-terminal chain elongation with Z-Arg(NO_2)-OH is followed by simultaneous hydrogenolytic N^{α}- and $N^{\omega'}$-deprotection to give the protonated arginine residues for the subsequent synthetic steps (Section 2.6.1.1). Hydrogenolysis over palladium black catalyst[35–37] or reduction with tin(II) chloride[38] or titanium(III) chloride[39] proceeds without any particular difficulty for short peptides containing single nitroarginine residues,

for references see p 330

whereas reduction with Zn[40] generally leads to N^{ω}-aminoarginine.[41] Similarly, with multiple nitroarginine residues, and additionally with sequence dependency, difficulties are encountered even with the best catalysts. These difficulties are mainly derived from the long reaction times,[42,43] resulting in partial hydrogenation of tryptophan and phenylalanine residues,[44] as well as reduction of the nitro to the amino group or partial conversion into citrulline and N^{δ}-(formimidoyl)ornithine (Scheme 5).[2,45] Catalytic transfer hydrogenation with cyclohexene,[46] formic acid,[47] or ammonium formate has been proposed as an alternative,[48] but partial hydrogenation of tryptophan is not prevented.[49]

Scheme 5 Side Reactions in the Hydrogenolytic Cleavage of $N^{\omega'}$-Nitroarginine[2,45]



For peptide-chain-elongation steps, the derivatives Z-Arg(NO_2)-OH[50] and Boc-Arg(NO_2)-OH[31,51] are mainly used, which are routinely prepared from commercially available H-Arg(NO_2)-OH by standard procedures using Z-Cl or Boc_2O. Acylation steps with these nitroarginine derivatives are generally performed using the mixed anhydride procedure according to Anderson et al.,[52] although the pentachlorophenyl ester has also been proposed as a useful alternative,[53] as well as DCC/HOBt[54] and the symmetrical anhydride method.[55]

Taking into account the low cost of the nitro derivative and the relatively clean removal of the nitro group by hydrogenolysis, this type of arginine protection still finds application in large-scale synthesis in solution of short arginine peptides and, particularly, of peptidyl arginals,[56,57] whereas alternative protecting groups are generally preferred for syntheses on solid supports.

Z-Arg(NO_2)-OH:[15]
To an ice-cold soln of H-Arg(NO_2)-OH (16.4 g, 75 mmol) in 1 M NaOH (75 mL) were added Z-Cl (13.4 g, 78 mmol) and 1 M NaOH (79 mL) in portions under vigorous stirring. Upon acidification, the precipitate was collected, dissolved in 5% aq $KHCO_3$ and extracted with EtOAc. The aqueous phase was acidified with 6 M HCl and the precipitate was filtered off and recrystallized (EtOH/H_2O); yield: 60–95%; mp 134–136 °C.

Boc-Arg(NO_2)-OH:[58]
Similar to the procedure reported for Boc-N_3,[15] Boc_2O (327 g, 1.5 mol) was added to a soln of H-Arg(NO_2)-OH (232 g, 1.06 mol) in 2 M NaOH (1 L) and dioxane (1 L). After 12 h at rt, the soln was neutralized, concentrated to a small volume, acidified to pH 3 with 1 M HCl, and extracted with EtOAc. The organic layers were washed with H_2O, dried (Na_2SO_4), and the solvent removed. The product was recrystallized (THF); yield: 350 g (84.5%); mp 111–113 °C.

2.6.1.3 N^{δ}- and/or N^{ω}-Alkoxycarbonyl Protection

Depending upon whether mono-N^{ω}-, bis-$N^{\delta,\omega}$-, or bis-$N^{\omega,\omega'}$-arginine derivatives are used, the efficiency of urethane protecting groups in terms of side reactions (see Section 2.6.1.2) varies to a great extent. Moreover, the orthogonality of this type of protection over N^{α}-protection is

generally deficient, a fact which is not as important in synthesis in solution compared to SPPS, where only Fmoc/tBu chemistry a priori is compatible with the (alkoxycarbonyl)-guanidino derivatives.

2.6.1.3.1 *tert*-Butoxycarbonyl Group

Maximal protective efficiency by spatial shielding is expected from bis(*tert*-butoxycarbonyl) derivatives of the guanidino group. Unfortunately, the reaction of N^{α}-protected arginine with excesses of *tert*-butoxycarbonylating agents leads to a mixture of the $N^{\omega,\omega'}$-bis- and $N^{\delta,\omega}$-bis-protected isomers,[59] the $N^{\omega,\omega'}$-derivative being the more stable. This derivative is better accessed via guanidinylation of N^{α}-protected ornithine with suitably substituted reagents (see Section 2.6.1.6).[60–64] Coupling steps with N^{α}-protected Arg(Boc)$_2$ derivatives are generally performed via active esters, e.g. HOBt esters, or by the mixed anhydride or carbodiimide procedure. However, relatively large excesses of the arginine derivative are required due to steric effects on the coupling rates.[66] Additionally, despite the bulkiness of N^{ω},$N^{\omega'}$-bis(*tert*-butoxycarbonyl) protection, δ-lactam formation is still observed.[67] Therefore, at least in SPPS with Fmoc/tBu chemistry, Fmoc-Arg(Boc)-OH is mainly suggested,[68] although this derivative leads to substantial N^{ω}-acylation in the repetitive acylation steps[8] and, thus, to increased conversion of arginine into ornithine peptides in the piperidine-mediated Fmoc-deprotection steps. The N^{ω}-Boc protection is removed by TFA treatments in the presence of scavengers[69,70] to capture the *tert*-butyl cations and *tert*-butyl trifluoroacetate.[71] In summary, the difficult preparation, and the instability and protective insufficiency, as well as the poor coupling rates, of the mono- and bis(Boc)arginine derivatives are not very supportive for their standard use.[7,8,72]

Z-Arg(Boc)-OH:[65]
A soln of Boc-N_3 (2.15 g, 15 mmol) in acetone (15 mL) was added to a soln of Z-Arg-OH (3.1 g, 10 mmol) in acetone (50 mL) and 4 M NaOH (10 mL). The mixture was stirred for 4 h at 40 °C and after 24 h at rt the bulk of the acetone was removed. Upon cooling to 0 °C, the soln was acidified with aq citric acid to pH 3, and the oily precipitate was extracted with $CHCl_3$ (100 mL). The combined $CHCl_3$ extracts were washed with H_2O, dried (Na_2SO_4), and concentrated to an oily residue which was solidified with petroleum ether or Et_2O. The solid was taken into MeOH and insoluble material was filtered off and washed with MeOH. The filtrate was removed and the residue solidified with petroleum ether or Et_2O; yield: 2.5 g (61%); mp 100 °C (dec); $[\alpha]_D^{20}$ −2.44 (*c* 2, MeOH).
After N^{α}-deprotection by standard catalytic hydrogenolysis, other N^{α}-protecting groups can be introduced using standard procedures (see Section 2.1.1).

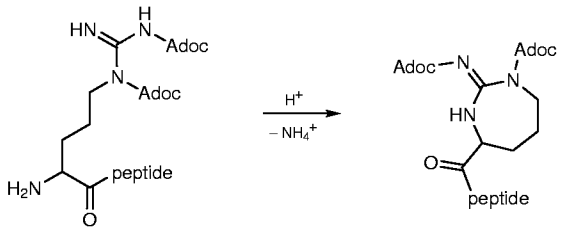
2.6.1.3.2 Adamantyloxycarbonyl Group

Many of the disadvantages of Boc protection (*vide supra*) are apparently avoided with the comparably acid-labile, but sterically more hindered, adamantyloxycarbonyl (Adoc) group. Due to its pronounced hydrophobic character, the solubility of the protected arginine derivatives and related peptides in organic solvents is significantly increased. In contrast to the formation of the bis(Boc)arginine derivative, reaction of N^{α}-protected arginine with adamantyloxycarbonyl chloride[6] or fluoride[73] leads exclusively to the $N^{\delta,\omega}$-bis(Adoc) derivative which provides full protection against δ-lactam formation, but not against $N^{\omega'}$-acylation when strong acylating agents, such as symmetrical anhydrides, are used.[7,8] This Adoc protection is stable toward catalytic hydrogenolysis[6] and 50% piperidine in DMF,[74] thus making this type of arginine derivative compatible with both Z/tBu chemistry in solution and Fmoc/tBu chemistry in SPPS. However, N^{α}-deprotected $N^{\delta,\omega}$-bis(Adoc)arginine is unstable in MeOH and cyclizes to the 4-carboxy-2-imino-1,3-diazacycloheptane compound (Scheme 6), particularly in the presence of weak acids as used for hydrogenolysis of the related N^{α}-benzyloxycarbonyl derivative.[6,75] The extent of this side reaction is strongly time dependent

for references see p 330

and, in the fast Fmoc-cleavage steps, is negligible; however, in the time-demanding hydrogenation steps in solution it occurs to an extent that considerably limits the usefulness of this approach.

Scheme 6 Cyclization of $N^{\delta,\omega}$-Bis(Adoc)arginine Derivatives to 4-Carboxy-2-imino-1,3-diazacycloheptane Compounds[6,75]



The strong solubilizing effect of the Adoc group can also be exploited by its introduction, with Adoc-F, into arginine residues of suitably protected peptide derivatives.[76] Generally, Adoc-Cl[6] and Adoc-F[73] are used for the preparation of the $N^{\delta,\omega}$-bis(Adoc)arginine derivatives, but Adoc-OSu has also been proposed.[6] The yields of the derivatives can certainly be improved if silylated N^{α}-protected arginine is employed as starting material in a way similar to that proposed for the synthesis of Z-Arg(Z)$_2$-OH (see Section 2.6.1.3.3).[77]

For chain-elongation steps with Fmoc-Arg(Adoc)$_2$-OH standard activation procedures are applied, such as active esters[6,78] and the symmetric anhydride method,[74] but even for the use of more powerful acylating agents, e.g. phosphonium and uronium salts, there are no contraindications. Final deprotection is performed using TFA under standard conditions of Fmoc/tBu chemistry. Properties similar to the Adoc group have been reported for isobornyloxycarbonyl protection,[79] whereas increased acid lability is obtained with the 1-(1-adamantyl)-1-methylethoxycarbonyl group.[80]

Z-Arg(Adoc)$_2$-OH:[73]
To a suspension of Z-Arg-OH (9.3 g, 30 mmol) in dioxane/H_2O (2:1, 20 mL) was slowly added Adoc-F[81] (23.8 g, 120 mmol) in dioxane (10 mL) over 2 h at rt. The mixture was stirred for an additional 2 h while keeping the pH at 12 with 4 M NaOH. The product was filtered, washed with a large amount of Et_2O, and suspended in H_2O. The suspension was acidified to pH 2.5 with 5 M citric acid and extracted with Et_2O. The combined extracts were washed with 0.1 M citric acid and H_2O, dried ($MgSO_4$), and the solvent removed. The residue was dissolved in hot EtOH, insoluble material was filtered off, and after addition of hot H_2O the product crystallized on cooling; yield: 17.2 g (86%); mp 118–120 °C; $[\alpha]_D^{22}$ +17.7 (*c* 1, $CHCl_3$).

H-Arg(Adoc)$_2$-OH:[73]
Z-Arg(Adoc)$_2$-OH (6.6 g, 10 mmol) was hydrogenated over palladium black (0.5 g) in CH_2Cl_2 containing 7% formic acid. After 3 h the catalyst was filtered off and the soln concentrated. The oily residue was solidified with Et_2O, and the product was redissolved in MeOH and immediately concentrated at rt. The precipitate was collected, and washed with Et_2O containing 2% MeOH, and Et_2O; yield: 4.4 g (82%); mp 174–177 °C; $[\alpha]_D^{22}$ –30.3 (*c* 1, $CHCl_3$).

Fmoc-Arg(Adoc)$_2$-OH:[73]
To a suspension of H-Arg(Adoc)$_2$-OH (2.65 g, 5 mmol) in MeOH (25 mL) containing TEA (0.76 mL, 5.5 mmol) was added dropwise Fmoc-OSu (1.69 g, 5 mmol) in dioxane (6 mL). After 1 h at rt, the solvent was removed and the residue dissolved in EtOAc. The soln was washed with 0.1 M citric acid and H_2O, dried ($MgSO_4$), and the solvent removed. The oily residue crystallized under petroleum ether; yield: 3.5 g (93%); mp 154–156 °C; $[\alpha]_D^{22}$ +19.2 (*c* 1, $CHCl_3$).

2.6.1.3.3 Benzyloxycarbonyl Group

Reaction of N^{α}-protected arginine with benzyl chloroformate leads to the $N^{\delta,\omega}$-bis(benzyloxycarbonyl) derivative from which the N^{ω}-mono(benzyloxycarbonyl) compound can be prepared by base treatment.[82,83] Since the latter derivative is very prone to δ-lactam formation,[82] it has not been used in peptide synthesis. Basicity and nucleophilicity of the bis-protected guanidino group are nearly completely masked,[83] but $N^{\omega'}$-acylation by symmetric anhydrides is not entirely prevented. Due to the instability of the $N^{\delta,\omega}$-bis(benzyloxycarbonyl) derivative upon N^{α}-deprotection, e.g. of Boc-Arg(Z)$_2$-OH, only the Z-Arg(Z)$_2$-OH derivative is used in chain-elongation steps where subsequent hydrogenolysis leads to the protonated guanidino group for further synthetic steps (see Scheme 4). This arginine derivative can be conveniently applied in the preparation of intermediate fragments with an unprotected C-terminus where aminoacylation has to be performed with active esters. For this purpose, the related *N*-hydroxysuccinimide ester[84,85] is generally used.

The difficult preparation of Z-Arg(Z)$_2$-OH, and thus the related low yields associated with earlier methods,[15,83,86] are conveniently bypassed if persilylated N^{α}-(benzyloxycarbonyl)arginine is acylated with benzyl chloroformate.[77] This procedure is also well-suited for the preparation of Boc-Arg(Z)$_2$-OH.[77] Alternatively, the $N^{\delta,\omega}$-bis-protected derivatives are also obtained in good yields via guanidinylation of N^{α}-protected ornithine with N,N′-bis-protected guanidinylating reagents (see Section 2.6.1.6).[60–64] For peptide acylation steps, both Z-Arg(Z)$_2$-OH and Boc-Arg(Z)$_2$-OH are mainly used as active esters,[15,87] mixed anhydrides or carbodiimides.[82]

Z-Arg(Z)$_2$-OH:[77]
To a suspension of Z-Arg-OH (15.0 g, 48.7 mmol) in 1,2-dichloroethane (125 mL) and DIPEA (25.5 mL, 146.1 mmol) was added TMSCl (18.7 mL, 146.1 mmol) under a N_2 atmosphere and the mixture was kept at 40 °C for 1.5 h. It was then cooled to 0 °C and additional DIPEA (25.5 mL, 146.1 mmol) was added, followed by Z-Cl (21.8 mL, 146.1 mmol). The soln was stirred at 0 °C for 20 min, after which the mixture was allowed to warm to rt. Stirring was continued for an additional 4 h, the soln was acidified with 1 M HCl to pH 2 and the product was extracted with CH_2Cl_2 (3 × 100 mL); the extracts were combined concentrated and the residue recrystallized (MeOH/H_2O 4:1); yield: 30.4 g (64%); mp 139 °C; $[\alpha]_D^{25}$ +17.1 (*c* 1, $CHCl_3$).
Z-Arg(Z)$_2$-OH is converted into the *N*-hydroxysuccinimide ester by reaction with HOSu (1.1 equiv) and DCC (1 equiv) in DMF/EtOAc (2:8), followed by recrystallization (EtOAc/petroleum ether); yield: 80–90%; mp 90–91 °C; $[\alpha]_D^{20}$ −10.2 (*c* 2, dioxane).[85]

Z-Arg(Z)$_2$-Glu(OtBu)-Arg-Lys(Boc)-OH (Scheme 4):[23]
To an ice-cold soln of H-Glu(OtBu)-Arg(HCl)-Lys(Boc)-OH (20.5 g, 32.8 mmol) in DMF (800 mL) was added Z-Arg(Z)$_2$-OSu (24.3 g, 36.1 mmol) and, after stirring at rt for 48 h, TEA (4.56 mL, 32.8 mmol) was added. The mixture was concentrated and the crude product was precipitated with H_2O and recrystallized (90% aq EtOH); yield: 27.5 g (71%, as the dihydrate); mp 200–202 °C (dec); $[\alpha]_D^{20}$ −11.1 (*c* 1, 80% AcOH).

H-Arg(HBr)-Glu(OtBu)-Arg(HBr)-Lys(Boc)-OH · HBr (Scheme 4):[23]
Z-Arg(Z)$_2$-Glu(OtBu)-Arg-Lys(Boc)-OH (25.4 g, 21.5 mmol) was hydrogenated over palladium black in DMF (600 mL) with concomitant titration at pH 4.0 with 1 M HBr. The catalyst was filtered off, the filtrate was removed, and the residue reprecipitated from EtOAc/Et_2O; yield: 23 g (99%); mp 90–95 °C; $[\alpha]_D^{20}$ +4.6 (*c* 1, DMF).

Z-Ala-Pro-Arg(HBr)-Glu(OtBu)-Arg(HBr)-Lys(Boc)-OH (Scheme 4):[23]
To an ice-cold soln of H-Arg(HBr)-Glu(OtBu)-Arg(HBr)-Lys(Boc)-OH · HBr (10 g, 10.1 mmol) in DMF (200 mL) were added TEA (1.4 mL, 10.1 mmol) and Z-Ala-Pro-OSu (4.3 g, 10.2 mmol). After 20 h at rt, the bulk of the solvent was removed and the residue was taken up in H_2O. The aqueous soln was extracted with EtOAc (2 ×) and then with BuOH (3 ×). The combined BuOH extracts were washed once with H_2O, then concentrated to a small volume. The product was precipitated with Et_2O; yield: 10.24 g (84%); mp 140–145 °C (dec); $[\alpha]_D^{20}$ −34.1 (*c* 1, DMF).

for references see p 330

2.6.1.3.4 Allyloxycarbonyl Group

Protection of the guanidino group of Fmoc-Arg-OH or Boc-Arg-OH as the bis(Aloc) derivative has been proposed.[166] However, the stability of the N^{ω}-Aloc groups is not satisfactory within the Fmoc/tBu strategy, since these are removed progressively in the repetitive piperidine treatments.[166]

Boc-Arg(Aloc)$_2$-OH·DCHA:[166]
To a stirred suspension of Boc-Arg-OH (22 g, 0.08 mol) in toluene (400 mL) was added Aloc-Cl (23 g, 0.19 mol) followed by BTMSA (63 g, 0.31 mol) and the mixture was then heated at 60 °C for 3 h to give a clear yellow soln. This soln was poured slowly into cold H_2O (500 mL) while the pH was maintained between 3 and 7 by addition of 1 M NaOH. The crude product was obtained by standard workup as a clear oil which was converted into the DCHA salt; yield: 30 g (60%); mp 84–86 °C; $[\alpha]_D^{20}$ +14.7 (*c* 1, MeOH).

2.6.1.4 N^{ω}-Arylsulfonyl Protection

Currently, arylsulfonyl-based guanidino protecting groups are the most widely applied, although such protection does not completely prevent δ-lactam formation on activation of the C-terminus for coupling steps. The common feature of these protecting groups is their acid lability which can be modulated by varying the pattern of electron-withdrawing and electron-donating substituents on the aryl moiety. As shown in Table 1, a large number of arylsulfonyl groups have been developed with acid labilities that range from TFA stable and HF cleavable to TFA cleavable.

Table 1 Arylsulfonyl Protecting Groups for the Arginine Guanidino Group[28,72,88–102]

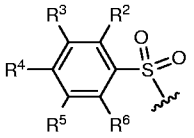
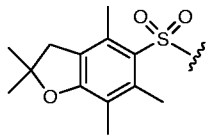
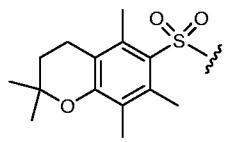
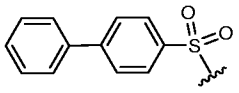
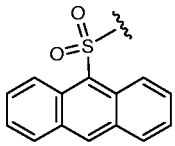
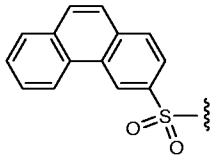
	Protecting Group	Ref
		
R^4 = Me	4-toluenesulfonyl (Tos)	[88,89]
$R^2 = R^4 = R^6$ = Me	2,4,6-trimethylbenzenesulfonyl (Mts)	[90,91]
$R^2 = R^3 = R^4 = R^5 = R^6$ = Me	2,3,4,5,6-pentamethylbenzenesulfonyl (Pme)	[92]
$R^2 = R^4 = R^6$ = iPr	2,4,6-triisopropylbenzenesulfonyl	[28]
R^2 = OMe; $R^4 = R^6$ = Me	2-methoxy-4,6-dimethylbenzenesulfonyl (iMds)	[93]
R^4 = OMe	4-methoxybenzenesulfonyl (Mbs)	[94]
R^4 = OMe; $R^2 = R^6$ = Me	4-methoxy-2,6-dimethylbenzenesulfonyl (Mds)	[92,95,96]
R^4 = OMe; $R^2 = R^3 = R^6$ = Me	4-methoxy-2,3,6-trimethylbenzenesulfonyl (Mtr)	[72]
R^4 = OMe; $R^2 = R^3 = R^5 = R^6$ = Me	4-methoxy-2,3,5,6-tetramethylbenzenesulfonyl	[92]
R^3 = tBu; R^4 = OMe	3-*tert*-butyl-4-methoxybenzenesulfonyl	[97]
$R^3 = R^5$ = tBu; R^4 = OMe	3,5-di-*tert*-butyl-4-methoxybenzenesulfonyl	[28]
$R^2 = R^4 = R^6$ = OMe	2,4,6-trimethoxybenzenesulfonyl	[92]
	2,2,4,6,7-pentamethyldihydrobenzofuran-5-sulfonyl (Pbf)	[98]
	2,2,5,7,8-pentamethylchroman-6-sulfonyl (Pmc)	[99–101]

Table 1 (cont.)

	Protecting Group	Ref
	biphenyl-4-sulfonyl (Bip)	[97]
	anthracene-9-sulfonyl (Ans)	[102]
	phenanthrene-3-sulfonyl (Phen)	[28]

Of the various arylsulfonyl derivatives proposed so far, only a restricted number have found widespread application, being commercially available and essentially fulfilling the requirements of the different synthetic strategies. This series of arylsulfonyl protecting groups began with the tosyl (Tos) derivative,[88] which is still largely used in the synthesis of peptides in solution[103,104] and on solid supports,[105] as well as in convergent strategies[106] when Boc/Bzl chemistry is applied. In fact, it is perfectly stable to repeated treatment with TFA, but is cleaved quantitatively upon exposure to HF/anisole[107] and TfOH/TFA/methyl phenyl sulfide.[108] Less common, is cleavage of the tosyl group using sodium in liquid ammonia,[88] because of serious side reactions,[1] and acidolytic removal with boron tris(trifluoroacetate).[109]

As shown in Table 2, the 4-methoxybenzenesulfonyl (Mbs) group[94] is similarly stable to TFA treatments, but quantitatively cleaved with HF, while the 2,4,6-trimethylbenzenesulfonyl (Mts) group[90,91] is less suitable for the synthesis of larger peptides due to its lower stability toward TFA treatment. Since the 4-methoxy-2,3,6-trimethylbenzenesulfonyl (Mtr) group[72] is even more acid labile and can be cleaved at acceptable rates by suitable combinations of TFA with scavengers,[115,116] it has been largely used in SPPS with the Fmoc/tBu strategy; however, in the case of multiple-arginine-containing peptides, quantitative cleavage is difficult.[99,116] Therefore, the 2,2,5,7,8-pentamethylchroman-6-sulfonyl (Pmc)[99–101] and 2,2,4,6,7-pentamethyldihydrobenzofuran-5-sulfonyl (Pbf)[98] groups are currently generally applied for synthesis on solid supports using the Fmoc/tBu strategy as these two protecting groups are significantly more acid labile than the Mtr group. The Pbf group gives rise to lower levels of tryptophan sulfonation (*vide infra*) than the Pmc group, particularly when used in combination with N^{in}-Boc-protected tryptophan and silane scavengers.[121] In this context, attempts to improve the properties of the Pmc group, by varying the substituents, were unsuccessful.[122]

for references see p 330

Table 2 Acidolytic Cleavage of N^{ω}-(Arylsulfonyl)arginine Derivatives

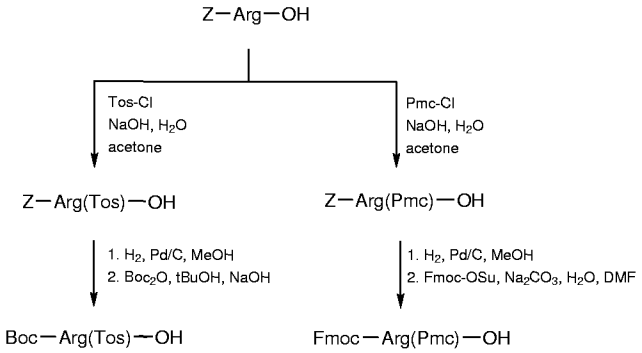
Tos	Mbs	Mts	Mtr	Pmc	Pbf
HF[107]	HF[94]	HF[90,91]	TFA/PhSMe[115]	TFA/anisole or PhSMe[100]	reagent B[121]
TfOH/TFA/ PhSMe[108]	MsOH/TFA/ anisole or MsOH/TFA/ anisole/PhSMe/*o*-cresol[111]	MsOH/ anisole[90,91]	reagent K[a][116]	TFA/5% H_2O/3% anisole/1% ethane-1,2-dithiol[101]	reagent K[121]
1M TMSOTf/ 1M PhSMe in TFA[110]	1M TMSOTf/ 1M PhSMe in TFA[110,112]	TfOH[90,91]	HBF_4/TFA[117]	TFA/10% MeOH/3% EtSMe/Et_3SiH (15 equiv)[69]	reagent R[98]
		TfOH/ TFA[90,111,113]		reagent K[a][118]	
		TMSBr/TFA/ PhSMe[114]		reagent B[b][119]	
		1M TMSOTf/ 1M PhSMe in TFA[110,112]		reagent R[c][120]	

[a] Reagent K: TFA/PhOH/H_2O/PhSMe/ethane-1,2-dithiol (82.5:5:5:5:2.5).[116]
[b] Reagent B: TFA/PhOH/H_2O/iPr_3SiH (88:5:5:2).[119]
[c] Reagent R: TFA/PhSMe/ethane-1,2-dithiol/anisole (90:5:3:2).[120]

Besides HF and TFA, the two extremes of acid treatment generally employed in the final deprotection step, alternative acids have been proposed, such as MsOH and TfOH, particularly in the context of arginine protection (Table 2). These acids, however, are known to generate serious side reactions such as aspartimide formation[92,123,124] and an N → O shift at serine or threonine residues.[125] Moreover, independent of the type of arylsulfonyl group employed, cations originating from the acid-mediated cleavage lead to alkylation and arylsulfonylation of reactive side chains, as present in tryptophan and tyrosine residues.[90,111,126–129] Additionally, sulfation of hydroxy groups is even observed,[130] as well as sulfated guanidino groups.[131] Thus, efficient scavengers are required as additives, not only to drive cleavage of the arylsulfonyl groups under mild conditions to completion, but also to capture reactive intermediates and, thus, suppress side reactions (see Section 4.3.4). Among the scavengers used, methyl phenyl sulfide and *o*- or *m*-cresol are more efficient than anisole,[90,91,111,123,126] and the addition of trialkylsilanes is highly recommended.[69] Similarly, N^{in}-protection efficiently serves to prevent alkylation of tryptophan residues.[121,132,133]

Since the N^{ω}-arylsulfonyl groups are resistant to catalytic hydrogenolysis,[88] N^{α}-protected N^{ω}-(arylsulfonyl)arginine derivatives are generally prepared according to Scheme 7, using N^{α}-(benzyloxycarbonyl)arginine as the key intermediate. This is obtained in good yield as a homogeneous material if care is taken to keep the pH within the range 9–11 during its preparation with benzyl chloroformate, to avoid concomitant acylation of the guanidino group (Section 2.6.1.3.2).[97,134]

Scheme 7 Synthesis of N^{α}-Protected N^{ω}-(Arylsulfonyl)arginine Derivatives



The arylsulfonyl groups are then introduced into Z-Arg-OH by reaction with the related arylsulfonyl chlorides where formation of side products, such as $N^{\omega,\omega'}$-bis-derivatives and δ-lactams, is known to be strongly pH dependent.[135,136] Purification is generally achieved by crystallization as the cyclohexylamine or dicyclohexylamine salt. After hydrogenolytic removal of the benzyloxycarbonyl group, the protection of choice of the N^{α}-amino group is carried out following standard procedures (see Section 2.1.1). The intermediates most commonly used are listed in Table 3; for subsequent coupling steps standard procedures are applied, such as DCC/HOSu or HOBt,[91,99,137,138] active esters, generally pentafluorophenyl esters,[98,138–140] and uronium and phosphonium salts.[98,140] Side reactions are not totally suppressed and partial conversion into δ-lactams occurs. Furthermore, the bulkiness of the side-chain protecting group generally substantially affects the coupling rates.

Table 3 N^{ω}-(Arylsulfonyl)arginine Derivatives Most Commonly Used in Peptide Synthesis[3,7,90,141]

Compound	mp (°C)	$[\alpha]_D^{25}$	Ref
Boc-Arg(Tos)-OH	96–97	−3.6 (*c* 4, DMF)	[141]
Boc-Arg(Mts)-OH[a]	122–125	+15.1 (*c* 0.5, MeOH)	[90]
Fmoc-Arg(Mtr)-OH	118–120	+7.9 (*c* 0.5, MeOH	[7]
Fmoc-Arg(Pmc)-OH	80 (dec)	+1.0 (*c* 1, MeOH)	[3]
Fmoc-Arg(Pbf)-OH[b]		−5.5 (*c* 1, DMF)	[c]

[a] Cyclohexylamine salt.
[b] Diisopropyl etherate.
[c] Commercial sample (Novabiochem).

Z-Arg(arylsulfonyl)-OH; General Procedure:[94]
To an ice-cold soln of Z-Arg-OH (55.5 g, 180 mmol) in 4 M NaOH (180 mL) and acetone (1.3 L) was added dropwise the arylsulfonyl chloride (360 mmol), dissolved in acetone (300 mL), over a period of 30 min under vigorous stirring. The mixture was stirred for 2 h at 0 °C, and for an additional 2 h at rt, then was acidified with citric acid, and concentrated. The aqueous layer was extracted with EtOAc and the combined extracts were washed twice with H_2O, and then re-extracted with 5% $NaHCO_3$ (3 × 400 mL). After acidification of the aq extracts with citric acid, the precipitated oil was dissolved into EtOAc. The organic layer was washed with H_2O, dried (Na_2SO_4), and the solvent removed. The oily residue was again dissolved in EtOAc (600 mL) and after addition of dicyclohexylamine (36.0 mL) or cyclohexylamine (20.7 mL) the product crystallized on standing in the cold; yield: 60–70%.

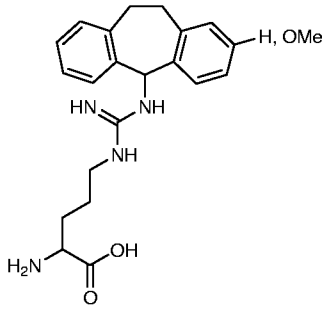
H-Arg(arylsulfonyl)-OH; General Procedure:[94]
Z-Arg(arylsulfonyl)-OH · DCHA (20 mmol) was partitioned between EtOAc and 5% citric acid and, after washing with H_2O, the organic layer was dried (Na_2SO_4) and concentrated. The residue was hydrogenated over Pd/C in MeOH. The catalyst was removed by filtration and the filtrate was concentrated. The product was precipitated with Et_2O and in some cases recrystallized (H_2O); yield: 80–95%.
N^{α}-Protection of H-Arg(arylsulfonyl)-OH was performed according to standard procedures (see Section 2.1.1).

for references see p 330

2.6.1.5 Other Protecting Groups

Based on early studies of the 2-carboxybenzoyl group,[42] the $N^{\omega,\omega'}$-bis[2-(*tert*-butoxycarbonyl)-3,4,5,6-tetrachlorobenzoyl]arginine derivative was proposed[142] which can be cleaved by a mild, two-stage acid treatment and has found application in particular cases. Among aryl-type protection, the trityl group has even been reexamined for use in the Fmoc/tBu strategy. The related Fmoc-Arg(Trt)-OH derivative is poorly soluble in the solvents generally employed in SPPS; thus, coupling rates are exceedingly slow and unsatisfactory for synthesis on solid supports.[1,143] In contrast, the 5*H*-dibenzo[*a*,*d*]cycloheptene-based groups, which have been recently proposed for N^{ω}-protection of arginine (Scheme 8) in combination with the N^{α}-Fmoc derivative, are more promising.[144] Coupling of the N^{α}-Fmoc-protected N^{ω}-(5*H*-dibenzo[*a*,*d*]cyclohepten-5-yl)-, N^{ω}-(10,11-dihydro-5*H*-dibenzo[*a*,*d*]cyclohepten-5-yl)-, or N^{ω}-(2-methoxy-10,11-dihydro-5*H*-dibenzo[*a*,*d*]cyclohepten-5-yl)arginine compound proceeds at an acceptable rate with phosphonium salts, while quantitative acidolytic cleavage of this aryl-type protection can be achieved with 25% TFA in CH_2Cl_2 within 30 minutes.[144] More experience, however, is required with these new aryl protecting groups in order to judge their usefulness in peptide synthesis.

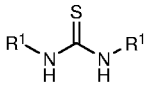
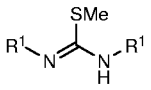
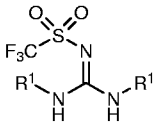
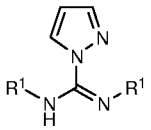
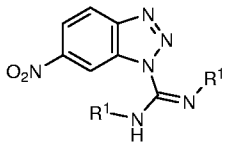
Scheme 8 5*H*-Dibenzo[*a*,*d*]cycloheptene-Derived N^{ω}-Protection of Arginine[144]



2.6.1.6 Arginine Precursors

The synthesis of arginine peptides by postsynthetic guanidinylation of the related ornithine precursors was previously a method of choice as N^{ω}-protection of arginine was not as advanced as nowadays.[145–148] For this purpose, the ornithine or lysine peptides are reacted at pH 10 with the guanidinylating reagent, leading to arginine or homoarginine peptides. This approach, which is also named the Orn → Arg strategy, is efficient in the case of short peptides containing single ornithine or lysine residues, but it exhibits serious limitations for the synthesis of multiple-arginine-containing peptides.[149] Moreover, if arginine has to be produced from ornithine in the presence of lysine residues, a selective protection of the side-chain groups of these two residues is essential. Among the known reagents are cyanamide, *O*-methylisourea hydrogen sulfate, *S*-methylisothiourea hydrogen sulfate, 3,5-dimethylpyrazole-1-carboximidamidium nitrate, and pyrazole-1-carboximidamide hydrochloride,[150] but only the latter reagent is sufficiently reactive to allow guanidinylation even on solid support.[149] Alternatively, the benzotriazole-1-carboximidamidium tosylate was proposed to exploit the good leaving group properties of benzotriazole.[167] In the context of peptide synthesis this precursor approach may be more useful for the preparation of suitably protected arginine, and arginine homologues and analogues. In fact, with protected guanidinylating reagents, such as those listed in Table 4, amino functions of natural and nonnatural amino acids are replaced with N-mono- or, even better, with N,N′-bis-protected guanidino groups to provide useful intermediates for synthesis in solution and on solid support.

Table 4 Guanidinylating Reagents[62–64,151–157,168]

Reagent	R^1	Ref
N,N′-bis(alkoxycarbonyl)thiourea		
	Boc Z	[151,152] [151]
N,N′-bis(alkoxycarbonyl)-*S*-methylisothiourea		
	Boc Z	[153] [62,151,153,154]
1,3-bis(alkoxycarbonyl)-2-triflylguanidine		
	Boc Z	[63,155] [63,155]
N,N′-bis(alkoxycarbonyl)pyrazole-1-carboximidamide		
	Boc Z	[156,157] [64,156]
N,N′-bis(alkoxycarbonyl)-6-nitrobenzotriazole-1-carboximidamide		
	Boc	[168]

2.6.1.6.1 *N,N′*-Bis(alkoxycarbonyl)pyrazole-1-carboximidamide

N-Mono- or N,N′-bis-protection of pyrazole-1-carboximidamide as the *tert*-butoxycarbonyl or benzyloxycarbonyl derivatives leads to reagents (Table 4)[64,156,157] that allow preparation of the related N^{ω}-mono- or $N^{\omega,\omega'}$-bis-protected arginine derivatives from the appropriate ornithine precursors. *N,N′*-Bis(*tert*-butoxycarbonyl)-4-nitro-1*H*-pyrazole-1-carboximidamide was found to be an even more efficient guanidinylating reagent because of the enhanced leaving group character of the 4-nitropyrazole.[169]

Z-Arg(Boc)$_2$-OH:[157]
To a stirred soln of *N,N′*-bis(*tert*-butoxycarbonyl)pyrazole-1-carboximidamide (1.55 g, 5 mmol) and DIPEA (1.05 mL, 6 mmol) in MeCN/H_2O (95:5, 25 mL) was added Z-Orn-OH (1.33 g, 5 mmol) and the mixture was stirred for 2 h under reflux. The solvent was removed under reduced pressure and the residue dissolved in EtOAc (50 mL). The organic phase was washed with 5% HCl and brine, then dried ($MgSO_4$). The solvent was removed; yield: 1.78 g (70%).

2.6.1.6.2 *N,N′*-Bis(alkoxycarbonyl)thiourea and *N,N′*-Bis(alkoxycarbonyl)-*S*-methylisothiourea

The reaction of *N,N′*-bis(*tert*-butoxycarbonyl)thiourea (see Table 4) with amines in the presence of mercury(II) chloride or water-soluble carbodiimide provides an efficient method for preparation of the corresponding bis-protected guanidines.[151,152,158] An adaptation of this

for references see p 330

reaction to the guanidinylation of Z-Orn-OMe or Z-Lys-OMe provides the corresponding $N^{\omega,\omega'}$-bis(Boc)arginine and -homoarginine derivatives in over 70% yield.[158] The reagent is also suitable for guanidinylation on solid support,[159] but for this purpose a more efficient reaction is obtained when the Mukaiyama reagent is present.[160]

Similarly active in terms of guanidinylation are *N,N'*-bis(benzyloxycarbonyl)-[62,151,153,154] and *N,N'*-bis(*tert*-butoxycarbonyl)-*S*-methylisothiourea[153] (Table 4) which can be used to generate Z-Arg(Z)$_2$-OH[62] and Z-Arg(Boc)$_2$-OH[60] from the corresponding ornithine derivative in good yield. Moreover, even these reagents are sufficiently reactive to guanidinylate amine groups in solid-phase synthesis.[161] In analogy to *N,N'*-bis(Boc)- and *N,N'*-bis(Z)-*S*-methylisothiourea, the *N*-Mtr- and *N*-Pmc-*S*-methylisothiourea derivatives have also been proposed for the conversion of side-chain amino groups into the related N^{ω}-Mtr- and N^{ω}-Pmc-protected guanidino groups.[162] This approach is useful in SPPS, using the Fmoc/tBu strategy, of compounds containing arginine analogues.

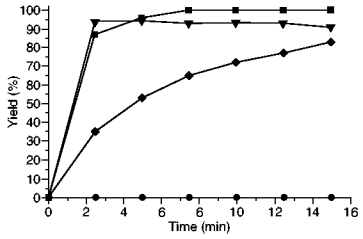
Z-Arg(Boc)$_2$-OH:[60]
A soln of *N,N'*-bis(*tert*-butoxycarbonyl)-*S*-methylisothiourea (17.76 g, 61.18 mmol) and Triton-B (10.77 g, 64.40 mmol) in MeOH (20 mL) was added under stirring to Z-Orn-OH (17.15 g, 64.64 mmol) in DMSO (200 mL). The mixture was kept at +10 °C under a N_2 atmosphere, and after 40 h a soln of $KHSO_4$ (48 g) and NaCl (semi-sat.) in H_2O (1.2 L) was added. The precipitate was partitioned between H_2O and EtOAc, and the organic layer was washed with sat. NaCl until pH 5–6. The organic layer was dried (Na_2SO_4) and concentrated. The residue was dissolved in Et_2O (80 mL). After addition of cyclohexylamine (3.95 mL, 34.46 mmol), the product crystallized in the cold; yield: 19.02 g (90%); mp 124–125 °C; $[\alpha]_D^{22}$ +9.34 (*c* 1, MeOH).
After hydrogenation in MeOH, the resulting H-Arg(Boc)$_2$-OH [yield: 99%; mp 230 °C (dec); $[\alpha]_D^{20}$ −12.44 (*c* 1, MeOH)] was converted into Fmoc-Arg(Boc)$_2$-OH by reaction with Fmoc-OSu under standard conditions; yield: 99.6%; mp 115 °C (dec); $[\alpha]_D^{20}$ +3.2 (*c* 1, MeOH).

Z-Arg(Z)$_2$-OH:[62]
To a suspension of Z-Orn-OH · HCl (1.21 g, 4 mmol) in CH_2Cl_2 (30 mL) was added TEA (0.56 mL, 4 mmol) followed by TMSCl (0.51 mL, 4 mmol) under vigorous stirring at 0 °C. After 1 h at rt, the soln was cooled in an ice bath, and TEA (1.12 mL, 8 mmol) was added followed by *N,N'*-bis(benzyloxycarbonyl)-*S*-methylisothiourea (1.79 g, 5 mmol). The mixture was refluxed for 12 h and, after cooling to rt, MeOH (5 mL) was added. The solvent was evaporated and the residue was partitioned between EtOAc and 5% aq $KHSO_4$. The organic layer was washed with H_2O, dried (Na_2SO_4), and concentrated. The product was precipitated from EtOAc with petroleum ether, and recrystallized (MeOH); yield: 2.01 g (87%); $[\alpha]_D^{25}$ +16.41 (*c* 1, $CHCl_3$); lit.[77] $[\alpha]_D^{25}$ +17.1 (*c* 1, $CHCl_3$).

2.6.1.6.3 1,3-Bis(alkoxycarbonyl)-2-triflylguanidine

A comparative kinetic analysis of the rate of guanidinylation of benzylamine with 1,3-bis(benzyloxycarbonyl)- or 1,3-bis(*tert*-butoxycarbonyl)-2-triflylguanidine[63,155] with the rates of those reagents discussed in Sections 2.6.1.6.1 and 2.6.1.6.2 clearly indicated that the triflyl derivatives are the most powerful reagents (Figure 1),[63] well-suited for guanidinylation of simple amino acid derivatives as well as of peptides in solution[63] or linked to resins.[155] In the latter case, a selective protection of the N^{δ}-amino group of the ornithine residues is required.

Figure 1 Rates of Guanidinylation of Benzylamine with *N,N'*-Bis(*tert*-butoxycarbonyl)-*S*-methylisothiourea (●), *N,N'*-Bis(*tert*-butoxycarbonyl)pyrazole-1-carboximidamide (▼), *N,N'*-Bis(*tert*-butoxycarbonyl)thiourea in the Presence of the Mukaiyama Reagent (♦), and 1,3-Bis(*tert*-butoxycarbonyl)-2-triflylguanidine (■)



To allow for alcoholic functions to be used as precursors of guanidino groups in arginine analogues, 1,3-bis(benzyloxycarbonyl)- and 1,3-bis(*tert*-butoxycarbonyl)guanidine,[163] as well as 1,2,3-tris(benzyloxycarbonyl)- and 1,2,3-tris(*tert*-butoxycarbonyl)guanidine,[155] are used under Mitsunobu conditions to convert alcoholic functions into protected guanidines, and thus to produce the related arginine analogues from hydroxyamino acids.

A highly reactive guanidinylating reagent has also been proposed with the traceless reagent derived from 1-(*tert*-butoxycarbonyl)-2-triflylguanidine which is linked at N3 via a benzyloxycarbonyl moiety to the resin.[170]

Boc-D-Val-Leu-Arg(Boc)$_2$-Gly-OEt:[63]
To an argon-sat. soln of Boc-D-Val-Leu-Orn-Gly-OEt (107 mg, 0.20 mmol) in CH_2Cl_2 (5 mL) were added TEA (28 μL, 0.20 mmol) and 1,3-bis(*tert*-butoxycarbonyl)-2-triflylguanidine (70 mg, 0.18 mmol). The reaction was allowed to stir at rt for 4 h. It was then washed with 1 M $NaHSO_4$, followed by 5% $NaHCO_3$ and H_2O. The organic layer was concentrated and chromatographed (silica gel, gradient from CH_2Cl_2 to 5% MeOH in CH_2Cl_2); yield: 120 mg (86%).

2.6.1.6.4 *N,N'*-Bis(alkoxycarbonyl)benzotriazole-1-carboximidamide

Because of the enhanced electrophilicity of the amidine carbon in *N,N'*-bis(alkoxycarbonyl)carboximidamide derivatives,[157] the *N,N'*-bis(*tert*-butoxycarbonyl)-1*H*-benzotriazole-1-carboximidamide and *N,N'*-bis(*tert*-butoxycarbonyl)-6-nitro-1*H*-benzotriazole-1-carboximidamide derivatives were proposed.[168] The latter compound proved to be the most reactive guanidinylating reagent for sterically hindered amines in solution and on solid support.

***N,N'*-Bis(*tert*-butoxycarbonyl)-6-nitro-1*H*-benzotriazole-1-carboximidamide:**[171]
To a stirred soln of 6-nitro-1*H*-benzotriazole (630 mg, 3.5 mmol), *N,N'*-di(*tert*-butoxycarbonyl)thiourea (980 mg, 3.5 mmol) and TEA (1.62 mL, 11.55 mmol) in dry DMF (12 mL) at 0 °C was added $HgCl_2$ (1.045 g, 3.85 mmol). After 4 h at rt, the reaction was concentrated under reduced pressure, diluted with EtOAc (100 mL), and filtered through a Celite pad. The filtrate was washed with H_2O (20 mL), 5% aq Na_2CO_3 (20 mL), H_2O (20 mL), brine (20 mL), and dried (Na_2SO_4). The solvent was removed and the product was purified by flash chromatography (90 g silica gel, hexane/*tert*-butyl methylether 3:1) and crystallized from *tert*-butyl methylether/hexane; yield: 700 mg (50%); mp 126–128 °C.

***N,N'*-Bis(*tert*-butoxycarbonyl)-*N''*-phenylguanidine:**[171]
Aniline (44 μL, 0.48 mmol) was added at rt to a stirred soln of *N,N'*-bis(*tert*-butoxycarbonyl)-6-nitro-1*H*-benzotriazole-1-carboximidamide (163 mg, 0.4 mmol) in dry CH_2Cl_2 (2 mL) at rt. After 2 h, the mixture was diluted with diisopropyl ether (12 mL) and washed with 0.25 M aq $KHSO_4$ (3 mL), H_2O (2×2 mL), and brine (1 mL). The organic phase was dried (Na_2SO_4), filtered, and concentrated. The crude product was purified by flash chromatography (18 g silica gel, hexane/*tert*-butyl methylether 4:1); yield: 134 mg (100%); mp 131–133 °C; lit.[155]; 132–134 °C.

for references see p 330

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2.6.2 Imidazole Group

J. H. Jones

The synthesis of histidine-containing peptides without protection of the imidazole side chains is feasible, but it can be inconvenient and unsatisfactory because of the basic, nucleophilic and hydrophilic character of such side chains. *im*-Nitrogens can react with electrophilic reagents such as *N,N'*-dicyclohexylcarbodiimide,[1] catalyze undesirable acyl-transfer reactions,[2] and cause loss of chiral integrity at the histidine α-carbon.[3] Side-chain protection has therefore generally been employed.

The principal desiderata of an ideal protecting group are obviously simplicity (and thus economy) of introduction, stability in use, prevention of *im*-nitrogen-mediated side reactions, and ease of removal under mild conditions. None of the many diverse N^{im}-protecting groups which have been employed over the years satisfy all these demands to perfection, and some of them bring their own problems with them. Clearly, a protecting group which entails elaborate extra manipulations, or actually creates new risks, is self-defeating. Furthermore, the early (pre-1975) literature on the subject is very confused, and some of the confusions in it linger on. Specifically, the significance of the difference between the two *im*-nitrogens was not recognized. Sometimes, even when the difference was acknowledged, confusion was compounded by arbitrary and incorrect structural assignments and by the use of inconsistent and contradictory ring-position numbering systems. Even *Houben–Weyl Synthese von Peptiden*,[4] an otherwise invaluable survey of the literature up to its time, falls into the trap of failing to appreciate the subtlety of the point, and displays incorrect structures for several key N^{im}-substituted histidine derivatives without any discussion. It therefore seems desirable to establish some relevant fundamentals at the outset.

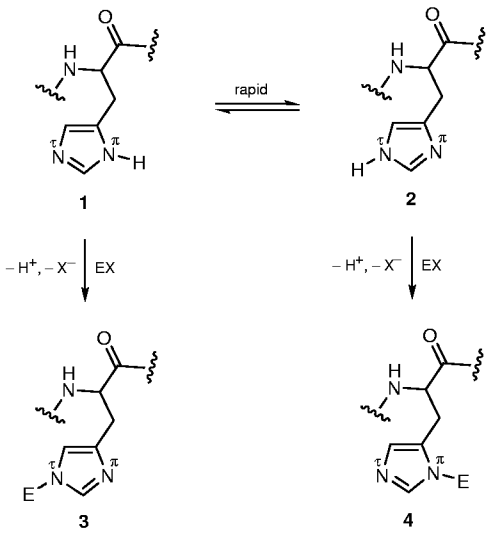
As different ring-numbering conventions have been used by different sectors in the past, the designation of the *im*-nitrogens as π (for *pros*, near the α-carbon) or τ (for *tele*, distant from the α-carbon), which has IUPAC-IUB approval, is greatly to be preferred over other systems in naming N^{im}-substituted derivatives. Thus, the N^{im}-methylhistidine which is still usually called 3-methylhistidine in biochemical circles actually has the methyl group on the *im*-nitrogen which is properly numbered 1. Named as an alanine derivative, this amino acid is 3-(1-methylimidazol-4-yl)alanine. This confusion is abolished if the name N^{π}-methylhistidine is used.

Although the basicities of the *im*-nitrogens are very similar (pK_a ca. 7), and rapid proton exchange makes the tautomers **1** and **2** inseparable, there is a significant difference between the *im*-nitrogens in their reactivity towards electrophiles, and in most cases interconversion between the two *im*-nitrogen-substitution products does not take place (Scheme 1).

In organic solvents alkyl halides (including benzyl halides, alkoxymethyl halides, and allyl halides), some acylating agents (alkoxycarbonyl chlorides, di-*tert*-butyl dicarbonate, carbamoyl chlorides), Michael electrophiles, and nitroaryl halides, among others, react with histidine side chains to give mixtures of τ-**3** and π-**4** derivatives in which the N^{τ}-product predominates, except when reactive alkylating agents are used in excess, where the disubstituted product is formed. Typically, the relative proportions of N^{π}- and N^{τ}-products are 10–20% and 80–90%, respectively. In some cases the predominant N^{τ}-product drops out of solution. The N^{α},N^{τ}-bis(*tert*-butoxycarbonyl)histidine methyl ester, a key compound for N^{π}-derivatization with reactive halides, for example, is easily isolated from the mixture resulting from treatment of histidine methyl ester with di-*tert*-butyl dicarbonate (see Section 2.6.2.2). But in many cases this is not so, and practically inseparable π/τ-mixtures are obtained more often than not. The situation is similar with other histidine derivatives reacting with electrophilic reagents in organic solvents, although in aqueous media under some circumstances[5] it appears that N^{π}-substitution can predominate. The extreme steric hindrance of

triphenylmethyl chloride results in the exclusive formation of the N^{τ}-derivative, which can then be used for controlled N^{π}-alkylation (see Section 2.6.2.5).

Scheme 1 The Reaction of Electrophiles with *im*-Nitrogens

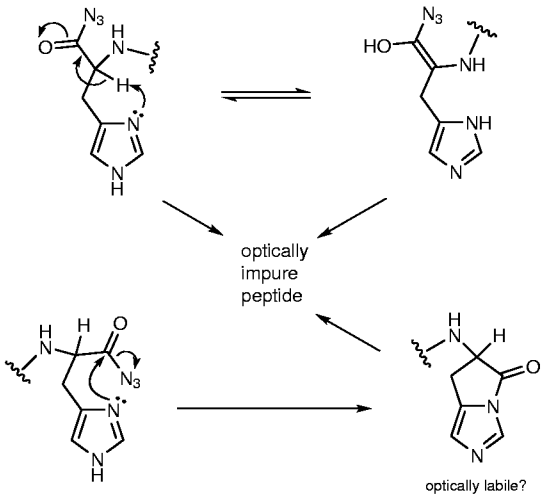


Trimethylsilyl chloride also gives the N^{τ}-product alone in simple models, but this is probably the consequence of thermodynamic control: in any event N^{τ}-trialkylsilyl derivatives are much too labile to be useful as stepping stones to N^{π}-alkylhistidines. The only thoroughly studied example of exclusive mono-N^{τ}-derivatization is the reaction of acetic anhydride with acylhistidine esters, when thermodynamic control appears to operate. Thus, N^{α}-(benzyloxycarbonyl)histidine methyl ester gives only the N^{τ}-acetyl derivative, providing a third kind of intermediate which is useful for N^{π}-alkylation with reactive halides (see Section 2.6.2.3).[6]

It is understandable that in the early days the τ/π distinction was neglected, because no general method of distinguishing between isomeric N^{τ}- and N^{π}-substituted structures in histidine derivatives was available. It is still the case that this distinction is nontrivial, but the problem has been solved in specific cases by crystallography,[7] chemical degradation,[8] or nuclear Overhauser effect measurements,[9] and no exception to the cross-ring coupling constant criterion of Matthews and Rapoport[10] is known.

Investigations of the extensive racemization which accompanies the coupling of, for example, *tert*-butoxycarbonylhistidine azide and the hindered nucleophile methyl pipecolate, led Veber[3] to implicate the π-nitrogen as catalyst. The π-nitrogen operates either by an intramolecular base-facilitating racemization by enolization or as an intramolecular nucleophile giving an optically labile cyclic imidazolide (Scheme 2). Analogous mechanisms could intervene with alkyl-type substituents at the τ-position (although in that case cyclic imidazolium intermediates would be involved, that makes no difference of principle), but not with any kind of N^{π}-substituent. Whatever the mechanism, blockade of the π-position should be helpful, Veber suggested.[3] Comparison[11] of the N^{im}-phenacyl isomers **5** and **6** (Scheme 3) confirmed this hypothesis, showing that activation and coupling of **5** gave optically pure products under conditions which caused gross racemization with **6**.

Scheme 2 Possible Mechanisms for Racemization during Coupling at Histidine by the Azide Method[3]



Similar comparisons of other N^{im}-substituted isomer pairs have confirmed the point and established that π-blockade does in fact provide a complete general solution to the problem.

Scheme 3 Activation and Coupling of the N^{π}-Pac Isomer Gives Optically Pure Products under Conditions Which Cause Gross Racemization with the N^{τ}-Pac Isomer[11]



5 **6**

The suppression of basic and nucleophilic properties at the π-position by location of a strongly electron-withdrawing group at the τ-position greatly reduces, but does not abolish, the danger of racemization.

The principal classical methods of N^{im}-protection all employ substituents located mainly (but generally not exclusively, resulting in characterization and purification problems) at the τ-position. Benzyl, 2,4-dinitrophenyl, 4-tosyl, 4-piperidinocarbonyl, and alkoxycarbonyl are the most significant. For details and access to the literature concerning them, see *Houben–Weyl Synthese von Peptiden*.[4] Occasional applications of these methods still appear, many of the relevant derivatives are commerically available and the alkoxycarbonyl theme is still under development,[12] but none of them can really be recommended for cases of any complexity. N^{τ}-Benzyl allows gross racemization on activation of histidine residues so substituted,[13] even if N^{α}-alkoxycarbonyl protection is in place. Experience has been especially bad with sluggish couplings to hindered amino components in solid-phase synthesis, and although racemization can be largely suppressed by addition of 1-hydroxybenzotriazole (HOBt),[14] the N^{τ}-benzyl tactic is clearly one of high risk. Furthermore, removal of N^{im}-benzyl groups is not easy, requiring sodium in liquid ammonia, or somewhat extended catalytic hydrogenolysis. Even N^{π}-benzyl protection, thus far unexplored (N^{α}-benzyloxycarbonyl-N^{π}-benzylhistidine has, however, been prepared[6] as a single fully characterized isomer), is also not likely to be very useful. Although the other classical N^{τ}-type protecting groups mentioned above are all electron withdrawing, so that racemization is a much less significant problem with them (except with N^{τ}-4-piperidinocarbonyl,[15] which is not electron

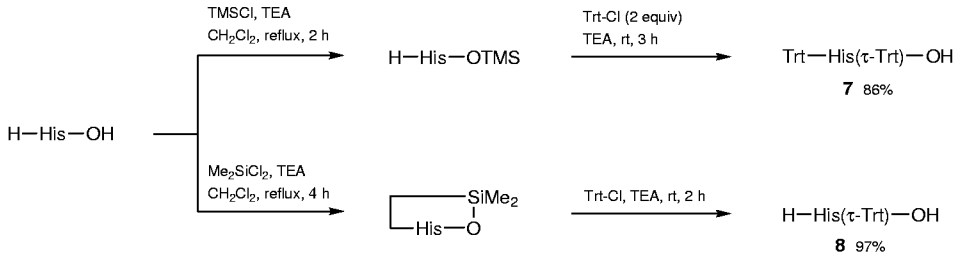
withdrawing enough), the very fact that they are electron withdrawing makes them susceptible to nucleophilic attack. This can lead to premature loss of N^{im}-protection[15] (which may take place by irreversible transfer to other functionalities[16] in the sequence being assembled) and open the way for other side reactions.[17,18]

N^{im}-Protecting groups are covered below in individual subsections (see Table 1 for a summary of some key N^{im}-protected histidine derivatives).

2.6.2.1 N^{τ}-Triphenylmethyl Protection

The N^{im}-triphenylmethyl (trityl, Trt) protection is easily introduced[19] invariably and exclusively at the τ-position (Scheme 4). Completely stable to basic conditions and nucleophiles, it is acid labile, but much less so than an N^{α}-trityl group, and the acidolysis is reversible. The reversibility has consequences in simple model compounds which at first seem curious: the N^{τ}-trityl group survives brief exposure to strong anhydrous acid, but warm aqueous acetic acid eventually removes it with precipitation of triphenylmethanol. These properties not only allow N^{α}-detritylation (5% HCO_2H in $ClCH_2CH_2Cl$, 8 min, 20 °C) and Boc cleavage (1 M HCl in 90% aq AcOH, 20 min, 20 °C) in the presence of a N^{τ}-trityl group, but also permit N^{α},N^{τ}-detritylation (90% aq AcOH, 1.5 h, 60 °C) without disturbing a Boc group.[20] The N^{τ}-trityl group is perfectly orthogonal with Fmoc[21] and is convenient for use in the Fmoc/tBu polyamide approach to solid-phase peptide synthesis: it is removed by the usual methods at the acidolytic final deprotection and resin-detachment stage. Many such applications have been reported, and N^{τ}-trityl (or a variant[22] with substituent-tuned acid lability) is the favored method in that approach; however, from the racemization point of view, N^{τ}-trityl is very risky in conjunction with N^{α}-Fmoc protection, especially if the amino component is sterically hindered. The received wisdom to the contrary which somehow became established in the late 1980s and early 1990s was without any foundation. In fact, the N^{τ}-trityl group is much the same as the N^{τ}-benzyl group in this respect,[23] and π-location of the trityl group has so far proved impossible to achieve.[24,57] When an N^{α}-trityl group is in place instead of N^{α}-alkoxycarbonyl protection, however, the electron-withdrawing influence impinging on the chiral center is much less, and the activation and coupling of N^{α},N^{τ}-bistritylhistidine (**7**) is practically free of racemization risk, even in the worst-case scenario of attachment to a resin hydroxy group.[21]

Scheme 4 Regioselective Protection of Histidine at the τ-Position[19]



Trt-His(τ-Trt)-OH (7):[19]

A mixture of H-His-OH (0.77 g, 5 mmol), TMSCl (0.63 mL, 5 mmol) and CH_2Cl_2 (15 mL) was refluxed for 2 h. After cooling, TEA (0.69 mL, 5 mmol) was added. After a further 5 min reflux and cooling, TEA (1.39 mL, 10 mmol), and Trt-Cl (2.79 g, 10 mmol) dissolved in CH_2Cl_2 (15 mL) were added in turn. The mixture was stirred for 3 h, and MeOH (excess) was added. The residue after removal of solvents was partitioned between $CHCl_3$ and 5% aq citric acid. The $CHCl_3$ layer was separated, washed with brine, and dried ($MgSO_4$). Concentration and addition of hexane gave, after 4 d, **7**; yield: 2.75 g (86%); mp 198 °C; $[\alpha]_D^{25}$ +3.6 (*c* 5, pyridine).

for references see p 345

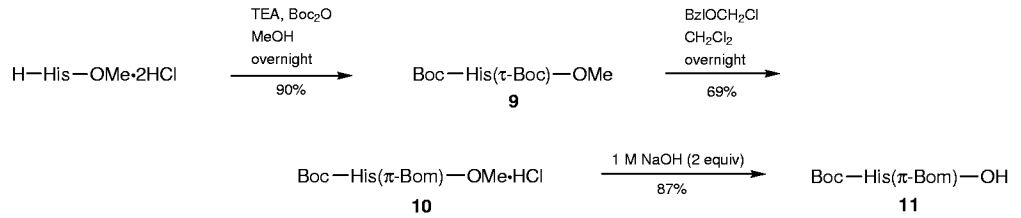
H-His(τ-Trt)-OH (8):[19]
The reagent Me_2SiCl_2 (1.21 mL, 10 mmol) was added to a stirred suspension of H-His-OH (1.55 g, 10 mmol) in CH_2Cl_2 (15 mL), and the mixture was refluxed for 4 h. TEA (2.79 mL, 20 mmol) was added and the mixture was refluxed for a further 15 min. After cooling, more TEA (1.39 mL, 10 mmol) was added with stirring, followed by a soln of Trt-Cl (2.79 g, 10 mmol), in CH_2Cl_2 (10 mL). After 2 h, MeOH (excess) was added, and solvents were removed. Water was added to the residue, and the pH was adjusted to 8–8.5 by dropwise addition of TEA. The slurry was shaken with $CHCl_3$. The insoluble material was separated by filtration and washed with water to give **8**; yield: 3.85 g (97%); after recrystallization from THF/H_2O (1:1), mp 220–222 °C; $[\alpha]_D^{25}$ −2.1 [*c* 1, THF/H_2O (1:1)].

Fmoc-His(τ-Trt)-OH:[21]
This compound was prepared from H-His(τ-Trt)-OH (**8**) in the usual way with Fmoc-OSu in aq dioxane and $NaHCO_3$ as base in 85% yield, but no further details have been published; mp 150 °C (dec); $[\alpha]_D^{ca.\,20}$ +86.6 (*c* 5, $CHCl_3$).

2.6.2.2 N^{π}-Benzyloxymethyl Protection

N^{π}-Benzyloxymethyl (Bom) protection[25,26] is necessarily introduced by convenient yet indirect means, employing the readily available, albeit hazardous reagent (**CAUTION:** *chloromethyl ethers are carcinogenic irritants*) benzyl chloromethyl ether (Scheme 5). The key building block **11** has been extensively used in stepwise classical and Merrifield solid-phase work with complete freedom from racemization. Final cleavage is by hydrogenolysis or strong acidolysis; the level of acidolysis required for N^{α}-Boc deprotection does not affect the Bom group, which is also completely stable to bases and nucleophiles. Both hydrogenolysis and strong acidolysis release formaldehyde, which can lead to side reactions, although these can be eased by use of appropriate scavengers.[27]

Scheme 5 Regioselective N^{π}-Benzyloxymethyl (Bom) Protection by Indirect Means[25,26]



N^{π}-2-Adamantyloxymethyl protection[28] is similar to N^{π}-Bom in both its protection and deprotection chemistry (except that it is hydrogenolysis resistant), but peptides protected with this variant may have solubility advantages.

Boc-His(π-Bom)-OH (11):[25]

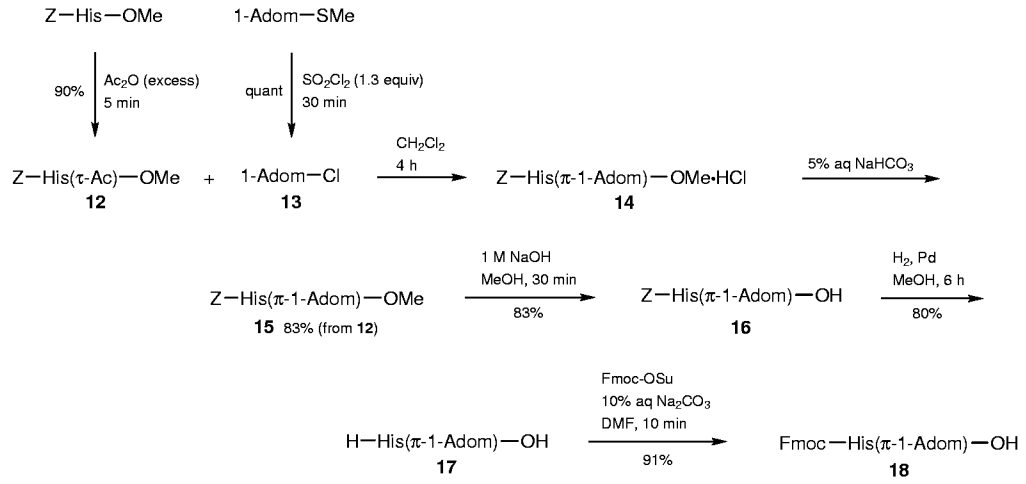
CAUTION: *Chloromethyl ethers are carcinogenic irritants. Proper safety precautions and procedures should be adopted at all stages of their preparation and use.*

TEA (70 mL, 0.5 mol), followed by Boc_2O (120 g, 0.55 mol), was added to a suspension of H-His-OMe·2 HCl (60.5 g, 0.25 mol) in MeOH (200 mL) and the mixture was set aside overnight. The solvent was removed, $CHCl_3$ (250 mL) was added, and the mixture was washed with 10% aq citric acid (2 × 50 mL). Drying ($MgSO_4$) and solvent removal gave an oil which solidified when triturated with light petroleum (bp 40–60 °C) to give **9**; yield: 83 g (90%); mp 96 °C; $[\alpha]_D^{25}$ +25.6 (*c* 1, CCl_4). A soln of **9** (32 g, 0.082 mol) and freshly redistilled $BzlOCH_2Cl$ (18 mL, 0.13 mol) in CH_2Cl_2 (200 mL) was set aside overnight. Solvent removal, dissolution in MeOH (30 mL), and addition of Et_2O (400 mL) gave a slightly turbid soln from which **10** crystallized overnight; yield: 24 g (69%); mp 152 °C; $[\alpha]_D^{25}$ −19.1 (*c* 1, MeOH). To a soln of this hydrochloride (22 g, 0.052 mol) in MeOH (50 mL) was added 1 M NaOH (120 mL). After 15 min, H_2O (1 L) was added and the pH was adjusted to 4.5 by the dropwise addition of 1 M HCl. The soln was extracted with $CHCl_3$ (3 × 100 mL) and the combined organic extracts were dried ($MgSO_4$). Removal of the solvent gave an oil which was dissolved in EtOAc (50 mL). Solvent removal gave **11**; yield: 17 g (87%); mp 155 °C; $[\alpha]_D^{20}$ +6.9 (*c* 0.5, MeOH); overall yield: 54%.

2.6.2.3 N^{π}-1-Adamantyloxymethyl Protection

The simplest trifluoroacetic acid (TFA) labile and hydrogenolysis-resistant analogue of and complement to the N^{π}-Bom group is the N^{π}-*tert*-butoxymethyl (Bum) group.[29] The requisite Bum derivatives have been made, and the related N^{α}-Fmoc acid is commercially available. It would perhaps be the ideal building block for Fmoc/tBu polyamide solid-phase synthesis, but is expensive. The main reason for this is the long and tricky synthesis; in particular, *tert*-butyl chloromethyl ether[30] is not an easy compound to prepare and handle (**CAUTION:** *chloromethyl ethers are carcinogenic irritants*). More elaborate alternatives which might be expected to be TFA labile have therefore been explored, and of these the N^{π}-1-adamantyloxymethyl (1-Adom) group (Scheme 6),[31] a Bum analogue in every respect, is the one which is recommended.

Scheme 6 Regioselective N^{π}-1-Adamantyloxymethyl Protection by Indirect Means[31]



Z-His(τ-Ac)-OMe (12):[30,31]
Z-His-OMe[31,32] (1.46 g, 4.23 mmol) was dissolved in excess Ac_2O (5 mL). After 5 min, the solvent was removed and anhyd Et_2O (8 mL) was added to give crystalline **12**; yield: 1.49 g (90%); mp 97–102 °C; $[\alpha]_D^{20}$ +31.0 (*c* 1, $CHCl_3$). After recrystallization from $CHCl_3$/hexane: mp 97–100.5 °C; $[\alpha]_D^{20}$ +35.5 (*c* 1, $CHCl_3$).

1-Adamantyl Chloromethyl Ether (13):[30,31]

CAUTION: *Chloromethyl ethers are carcinogenic irritants. Proper safety precautions and procedures should be adopted at all stages of their preparation and use.*

SO_2Cl_2 (2.88 g, 21.3 mmol) in CH_2Cl_2 (16 mL) was added over 10 min to a soln of 1-adamantyloxymethyl methyl sulfide (3.48 g, 16.4 mmol; prepared by reaction of adamantan-1-ol with Ac_2O/AcOH/DMSO), and the mixture was stirred for a further 20 min. Removal of the solvent gave **13**; yield: 3.29 g (quant); ^1H NMR ($CDCl_3$, δ): 5.67 (OCH_2Cl);[24] it was used without purification or delay.

Fmoc-His(π-1-Adom)-OH (18):[31]
A soln of **12** (3.45 g, 10 mmol) and **13** (3.0 g, 15 mmol) in CH_2Cl_2 (20 mL) was stirred at rt for 4 h. After removal of the solvent the residue was triturated with anhyd Et_2O to give **14**; mp 88–90 °C; $[\alpha]_D^{25}$ –27.1 (*c* 1, MeOH). A soln of this ester in $CHCl_3$ (50 mL), was treated with 5% aq $NaHCO_3$, H_2O, dried (Na_2SO_4) and taken to dryness. The residue, in $CHCl_3$ (5 mL), was applied to a silica gel column (3 × 25 cm), equilibrated and eluted with $CHCl_3$/MeOH (30:1). The eluent containing the desired product was collected and the solvents were removed to give **15** as an oil; yield: 3.9 g (83% from **12**). To a soln of this ester (3.40 g, 7.52 mmol) in MeOH (24 mL) was added with stirring 1 M NaOH (7.6 mL, 7.6 mmol) and the mixture was stirred for 30 min. After removal of the MeOH below 15 °C, the residue was dissolved in H_2O (100 mL). The pH was adjusted to 7.0–7.5 with 1 M HCl. The precipitate was

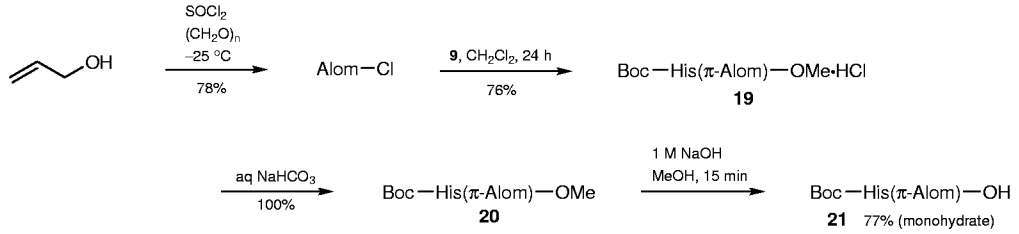
for references see p 345

collected by filtration and dried over KOH pellets to give **16**; yield: 2.83 g (83%); mp 96–98 °C; $[\alpha]_D^{25}$ −2.7 (*c* 1, MeOH). This acid (1.36 g, 3.00 mmol) in MeOH (20 mL) was hydrogenated over a Pd catalyst for 6 h. After removal of the catalyst and solvent, Et_2O was added to the residue. Filtration and drying over KOH pellets gave **17**; yield: 0.76 g (80%); mp 228–229 °C (dec); $[\alpha]_D^{25}$ −8.3 (*c* 1, MeOH). To an ice-cooled soln of this amino acid (0.63 g, 2.0 mmol) in 10% aq Na_2CO_3 (5 mL) was slowly added a soln of Fmoc-OSu (0.75 g, 2.2 mmol) in DMF (5 mL). The mixture was stirred for a further 10 min. After dilution of the mixture with H_2O (50 mL), it was washed with Et_2O and EtOAc. The pH of the water layer was adjusted to 6 with 1 M HCl. Filtration and drying over KOH pellets gave **18**; yield: 0.97 g (91%); mp 160–161 °C; $[\alpha]_D^{25}$ +5.4 (*c* 1, MeOH).

2.6.2.4 Allyl-Based N^{π}-Protection

Allyl-based N^{π}-protection, stable to base and moderate acid, can be achieved with the allyl group itself,[33] or with an allyloxymethyl (Alom) group (Scheme 7).[24,34,58] In either case cleavage by Pd(0)-catalyzed allyl transfer to *N,N'*-dimethylbarbituric acid[35] (which probably also scavenges the formaldehyde released from the latter) proceeds well, provided care is taken over the integrity of the catalytic complex.

Scheme 7 Regioselective N^{π}-Allyloxymethyl Protection by Indirect Means[24,33,34,58]



Allyl Chloromethyl Ether:[24,36,58]

CAUTION: *Chloromethyl ethers are carcinogenic irritants. Proper safety precautions and procedures should be adopted at all stages of their preparation and use.*

Freshly distilled $SOCl_2$ (10.43 g, 94 mmol) was added dropwise to a stirred mixture of distilled allyl alcohol (8.48 g, 146 mmol) and anhyd paraformaldehyde (4.38 g, 146 mmol) which had been cooled to −25 °C. When the mixture had clarified, the upper layer was separated, dried ($CaCl_2$) and filtered. The filtrate was distilled under reduced pressure to give allyl chloromethyl ether; yield: 8.32 g (78%); colorless liquid; bp 42–44 °C/60 Torr; ^1H NMR ($CDCl_3$, δ): 5.52 (OCH_2Cl).

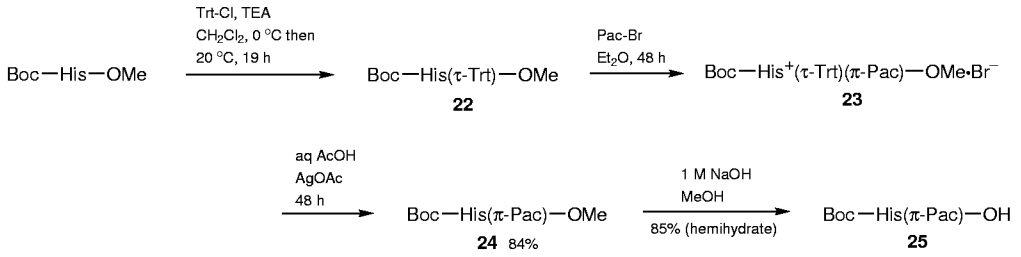
Boc-His(π-Alom)-OH (21):[24,58]

Allyl chloromethyl ether (1.05 g, 9.9 mmol) was added to a soln of **9** (3.63 g, 9.3 mmol) in CH_2Cl_2 (30 mL). The soln was stirred for 24 h. The solvent was removed. MeOH (20 mL) was added, mostly removed and Et_2O (50 mL) was added. The resulting colorless solid was recrystallized from MeOH/Et_2O to give **19**; yield: 2.64 g (76%); mp 129–130 °C; $[\alpha]_D^{24}$ +13.0 (*c* 0.1, MeOH). This hydrochloride (1.33 g, 3.5 mmol) was dissolved in CH_2Cl_2 (50 mL). The soln was washed with aq $NaHCO_3$ (50 mL), dried and the solvent removed to give **20**; yield: 1.20 g (100%); colorless oil. Ester **20** (0.69 g, 2.0 mmol) was dissolved in MeOH (8 mL) and 1 M NaOH (2.3 mL, 2.3 mmol) was added. The MeOH was removed after 15 min and 1 M HCl (2.3 mL, 2.3 mmol) was added. The mixture was taken to dryness under reduced pressure using a cool water bath. The mixture was extracted with $CHCl_3$ (2 × 50 mL). The solvent was removed to give **21**; yield: 0.50 g (77%) as a monohydrate; mp 113–115 °C; $[\alpha]_D^{28}$ +5.6 (*c* 1, MeOH).

2.6.2.5 N^{π}-Phenacyl Protection

N^{π}-Phenacyl (Pac) protection,[32] which can be introduced as shown in Scheme 8, in is completely acid stable. Cleavage of the N^{im}-Pac group with zinc dust and acetic acid is rapid, although occasionally the quantitative removal of zinc traces afterwards is troublesome. However, the group is not completely immune to basic and nucleophilic reagents. Nevertheless, the racemization-free formation of a peptide bond with a N^{π}-Pac group in place, followed by regiospecific N^{τ}-alkylation and Pac removal are pivotal stages in the synthesis[37] of a cyclic TRH analogue.

Scheme 8 Regioselective N^{π}-Phenacyl Protection by Indirect Means[37]



Boc-His(τ-Trt)-OMe (22):[32,38]
The amino acid derivative Boc-His-OMe[24,39] (5.24 g, 20 mmol) was dissolved in CH_2Cl_2 (50 mL), and TEA (3.12 mL, 22 mmol) was added with stirring at 0 °C followed by Trt-Cl (10.5 g, 38 mmol). Stirring was continued at 20 °C for 19 h. The mixture was diluted with CH_2Cl_2 (60 mL), washed with H_2O (20 mL), sat. aq citric acid (2 × 20 mL), H_2O (20 mL), sat. aq $NaHCO_3$ (5 × 20 mL), and H_2O (20 mL), and dried ($MgSO_4$). The CH_2Cl_2 was removed and the residue was dissolved in MeOH and refrigerated for 1 d. Filtration and solvent removal gave **22** as a foam contaminated with a trace of Trt-OH.

Boc-His(π-Pac)-OH (25):[32]
Ester **22** prepared as above (6.0 g, 12 mmol) was dissolved in Et_2O (10 mL), and Pac-Br (2.39 g, 12 mmol) was added. After 48 h the imidazolium bromide **23** (8.10 g, 95% purity) was separated and washed with Et_2O. This bromide (7.10 g, 10 mmol) was dissolved in 80% aq AcOH (20 mL), and AgOAc (1.67 g, 10 mmol) was added with stirring. The precipitated AgBr was removed by filtration. After 48 h the solvent was removed and sat. aq $NaHCO_3$ (100 mL) was added. The mixture was extracted with Et_2O (5 × 50 mL). The organic extract was dried briefly and then allowed to stand at rt for 48 h, when methyl ester **24** crystallized; yield: 3.25 g (84%); mp 139–141 °C; $[\alpha]_D^{20}$ −9.3 (*c* 1, MeOH). This methyl ester (3.0 g, 7.8 mmol) was dissolved in MeOH (3 mL), and 1 M NaOH (7.8 mL) was added with stirring. After 1 h H_2O (25 mL) was added, the MeOH was removed, and the pH was adjusted to 3.8 with 1 M HCl. Then, NaCl was added to saturate the aqueous phase, which was extracted exhaustively with $CHCl_3$ (6 × 50 mL). Drying ($MgSO_4$), solvent removal, trituration with Et_2O and reprecipitation from MeOH/Et_2O gave **25** as a hemihydrate; yield: 2.57 g (85%); mp 181–183 °C; $[\alpha]_D^{20}$ +9.4 (*c* 1, MeOH).

Table 1 Summary Table Showing Some Key N^{im}-Protected Histidine Derivatives[6,19,21,24,25,31,32]

Starting Material	Product	Yield (%)	$[\alpha]_D$	mp (°C)	Ref
H-His-OH	Trt-His(τ-Trt)-OH (**7**)	86	+3.6 (*c* 5, pyridine)	198	[19]
H-His-OH	H-His(τ-Trt)-OH (**8**)	97	−2.1 (*c* 1, THF/H_2O 1:1)	220–222	[19]
H-His(τ-Trt)-OH (**8**)	Fmoc-His(τ-Trt)-OH	85	+86.6 (*c* 5, $CHCl_3$)	150[a]	[21]
H-His-OMe[b]	Boc-His(τ-Boc)-OMe (**9**)	90	+25.6 (*c* 1, CCl_4)	96	[25]
Boc-His(τ-Boc)-OMe (**9**)	Boc-His(π-Bom)-OH (**11**)	60	+6.9 (*c* 0.5, MeOH)	155	[25]
Z-His-OMe	Z-His(τ-Ac)-OMe (**12**)	90	+31.0 (*c* 1, $CHCl_3$)	97–102	[6,31,32]
Z-His(τ-Ac)-OMe (**12**)	Z-His(π-1-Adom)-OMe (**15**)	83	−27.1 (*c* 1, MeOH)	88–90	[31]
H-His(π-1-Adom)-OH (**17**)	Fmoc-His(π-1-Adom)-OH (**18**)	91	+5.4 (*c* 1, MeOH)	160–161	[31]
Boc-His(τ-Boc)-OMe (**9**)	Boc-His(π-Alom)-OMe (**20**)[c]	76	+13.0 (*c* 0.1, MeOH)	129–130	[24]
Boc-His-OMe	Boc-His(π-Pac)-OMe (**24**)	84	−9.3 (*c* 1, MeOH)	139–141	[32]
Boc-His(π-Pac)-OMe (**24**)	Boc-His(π-Pac)-OH (**25**)[d]	85	+9.4 (*c* 1, MeOH)	181–183	[32]

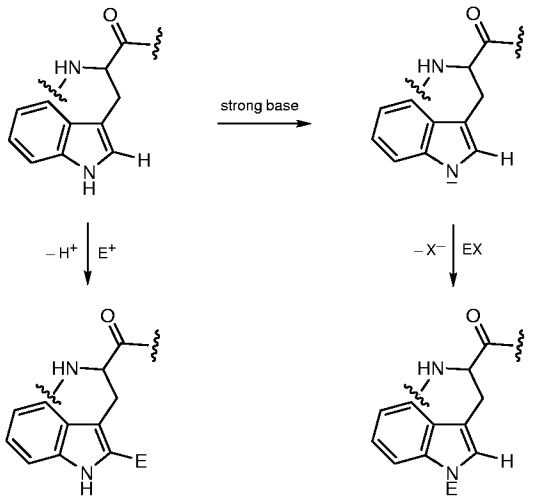
[a] With decomposition. [b] Bis-hydrochloride. [c] Hydrochloride. [d] Hemihydrate.

for references see p 345

2.6.3 Indole Group

Tryptophan has a 3-alkylindole side chain which is easily attacked by electrophiles (Scheme 9) under acidic conditions, especially at the 2-position, but also elsewhere, and is susceptible to oxidative degradation.[40] In the presence of strong bases the indole system reacts as an anion, mainly at the *in*(1)-position (pK_a 17).

Scheme 9 Electrophilic Attack on the Indole Side Chain of Tryptophan[40]



The conditions and electrophiles encountered in peptide-coupling procedures do not seem to cause much difficulty (except for side-chain nitrosation which has been observed[41] during the generation of acyl azides). Protected tryptophan-containing sequences can be assembled without special attention to the indole side chain, provided exposure to acidic conditions is avoided or minimized. But, at the final deprotection stage the principal conventional strategies of peptide synthesis employ acidolysis for the release of unprotected side chains, generating electrophiles which remain at large until workup, and extended treatment is often required. The electrophiles which can be present and cause trouble include *tert*-butyl carbocations (from Boc groups, etc.), arylsulfonyl cations (from Pmc groups, etc., on arginine side chains), acylium ions (from carboxy groups), and resin-bound modified benzyl carbocations from scission of the anchor link. Reaction of the tryptophan side chain with these can result in irreversible peptide capture by the resin. Protonated indole side chains themselves can attack scavengers such as anisole or thiols, or other indole rings giving dimers. This is the stage at which serious difficulties (which may be sequence dependent[42]) can arise. A great deal of effort has been expended optimizing the composition of final deprotection cocktails comprising acids and various scavengers,[43,44] and tuning side-chain protection for other residues (especially arginine)[42] so that shorter exposure times suffice, but the problem remains. Clearly what is required is a conveniently introduced means of reducing electron availability in the heteroaromatic system, which survives to exert that influence throughout the acidolysis of other protecting groups, but is amenable to smooth removal later. Various types of electron-withdrawing groups, e.g. acyl (*vide infra*), sulfonyl (such as mesitylenesulfonyl,[45] resistant even to hydrazine and HF, but cleaved by MsOH/anisole in TFA), and phosphorus-containing (such as diphenylphosphinothioyl,[46] base resistant, but cleaved by TFA and anisole in TfOH, or fluoride ion in dimethylformamide) groups, have been investigated. A clear consensus on N^{in}-protection, whether it is worthwhile to protect routinely

or not, and if so with what, has yet to emerge, but the trend is very much towards protection, and it appears that the acyl (especially alkoxycarbonyl) type is the most promising. Table 2 presents a summary of some key N^{in}-protected tryptophan derivatives.

2.6.3.1 N^{in}-Formyl Protection

The N^{in}-formyl protecting group is easily introduced; it is very stable to acidic reagents, even pure hydrofluoric acid; it is cleaved by strong acid–soft nucleophile cocktails (such as HF/thiols), and by nucleophiles in basic conditions (hydrazine, alkali, piperidine in DMF).[44,47] It provides good protection against electrophilic attack on the indole system. Orthogonality with Boc makes N^{in}-formyl appropriate for Boc/Bzl solid-phase assembly (although incomplete final deprotection is a common experience[48]), but not for Fmoc/tBu solid-phase assembly, as Fmoc and N^{in}-formyl cleavage by nucleophiles occur under similar conditions.

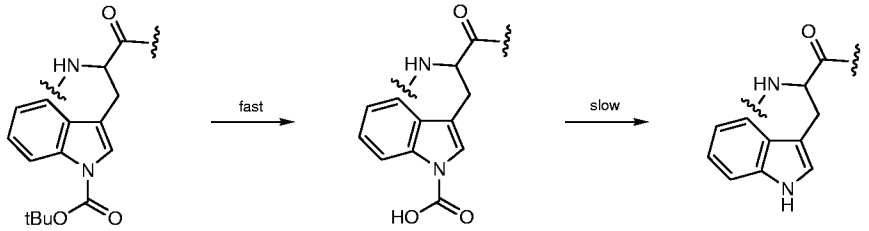
H-Trp(For)-OH:[49]
Anhyd HCl(g) was passed through a soln of H-Trp-OH (20.4 g, 100 mmol) in HCO_2H (300 mL) until there was no further increase in the UV maximum at 298 nm. Removal of the solvent and addition of Et_2O gave the title compound; yield: 26.8 g (quant); mp 218–220 °C (dec); $[\alpha]_D^{23}$ −4.7 (*c* 2, H_2O).

Boc-Trp(For)-OH:
A procedure using Boc-N_3/DMSO, which gave a 60% yield of the corresponding DCHA salt, mp 121–124 °C (dec), $[\alpha]_D^{24}$ +35.5 (*c* 2, EtOH), has been published.[47] The protected acid itself was liberated from the salt in the usual way. The reagent Boc_2O would doubtless be safer and better: Boc-Trp(For)-OH is commercially available.

2.6.3.2 N^{in}-*tert*-Butoxycarbonyl Protection

Protection by the N^{in}-*tert*-butoxycarbonyl group is acid labile, but somewhat less so than N^{α}-Boc. Selective N^{α}-Boc deprotection in the presence of N^{in}-Boc is possible in simple cases. Reliance on the difference in complex or repetitive operations would not be wise, however. Of mechanistic interest is the fact that the cleavage of N^{in}-Boc by trifluoroacetic acid is detectably stepwise (Scheme 10). Whereas Boc-protected amino group acidolysis leads directly to an amine salt product because the intermediate carbamic acid rapidly decarboxylates, with N^{in}-Boc the carbamic acid lingers.[50] This provides extended protection against electrophilic attack on the ring. The N^{in}-Boc group is stable to bases and nucleophiles and is perfectly orthogonal with Fmoc, and thus very suitable for use in the Sheppard approach to solid-phase peptide synthesis.[51,52] The amino acid derivative Fmoc-Trp(Boc)-OH is commercially available.

Scheme 10 The Stepwise Cleavage of N^{in}-Boc Protection[50]



for references see p 345

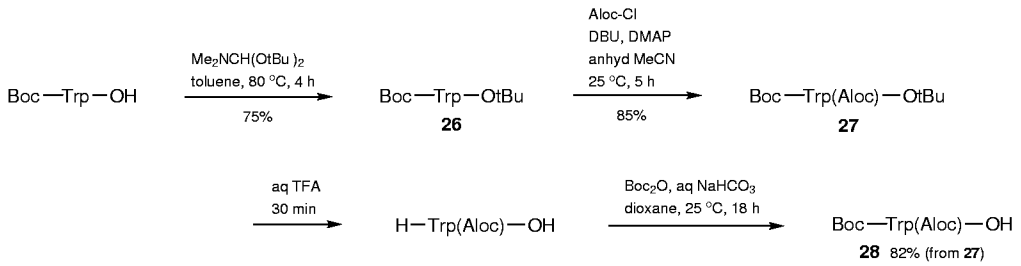
2.6.3.3 N^{in}-Cyclohexyloxycarbonyl Protection

Protection by the N^{in}-benzyloxycarbonyl group[53] is the most obvious complement to N^{in}-Boc, but it is not sufficiently resistant to trifluoroacetic acid to be used in conjunction with temporary N^{α}-Boc protection. This objection can be met by substituent tuning,[53] but there is another problem. Alkoxycarbonylindoles based on primary alcohols are considerably more reactive towards nucleophiles than are primary alkoxycarbonyl derivatives of amino compounds: N^{in}-Z groups are insufficiently stable to nucleophilic media for convenient use. The niche is filled by N^{in}-alkoxycarbonyl groups based on sterically hindered secondary alcohols,[54,55] especially the N^{in}-cyclohexyloxycarbonyl (Hoc) group,[54] which is stable to trifluoroacetic acid and piperidine in dimethylformamide but is cleaved by hydrogen fluoride/*p*-cresol without the need for thiol additives. The derivative Boc-Trp(Hoc)-OH has been prepared in two stages from Boc-Trp-OBzl or Boc-Trp-OPac {mp 119–121 °C; $[\alpha]_D^{25}$ −73.8 (*c* 1, DMF)}.[54]

2.6.3.4 N^{in}-Allyloxycarbonyl Protection

Protection by the N^{in}-allyloxycarbonyl (Aloc) group (Scheme 11) is completely orthogonal with N^{α}-Boc but not with N^{α}-Fmoc because N^{in}-Aloc, like N^{in}-Z (see Section 2.6.3.3), is cleaved by nucleophiles such as piperidine: it is stable to trifluoroacetic acid but is cleaved by Pd(0)-catalyzed allyl transfer to methylaniline in tetrahydrofuran/dimethyl sulfoxide or piperidine/dimethylformamide (1:1).[56]

Scheme 11 The Introduction of N^{in}-Allyloxycarbonyl Protection[56]



Boc-Trp(Aloc)-OH (28):[56]
The amino acid derivative Boc-Trp-OH (18.3 g, 60 mmol) was dissolved in toluene (100 mL), and $Me_2NCH(OtBu)_2$ (70 mL, 290 mmol) was added dropwise over 10 min. The mixture was stirred at 80 °C for 4 h, the solvent was removed and the residue was partitioned between EtOAc and H_2O. The organic layer was washed with sat. aq $NaHCO_3$, 5% aq $KHSO_4$/10% aq K_2SO_4 and sat. aq NaCl, dried (Na_2SO_4), filtered, and the solvent was removed. The residue was crystallized from Et_2O/hexane to give **26**; yield: 16 g (75%); mp 130–131 °C; $[\alpha]_D^{25}$ −21.7 (*c* 1, DMF). This ester (15.9 g, 44 mmol) was dissolved in anhyd MeCN (100 mL), and treated with Aloc-Cl (5.5 mL, 52 mmol) in the presence of DBU (7.75 mL, 52 mmol) and DMAP (488 mg, 4 mmol). The mixture was stirred at 25 °C for 5 h and the solvent was removed. The residue was taken up in EtOAc and washed with H_2O, sat. aq $NaHCO_3$, 5% aq $KHSO_4$/10% aq K_2SO_4, and sat. aq NaCl, dried (Na_2SO_4), and the solvent was removed. The residue was crystallized from Et_2O/hexane to give **27**; yield: 16.5 g (85%); mp 86 °C; $[\alpha]_D^{25}$ −14.3 (*c* 1, DMF). This ester (2.7 g, 6 mmol) was dissolved in a mixture of TFA (10 mL) and H_2O (0.2 mL) and stirred for 30 min under N_2. The solvent was removed and the residue was concentrated three times from toluene, suspended in a mixture of dioxane (20 mL) and H_2O (20 mL), and treated with Boc_2O (1.3 g, 6 mmol) in the presence of $NaHCO_3$ (2.0 g, 24 mmol) at 25 °C for 18 h. The resulting soln was diluted with EtOAc and cautiously acidified with 1 M HCl to pH 2. The organic phase was washed with 5% aq $KHSO_4$/10% aq K_2SO_4, and sat. aq NaCl, dried (Na_2SO_4), and concentrated. The residue was crystallized from Et_2O/hexane, to give **28**; yield: 1.9 g (82%); mp 123–124 °C; $[\alpha]_D^{25}$ −16.7 (*c* 1, DMF).

Table 2 Summary Table Showing Some Key N^{in}-Protected Tryptophan Derivatives[49,56][a]

Starting Material	Product	Yield (%)	$[\alpha]_D$	mp (°C)	Ref
H-Trp-OH	H-Trp(For)-OH	quant	−4.7 (*c* 2, H_2O)	218–220	[49]
Boc-Trp-OH	Boc-Trp-OtBu (**26**)	75	−21.7 (*c* 1, DMF)	130–131	[56]
Boc-Trp-OtBu (**26**)	Boc-Trp(Aloc)-OtBu (**27**)	85	−14.3 (*c* 1, DMF)	86	[56]
Boc-Trp(Aloc)-OtBu (**27**)	Boc-Trp(Aloc)-OH (**28**)	82	−16.7 (*c* 1, DMF)	123–124	[56]

[a] Not by any means a full list of important derivatives; several others which are commercially available are in use but preparative and physical data have not been released to the academic literature.

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2.6.4 Hydroxy Group

S. Reissmann, T. Steinmetzer, G. Greiner, L. Seyfarth, D. Besser, and C. Schumann

Hydroxy amino acids exert special functions in biologically active peptides and proteins. In serine and threonine proteases, for example, these two amino acids are involved in the catalytic mechanism. Protein kinases and phosphatases regulate the activity of numerous intracellular enzymes through phosphorylation or dephosphorylation of the hydroxy groups of tyrosine, serine, and threonine, and in some peptide hormones the aromatic hydroxy group of tyrosine is sulfated. Besides the three proteinogenic amino acids (serine, threonine, and tyrosine), a variety of noncoded amino acids, such as hydroxyproline, β-hydroxylysine, β-hydroxynorvaline, β-hydroxyisoleucine, and β-phenylserine, contain hydroxy groups in their side chains. Thus, in fibrillar proteins such as collagen, hydroxyproline, and hydroxylysine contribute to the stability and elasticity of the fibres. Some antibiotics, e.g. valinomycin and enniatin, as well as some toxins, e.g. didemnins, are biologically highly active depsipeptides containing hydroxy acids as building units.

Depending on the reactivity of the hydroxy group and on the synthetic strategy used, syntheses of peptides have been performed with or without hydroxy protection. While peptide assembly without protection of the hydroxy group is limited to special conditions, the use of hydroxy protection allows application of a wide range of coupling reagents and even an excess of acylating components. The reactivity of hydroxy groups and the protection strategies applied to alcohols, carbohydrates, and phenols has been reviewed[1] from a more general organic chemistry perspective.

Under the conditions used in peptide synthesis, unprotected aliphatic hydroxy groups can undergo two types of side reactions: they can be acylated or dehydrated, the latter leading to dehydroamino acids. The hydroxy group of serine is a primary alcoholic function and therefore exhibits the highest reactivity. The secondary alcoholic functions of threonine, hydroxyproline, β-phenylserine, hydroxynorvaline, and hydroxynorleucine, as well as of other noncoded amino acids, are less reactive and thus more suited for use in the unprotected form. The aromatic hydroxy group of tyrosine is more acidic than the aliphatic hydroxy groups; nevertheless, it can be acylated to form esters. These are active esters which in turn can react with primary amines to form amide bonds.

The synthetic strategy based on the "minimum protection philosophy"[2] (Section 4.1.2) favors the use of unprotected hydroxy groups, which leads to more soluble peptides and milder deprotection procedures,[3,4] but also to a certain degree to side reactions involving the free hydroxy groups. The resulting side-products have to be removed by careful purification of the intermediate fragments.

In the contrasting philosophy of "global protection"[2] (Section 4.1.3) all side-chain functions are protected throughout the synthesis to allow for more vigorous reaction conditions in the coupling step, as required for higher yields and shorter reaction times.[5,6] In some cases the numerous hydrophobic protecting groups form clusters, leading to a very poor solubility. To circumvent this problem the use of different types of protecting groups is recommended. The global protection strategy requires either the use of relatively labile side-chain protecting groups for sensitive peptides or the application of more drastic cleavage conditions, such as HF and TfOH, only applicable for more stable molecules. The protection of all hydroxy groups is recommended in solid-phase synthesis because of the large excesses of acylating amino acid derivatives used. In contrast to this view,[2] some attempts have been made to perform solid-phase synthesis without hydroxy protection.[7,8] In such solid-phase syntheses, amino acids with a secondary hydroxy group, e.g. hydroxyproline, were successfully employed.[9] Thus, relatively large collagen-type peptides have been obtained in satisfactory yields and as highly homogeneous compounds, without side-chain protection of hydroxyproline.[10]

for references see p 375

Among the many protecting groups proposed so far, only a restricted set has found widespread application. Sufficient experimentation means that they can now be regarded as established procedures; however, few groups and combinations have been tried and tested for large-scale synthesis. Excellent reviews of the protection of hydroxy groups in peptide synthesis are available.[2,11,12]

Table 1 lists the commonly used hydroxy protecting groups, together with their stability, selectivity to amino and carboxy protecting groups, and cleavage conditions.

Table 1 Cleavage Conditions and Stability of Commonly Used Hydroxy Protecting Groups[2,13–43]

Type of Protecting Group	Cleavage Conditions	Stability[a]	N- and C-Protecting Groups for Selective Removal
Protection of Aliphatic Hydroxy Groups			
Alkyl Ethers:			
Bzl	catalytic hydrogenolysis (H_2/Pd)[13] super acids (HF, TfOH)[2]	TFA,[14] piperidine, NaOH	Boc, OtBu; Aloc, OAl; Fmoc, OFm
tBu	acidolysis (TFA,[2] 1 M HCl[15,16]) super acids (HF, TfOH)[2]	catalytic hydrogenolysis (H_2/Pd),[2] piperidine, NaOH	Z, OBzl; Aloc, OAl; Trt; Fmoc, OFm
Trt	mild acidolysis (5% TFA,[17] 20% $CHCl_2CO_2H$[18])	piperidine[17]	Trt, Fmoc
MsOb	1 M $SiCl_4$ (10 equiv)/TFA–anisole (10 equiv),[19] after reduction to Mtz with TFA[20]	TFA,[19] piperidine,[20] 1 M NaOH[19]	Boc, OtBu; Fmoc
Acyl Esters:			
Aloc	formation of a π-allyl palladium complex[21,22]	TFA,[22] piperidine[b][22]	Boc, OtBu; Fmoc, OFm
Silyl Ethers:			
TBDMS	fluorogenolysis (TBAF)[23] mild acidolysis (AcOH)[23] 1% iodine[24]	piperidine,[23] catalytic hydrogenolysis (H_2/Pd),[25] 5% NaOH[26]	Fmoc, OFm; Z, OBzl
TBDPS	fluorogenolysis (TBAF)[26,27]	catalytic hydrogenolysis (H_2/Pd),[27] 80% AcOH,[27] 50% TFA[27]	Bpoc; Trt; Z, OBzl
Protection of Aromatic Hydroxy Groups			
Alkyl Ethers:			
Bzl	catalytic hydrogenolysis (H_2/Pd)[13] super acids (HF,[28] TMSOTf[40])	piperidine,[30,31] partial cleavage with TFA[29]	Fmoc, OFm; Aloc, OAl; Boc[c]
2,6-$C_6H_3CH_2$	super acids (HF,[32,33] TMSOTf[40])	TFA,[32,33] piperidine[30,31]	Boc, OtBu; Fmoc, OFm; Aloc, OAl
tBu	acidolysis (TFA,[30] strong inorganic acids), super acids (HF, TfOH)[30] TMSBr[41]	catalytic hydrogenolysis (H_2/Pd),[34] NaOH,[30] piperidine[31]	Aloc, OAl; Fmoc, OFm; Z, OBzl
Trt	mild acidolysis (AcOH/TFE/CH_2Cl_2)[35]	piperidine[35]	Fmoc, OFm; Aloc, OAl
allyl	formation of a π-allyl palladium complex[21,22]	TFA,[21] piperidine[22]	Boc, OtBu; Aloc, OAl; Fmoc, OFm

Table 1 (cont.)

Type of Protecting Group	Cleavage Conditions	Stability[a]	N- and C-Protecting Groups for Selective Removal
3-Pn	HF[31]	TFA, piperidine[31]	Boc, OtBu; Aloc, OAl; Fmoc, OFm; Z, OBzl
Aryl Ethers:			
Dnp	thiolysis (β-mercaptoethanol),[27] piperidine,[38] DBU[38]	TFA, HF[38]	Boc, OtBu; Aloc, OAl; Fmoc, OFm
Acyl Esters:			
Z(2-Br)	super acids (HF,[36] TfOH, TMSOTf[40]) piperidine[31]	TFA[36]	Boc, OtBu; Aloc, OAl
2-Adoc	super acids (HF, TfOH)[37] 2 M NaOH[37]	TFA, 5 M HCl/dioxane,[37] catalytic hydrogenolysis (H_2/Pd)[37]	Boc, OtBu; Aloc, OAl; Z, OBzl
Doc	HF[39]	TFA,[39] partial cleavage with piperidine[39]	Boc, OtBu; Aloc, OAl
Msz	reductive acidolysis ($SiCl_4$/TFA)[19] 1 M NaOH[19]	TFA[19]	Boc, OtBu
Silyl Ethers:			
TBDMS	acidolysis (TFA)[23] fluorogenolysis (TBAF)[23]	catalytic hydrogenolysis (H_2/Pd),[27] piperidine,[23] mild acidolysis (1% TFA)[23,43]	Z, OBzl; Fmoc, OFm
TMSE	acidolysis (TFA)[42] fluorogenolysis (TBAF)[42]	catalytic hydrogenolysis (H_2/Pd),[42] mild acidolysis (0.5% TFA)[42]	Z, OBzl; Trt

[a] Protecting group is stable under the given conditions.
[b] O→N acyl migration possible.
[c] TFA removal of Boc protecting groups results in partial cleavage of the benzyl ethers.

2.6.4.1 Aliphatic Hydroxy Groups

Among the proteinogenic amino acids only serine contains a primary hydroxy group. This group is readily accessible to acylating agents, e.g. to activated amino acid or peptide derivatives. Thus, the hydroxy group should not be exposed to an excess of activated acylating agent, as used in solid-phase synthesis.[2] In the case of sterically hindered coupling reactions, the serine hydroxy group may prove to be a better nucleophile than the exposed amino group. The resulting *O*-acylserine residues may undergo β-elimination with formation of dehydroalanine peptides. Furthermore, widespread and serious problems can occur during the removal of the protecting groups in peptide synthesis when serine is unprotected. Thus, acylation of unprotected serine hydroxy groups can be found during the removal of N-blocking groups, mostly by a N→O acyl shift. In addition, trifluoroacetylation of the N-terminal serine has been observed during deprotection with TFA.[44,45] Oxazolid-2-one derivatives can be formed upon alkaline treatment of urethane-type N-protected peptides with N-terminal serine or threonine with an unprotected hydroxy group.[46]

Serine reduces the stability of the peptide backbone toward strong acids. Thus, under the influence of super acids, especially HF, a N→O acyl shift occurs, also when the acyl group

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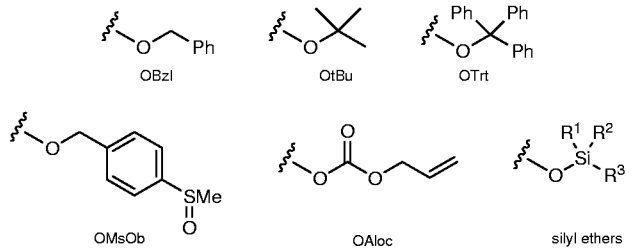
is a part of the peptide chain.[47] Treatment with aqueous bicarbonate leads to complete conversion into the corresponding *N*-acyl derivative.[47] Thus, TfOH and HF should be used with caution for serine-containing peptides, and preferably for short times and below room temperature. It should be pointed out that the last step in the sequence of reactions that lead to protein splicing is the intramolecular O→N or S→N acyl rearrangement.[48]

Despite the relatively high reactivity of the serine hydroxy group, serine esters and serine-containing peptides with free hydroxy groups have been used as amine components in solution synthesis. Syntheses with free serine hydroxy groups can be performed with azides, without the risk of O-acylation.[49,50] Mixed anhydrides, active esters, and carbodiimides have also been successfully used for the coupling reactions. Therefore, for this coupling reaction, any "over activation" has to be avoided.

Threonine contains a sterically hindered and therefore less reactive hydroxy group. The O-acylation of unprotected threonine during coupling reactions is therefore less problematic, but can nevertheless occur. Noncoded hydroxy amino acids containing a secondary hydroxy group have a reactivity similar to threonine. These hydroxy amino acids contain an additional asymmetric center at the β-carbon atom. In the case of reactions on the secondary hydroxy group, this center is accessible to racemization, e.g. threonine can be converted into allo-threonine.

The protecting groups of hydroxy moieties can be classified into three different types, namely alkyl ethers, carbonic acid esters, and silyl ethers (Scheme 1). Pseudoprolines derived from serine and threonine have also been suggested as a form of temporary protection for both of these amino acids.

Scheme 1 Protecting Groups for Aliphatic Hydroxy Groups



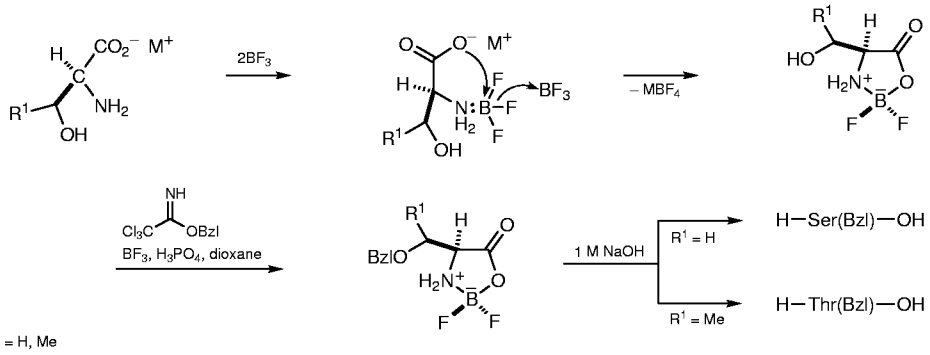
2.6.4.1.1 Alkyl Ethers

The protection of hydroxy groups as ethers of different structural types is the most frequently applied protection method. To facilitate the application of a variety of combinations with amino, carboxy, and side-chain protecting groups, residues requiring different removal conditions were needed. Therefore, ethers of different structures were developed which could be removed under very strong, normal, or very mild acidic conditions, as well as by hydrogenolysis. Of the numerous hydroxy protecting groups reported, few are well established. Apart from this type of protection, two groups with interesting chemical properties are worth mentioning. The 1-(benzyloxycarbonylamino)-2,2,2-trifluoroethyl group for the protection of serine and threonine[51] can be removed by catalytic hydrogenation, but this group has been used only sporadically.[52] The second group is a safety-catch-type hydroxy protecting group, the 4-(methylsulfinyl)benzyl (MsOb) ether, which is stable under both acidic and basic conditions and can be removed in a one-pot reaction involving reductive acidolysis (see Section 2.6.4.1.1.4).[19]

2.6.4.1.1.1 *O*-Benzyl Protection

One of the earliest forms of serine and threonine side-chain protection was use of the benzyl ether. The derivatives Boc-Ser(Bzl)-OH, Boc-Thr(Bzl)-OH, Fmoc-Ser(Bzl)-OH, and Fmoc-Thr(Bzl)-OH have been widely employed in solution and solid-phase peptide synthesis; however, the synthesis of these benzyl ethers is complicated. The compound Boc-Ser(Bzl)-OH is obtained via direct benzylation of Boc-Ser-OH in sodium/liquid ammonia at −30 °C.[53] According to Sugano and Miyoshi,[54] this method is suitable for large-scale preparations and offers high efficiency, procedural simplicity, and mild reaction conditions; however, this procedure is not suitable for threonine protection. H-Thr(Bzl)-OH is obtained by cleavage of the hemioxalate of H-Thr(Bzl)-OBzl, which itself is prepared by benzylation of H-Thr-OH.[14] During this conversion the optical purity is maintained, but the overall yield is very low. A method has since been developed which allows the direct preparation of H-Ser(Bzl)-OH and H-Thr(Bzl)-OH from the amino acids, in good yield and purity. According to this method, salts of serine and threonine are converted with $BF_3 \cdot OEt_2$ into 4-substituted 2,2-difluoro-1,3,2-oxazaborolidin-5-ones and benzylated with benzyl trichloroacetimidate in dioxane with BF_3 as catalyst (Scheme 2).[55] Treatment with 1 M NaOH provides the required benzyl ethers.

Scheme 2 Synthesis of H-Ser(Bzl)-OH and H-Thr(Bzl)-OH via 2,2-Difluoro-1,3,2-oxazaborolidin-5-ones[55]



The benzyl ethers of serine and threonine are stable under the acidic conditions used in the Boc strategy.[56] Cleavage of the benzyl ethers is performed with HF under normal conditions or with mixtures of super acids. Especially in the case of large peptides with many benzyl ether groups, side reactions result from the intermediate benzyl halides that can react with unprotected functionalized side chains; scavengers are added to avoid such side reactions. A modified TMSBr method has also been developed.[57]

Table 2 lists *O*-benzyl-protected serine and threonine derivatives that have been synthesized, along with their physical data.

Table 2 *O*-Benzyl-Protected Serine and Threonine Derivatives[14,53–55,58]

Starting Material	Product	Yield[a] (%)	$[\alpha]_D$[a]	mp[a] (°C)	Ref
H-Ser-OH	H-Ser(Bzl)-OH	95	+6.0 (*c* 1.0, 3 M HCl)	221–222 (dec)	[55]
H-Thr-OH	H-Thr(Bzl)-OH	91	−30.8 (*c* 1.0, AcOH)	203–204	[55]
Boc-Ser-OH	Boc-Ser(Bzl)-OH	45	+19.8 (*c* 2.0, 80% EtOH)	56–58	[53]
Boc-Ser-OH	Boc-Ser(Bzl)-OH · CHA	56	+29.0 (*c* 1.0, MeOH)	159–160	[54]
H-Thr-OH	H-Thr(Bzl)-OBzl · $0.5HO_2CCO_2H$	23	−44.6 (*c* 1.2, MeOH)	165–167	[14]
H-Thr(Bzl)-OBzl · $0.5HO_2CCO_2H$	H-Thr(Bzl)-OH	72	−30.4 (*c* 1.1, AcOH)	n.r.	[14]

for references see p 375

Table 2 (cont.)

Starting Material	Product	Yield[a] (%)	$[\alpha]_D$[a]	mp[a] (°C)	Ref
H-Thr(Bzl)-OBzl · $0.5HO_2CCO_2H$	Boc-Thr(Bzl)-OH	48	n.r.	115–116	[14]
H-Thr(Bzl)-OH	Boc-Thr(Bzl)-OH	75	+15.8 (*c* 1.1, MeOH)	115–116	[14]
Boc-Ser-ONbz	Boc-Ser(Bzl)-ONbz	89	−17.5 (*c* 0.8, MeOH)	n.r.	[58]
Boc-Ser(Bzl)-ONbz	Boc-Ser(Bzl)-OH · CHA	n.r.	n.r.	158–160	[58]

[a] n.r. = not reported.

H-Ser(Bzl)-OH:[55]

To the salt H-Ser-OLi (10 mmol) suspended in THF (15 mL) was added $BF_3 \cdot OEt_2$ (6.0 mL). The mixture was stirred at rt for 6 h, and then at 40–45 °C for an additional 2 h, to give 2,2-difluoro-4-(hydroxymethyl)-1,3,2-oxazaborolidin-5-one in quantitative yield. The THF was removed under reduced pressure at rt and dried (1.0 Torr) for an additional 0.5 h. The residue was dissolved in dioxane (30 mL) and then treated with benzyl trichloroacetimidate (2.15 mL, 11.5 mmol). After 2 h, the mixture was treated with anhyd MeOH (5 mL), stirred for 10 min, and then heated with 1 M NaOH (30 mL); after stirring for 30 min and concentration, the residue was dissolved in H_2O (200 mL) and the resulting soln was washed with Et_2O (3 × 15 mL). The aqueous phase was adjusted to pH 6.0 and applied to an Amberlite XAD-4 resin column. The column was washed with H_2O and then with 50% EtOH. The latter eluent was collected and concentrated; yield: 1.85 g (95%); mp 221–222 °C (dec); $[\alpha]_D^{25}$ +6.0 (*c* 1.0, 3 M HCl).

Boc-Ser(Bzl)-OH · CHA:[54,58]

Method A:[54] To Boc-Ser-OH (20.5 g, 100 mmol) in DMF (500 mL) was added NaH (220 mmol) at 0 °C. When no more gas was evolved, benzyl bromide (18.8 g, 13.1 mL, 110 mmol) was added and the mixture was stirred at 25–30 °C for ca. 5 h. The solvent was removed and the residue was distributed between H_2O (500 mL) and Et_2O (2 × 200 mL). The aqueous layer was acidified to pH 3.5 with 3 M HCl and extracted with EtOAc (5 × 200 mL). The extracts were combined, washed with H_2O (2 × 200 mL), and dried ($MgSO_4$). The solvent was removed in vacuo and the residue was precipitated from Et_2O (300 mL) as the cyclohexylamine salt (10.4 mL, 90 mmol). The product was recrystallized (EtOAc); yield: 22 g (56%); mp 159–160 °C; $[\alpha]_D^{25}$ +29 (*c* 1.0, MeOH).

Method B:[58] Boc-Ser-ONbz (0.68 g, 2 mmol) was added to a mixture of Ag_2O (1.39 g, 6 mmol) and benzyl bromide (0.38 g, 2.2 mmol) in DMF. The mixture was stirred at rt for 10 h, then the solid was filtered off and rinsed with DMF. After the removal of DMF, the residue was purified by chromatography (silica gel, MeOH/$CHCl_3$ 2:98) to give the intermediate Boc-Ser(Bzl)-ONbz; yield: 89%; $[\alpha]_D^{25}$ −17.5 (*c* 0.8, MeOH).

A soln of Boc-Ser(Bzl)-ONbz in THF was treated with 1 M TBAF · $3H_2O$ (2.4 equiv) at rt. After completion of the reaction (TLC), cold H_2O was added and the soln was concentrated under reduced pressure. The residue was dissolved in EtOAc and washed with 5% $KHSO_4$ and with brine. The organic layer was dried (Na_2SO_4) and concentrated to dryness. The residue was converted into the cyclohexylamine salt; yield: not reported; mp 158–160 °C.

H-Thr(Bzl)-OH:[14,55]

Method A:[14] A mixture of benzyl alcohol (100 mL), toluene (200 mL), H-Thr-OH (11.9 g, 0.1 mol), and TosOH · H_2O (24.7 g) was refluxed using a Dean–Stark trap until no more H_2O was collected (18–22 h). The mixture was chilled, diluted with EtOAc (150 mL), and extracted with sufficient cold 0.5 M Na_2CO_3 to bring the aqueous phase to pH 9. The combined organic phases were washed with H_2O and dried ($MgSO_4$). Oxalic acid dihydrate (12 g, 0.1 mol) in MeOH (60 mL) was added and the hemioxalate salt was collected by filtration then washed with cold EtOH to give colorless crystals; yield: 9.1 g (23%); mp 165–167 °C; $[\alpha]_D^{22}$ −44.6 (*c* 1.2, MeOH).

H-Thr(Bzl)-OBzl · $0.5HO_2CCO_2H$ (3.9 g, 0.01 mol) was converted into the free base by partitioning between 1 M K_2CO_3 and EtOAc. The organic layer was dried ($MgSO_4$) and concentrated. The oil was saponified with 1 M NaOH (12 mL) in MeOH (50 mL) at rt for 2 h. The amino acid derivative was isolated by chromatography (Dowex 1, 1 M AcOH) and was recrystallized from H_2O by the addition of PrOH; yield: 1.5 g (72%); $[\alpha]_D^{22}$ −30.4 (*c* 1.1, AcOH).

Method B:[55] To a suspension of H-Thr-ONa (10 mmol) in THF (15 mL) was added $BF_3 \cdot OEt_2$ (6.0 mL). The mixture was stirred at rt for 6 h, and then at 40–45 °C for an additional 2 h, to give 2,2-difluoro-4-(1-hydroxyethyl)-1,3,2-oxazaborolidin-5-one in quantitative yield. H-Thr(Bzl)-OH was then prepared as per H-Ser(Bzl)-OH above; yield: 1.9 g (91%); mp 203–204 °C; $[\alpha]_D^{25}$ −30.8 (*c* 1.0, AcOH).

2.6.4.1.1.2 *O-tert*-Butyl Protection

Besides the benzyl ethers (Section 2.6.4.1.1.1), which are stable toward normal acidic conditions, acid-labile ethers have been developed. *tert*-Butyl ethers can be cleaved under conditions used for cleavage of *tert*-butyl ester and Boc groups, while triphenylmethyl ethers can be cleaved under very mild acidic conditions.

The *tert*-butyl ethers of serine and threonine are available by *tert*-butylation of various starting materials, e.g. Z-Ser-OMe/Z-Thr-OMe,[59] Z-Ser-ONbz/Z-Thr-ONbz,[34] and H-Ser-OMe · TosOH/H-Thr-OMe · TosOH[60] (see also Table 3). Analogous to benzyl ether formation, the *tert*-butyl ethers can also be produced via 4-substituted 2,2-difluoro-1,3,2-oxazaborolidin-5-ones.[55] In most cases, isobutylene with 4-toluenesulfonic acid, or a concentrated inorganic acid is used as catalyst for *tert*-butylation. The use of Fmoc-Ser(tBu)-OH and Fmoc-Thr(tBu)-OH in solid-phase peptide synthesis is very well established. These amino acid derivatives can be synthesized either by introduction of the Fmoc group into H-Ser(tBu)-OH and H-Thr(tBu)-OH or by *tert*-butylation of the Fmoc-protected serine and threonine (Table 3).[61]

tert-Butyl ethers are stable toward hydrogenolysis and the cleavage conditions used for Fmoc/OFm, Dde/ODmab, and Aloc/OAl, but are hydrolyzed with TFA, 1 M HCl, other strong inorganic acids and super acids, such as HF and TfOH, or related deprotection cocktails. Since the *tert*-butyl ethers are to some degree more stable toward acidic conditions than the Boc group, many attempts have been made to find conditions for selective cleavage of Boc groups in the presence of *tert*-butyl ethers; e.g. Gibson et al.[62] have described the selective removal of the Boc group from Boc-Ser(tBu)-OtBu with 1 M HCl in EtOAc. On the other hand, selective cleavage of *tert*-butyl esters in the presence of *tert*-butyl ether groups is possible using trimethylsilyl trifluoromethanesulfonate.[63] This procedure has been exploited for a more efficient preparation of Z-Thr(tBu)-OH, Z-Ser(tBu)-OH, and Z-Hyp(tBu)-OH. Protecting groups such as Ddz[64] or Bpoc,[65] with a higher acid sensitivity than the *tert*-butyl ethers, are removed selectively. The combination of Ddz for α-nitrogen with *O-tert*-butyl protection has been explored for large-scale synthesis.[66] Derivatives of hydroxyproline[10] (Table 3) have been prepared using methods similar to those for serine and threonine.

Table 3 *O-tert*-Butyl-Protected Serine, Threonine, and Hydroxyproline Derivatives[10,11,34,55,59–62,67]

Starting Material	Product	Yield (%)	$[\alpha]_D^a$	mp[a] (°C)	Ref
H-Ser-OH	H-Ser(tBu)-OH	86–92	−17.0 (*c* 1.0, H_2O)	225–226 (dec)	[55]
H-Thr-OH	H-Thr(tBu)-OH	74–87	−44.6 (*c* 1.0, MeOH)	234–236 (dec)	[55]
Z-Ser-OH	Boc-Ser(tBu)-OtBu	67 (Ser)	+5.0 (*c* 1.4, $CHCl_3$)	oil	[62]
Boc-Ser(tBu)-OtBu	H-Ser(tBu)-OtBu · HCl	85	−8.7 (*c* 1.4, $CHCl_3$)	149–150	[62]
Z-Thr(tBu)-ONbz	Z-Thr(tBu)-OH · DCHA	80	+9.93 (*c* 1.0, EtOH)	146–147	[34]
Z-Thr(tBu)-ONbz	H-Thr(tBu)-OH	85	−42.11 (*c* 2.0, MeOH)	259–260 (dec)	[34]
Z-Ser(tBu)-ONbz	Z-Ser(tBu)-OH · DCHA	85	+22.13 (*c* 1.0, EtOH)	149–150	[34]
Z-Ser(tBu)-ONbz	H-Ser(tBu)-OH · 0.25H_2O	90	−17.69 (*c* 2.0, H_2O)	250 (dec)	[34]
Z-Ser-OMe	Z-Ser(tBu)-OMe	95	+3.1 (*c* 1.0, MeOH)	32–33	[59]
Z-Thr-OMe	Z-Thr(tBu)-OMe	100	+6.3 (*c* 1.0, DMF)	oil	[59]
Z-Ser(tBu)-OMe	Z-Ser(tBu)-OH	66	+18.4 (*c* 1.0, MeOH)	83.5–85	[59]
Z-Thr(tBu)-OMe	Z-Thr(tBu)-OH	75	+8.4 (*c* 1.0, DMF)	oil	[59]
Z-Thr(tBu)-OH	Z-Thr(tBu)-OH · DCHA	92	+8.2 (*c* 1.0, DMF)	126–126.5	[59]
Z-Ser(tBu)-OMe	H-Ser(tBu)-OMe · HCl	70	+7.7 (*c* 1.0, DMF)	171–172	[59]
Z-Thr(tBu)-OMe	H-Thr(tBu)-OMe · HCl	67	+12.7 (*c* 0.5, DMF)	153–153.5	[59]

for references see p 375

Table 3 (cont.)

Starting Material	Product	Yield (%)	$[\alpha]_D$[a]	mp[a] (°C)	Ref
H-Ser(tBu)-OMe	Boc-Ser(tBu)-OMe	73	n.r.	oil	[59]
H-Thr(tBu)-OMe	Boc-Thr(tBu)-OMe	75	n.r.	oil	[59]
H-Ser-OMe · HCl	Fmoc-Ser(tBu)-OH	85	+25.9 (*c* 1.0, EtOAc)	129–130.5	[60]
H-Thr-OMe · HCl	Fmoc-Thr(tBu)-OH	61	+15.3 (*c* 1.0, EtOAc)	131–132	[60]
Fmoc-Ser-OBzl	Fmoc-Ser(tBu)-OBzl	90	−5.2 (*c* 1.0, EtOAc)	70–71	[61]
Fmoc-Ser(tBu)-OBzl	Fmoc-Ser(tBu)-OH	95	+25.4 (*c* 1.0, EtOAc)	126–129	[61]
Fmoc-Thr-OBzl	Fmoc-Thr(tBu)-OBzl	72	+6.65 (*c* 1.0, EtOAc)	69–70	[61]
Fmoc-Thr(tBu)-OBzl	Fmoc-Thr(tBu)-OH	73	+15.5 (*c* 1.0, EtOAc)	129–132	[61]
–	Z-Hyp(tBu)-OH	–	−26.5 (*c* 1.0, MeOH)	80–81	[11]
–	Z-Hyp(tBu)-OH · DCHA	–	−6.9 (*c* 1.0, EtOH)	149–150	[11]
Z-Hyp(tBu)-OBzl	Fmoc-Hyp(tBu)-OH	89	n.r.	84–87	[10]
Z-Hyp-OH	Z-Hyp(tBu)-OtBu	79	−37.6 (*c* 2.56, EtOH)	n.r.	[67]

[a] n.r. = not reported.

H-Thr(tBu)-OH:[34,68]

To Z-Thr-OH (50.6 g, 200 mmol) in EtOAc (200 mL) were added TEA (42 mL, 300 mmol) and 4-nitrobenzyl bromide (64.8 g, 300 mmol). The mixture was kept at 80 °C for 8–9 h and then cooled to rt. The separated triethylammonium bromide was removed by filtration and the soln was washed with 2 M HCl (200 mL), H_2O (200 mL), 10% $NaHCO_3$ (200 mL), and again with H_2O (200 mL), then dried (Na_2SO_4) and concentrated. The residue was crystallized (EtOAc/hexane) to give Z-Thr-ONbz; yield: 75.4 g (97%); mp 114–115 °C; $[\alpha]_D^{20}$ −14.0 (*c* 2.0, MeOH).

To a soln of Z-Thr-ONbz (38.8 g, 100 mmol) in CH_2Cl_2 (400 mL) in a thick-walled round-bottomed flask, cooled in an ice–water bath, were cautiously added isobutylene (350 mL) and concd H_2SO_4 (5 mL). The flask was stoppered, wrapped in a towel, and kept at rt for 4 d. The mixture was cooled again, then washed with ice-cold 5% Na_2CO_3. The aqueous extracts were re-extracted with CH_2Cl_2 and the combined organic phases were washed with H_2O, dried (P_2O_5), and concentrated. The residue was crystallized (EtOAc/hexane) to Z-Thr(tBu)-ONbz; yield: 36 g (81%); mp 55–56.5 °C.

The fully protected Z-Thr(tBu)-ONbz (22.3 g, 50 mmol) was dissolved in MeOH (150 mL), H_2O (50 mL) and AcOH (4 mL), and hydrogenated over 10% Pd/C (4.5 g). The catalyst was filtered off, the solvent was removed under reduced pressure, and the residue was recrystallized (MeOH/acetone) to furnish the title product; yield: 7.5 g (85%); mp 259–260 °C (dec); $[\alpha]_D^{20}$ −42.1 (*c* 2.0, MeOH).

Z-Xaa(tBu)-OH · DCHA (Xaa = Ser, Thr, Hyp); General Procedure:[63]

tert-Butylation of Z-amino acids was performed according to Roeske[69] and Callahan et al.[67] To a soln of Z-Xaa(tBu)-OtBu (500 mmol) in dioxane (600 mL) and TEA (750 mmol) was added TMSOTf (750 mmol) under argon, and the mixture was refluxed for 90 min. After cooling to rt, the mixture was poured into H_2O (1.5 L) and extracted with Et_2O. The extract was washed with 5% $KHSO_4$, 10% K_2SO_4, H_2O, and brine. The organic phase was dried (Na_2SO_4) and concentrated. The residue was taken up in Et_2O (600 mL) and DCHA (500 mmol) was added at 4 °C. The resulting crystals of Z-Xaa(tBu)-OH · DCHA were filtered off, washed with Et_2O, and dried at 40 °C; yield: 53–90%.

H-Thr(tBu)-OH via the 2,2-Difluoro-1,3,2-oxazaborolidin-5-one Derivative:[55]

A soln of 2,2-difluoro-4-(1-hydroxyethyl)-1,3,2-oxazaborolidin-5-one [obtained from 10 mmol of H-Thr-OH; see Section 2.6.4.1.1.1, experimental procedure for H-Thr(Bzl)-OH, Method B] in THF was concentrated to dryness, and the drying procedure was continued for an additional 0.5 h. The residue was dissolved in dioxane (30 mL) and stirred for 15 min, then 98% H_3PO_4 (0.4 mL) was added, and stirring was continued for an additional 15 min. After cooling to −20 °C, isobutylene (25 mL) was added at −20 °C; the flask was securely stoppered to ensure no leakage of isobutylene. The resulting soln was stirred at rt for 2.5–3 h. After discharge of isobutylene, the soln was poured into a stirred soln of 1 M aq NaOH (80 mL) at 0 °C, and stirring was continued for 0.5 h. The solvent was then removed in vacuo, the residue was dissolved in H_2O (30 mL), and the soln was extracted with Et_2O (3 × 15 mL). The aqueous soln was adjusted to pH 6.0 with 1 M HCl, after which the soln was chromatographed on Amberlite XAD-4 resin (2.5 × 34 cm). The column was washed with H_2O and then with 80% EtOH. The latter eluent, containing H-Thr(tBu)-OH, was collected and concentrated to give a white solid; yield: 1.29–1.52 g (74–87%); mp 234–236 °C (dec); $[\alpha]_D^{25}$ −44.6 (*c* 1.0, MeOH).

H-Ser(tBu)-OH via the 2,2-Difluoro-1,3,2-oxazaborolidin-5-one Derivative:[55]
H-Ser(tBu)-OH was prepared from 2,2-difluoro-4-(hydroxymethyl)-1,3,2-oxazaborolidin-5-one [obtained from 10 mmol of serine; see Section 2.6.4.1.1.1, experimental procedure for H-Ser(Bzl)-OH] using a procedure similar to that described for H-Thr(tBu)-OH (above), except for the amount of isobutylene (20 mL) and the reaction time for *tert*-butylation (2–2.5 h). After isolation on Amberlite XAD-4 resin, H-Ser(tBu)-OH was obtained as a white solid; yield: 1.38–1.49 g (86–92%); mp 225–226 °C (dec); $[\alpha]_D^{25}$ −17.0 (*c* 1.0, H_2O).

Fmoc-Ser(tBu)-OH:[60]
A soln of H-Ser-OMe · HCl (10.2 g, 65.7 mmol) and TosOH · H_2O (25.0 g, 131 mmol) in CH_2Cl_2 (500 mL) was kept under isobutylene gas (260 Torr) for 72 h. Alternatively, liquid isobutylene, equal to ca. one-fifth the volume of CH_2Cl_2, can be used in a thick-walled, well-stoppered flask. Careful degassing, followed by concentration to one-third volume and addition of Et_2O (2.5 L), gave the *tert*-butylated product; yield: 20.2 g (89%); mp 141–142 °C; $[\alpha]_D^{20}$ +13.0 (*c* 1.0, DMF).

A soln of H-Ser(tBu)-OMe · TosOH (2.0 g, 5.75 mmol) and NaOH (0.46 g, 11.5 mmol) in H_2O (50 mL) was stirred at 0 °C for 2 h and then neutralized with concd HCl. Then, 10% Na_2CO_3 soln was added to the mixture. The mixture was chilled (0 °C) and Fmoc-OSu (2.04 g, 6.04 mmol) in dioxane was added dropwise. After stirring (24 h) and gradual warming to rt, the mixture was washed with Et_2O, and the aqueous phase chilled (0 °C), acidified with concd HCl to pH 2, and then extracted with Et_2O. The combined Et_2O extracts were dried (Na_2SO_4) and concentrated to an oil, which gave white crystals of Fmoc-Ser(tBu)-OH from $MeNO_2$; yield: 2.1 g (95%); mp 129–130.5 °C; $[\alpha]_D^{20}$ +25.9 (*c* 1.0, EtOAc).

2.6.4.1.1.3 *O*-Trityl Protection

Triphenylmethyl (trityl, Trt) ethers of aliphatic hydroxy groups are much more sensitive to acidic conditions than the corresponding *tert*-butyl ethers. They allow deprotection under very mild conditions, as required for very sensitive sequences or for selective deprotection of specific hydroxy groups. The trityl ethers are prepared via the hydroxy amino acid *N,O*-ditrityl esters. The trityl ester is cleaved upon hydrolysis with methanol and the N^{α}-trityl group is removed with 1% TFA in dichloromethane. While the hydroxy group of serine reacts with trityl chloride rapidly, tritylation of threonine proceeds very slowly and thus needs catalysis by DMAP.[17] O-Tritylated amino acids (Table 4) are used in solution or solid-phase synthesis mostly as Fmoc derivatives. The trityl ether group is stable under basic conditions and cleaved under acidic conditions. For selective removal of the *O*-trityl group in combination with N^{α}-Boc, tBu, and OtBu, 20% dichloroacetic acid in dichloromethane is used. This method has been applied in the synthesis of phosphopeptides for phosphorylation of distinct side-chain deprotected serine residues.[18] The possibility of selective removal of N^{α}-trityl protection with 1% TFA in dichloromethane enables the use of Trt-Ser(Trt)-OH and Trt-Thr(Trt)-OH in solid-phase peptide synthesis where N^{α}-deprotection is achieved without affecting the hydroxy protection.[70] The trityl ether group can be quantitatively cleaved with 5% TFA in TFE/CH_2Cl_2 (5:95) in 2–5 minutes at room temperature.[17] The trityl ether derivative of serine can be used as an amino acid derivative or incorporated into oligopeptides for direct glycosylation by cyanoethylidene condensation in the presence of $TrtClO_4$.[71]

for references see p 375

Table 4 *O*-Trityl-Protected Serine and Threonine Derivatives[17,70,71]

Starting Material	Product	Yield[a] (%)	$[\alpha]_D$[a]	mp (°C)	Ref
Trt-Ser-OH	Trt-Ser(Trt)-OTrt	91	−43.4 (*c* 2.0, $CHCl_3$)	201–202	[70]
Trt-Ser(Trt)-OTrt	Trt-Ser(Trt)-ONa	72	+32.3 (*c* 1.0, $CHCl_3$)	209	[70]
Trt-Ser(Trt)-OH	H-Ser(Trt)-OH	66	−4.6 (*c* 2.0, 1 M NaOH/HMPT 1:1)	192	[70]
H-Ser(Trt)-OH	Fmoc-Ser(Trt)-OH	77	+2.2 (*c* 2.0, $CHCl_3$)	195–197	[70]
H-Thr(Trt)-OH	Fmoc-Thr(Trt)-OH	n.r.	−54.5 (*c* 1.0, DMF)	109–110	[17]
H-Ser-OMe · HCl	Trt-Ser(Trt)-OMe	53	n.r.	130–135	[71]
Trt-Ser(Trt)-OMe	H-Ser(Trt)-OMe	41	n.r.	oil	[71]
H-Ser(Trt)-OMe	Fmoc-Ser(Trt)-OMe	69	+3.6 (*c* 2.0, $CHCl_3$)	oil	[71]
H-Ser-OBzl · HCl	Trt-Ser(Trt)-OBzl	94	n.r.	156–157	[71]
Trt-Ser(Trt)-OBzl	H-Ser(Trt)-OBzl	91	n.r.	oil	[71]
H-Ser(Trt)-OBzl	Fmoc-Ser(Trt)-OBzl	98	+1.2 (*c* 2.0, $CHCl_3$)	69–73 (foam)	[71]

[a] n.r. = not reported.

Trt-Ser(Trt)-OTrt:[70]
To a soln of the triethylammonium salt of Trt-Ser-OH (26.0 g, 58 mmol) in $CHCl_3$ (115 mL) was added TEA (7 mL, 50 mmol) and Trt-Cl (17.0 g, 61 mmol), and the mixture was stirred for 20 min at rt. Then, additional TEA (7 mL) and Trt-Cl (17.0 g) were added. After stirring at rt for 3 h, the mixture was concentrated and the residue partitioned between Et_2O and 5% citric acid. The precipitated (light yellow) product was washed with H_2O, MeOH, and Et_2O; yield: 43.7 g (91%); mp 201–202 °C; $[\alpha]_D^{25}$ −43.4 (*c* 2.0, $CHCl_3$).

Trt-Ser(Trt)-ONa:[70]
A suspension of Trt-Ser(Trt)-OTrt (43.7 g, 52.5 mmol) in $CHCl_3$ (80 mL), MeOH (25 mL), and AcOH (2 mL) was warmed to 40–45 °C and stirred for 5–10 min until complete dissolution occurred. Then, the mixture was concentrated and partitioned between Et_2O and 5% citric acid. The organic layer was washed with H_2O. After addition of 1 M NaOH, the sodium salt of ditrityl serine was filtered off, washed with 1 M NaOH, H_2O, and Et_2O, and recrystallized (acetone); yield: 23.05 g (72%); mp 209 °C; $[\alpha]_D^{25}$ +32.3 (*c* 1.0, $CHCl_3$).

H-Ser(Trt)-OH:[70]
Trt-Ser(Trt)-OH (2–5 mmol) was stirred in 1% TFA in CH_2Cl_2 (20–50 mL) at 0 °C for 3 min, then the volatiles were removed at 0 °C. The residue was partitioned between 1 M NaOH and Et_2O and the aqueous layer was acidified to pH 4.8–5.0 with AcOH. After 1 h, the precipitate was separated, washed with H_2O, acetone, and Et_2O, and recrystallized (EtOH/H_2O 1:1); yield: 66%; mp 192 °C; $[\alpha]_D^{25}$ −4.6 (*c* 2.0, 1 M NaOH/HMPT 1:1).

2.6.4.1.1.4 *O*-[4-Methylsulfinylbenzyl] Protection

The synthesis of peptides with sulfated or otherwise modified tyrosine residues requires selective protection of the hydroxy groups of serine and threonine, on the one hand, and of the hydroxy group of tyrosine on the other hand (see Vol. E 22b, Section 6.6). Futaki et al.[20] have used *O*-[4-methylsulfinylbenzyl] (MsOb) protection for serine in order to perform selective sulfation of tyrosine in leucosulfakinin and cholecystokinin fragment CCK-12. The MsOb group was originally introduced as a carboxy protecting group and is stable to TFA, but its reduced form, i.e. the 4-(methylthio)benzyl group (Mtz) is readily removed with TFA; the same situation is true for protection of aliphatic hydroxy groups as MsOb derivatives. In fact, these derivatives are 10 times more stable toward TFA than the corresponding benzyl ethers. The MsOb ethers of serine and threonine can be cleaved with 1 M $SiCl_4$ (10 equiv)/TFA–anisole (10 equiv);[19] however, in the reduced form the ethers can be cleaved with TFA, 90% aqueous TFA, or 10% EDT in TFA at 0 °C. The reduction of sulfinyl to the corresponding thio derivatives is undertaken with a DMF–SO_3 complex/EDT system.[20]

Z(OMe)-Ser(MsOb)-OH · CHA:[20]
To a soln of Z(OMe)-Ser-OH (0.95 g, 3.53 mmol) in DMF (40 mL) at −20 °C was added NaH (7.77 mmol) in 30 min, followed by MsOb-Br (0.99 g, 4.24 mmol). The mixture was stirred at rt for 3 h, then the solvent was removed, and the residue was dissolved in 5% $NaHCO_3$ (25 mL) and extracted with Et_2O. The aqueous layer was acidified with citric acid, the product was extracted with EtOAc (30 mL), and the extract was washed with H_2O, dried (Na_2SO_4), and concentrated to 5 mL. Cyclohexylamine (0.35 mL) was added to the ice-chilled soln, then the solvent was removed. The residue was treated with Et_2O and the solid was recrystallized (MeOH/Et_2O); yield: 0.82 g (44%); mp 135–138 °C; $[\alpha]_D^{25}$ +12.0 (*c* 1.0, MeOH).

2.6.4.1.1.5 Other Ether-Type Protection

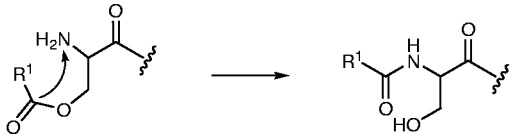
To improve preparation, stability, and cleavage properties, new alkyl protecting groups with modified chemical properties have been developed. Among these, the methoxymethyl (Mom) group is stabile to acidolysis and hydrogenolysis, and can be cleaved with HF.[72] The 1-(benzyloxycarbonylamino)-2,2,2-trifluoroethyl group, an amino acetal group, is used for hydroxy protection of serine and threonine;[45] this protecting group is easily introduced into the hydroxy groups of serine and threonine, can be removed smoothly by catalytic hydrogenation, and has been used for solid-phase synthesis.[46]

Since the benzyl group is not completely adequate for Boc-based solid-phase synthesis of longer peptides and proteins, the cyclohexyl group has been proposed for the side-chain protection of serine.[73] This alkyl group is highly stable toward 50% TFA in CH_2Cl_2, with no cleavage occurring over a 96-hour period at room temperature. The cyclohexyl group can be quantitatively removed by treatment with 1 M TfOH/thioanisole in TFA at room temperature for 30 minutes. As it is not possible to introduce the cyclohexyl group with cyclohexyl bromide, 3-bromocyclohexene is used for this purpose, followed by hydrogenation over PtO_2.

2.6.4.1.2 Acyl Esters

The chemical reactivity of aliphatic hydroxy groups is reduced by acylation, but the resulting esters are reactive toward amino groups. Thus, under basic conditions, as commonly used for Fmoc deprotection, an intramolecular O→N migration of the acyl residue can occur.[11,12,74] This O→N shift, which produces N-acylated derivatives (Scheme 3), is nearly irreversible and further elongation of the peptide is prevented. The same reaction can occur during acidolytic deprotection of final peptides with TFA via intermediate formation of the trifluoroacetyl ester of N-terminal hydroxy amino acids.[44,45]

Scheme 3 O→N Shift of Acyl Residues in Hydroxy Amino Acids[11,12,74]

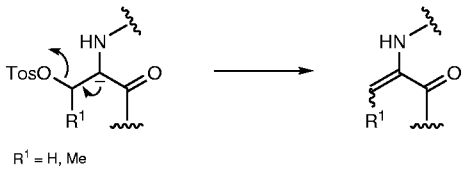


Another side reaction described for *O*-acyl derivatives of serine and threonine is base-catalyzed β-elimination, particularly if the acyl residue provides a stable leaving group. The formation of dehydroalanine and dehydrobutyrine from *O*-tosylserine and *O*-tosylthreonine, respectively, is shown in Scheme 4. Because of the serious problems encountered, *O*-acetyl, *O*-benzyloxycarbonyl, and *O*-tosyl derivatives of serine and threonine, in general, are no longer used in peptide synthesis.[2,12,75] As a selective protection for the hydroxy group of serine, the phenoxyacetyl Ph(OAc) residue has also been used.[76] This group was chosen because it is stable to HF and TFA treatment, but is readily removed in 0.1 M aqueous

for references see p 375

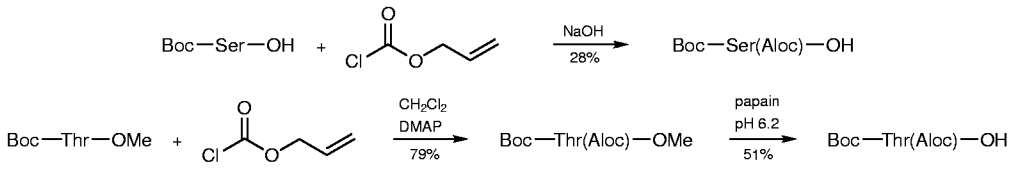
ammonia within 30 minutes at room temperature. To avoid O→N migration, the serine derivatives were initially incorporated into peptide segments without protection of the hydroxy group and then subsequently protected at an appropriate stage with the phenoxyacetyl group by treatment with phenoxyacetic anhydride in pyridine at 0 °C for 30 minutes.[76]

Scheme 4 Formation of Dehydroalanine and Dehydrobutyrine from *O*-Tosylserine and *O*-Tosylthreonine[12]



From the large variety of acyl residues proposed as protecting groups, only the allyloxycarbonyl (Aloc) group is apparently less problematic. The allyl ether group itself cannot be used for hydroxy protection of serine and threonine because allyl ethers do not form π-allyl complexes in the presence of zero-valent palladium catalysts.[21,22] Introduction of the Aloc protecting group into N^{α}-protected serine and threonine is performed with allyl chloroformate (Scheme 5). N^{α}-Fmoc-protected amino acids can be obtained by conversion of the related Boc-protected derivatives.

Scheme 5 Synthesis of *O*-Allyloxycarbonyl-Protected Serine and Threonine Derivatives[22]



Final products isolated as DCHA salts

The great advantage of the allyl-based protecting groups is their selective and very smooth removal through formation of a π-allyl palladium complex. This complex reacts with a nucleophile Nu to give Nu^- and the regenerated catalyst. The deprotection is usually performed with palladium(0) under neutral conditions.[21,22] The variety of catalysts and the conditions used for the removal of the Aloc group from the α-amino nitrogen can also be applied to the cleavage of Aloc-protected hydroxy groups (see Section 2.1.1; for a review see[21]). The Aloc side-chain protecting group is stable under the basic and acidic conditions used for Fmoc and Boc deprotection. The derivatives are thus compatible with most of the current strategies used in peptide synthesis. Furthermore, Aloc side-chain-protected serine and threonine derivatives (Boc, Fmoc; see Table 5) can be produced economically in high purity.[22] Because of the very smooth removal, Aloc protection has been used for the synthesis of glycopeptides (see Vol. E 22b, Section 6.3)[77,78] and peptide derivatives which are very sensitive toward acidic or basic conditions.[79] Despite all of these advantages of allyloxycarbonyl protection, there is still the risk of an O→N migration, as well as the formation of dehydroamino acids (see Scheme 4).

Table 5 *O*-Allyloxycarbonyl-Protected Serine and Threonine Derivatives[22]

Starting Material	Product	Yield (%)	$[\alpha]_D$	mp (°C)	Ref
Boc-Ser-OH	Boc-Ser(Aloc)-OH · DCHA	28	+19.6 (*c* 1.0, MeOH)	111–113	[22]
Boc-Thr-OMe	Boc-Thr(Aloc)-OH · DCHA	51	+22.0 (*c* 1.0, MeOH)	108–109	[22]
Boc-Ser(Aloc)-OH · DCHA	Fmoc-Ser(Aloc)-OH	24	−10.6 (*c* 1.0, DMF)	67–70	[22]
Boc-Thr(Aloc)-OH · DCHA	Fmoc-Thr(Aloc)-OH	63	+10.5 (*c* 1.0, MeOH)	oil	[22]

Boc-Ser(Aloc)-OH · DCHA:[22]
To a soln of Boc-Ser-OH (10.2 g, 0.05 mol) in 1 M NaOH (50 mL) were added allyl chloroformate (20 mL, 0.188 mol) and 2 M NaOH (200 mL) alternately in portions of 2 and 20 mL, respectively. The pH of the mixture was kept at 9–13. The mixture was stirred for 1 h at rt, then the pH was adjusted to 3.0 with 12 M HCl. The suspension was extracted with EtOAc (3 × 100 mL), and the combined organic phases were washed with H_2O and dried ($MgSO_4$). The solvent was removed and the crude product was converted into the DCHA salt, which crystallized from Et_2O/hexane; yield: 6.7 g (28%); mp 111–113 °C; $[\alpha]_D^{25}$ +19.6 (*c* 1.0, MeOH).

Boc-Thr(Aloc)-OH · DCHA:[22]
Boc-Thr-OMe (7.1 g, 0.03 mol) and allyl chloroformate (4.3 g, 0.036 mol) were dissolved in CH_2Cl_2 (50 mL). The soln was cooled at 0 °C and stirred while DMAP (4.4 g, 0.036 mol) was added. The mixture was then stirred for 20 h at rt. The solvent was removed and the residue was dissolved in Et_2O (100 mL). The organic phase was washed with 0.1 M HCl, 10% $NaHCO_3$, and brine, then dried ($MgSO_4$). The solvent was removed to give Boc-Thr(Aloc)-OMe as a clear oil; yield: 7.5 g (79%).

The crude product (3.77 g, 0.015 mol) was dissolved in EtOH (50 mL). To this soln was added H_2O (100 mL) and cysteine hydrochloride monohydrate (150 mg). The resulting mixture was adjusted to pH 6.2 with 0.1 M NaOH and heated to 35 °C while crude papain (200 mg) was added. The pH was adjusted to 6.2 by the addition of 0.1 M NaOH over a period of 6 to 10 h at 35 ± 2 °C. The mixture was stirred for 1 h at rt, after which the pH was adjusted to 3.0 with 12 M HCl. This suspension was extracted with EtOAc (3 × 100 mL), and the combined organic phases were washed with H_2O and dried ($MgSO_4$). The solvent was removed and the crude product was converted into the DCHA salt, which crystallized from Et_2O/hexane; yield: 3.7 g (51%); mp 108–109 °C; $[\alpha]_D^{25}$ +22.0 (*c* 1.0, MeOH).

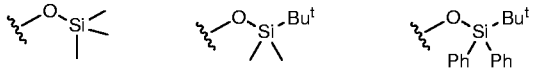
Removal of the Aloc Group from *O*-Aloc-Protected Serine and Threonine; General Procedure:[22]
To a soln of the Aloc-protected amino acid (1 mmol) in CH_2Cl_2 (5 mL) were added $PdCl_2(PPh_3)_2$ (0.04 mmol) and AcOH (2.5 mmol) followed by Bu_3SnH (2 mmol). The mixture was stirred for 10 min at rt. In most cases a color change of the soln from yellow to purple indicated the end of the reaction.

2.6.4.1.3 Alkylsilyl and Alkyl(aryl)silyl Ethers

For the synthesis of peptides containing a variety of functional side chains, or special non-proteinogenic amino acids requiring selective protection and deprotection, more than two or three different types of protecting groups are needed. In addition to alkyl ethers and acyl esters (see Sections 2.6.4.1.1 and 2.6.4.1.2), protecting groups based on the Si—O bond increase the versatility of protection strategies. Depending on the silyl substituents, the stability of the silyl ethers as well as the cleavage conditions differ significantly. A general advantage of silyl-based protection compared to use of alkyl ether groups is their smooth and nearly quantitative introduction, even in the case of the secondary hydroxy group, as present in threonine. Of the various alkylsilyl and alkyl(aryl)silyl ethers (for a review see ref[27]), only three have been fully established in peptide chemistry for the protection of aliphatic hydroxy groups: these are the ethers derived from the trimethylsilyl (TMS), the *tert*-butyldimethylsilyl (TBDMS), and the *tert*-butyldiphenylsilyl (TBDPS) groups (Scheme 6).

for references see p 375

Scheme 6 Silyl Protecting Groups for Aliphatic Hydroxy Groups



TMS TBDMS TBDPS

The trimethylsilyl group was the first to be developed and is widely used for the protection of serine and threonine (Table 6). Chlorotrimethylsilane, 1,1,1,3,3,3-hexamethyldisilazane, and *N,O*-bis(trimethylsilyl)acetamide are commercially available reagents used for the conversion of alcohols into the corresponding trimethylsilyl derivatives.[80,81] Furthermore, trimethylsilyl cyanide has been used to protect the side chains of serine, threonine, and tyrosine.[82] This silyl protection allows the formation of *N*-carboxyanhydrides from H-Ser(TMS)-OH and H-Thr(TMS)-OH, and their application in peptide synthesis in the aqueous phase.[83] The TMS group can be removed under various conditions, depending on the kind of functional group to which it is bound; the TMS ethers are more stable than related amino or carboxy derivatives.[81] These differences in stability allow the direct application of completely silylated hydroxy amino acids in peptide synthesis.[84]

Table 6 *O*-Trimethylsilyl-Protected Serine, Threonine, and Hydroxyproline Derivatives[80,81,83,85]

Starting Material	Product	Yield[a] (%)	$[\alpha]_D$[a]	mp[a] (°C) or bp (°C/Torr)	Ref
H-Ser-OH	TMS-Ser(TMS)-OTMS	97	n.r.	73/0.4	[85]
TMS-Ser(TMS)-OTMS	H-Ser(TMS)-OTMS	45	n.r.	120; 115–116/17	[80]
H-Ser(TMS)-OTMS	H-Ser(TMS)-OH	n.r.	n.r.	n.r.	[81]
H-Ser-NCA	H-Ser(TMS)-NCA	49	−46.7 (*c* 1.1, $CHCl_3$)	n.r.	[83]
H-Thr-OH	TMS-Thr(TMS)-OTMS	90	n.r.	75–78/0.2	[80]
TMS-Thr(TMS)-OTMS	H-Thr(TMS)-OTMS	84	n.r.	173–174; 116–118/16	[80]
H-Thr-NCA	H-Thr(TMS)-NCA	65	−50.5 (*c* 1.9, EtOAc)	76	[83]
H-Hyp-OH	TMS-Hyp(TMS)-OTMS	93	n.r.	75/0.001	[85]

[a] n.r. = not reported.

Cleavage of silyl ethers is usually performed with fluoride ions, using tetrabutylammonium fluoride in tetrahydrofuran.[23,26] While the TMS group is used only as a temporary hydroxy protection during peptide assembly, the *tert*-butyldimethylsilyl and *tert*-butyldiphenylsilyl groups are more stable to hydrolysis, but are still readily cleaved under a variety of selective conditions. Table 7 lists the stability characteristics of these silyl ethers, and indicates that all three protecting groups are stable under the general conditions of peptide synthesis. Bulky substituents at the silicon atom increase the stability of the ether bond (TMS<TBDMS<TBDPS); TBDPS ethers are unaffected under conditions used for the cleavage of *N*-trityl, *O*-trityl, *O-tert*-butyldimethylsilyl, and Boc groups. Thus, its advantage over other silyl-based protecting groups comes from the possibility of selective protection of hydroxy groups and from its much higher stability under acidic conditions. The TBDMS and TBDPS groups, in particular, are stable toward conditions used for the cleavage of other protecting groups, as well as for the preparation of aldehydes from amino acid derivatives. Furthermore, TBDMS ethers are stable to catalytic hydrogenolysis (H_2/Pd) and mild chemical reduction (Zn, DIBAL-H). There are, however, contradictory reports about their stability under alkaline conditions. In fact, under alkaline conditions, side reactions may occur as a result of a possible intramolecular migration of the silyl moiety from the hydroxy to the carboxy group, thus leading to its removal.[26]

The TBDMS ethers are prepared with *tert*-butyldimethylsilyl chloride and imidazole in tetrahydrofuran. The usual reagent for cleavage of this type of silyl ether is also tetrabutylammonium fluoride; another mild and efficient method for the deprotection is the use of iodine in methanol.[24]

Table 7 Stability Characteristics of Trimethylsilyl, *tert*-Butyldimethylsilyl, and *tert*-Butyldiphenylsilyl Ethers of Serine, Threonine, and Hydroxyproline[23–27,80,81,84,86–89]

Protecting Group	Conditions	Time (min)	Deprotection (%)	Ref
TMS	aq basic and acidic extraction	–	100	[84]
	gaseous NH_3	–	0	[80]
	HCl, Et_2O	–	0	[81]
	alcoholysis, 25 °C	–	0	[81]
	1% HCl/95% MeOH	< 1	50	[26]
TBDMS	1% HCl/95% MeOH	< 1	50	[26]
	neat TFA	15	100	[23]
	AcOH/THF/H_2O (3:1:1), 60 °C, 2 h	120	100	[23]
	AcOH/THF/H_2O (3:1:1), 22 °C, 18 h	1080	100	[23]
	aq basic and acidic extraction	–	0	[25,27]
	piperidine/DMF (1:4), DMAP, DIPEA, DBU	1080	0	[23]
	DIBAL-H, −78 °C	60	0	[86]
	0.1 M TBAF, THF, H_2O, 60 °C	30–1080	100	[23,88]
	1% I_2, MeOH, 25 °C	120	92	[24]
	MeI, Ag_2O	–	0	[27]
	Na/liq NH_3	–	0	[27]
	CrO_3, Py	–	0	[27]
	hydrogenation (Pd/C, MeOH)	–	0	[25]
TBDPS	80% AcOH	–	0	[27]
	hydrogenation (Pd/C, MeOH)	–	0	[27]
	DIBAL-H, −78 °C	–	0	[27]
	traces NaOMe, MeOH, 25 °C	1440	0	[27]
	9 M NH_3, dioxane, 60 °C	120	0	[27]
	1 M TBAF (2–3 equiv), THF, 25 °C	60–300	100	[26,27,89]
	TBAF (20 equiv), H_2O, EDT, 30 °C	120	100	[87]
	1% HCl/95% MeOH	41–225	50	[26]
	2 M NaOH, 50% EtOH, 25 °C	420	100	[27]
	50% aq TFA, dioxane, 25 °C	15	0	[27]

Silyl ethers of hydroxy amino acids have been employed in both solid-phase and solution peptide synthesis. Since the TBDMS and TBDPS ether groups are stable to piperidine and other basic reagents, these derivatives of Ser and Thr (Tables 8 and 9) can be used in the Fmoc strategy.[23,26] Homodetic peptides and depsipeptides have also been synthesized using TBDMS hydroxy protection for phenyllactic acid and hydroxyisovaleric acid, respectively.[90]

for references see p 375

Table 8 *O-tert*-Butyldimethylsilyl-Protected Serine, Threonine, and Hydroxyproline Derivatives[23,88,91–107]

Starting Material	Product	Yield[a] (%)	$[\alpha]_D$[a]	mp (°C) or bp (°C/Torr)	Ref
H-Ser-OMe	H-Ser(TBDMS)-OMe	96	+7.0 (*c* 2.1, $CHCl_3$)	112–114/3	[88,91]
H-Ser-OEt	H-Ser(TBDMS)-OEt	98	n.r.	oil	[92]
Boc-Ser-OMe	Boc-Ser(TBDMS)-OMe	70	+21.6 (*c* 2.1, $CHCl_3$)	125/1	[93]
H-Ser-OMe·xHCl	Boc-Ser(TBDMS)-OMe	90	+28.0 (*c* 1.6, $CHCl_3$)	oil	[94]
Boc-Ser-OMe	Boc-Ser(TBDMS)-OMe	89	+7.5 (*c* 3.2, $CHCl_3$)	oil	[95]
Z-D,L-Ser-OH	Z-D,L-Ser(TBDMS)-OH	86	n.r.	65–66	[96]
Z-Ser-OH	Z-Ser(TBDMS)-OH	n.r.	n.r.	oil	[23]
Z-Ser-OMe	Z-Ser(TBDMS)-OMe	90	+5.0 (*c* 1.5, $CHCl_3$)	oil	[97]
Z-Ser-OMe	Z-Ser(TBDMS)-OMe	96	+18.6 (*c* 2.3, $CHCl_3$)	oil	[98]
Z-Ser-OH	Z-Ser(TBDMS)-OTMSE	42	+8.6 (*c* 1.0, $CHCl_3$)	oil	[99]
Fmoc-Ser-OH	Fmoc-Ser(TBDMS)-OH	n.r.	n.r.	amorphous	[23]
Aloc-D-Ser-OMe	Aloc-D-Ser(TBDMS)-OMe	73	n.r.	oil	[100]
H-D-Thr-OMe·xHCl	H-D-Thr(TBDMS)-OMe	94	+18.5 (*c* 1.3, $CHCl_3$)	oil	[101]
Z-Thr(TBDMS)-OH	H-Thr(TBDMS)-OH	100	−28.7 (*c* 1.0, MeOH)	152–169	[102]
Z-Thr-OH	Z-Thr(TBDMS)-OH	89	+13.2 (*c* 1.0, $CHCl_3$)	151–153	[102]
Z-Thr-OH	Z-Thr(TBDMS)-OH	73	n.r.	157–158	[103]
H-Thr-OMe·xHCl	Z-Thr(TBDMS)-OMe	99	−7.3 (*c* 3.6, $CHCl_3$)	oil	[104]
Z-Thr-OMe	Z-Thr(TBDMS)-OMe	94	−5.8 (*c* 1.1, $CHCl_3$)	oil	[105]
Z-Thr(TBDMS)-OMe	Z-Thr(TBDMS)-OH	65	+15.1 (*c* 1.0, $CHCl_3$)	158–160	[105]
Fmoc-Thr-OH	Fmoc-Thr(TBDMS)-OH	n.r.	n.r.	amorphous	[23]
H-thr-OMe · xHCl	Boc-Thr(TBDMS)-OMe	94	−12.0 (*c* 0.8, $CHCl_3$)	oil	[94]
Z-Hyp(TBDMS)-OMe	Z-Hyp(TBDMS)-OH	99	−47.8 (*c* 1.5, $CHCl_3$)	oil	[106]
Z-Hyp-OMe	Z-Hyp(TBDMS)-OMe	89	−39.0 (*c* 0.8, $CHCl_3$)	oil	[106]
Boc-Hyp-OMe	Boc-Hyp(TBDMS)-OMe	98	n.r.	137–150/1.1	[107]

[a] n.r. = not reported.

Table 9 *O-tert*-Butyldiphenylsilyl-Protected Serine and Threonine Derivatives[26,89,108–111]

Starting Material	Product	Yield[a] (%)	$[\alpha]_D$[a]	mp (°C)	Ref
Z-Ser(TBDPS)-OMe	H-Ser(TBDPS)-OMe	65	n.r.	oil	[26]
Z-Ser(TBDPS)-OMe	Z-Ser(TBDPS)-OH	35	n.r.	146–148	[26]
Z-Ser-OMe	Z-Ser(TBDPS)-OMe	61	n.r.	oil	[26]
Z-Ser-OMe	Z-Ser(TBDPS)-OMe	100	+8.2 (*c* 1.2, CH_2Cl_2)	oil	[108]
Boc-Ser(TBDPS)-OBzl	Boc-Ser(TBDPS)-OH	90	n.r.	155–156	[26]
Boc-D-Ser-OH	Boc-D-Ser(TBDPS)-OH	n.r.	−13.4 (*c* 1.0, HCl_3)	150–152	[109]
Boc-Ser-OBzl	Boc-Ser(TBDPS)-OBzl	79	n.r.	oil	[26]
Boc-Ser-OMe	Boc-Ser(TBDPS)-OMe	95	+14.3 (*c* 1.2, $CHCl_3$)	oil	[110]
Boc-Ser-OMe	Boc-Ser(TBDPS)-OH[b]	90	+4.1 (*c* 2.8, $CHCl_3$)	72–73	[89]
Z-*erythro*-D-Thr-OMe	Z-*erythro*-D-Thr(TBDPS)-OMe	90	+15.0 (*c* 1.0, $CHCl_3$)	oil	[111]
Z-*threo*-D-Thr-OMe	Z-*threo*-D-Thr(TBDPS)-OMe	92	−9.1 (*c* 2.0, $CHCl_3$)	oil	[111]
Boc-Thr-OMe	Boc-Thr(TBDPS)-OMe	92	−4.6 (*c* 1.0, $CHCl_3$)	oil	[110]

[a] n.r. = not reported.
[b] Serinol derivative.

Z-Thr(TBDMS)-OH; Typical Procedure for O-*tert*-Butyldimethylsilyl Protection:[105]

A mixture of Z-Thr-OMe (5.06 g, 20 mmol), TEA (4.12 g, 29.6 mmol), TBDMSCl (4.34 g, 28.8 mmol), and DMAP (200 mg, 1.6 mmol) as catalyst in DMF (40 mL) was heated at 40 °C for 8.5 h under an argon atmosphere. The mixture was poured into ice water, and the product was extracted with Et_2O. The organic soln was washed with H_2O and brine, and dried (Na_2SO_4). The solvent was removed and the residue chromatographed (silica gel) to give the silyl derivative Z-Thr(TBDMS)-OMe; yield: 7.14 g (94%); $[\alpha]_D^{20}$ −5.84 (*c* 1.08, $CHCl_3$).

To a soln of this ester (7.0 g, 18.3 mmol) in MeOH (60 mL) was added 1 M NaOH (29 mL, 29 mmol) at 0 °C, and the mixture was stirred at rt for 5 h. Most of the MeOH was removed, and the aqueous soln was extracted with Et_2O to remove unhydrolyzed ester (0.9 g, 13%). The aqueous soln was acidified with 2 M HCl, and NaCl was added to saturation. The product was extracted with EtOAc, and the organic layer was washed with brine, and dried (Na_2SO_4). The soln was taken to dryness to give Z-Thr(TBDMS)-OH; yield: 4.4 g (65%); mp 158–160 °C; $[\alpha]_D^{20}$ +15.1 (*c* 1.0, $CHCl_3$).

Boc-Ser(TBDPS)-OH; Typical Procedure for *O-tert*-Butyldiphenylsilyl Protection:[26]
To a soln of Boc-Ser-OBzl (200 mg, 0.68 mmol) and imidazole (116 mg, 1.70 mmol, 2.5 equiv) in dry THF was added TBDPSCl (206 mg, 0.75 mmol, 1.1 equiv) and the soln was stirred overnight at rt. After filtration and removal of the solvent, the silyl derivative was purified by chromatography (silica gel, CH_2Cl_2) to give Boc-Ser(TBDPS)-OBzl as an oil; yield: 290 mg (79%).

Boc-Ser(TBDPS)-OBzl (171 mg, 0.32 mmol) was hydrogenated in MeOH (10 mL) over Pd/C (17 mg). The catalyst was filtered off and, on removal of the solvent, Boc-Ser(TBDPS)-OH was obtained as a syrup; yield: 113 mg (90%). The syrup crystallized on trituration with Et_2O to give a solid; mp 155–156 °C.

2.6.4.1.4 Pseudoprolines (Synthesis of Dipeptide Building Units from Serine and Threonine)

A newer method for protection of the hydroxy groups of serine and threonine during peptide assembly consists of the formation of oxazolidine derivatives, and their use as intermediates, from which the hydroxy groups are restored after ring opening.

Mutter et al.[112-115] opened the way for the use of pseudoprolines as a temporary protection technique for serine, threonine, and also cysteine side chains. The incorporation of pseudoproline moieties into peptides results in a kinked conformation, thus preventing peptide aggregation or β-structure formation. Thus, the pseudoprolines lead to an increased solubility of the growing protected peptide chain. Because the oxazolidine ring needs to be stabilized by acylation, dipeptide building units have been synthesized and used for peptide assembly. The two alternative routes for their synthesis are given in Scheme 4 of Section 2.3.2.4.

In recent years many pseudoproline-containing dipeptide building units have been described (see below). Due to the differences in the chemical stability as a result of modifications at the C2 position of the oxazolidine ring, this type of hydroxy group protection is compatible with various common strategies in peptide synthesis. Most of the building units are N^{α}-Fmoc protected and are used in Fmoc-based solid-phase synthesis because the pseudoprolines are stable under the conditions used for Fmoc removal. The conditions for the ring-opening reaction depend on the substituents at the C2 position. Thus, dimethylated pseudoprolines can be opened with TFA in dichloromethane or in other aprotic solvents; however, they are stable toward the acidic conditions used for removal from 2-chlorotrityl resin and SASRIN resin, allowing the cleavage of pseudoproline-containing peptides from these very acid-labile resin linkers. Unsubstituted oxazolidines are more stable toward acidolysis, and their ring opening requires exposure to TfOH.

The pseudoproline technique has been applied to the synthesis of sparingly or completely insoluble peptides, e.g. for the synthesis of Aib-rich peptides.[113] The temporary protection of serine and threonine hydroxy groups is now well established in peptide chemistry and has been strongly facilitated by the variety of corresponding dipeptide building units now commercially available. The general methods of preparation of the dipeptide units Fmoc-Xaa-Thr-OH, Fmoc-Xaa-Oxa(2,2,5-Me_3)-OH, Fmoc-Xaa-Thr-OH, and Fmoc-Xaa-Oxa(5-Me)-OH are described in Section 2.3.2.4; here an illustrative example is given of the synthesis of a difficult peptide sequence using temporary serine protection via this pseudoproline approach.

Fmoc-Ala-Thr-OH:[112]
A soln of formaldehyde (37% in H_2O; 256 mmol, 20 equiv) was added to a soln of serine (10.8 g, 103 mmol, 8 equiv) and 2.5 M Na_2CO_3 (32 mL) at rt. The clear soln was left to stand overnight at 4 °C, after which the pH was 7.8. 1 M Fmoc-Ala-F (4.0 g, 12.8 mmol) in acetone was added dropwise over a period of 70 min while the pH was adjusted to between 7.8 and 8.2 with Na_2CO_3 (ca. 1 g). The mixture was stirred for another 30 min at rt and then cooled to 0 °C. The soln was acidified with concd HCl (16 mL), to give a white suspension at pH 3.0 to 4.0, which was extracted with EtOAc (4 × 50 mL). The combined organic phases were washed with brine (3 × 80 mL) then dried ($MgSO_4$). After removal of the solvent, the crude product was purified by flash chromatography ($CHCl_3$/MeOH/AcOH 100:3:3) and was lyophilized from MeCN/H_2O to give a white powder; yield: 3.94 g (75% from the amino acid fluoride); mp 115–120 °C.

for references see p 375

Fmoc-Ala-Oxa(5-Me)-OH:[112]
Fmoc-Ala-Oxa(5-Me)-OH was synthesized according to the procedure described above for Fmoc-Ala-Oxa-OH, except that the acylation reaction was performed with commercially available Fmoc-Ala-NCA, instead of the less accessible amino acid fluoride; yield: 50%; mp 103–108 °C.

Reconversion of Pseudoproline Moieties into Hydroxy Amino Acid Derivatives:[113]
The peptide Ac-A-S-A-S-A-S-A-S-A-S-C-NH_2 was obtained from Ac-A-S-A-Oxa-A-S-A-Oxa-A-S-C-NH_2 by treatment with 10% TfOH in TFA for 4.5 h. The soln was then concentrated in vacuo, and the product was precipitated with dry Et_2O then washed three times with Et_2O. Characterization of peptide Ac-A-S-A-S-A-S-A-S-A-S-C-NH_2 was rendered extremely difficult due to its insolubility in solvents other than TFA and HFIP.

2.6.4.2 Aromatic Hydroxy Groups

The phenolic function of tyrosine differs in its chemical properties to some degree from aliphatic hydroxy groups. It is more acidic, thus leading to reduced stability of alkyl protecting groups toward acids. Moreover, the aromatic ring is susceptible to electrophilic substitutions in the position *ortho* to the hydroxy group by carbocations formed in acidic deprotection steps. Therefore, in addition to the protecting groups established for aliphatic hydroxy groups, some special derivatives have been developed for tyrosine to improve the stability and to minimize possible side reactions. Based on their structure, the tyrosine protecting groups can also be classified as alkyl, acyl, and alkylsilyl derivatives.

The choice of the most useful tyrosine protection depends on the general synthetic strategy employed, as well as on the presence of additional sensitive residues in the peptide, the position of the tyrosine residue in the sequence, the use of unusual reaction conditions, and the final deprotection method chosen in solution or solid-phase synthesis. For standard SPPS according to the Boc/Bzl and Fmoc/tBu strategies, Boc-Tyr[Z(2-Br)]-OH and Fmoc-Tyr(tBu)-OH, respectively, represent the most suitable tyrosine derivatives;[30,116] however, an orthogonal protection of the phenolic side chain is required if additional chemistry is to be performed thereafter, such as cyclization, phosphorylation, sulfation, or alkylation. Table 1 lists the most commonly used protecting groups, together with the cleavage methods, which range from strongly acidic to very mild and selective conditions.

Despite the reactivity of the phenolic group, successful syntheses with unprotected tyrosine have repeatedly been reported.[4] In such cases, special coupling conditions are required, such as weak carboxy-activating reagents and avoidance of excess acylating agent. Therefore, unprotected tyrosine is mainly used in solution synthesis, preferably for short peptides or in fragment condensation steps where purification of intermediates is relatively simpler. In stepwise syntheses, excesses of auxiliary base should be avoided to minimize formation of reactive phenolate moieties, which can result in several side reactions.[2]

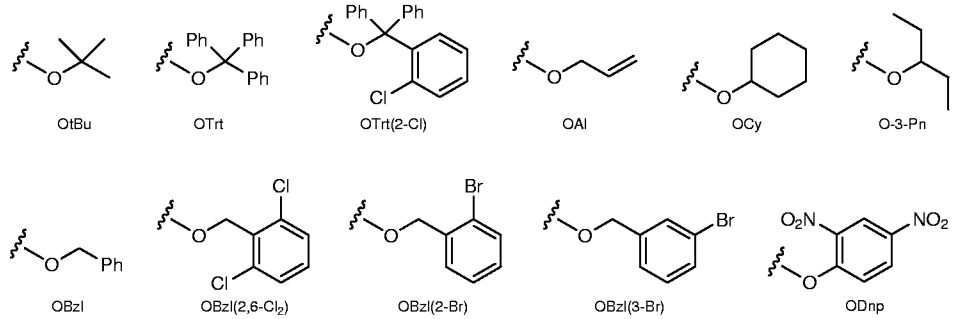
An additional reason for the higher reactivity of the phenolic group compared to the sterically hindered threonine hydroxy group may be its free accessibility. In contrast, the reactivity of ring-substituted tyrosines, such as 3-iodo- or 3,5-diiodotyrosine, is relatively low, thus allowing the use of these derivatives without side-chain protection, even in solid-phase synthesis.[117]

2.6.4.2.1 Alkyl and Aryl Ethers

Alkyl ethers of tyrosine were originally prepared by treating the copper(II) complex of tyrosine with alkyl halides under basic conditions. Although many different alkyl and aryl ethers of tyrosine have been prepared, few of them have found practical application as protecting groups. The ethers differ widely in their stability toward acids and hydrogenolysis, thus requiring different cleavage procedures. Protecting groups of the benzyl, cyclohexyl, pent-3-yl, allyl, and 2,4-dinitrophenyl type are sufficiently stable under the acidic conditions

used in Boc/Bzl chemistry, whereas the *tert*-butyl ether is smoothly cleaved with 50% TFA. Among the structures shown in Scheme 7, the triphenylmethyl (Trt) ether is the most acid-sensitive group as it can be hydrolyzed in 1% TFA in CH_2Cl_2 or by other mild acidic reagents. Conversely, ethers of the benzyl type can be cleaved by hydrogenolysis, and allyl ethers with (π-allyl)palladium(0) procedures. Moreover, acidic cleavage of the benzyl group, as well as the cyclohexyl and pent-3-yl groups, occurs with the strong acids HF, TfOH, and TMSOTf.

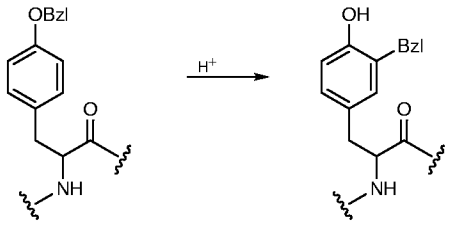
Scheme 7 Alkyl and Aryl Protecting Groups



The phenolic benzyl ether is much more acid labile than the benzyl ethers of serine and threonine. Therefore, *O*-benzyltyrosine is not adequately stable for multistep SPPS with the Boc/Bzl strategy. To prevent loss of the *O*-benzyl protection, halogenated benzyl derivatives[29,32,33,118] have been developed (Scheme 7); the most often used derivative, H-Tyr[Bzl(2,6-Cl_2)]-OH, is ~5000 times more stable to 50% TFA/CH_2Cl_2 treatment than the unsubstituted benzyl compound.

Due to the nucleophilic properties of the aromatic side chain of tyrosine, ring alkylation may occur, either during the introduction of the alkyl group or during the acidolytic cleavage of the protecting groups. This side reaction has mainly been studied with the *O*-benzyl-type protection. In fact, formation of 3-benzyltyrosine (Scheme 8) has been detected in the acidic deprotection step of tyrosine-containing peptides carrying benzyloxycarbonyl, benzyl ester, or benzyl ether groups with strong acids or even upon exposure to TFA.[119]

Scheme 8 Formation of 3-Benzyltyrosine[119]



Although this side reaction is minimized when TFA/H_2O (7:3) in the presence of scavengers is used, it remains a serious problem in the final HF treatment. Regarding the mechanism of the ring alkylation, an acid-catalyzed rearrangement of an aryl ether to a ring-substituted phenol can proceed via an intermolecular or intramolecular pathway.[120-122] The intermolecular pathway can be suppressed by scavengers which compete with the starting material and reaction products for the alkylating species. However, it is more difficult to interfere with the intramolecular pathway. It should be noted that steric effects might also play an important role in the ratio of the intra- to intermolecular reaction. In view of this side

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reaction, a large number of additional *O*-alkyl protecting groups have been evaluated. Among them, the cyclohexyl and pent-3-yl ethers have proven to be the most useful derivatives for avoiding ring alkylation. The cyclohexyl and pent-3-yl ethers are sufficiently stable under Boc-deprotection conditions to allow for the synthesis of larger peptides and proteins, because after final HF cleavage of the synthetic products in the presence of 10% anisole[122] less than 0.5% of the 3-rearrangement product is detected.

2.6.4.2.1.1 *O*-Benzyltyrosine and Halogenated Derivatives

O-Benzyl-protected tyrosine residues are stable toward the basic conditions used for the Fmoc strategy, but not stable enough for Boc chemistry if 50% TFA in CH_2Cl_2 is used for the repeated N^{α}-Boc deprotection. Thus, during 15 minutes with TFA/CH_2Cl_2 more than 0.5% of the benzyl protection is removed.[29] However, a more recently described cleavage procedure[123] for the N^{α}-Boc group using 1 M TMSCl/phenol in CH_2Cl_2 allows the use of *O*-benzyl-protected tyrosine, even in SPPS of longer peptides, as demonstrated with the synthesis of glucagon. The benzyl protection is finally removed by treatment with HF and other super acids or strongly acidic cocktails in the presence of scavengers. In solution synthesis quantitative cleavage of the benzyl ether is preferably achieved by catalytic hydrogenation, as ring alkylation is fully prevented. The most often used *O*-benzyltyrosine derivatives are listed in Table 10.

Table 10 *O*-Benzyl-Protected Tyrosine Derivatives[64,124–131]

Starting Material	Product	Yield[a] (%)	$[\alpha]_D^a$	mp (°C)	Ref
H-Tyr-OH	H-Tyr(Bzl)-OH	64	−9.5 (*c* 1, 80% AcOH)	260–270 (dec)	[124,125]
H-Tyr(Bzl)-OH	Boc-Tyr(Bzl)-OH	80	+27.4 (*c* 2, EtOH)	109–114	[126,127]
H-Tyr(Bzl)-OH	Fmoc-Tyr(Bzl)-OH	n.r.	−15.8 (*c* 1, DMF)	164–166	[128]
H-Tyr(Bzl)-OH	Z-Tyr(Bzl)-OH	82	+11.5 (*c* 0.5, AcOH)	110–116.5	[124,129]
H-Tyr(Bzl)-OH	Ddz-Tyr(Bzl)-OH	88	n.r.	69–72	[64]
H-Tyr(Bzl)-OH	H-Tyr(Bzl)-OMe · HCl	93	+11.5 (*c* 4, MeOH)	181	[124,130]
H-Tyr-OH	H-Tyr(Bzl)-OBzl · HCl	20	−17.3 (*c* 2, MeOH)	222	[131]

[a] n.r. = not reported.

Because of the acidic lability of *O*-benzyl protection, some attempts have been made to stabilize the group by introducing electron-withdrawing halogen substituents. The resulting *O*-(2,6-dichlorobenzyl)-, *O*-(3-bromobenzyl)-, and *O*-(2-bromobenzyl)tyrosine derivatives[29,32,33,118] are significantly more stable than the benzyl derivative and therefore well suited for peptide synthesis using the Boc strategy. However, these groups still show some tendency to undergo 3-alkylation during the deprotection step. Only *O*-(2,6-dichlorobenzyl) protection has found wide application, and the related derivatives are listed in Table 11.

Table 11 *O*-(2,6-Dichlorobenzyl)-Protected Tyrosine Derivatives[32,33,132–138]

Starting Material	Product	Yield[a] (%)	$[\alpha]_D$[a]	mp[a] (°C)	Ref
H-Tyr-OH	H-Tyr[Bzl(2,6-Cl_2)]-OH	43	−10.6 (*c* 2, 80% AcOH)	211–212	[32,33]
H-Tyr[Bzl(2,6-Cl_2)]-OH	Boc-Tyr[Bzl(2,6-Cl_2)]-OH	72	+21 (*c* 2, MeOH)	108–110	[32,33]
H-Tyr[Bzl(2,6-Cl_2)]-OH	Fmoc-Tyr[Bzl(2,6-Cl_2)]-OH[b]	n.r.	−17.8 (*c* 1, DMF)	160–163	[132,133]
H-Tyr[Bzl(2,6-Cl_2)]-OH	Z-Tyr[Bzl(2,6-Cl_2)]-OH[c]	n.r.	n.r.	n.r.	[134]
–	H-Tyr[Bzl(2,6-Cl_2)]-OBzl[c]	–	n.r.	n.r.	[135]
Boc-Tyr[Bzl(2,6-Cl_2)]-OH	Boc-Tyr[Bzl(2,6-Cl_2)]-OMe[d]	n.r.	n.r.	n.r.	[136]
Boc-Tyr[Bzl(2,6-Cl_2)]-OH	Boc-Tyr[Bzl(2,6-Cl_2)]-OPac	n.r.	−21 (*c* 1, DMF)	102–104	[137]
Boc-Tyr[Bzl(2,6-Cl_2)]-OPac	H-Tyr[Bzl(2,6-Cl_2)]-OPac · TFA	n.r.	n.r.	n.r.	[137]
Boc-Tyr[Bzl(2,6-Cl_2)]-OH	H-Tyr[Bzl(2,6-Cl_2)]-OFm · HCl	63	n.r.	190–194	[138]

[a] n.r. = not reported.
[b] Analytical data taken from ref[133].
[c] The compound was used in peptide synthesis; however, no information about the preparation was given.
[d] Used as an intermediate in the synthesis of the thrombin inhibitor cyclotheonamide B. Prepared from the starting compound by treatment with MeI and $KHCO_3$ in DMF at rt for 4 h.

H-Tyr(Bzl)-OH:[68]
To H-Tyr-OH (18.1 g, 100 mmol) in 2 M NaOH (100 mL) was added a soln of $CuSO_4 \cdot 5H_2O$ (12.5 g, 50 mmol) in H_2O (50 mL). Upon dissolution of the precipitate the mixture was heated to 60 °C; it was then cooled to rt, diluted with MeOH (350 mL) and made more alkaline with 2 M NaOH (15 mL). Then, benzyl bromide (12 mL, 100 mmol) was added and the soln was vigorously stirred at 25–30 °C for about 1.5 h. The purple-blue precipitate was collected, and washed with a mixture of MeOH (50 mL) and H_2O (175 mL), and then with MeOH (25 mL). The copper complex was triturated and washed with 1 M HCl (5×50 mL), H_2O (2×25 mL), 1.5 M NH_4OH (5×25 mL), and finally H_2O (2×25 mL); yield: 17.5 g (64%); mp 260–270 °C (dec); $[\alpha]_D^{20}$ −9.5 (*c* 1.0, 80% AcOH).

H-L-Tyr[Bzl(2,6-Cl_2)]-OH:[32,33]
$CuCl_2 \cdot 2H_2O$ (1.70 g, 10.0 mmol) and MeOH (20 mL) were added to a soln of H-Tyr-OH (3.60 g, 20.0 mmol) in 1 M NaOH (40.0 mL, 40.0 mmol). Upon dissolution of the solid a soln of 2,6-dichlorobenzyl bromide (4.50 g, 18.8 mmol) in MeOH (25 mL) was added. The mixture was stirred at rt for 13 h, then filtered. The filter cake was washed with MeOH/H_2O (1:1), MeOH, and Et_2O, and then dissolved in a hot mixture of AcOH (50 mL), 1 M HCl (17 mL), and EDTA (2.92 g, 10.0 mmol). The hot soln was filtered to remove undissolved EDTA and diluted with cold H_2O (400 mL). The gelatinous solid was collected and washed with H_2O; yield: 2.64 g (43%); mp 211–212 °C (dec); $[\alpha]_D^{24}$ −10.6 (*c* 2.0, 80% AcOH).

2.6.4.2.1.2 *O-tert*-Butyltyrosine

In analogy to the *tert*-butyl ethers of hydroxy amino acids (see Section 2.6.4.1.1.2), the *tert*-butyl ether of tyrosine is a well-established derivative which is stable under basic and mild acidic conditions, as used for removal of the Fmoc, Trt, Bpoc, and Ddz groups; it is even stable toward hydrogenolysis, palladium(0)-catalyzed π-allyl rearrangement, and fluoride ions (TBAF). Thus, *tert*-butyl protection allows for the selective removal of different types of protecting groups. *tert*-Butylation of the phenolic group is performed with isobutylene in the presence of an inorganic acid, generally H_2SO_4, as catalyst. The most common *O-tert*-butyltyrosine derivatives are listed in Table 12.

for references see p 375

Table 12 O-*tert*-Butyl-Protected Tyrosine Derivatives[11,34,59–61,67,139–143]

Starting Material	Product	Yield[a] (%)	$[\alpha]_D$[a]	mp (°C)	Ref
Z-Tyr-ONbz	H-Tyr(tBu)-OH	87	−25.8 (*c* 1, H_2O)	248–249.5	[34,59]
H-Tyr(tBu)-OH	Boc-Tyr(tBu)-OH	76	+15.9 (*c* 1.03, MeOH)	112–115	[139]
Fmoc-Tyr(tBu)-OMe	Fmoc-Tyr(tBu)-OH	74	−28.0 (*c* 1, DMF)	150–151	[60,61]
Z-Tyr(tBu)-ONbz	Z-Tyr(tBu)-OH	93	+9.55 (*c* 2, AcOH)	76–78	[11,34]
H-Tyr(tBu)-OH	Ddz-Tyr(tBu)-OH · CHA	70	+64.6 (*c* 1, MeOH)	153–159	[142]
Fmoc-Tyr-OMe	Fmoc-Tyr(tBu)-OMe	83	−22.1 (*c* 1, DMF)	90–92	[60]
Z-Tyr-OMe	Z-Tyr(tBu)-OMe	58	−29.3 (*c* 1, DMF)	51–53.5	[140]
Z-Tyr(tBu)-OMe	H-Tyr(tBu)-OMe · HCl[b]	n.r.	+39.7 (*c* 1, DMF)	146–152	[11,140,143]
Z-Tyr(tBu)-OEt	H-Tyr(tBu)-OEt	90	+22.2 (*c* 0.5, DMF)	oil	[59]
H-Tyr(tBu)-OEt	Boc-Tyr(tBu)-OEt	82	n.r.	oil	[59]
Z-Tyr-OH	H-Tyr(tBu)-OtBu · HCl	45	+37.6 (*c* 3.02, EtOH)	159–160	[67,141]
Fmoc-Tyr(tBu)-OAl	H-Tyr(tBu)-OAl · HCl	n.r.	n.r.	121–124	[143]

[a] n.r. = not reported.
[b] Originally H-Tyr(tBu)-OMe was prepared as the free base, which has to be used immediately after preparation because of 2,5-dioxopiperazine formation;[11,140] however, it is stable as the HCl salt.[143]

H-Tyr(tBu)-OH:[68]
tert-Butylation was carried out under similar conditions as described for threonine (see Section 2.6.4.1.1.2), the reaction sequence starting from Z-Tyr-OH, via Z-Tyr-ONbz and Z-Tyr(tBu)-ONbz, followed by hydrogenation to give H-Tyr(tBu)-OH.

Fmoc-Tyr(tBu)-OMe:[60]
Fmoc-Tyr-OMe (5.00 g, 12.0 mmol), concd H_2SO_4 (0.33 mL, 6.0 mmol), and CH_2Cl_2 (100 mL) were stirred under isobutylene gas (260 Torr) for 6 h at rt. The soln was washed with cold 10% $NaHCO_3$ (2 × 100 mL) and brine (100 mL), dried (Na_2SO_4), and concentrated. The residue was dissolved in MeOH/CCl_4 (1:1, 400 mL), washed with H_2O (300 mL) and additionally with MeOH/H_2O (1:1, 2 × 200 mL), dried (Na_2SO_4), and the solvent removed to give a white solid which was recrystallized (CH_2Cl_2/hexane); yield: 4.70 g (83%); mp 90–92 °C; $[\alpha]_D^{20}$ −22.1 (*c* 1.0, DMF).

Fmoc-Tyr(tBu)-OH:[60]
A mixture of Fmoc-Tyr(tBu)-OMe (2.00 g, 4.22 mmol) in MeCN (250 mL) and 3% Na_2CO_3 (375 mL) was stirred for 15 h, then washed with hexane (3 × 500 mL), acidified with 2 M HCl to pH 3–4, and extracted with $CHCl_3$ (2 × 600 mL). The combined $CHCl_3$ fractions were washed with brine (500 mL), dried (Na_2SO_4), and the solvent removed to provide an oil which gave white crystals from EtOAc/hexane; yield: 1.43 g (74%); mp 150–151 °C; $[\alpha]_D^{20}$ −28.0 (*c* 1.0, DMF).

2.6.4.2.1.3 *O*-Trityltyrosine

Triphenylmethyl (trityl) groups can be cleaved under very mild acidic conditions and the resulting trityl cations display a lower electrophilicity than *tert*-butyl cations; therefore, trityl side-chain protection is especially useful in SPPS of sensitive peptides containing Met and Trp. In a recent comparative study, crude products of higher purity were obtained using side-chain-trityl-protected amino acids compared to standard Fmoc/tBu amino acid derivatives.[144]

The *O*-trityl protection can be removed under the same conditions as used for cleavage of peptides from 2-chlorotrityl resin, i.e. with AcOH/TFE/CH_2Cl_2 (2:2:6, rt, 1 h) as the cleavage mixture or with 1% TFA without affecting *tert*-butyl side-chain protection.[17,35,144] Silanes are added as hydride donors to scavenge the trityl cations and to avoid retritylation. Most commonly, the trityl derivatives of tyrosine, serine, and threonine are used for the synthesis of phosphopeptides by postsynthetic methods on a solid support.[145,146] In contrast to the use of protected phosphoamino acid derivatives, such as Fmoc-Tyr(POOBzl)-OH,[147] both the phosphorylated and nonphosphorylated peptides can be prepared from one syn-

thetic batch using trityl protection. In addition, O-tritylated tyrosine may be useful for the synthesis of side-specific sulfated, glycosylated, and alkylated peptides.

The *O*-trityltyrosine derivative is more acid labile than the related *O*-trityl derivatives of serine and threonine. Therefore, the more stable *O*-(2-chlorotrityl)tyrosine derivative has been introduced to ensure better stability during coupling reactions in the presence of the weakly acidic HOBt.[148] Selected *O*-trityltyrosine derivatives are listed in Table 13.

Table 13 *O*-Trityl-Protected Tyrosine Derivatives[17,148,149]

Starting Material	Product	Yield (%)	$[\alpha]_D$	mp (°C)	Ref
H-Tyr-OBzl · TosOH	H-Tyr(Trt)-OBzl · HCl	72	−24.2 (*c* 6.6, MeOH)	203–205	[149]
Trt-Tyr-OH	Fmoc-Tyr(Trt)-OH	[a]	−23.5 (*c* 1, DMF)	168–170	[17]
Trt-Tyr-OH	Fmoc-Tyr[Trt(2-Cl)]-OH	[a]	−27.0 (*c* 1, EtOAc)	148–149	[148]

[a] Multistep reaction, yield not reported.

So far, no detailed synthetic procedure has been reported for the most commonly used derivative, Fmoc-Tyr(Trt)-OH.[17] Due to the increased acid sensitivity, some modifications to the procedure for the preparation of Fmoc-Ser(Trt)-OH (see Section 2.6.4.1.1.3, Table 4) have to be performed. The mixture comprising the intermediates Trt-Tyr(Trt)-OH and Trt-Tyr(Trt)-OTrt, obtained on treatment of Trt-Tyr-OH · TEA with trityl chloride, can be selectively detritylated at the amino and carboxy groups by the addition of AcOH/TFE/CH_2Cl_2 (1:2:7) or with HOBt in TFE/CH_2Cl_2 (1:3) at 0 °C. No information concerning reaction time, HOBt amount, or workup procedures was reported.

2.6.4.2.1.4 *O*-Allyltyrosine

In contrast to serine and threonine, the allyl ether of tyrosine can be cleaved using π-allyl palladium(0) technology.[150,151] This allyl ether is stable under the acidic or basic conditions commonly used in the Boc and Fmoc strategies, and is cleaved under very mild conditions that are compatible with a range of reducible functional groups, such as nitro, acetals, carboxylic acid amides, nitriles, carbamates, and imides.[152] Therefore, the orthogonally protected *O*-allyltyrosines[22] (Table 14) are useful derivatives in peptide synthesis. It should be noted that this allyl protection is not compatible with hydrogenation. Many procedures are described in the literature[21] for removal of the Aloc/OAl protecting groups, which differ in the palladium(0) species, the nucleophilic allyl acceptor, the hydride donor, and the solvent. Most of the procedures are suitable for selective removal of the allyl ether of tyrosine, both in solution and on solid supports. An important advantage of allyl ether protection of tyrosine is its stability and the absence of an O→N acyl shift that can occur upon Aloc protection of serine and threonine.[22]

Table 14 *O*-Allyl-Protected Tyrosine Derivatives[22,150,153–156]

Starting Material	Product	Yield (%)	$[\alpha]_D$[a]	mp[a] (°C)	Ref
H-Tyr-OH	H-Tyr(Al)-OH	46	−6.5 (*c* 2.0, AcOH)	250	[22]
H-Tyr(Al)-OH	Aloc-Tyr(Al)-OH	20[b]	+60.25 (*c* 0.8, $CHCl_3$)	97–97.5	[150]
Boc-Tyr-OMe	Boc-Tyr(Al)-OH[c]	75	+17.7 (*c* 1.0, MeOH)	67–68	[22,153,154]
H-Tyr(Al)-OH	Fmoc-Tyr(Al)-OH	86	−24.4 (*c* 1.0, DMF)	143–144	[22]
Tos-Tyr-OH	Tos-Tyr(Al)-OH	50	n.r.	n.r.	[153]
Trt-Tyr-OH	Trt-Tyr(Al)-OH · DEA	45	+2.1 (*c* 2, MeOH)	124–125	[155]
Boc-Tyr-OMe	H-Tyr(Al)-OMe	95	n.r.	waxy solid	[156]

[a] n.r. = not reported.
[b] Yield calculated from H-Tyr-OH as starting material.
[c] In[153,154] Boc-Tyr-OH was used as starting material (yield: 60%).

for references see p 375

H-Tyr(Al)-OH:[22]
This compound was prepared from the copper(II) complex of tyrosine by reaction with allyl chloride using the procedure described for the preparation of H-Tyr[Bzl(2,6-Cl_2)]-OH[33] (see Section 2.6.4.2.1.1); yield: 46%; mp 250 °C; $[\alpha]_D$ −6.5 (*c* 2.0, AcOH).

Boc-Tyr(Al)-OH:[22]
To a soln of Boc-Tyr-OMe (14.8 g, 0.05 mol) in acetone (30 mL) were added K_2CO_3 (9.3 g, 0.055 mol) and allyl bromide (6.6 g, 0.055 mol). This soln was stirred for 48 h at rt, then the solvent was removed. The residue was distributed between 5% NaOH (100 mL) and $CHCl_3$ (300 mL), and the organic phase was dried ($MgSO_4$). The solvent was removed to give the crude ester product as a yellow oil; yield: 16 g (ca. 100%). A soln of the crude ester (9.6 g, 0.03 mol) in EtOH (30 mL) was cooled to 0 °C and saponified with 1 M NaOH (60 mL). After 1.5 h at rt, the soln was concentrated, extracted with $CHCl_3$ (50 mL) and acidified to pH 2–3 with 12 M HCl. This suspension was extracted with $CHCl_3$ (3 × 50 mL), and the combined organic phases were dried ($MgSO_4$) and the solvent removed. The crude product was crystallized (Et_2O/hexane, −10 °C); yield: 7.2 g (75%); mp 67–68 °C; $[\alpha]_D$ +17.7 (*c* 1.0, MeOH).

Fmoc-Tyr(Al)-OH:[22]
H-Tyr(Al)-OH (6.63 g, 0.03 mol) was suspended in H_2O (150 mL) and dioxane (10 mL), and the pH was adjusted to 11.5 by the addition of 2 M NaOH. To the resulting soln was added dropwise Fmoc-F (0.027 mol) in DME over a period of 30 min while maintaining the pH at 11.5 by the addition of 2 M NaOH. The mixture was further stirred for 1 h, then washed twice with Et_2O. The pH was adjusted to 3.0 with 12 M HCl, the resulting suspension was extracted with EtOAc (3 × 100 mL), and the combined organic phases were washed with 1 M HCl, aq NaCl, and H_2O, then dried ($MgSO_4$) and the volatiles removed. The crude product was crystallized (Et_2O/hexane); yield: 11.4 g (86%); mp 143–144 °C; $[\alpha]_D$ −24.4 (*c* 1.0, DMF).

2.6.4.2.1.5 *O*-Cyclohexyl- and *O*-(Pent-3-yl)tyrosine

O-Cyclohexyltyrosine is only obtained as a mixture of four reaction products; laborious purification using dry silica gel chromatography gives a yield of about 50%.[122] As an alternative, Boc-Tyr(3-Pn)-OH (3-Pn = pent-3-yl) has been proposed as an open-chain analogue of the *O*-cyclohexyl derivative.[31] The *O*-(pent-3-yl) protection is base resistant (completely stable to 20% piperidine in DMF) and also sufficiently stable to 50% TFA/CH_2Cl_2 (0.3% cleavage in 24 h). It is readily cleaved using the standard HF procedure, in the presence of anisole or cresole as scavenger, without significant formation of the alkyltyrosine rearrangement product (only 0.5% in HF at −5 °C for 60 min). The *O*-(pent-3-yl)tyrosine derivative is prepared in a one-step reaction by chemoselective monoalkylation of the N^{α}-protected tyrosine disodium salt using 3-bromopentane in DMF.

Boc-Tyr(3-Pn)-OH:[31,157]
A soln of Boc-Tyr-OH (5 g, 17.8 mmol) in dry DMF (75 mL) was cooled to 10 °C, and a NaH suspension (41 mmol) was added all at once. After stirring for 1 h at 10 °C, 3-bromopentane (2.21 mL, 17.8 mmol) was added. After stirring the mixture for 3 h at 10 °C, ice water (150 mL) and EtOAc (400 mL) were added. The aqueous phase was separated, extracted with EtOAc (2×75 mL), and acidified with 6 M HCl with cooling. The product was extracted into EtOAc (3×75 mL), which was washed with H_2O, dried ($MgSO_4$), and concentrated. The compound was crystallized as the DCHA salt; yield: not reported; mp 107–108 °C; $[\alpha]_D$ +28.8 (*c* 1.0, MeOH).

2.6.4.2.1.6 *O*-(2,4-Dinitrophenyl)tyrosine

O-(2,4-Dinitrophenyl)tyrosine[38] is a very acid-stable derivative and therefore fully compatible with the Boc strategy in SPPS. 2,4-Dinitrophenyl (2,4-Dnp) protection is also very resistant toward anhydrous HF, and can be selectively removed by thiolysis prior to or following the final deprotection/resin-cleavage step, making this type of protection useful for further modifications of the tyrosine residue. The thiolysis can be monitored spectrophotometrically, e.g. for *S*-Dnp-β-ME at 341 nm (ε ~12500 $M^{-1} \cdot cm^{-1}$). The Dnp group is also

readily cleaved with 20% piperidine or with 2% DBU in DMF, and thus is not compatible with the Fmoc strategy, except for N-terminal residues.

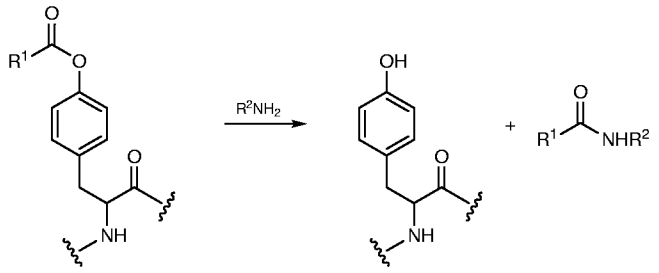
Boc-Tyr(Dnp)-OH:[38]
A soln of Boc-Tyr-OH (3.0 g, 10.6 mmol) in H_2O (32 mL) containing $NaHCO_3$ (31.9 mmol) was cooled in an ice bath, and 1-fluoro-2,4-dinitrobenzene (2.04 mL, 16.17 mmol) in dioxane (32 mL) was added. The mixture was stirred for 18 h at 4 °C and then concentrated under vacuum to ca. 20 mL. Upon acidification with ice-cold 0.5 M H_2SO_4, a yellow precipitate formed which was extracted with EtOAc. The organic layer was washed with H_2O and a sat. soln of NaCl, then dried (Na_2SO_4) and the solvent removed. The oily residue was dissolved in EtOAc. Addition of petroleum ether (40–60 °C) gave yellow-white crystals which were collected; yield: 3.04 g (64%); mp 89–92 °C; $[\alpha]_D^{25}$ +30 (*c* 1.0, $CHCl_3$).

Alternatively, the Dnp group can be introduced in good yield by the reaction of 1-fluoro-2,4-dinitrobenzene with Boc-Tyr-OH or Fmoc-Tyr-OH in CH_2Cl_2 in the presence of TEA.[38]

2.6.4.2.2 Acyl Esters

O-Acyltyrosine derivatives bear the disadvantage of being active esters capable of acylating amino groups or other nucleophiles in a particularly efficient manner by intermolecular O→N transfer (Scheme 9). Due to the reactivity of these phenyl esters, acyl groups such as acetyl or benzoyl, which are commonly used in carbohydrate chemistry, cannot be recommended for the protection of the tyrosine hydroxy group.[2]

Scheme 9 O→N Acyl Migration in *O*-Acyltyrosines Catalyzed by Nucleophiles



O-(Benzyloxycarbonyl)tyrosine is considerably more labile than the related N^{α}-protected amino derivatives, although it is about sevenfold more stable than the benzyl ether in 50% TFA, and no 3-benzyl alkylations have been detected after HF cleavage.[33] It can also be removed by nucleophilic displacement, therefore loss of the O-protection under such conditions is to be anticipated.[2,33] The drawback of the acid lability can be eliminated by the use of the corresponding *O*-(2-bromobenzyloxycarbonyl) group.[36] Other ester-type protecting groups have been proposed, such as the 2-adamantyloxycarbonyl (2-Adoc) and 2,4-dimethylpent-3-yloxycarbonyl (Doc) groups.[37,39] These ester-type protecting groups can be cleaved by standard treatment with HF or TfOH[158,159] in the presence of thioanisole, anisole, or dimethyl sulfide as scavenger. Furthermore, N^{α},*O*-bis(benzyloxycarbonyl)-[160,161] and N^{α},*O*-bis(*tert*-butoxycarbonyl)tyrosine[162,163] have been prepared and represent suitable derivatives for the introduction of an N-terminal tyrosine residue in synthesis. The 4-(methylsulfinyl)benzyloxycarbonyl (Msz) group[19] is the most stable among several phenolic protecting groups tested [Bzl; Z(2-Br); Bzl(2,6-Cl_2)]; therefore, this protection seems to be advantageous for the synthesis of long peptide sequences. The Msz group can be removed by mild reductive acidolysis. The structures of the commonest ester-type protecting groups are shown in Scheme 10, while various O-acyl-protected tyrosine derivatives are listed in Table 15.

for references see p 375

Scheme 10 Selected Acyl Protecting Groups

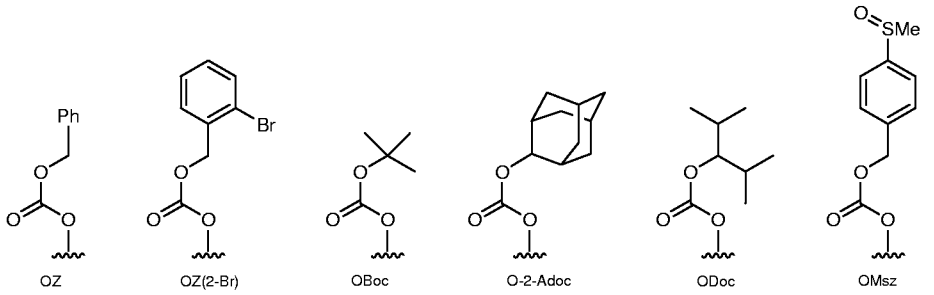


Table 15 *O*-Acyl-Protected Tyrosine Derivatives[19,36,37,39,133,160–167]

Starting Material	Product	Yield[a] (%)	$[\alpha]_D$[a]	mp[a] (°C)	Ref
H-Tyr-OH	H-Tyr[Z(2-Br)]-OH · HCl	50	−4 (*c* 2, 80% AcOH)	203–208 (dec)	[36]
H-Tyr[Z(2-Br)]-OH · HCl	Boc-Tyr[Z(2-Br)]-OH[b]	75	+17.5 (*c* 1, MeOH)	103–105	[36,133]
H-Tyr[Z(2-Br)]-OH · HCl	Fmoc-Tyr[Z(2-Br)]-OH	n.r.	−0.9 (*c* 1, MeOH)	100–105	[133]
Boc-Tyr[Z(2-Br)]-OH	Boc-Tyr[Z(2-Br)]-OBzl	89	−10.5 (*c* 0.4, DMF)	oil	[164]
Boc-Tyr[Z(2-Br)]-OBzl	H-Tyr[Z(2-Br)]-OBzl · TFA[c]	n.r.	n.r.	n.r.	[164,165]
Boc-Tyr[Z(2-Br)]-OH	Boc-Tyr[Z(2-Br)]-OPac	81	−1.4 (*c* 1, $CHCl_3$)	106–107	[166]
Boc-Tyr[Z(2-Br)]-OPac	H-Tyr[Z(2-Br)]-OPac · TFA[d]	n.r.	n.r.	n.r.	[166]
H-Tyr-OH	Boc-Tyr(Boc)-OH	94–99	+29.3 (*c* 1, dioxane)	95–97	[162,163]
Z-Tyr-OH	Z-Tyr(Boc)-OH	92	−5.0 (*c* 1, EtOH)	48–50	[167]
H-Tyr-OH	Z-Tyr(Z)-OH	n.r.	−5.0 (*c* 1, EtOH)	117	[160,161]
H-Tyr-OH	H-Tyr(2-Adoc)-OH	63	−6.2 (*c* 1, AcOH)	213–215	[37]
H-Tyr(2-Adoc)-OH	Boc-Tyr(2-Adoc)-OH	86	−11.5 (*c* 1, DMF)	95–98	[37]
Boc-Tyr(Doc)-OBzl	Boc-Tyr(Doc)-OH	n.r.	+37 (*c* 1, $CHCl_3$)	n.r.	[39]
H-Tyr(Msz)-OH	Boc-Tyr(Msz)-OH	50	−167.8 (*c* 0.44, MeOH)	61–64	[19]

[a] n.r. = not reported.
[b] Yield of the DCHA salt;[36] other analytical data taken from ref[133].
[c] Prepared by TFA treatment of the starting material; no analytical data given in ref[164]. In ref[165] H-Tyr[Z(2-Br)]-OBzl · TosOH was used for peptide synthesis; no further information about the preparation was provided.
[d] Prepared by TFA treatment of the starting material; no analytical data given.[166]

2.6.4.2.2.1 *O*-(2-Bromobenzyloxycarbonyl)tyrosine

The 2-bromobenzyloxycarbonyl [Z(2-Br)] group is very stable toward the acidic conditions commonly used in Boc chemistry. In addition, only minor amounts of rearrangement products can be detected after the final acidic cleavage from the resin. Therefore, this protection is even superior to the Bzl(2,6-Cl_2) group and is now the recommended *O*-tyrosine protection in the synthesis of large peptide sequences using the Boc/Bzl strategy in SPPS. A crucial drawback of Z(2-Br) protection is its sensitivity to nucleophiles ($t_{1/2}$ 30 sec in 20% piperidine in DMF).[39]

H-Tyr[Z(2-Br)]-OH · HCl:[36]
A soln of H-Tyr-OH (13.9 g, 77 mmol) in 2 M NaOH (78 mL) was mixed with a soln of $CuSO_4 \cdot 5H_2O$ (9.4 g) in H_2O (39 mL), heated to 60 °C, and then cooled to 24 °C. The pH was adjusted to ca. 5 with glacial AcOH; the solid was collected and washed with H_2O and acetone. This solid was suspended in 70% aq DMF (1100 mL), and 2-bromobenzyl 4-nitrophenyl carbonate (41.5 g, 118 mmol) and $NaHCO_3$ (14.2 g) were added. The mixture was stirred for 20 h, and the product was collected and washed with 75% aq DMF, H_2O, and acetone; yield: 21.4 g. The copper complex was stirred in 1 M HCl (300 mL) for 1 h, filtered, and washed with 1 M HCl (250 mL), H_2O, and acetone; yield: 15.9 g (50%). For analysis, a sample was recrystallized (50% AcOH); mp 203–208 °C (dec).

2.6.4.2.2.2 *O*-(2,4-Dimethylpent-3-yloxycarbonyl)tyrosine

As noted in Section 2.6.4.2.2.1, H-Tyr[Z(2-Br)]-OH is sensitive to nucleophiles, which limits its application in Boc chemistry when nucleophiles should be used. Therefore, the 2,4-dimethylpent-3-yloxycarbonyl (Doc) group[39] has been proposed as a new, nucleophile-resistant protection for the side chain of tyrosine in SPPS. This Doc protection is 1000-fold more stable toward the nucleophile piperidine (2% cleavage in 10 min with 20% piperidine in DMF) than the commonly used Z(2-Br) group, but it is quantitatively cleaved with HF. On the other hand, the Doc group is more acid-labile than the Z(2-Br) group, but its stability is sufficient for most applications.

No detailed synthetic protocol has been reported for the preparation of Boc-Tyr(Doc)-OH. However, this derivative has been obtained from Boc-Tyr-OH by treatment with benzyl bromide in acetonitrile in the presence of DIPEA to yield the intermediate Boc-Tyr-OBzl, which further reacts with Doc-Cl under similar conditions (MeCN/DIPEA), followed by final hydrogenolytic removal of the benzyl ester and crystallization from EtOH/H_2O. Boc-Tyr(Doc)-OH {$[\alpha]_D$ +37 (*c* 1.0, $CHCl_3$)} exists in two isomeric forms with the Boc group in the *Z*- or *E*-position with respect to the side chain of tyrosine.[39]

2.6.4.2.2.3 *O*-(2-Adamantyloxycarbonyl)tyrosine

The 2-adamantyloxycarbonyl (2-Adoc) group, which was originally developed to protect the ε-amino function of lysine, has more recently been proposed as a useful protecting group for the side-chain hydroxy group of tyrosine.[37] This protection allows peptide synthesis in combination with C-terminal benzyl esters, which can be cleaved by hydrogenolysis without affecting the 2-Adoc protection. H-Tyr(2-Adoc)-OH is stable to TFA, 5.0 M HCl/dioxane, 25% HBr/AcOH, and H_2/Pd, and is cleaved by TfOH/thioanisole/TFA (100% in 10 min) or HF (100% in 30 min). It is stable to tertiary amines; however, it is also cleaved by 2 M NaOH (100% after 10 min), and partially cleaved by 20% piperidine in DMF (2% in 10 min, 18% in 1 h, 45% in 2 h). No ring alkylation has been observed during cleavage of the 2-Adoc group with HF containing the standard scavengers.

H-Tyr(2-Adoc)-OH:[37]
The copper(II) complex of tyrosine prepared from H-Tyr-OH (15.6 g, 86 mmol) and $CuSO_4 \cdot 5H_2O$ (10.75 g, 43 mmol) was suspended in a mixture of dioxane (50 mL), H_2O (50 mL), and 1 M NaOH (86 mL). 2-Adoc-Cl (21.9 g, 103.1 mmol) in dioxane (50 mL) was added under cooling at 0 °C and the mixture was stirred at 0 °C for 3 h. The precipitate was collected by filtration and the solid was washed with dioxane/H_2O (1:1). The copper was removed with sat. EDTA soln and the product was washed with MeOH; yield: 19.3 g (63%); mp 213–215 °C; $[\alpha]_D^{26}$ −6.2 (*c* 1.0, AcOH).

Boc-Tyr(2-Adoc)-OH:[37]
Boc-Tyr(2-Adoc)-OH was prepared from H-Tyr(2-Adoc)-OH (see above) by standard procedures using $(Boc)_2O$ in DMF containing TEA. The product was recrystallized ($CHCl_3$/acetone); yield: 86%; mp 95–98 °C; $[\alpha]_D^{26}$ −11.5 (*c* 1.0, DMF).

2.6.4.2.2.4 *O*-[4-(Methylsulfinyl)benzyloxycarbonyl]tyrosine

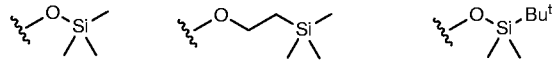
The 4-(methylsulfinyl)benzyloxycarbonyl (Msz) group[19] is a hydroxy protecting group of the safety-catch type, which is stable under both acidic (TFA/anisole or 0.5 M MsOH) and basic (5% TEA/DMF or 5% DIPEA/CH_2Cl_2) conditions. The Msz group can be removed by a one-pot reaction involving reductive acidolysis using $SiCl_4$/TFA in the presence of a scavenger. The *O*-Msz group of tyrosine has also been cleaved by saponification with 1 M NaOH/MeOH at 25 °C for 24 hours; under these conditions the MsOb ethers of serine and threonine are fully resistant. Boc-Tyr(Msz)-OH can be prepared from H-Tyr(Msz)-OH by *tert*-butoxycarbonylation with $(Boc)_2O$, followed by oxidation with sodium bromite trihydrate in EtOAc/H_2O at 25 °C for 30 minutes.[19]

for references see p 375

2.6.4.2.3 Alkylsilyl Ethers

The hydroxy group of tyrosine can also be protected as a silyl ether, as described for serine and threonine (see Section 2.6.4.1.3). Preparation of the *O*-silyltyrosine derivatives is performed using similar procedures, except for *tert*-butyl(diphenyl)silylation, which proceeds very slowly. Thus, serine is silylated quantitatively with *tert*-butyldimethylsilylchloride in the presence of imidazole within 30 minutes, while tyrosine is only silylated to about 40% after 4 hours under such conditions.[87] Because of this difficult preparation, *O*-[*tert*-butyl(diphenyl)silyl]tyrosine derivatives are not used. As well as tyrosine derivatives with trimethylsilyl (TMS) and *tert*-butyl(dimethyl)silyl (TBDMS) protection of the phenolic group, the 2-(trimethylsilyl)ethyl (TMSE) group[42] has also been used which, however, can be classified with the alkyl-type protecting groups (Scheme 11). The stability of silyl ethers under acidic and basic conditions is described in Section 2.6.4.1.3. The 2-(trimethylsilyl)ethyl group is 3–4 times more stable toward 0.5% TFA than the *tert*-butyl ether group. This stability facilitates the selective removal of the Bpoc group in the presence of the TMSE group; the latter is resistant to a variety of reagents used in peptide synthesis, such as trialkylamines, hydroxybenzotriazole, trialkylphosphines, and nucleophiles, as well as hydrogenation, and is readily removed in neat TFA within 5–20 minutes in the absence of cation scavengers. The deprotection reaction proceeds without any detectable alkylation of the aromatic ring.[42] A successful application of this protecting group has been demonstrated in the synthesis of the C-terminal 29-residue peptide of the basic pancreatic trypsin inhibitor.[42] Selected tyrosine derivatives with silicon-based side-chain protection are listed in Table 16. These derivatives have been chemically characterized and seem to be useful for application in peptide synthesis.

Scheme 11 Silyl Protecting Groups for Aromatic Hydroxy Groups



TMS TMSE TBDMS

Table 16 *O*-Trimethylsilyl-, *O*-[2-(Trimethylsilyl)ethyl]-, and *O*-*tert*-Butyldimethylsilyl-Protected Tyrosine Derivatives[23,42,43,80,85,168–172]

Starting Material	Product	Yield[a] (%)	$[\alpha]_D$[a]	mp (°C) or bp (°C/Torr)	Ref
H-Tyr-OH	TMS-Tyr(TMS)-OTMS	82	n.r.	103/0.0001	[85,168]
TMS-Tyr(TMS)-OTMS	H-Tyr(TMS)-NCA	86	n.r.	75–77	[169]
H-Tyr-OH	H-Tyr(TMS)-OTMS	61	n.r.	170; 194–196/21	[80]
Z-Tyr-ONbz	Z-Tyr(TMSE)-ONbz	35	n.r.	oil	[42]
Z-Tyr(TMSE)-ONbz	H-Tyr(TMSE)-OH	100	n.r.	amorphous	[42]
H-Tyr(TMSE)-OH	Bpoc-Tyr(TMSE)-OH · DCHA	70	n.r.	141–145	[42]
Boc-Tyr-OH	Boc-Tyr(TBDMS)-OH	90	+45.3 (*c* 1.0, $CHCl_3$)	glass	[170]
Boc-Tyr-OH	Boc-Tyr(TBDMS)-OMe	93	−6.2 (*c* 1.0, MeOH)	oil	[43,171]
Fmoc-Tyr-OH	Fmoc-Tyr(TBDMS)-OH	n.r.	n.r.	208–210	[23]
Fmoc-Tyr-OMe	Fmoc-Tyr(TBDMS)-OMe	86	+31.0 (*c* 1.0, $CHCl_3$)	47–49	[172]

[a] n.r. = not reported.

H-Tyr(TMS)-OTMS:[80]

Unprotected, dry tyrosine (3.62 g, 20 mmol) was suspended in *N*,*N*-diethyl-*N*-(trimethylsilyl)amine (15 mL, 79 mmol) at 120 °C until all of the formed DEA was removed by distillation. After cooling, the mixture was saturated with dry NH_3. The resulting solid compound was purified by distillation (bp 194–196 °C/21 Torr); yield: 3.96 g (61%); mp 170 °C.

Boc-Tyr(TBDMS)-OH:[170]
Boc-Tyr-OH (14.0 g, 50 mmol), TBDMSCl (17.0 g, 110 mmol), and imidazole (15.0 g, 220 mmol) were stirred in DMF (75 mL) at rt for 48 h under an argon atmosphere. After the mixture was diluted with Et_2O (500 mL), it was washed with H_2O (5 × 100 mL), dried (Na_2SO_4), and concentrated to yield a pale yellow oil (25.4 g). This material (16.3 g) was chromatographed (Waters Prep 500, 50 × 300 mm Porasil column, EtOAc/CH_2Cl_2/AcOH 10:90:1); yield: 11.4 g (90%); $[\alpha]_D$ +45.3 (*c* 1.0, $CHCl_3$).

Z-Tyr(TMSE)-ONbz:[42]
To a cooled soln of Ph_3P (3.72 g, 14 mmol) and DEAD (2.3 mL, 14 mmol) in THF (25 mL) in a flame-dried round-bottomed flask, was added Z-Tyr-ONbz (14 mmol), followed by 2-(trimethylsilyl)ethanol (2.14 mL, 14 mmol). Soon after the addition was complete, the evolution of N_2 was noted. The ice bath was removed and the soln was stirred for 10 min. The solvent was removed, the resulting syrup was dissolved in MeOH (1 L), and the soln was allowed to stand at rt for at least 2 d. The MeOH was then removed and the residue was subjected to flash chromatography (10-cm column, CH_2Cl_2/Et_2O 98:2), which gave an oil; yield: 2.65 g (35%).

H-Tyr(TMSE)-OH:[42]
A soln of Z-Tyr(TMSE)-ONbz (430 mg, 0.78 mmol) in MeOH (25 mL) was hydrogenated in the presence of Pd/C at 3.4 atm for 1 h. The catalyst was retained over a bed of Celite and washed with MeOH. The solvent from the combined filtrates was removed to give an amorphous white solid; yield: 230 mg (100%).

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2.6.5 Thioether Group

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The main concern in the synthesis of methionine-containing peptides, both by methods in solution and on solid supports, are alkylation and oxidation at the sulfur atom. Carbocation formation during deprotection steps provokes thioether alkylation reactions to give sulfonium salts which may undergo subsequent degradation reactions.[1–3] Oxidation of methionine to the corresponding sulfoxide Met(O) occurs in a multistep peptide synthesis unless precautions are taken to operate in degassed, peroxide-free solvents under a nitrogen or argon atmosphere. In a multiple peptide synthesis where tea-bags or open multiple synthesizers are used and air oxygen is repeatedly sucked through the resin, high percentages of Met(O) peptides have been reported.[4] Acid-catalyzed oxidation of methionine[5] may also occur in peptide deprotection and resin-cleavage steps with strong acids, as well as during disulfide bridge formation with more or less mild oxidants (see Vol. E 22b, Section 6.1). Formation of Met(O) peptide derivatives is detected and quantified by amino acid analysis upon acid hydrolysis, which can alter the oxidation state of methionine, and thus more properly upon enzymatic digestion. Alternatively, NMR spectroscopy has also proven to be a method of choice for the detection and quantification of Met(O) due to the high-field shift of the sulfoxide methyl signal.[6] Moreover, Met(O) side products exhibit shorter retention times than the nonoxidized peptides in RP-HPLC. For polymer-bound peptides, Met(O) can be detected with acetyl cloride and sodium iodide in DMF by the release of iodine which can be titrated for quantitative assays.[7]

2.6.5.1 Protection of Methionine as Sulfoxide

In order to prevent S-alkylation in the repetitive acid-cleavage steps required in a multistep peptide synthesis, the protection of methionine residues as sulfoxides was proposed by Iselin.[8] This approach has since been widely used for peptide synthesis in solution and on solid supports.[2,3] Generally, oxidation of L-methionine yields a mixture of *S*- and *R*-sulfoxides and, depending on the conditions used, even a mixture of the related sulfone. Thereby, at least by oxidation with hydrogen peroxide the *R*-sulfoxide is formed in a preferred manner (80%),[8] whereas oxidation of L-methionine by tetrachloroauric(III) acid has been reported to produce stereospecifically the *S*-sulfoxide diastereoisomer.[9] Isomeric pure Met(O) derivatives are obtained by isolation of the isomers from the *S,R*-sulfoxide mixture, taking advantage of the differences in solubility of the picrate salts.[10]

A comparative analysis of oxidants used for the preparation of Met(O) derivatives from the related N^{α}-protected amino acid (Table 1) indicates that sodium perborate is apparently the most convenient reagent, since only negligible amounts of the sulfones are formed.[11] In all cases a heterogeneous mixture of the two sulfoxide diastereoisomers is obtained with an (*S,S*/*R,R*) ratio of ca. 1 in the case of the perborate oxidation.[11]

Table 1 Synthesis of Z(OMe)-Met(O)-OH with Various Oxidants[11]

Reagent	mp (°C)	$[\alpha]_D^{22}$ in AcOH[a]	Yield (%)
H_2O_2	97–100	+6.4 *c* 2	81
$NaIO_4$	97–99	+2.9 *c* 2.1	72
$NaBO_3$	98–101	+8.4 *c* 1.9	79

[a] The differences in the values of optical rotation derive from different ratios of the two diastereoisomers.

for references see p 383

Z(OMe)-Met(O)-OH:[11]
A mixture of Z(OMe)-Met-OH (15.67 g, 50 mmol) in EtOAc (100 mL) and $NaBO_3$ (9.23 g, 1.2 equiv) in H_2O (50 mL) was stirred at rt for 7 h until the starting material disappeared on TLC. The soln was acidified with citric acid, the organic phase was washed with brine, dried (Na_2SO_4), and concentrated. The residue was triturated with EtOAc and the solid was recrystallized (MeOH/EtOAc); yield: 13 g (79%); mp 98–101 °C; $[\alpha]_D^{22}$ +8.4 (*c* 1.9, AcOH); TLC ($CHCl_3$/MeOH/AcOH 9:1:0.5) R_f 0.39; ($CHCl_3$/MeOH/ H_2O 8:3:1) R_f 0.61.

2.6.5.2 Reduction of Methionine Sulfoxide

Independently whether partial oxidation of Met residues occurs during the synthesis or whether Met(O) derivatives are purposely used, the sulfoxide has to be quantitatively reduced in the final steps to regenerate the methionine peptides. For this purpose, various reagents were proposed for the reduction in aqueous solution upon final deprotection and in organic solvents generally prior to the deprotection step in the presence or absence of cystine disulfides.

2.6.5.2.1 With Thiols

Among the thiols used for reduction of Met(O) in fully protected or deprotected peptides, mainly 2-sulfanylethanol, dithiothreitol, or thioglycolic acid are used;[8,12–20] these reductants, however, may require long reaction times and even elevated temperatures, which can lead to undesired side reactions like hydrolysis of labile peptide bonds.[21,22] To optimize this type of reduction in peptides and proteins, a systematic study with different thiols was conducted focusing on temperature, pH of the solution, and concentration of the reducing agent.[23,24] From this detailed comparative analysis, *N*-methylsulfanylacetamide emerged as the reagent of choice owing to the lack of side reactions and the greater reactivity at neutral and basic pH compared to thioglycolic acid. The latter reagent is of similar efficiency in acidic media like aqueous acetic acid, as it may be required to solubilize the peptides.

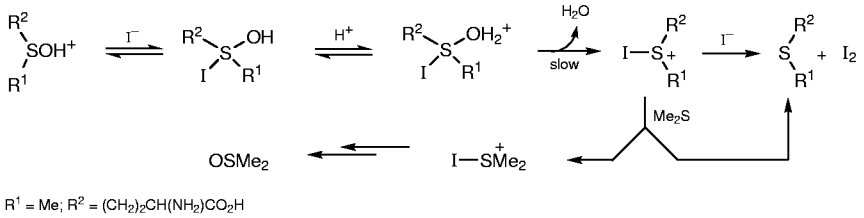
H-Lys-Glu-Thr-Ala-Ala-Ala-Lys-Phe-Glu-Arg-Gln-His-Met-Asp-Ser-OH:[19]
The pentadecapeptide Met-*R*-sulfoxide (20 mg, 0.01 mmol) was incubated in 1% aq thioglycolic acid (25 mL) at 45 °C for 24 h, the thioglycolic acid was then exchanged for acetate on an IRA-400 column by eluting with H_2O. Freshly distilled thioglycolic acid was added to make a 1% soln. The incubation and ion exchange steps were carried out three times and the final aq soln was lyophilized; yield: 20 mg (92 %); amino acid analysis of the aminopeptidase M digest confirmed quantitative reduction of the Met(O) residue.

2.6.5.2.2 With Ammonium Iodide and Trifluoroacetic Acid

Reduction with iodide and TFA is a mild procedure[25] that may be particularly useful for concomitant cleavage of TFA-labile protecting groups. Acid catalysis causes protonation of the sulfoxide, promoting attack of iodide at the sulfur (Scheme 1). Further acid-catalyzed dehydration of the species thus obtained, followed by reaction with iodide, yields the sulfide and iodine. The reaction rates can be increased by addition of dimethyl sulfide.[26,27] This result was explained in terms of competition of dimethyl sulfide with iodide for the attack at the iodosulfonium cation. The iodide/TFA method is mild enough to reduce Met(O) in the presence of disulfide bridges.[26–28] This was further confirmed in the synthesis of β-human calcitonin gene related peptide (β-hCGRP), where trimethylsilyl triflate/TFA in the presence of diphenyl sulfide and ammonium iodide was used in the deprotection and Met(O) reduction step without scrambling of the disulfide.[29] However, in this procedure, temporary formation of iodosulfonium ions and release of iodine occurs, two species that can lead to side reactions via electrophilic substitution or oxidation, respectively. This aspect has been carefully analyzed[30] and the study confirmed that free cysteine (Cys) residues undergo oxidation to yield disulfide derivatives. Moreover, while the histidine (His) and tyrosine (Tyr) residues remain intact under these conditions, overexposure of tryptophan (Trp)

residues to the acidic conditions, that are needed to carry out the reduction, leads to 2,2′-dimerization to indolylindole derivatives,[30] thus confirming previous results.[25] Therefore, by enhancing the reaction rate with addition of dimethyl sulfide and using smaller excesses of iodide,[28] this side reaction can be largely prevented. Additionally, protection of the Trp residue as the N^{in}-formyl derivative is advisable.[30]

Scheme 1 Reduction of Methionine Sulfoxide with Ammonium Iodide/Dimethyl Sulfide in Trifluoroacetic Acid[26]



R^1 = Me; R^2 = $(CH_2)_2CH(NH_2)CO_2H$

Taking advantage of the experience gained with ammonium iodide/TFA for reduction of Met(O) residues, a new resin-cleavage-deprotection cocktail has been used that consists of TFA (81%), phenol (5%), thioanisole (5%), ethane-1,2-dithiol (2.5%), water (3%), dimethyl sulfide (2%), and ammonium iodide (1.5% w/w).[31] It was found to prevent oxidation of Met residues in the acid cleavage step that was observed to occur in considerable extents with other deprotection cocktails.

As an alternative to ammonium iodide/TFA, the use of titanium(IV) chloride/sodium iodide in TFA has been proposed for clevage of the acid-labile protecting groups and reduction of Met(O) residues.[32] However, this reagent, besides bearing all the disadvantages of the use of heavy metals in peptides, leads to time-dependent reduction of disulfide bridges as well as to side products at the level of Trp residues.[33]

H-Arg-Pro-Lys-Pro-Gln-Gln-Phe-Phe-Gly-Leu-Met-NH_2; Substance P:[25]
To a soln of H-Arg-Pro-Lys-Pro-Gln-Gln-Phe-Phe-Gly-Leu-Met(O)-NH_2 (35 mg, 0.026 mmol) in TFA (3 mL) was added, at 0 °C, NH_4I (20 mg, 0.138 mmol) in H_2O (0.5 mL). After 10 min, H_2O (10 mL) was added and the I_2 was extracted with CCl_4. The aqueous phase was treated with an ion-exchange resin (BioRad AG 2-X8, acetate) and the product was eluted with 0.1 M AcOH. The fractions containing the product with R_f 0.65 (EtOAc/pyridine/AcOH/H_2O 5:5:3:1) were collected and lyophilized. The product was purified by gel filtration (Sephadex G-25, 0.1 M AcOH); fractions of the main peak were combined and lyophilized; yield: 20 mg (60%).

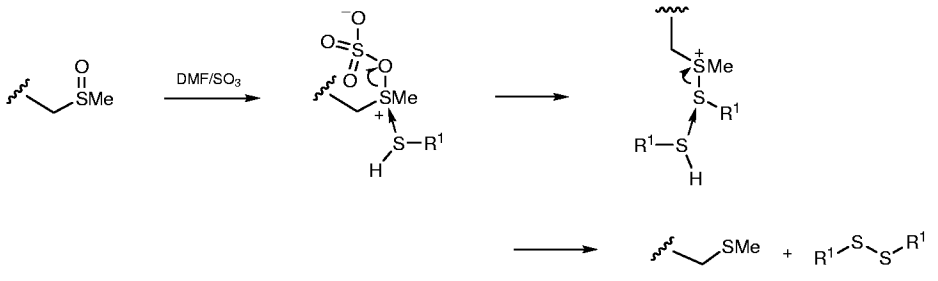
H-(β-CGRP-1–37)-NH_2:[29]
Protected β-hCGRP (50 mg, 9 μmol) in TFA (5.0 mL) was treated with $(CF_3CO_2)_3Tl$ (6.23 mg, 1.2 equiv) in the presence of anisole (50 μL) in an ice bath for 60 min; then TFA was removed and dry Et_2O was added. The residue was treated with 0.5 M Ph_2S in 1 M TMSOTf/TFA (4.6 mL) containing *m*-cresol (234 μL) in an ice bath for 180 min. Then NH_4I (27 mg, 20 equiv) and Me_2S (14 μL, 20 equiv) were added and treatment was continued for an additional 30 min. After addition of dry Et_2O, the precipitate was collected by centrifugation. The crude product was purified by gel filtration on Sephadex G-15 followed by ion exchange chromatography on CM-Trisacryl and RP-HPLC; the final product was isolated by lyophilization; yield: 4.4 mg (12.2%); amino acid analysis of the leucine-aminopeptidase digest confirmed the quantitative reduction of the Met(O).

2.6.5.2.3 With Sulfur Trioxide/Thiol

Reduction of Met(O) residues may also be performed prior to the acid deprotection step by use of the SO_3/DMF complex in pyridine in the presence of a thiol.[34,35] The reduction proceeds with the initial formation of a sulfonium ion and subsequent nucleophilic attack of the thiol (Scheme 2). The reaction is performed at room temperature and is generally completed within one hour.

for references see p 383

Scheme 2 Reduction of Methionine Sulfoxide with Sulfur Trioxide/Thiol in Organic Solvents[34]

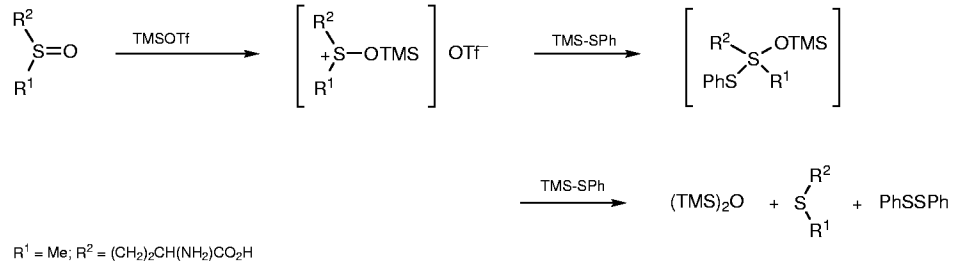


Z(OMe)-Phe-Met-OMe:[35]
Z(OMe)-Phe-Met(O)-OMe (491 mg, 1.0 mmol) in DMF/pyridine (4:1, 5 mL) was treated with DMF/SO_3 (766 mg, 5 equiv) and ethane-1,2-dithiol (0.47 mL, 5 equiv). After stirring at 20 °C for 30 min, the solvent was removed. The residue was treated with H_2O to afford a powder which was recrystallized (DMF/Et_2O); yield: 402 mg (85%); mp 118–120 °C; $[\alpha]_D^{20}$ −16.5 (*c* 1.0, DMF).

2.6.5.2.4 With Trimethylsilyl Compounds

Selenophenol has been proposed as an effective Met(O) reducing agent in organic solvents.[36] This procedure has so far not found wider application. More attention has been paid to the use of (phenylsulfanyl)trimethylsilane, which was reported to deoxygenate various sulfoxides in the presence of the catalyst tetrabutylammonium bromide.[37] This reagent was found to efficiently reduce Met(O) even in the absence of a catalyst,[36] but addition of a small amount of the oxygenophilic reagent trimethylsilyl trifluoromethanesulfonate[38] was found to markedly enhance the reduction rates.[39–41] Alternatively, bromotrimethylsilane–thioanisole in TFA has been proposed[4] to afford peptide deprotection and reduction of Met(O). The efficiency of this procedure has been documented in the synthesis of phosphopeptides, biotinylated peptides, and peptide libraries.

Scheme 3 Reduction of Methionine with (Phenylsulfanyl)trimethylsilane/Trimethylsilyl Trifluoromethanesulfonate in Organic Solvents[39]



R^1 = Me; R^2 = $(CH_2)_2CH(NH_2)CO_2H$

Z(OMe)-Val-Pro-Leu-Pro-Ala-Gly-Gly-Gly-Thr-Val-Leu-Thr-Lys(Z)-Met-Tyr-Pro-Arg(Mts)-Gly-Asn-His-Trp(Mts)-Ala-Val-Gly-His-Leu-Met-NH_2:[39]
A soln of Z(OMe)-Val-Pro-Leu-Pro-Ala-Gly-Gly-Gly-Thr-Val-Leu-Thr-Lys(Z)-Met(O)-Tyr-Pro-Arg(Mts)-Gly-Asn-His-Trp(Mts)-Ala-Val-Gly-His-Leu-Met(O)-NH_2 (150 mg, 0.04 mmol) (protected hGRP) in DMF (3 mL) was treated under an argon atmosphere with TMS-SPh (0.77 mL, 100 equiv) and TMSOTf (37.6 µL, 4 equiv) at 25 °C for 30 min. The soln was neutralized with Et_3N, concentrated, and the residue was then treated with Et_2O; yield: 134 mg (90%); negligible amounts of Met(O) were detected by amino acid analysis.

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2.6.6 Thiol Group

L. Moroder, H.-J. Musiol, N. Schaschke, L. Chen, B. Hargittai, and G. Barany

The synthesis of peptides containing cysteine and/or cystine residues may pose a number of intricacies beyond the routine chain assembly and final deprotection. The focus of this section is on various options for protection of the β-thiol side chain, along with methods to remove such protecting groups either globally, i.e. concurrent with all other side-chain protecting groups, or selectively, particularly for regioselective schemes as discussed in Vol. E 22b, Section 6.1. Because some deprotection methods are designed to provide disulfides directly, there is inevitably some overlap with Vol. E 22b, Section 6.1. The reader is directed further to a number of excellent reviews,[1–9] each of which cover cysteine protection and/or disulfide formation in a manner that was state-of-the-art as of when that review was written, and that also refer to methods which today can be considered to be primarily of historical interest.

For selected cases, derivatives of cystine itself can be used as a "self-protected" starting material that can be extended further at the N- and/or C-termini to give either symmetric dimers (which can then be reduced to provide the corresponding cysteine-containing peptide if needed) or asymmetric cystine peptides. Alternative strategies which are considerably more general, and also test the versatility of the repertoire of sulfur protecting groups, involve sequential S-deprotection and regioselective disulfide-forming steps. Finally, it must be recognized that all decisions related to cysteine protection and the panoply of further manipulations must be made in the context of the overall protection strategy for synthesis (be it in solution or polymer supported), and then later, must take into account the ready tendency of disulfides to disproportionate and/or scramble upon exposure to a variety of experimental conditions. For every set of procedures that appears to be reasonably reliable for the majority of routine applications, many more have been tried and found wanting.

The native chain-ligation methods,[10] in which one component of the couplings is often a peptide containing an N-terminal cysteine with an unprotected β-thiol, is covered in Section 4.1.5.

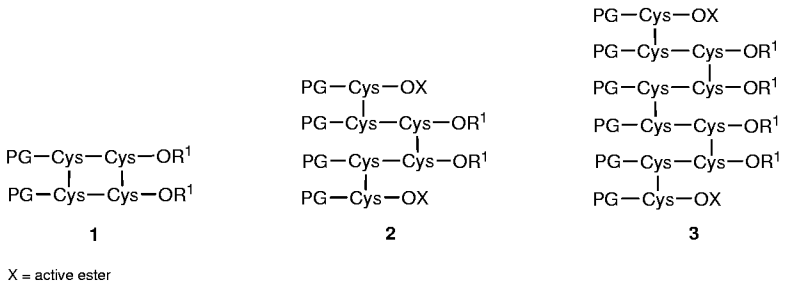
2.6.6.1 Peptide Synthesis with Cystine Derivatives

The nominally simplest approach to synthesize cysteine peptides is by direct acylation of cysteine.[11–13] Interestingly, it is the side-chain sulfur that becomes acylated, but this is quickly followed by a sulfur to nitrogen migration of the acyl moiety. It is also possible to convert *N*-acylcysteines into the corresponding four-membered thiolactones by the action of DCC in dioxane, followed by aminolysis to open the ring, create a peptide bond, and restore the free thiol of cysteine.[14] Because the aforementioned methods are inconvenient and not overly reliable, it is more prudent to start with a suitable cystine derivative that can be extended further while the side-chain sulfur is "self-protected" as a disulfide. Further reaction conditions must be such as to maintain the integrity of the disulfide. In brief, oxidizing and reducing agents are to be avoided, and strongly acidic or mildly alkaline solutions may promote disulfide exchange in asymmetric cystine peptides.[4] Peptides with preformed intra- and intermolecular disulfide bridges are routinely treated with TFA, or even hydrogen fluoride, in deprotection steps without significant scrambling effects. Conversely, the presence of even a small amount of free thiol will catalyze rapid disulfide interchange at a pH above 7. Similarly, saponification or hydrazinolysis of ester derivatives should be avoided.

2.6.6.1.1 Symmetric Cystine Derivatives

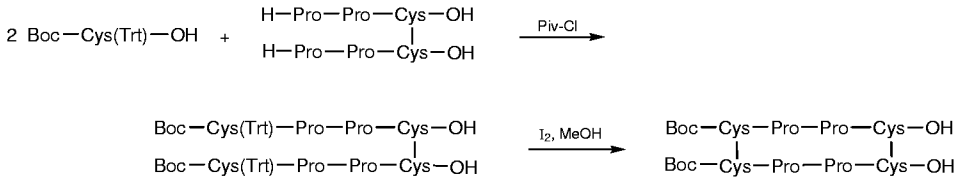
If the desired final product contains a single cysteine, or is a symmetric homodimer cross-linked by a single cystine, a symmetrically protected cystine is a useful starting material for further synthetic steps in solution. When needed, the final peptide can be reduced to the free thiol form, and further oxidations can be carried out subsequently. This simple and economic approach, which has been applied in the classical synthesis of glutathione,[11] may be recommended for the preparation of small cystine peptides. It may also be used for the synthesis of some S-protected peptide intermediates since reaction of the peptide dimer with 2-methylpropane-2-thiol under air oxidation conditions leads to the monomeric *S*-(*tert*-butylsulfanyl) derivative.[15] On-resin synthesis using this approach is problematic since acylation of resin-bound peptide chains with the N,N′-bis-protected cystine leads not only to the desired homodimer; rather mono-aminolysis of the activated cystine derivative can also occur leading to asymmetric cystine peptides.[16] Similar difficulties are anticipated in the synthesis of homodimers containing two cystine residues, since acylation of the mono-cystine peptide with the second N,N′-bis-protected cystine can produce not only the desired dimer **1**, but also compounds **2** and **3** as observed in the synthesis of oxytocin (Scheme 1).[17] For the latter goal, a combination of the cystine approach with the use of S-protected cysteines (Scheme 2), and subsequent formation of the second intrachain disulfide bridge, is much more advisable.[18]

Scheme 1 Products Resulting from Acylation of a Mono-cystine Peptide Dimer with N,N′-Bis-Protected Cystine[17]



X = active ester

Scheme 2 Synthesis of a Bis-cystine Peptide Dimer by a Combination of the Cystine Approach with S-Protected Cysteine Derivatives[18]



[Z-Pro-Pro-Cys-OH]$_2$:[18]
Cystine (1.44 g, 6 mmol) was dissolved at 50 °C in MeOH (10 mL) by addition of 2.2 M Triton B in MeOH (5.4 mL, 12 mmol). The soln was concentrated and the residue redissolved in DMF (10 mL). To Z-Pro-Pro-OH (4.56 g, 13.2 mmol) in DMF (60 mL) was added at −10 °C NMM (1.46 mL, 13.2 mmol) followed by Piv-Cl (1.52 mL, 12.4 mmol). After 5 min, the mixed anhydride was added to the precooled cystine soln. The mixture was allowed to reach rt over 2 h and was then stirred overnight. The solvent was removed and the residue in EtOAc was washed with 1% aq $KHSO_4$ and H_2O. The soln was dried (Na_2SO_4) and the volatiles were removed. The product was precipitated from MeOH/Et_2O and EtOH/iPr_2O; yield: 4.2 g (78%); mp 110–115 °C; $[\alpha]_D^{20}$ −123.6 (*c* 1, MeOH).

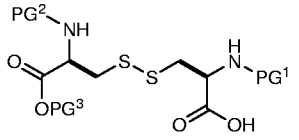
[H-Pro-Pro-Cys-OH]$_2$ · 2HBr:[18]
To a soln of [Z-Pro-Pro-Cys-OH]$_2$ (2.2 g, 2.45 mmol) in AcOH (9 mL) were added 3.6 M HBr in AcOH (11.1 mL) and some crystals of phenol. After 1 h at rt, the oily precipitate was solidified with Et_2O and reprecipitated from AcOH/Et_2O; yield: 1.69 g (89%).

[Boc-Cys(Trt)-Pro-Pro-Cys-OH]$_2$ · 2DCHA:[18]
To Boc-Cys(Trt)-OH [prepared from its DCHA salt (3.7 g, 5.7 mmol) by washing in EtOAc with 5% aq $KHSO_4$] in DMF (50 mL) was added at −15 °C NMM (0.56 mL, 5.1 mmol) followed by Piv-Cl (0.6 mL, 4.9 mmol). After 5 min, the soln was poured into a precooled soln of [H-Pro-Pro-Cys-OH]$_2$ · 2HBr (1.5 g, 1.9 mmol) and TEA (1.06 mL, 7.6 mmol) in DMF (30 mL). The mixture was allowed to reach rt and, after 12 h, the solvent was removed. The residue in EtOAc was washed with 1% aq $KHSO_4$ and H_2O, and the soln was dried (Na_2SO_4) and the volatiles were removed. The product was precipitated from EtOAc as the DCHA salt with Et_2O; yield: 2.87 g (82%); mp 160–163 °C; $[\alpha]_D^{20}$ −91.7 (*c* 1, MeOH).

2.6.6.1.2 Asymmetric Cystine Derivatives

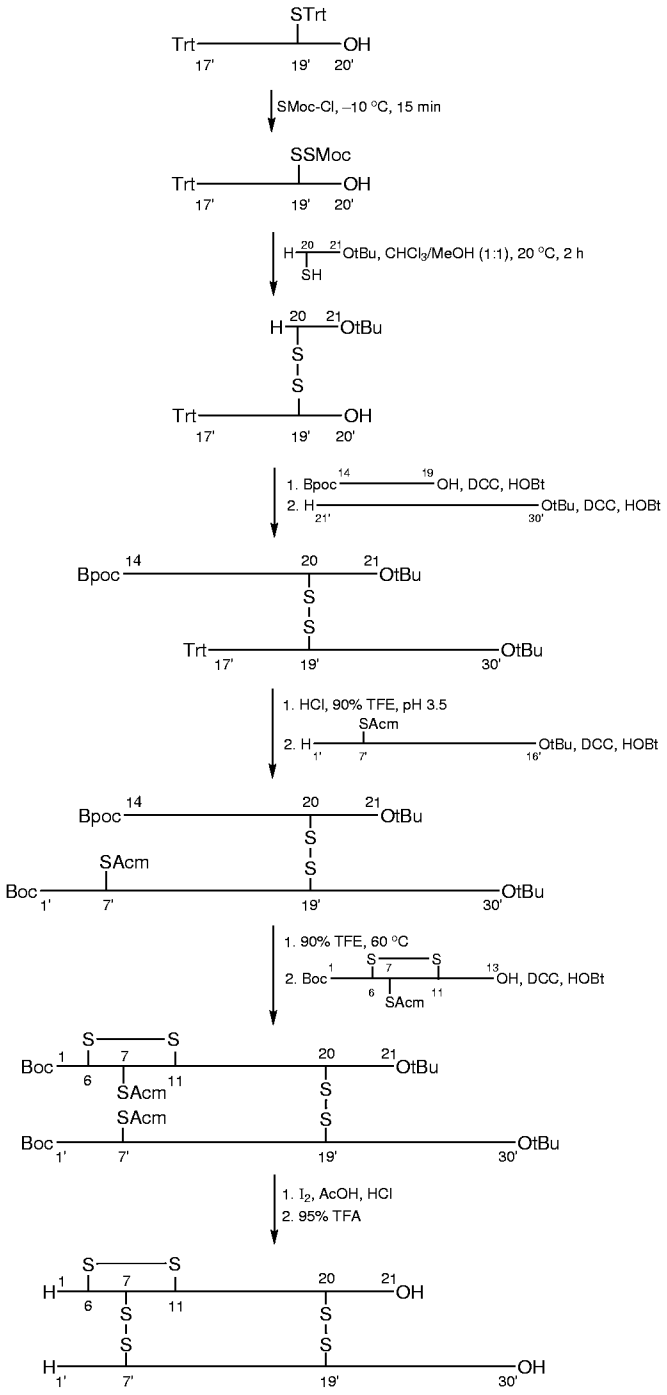
Asymmetric orthogonally protected cystine derivatives **4** (see Scheme 3) are prepared by site-directed disulfide formation between two appropriate cysteine derivatives with different N^{α}-, C^{α}- and S-protection via methods of selective disulfide bridging as described in Vol. E 22b, Section 6.1.[19] It is not necessary to start per se at the amino acid level; thus, relatively short heterodimers can be constructed and used for further synthetic steps. Another variation starts with a single peptide chain that includes an already preformed intramolecular disulfide bridge. The overall principle is to carry out further manipulations with one or more of the needed disulfides already in place; this requires precautions at all further deprotection steps, and at the acylations, in order to avoid disproportionation and other side reactions involving the preexisting disulfides. This synthetic strategy was elegantly demonstrated in the synthesis of insulin by Sieber et al.[20,21] (Scheme 4; see also Vol. E 22b, Section 6.1.4) and of related analogues with nonnatural cystine networks.[22] The overall protection scheme was based on *tert*-butanol-derived, acid-labile side-chain protections, in combination with more acid-labile and, thus, largely orthogonal N^{α}-trityl or N^{α}-Bpoc protections, which made it possible to work under relatively weak acid conditions that prevent disulfide scrambling.

Scheme 3 Asymmetric Cystine Derivatives Suitable for the Synthesis of Asymmetric Cystine Peptides[19]



4 PG^1 = Fmoc; PG^2 = Boc; PG^3 = $(CH_2)_2TMS$
PG^1 = Dde; PG^2 = Fmoc; PG^3 = $(CH_2)_2TMS$
PG^1 = Boc; PG^2 = Fmoc; PG^3 = Dmab

Scheme 4 Synthesis of Insulin[21]



2.6.6.2 Peptide Synthesis with S-Protected Cysteine Derivatives

In Table 1, the most common S-protecting groups of cysteine are compiled according to their compatibility with either of the two major strategies used in stepwise solid-phase peptide synthesis: the Boc/Bzl schemes based on acidolyzable *N*-Boc derivatives for temporary

for references see p 418

protection of the α-amino group, and the Fmoc/tBu schemes that rely on the base-labile *N*-Fmoc derivatives of the α-amino group. To extend the information in Table 1 to synthetic strategies in solution, most of which were developed chronologically before the solid-phase methodologies, it can be recognized that all of the appropriate S-protecting groups that are stable to moderate strength acid can be used in combination with N^{α}-Boc or, equivalently, with N^{α}-Z(OMe) protection which is removed under comparable conditions. In addition, the required chemical selectivity can be achieved when combining *S*-Trt with more acid-labile N^{α}-amino protecting groups such as Trt, Bpoc, or Nps. For the Nps strategy to be successful, acidolytic deprotection requires the inclusion of scavengers such as 2-methylindole,[23–25] or alternative Nps cleavage reagents, such as thioacetamide[26] or phosphines,[24] can be applied. S-Protected cysteine residues, as well as methionine, prevent the use of the popular Z/tBu strategy in solution due to poisoning of the catalysts that are used for hydrogenolytic N^{α}-deprotection. In this latter regard, the discovery that catalytic hydrogenolysis is successful when the solvent is liquid ammonia provides a useful way out of this dilemma.[27,28]

Table 1 emphasizes those protecting groups of value that have emerged as state-of-the-art within the context of their relative labilities; the remainder of this section, however, is more comprehensive and organized according to structural classes, i.e. the chemical functional group represented by the S-protected cysteine derivative.

Table 1 The Most-Common Thiol Protecting Groups Used in Peptide Synthesis[a]

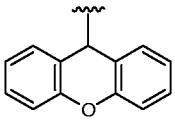
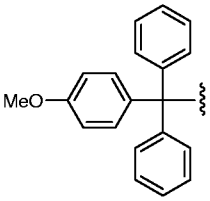
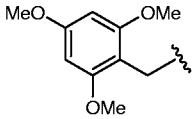
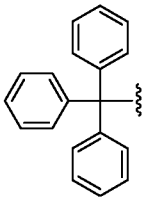
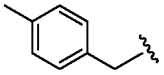
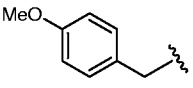
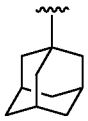
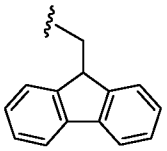
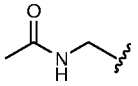
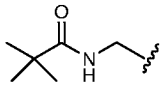
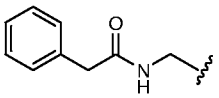
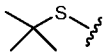
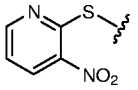
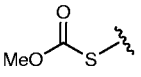
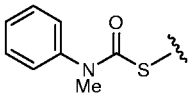
Protecting Group	Structure	Lability	Stability	Removal[b]
9*H*-xanthen-9-yl (Xan)		dil acid	base, nucleophiles	dil TFA/scavenger, I_2, Tl(III)
4-methoxytrityl (Mmt)		dil acid	base, nucleophiles	dil TFA/scavenger, I_2, Ag(I), Hg(II), Tl(III), R^1SCl
2,4,6-trimethoxybenzyl (Tmb)		dil acid	base, nucleophiles	dil TFA/scavenger, I_2, Tl(III)
triphenylmethyl (trityl) (Trt)		moderate acid	base, nucleophiles	dil TFA/scavenger, I_2, Ag(I), Hg(II), Tl(III), R^1SCl
4-methylbenzyl [Bzl(4-Me)]		concd acid	TFA, Ag(I), base, R^1SCl	HF (0 °C), Tl(III), $Ph_2SO/MeSiCl_3$
4-methoxybenzyl (Mob)		concd acid	TFA, base, R^1SCl, I_2	Ag(I), Hg(II), Tl(III), $Ph_2SO/MeSiCl_3$, HF (0 °C)

Table 1 (cont.)

Protecting Group	Structure	Lability	Stability	Removal[b]
1-adamantyl (1-Ada)		concd acid	HF (0 °C), Ag(I), base, I_2	Hg(II), Nps-Cl, Ph_2SO/ $MeSiCl_3$, HF (20 °C)
tert-butyl (tBu)		concd acid	HF (0 °C), Ag(I), base, I_2	Hg(II), Nps-Cl, Ph_2SO/ $MeSiCl_3$, HF (20 °C)
9-fluorenyl-methyl (Fm)		base	HF, I_2	NH_3 in MeOH, piperidine, dil DBU
acetamidomethyl (Acm)		–	HF (0 °C), TFA, base	I_2, Ag(I), Hg(II), Tl(III), R^1SCl, Ph_2SO/ $MeSiCl_3$
trimethylacet-amidomethyl (Tacm)		–	HF (0 °C), TFA, base	I_2, Ag(I), Hg(II), Tl(III), R^1SCl, Ph_2SO/ $MeSiCl_3$
phenylacet-amidomethyl (Phacm)		–	HF, base	I_2, Hg(II), Tl(III), penicillin amidohydrolase
tert-butylsulfanyl (StBu)		reducing agents	TFA, HF (partial), base, R^1SCl	R^1SH, Bu_3P, TCEP, other reducing agents
3-nitro-2-pyr-idylsulfanyl (Npys)		reducing agents	HF	R^1SH, Bu_3P, TCEP
methoxycarbon-ylsulfanyl (SMoc)		reducing agents	HF	R^1SH, other reducing agents
N-methyl-*N*-phen-ylcarbamoyl-sulfanyl (SMpc)		reducing agents	HF	R^1SH, specifically DTT or pyridine-2-thiol

[a] Adapted from similar compilations in refs[6,8,9].

[b] Conditions or reagents listed for the removal of the protecting groups are intended for quantitative deprotection and/or oxidative cleavage; the products from these reactions may be free thiols, thiolates, or disulfides. For metal-mediated removals, the counterion and solvent are sometimes critical; for acidolytic cleavages the nature and amount of added scavengers may significantly affect the stability and/or lability of the protecting group.

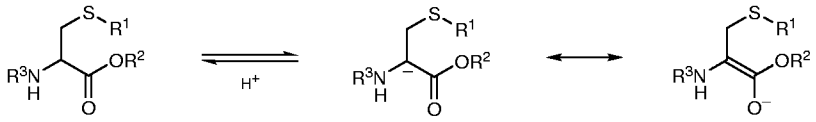
Racemization of S-Protected Cysteines

Independent of the thiol protecting groups used in the synthesis of cysteine peptides, the facile racemization of the Cys residue has been recognized as a serious problem since the early days of peptide chemistry (see Section 1.2.1.2 and Vol. E 22b, Section 7.4).[29] It is well established that N,S-protected cysteine active esters are unusually prone to racemization via a C^{α}-proton-abstraction mechanism in the presence of excess amines (Scheme 5).[30,31] More recently, unexpectedly high rates of racemization were detected for urethane-protected cysteine derivatives, even when fast coupling reactions were performed with acylating agents such as TBTU/HOBt in the presence of DIPEA as added base.[32–35] In contrast, racemization

for references see p 418

of the cysteine residues is almost totally suppressed when the condensations are performed in the absence of base, e.g. via the pentafluorophenyl ester,[32,33,35] or the symmetric anhydride,[34] or even via TBTU/HOBt but with significantly decreased amounts of the added base.[36] A systematic study of factors that affect rates of Cys racemization via proton abstraction on the α-carbon has been undertaken.[35]

Scheme 5 Proposed Mechanism of Racemization of N,S-Protected Cysteine Esters[29–31]



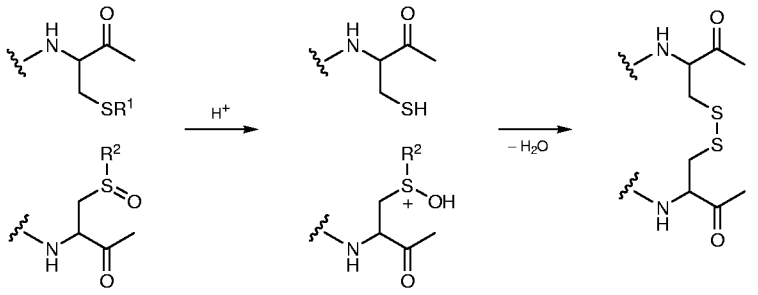
Racemization occurring during esterification of protected amino acids to hydroxymethyl polymers is a well-recognized problem.[3,37] This reaction is promoted by the use of DMAP as catalyst,[38] and racemization is particularly pronounced in the case of cysteine.[38–44] By the use of optimized esterification methods racemization can be reduced, but not totally suppressed.[41] When starting with halomethyl-functionalized resins, such as 2-chlorotrityl chloride resin,[43,45] bromo-SASRIN resin,[46,47] or 4-(bromomethyl)phenoxyacetamide resin,[48] racemization levels are significantly lowered. Alternatively, racemization is largely prevented by using the TDO esters without DMAP,[49,50] or PyAOP in the presence of DIPEA at low temperature.[51]

In agreement with the facile racemization of cysteine esters via base-induced formation of the carbanion of the S-protected Cys residue, and its stabilization by enolization of the carbonyl (see Scheme 5),[29] epimerization of cysteine residues linked to a resin via ester bonds occurs even during peptide-chain elongation using the Fmoc/tBu strategy, i.e. on repeated exposure to piperidine.[38] Detailed studies clearly reveal that this epimerization also occurs when bases other than piperidine for Fmoc cleavage are used.[40,42,43] Moreover, the thiol protecting group strongly affects the epimerization rate, with a rank order of *S*-StBu>*S*-Trt>>*S*-Tacm>*S*-Acm>*S*-MeBzl>*S*-tBu; the type of ester linkage is also a factor with the lowest rates determined for the trityl ester linkage.[43,44] There is no apparent epimerization when the C-terminal cysteine residue is anchored as amide to the resin.[40,44]

Side Reactions with S-Protected Cysteines

As precedented for methionine residues, cysteine residues S-protected as sulfides,[52,53] thioacetals,[54] or asymmetric disulfides[55] can undergo oxidation to the corresponding sulfoxides unless precautions are taken to avoid contact with oxygen or peroxide contaminants in solvents. Oxidation of the disulfide group of *S*-(alkylsulfanyl)cysteine residues leads to thiolsulfinates, species that react readily with thiols to form disulfides.[56,57] The fate of sulfoxides corresponding to sulfide- and thioacetal-type protections is largely unknown, although it is worth noting that such sulfoxides, e.g. S(O)Acm, S(O)Mob, or S(O)(1-Ada), react with cysteine or even S-protected cysteine, such as Cys(Mob), Cys(tBu), or Cys(1-Ada), under strong acidic conditions to form the related disulfides (Scheme 6) in high yields.[58,59] This sulfoxide-directed disulfide formation has been exploited for regioselective disulfide bridgings (see Vol. E 22b, Section 6.1.3.4). Therefore, partial oxidation of S-protected cysteine residues could become problematic precisely in the context of selective disulfide formation in multiple cysteine peptides.

Scheme 6 Sulfoxide-Mediated Disulfide Formation as a Potential Side Reaction[58]

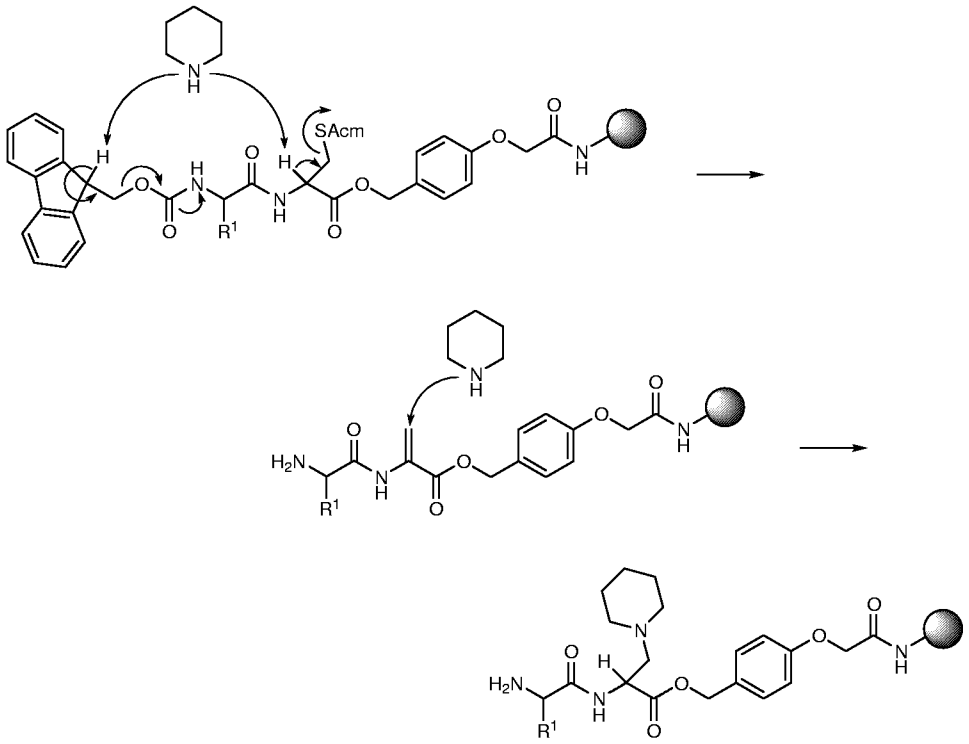


While it is reasonable to assume that steric effects may markedly affect the rate of sulfoxide formation, a detailed study has been performed only for *S*-(acetamidomethyl) and related protecting groups.[60] The *S*-(trimethylacetamidomethyl) group clearly shows enhanced resistance to oxidation, compared to *S*-(acetamidomethyl), implying a steric shielding of the sulfur by the bulkier trimethylacetamidomethyl group. Since reduction of sulfoxides to sulfides is generally difficult,[53,54,60,61] the presence of oxidized forms may explain formation of various side products in acidolytic deprotection/resin-cleavage steps, from which the cysteine residues can no longer be recovered.[52–54]

Exposure of S-protected cysteine peptides to strong bases such as sodium in liquid ammonia[62,63] (as required for deprotection of *S*-Bzl derivatives), to alkaline conditions,[64–68] or to hydrazinolysis,[64] but also to strong acids such as hydrogen fluoride,[69] is accompanied by some β-elimination. With good leaving groups, the rates of this side reaction are markedly enhanced.[70–72] More recently, it has been observed that use of Fmoc/tBu chemistry for the solid-phase synthesis of peptides containing a C-terminal cysteine residue led to an initial β-elimination, with subsequent addition of the piperidine moiety to form a C-terminal β-(1-piperidyl)alanine residue. This side reaction occurs with *S*-Trt,[73,74] *S*-Acm,[74] and *S*-StBu[44,75,76] in the rank order of *S*-StBu ≥ *S*-Acm >> *S*-Trt. A plausible mechanism involves base-induced formation of the C^{α}-carbanion of the S-protected cysteine ester, which is stabilized by enolization and results in epimerization (*vide supra*), and then β-elimination of the alkylthiolates or alkyl disulfide moieties (Scheme 7). Nucleophilic addition of piperidine to the dehydroalanine intermediate results in β-(1-piperidyl)alanine peptides as the main side products. Neither epimerization, nor such side products, are observed in the synthesis of peptides with C-terminal Cys(StBu) amides or with Cys(StBu) residues in internal sequence positions. The use of other bases or of lower base concentrations can reduce the rate of this side reaction, as shown in the case of C-terminal Cys(Trt) esterified on resin.[73]

for references see p 418

Scheme 7 β-Elimination of an S-Protecting Group with Formation of Dehydroalanine, Followed by the Formation of the Corresponding Piperidine Adduct, β-(1-Piperidyl)alanine.[74]



Histidine is frequently protected as its N^{π}-benzyloxymethyl derivative for Boc/Bzl strategies. Cleavage of this protecting group with hydrogen fluoride produces one equivalent of formaldehyde,[77] which under the acidic conditions reacts with N-terminal cysteine residues to produce the related Thz-containing peptide.[78,79] This side reaction is suppressed by the addition of excess cysteine, as a scavenger, to the hydrogen fluoride mixture.[79]

The products of cysteine deprotection reactions can be either the free thiol, the thiolate metal salts, or the disulfide, depending on the precise choice of reagents and conditions. The acidolytic cleavage mechanisms generate carbocations which can readily reattach to the cysteine thiol group unless carefully optimized scavengers are included in the cleavage cocktail. Model studies on TFA-mediated deprotection of Boc groups in the presence of various S-protecting groups clearly reveal formation of Cys(tBu) at extents depending upon the deprotection conditions.[80] Moreover, acid cleavage of peptides from resins generates resin-bound carbocations that lead to reattachment not only via the free cysteine thiols, but also via the S-protected cysteine side chains, as well documented in a comparative analysis with *S*-(*tert*-butylsulfanyl) and *S*-(acetamidomethyl) peptides.[33]

2.6.6.2.1 *S*-Alkylcysteine Derivatives

Historically, the first important cysteine protecting group was the *S*-benzyl group, developed by du Vigneaud et al. and used in the classical synthesis of oxytocin,[81] as well as in the syntheses of a number of cysteine-containing peptides in the first years of solid-phase synthetic methodology.[82,83] This *S*-Bzl protection is nowadays used infrequently due to the fact that its removal involves harsh conditions, e.g. sodium in liquid ammonia[84] or anhydrous hydrogen fluoride at 20 °C,[85] conditions which may lead to side reactions such as elimination

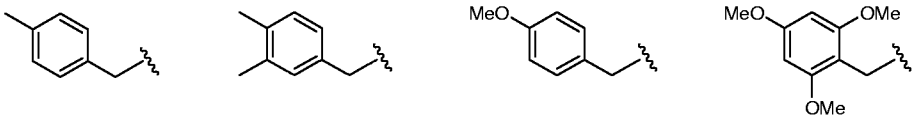
of phenylmethanethiol to furnish the dehydroalanine residue.[62,63,68,69,71] Similar considerations apply to the *S*-diphenylmethyl group.[86–89]

To modulate the acid stability/lability of sulfide-type protecting groups for their compatibility with various overall protection schemes, the enhanced stability of the carbocations resulting from acid and metal ion mediated cleavage of substituted *S*-benzyl, *S*-(*tert*-alkyl), *S*-(*sec*-aralkyl), and *S*-(*tert*-aralkyl) derivatives can be exploited. Among the large number of potential S-protecting groups that have been proposed, only a relatively small number have found general application.

2.6.6.2.1.1 Substituted *S*-Benzylcysteine Derivatives

The search for TFA stable, but HF cleavable, benzyl-type sulfides led to the 4-methylbenzyl [Bzl(4-Me)],[90] 3,4-dimethylbenzyl,[91] and 4-methoxybenzyl (Mob) derivatives[85,92] (Scheme 8), while further substitutions of the benzene ring with electron-donating groups, as in the 2,4,6-trimethoxybenzyl (Tmb) group,[93] led to very acid-labile S-protection.

Scheme 8 The Structures of *S*-Aryl-type Thiol Protecting Groups[85,90–93]



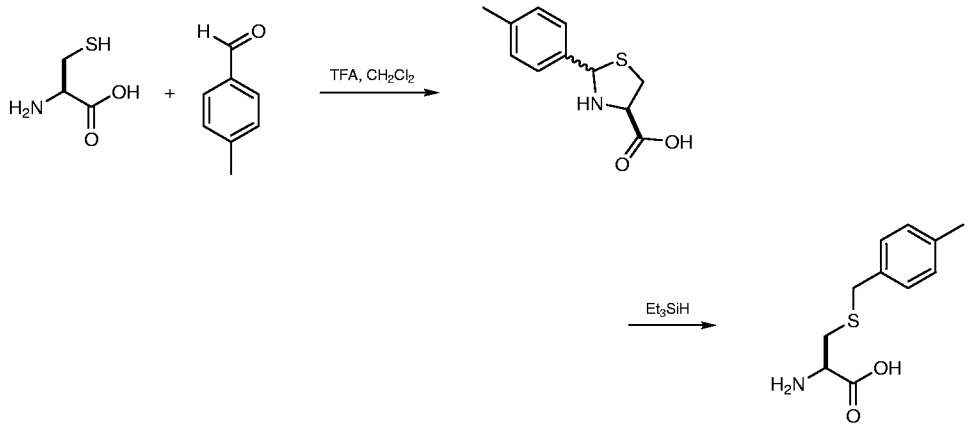
pMeBzl Mob Tmb

2.6.6.2.1.1.1 *S*-(4-Methylbenzyl)cysteine

Of the proposed methyl-substituted *S*-benzyl derivatives, *S*-(4-methylbenzyl)[90] protection is used mainly in the synthesis of cysteine peptides by the Boc/Bzl strategy in solution and on solid support. The S-protected cysteine derivative is obtained by alkylation of the thiol group with 4-methylbenzyl bromide in the presence of bases.[90] A highly efficient alternative procedure is based on reductive S-alkylation via reaction of cysteine in TFA with 4-methylbenzaldehyde followed by in situ reduction with triethylsilane and N^{α}-Boc protection with Boc_2O (Scheme 9).[94] The *S*-(4-methylbenzyl) protection is generally removed in the final acidolytic deprotection/resin-cleavage step with hydrogen fluoride containing 10–20% anisole or *p*-cresol[95] or 5% *p*-cresol with 5% *p*-thiocresol[53] at −5 to 0 °C for one hour, leading to peptides with all cysteine residues S-deprotected for the oxidative refolding step[85,96] or only partially deprotected for selective disulfide formation (for experimental procedures, see Vol. E 22b, Sections 6.1.2–6.1.4). Cleavage of the *S*-(4-methylbenzyl) group has also been achieved with thallium(III) trifluoroacetate, but not as cleanly as for other sulfides such as *S*-Mob, *S*-(1-Ada), or *S*-Trt.[97]

for references see p 418

Scheme 9 Reductive S-Alkylation of Cysteine by Reaction with 4-Methylbenzaldehyde, Followed by Treatment with Triethylsilane[94]



H-Cys[Bzl(4-Me)]-OH:[90]
[Bzl(4-Me)]-Br (3.42 g, 18.5 mmol) and TEA (7.75 ml, 55.5 mmol) were added to a soln of H-Cys-OH · HCl · H_2O (3.25 g, 18.5 mmol) in EtOH/H_2O (2:1, 30 mL). The mixture was stirred at rt for 12 h and then filtered. The solid was washed with H_2O and crystallized (EtOH/H_2O 3:2); mp 209–211 °C (dec).

Boc-Cys[Bzl(4-Me)]-OH:[94]
To a soln of H-Cys-OH (363 mg, 3.0 mmol) in TFA/CH_2Cl_2 (1:1, 10 mL), 4-methylbenzaldehyde (397 mg, 3.3 mmol) was added dropwise. After stirring for 90 min at rt, Et_3SiH (698 mg, 6.0 mmol) was added dropwise at 0 °C. After stirring for 16 h at rt, MeOH (10 mL) was added, the crude mixture was poured into H_2O/CH_2Cl_2 (4:3, 70 mL), and the organic layer was extracted with H_2O. The combined aqueous layers were taken to dryness, and the product was dissolved in MeOH/H_2O, neutralized with aq $NaHCO_3$, and recrystallized (MeOH/H_2O 1:3). The intermediate was reacted with Boc_2O in THF/H_2O (1:1, 80 mL) at pH 8.5 and rt for 2 h; then the mixture was acidified to pH 3–4 with dil H_2SO_4, and extracted with Et_2O. The combined organic layers were washed with brine and dried. The soln was concentrated and the product was precipitated with petroleum ether and recrystallized (EtOAc/hexane); yield: 0.8 g (82%); mp 81 °C; $[\alpha]_D^{20}$ –47.1 (*c* 1, AcOH).

Cleavage of the *S*-(4-Methylbenzyl) Group with Hydrogen Fluoride/Anisole; General Procedure:[53]
The correct volume of HF was distilled into a Teflon vessel containing the mixture of the protected peptidyl resin and dry anisole, precooled in a dry ice/EtOH bath. The reaction vessel was allowed to reach 0 °C in an ice bath, and stirred for 45–60 min. Subsequently, HF was removed under water aspiration until the resin was dry. The peptide–resin mixture was transferred to a fritted funnel using EtOAc (if the peptide is very hydrophobic, Et_2O should be used for the transfer) and washed with EtOAc (or Et_2O). The crude peptide was extracted from the resin using 10% aq AcOH for water-soluble peptides, and glacial AcOH or TFA for more hydrophobic peptides. Finally, the peptide was lyophilized.

2.6.6.2.1.1.2 *S*-(4-Methoxybenzyl)cysteine

S-(4-Methoxybenzyl)-protected cysteine is prepared by reaction of cysteine with 4-methoxybenzyl chloride in ammonia[92] or in TFA/CH_2Cl_2,[94] and by reductive alkylation with 4-methoxybenzaldehyde in TFA/CH_2Cl_2,[94] and is then further converted into the N^{α}-protected derivatives by standard procedures. Although this group is too TFA-labile for the synthesis of large peptides according to the Boc/Bzl strategy,[80,90,91] it has found successful application in many syntheses in solution and on resin, e.g. refs[98,99]. The *S*-(4-methoxybenzyl)cysteine derivative is cleaved with hydrogen fluoride at 0 °C in the presence of scavengers[85,90,100] or with TfOH/TFA.[101] It is also smoothly removed with one equivalent of mercury(II) acetate in TFA (0 °C, 15 min) and less rapidly with mercury(II) trifluoroacetate in 80% aqueous acetic acid (20 °C, 60 min).[102] Similarly efficient is the cleavage of the *S*-(4-methoxybenzyl) group with silver(I) trifluoromethanesulfonate (5 equiv) in TFA (0 °C,

60 min), whereas the *S*-(4-methylbenzyl) group is not cleaved under these conditions.[103] More important in terms of selective intramolecular disulfide pairing is the oxidative cleavage of the 4-methoxybenzyl sulfide with thallium(III) trifluoroacetate in TFA[97] (for experimental procedures, see Vol. E 22b, Sections 6.1.2 and 6.1.4). Cleavage and concomitant oxidation to intramolecular disulfide formation between two Cys(Mob) residues is readily achieved with trichloro(methyl)silane or tetrachlorosilane in TFA in the presence of diphenyl sulfoxide at 4 °C in 30 minutes.[104] By varying the chlorosilane component as well as the sulfoxide, partial selectivities in this oxidative deprotection of various S-protecting groups can be achieved; however, these are not sufficient for regioselective disulfide formation.[105]

H-Cys(Mob)-OH:[94]
To a soln of H-Cys-OH (363.5 mg, 3 mmol) in TFA/CH_2Cl_2 (3:18, 21 mL), 4-methoxybenzyl chloride (470 mg, 3.0 mmol) in CH_2Cl_2 (18 mL) was added dropwise in 1 h at rt. Then MeOH (10 mL) and H_2O (40 mL) were added, the layers were separated, and the organic layer was re-extracted with H_2O (30 mL). The combined aqueous layers were washed with CH_2Cl_2 (30 mL), filtered and concentrated. The residue was dissolved in H_2O, neutralized to pH 6 with aq $NaHCO_3$, and the product was recrystallized (MeOH/H_2O 3:1); yield: 586 mg (81%); mp 208–209 °C; $[\alpha]_D^{25}$ +24.5 (*c* 1.0, 1 M NaOH).

Removal of *S*-(4-Methoxybenzyl) with Silver(I) Trifluoromethanesulfonate; General Procedure:[103]
The Cys(Mob)-containing peptide (1 equiv) was dissolved in TFA/anisole (19:1), cooled to 0 °C, and AgOTf (10 equiv) was added. After stirring at 0 °C for 1 h, the peptide was precipitated with cold Et_2O. To release the free thiol from its silver(I) salts, the peptide was dissolved in AcOH/H_2O (1:1), and treated with DTT (50 equiv) at 25 °C for 2 h.

2.6.6.2.1.1.3 *S*-(2,4,6-Trimethoxybenzyl)cysteine

The *S*-(2,4,6-trimethoxybenzyl)cysteine derivative is obtained by S-alkylation of cysteine with 2,4,6-trimethoxybenzyl alcohol under acid catalysis.[93] The 2,4,6-trimethoxybenzyl group is rapidly removed by TFA-promoted acidolysis and the resulting carbocation has to be scavenged efficiently to prevent reattachment on the cysteine thiol group and tryptophan alkylation.[93] Among the various cleavage cocktails tested, a combination of water, phenol, and thioanisole (5% each) in TFA/CH_2Cl_2 (3:7), or triethylsilane in TFA/CH_2Cl_2 (6:94), allows complete deblocking at 25 °C in five minutes; substitution of triisopropylsilane for triethylsilane is less efficient.[93] Compared to other highly acid-labile S-protecting groups, such as those described in Section 2.6.6.2.1.3, *S*-(2,4,6-trimethoxybenzyl) exhibits properties similar to the *S*-(4-methoxytrityl) derivatives. Thus, with appropriate handles, selective *S*-(2,4,6-trimethoxybenzyl) deprotection can be achieved while retaining the peptide on the support for subsequent on-resin steps such as oxidations and alkylations.[106] The *S*-(2,4,6-trimethoxybenzyl) derivatives are also readily cleaved by metal ions and are therefore suitable for oxidative cleavage with thallium(III) trifluoroacetate to form, directly on resin or in solution, the desired intramolecular disulfides; similarly, oxidative cleavage with iodine can be applied for the same purpose and thus for selective disulfide formation in multiple cysteine peptides[106] (for experimental procedures, see Vol. E 22b, Sections 6.1.2.1 and 6.1.2.2).

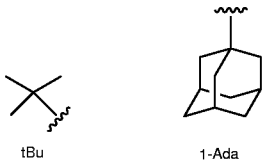
H-Cys(Tmb)-OH Trifluoroacetate:[93]
To a suspension of H-Cys-OH (0.9 g, 7.4 mmol) in CH_2Cl_2 (60 mL) at <5 °C, TFA (12.0 mL, 156 mmol) was added dropwise to dissolve the amino acid completely. A soln of 2,4,6-trimethoxybenzyl alcohol[93,107] (1.5 g, 7.5 mmol) in CH_2Cl_2 (30 mL) was added dropwise over 10 min, the volatiles were removed, and the resultant solid was triturated with Et_2O. The product was washed with anhyd Et_2O; yield: 2.9 g (84%); mp 175–185 °C.
The related Fmoc-Cys(Tmb)-OH was prepared via Fmoc-OSu by standard protocols; yield: 80%; mp 98–101 °C; $[\alpha]_D^{20}$ +6.6 (*c* 1, $CHCl_3$).[93]

for references see p 418

2.6.6.2.1.2 *S*-(*tert*-Alkyl)cysteine Derivatives

The cysteine *S*-(*tert*-butyl) derivative is obtained by reaction of cysteine with isobutene[108] or *tert*-butyl acetate[109] under acid catalysis. Under both of these reaction conditions, the carboxy group is esterified and has to be deprotected by TFA treatment. A more general procedure, applicable for the synthesis of both *S*-(*tert*-butyl) (tBu) and *S*-(1-adamantyl) (1-Ada) derivatives (Scheme 10), is based on S-alkylation of cysteine with the *tert*-alcohols under acidic conditions.[102,110] The related N^{α}-Boc and N^{α}-Fmoc derivatives are obtained by standard procedures.

Scheme 10 The Structures of *S*-(*tert*-Alkyl) Thiol Protecting Groups[102,108]



Both *S*-(*tert*-butyl)- and *S*-(1-adamantyl)cysteine derivatives are fully stable to TFA treatment and thus compatible with the Boc/Bzl and Fmoc/tBu strategies. Moreover, at least the *S*-(1-adamantyl) group is more resistant to sulfoxide formation than other sulfide-type protecting groups.[111] The extent of cleavage by hydrogen fluoride[80,112–114] and TfOH/TFA[115] is strongly dependent on the scavengers used. The *S*-(*tert*-butyl) group is stable to short treatment with hydrogen fluoride/*m*-cresol at 0 °C, but is removed with neat hydrogen fluoride at 20 °C or at 0 °C in the presence of anisole and/or sulfur-containing scavengers such as benzenethiol and thioanisole.[59,80,112–115] Additionally, cleavage of the *S*-(*tert*-butyl) group is achievable with tetrafluoroboric acid[116] and with 1 M trimethylsilyl trifluoromethanesulfonate in TFA,[59] as well as with 1 M bromotrimethylsilane in TFA in the presence of thioanisole/ethanedithiol as scavengers.[117] The *S*-(1-adamantyl) group is also removed with 1 M TfOH in TFA in the presence of thioanisole.[97,118]

Both the *S*-(*tert*-butyl) and *S*-(1-adamantyl) groups are cleaved with mercury(II) acetate in TFA at 20 °C (60 min),[102,119,120] but not with silver(I) trifluoromethanesulfonate in TFA at 0 °C (60 min), where the *S*-(4-methylbenzyl) group is quantitatively removed.[103] Similarly, both S-protecting groups are oxidatively removed by the thallium(III) trifluoroacetate procedure to yield the disulfide directly.[59] Cleavage of *S*-(*tert*-butyl) by trichloro(methyl)-silane/diphenyl sulfoxide in TFA[104] was elegantly exploited for selective cysteine pairings in a synthesis of insulin (see Vol. E 22b, Section 6.1.4.2).[112] The *S*-(*tert*-alkyl) derivatives are stable toward iodine oxidation,[121] a property well suited for selective S-protection schemes as required in site-directed disulfide formation. Similarly, their reactivity toward sulfenyl chlorides can be exploited for their deprotection[26,110,122,123] or concurrent activation of the related cysteine residue as asymmetric disulfide for selective interchain disulfide formation (see Vol. E 22b, Section 6.1.3). The sensitivity toward sulfenyl chlorides limits their use in the Nps/tBu protection scheme for the synthesis of cysteine peptides in solution, unless 2-methylindole is used to capture Nps-Cl in the HCl-mediated N^{α}-deprotection,[25] or alternative Nps-cleavage procedures are applied (see Section 2.1.1.4.1).

H-Cys(tBu)-OH · HCl:[110]
A soln of H-Cys-OH · HCl · H_2O (175.6 g, 1 mol) in 2 M HCl (450 mL) and tBuOH (123 mL, 1.3 mol) was refluxed for 10–12 h. The soln was concentrated, and the resultant solid was filtered off, washed with dry acetone and recrystallized (4 M HCl); yield: 209 g (90%); mp 204 °C; $[\alpha]_D^{20}$ +6.35 (*c* 2, 5 M HCl).

H-Cys(1-Ada)-OH:[102]
A soln of H-Cys-OH (12.1 g, 0.1 mol) and adamantan-1-ol (15.2 g, 0.1 mol) in TFA (120 mL) was stirred at rt for 12 h. The TFA was removed and the oily residue was dissolved in cold H_2O (200 mL) and

neutralized to pH 6.0 with NH_3. The crystalline compound was filtered off, washed with EtOH and Et_2O, and then recrystallized (H_2O); yield: 23.0 g (90%); mp 227–228 °C (dec); $[\alpha]_D^{26}$ −14.5 (*c* 0.51, AcOH). Using this procedure, H-Cys(tBu)-OH was obtained in 30% yield; mp 244–245 °C; $[\alpha]_D^{26}$ −15.4 (*c* 0.57, AcOH).

Removal of *S*-(*tert*-Butyl) with Mercury(II) Acetate; General Procedure:[102,119,120]
The Cys(tBu)-containing peptide was dissolved in cold TFA (10 mL/1 mmol peptide), following which anisole (0.2 mL/1 mmol peptide) and $Hg(OAc)_2$ (1 equiv) were added. The mixture was stirred at 0 °C for 15 min, and then the solvent was removed by evaporation using a constant stream of N_2, while the temperature remained at <25 °C. The resultant peptide was dissolved in H_2O and washed with cold anhyd Et_2O. Next, H_2S was bubbled through the aqueous soln for 15–30 min to precipitate HgS. The filtrate was concentrated and dissolved in H_2O (this procedure was repeated 3 times), giving the desired deprotected peptide.

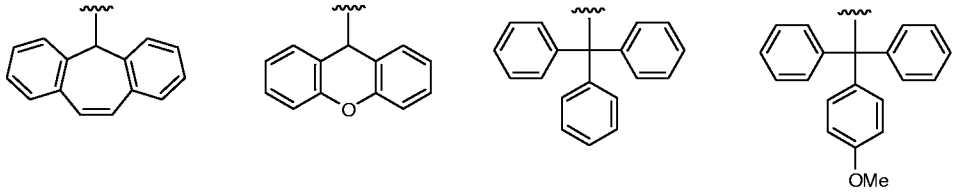
Removal of *S*-(*tert*-Butyl) with Nps-Cl; General Procedure:[110]
The Cys(tBu)-containing peptide was dissolved in AcOH and Nps-Cl (1 equiv) was added. The mixture was stirred at 25 °C for 2 h or until all of the Nps-Cl dissolved. The product was isolated either by filtration, and removal of the solvent, or precipitation by cold Et_2O, depending on the peptide system, to provide the *S*-Nps derivative. Transformation to the free thiol derivatives can be achieved by treatment with either $NaBH_4$, β-ME, or sulfanylacetic acid.[124]

2.6.6.2.1.3 *S*-(*sec*-Aralkyl)- and *S*-(*tert*-Aralkyl)cysteine Derivatives

Attempts to increase the acid sensitivity of aralkyl sulfides by introduction of a second phenyl group on the C^{α}-benzyl moiety, i.e. with *S*-diphenylmethyl[86] and its related di- and tetramethoxy derivatives,[87] or by bridging the phenyl groups such as in 10,11-dihydro-5*H*-dibenzo[*a*,*d*]cyclohepten-5-yl (5*H*-dibenzosuberyl),[125] did not lead to the expected effects. More recently, 5*H*-dibenzo[*a*,*d*]cyclohepten-5-yl (5*H*-dibenzosuberenyl, Dbs)[126] and 9*H*-xanthen-9-yl (Xan)[127] (Scheme 11) have emerged as two highly acid-labile S-protecting groups.

As a *tert*-aralkyl sulfide, *S*-trityl (Trt)[128,129] protection became important in peptide chemistry only upon discovery of its quantitative cleavage with metal ions,[86,130] or by oxidative iodinolysis,[131] as well as upon the development of optimized scavengers capable of preventing retritylation of the thiol group under acidic conditions,[132,133] particularly when trialkylsilanes are included as scavengers.[134,135] Further enhancement of the acid lability of *S*-trityl was obtained, as expected, with the 4-methoxytrityl (Mmt) group (Scheme 11).[2,136]

Scheme 11 The Structures of Highly Acid-Sensitive Thiol Protecting Groups[2,126–129,136]



All of these *S*-cysteine derivatives are stable to moderate basic conditions, and therefore fully compatible with the Fmoc/tBu strategy in which a single acidolytic step removes all protecting groups simultaneously to release the peptide from the resin. By comparing *S*-(2,4,6-trimethoxybenzyl) (see Section 2.6.6.2.1.1.3) with these S-protecting groups, an acid lability rank order of Xan>Mmt~Tmb>Trt is obtained. The mechanism of acidolysis of these protecting groups involves generation of stable carbocations, which normally realkylate the cysteine thiol group unless scavenged by suitable aromatic ethers and sulfides, phenol, aliphatic thiols, aliphatic sulfides and, most efficiently, by trialkylsilanes. Effective

for references see p 418

deprotection/resin-cleavage cocktails are reagent K (TFA/phenol/H_2O/thioanisole/EDT 82.5:5:5:5:2.5),[132] reagent R (TFA/thioanisole/EDT/anisole 90:5:3:2),[133] and reagent B (TFA/phenol/H_2O/iPr_3SH 88:5:5:2).[135,137] Due to the high acid-lability of the *S*-(9*H*-xanthen-9-yl) and *S*-(2,4,6-trimethoxybenzyl) groups these can be cleaved with the same scavenger concentrations, but using a much lower concentration of TFA (1–7%). In this way selective deprotection of the cysteine residues is achieved in regard to the remaining functionalities for additional chemistry to be performed on resin or in solution depending upon the acid lability of the resin linker.[106,138] Despite their pronounced acid sensitivity, these S-protecting groups are fully stable to HOBt, often used as an additive in amino acid coupling steps. Finally, the groups are cleaved orthogonally in solution or on resin by their oxidation with reagents such as iodine or thallium(III) trifluoroacetate to directly generate the disulfides, or by treatment with sulfenyl chlorides to convert them into the appropriate mixed disulfides for regioselective disulfide formation.

2.6.6.2.1.3.1 *S*-(9*H*-Xanthen-9-yl)cysteine

The 9*H*-xanthen-9-yl (Xan) group, as well as the related 2-methoxy derivative, are stable to the usual conditions of N^{α}-Fmoc cleavage and thus are fully compatible with the Fmoc/tBu chemistry.[127] These two cysteine derivatives are obtained directly from cysteine or N^{α}-Fmoc-cysteine by reaction with the corresponding alcohols under acid catalysis. Using thiols or phenol/thioanisole to capture the carbocations, somewhat high concentrations of TFA in CH_2Cl_2 are required for removal of both S-protecting groups. Conversely, with triethylsilane, or preferably triisopropylsilane, in solution quantitative cleavage is achieved with 0.1% TFA in CH_2Cl_2 containing 0.5% silane, whereas on resin at least 1% TFA is required. The silanes drive the reaction to completion by reducing the carbocations irreversibly to the xanthene derivatives and thus preventing realkylation,[127] as demonstrated in the cleavage of *S*-trityl (see Scheme 12).[134] *S*-Xan cleavage in the solid-phase mode requires a higher TFA concentration, i.e. >1% TFA, and longer reaction times (2 h vs 30–60 min in solution); these conditions do not affect the peptide linkage to the acid-labile PAL handle. In the case of more acid-labile resin handles, the use of *S*-(2-methoxy-9*H*-xanthen-9-yl) could be advisable. Both groups are smoothly removed oxidatively by iodine or thallium(III) trifluoroacetate, with concomitant disulfide formation.[127] These properties are well suited for use of the *S*-xanthenyl-type protection in combination with other S-protecting groups for stepwise regioselective disulfide formation in multiple cysteine peptides.[138]

H-Cys(Xan)-OH:[127]
A suspension of H-Cys-OH (0.61 g, 5.0 mmol) in TFA/DME (1:50, 100 mL) was stirred at 25 °C under N_2, until a clear soln was obtained. Next 9*H*-xanthen-9-ol (1.0 g, 5.1 mmol) was added in one portion to this soln. The mixture was stirred at 25 °C for 30 min, neutralized to pH 7 by the addition of sat. aq Na_2CO_3, concentrated and diluted with H_2O (50 mL). The pH was adjusted to 6 with 10% aq citric acid and stirred at 25 °C for 1 h. The resultant white solid was filtered off, and washed with H_2O and EtOAc; yield: 1.33 g (88%); mp 165 °C.

The corresponding 2-methoxy derivative was prepared from 2-methoxy-9*H*-xanthen-9-ol using a similar procedure; yield: 97%; mp 185 °C.

Fmoc-Cys(Xan)-OH:[127]
9*H*-Xanthen-9-ol (1.45 g, 7.3 mmol) was added in 5 portions to a soln of Fmoc-Cys-OH (2.52 g, 7.3 mmol) and TFA (0.56 mL, 7.3 mmol) in CH_2Cl_2 (100 mL). After stirring for 30 min at rt under N_2, the product was precipitated with petroleum ether. The solid was filtered off and washed with CH_2Cl_2/petroleum ether; yield: 3.54 g (90%); mp 70–72 °C.

The corresponding 2-methoxy derivative was prepared from 2-methoxy-9*H*-xanthen-9-ol using a similar procedure; yield: 83%; mp 63–65 °C.

On-Resin Cleavage of Cys(Xan) Under "Low TFA" Conditions; General Procedure:[127,138]
The Cys(Xan)-peptidyl resin was swollen in CH_2Cl_2 (2 mL/100 mg peptidyl resin) at 25 °C for 10 min. Cleavage was accomplished with TFA/Et_3SiH/CH_2Cl_2 (2:1:97, cocktail X) (1 mL/100 mg peptidyl resin) at 25 °C for 2 × 1 h under occasional manual agitation, followed by washing with CH_2Cl_2 (3 × 2 mL/100 mg peptidyl resin).

2.6.6.2.1.3.2 *S*-(5*H*-Dibenzo[*a*,*d*]cyclohepten-5-yl)cysteine

By replacing *S*-(10,11-dihydro-5*H*-dibenzo[*a*,*d*]cycloheptene-5-yl) (5*H*-dibenzosuberyl)[125] with the side-chain conjugated aromatic system 5*H*-dibenzo[*a*,*d*]cyclohepten-5-yl (5*H*-dibenzosuberenyl, Dbs, see Scheme 11) a significantly enhanced acid sensitivity, compared to *S*-trityl, is achieved.[126] The S-protected cysteine derivative is synthesized by the procedure analogous to the related *S*-(5*H*-dibenzosuberyl) derivative,[125] i.e. by reaction of the commercially available 5*H*-dibenzosuberenol with cysteine in the presence of boron trifluoride–diethyl ether complex in acetic acid. The *S*-Dbs group is stable under standard conditions of peptide synthesis using Fmoc/tBu chemistry, as well as to the very weak acid treatment with AcOH/TFE/CH_2Cl_2 as required for a chlorotrityl resin-cleavage step.[139] A comparative analysis of its acid stability/lability relative to *S*-(2,4,6-trimethoxybenzyl), *S*-(9H-xanthen-9-yl), and *S*-(4-methoxytrityl) is not yet available.

2.6.6.2.1.3.3 *S*-Tritylcysteine

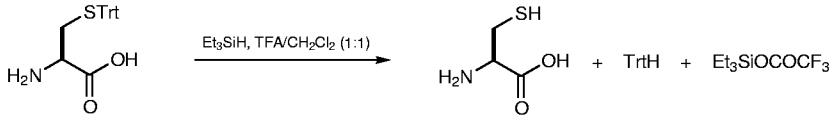
The *S*-trityl group is one of the most widely used for thiol protections of cysteine in synthesis in solution and on solid supports, in the latter case in combination with Fmoc/tBu chemistry. The derivative H-Cys(Trt)-OH is obtained by N,S-tritylation of cysteine,[128,129] followed by selective N^{α}-detritylation by means of hydrochloric acid,[128,129,140] or by direct S-tritylation of cysteine with triphenylmethanol in the presence of acids as catalysts (TosOH[2] or $BF_3 \cdot OEt_2$[141]). *N*,*S*-Ditritylcysteine represents a key intermediate for synthesis in solution, since N^{α}-detritylation is achieved selectively with acetic acid[142] or pyridine hydrochloride,[143] as well as in aqueous trifluoroethanol at apparent acidic pH values which can be adjusted with hydrochloric acid to obtain desired selectivities even versus other highly acid-labile N-protecting groups such as *N*-Bpoc.[144] This property is of decisive importance in complex syntheses, such as that of insulin.[20] Thus, chain elongation of protected intermediate fragments containing Cys(Trt) residues is generally performed along with use of N^{α}-Bpoc protection, which is removed in aqueous acetic acid or in 90% aqueous trifluoroethanol at defined apparent pH values, generally pH 4. Alternatively, N^{α}-Nps protection is used in combination with Cys(Trt); Nps removal occurs with 1–2 equivalents of hydrochloric acid in alcohols or dimethylformamide, and in the presence of 2-methylindole to capture the Nps-Cl intermediate.[25]

S-Detritylation by acids is an equilibrium reaction,[89,145] therefore the reaction needs to be driven to completion by capture of the formed carbocation. Capture is efficiently achieved by addition of water and/or excess thiols;[2] however, for quantitative removal, the best strategy is addition of silanes to TFA,[134] since the trityl cation is reduced to the inert triphenylmethane and thus the equilibrium is shifted to the free thiol (Scheme 12). Triethylsilane is more efficient, but due to its reducing power, partial reduction of the tryptophan indole function may occur under the reaction conditions. Therefore, when tryptophan is present, the use of triisopropylsilane is recommended. In standard Fmoc/tBu syntheses on solid support with Cys(Trt) residues, resin cleavage/deprotection is generally performed with TFA/H_2O/triisopropylsilane (95:5:1.5) or TFA/CH_2Cl_2/triisopropylsilane (95:5:1.5). *S*-Trityl groups are also readily removed by oxidative cleavage with iodine;[131] this reaction has an extraordinary solvent dependency which allows selective conversion of *S*-trityl into disulfides even in the presence of other S-protecting groups, such as *S*-(acetamidomethyl) (see Vol.

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E 22b, Section 6.1.1.2.1).[146] For stepwise regioselective disulfide formation, the iodine-mediated oxidative deprotection of a pair of Cys(Trt) residues is a widely applied procedure (see Vol. E 22b, Sections 6.1.2–6.1.4). Finally, *S*-trityl groups are efficiently cleaved with metal ions such as mercury(II) and silver(I),[86,89,145] or with thallium(III) trifluoroacetate.[59]

Scheme 12 Acidolytic S-Detritylation in the Presence of a Trialkylsilane[134]



In the solid-phase synthesis of peptides with C-terminal Cys(Trt) residues attached via ester linkages to the resin, the Fmoc/tBu strategy with repetitive base treatments leads to epimerization, which can be reduced by the use of DBU and shortened Fmoc-cleavage cycles, but is not fully suppressed.[73] Concomitant β-elimination and formation of β-(1-piperidyl)alanine is observed, although to a lesser extent than with the *S*-(acetamidomethyl)[74] or *S*-(*tert*-butylsulfanyl) group.[44]

Trt-Cys(Trt)-OH · DEA:[129]
To a soln of H-Cys-OH · HCl (25 g, 159 mmol) in H_2O (480 mL) under N_2, DEA (80 mL, 770 mmol) in Et_2O (480 mL) was added at 0 °C, followed by Trt-Cl (120 g, 430 mmol) at −5 °C. After 3 h, the mixture was extracted with $CHCl_3$. The organic layers were washed with H_2O, dried (Na_2SO_4), and the volatiles were removed. The residue was dissolved in EtOH, DEA (1 mL) was added, and the product was precipitated with Et_2O; yield: 76 g (70%); $[\alpha]_D^{20}$ +71 (*c* 2, $CHCl_3$).

Trt-Cys(Trt)-OH:[147]
Trt-Cys(Trt)-OH · DEA (54 g, 79 mmol) was distributed between EtOAc and 10% aq citric acid; the organic layer was washed with H_2O, dried, and the volatiles were removed. The residue was crystallized (benzene/petroleum ether); yield: 45 g (94%); mp 125–128 °C (dec); $[\alpha]_D^{20}$ +70 (*c* 1, MeOH).

H-Cys(Trt)-OH:[2,89,141]
Method A:[2] A soln of H-Cys-OH · HOTos (14.7 g, 50 mmol) and Trt-OH (13 g, 50 mmol) was stirred in AcOH (60 mL) at 45–50 °C for 4 h. The AcOH was removed and the residue was triturated with Et_2O. The residue was dissolved in H_2O and an aq soln of NaOAc was added to precipitate the compound. The precipitate was filtered off, washed with H_2O, acetone, and Et_2O, and recrystallized (AcOH/Et_2O); yield: 13.6 g (75%); mp 179–180 °C; $[\alpha]_D^{27}$ −15.7 (*c* 2, 0.1 M NaOH).

Method B:[89] An equimolar soln of H-Cys-OH and Trt-OH was allowed to stand in TFA (1.8 mL/1 mmol amino acid) at 25 °C for 15 min, and the TFA was removed. Anhyd Et_2O was added to the resultant solid, and the product was dissolved in H_2O and neutralized by the addition of 10% aq NaOAc. The precipitate was collected and washed with H_2O; yield: 85–90%.

Method C:[141] To a soln of H-Cys-OH (102.5 g, 0.847 mol) and Trt-OH (220 g, 0.845 mol) in AcOH (850 mL), $BF_3 \cdot OEt_2$ (119 mL, 0.935 mol) was added, and the mixture was stirred at 40–50 °C for 30 min. After an additional 45 min at rt, the soln was poured into EtOH (1.3 L), diluted with H_2O, and neutralized with NaOAc (254 g, 3.1 mol). Upon addition of H_2O (3.3 L), the precipitate was collected, and washed with H_2O, acetone, and Et_2O; yield: 289 g (93%).

N^{α}-Detritylation of Cys(Trt)-Containing Peptides; Typical Procedure:[148]
To a soln of Trt-Cys(Trt)-Cys(Acm)-Thr(tBu)-Ser(tBu)-Ile-Cys(Trt)-Ser(tBu)-Leu-OH (27.0 g, 15 mmol) in AcOH (192 mL), was added H_2O (48 mL). After 1 h at 20 °C the soln was concentrated to ca. 100 mL, and lyophilized. The residue was triturated with Et_2O/petroleum ether (1:1) and crystallized ($CHCl_3$/MeOH); yield: 16.3 g (70%); mp 207–210 °C.

***N*$^{\alpha}$-Bpoc Cleavage of Cys(Trt)-Containing Peptides; Typical Procedure:**[148]
A soln of Bpoc-Ile-Cys(Trt)-Ser(tBu)-Leu-OH (2.91 g, 3.0 mmol) in AcOH/H_2O (8:2) was kept for 1 h at 50 °C. The mixture was diluted with AcOH (100 mL), lyophilized, and triturated with Et_2O. The residue was crystallized (MeOH/H_2O); yield: 1.62 g (70%); mp 191–194 °C; $[\alpha]_D^{20}$ −71 (*c* 1, $CHCl_3$).

N^{α}-Desulfenylation of Cys(Trt)-Containing Peptides; Typical Procedure:[149]
To an ice-cold soln of Nps-Val-Cys(Trt)-Gly-Glu(tBu)-OH (22.9 g, 26.7 mmol) in DMF (200 mL), 0.55 M HCl in dioxane (97.1 mL, 53.4 mmol) was added dropwise over 30 min. The mixture was kept for 1 h at rt and then concentrated. Upon addition of iPr_2O, the oil was reprecipitated from EtOAc with Et_2O/iPr_2O. The solid hydrochloride was dissolved in MeOH (400 mL) and H_2O (20 mL), and neutralized with 1 M NaOH (1 equiv). The precipitate was collected and washed with MeOH; yield: 18.8 g (82%); mp 205–208 °C (dec); $[\alpha]_D^{20}$ −14.2 (*c* 1, DMF).

Acidolytic Deprotection and Cleavage from Peptidyl Resin of Cys(Trt)-Containing Peptides under "High TFA" Conditions; General Procedure:[36,132,135,137]
The peptidyl resin was swollen in CH_2Cl_2 (2 mL/100 mg peptidyl resin) at 25 °C for 10 min. Cleavage was accomplished with freshly prepared TFA/phenol/H_2O/thioanisole/EDT (82.5:5:5:5:2.5, reagent K),[132] TFA/phenol/H_2O/iPr_3SiH (88:5:5:2, reagent B)[135,137] (1 mL/100 mg peptidyl resin), or TFA/H_2O/iPr_3SiH (95:2.5:2.5)[36] at 25 °C for 90 min, under occasional manual agitation. The cleavage mixture was filtered, and the resin was washed with neat TFA (3 × 2 mL/100 mg original peptidyl resin). The washes were combined with the original filtrate, TFA was evaporated under a constant stream of N_2, and the deprotected peptide was precipitated with cold anhyd Et_2O (15 mL/100 mg original peptidyl resin).

2.6.6.2.1.3.4 *S*-(4-Methoxytrityl)cysteine

The *S*-(4-methoxytrityl)cysteine derivative[2,136] is stable to weak bases such as piperidine used for Fmoc cleavage, and to very weak acids such as AcOH/TFE/CH_2Cl_2 (1:2:7) used to cleave fully protected peptides from the chlorotrityl resin linker.[139] *S*-Mmt is quantitatively cleaved upon exposure at 22 °C for 30 minutes to 1.0% TFA in CH_2Cl_2/EDT (95:5) or 0.5–1% TFA in CH_2Cl_2/triethylsilane (95:5), i.e. under acidic conditions that essentially do not affect acid-labile protecting groups of other functionalities used in the Fmoc/tBu chemistry. This combination of properties allows the cleavage of peptides that are fully protected, even at the thiol functions, from the chlorotrityl resin, or to achieve selective deprotection at the thiol functions on resin (with a more acid-stable linker) or in solution. Addition of iodine to the cleavage solution leads to concurrent oxidative cleavage of *S*-(4-methoxytrityl) groups with direct formation of the disulfide.[150] In contrast to the fast reduction of tryptophan residues observed with triethylsilane (5 equiv), and to lesser extents with triisopropylsilane (5 equiv) in 90% TFA/CH_2Cl_2,[134] exposure of Trp and Trp-containing peptides to cleavage cocktails with low TFA content does not affect the indole moiety. The mild acid cleavage of the *S*-(4-methoxytrityl) groups on resin can be exploited along with further combinations of S-protecting groups in the stepwise regioselective formation of multiple disulfide bonds.[151]

Fmoc-Cys(Mmt)-OH:[136]
To a suspension of H-Cys-OH · HCl · H_2O (175 g, 1 mol) in DMF (1 L) was added Mmt-Cl (308 g, 1 mol) and the mixture was stirred at rt for 3 h, and then concentrated. The mixture was diluted with H_2O, and the pH was adjusted to 5 with 4 M NaOH. The precipitate was filtered off, and washed with H_2O and Et_2O; yield: 291 g (74%). The crude H-Cys(Mmt)-OH was dissolved in 10% aq Na_2CO_3 (2 L) and dioxane (2 L). A soln of Fmoc-OSu (253 g, 0.75 mol) in dioxane (1 L) was added dropwise and after 2 h at rt, the bulk of the dioxane was removed and the aqueous soln was extracted with EtOAc. The aqueous phase was acidified and extracted with EtOAc, and the organic layer was washed with H_2O and brine, dried (Na_2SO_4), and the volatiles were removed. The residue was crystallized (EtOAc); yield: 314 g (69%); mp 124–126 °C; $[\alpha]_D^{22}$ +16.7 (*c* 1, DMF).

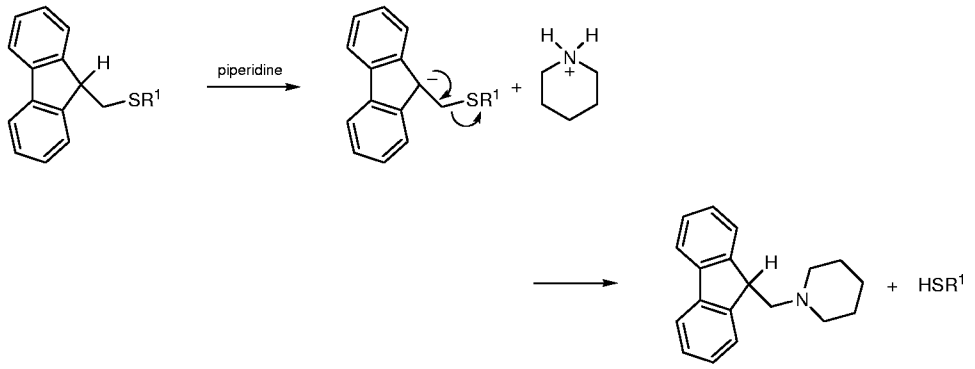
2.6.6.2.1.4 *S*-(9-Fluorenylmethyl)cysteine

The strong leaving group character of the 9-fluorenylmethyl (Fm) group, which derives from the aromaticity effect of a base-induced proton abstraction to form the dibenzocyclopentadienyl anion,[152,153] is exploited in the *S*-(9-fluorenylmethyl) group[154] to obtain a cysteine derivative fully stable even to hydrogen fluoride,[155] but cleavable under relatively weak basic conditions such as piperidine (10–50%) or DBU (2%) in DMF (Scheme 13).[154,155] This S-protection is thus compatible with the Boc/Bzl chemistry and fully or-

for references see p 418

thogonal versus the *S*-(4-methylbenzyl) and *S*-(acetamidomethyl) groups generally applied in this type of synthetic strategy. The *S*-(9-fluorenylmethyl)-protected cysteine is conveniently prepared by reaction of cysteine with fluorenylmethyl *p*-toluenesulfonate,[156] the preferred reagent since the corresponding chloride is available in relatively low yields.[153] *S*-(9-Fluorenylmethyl) can be removed while the peptide chain remains anchored to the support to directly produce on-resin disulfides,[157,158] or after resin cleavage and purification of the *S*-Fm-protected intermediate. Free thiol functions are obtained when removal of the S-protecting group is carried out in the presence of reducing thiols; in the absence of reducing agents the disulfides are obtained directly via air oxidation (for experimental procedures, see Vol. E 22b, Section 6.1.1.3.1).[155,159] The high resistance of *S*-(9-fluorenylmethyl) to strong acids, as well as to iodine- or thallium(III)-mediated oxidative cleavage of other S-protecting groups, and the quantitative removal of *S*-(9-fluorenylmethyl) by base can be exploited for stepwise interchain and intrachain disulfide formation in multiple-cysteine-containing peptides (see Vol. E 22b, Section 6.1.2.1.1).

Scheme 13 Base-Mediated S-Deprotection of Cys(Fm) Residues



The *S*-[2-(2,4-dinitrophenyl)ethyl] group represents an additional base-labile alternative to *S*-(9-fluorenylmethyl) that has some utility, but has no obvious advantages.[160,161]

9-Fluorenylmethyl 4-Toluenesulfonate (Fm-OTos):[156]
To an ice-cold soln of 9-fluorenemethanol (19.6 g, 0.1 mol) in $CHCl_3$ (100 mL) was added Tos-Cl (19.6 g, 0.1 mol) in dry pyridine (16.1 mL, 0.2 mol). After stirring for 2 h, the soln was washed with 10% aq $NaHCO_3$ (2 × 25 mL) and brine (2 × 25 mL), and dried (Na_2SO_4). The solvent was removed and the product recrystallized ($CHCl_3$/hexane); yield: 28.9 g (83%); mp 115 °C.

H-Cys(Fm)-OH:[156]
To a soln of H-Cys-OH · HCl (5.3 g, 34 mmol) and Fm-OTos (15 g, 43 mmol) in DMF (150 mL), was added at 0 °C DIPEA (17 mL, 102 mmol) in portions. The resulting suspension was stirred at rt for 16 h and then EtOAc (150 mL) was added. The precipitate was filtered off, washed with EtOAc and recrystallized (1 M HCl); yield: 8.1 g (71%); mp 210 °C.

The related Boc-Cys(Fm)-OH derivative was obtained by standard procedures using Boc_2O; yield: 74%; mp 74–75 °C; $[\alpha]_D^{20}$ −10.1 (*c* 1, DMF).[158]

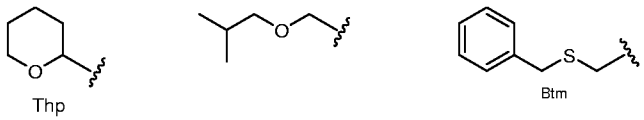
[H-Gly-Cys-Phe-Val-Pro-Cys(Acm)-Gly-OH]$_2$; Typical Procedure for *S*-(9-Fluorenylmethyl) Deprotection:[162]
A soln of H-Gly-Cys(Fm)-Phe-Val-Pro-Cys(Acm)-Gly-OH (279 mg, 0.3 mmol) in piperidine/DMF (1:1, 15 mL) was stirred at rt for 2 h. After removal of the solvents, the residue was washed extensively with CH_2Cl_2; yield: 180 mg (82%).

On-Resin *S*-(9-Fluorenylmethyl) Cleavage; General Procedure:[157]
The peptidyl resin (50–60 mg, 0.35 mmol/g, ~19 μmol) was treated with freshly prepared piperidine/DMF/β-ME (10:10:0.7, 4 mL/100 mg resin) at 25 °C for 3 h under argon. The resin was then washed with CH_2Cl_2 (3 × 2 mL), DMF (3 × 2 mL) and CH_2Cl_2 (5 × 2 mL). This was followed by additional on-resin transformations, including oxidation of the free thiols to form a peptide disulfide. Finally, the peptide was removed from the resin using conditions appropriate for the peptide synthesis chemistry used.

2.6.6.2.2 *S,O*- and *S,S*-Acetal Derivatives of Cysteine

A number of mono- and dithioacetals, including *S*-(tetrahydropyranyl), (ThP)[163] *S*-(isobutyloxymethyl),[164,165] and *S*-(benzylsulfanylmethyl) (Btm)[164] derivatives of cysteine (Scheme 14), have been proposed and studied (for reviews, see refs[1,5]), but are now superseded by thiol protecting groups with more advantageous lability/stability characteristics. The *S*-(tetrahydropyranyl) derivative is produced by acid-catalyzed addition of the β-thiol of cysteine methyl ester to the double bond of 3,4-dihydro-2*H*-pyran; the *S*-(isobutyloxymethyl) and *S*-(benzylsulfanylmethyl) derivatives are prepared by base-catalyzed thiol alkylations involving the corresponding chlorides. The benzylsulfanylmethyl group can be removed by metal ions, e.g. mercury(II), whereas the base-stable tetrahydropyranyl and isobutyloxymethyl groups are removed by acidolysis, e.g. HBr in TFA[166] (this means that *N*-Trt and *N*-Boc groups can be cleaved selectively while the S-protecting groups are stable), as well as by heavy metal ions and electrophiles such as thiocyanogen.

Scheme 14 Examples of Acetal-type Thiol Protection[163–165]



2.6.6.2.3 Cyclic *S,N*-Acetal Derivatives of Cysteine

Thiazolidine-4-carboxylic acids (Thz), i.e. cyclic *S,N*-acetals, formed by reaction of cysteine with aldehydes or ketones, have been proposed since the early days of peptide synthesis for cysteine protection.[167,168] These types of proline-like cysteine derivatives favorably disrupt ordered secondary structures of synthesized peptides with a concomitant increase in their solvation and solubility[169,170] and, furthermore, as proline analogues exhibit high resistance to epimerization compared to other S-protected cysteine derivatives.[171–173] Unsubstituted Thz is resistant to acids, while the 2-monosubstituted and disubstituted Thz derivatives are cleaved by moderate acid treatment, e.g. TFA/H_2O (49:1). All Thz derivatives are ring-opened with mercury(II) in acetic acid.[168,172,173] Ring opening can also be accomplished with sulfenyl halides such as SMoc-Cl[171,174] or SPy-Cl;[172] however, reaction with Nps-Cl under acidic conditions produces, as side product, *N*-(hydroxymethyl)-*S*-(2-nitrophenylsulfanyl)-cysteine derivatives.[175] The resulting mixed disulfides are readily converted into cysteine residues by reduction with thiols or phosphines.[174] Iodine-mediated oxidation of Thz residues, as advantageously applied for the linear *S,N*-acetals, i.e. *S*-(acylaminoalkyl) derivatives (see Section 2.6.6.2.4), does not yield the expected cystine peptides.[172]

The synthesis of these cyclic *S,N*-acetals, their application as solubilizing building blocks, and their deprotection is described in Section 2.3.2.4.

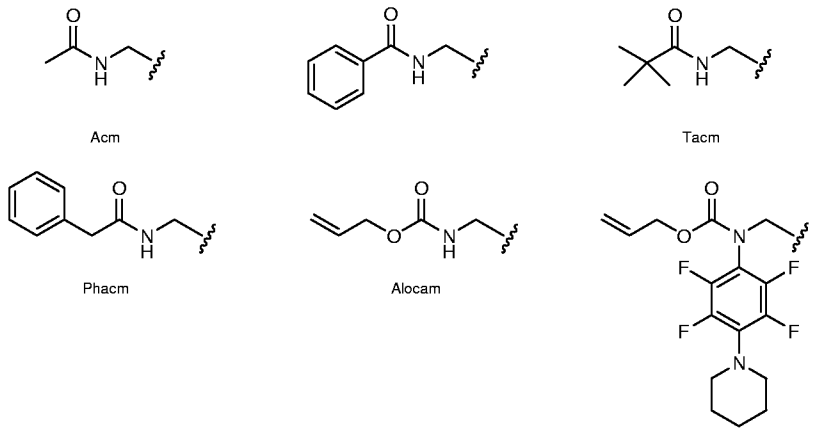
for references see p 418

2.6.6.2.4 *S*-(Acylaminoalkyl)cysteine Derivatives

Protection of the cysteine thiol function via formation of acylaminoalkyl hemithioacetals was first introduced by Weygand et al.[176] with the *S*-[1-(benzoylamino)-2,2,2-trifluoroethyl]- and *S*-[1-(benzyloxycarbonylamino)-2,2,2-trifluoroethyl]cysteine derivatives. The real breakthrough, however, was obtained by Veber and et al.[177,178] with the acetamidomethyl (Acm) group; it is one of the most valuable protecting groups for essentially all synthetic strategies in solution and on solid supports because of its unique stability and the various alternatives for its removal. In fact, it survives the conditions for acidolytic or base-promoted N^{α}-amino protection, as well as the acidic conditions for removal of most of the S-protecting groups discussed in the previous sections.

Additional members of this family of S-protecting groups have been proposed, as shown in Scheme 15, i.e. the analogous benzamidomethyl,[179,180] trimethylacetamidomethyl (Tacm),[60,181] phenylacetamidomethyl (Phacm),[182–184] and more recently the allyloxycarbonylaminomethyl (Alocam),[185,186] *N*-(allyloxycarbonyl)-*N*-(2,3,5,6-tetrafluoro-4-piperidinophenyl)-aminomethyl,[187] and *N*-(allyloxycarbonyl)-*N*-[2,3,5,6-tetrafluoro-4-(phenylsulfanyl)phenyl]aminomethyl[299] groups. The allyl-based S-protecting groups can be removed by palladium-catalyzed allyl cleavage in the presence of nucleophiles. Particularly, the last group is apparently easy to handle, fully stable to basic and acidic conditions, and thus perfectly compatible with both the Boc/Bzl and Fmoc/tBu chemistries. It is readily removed with $PdCl_2(PPh_3)_2/Bu_3SnH$, as well as with $Pd(PPh_3)_4$ in the presence of $PhSiH_3$ or *N*,*N*′-dimethylbarbituric acid.[299]

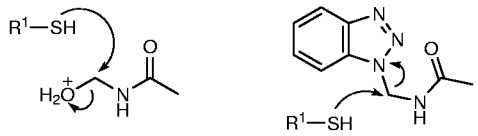
Scheme 15 The Structures of Acylaminoalkyl Thiol Protecting Groups[60,177–187]



2.6.6.2.4.1 *S*-(Acetamidomethyl)cysteine

The *S*-(acetamidomethyl)cysteine derivative is prepared by straightforward acid-catalyzed reaction of cysteine with *N*-(hydroxymethyl)acetamide (Scheme 16),[177] which in turn is accessed from acetamide and formaldehyde.[188,189] The reaction is advantageously performed in aqueous hydrochloric acid,[177] TFA,[190] or TfOH/TFA (1:19).[189] The thiazolidine-2-carboxylic acid formed as a side product is removed by recrystallization of H-Cys(Acm)-OH as its hydrochloride, whereas unreacted cysteine, when still present, is removed by recrystallization of the free base obtained by addition of silver(I) oxide,[177,178] pyridine,[191] triethylamine,[192] or ion-exchange resins.[193] Conversion into the N^{α}-Boc[177,193] and N^{α}-Fmoc[189] derivatives is achieved using standard protocols.

Scheme 16 Synthesis of *S*-Acetamidomethyl Derivatives[177]



A milder method for the preparation of H-Cys(Acm)-OH is based on benzotriazole as a synthetic auxiliary.[194,195] The acetamidomethylating agent is prepared from benzotriazole, formaldehyde, and acetamide, and is then reacted with cysteine (Scheme 16) in aqueous dioxane at 50 °C for three to five hours in the presence of sodium hydroxide; the desired product is obtained free of thiazolidine-2-carboxylic acid.

There are contradictory reports about the stability of the *S*-(acetamidomethyl) group upon exposure to strong acids. Partial cleavage has been reported after treatment with hydrogen fluoride[196–199] and even with TFA.[200,201] The type of scavenger used, e.g. anisole or thioanisole as additive to hydrogen fluoride,[196,199] or thiols added to TFA,[201] apparently plays an important role; for example, use of triisopropylsilane as a scavenger prevents partial cleavage of *S*-(acetamidomethyl) by TFA.[201] The observed acidolytic cleavage could well derive from partial oxidation of Cys(Acm) residues to the related sulfoxide derivatives during the peptide chain elongation steps, since exposure of S-protected cysteine sulfoxide derivatives to strong acids in the presence of anisole and thiols leads to new cysteine derivatives and asymmetric disulfides, respectively.[58] Nevertheless, the many successful syntheses of multiple cysteine peptides in solution and on solid support confirm that, under optimized conditions, the *S*-(acetamidomethyl) group is resistant to acid- or base-promoted N^{α}-deprotections, as well as to the final TFA- or HF-mediated resin-cleavage and/or deprotection steps. *S*-(Acetamidomethyl) is therefore compatible with Boc/Bzl[95,196] and Fmoc/tBu chemistry.[120,199] In the case of multiple cysteine peptides it offers the advantage of producing intermediates deprotected at all functionalities, except the cysteine thiol functions, thus allowing for purification prior to the Cys(Acm) deprotection and subsequent oxidative refolding step (see Vol. E 22b, Section 6.1.5). Moreover, the *S*-(acetamidomethyl) group is stable under conditions of reductive cleavage of *S*-(alkylsulfanyl) groups, base-mediated cleavage of *S*-(9-fluorenylmethyl) derivatives, and acidic removal of most of the *S*-alkyl- and *S*-aryl-type protections; thus, it is also ideally suited for combinations with other thiol protecting groups for selective intrachain and interchain disulfide formation in multiple cysteine peptides (see Vol. E 22b, Sections 6.1.2–6.1.4).

In addition to the problems of racemization of S-protected cysteine residues during both their anchoring to hydroxymethyl-type resins and formation of peptide bonds (see Section 2.6.6.2), the synthesis of C-terminal Cys(Acm)-containing peptides using Fmoc/tBu chemistry has been observed to produce peptides with C-terminal β-(1-piperidyl)alanine (see Section 2.6.6.2).[74] This type of side reaction can be reduced by minimal exposure to piperidine, but it can not be suppressed completely.

The *S*-(acetamidomethyl) group is not necessarily compatible with Nps/tBu chemistry in solution, since it is cleaved by Nps-Cl, formed as an intermediate in the N^{α}-deprotection steps with hydrochloric acid.[26] To bypass this problem, removal of the Nps group can be achieved with alternative reagents such as thiols[26] or phosphines,[24] or with hydrochloric acid, in this case operating in the presence of 2-methylindole to capture the Nps-Cl.[25]

Cleavage of the *S*-(acetamidomethyl) group is achieved with metal ions such as mercury(II) in aqueous acetic acid[95,178] or silver(I) in TFA.[103] In the latter case, the trifluoromethanesulfonate salt was found to be superior to the trifluoroacetate, and silver tetrafluoroborate is an efficient alternative.[202] While sensitive residues such as Met, Trp, and Tyr are largely unaffected by cleavage with silver(I) salts,[103] mercury(II)-mediated cleavage has been reported to give side reactions at Trp residues,[196,203] which can be suppressed by operating in the

for references see p 418

presence of excess free tryptophan as a scavenger.[203] For the subsequent oxidation steps to the desired cystine peptides, the thiolates are converted into the free thiols by treatment with excess thiols such as dithiothreitol or β-sulfanylethanol. For silver thiolates, oxidation can be performed directly with dimethyl sulfoxide/1 M hydrochloric acid.[204,205] When the mercury(II)/β-sulfanylethanol deprotection procedure was used, side products were detected in which Trp residues were converted into 2- and 7-(hydroxyethylsulfanyl)tryptophan or even to the 2,7-bis(hydroxyethylsulfanyl)tryptophan derivatives.[206] This side reaction is avoided completely when 1.1 equivalents of mercury(II) per *S*-(acetamidomethyl) group are used, and the concentration of acetic acid is increased to 50%. Mercury(II)-mediated *S*-(acetamidomethyl) cleavage can be successfully achieved, even on resin.[161,207,208]

Cleavage of *S*-(acetamidomethyl) groups is also possible with sulfenyl halides, e.g. Nps-Cl,[26,123] SMoc-Cl,[209] SPy-Cl,[210] or Npys-Cl[211] in acidic solutions such as acetic acid or acetic acid/TFA mixtures in order to prevent sulfenamide formation at free amino groups (for related experimental procedures, see Vol. E 22b, Sections 6.1.3.1 and 6.1.3.2). Although the resulting asymmetric disulfides are readily reduced to the free thiols with thiols or phosphines, this type of deprotection/conversion into activated disulfides is used mainly for site-directed disulfide formation with a second cysteine residue (see Vol. E 22b, Sections 6.1.3.1 and 6.1.3.2). This method, however, cannot be used in the presence of Trp residues, due to the high reactivity of the sulfenyl halides with the indole moiety.[212]

In addition, *S*-(acetamidomethyl) groups are oxidatively cleaved by iodine[213] or by thallium(III) trifluoroacetate[59,97,214] to directly form the corresponding cystine peptides (for experimental procedures in solution and on solid support, see Vol. E 22b, Sections 6.1.1.2.1 and 6.1.1.2.2). For oxidative cleavage with iodine, solvent effects must be taken into consideration in order to minimize iodination at sensitive amino acid residues, such as Tyr, His, Met, and Trp.[146] Moreover, the reactive sulfenyl iodide formed as an intermediate can react with the indole moiety of Trp residues, particularly if the ring size of the resulting sulfide compound favors such cyclization.[215] Since thallium(III) trifluoroacetate acts first as a soft acid to cleave the *S*-(acetamidomethyl) group, then as a mild oxidant to form the disulfide bond, sensitive amino acids such as Met and Trp must be protected; Met(O) and Trp(Mts) derivatives, respectively, have been recommended.[59]

It is important to note that cleavage of the *S*-(acetamidomethyl) group with metal ions, such as thallium(III) trifluoroacetate, or with iodine may lead to alkylation of hydroxy groups,[201,216,217] as well as of amide groups,[218,219] via reactive acetamidomethanol intermediates. These side reactions are minimized by the use of glycerol as a scavenger.

H-Cys(Acm)-OH · HCl:[178]
To an ice-cold soln of H-Cys-OH · HCl · H_2O (228 g, 1.3 mol) in H_2O (350 mL) and concd HCl (50 mL) was added commercial *N*-(hydroxymethyl)acetamide (127 g, 1.43 mol). The mixture was kept at rt for 48 h and then taken to dryness by coevaporation with EtOH. The product was crystallized (MeOH/Et_2O); yield: 158 g (52%); mp 159–163 °C (dec); $[\alpha]_D^{25}$ −30.7 (*c* 1, H_2O).

H-Cys(Acm)-OH · H_2O:[192]
To a soln of H-Cys(Acm)-OH · HCl (100 g, 0.438 mol) in H_2O (125 mL), was added TEA (60.1 mL, 0.438 mol) in EtOH (1.25 L). After 2 h at 0 °C, the product was filtered off and washed extensively with EtOH; yield: 75.5 g (81%); mp 194 °C (dec); $[\alpha]_D^{20}$ −41.9 (*c* 1, H_2O).

Fmoc-Cys(Acm)-OH:[189]
To an ice-cold soln of H-Cys-OH · HCl (3.8 g, 24 mmol) and *N*-(hydroxymethyl)acetamide (3.0 g, 29 mmol) in H_2O (5.6 mL), was added a mixture of TfOH/TFA (1:19, 37 mL). After stirring for 90 min under argon, the solvent was removed and the residue was reconcentrated from Et_2O (5 × 20 mL). The resulting oil was dissolved in 10% aq Na_2CO_3 (50 mL), and the soln was adjusted to pH 10 with additional 10% aq Na_2CO_3 and reacted with Fmoc-N_3 (5.8 g, 21 mmol) in dioxane (62 mL). The reaction was kept in an ice bath for 2 h and at rt for 48 h. The mixture was diluted with H_2O and extracted with Et_2O. The aqueous phase was acidified and extracted with EtOAc, and the organic layer was dried ($MgSO_4$) and the volatiles were removed. The product was crystallized (CH_2Cl_2/hexane); yield: 5.35 g (62%); mp 147–148 °C; $[\alpha]_D$ −44 (*c* 1, DMF).

Cleavage of the *S*-(Acetamidomethyl) Group with Mercury(II) Acetate; Typical Procedure:[220]
To a soln of the hexa(*S*-Acm) derivative of the 60-mer dendrotoxin-I (155 mg, 20.5 μmol) in 50% AcOH (20 mL), was added $Hg(OAc)_2$ (43 mg, 135.2 μmol), and the mixture was stirred for 2 h at rt. β-ME (0.32 mL, 4.6 mmol) was added and the soln was stirred for 2 h at rt. The soln was applied to a Sephadex G-25 column and eluted with 1 M AcOH. The principal fraction was collected and lyophilized and then treated with DTT (22 mg, 0.14 mmol) in 0.1 M NH_4OAc (pH 7.8). Further purification was achieved by RP-HPLC using a linear gradient (20–40% MeCN in 0.1% TFA/H_2O for 60 min); yield of the hexathiol peptide: 115 mg (78%).

Cleavage of the *S*-(Acetamidomethyl) Group with Silver(I) Trifluoromethanesulfonate; Typical Procedure:[221]
To an ice-cold soln of the mono(*S*-Acm) derivative of a 30-mer fragment of laminin (30 mg, 8.73 μmol) in TFA (1.0 mL), were added AgOTf (112 mg, 437 μmol) and anisole (100 μL). After 1 h, Et_2O was added and the precipitate was collected by centrifugation. The precipitate was dissolved in 0.1 M Tris · HCl (pH 8.5, 10 mL) containing 6 M Gn-Cl, and then DTT (269 mg, 200 equiv) was added and the soln was kept at rt for 12 h. The silver salt was removed by centrifugation and the soln was applied to a Sephadex G-10 column and eluted with 1 M AcOH. The principal fraction was collected and lyophilized; yield: 24 mg (82%).

On-Resin Removal of *S*-(Acetamidomethyl) with Mercury(II) Acetate; General Procedure:[161,208]
The *S*-Acm-containing peptidyl resin (1 equiv) was swelled in CH_2Cl_2 (5 min) and DMF (5 min), and a 0.06 M soln of $Hg(OAc)_2$ in DMF was added. After stirring for 3 h at rt, the peptidyl resin was washed with DMF (3 × 2 min) and β-ME/DMF (1:9, 3 × 2 min) to remove the mercury(II) salts.

2.6.6.2.4.2 *S*-(Trimethylacetamidomethyl)cysteine

Various alternatives to the *S*-(acetamidomethyl) group (see Section 2.6.6.2.4.1) have been proposed to avoid the formation of thiazolidine-2-carboxylic acid as a byproduct. Among these, the *S*-(benzamidomethyl) group[179,180] is significantly less stable than the *S*-(acetamidomethyl) group to strong acid treatment (Table 2) and thus is not compatible with the Boc/Bzl strategy.[60] In contrast to the initial report,[180] *S*-(benzamidomethyl) also appears to undergo significant decomposition under alkaline conditions, such as 0.05 M sodium hydroxide, or upon exposure to hydrazine hydrate (Table 2).[60] Similarly, the *S*-(phthalimidomethyl) derivative is not a suitable S-protecting group, since ring opening with hydrazine hydrate is required prior to its removal with mercury(II),[222] and hydrazine hydrate is known to provoke, in sequence dependency, various side reactions (see Section 3.1.1). The trimethylacetamidomethyl (Tacm) group (see Scheme 15), which is readily prepared without side reactions from *N*-(hydroxymethyl)trimethylacetamide and cysteine in TFA,[60,181] appears to be highly promising. *S*-(Trimethylacetamidomethyl) offers the advantage of greater stability to strong acids, and of a higher resistance to air oxidation, to give the related sulfoxide, than the *S*-(acetamidomethyl) group.[60] In this context, it is noteworthy that the stabilities determined for the S-protected cysteine compounds in model experiments may not necessarily reflect the stability of these derivatives once incorporated into peptides. The higher resistance to air oxidation of the *S*-(trimethylacetamidomethyl) group is particularly important, since mercury(II) or iodine fail to cleave Cys(Acm) in the sulfoxide form.[54] The *S*-(trimethylacetamidomethyl) group is removed with mercury(II) or iodine, as reported for the *S*-(acetamidomethyl) group,[60,181] but is also very efficiently removed with silver tetrafluoroborate in TFA.[202,223]

Table 2 Recovery of S-Protected Cysteine Derivatives (%) after Hydrogen Fluoride or Alkaline Treatment[60]

Cys Derivative	HF (0 °C, 1 h)	HF (rt, 1 h)	0.05 M NaOH (0 °C, 1 h)	H_2NNH_2 (10 equiv)	Ref
H-Cys(Acm)-OH	92.0	85.1	95.8	95.2	[60]
H-Cys(Bam)-OH[a]	50.8	26.5	56.4	38.0	[60]
H-Cys(Tacm)-OH	98.1	86.8	96.7	95.7	[60]

[a] Bam = benzamidomethyl.

for references see p 418

N-(Hydroxymethyl)trimethylacetamide:[60]
Trimethylacetamide (5.0 g, 49.4 mmol) and KOH (0.30 g, 5.0 mmol) were dissolved in a soln of formaldehyde/H_2O (7:13; 3.0 mL, 34.9 mmol) at 80 °C, and then stirred at rt overnight. The soln was acidified with 6 M HCl and the volatiles were removed. The residue was extracted with acetone (50 mL), the extracts were dried (Na_2SO_4), and the volatiles were removed to provide a colorless oil that solidified on standing.

H-Cys(Tacm)-OH:[60]
A mixture of H-Cys-OH · HCl · H_2O (3.35 g, 19.1 mmol), *N*-(hydroxymethyl)trimethylacetamide (2.76 g, 21.0 mmol), and TFA (15 mL) was stirred at rt for 1 h. The TFA was removed, the residue was dissolved in EtOH (20 mL), and the concentration was repeated. The product was triturated with Et_2O; yield: 4.1 g (94%). The product can be recrystallized from H_2O; mp 188 °C (dec); $[\alpha]_D^{17}$ −20.8 (*c* 0.5, H_2O).

Boc-Cys(Tacm)-OH · CHA:[60]
This compound was prepared by standard protocols using Boc_2O, followed by recrystallization (MeOH/Et_2O); yield: 94%; mp 142–144 °C; $[\alpha]_D^{20}$ −14.7 (*c* 0.7, MeOH).

Fmoc-Cys(Tacm)-OH:[60]
This compound was prepared by standard protocols with Fmoc-Cl, and was isolated by silica gel chromatography using $CHCl_3$/MeOH (10:0.5) as eluent, followed by precipitation with hexane; yield: 64%; mp 62–65 °C; $[\alpha]_D^{29}$ −40.8 (*c* 0.5, MeOH).

Cleavage of the *S*-(Trimethylacetamidomethyl) Group; Typical Procedure:[223]
To an ice-cold soln of [Cys(Tacm)^{10,26}]-porcine brain natriuretic peptide-32 (27.6 mg, 7.2 μmol) and anisole (32 μL, 40 equiv) in TFA (4 mL), was added $AgBF_4$ (56 mg, 40 equiv). After stirring for 60 min in an ice bath, Et_2O was added, and the precipitate was collected and dissolved in 1 M AcOH (4 mL) containing DTT (90 mg, 80 equiv). The soln was stirred at rt for 3 h, and then the supernatant was applied to a Sephadex G-25 column which was eluted with 1 M AcOH. The peptide fractions were collected, adjusted to pH 7.5 with 5% aq NH_3, and subjected to air oxidation.

2.6.6.2.4.3 *S*-(Phenylacetamidomethyl)cysteine

The phenylacetamidomethyl (Phacm) group (see Scheme 15) is particularly attractive for regioselective multiple disulfide formation strategies because it can be removed orthogonally by penicillin G acylase from *Escherichia coli*, an enzyme with P_1 specificity for phenylacetyl moieties.[182–184] The *S*-(phenylacetamidomethyl) group is introduced by methods analogous to those for *S*-(acetamidomethyl) (see Section 2.6.6.2.4.1), i.e. via reaction of cysteine with *N*-(hydroxymethyl)phenylacetamide under acid catalysis, and the related N^{α}-Boc and N^{α}-Fmoc derivatives are obtained by standard protocols.[184] Due to stability properties which are similar to those of the *S*-(acetamidomethyl) derivative, *S*-(phenylacetamidomethyl) is compatible with both the Boc/Bzl and Fmoc/tBu synthetic strategies, and is fully orthogonal to the base-labile *S*-(9-fluorenylmethyl) group, as well as to the acid-labile *S*-Bzl(4-Me), *S*-Mob, *S*-Trt, *S*-Tmb, *S*-Mmt, and *S*-Xan groups. The *S*-(phenylacetamidomethyl) protection can also be applied in combination with *S*-(acetamidomethyl), since the latter is resistant to the enzymatic cleavage of the former;[184] thus, the enzymatic approach may provide an additional level of orthogonality.[224]

Enzymatic cleavage of the *S*-(phenylacetamidomethyl) group is typically carried out at 35 °C using the penicillin G acylase, either in solution, or immobilized on dry fibre or on acrylic beads (Eupergit C); the *S*-(phenylacetamidomethyl)-protected peptide is dissolved in phosphate buffer (pH 7.8). In the presence of β-sulfanylethanol (2% v/v), the peptide product of the enzymatic hydrolysis is obtained in the free thiol form, whereas in the absence of the reducing agent, air oxygen under the slightly alkaline conditions leads to the symmetric homodimers if there is a single Cys residue in the sequence, and to an intramolecular disulfide if the product has two appropriately positioned Cys residues. In the case of N-terminal Cys residues, aminolysis of the intermediate *S*-(aminomethyl) derivative readily leads to Thz-containing peptides.[183]

N-(Hydroxymethyl)phenylacetamide:[184]
Phenylacetamide (10.0 g, 74 mmol) and KOH (0.5 g, 8.7 mmol) were dissolved in 35% aq formaldehyde (5.9 mL, 75 mmol). After 5 min at 70 °C and 12 h at rt, the mixture was acidified to pH 7 with 1 M HCl. The soln was concentrated and the residue was dissolved in acetone (50 mL), dried ($MgSO_4$), and the volatiles were removed; yield: 12.2 g (99%); mp 75–77 °C.

Boc-Cys(Phacm)-OH:[184]
Under ice-bath cooling *N*-(hydroxymethyl)phenylacetamide (5 g, 30.3 mmol) and H-Cys-OH · HCl (3.9 g, 25.0 mmol) in H_2O (30 mL) were treated with TfOH/TFA (1:19, 39 mL). After 6 h at rt under argon, the solvent was removed and the residue was reconcentrated from Et_2O (8 × 50 mL). The oil was suspended in tBuOH/H_2O (2:1, 150 mL), cooled in an ice bath and reacted with Boc_2O (6 g, 27.5 mmol) in tBuOH (49 mL). After 2 h in an ice bath, the mixture was stirred at rt for 24 h while maintaining the pH at 9.0–9.5 by the addition of aq Na_2CO_3. The mixture was washed with hexane, acidified with aq HCl to pH 2, and extracted with EtOAc. The extracts were washed with brine, dried ($MgSO_4$), and the volatiles were removed. The product was crystallized (EtOAc/hexane); yield: 5.2 g (56%); mp 100–101 °C.

Fmoc-Cys(Phacm)-OH:[184]
This compound was prepared analogously to the Boc derivative by acylating the intermediate H-Cys(Phacm)-OH with Fmoc-OSu in dioxane/H_2O (pH 9.0–9.5). The product was crystallized (aq MeOH); yield: 67%; mp 108–110 °C.

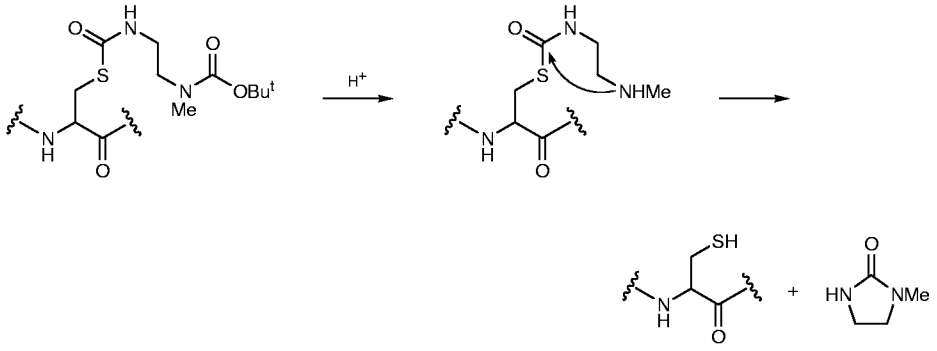
Ac-Cys-Pro-D-Val-Cys-NH_2; Typical Procedure for Enzymatic Cleavage of the *S*-(Phenylacetamidomethyl) Group:[184]
The tetrapeptide Ac-Cys(Phacm)-Pro-D-Val-Cys(Phacm)-NH_2 was incubated at 1.25 mM concentration in 0.05 M NH_4OAc buffer (pH 7.8) at 35 °C with penicillin amidohydrolase immobilized on acrylic beads (Eupergit C; 2.12 EU/μmol of *S*-Phacm); the reaction was followed by HPLC. After completion of deprotection and oxidation, AcOH/H_2O (9:1) was added, and the mixture was filtered through a disposable pipet with a glass wool plug, then lyophilized; yield: 95%.

2.6.6.2.5 *S*-Acylcysteine Derivatives

S-Acyl derivatives of cysteine, in particular *S*-benzoyl,[2,84,225] are active esters which undergo facile nucleophilic attack from amino components as well as intramolecular S→N shift under alkaline conditions.[226] The *S*-(alkyloxycarbonyl) protecting groups, such as *S*-(benzyloxycarbonyl),[227] *S*-(*tert*-butoxycarbonyl),[228] or *S*-(allyloxycarbonyl),[229] are less activated derivatives which, however, still suffer from the inherent ability to undergo intramolecular N-acylation in alkaline media. Nevertheless, *N*,*S*-diurethane-protected cysteines represent very economical intermediates for the synthesis of N-terminal cysteine peptides.[230] In *S*-(alkylaminocarbonyl) derivatives such as *S*-(ethylcarbamyl),[231] the carbonyl is resonance-stabilized and thus the tendency for S→N shifts is significantly reduced. A further development of this theme is the *S*-{[2-(*N*-*tert*-butoxycarbonyl-*N*-methylamino)ethyl]aminocarbonyl} derivative, with its built-in "safety-catch" principle, as shown in Scheme 17.[232]

for references see p 418

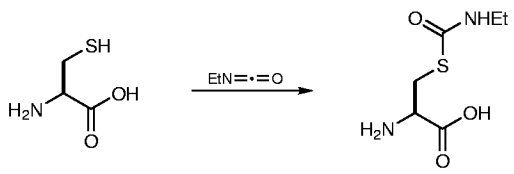
Scheme 17 Cleavage of the *S*-{[2-(*N*-*tert*-Butoxycarbonyl-*N*-methylamino)ethyl]aminocarbonyl} Group by Acids and Neutralization of the Intermediate β-Aminoethylcarbamyl Derivative[232]



2.6.6.2.5.1 *S*-(Ethylaminocarbonyl)cysteine

The ethylaminocarbonyl (ethylcarbamyl, EC) derivatives of N-protected cysteine or cysteine are readily obtained by reaction with ethyl isocyanate under anhydrous conditions (Scheme 18).[231,233] This S-protection is stable toward acids such as HBr/AcOH or TFA, as well as to the standard conditions of peptide synthesis. Deprotection is achieved by treatment with 1 M aqueous sodium hydroxide, hydrazine, or ammonia in methanol,[231] or by exposure to silver(I) or mercury(II) ions.[233]

Scheme 18 Synthesis of H-Cys(EC)-OH[231]



H-Cys(EC)-OH:[231]
To a precooled soln of anhyd H-Cys-OH · HCl (158 g, 1 mol) in DMF (1.5 L) was added freshly distilled ethyl isocyanate (78.3 g, 1.1 mol). After 70 h at rt, the mixture was concentrated to dryness, and the residue was triturated with Et_2O and then dissolved in H_2O (2 L). The pH was adjusted to 6.5, and then the soln was concentrated. The crystalline product was filtered, and washed with cold H_2O and alcohol/Et_2O; yield: 131 g (76%); mp 219 °C; $[\alpha]_D^{22}$ +91.1 (*c* 0.8, 95% AcOH).

2.6.6.2.6 *S*-(Alkylsulfanyl)- and *S*-(Arylsulfanyl)cysteine Derivatives

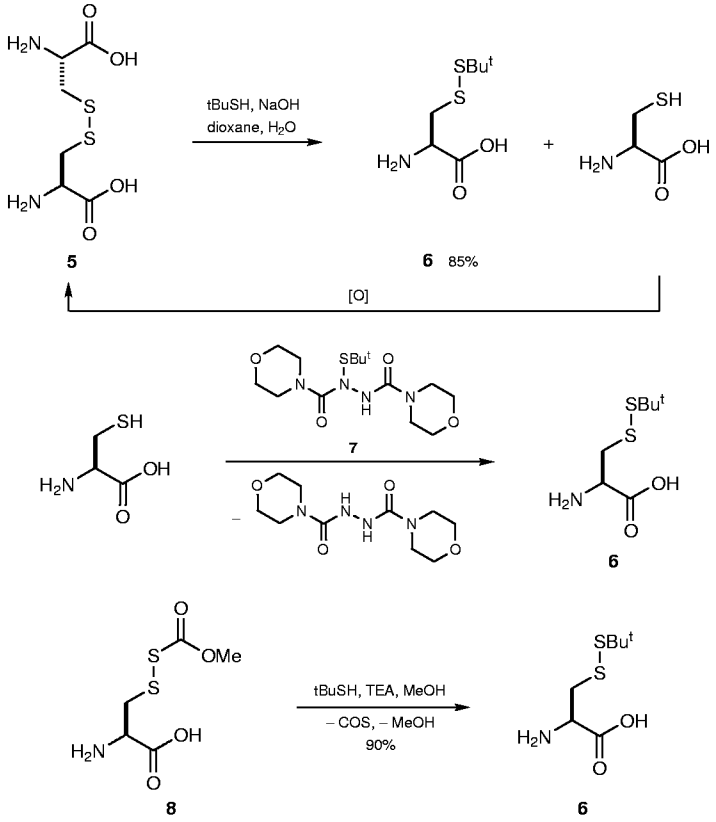
The usefulness of *S*-(alkylsulfanyl) or *S*-(arylsulfanyl) derivatives of cysteine as thiol protecting groups in a multistep peptide synthesis depends strongly upon their resistance toward disproportionation under acidic and, particularly, basic conditions. The asymmetric disulfides are generally prepared by reacting cysteine with thiosulfinates or *S*-alkyl thiosulfates, or by reacting cystine *S*-dioxide or *S*-phthalimidocysteine with thiols.[1] Although the *S*-(phenylsulfanyl)[234] and *S*-(ethylsulfanyl) groups[196,235–237] have found some limited application in peptide synthesis, most of the mixed disulfides undergo rather facile disulfide exchange reactions, i.e. disproportionation.[236,238] An exception to this pattern is the *S*-(*tert*-butylsulfanyl) (StBu) derivative of cysteine since steric hindrance prevents formation of the symmetrical di-*tert*-butyl disulfide.[1,239] The *S*-(tritylsulfanyl) (STrt) group is similarly stable;[44] however, *S*-(tritylsulfanyl) has not seen extensive use in the way that *S*-(*tert*-butylsulfanyl) has seen. A dual-purpose S-protecting group is the 3-nitro-2-pyridylsulfanyl

(Npys) group;[240,241] the *S*-Npys derivative of cysteine is an activated asymmetric disulfide that is sufficiently acid-stable to allow for its incorporation into peptide sequences using Boc/Bzl chemistry. *S*-(3-Nitro-2-pyridylsulfanyl) then serves for regioselective disulfide formation in a subsequent synthetic step via thiol/disulfide exchange reactions with a second unprotected cysteine residue (see Vol. E 22b, Section 6.1.3.2).

2.6.6.2.6.1 *S*-(*tert*-Butylsulfanyl)cysteine

Cysteine is converted into the *S*-(*tert*-butylsulfanyl) derivative by reaction with *tert*-butylsulfinic acid *tert*-butyl thioester,[235] or preferably by exploiting the steric barriers toward dimerization of 2-methylpropane-2-thiol by air oxygen. The latter approach makes it possible to drive the thiol/disulfide exchange reaction between cystine (**5**) and 2-methylpropane-2-thiol in favor of the desired product, i.e. H-Cys(StBu)-OH (**6**), in aqueous alkaline media because the cysteine intermediate is reoxidized to cystine and transformed further (Scheme 19).[1,239] The same principle has also been conveniently exploited for the conversion of short dimeric cystine peptides, prepared directly from cystine, into the related S-protected monomers.[15]

Scheme 19 Syntheses of H-Cys(StBu)-OH[1,239,242,243]



To reduce exposure to the disagreeable odor of the thiol, two alternative methods have been proposed. In one approach, 1-(*tert*-butylsulfanyl)-1,2-hydrazinedicarboxylic acid esters, e.g. the di-*tert*-butyl ester, or amides, e.g. the dimorpholide **7**, are used as efficient *tert*-butylsulfanyl donors for the preparation of *S*-(*tert*-butylsulfanyl) derivatives of cysteine or

for references see p 418

cysteine peptides (Scheme 19).[242] In a second approach, the activated *S*-(methoxycarbonylsulfanyl)cysteine species (**8**) (see Section 2.6.6.2.7.1) reacts rapidly and quantitatively with thiols to produce the asymmetric disulfides, such as the *S*-(*tert*-butylsulfanyl) and *S*-(tritylsulfanyl) derivatives (Scheme 19).[243]

Although it has been reported that the protecting group of the *tert*-butyl disulfide type is partly cleaved by hydrogen fluoride,[244,245] it is fully stable when exposed to TFA for several hours,[23,199] even in the presence of cation scavengers such as ethanedithiol.[44] *S*-(*tert*-Butylsulfanyl) is stable to bases such as piperidine,[44,199] and thus compatible in all aspects with Fmoc/tBu solid-phase chemistry.[120,199] In solution synthesis, *S*-(*tert*-butylsulfanyl) has been successfully applied in combination with Ddz/tBu[246] and Nps/tBu strategies.[23,247,248] In the latter case, it is advisable to use 2-methylindole to capture the reactive Nps-Cl intermediate resulting from HCl-mediated N^{α}-deprotection,[23] in order to avoid a potential exchange reaction of sulfenyl chlorides with disulfides.[249]

The stability of the *tert*-butyl disulfide toward the final acidolytic deprotection and/or resin-cleavage procedures makes it possible to obtain peptides in the cysteine-protected form for purification prior to their conversion into the free thiol form. Alternatively, fully protected peptides may be selectively unmasked at their thiol functions for further chemistry. Cleavage of the *S*-(*tert*-butylsulfanyl) group is achieved in aqueous or aqueous organic media under slightly alkaline conditions by reduction with thiols (e.g., benzenethiol, β-sulfanylethanol, dithiothreitol), by reduction with sodium borohydride, or by sulfitolysis.[1,239] Reduction with benzenethiol and sodium borohydride are reported to be very slow, whereas propane-1,3-dithiol is a much more efficient reductant.[68] Efficient reduction of the asymmetric disulfide with phosphines occurs under slightly alkaline (NH_4OAc or Na_2CO_3 buffer),[120,199] acidic (aq AcOH),[23] or neutral conditions.[247] Mindful of the known possible desulfurization of disulfides with phosphines,[250,251] trialkylphosphines are the preferred reagents to minimize side reactions.[252] Nonetheless, desulfurization of Cys(StBu) to alanine has been observed, unless fresh tributylphosphine is used.[253] A comparative analysis suggested, as optimal reaction conditions, the use of about five equivalents of tributylphosphine per *S*-(*tert*-butylsulfanyl) group in aqueous alcohols such as propan-2-ol and trifluoroethanol;[252] the latter solvent is particularly advantageous for deprotection of protected peptides due to its strong solubilizing properties. For deprotection of Cys(StBu)-containing peptides in aqueous media, the water-soluble tris(2-carboxyethyl)phosphine hydrochloride[254] is the reagent of choice, and it can be used at the stoichiometric reductant/*S*-(*tert*-butylsulfanyl) ratio of 1:1.[255,256] On-resin deprotection of Cys(StBu) residues by treatment with β-sulfanylethanol has been reported,[75,257] although in some cases it could be achieved only at elevated temperatures, and not at all with tributylphosphine.[258] The reductive cleavage of *S*-(*tert*-butylsulfanyl) groups is orthogonal to the deprotection procedures used for *S*-sulfide- and *S*-thioacetal-type protections of cysteine residues, and thus it is well suited for the design of regioselective disulfide-pairing strategies for multiple-cysteine-containing peptides (see Vol. E 22b, Sections 6.1.2 and 6.1.4).

As noted in Section 2.6.6.2, Cys(StBu) can epimerize during coupling steps,[32–35] particularly when it is anchored to resins as an ester and exposed to repetitive piperidine treatments used in Fmoc/tBu chemistry.[38–44] Moreover, C-terminal Cys(StBu) esters to the resin undergo base-induced β-elimination of the alkyl disulfide moiety, with subsequent nucleophilic addition of piperidine or of an alkyl sulfide moiety to the dehydroalanine with formation of a β-(piperidyl)alanine derivative as the major side product.[44,75,76] Neither the piperidine-induced epimerization nor the side products are observed in the synthesis of peptides with C-terminal Cys(StBu) amides or with Cys(StBu) residues in internal sequence positions. Additionally, during acid-mediated resin cleavage of Cys(StBu) peptides, reattachment to the resin-bound carbocations occurs at levels comparable to those of other S-protecting groups, e.g. *S*-(acetamidomethyl), despite the bulky *tert*-butyl group that would be expected to sterically hinder such reactions of the disulfide with carbocations.[33]

H-Cys(StBu)-OH via Thiol/Disulfide Exchange Reactions:
Method A:[1] Cystine (48 g, 200 mmol) in 2 M NaOH (200 mL) and dioxane (300 mL) was reacted with tBuSH (36 g, 400 mmol) at rt for 12 h. Air was bubbled slowly through the mixture. After completion of the reaction (5–7 d), as monitored by TLC (disappearance of H-Cys-OH), the soln was acidified with AcOH (50 mL) and taken to dryness. The residue was washed with H_2O and extracted with hot 50% AcOH (2 × 300 mL). The combined extracts were concentrated to 100 mL and poured into H_2O (500 mL). The product crystallized in the cold; yield: 70 g (85%); $[\alpha]_D^{20}$ −84 ± 1 (*c* 1, 1 M HCl).

Method B:[243] To a soln of tBuSH (2.76 mL, 24.3 mmol) and TEA (1.71 mL, 10.9 mmol) in MeOH (14 mL), a soln of H-Cys(SMoc)-OH · HCl (3.0 g, 12.2 mmol) in MeOH (9 mL) was added dropwise with stirring at rt. A gel was formed which was triturated with Et_2O to form a solid. The product was filtered and washed with MeOH and Et_2O; yield: 2.29 g (90%); mp >159 °C (dec); $[\alpha]_D^{25}$ −86.4 ± 1 (*c* 1, 1 M HCl).

1-(*tert*-Butylsulfanyl)-1,2-hydrazinedicarboxymorpholide:[242]
To a soln of azodicarboxymorpholide[259] (5.12 g, 20 mmol) and tBuSH (2.3 mL, 20 mmol) in THF (150 mL) was added a catalytic amount of NaOEt. The mixture was treated with charcoal, filtered, and the filtrate was concentrated. Upon addition of petroleum ether, the product was obtained as a crystalline material; yield: 6.44 g (93%); mp 152–154 °C.

H-Cys(StBu)-OH via the Sulfenohydrazide Method:[242]
To a suspension of 1-(*tert*-butylsulfanyl)-1,2-hydrazinedicarboxymorpholide (3.46 g, 10 mmol) in argon-sat. H_2O (40 mL), H-Cys-OH (0.6 g, 5 mmol) in argon-sat. H_2O (20 mL) was added dropwise at rt under stirring. A clear soln was formed, from which the product crystallized on standing; yield: 0.92 g (84%).

H-Thr-Ala-Cys(StBu)-Gly-Gln-Lys-Ser-Pro-OH:[242]
To a stirred soln of 1-(*tert*-butylsulfanyl)-1,2-hydrazinedicarboxymorpholide (0.22 g, 0.64 mmol) in argon-sat. H_2O (10 mL), H-Thr-Ala-Cys-Gly-Gln-Lys-Ser-Pro-OH (0.3 g, 0.32 mmol; peptide content: 84.9%) in argon-sat. H_2O (5 mL) was added dropwise at rt. After 24 h, the soln was concentrated to a small volume and the product was precipitated from EtOH/EtOAc; yield: 0.26 g (83%); peptide content: 87.8%; amino acid analysis: Cys(StBu) 0.98 (1).

H-Thr(tBu)-Cys(Acm)-Pro-Pro-Cys-Pro-Ala-Pro-OH; Typical Procedure for Reductive *S*-(*tert*-Butylsulfanyl) Cleavage with Phosphines:[24]
To a soln of H-Thr(tBu)-Cys(Acm)-Pro-Pro-Cys(StBu)-Pro-Ala-Pro-OH (11.7 g, 10.0 mmol) in 95% aq TFE (300 mL), was added Bu_3P (12.3 mL, 50.0 mmol). After 3 h at rt, the soln was concentrated and the product was precipitated with EtOAc/MTBE (1:3). To quantitatively remove the phosphine, the product was reprecipitated from the same solvent system; yield: 8.5 g (94%); $[\alpha]_D^{20}$ −216.2 (*c* 1, MeOH).

On-Resin Cleavage of *S*-(*tert*-Butylsulfanyl) with Phosphines; General Procedure:[257]
The peptidyl resin was swelled in DMF for 5 min and treated twice at 25 °C for 1 h with Bu_3P (50 equiv) in NMP/H_2O (9:1). Progress of the deprotection was monitored by analytical HPLC on small aliquots of the fully deprotected (except for the StBu group) and cleaved peptide. After completion of the reaction, the resin was washed with NMP.

On-Resin Cleavage of *S*-(*tert*-Butylsulfanyl) with Thiols; General Procedure:[75]
The peptidyl resin was swelled in DMF for 5 min and treated with β-ME/DMF (1:1). After stirring at 25 °C for 5 h, the peptidyl resin was washed with DMF, CH_2Cl_2, DMF, EtOH, and DMF (each 2 × 2 min), followed by further on-resin manipulations and cleavage from the resin.

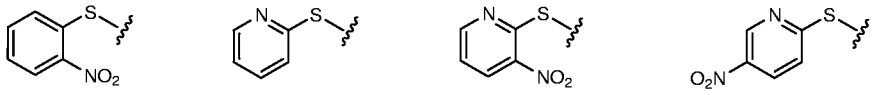
2.6.6.2.6.2 *S*-(3-Nitro-2-pyridylsulfanyl)cysteine

One of the most successful approaches for site-directed disulfide bond formation involves the application of aromatic sulfanyl protecting/activating groups, i.e. of 2-nitrophenylsulfanyl (Nps),[26] 2-pyridylsulfanyl (SPy),[210] 3-nitro-2-pyridylsulfanyl (Npys),[211,240,241] and 5-nitro-2-pyridylsulfanyl[260] derivatives of cysteine (Scheme 20), which subsequently react with a second cysteine residue that has a free thiol to form the desired disulfide (see Vol. E 22b, Section 6.1.3.2). Although all of these *S*-(arylsulfanyl) derivatives are sufficiently stable toward acids such as TFA or hydrogen fluoride,[240] and thus in principle are compatible with the Boc/Bzl chemistry, only the Cys(Npys) derivative has found extensive application in

for references see p 418

solution and solid-phase syntheses.[158,241,261–265] The activated disulfides are not compatible with Fmoc/tBu chemistry, due to the fast disproportionation under the basic conditions required for Fmoc cleavage.[263,264] Even with the Boc/Bzl strategy, the choice of compatible coupling reagents and scavengers for the acidolytic deprotection is limited.[263] The Cys(Npys) derivatives are prepared by reaction of the thiol group with Npys-Cl,[240,241] which is in turn obtained by reaction of the corresponding disulfide with gaseous chlorine[211,266] or, more efficiently, with sulfuryl chloride.[267] Due to the instability of Npys-Cl, direct reaction of cysteine derivatives with the disulfide, in analogy to the preparation of the *S*-(2-pyridylsulfanyl)[210] and *S*-(5-nitro-2-pyridylsulfanyl) derivatives,[260] may be more advantageous.[268] The *S*-(3-nitro-2-pyridylsulfanyl) group is reduced with thiols or phosphines;[240] aliphatic thiols such as sulfanylacetic acid or β-sulfanylethanol give a rapid reaction, while aromatic thiols such as 1-methyl-1*H*-imidazole-2-thiol and pyridine-2-thiol do not promote significant cleavage.[269] However, the main way to cleave the *S*-(3-nitro-2-pyridylsulfanyl) group is by thiol/disulfide exchange reactions with a second cysteine residue to form the desired intrachain and interchain disulfide bonds, respectively.

Scheme 20 The Structures of Arylsulfanyl Thiol Protecting Groups[26,210,211,240,241,260]



Nps SPy Npys

Boc-Cys(Npys)-OH:[240,241]

Method A:[241] To H-Cys-OH (1.21 g, 10 mmol) in 90% formic acid (10 mL) was added Npys-Cl (2.10 g, 11 mmol). After 1 h at rt, insoluble material was filtered off, and the crude product was precipitated with Et_2O and reprecipitated from MeOH with Et_2O; yield: 2.23 g (81%) of H-Cys(Npys)-OH as the hydrochloride hemihydrate (determined by CHN analysis); mp 188–190 °C (dec); $[\alpha]_D^{22}$ +140.5 (*c* 1, MeOH).

The corresponding Boc-Cys(Npys)-OH derivative was obtained by the standard Boc-ON procedure and the product was isolated by precipitation from EtOAc with petroleum ether; yield: 1.29 g (69%); mp 153–155 °C (dec).

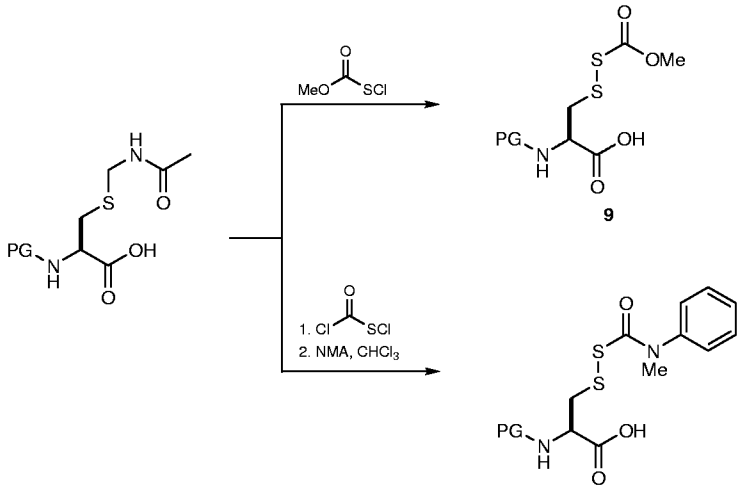
Method B:[240] A suspension of Boc-Cys-OH · DCHA[240] (2.01 g, 5 mmol) in CH_2Cl_2 (200 mL) was washed with 5% aq citric acid and dried (Na_2SO_4). To this soln, Npys-Cl (1.04 g, 5.5 mmol) was added in 4 portions while a soln of TEA (1.11 g, 11 mmol) in CH_2Cl_2 (30 mL) was added dropwise under stirring at 0 °C. After 2 h at rt, the solvent was removed and the residue was dissolved in acetone (150 mL). On addition of DCHA (1 mL, 5 mmol) the salt was filtered off and redissolved in MeOH. Insoluble material was filtered off and the filtrate was concentrated. The residue was crystallized (EtOAc/petroleum ether) to give the DCHA salt; yield: 1.84 g (66%); mp 150–152 °C; $[\alpha]_D^{22}$ −86.3 (*c* 1, MeOH).

Boc-Cys(Npys)-OH · DCHA was desalted in EtOAc by washing with aq 5% citric acid. Boc-Cys(Npys)-OH was obtained by precipitation with petroleum ether; yield: quant; mp 153–155 °C (dec); $[\alpha]_D^{22}$ −88.7 (*c* 1, MeOH).

2.6.6.2.7 *S*-(Acylsulfanyl)cysteine Derivatives

Methoxycarbonylsulfenyl chloride (SMoc-Cl)[270] is a valuable reagent for the synthesis of asymmetric disulfides via reasonably stable, isolable *S*-(methoxycarbonylsulfanyl) derivatives.[271] This procedure has been successfully transferred to cysteine peptides for selective intrachain and interchain disulfide formation (see Vol. E 22b, Section 6.1.3.1).[209,272] Due to the high reactivity of *S*-(methoxycarbonylsulfanyl)cysteine derivatives **9** (Scheme 21),[209] and complications arising from their base lability and the risk of S→N migration,[226,231,273] this type of protection is not suitable for multistep peptide synthesis. The more recently proposed *N*-methyl-*N*-phenylcarbamoylsulfanyl (SMpc) derivative of cysteine (Scheme 21) is apparently less prone to sulfur to nitrogen shift.[274]

Scheme 21 Synthesis of N^{α}-Protected H-Cys(SMoc)-OH[209,243] and H-Cys(SMpc)-OH[274]



2.6.6.2.7.1 *S*-(Methoxycarbonylsulfanyl)cysteine

The *S*-(methoxycarbonylsulfanyl) derivatives of cysteine and cysteine peptides are prepared from *S*-(acetamidomethyl)- and *S*-trityl-protected precursors by reaction with SMoc-Cl to produce transiently protected/activated intermediates for further chemical manipulations.[209,243,272,274–276] According to prior experience with Nps-Cl,[26] even other *S*-aralkyl derivatives such as *S*-Xan or *S*-Tmb are converted into the *S*-SMoc compounds; similarly, reaction of 2,2-dimethoxythiazolidine with SMoc-Cl leads to the *S*-SMoc derivatives.[171] By reaction of cysteine hydrochloride with SMoc-Cl, the H-Cys(SMoc)-OH derivative is obtained as a crystalline compound, which is used as the active intermediate for the synthesis of *S*-(arylsulfanyl) and *S*-(alkylsulfanyl) derivatives (see Section 2.6.6.2.6).[243]

H-Cys(SMoc)-OH · HCl:[243]
A soln of H-Cys-OH · HCl (4.46 g, 28.5 mmol) in MeOH (38 mL) was treated briefly with gaseous HCl bubbled through the soln. Next, an ice-cold soln of SMoc-Cl (5.0 mL, 55.0 mmol) in MeOH (40 mL) was added dropwise. After 1 h at 0 °C, the MeOH was removed (<30 °C) and the product was obtained as a crystalline compound; yield: 7.01 g (99%); mp 168 °C (dec); $[\alpha]_D^{25}$ +28.5 (*c* 1, AcOH).

Boc-Cys(SMoc)-OH · DCHA:[274]
A hazy soln of Boc-Cys(Acm)-OH (350 mg, 1.2 mmol) in $CHCl_3$ (6 mL) was treated at 5 °C with SMoc-Cl (0.12 mL, 1.3 mmol). After 10 min, the volatiles were removed and the residue in $CHCl_3$ was washed with 1 M HCl and H_2O, then dried ($MgSO_4$). The soln was concentrated and the crude product in EtOAc (2 mL) was treated at 5 °C with 2.5 M DCHA in EtOAc (0.40 mL, 1.0 mmol). The soln was concentrated and the precipitate was collected; yield: 440 mg (89%); mp 140–145 °C.

2.6.6.2.7.2 *S*-(*N*-Methyl-*N*-phenylcarbamoylsulfanyl)cysteine

A protecting/activating group that gives higher yields and purities in disulfide formation by the thiol/disulfide exchange reaction, compared to *S*-(methoxycarbonylsulfanyl) derivatives, is the *S*-(*N*-methyl-*N*-phenylcarbamoylsulfanyl) function.[274,277–279] The *S*-(*N*-methyl-*N*-phenylcarbamoylsulfanyl)cysteine derivatives are produced from the *S*-(acetamidomethyl) compounds in two steps (see Scheme 21).

Boc-Cys(SMpc)-OH:[274]
To a hazy soln of Boc-Cys(Acm)-OH (0.84 g, 2.9 mmol) in $CHCl_3$ (14 mL), a soln of chlorocarbonylsulfenyl chloride[280] (0.24 mL, 2.9 mmol) in $CHCl_3$ (2.9 mL) was added dropwise over 5 min at 5 °C. After 5 min, the relatively clear soln was filtered through glass wool into a soln of 2.0 M NMA in $CHCl_3$

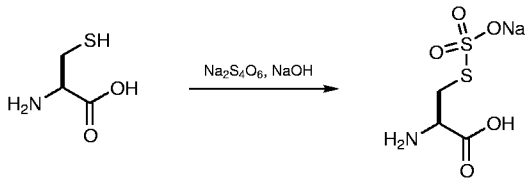
for references see p 418

(14 mL, 28 mmol) at 5 °C. After 5 min, the mixture was washed with 1 M HCl and H_2O, then dried ($MgSO_4$). The residue was crystallized twice (CH_2Cl_2/petroleum ether); yield: 0.87 g (76%) as the hemihydrate (determined by CHN analysis); mp 145–148 °C.

2.6.6.2.8 *S*-Sulfonate Derivatives of Cysteine

S-Sulfonate derivatives of cysteine peptides, also referred to as Bunte salts, are prepared by oxidative sulfitolysis of cysteine or cystine peptides (Scheme 22).[281,282] These derivatives have been used in synthetic efforts on insulin and insulin analogues,[283–288] proinsulin,[289] and ribonuclease A,[83] mainly for the purpose of purification, solubilization, and storage. Thiolysis of the *S*-sulfonate derivatives with a second cysteine component can be used for disulfide formation, but this reaction leads to an equilibrium with a product distribution similar to the mixture of cystine peptides obtained by random air oxidation.[284,288]

Scheme 22 Synthesis of H-Cys(SO_3Na)-OH[290]



Based on the stability of *S*-sulfocysteine peptides toward acid and alkaline conditions,[283] the usefulness of the *S*-sulfonate moiety for peptide synthesis in solution and by solid-phase Fmoc/tBu chemistry warranted attention.[290] In fact, Boc-Cys(SO_3Na)-ONa and Fmoc-Cys(SO_3Na)-ONa are efficiently coupled in organic solvents such as DMF to amino components in solution and in the solid phase using the TBTU procedure but, in contrast to similar tyrosine-*O*-sulfate derivatives (see Vol. E 22b, Section 6.6), not using the DCC/HOBt method.[290] The *S*-sulfonate group is stable to 20% piperidine in DMF or 20% piperidine/2% DBU in DMF, which are standard reagents for Fmoc cleavage. Satisfactory stability of the *S*-sulfonate is observed in TFA-mediated deprotection and resin-cleavage steps using water or *p*-cresol, as well as triethylsilane, as scavengers. Cleavage of the *S*-sulfonate group occurs quantitatively with thiols[283,288] or upon reduction with phosphines.[290,291]

H-Cys(SO_3Na)-OH · 2H_2O:[290]
To a soln of H-Cys-OH · HCl · H_2O (10.54 g, 60.0 mmol) in H_2O (100 mL), were added successively 1 M NaOH (60 mL) and $Na_2S_4O_6 \cdot 2H_2O$ (18.38 g, 60.0 mmol). After 1 h at rt, the mixture was concentrated and the solid formed was filtered off and discarded. Sulfur that had precipitated from the filtrate was removed by filtration after several h at 5 °C. Then the soln was concentrated to 100 mL; the product crystallized upon addition of EtOH (800 mL); yield: 12.14 g (78%); $[\alpha]_D^{20}$ −89.32 (*c* 1, H_2O).

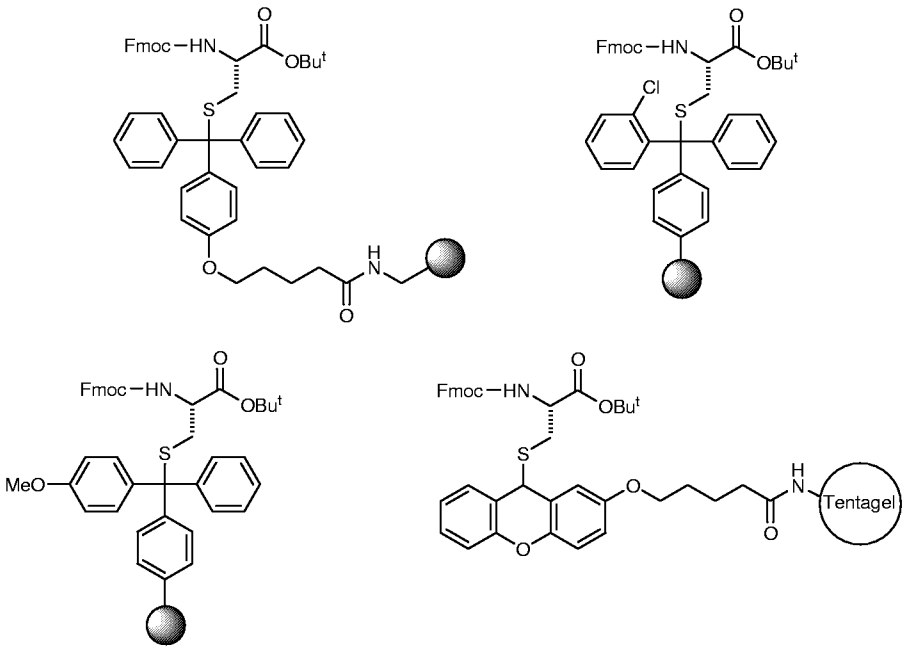
Boc-Cys(SO_3Na)-ONa · 2H_2O:[290]
To H-Cys(SO_3Na)-OH · 2H_2O (12.0 g, 46.3 mmol) in H_2O (400 mL), 1 M NaOH (73 mL), and dioxane (600 mL), was added Boc_2O (53.3 g, 244.2 mmol). After 6 h at rt, the soln was taken to dryness and the residue was distributed between H_2O and hexane. The aqueous phase was concentrated to dryness; yield: 16.8 g (95%); mp 153 °C (dec); $[\alpha]_D^{20}$ −40.8 (*c* 1, H_2O).

Fmoc-Cys(SO_3Na)-ONa · 0.5H_2O:[290]
To H-Cys(SO_3Na)-OH · 2H_2O (4.54 g, 17.51 mmol) in H_2O (20 mL) and 1 M NaOH (17.5 mL), was added Fmoc-OSu (11.81 g, 35.01 mmol) in dioxane (100 mL). After 12 h at rt, the soln was concentrated, diluted with H_2O, and extracted with EtOAc. The organic layer was extracted with H_2O and the aqueous phase was re-extracted with EtOAc. The combined aqueous phases were concentrated and coevaporated with toluene. The solid was dried and suspended in EtOH for 1 h at 60 °C. Upon cooling, the precipitate was collected by filtration; yield: 7.26 g (87%); mp 168 °C (dec); $[\alpha]_D^{20}$ −36.8 (*c* 1, H_2O).

2.6.6.2.9 Resin-Linked Thiol Protecting Groups

Racemization of C-terminal cysteine residues that can occur in the esterification step of solid-phase synthesis is largely minimized with optimized procedures (see Section 2.6.6.2).[49,50] Similarly, epimerization that is due to repetitive exposure to base during Fmoc cleavage can be minimized using chlorotrityl resin.[42,43] An alternative way to avoid racemization is the use of linkers to the resin that act as thiol protecting groups. A further advantage of this approach is the subsequent exploitation of pseudo-dilution on resin for intramolecular disulfide formation with a second suitably protected cysteine residue. Side-chain resin linkage of cysteine has been proposed with trityl-type protections, i.e. 5-[4-(hydroxydiphenylmethyl)phenoxy]pentanoic acid linked to aminomethylpolystyrene,[292] 2-chlorotrityl resin,[293,294] and 4-methoxytrityl resin,[293] as well as with a xanthenyl handle (Scheme 23);[295] all of these are fully compatible with Fmoc/tBu chemistry. Moreover, this type of side-chain resin linkage has been proposed for the phase-change approaches in convergent synthetic strategies.[296,297] Compared to Cys(Trt) derivatives, the chlorotrityl group on resin is significantly more acid-resistant. It requires several hours treatment with TFA/triethylsilane (95:5), while the resin-bound *S*-(4-methoxytrityl) derivatives are cleaved selectively in the presence of *tert*-butyl-type protecting groups, and of *S*-trityl derivatives, using 1% TFA in dichloromethane/triethylsilane (95:5) within 30 minutes.[293] Due to the acid stability of alkylsulfanyl-type protecting groups (see Section 2.6.6.2.6), the sulfanylpropanoyl-MBHA resin was used for side-chain protection/resin linkage of cysteine residues in synthesis on solid support according to the Boc/Bzl strategy.[298]

Scheme 23 Resin-Linked Thiol Protecting Groups[292–295]



for references see p 418

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3 Peptide Bond Formation

L. Moroder

In principle, peptide bond formation can be accomplished using all the procedures available in organic chemistry for the synthesis of carboxylic acid amides. However, due to the presence of various functional groups in natural and nonnatural amino acids, and particularly in view of the requirement for full retention of chiral integrity of the constituent amino acids during amide bond formation, the coupling of amino acids and peptides is significantly more challenging. In addition to these inherent problems associated with peptide bond formation, there is the polyamide structure of peptides which requires the repetitive stepwise acylation of the growing peptide chain with multiple amino acids in the correct sequence; therefore, the demands on coupling procedures in terms of efficiency are markedly enhanced. In solution syntheses, the isolation of intermediates allows for their purification at least to some extent, on a resin, however, such operations are prevented. Thus, very high if not quantitative acylation yields in each step become very important for the success of peptide-chain assembly, particularly if larger polypeptides are the synthetic targets. In view of these severe limitations, coupling methods available either from organic chemistry or purposely elaborated for peptide chemistry were investigated for their usefulness in peptide synthesis in solution and on solid support. Several of the earlier methods (for an exhaustive review, see ref[1]) have been superseded by more recently developed highly efficient procedures, often as a result of the special requirements for automation in solid-phase synthesis. In the following chapters the basic methods that are more routinely applied are classified and discussed according to the type of activation involved. For additional reviews regarding the various methods available in peptide chemistry, the reader is directed to refs[1–4].

The acid azide and acid chloride methods were the first methods to be employed by Curtius[5] and Fischer,[6] respectively, in peptide synthesis. The azide method (Section 3.1) has been widely applied for the assembly of suitably protected peptide segments in solution, even applying the strategy of minimum protection (Section 4.1.2). For this purpose, the method has been optimized by increasing the acylating potency via in situ transformation into active esters. Similarly, the acid chloride[7] and particularly the acid fluoride method[8] have regained increasing importance for sterically demanding couplings (Section 3.3). In this context, even the Leuchs' anhydrides[9] (Section 3.4.3) that are used mainly for the preparation of poly-α-amino acids[10] (although under optimized conditions these were also successfully used for peptide synthesis)[11] have become attractive reagents in the form of their urethane-protected species, i.e. the UNCAs,[12] due to their high reactivity and to the fact that carbon dioxide is the sole byproduct (Section 3.4.4). Conversely, the use of symmetric and various types of mixed anhydrides is restricted mostly to solution synthesis where the byproducts and potential troublesome side products can be removed more easily by intermediate purification steps (Sections 3.4.1 and 3.4.2). A large variety of active esters are available for amide bond formation (Section 3.2) which *per se* represent ideal acylating agents due to their stability on storage. However, active esters are generally not as reactive as other acylating species. Therefore, with the exception of the pentafluorophenyl (OPfp) and 4-oxo-3,4-dihydro-1,2,3-benzotriazin-3-yl (ODhbt) esters,[13] active esters are mainly employed for peptide synthesis in solution, although in most cases they represent the reactive species in the postulated mechanistic pathways of amide bond formation using coupling reagents such as the carbodiimides with 1,2-dinucleophiles as additives (Section 3.5), the phosphonium salts (Section 3.7), and the uronium/guanidinium salts (Section 3.8). The latter two types of reagent are the most commonly used in coupling reactions nowadays for peptide synthesis in the solid-phase mode,[14] whereas the carbodiimides are efficiently used for the condensation of fully protected peptides prepared in solution or on solid phase[15] (Sections 4.1.3 and 5.2). While the carbodiimides and phosphonium salts do not react with

for references see p 426

amino groups and, thus, can be added directly to the amino and carboxy components that are to be coupled, the uronium/guanidinium salts react with amino groups forming tetramethylguanidinium derivatives.[16] Correspondingly, preactivation of the amino acid to be coupled is required prior to its addition to the amino component.

Besides a few additional coupling procedures reported in Sections 3.6 and 3.9 that illustrate the occasional interesting application, the oxidation–reduction condensation with phosphines in the presence of disulfides (in which acyloxyphosphonium species are generated as reactive intermediates),[17] is discussed in the context of the sulfenamide-type protection of amino groups (Section 2.1.2.2.4), since their deprotection with phosphines and coupling with the amino acid occurs concurrently. However, methods used for orthogonal ligation of unprotected peptide fragments via amide bond formation are presented separately in Section 4.1.5.

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3.1 Acid Azides

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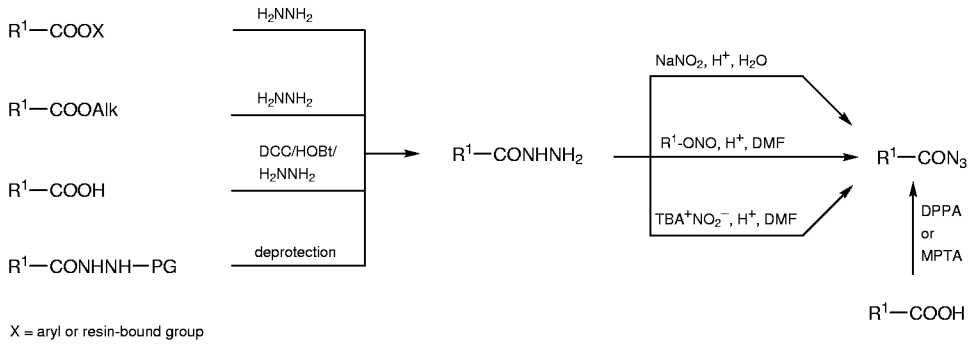
Almost at the same time as E. Fischer proposed the use of acid chlorides for peptide bond formation,[1,2] T. Curtius introduced the azide procedure for acylation of amino acids under Schotten–Baumann conditions with *N*-acylamino acid azides.[3–12] The azide procedure has found widespread application for coupling of single amino acid residues, e.g. histidine, serine, threonine, and particularly for the condensation of peptide fragments using both the maximum and minimum protection strategy for synthesis of peptides in solution (see Section 4.1). Comprehensive reviews have been published.[13,14] The main advantage of the azide procedure over the chloride method, which readily leads to side reactions and racemization (see Section 3.3), is the low tendency for oxazol-5(4*H*)-one formation and consequent racemization, even of the acyl-amino acid azides, due to their moderate activation and thus low affinity of the α-carbonyl towards nucleophilic oxygens.[15,16] It was this latter observation that allowed condensations of peptide fragments with C-terminal amino acid residues other than glycine and proline and therefore allowed a less restricted choice of fragments in the design of synthetic routes for peptides and proteins, e.g. ACTH,[17] β_1-MSH,[18]secretin,[19] insulin A and B chains,[20–22] proinsulin,[23] ribonuclease A S-protein,[24] ribonuclease T_1,[25] ribonuclease A,[26] iso-1-cytochrome c.[27] There were, however, cases where racemization had been observed to occur even with the azide method[28] and up to 40 % racemization was reported for a fragment coupling step related to the synthesis of human calcitonin M and involving as C-terminal amino acid phenylalanine.[29] Nonetheless, extensive model studies clearly have confirmed that the azide method leads to racemization in peptide couplings at levels significantly lower than other procedures and that high percentages of racemization occurs only in special cases, and particularly under inappropriate conditions such as excess of base.[30–35]

An additional advantage of the azide procedure derives from the possibility to prepare peptide fragments as C-terminal alkyl or aryl esters, which generally are readily converted into hydrazides and subsequently into the reactive azides for segment condensations. However, careful planning of the fragments is required for the synthesis of large peptides, as intermediate peptide esters or hydrazides are not usually suitable for alternative coupling procedures. Moreover, to avoid side reactions such as the Curtius rearrangement[36–39] which leads to the undesired ureido bond formation, the reaction conditions of the acylation step have to be carefully controlled. With new developments in the preparation of azides, e.g. via diphenyl phosphorazidate (DPPA),[40–43] this procedure has regained popularity as a racemization-free and efficient fragment condensation and peptide cyclization procedure, even in large scale synthesis due to its cost effectiveness and ready execution.[44–46]

As outlined in Scheme 1, azides are obtained by reaction of suitably protected amino acid or peptide hydrazides with nitrous acid or directly from the related carboxy groups by azide-transfer reagents. The hydrazides are generated by hydrazinolysis of esters, preferentially alkyl esters or from protected peptide hydrazide precursors. The strategy via the hydrazides has the advantage of allowing the minimum protection strategy to be applied. These synthetic routes, originally developed for synthesis of peptides in solution, can also be applied to synthesis on resins.

for references see p 440

Scheme 1 Different Routes to Peptide Hydrazides and Peptide Azides



3.1.1 Synthesis of Peptide Hydrazides

Fully or minimally protected peptide hydrazides are stable on storage, and the most common procedure for their preparation is hydrazinolysis of the corresponding methyl, ethyl or more rarely benzyl esters with hydrazine hydrate. Conversion into the hydrazides is often facilitated by their tendency to precipitate from the mixture if suitable solvents or solvent mixtures are used. Alternatively, hydrazides are obtained by hydrazinolysis of C-terminally activated amino acids and peptides. Hydrazinolysis occurs with similar efficiency when peptides are bound to resins via suitable linkers.[47–52] In the case of larger peptide fragments, longer reaction times and higher excesses of hydrazine hydrate are needed, which may lead to side reactions at the many different side-chain functionalities as a result of insufficiently stable side-chain protecting groups. To bypass such problems, a synthetic strategy was developed that is based on the use of N′-protected amino acid hydrazides.[53,54] In this procedure the peptides are assembled from C-terminal N′-protected amino acid hydrazides. Following cleavage of the hydrazide protecting group and conversion to the azide, the peptide fragments are ready for use in segment condensations. An identical principle can be applied to the solid-phase synthesis of peptide hydrazides, providing the C-terminal amino acid residues are attached to special hydrazide linkers.[55–57]

3.1.1.1 Hydrazinolysis of Amino Acid and Peptide Esters

In general the synthesis of suitably protected amino acid and peptide hydrazides from the corresponding alkyl or aryl esters by reaction with hydrazine hydrate is carried out in alcohols or DMF at ambient temperature.[58,59] However, with C-terminal branched amino acids such as valine or isoleucine, or larger peptides, more vigorous conditions are required, i.e. higher temperature (25–80 °C) and longer reaction times. To avoid formation of symmetrical bis(*N*-acyl-aminoacyl)hydrazides an excess of hydrazine hydrate or hydrazine is recommended, which also enhances the reaction rates. In this context, the solvent is known to play an important role, with alcohols being more appropriate than DMF[58,60,61] or solvent mixtures such as dioxane/methanol.[62] The use of hydrazine hydrate without any solvent has also been reported.[63]

Stability of Protecting Groups to Hydrazinolysis

In general the *tert*-butyl esters, as used in the Z/tBu strategy in solution, are stable towards hydrazinolysis. Therefore, peptide intermediates containing Glu(OtBu) residues are readily converted into the corresponding hydrazides,[64,65] although in the case of Asp(OtBu) residues controversial results were reported depending upon the reaction conditions and apparently even upon the peptide sequence.[66–69] In the case of the base-sensitive Asp-Gly and Asp-Ser sequences the *tert*-butyl ester protection does not prevent aspartimide formation;[70] similarly, in cases of Asn-Gly sequences $\alpha \rightarrow \beta$ transpeptidation on treatment with hydrazine was found to occur although to a lower extent.[68,71]

Hydrazinolysis is not compatible with the Boc/Bzl chemistry, as benzyl esters react readily with hydrazine. Regarding the stability of other protecting groups towards hydrazine hydrate, the urethane-based protecting groups of amino functions Z, Boc and other related *tert*-alkyl urethane-derivatives are stable, together with the Trt and Tos derivatives. Conversely, the base-labile amino protecting groups Phth, Tfa, For, Fmoc and 1-(4,4-dimethyl-2,6-dioxocyclohexylidene)ethyl (Dde)[72] and Nps-derivatives are unstable. The ether-type protecting groups of hydroxyl functions, e.g. *tert*-butyl, benzyl, halobenzyl ethers, are stable, whereas the *S-tert*-butyl, *S*-acetamidomethyl and *S*-benzyl cysteine derivatives are partially sensitive to hydrazine treatments. During hydrazinolysis at elevated temperatures *S*-benzylcysteine peptides can undergo racemization and elimination of benzylthiol.[73] Similarly, nitroarginine-containing peptides are not suitable for hydrazinolysis, as this residue is converted at least partially into aminoguanidine derivatives and ornithine.[74,75] Similar side reactions are reported for N^{ω}-tosylarginine-containing peptides.[76] Thus, special reaction conditions are required for arginine-peptides.[59] Moreover, the side-chain carboxamide of glutamine is more readily cleaved by hydrazine than peptide bonds.[73]

Hydrazinolysis and Racemization

Only a few cases of racemization occurring during hydrazinolysis are known.[34,77] An explanation for the low levels of racemization was given by Goodman et al.[78] by showing that a chirally pure hydrazide was obtained from the chirally pure oxazol-5(4*H*)-one of Z-Aib-L-Phe-OH. This observation proves that hydrazide formation from oxazol-5(4*H*)-one may be faster than the oxazol-5(4*H*)-one racemization process.

Z-His-NHNH$_2$:[79,80]

CAUTION: *Hydrazine hydrate ($H_2NNH_2 \cdot H_2O$) is a poison upon ingestion and by intravenous routes. It is an experimental carcinogen and is a corrosive irritant to eyes, skin, and mucous membranes. Appropriate safety precautions should be taken during all stages of its handling and disposal.*

To an ice-cold soln of H-His-OMe·2HCl (4.85 g, 20 mmol) in $CHCl_3$ (30 mL) TEA (5.6 mL, 40 mmol) was added followed by Z-Cl (3 mL, 21 mmol) and TEA (2.8 mL, 20 mmol). After 30 min at rt, the soln was washed with H_2O, evaporated and reevaporated from EtOH. The residue was dissolved in EtOH (10 mL), 80 % hydrazine hydrate (2.5 mL, 50 mmol) was added and the crystalline product formed on standing at rt was collected and recrystallized (EtOH); yield: 4.1 g (67.5%); mp 172–173 °C; $[\alpha]_D^{30}$ −35.7 (*c* 2.0, 1 M HCl).

Z-Thr-NHNH$_2$:[81]

CAUTION: *Hydrazine hydrate ($H_2NNH_2 \cdot H_2O$) is a poison upon ingestion and by intravenous routes. It is an experimental carcinogen and is a corrosive irritant to eyes, skin, and mucous membranes. Appropriate safety precautions should be taken during all stages of its handling and disposal.*

To a soln of Z-Thr-OMe (29 g, 0.108 mol) in MeOH (150 mL), hydrazine hydrate (11 mL, 0.217 mol) was added. After 20 h the precipitate was filtered off, suspended in MeOH (600 mL) at 60 °C and on cooling the product was collected; yield: 26.5 g (91.5%); mp 190–191.5 °C; $[\alpha]_D^{20}$ −10.9 (*c* 1, AcOH).

for references see p 440

Z-Pro-Ser-$NHNH_2$:[82]

CAUTION: *Hydrazine hydrate ($H_2NNH_2 \cdot H_2O$) is a poison upon ingestion and by intravenous routes. It is an experimental carcinogen and is a corrosive irritant to eyes, skin, and mucous membranes. Appropriate safety precautions should be taken during all stages of its handling and disposal.*

To Z-Pro-Ser-OMe (7.0 g, 20 mmol) in MeOH (20 mL), hydrazine hydrate (4 mL, 80 mmol) was added. After 1 h at rt, the precipitate was collected and from the mother liquid a second fraction was obtained by precipitation with Et_2O. The combined fractions were recrystallized (EtOH/Et_2O); yield: 7.0 g (100%); mp 185–186 °C; $[\alpha]_D^{22}$ −76.8 (*c* 1, AcOH).

Boc-Lys(Boc)-Glu(OtBu)-Thr-Ala-Ala-Ala-Lys(Boc)-$NHNH_2$:[83]

CAUTION: *Hydrazine hydrate ($H_2NNH_2 \cdot H_2O$) is a poison upon ingestion and by intravenous routes. It is an experimental carcinogen and is a corrosive irritant to eyes, skin, and mucous membranes. Appropriate safety precautions should be taken during all stages of its handling and disposal.*

To a soln of Boc-Lys(Boc)-Glu(OtBu)-Thr-Ala-Ala-Ala-Lys(Boc)-OMe (1.5 g, 14 mmol) in MeOH (100 mL) hydrazine hydrate (1 mL, 20 mmol) was added. After 20 h at rt, the mixture was cooled and the precipitate was collected, washed with ice-cold MeOH and dried over concd H_2SO_4 and KOH; yield: 1.2 g (80%); mp 229–231 °C.

Boc-Ala-Ala-Ala-Lys(Boc)-Phe-Glu(OtBu)-Orn(Boc)-Gln-His-$NHNH_2$:[84]

CAUTION: *Hydrazine hydrate ($H_2NNH_2 \cdot H_2O$) is a poison upon ingestion and by intravenous routes. It is an experimental carcinogen and is a corrosive irritant to eyes, skin, and mucous membranes. Appropriate safety precautions should be taken during all stages of its handling and disposal.*

To a soln of the corresponding nonapeptide methyl ester (0.340 g, 0.245 mmol) in DMF (15 mL) hydrazine hydrate (0.40 mL, 8 mmol) was added. The mixture was kept for 1 h at 50 °C and for additional 7 days at rt. After the addition of Et_2O, the resulting precipitate was collected washed with Et_2O and dried; yield: 0.312 g (92%); mp 224–226 °C (dec); $[\alpha]_D^{20}$ −24.3 (*c* 1, AcOH).

3.1.1.2 Peptide Hydrazides from Activated Carboxylic Acids

Peptides with unprotected C-terminal carboxy groups can be converted to the related hydrazides after esterification, e.g. with diazomethane[85] or following activation with 1-chloro-2,3-epoxypropane followed by hydrazinolysis.[86] Peptide hydrazides are also obtained by coupling of free carboxy groups via DCC/HOBt[87] or mixed anhydrides[88] with hydrazine or better with a suitably N′-protected hydrazine.[89] The latter procedures are particularly recommended for peptides containing base-labile protecting groups or base-sensitive sequences (see Section 3.1.1.1).

Boc-Asp(OBzl)-Leu-Lys(Boc)-$NHNH_2$:[87]

CAUTION: *Hydrazine (H_2NNH_2) is a highly toxic and potentionally explosive compound. The operation should only be conducted in a well-ventilated hood and behind a safety shield.*

To Boc-Asp(OBzl)-Leu-Lys(Boc)-OH (7.0 g, 10 mmol) in DMF (25 mL) anhyd hydrazine (0.38 mL, 12 mmol), HOBt·H_2O (3.4 g, 22 mmol) and finally DCC (2.3 g, 11 mmol) were added at 0 °C. The mixture was stirred in ice water for 2 h and at rt overnight. The DCU was filtered off and the solvent was removed from the soln. The residue was dissolved in EtOAc and washed with H_2O, the organic layer dried (Na_2SO_4) and the solvent removed. The product was recrystallized (MeOH/Et_2O); yield: 5.1 g (71%); mp 133–137 °C; $[\alpha]_D^{25}$ −19.4 (*c* 1, THF).

Z-Glu(OtBu)-Lys(Tfa)-Asp(OtBu)-$NHNH_2$:[89]

To a soln of Z-Glu(OtBu)-Lys(Tfa)-Asp(OtBu)-OH (2.0 g, 2.73 mmol) and Trt-$NHNH_2$·HCl[90] (0.93 g, 3.0 mmol) and TEA (0.42 mL, 3.0 mmol) in THF at −10 °C was added HOSu (0.645 g, 5.6 mmol) and DCC (0.58 g, 2.8 mmol). The mixture was kept at 4 °C for 12 h. The precipitate was filtered off and the solvent was removed from the soln. The residue was taken up in EtOAc and washed with aq citric acid, aq Na_2CO_3 and H_2O; the soln was dried (Na_2SO_4) and the solvent removed. The product was precipitated from EtOAc with Et_2O; yield: 1.9 g (70%); mp 177–179 °C; $[\alpha]_D^{25}$ −21.5 (*c* 1, MeOH).

The trityl-derivative (1.5 g, 1.52 mmol) was suspended in 50% aq AcOH and heated for 3 min in a boiling water bath. The soln was cooled to rt and the solvent removed. The residue was dissolved in EtOAc and washed with aq Na_2CO_3 and H_2O. The soln was dried (Na_2SO_4) and concentrated, and the product precipitated with Et_2O; yield: 0.7 g (62%); mp 191–192 °C; $[\alpha]_D^{25}$ −28.3 (*c* 1, MeOH),

3.1.1.3 Hydrazinolysis of Resin-Bound Peptides

Peptide hydrazides are readily obtained by hydrazinolysis of suitably protected peptides linked to the resin via a benzyl ester.[47,48] Alternative to the hydroxymethyl ester linkage of the Merrifield resins, 1-methyl-2-oxo-phenylethyl ester,[91] 2-nitrobenzyl ester[51] or 4-methoxybenzyl ester type linkages are also utilized.[50,92] Recently, fully protected hydrazides were obtained in good yields on 2-methoxy-4-alkoxybenzyl alcohol resin (Sasrin™) by hydrazinolysis with 20% anhydrous hydrazine in DMF; using DMA as solvent the reaction was significantly slower, but with less side products.[93,94]

H-D-Phe-Ile-D-Pip-Pip-D-(Me)Phe-Pro-NHNH$_2$:[91]

CAUTION: *Hydrazine (H_4N_2) is a highly toxic and potentionally explosive compound. The operation should only be conducted in a well-ventilated hood and behind a safety shield.*

The Fmoc-D-Phe-Ile-D-Pip-Pip-D-(Me)Phe-Pro-(PAM)-resin (1.5 g) with a peptide loading of 0.59 mmol/g was stirred with 1:1 MeOH/hydrazine (60 mL) under N_2 at rt for 1 h and at 50 °C for 30 min. Excess hydrazine and MeOH were removed at 50 °C, and MeOH (60 mL) was added and removed; the process was repeated three times. The residue was then suspended in MeOH (60 mL) and filtered. The filtrate was removed and the residue dissolved in BuOH (100 mL) and extracted with H_2O (3×50 mL) to remove traces of hydrazine. The organic layer was removed and the residue reevaporated from MeOH (100 mL); yield: 0.58 g (75%).

3.1.1.4 Protected Hydrazides

The synthesis of peptide hydrazides by hydrazinolysis of suitable esters has some disadvantages derived from base-sensitive side-chain protecting groups, e.g. benzyl esters, and base-sensitive amino acid sequences. By the use of N′-protected hydrazides a strategy was developed to circumvent these problems.[53,54] For this synthetic approach N′-protected amino acid hydrazides are generally used as starting compounds for the synthesis of suitable fragments, although even smaller subfragments may be conveniently converted into the N′-protected hydrazides. The N′-protected amino acid hydrazides are prepared by reaction of N^{α}-protected amino acids with mono-substituted hydrazines using the mixed anhydride or DCC in presence of 1,2-dinucleophiles as coupling reagents.[27,53,54] For histidine residues generally the azide procedure was used.[95] After selective N^{α}-deprotection, the peptide segments are assembled by standard peptide chemistry followed by cleavage of the hydrazide protecting group and conversion into the azide.

For the design of synthetic routes based on protected hydrazides the appropriate choice of the protecting groups for the α-amino, *N′*-hydrazide and side-chain functionalities is of critical importance, although the possibility offered by the azide procedure to apply the strategy of minimum protection significantly facilitates the problem of an orthogonal protection scheme. In Table 1 different examples of such protection schemes are listed.

for references see p 440

Table 1 Protection Schemes for Synthesis of Peptides with N′-Protected Peptide Hydrazides

N′-Hydrazide-Protection	N^{α}-Amino-Protection	Side-Chain Amino-, Hydroxy- and Carboxyl-Protection	Ref
Z	Boc, Bpoc	For, Phth, OtBu	[53,54,96,97]
Boc	Z	For, OBzl, OtBu	[98–101]
Trt	Bpoc, Z	Tfa, OtBu	[90]
Boc	Z	Tfa, OtBu	[27]
Tfa	Nps, Z	–	[102]
For	Z	–	[103]
Troc	Z(OMe), Boc	Z, OBzl, OtBu	[104–106]
iNoc	Boc	OBzl, OPic	[107]

In cases where azide coupling fails due to unsatisfactory yields, slow reactions or formation of side products, conversion of the hydrazide into the corresponding free carboxy group can be achieved with *N*-bromosuccinimide.[108] This procedure cannot be used in the presence of sensitive amino acids such as tryptophan, methionine or tyrosine. A milder procedure was proposed by Greenlee et al.[109] which is based on the hydrolysis of unsubstituted carboxamides and hydrazides into the corresponding carboxylic acids by the use of acidic resins under mild conditions. Conversely, there are cases where conversion of esters into N′-protected hydrazides may be convenient; for example, in the total synthesis of 4-epi-A83586C a diphenylmethyl ester group was temporarily converted into a Boc-hydrazide, to avoid diketopiperazine formation during deprotection of the Fmoc-protected amine.[110] This result agrees with the observation of Ward et al.[111] that formation of diketopiperazine can be avoided in the synthesis of H-(Me)Val-(Me)Ala-OBzl by converting the compound into the corresponding Boc-hydrazide.

Z-Lys(Tfa)-NHNH-Boc:[27]
Method A: With DCC: To Z-Lys(Tfa)-OH (2.0 g, 5.3 mmol) and Boc-$NHNH_2$[112] (0.7 g, 5.3 mmol) in EtOAc (50 mL) was added DCC (1.0 g, 5.3 mmol) at 0 °C. After 12 h at 0 °C, DCU was filtered off and the soln washed with 0.5 M citric acid, 0.5 M Na_2CO_3 and H_2O. The organic layer was dried (Na_2SO_4), concentrated and the product precipitated with petroleum ether and reprecipitated from EtOH/petroleum ether; yield: 2.4 g (92%); mp 81–82 °C; $[\alpha]_D^{20}$ −24.0 (*c* 1, MeOH).

Method B: With Mixed Anhydrides: A mixed anhydride, prepared from Z-Lys(Tfa)-OH (15.0 g, 40 mmol) in THF (200 mL) with NMM (4.35 mL, 40.0 mmol) and isobutyl chloroformate (5.2 mL, 40.0 mmol) at −20 °C in 2 min, was added with stirring to an ice-cold soln of Boc-$NHNH_2$ (7.93 g, 60.0 mmol) in THF (50 mL). Stirring was continued in an ice bath for 30 min and at rt for 1 h. The bulk of the solvent was removed and the product worked up as described above; yield: 18.5 g (94%).

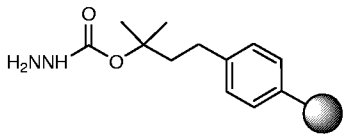
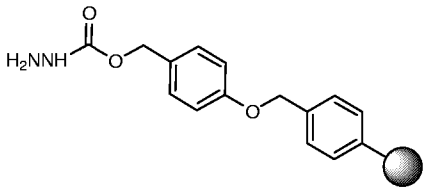
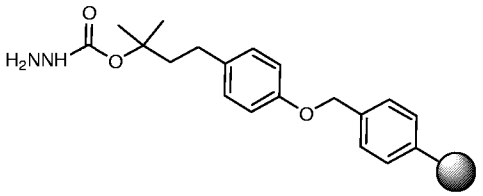
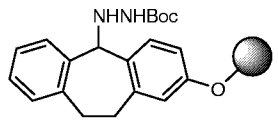
Z-His-NHNH-Boc:[95]
To a soln of Z-His-$NHNH_2$[79] (15.1 g, 49.8 mmol) in DMF (200 mL) and 4.8 M HCl in dioxane (41.50 mL, 199.12 mmol), *tert*-butyl nitrite (6.52 mL, 54.76 mmol) was added at −20 °C. The mixture was stirred for 10 min at −20 °C, cooled to −60 °C and then neutralized with TEA (28.1 mL, 200 mmol). To the azide soln Boc-$NHNH_2$ (13.0 g; 100 mmol) was added. After 24 h at 5 °C, the bulk of the solvent was evaporated and the residue distributed between EtOAc and H_2O. The organic layer was washed with 0.5 M Na_2CO_3 and H_2O, dried (Na_2SO_4) and the solvent was removed. The residue was recrystallized (EtOAc/pentane); yield: 12.6 g (61.5%); mp 90–93 °C; $[\alpha]_D^{25}$ −28.4 (*c* 1, MeOH).

Z-Pro-Gly-NHNH-Boc[99,113]
A soln of Z-Pro-Gly-OH (30.6 g, 100 mmol) and TEA (14.0 mL, 100 mmol) in THF (250 mL) was cooled to −10 °C and treated with ethyl chloroformate (9.6 mL, 100 mmol). After 10 min Boc-$NHNH_2$ (13.2 g, 100 mmol) was added and the mixture stirred for 5 h at rt. The solvent was removed, the residue dissolved in EtOAc (300 mL) and H_2O (150 ml), the organic layer washed with H_2O, 1 M NH_4OH and dried (Na_2SO_4). The solvent was removed and the residue recrystallized [Et_2O (75 mL)]; yield: 19.5 g (46%); mp 119 °C; $[\alpha]_D^{21}$ −30 (*c* 2, DMF).

3.1.1.5 Solid-Phase Synthesis of Peptide Hydrazides

To utilize the N′-protected hydrazide approach for solid-phase synthesis, different hydrazide linkers for polystyrene resins were developed by Wang and Merrifield[55–57] (Table 2). Generally, the first N^{α}-protected amino acid is attached to the linker by treatment with DCC in presence of additives (HOBt, HOSu). The Bpoc group[114,115] which is readily cleaved with 0.5% TFA in dichloromethane in 15 minutes at room temperature, was used as temporary N^{α}-protection. After completion of the peptide synthesis on the solid support, the hydrazides are cleaved from the resin with 50% TFA in 30 minutes. The dibenzocycloheptadienyl-hydrazide linker allows the synthesis of protected hydrazides by the Fmoc/tBu strategy,[116,117] as the hydrazides are cleaved from the resin with 1% TFA in dichloromethane in less than 3 hours.

Table 2 Hydrazide Linkers for Solid-Phase Synthesis of Peptide Hydrazides

Structure	Linker Name	Ref
	1,1-dimethylpropyloxycarbohydrazide resin	[54]
	*p*-methoxybenzyloxycarbohydrazide resin	[55]
	3-(*p*-benzyloxyphenyl)-1,1-dimethylpropyloxycarbohydrazide resin	[56]
	dibenzocycloheptadienyl-*tert*-butyloxycarbohydrazide resin	[116]

3.1.2 Peptide Synthesis by the Azide Procedure

The first *N*-acylamino acid azides were synthesized according to procedures developed by Curtius.[4] The reactions were carried out under acidic conditions, such as aqueous acetic acid, hydrochloric acid or mixtures of both. Low temperature solutions of sodium nitrite were used as oxidizing agents.[118] These older procedures have been comprehensively described[119] and are presently used mainly in the preparation of reagents like Boc-N_3 (see Section 2.1.1). Since the sodium nitrite procedure when applied to peptide hydrazides, is cumbersome and readily accompanied by side reactions, alternative methods were developed by Honzl and Rudinger[120] which are based on the use of alkyl nitrites. With these

for references see p 440

reagents conversion of hydrazides to azides is performed in organic solvents, thus even the solubility problems are largely bypassed.

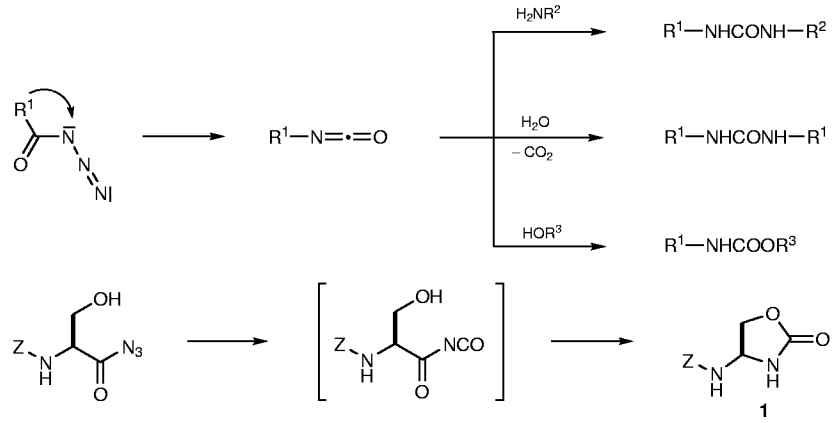
Generally, the acylation step is carried out immediately after production of the azide without its isolation, at low temperature (0–5 °C) by maintaining the pH of the mixture between 7 and 8 with amines such as TEA and DIPEA. Due to the relatively low reactivity of the azides it is advisable to carry out the fragment couplings at the highest possible concentration, while cyclization reactions require the principle of high dilution to be applied. DMF and DMA are used as solvents, and in cases of poor solubility HMPA and DMSO or related mixtures are employed. In cases where the minimum protection strategy is applied, water/organic solvent mixtures are used due to the high stability of azides towards hydrolysis. Alcohols and phenols are not suitable solvents for azide couplings, as urethanes (from Curtius rearrangement to isocyanates) or esters can be formed.[28] Due to the slow reaction rates during segment condensations, it is essential to use highly purified solvents such as DMF to avoid the presence of amines which would act as competitors in the reaction.

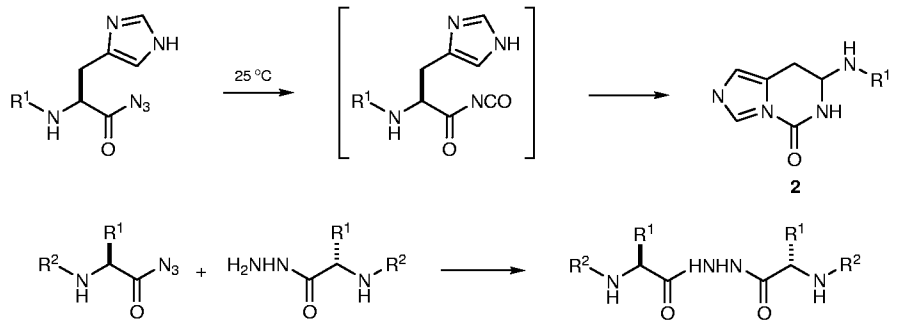
The mild acylation conditions enable condensation of segments with minimal side chain protection. Side-chain protection of serine, threonine, tyrosine, histidine, arginine, aspartic and glutamic acid is not required; similarly the C-terminal carboxy group of the amino component does not need any protection. Only side-chain amino and thiol groups together with the α-amino group of the azide component have to be protected.[121]

Side Reactions by the Azide Coupling Procedure

The isocyanate formation via Curtius rearrangement is one of the major side reactions which may occur in azide coupling steps.[36–39] As shown in Scheme 2, the isocyanates react further to form secondary contaminants, e.g. with amines ureido bonds are formed, and urethanes originate from the reaction with hydroxy groups.[37] Symmetrical ureas are generated by hydrolysis of the isocyanates to amines and further reaction with isocyanides. Moreover, Z-Ser-N_3 may generate the 2-oxotetrahydro-1,3-oxazole **1**,[122,123] and Z-His-N_3 via Curtius rearrangement yields a 7-benzyloxycarbonylamino-7,8-dihydroimidazo[1,5-*c*]pyrimidin-5(6*H*)-one **2**.[124,125] However, by maintaining the reaction conditions below 0 °C, this undesired side reaction can be greatly reduced. Since isocyanates exhibit a characteristic IR-band at 2220 cm^{-1} and azides at 2100 cm^{-1}, the side reactions which are provoked by a Curtius rearrangement can readily be monitored.[126] If conversion of hydrazides into the corresponding azides is not quantitative, symmetrical bis-acylhydrazides are generated.[127,128]

Scheme 2 Side Reactions of the Azide Coupling Procedure[37,122–125]



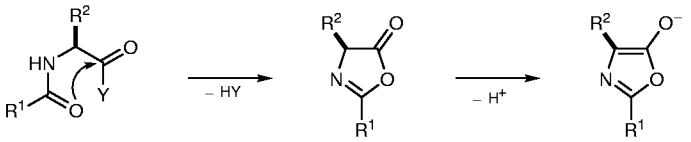


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Racemization by the Azide Coupling Procedure

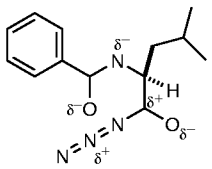
The azide procedure for peptide synthesis and particularly for fragment condensations is considered to be a mainly racemization free method. This low racemization tendency of azides was explained by several theories, which have been reviewed.[129] The most plausible cause of racemization is the formation of oxazoles (Scheme 3) and the related enolization. In presence of bases the α-carbon proton is readily abstracted to form an anionic oxazol-5(4*H*)-one resonance system.[15] For the formation of the oxazol-5(4*H*)-one the influence of the substituent Y on the α-carbonyl is essential. Since the α-carbonyl group of amino acid azides are less activated and thus relatively insensitive to oxygen containing nucleophiles such as water and alcohols, oxazol-5(4*H*)-one formation is largely prevented. It was proposed that the soft electron shell of the azide shields the α-carbonyl atom, so that only strong nucleophiles can attack it.[15,16] The reactivity towards amines can be explained in a manner analogous to the aminolysis of anchimerically assisted active esters.[130–132]

Scheme 3 Formation of the Oxazole Ring System[15]



An alternative explanation was derived from the model compound, *N*-benzoylleucyl azide **3** in which the oxygen anion of the benzoyl-carbonyl group interacts electrostatically with the central nitrogen atom of the azide group (Scheme 4).[133] With this conformation the oxazol-5(4*H*)-one formation is hindered, as in order for nucleophilic attack to occur at the α-carbonyl group, the bond between the α-carbonyl and the α-carbon atom has to be rotated by 90°.

Scheme 4 Proposed Electrostatic Stabilization of *N*-Benzoylleucyl Azide (**3**) which Prevents Oxazol-5(4*H*)-one Formation[133]



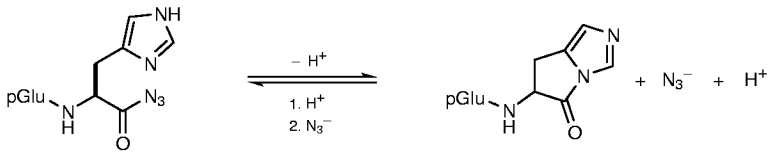
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for references see p 440

Despite the general observation that the azide procedure is almost racemization free, examples were reported where racemization occurred to some extent, particularly in presence of excess of base.[28,134] Contrary to reports which proved that no racemization occurs by treatment with excess of base, it is recommended to avoid any additional base.[28,31,33,135] This was confirmed by detailed studies on model compounds.[30,31,136] Aside from the amount of base, its type plays a significant role in racemization, as is well demonstrated in model studies.[29] Although *N*-methyl- and *N*-ethyl-morpholine cause the least amounts of racemization, the reaction rates were three times slower than in the presence of TEA or DIPEA; therefore these bases are not suitable for large segment couplings in spite of their good racemization properties. It was also reported that 1-(diethylamino)propan-2-ol is an appropriate base for azide coupling.[34] Thus, DIPEA and 1-(diethylamino)propan-2-ol are the most suitable bases for the azide method. Even the slightest excess of base has to be avoided during peptide couplings at any time. For the first neutralization step the use of stoichiometric amounts of a strong base is recommended, whereas for maintaining a pH of 7 to 7.5 during the coupling reaction weaker bases such as NMM are advisable. Despite the reports of racemization in azide coupling steps, this method when applied with the due precautions still represents the procedure with the least racemization.[137]

For decades, the histidine residue has been incorporated into peptides mainly via the azide procedure, and consequently its chemistry has been extensively studied. During azide couplings with unprotected histidine under basic conditions unusually high racemization of the histidine was observed.[138,139] The azide may react with the imidazole nitrogen to form an equilibrium with an imidazolide as shown in Scheme 5, with loss of chirality. In the synthesis of thyroliberin analogues the equilibrium between pGlu-His-N_3 and the corresponding imidazolide was analyzed by Veber[139] and it was found to be promoted by bases and reversed by weak acids such as HOSu.

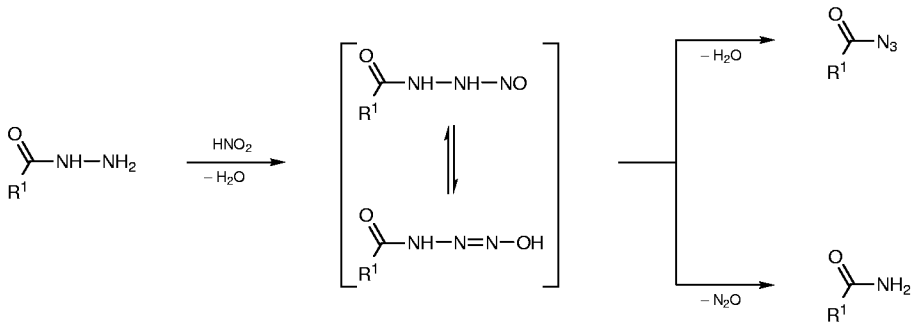
Scheme 5 Equilibrium between pGlu-His-N_3 and the Imidazolide Compound[139]



3.1.2.1 Azide Formation with Alkyl Nitrites

A major side reaction in the synthesis of azides is the formation of amides. This was detected for the first time in the preparation of Z-Lys(Z)-N_3 via the $NaNO_2$/HCl procedure where Z-Lys(Z)-NH_2 was obtained as a side product.[140,141] Similarly, in the case of Z-Cys(Bzl)-N_3 the related amide is readily formed.[142] As this side reaction does not occur when the azide is prepared from the acid chloride and sodium azide, it was proposed that the mechanism involves cleavage of the nitrosated hydrazide as shown in Scheme 6.[120]

Scheme 6 Amide Formation during the Synthesis of Azides[120]



It was shown that amide formation can be suppressed by the appropriate choice of reaction conditions, such as homogeneous solutions, e.g. in DMF, DMSO, HMPA, high acidity and low temperatures (−30 to −5 °C). If the peptide hydrazide contains acid-labile protecting groups, e.g. trityl, 2-(4-biphenyl)propyloxycarbonyl, *tert*-butyloxycarbonyl, *tert*-butyl esters and ethers, the temperature has to be kept below −20 °C. Generally, *tert*-butyl or butyl nitrite is used as the organic nitrite, and for sterically hindered peptides amyl nitrite is employed. The time required for full conversion of the hydrazide into the azide may vary between 5 to 30 min and can be monitored by spray reagents (see experimental procedure below).

Besides amide formation and partial removal of acid-labile protecting groups, nitration of sensitive amino acid residues such as tyrosine and tryptophan has been reported.[143,144] Sulfoxide formation in *S*-benzylcysteine and methionine residues was also detected.[120]

Z-Thr-Glu-Phe-Lys(Tfa)-Ala-Gly-Ser-Ala-Lys(Tfa)-Lys(Tfa)-NHNH-Boc:[27]
To Z-Thr-Glu-Phe-Lys(Tfa)-$NHNH_2$ (1.18 g, 1.5 mmol) in DMF (30 mL) at −15 °C, was added 5.8 M HCl in dioxane (1.0 mL, 5.8 mmol) followed by *tert*-butyl nitrite (0.19 mL, 1.6 mmol). The reaction mixture was stirred for 20 min at −15 °C, cooled to −60 °C and neutralized with TEA (1.26 mL, 9.0 mmol). To this soln of the azide, an ice cold soln of H-Ala-Gly-Ser-Ala-Lys(Tfa)-Lys(Tfa)-NHNH-Boc acetate [obtained by hydrogenation of the related Z-derivative (1.0 g, 1.0 mmol) in DMF in presence of acetic acid and precipitation with Et_2O] in DMF was added and the resulting mixture was kept at 5 °C for 48 h at pH 7–7.5. The bulk of the solvent was removed and the crude product was distributed in a countercurrent manner between BuOH equilibrated with 5% AcOH (6×100 mL), and 5% AcOH equilibrated with BuOH (12×100 mL). The BuOH layers were combined and evaporated. The product was precipitated twice from DMF with Et_2O; yield: 1.24 g (77.4%); mp 215–216 °C; $[\alpha]_D^{25}$ −10.65 (*c* 1, DMF).

(The quantitative conversion of hydrazides into azides can be assessed by spraying spots of the reaction mixture on paper taken at time intervals with the hydrazide test solution, which is a freshly prepared 1:1 mixture of 0.3 M $FeCl_3$ in 0.1 M AcOH and 0.2 M potassium ferricyanide in 0.1 M AcOH. Hydrazides give an intensive blue color and brown colored spots on quantitative formation of the azides. Phenols, i.e. unprotected tyrosine, also yields blue colored spots.)

3.1.2.2 Azide Formation with Tetrabutylammonium Nitrite

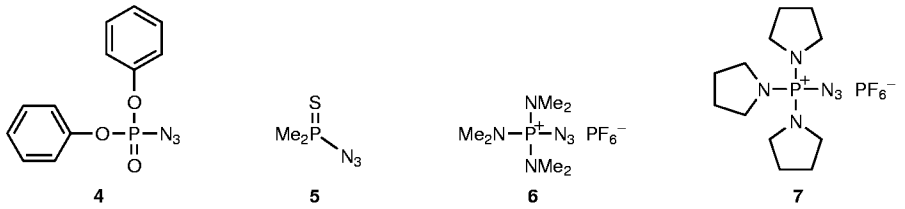
More recently another modification for the preparation of peptide azides was introduced by Alfeeva et al.[145] using tetrabutylammonium nitrite as auxiliary reagent. In contrast to the alkyl nitrites which are relatively unstable and therefore have to be purified prior to use by distillation, tetrabutylammonium nitrite is a crystalline and stable compound, which is soluble in anhydrous dipolar aprotic solvents. Moreover, in this procedure the acidity of the reaction mixture is adjusted with anhydrous *p*-toluenesulfonic acid instead of HCl in anhydrous organic solvents. These conditions are experimentally convenient and more easily controlled than those of the Honzl–Rudinger method. Comparative model reactions performed with *tert*-butyl nitrite and tetrabutylammonium nitrite produced nearly identical peptide yields. To date, there are no reports of the condensation of larger fragments and peptide cyclization by this azide procedure.

for references see p 440

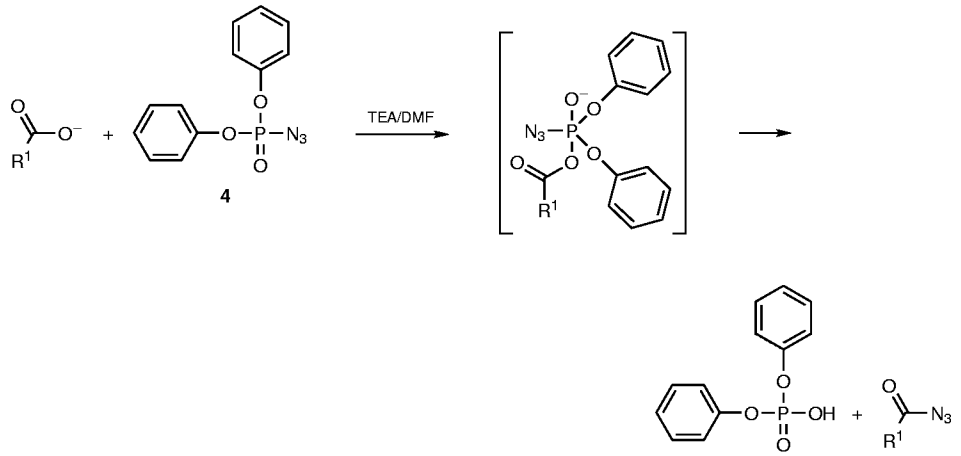
3.1.2.3 Azide Formation with Phosphorus Compounds

A very convenient way to apply the azide coupling method and bypass the intermediate step of hydrazide formation is the use of phosphorus azide reagents for in situ conversion of carboxylic acids into azides (Scheme 7). The most commonly used reagent of this class of compounds is the diphenyl phosphorazidate (DPPA, **4**) developed by Shioiri and Yamada.[40–43] The reaction involves the conversion of a carboxylic group into an azide by treatment with DPPA in the presence of a base (Scheme 8). In a detailed study of the synthesis of model dipeptides with DPPA practically no racemization could be detected using the Youngs racemization test (see Section 7.3.1).[146,147] In addition the yields for the dipeptides were high and side reactions were not observed.[40] DPPA is also useful in the preparation of urethanes by reaction with alcohols.

Scheme 7 Structures of Diphenyl Phosphorazidate (DPPA, **4**),[40] Dimethylphosphinothioylazide (MPTA) (**5**),[153] Azido tris(dimethyl)phosphonium Hexafluorophosphate (**6**)[156] and Azidotrispyrrolidinophosphonium Hexafluorophosphate (**7**)[155]



Scheme 8 Preparation of Peptide Azides by the DPPA Method[41]



Due to the facile application and the very low degree of racemization, the DPPA method is particularly useful in cyclization reactions of peptides; in some cases an inorganic auxiliary base K_2HPO_4,[39] $NaHCO_3$, or Na_2CO_3[46,148] was found to be more convenient than organic bases. Comparative studies on different coupling reagents for cyclization reactions[149,150] and for sterically hindered amide bond formation[151] clearly confirmed the efficiency of DPPA. However, in comparison to the recently developed HOAt-derived coupling reagents, the DPPA method is less efficient.[152]

The coupling reagent dimethylphosphinothioylazide (MPTA, **5**) was introduced by Ueki and co-workers[153,154] In contrast to DPPA which is an oil and somewhat difficult to handle, MPTA is a stable crystalline powder (Scheme 7). Compared to DPPA, MPTA leads to considerably faster reaction rates, but retains the same properties as DPPA in terms of lack

of racemization. However, BOP leads to much faster reaction rates, but also to higher amounts of racemization. Although MPTA exhibits more convenient properties than DPPA as a coupling reagent, it has not been widely utilized.

The compound azidotris(dimethyl)phosphonium hexafluorophosphate (**6**) and the corresponding pyrrolidino homologue **7** (Scheme 7) are reported to have a similar coupling potential as DPPA. However, not many applications of these compounds are reported to date.[155,156]

Z-Ser-Asp(OtBu)-Ala-Val-Phe-OMe:[157]

CAUTION: *Benzene is carcinogenic!*

To a mixture of Z-Ser-Asp(OtBu)-OH (548 mg, 1.28 mmol) and H-Ala-Val-Phe-OMe (405 mg, 1.16 mmol) in DMF (24 mL) was added DPPA (383 mg, 1.39 mmol) in DMF (3 mL) with ice cooling, followed by the addition of TEA (156 mg, 1.54 mmol) in DMF (3 mL). The mixture was stirred with ice-cooling for 6 h, and then at rt for 20 h. EtOAc/benzene (4:1; 1 L) was added, and the mixture was washed with 10 % citric acid (50 mL), H_2O (50 mL) and brine (50 mL). Drying followed by removal of solvent afforded a colorless solid, which was washed with Et_2O (50 mL) and recrystallized (MeOH); yield: 655 mg (77%); mp 211–213 °C; $[\alpha]_D^{20}$ −16.4 (*c* 1, DMF).

Z-Leu-Leu-Val-Phe-OMe:[158]

To a stirred soln of Z-Leu-Leu-OMe (0.39 g, 1 mmol) in DMF (10 mL) was added 2 M NaOH (0.5 mL, 1 mmol) with ice-cooling. The mixture was stirred at 0 °C for 2 h. To this mixture was successively added DPPA (0.30 g, 1.1 mmol), TEA (0.105 g, 1 mmol) and H-Val-Phe-OMe·HCl (0.32 g, 1 mmol) in DMF (5 mL). The mixture was stirred at 0 °C for 3 h and then at rt for 16 h. The mixture was washed with a mixture of Et_2O and hexane and the residue was recrystallized (EtOAc); yield: 401 mg (63%); mp 207–209 °C; $[\alpha]_D^{20}$ −25 (*c* 1.1, DMF).

***c*[Phe-D-Trp-Lys(Boc)-Thr(tBu)-Phe-Pro-]:**[39]

The hexapeptide H-Phe-D-Trp-Lys(Boc)-Thr(tBu)-Phe-Pro-OH (3.0 g, 3.06 mmol) in DMF (6 L) was reacted at 4 °C with TEA (0.44 mL, 3.16 mmol) and DPPA (1.09 g, 3.98 mmol). After 12 h at 4 °C, the mixture was taken to dryness and the residue precipitated from MeOH/H_2O. The product was dissolved in 99% MeOH, applied to a DEAE-Sephadex column equilibrated with 99 % MeOH containing 0.1% conc. NH_3. The product eluted with the solvent front; the solvent was removed from the eluate and the solid residue was washed with Et_2O; yield: 2.84 g (95%); mp 142–144 °C; $[\alpha]_D^{20}$ −46.3 (*c* 1, MeOH).

3.1.2.4 Transfer Active Ester Condensation Procedure (TAEC)

To enhance the reaction rate of azides independently of their method of preparation and to prevent related side reactions, in situ conversion of peptide azides into their corresponding *N*-hydroxysuccinimide esters was introduced by Hirschmann et al.[159] This approach was further extended by Wang et al.[116] by the use ethyl 1-hydroxy-1*H*-1,2,3-triazole-4-carboxylate (HOCt) or 7-aza-1,2,3-benzotriazol-1-ol (HOAt) as additives, to enhance yields and reaction rates while maintaining the advantages of minimal side chain protection. In this study it was shown that HOBt is not suitable for the TAEC method, although previous reports indicated that HOBt could catalyze cyclization reactions, which were carried out by the azide method.[160]

for references see p 440

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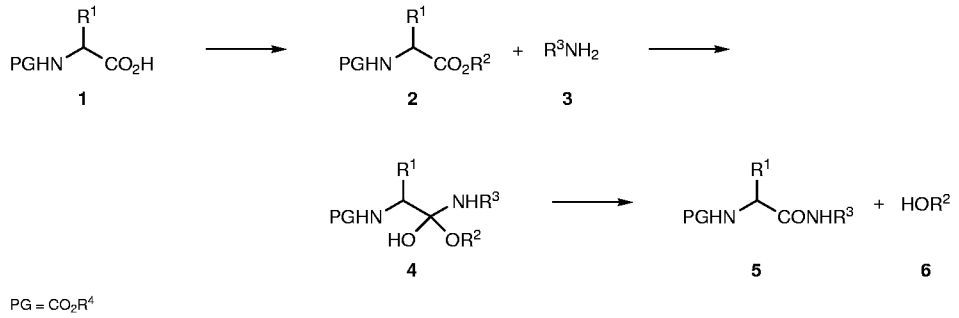
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3.2 Active Esters

N. L. Benoiton

Alkyl esters of carboxylic acids **1** are inert to amino groups. However, substitution of the alkoxy group with an electron-withdrawing moiety renders the carbonyl susceptible to nucleophilic attack by an amino-bearing molecule **3** because of its augmented electrophilicity. Ester **2** of enhanced reactivity is said to be an active ester. Most active esters are stable enough to be purified and to survive storage. They exhibit varying resistance to hydrolysis, some sufficient to allow their purification by aqueous washes or their use in water-containing solvents. Aminolysis of active esters occurs by a two-step process through formation of tetrahedral intermediate **4** whose breakdown is the rate-limiting step. The products of the reaction are amide **5** and alcohol **6** (Scheme 1).

Scheme 1 Peptide Bond Formation via Active Esters

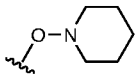
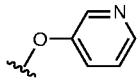
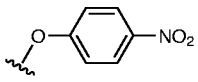
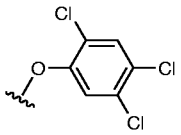
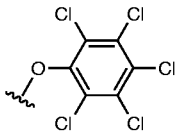
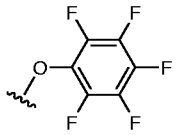
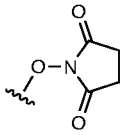
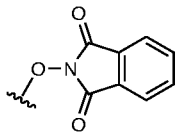
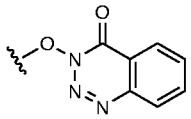
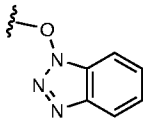


Recognition of the value of active esters for peptide bond formation emerged from work with vinyl esters **7**,[1] cyanomethyl esters **8**,[2] *S*-phenyl thioesters **9**,[3] piperidino esters **10**,[4] 3-pyridyl esters **11**,[5] 4-nitrophenyl esters **12**,[6] 2,4,6-trichlorophenyl esters,[7] and phthalimido esters **17**.[8] Many activating moieties have surfaced over the years but only a limited number have survived the test of time and are in use today. These include 4-nitrophenyl **12**,[6] 2,4,5-trichlorophenyl **13**,[9] pentachlorophenyl **14**,[10,11] pentafluorophenyl **15**,[12] succinimido **16**,[13] and 4-oxo-3,4-dihydrobenzotriazin-3-yl esters **18**[14] (Table 1). Benzotriazolyl esters **19**[15] are also well-known but they are too reactive for routine use.

Thus, two types of active esters are of interest: those formed from an acid and a substituted phenol (**12–15**) and those formed from an acid and a substituted hydroxylamine (**16–19**). Both types are reactive by virtue of the electron-withdrawing properties of the OR^2 moiety in **2**. The level of activation of the substituted phenyl esters varies directly with the electronic effect going from 4-nitrophenyl to 2,4,5-trichlorophenyl, pentachlorophenyl, and pentafluorophenyl, which corresponds with the increasing acidity of the phenols. A diminution in the rate of aminolysis is caused by the presence of a substituent in the *ortho* position of the ring.[9] An additional phenomenon contributes to the reactivity of the esters formed from substituted hydroxylamines, namely anchimeric assistance. Since the aminolysis of active esters is a bimolecular reaction, it is dependent on concentration and can be forced to completion by an excess of one of the reactants. Aminolysis is also characterized by a pronounced dependence on the polarity of the solvent in particular for the esters formed from substituted phenols, the half-life of a 2,4,5-trichlorophenyl ester in the presence of benzylamine being one hundred times less in dimethylformamide than in benzene.[9] Furthermore, aminolysis is catalyzed by mild acid such as acetic acid. The rate of aminolysis is slowed if the side chain of the active ester contains a β-methyl substituent.

for references see p 472

Table 1 Activating Moieties Found in Active Esters $PGHNCHR^1CO_2R^2$[1–6,8–15]

Compound Number	OR^2	Abbreviated Form	Ref
7	$OCH{=}CH_2$		[1]
8	OCH_2CN		[2]
9	SPh		[3]
10		OPip	[4]
11		O3Py	[5]
12		ONp	[6]
13		OTcp	[9]
14		OPcp	[10,11]
15		OPfp	[12]
16		OSu	[13]
17		ONPhth[a]	[8]
18		ODhbt	[14]
19		OBt	[15]

[a] NPhth = phthalimido.

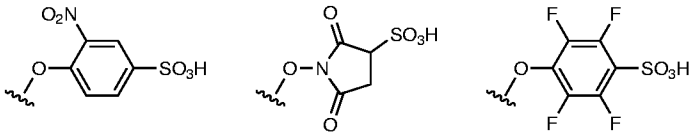
Active esters are chirally stable under the usual conditions of coupling in peptide synthesis, but with the single exception of piperidino esters[4] they may undergo isomerization if left in the presence of tertiary amines. In addition to their role as shelf-stable reagents, active esters are postulated as intermediates in carbodiimide-mediated reactions where a substituted hydroxylamine is added in order to suppress the side reaction of epimerization in the

coupling of peptides. The most common of these additives are *N*-hydroxysuccinimide,[16–18] 1,2,3-benzotriazol-1-ol,[15] 3-hydroxy-3,4-dihydrobenzotriazin-4-one,[14] and more recently 7-aza-1,2,3-benzotriazol-1-ol.[19,20] Active esters may also play a role as intermediates in coupling reactions mediated by phosphonium and uronium salts that contain the substituted hydroxylamine incorporated into the reagent and especially when the substituted hydroxylamine is added to the coupling mixture. The issue of the intermediacy of active esters in these reactions is addressed in Sections 3.7 and 3.8. A comprehensive account of the development of the active ester method of coupling including an exhaustive list of references is available in a review by Bodanszky.[21] Illustrative examples with experimental details for the preparation and use of active esters are available in the compendium edited by Wünsch.[22] References relating to the use of Fmoc derivatives are available in a review.[23]

3.2.1 Methods for Preparing Active Esters of *N*-Alkoxycarbonyl Amino Acids

Active esters of *N*-alkoxycarbonyl amino acids are prepared by two general approaches: activation of the acid by routine procedures followed by reaction with hydroxy compound HOR^2 that is present during the activation or added afterwards, and transesterification of the oxy moiety OR^2 from a carbonic, sulfinic, or trifluoroacetic acid ester to the *N*-alkoxycarbonyl amino acid. Activation is effected using a carbodiimide, a chloroformate that produces the mixed anhydride, or a chlorinating reagent that generates the acid chloride. Only carbodiimides and chloroformates are compatible with derivatives possessing acid-sensitive protecting groups. Products that crystallize readily are purified satisfactorily by crystallization. Others are routinely purified by passing them through a column of silica gel. Only in a few cases does the protocol involve preliminary purification by washing the compounds with aqueous solutions; this is puzzling, since similarly activated species such as symmetrical and mixed anhydrides and oxazol-5(4*H*)-ones are resistant enough to water to be purified by aqueous washes. There are conflicting reports concerning the stability to water of pentafluorophenyl esters, indeed, some have been purified by aqueous washes. Some esters, succinimido in particular, are used for acylation in partially aqueous solution so they have to have reasonable stability to water. A few active esters have been made soluble in water by incorporating sulfonic acid into the alcohol moieties, examples being 2-nitro-4-sulfophenyl **20**,[24,25] sulfosuccinimido **21**,[26] and tetrafluoro-4-sulfophenyl **22**[27] esters (Scheme 2). Other active esters have been made insoluble in all solvents by attaching *N*-hydroxysuccinimide[28] or 1,2,3-benzotriazol-1-ol[29] to a solid support, the support-bound alcohols HOR^2 liberated by aminolysis being readily disposed of by filtration.

Scheme 2 Activating Moieties OR^2 Rendering Esters $PGHNCHR^1CO_2R^2$ Soluble in Water[24–27]



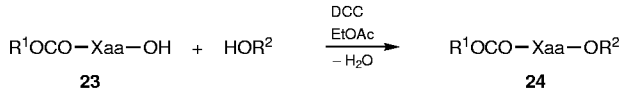
20 **21** **22**

for references see p 472

3.2.1.1 Synthesis of Active Esters with Carbodiimides

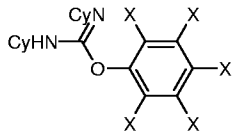
Active esters **24** are most frequently prepared from acid **23** and a hydroxy compound by straightforward esterification using *N,N'*-dicyclohexylcarbodiimide (Scheme 3).[30] The reagent is added to a mixture of the two reactants generally in ethyl acetate or dimethylformamide at 0–5 °C. The two reactants can compete for the diimide; this occasionally leads to generation of an activated side product that contaminates the desired ester. Cold dimethylformamide is used for asparagine and glutamine derivatives to minimize cyclization to the imide. Pyridine is used as the solvent for the *ortho*-substituted nucleophiles 2-nitrophenol[31] and 2-hydroxypyridine;[32] if pyridine is not used then the product is almost completely the *N*-acyl-*N,N'*-dicyclohexylurea resulting from oxygen-to-nitrogen acyl migration at the *O*-acylisourea stage. A variant of the procedure can be used to prepare pentachloro- and pentafluorophenyl esters; acid **23** is mixed with complex **25** comprised of *N,N'*-dicyclohexyl-*O*-pentahalophenylisourea and 2 equivalents of the pentahalophenol.[33]

Scheme 3 Preparation of Active Esters Using *N,N'*-Dicyclohexylcarbodiimide[30,33]



23 **24**

R^1 = tBu, Bzl, Fm; R^2 = Np, Su, Pfp, Dhbt



2 HOC_6X_5

25

X = Cl, F

The occasional preliminary purification of products by aqueous washes usually coincides with the use of pyridine or dimethylformamide as solvent. Esters such as succinimido and benzotriazolyl are often prepared in situ, meaning that they are not isolated but subjected to aminolysis immediately after the esterification reaction.[34–36] Slight decomposition can accompany the esterification of Boc amino acids, in particular if dichloromethane is used as solvent and the acidic EDC is used as reagent.[37] Publications with a major focus on the synthesis of active esters using DCC are listed in Table 2. A list of Fmoc-Xaa-OR^2 active esters that had been described to 1987 has been compiled.[53]

Table 2 Active Esters Prepared Using *N,N'*-Dicyclohexylcarbodiimide

Compound	Ref	Compound	Ref
Z-Xaa-ONp	[38,39]	Boc-Xaa-O2Py	[32]
Boc-Xaa-ONp	[40,41]	Z-Xaa-OPh(X)	[9]
Fmoc-Xaa-ONp	[42]	$(NO_2)_x$Ph-Ahx-ONs	[25]
Bpoc-Xaa-ONp	[43]	Boc-Xaa-OTcp	[40,44]
Boc-Xaa-O2Np	[31]	Z-Xaa-OPcp	[11]
Z-Xaa-OCH_2CN	[2]	Z-Xaa-OPfp	[45]
Z-Xaa-ONPhth[a]	[8]	Boc-Xaa-OPfp	[46]
Z-Xaa-OSu	[13]	Fmoc-Xaa-OPip	[47]
Boc-Xaa-OSu	[13]	Fmoc-Xaa-OPfp	[48,49]
Z-Xaa-OBt	[15]	Fmoc-(Fmoc-Hmb)Xaa-OPfp	[50]
Z-Xaa-OPip	[4]	Fmoc-Xaa-ODhbt	[51]
Z-Xaa-O3Py	[5]	Fmoc-Val-OR^2[b]	[52]

[a] NPhth = phthalimido.
[b] R^2 = Np, Su, Pfp, Dhbt.

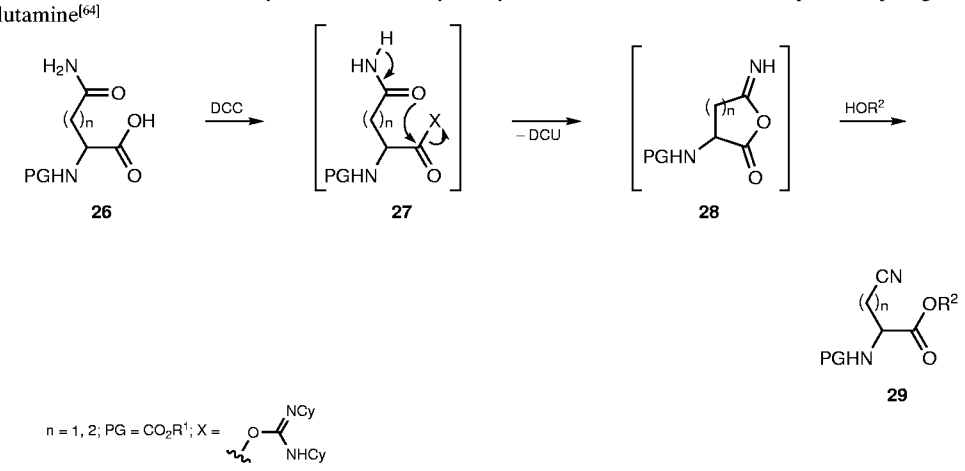
3.2.1.1.1 Nitrophenyl Esters

Amongst the 2- and 4-nitrophenyl, and dinitrophenyl esters examined, 4-nitrophenyl esters have been recommended as superior and are the only ones that are still in use. The corresponding 4-nitrophenyl Boc amino acid esters were used in the ground-breaking synthesis of oxytocin[38] as well as for the preparation of protected segments of secretin.[54,55] The former provided the first demonstration of the value of synthesis by the stepwise approach and was the procedure employed by Merrifield to acquire the tetrapeptide that served as the reference compound in the development of the solid-phase method of peptide synthesis.[56] The esters were used successfully in the solid-phase synthesis of several peptides,[57,58] including the A-chain of insulin.[59] Various 2-nitrophenyl esters, which are more reactive but with reaction rates that are less solvent-dependent, have also been used for solution[31] and solid-phase synthesis.[60] 4-Nitrophenyl esters were the first commercially available (1961) active esters. The 4-nitrophenyl esters of N^{α}-protected glutamine and asparagine are of particular interest, as their use circumvents the dehydration that occurs at the carboxamide groups in DCC-mediated couplings of the protected acids. These two 4-nitrophenyl esters are more readily purified by crystallization than the corresponding succinimido esters. 4-Nitrophenyl esters are usually prepared in ethyl acetate, pyridine, or dimethylformamide. Most are crystalline (see Table 3); removal of final traces of 4-nitrophenol occasionally presents a problem as it has a tendency to adhere to peptides. A variety of bis-4-nitrophenyl esters,[61] such as that of bis-Z-cystine,[62] have been used for cross linking peptides and proteins. 2-Nitro-4-sulfophenyl esters **20**, which are soluble in water, have been used for assembling a peptide chain[24] and for conjugation of monomers to biomaterials.[25,63] The sulfo-containing esters are more stable to hydrolysis than 4-nitrophenyl and succinimido esters.[25]

Side Reaction of Dehydration Involving Side Chains of Activated Residues

The carboxamide groups of asparagine and glutamine derivatives **26** are susceptible to dehydration during carbodiimide-mediated esterification at the α-carboxy groups (Scheme 4).[64] The reaction probably involves formation of aspartyl- or glutamylisoimide **28** by attack of the carboxamide carbonyl at the carbonyl of *O*-acylisourea **27** followed by a ring-opening rearrangement.[65] The resulting β/γ-cyano-substituted amino acid esters **29** are acylating agents and consequently will be incorporated into the chain that is being assembled. The impurities should be removed by repeated crystallizations of the parent esters from methanol or ethanol,[66] but caution should be exercised as the manipulation can generate the impurity. A cyclodehydration to piperidone **31** can also occur during the preparation of activated esters of arginine derivatives **30** (Scheme 5).[67]

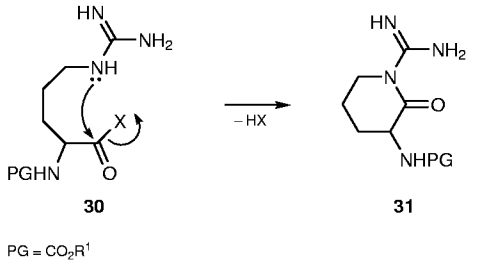
Scheme 4 Side Reaction of Cyano Formation by Dehydration of Carboxamide Groups of Asparagine and Glutamine[64]



26 **27** **28** **29**

for references see p 472

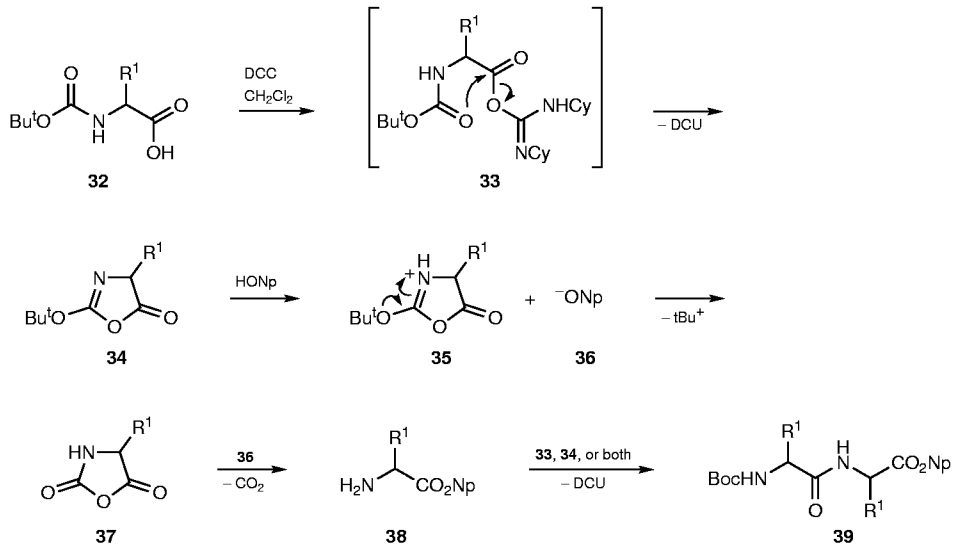
Scheme 5 Side Reaction of Piperidone Formation by Cyclodehydration of Activated Arginine during the Preparation of Active Esters[67]



Side Reaction of Decomposition in the Esterification of Boc Amino Acids

Slight decomposition at the Boc group can occur during esterification of Boc amino acids **32** with slower reacting phenols such a 4-nitrophenol. This occurs in particular if the solvent is dichloromethane and the reagent is EDC.[37] The decomposition is due to formation of a 2-*tert*-butoxyoxazol-5(4*H*)-one **34** from *O*-acylisourea **33**. Oxazolone **34** is sensitive to acid and undergoes protonation (to give **35**) and acidolysis generating *N*-carboxyanhydride **37** that reacts with nitrophenoxy anion **36** to give amino acid 4-nitrophenyl ester **38**, which is a good acceptor of activated Boc amino acids **33** and **34**. The result is formation of some Boc dipeptide ester **39** (Scheme 6). Decomposition does not occur in DCC-mediated reactions effected in the presence of an equivalent of *N*-methylmorpholine and when the carbodiimide hydrochloride mediated reactions are carried out in pyridine.[37]

Scheme 6 Boc Decomposition and Dimerization in the Esterification of Boc Amino Acids Using *N,N'*-Dicyclohexylcarbodiimide[37]



Z-Xaa-ONp; General Procedure:[38]

To a 0.2–0.5 M soln of Z-Xaa-OH in EtOAc was added a 20% excess of HONp. The calculated amount of DCC was added to the soln at 0 °C, which was stirred for 30 min. After 1 h at rt, the DCU was removed by filtration, the precipitate was washed with EtOAc, the filtrate and washings were combined and the solvent was removed. The residue was dissolved in hot EtOH and the crystals that formed on cooling were collected, washed with EtOH, and recrystallized (hot EtOH); yield: 85–90%. Lower yields were obtained for Z-Asn-ONp and Z-Gln-ONp, which were prepared in DMF and crystallized by the addition of water to the EtOH. Fmoc-Ser(tBu)-ONp and Fmoc-Thr-ONp required purification on a silica gel column.[42]

Table 3 R^1OCO-Xaa-ONp Prepared Using *N,N'*-Dicyclohexylcarbodiimide[40,42,43,66,68]

Compound	Crystallization Solvent[a]	Yield (%)	mp (°C)	Ref
Z-Gly-ONp	EtOH	87	127–128	[66]
Z-Lys(Boc)-ONp	Et_2O	85	88–90	[40]
Boc-Phe-ONp	EtOH/H_2O (4:1)	82	132	[40]
Boc-Asn-ONp	EtOH	69	178	[40]
Boc-His(Boc)-ONp	CH_2Cl_2/PE	65	142–144	[68]
Bpoc-Gln-ONp	acetone/PE	63	142–144	[43]
Bpoc-Tyr(Bzl)-ONp	acetone/PE	81	147–149	[43]
Fmoc-Leu-ONp	EtOH/H_2O/AcOH (94:5:1)	85	115–116	[42]
Fmoc-Lys(Boc)-ONp	EtOH/H_2O/AcOH (94:5:1)	71	105–106	[42]
Fmoc-Phe-ONp	EtOH/H_2O/AcOH (94:5:1)	76	207–209	[42]

[a] PE = Petroleum ether.

3.2.1.1.2 Succinimido Esters

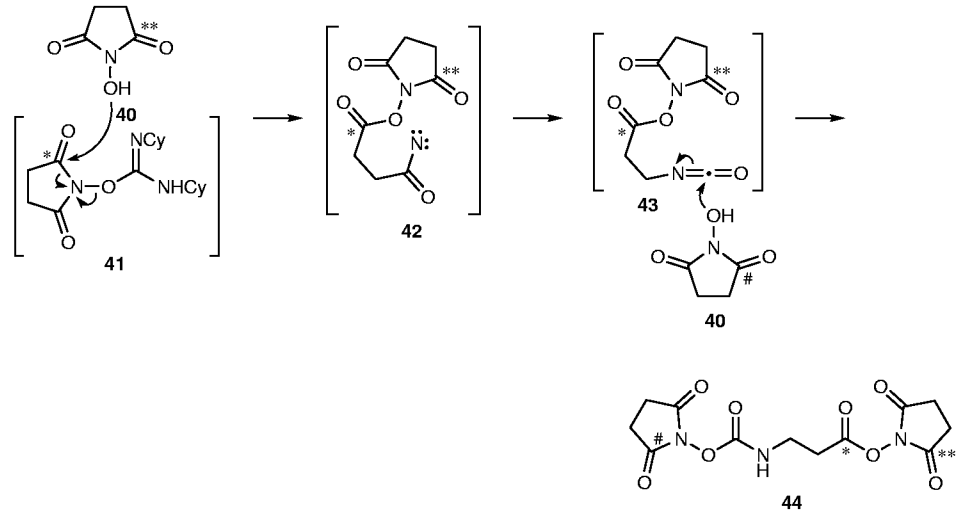
Succinimido esters (see Table 4) are the most popular of the hydroxylamine-based esters. Their development resulted from recognition by Anderson[13] of the good reactivity but decreased performance in partially aqueous solution of phthalimido esters[8] as the phthalimide liberated by aminolysis is insoluble in water. The esters have found large application due to their ease of preparation and purification (with the exception of derivatives of glutamine and asparagine), good reactivity, and ease of removal of the liberated *N*-hydroxysuccinimide that is soluble in water. They are prepared in dioxane,[13] 1,2-dimethoxyethane,[13] or acetonitrile[69] and generally recrystallized from propan-2-ol. *N*-Hydroxysuccinimide is neither carcinogenic, teratogenic, nor mutagenic. The esters are employed primarily for chain assembly of segments by stepwise synthesis in solution,[18,34] and are particularly suited for reaction with amino acids or small peptides with unprotected α-carboxy groups.[13,34,69] They have been used successfully for solid-phase synthesis[35] but are not generally suited for this approach.[70,71] Water-soluble sulfo-containing esters of dicarboxylic acids have been used for cross linking proteins.[26,72]

Side Reaction of Trimerization of *N*-Hydroxysuccinimide to an Active β-Alanine Derivative

DCC in tetrahydrofuran reacts with three molecules of *N*-hydroxysuccinimide (**40**) to give *N*-succinimidooxycarbonyl-β-alanine succinimido ester (**44**, >80% yield), which is a good acylating agent.[73] This compound occurs as a contaminant in succinimido esters that have been prepared by improper protocols and/or inadequately purified. Use of such products generates peptides contaminated with β-alanine. Formation of the undesired ester is explained by reaction of the carbodiimide with *N*-hydroxysuccinimide (**40**) giving nitrogen-linked isourea **41** that undergoes ring opening by attack by a second molecule of **40** to give acylnitrene **42**. The latter rearranges to isocyanate **43**, which combines with the third molecule of **40** producing the succinimidooxy moiety of β-alanine ester **44** (Scheme 7).[73]

for references see p 472

Scheme 7 Side Reaction of *N*-Succinimidooxycarbonyl-β-alanine Succinimido Ester Formation by Reaction of One Molecule of *N,N'*-Dicyclohexylcarbodiimide with Three Molecules of *N*-Hydroxysuccinimide[73]



Boc-Ala-OSu; Typical Procedure:[13]
To a soln of Boc-Ala-OH (1.32 g, 7 mmol) and HOSu (0.805 g, 7 mmol) in anhyd DME (10 mL) was added DCC (1.59 g, 0.77 mmol) with stirring and the soln was left at 0–5 °C for 20 h. The mixture was filtered and the filtrate was removed. The crystalline residue was recrystallized twice (iPrOH); yield: 1.42 g (71%); mp 143–144 °C.

Table 4 Z- and Boc-Xaa-OSu Prepared Using *N,N'*-Dicyclohexylcarbodiimide[13]

Compound	Crystallization Solvent[a]	Yield (%)	mp (°C)	$[\alpha]_D^{25}$ (*c* 2, dioxane)	Ref
Z-Phe-OSu	$CHCl_3$/PE	76	140–140.5	−17.3	[13]
Z-Leu-OSu	iPrOH	51	116–117	−33.6	[13]
Z-Glu(OMe)-OSu	iPrOH	53	107–108	−23.3	[13]
Z-Arg(Z)$_2$-OSu	EtOAc/PE	80	85–86	−9.7	[13]
Boc-Phe-OSu	iPr_2O/iPrOH	81	152–153	+20.9	[13]
Boc-Ile-OSu	iPrOH	61	91–92	−26.5	[13]
Boc-Cys(Bzl)-OSu	iPrOH	49	117–117.5	−54.0	[13]
Boc-Phe-OSu	iPr_2O/iPrOH	81	152–153	−19.0	[13]

[a] PE = petroleum ether.

3.2.1.1.3 Halogenated Phenyl Esters

Trichloro-[9,40,44] and pentachlorophenyl[11] esters (see Tables 5 and 6) have been used sporadically. The trichlorophenyl esters are slightly more reactive than 4-nitrophenyl esters and have the advantage that the liberated phenol is easier to remove; they are also resistant to catalytic hydrogenation. Pentafluorophenyl[12] esters (see Tables 7–9) are the most highly reactive and remain in use for specific purposes. They have the feature that their aminolysis is not impeded by the *ortho* substituent whose van der Waals radius is much smaller than that in the chlorophenyl esters. Boc derivatives have been used successfully for assembling decamers in solution.[74,75] Fmoc derivatives are suitable for solid-phase synthesis.[71] The monomers are prepared in dichloromethane,[11] ethyl acetate,[44,45] or dioxane/dimethylformamide.[46] Halophenyl esters find use for the temporary protection of a carboxy group during the synthesis of peptides by the backing-off procedure[11] and the O-glycosylation of hydroxyamino acid derivatives.[76] Pentafluorophenyl esters of appropriate derivatives **45**

(Scheme 8) are employed for the introduction of nitrogen alkylation into a peptide chain in order to suppress aggregation that interferes with chain assembly.[50] Pentafluorophenyl esters of Fmoc-cysteine derivatives are employed in solid-phase synthesis[77] to circumvent the enantiomerization that occurs during certain onium salt mediated couplings of the corresponding acids[77,78] which have to be carried out in the presence of tertiary amines. The same phenomenon may apply to serine derivatives.[79] Water-soluble tetrafluoro-4-sulfophenyl esters **22** are available.[27]

Table 5 Boc-Xaa-OTcp Prepared Using *N,N'*-Dicyclohexylcarbodiimide[40,44]

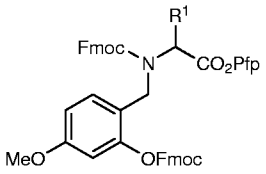
Compound	Crystallization Solvent	Yield (%)	mp (°C)	Ref
Boc-Asp(OBzl)-OTcp	EtOH/H_2O (24:1)	85	92	[40]
Boc-Asn-OTcp	EtOAc/Et_2O	69	182	[40]
Boc-Phe-OTcp	EtOH/H_2O (4:1)	73	122	[40]
Boc-Ala-OTcp	petroleum ether	76	85–86	[44]
Boc-Gln-OTcp	benzene	45	161–162	[44]
Boc-Lys(Z)-OTcp	EtOH	86	98–99	[44]
Boc-Met-OTcp	cyclohexane	82	91–91.5	[44]

Table 6 Z-Xaa-OPcp[a] Prepared Using *N,N'*-Dicyclohexylcarbodiimide[11]

Compound	mp (°C)	Ref
Z-Ala-OPcp	172–173	[11]
Z-Ser-OPcp	192–193	[11]
Z-Gln-OPcp	186–187	[11]
Z-Asp(OtBu)-OPcp	151–152	[11]
Z-Met-OPcp	132–133	[11]
Z-Phe-OPcp	158	[11]
Fmoc-Tyr(Bzl)-OPcp[39]	182–184	[11]

[a] Pcp = pentachlorophenyl.

Scheme 8 Pentafluorophenyl Esters for the Introduction of Nitrogen Alkylation into a Peptide Chain[50]



45

Boc-Xaa-OPfp; General Procedure:[46]

Boc-Xaa-OH (5 mmol) and HOPfp (5.5 mmol) were dissolved in DMF/dioxane (1:1) and the soln was cooled to 0 °C. DCC (5.5 mmol) was added and the mixture was stirred at 0 °C. After 1 h, the DCU was removed by filtration, the filtrate was removed, and the residue was crystallized (hexane, EtOAc/hexane, or Et_2O/hexane). The compounds were stable for several weeks in a desiccator.

Fmoc-Xaa-OPfp; General Procedure:[48]

To a stirred ice-cold soln of Fmoc-Xaa-OH (2.0 mmol) and HOPfp (2.0 mmol) in dry dioxane, EtOAc, or dioxane/DMF (5–10 mL), DCC (2.0 mmol) was added and stirring was continued for 1 h at 0 °C and 1 h at rt. The DCU was removed by filtration, the solvent was removed, the residue was triturated with hexane, and the product was recrystallized (EtOAc or EtOAc/hexane). Fmoc-Asn-OPfp was prepared in dioxane/DMF (24:1) using HOPfp (6 mmol); yield: 90%; mp 164–165 °C (EtOAc/hexane). Fmoc-Ser-OPfp was purified by chromatography (silica gel, dried at 120 °C for 24 h, 33% EtOAc/petroleum ether).[76] Fmoc-Ser(tBu)-OPfp[76] and Z-Lys(Z)-OPfp[80] also required purification on silica gel.

for references see p 472

Table 7 Fmoc-Xaa-OPfp Prepared Using *N,N'*-Dicyclohexylcarbodiimide[48,49,76,80]

Compound[a]	Crystallization Solvent	Yield (%)	mp (°C)	$[\alpha]_D^{25}$ (*c* 2, $CHCl_3$)	TLC R_f[b]	Ref
Fmoc-Ala-OPfp	EtOAc/hexane	93	171–173	−22.7	0.79	[48]
Fmoc-Phe-OPfp	EtOAc	93	154–157	−20.3	0.81	[48]
Fmoc-Asp(OtBu)-OPfp	hexane	89	98–100	−2.5	0.78	[48]
Fmoc-Cys(Acm)-OPfp	EtOAc	78	157–158	−32.6	0.62	[48]
Fmoc-Gln-OPfp	EtOAc	97	151–153	−19.8	0.55	[48]
Fmoc-Ser(tBu)-OPfp		67	67–71			[49]
Fmoc-Lys(Fmoc)-OPfp[c]		77	171–173			[80]
Fmoc-Ser-OPfp[c]		54	144–145			[76]

[a] A complete list has been compiled.[81]
[b] $CHCl_3$/MeOH/AcOH (80:10:1).
[c] Purified by chromatography on silica gel.

Table 8 Z-Xaa-OPfp Prepared Using *N,N'*-Dicyclohexylcarbodiimide[45]

Compound	Crystallization Solvent	mp (°C)	$[\alpha]_D^{22}$ (*c* 2, EtOAc)	Ref
Z-Ala-OPfp	hexane	84–85	−22.4	[45]
Z-Lys(Z)-OPfp	EtOH/H_2O/iPrOH	115–117	+8.58	[45]
Z-Glu(OtBu)-OPfp	hexane/benzene	85–87	−16.5	[45]
Z-Gln-OPfp	EtOH/H_2O	116–118	−10.9	[45]
Z-Cys(Bzl)-OPfp	EtOAc/pentane	85–86	−25.6	[45]

Fmoc-(Fmoc-Hmb)Xaa-OPfp 45; **General Procedure:**[50]
Fmoc-(Fmoc-Hmb)Xaa-OH (10 mmol) was dissolved in dry THF (30 mL) with stirring and the soln was cooled in ice. HOPfp (11 mmol) in THF (10 mL) was added followed by DCC (10.5 mmol) in THF (10 mL). The mixture was stirred overnight and filtered, the DCU was washed with THF (2 × 10 mL) and the filtrate was removed. The residue was chromatographed (silica gel packed and eluted with $CHCl_3$). Appropriate fractions were pooled, the solvent was removed, and the ester was triturated with cold hexane (3 × 10 mL) and dried under reduced pressure; yield: 67–93%.

Table 9 Fmoc-(Fmoc-Hmb)Xaa-OPfp Prepared Using *N,N'*-Dicyclohexylcarbodiimide[50]

Xaa	Yield (%)	mp (°C)	$[\alpha]_D^{20}$ (*c* 1, DMF)	Ref
Ala	93	67[a], 73–75[b]	−7.8	[50]
Asp(OtBu)	84	50–53[a], 68–71[b]	−21.1	[50]
Gly	76	73–75[b]		[50]
Ile	93	70[a], 75–77[b]	−22.8	[50]
Lys(Boc)	89	61–62[a], 67[b]	−14.6	[50]
Thr(tBu)	67	76[a], 81–86[b]	−39.6	[50]
Tyr(tBu)	91	64[a],67–68.5[b]	−31.6	[50]
Val	88	68[a], 73–76[b]	−21.0	[50]

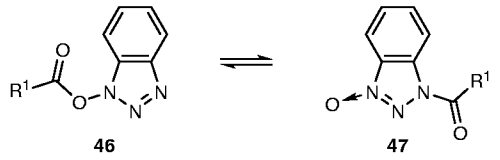
[a] Shrinks.
[b] Clear glass.

3.2.1.1.4 Benzotriazolyl and 4-Oxo-3,4-dihydrobenzotriazin-3-yl Esters

These esters (see Tables 10 and 11) are obtained by the usual procedure. The benzotriazolyl esters are rarely isolated because they are too reactive for routine use but are often prepared and subjected to aminolysis immediately.[82] The latter is the case in solid-phase synthesis where DIC is used in preference to DCC because the urea is soluble.[36] Benzotriazolyl esters possess the unique feature of existing in two forms in solution, ester **46** being in equilibrium with N^3-acylated N^1-oxide **47** (Scheme 9), readily distinguishable by their absorbances in the infrared spectrum at 1827 and 1725 cm^{-1}, respectively.[14] The two forms are stable, the form

crystallizing out of solution usually being the ester. HPLC of a solution of the Z-valine adduct showed two well-separated peaks.[83] Single crystal X-ray analysis has shown that the crystallized benzotriazolyl adduct of the *N*-trityl-L-methionine is the N^3-acylated N^1-oxide form **47**.[84] 7-Azabenzotriazolyl esters are used in the same manner and probably exhibit the same phenomenon.[85] 4-Oxo-3,4-dihydrobenzotriazin-3-yl esters are very reactive but stable enough for routine use.[86] Precautions must be taken to avoid dehydration to cyano derivatives (Scheme 4) when preparing the esters of asparagine and glutamine derivatives.

Scheme 9 The Equilibrium of Benzotriazolyl Esters in Solution



46 **47**

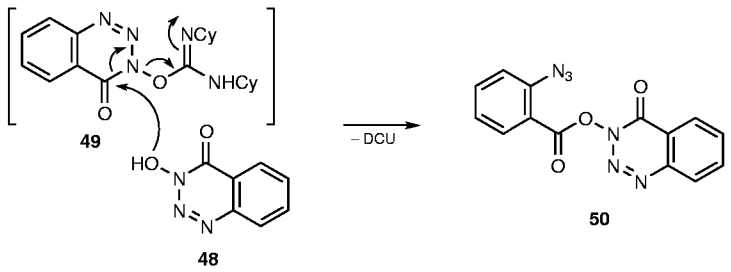
Table 10 Benzotriazolyl Esters Prepared Using *N,N'*-Dicyclohexylcarbodiimide[15,87]

Compound	Yield (%)	mp (°C)	Ref
Z-Phe-OBt	70	120–122	[15]
Z-Gly-OBt	49	147	[15]
Z-Thr-OBt	68	150–153	[15]
Boc-Cys[Bzl(4-Me)]-OBt	89	80–81	[87]
Boc-Cys[Bzl(4-Me)]-OAt	68	90–91	[87]

Side Reaction of Dimerization of 3-Hydroxy-4-oxo-3,4-dihydrobenzotriazine

The reaction in Scheme 3 for HOR^2 = HODhbt is accompanied by a side reaction giving rise to the formation of oxobenzotriazinyl 2-azidobenzoate **50** (Scheme 10).[14] The 2-azidobenzoate originates from reaction of HODhbt (**48**) with addition product **49** formed by reaction of the carbodiimide with **48**. It is detectable by HPLC and by the azide absorbance at 2120 cm^{-1} in the IR spectrum. It is an acylating agent and thus causes chain termination if left as a contaminant of the ester. The side reaction is completely eliminated by allowing a 5-minute activation time before addition of the hydroxy compound.[14,51]

Scheme 10 Side Reaction of 4-Oxo-3,4-dihydrobenzotriazin-3-yl 2-Azidobenzoate Formation by Reaction of One Molecule of *N,N'*-Dicyclocarbodiimide with Two Molecules of 3-Hydroxy-4-oxo-3,4-dihydrobenzotriazine[14]



49 **48** **50**

Fmoc-Lys(Boc)-ODhbt; Typical Procedure:[51]

Fmoc-Lys(Boc)-OH (11.7 g, 25 mmol) was dissolved in dry THF (60 mL) and the soln was cooled to −15 °C. DCC (5.15 g, 25 mmol) dissolved in THF (6 mL) was added followed by solid HODhbt (4.0 g, 25 mmol) after 5 min. The mixture was stirred at −15 °C for 20 min and then at 4 °C (cold room) for 3.5 h. The mixture was filtered and the solvent was removed. The residue was dissolved in Et_2O (100 mL) and hexane (100 mL) was added. The product was collected after 15 h at 4 °C, washed with petroleum ether (bp 40–60 °C), and dried (silica gel) under reduced pressure; yield: 14.5 g (95%); mp 157–159 °C. For Fmoc-Cys(Acm)-ODhbt the Fmoc-Cys(Acm)-OH was dissolved in DMF prior to addition of THF.

for references see p 472

Table 11 Fmoc-Xaa-ODhbt Prepared Using *N,N'*-Dicyclohexylcarbodiimide[51]

Compound[a]	Crystallization Solvent	Yield (%)	mp (°C)	$[\alpha]_D^{25}$ (DMF)	Ref
Fmoc-Val-ODhbt	Et_2O	80	116–118	−125.5	[51]
Fmoc-Phe-ODhbt	CH_2Cl_2/hexane	94	194–199	−110.9	[51]
Fmoc-Ser(tBu)-ODhbt	Et_2O/hexane	96	65–75	−29.3	[51]
Fmoc-Trp-ODhbt	THF/Et_2O	88	144–146	−79.6	[51]
Fmoc-Ile-ODhbt	dioxane/Et_2O	83	122–125	−111.0	[51]

[a] A complete list has been compiled.[81]

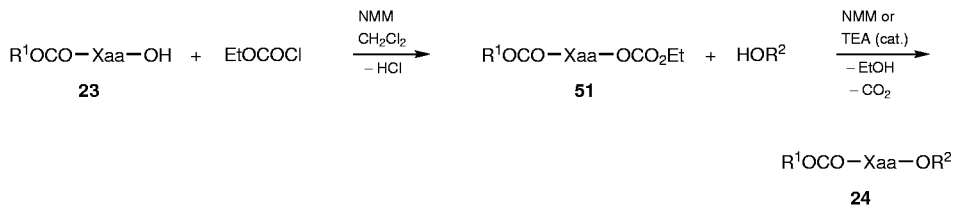
Fmoc-Asn-ODhbt:[51]
Powdered Fmoc-Asn-OH (3.9 g, 11 mmol) was dissolved in warm DMF (43 mL) and the soln was cooled to −50 °C. HODhbt (1.97 g, 12 mmol) in DMF (7 mL) was added followed by DCC (2.25 g, 10.9 mmol). The mixture was allowed to warm to 0 °C over 1 h, stirred at 0 °C for 1 h and left overnight at 5 °C. The precipitate was filtered off and the DMF was removed at 15 °C/0.05 Torr until crystals appeared. Et_2O (125 mL) was added and the suspension was stirred at 0 °C for 1 h. The product was washed with Et_2O and hexane, dried under reduced pressure, dissolved in DMF (25 mL) and the soln was filtered to remove residual DCU. Et_2O (300 mL) was added and the mixture was left at 5 °C for 18 h; yield: 4.07 g (86%); mp 129–132 °C. The cold DMF serves to suppress β-cyanoalanine formation (Scheme 4) that was observed for reactions in dioxane.

Fmoc-His(Boc)-ODhbt was crystallized by adding a soln in THF/Et_2O (1:4) to 5 vols of petroleum ether (bp 40–60 °C) with stirring. Fmoc-Arg(Mtr)-ODhbt was obtained free from lactam (Scheme 5) if the solvent was THF, the reaction temperature was −25 °C for 1 h followed by 4 h at ice–salt temperature, and crystallization was at −15 °C. In soln Fmoc-Arg(Mtr)-ODhbt was gradually converted into the lactam, with a half-life of 10 h in DMF and 15 min in CH_2Cl_2. Fmoc-Asp(OtBu)-ODhbt was optically stable over 3.5 h in DMF, but generated 20% of the D-isomer in the presence of 1% DIEA.

3.2.1.2 Synthesis of Active Esters via Mixed Anhydrides

Active esters can be prepared via the mixed anhydrides, but the method has not been popular probably due to its unreliability and the moderate yields obtained.[4,88–90] A better understanding of the chemistry of mixed anhydrides[91] has led to a simple general procedure that gives good yields in most cases (see Table 12).[92] Mixed anhydride **51** is generated from acid **23** in dichloromethane at room temperature by the addition of ethyl chloroformate in the presence of *N*-methylmorpholine (Scheme 11). Reaction of **51** with a 50% excess of the hydroxy compound in the presence of an equal amount of *N*-methylmorpholine or a catalytic amount of triethylamine generates active ester **24**.

Scheme 11 Preparation of Active Esters via Mixed Anhydrides[92]



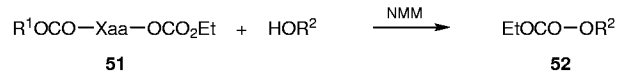
Formation of the mixed anhydride is slow in dichloromethane if the base is triethylamine. Esterification is inefficient in the absence of tertiary amine. Products prepared using *N*-methylmorpholine were enantiomerically pure. Products prepared using triethylamine should be isolated within 30 minutes to avoid enantiomerization of the ester that occurs in the presence of tertiary amine; Z-Phe-ONp obtained using 0.2 equivalents of triethylamine contained 0.35% of the D-isomer. Yields are lower for valine because of its β-methyl substituent. Higher yields are obtained if the reagent is isopropyl chloroformate. The method worked well for fatty acids and Z-γ-Abu-OSu that was not accessible using DCC, but is not

applicable to derivatives of serine, asparagine and glutamine with unprotected functionalities. Use of symmetrical anhydrides was found to be superior to the DCC method for preparing water soluble 2-nitro-4-sulfophenyl esters **20**.[24]

Side Reaction of Esterification at the Carbonate Moiety of a Mixed Anhydride

Active ester formation by the mixed anhydride method is accompanied by the side reaction of esterification at the carbonate moiety of mixed anhydride **51** which generates mixed carbonate **52** (Scheme 12).[92] This decreases the yields, but is more of a nuisance than an obstacle as the side products do not interfere with crystallization of the esters as the former are soluble in the crystallizing solvent. More mixed carbonate is formed from derivatives of the hindered amino acids and proline; none is formed from α-unsubstituted acids. *N*-Hydroxysuccinimide gives rise to much less byproduct than 4-nitrophenol; other phenols generate intermediate amounts. Less byproduct is generated when the reagent is isopropyl chloroformate. The impurity can be readily removed from a solution of the ester by adsorption of the compounds on reverse-phase chromatography beads followed by separation by selective displacement.[93]

Scheme 12 Side Reaction of Esterification at the Carbonate Moiety of a Mixed Anhydride[92]



R^1 = tBu, Bzl, Fm; R^2 = Np, Su, Pfp, Dhbt

R^1OCO-Xaa-OR^2; General Procedure:[92]
To R^1OCO-Xaa-OH (1.0 mmol) and NMM (1.0 mmol) in CH_2Cl_2 (10 mL) at rt was added EtOCOCl (1.0 mmol). After stirring for 2 min, a soln of R^2OH (1.5 mmol) and NMM (1.5 mmol) in CH_2Cl_2 (10 mL) was added and the soln was stirred for 2 h. The soln was washed with 5% aq citric acid and 5% aq $NaHCO_3$, dried ($MgSO_4$), and the solvent was removed. The solid residue was crystallized (CH_2Cl_2/petroleum ether); yield: 65–90%. An additional 10–20% yield was obtained in difficult cases when the reagent was iPrOCOCl; a 5% excess of NMM was used to prepare the mixed anhydride and 0.2 equiv of TEA was used instead of the NMM to catalyze the esterification. The tertiary amine is necessary for efficient reaction and to prevent decomposition of the anhydrides of Boc amino acids that result from acidolysis at the *tert*-butoxy function of the 2-*tert*-butoxyoxazol-5(4*H*)-one that is formed when the esterification is slow (Scheme 6).

Table 12 R^1OCO-Xaa-OR^2 Prepared through Mixed Anhydrides[92]

Compound[a]	Yield (%)	mp (°C)	Ref
Z-Phe-ONp	70	126–127	[92]
Boc-Pro-OSu	62	135–136	[92]
Fmoc-Leu-OPfp	80	114–115	[92]
Boc-Asp(OBzl)-ONp	60	81–82	[92]
Boc-Leu-OSu	80	114–115	[92]
Boc-Val-OPfp	50	63–64	[92]
Boc-Phe-OPfp	80	112–114	[92]
Fmoc-Phe-ODhbt	80	192–193	[92]
Z-γAbu-OSu	80	65–68	[92]
$C_{17}H_{35}CO_2Su$	86	90–91	[92]

[a] Crystallized from CH_2Cl_2/petroleum ether.

Z-(Me)Leu-OSu; Typical Procedure:[93]
To Z-(Me)Leu-OH (0.265 g, 1.0 mmol) and NMM (0.10 g, 1.0 mmol) in CH_2Cl_2 (10 mL) was added iPrOCOCl (0.12 g, 1.0 mmol) in toluene (1 mL) and the mixture was stirred for 2 min. To this was added HOSu (0.24 g, 1.5 mmol) and NMM (0.15 g, 1.5 mmol) in CH_2Cl_2 (10 mL) and the soln was stirred for 15 h. The soln was washed with 5% aq citric acid and 5% aq $NaHCO_3$, dried ($MgSO_4$), and the solvent

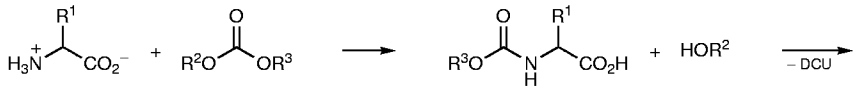
for references see p 472

was removed. The residue containing Z-(Me)Leu-OSu/iPrOCO$_2$Su (19:1) was crystallized (CH_2Cl_2/petroleum ether); yield: 80%; mp 81–83 °C; $[\alpha]_D$ −51.2 (*c* 1.5, $CHCl_3$). For Z-(Me)Val-OSu that did not crystallize, the residue was dissolved in MeOH (1 mL) and the soln was deposited on a Waters Sep-Pak C-18 cartridge (6 $cm^3 \cdot g^{-1}$ of peptide). The iPrOCO$_2$Su was washed through with MeOH/H_2O (3:7, 100 mL) under suction (0.5 h), the active ester was eluted from the column with MeOH (50 mL) and the soln was removed to leave an oil; yield: 80%; $[\alpha]_D$ −104.2 (*c* 1.05, $CHCl_3$). The products were pure based on analysis by HPLC and NMR spectroscopy. The method is inapplicable to Boc-(Me)Xaa-OH because the activated derivatives decompose too readily by the mechanism described in Scheme 6.

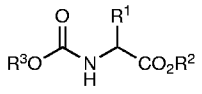
3.2.1.3 Synthesis of Active Esters with Mixed Carbonates

Compounds such as 4-nitrophenyl and 2,4,5-trichlorophenyl esters of N-protected amino acids were originally prepared also by reaction of the N-protected acids with di- or trisubstituted esters of carbonic,[94] sulfinic,[95] or phosphinic acids[9] in, or with addition of, pyridine. This was followed by the use of mixed carbonates **54** that served both for acylation of amino acid **53** to N-protected acid **55** and subsequent esterification to **56** using carbodiimide (Scheme 13).[4,49] A variant of this was the preparation of the unusual 1,1,4-trioxo-2,5-diphenyl-4,5-dihydro-3-thienyl **57** esters of Boc amino acids **56c**, which avoided the use of carbodiimide since esterification proceeds in the presence of tetramethylguanidine.[96]

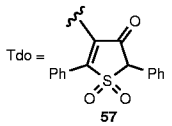
Scheme 13 Preparation of Active Esters via Mixed Carbonates[44,49,96]



53 54 55



56

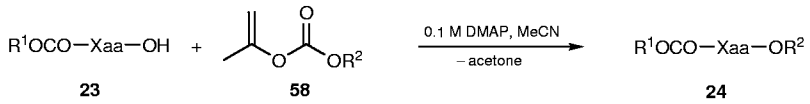


57

56	OR^2	$C(O)OR^3$	Conditions	Ref
a	OTcp	Boc	DCC	[44]
b	OPfp	Fmoc	DCC	[49]
c	OTdo	Boc	$(Me_2N)_2C{=}NH$	[96]

None of these methods has achieved popularity, but a unique variant of the carbonates provides a general method that is simple and efficient. Mixed carbonates **58** formed from isopropenyl chloroformate and substituted phenols or hydroxylamines react at room temperature with N^{α}-protected acids **23** in the presence of *N*-methylmorpholine[97] or a catalytic amount of 4-(dimethylamino)pyridine[98] giving active esters **24** in exceptionally high yield (see Table 13) with concomitant liberation of acetone (Scheme 14). Compounds prepared using the non-aromatic base require purification on a silica gel column.

Scheme 14 Preparation of Active Esters via Mixed Carbonates Obtained from Isopropenyl Chloroformate[97,98]

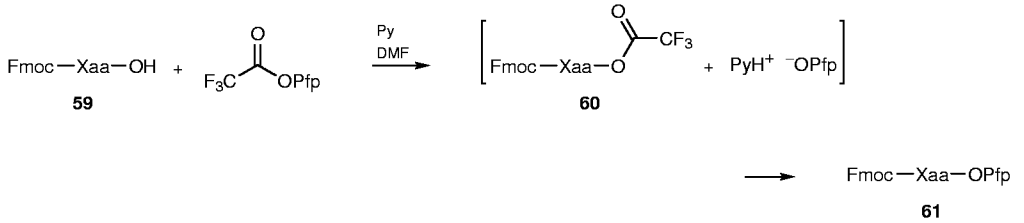


23 58 24

R^1 = tBu, Bzl; R^2 = Su, phthalimido, S-norbornene-2,3-dicarboximido

In a related approach, pentafluorophenyl esters of Fmoc amino acids **61** (see Table 14) were prepared from a mixture of acid **59** and pentafluorophenyl trifluoroacetate in dimethylformamide in the presence of pyridine,[99] which is a transesterification method developed by Sakakibara (Scheme 15).[100,101] The reaction involves formation of mixed carboxylic acid anhydride **60** that is then attacked by the pentafluorophenoxy anion generated by the pyridine. The products obtained do not require recrystallization. The reaction is sufficiently clean that purification is satisfactorily achieved by washing with aqueous solutions.

Scheme 15 Pentafluorophenyl Esters of Fmoc Amino Acids Using Pentafluorophenyl Trifluoroacetate[99]



59 60 61

Mixed Carbonates Formed from Isopropenyl Carbonate:
R^1OCO-Xaa-OR^2; General Procedure:[98]

A mixture of R^1OCO-Xaa-OH (0.5 mmol), CH_2=C(Me)OCO_2Su (0.5 mmol), and DMAP (0.05 mmol) in MeCN (3 mL) was stirred at rt for 4 h. The solvent was removed, the residue was dissolved in EtOAc (100 mL), and the organic phase was washed successively with 1 M HCl (50 mL), H_2O (50 mL), 5% aq $NaHCO_3$, and brine (50 mL), and then dried (Na_2SO_4). The solvent was removed and the products were crystallized (MeCN or dioxane); yield: 88–98%.

Table 13 R^1OCO-Xaa-OR^2 Prepared Using an Isopropenyl Carbonate[97,98]

Compound	Yield (%)	mp (°C)	$[\alpha]_D^{26}$ (*c* 2, dioxane)	Ref
Z-Val-OSu	99	119–121	–25.3	[98]
Boc-Phe-OSu	88	156–157	–18.8	[98]
Z-Val-ONPhth[a]	96	141–142	–31.6	[98]
Z-Val-ONdc	99	92–94	–25.5	[98]
Boc-Phe-OTcp	79			[97]
Boc-Arg(NO_2)-OTcp	51			[97]
Boc-Pro-ODnp	94			[97]

[a] NPhth = phthalimido.

Use of Pentafluorophenyl Trifluoroacetate:[100]
Fmoc-Xaa-OPfp; General Procedure:[99]

To a soln of Fmoc-Xaa-OH (0.50 mmol) in DMF (1 mL) was added pyridine (0.55 mmol) followed by CF_3CO_2Pfp (0.58 mmol) [obtained from HOPfp and TFAA (1.5 equiv), stirred at 40 °C and then distilled, bp 122 °C]. The mixture was stirred at rt for 45 min and diluted with EtOAc (100 mL); the soln was washed with 0.1 M HCl (3 × 100 mL) and 5% aq $NaHCO_3$ (3 × 100 mL), dried ($MgSO_4$), and the solvent was removed.

for references see p 472

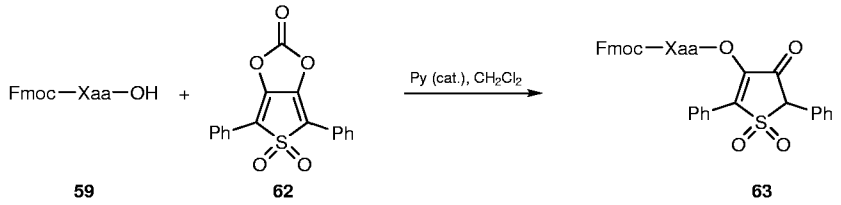
Table 14 Fmoc-Xaa-OPfp Prepared Using Pentafluorophenyl Trifluoroacetate[99]

Compound	Yield (%)	mp (°C)	$[\alpha]_D^{25}$ (*c* 1, $CHCl_3$)	TLC R_f (hexane/EtOAc 3:1)	Ref
Fmoc-Phe-OPfp	99	150–153	−20.8	0.68	[99]
Fmoc-Tyr(tBu)-OPfp	99	61–64	−13.8	0.63	[99]
Fmoc-Ile-OPfp	99	73–75	−13.1	0.66	[99]
Fmoc-Cys(Trt)-OPfp	92	71–73	+15.0	0.77	[99]
Fmoc-Lys(Boc)-OPfp	98	99–101	−14.1	0.35	[99]
Fmoc-Asp(OtBu)-OPfp	99	54–55	−2.5	0.54	[99]
Fmoc-Trp-OPfp	96	183–185	−42.6	0.33	[99]

3.2.1.3.1 1,1,4-Trioxo-2,5-diphenyl-4,5-dihydro-3-thienyl Esters of Fmoc Amino Acids

A cyclic carbonate that serves as the equivalent of a mixed carbonate is 4,6-diphenylthieno[3,4-*d*]-1,3-dioxol-2-one 5,5-dioxide (**62**).[96] It reacts with Fmoc amino acids **59** in dichloromethane at room temperature in the presence of a catalytic amount of pyridine to give 1,1,4-trioxo-2,5-diphenyl-4,5-dihydro-3-thienyl esters **63** in high yield (Scheme 16 and Table 15).[102] The compounds are stable for at least a year.

Scheme 16 1,1,4-Trioxo-2,5-diphenyl-4,5-dihydro-3-thienyl Esters of Fmoc Amino Acids Using a Cyclic Carbonate[102]



Fmoc-Xaa-OTdo; General Procedure:[102]
Fmoc-Xaa-OH (20 mmol) and carbonate **62** (7.18 g, 22 mmol) were stirred in dry CH_2Cl_2 (200 mL) in a flask with a condenser and drying tube. Pyridine (0.16 mL, 2 mmol) was added and the soln that formed generally became clear within 20 min and was stirred an additional 1.5 h. (A) When crystallization had occurred, the solvent was reduced to half-volume, hexane (20% vol) was added, and the crystals were collected with the help of a mixture of hexane/EtOAc (2:1). (B) When the ester remained in soln, the yellow soln was washed with 20% aq citric acid (3×) followed by sat. aq $NaHCO_3$ until it was nearly colorless. The soln was washed again with 15% aq citric acid and H_2O (2×), dried ($MgSO_4$), and the solvent was removed. The esters were recrystallized (EtOAc/petroleum ether or CH_2Cl_2/petroleum ether).

Table 15 Fmoc-Xaa-OTdo Prepared Using 4,6-Diphenylthieno[3,4-*d*]-1,3-dioxol-2-one 5,5-Dioxide[102]

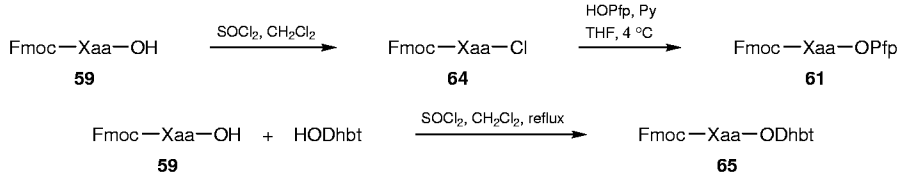
Compound	Method	Crystallization Solvent[a]	Yield (%)	mp (°C)	Ref
Fmoc-Asp(OtBu)-OTdo	B	EtOAc/PE	91	178–179	[102]
Fmoc-Arg(Mtr)-OTdo	B		80	138–141	[102]
Fmoc-Asn(Mbh)-OTdo	B	EtOAc/PE	74	174–176	[102]
Fmoc-Leu-OTdo	A	EtOAc/PE	85	183–186	[102]
Fmoc-Lys(Boc)-OTdo	A	CH_2Cl_2/PE	88	153–155	[102]
Fmoc-Met-OTdo	B		86	148–151	[102]
Fmoc-Pro-OTdo	A	CH_2Cl_2/PE	81	153–154	[102]
Fmoc-Trp-OTdo	B	EtOAc/PE	78	226–228	[102]

[a] PE = Petroleum ether.

3.2.1.4 Fmoc Amino Acid Esters from Acid Chlorides

Pentafluorophenyl **61** and 4-oxo-3,4-dihydrobenzotriazin-3-yl **65** esters (see Table 16) can be obtained from acid **59** and hydroxy compounds using thionyl chloride;[103] the intermediate is the Fmoc amino acid chloride **64** (Scheme 17). In the case of the pentafluorophenyl esters **61**, the acid chloride **64** is generated in the absence of the nucleophile and acylation is promoted in the presence of pyridine. While in the case of the 4-oxo-3,4-dihydrobenzotriazin-3-yl esters **65**, condensation is achieved by heating the reactants in refluxing dichloromethane. The method is not applicable to derivatives with acid-sensitive side-chain protectors.

Scheme 17 Pentafluorophenyl Esters via Fmoc Amino Acid Chlorides, and 4-Oxo-3,4-dihydrobenzotriazin-3-yl Esters of Fmoc Amino Acids Using Thionyl Chloride[103]



Fmoc-pGlu-OSu has been obtained by reaction of Fmoc-pGlu-Cl with *N*-hydroxysuccinimide, the chloride being obtained by slow cyclization of Fmoc-glutamic acid dichloride which is spontaneous.[104] Residual dichloride was destroyed by washing the Fmoc-pGlu-Cl with water. This approach guarantees preservation of chiral integrity of the activated forms of this substituted lactam. In contrast, Z- and Boc-pyroglutamic acids are prepared by treating the corresponding N-protected glutamic acid anhydrides with dicyclohexylamine,[105,106] a procedure that could be suspect in this regard and is not applicable to the Fmoc derivative. N-Protection is not a requirement for coupling pyroglutamic acid, but the derivatives are more soluble in the usual solvents. A few piperidino[4] and 4-nitrophenyl[6] esters of acid-stable derivatives were prepared from the chlorides during earlier times.

Fmoc-Xaa-OPfp; General Procedure:[103]
A mixture of Fmoc-Xaa-OH (2 mmol) and $SOCl_2$ (20 mmol) in CH_2Cl_2 (12 mL) was heated at reflux until conversion into the acid chloride was complete as indicated by TLC (petroleum ether/EtOAc/AcOH 15:5:1) of the methyl ester derived by adding MeOH to an aliquot. Residual $SOCl_2$ was removed in vacuo, the Fmoc-Xaa-Cl was dissolved in THF (10 mL), PfpOH (2.1 mmol) was added, the soln was cooled and pyridine was added to the stirred soln. After completion of the reaction, CH_2Cl_2 (50 mL) was added, the soln was washed with cold brine (2×), dried ($MgSO_4$), and the solvent was removed. Trituration with petroleum ether afforded the products; contaminating Fmoc-Xaa-OH was removed by recrystallization.

Fmoc-Xaa-ODhbt; General Procedure:[103]
A mixture of Fmoc-Xaa-OH (2 mmol), HODhbt (2.2 mmol), and $SOCl_2$ (20 mmol) in CH_2Cl_2 (12 mL) was heated at reflux until the esterification was complete as indicated by TLC as described above. Residual $SOCl_2$ was removed in vacuo and the products were crystallized by adding EtOAc to the resulting oils or by adding a soln of the products in EtOAc to petroleum ether while stirring. Some products were purified by passing them through a column of silica gel using EtOAc or EtOAc/petroleum ether as solvent.

Table 16 Fmoc Amino Acid Esters Prepared through Acid Chlorides[4,103,104]

Compound	Yield (%)	mp (°C)	$[\alpha]_D$ (*c* 1)	Ref
Fmoc-Ala-OPfp	92	178–179	−22.2 ($CHCl_3$)	[103]
Fmoc-Met-OPfp	90	109–112	−13.1 ($CHCl_3$)	[103]
Fmoc-Val-OPfp	91	120–121	−23.1 ($CHCl_3$)	[103]
Fmoc-Ile-ODhbt	89	121–123	−109.5 (DMF)	[103]
Fmoc-Phe-ODhbt	97	196–199	−109.7 (DMF)	[103]
Fmoc-Val-ODhbt	94	116–117	−124.8 (DMF)	[103]
Fmoc-pGlu-OSu	80	105–108	−43.5 ($CHCl_3$)	[104]
Phth-Phe-OPip	46	128–130	−184 (EtOAc)	[4]

for references see p 472

Fmoc-pGlu-OSu:[104]
To freshly prepared Fmoc-pGlu-Cl (from 1 mmol of Fmoc-pGlu-OH) in CH_2Cl_2 (10 mL) was added a soln of HOSu (0.16 g, 1 mmol) and NMM (0.10 g, 1 mmol) in CH_2Cl_2 (10 mL). The mixture was stirred for 24 h, the soln was washed with 1 M HCl and aq $NaHCO_3$, dried ($MgSO_4$), and the residue was crystallized from CH_2Cl_2/EtOH by slowly removing the CH_2Cl_2 with a rotary evaporator; yield: 80%; mp 105–108 °C.

Fmoc-pGlu-Cl (0.33 g, 0.85 mmol), mp 40–50 °C, was obtained from a mixture of Fmoc-pGlu-OH (0.35 g, 1 mmol), $SOCl_2$ (2 mL), DMF (1 drop) and CH_2Cl_2 (20 mL) that was left for 3 d, after which the solvent was replaced with CH_2Cl_2 (20 mL), the soln was washed with dil acid and aq $NaHCO_3$, dried, and the solvent was removed.

3.2.2 Methods for Preparing Active Esters of Peptides

Active esters of N^{α}-protected peptides are obtained by two general approaches, esterification of the N^{α}-protected peptide and chain buildup starting with the active ester of an amino acid. Esterification of an N^{α}-protected peptide can be effected by the usual procedures for esterifying monomers (Section 3.2.1). However, except when the carboxy terminal residue is glycine or proline there is a danger that the carboxy terminal residue will enantiomerize during esterification[21] and this is confirmed by experimental data.[101,107–109] Moreover, there are methods for coupling acids that are simpler than coupling through active esters. In view of this, N^{α}-protected peptides are not routinely converted into active esters for subsequent use. However, some pentachlorophenyl[11] and pentafluorophenyl[46] esters **67** of peptides of acceptable enantiomeric purity were obtained from protected peptides **66** using *N,N'*-dicyclohexylcarbodiimide or adduct **25** formed from it and the halogenated phenol, but this method has not been generally adopted (Scheme 18).

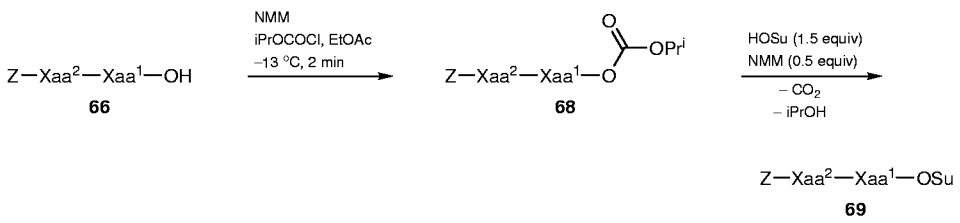
Scheme 18 Active Esters of *N*-Protected Dipeptides Using Carbodiimides



OR^1	Conditions	Ref
OPcp	HOPcp, DCC	[11]
OPfp	HOPfp, DCC	[46]
OPcp or OPfp	adduct **25** (X = Cl or F)	[46]

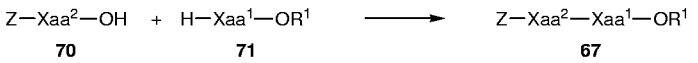
A variant of the mixed anhydride method used for esterification of monomers that appears promising gave enantiomerically pure succinimido esters of protected dipeptides **69**.[109] Isomerization was avoided by generating anhydride **68** at −13 °C using isopropyl chloroformate and *N*-methylmorpholine (Scheme 19). On the other hand, succinimido esters, in particular of di- and tripeptides with glycine or proline at the carboxy terminus, have been extensively used for the preparation of larger segments in solution.[110] The approach is successful when the reactants and product show differential solubilities allowing purification of the latter.

Scheme 19 Succinimido Esters of N-Protected Dipeptides through Mixed Anhydrides[93,109]



The second approach to peptide active esters **67** is to assemble **67** starting with the active ester of carboxy terminal amino acid **71** (Scheme 20). Choice of a solvent such as ethyl acetate, in which aminolysis of active esters is slow, favors bond formation between the amino group of the amino acid or peptide ester **71** and the carboxy group of the next residue **70**. Coupling with active ester **71** can be achieved using *N,N'*-dicyclohexylcarbodiimide[111] or the acyl azide[11] or mixed anhydride[45,112] methods. The required substituted phenyl esters of amino acids are obtainable from the *tert*-butoxycarbonyl derivatives by treatment with trifluoroacetic acid. Pentachlorophenyl esters of amino acids are obtainable also from the benzyloxycarbonyl derivatives by catalytic hydrogenation in methanol in the presence of 1 equivalent of hydrogen chloride, which suppresses dechlorination of the ester moiety.[11] Extension to N-protected tri- and tetrapeptide pentachlorophenyl esters has been achieved,[113] but there is an obstacle in forming the tripeptide ester because of the tendency for dioxopiperazine formation when the dipeptide ester is deprotonated for reaction with the third residue. Nevertheless, a variety of tripeptide esters unprotected at the amino terminus have been used for synthesizing polymers of repeating sequence.[11,114,115]

Scheme 20 Active Esters of N-Protected Dipeptides by Acylation of an Amino Acid Active Ester



70 **71** **67**

OR^1	Conditions	Ref
ONp	DCC, MeCN, 0 °C	[111]
OPfp	iBuOCOCl, NMM, EtOAc, −20 °C	[45]
OPcp	iBuOCOCl, TEA, CH_2Cl_2, −10 °C	[112]

3.2.2.1 Esterification of an N^{α}-Protected Peptide

3.2.2.1.1 Pentachlorophenyl Esters Using *N,N'*-Dicyclohexylcarbodiimide and Pentachlorophenol

See the discussion in Section 3.2.1.1.3.

N^{α}-Protected Peptide Pentachlorophenyl Esters:[11]
Z-Glu(OtBu)-Ala-Glu(OtBu)-OPcp [(from EtOAc); yield: 78%; mp 132–133 °C; $[\alpha]_D$ −13.2 (*c* 2, $CHCl_3$)] and Z-Glu(OtBu)-Ala-OPcp [(from MeOH); yield: 64%; mp 171–172 °C; mp 173–174 °C after three crystallizations, $[\alpha]_D$ −18.5 (*c* 1.33, $CHCl_3$)] were prepared from the acid and HOPcp using DCC according to Section 3.2.1.1.3.[11] The protected tripeptide ester had the same specific rotation as that prepared by coupling the protected dipeptide hydrazide and the amino acid ester by the acyl azide method. The two esters prepared by direct esterification were demonstrated by indirect optical rotation measurements to be >99.6 and 98% enantiomerically pure, respectively.

3.2.2.1.2 Pentafluorophenyl Esters Using an *N,N'*-Dicyclohexylcarbodiimide/ Pentafluorophenol Adduct

Discussion of the use of the *N,N'*-dicyclohexylcarbodiimide/pentafluorophenol adduct **25** in the preparation of pentafluorophenyl esters of *N*-alkoxycarbonyl amino acids can be found in Section 3.2.1.1.

R^1OCO-Xaa¹-Xaa²-OPfp; General Procedure:[46]
Crystalline adduct **25** (2.4 mmol) obtained from DCC (1 mmol) and HOPfp (3 mol) was added to a soln or suspension of R^1OCO-Xaa¹-Xaa²-OH (2 mmol) in EtOAc (4–5 mL) while stirring. After 10 min, the temperature was brought to 0 °C and stirring was continued for 10 min. The mixture was filtered, the solvent was removed, the residue was triturated in hexane, and the crystals were collected. The products were recrystallized (hexane or EtOAc/hexane). The products were enantiomerically pure based on analysis for isoleucine/alloisoleucine after hydrolysis and comparison of the specific rotation of a model peptide prepared by the acyl azide method.

for references see p 472

3.2.2.1.3 Succinimido Esters via Mixed Anhydrides

See the discussion in Section 3.2.2.

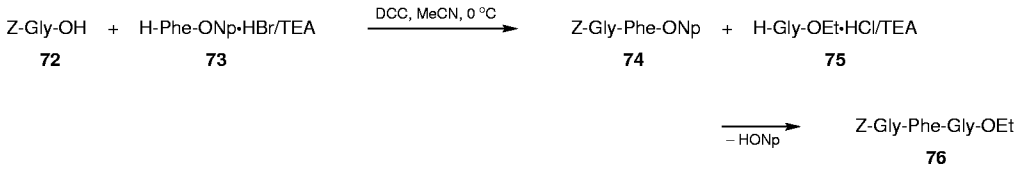
Z-Leu-Val-OSu; Typical Procedure:[109]
To Z-Leu-Val-OH (0.36 g, 1.0 mmol) and NMM (0.1 g, 1.0 mmol) in EtOAc/DMF (97:3, 10 mL) at −13 °C was added iPrOCOCl (0.12 g, 1.0 mmol). After stirring for 2 min, a soln of HOSu (0.24 g, 1.5 mmol) and NMM (0.05 g, 0.5 mmol) in EtOAc (10 mL) was added and the soln was stirred at −13 °C for 2 h. The soln was washed with 5% aq citric acid and 5% aq $NaHCO_3$, and dried ($MgSO_4$); the solvent was removed and the solid residue was crystallized (CH_2Cl_2/petroleum ether); yield: 77%; mp 60–65 °C; $[\alpha]_D$ −40.4 (*c* 1.0, $CHCl_3$). The product contained <0.5% of the LD-isomer. Several esters containing glycine contained <1% of the D-isomer based on analysis by HPLC of epimers generated by reaction with an amino acid ester.

3.2.2.2 Acylation of an Amino Acid Active Ester, "Backing Off Procedure"

3.2.2.2.1 Nitrophenyl Ester via a Carbodiimide, Followed by Its Aminolysis

See the discussion in Section 3.2.2; Scheme 21 shows the synthesis of the protected tripeptide **76** from the amino acid nitrophenyl ester, followed by aminolysis.[111]

Scheme 21 Synthesis of a Protected Tripeptide Starting with the Active Ester of the Central Residue[111]



Z-Gly-Phe-Gly-OEt (76) from Z-Gly-Phe-ONp (74); Typical Procedure:[111]
To a soln of Z-Gly-OH (**72**; 0.21 g, 1 mmol) in MeCN (12 mL) at 0 °C was added H-Phe-ONp · HBr (**73**; 0.37 g, 1 mmol), prepared from Z-Phe-ONp using HBr in AcOH,[116] followed by TEA (0.14 mL, 1 mmol). The mixture was stirred and when dissolution was complete, DCC (0.21 g, 1 mmol) was added and the mixture was stirred at 0 °C for 1 h and rt for 5 h to give Z-Gly-Phe-ONp (**74**). H-Gly-OEt · HCl (**75**; 0.14 g, 1 mmol) and TEA (0.15 mL, 1 mmol) were added and the mixture was left at rt for 15 h. The solvent was replaced with EtOAc, the soln was washed with 2 M HCl, and 5% aq Na_2CO_3 until the HONp had been removed, dried ($MgSO_4$), and the solvent was removed. Crystallization (EtOAc/petroleum ether) gave Z-Gly-Phe-Gly-OEt (**76**); yield: 0.35 g (80% overall); mp 118–119 °C; $[\alpha]_D$ −12.4 (*c* 2, EtOH).

3.2.2.2.2 Pentachlorophenyl Esters via an Acyl Azide

For discussion see Sections 3.2.1.1.3 and 3.2.2.

Z-Glu(OtBu)-Ala-Glu(OtBu)-OPcp; Typical Procedure:[11]
To a suspension of Z-Glu(OtBu)-Ala-N_2H_3 (0.35 g, 0.83 mmol) in ice-cold H_2O (50 mL) was added concd HCl (0.3 mL) and AcOH (0.75 mL). When the soln was clear, $NaNO_2$ (0.13 g, 1.6 mmol) in cold H_2O (3 mL) was added and the gummy material that separated out was extracted into cold EtOAc (2 × 35 mL). The extract was washed with 5% aq $NaHCO_3$ (2 × 10 mL) and H_2O (3 × 20 mL), dried (Na_2SO_4), and H-Glu(OtBu)-OPcp · HCl (0.405 g, 0.83 mmol) and TEA (0.116 mL, 0.83 mmol) were added. The mixture was stirred at rt for 24 h, after which it was washed with 1 M HCl (2 × 20 mL), H_2O, 5% aq $NaHCO_3$, and H_2O. The EtOAc phases were dried (Na_2SO_4), the solvent was removed, and the residue was crystallized twice (EtOAc/petroleum ether); yield: 0.320 g (46%); mp 132–133 °C. Crystallization (MeOH) gave material with mp 151–152 °C, the two forms being interconvertible.

H-Glu(OtBu)-OPcp · HCl; Typical Procedure:[11]
Pd/C catalyst (10%, 0.5 g) suspended in AcOH (1 mL) and MeOH (20 mL) was hydrogenated at rt. MeOH containing 0.150 g of HCl · mL^{-1} (0.29 mL, 12 mmol) was added and hydrogenation was continued until uptake of gas had ceased. Z-Glu(OtBu)-OPcp (4.0 g, 11.4 mmol) in MeOH (10 mL) was added and hydrogenation was continued until there was no uptake (110 mL in 7 min). The catalyst and

then the solvent were removed and Et_2O was added; yield: 2.55 g (88%); mp 125–126 °C; $[\alpha]_D$ +23.9 (*c* 1.02, MeOH).

3.2.2.2.3 Substituted Phenyl Esters Using the Mixed Anhydride Method

Discussion of the use of the mixed anhydride method in the preparation of active esters of *N*-alkoxycarbonyl amino acids can be found in Section 3.2.1.2. See also the discussion in Section 3.2.2.[45,111]

Z-Gly-Cys(Bzl)-OPfp; Typical Procedure:[45]
To a soln of Z-Gly-OH (1.045 g, 5 mmol) and NMM (0.55 mL, 5 mmol) in EtOAc (13 mL) at −20 °C was added iBuOCOCl (0.7 mL, 5.3 mmol). After stirring for 15 min, H-Cys(Bzl)-OPfp · HBr (2.29 g, 5 mmol), prepared from Z-Cys(Bzl)-OPfp using HBr in AcOH,[116] and TEA (0.7 mL, 5 mmol) were added. The mixture was stirred at −20 °C for 30 min and at 0 °C for 1 h, diluted with EtOAc (13 mL), washed with 1 M HCl (13 mL), 5% aq $NaHCO_3$ (2 × 13 mL), 1 M HCl (2 × 20 mL), and H_2O (3 × 20 mL), and dried (Na_2SO_4). The solvent was removed, and the oily residue (3.39 g, 60%) was dissolved in Et_2O and placed in a freezer. The crystalline product was washed with Et_2O and pentane; yield: 2.50 g (44%); mp 74–80 °C; after recrystallization (Et_2O/pentane) mp 84–85 °C; $[\alpha]_D$ −30.7 (*c* 2.04, EtOAc).

3.2.3 Peptide Bond Formation by Aminolysis of Active Esters

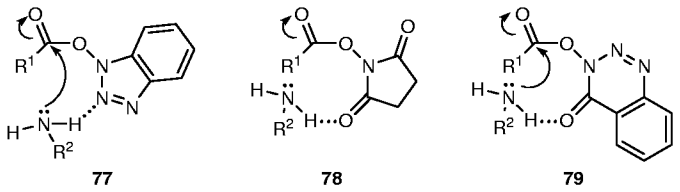
Peptide bond formation between an active ester and an amino group is a simple nucleophilic displacement reaction that occurs spontaneously at room temperature through formation of tetrahedral intermediate **4** (Scheme 1) whose breakdown is the rate-limiting step. The rates of aminolysis of substituted phenyl esters vary directly with the electron-withdrawing strengths of the moieties, which correspond with the relative acidities of the parent phenols, in the order pentafluorophenyl **15** > pentachlorophenyl **14** > 2,4,5-trichlorophenyl **13** > 4-nitrophenyl **12**.[9,11,12] 4-Oxo-3,4-dihydrobenzotriazine-3-yl esters **18** are five times as reactive as pentafluorophenyl esters.[86] The following order of activity was found for acylation in dimethylformamide of γ-*N*-trimethoxybenzylglutaminylalanyl–polystyrene by a series of active esters of Fmoc-valine: 4-oxo-3,4-dihydrobenzotriazin-3-yl > pentafluorophenyl > phthalimido > 2,4,5-trichlorophenyl > pentachlorophenyl = succinimido > 5-norbornene-2,3-dicarboxyimido > 4-nitrophenyl > 2-pyridyl > 2-nitrophenyl > piperidyl.[52] 2-Nitro-4-sulfophenyl esters have the same reactivity as 2-nitrophenyl esters in dimethylformamide.[24] The rates are much higher in polar or non-anhydrous solvents,[9,11] the solvent effect is less pronounced for succinimido esters, and aminolysis is catalyzed by mild acids such as acetic acid, pyridine,[117] and 1,2,3-benzotriazol-1-ol,[42,86,118,119] the latter a catalyst even in water.[24] The effect of 1,2,3-benzotriazol-1-ol is increased in the presence of tertiary amine.[120]

In the absence of base, active esters of both *N*-alkoxycarbonyl amino acids and N^{α}-protected peptides undergo aminolysis with preservation of chiral integrity. There is ample evidence to substantiate this statement[45,46,109,111] and no evidence to the contrary. But there is a wealth of accumulated evidence that active esters undergo isomerization when left in the presence of tertiary amine.[92,121] In this regard, 4-nitrophenyl esters are much more sensitive than trichlorophenyl esters[122] and piperidino esters are exceptions in that they are unaffected by tertiary amine.[4] The isomerization recently observed in the TBTU-mediated couplings of Fmoc-Cys(R^1)-OH in the presence of 1,2,3-benzotriazol-1-ol in continuous flow solid-phase synthesis is a further example of this phenomenon.[77] The result can be attributed to the effect of tertiary amine on the benzotriazolyl ester that is formed by capture of the acyluronium intermediate before it has time to be aminolyzed.

for references see p 472

The high reactivity that characterizes active esters containing a N-substituted hydroxylamine moiety is explained on the basis of anchimeric assistance provided by a neighboring electron-rich atom.[123,124] Unshared electrons of the nitrogen or oxygen atoms of esters **77**, **78**, and **79** (Scheme 22) form a hydrogen bond with the hydrogen atom of the amino group of an incoming nucleophile thus favoring combination of the two reactants. In the case of 7-azabenzotriazole esters, the unshared electrons may be provided by the N7 atom.[19]

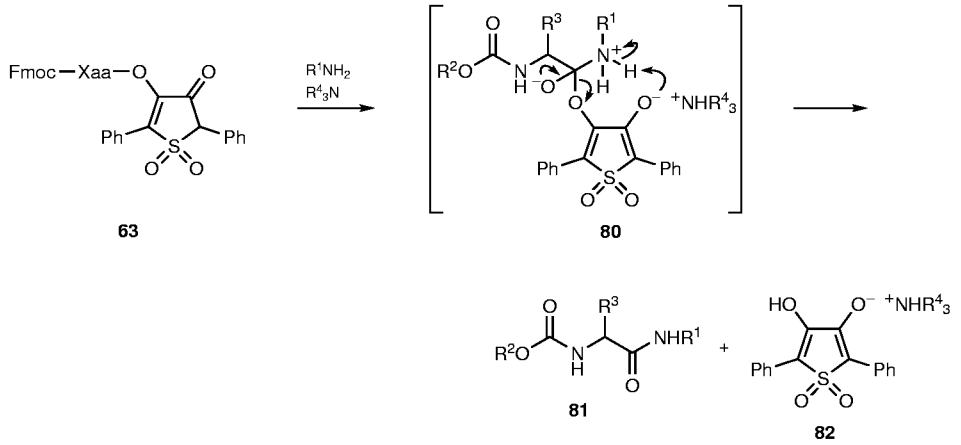
Scheme 22 Anchimeric Assistance in the Aminolysis of Active Esters Containing a Substituted Hydroxylamine Moiety[124]



77 **78** **79**

A third phenomenon is implicated in the aminolysis of 1,1,4-trioxo-2,5-diphenyl-4,5-dihydro-3-thienyl esters **63** (Scheme 23). Decomposition of the tetrahedral intermediate is promoted by internal base catalysis by the anion **80** that is formed by enolization of the 4-oxo function of **63** that is effected by a tertiary amine, generating peptide **81** and the anion of hexasubstituted thiophene **82**.[125]

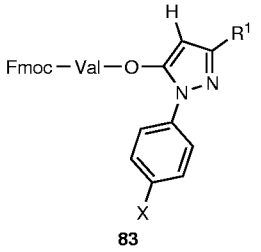
Scheme 23 Mechanism of Aminolysis of 1,1,4-Trioxo-2,5-diphenyl-4,5-dihydro-3-thienyl Esters[125]



63 **80** **81** **82**

With respect to the influence on reactivity of the nature of the substituents and the interacting residues, Z and Boc amino acid derivatives exhibit similar reactivities,[9,126] esters of hindered residues valine and isoleucine are much less reactive than those of other residues,[42,51,117,126] and *tert*-butyl esters are significantly better nucleophiles than ethyl esters.[126] Benzotriazolyl esters of *N*-(alkoxycarbonyl)-*N*-methyl amino acids have greatly diminished reactivity relative to those of unmethylated residues,[93,127] however the succinimido esters react sufficiently to be of practical use.[93] A variant of esters of substituted hydroxylamines that can be used to generate amide and peptide bonds is the acyloxime resin.[128,129] A new type of active ester has been developed based on the reaction of acids with substituted pyrazolinones, the products being 1-phenylpyrazolinone enol esters **83** (Scheme 24).[130,131] More than a dozen of these esters of Fmoc-valine have been characterized. They are about as reactive as esters **63**, are self indicating, and show promise for the future.

Scheme 24 1-Phenylpyrazolinone Enol Esters[130,131]



83

R^1 = Me, Ph; X = Cl, NO_2

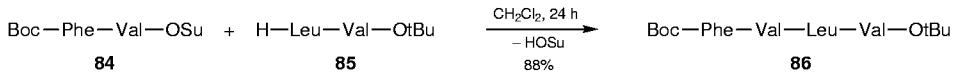
3.2.3.1 Aminolysis by an Amino Acid or Peptide Ester or Amide

Active esters are used routinely for single residue addition in solution synthesis at room temperature.[11,13,42,46,64,75] The 4-nitrophenyl and pentafluorophenyl esters of asparagine and glutamine derivatives are used routinely instead of DCC and the acids for coupling these residues in solid-phase synthesis to avoid the dehydration of carboxamido groups (Section 3.2.1.1.1) that accompanies use of the carbodiimide. While succinimido esters of Fmoc amino acids have been found to be unsatisfactory, nitrophenyl,[57–59] pentafluorophenyl[71] and 4-oxo-3,4-dihydrotriazin-3-yl esters[86] have been used successfully for solid-phase synthesis. The latter have the unique property of generating a yellow color during aminolysis due to ion pairing between the anion that is generated by abstraction by the amino group of the proton of the 3-hydroxy-4-oxo-3,4-dihydrobenzotriazine that is liberated and the protonated amino group.[14,86] The color disappears once all the amino groups have been consumed by peptide bond formation. This phenomenon has been utilized for monitoring the course of coupling reactions in continuous-flow solid-phase synthesis.[86] Pentafluorophenyl esters of Fmoc amino acids bearing an N^{α}-(2-fluorenylmethoxy-4-methoxybenzyl) substituent **45** have been used to introduce a *N*-alkyl group into a peptide chain in order to prevent secondary structure formation that impedes chain assembly.[132] Active esters of *N*-trimethylsilylethoxycarbonyl and Boc amino acids have been used for coupling to dipeptide methyl esters that were generated by *N*-benzyloxycarbonyl removal by hydrogenation in the presence of the active esters in order to avoid dioxopiperazine formation,[133] which can be avoided also by use of the dipeptide *tert*-butyl esters.[133,134] Pentafluorophenyl esters have been used for both temporary protection and then activation during glycosylation of the side chains of Fmoc-aspartic acid[135] and serine[80] derivatives prior to chain extension at the α-carboxy groups in the usual manner. Protected tripeptide Z-Pro-Val-Gly-OEt was prepared in 88% yield in water by acylation of the dipeptide ester with Z-Pro-ONs.[24] In order to speed up a synthesis, 4-nitrophenyl esters of monomers were aminolyzed in the presence of an equivalent of 1,2,4-tetrazole that converted them into the more reactive tetrazolides.[41]

Aminolysis of peptide esters occurs uneventfully but this approach to peptide bond formation is not used routinely because access to enantiomerically pure active esters of peptides is straightforward only when the activated residue is glycyl or prolyl (Section 3.2.2). Succinimido esters of small peptides with glycine or proline at the carboxy terminus have been used extensively for the preparation of larger segments in solution.[108] Aminolysis by dipeptide ester **85** of dipeptide succinimido ester **84**, obtained through the mixed anhydride (Scheme 19), gave enantiomerically pure (<0.5% L-D-L-L isomer) protected tetrapeptide **86** in 88% yield (Scheme 25).

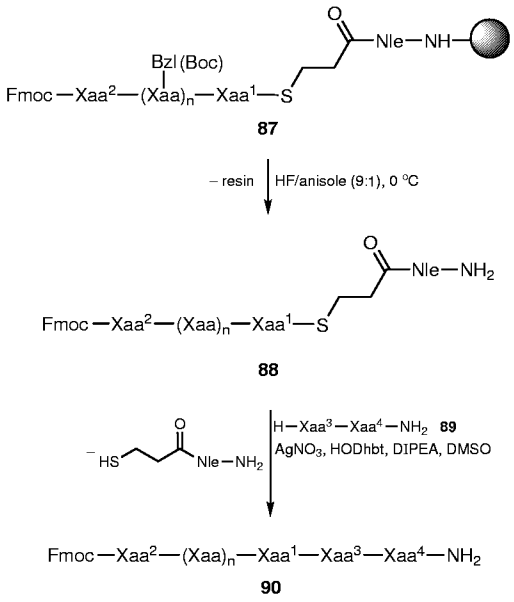
for references see p 472

Scheme 25 Synthesis of a Protected Tetrapeptide by Aminolysis of the Succinimido Ester of a Dipeptide[93,109]



A recent novel approach to the use of active esters is assembly of a segment utilizing Boc/benzyl chemistry of a chain **87** that is attached to a resin-bound β-mercaptopropionyl residue through a thioester linkage.[149,151] The last residue is incorporated as the Fmoc derivative. Benzyl side-chain protectors are removed by HF at the same time as the chain is released from the resin giving N^{α}-Fmoc-protected peptide thioester **88**. Aminolysis of active ester **88** is then effected by peptide amide **89** giving N^{α}-Fmoc-substituted peptide amide **90** (Scheme 26). Thioesters are also one of the activated forms involved in segment condensation by the procedure known as chemical ligation[150] (see Section 4.1.5).

Scheme 26 Segment Coupling via Active Thioesters Obtained by Solid-Phase Synthesis



Z-Asn-Cys(Bzl)-OMe via the Nitrophenyl Ester; Typical Procedure:[64]
Z-Asn-ONp (0.78 g, 2 mmol) and H-Cys(Bzl)-OMe · HCl (0.66 g, 2.5 mmol) were dissolved in DMF (2 mL) and TEA (0.38 mL, 2.7 mmol) was added. After 1 h, the solid product was triturated with H_2O (50 mL), collected by filtration, and washed with H_2O and acetone; yield: 0.76 g (80%); mp 199–201 °C; $[\alpha]_D$ −31.5 (*c* 1, pyridine)

Boc-Gly-Phe-Gly-OEt via the Succinimido Ester; Typical Procedure:[13]
A soln of Boc-Gly-OSu (2.04 g, 75 mmol), H-Phe-Gly-OEt · HBr (2.48 g, 75 mmol), and TEA (1.05 mL, 75 mmol) in DME (25 mL) was stirred for 20 min. Cold H_2O (100 mL) was added and the crystals that formed (2.58 g; mp 98–99 °C) were recrystallized (EtOH/H_2O); yield: 79%; mp 100.5–101.5 °C; $[\alpha]_D$ −9.75 (*c* 2.05, MeOH).

Z-Glu(OtBu)-Ala-OMe via the Pentachlorophenyl Ester; Typical Procedure:[11]
To a soln of Z-Glu(OtBu)-OPcp (6 g, 10.2 mmol) in CH_2Cl_2 (50 mL) was added H-Ala-OMe · HCl (11.6 g, 11.5 mmol) and TEA (1.6 mL, 11.5 mmol) in CH_2Cl_2 (10 mL) with stirring. 2-Hydroxypyridine (0.15 g) was added[117] and stirring was continued at rt for 24 h. The TEA · HCl was removed by filtration, the solvent was removed, the residue was taken up in EtOAc (100 mL) which was filtered, the filtrate was washed with 1 M HCl (3 × 30 mL), aq $NaHCO_3$, and H_2O, dried (Na_2SO_4) and the solvent was removed. The residue was crystallized [Et_2O/petroleum ether (bp 30–60 °C)]; yield: 3.1 g (72%); mp 97–98 °C; $[\alpha]_D$ −6.7 (*c* 3.24, $CHCl_3$).

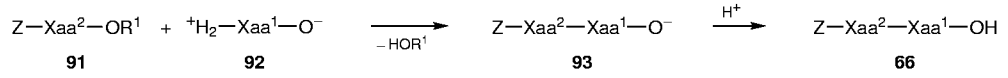
Z-Asn-Gly-OtBu via the Pentafluorophenyl Ester; Typical Procedure:[46]
To a soln of Z-Asn-OPfp (0.43 g, 1 mmol) in dioxane (10 mL) at 10 °C was added H-Gly-OtBu · HCl (0.185 g, 1.1 mmol) followed by TEA (0.15 mL, 1.1 mmol) while stirring. After stirring at rt for 30 min, the mixture was filtered, the solvent was removed, and the residue was triturated in Et_2O; yield: 0.34 g (90%); mp 150–151 °C.

Z-(Me)Xaa-Val-OMe via the Succinimido Ester; General Procedure:[93]
Solid Z-(Me)Leu-OSu (2.04 g, 0.5 mmol) was added to a soln of H-Val-OMe in dried CH_2Cl_2 (10 mL) that had been obtained by washing a soln of H-Val-OMe · HCl (0.84 g, 0.5 mmol) with aq $NaHCO_3$. After 24 h at rt the soln was washed with 1 M HCl, aq $NaHCO_3$, and H_2O, dried ($MgSO_4$) and the solvent was removed to give an oil; yield: 90%; $[\alpha]_D$ −71.8 (*c* 1.34, $CHCl_3$). Z-(Me)Val-Val-OMe was obtained as an oil; yield: 75%; $[\alpha]_D$ −92.0 (*c* 1.0, $CHCl_3$). The compounds were pure based on analysis by HPLC and NMR spectroscopy.

3.2.3.2 Aminolysis by an Amino Acid or Peptide Anion

Aminolysis of active ester **91** can be effected by the amino group of an amino acid or peptide that is not protected at the terminal carboxy group (Scheme 27). Zwitterionic amino acid **92** or peptide is deprotonated by the addition of a base that generates the anionic form of the reactant. A partially aqueous milieu containing a water-miscible solvent such as acetone, dioxane, or acetonitrile is best for solubilizing the reactants. Esters that are hydrolyzed slowly or not at all are preferred because the acid produced is not readily separated from target molecule **66** (obtained from anion **93**) because of their similar properties. Piperidino esters do not undergo hydrolysis but their aminolysis is slow.[4] 4-Nitrophenyl esters react efficiently but the nitrophenol is difficult to remove. Succinimido esters[13] are employed most frequently and are particularly useful in the preparation of N^{α}-protected dipeptides. Use of an excess of amino acid **92** favors complete consumption of active ester **91** by aminolysis before any of it can hydrolyze.

Scheme 27 Aminolysis of an Active Ester by an Amino Acid Anion



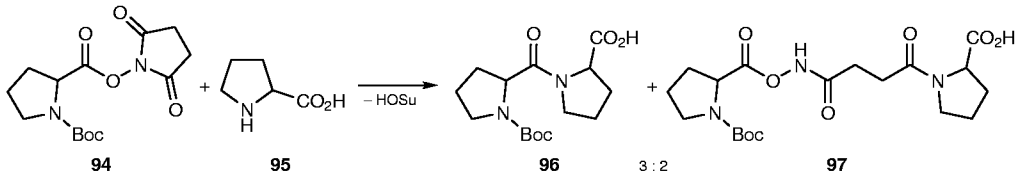
R^1	Conditions	Ref
OSu	$NaHCO_3$, EtOH/H_2O (1:1)	[13]
OSu	aq Na_2CO_3, acetone	[69]
OPip	1 M NaOH, dioxane	[4]
OTcp	DCHA, CH_2Cl_2	[112]
OSu	Na_2CO_3, DMF/H_2O (4:1)	[92]

Successive aminolysis of pentachlorophenyl esters of Z amino acids in the presence of dicyclohexylamine, the protectors being removed by catalytic hydrogenolysis, has been used to prepare Z-tetrapeptides; however the yields diminished from 65–70% initially to the 40% range for aminolysis of the di- and tripeptide esters.[136] Successive aminolysis of succinimido esters of Boc amino acids in the presence of diisopropylethylamine has been used to synthesize Boc-Thr(Bzl)-Asn-Cys(Bzl)-Pro-OH in kilogram amounts.[137] Attempted reaction of the free tetrapeptide with Boc-Ile-OSu then failed. A modification of this approach where the reaction was carried out in dimethylformamide without the addition of water or base proved efficient for the preparation of several peptides on a multigram scale.[34] Reactants that were initially insoluble gradually went into solution. Because the pK_a of the amino group of a peptide is 1–1.5 pH units lower than the pK_a of the amino acid and close to neutrality there is sufficient unprotonated primary amine in equilibrium with the zwitterion for aminolysis to take place without the addition of base. The acetate salt of an arginyl-containing tetrapeptide also reacted with an active ester without the addition of

for references see p 472

base.[138] A side reaction in the synthesis of Boc-dipeptide **96** involving aminolysis at the carbonyl of the succinimido moiety of Boc-proline **94**, giving succinic acid diamide **97**, has been observed in a coupling with proline **95** (Scheme 28).[139] The side reaction can be avoided by first converting **95** or hydroxyproline into the benzyltrimethylammonium salt and carrying out the reaction in dimethylformamide.[140]

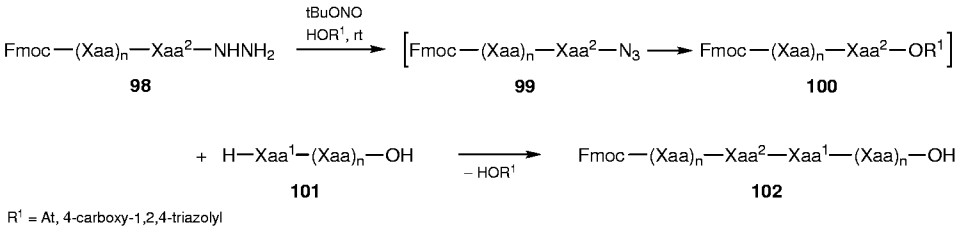
Scheme 28 Aminolysis at the Carbonyl of the Succinimido Moiety of an Ester in the Reaction between Two Prolyl Residues[139]



94 **95** **96** **97**

The disadvantage of slow coupling of peptide azides **99** has been circumvented by transesterifying them into the 7-azabenzotriazolyl or 4-carboxy-1,2,4-triazolyl esters **100** simultaneously with their generation from the hydrazides **98** (Scheme 29).[141] Azides **99** are formed by reaction of hydrazides **98** with tertiary butyl nitrite in the presence of substituted hydroxylamines that serve as acids. The products of aminolysis by peptides **101** are N^{α}-protected peptides **102**.

Scheme 29 Transfer Active Ester Condensation[141]



98 **99** **100**

101 **102**

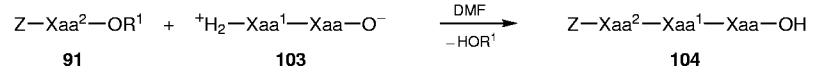
R^1 = At, 4-carboxy-1,2,4-triazolyl

Side Reaction of Enantiomerization during Aminolysis by an Amino Acid Anion

Active esters **2**, with piperidino esters **10** being an exception,[4] enantiomerize when left in the presence of tertiary amine.[92,121] Aminolysis of active esters by a carboxy-substituted residue is not carried out in the presence of tertiary amine so the esters do not isomerize under normal operating conditions. However, a base is present when active esters are aminolyzed by an amino acid anion and this can result in enantiomerization.[142] A water-miscible solvent and water are employed in order to solubilize the reactants. The degree of enantiomerization depends on many factors that include the nature of the organic cosolvent, dimethylformamide leading to more than *N*-methylpyrrolidone or tetrahydrofuran. But the cyclic solvents produce too much hydrolysis so the best cosolvent is dimethylformamide. Contrary to what might be expected more isomerization occurs in the presence of weaker bases such as sodium hydrogencarbonate and *N*-methylmorpholine; the latter is also unsatisfactory because it promotes hydrolysis. Less isomerization occurs when diisopropylethylamine or sodium carbonate are employed because they induce quicker consumption of the active ester due to their efficiency as deprotonators. The chiral sensitivity is in the order pentafluorophenyl > 4-nitrophenyl > succinimido. This suggests that something else besides contact of the ester with base is harmful since the fluorinated ester is the most active of the three. The isomerization was significant for aminolysis by D-valine neutralized by an equivalent of sodium hydrogencarbonate, ~10% of epimer being produced from the

4-nitrophenyl ester of Z-phenylalanine. But negligible amounts of epimer were produced by aminolysis with D-leucine or D-phenylalanine. The limited data suggest that the isomerization is related to bulkiness in the side chain of the aminolyzing residue. Isomerization was effectively suppressed by use of sodium carbonate in 50% excess as base combined with a 50% excess of the amino acid **92**, the latter being the modification with the greatest impact. On the other hand, if aminolysis is effected by a dipeptide or larger peptide **103** without the addition of base, peptide **104** is obtained without enantiomerization (Scheme 30).[34]

Scheme 30 Aminolysis of an Active Ester by a Peptide without the Addition of Base[34]

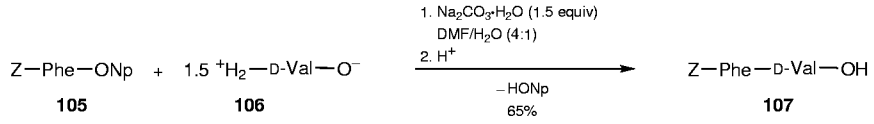


Z-Arg(Z)$_2$-Arg-Ala-Gln-Asp(OtBu)-OH via the 4-Nitrophenyl Ester:[138]
Z-Arg(Z)$_2$-ONp (40.0 g, 57 mmol) was added to H-Arg-Ala-Gln-Asp(OtBu)-OH · AcOH (29.6 g, 49 mmol) in DMF (500 mL) at −15 °C and the mixture was stirred at 0 °C for 30 h and rt for 20 h. A small amount of solvent was removed under reduced pressure, the remainder was poured into warm EtOAc with stirring, the crystals were collected, washed with EtOAc and Et_2O, triturated in H_2O, and recrystallized (DMF/EtOAc); yield: 48.7 g (93%); mp 230 °C (dec); $[\alpha]_D$ +15.5 (*c* 1.5, 80% AcOH).

Z-Gly-Xaa-OH via the Succinimido Ester; General Procedure:[69]
A mixture of H-Xaa-OH in H_2O (4 mL/mmol) containing Na_2CO_3 (2 equiv) was stirred until dissolution or for 1 h. A soln of Z-Gly-OSu (0.95 equiv) in acetone (3–5 mL/mmol) was added dropwise over 1.5 h while stirring. After further stirring for 16 h, the acetone was removed, the soln was diluted with H_2O, brought to pH 3.2 with 2 M HCl and extracted twice with EtOAc. The combined extracts were washed with 1 M HCl, dried ($MgSO_4$), and concentrated to an oil. For Xaa = Ser, the aqueous phase was not diluted but saturated with NaCl before extraction. For Xaa = Tyr(Bzl) the foam obtained was purified by column chromatography [Kieselgel, $CHCl_3$/MeOH/AcOH (45:4:1) then $CHCl_3$/MeOH/AcOH (17:2:1)]. Z-Asp(OBzl)-Gly-OH · DCHA (mp 137–138 °C) was obtained similarly from Z-Asp(OBzl)-OSu after purification on a column of Kieselgel using the latter solvent followed by addition of DCHA, removal of the solvent and trituration of the residue with Et_2O.

Z-Gly-Val-OH via the Succinimido Ester Prepared In Situ; General Procedure:[92]
A soln of Z-Gly-OSu was prepared from Z-Gly-OH (1 mmol) through the mixed anhydride (Section 3.2.1.2) and the solvent was removed. The residue was dissolved in DMF/H_2O (4:1, 10 mL), a soln of valine (1.5 equiv) and Na_2CO_3 (1.45 equiv) in DMF/H_2O (4:1) was added and the mixture was stirred for 8 h. CH_2Cl_2 (10 mL) and 5% aq $NaHCO_3$ (10 mL) were added, the mixture was shaken and the aqueous layer was acidified with 12 M HCl. The soln was extracted with CH_2Cl_2 (2 × 10 mL), the extract was dried ($MgSO_4$), the solvent was removed and residual DMF was removed under reduced pressure at 70 °C; yield: 81% containing a trace of HOSu; after recrystallization (CH_2Cl_2/petroleum ether); mp 112–113 °C.

Scheme 31 Aminolysis of a 4-Nitrophenyl Ester by an Amino Acid Anion Under Conditions that Suppress Enantiomerization[142]



Z-Phe-D-Val-OH (107) under Conditions that Suppress Enantiomerization; Typical Procedure (Scheme 31):[142]
D-Valine (**106**; 176 mg, 1.5 mmol) was dissolved in H_2O (3 mL) containing $Na_2CO_3 \cdot H_2O$ (186 mg, 1.5 mmol), DMF (6 mL) was added and the soln was poured into DMF (6 mL) containing Z-Phe-ONp (**105**; 420 mg, 1 mmol). The mixture was stirred at rt for 15 min, EtOAc (60 mL) and 1 M HCl (30 mL) were added, the mixture was shaken and the organic layer was collected. The aqueous layer was washed with EtOAc (60 mL) and the combined organic extracts were washed with H_2O (4 × 20 mL) to remove DMF, 5% aq $NaHCO_3$ (10 mL) was added, and the aqueous layer was taken and washed with EtOAc (10 mL). The soln was acidified with 1 M HCl and extracted with EtOAc (2 × 60 mL), the extract was dried ($MgSO_4$), the solvent was removed and the residue was crystallized (CH_2Cl_2/petroleum ether).

for references see p 472

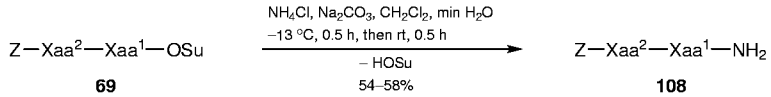
Compound **107** was recrystallized (EtOAc); yield: 65%; mp 152–153 °C; $[\alpha]_D$ −16.5 (*c* 1.02, EtOH). It contained <0.1% of Z-D-Phe-D-Val-OH and 0.2% of Z-Phe-OH. Material prepared using equivalent amounts of D-valine and $NaHCO_3$ or Na_2CO_3 contained 10.2 and 4.7% of epimer, respectively.

Z-Asp(OtBu)-Phe-Ser(tBu)-OH via the Succinimido Ester without Base; Typical Procedure:[34]
A soln of Z-Asp(OtBu)-OSu (62.2 g, 0.148 mol) and H-Phe-Ser(tBu)-OH · H_2O (48 g, 0.147 mol) in DMF was left at rt for 12 h. The bulk of the DMF was removed and the residue was partitioned between EtOAc and 1% aq $KHSO_4$. The organic layer was dried (Na_2SO_4), the solvent was removed and the residue was crystallized (EtOAc/Et_2O/petroleum ether); yield: 87% (97%); mp 134–136 °C; $[\alpha]_D^{20}$ −8.5 (*c* 1, MeOH).

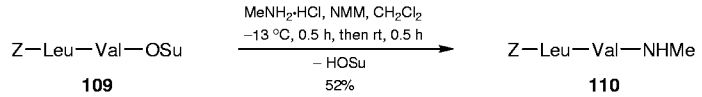
3.2.3.3 Aminolysis by Ammonia or Substituted Amines

Active esters of *N*-alkoxycarbonyl amino acids such as 4-nitrophenyl and succinimido react efficiently with ammonia to give the amides.[143,152] However, there are several simpler techniques such as the mixed anhydride method for obtaining amides from *N*-alkoxycarbonyl amino acids. A convenient approach involves in situ preparation of benzotriazolyl esters using DCC followed by reaction with ammonium hydroxide.[144] A variant utilizes diaminomethane dihydrochloride in the presence of triethylamine instead of aqueous ammonia.[145] An approach analogous to the former method involves a DCC-mediated reaction with salts formed from *N*-hydroxysuccinimide or 3-hydroxy-3,4-dihydrobenzotriazin-4-one and ammonia or alkylamines.[146] The products obtained from monomers are enantiomerically pure. However, the challenge is to make enantiomerically pure amides from N^{α}-protected peptides. No method is yet recognized as generally applicable because of the danger of enantiomerization of the activated residue during amidation. The two peptide amides obtained from benzotriazolyl esters prepared in situ had low specific rotations and unestablished enantiomeric purities.[144] In our hands, reaction of Z-Leu-Val-OSu (**109**) with diaminomethane hydrochloride/triethylamine[145] gave a product that was highly (45% L-D) epimerized.[93] Peptide amides obtained from 3-hydroxy-3,4-dihydrobenzotriazin-4-one and ammonia/alkylamine salts were not enantiomerically pure.[146] On the other hand, a preliminary study has demonstrated that aminolysis and methylaminolysis of succinimido esters can be effected without loss of chirality if stringent conditions are respected.[93,109] Reactions of Z-Leu-Val-OSu (**109**) involving a reactive hindered residue gave products containing <0.5% of epimeric contaminants. Thus, Z-dipeptide esters **69** in dichloromethane with ammonium chloride in the presence of sodium carbonate in the minimum amount of water required for dissolution gave amides **108** that precipitated out of solution (Scheme 32 and Table 17). A critical feature to suppress isomerization was the low temperature. The analogous reaction of ester **109** with methylamine hydrochloride in dichloromethane in the presence of *N*-methylmorpholine gave methylamide **110** (Scheme 33). It is likely that the method would be applicable for the preparation other alkylamides.

Scheme 32 Amides of N-Protected Dipeptides by Reaction of Succinimido Esters with Ammonium Chloride Neutralized with Sodium Carbonate[93,109]

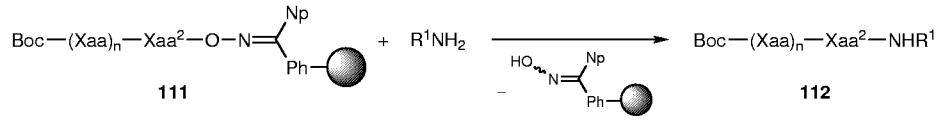


Scheme 33 Synthesis of a Peptide Methylamide by Reaction of a Succinimido Ester with Methylamine Hydrochloride Neutralized with *N*-Methylmorpholine[93,109]



Peptide amides with glycine at the carboxy terminal have routinely been obtained by aminolysis of a benzyl ester that is linked to a resin.[147] A more recent approach that involves an active ester is aminolysis of peptidyloxime resin **111** by a substituted amine giving amide **112** (Scheme 34).[128,129] A series of alkyl amides of Boc-trialanine have been prepared[148] but the issue of enantiomeric purity was not addressed.

Scheme 34 Amide Bond Formation by Aminolysis of Oxime Resin[128,129,148]



R^1 = H, alkyl, $CHR^2CO_2R^3$

Z(OMe)-Ala-NH_2; Typical Procedure:[144]
To a soln of Z(OMe)-Ala-OH (2.53 g, 10 mmol) in CH_2Cl_2/DMF (4:1, 50 mL) was added HOBt (1.37 g, 11 mmol) and DCC (2.17 g, 10.5 mol) and the mixture was stirred for 0.5 h. 25% Aq NH_3 (1 mL, 13 mmol) was added and stirring was continued for 3–5 h. The precipitate was removed by filtration, the filtrate was diluted with CH_2Cl_2 (100 mL), and the soln was washed with 5% aq $NaHCO_3$ (3 × 30 mL), H_2O (30 mL), 5% aq citric acid (2 × 20 mL), and H_2O (30 mL), and dried (Na_2SO_4). The solvent was removed and the product was recrystallized (MeOH/Et_2O 1:5); yield: 2.09 (83%); mp 134–136 °C; $[\alpha]_D$ −2.0 (*c* 2.0, MeOH).

Z-Leu-Val-NH_2; Typical Procedure:[109]
A soln of Z-Leu-Val-OSu (1.02 g, 2 mmol) in CH_2Cl_2 at −13 °C was added to a cold soln of NH_4Cl (0.13 g, 2.5 mmol) and Na_2CO_3 (0.15 g, 1.25 mmol) in a minimum of H_2O that had been warmed to facilitate dissolution. The mixture was stirred at −13 °C for 0.5 h and at rt for 0.5 h. The white precipitate was collected, washed with cold CH_2Cl_2 and recrystallized (MeOH); yield: 58%; mp 197–199 °C; $[\alpha]_D$ −30.1 (*c* 0.92, MeOH). The product contained <0.5% of L-D isomer based on analysis by RP-HPLC [C-18 column, MeCN/H_2O (1:3) 2 mL · min^{-1}]; t_R L-L 27.0 min, t_R L-D 31.3 min.

Z-Leu-Val-NHMe; Typical Procedure:[109]
To Z-Leu-Val-OSu (0.51 g, 1 mmol) in CH_2Cl_2 (10 mL) at −13 °C was added $MeNH_2 \cdot HCl$ (0.08 g, 1.25 mmol) and NMM (0.126 g, 1.25 mmol) in cold CH_2Cl_2 (5 mL), and the soln was stirred at −13 °C for 0.5 h and rt for 0.5 h. The white precipitate was collected, washed with cold CH_2Cl_2, and recrystallized (MeOH); yield: 52%; mp 202–203 °C; $[\alpha]_D$ −35.6 (*c* 0.84, MeOH). The product contained <0.4% of L-D isomer based on analysis by RP-HPLC [C-18 column, MeCN/H_2O (1:3) at 2 mL · min^{-1}]; t_R L-L 34.5 min, t_R L-D 40.2 min.

Table 17 Z-Xaa^2-Xaa^1-NH_2 Prepared by Reaction of Succinimido Esters with Ammonium Chloride[93,109]

Compound	Crystallization Solvent	Yield (%)	mp (°C)	$[\alpha]_D$ (*c* 1, 95% MeOH)	Ref
Z-Val-Leu-NH_2	MeOH	56	197–199	−30.1	[93]
Z-Gly-Val-NH_2	CH_2Cl_2	54	190–192	−8.2	[109]
Z-Gly-Phe-NH_2	MeOH/Et_2O	56	134–136	+11.5[a]	[93]

[a] (*c* 1.85).

for references see p 472

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3.3 Acid Halides

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The exploration of the simple acid chloride activation for linking amino acids through amide bonds goes back to the very first attempts of peptide synthesis. In 1903 Fischer described the preparation of *N*-ethoxycarbonylglycine chloride and its reaction with amino acid esters to produce dipeptides.[1] Unfortunately, it was not possible to remove the *N*-ethoxycarbonyl group without destroying the newly formed peptide bond. Subsequently, new α-amino protecting groups were frequently studied with respect to their compatibility with acid chloride activation.[2] Real application of protected amino acid chlorides to peptide synthesis had to await the discovery of the first amide-compatible amino protecting group, the benzyloxycarbonyl (Z) group, by Bergmann and Zervas[3] who applied the technique in a number of peptide syntheses. It soon became clear that N^{α}-Z-protected amino acid chlorides show only limited shelf stability, due to spontaneous decomposition to the corresponding amino acid *N*-carboxyanhydrides with loss of benzyl chloride.[4] The more recently examined *tert*-butoxycarbonyl amino acid derivatives[5] were found to be even less stable. Another serious side reaction of Z-protected amino acid chlorides, greatly facilitated by the presence of two geminal alkyl groups, was observed for α-aminoisobutyric acid (Aib). Attempts to convert Z-Aib-OH into the acid chloride gave only the corresponding oxazol-5(4*H*)-one. Conversely, the *N*-tosyl derivative, Tos-Aib-Cl, which is unable to form an oxazol-5(4*H*)-one, was readily obtained and so represented an efficient acylation reagent. This did not prove to be a general solution, however, as deblocking of the N^{α}-tosyl group[6] cannot be achieved under mild conditions.[7] Other derivatives, such as the N^{α}-phthaloyl amino acid chlorides[8] or {[2-(triphenylphosphonio)ethyl]oxy}carbonyl amino acid chlorides,[9] however have successfully been used. Development of the 9-fluorenylmethoxycarbonyl (Fmoc) group[10] as N^{α}-protectant allowed for the first time both the preparation of the corresponding shelf-stable acid chlorides[11,12] from the commercially available amino acid derivatives and deprotection under mild conditions after their incorporation into peptides for stepwise synthesis. Successful use of Fmoc amino acid chlorides was reported for the synthesis of short peptides in a two-phase solvent system.[11–14] The high degree of activation associated with an acid chloride makes couplings of very hindered or weakly nucleophilic systems possible where other methods give only poor results. For reactions carried out in homogeneous solution, the addition of tertiary amines, required for capture of the hydrogen chloride liberated during the coupling process, causes the Fmoc-protected amino acid chlorides to be converted into oxazol-5(4*H*)-ones, and, therefore, limits drastically the application of this method.[15] If reactions of these acid chlorides are carried out with additives such as silver cyanide[16,17] or ammonium[15,18] and potassium[18–22] salts of HOBt, they can be successfully used in solution and solid-phase synthesis for coupling of sterically hindered amino acids. In contrast, Fmoc amino acid fluorides, relative to the corresponding chlorides, show only a very slow rate of oxazol-5(4*H*)-one formation in the presence of a tertiary amine and are equally stable in the case of amino acids bearing tBu, Boc, or *N*-Trt side-chain protections.[95,102] These properties of Fmoc amino acid fluorides and their high reactivity have made them ideal intermediates for both solution[23,24] and solid-phase peptide synthesis.[23–27] In contrast to all other common coupling systems, the fluorides are useful even in the case of hindered amino acids such as Aib[25–28] and very weak nucleophiles.[29,30] Due to their reduced activation relative to the corresponding acid chlorides, acid fluorides of Z-[24,31–33] and Boc-protected amino acids[31] are readily prepared. For *N*-urethane protected amino acid fluorides, which are prone to ready cyclization, e.g. Fmoc-$C^{\alpha,\alpha}$-diphenylglycine fluoride, conversion into oxazol-5(4*H*)-one occurs within minutes, which limits their application for coupling to more sterically hindered amino components. In this case, incorporation via the corresponding α-azido acid chlorides followed by reduction of the azido to the amino function has been successfully used (see

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Section 2.1.1.10.2).[34] For coupling of extremely hindered systems, such as N-protected Aib or (Me)Aib to amine-free (Me)Aib derivatives, arylsulfonic-protected amino acid chlorides proved to be the method of choice, while the corresponding acid fluorides do not react at all.[35,36]

3.3.1 N^{α}-9-Fluorenylmethoxycarbonyl Amino Acid Chlorides

Conditions for the efficient use of Fmoc-protected amino acid chlorides in stepwise peptide synthesis have been delineated. The acid chlorides are reasonably stable solids that are readily aminolyzed if a base is added to neutralize the hydrogen chloride liberated in the reaction. They form the corresponding oxazol-5(4*H*)-ones and undergo stereomutation if left in the presence of tertiary amines, however, amide formation without stereomutation can be obtained under appropriate conditions. The acid chlorides are advantageously applied to acylate weak nucleophiles and sterically hindered amino components, but their application may be limited by the lack of stability of *tert*-butyl-type side-chain protecting groups toward conditions of acid chloride preparation.

3.3.1.1 Synthesis of N^{α}-9-Fluorenylmethoxycarbonyl Amino Acid Chlorides

Following the first report on the in situ generation and use of an N^{α}-protected amino acid chloride,[37] all of the bifunctional amino acids as well as those bearing benzyl- or allyl-based side-chain protection were converted into stable, crystalline acid chlorides.[11] As expected, amino acids bearing *tert*-butyl-type protected side-chains cannot be transformed into the chloride. In some cases, e.g. Asp and Glu, the acid chlorides of the side-chain protected amino acids are not obtained due to their facile conversion to cyclic anhydrides with loss of the *tert*-butyl protection. In other cases, e.g. Tyr, Ser, and Thr, the related chlorides are obtained but their shelf stability appears insufficient for practical utilization.[23] Whether the slow degradation of these derivatives on storage is due to the presence of impurities resulting from their synthesis or to some inherent instability is not yet clear. Fmoc amino acid chlorides are conveniently prepared from the corresponding acids by treatment with thionyl chloride[11,12,19,21,22] or alternatively by reaction of the Fmoc amino acid mixed anhydride with hydrogen chloride (Table 1).[38]

In addition to the acid chlorides listed in Table 1, Fmoc-protected amino acid chlorides of δ-aminovaleric acid[13] and Tyr(tBu)[15] were prepared and used in peptide synthesis, but these derivatives were contaminated with impurities due to their low stability.

Table 1 N^{α}-9-Fluorenylmethoxycarbonyl Amino Acid Chlorides[19,21,22]

Xaa	Yield (%)	mp (°C)	Ref	Xaa	Yield (%)	mp (°C)	Ref
Gly	95	108–109	[11]	Aib	84	164–165	[21]
Ala	81	112–114	[11]	(Me)Phe	92	103–105	[19]
Ile	87	103–114	[11]	(Me)Val	79	95–98	[19]
Pro	79	93–94	[11]	(Me)Met	82	66–68	[19]
Val	85	111–112	[11]	(Me)Ala	87	92–95	[19]
Met	90	118–119	[11]	(Me)Leu	76	87–89	[19]
Cys(Bzl)	95	104–105	[11]	Sar	72	158–160	[19]
Lys(Z)	88	67–68	[11]	(Me)Phg	83	137–139	[19]
Phe	96	120–121	[11]	PheGly	86	148–150	[19]
Leu	96	80–81	[11]	Ac_5c	85	122–125	[22]
Tyr(Bzl)	91	114–115	[11]	Ac_6c	87	172–173	[22]
Ser(Bzl)	91	96–98	[11]	Ac_7c	80	143–145	[22]
Glu(OBzl)	84	102–103	[13]	Deg	75	58–59	[22]
Asp(OBzl)	89	50–53	[11]	Dpg	83	62–64	[22]
Phg	76	138–140	[12]	Dbug	84	65–66	[22]

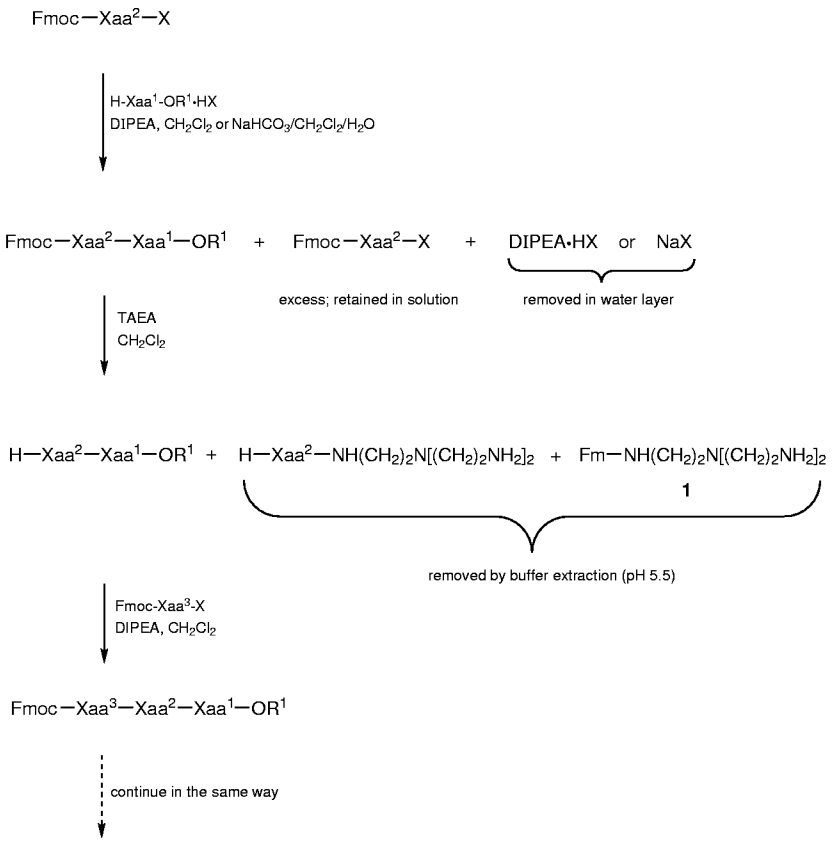
Fmoc-Val-Cl; Typical Procedure:[11]
A soln of Fmoc-Val-OH (0.3 g, 0.88 mmol) in CH_2Cl_2 (5 mL) was treated with $SOCl_2$ (1.05 g, 8.8 mmol) and the mixture refluxed for 15 min under N_2 atmosphere. In other cases up to 1–2 h refluxing was required to complete the reaction. In the presence of DMF (0.1 equiv) as catalyst, complete conversion into the acid chloride occured within 1 h at rt. Solvent and excess $SOCl_2$ were removed under reduced pressure and the residue was dissolved in CH_2Cl_2 (1 mL). After addition of hexane (10 mL) the resulting precipitate was collected and recrystallized (CH_2Cl_2/hexane 1:10); yield: 0.27 g (85%); mp 111–112 °C.

3.3.1.2 Peptide Synthesis with N^{α}-9-Fluorenylmethoxycarbonyl Amino Acid Chlorides

3.3.1.2.1 Synthesis in a Two-Phase System

A convenient method of coupling Fmoc amino acid chlorides involves a two-phase system with a mild inorganic base in the aqueous layer to minimize contact with the acid chloride. Under these conditions peptide bond formation is completed within a few minutes and occurs without loss of configuration at the carboxylic component.[11–14] By following the very rapid coupling via Fmoc amino acid chlorides with an equally rapid deblocking step involving the dibasic amine AMP[11,13] or preferably tris(2-aminoethyl)amine (TAEA)[14], it is possible to devise a rapid, stepwise solution synthesis of short peptides without isolation of intermediates (Scheme 1). Deblocking is over in a few minutes, and the DBF-adduct **1** can be selectively extracted by means of a buffer at pH 5.5, leaving in the organic phase only the growing peptide ready for the next coupling step.

Scheme 1 Rapid Continuous Solution Synthesis



for references see p 492

In order to assure high coupling yields, a 10% or greater excess of Fmoc amino acid chloride is used over the initial amino acid ester or growing peptide ester. Removal of excess acid chloride after coupling is achieved by using a large excess of the polybasic amine, which forms the corresponding amide and, at the same time, acts as deblocking reagent of both the Fmoc amino acid amide and growing peptide chain. Considering the easy access, low cost, shelf-stability and reaction rates, Fmoc amino acid chlorides are among the most convenient reagents for multigram synthesis of short peptide fragments. Application of Fmoc amino acid chlorides in the two-phase system for very sluggish couplings may be limited by their hydrolysis, premature deblocking or formation of the corresponding oxazol-5(4*H*)-one. With certain sequences, especially if dichloromethane is used as solvent, separation of the growing peptide from the AMP–DBF adduct can be complicated by formation of emulsions or separation of a voluminous precipitate during buffer extraction. Formation of such precipitates can often be avoided by using TAEA rather than AMP as the deblocking/scavenging agent.[14]

H-Lys(Z)-Pro-Lys(Z)-Pro-NHC$_{12}$H$_{25}$·HCl:[13]
To a mixture of H-Pro-NHC$_{12}$H$_{25}$·HCl (319 mg, 1 mmol) in $CHCl_3$ (10 mL) and 5% aq Na_2CO_3 (10 mL) was added a soln of Fmoc-Lys(Z)-Cl (782 mg, 1.5 mmol) in $CHCl_3$ (10 mL). The mixture was stirred vigorously for 10 min. The separated organic layer was treated with AMP (5 mL) and after 30 min $CHCl_3$ (40 mL) was added. The organic phase was washed with phosphate buffer (5×25 mL, pH 5.5) and then concentrated to about 10 mL. Chain extension was continued in the same manner with Fmoc-Pro-Cl and Fmoc-Lys(Z)-Cl. The product H-Lys(Z)-Pro-Lys(Z)-Pro-NHC$_{12}$H$_{25}$·HCl was isolated from the dried $CHCl_3$ layer by addition of 4 M HCl/EtOAc, removal of solvent, and precipitation ($CHCl_3$/petroleum ether); yield: 440 mg (47%).

H-Phe-Phe-Val-Gly-Leu-Met-OBzl:[14]

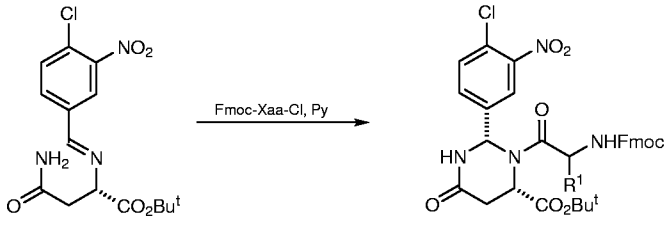
CAUTION: *Benzene is carcinogenic! CCl_4 is a carcinogen!*

To H-Met-OBzl·TosOH (206 mg, 0.5 mmol) in 5% aq Na_2CO_3 (5 mL) and $CHCl_3$ (5 mL) Fmoc-Leu-Cl (223 mg, 0.6 mmol) in $CHCl_3$ (5 mL) was added dropwise with vigorous stirring. After 15 min, the organic layer was separated and treated with TAEA (3.5 mL). After a further 15 min, the soln was extracted with sat. NaCl soln and phosphate buffer (pH 5.5), the organic layer was concentrated to 5 mL and the next four cycles with Fmoc-Gly-Cl, Fmoc-Val-Cl and Fmoc-Phe-Cl were carried out in the same manner, but with increased deblocking times of 30 min. After the final cycle, the organic layer was concentrated, the residual oil was triturated with benzene and CCl_4, and the solvent was removed. Precipitation from $CHCl_3$ by Et_2O followed by recrystallization (MeOH) gave the peptide benzyl ester as a white solid; yield: 160 mg (40%); mp 210–216 °C; $[\alpha]_D^{23}$ −43.8 (*c* 0.5, $CHCl_3$).

3.3.1.2.2 Synthesis in Homogeneous Systems with Hydrogen Chloride Acceptors

For reactions carried out in homogeneous solution or under solid-phase conditions the use of Fmoc amino acid chlorides is limited by the competition between their aminolysis and the formation of the less reactive oxazol-5(4*H*)-ones in the presence of tertiary amines,[15] which are essential components of such reaction systems. To improve the results under these conditions a hindered base, e.g. 2,6-di-*tert*-butylpyridine, can be used as a hydrogen chloride acceptor since conversion to oxazol-5(4*H*)-one is slow with such bases. Although shown to be advantageous in certain cases, Fmoc amino acid chlorides are used in homogeneous solution synthesis only in particular cases. They react efficiently in the presence of pyridine with weak nucleophiles such as imine **2**[39] (Scheme 2) where other activated species such as an active ester, anhydride, acyl fluoride, and acyl imidazolide fail.

Scheme 2 Acylation of the Imine **2** by the Acid Chloride/Pyridine Method[39]



2

Fmoc amino acid chlorides are used in the presence of DIPEA for the acylation of hindered secondary amines such as the tetrahydro-β-carbolines,[40] where other methods including PyBroP, PyBroP/HOAt, HATU, and amino acid fluorides only led to poor results. In the case of difficult acylations, e.g. of pipecolic acid and other secondary amino acids under solid-phase conditions, Fmoc amino acid chlorides can be successfully applied, by shaking a solution of the acid chloride in dichloromethane with the peptide resin for 5 min, while adding a tertiary amine, e.g. DIPEA, and adjusting the supernatant to pH 8.[16,17] An alternative technique, useful for both solution and solid-phase syntheses, involves the use of ammonium[15,18] or potassium salts of HOBt[19–22] or HOAt[41,42] as bases. With KOBt, acylation of highly hindered amino groups including *N*-methyl amino acids and $C^{\alpha,\alpha}$-dialkyl amino acids can be accomplished (Table 2).

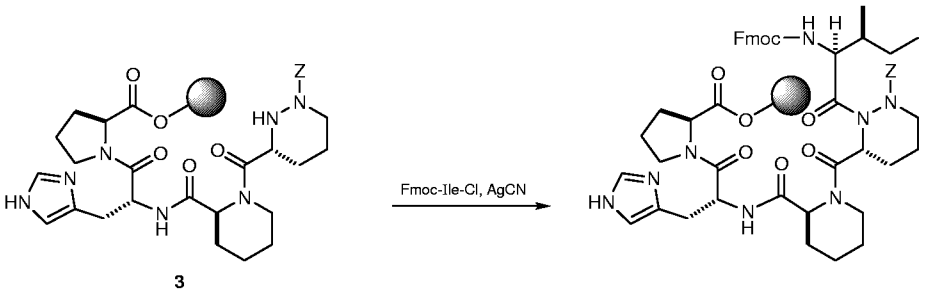
Table 2 Dipeptides Synthesized Using the 1,2,3-Benzotriazol-1-yloxypotassium/N^{α}-9-Fluorenylmethoxycarbonyl Amino Acid Chloride Method[22]

Compound	Yield (%)	mp (°C)	Ref
Fmoc-Ac_6c-Ac_6c-OMe	70	164–165	[22]
Fmoc-Dbug-Ac_6c-OMe	80	89–90	[22]
Fmoc-Ac_7c-Dbug-OMe	75	130–132	[22]
Fmoc-Ac_7-Ac_5c-OMe	70	207–209	[22]
Fmoc-Dpg-Dpg-OMe	80	128–129	[22]
Fmoc-Dpg-Dbug-OMe	74	Gum	[22]

An especially difficult acylation of the resin-bound, sterically hindered D-Piz(Z) derivative **3** (Scheme 3) was achieved by using silver cyanide as additive. While Fmoc-Ile-Cl itself led to a yield of only 50%, the efficiency was increased when the reaction was carried out in the presence of silver cyanide.[16] The exact mechanism of silver cyanide catalysis is not clear, but it was shown that silver cyanide effects conversion of the acid chloride to the oxazol-5(4*H*)-one. Although the hydrogen cyanide formed during this process may catalyze the reaction of the oxazol-5(4*H*)-one, the results of separate hydrogen cyanide catalyzed coupling of preformed oxazol-5(4*H*)-one did not match exactly the silver cyanide catalyzed reaction of the acid chloride.[16]

for references see p 492

Scheme 3 Acylation of the Sterically Hindered Piperazic Acid Derivative **3** by the Amino Acid Chloride/Silver Cyanide Method[16]



Silver cyanide was also used in efficient fragment couplings involving protected peptide acid chlorides.[43] For such couplings the standard two-phase methodology ($NaHCO_3$, H_2O, CH_2Cl_2)[43] and the KOBt technique[44,45] were also effective. The coupling of Fmoc amino acid chlorides in the presence of zinc dust as a hydrogen chloride acceptor[46] has been reported.

To overcome the major limitation of Fmoc amino acid chlorides which arises from the instability of acid sensitive side-chain protecting groups, an in situ activation technique offers advantages. In situ activation can be carried out with thionyl chloride,[37,47] oxalyl chloride,[48] bis(trichloromethyl) carbonate,[49] or 1-chloro-*N*,*N*,2-trimethylprop-1-en-1-amine.[50] Using in situ activation N^{α}-Z-[48,51,52] and $N^{\alpha,\alpha}$-di-Z-protected amino acid chlorides[53] can be successfully used in peptide coupling reactions. Fmoc amino acid chlorides, even those bearing *tert*-butyl-type side-chain protecting groups such as Asp(tBu), Cys(Trt), Lys(Boc), Ser(tBu), Thr(tBu), Trp(Boc), Tyr(OtBu), and Arg(Pmc), can be used for efficient coupling to hindered *N*-alkylamino acid units in inert solvents using solid-phase techniques and in situ activation via bis(trichloromethyl) carbonate.[49]

The acid chloride method has been successfully used for the synthesis of depsipeptides by O-acylation of hydroxy acids.[9] Although Fmoc amino acid chlorides were recommended for such reactions,[54] and can be used for the introduction of the first amino acid onto hydroxy resins for SPPS,[55] oxazol-5(4*H*)-one formation promoted by the presence of the required tertiary amine leads to relatively low yields compared to other activation methods such as those involving UNCA groups.[54]

Coupling with KOBt/Fmoc Amino Acid Chlorides; General Procedure:[22]
To a soln of amino acid ester hydrochloride (1 mmol) and KOBt (1 mmol) in CH_2Cl_2 (3 mL) was added a soln of Fmoc amino acid chloride (1 mmol) and KOBt (1 mmol) in CH_2Cl_2 (3 mL). The mixture was stirred for 3–4 h then washed with aq 5% $NaHCO_3$, 5% HCl, H_2O, sat aq NaCl soln, and dried. The organic layer was concentrated and the residue recrystallized (CH_2Cl_2/hexane); yields: 70–80% (Table 2).

3.3.2 α-Azido Acid Chlorides

As α-azido acids are readily reduced to α-amino acids, the problem of α-amino protection is fully bypassed with the highly reactive α-azido acid chlorides (see also Section 2.1.1.10.2). The quantitative reduction of α-azido acids to the corresponding amino acids is accomplished by treatment with DTT/DIPEA/DMF at 50 °C.[34] Chiral α-azido acid chlorides can be used under solid-phase conditions without loss of configuration, but since the reduction process is slow, the method is not attractive for standard peptide synthesis. Compatibility of the α-azido acids with Fmoc- and Boc-chemistry and acid chloride activation, however, makes α-azido amino acid chlorides highly useful for incorporation of very sterically hindered amino acids such as $C^{\alpha,\alpha}$-disubstituted glycines.

SAFETY: Most azides are explosive and toxic and should be handled with care at temperatures above 50 °C.

Such α-azido acids are prepared by α-bromination of α-branched acids using *N*-bromosuccinimide followed by substitution with sodium azide (see also Section 2.1.1.10.1). Treatment with thionyl chloride in an inert solvent gives the corresponding acid chloride in solution, which can then be used without isolation. Reduction of α-azido acids by DTT is compatible with both base- and acid-sensitive linkers used for SPPS and most side-chain protecting groups. Therefore, a combination of α-azido acids with Fmoc/tBu chemistry offers an easy access to sterically demanding peptides. As the reduction time strongly depends on steric effects, completion of the reduction has to be checked by IR prior to the next step. In this manner, a peptide bearing four consecutive $C^{\alpha,\alpha}$-diphenylglycine residues was synthesized,[34] but prolonged coupling and reduction times were required due to increased steric crowding.

2-Azido-2-propylpentanoic Acid Chloride; Typical Procedure for α-Azido Acids:[56]

CAUTION: *CCl_4 is a carcinogen!*

2-Propylpentanoic acid (12.64 g, 87.7 mmol) and NBS (15.8 g, 88.8 mmol) were refluxed in CCl_4 (100 mL) under argon for 2 d, filtered, and the bulk of solvent was removed. The residue was distilled under reduced pressure to give 2-bromo-2-propylpentanoic acid (6.96 g, 36%); bp 126–130 °C/5 Torr. 2-Bromo-2-propylpentanoic acid (3.9 g, 17.5 mmol) and NaN_3 (1.74 g, 26.8 mmol) were stirred in dry DMF (50 mL) at rt and after 8 d the solvent was removed. The residue was taken up in toluene (50 mL) and then partitioned between EtOAc and H_2O. The aqueous layer was extracted with EtOAc and the organic layer concentrated to give the crude α-azido acid which was purified by HPLC; yield: 1.56 g (48%).

SAFETY: If it is necessary to acidify the aqueous solution following removal of DMF, perform with care, due to possible evolution of toxic gaseous HN_3.

2-Azido-2-propylpentanoic acid (95.6 mg, 0.516 mmol) was dissolved in freshly distilled thionyl chloride/dry CH_2Cl_2 (0.8 mL, 1:1) and heated at 50 °C. After 3 h, the soln was taken up in toluene (3 mL) in an inert atmosphere and the α-azido acid chloride was diluted with anhydr CH_2Cl_2 (1 mL).

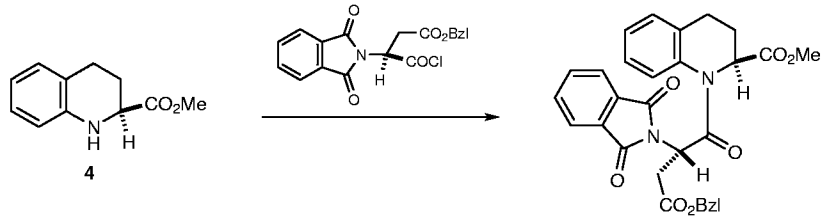
Methionyl-α,α-dipropylglycinylphenylalanine Methyl Ester; Typical Procedure for α-Azido Acids:[56]
A PEG-acrylamide resin (0.102 mmol) was derivatized with 4-(hydroxymethyl)benzoic acid by activation with TBTU and *N*-ethylmorpholine. Fmoc-Phe-OH was coupled to the linker and the Fmoc group removed with piperidine. The resin was washed with anhydr CH_2Cl_2 (6×) and then *N*-ethylmorpholine and the freshly prepared soln of 2-azido-2-propylpentanoic acid chloride prepared as above, was added to the resin. After 24 h, IR analysis of a bead gave a strong absorption at 2119.1 cm^{-1} (azide band). The resin was washed with CH_2Cl_2 and DMF and a freshly prepared soln of 2 M DTT and 1 M DIPEA in DMF was added and the mixture heated under argon at 50 °C. After 24 h, the reduction was complete. Coupling with Fmoc-Met-OPfp/HODhbt followed by piperidine deblocking gave the peptide which was cleaved from the resin by 0.1 M NaOMe in MeOH for 2 h. Purification was carried out by HPLC; yield: 41.1 mg (84%). (H_2O from the air is trapped by the resin when it is sucked dry from CH_2Cl_2. To minimize H_2O absorption, and resulting competing side reactions, a minimum of solvent was kept on the resin).

3.3.3 N^{α}-Arylsulfonic Amino Acid Chlorides

N^{α}-Protected amino acid chlorides which cannot convert to oxazol-5(4*H*)-ones are very efficient agents for acylation of extremely hindered amino groups or very weak nucleophiles. Thus, Phth-Asp(OBzl)-Cl can be coupled with the amino acid ester **4**,[57] a reaction which fails with the corresponding acid fluoride (Scheme 4).

for references see p 492

Scheme 4 Acylation of a Sterically Hindered Weak Nucleophile with Phth-Asp(OBzl)-Cl[57]



Amide bond formation with extremely hindered components such as those present in H-Aib-(Me)Aib-OH or especially in H-(Me)Aib-(Me)Aib-OH, is not possible with Fmoc amino acid chlorides or fluorides.[35,36] With steric hindrance of this magnitude the practical limit for urethane-based amino acid coupling appears to have been reached. This led to the consideration of other N-protecting groups for which oxazol-5(4*H*)-one formation, even in the case of acid chlorides, is precluded. Indeed, early studies showed that Tos-Aib-Cl was effective in the production of Aib-Aib peptide bonds[58] provided that nonaqueous media was used to avoid degradation of the acid chloride. In the absence of other influences, the greater inductive effect of the sulfonyl group relative to that of a carbonyl group was expected to increase the reactivity of a sulfonyl substituted amino acid halide over that of its carbamoyl counterpart. In line with this expectation, Tos-Aib-Cl reacts readily with H-(Me)Aib-OMe. Conversely, the reactivity of Tos-Aib-F was insufficient to acylate H-(Me)Aib-OMe.[35] Substitution of the more easily cleaved Pbf residue for the Tos group makes practical use of this observation. Under anhydrous conditions with toluene as the solvent even H-(Me)Aib-(Me)Aib-OMe can be prepared by this procedure.

Using the 2-nitrobenzenesulfonyl group as N-protectant[59] opens an alternative approach, in that deblocking of 2-nitrobenzenesulfonyl amino acid derivatives is achieved by treatment with thiol/base systems which are compatible with Fmoc/tBu chemistry. Shelf-stable 2-nitrobenzenesulfonyl amino acid chlorides can be prepared and, as they cannot be converted into oxazol-5(4*H*)-ones, they can be used for coupling to very hindered amines, such as *N*-allyl amino acids under solid-phase conditions.[60] By substitution of the N^{α}-2-nitrobenzenesulfonyl for the Pbf group, coupling of (Me)Aib to Aib in SPPS can be accomplished with subsequent removal of the 2-nitrobenzenesulfonyl group by use of β-ME/DBU/DMF.[61]

H-(Me)Aib-(Me)Aib-OMe · TFA:[35]
To a soln of H-(Me)Aib-OMe · HCl (100.2 mg, 0.6 mmol) in toluene (900 μL) was added DIPEA (102 μL, 0.6 mmol). The mixture was filtered and added to Pbf-(Me)Aib-Cl (58.05 mg, 0.15 mmol). The mixture was stirred for 2.5 h and the crude protected dipeptide ester was purified by preparative HPLC (38 mg, 52%) of the homogeneous Pbf-(Me)Aib-(Me)Aib-OMe. After cleavage of the Pbf group with 10% DMS in TFA (1 mg · 100 μL^{-1}) for 1 h, the product was precipitated with Et_2O; yield: 15.6 mg (86%); mp 162–164 °C.

3.3.4 N^{α}-9-Fluorenylmethoxycarbonyl Amino Acid Fluorides

Two serious deficiencies limit the application of Fmoc amino acid chlorides; their instability in the cases of amino acids bearing acid-labile *tert*-butyl-type side-chain protecting groups and their rapid conversion into oxazol-5(4*H*)-ones by tertiary amines. In contrast, the corresponding acid fluorides suffer no such limitations, owing to their stability even when carrying tBu, Boc, or *N*-Trt side-chain protecting groups.[62] Another advantage of the acid fluorides relative to the chlorides includes their greater stability to water,[63] including moisture in the

air. Extensive mechanistic studies have suggested that reactions of acid fluorides resemble those of active esters rather than those of acid chlorides, bromides, or iodides.[64]

3.3.4.1 Synthesis of N^{α}-9-Fluorenylmethoxycarbonyl Amino Acid Fluorides

The acid fluorides are generally available via reaction of the acid with cyanuric fluoride in the presence of pyridine according to the general technique of Olah.[65] An alternative method for the synthesis of Fmoc amino acid fluorides involves the use of DAST (*N,N*-diethylaminosulfur trifluoride), which is advantageous relative to cyanuric fluoride owing to the formation of only water-soluble compounds as byproducts, thus, facilitating the isolation of the acid fluorides.[66] Examples can be found in Table 3.

Table 3 N^{α}-9-Fluorenylmethoxycarbonyl Amino Acid Fluorides[23,24,28–30,63]

Xaa	Yield (%)	mp (°C)	Ref	Xaa	Yield (%)	mp (°C)	Ref
Gly	81	140–141	[23]	Glu(OtBu)	72	80–82	[23]
Ala	76	111–112	[23]	Tyr(tBu)	67	97–99	[23]
Val	70	113–114	[23]	Tyr(PO_3Me_2)	80	foam	[81]
Leu	75	95–96	[23]	Tyr(OBzl)	80	128–130	[24]
Ile	73	115–116	[23]	Gln(Trt)	84	98–99	[63]
Pro	78	88–89	[23]	Asn(Trt)	81	123–124	[63]
Phe	64	118–120	[23]	(Me)Gly	68	94–96	[28]
Trp	71	125–128	[23]	(Me)Val	73	158–160	[28]
Ser(tBu)	73	89–91	[23]	Deg	74	solid	[28]
Thr(tBu)	73	53–54	[23]	Phg	75	144–146	[23]
Lys(Boc)	80	128–130	[23]	Iva	78	131	[29]
Asp(OtBu)	68	74–75	[23]	Aib	83	118–120	[28]
Met	72	137–139	[23]	(Me)Aib	64	174–178	[30]

Fmoc-Val-F with Cyanuric Fluoride; Typical Procedure:[23]
Fmoc-Val-OH (0.339 g, 1 mmol) in CH_2Cl_2 (5 mL) was refluxed under N_2 with cyanuric fluoride (700 µL, 8 mmol) and pyridine (81 µL, 1 mmol) for 2 h. The mixture, from which a white water-soluble precipitate had settled, was extracted with ice water (2×15 mL). Removal of the solvent from the dry ($MgSO_4$) organic layer gave a white solid which was recrystallized (CH_2Cl_2/hexane) to give the pure acid fluoride; yield: 239 mg (70%); mp 113–114 °C.

Fmoc-Val-F with DAST; Typical Procedure:[66]
To a stirred soln of Fmoc-Val-OH (0.339 g, 1 mmol) in CH_2Cl_2 (10 mL) DAST (1.2 mmol) was added at rt. After 10 min the mixture was extracted with ice water, the organic layer dried ($MgSO_4$) and molecular sieves (10 Å). The solvent was removed and the residue recrystallized (CH_2Cl_2/hexane); yield: 259 mg (76%); mp 113–114 °C.

3.3.4.2 Peptide Synthesis on Solid Support with N^{α}-9-Fluorenylmethoxycarbonyl Amino Acid Fluorides

While there is evidence from IR investigations for the formation of traces of oxazol-5(4*H*)-one by treatment of Fmoc-protected amino acid fluorides, e.g. Fmoc-Val-F, with DIPEA, this reaction is not rapid enough to compete with the acylation process. In contrast to Fmoc amino acid chlorides, the related fluorides can be used in homogeneous systems with tertiary amines as efficient agents in peptide synthesis.

3.3.4.2.1 Synthesis of Difficult Sequences with N^{α}-9-Fluorenylmethoxycarbonyl Amino Acid Fluorides

The first step in the synthesis of peptide acids on solid supports involves anchoring the C-terminal amino acid to a resin-linker via an ester bond. Acid fluorides have been shown to

for references see p 492

accomplish even difficult esterifications such as between a weakly nucleophilic secondary alcohol and a cyclohexyl amino acid,[67] where other activation methods such as DCC/DMAP, mixed anhydrides, BOP-Cl, or BOP have failed. Indeed, the fluoride method can be used for efficient anchoring of Fmoc amino acids to 4-alkoxybenzyl alcohol resin (Wang resin),[68,69] particularly in nonpolar solvents such as toluene or dichloromethane using DMAP as catalyst.[68]

Successful syntheses using Fmoc amino acid fluorides in SPPS were reported for ACP (65–74) (a decapeptide sequence of the acyl carrier protein), magainin-II-amide and h-CRF (human corticotropin-releasing factor); three peptides which are considered to be difficult sequences.[26] Moreover, the SPPS of the major amyloid deposit associated with Alzheimers disease Aβ (1–42) which is also known as a difficult sequence due to its strong self-association properties, could be remarkably improved by the use of acid fluorides rather than the BOP/HOBt method.[70] Efficient syntheses have been reported using other appropriate coupling reagents.[71] The high reactivity of acid fluorides was further demonstrated by successful couplings to very weak amines. Thus, N-acylation of 2-aminobenzophenones can be achieved with Fmoc amino acid fluorides and 2,6-di-*tert*-butyl-4-methylpyridine as base, whereas the same amide bond formation does not occur when standard activation methods are used (e.g., carbodiimides/HOBt or pentafluorophenyl ester).[30]

Very difficult peptide syntheses are those which contain sterically hindered $C^{\alpha,\alpha}$-dialkylamino acids such as Aib residues, especially if these residues are found next to each other. Due to their influence on the stabilization of helical structures[72] in small peptides, the introduction of such amino acids is of interest. Various protocols for the coupling of hindered systems using UNCAs,[73] symmetric anhydrides,[74] HATU,[75] CIP/HOAt,[76] and PyBroP[77] activation have been reported. In a comparative analysis of activation methods for the synthesis of H-Aib-Aib-Aib-Aib-OH as a model peptide which included the symmetric anhydride, PyBroP, acid fluoride, and UNCA methods, the acid fluoride technique was shown to be exceptionally well suited.[25,26]

The efficiency of the fluoride method was further confirmed by the synthesis of peptaibols. These peptides are characterized by a high content of the $C^{\alpha,\alpha}$-dialkylamino acids Aib or isovaline and an amino alcohol at the C-terminus such as phenylalaninol (Pheol) or valinol (Valol). Degradation of the peptides in the final cleavage step due to the acid-labile Aib-Pro linkages,[78] can be prevented by the use of the 2-chlorotrityl-resin.[79] This achieves direct anchoring of Fmoc-protected amino alcohols which allows the final cleavage to occur under very mild conditions.[27]

Besides coupling of adjacent Aib residues[80], the incorporation of other hindered amino acids into peptides such as 1-aminocyclohexane-1-carboxylic acid (Ac_6c)[81,82] and 4-aminopiperidine-4-carboxylic acid,[82] as well as the coupling of *O*,*O*-(dimethylphosphoro)tyrosine to these amino acids[81] was achieved by the use of the corresponding acid fluorides with solid-phase methods.

Fmoc-Val-Wang Resin; Typical Procedure:[68]
To Wang resin (0.5 g, 0.38 mmol) which had been washed with CH_2Cl_2 a soln of Fmoc-Val-F (0.341 g, 1 mmol) in CH_2Cl_2 (7.5 mL) and DMAP (92.7 mg, 0.76 mmol) was added with stirring. After 10 min, the resin was washed with CH_2Cl_2 to yield resin with substitution: 0.76 mmol·g^{-1} (100%), stereomutation: <0.01% [For Fmoc-Cys(tBu)-F 2,6-di-*tert*-butyl-4-methylpyridine was used instead of DMAP to reduce stereomutation from 5.9% to 0.05%].

Magainin-II-amide, H-GIGKFLHGAKKFGKAFVGEIMNS-NH_2·2TFA; Typical Procedure:[26]
The synthesis was carried out manually in a batch reactor on a TentaGel support (S RAM, 0.24 mmol·g^{-1}). The standard protocol involved double couplings (2×15 min) with Fmoc amino acid fluorides (3 equiv, 0.3 M) in DMF with DIPEA (3 equiv), piperidine/DMF (1:3, v/v) deblocking for 10 min, with DMF washes between each step. For Fmoc-His(Trt)-OH (3 equiv, 0.3 M), coupling via activation with TOPPipU (3 equiv) and DIPEA (6 equiv) was used. The peptide resin cleavage was carried out with TFA/H_2O/phenol/thioanisole/ethanedithiol (82.5:5:5:2.5) for 2 h. After removing the

resin precipitation, by Et_2O gave the crude product which contained, according to HPLC, >75% of the desired peptide; yield: 82%.

H-Aib-Aib-Aib-Aib-Lys-Leu-Met-Phe-Ile-Ile-NH_2·2TFA; Typical Procedure:[26]
The C-terminal hexapeptide was automatically synthesized by standard procedures using TOPPipU/DIPEA in DMF on a TentaGel S RAM resin (0.22 mmol·g^{-1}). Further elongation of the peptide was performed by couplings (2×15 min) with Fmoc-Aib-F (3 equiv) in DMF (0.3 M) in the presence of DIPEA (3 equiv). The peptide resin cleavage was carried out with TFA/H_2O/phenol/thioanisole/EDT (82.5:5:5:2.5) for 2 h. After removing the resin, precipitation by Et_2O gave a crude product which contained, according to HPLC, >80% of the desired peptide; yield: 74%.

Alamethicin F 30; Typical Procedure:[27]
Linkage of the amino alcohol to Trt(2-Cl) resin (1.6 mmol·g^{-1}) was performed by stirring the resin (0.32 mmol), Fmoc-Pheol (0.96 mmol), and pyridine (1.92 mmol) in dry CH_2Cl_2/DMF at rt. After 6 h, MeOH (4 mL) was added, and the mixture stirred for 30 min to terminate the reaction and to remove any remaining reactive chlorotrityl functionality. After washes with DMF the Fmoc group was cleaved upon treatment with piperidine/DMF (1:4) for 15 min, and the loading determined by UV (0.24 mmol·g^{-1}, 15%). Fmoc amino acid fluorides (3 equiv, 0.3 M) were coupled in DMF soln in the presence of DIPEA (3 equiv) for 15 min. After the final deprotection, the N-terminus was acetylated with a soln of $(AcO)_2O$/DIPEA/DMF (2:1:7) for 15 min. Cleavage of the peptide resin and deprotection of the peptide with a mixture consisting of TIS/phenol/H_2O/TFA/CH_2Cl_2 (2:5:5:44:44) for 30 min gave after precipitation with Et_2O, the crude product which contained, according to HPLC, >70% of the desired peptide; yield: 84%. Examination of the product for the presence of D-amino acids showed that loss of stereochemical integrity could not have exceeded 0.8% for any of the chiral amino acids.

3.3.4.2.2 Acylation with N^{α}-9-Fluorenylmethoxycarbonyl Amino Acid Fluorides in the Absence of Base

Another unique difference between acid chlorides and acid fluorides is the ability of the latter compounds to undergo efficient aminolysis in the absence of base. Whilst reaction of Fmoc amino acid chlorides with free amino acid esters proceeds only to an extent of 50%, since half of the amino acid ester is used up by formation of the unreactive hydrochloride salt,[15] the corresponding reaction with Fmoc amino acid fluorides goes to completion in the presence of 0.5 equiv of DIPEA. More remarkably, the reaction proceeds only slightly less rapidly in the total absence of any auxiliary tertiary amine.[83] Thus, alamethicin F 30 was prepared in a quality comparable to that obtained by the method described above even in the absence of DIPEA.[84] With their exceptional reactivity, and high solubility and shelf stability in organic solvents such as DMF, the Fmoc amino acid fluorides clearly satisfy the requirements of automated instrumentation for multiple peptide synthesis as demonstrated by the multiple syntheses of alamethicin analogues using the no-base approach.[84]

3.3.4.3 Synthesis in Solution with N^{α}-9-Fluorenylmethoxycarbonyl Amino Acid Fluorides

Fmoc amino acid fluorides can be used in a two-phase solvent system for stepwise peptide assembly[23] in a manner analogous to that described for Fmoc amino acid chlorides (Scheme 1). Since coupling reactions compete efficiently with oxazol-5(4*H*)-one formation, synthesis of peptides using Fmoc-protected amino acid fluorides may be efficiently performed in homogeneous systems in the presence of tertiary amines. Following the no-base approach, base-catalyzed side reactions such as oxazol-5(4*H*)-one formation, epimerization and premature deblocking during slow couplings are largely prevented. Therefore, the method is well suited for the rapid solution synthesis of short peptides in multigram quantities.

Fmoc-Val-Gln(Trt)-Ala-Ala-Ile-ODicycpropme (No-Base Coupling):[83]
A soln of Fmoc-Ile-ODicycpropme (965 mg, 2.0 mmol) in MeCN/Et_2NH (1:1; 20 mL) was stirred at rt for 40 min, the solvent and reagent were removed and the residue dissolved in CH_2Cl_2 (10 mL). After

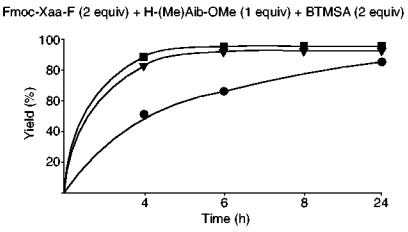
for references see p 492

addition of Fmoc-Ala-F (814 mg, 2.6 mmol) the soln was stirred for 30 min and TAEA (10 mL) was added. The soln was stirred for 15 min and washed with sat. NaCl soln (2 × 7 mL) and phosphate buffer (pH 5.5, 3×7 mL). To the organic layer Fmoc-Ala-F (814 mg, 2.6 mmol) was added and the soln stirred for 30 min. TAEA (10 mL) was added, stirring was continued for 15 min and the reaction worked up as in the first cycle. Reaction cycles with Fmoc-Gln(Trt)-F (2.6 mmol) and Fmoc-Val-F (2.6 mmol) were performed in the identical manner. After the final coupling the organic soln was washed with H_2O, $NaHCO_3$ soln, sat. NaCl soln, dried ($MgSO_4$), the solvent removed and the residue recrystallized (EtOAc/hexane) to give the protected peptide; yield: 1.62 g (74%).

3.3.4.4 Synthesis of Sterically Hindered Peptides

Coupling of amino acids more hindered than those involving adjacent Aib residues, can also be accomplished using Fmoc amino acid fluorides under appropriate conditions. In a polar solvent such as DMF, a coupling in absence of base of Fmoc-Aib-F or Fmoc-(Me)-Gly-F with H-Aib-OMe proceeds without difficulties, however problems arise when even more hindered amino acids are used. Thus, the speed of acylation was significantly decreased when Fmoc-Iva-F and Fmoc-(Me)Val-F were used. The still more hindered compounds Fmoc-Deg-F and Fmoc-(Me)Aib-F gave only small amounts of the desired dipeptides. When the coupling rate is particularly low as in these two cases, premature deblocking of the Fmoc group becomes a prominent side reaction. In addition, IR studies revealed that Fmoc amino acid fluorides derived from α,α-dialkylated species are converted slowly into the corresponding oxazol-5(4*H*)-ones when tertiary amines are present.[28] Premature Fmoc-cleavage was significantly reduced when nonpolar dichloromethane was used as solvent.[28] The most efficient method of performing such difficult acylations involves silylation of the amino component, which leads to the formation of Fmoc-(Me)Aib-Aib-OMe in 34% yield after 2 h, and almost no dibenzofulvene (0.3% after 4 h).[28] Using this silylation procedure Fmoc-protected amino acid fluorides can be coupled efficiently even to very hindered amino components such as H-(Me)Aib-OMe (Figure 1).[28] Similarly, significant improvement of coupling yields of α-azidophenylacetyl fluoride to a silylated sterically hindered amine was observed.[85]

Figure 1 Coupling of N^{α}-9-Fluorenylmethoxycarbonyl Amino Acid Fluorides with H-(Me)Aib-OMe[28] (■ Phe, ▼ Ala, ● Val)[28]



A special one-pot deprotection–transacylation method involves coupling of activated Fmoc amino acids with Aloc-protected amino acid esters.[86] The Aloc-deprotection proceeds with palladium/phenylsilane in the presence of the acylating species. Using Fmoc-Phe-F the synthesis of the sterically demanding dipeptide Fmoc-Phe-(Me)Aib-OMe was accomplished in a yield of 65%.[87] A similar one-pot approach to the acylation of the even more difficult hindered and weakly nucleophilic α-trifluoromethyl amino acid esters involves the intermediate *N*-Teoc protection. Thus treatment of (±)-Teoc-α-(α-CF_3)Leu-OMe with Fmoc-Gly-F and a catalytic amount of tetraethylammonium fluoride in acetonitrile at 50 °C for 1–2 weeks gave (±)-Fmoc-Gly-(α-CF_3)Leu-OMe (77%).[87]

3.3.4.5 In Situ Formation of N^{α}-9-Fluorenylmethoxycarbonyl Amino Acid Fluorides

In order to take advantage of the excellent properties of Fmoc amino acid fluorides in routine solid-phase and solution peptide synthesis, reagents for their in situ generation have been developed. Among these are the fluoroformamidinium salts such as tetramethylfluoroformamidinium hexafluorophosphate (TFFH) and bis(tetramethylene)fluoroformamidinium hexafluorophosphate (BTFFH)[88] (see also Section 3.8). For rapid solution phase peptide assembly (Scheme 1), TFFH can be added directly to the one- or two-phase coupling medium instead of the preformed acid fluoride. No modification in the subsequent work up procedure is required. The pentapeptide leucine enkephalin was assembled by this method in high purity.[88] The Mukaiyama[89] reagent 2-fluoro-1,3-dimethylpyridinium 4-toluenesulfonate has also been used for in situ activation.[90] In addition, TFFH can be used advantageously as a coupling reagent in SPPS for difficult sequences, e.g. the pentapeptide H-Tyr-Aib-Aib-Phe-Leu-NH_2,[88] alamethicin acid peptide,[91] and a 26mer peptide of a CTL epitope.[92] None of these three peptides are readily assembled via standard solid-phase protocols. Moreover, in situ activation with TFFH allows coupling of the Fmoc amino acid derivatives of His and Arg from which no stable acid fluorides could be prepared.[88] A number of polypeptides, including ACP(65–74) and magainin-II-amide, were synthesized in high yield and quality using BTFFH in SPPS.[93]

3.3.5 N^{α}-Benzyloxycarbonyl and N^{α}-*tert*-Butoxycarbonyl Amino Acid Fluorides

N^{α}-Boc- and N^{α}-Z-protected amino acid fluorides are easily prepared, and are often in crystalline form. However, they differ from their Fmoc analogues, especially in the Boc series, in their stability, requiring refrigerator temperatures for storage. They are rapidly aminolyzed without loss of configuration under appropriate conditions.

3.3.5.1 Synthesis of N^{α}-*tert*-Butoxycarbonyl and N^{α}-Benzyloxycarbonyl Amino Acid Fluorides

N^{α}-Boc- and N^{α}-Z-protected amino acid fluorides are prepared from the corresponding acids employing cyanuric fluoride at low temperature. Interestingly, when either Boc-Phe-F or Z-Phe-F are refluxed in anhydrous tetrahydrofuran for 2 h the crystalline protected amino acid fluorides are recovered after removal of solvent, demonstrating their increased stability relative to the corresponding acid chlorides which undergo conversion into the Leuchs' anhydrides.[31] Examples can be found in Table 4.

for references see p 492

Table 4 *tert*-Butoxycarbonyl and Benzyloxycarbonyl Amino Acid Fluorides[24,31–33,94]

Compound	Yield (%)	mp (°C)	Ref	Compound	Yield[a] (%)	mp (°C)	Ref
Boc-Gly-F	89	52–54	[31]	Boc-D-Phg-F	73	56–58	[31]
Boc-Ala-F	65	62–64	[31]	Bz(Boc)-Leu-F	91	38–40	[94]
Boc-Val-F	87	36–38	[31]	Z-Gly-F	83/89	45–47/50–51	[31,24]
Boc-Leu-F	68	55–57	[31]	Z-Ala-F	78/80	39–41/liq	[31,24]
Boc-Ile-F	77	oil	[31]	Z-Val-F	81/75	oil	[31,24]
Boc-Pro-F	88	oil	[31]	Z-Leu-F	80	oil	[24]
Boc-Phe-F	76	66–68	[31]	Z-Pro-F	83	oil	[31]
Boc-D-Phe-F	74	67–69	[31]	Z-Phe-F	73/70	87–88/80–83	[31,24]
Boc-Trp-F	78	114 dec	[31]	Z-Lys(Boc)-F	88	oil	[31]
Boc-Ser(Bzl)-F	66	37–38	[31]	Z-Asp(OtBu)-F	70	42–44	[31]
Boc-Thr(Bzl)-F	93	oil	[31]	Z-Glu(OtBu)-F	83	oil	[31]
Boc-Lys(Z)-F	78	84–86	[31]	Z-Met-F	83	oil	[31]
Boc-Asp(OtBu)-F	89	52–54	[31]	Z-Ser(tBu)-F	90	oil	[31]
Boc-Met-F	60	32–34	[31]	Z-Phg-F	81	75–77	[31]
Boc-Glu(OBzl)-F	81	50–51	[31]	Z-Val(αMe)-F	n.r.	oil	[32]
Boc-Tyr(Bzl)-F	80	58–60	[31]	Z-Iva-F	n.r.	oil	[33]
Boc-Cys(Bzl)-F	67	52–53	[31]	Z-(Me)Phe-F	n.r.	oil	[33]
Boc-Phg-F	75	54–56	[31]				

[a] n.r. = not reported.

Boc and Z Amino Acid Fluorides; General Procedure:[31]
To a stirred soln of the Boc or Z amino acid (2 mmol) in dry CH_2Cl_2 (5 mL) and pyridine (2 mmol, 162 μL) kept under a N_2 atmosphere at −20 to −10 °C was added cyanuric fluoride (10 mmol, 900 μL). A precipitate or emulsion was formed which gradually increased in amount. After stirring at −10 °C for 1 h, crushed ice was added to the mixture along with additional CH_2Cl_2 (10 mL). The organic layer was separated and the aqueous layer extracted once with CH_2Cl_2 (5 mL). The combined CH_2Cl_2 layers were extracted once with ice cold H_2O (10 mL), dried ($MgSO_4$), and the solvent was removed under reduced pressure at rt. Recrystallization (CH_2Cl_2/hexane or hexane) gave the pure acid fluoride; yield: 60–90% (see Table 4).

3.3.5.2 Peptide Synthesis with N^{α}-Benzyloxycarbonyl and N^{α}-*tert*-Butoxycarbonyl Amino Acid Fluorides

Although their suitability for peptide synthesis has been demonstrated,[31] Boc amino acid fluorides have not yet found widespread application in polypeptide synthesis. Conversely, Z-amino acid fluorides have been used in the solution synthesis of sterically demanding peptides consisting solely of $C^{\alpha,\alpha}$-dialkylated amino acids, such as Z-[Val(αMe)]$_8$-OtBu[32] and Z-Iva-Val(αMe)-Iva-Val(αMe)-MePhe-Val(αMe)-Iva-NHMe.[33]

3.3.6 *N,N*-Bis(alkoxycarbonyl) Amino Acid Fluorides

It is assumed that N^{α}-bis(alkoxycarbonyl) protection of amino acids precludes oxazol-5(4*H*)-one formation and thus could be advantageous with regard to potential epimerization. On the other hand, the presence of a second electron-withdrawing carbonyl function might be expected to enhance any component of stereomutation via C^{α}-proton abstraction. Bis(Boc) amino acids, presumably because of severe steric hindrance, undergo coupling reactions very sluggishly by ordinary procedures (active ester, mixed anhydride, or carbodiimide methods).[96] These coupling steps can also be accompanied by side reactions such as hydantoin formation,[94,97] stereomutation,[94] and formation of a Dakin–West-type product.[94] With bis(Boc) amino acid fluorides, however, such problems were not encountered.[94,97]

3.3.6.1 Synthesis of *N,N*-Bis(alkoxycarbonyl) Amino Acid Fluorides

N,N-Bis(alkoxycarbonyl) amino acid fluorides are prepared from the corresponding amino acid derivatives in a manner similar to that described for Boc amino acid fluorides.

Table 5 *N,N*-Bis(alkoxycarbonyl) Amino Acid Fluorides[94,97,98]

Compound	Yield (%)	mp (°C)	Ref
$(Boc)_2$Asp(OBzl)-F	62	oil	[94]
$(Boc)_2$-Lys(Boc-Z)-F	85	oil	[94]
Z-Boc-Gly-F	76	36–38	[94]
Z-Boc-Glu(OtBu)-F	89	oil	[94]
$(Boc)_2$-Gly-F	80/76	42–44/50–52	[97,98]
$(Boc)_2$-Tyr(Bzl)-F	65	oil	[97]
$(Boc)_2$-Phe-F	80/82	36–38/43–45	[97,98]
$(Boc)_2$-D-Phe-F	70	36–38	[97]
Z-Boc-Phe-F	79	oil	[98]

***N,N*-Bis(Boc) and *N*-Boc, *N*-Z Amino Acid Fluorides; General Procedure:**[94]
N,N-Bis(alkoxycarbonyl) amino acid (10 mmol) dissolved in dry CH_2Cl_2 (25 mL) in the presence of dry pyridine (10 mmol) was cooled to −50 °C under argon. Cyanuric fluoride (10 mmol) was added in 5 portions over 5 min. The temperature of the stirred mixture was allowed to rise and was kept at −30 °C for 1 h and then at −10 to 0 °C for 1 h. The mixture which contained a precipitate was diluted with cold CH_2Cl_2 and washed three times with ice-cold H_2O. After drying, the solvent was removed and the residual oil was kept under high vacuum for 1 h. When possible, the fluoride was crystallized by leaving the product in an appropriate solvent mixture for 2 d in the deep-freeze, decanting the solvent at −30 °C and drying the resulting crystals. Examples can be found in Table 5.

3.3.6.2 Synthesis of Peptides with *N,N*-Bis(alkoxycarbonyl) Amino Acid Fluorides

N,N-Bis(alkoxycarbonyl) amino acid fluorides have been used in stepwise peptide synthesis without loss of configuration.[97] An impressive example of the application of a bis(alkoxycarbonyl)-protected amino acid fluoride is the coupling of the phenylalanine derivative to 2-pyrrolecarboxylic acid *tert*-butyl ester. Coupling to the pyrrole nitrogen failed with Boc-Phe-OH using standard coupling procedures which included symmetrical and mixed anhydrides, DCC, and UNCA[98] reagents.

$(Boc)_2$-Tyr(Bzl)-Gly-Gly-Phe-Leu-OBzl:[97]
To a stirred soln of H-Leu-OBzl·TosOH (1.27 g, 3.2 mmol) and DIPEA (1.1 mL, 3.2 mmol) in CH_2Cl_2 (7.5 mL) a soln of $(Boc)_2$-Phe-F (1.30 g, 3.4 mmol) in CH_2Cl_2 (7.5 mL) was added over a period of 1 min at rt. The mixture was stirred for 2 h and CH_2Cl_2 (10 mL) was added. The soln was washed three times each with 1 M $KHSO_4$, 10% Na_2CO_3, and sat. NaCl, dried ($MgSO_4$) and concentrated. The crude product (1.77 g, 96%) was recrystallized (hexane) to give the pure protected dipeptide ester (1.47 g, 80%). The dipeptide ester (1.14 g, 2 mmol) was treated with TFA at 0 °C for 1 h. After removal of TFA, the residue was dissolved in a mixture of EtOAc (75 mL) and 1 M $NaHCO_3$ soln (25 mL). The organic layer was washed with 1 M $NaHCO_3$ soln and sat. NaCl soln, dried ($MgSO_4$), and concentrated to give the free dipeptide ester. Further peptide assembly was carried out as described for the dipeptide to yield, after column chromatography (CH_2Cl_2/acetone, 70:30), the protected pentapeptide ester as an oil; yield: 1.83 g (62%).

3.3.7 N^{α}-Trityl Amino Acid Fluorides

The non-urethane type N^{α}-Trt amino acid fluorides have been synthesized and were shown to undergo clean reaction with amino acid esters without loss of configuration.

for references see p 492

3.3.7.1 Synthesis of N^{α}-Trityl Amino Acid Fluorides

N^{α}-Trt-protected amino acid fluorides can be prepared from the corresponding amino acid derivatives via cyanuric fluoride.[99]

Table 6 N^{α}-Trityl Amino Acid Fluorides[99]

Xaa	Yield (%)	mp (°C)	Ref
Phe	68	149–152	[99]
D-Phe	65	148–151	[99]
Ile	70	170–172	[99]
Val	65	oil	[99]
Leu	75	oil	[99]
Pro	60	165–168	[99]
pGlu	55	159–162	[99]

N^{α}-Trt-Xaa-F; General Procedure:[99]
To a cold soln (−10 °C) of Trt-Xaa-OH (6 mmol) and anhyd pyridine (3.4 mL, 47 mmol) in CH_2Cl_2 (8 mL), a soln of cyanuric fluoride (12 mmol) in CH_2Cl_2 (4 mL) was added dropwise over a period of 15 min and the resulting mixture was stirred at −10 °C for a further 45 min. Glacial acetic acid (1.71 mL, 30 mmol) was added and the mixture stirred for 15 min, then diluted with CH_2Cl_2 (30 mL) and washed once with ice-cold 5% $NaHCO_3$ soln and twice with ice-cold H_2O. The organic layer was dried (Na_2SO_4) and concentrated to a yellowish residue. Alternatively, the mixture was triturated with Et_2O (60 mL) and the resulting yellowish precipitate filtered off and washed with Et_2O. The combined filtrates were then concentrated and the residue was purified by flash chromatography on silica gel with toluene as eluent. The ninhydrin-positive fractions were pooled and concentrated to an oil which was used for peptide synthesis. Addition of hexane or Et_2O caused crystallization of the fluorides in most cases. Examples can be found in Table 6.[99]

3.3.7.2 Peptide Synthesis with N^{α}-Trityl Amino Acid Fluorides

All of the N^{α}-Trt amino acid fluorides listed in Table 6 are soluble in diethyl ether, with the exception of Trt-pGlu-F.[99] The fluorides undergo aminolysis with amino acid esters and amides without loss of configuration, although relatively slowly due to the steric hindrance resulting from the bulky Trt group. For example, the reaction of Trt-pGlu-F with H-His-OMe and Trt-Ile-F with H-Val-OBzl required 30 minutes and 6 hours respectively.[99]

3.3.8 N^{α}-1,1-Dioxobenzo[*b*]thiophen-2-ylmethoxycarbonyl Amino Acid Fluorides

Cleavage of the 1,1-dioxobenzo[*b*]thiophen-2-ylmethoxycarbonyl (Bsmoc) group, as N-protection, is mechanistically different from that of the Fmoc group. Under appropriate conditions the two protecting groups can be selectively removed (Section 2.1.1.1.1.4). Due to greater hydrophilicity of the side products, rapid continuous solution syntheses (Scheme 1) of short peptides is facilitated by Bsmoc protection.[100,101]

3.3.8.1 Synthesis of N^{α}-1,1-Dioxobenzo[*b*]thiophen-2-ylmethoxycarbonyl Amino Acid Fluorides

The preparation of N^{α}-Bsmoc amino acid fluorides follows that of the Fmoc analogues. Examples are given in Table 7.

Table 7 1,1-Dioxobenzo[*b*]thiophen-2-ylmethoxycarbonyl Amino Acid Fluorides[101]

Xaa	Yield (%)	mp (°C)	Ref	Xaa	Yield (%)	mp (°C)	Ref
Gly	83	132–133	[101]	Ile	85	68–70	[101]
Ala	80	118–120	[101]	Pro	95	oil	[101]
Phe	86	foam or amorphous	[101]	Thr(tBu)	78	118–120	[101]
D-Phe	80	foam or amorphous	[101]	Asn(Trt)	89	foam	[101]
Val	82	114–115	[101]	Gln(Trt)	87	foam	[101]
Asp(OtBu)	80	107–110	[101]	Trp	80	foam	[101]
Glu(OtBu)	75	amorphous	[101]	Phg	78	foam	[101]
Lys(Boc)	93	foam	[101]	Ser(tBu)	82	amorphous	[101]
Tyr(tBu)	80	foam	[101]	Met	75	amorphous	[101]
Aib	67	126–127	[101]	Asn(Dmcp)	85	amorphous solid	[101]
Leu	90	oil	[101]	Gln(Dmcp)	85	amorphous solid	[101]

3.3.8.2 Peptide Synthesis with N^{α}-1,1-Dioxobenzo[*b*]thiophen-2-ylmethoxycarbonyl Amino Acid Fluorides

N^{α}-Bsmoc amino acid fluorides are useful for solid phase and, in particular, rapid continuous solution synthesis of peptides. An example of the latter procedure is given.

Bsmoc-Leu-Thr(tBu)-Phe-Thr(tBu)-Pro-Asn(Trt)-Trp-NH(Dmcp):[101]
To a soln of Dmcp 4-toluenesulfonate (135.5 mg, 0.5 mmol) in CH_2Cl_2 (10 mL) was added DIPEA (0.192 mL, 1.0 mmol) followed by Bsmoc-Trp-F (258.5 mg, 0.51 mmol). After 10 min, the reaction was complete and TAEA (1.5 mL, 10 mmol) was added. Bsmoc-cleavage was complete in 10 min. The mixture was diluted with CH_2Cl_2 (10 mL) and washed with sat. aq NaCl soln (2×20 mL). After drying the organic layer ($MgSO_4$), concentration gave an oil which was treated in the same way with Bsmoc-Asn(Trt)-F (310.96 mg, 0.52 mmol) and then Bsmoc-Pro-F (176.18 mg, 0.52 mmol). Although it was not essential, the crude product was partially purified by flash chromatography (CH_2Cl_2/acetone to give the protected tripeptide (230 mg, 60%). The soln synthesis was then continued with this material (200 mg, 0.208 mmol), in cycles with Bsmoc-Thr(tBu)-F, Bsmoc-Phe-F, Bsmoc-Thr(tBu)-F, and Bsmoc-Leu-F (0.208 mmol). After flash chromatography (CH_2Cl_2/acetone) the protected heptapeptide was obtained as a homogeneous material; yield: 200 mg (63% based on the tripeptide).

for references see p 492

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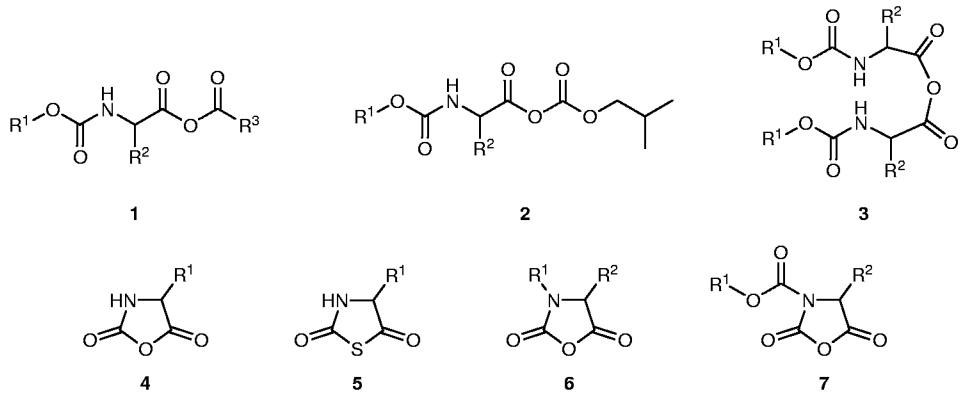
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3.4 Acid Anhydrides

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The use of anhydrides for the formation of peptide bonds dates to the very beginning of synthetic peptide chemistry. The mixed carboxylic acid anhydride derived from *N*-benzoylglycine and benzoic acid was first isolated in 1865 by Kraut and Hartmann[1] by reaction of *N*-benzoylglycine silver salt and benzoyl chloride in boiling diethyl ether. Curtius,[2] in 1881, apparently observed the reaction of this mixed anhydride with glycine silver salt when he isolated *N*-benzoylglycylglycine, together with *N*-benzoylglycine, from the reaction of benzoyl chloride with silver glycinate in boiling benzene. In 1906, Leuchs[3] reported the synthesis of amino acid *N*-carboxyanhydrides and their polymerization to give homo-polyamino acids. The historical aspects of anhydrides in peptide synthesis were thoroughly reviewed by Greenstein and Winitz[4] in 1961. Such early work provided the basis for the development of the three most popular types of anhydrides used in peptide synthesis (Scheme 1): (1) the mixed anhydrides **1** and **2**, (2) the symmetrical anhydrides **3**, and (3) the α-amino acid *N*-carboxyanhydrides **4**. Amino acid anhydrides constitute some of the most highly activated amino acid derivatives and as such are exceedingly efficient, when used correctly, for peptide bond formation.

Scheme 1 Mixed Carboxylic Acid Anhydrides **1**, Mixed Carbonic Acid Anhydrides **2**, Symmetrical Anhydrides **3**, *N*-Carboxyanhydrides **4**, *N*-Thiocarboxyanhydrides **5**, *N*-Substituted *N*-Carboxyanhydrides **6**, and Urethane-Protected *N*-Carboxyanhydrides **7**

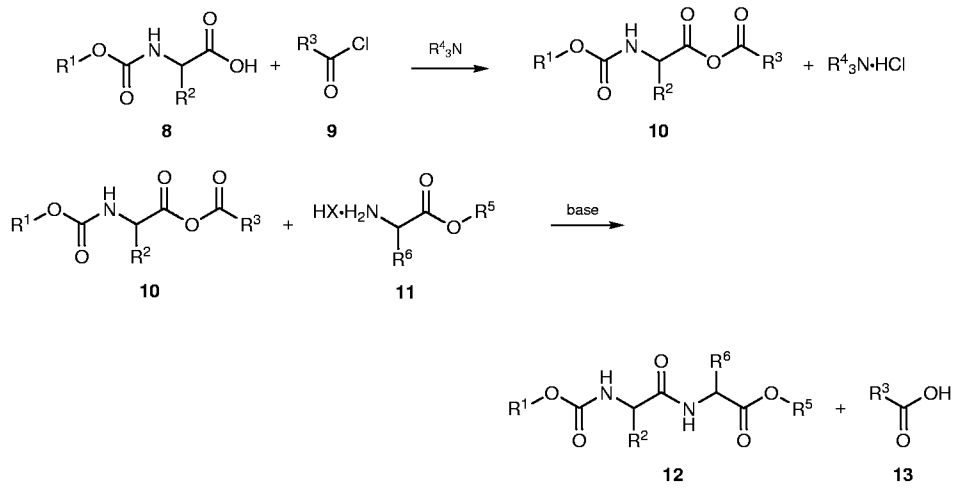


3.4.1 Mixed Anhydrides

All anhydrides derived from an amino acid, other than a symmetrical anhydride, are "mixed anhydrides". However, in the field of peptide chemistry, the term "mixed anhydride" is commonly used to refer to anhydrides derived from an amino acid and either a carboxylic acid **1** or a carbonic acid **2** (Scheme 1). Other mixed anhydrides will be addressed separately. The use of mixed anhydrides **1** and **2** for peptide synthesis encompasses two distinct reaction steps, as outlined in Scheme 2. The first step is the formation of the mixed anhydride, the activation step, which is the reaction of an amino acid derivative **8** with an acid chloride **9**, in the presence of a tertiary amine, in an inert solvent to give the mixed anhydride **10**. The second step is the condensation of the mixed anhydride **10** with the amine of an amino acid derivative **11** to give peptide product **12** and the carboxylic acid **13**.

for references see p 513

Scheme 2 Preparation of an Amino Acid Mixed Anhydride and Its Reaction with an Amino Acid Derivative To Form a Peptide



3.4.1.1 Peptide Synthesis with Mixed Anhydrides in Solution

Wieland et al.[5] in 1950 reported the isolation of anhydrides of the type **1** (Scheme 1) derived from *N*-acyl amino acids and acetic or benzoic acid chloride, their reactivity with amines to give peptides, and a disproportionation reaction which occurred on mild heating, either in solution or in the solid state, to give the two symmetrical anhydrides of the type **3** (Scheme 1). In a simultaneous communication[6] the in situ preparation of the mixed anhydride at 0 °C in an inert organic solvent was described, followed by direct reaction with an amino acid derivative (alkali metal salt or ester) in aqueous solution to give the dipeptide derivative. Shortly thereafter, the use of anhydrides derived from N-protected amino acids and alkyl chloroformates was independently described by Wieland and Bernard,[7] Boissonnas,[8] and Vaughan.[9] This procedure, known as the mixed carbonic acid anhydride method of peptide synthesis, was developed in order to increase the electronegativity of the amino acid carbonyl relative to the activating acid carbonyl in order to direct nucleophilic attack of an amine to the amino acid.

Isobutyl chloroformate has become the most widely used reagent for the preparation of mixed anhydrides for peptide synthesis. The method provides rapid reaction rates at low temperatures, high yields of readily purifiable products, easily removable side products, high purity of crude products, and low cost reagents which are commercially available. These characteristics result in a routine and widely used technique, which can be easily scaled up to the industrial level.

Trimethylacetyl chloride (pivaloyl chloride) has also found widespread use for mixed anhydride formation. This reagent is frequently superior to isobutyl chloroformate where condensation reactions involve highly hindered amines. Mixed anhydrides derived from pivaloyl chloride may be generated in the same manner as from isobutyl chloroformate.

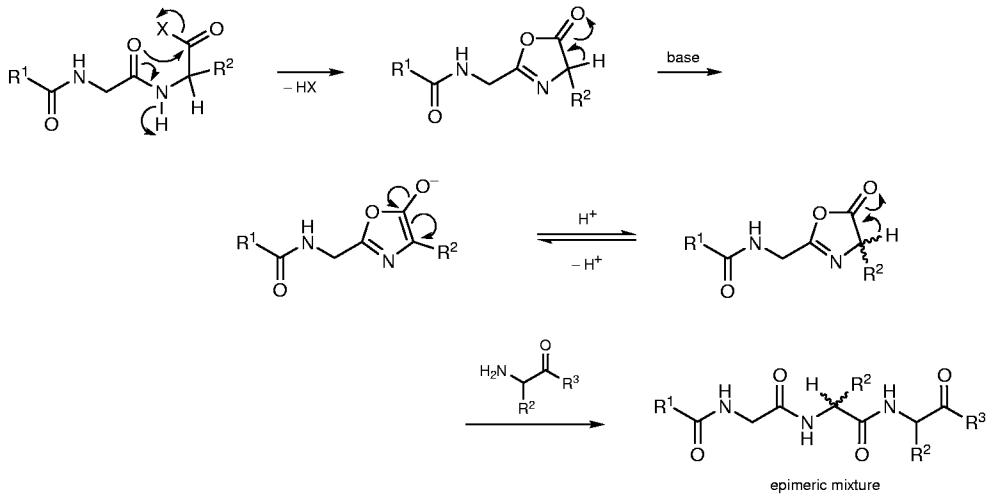
The advent of numerous new coupling reagents and the drawbacks discussed below have substantially decreased the popularity of mixed anhydrides for peptide synthesis in the laboratory. The mixed anhydride method is, therefore, a frequently overlooked technique which is an effective tool for peptide synthesis. This procedure has been comprehensively reviewed[10–12] and, as stated by Meienhofer,[11] the mixed anhydride method, when carried out under well-defined reaction conditions, is considerably less afflicted with undesired side reactions than, for example, the azide or carbodiimide procedures.

All carboxylic acid anhydrides have two carbonyl groups, either of which can theoretically participate in an aminolysis reaction to give an amide bond. One of the keys to successful peptide synthesis using mixed anhydrides is the design of the reaction conditions such that only the desired carbonyl participates in the condensation reaction. Among the factors that influence which of the two carbonyls will undergo aminolysis are the relative electrophilicity of the carbonyl groups and the steric hindrance of the activating acid. Thus the carbonyl with the lowest electron density and least steric hindrance will be the carbonyl to undergo nucleophilic attack.[13,14] Pivaloyl chloride and isovaleroyl chloride have both become very popular reagents for mixed anhydride formation and peptide synthesis. Similarly, the mixed anhydrides derived from *sec*-butyl chloroformate and isobutyl chloroformate were shown to be superior to mixed anhydrides derived from a number of other alkyl chloroformates for providing high yields of amides.[15]

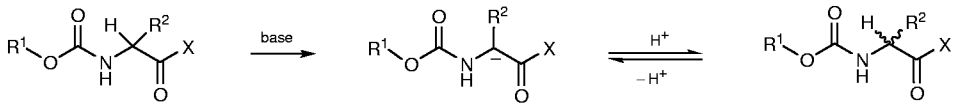
The factors which affect racemization and yield in peptide synthesis via the mixed anhydride method were thoroughly investigated by Anderson et al.,[16] to optimize conditions for activation and coupling of optically active *N*-acyl amino acids (a protected dipeptide) with an amino acid ester to obtain high yields of fully protected tripeptides with retention of optical purity. The reaction conditions during the formation of the mixed anhydride and the time prior to condensation are the critical stages of the synthesis. The solvent, pK_a of the tertiary amine, steric hindrance of the tertiary amine, reaction time, and the temperature all play a key role in successful peptide synthesis. Racemization can occur by either of two mechanisms. One is the intramolecular condensation of the enol form of the *N*-acyl group with the mixed anhydride to give an oxazol-5(4*H*)-one (Scheme 3). The second is by direct proton extraction from the α-carbon of the mixed anhydride (Scheme 4). Thus, base is a major factor in racemization, and racemization begins immediately upon formation of the mixed anhydride in the presence of base. Anderson et al. established that for mixed anhydride formation: (1) THF and ethyl acetate are the best solvents, although DMF is usually acceptable, (2) dichloromethane is a bad solvent, (3) *N*-methylmorpholine (NMM) is one of the best tertiary amines, (4) TEA is not a good base, (5) activation times should be short, of the order of 30 s to 5 min, (6) the tertiary amine should contain at least one methyl group, (7) the tertiary base should have a relatively low pK_a, (8) the temperature of the reaction should be kept at −15 to −5 °C, and (9) that DIPEA, which does not complex with acid chlorides owing to steric bulkiness, essentially does not promote anhydride formation. These factors directly apply when forming mixed anhydrides from the significantly less sensitive urethane-protected amino acids and should always be taken into consideration when using mixed anhydrides in peptide synthesis. For example, mixed anhydrides derived from many N^{α}-9-fluorenylmethyloxycarbonyl amino acids and isobutyl chloroformate, which form instantly in THF with NMM as base, either do not form or form slowly when dichloromethane is the solvent and TEA the base. If dichloromethane must be used as the solvent, then NMM or *N*-methylpyridine should be used as base.[17]

for references see p 513

Scheme 3 Racemization of a *N*-Acyl Amino Acid Mixed Anhydride by Oxazol-5(4*H*)-one Formation



Scheme 4 Racemization of a *N*-Acyl Amino Acid Mixed Anhydride by Proton Extraction from the α-Carbon



Extensive additional optimization experiments have been performed to suppress racemization and side products, which allow their use even for coupling of peptides.[18–24] Taking into account the results of these studies, the mixed anhydride formation reaction is generally carried out with equivalent amounts of amino acid derivative, chloroformate, and tertiary amine or preferably with a slight excess of amino acid component. Optimum reaction conditions should not allow the mixed anhydride formation reaction to be basic. An unhindered weak base should be used and the preferred solvent is THF or ethyl acetate. Once the mixed anhydride is formed the solvent in which the amine component is added is much less critical (DMF, H_2O, CH_2Cl_2). The base used for neutralization of the amine component, if required, is also less critical. Excess base, however, should always be avoided because strong activation of an amino acid greatly increases the acidity of the proton on the α-carbon.

Most reports on mixed anhydride formation consist of the addition of isobutyl chloroformate to a cold mixture of the amino acid derivative and NMM. When mixed anhydride reactions are done on a large scale, where addition of the isobutyl chloroformate may take a substantial amount of time (in order to control the temperature), it is advantageous to add the NMM to the solution of the amino acid derivative and isobutyl chloroformate. This prevents the mixed anhydride from spending a significant amount of time in the presence of base.

Although mixed anhydrides are often generated and used in condensation reactions in situ, mixed anhydrides **1** derived from carboxylic acids (Scheme 1) are frequently sufficiently stable for isolation and storage if required.[12,25] Mixed carbonic anhydrides **2** (Scheme 1) are generally considered too unstable at room temperature for isolation.

Mixed anhydrides are also useful reagents for the generation of *N*-hydroxysuccinimide esters,[26] as well as HOBt and HOAt esters. After formation of the mixed anhydride, as described in the experimental procedures below, typically, 2 equivalents of the *N*-hydroxy component are added followed by 2 equivalents of NMM. The active ester is rapidly formed and in about 10 min the amine component to be coupled is added. The procedure is useful for the condensation of hindered amines where wrong-way opening would otherwise be a problem. Prolylprolyl dipeptides have been obtained rapidly and in high yield with less than 2% wrong opening.

Boc-Asn-Pro-OBzl; Typical Procedure for Mixed Anhydrides with Pivaloyl Chloride:[27]
To a chilled soln (−5 to −10 °C) of Boc-Asn-OH (2.32 g, 10 mmol), pyridine (0.8 mL), and TEA (1.44 mL, 10 mmol) in $CHCl_3$ (50 mL), pivaloyl chloride (1.22 g, 11 mmol) in $CHCl_3$ (10 mL) was added at rates to maintain the temperature below 10 °C. After 5 min at 5–10 °C, the soln was cooled to −10 °C and H-Pro-OBzl · HCl (2.4 g, 10 mmol) and TEA (1.44 mL, 10 mmol) in $CHCl_3$ (50 mL) were added. After 12 h at rt, the $CHCl_3$ was evaporated, and the residue was washed with 5% aq Na_2CO_3, 5% aq citric acid, and H_2O, and recrystallized (EtOAc/petroleum ether); yield: 3.6 g (85%); mp 115–116 °C; $[\alpha]_D^{27}$ −53.8 (*c* 1, MeOH).

Z-Gly-Phe-Gly-OEt; Typical Procedure for Mixed Anhydrides with Isobutyl Chloroformate:[16]
To a chilled soln (−15 °C) of Z-Gly-Phe-OH (1.85 g, 5 mmol) and NMM (0.55 mL, 5 mmol) in THF (25 mL), isobutyl chloroformate (0.67 mL, 5 mmol) was added under stirring. After about 45 s, a soln of H-Gly-OEt · HCl (0.77 g, 5 mmol) and TEA (0.70 mL, 5 mmol) in dry DMF (10 mL) was added, using DMF for rinsing. After 5 min, the bath temperature was 5 °C. The TEA · HCl salt was filtered off; the filtrate was concentrated and, upon addition of H_2O (25 mL), immediate crystallization of the tripeptide occurred. After filtration and washing with H_2O, dil aq $NaHCO_3$, and H_2O, the product was dried; yield: 2.0 g (91%); mp 118–120 °C.

Note: when urethane-protected amino acids are used the activation time should be increased to 4–8 min.

Z-Asp(OBzl)-D-Ala-OH:[28]
To H-D-Ala-OH (20 g, 0.225 mol) in DMF (400 mL), TMSCl (26.8 g, 0.250 mol) was added and the mixture was stirred at rt until dissolution (ca. 45 min). To a chilled (−15 °C) soln of Z-Asp(OBzl)-OH (72 g, 0.200 mol) in DMF/THF (1:1, 880 mL), NMM (22.4 mL, 0.200 mol) and isobutyl chloroformate (26.2 mL, 0.200 mol) were added. After 8 min at −15 °C, the precooled soln of H-D-Ala-OH silyl ester from above was added, followed by the dropwise addition of NMM (22.4 mL, 0.200 mol), ensuring that the temperature of the mixture was maintained at −15 °C. The soln was allowed to warm to rt slowly and stirred for several h before acidifying to pH 1–2 (with cooling) using aq HCl. Then $CHCl_3$ was added, the phases were separated and the aqueous layer was reextracted with $CHCl_3$. The combined organic extracts were washed with 1 M HCl (3×), and sat. aq NaCl, and dried ($MgSO_4$). After removal of the solvent, the oily residue was triturated with Et_2O. The resulting solid was filtered; yield: 67 g (78%); mp 158–159 °C.

(This method has been scaled up to the multikilogram level by adding the NMM after the isobutyl chloroformate. In order to control the temperature, the addition takes several min and the mixed anhydride, which is rapidly forming, is never in contact with large amounts of NMM.)

3.4.1.2 Peptide Synthesis in Solution by Repetitive Excess Mixed Anhydride Method

Tilak et al.[29] described the use of excess mixed carbonic anhydrides to force condensation reactions to completion followed by the destruction of the excess mixed anhydride via the addition of aqueous potassium hydrogencarbonate. Hydrolysis of the mixed anhydride was rapid and the resulting protected dipeptide could be extracted into ethyl acetate in a high state of purity, leaving the excess amino acid derivative and the salts in the aqueous phase. Without further purification the protected dipeptide was N^{α}-deprotected and reacted with the next mixed anhydride, and the process repeated until the desired peptide was obtained. Beyerman et al. substantially expanded the scope of this procedure and named it the REMA method for peptide synthesis (Repetitive Excess Mixed Anhydride).[30] These reaction conditions provide an excellent method to ensure complete reaction of the amine component as well as rapid reaction rates and minimal side products. However, care must be taken to ensure that the excess carboxylic acid component is soluble in sodium hydrogencarbonate solution, e.g. when Z-Asp(OBzl)-OH is the acid component, it is extracted into the ethyl acetate as the sodium salt along with the product. With the due precautions the yields of small peptides are so high that the method could be applied without purification of the intermediate products, that is, in a repetitive way.

for references see p 513

Boc-Leu-Phe-OMe:[31]
To a stirred soln of Boc-Leu-OH · H_2O (3.98 g, 16 mmol) and NMM (1.76 mL, 16 mmol) in THF (25 mL), isobutyl chloroformate (2.00 mL, 15 mmol) was added at −15 °C. After 2 min, a precooled (−15 °C) soln of H-Phe-OMe · HCl (2.16 g, 10 mmol) and NMM (1.10 mL, 10 mmol) in THF (25 mL) was added. After 3 h at −15 °C the excess of mixed anhydride was hydrolyzed with 2.5 M $KHCO_3$ (10 mL). After 30 min stirring at 0 °C, the peptide was precipitated with 50–90% sat. aq NaCl and extracted with EtOAc. The extracts were washed with sat. aq NaCl, dried ($MgSO_4$), and concentrated. The residue was recrystallized (EtOAc/petroleum ether); yield: 3.65 g (93%); mp 86–88 °C; $[\alpha]_D^{25}$ −18.6 (*c* 1, DMF).
After N^{α}-deprotection the cycle is repeated with average yields >90%.

3.4.1.3 Peptide Synthesis with Mixed Anhydrides on Solid Supports

Mixed anhydrides have found limited applicability to peptide synthesis on a solid support. The reported instability of mixed carbonic anhydrides at room temperature, where solid-phase synthesis is generally conducted, and the potential for "wrong way" opening, thus terminating the growing peptide chain, has discouraged their development for SPPS. However, there are several reports in the literature on the use of mixed anhydrides in SPPS.[32–35] Merrifield et al.[33] reported the synthesis of H-Leu-Ala-Gly-Val-OH on a solid support via Bpoc-protected amino acid mixed anhydrides at room temperature. The activation step involved the protected amino acid in dichloromethane at low temperature, addition of ethyl chloroformate followed by TEA with subsequent reaction with an amine on the solid support. Under these conditions (see discussion in Section 3.4.1.1 on TEA in CH_2Cl_2), a substantial amount of H-Leu-Ala-Gly-Gly-Val-OH was obtained as a side product. Changing the base to NMM reduced this side product but changing to isobutyl chloroformate had little effect. This product increased in proportion to the activation time prior to addition to the amine resin and appeared to arise from acylation of Bpoc-Gly-OH prior to coupling to the amine on the resin. The side reaction was shown to be the result of disproportionation of the mixed anhydride to the symmetrical anhydride, followed by a rearrangement to Bpoc-Gly-(Bpoc)Gly-OH and subsequent activation.

The use of Fmoc-protected amino acid mixed anhydrides for SPPS in a pressurized flow reactor has been examined in a detailed manner.[34,35] These studies indicated that N^{α}-Fmoc amino acid mixed anhydrides obtained by reaction of Fmoc-protected amino acids, isobutyl chloroformate, and NMM in THF (without excess base) were stable at room temperature for several hours. When H-Leu-Ala-Gly-Val-OH was synthesized on the solid support in DMF (with activation in DMF), the Gly-Gly side product was not detected. Additionally, no "wrong way" opening was observed. Numerous additional peptides were synthesized via this procedure with good results. Care must be exercised when coupling to proline, especially proline to proline, as total "wrong way" opening of the anhydride can occur.[30] This problem was overcome by reacting the N-protected proline mixed anhydride with excess HOBt and NMM, as previously proposed for acylation of hindered amino acids in solution.[36] The active ester coupling proceeded without difficulty on a solid support. A slight excess of amino acid component over chloroformate and NMM is always maintained, thus preventing base-catalyzed side reactions and inadvertent acylation of the growing peptide chain on the solid support by the chloroformate. Owing to lack of a good way to monitor SPPS, and fear of side reactions of the mixed anhydrides during slow couplings and possible "wrong way" opening due to hindered peptides or mixed anhydrides, a routine use of the HOBt procedure, i.e. conversion into the related active ester, for SPPS is recommended.

An increased stability of mixed anhydrides derived from isopropyl chloroformate and their lower propensity for "wrong way" opening has been reported by Benoiton and coworkers[37] Possibly this type of mixed anhydride, in conjunction with Fmoc protection, may be useful for SPPS. Nonetheless, at the current time, the mixed anhydride method of peptide synthesis seems to have little applicability to SPPS.

The use of mixed anhydrides derived from inorganic acids[38–40] has been reported for SPPS, but condensation reactions are frequently slow and the procedures have not found general application.

3.4.1.4 Other Mixed Anhydrides

Although N-protected amino acid mixed anhydrides, and particularly mixed carbonic anhydrides, have been by far the most commonly used mixed anhydrides in peptide synthesis, a number of additional mixed anhydrides derived from acids of phosphorus, arsenic, and sulfur have been reported and used in peptide synthesis;[10,12] however, these reagents have not found general applicability in routine peptide synthesis.

The intramolecular cyclic anhydrides of unprotected aspartic and glutamic acids[41,42] (amine as a salt) and the corresponding N-protected anhydrides have been prepared and used in peptide synthesis.[4,10,43–45] Nucleophilic condensation with these anhydrides can frequently be directed to one or the other of the carbonyls by appropriate selection of N-protecting group, solvent, and reaction conditions; however, frequently a mixture of the two possible products is obtained. These anhydrides offer excellent applicability to large-scale industrial reactions where the exact reactions conditions can be thoroughly worked out to provide high yields and purity of the desired product. The well-known sweet peptide, H-Asp-Phe-OMe, has been prepared industrially from an intramolecular aspartic acid anhydride.[46]

Z-Glu-Anhydride:[45]
A suspension of Z-Glu-OH (20 g, 70 mmol) in Ac_2O (50 mL) was kept at rt with occasional shaking until dissolution. The mixture was concentrated at 50–60 °C and the residue was recrystallized ($CHCl_3/Et_2O$); yield: 13.0 g (70%); mp 93–94 °C; $[\alpha]_D$ −45.1 (*c* 10, AcOH).

3.4.2 Symmetrical Anhydrides

Symmetrical anhydrides **3** (Scheme 1) derived from N-protected amino acids have found widespread use for the synthesis of peptides, both in solution and particularly on a solid phase. They react rapidly with amino acid esters or salts, and with the free amino group of an otherwise protected peptide, to give high acylation yields. Numerous methods for the preparation of N-protected amino acid anhydrides have been reported, including disproportionation of a mixed anhydride,[5,47,48] reaction of a mixed anhydride with a second equivalent of the N-protected amino acid, and reaction of 2 equivalents of the N-protected amino acid with 1 equivalent of either ethoxyacetylene[49] or methylethynyldiethylamine.[29,30] They are also prepared from the N-protected amino acid and 1,1′-carbonyldiimidazole followed by a second equivalent of N-protected amino acid or by the action of phosgene on the N-protected amino acid salts.[50,51] However, by far the most commonly used method for the generation of a symmetrical anhydride for use in peptide synthesis, either in solution or on a solid phase, is the reaction of 2 equivalents of a N-protected amino acid with 1 equivalent of a carbodiimide.[52–55] Thereby, the symmetrical anhydrides may be generated in situ with DCC or DIC for use in SPPS.[56]

Symmetrical anhydrides are stable to washing by dilute acid or aqueous sodium hydrogencarbonate at room temperature and, consequently, can readily be obtained pure using water-soluble carbodiimide.[57] Furthermore, when pure, most of the crystalline products are stable at room temperature. The symmetrical anhydride method was reported by Chen et al.[57] to give superior yields of peptides as compared to the normal carbodiimide coupling procedure on a solid phase and to be superior to the mixed anhydride method in solution.

for references see p 513

3.4.2.1 Peptide Synthesis with Symmetrical Anhydrides in Solution

The use of symmetrical anhydrides for the synthesis of peptides in solution has not found widespread application, although a number of efficient peptide syntheses have been performed using this procedure.[20,21] Preparation and use of symmetrical anhydrides for peptide synthesis in solution has been comprehensively reviewed.[58]

Symmetrical anhydride couplings with amino acid or peptide esters are complete within a few minutes and are quantitative when 1.4 equivalents of anhydride is used. The excess anhydride can be removed by treatment of the mixture with *N*-(2-aminoethyl)piperazine.

However, the method is not substantially better than the mixed anhydride method, the standard DCC method, the active ester method, or the newer procedures (HBTU, BOP, PyBroP, etc.), and it uses double the amount of N-protected amino acid to accomplish the condensation reaction. Although many symmetrical anhydrides are isolable and stable,[58] they are not commercially available. On the other hand, the use of preformed symmetrical anhydrides is very convenient. One needs only to add an amino acid ester to the protected amino acid anhydride in a nonreactive solvent and, upon completion of the condensation, destroy the excess anhydride and remove the acid components by extraction.

(Fmoc-Xaa)$_2$O; General Procedure:[59]
To an ice-cold soln of the Fmoc amino acid (20 mmol) in EtOAc (250 mL), EDC (2.11 g, 11 mmol) was added with stirring. Stirring was continued at 0 °C for 1 h and at rt for an additional 1 h. The mixture was poured into ice water (250 mL), more EtOAc was added, and the organic phase was washed with 0.1 M HCl, H_2O, 0.05 M $KHCO_3$, and H_2O, and dried ($MgSO_4$). The solvent was removed and the residue crystallized (EtOAc/hexane); yield: 50–86%. The Fmoc amino acid anhydrides have characteristic IR bands at 1830 and 1750 cm^{-1}.

(Boc-Phe)$_2$O; Typical Procedure:[57,59]
To an ice-cold soln of Boc-Phe-OH (5.30 g, 20 mmol) in CH_2Cl_2 (200 mL), EDC (1.92 g, 10 mmol) was added and the mixture was stirred at ice-bath temperature for 2 h. The solvent was removed at a bath temperature of 0 °C and the residue was dissolved in an ice-cold mixture of EtOAc (200 mL) and H_2O (100 mL). The organic layer was washed with ice water (2×100 mL), ice-cold 0.25 M $KHCO_3$, and brine. The soln was dried ($MgSO_4$) and the solvent was removed at a bath temperature of 0 °C. The residue crystallized on storage in the cold and was washed with hexane/Et_2O (20:1); yield: 3.9 g (76%); mp 95–97 °C; $[\alpha]_D^{23}$ −32.8 (*c* 2, $CHCl_3$).

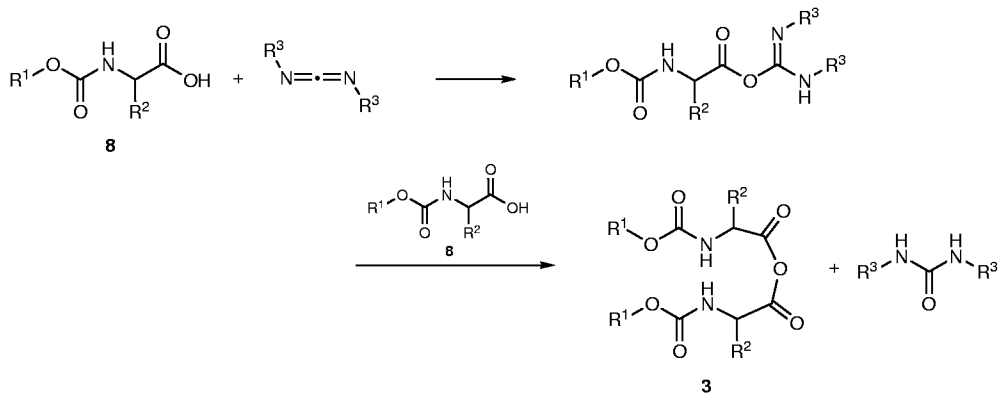
Boc-Ala-Ala-OBzl; Typical Procedure for Symmetrical Anhydride Preparation via Methylethynyldiethylamine:[29,58]
To a soln of Boc-Ala-OH (2.84 g, 15 mmol) in CH_2Cl_2 (15 mL), a soln of methylethynyldiethylamine (0.78 g, 7 mmol) in CH_2Cl_2 (5 mL) was added dropwise over a period of 30 min at 0 °C. Then a soln of H-Ala-OBzl · TosOH (1.75 g, 5 mmol) and DIPEA (0.71 g, 5.5 mmol) in CH_2Cl_2 (5 mL) was added. After 15 min, *N*-(2-aminoethyl)piperazine (0.258 g, 2 mmol) was added and the mixture was stirred for an additional 10 min. After extraction with dil HCl (3×), 0.05 M $NaHCO_3$ (3×), and H_2O, the soln was dried (Na_2SO_4) and the solvent was removed, and the residue was recrystallized (EtOAc/petroleum ether); yield: 1.68 g (96%); mp 68–70 °C.

3.4.2.2 Peptide Synthesis with Symmetrical Anhydrides on a Solid Support

Symmetrical anhydrides have been especially useful in SPPS. Following its introduction in 1955,[60] DCC became not only the most commonly used reagent for the condensation of amino acids but also a convenient reagent for the preparation of symmetrical anhydrides.[30] Since numerous reports[51,61–63] indicated that the use of symmetrical anhydrides gave higher yields of peptides with greater purity than the normal carbodiimide procedure, Hagenmeier and Frank[56] proposed the in situ generation of symmetrical anhydrides from protected amino acids and DCC (Scheme 5) and their use in SPPS. This "modified" (symmetrical anhydride, protected amino acid/DCC ratio 2:1) procedure gave higher coupling yields at each step than the standard DCC procedure (protected amino acid/DCC ratio 1:1). While

the mechanism for condensation of an amine with a DCC-activated amino acid in solution involves direct reaction of the amine with the *O*-acylisourea (Scheme 5),[64,65] condensation of an amine on a solid support with a DCC-activated amino acid proceeds by formation of the symmetrical anhydride followed by acylation of the growing peptide chain on the solid support by the symmetrical anhydride.[61] The excess of DCC relative to the anhydride during "standard" solid-phase synthesis may well promote side reactions.[62,63] For optimal results in SPPS a protected amino acid/DCC ratio of 2:1 is recommended, or the use of preformed symmetrical anhydrides.

Scheme 5 Reaction of a N-Protected Amino Acid with a Carbodiimide To Give an *O*-Acylisourea, Followed by Condensation with a Second N-Protected Amino Acid To Give a Symmetrical Anhydride and a Urea[64,65]



The N-protected amino acid symmetrical anhydrides have the disadvantage of providing only one of the two amino acid units of the anhydride for peptide chain elongation. For a given excess of acylating reagent, a two-fold excess of the amino acid derivative is therefore required, making this procedure costly in comparison to other reagents. N-Protected amino acid symmetrical anhydrides are susceptible to racemization by direct proton abstraction from the α-carbon with base as described for mixed anhydrides (Section 3.4.1.1) and, thus, care should be taken to use these reagents under neutral or slightly acidic conditions or at least under the weakest possible basic conditions. For the generation of symmetrical anhydrides, DCC is frequently replaced by DIC or EDC. The urea formed from these reagents is typically soluble in the solvent system being used and does not have to be removed by filtration prior to the use of the symmetrical anhydrides.

Symmetrical Anhydrides of Boc Amino Acids Generated In Situ for SPPS; General Procedure:[56]
The Boc-protected amino acid (2 equiv) and DCC (1 equiv) were dissolved separately in a minimum amount of CH_2Cl_2, cooled to 0 °C, mixed, and allowed to stand for 1 h at 0 °C with occasional shaking of the mixture. Subsequently, DCU was filtered off and the filtrate immediately used in the coupling reaction.

Symmetrical Anhydrides of Fmoc Amino Acids Generated In Situ for SPPS; General Procedure:[66]
The Fmoc amino acids (3.2 mmol, 6 equiv) in CH_2Cl_2/DMF (2:1, 6 mL) were reacted with DCC (330 mg, 1.6 mmol, 3 equiv) in CH_2Cl_2 (4 mL) at 0 °C. After 30 min the DCU was filtered off and washed with CH_2Cl_2/DMF (2:1, 2 mL). The combined filtrates and washings were used directly for coupling.

for references see p 513

3.4.3 α-Amino Acid *N*-Carboxyanhydrides (Leuchs' Anhydrides)

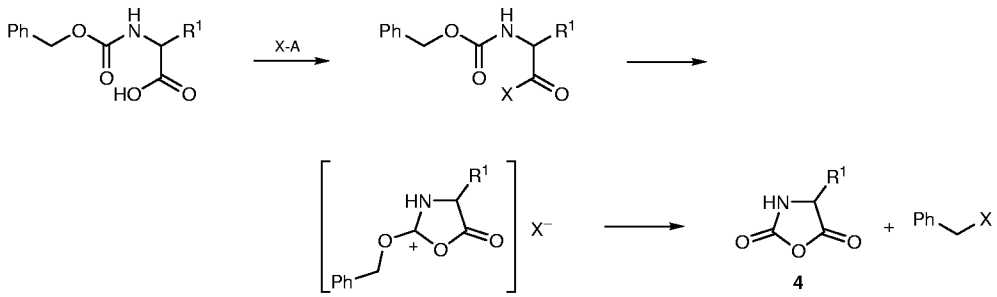
α-Amino acid *N*-carboxyanhydrides **4** (see Scheme 1), named NCAs, Leuchs' anhydrides, or (systematically) oxazolidine-2,5-diones, have found continued widespread interest for peptide synthesis and for preparation of high molecular weight poly-α-amino acids. The NCA chemistry has been comprehensively reviewed in the past.[4,67,68] For the synthesis of poly-α-amino acids by the use of NCAs, see Vol. E 22d, Section 14.2.

Amino acid *N*-thiocarboxyanhydrides **5** (Scheme 1) participate in similar reactions as NCAs, but these anhydrides are not as commonly used as NCAs and will not be discussed here. The reader is again directed to previous reviews.[67,68]

3.4.3.1 Synthesis of α-Amino Acid *N*-Carboxyanhydrides

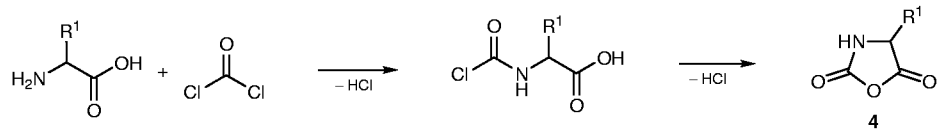
Leuchs obtained NCAs, along with chloroethane, when attempting to distill *N*-ethoxycarbonyl amino acid chlorides obtained by reaction with thionyl chloride.[3] Subsequently, the observation was made that methoxycarbonyl amino acid chlorides cyclize to NCAs at even lower temperature and in higher yields.[69,70] This cyclization of *N*-alkoxycarbonyl amino acid halides to NCAs (Scheme 6), commonly known as the Leuchs method, has been extensively investigated and various acyl halide producing reagents have been proposed, such as thionyl chloride,[3,68–71] phosphorus pentachloride,[72] oxalyl chloride,[73] phosphorus trichloride,[3,4,74,75] dichloromethyl methyl ether,[76] and phosphorus tribromide.[75] Katchalski and Ben-Ishai[77] reported that *N*-benzyloxycarbonyl amino acid halides form NCAs more readily than either the ethoxy- or methoxycarbonyl derivatives and that the acid bromides cyclize to NCAs more readily than the acid chlorides; thus, *N*-benzyloxycarbonyl amino acid bromides give NCAs even below room temperature.[75] *N*-Boc-Protected amino acids are converted into NCAs by reaction with triphosgene[78] or even with small excesses of thionyl chloride in THF.[71] These reactions generally provide complete formation of NCAs at 0 °C in high rates.

Scheme 6 The Leuchs Method for Preparation of a *N*-Carboxyanhydride: Reaction of a *N*-Benzyloxycarbonyl-Protected Amino Acid with an Acid Halide Generating Agent (X-A) Followed by Cyclization To Give the NCA and a Benzyl Halide[69,70]



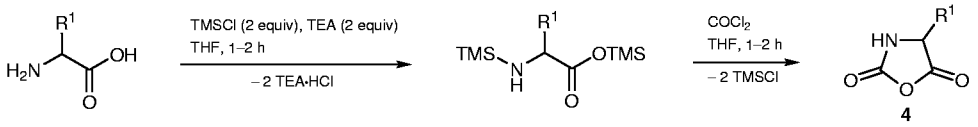
The most economical synthesis of NCAs involves the direct reaction of amino acids in an inert solvent such as ethyl acetate, dioxane, toluene, chlorinated hydrocarbons, acetonitrile, or THF with phosgene according to Scheme 7. Fuchs reported on the synthesis of NCAs by bubbling phosgene gas into a cold aqueous alkaline solution of an amino acid;[79] this method was refined by Farthing[80–82] so that NCAs are obtained by the slow bubbling of phosgene through a hot suspension of an amino acid in an inert solvent such as dioxane (Fuchs–Farthing method). This procedure was further refined[71,83] by the use of limited excesses of phosgene, added as a previously prepared solution in benzene, toluene, or THF, and performing the reaction at about 60 °C. THF is the most generally applied solvent.[83]

Scheme 7 Reaction of an Amino Acid with Phosgene To Give a *N*-Carboxyanhydride and Hydrogen Chloride



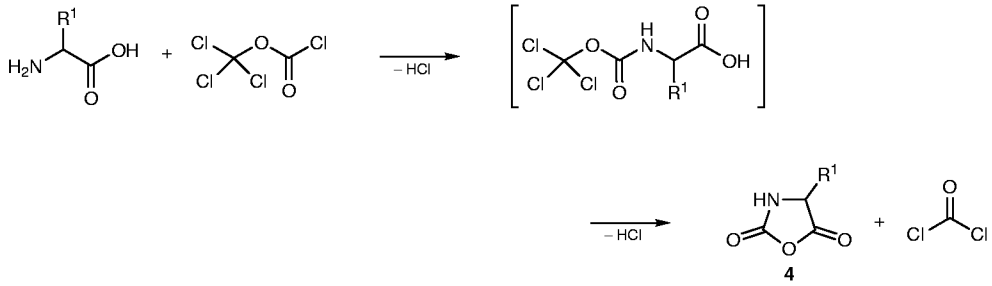
Additionally, for amino acids containing acid-labile side-chain protection, the amine and the carboxy groups may be trimethysilylated and subsequently reacted with phosgene to give the NCA and trimethylsilyl chloride as the leaving group, instead of HCl, thus preventing acidolysis of the side-chain protecting group (Scheme 8).[68,71]

Scheme 8 Reaction of *N*- and *O*-Trimethylsilylated Amino Acids with Phosgene To Produce *N*-Carboxyanhydrides[68,71]



Trichloromethyl chloroformate (diphosgene) and bis(trichloromethyl) carbonate (triphosgene) can both be used essentially as direct replacements for phosgene in most reactions.[84–86] Both compounds offer safety advantages, particularly in the laboratory setting, compared to the use of phosgene gas. Diphosgene is a moderate boiling liquid and triphosgene is a crystalline solid. Diphosgene reacts with an amino acid, as shown in Scheme 9, to give the NCA and one equivalent of phosgene, which then reacts with a second equivalent of amino acid to give a second equivalent of NCA. One molecule of triphosgene reacts with an amino acid to give an NCA and one molecule of diphosgene, which then reacts with a second and third molecule of amino acid as outlined in Scheme 10 to give a total of three molecules of NCA. These reagents are frequently used in excess and the researcher must keep in mind that phosgene is produced in these reactions and that appropriate care must be exercised during their use.

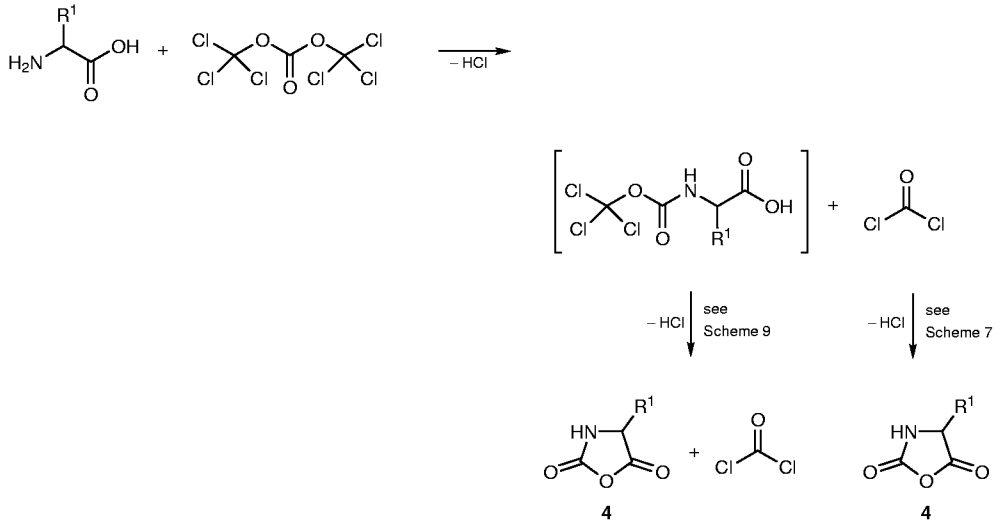
Scheme 9 Reaction of an Amino Acid with Trichloromethyl Chloroformate (Diphosgene) To Produce *N*-Carboxyanhydrides



Of all the methods described, the most convenient laboratory preparation of NCAs is the reaction of Boc-protected amino acids with thionyl chloride in THF. The method provides NCAs in good yields, which may then be purified by crystallization. Typically, a urethane-protected amino acid in THF is treated with 1.1–4 equivalents of thionyl chloride at 0 °C. The NCA is formed in 30 minutes to 24 hours, depending on the amino acid and the urethane protecting group. Conversely, direct phosgenation of an amino acid is the preferred industrial process and modifications, such as trimethylsilylation, allow preparation of NCAs

for references see p 513

Scheme 10 Reaction of an Amino Acid with Bis(trichloromethyl) Carbonate (Triphosgene) To Produce *N*-Carboxyanhydrides



with acid-labile side-chain protecting groups. The insolubility of glycine prevents the practical preparation of the NCA by direct phosgenation, but glycine NCA is readily prepared from Z-protected glycine in neat thionyl chloride. Proline, on direct phosgenation, forms the carbamoyl chloride but does not cyclize to the NCA. To obtain the NCA, base must be added; however, proline NCA is readily prepared from urethane-protected proline and thionyl chloride.

Gly-NCA with Thionyl Chloride; Typical Procedure:[3,71]
A mixture of Z-Gly-OH (100.0 g, 0.48 mol) and $SOCl_2$ (783.0 g, 6.58 mol) was stirred overnight at 20 °C. The soln was evaporated to an oil and MTBE (1.5 L) was added. The crystalline product was collected and washed with MTBE (4 × 250 mL); yield: 44.11 g (91%); IR (THF) ν_{max}: 1866, 1891 cm^{-1} (C=O). (In many cases the reaction mixture is preferably diluted with THF.)

Leu-NCA by Direct Phosgenation; Typical Procedure:[71,81]

CAUTION: *Phosgene is a colorless, poisonous gas or volatile liquid. This reagent should be handled with the utmost care.*

The reaction was performed in a 3-L, three-necked, round-bottomed flask equipped with an overhead mechanical stirrer, a glass stopper with a N_2 inlet, and a side-arm adapter with a thermometer and a dry ice condenser with a N_2 outlet. The flask was set up with a heating mantle and a variac controller. To a stirred suspension of H-Leu-OH (80.0 g, 0.61 mol) in dry THF (1.5 L) was added 3.3 M phosgene in THF (240 mL, 0.79 mol) in one portion. The resulting mixture was heated to 60 °C until dissolution. After 1.5 h, the soln was cooled to rt and completion of the reaction was assesssed by IR spectroscopy. The solvents and excess phosgene were removed. Hexane (1.8 L) was added to the oil and the resulting soln was cooled to −15 °C. The crystalline product was filtered in a specially designed filtration funnel equipped with a standard taper joint and stopcocks (Schlenck tube) under N_2, washed with hexane (3 × 50 mL), and dried; yield: 81.8 g (88%); IR (THF) ν_{max}: 1858, 1790 cm^{-1} (C=O).

Glu(OBzl)-NCA by Direct Phosgenation:[83]

CAUTION: *Phosgene is a colorless, poisonous gas or volatile liquid. This reagent should be handled with the utmost care.*

To a suspension of H-Glu(OBzl)-OH (50 g, 210 mmol) in dry THF (500 mL), 3 M phosgene in benzene (100 mL) was added. After 1.5 h at 65 °C, the clear soln was poured into petroleum ether (1 L). The product crystallized on storage for 24 h at −15 °C. The product was rapidly filtered and washed with petroleum ether; yield: 50 g (91%); IR ν_{max}: 1865, 1790 cm^{-1} (C=O).

Glu(OtBu)-NCA by Silylation Followed by Phosgenation; Typical Procedure:[71]

CAUTION: *Phosgene is a colorless, poisonous gas or volatile liquid. This reagent should be handled with the utmost care.*

In a three-necked, round-bottomed flask (2 L) equipped with a thermometer, a mechanical stirrer, and a Claisen adapter, which was equipped with a dropping funnel and a N_2 inlet, were placed H-Glu(OtBu)-OH (53.2 g, 0.26 mol) and dry THF (900 mL). The slurry was cooled to −2 °C and TMSCl (60.1 g, 0.55 mol) was added in one portion with stirring. After 20 min the soln clarified and TEA (55.6 g, 0.55 mol) was added dropwise from the dropping funnel over 17 min. The mixture was stirred for 2 h at 0 °C, warmed to rt, and filtered on Celite 521 (20.0 g) in a Schlenck tube under N_2 pressure into 3.51 M phosgene in THF (95.1 mL, 0.33 mol). The soln was stirred for 100 min at rt (IR monitoring) and concentrated to an oil. The oil was redissolved in THF (100 mL) and diluted with hexane (100 mL). Crystals formed rapidly, and upon storage at −15 °C overnight, the product was collected by filtration through a Schlenck tube under N_2 and dried; from the mother liquor a second fraction was obtained; yield: 38.8 g (65%); IR (THF) ν_{max}: 1861, 1790, 1730 cm^{-1} (C=O).

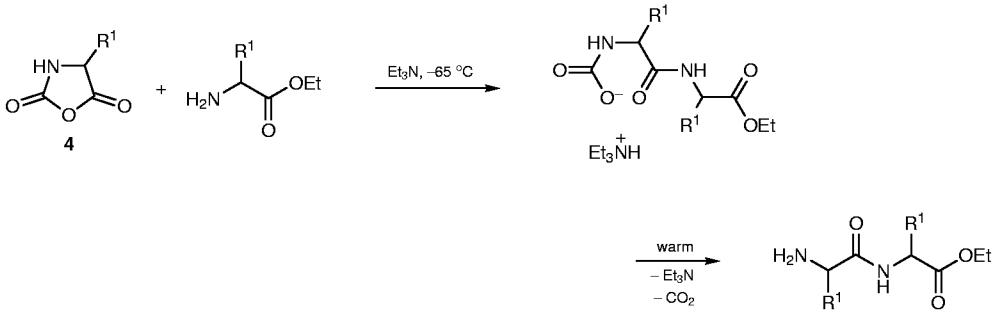
Asn-NCA from Boc-Asn-OH and Thionyl Chloride:[71]

To a suspension of Boc-Asn-OH (88.4 g, 0.38 mol) in THF (1.0 L) at 0–5 °C, $SOCl_2$ (45.3g, 0.38 mol) was added dropwise over 1 h. After stirring at 0–5 °C for 2 h and at rt for 1 h, the precipitate was collected by filtration, washed with Et_2O (4 × 200 mL), and dried; a second crop was isolated from the mother liquors by concentration; yield: 55.9 g (93%); IR (THF) ν_{max}: 1860, 1790, 1694 cm^{-1} (C=O).

3.4.3.2 Peptide Synthesis with *N*-Carboxyanhydrides

The propensity of NCAs to polymerize has substantially limited their routine use for sequential peptide synthesis. Upon first reports on the reaction of NCAs with amines or amino acid esters,[87] Bailey[88,89] provided a definitive study of the reaction of amino acid esters with a variety of NCAs at −65 °C to give *N*-carbamoyl dipeptides which on warming to room temperature liberated carbon dioxide to give the free amino dipeptide. The mixture is then recooled to −65 °C and condensed with a second NCA to give the tripeptide (Scheme 11).

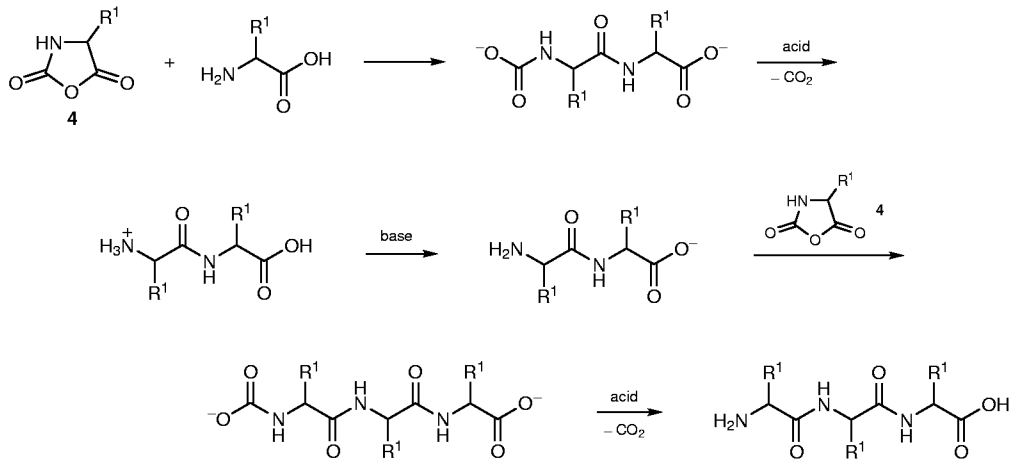
Scheme 11 The Bailey Method for the Synthesis of Peptides from *N*-Carboxyanhydrides at Low Temperature[88,89]



The controlled synthesis of peptides with NCAs in aqueous solution was also examined.[90,91] By this procedure, outlined in Scheme 12, two side reactions which were difficult to control were noted. The first is the premature loss of carbon dioxide from the carbamoyl dipeptide, which may result in oligomerization, and the second is hydrolysis of the NCA to give the amino acid or carbamoyl amino acid. These side reactions are best controlled by operating at pH 10.2–10.5, 0–2 °C, and with strong mixing.[92,93] Alternatively, the synthesis is performed in a biphasic, aqueous sodium carbonate/acetonitrile solvent system.[94] The sodium carbonate stabilizes the carbamate intermediates in the aqueous phase while keeping the NCA in the organic phase, thus minimizing polycondensation and hydrolysis of the NCA. This procedure was subsequently refined[67,95–97] by the use of potassium carbonate as the preferred base.

for references see p 513

Scheme 12 The Synthesis of a Tripeptide from *N*-Carboxyanhydrides in Aqueous Solution[90,91]



The routine use of NCAs in sequential polypeptide synthesis has not been realized even though they appear to be the ideal coupling reagents as both protection and activation are achieved in one step.[67] Moreover, it represents a racemization-free condensation procedure.[98] Clearly this method should supersede any other coupling method. Some peptides, and particularly the enzyme ribonuclease S, have been prepared using this technique, where about 40% of the peptide bonds in the 19-peptide segments were formed through the use of NCAs and *N*-thiocarboxyanhydrides.[99] This work also showed that because of side reactions, extensive purification of intermediates was required to attain high purity, and that each step had to be optimized to minimize other reaction products. Unfortunately, NCAs are prone to self-polymerization when mishandled. If the purity of the starting reagents is not carefully controlled, and the reaction is not optimized for the particular coupling being undertaken, the yield of product will be disappointing. Thus it is desirable to crystallize or by some other means purify the peptide intermediates after only two or three chain elongations.

A primary value of the method is to provide economical large-scale preparations of small peptides, especially dipeptides. Two of the best examples are the industrial preparations of the dipeptides H-Ala-Pro-OH and H-Lys(Tfa)-Pro-OH as intermediates in the synthesis of the angiotensin-converting enzyme inhibitors enalapril and lysinopril.[96]

The reaction of small amounts of an alcohol with an NCA results in initiation of polymerization and even with large amounts of alcohol high yields of esters are not obtained, since the amine formed from esterification is nucleophilic and reacts preferentially with another NCA. Thus the preparation of esters from NCAs requires at least one equivalent of HCl. High yields of esters are obtained from the reaction of an alcohol and an NCA under anhydrous conditions and saturation with HCl.[100–104]

H-Ala-Ala-Gly-OH by the Leggett–Bailey Method:[4,88,89]

To a 2% soln of NH_3 in $CHCl_3$ (1.2 mL) was added H-Gly-OMe · HCl (0.28 g, 2.2 mmol) and the precipitated NH_4Cl was removed by filtration. The soln was evaporated to dryness to remove the slight excess of NH_3. The residue was taken up in $CHCl_3$ (2 mL) and the soln was cooled to −65 °C. After the addition of methyldioctylamine (0.40 mL, 1.25 mmol), a precooled (−65 °C) soln of Ala-NCA (0.23 g, 2.0 mmol) in THF (1.5 mL) was added slowly to the mixture. After 3 h, the mixture was allowed to warm to rt and a vacuum was applied. The soln, which now contained a mixture of H-Ala-Gly-OMe and methyldioctylamine, was again cooled to −65 °C and reacted with a second equiv of Ala-NCA as before. After 3 h, the soln was permitted to warm to rt and then concentrated to dryness. The residue was saponified with 1.2 equiv of 0.15 M $Ba(OH)_2$ at rt for 15 min and the soln was then neutralized with 0.15 M H_2SO_4, filtered, and the filtrate was taken to dryness. The residue was dissolved in a few drops of H_2O and the product was precipitated with acetone; yield: 0.405 g (85%); $[\alpha]_D^{22}$ −48.1 (*c* 2, H_2O).

H-Ala-Pro-OH by the Biphasic Aqueous Carbonate/MeCN Method:[67,96]
A two-phase mixture of H_2O (500 mL), H-Pro-OH (31.5 g, 0.274 mol), KOH (15.6 g, 0.274 mol; corrected for H_2O and K_2CO_3 content), K_2CO_3 (36.1 g, 0.261 mol), and MeCN (500 mL) was prepared and cooled to 0 °C. A soln of Ala-NCA (30.0 g, 0.261 mol) in sieve-dried MeCN (40 mL) was cooled between −10 and −15 °C. This soln was rapidly added to the stirred, two-phase mixture, during which time the temperature of the mixture rose to 5 °C. The mixture was stirred until the reaction was complete, as determined by HPLC, while its temperature was maintained at about 0–5 °C. The temperature was then raised to 10 °C and the layers were permitted to separate. The aqueous phase (pH 9.4–10.2) was removed, weighed, and assayed against an external standard by HPLC for H-Ala-Pro-OH content; yield: 43.9 g (90%).

3.4.4 N-Protected Amino Acid *N*-Carboxyanhydrides

Owing to the difficulty in controlling NCA condensation reactions so as not to obtain oligomers or hydrolysis, various N-substituted NCAs **6** (Scheme 1) were proposed, with substituents such as 2-nitrophenylsulfanyl,[105] trityl (see Section 2.1.1.6.3.1),[106] and 9-xanthenyl,[107,108] for use in stepwise peptide synthesis. These N-protected NCAs have not found significant use in peptide synthesis either in solution or on a solid support.

Urethane-protected amino acid *N*-carboxyanhydrides **7** (UNCAs, Scheme 1) are the most recently developed type of anhydride used in peptide synthesis.[109,110] First recognized as potentially useful reagents for peptide synthesis by Kricheldorf,[111] they should exhibit all of the positive features of an NCA without the uncontrollable side reactions, while providing the peptide chemist with the commonly used urethane protecting groups. UNCAs generate a urethane-protected peptide upon coupling to the free amine of an amino acid or peptide and are resistant to active monomers or nucleophilic polymerization. UNCAs are generally stable, crystalline compounds which can be stored for extended periods of time (Table 1). For a comprehensive review of the synthesis and use of UNCAs in peptide synthesis, see ref[71].

Table 1 Urethane-Protected α-Amino Acid *N*-Carboxyanhydrides[73]

Amino Acid	N^{α}-Protection	N^{ω}-Protection	mp (°C)	$[\alpha]_D^{25}$ (*c* 2.0, THF)	Ref
Ala	Boc	–	103–104.5	+56.0	[73]
	Z	–	113–115	+51.5	[73]
	Fmoc	–	106–107	+28.7	[73]
D-Ala	Boc	–	104–105.5	−56.3	[73]
	Fmoc	–	109–113	−28.7	[73]
Aib	Boc	–	112–113.5	–	[73]
	Z	–	88–90	–	[73]
	Fmoc	–	96–98	–	[73]
Asp	Boc	Bzl	106.5–107.5	+33.5	[73]
	–	tBu	133–135	+43.8	[73]
	–	Cy	103–104	+44.3	[73]
	Z	Bzl	69–70	+43.7	[73]
	–	tBu	127	+51.3	[73]
	Fmoc	tBu	65–70	+22.4	[73]
Asn	Boc	Trt	225	+64.3	[73]
	Fmoc	Trt	134–137	+29.1	[73]
	Fmoc	–	57–58	+3.4	[73]
Arg	Fmoc	Mtr	102	+10.2	[73]
	–	Pmc	100	+120.4	[73]
Cys	Boc	tBu	127	−62.4	[73]
	–	Mob	87–88	+53.7	[73]
	Fmoc	Trt	70–80	+32.3	[73]

for references see p 513

Table 1 (cont.)

Amino Acid	N^{α}-Protection	N^{ω}-Protection	mp (°C)	$[\alpha]_D^{25}$ (*c* 2.0, THF)	Ref
Gln	Boc	Trt	123–126	+19.4	[73]
	Z	Trt	107.5–109.5	+20.6	[73]
Glu	Boc	Bzl	75–76.5	+30.1	[73]
	Z	Bzl	85–86	+41.3	[73]
	–	tBu	73–75	+48.8	[73]
	Fmoc	tBu	120–123	+29.3	[73]
Gly	Boc	–	160	–	[73]
	Z	–	127–128	–	[73]
	Fmoc	–	156	–	[73]
His	Boc	Dnp	83	+66.2	[73]
	Fmoc	Trt	110–112	+53.1	[73]
Ile	Boc	–	106–107	+59.3	[73]
	Z	–	99–100	+63.7	[73]
	Fmoc	–	117–118	+25.9	[73]
Leu	Boc	–	56.5–57.5	+66.8	[73]
	Z	–	75–76	+66.2	[73]
	Fmoc	–	118–120	+38.0	[73]
Lys	Boc	Tfa	115–116	+38.8	[73]
	Fmoc	Z	80–82	+24.2	[73]
	–	Tfa	70–72	+22.6	[73]
	–	Boc	81–85	+25.3	[73]
	Z	Z	59–61	+36.7	[73]
Met	Boc	–	97–98.5	+93.2	[73]
	Z	–	97–98	+95.2	[73]
	Fmoc	–	74–75	+69.3	[73]
Nle	Boc	–	72–73	+56.6	[73]
Phe	Boc	–	101.5–103	+115.2	[73]
	Fmoc	–	59–61	+101.9	[73]
D-Phe	Boc	–	105–106.5	–117.3	[73]
	Z	–	105–106	–130	[73]
Ser	Boc	Bzl	100–101.5	+52.0	[73]
	Z	Bzl	89–90	+54.0	[73]
	Z	tBu	88	+55.6	[73]
	Fmoc	tBu	54–57	+27.5	[73]
Thr	Boc	Bzl	80–81	+72.7	[73]
	Z	tBu	140	+67	[73]
	Fmoc	tBu	124–127	+31.2	[73]
Trp	Boc	For	87–89	+73.9	[73]
	Z	–	72–74	+96.8	[73]
	Fmoc	For	108	+87.9	[73]
Tyr	Boc	Bzl	124.5–125	+98.0	[73]
	Boc	Bzl(2,4-Cl_2)	149–150	+85.8	[73]
	Z	Bzl	124	+119.5	[73]
	Z	tBu	85–86	+120.4	[73]
	Fmoc	tBu	122–124	+110.6	[73]
Val	Boc	–	116.5–118	+58.9	[73]
	Fmoc	–	83.5–87	+14.8	[73]
D-Val	Z	–	82–84	–58.1	[73]

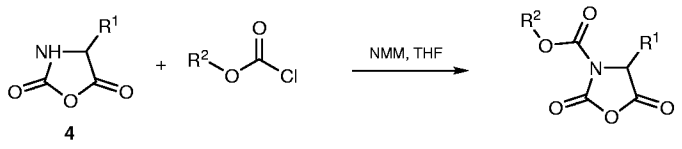
3.4.4.1 Synthesis of N-Protected α-Amino Acid *N*-Carboxyanhydrides

The base NMM is particularly useful for promoting the condensation of a chloroformate with amino acid *N*-carboxyanhydrides because of its relative inability to initiate NCA polymerization compared to many tertiary amines. The choice of NMM as the base is a key factor in the successful synthesis of UNCAs. When the condensation reaction between the NCA and the chloroformate (Scheme 13) is complete, the excess base is precipitated by titration of the reaction mixture to pH 3–5 with anhydrous HCl in dioxane. If an acid-labile side-chain protecting group is present in the UNCA, care must be exercised during the HCl

addition to ensure that acidolysis does not occur. Although UNCAs may be isolated without the neutralization procedure, removal of the excess base greatly facilitates purification and crystallization of the desired product. Excess NMM not removed by neutralization can be carried through subsequent steps to the final crystalline product and small amounts of NMM can catalyze nucleophilic ring opening of the UNCAs.

The synthesis of UNCAs is conveniently monitored by IR spectroscopy. The shift in the NCA ring carbonyl frequencies at about 1790 and 1850 cm^{-1} to the UNCA ring carbonyl frequencies at about 1810 and 1865 cm^{-1} along with the appearance of the urethane carbonyl at about 1740 cm^{-1} provide characteristic indicators of the progress of the reaction. The UNCAs frequently have a shoulder on the carbonyl peak at 1810 cm^{-1} which may resolve into a well-defined peak at about 1820 cm^{-1}, depending on solvent and concentration. Quantitation of both NCA and UNCA homogeneity may be obtained using gel-permeation chromatography (GPC) on small-pore polystyrene columns with THF as eluent.[71]

Scheme 13 Synthesis of Urethane-Protected *N*-Carboxyanhydrides[71]



R^1 = amino acid side chain; R^2 = Fm, Bzl, tBu

Fmoc-Leu-NCA:[71]
To a cooled (−20 °C) soln of Leu-NCA (54.1 g, 0.344 mol) and Fmoc-Cl (93.43 g, 0.361 mol) in dry THF (1.3 L) was added dry NMM (51.33 g, 0.508 mol) under N_2 dropwise over 15 min. The slurry which formed was stirred at −20 °C for 90 min. After IR monitoring, the excess NMM was neutralized by adjusting the pH to between 3–5 by the addition of 4.68 M HCl in dioxane (35 mL, 0.16 mol). The slurry was transferred to a sealed filtration funnel (Schlenck tube) containing Celite (10 g) via a double-tipped transfer needle. The precipitate was washed with THF (2 × 20 mL) and the washes were added to the clear yellow filtrate. After concentration of the filtrate, the resulting oil was crystallized from THF (150 mL) and iPr_2O (500 mL); from the mother liquors a second crop was collected; yield: 100.2 g (77%); mp 118–120 °C; $[\alpha]_D^{25}$ +38.0 (*c* 2.0, THF); IR (THF) ν_{max}: 1870, 1810, 1740 cm^{-1} (C=O).

Z-Leu-NCA:[71]
To Leu-NCA (7.10 g, 0.045 mol) in THF (110 mL) at −25 °C, Z-Cl (8.89 g, 0.050 mol; 95% pure) was added in one portion. Then NMM (5.23 g, 0.052 mol) in THF (20 mL) was added dropwise over 30 min and the mixture was stirred for 1 h at −25 °C, after which time IR and GPC indicated complete reaction. The mixture was acidified to pH 3 by the addition of 5.0 M HCl in dioxane and allowed to warm to rt. The NMM · HCl was removed by filtration under N_2 with the aid of Celite 521 (10.0 g) in a Schlenck tube and washed with THF (2 × 20 mL). The resulting soln was concentrated to an oil, dissolved in MTBE (100 mL), and diluted with hexane (500 mL). The crystals were collected by filtration under N_2, washed with hexane (2 × 100 mL), and dried; yield: 9.65 g (74%); mp 75–76 °C; $[\alpha]_D^{25}$ +66.2 (*c* 2.0, THF); IR (THF) ν_{max}: 1869, 1807 (sh), 1743 cm^{-1} (C=O).

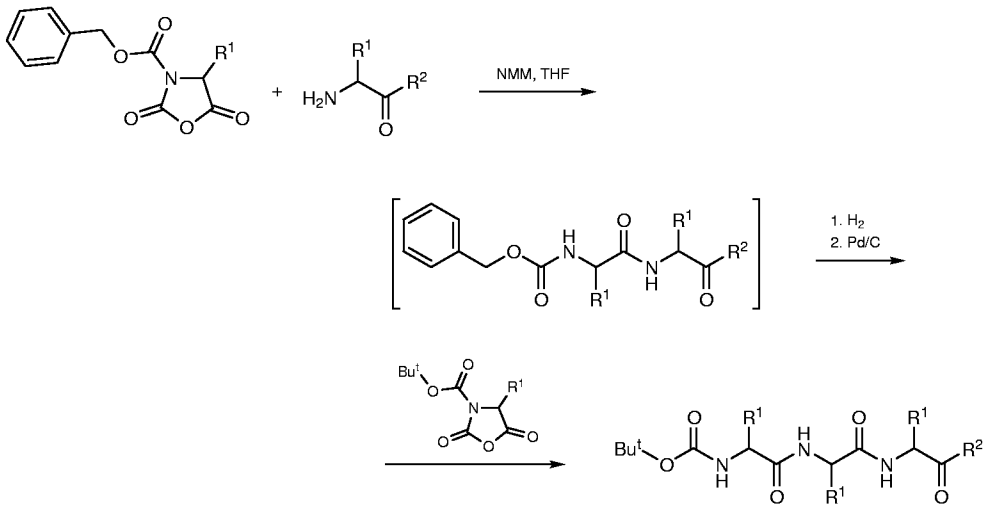
Boc-Leu-NCA:[71]
To Leu-NCA (14.1 g, 0.090 mol) in THF (250 mL) at −15 °C under N_2 was added Boc_2O (23.5 g, 0.108 mol), pyridine (14.2 g, 0.180 mol), and powdered 4 Å molecular sieves (2.0 g). The flask was sealed and stored at −15 °C for 3 d. The mixture was acidified to pH 3–4 by the slow addition of 4 M HCl in dioxane at −15 °C. The precipitate and molecular sieves were removed by filtration with the aid of Celite (10.0 g) and a Schlenck tube with N_2 pressure. The filtrate was concentrated and the resulting oil was crystallized from THF (20 mL) and hexane (300 mL).The product was collected in a Schlenck tube under N_2 pressure, washed with hexane (100 mL), and dried; yield: 14.4 g (82%); mp 56.5–57.5 °C; $[\alpha]_D^{25}$ +66.8 (*c* 2.0, THF); IR (CH_2Cl_2) ν_{max}: 1875, 1820, 1810, 1740 cm^{-1} (C=O).

for references see p 513

3.4.4.2 Peptide Synthesis with N-Protected α-Amino Acid *N*-Carboxyanhydrides in Solution

Urethane-protected amino acid *N*-carboxyanhydrides are highly effective reagents for various reactions involved in peptide synthesis. Acylation of suitably protected amino acids or peptides occurs rapidly in a variety of solvents without detectable racemization.[71,109,112] Owing to their high reactivity, UNCAs can be efficiently used for acylation of sterically hindered amino groups.[113] The lack of contaminating byproducts makes these intermediates attractive for the synthesis of peptides in solution, particularly in large-scale approaches. Besides their application in classical solution techniques,[114,115] a one-pot procedure of peptide synthesis can be performed, as demonstrated by Zhu and Fuller[116] with the synthesis of tripeptides consisting of three consecutive steps, i.e. coupling, deprotection, and coupling without intermediate purification, as shown in Scheme 14. The one-pot procedure was extended to the synthesis of oligopeptides of up to six amino acids in length.[71] The reactions can be conducted in a variety of solvents such as THF, DMF, toluene, DMA, NMP, acetonitrile, DMSO, and glacial acetic acid or mixtures of these solvents. Interestingly, the acylation reactions with UNCAs proceed smoothly in glacial acetic acid, although at somewhat slower rates than in THF. As with most peptide syntheses in solution, solubility frequently becomes the limiting factor; the variety of solvents which can be used with UNCAs, including glacial acetic acid, may help to overcome this difficulty.

Scheme 14 One-Pot Synthesis of a Tripeptide by the Use of a Z-Protected Amino Acid *N*-Carboxyanhydride, Followed by Hydrogenolysis in the Presence of a Boc-Protected Amino Acid *N*-Carboxyanhydride[116]



COR^2 = ester, amide

The UNCA-based peptide synthesis in solution is ideally suited for scale-up. In fact, a multikilogram synthesis of a pentapeptide in solution was achieved using Boc isoleucine *N*-carboxyanhydride as a key intermediate.[117]

Boc-Asp(OBzl)-Phe-OBzl:[71]
To a suspension of H-Phe-OBzl · HCl (10.3 g, 35.3 mmol) and Boc-Asp(OBzl)-NCA (12.3 g, 35.3 mmol) in toluene (200 mL), NMM (4.28 g, 42 mmol) was added in one portion with stirring. After 1 h, the slurry was concentrated and the residue was partitioned between EtOAc and H_2O. The organic layer was dried ($MgSO_4$) and the solvent was removed. The resulting oil was crystallized from iPr_2O/petroleum ether; yield: 18.5 g (93%); mp 91 °C.

Boc-Ser(Bzl)-Ala-Ser(tBu)-OMe; One-Pot Procedure:[71,116]
To a stirred soln of Z-Ala-NCA (5.2 g, 21 mmol) in THF (100 mL), H-Ser(tBu)-OMe · HCl (4.23 g, 20 mmol) and NMM (2.6 mL, 24 mmol) were added under a N_2 atmosphere. After 30 min (monitoring of the reaction by IR and TLC) were added Boc-Ser(Bzl)-NCA (6.43 g, 20 mmol) and Pd/C (10%; 3.8 g), followed by the dropwise addition of cyclohexa-1,4-diene (19 mL, 200 mmol) over 30 min, while maintaining the temperature at about 37 °C. The hydrogenation was continued for 10 min after complete addition of the cyclohexa-1,4-diene. The catalyst and salts were filtered off, the solvent was evaporated, and the product was precipitated (EtOAc/iPr_2O); yield: 9.5 g (91%); mp 104–105 °C.

3.4.4.3 Peptide Synthesis with N-Protected α-Amino Acid *N*-Carboxyanhydrides on a Solid Support

Peptide bond formation using UNCAs occurs at rates at least equivalent to and usually faster than the rates observed using BOP, PyBroP, or HBTU as the condensing reagent.[108] Kinetic studies to form dipeptides from amino acids linked to solid supports showed complete coupling with 1.2–1.3 equivalents of UNCA in DMF in 5–15 min.[118] Therefore, UNCAs may conveniently be used in any standard protocol for SPPS. No activation step is required and special handling is unnecessary. Coupling reactions with UNCAs generate no byproducts other than carbon dioxide, which simplifies operating procedures. Their ease of manipulation is similar to isolated symmetrical anhydrides but without the waste of one equivalent of protected amino acid. UNCAs have been used for the successful solid-phase syntheses of numerous peptides.

Classically, protected amino acids are coupled to solid supports such as 4-alkoxybenzyl alcohol (Wang) resin or 4-alkoxy-2′,4′-dimethoxybenzhydryl (Rink) resin by using DMAP-catalyzed DCC-mediated esterification. Low yields, significant racemization, and dipeptide formation have plagued this approach.[119–122] Conversely, Fmoc-protected NCAs react smoothly with hydroxy resins with minimal racemization. In the case of Rink resin,[123] a 45 min reaction with Fmoc-Leu-NCA (3 equiv) in toluene in the presence of NMM (0.02 equiv) resulted in high substitution levels and with no detectable racemization (<0.1%).[71] Loading of Fmoc-His(Trt)-OH on Wang resin with a variety of activating reagents has been associated with poor yields and high levels of racemization.[97,123] Esterification with Fmoc-His(Trt)-NCA, in the presence of NMM in toluene as solvent, produced 70–100% of theoretical loading without detectable racemization (<0.3%).

Fmoc-His(Trt)-Wang-Resin:[71]
Wang resin (10.5 g, 0.56 mmol OH · g^{-1}), Fmoc-His(Trt)-NCA (0.87 g, 5.0 equiv), and NMM (1.0 equiv) in toluene (10 mL) were shaken for 12 h at rt. The resin was then filtered off and washed with toluene (50 mL), CH_2Cl_2 (2 × 50 mL), and THF (50 mL), and dried; a substitution of 0.42 mmol · g^{-1} (75% yield) was obtained; racemization <0.3% as determined by HPLC, on a chiral column, of Fmoc-His-OH cleaved from the resin.

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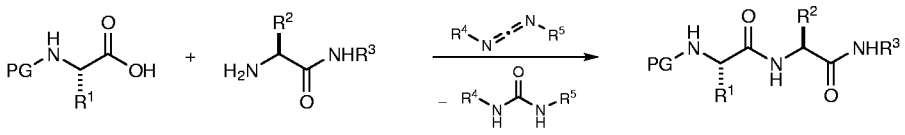
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3.5 Carbodiimides

J. Podlech

Among the plethora of methods available for the linkage of amino acids and peptides, the coupling with carbodiimides is one of the most frequently used, and the peptide-related chemistry has been repeatedly summarized and reviewed.[1–9] The activation of carboxy groups of amino acids and peptides with carbodiimides for the preparation of esters and amides[10] was adopted by Sheehan et al. in 1955.[11] The major advantage of this method is that the reagent can be added directly to a mixture of a carboxylic acid and an amino compound, and therefore preactivation of the carboxy group is not required as activation and coupling proceed concurrently. As an inexpensive and readily available reagent, *N,N′*-dicyclohexylcarbodiimide (**1**, DCC) has predominantly been used. However, carbodiimides not only generate peptide bonds, but also act as dehydrating agents giving rise to the respective urea derivatives as byproducts (Scheme 1). It has been recognized that the ureas and the generated peptide fragments often exhibit similar solubilities in the solvents frequently used for peptide coupling. Consequently, Sheehan and co-workers prepared and used water-soluble carbodiimides, as the corresponding ureas are more readily separated by extraction with water.[12] The most popular carbodiimide of this kind is *N*-[3-(dimethylamino)propyl]-*N′*-ethylcarbodiimide (**2**), and its corresponding hydrochloride **3** (EDC or WSCI).

Scheme 1 General Scheme for Peptide Coupling Using Carbodiimides



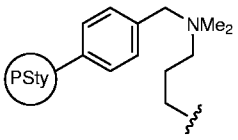
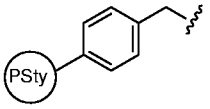
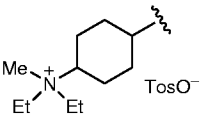
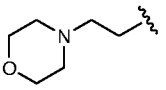
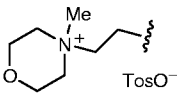
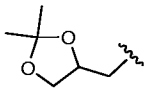
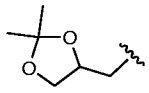
The carbodiimide derivatives most frequently used in peptide chemistry are listed in Table 1. The applicability of these reagents is ruled by various properties. Aside from their commercial availability, price, and stability (hygroscopicity), the solubility properties of the carbodiimides and of their respective ureas are of great practical importance. Further crucial factors are the degree of epimerization of the activated fragment occurring during the coupling procedure, as well as the amount and the separability of unwanted byproducts.

The considerable amount of byproducts and the racemization occurring with the originally developed DCC-coupling procedure can partly be overcome by using specific solvent systems, by changing the type of carbodiimide, or by utilizing special protecting groups, such as the otherwise interfering side-chain amide groups of glutamine and asparagine. The use of protic additives such as *N*-hydroxysuccinimide (HOSu) and 1,2,3-benzotriazol-1-ol (HOBt) is especially advantageous. The latter exert a positive effect due to their acidic properties which reduce the nucleophilicity and basicity of potentially reactive functionalities. Moreover, the additives lead to intermediate active esters which are potent acylating agents and do not possess interfering nucleophilic centers (*vide infra*).

In addition to the direct peptide bond formation, the DCC method has found widespread application in peptide chemistry for the clean and fast production of symmetrical anhydrides (Section 3.4.2) and of active esters (Section 3.2).

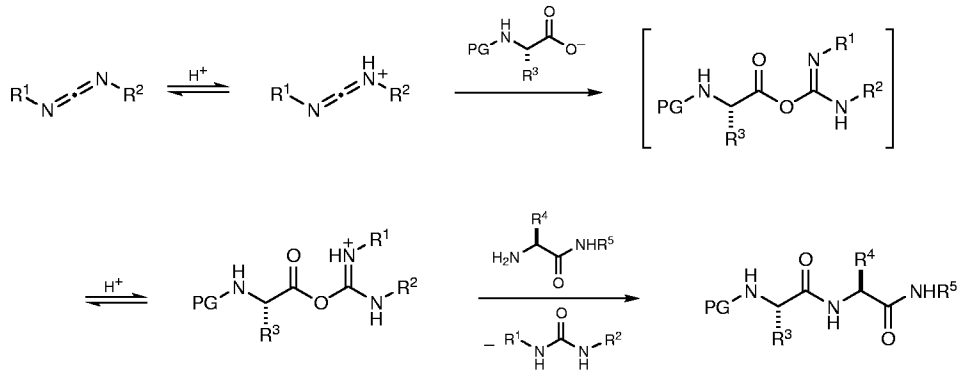
for references see p 531

Table 1 Carbodiimides $R^1–N=C=N–R^2$ Frequently Used in Peptide Chemistry

R^1	R^2	Compound	Solubility	Ref
Et	$(CH_2)_3NMe_2$	**2**	dilute acid, H_2O, CH_2Cl_2, DMF, THF	[13,14]
Et	$(CH_2)_3NMe_2$·HCl	**3** (EDC or WSCI)	H_2O, CH_2Cl_2, DMF, THF	[15]
Et		**4**	insoluble	[16]
iPr	iPr	**5** (DIC)	DMF	[17]
iPr	Cy	**6**	CH_2Cl_2	[18]
iPr		**7**	insoluble	[19]
Cy	Cy	**1** (DCC)	CH_2Cl_2, THF, MeCN, DMF	[17]
Cy		**8**	H_2O	[12]
Cy		**9**	H_2O, DMF, CH_2Cl_2, dioxane, xylene	[12]
Cy		**10**	H_2O, DMF, MeCN	[12]
		**11**	DMF, THF, toluene	[20]
$[(CH_2)_6N=C=N]_nR^3$	$[(CH_2)_6N=C=N]_nR^3$	**12**	insoluble	[21]

The mechanism of the carbodiimide-mediated coupling of peptides is well understood.[1] The first step involves protonation of the carbodiimide moiety by the carboxylic acid. The resulting activated reagent is then attacked by the carboxylate leading to the highly reactive *O*-acylisourea species. This is cleaved by a nucleophile, the amine group of an amino acid or peptide, thus generating the desired product and the urea derivative (Scheme 2). Although the *O*-acylisourea has not been isolated or detected, its intermediate formation is postulated by analogy to stable products.[22]

Scheme 2 Mechanism of Peptide Coupling Using Carbodiimides[1]

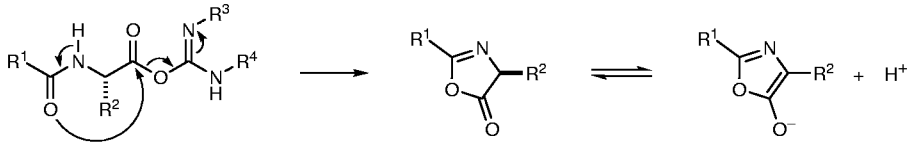


The high efficiency of the carbodiimide-mediated coupling is, however, accompanied by a high tendency for epimerization and for formation of byproducts as discussed below.

Racemization

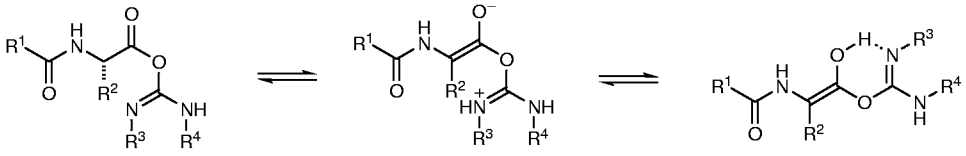
The strong activation of carboxy groups by carbodiimides sometimes leads to an overactivation, especially when peptide fragments are coupled.[23] Thus, competing intramolecular attack of the carbonyl function of the adjacent amide group at the activated carboxy group can lead, upon tautomerization, to the 4,5-dihydrooxazol-5(4*H*)-one (azlactone) intermediate (see Section 7.3). This intermediate, after deprotonation, gives rise to an aromatic species and thus to epimerization of the C-terminal amino acid residue (Scheme 3). The oxazol-5(4*H*)-one intermediate can be considered as a cyclic iminoanhydride, which is also attacked by nucleophiles to produce peptide bonds. However, racemization of the oxazol-5(4*H*)-ones occurs at much faster rates than aminolysis.[24] Although the oxazol-5(4*H*)-one formation takes place even with urethane-protected amino acids, their deprotonation is less likely.[2] Consequently, racemization is observed predominantly when acyl-protected amino acids and thus peptide fragments are coupled by the carbodiimide method.

Scheme 3 Racemization of *O*-Acylisoureas via Oxazol-5(4*H*)-one Formation



Though less attention has been paid to the intramolecular proton abstraction from the intermediate *O*-acylisoureas, it represents an obvious alternative racemization pathway (Scheme 4). The resulting isouronium–enolate betaine might be further stabilized by formation of the corresponding enol.[25]

Scheme 4 Racemization of *O*-Acylisoureas via Enolate Formation



for references see p 531

Racemization via either mode of reaction is efficiently suppressed when acidic additives such as HOSu or HOBt are added (*vide infra*). Even lower amounts of racemization are observed when the coupling reaction is carried out in the presence of copper(II) chloride.[26–29]

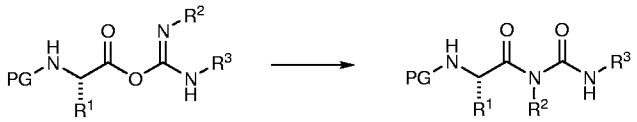
If the amino component is used as a salt such as the hydrochloride, it has to be liberated by means of an auxiliary base which is known to further enhance the rates of racemization.[30] With bulky tertiary amines such as ethyldiisopropylamine, *N*-ethylpiperidine, or *N*-ethylmorpholine rather than triethylamine, this problem is partly circumvented (see Section 7.3),[2] however it is advisable to avoid the use of bases whenever possible by isolating the free amino component prior to the coupling step. Moreover, detailed investigations clearly revealed that racemization is favored at higher temperatures (rt) when the reaction is carried out in polar solvents like DMF,[31] or when the polarity of the solvent is further enhanced by the presence of salts such as LiBr or triethylammonium chloride.[30]

Formation of *N*-Acylureas

A severe problem in carbodiimide-mediated coupling is the formation of *N*-acylureas due to the nucleophilic imino moiety of the isourea competing with the amine nucleophile leading to an intramolecular acyl migration (Scheme 5).[32] This type of side reaction not only reduces the coupling yields, but the resulting *N*-acylureas usually exhibit very similar solubilities to the desired products, thus making their removal difficult to achieve. The amount of this side product is significantly affected by the experimental conditions; *N*-acylurea formation is less likely at lower temperatures[33,34] and when solvents with low polarity are used (CCl_4, CH_2Cl_2, benzene), the amount increases remarkably in polar solvents such as DMF, acetonitrile, DMSO, or water.[11,23] As the acyl migration is initiated by an intramolecular nucleophilic attack, it is again efficiently suppressed by protic additives such as HOSu or HOBt, which reduce the nucleophilic character of the imino moiety by partial protonation.

The difficult separation of the *N*-acylurea byproduct from the desired coupling product can be partly overcome by utilizing water-soluble carbodiimides such as **2**, as not only the resulting urea, but also the *N*-acylurea are water soluble and thus easy to extract.

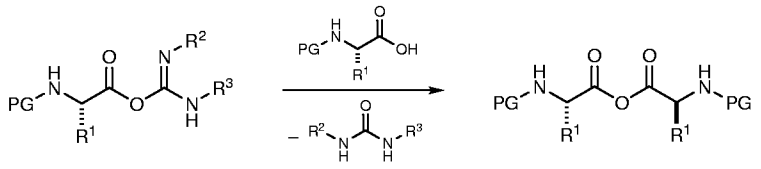
Scheme 5 [*O*→*N*]-Acyl Migration in *O*-Acylisoureas[32]



Formation of Symmetric Anhydrides

In the presence of unreacted carboxylic acid, protonation of the *O*-acylisourea can occur forming a highly activated acyl species (more electrophilic than carbodiimide itself). Consequently, formation of a symmetric anhydride by extrusion of urea may become dominant (Scheme 6).[35] Urethane-type protected amino acids are known to form particularly stable anhydrides (see Section 3.4.2).[36] However, this type of side reaction does not present any drawback to the method, as an attack of the amine nucleophile on the symmetrical anhydride generates the desired peptide bond together with one equivalent of unreacted carboxylate, which further reacts with carbodiimide until full consumption of this reagent has occurred. Detailed investigations have shown that the symmetrical anhydrides are formed spontaneously,[37] being the actual acylating species, which under longer equilibrating conditions are transformed into *O*-acylisoureas.[38] Via this reaction pathway a part of the carboxy component (theoretically one-half) is not consumed when only half an equivalent of carbodiimide is used. To circumvent this nonquantitative consumption of the carboxy component, a double coupling procedure has been developed based on the use of preformed symmetrical anhydrides followed by conversion of the unreacted carboxy component into the active ester via addition of carbodiimide/HOBt.[39]

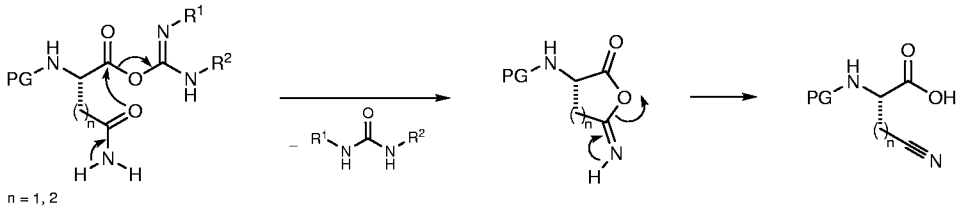
Scheme 6 Formation of Symmetrical Anhydrides as a Side Reaction in *N,N'*-Dicyclohexylcarbodiimide Coupling Reactions[35]



Dehydration of Glutamine and Asparagine

Amino acids with amide-containing side chains such as glutamine or asparagine cannot be activated using carbodiimides, as the intermediate *O*-acylisourea is instantaneously attacked by the amide carbonyl group leading to an imino succinic anhydride and urea.[40–42] Intramolecular elimination of the carboxy group gives rise to a nitrile [i.e., the dehydrated amide (Scheme 7)]. This dehydration reaction can be suppressed to some extent when a fast reacting nucleophile is added (e.g., HOBt) leading to an intermediate active ester.[43] The active ester is thus no longer prone to intramolecular nucleophilic attack of the amide moiety. However, the dehydration problem is most efficiently bypassed by the use of suitable protection of the side-chain amide function such as the ω-trityl[44] and ω-9*H*-xanthen-9-yl derivative[106] (see Section 2.3.3). In this context it is worth noting that asparagine or glutamine residues located in internal positions of the peptide chain do not undergo the dehydration process by carbodiimides. The formation of γ-methyl glutamate has been reported as a side reaction when C-terminal glutamine is activated with DCC in methanol as a solvent with or without the addition of HOBt or HOSu.[45,46]

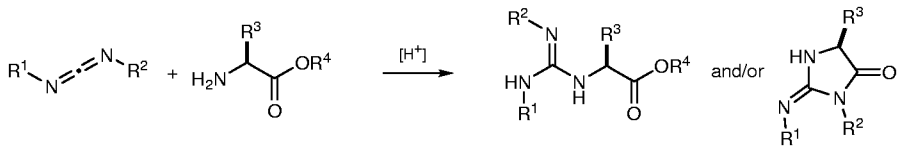
Scheme 7 Dehydration of Activated Asparagine and Glutamine Residues[4]



Reaction of Amino Acid Esters with Carbodiimides

Carbodiimides are known to react with primary amines to form guanidines.[47] In the case of amino acid esters, a subsequent elimination of alcohol is possible with formation of an iminohydantoin species (Scheme 8). However, these reactions usually require higher temperatures or the occurrence of acid catalysis and therefore can be neglected when unprotonated amino acid derivatives are acylated using carbodiimides. On the other hand, this side reaction becomes more important when protonated amino acids (i.e., amino acid ester hydrochlorides) are coupled[48] and these byproducts may occur.

Scheme 8 Addition of Amino Compounds to Carbodiimides[48]



for references see p 531

3.5.1 Peptide Synthesis in Solution with Carbodiimides

3.5.1.1 Carbodiimides without 1,2-Dinucleophiles

Peptide couplings with the sole use of carbodiimides and in the absence of a protic additive are usually restricted to carbamate- or Nps-protected[49] amino acids or to peptide fragments bearing a nonracemizing glycine or proline residue at the C-terminus. In these cases *N,N'*-dicyclohexylcarbodiimide (**1**, DCC) may be employed and is a very cheap reagent, which is also used on an industrial scale.

Best results are obtained when the reaction is run in CH_2Cl_2 or $CHCl_3$; THF and DMF may be used, but are reported to reduce reaction rates, to enhance formation of the *N*-acylurea byproduct and to increase the rate of epimerization. In CH_2Cl_2 the resulting *N,N'*-dicyclohexylurea (DCU) is almost insoluble and can be removed by filtration, although minor amounts may still remain in solution and have to be separated by crystallization or chromatography. Reactions should preferably be carried out at lower temperatures and if possible, in the absence of tertiary bases. In most cases the exclusion of water is not necessary as hydrolysis of the intermediate *O*-acylisourea is very slow.

The unfavorable solubility properties of DCU can be overcome by the use of *N,N'*-diisopropylcarbodiimide (**5**, DIC) or *N*-cyclohexyl-*N'*-isopropylcarbodiimide (**6**).[18,31] Due to the high solubility of the corresponding ureas, these reagents are particularly useful in solid-phase synthesis where excess reagent and ureas may be readily separated from the insoluble resin-bound peptide derivatives by washing steps. Alternatively, the water-soluble *N*-[3-(dimethylamino)propyl]-*N'*-ethylcarbodiimide (**2**) or its corresponding hydrochloride (**3**)[12] can be used; the latter reagent was found to give somewhat cleaner reactions.[31] Carbodiimides **2** and **3**, the corresponding urea, and *N*-acylurea byproducts are readily removed by aqueous workup procedures. Reactions with **2** are carried out in organic solvents and also in water.[50,51] Other water-soluble carbodiimides used in peptide chemistry include *N*-cyclohexyl-*N'*-(2-morpholinoethyl)carbodiimide (**9**), *N*-cyclohexyl-*N'*-[2-(4-methylmorpholin-4-ium-4-yl)ethyl]carbodiimide tosylate (**10**), and *N*-cyclohexyl-*N'*-[4-(diethylmethylammonio)cyclohexyl]carbodiimide tosylate (**8**).[12] The 4-toluenesulfonates proved to be advantageous due to lower hygroscopicity and thus better storage characteristics. *N,N'*-Bis(2,2-dimethyl-1,3-dioxolan-4-yl)methyl]carbodiimide (**11**) is a distillable, oily carbodiimide, which is soluble in a variety of organic solvents such as THF, DMF, and toluene.[20] On completion of the reaction, the acetal moieties are cleaved by dilute acids (0.1 M HCl or phosphoric acid), whereupon it becomes water soluble.

Comparative investigations on various carbodiimides did not reveal significant differences in the coupling yields when DCC, DIC, *N-tert*-butyl-*N'*-methylcarbodiimide, or *N-tert*-butyl-*N'*-ethylcarbodiimide were used as coupling reagents.[52]

An inverted strategy involving the utilization of insoluble polymeric carbodiimides or carbodiimides linked to solid supports has been investigated. For this purpose polyhexamethylenecarbodiimide (**12**)[21] or *N*-benzyl-*N'*-isopropylcarbodiimide (**7**) in which the benzyl residue is part of a polystyrene framework[19] were synthesized (Table 1). Using this approach, the products were separated from excess reagent and from the resulting urea by simple filtration. Similarly, the otherwise difficult to remove *N*-acylureas remain grafted to the polymer. A polymer-bound derivative of **2**, compound **4**, has also been prepared. Although **4** is no longer water soluble due to its linkage to a resin, it is preferable to other polymer-bound carbodiimides, since it is easily prepared from **2** and chloromethylated polystyrene–divinylbenzene resin, which are both commercially available.[16]

The problem of epimerization has not been examined with the insoluble carbodiimides, thus addition of a protic additive might be imperative.

For difficult couplings, as in the case of sterically hindered amino acids or peptide derivatives, it is advantageous to add catalytic amounts of DMAP[53,54] or 4-pyrrolidinopyr-

idine.[55] However, epimerization occurs at higher degrees under these reaction conditions. For example Boc-Phe-OH is coupled to H-Glu(OBzl)-OCH_2-resin with DCC/DMAP with 3–6% racemization, while acylation with Boc-Ile-OH under similar conditions occurs without detectable racemization.[56] Nps-protected amino acids can be coupled in the presence of DMAP without racemization.[49]

Boc-Ala-Val-OMe:[57]
To a soln of Boc-Ala-OH (1.32 g, 6.98 mmol) in CH_2Cl_2 (10 mL) at 0 °C was added H-Val-OMe [obtained from the hydrochloride (2.33 g, 14.0 mmol) by neutralization, subsequent extraction with EtOAc and concentration (5 mL)] followed by DCC (1.43 g, 6.98 mmol), and the mixture was allowed to warm to rt. After stirring for 12 h, CH_2Cl_2 was evaporated and EtOAc added. Insoluble DCU was removed by filtration, the filtrate was washed with 1 M Na_2CO_3, 1 M HCl, and brine (3×30 mL), dried (Na_2SO_4) and the solvent removed under reduced pressure to give the title compound; yield: 2.42 g (96%, crystalline).

Nps-Leu-Gly-OBzl:[58]
To a soln of H-Gly-OBzl·HCl (4.03 g, 20.0 mmol) in dry EtOAc (200 mL) was added Nps-Leu-OH·D-CHA (9.30 g, 20 mmol). At 0 °C DCC (4.53 g, 22 mmol) was added with stirring. Stirring was continued for 2 h at 0 °C and for 12 h at rt. DCU and DCHA·HCl were removed by filtration and the filtrate was successively extracted with aq $NaHCO_3$, H_2O, 5% $KHSO_4$, and H_2O, dried (Na_2SO_4), and the solvent removed under reduced pressure to give the product; yield: 7.78 g (90%, oil); R_f ($CHCl_3$/MeOH, 8:1) 0.85.

Boc-Val-Trp-OMe:[59]
To a soln of H-Trp-OMe (1.51 g, 6.90 mmol) in dry CH_2Cl_2 (50 mL) was added Boc-Val-OH (1.50 g, 6.90 mmol) and **2** (1.32 g, 6.90 mmol) under an argon atmosphere. The flask was protected from light and stirred at rt for 24 h. The soln was washed with 1 M HCl (7 mL) and 1 M $NaHCO_3$ (7 mL), dried, and concentrated. The residue was chromatographed on silica gel (EtOAc) to yield the crystalline title compound; yield: 1.55 g (70%); mp 136–138 °C; $[\alpha]_D^{25}$ +51.6 (*c* 0.5, $CHCl_3$).

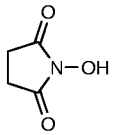
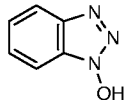
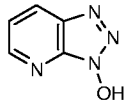
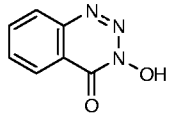
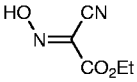
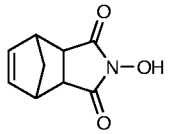
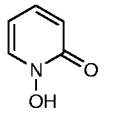
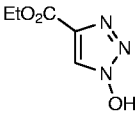
CAUTION: *DCC and other carbodiimides are acute skin irritants and allergenic in susceptible individuals. The carbodiimides should be handled with gloves in a fumehood. Since DCC has a low melting point (mp 34–35 °C), it can also be handled as a liquid after gentle warming in a water bath.*

3.5.1.2 Carbodiimides with 1,2-Dinucleophiles

The drawbacks of the DCC-coupling method, racemization and formation of *N*-acylureas, can be overcome by the addition of protic 1,2-nucleophiles. As reported by Wünsch and Weygand, the addition of two equivalents of *N*-hydroxysuccinimide (**13**, HOSu) and one equivalent of DCC to a mixture of an *N*-acylamino acid and a peptide ester efficiently suppresses racemization and byproduct formation.[60,61] This discovery made an otherwise rather racemization-prone fragment condensation possible and thus allowed for more convergent strategies of peptide synthesis in solution. Further additives have been subsequently analyzed for this purpose and a selection of the most efficient ones is listed in Table 2.[4] One major advantage of these additives is their protic character, which reduces the nucleophilicity of the intermediate *O*-acylisourea thereby preventing *O*→*N*-acyl migration and proton abstraction. The protic character of these additives allows deprotonation of amino groups even with excess tertiary base which otherwise can lead to considerable racemization.

for references see p 531

Table 2 Acidic Additives

Compound	Structure	Ref
N-hydroxysuccinimide (**13**, HOSu)		[60–62]
1,2,3-benzotriazol-1-ol (**14**, HOBt)		[43,63,64]
7-aza-1,2,3-benzotriazol-1-ol (**15**, HOAt)		[65]
3-hydroxy-1,2,3-benzotriazin-4(3*H*)-one (**16**, HODhbt)		[43,63,64]
ethyl cyano(hydroxyimino)acetate (**17**)		[66]
N-hydroxy-5-norbornene-2,3-dicarboximide (**18**)		[67]
1-hydroxypyridin-2(1*H*)-one (**19**)		[68]
ethyl 1-hydroxy-1*H*-1,2,3-triazole-4-carboxylate (**20**)		[69]

When activation is carried out in the presence of a protic additive, an active ester intermediate is formed, which though less reactive than the *O*-acylisourea, is reactive enough for efficient acylation of amino groups. However, some mechanistic details of this reaction are still a puzzle, as it is not yet clear at which step of the reaction a possible racemization is actually taking place. It has been argued that oxazol-5(4*H*)-one formation (*vide supra*) is less likely for active esters and thus the racemization should occur at the level of the intermediate *O*-acylisourea. Investigations suggested that even the active esters racemize to some extent. When BOP is used for activation racemization occurs, although the activation is exclusively due to active ester formation (see Section 3.8).

Not all additives investigated showed improved yields through reduction of the amount of byproducts or the extent of racemization. For example, 5-chlorohydroxytriazole is a highly activating additive enabling high acylation yields, but concomitantly leads to high levels of racemization of the activated amino acid residue.[70]

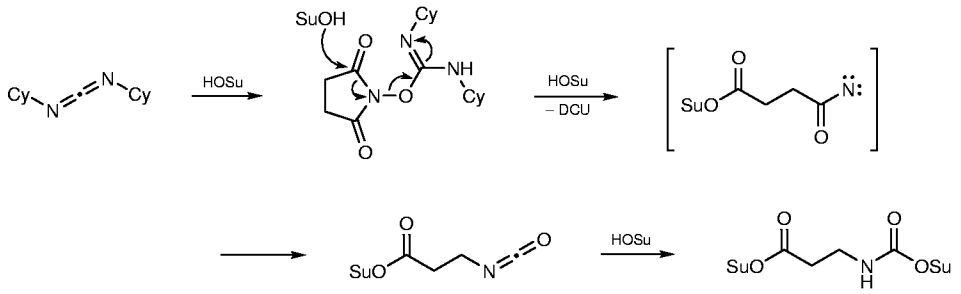
3.5.1.2.1 *N*-Hydroxysuccinimide as an Additive

The DCC/HOSu procedure[60,61] has been extensively applied to fragment condensations in solution. In racemization-prone coupling steps this procedure was generally found to be superior to the DCC/HOBt method (see Section 3.5.1.2.2) even if HOBt (**14**) is more efficient as an additive in terms of acylation rates.[52,71]

The use of HOSu as an additive is imperative when Nps-protected fragments are to be coupled, as other additives such as HOAt (**15**, pK_a 3.47), HODhbt (**16**, 3.97), ethyl cyano(hydroxyimino)acetate (**17**, 4.60) or HOBt (4.60) are too acidic[72] and may lead to cleavage of acid-sensitive protecting groups. The 2-nitrophenylsulfanyl group (Nps) is partially cleaved, when HOBt is used,[73] while no sulfenamide cleavage was observed with HOSu.[74] Further sensitive protecting groups include Trt and Bpoc.[75] The nucleophilicity of the additives is also disadvantageous when certain protecting groups are used, e.g. the N^{im}-tosyl protection of histidine is partially removed by HOBt,[76,77] whereas no displacement was observed with HOSu.[78]

When sterically demanding peptide fragments are coupled in the presence of HOSu, a side reaction that leads to a β-alanine derivative becomes dominant (Scheme 9).[75] In this side reaction, HOSu is activated by DCC and attacked by a second equivalent of HOSu. The nitrene is probably not a real intermediate as it is rearranged concomitantly to its formation to an isocyanate moiety, which is trapped by a third equivalent of HOSu.[79] By replacing the HOSu additive with *N*-hydroxy-5-norbornene-2,3-dicarboximide (**18**)[67] or other dinucleophiles, this side reaction is efficiently bypassed.

Scheme 9 Formation of a β-Alanine Derivative by the Addition of *N*-Hydroxysuccinimide to Carbodiimides[79]



Nps-Lys(Adoc)-Asn-Phe-Phe-Trp-Lys(Boc)-Thr(tBu)-Phe-Thr(tBu)-Ser(tBu)-Cys(StBu)-OtBu, Somatostatin Fragment [4–14]:[80]

To a soln of Nps-Lys(Adoc)-Asn-Phe-Phe-Trp-OH (23.5 g, 21.9 mmol), H-Lys(Boc)-Thr(tBu)-Phe-Thr(tBu)-Ser(tBu)-Cys(StBu)-OtBu (21.9 g, 19.9 mmol) and HOSu (2.75 g, 23.9 mmol) in DMF (450 mL) was added DCC (4.52 g, 21.9 mmol) at −10 °C. The mixture was stirred at 4 °C for 12 h and at rt for an additional 48 h. It was then filtered to remove insoluble DCU and concentrated to half volume. The soln was poured into water-saturated EtOAc. The gelatinous precipitate was collected by centrifugation and reprecipitated twice from DMF/MeCN to give the title compound; yield: 42 g (98%); mp 218–221 °C.

Adoc-His(Adoc)-Ser(tBu)-Asp(OtBu)-Gly-Thr(tBu)-Phe-Thr(tBu)-Ser(tBu)-Glu(OtBu)-Leu-Ser-(tBu)-Arg(HBr)-Leu-Arg(HBr)-Asp(OtBu)-Ser(tBu)-Ala-Arg(HBr)-Leu-Gln-Arg(HBr)-Leu-Leu-Gln-Gly-Leu-Val-NH_2, Fully Protected Secretin:[81]

To Adoc-His(Adoc)-Ser(tBu)-Asp(OtBu)-Gly-Thr(tBu)-Phe-OH (4.50 g, 3.79 mmol) and H-Thr(tBu)-Ser(tBu)-Glu(OtBu)-Leu-Ser(tBu)-Arg(HBr)-Leu-Arg(HBr)-Asp(OtBu)-Ser(tBu)-Ala-Arg(HBr)-Leu-Gln-Arg(HBr)-Leu-Leu-Gln-Gly-Leu-Val-NH_2·HBr (4.00 g, 1.27 mmol) in DMF/DMA/HMPA (280 mL, 1:4:2) were added at −10 °C TEA (175 µL, 1.26 mmol), HOSu (575 mg, 5.00 mmol), and DCC (775 mg, 3.76 mmol). After stirring at 4 °C for 48 h, additional portions of HOSu (145 mg, 1.26 mmol) and DCC (254 mg, 1.23 mmol) were added. After further stirring at rt for 60 h, the soln was reduced to 80 mL and poured with stirring into hot EtOAc (3 L). After standing overnight, the precipitate was filtered off,

for references see p 531

washed thoroughly with hot EtOAc and triturated several times with H_2O/MeOH (3:1) to give the title compound; yield 4.50 g (83%).
Detailed investigation showed that under these coupling conditions about 2% epimerization occurred at the Phe residue, whereas when HOBt was used as additive (*vide infra*), about 15% epimerization was observed.[46]

3.5.1.2.2 1,2,3-Benzotriazol-1-ol as an Additive

König and Geiger found that besides HOSu, 1,2,3-benzotriazol-1-ol (**14**, HOBt) can be used as an additive to suppress racemization and formation of the *N*-acylurea byproduct during peptide fragment coupling with DCC.[43] A comparative analysis of the use of the DCC/HOBt method for condensation of various peptide fragments in DMF with other procedures revealed that this combination of reagents leads to particularly low racemization when Z-Gly-Xaa-OH [with Xaa = Val, Ile, Tyr, Ser(Bzl), or Lys(Z)] was coupled. Higher degrees of racemization have been obtained with Xaa = Ala, Thr, Glu(OBzl), Ser(tBu), or Arg(Mtr). In these cases BOP/HOBt or HBTU/HOBt led to much better results.[82] The additive HOBt has been successfully used in many syntheses of bioactive peptides in solution including the final fragment coupling steps of the fully protected human insulin, in which racemization was not observed to occur under the chosen experimental conditions.[83]

Ddz-Tyr(tBu)-Gln-Leu-Glu(tBu)-Asn-Tyr(tBu)-Cys(Trt)-Asn-OtBu:[107]
To an ice-cold soln of Ddz-Tyr(tBu)-Gln-OH (9.7 g, 16.5 mmol), H-Leu-Glu(tBu)-Asn-Tyr(tBu)-Cys(Trt)-Asn-OtBu·TFA (19.2 g, 15 mmol) and HOBt (2.025 g, 15 mmol) in DMF (30 mL), *N*-ethylmorpholine (1.95 mL, 15 mmol) and a soln of DCC (3.3 g, 16 mmol) in DMF (9 mL) were added. The mixture was stirred at 0 °C for 1 h, and at rt for 20 h. The urea was filtered off, the filtrate was poured into sat. $NaHCO_3$ (159 mL), and stirring was continued until a precipitate was formed. This was collected, washed with citrate buffer (pH 3) and H_2O, and dried. The crude product was suspended in hot EtOAc and filtered after cooling, then washed with EtOAc and Et_2O; yield: 20 g (77%); CHN, amino acid analysis.

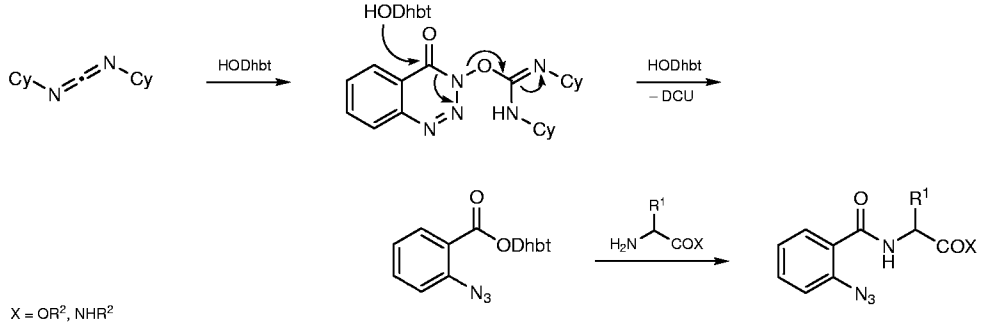
Antrimycin D_V Fragment: Allyl (*S*)-2-{2-[(*S*)-2-*tert*-Butoxycarbonylamino-4-methylpentanoylamino]-3-methylbut-2-enoylamino}-3-hydroxypropanoate:[84]
To a soln of Boc-Leu-ΔVal-OH (1.76 g, 5.40 mmol), H-Ser-OAl·HCl (1.17 g, 6.45 mmol), HOBt (1.09 g, 8.10 mmol), and TEA (1.65 mL, 7.50 mmol) in anhyd DMF (15 mL) at −20 °C, was added EDC (1.70 g, 5.90 mmol). The mixture was allowed to warm to rt over 16 h. After diluting with EtOAc (100 mL), the organic layer was successively washed with H_2O (80 mL), 0.5 M H_2SO_4 (3×80 mL), 1 M $NaHCO_3$ (80 mL), and brine (50 mL), dried ($MgSO_4$), and the solvent removed under reduced pressure. Unreacted Boc-Leu-ΔVal-OBt was separated from the required tripeptide derivative by flash chromatography (hexane/EtOAc 6:4 → 2:8) to give the title compound; yield: 1.26 g (52%); R_f (hexane/EtOAc, 1:3) 0.28; mp 142–143 °C; $[\alpha]_D^{20}$ −32.4 (*c* 0.78, CH_2Cl_2).

3.5.1.2.3 3-Hydroxy-1,2,3-benzotriazin-4(3*H*)-one as an Additive

3-Hydroxy-1,2,3-benzotriazin-4(3*H*)-one (**16**, HODhbt) is another additive, which is known to produce excellent results in terms of condensation yields.[63,85,86] Almost no *N*-acylurea formation or racemization occurs with this reagent and the progress of the acylation reaction is monitored by a color change. Unfortunately, in the presence of carbodiimide a partial ring opening side reaction may occur leading to an activated 2-azidobenzoic ester which is subsequently attacked by the amine present in the reaction mixture (Scheme 10).[64] Comparative studies of the use of **2** in the presence of HOBt, HOAt, and HODhbt clearly confirmed the latter additive as superior in terms of yields and racemization, but also in terms of ester formation when alcohols are used as the solvent for solubility reasons.[86–88] To overcome such solubility problems, solvent mixtures such as TFE/$CHCl_3$ or TFE/CH_2Cl_2 are used successfully in combination with **2**/HODhbt in which the carbodiimide is usually added as the free base to capture the counterion of the amino component. The combination of solvent mixtures and coupling reagents was routinely used by Sakakibara et al. in many successful syntheses of larger peptides and proteins (see Section 5.2).[108] When chlorinated

solvents like CH_2Cl_2 or 1,2-dichloroethane are used with HOBt, HODhbt, or HOSu as additives in the presence of tertiary amines, nucleophilic attack of the hydroxy group on the solvent is observed and thus has to be considered as a potential side reaction.[89] Similarly good results as with HODhbt have been reported for 1-hydroxypyridin-2(1*H*)-one (**19**) as an additive to carbodiimide in CH_2Cl_2/H_2O or isopropyl acetate/H_2O as solvents.[90]

Scheme 10 Formation of 2-Azidobenzamides as Side Products when Using 3-Hydroxy-1,2,3-benzotriazin-4(3*H*)-one as an Additive[64]



The **2**/HODhbt procedure has found widespread application for fragment condensation steps in the synthesis of proteins in solution (see Section 5.2).[91–93]

Protected Human Parathyroid Hormone [1–84]:[86]
Protected Boc-Trp-Leu-Arg(Tos)-Lys[Z(2-Cl)]...Ala-Lys[Z(2-Cl)]-Ser(Bzl)-Gln-OBzl [23–84] (4.07 g, 0.4 mmol) was treated with TFA (30 mL) in an ice bath for 60 min. After removal of excess TFA, 5.7 M HCl in dioxane (0.14 mL, 0.8 mmol) was added and the product precipitated with Et_2O. The product was washed with Et_2O and dried over NaOH pellets under reduced pressure. This material was dissolved together with Boc-Ser(Bzl)-Val-Ser(Bzl)-Glu(OBzl)...Glu(OBzl)-Arg(Tos)-Val-Glu(OBzl)-OH [1–22] (1.68 g, 0.48 mmol) in $CHCl_3$/TFE (120 mL, 3:1), and then HODhbt (78 mg, 0.48 mmol) and **2** (0.09 mL, 0.48 mmol) were added. After stirring for 5 h, the mixture was concentrated and chilled, excess aq 1% $NaHCO_3$ was added, and the precipitate was collected and washed with 1 M HCl and H_2O. The product was dissolved in $CHCl_3$/TFE (3:1) and after concentration, precipitated with MeCN. The precipitate was filtered off and washed with MeOH, EtOAc, and hexane to give the product; yield: 4.63 g (93%); amino acid analysis.

3.5.1.2.4 7-Aza-1,2,3-benzotriazol-1-ol as an Additive

7-Aza-1,2,3-benzotriazol-1-ol (**15**, HOAt) was also found to be generally superior compared to HOBt, as it improves yield and suppresses racemization.[31,65] In addition, the completion of the acylation is indicated by the reaction mixture color which turns from yellow to colorless. The higher reaction rates and lower amounts of byproducts were explained by a precomplexation of the nucleophilic amine by the aza moiety in the vicinity of the activated carboxylic acid, allowing the coupling even of sterically very demanding amino compounds.[65] In addition, the electron-withdrawing effect of the nitrogen atom stabilizes the leaving group, leading to a greater reactivity of the intermediate active ester. It has been reported that when CH_2Cl_2 is used as the solvent, almost no racemization is observed with this additive.[31]

Fully Protected Consensus Sequence of Raf-1 Kinase; Aloc-Ser[PO(OAl)$_2$]-Thr-Ser-OHep Ester:[94]
To a soln of Aloc-Ser[PO(OAl)$_2$]-OH (75 mg, 2.14 μmol), **2** (83 mg, 4.28 mmol), and HOAt (59 mg, 4.28 mmol) in anhyd CH_2Cl_2 (10 mL) was added at 0 °C a soln of H-Thr-Ser-OHep·HCl (73 mg, 2.14 mmol) and TEA (22 mg, 2.14 mmol) in anhyd CH_2Cl_2 (5 mL). After stirring for 1 h at this temperature, the mixture was left overnight at rt and filtered. The organic layer was extracted with 0.5 M HCl (3×10 mL), dried, and concentrated. The crude product was purified by flash chromatography (EtOAc/hexane, 4:1) to yield the title product; yield: 126 mg (93%); $[\alpha]_D^{20}$ −21.6 (*c* 0.06, CH_2Cl_2); HRMS, ^1H NMR; ^{13}C NMR; CHN.

for references see p 531

3.5.1.2.5 Other Additives

Ethyl 1-hydroxy-1*H*-1,2,3-triazole-4-carboxylate (**20**) has been investigated as an additive to carbodiimides particularly for solid-phase peptide synthesis. No racemization was observed when Fmoc-protected amino acids were coupled to the resin-bound peptides, except for histidine residues which under these reaction conditions racemized extensively unless the activation time was kept to one minute and three equivalents of ethyl 1-hydroxy-1*H*-1,2,3-triazole-4-carboxylate (**20**) were added. Under the latter conditions no racemization was detected.[69] The efficiency of this newly developed method has been well documented by the successful synthesis of deglycosylated human erythropoietin.[95]

Ethyl cyano(hydroxyimino)acetate (**17**) leads to less racemization than HOBt when Ac-Ile-OH is coupled with H-Gly-OEt·HCl in the presence of triethylamine. Epimerization to Ac-D-aIle-Gly-OEt was only 1.8%, as opposed to 2.7% with HOSu and 8.8% with HOBt.[66] Further detailed studies exemplifying the versatility of this additive have not been reported.

Ac-Ile-Gly-OEt:[66]

To H-Gly-OEt·HCl (140 mg, 1.00 mmol) in dry DMF (3 mL) TEA (0.14 mL, 1.00 mmol), ethyl cyano-(hydroxyimino)acetate (170 mg, 1.20 mmol), Ac-Ile-OH (173 mg, 1.00 mmol) and a soln of DCC (206 mg, 1.00 mmol) in DMF (2 mL) were added successively at 0 °C. After stirring for 3 h at +5 °C and overnight at rt, the solvent was removed under reduced pressure and the residue extracted with EtOAc. The filtered extract was washed with H_2O, 1 M $NaHCO_3$ (4 ×), H_2O, 1 M HCl, and H_2O and dried ($MgSO_4$). After removal of the solvent under reduced pressure, the product was precipitated with Et_2O/petroleum ether; yield: 178 mg (69%). A detailed analysis showed that 1.8% racemization occurred during the coupling procedure.

3.5.1.2.6 Copper(II) Chloride as an Additive

Addition of copper(II) salts during carbodiimide couplings has been reported to lead to further suppression of racemization[26–28] even when the very racemization-prone *N*-methylalanine is the carboxy-terminal residue.[29] However, in some cases yields drop significantly with this additive. Lowest amounts of racemization have been observed when HOSu/$CuCl_2$ were used as additives, whereas yields were somewhat higher with HOAt/$CuCl_2$.[29]

Further investigations have shown that addition of the corresponding preformed copper salts [$Cu(OBt)_2$, $Cu(OAt)_2$, $Cu(ODhbt)_2$, or $Cu(OSu)_2$] is even more advantageous, since it gives substantial suppression of racemization combined with acceptable yields.[96] DIC activation of a C-terminal Phe or Tyr(OH) residue, in the presence of 0.5 equivalents of $Cu(OBt)_2$, resulted in 80–90% isolated yields of peptide products contaminated with <1% of the corresponding D-diastereomer. Segment coupling reactions do not require a tertiary amine to neutralize the protonated amino component, and byproducts are easily removed by washing with dilute aqueous base containing EDTA, followed by aqueous acid.

Boc-Tyr(Bzl)-Gly-Gly-Phe-Leu-OMe:[27]

CAUTION: *HOBt is commercially available with or without water of crystallization; the hydrated form can be used for peptide coupling by the DCC/HOBt method. However, when water-free material is required, drying should be carried out carefully, as HOBt (and HOAt) decompose in a rapid exothermal reaction above 180 °C. Water-free HOBt can also be obtained by recrystallization from EtOH or EtOH/Et_2O.*

To a soln of Boc-Tyr(Bzl)-Gly-Gly-Phe-OH (105 mg, 166 μmol), H-Leu-OMe·HCl (30.2 mg, 166 μmol), TEA (16.8 mg, 166 μmol), HOBt (44.8 mg, 332 μmol), $CuCl_2$ (11.2 mg, 84 μmol) in DMF (13 mL) was added EDC (31.7 mg, 166 μmol) and stirred for 24 h at 5 °C. Citric acid (10%, 10 mL) was added and the soln extracted with EtOAc (60 mL). The organic extract was washed with citric acid (2×10 mL), brine (10 mL), sat. $NaHCO_3$ (3×10 mL), and brine again (2×10 mL), dried (Na_2SO_4), and the solvent removed under reduced pressure. The crude crystalline residue was recrystallized (EtOAc/MeOH/petroleum ether) to give the product; yield: 108 mg (86%); mp 152–153 °C; $[\alpha]_D^{25}$ −9.9 (*c* 1.0, MeOH).

3.5.2 Solid-Phase Peptide Synthesis with Carbodiimides

In SPPS a large excess of reagents is generally used, therefore the coupling reactions are much faster and the amount of racemization is considerably lower. Nevertheless, especially for certain sensitive amino acids such as cysteine or histidine, the possibility of loss of configuration should be kept in mind. When the SPPS is carried out by the Boc/Bzl strategy, DCC is well suited as the coupling reagent, as DCU is cleanly removed owing to its solubility in TFA/CH_2Cl_2 used in the Boc cleavage steps. When the Fmoc/tBu strategy is applied, DIC is the carbodiimide of choice, due to the high solubility of the corresponding urea in the solvents generally used. Otherwise, **2** is the reagent of choice for peptide synthesis on solid supports.[97] Generally preactivation of the amino acid is carried out with the production of the symmetrical anhydrides or of active esters, in the latter case by operating in the presence of protic additives such as HOBt, HOAt, or ethyl 1-hydroxy-1*H*-1,2,3-triazole-4-carboxylate (**20**). This activation is usually carried out in CH_2Cl_2, although when required mixtures with DMF are used. If desired, the urea derivative is removed by filtration and the CH_2Cl_2 replaced by DMF after evaporation.

To prevent racemization of sensitive amino acid derivatives, the activation time should be kept as short as possible (5 min) and the activation should be run at 4 °C rather than room temperature. Optimized conditions were developed for intermediate formation of active esters (DIC/HOBt or DIC/HOAt in DMF/CH_2Cl_2, 1:1, for 5 min) and symmetrical anhydrides (activation in DMF/CH_2Cl_2, 9:1, and coupling in DMF/CH_2Cl_2, 1:1).[97]

As mentioned above, in the case of glutamine or asparagine residues the side-chain amide groups have to be protected or alternatively, preformed and purified active esters have to be used. When unprotected side-chain threonine derivatives are activated, a partial β-elimination to α-aminocrotonic acid has been reported,[98] whereas arginine, either unprotected and protonated or as ω-tosyl derivative may cyclize to a δ-lactam during activation.[97]

SPPS Coupling with Boc Amino Acids via Symmetrical Anhydrides:[97]
To a soln of Boc amino acid (8 equiv) in CH_2Cl_2 at 4 °C DCC (4 equiv) was added and the mixture left for 5–15 min at 4 °C. The soln was filtered and the filtrate diluted with DMF and added to the resin-bound amino component. After 30 min, the resin was filtered and washed with DMF.

SPPS Coupling with Fmoc Amino Acids via Symmetric Anhydrides:[97]
To a soln of Fmoc amino acid (8 equiv) in CH_2Cl_2 containing a minimum of DMF as required to dissolve the Fmoc amino acid, at 4 °C DIC (4 equiv) was added and the mixture left for 5–15 min at 4 °C. The soln was diluted with DMF and added to the resin-bound amino component. After 30 min, the resin was filtered and washed with DMF.

SPPS Coupling with Fmoc Amino Acids via Active Esters:[97]
Fmoc amino acid (4 equiv) was dissolved at 4 °C in CH_2Cl_2 with a minimum amount of DMF, then HOAt or HOBt (4 equiv) and DIC (4 equiv) was added and the mixture left for 5–15 min at 4 °C. The soln was diluted with DMF and added to the resin-bound amino component. After 30 min the resin was filtered and washed with DMF.

SPPS Coupling of Fmoc Amino Acids via 1-Hydroxy-1*H*-1,2,3-triazole-4-carboxylic Acid Esters:[95]
Fmoc amino acid (1 mmol), 1-hydroxy-1*H*-1,2,3-triazole-4-carboxylic acid ester (1 mmol) and DIC (1 mmol) were dissolved in DMF and sonicated for 10 min. The mixture was then added to the resin-bound amino component and the reaction was left for 30–60 min.

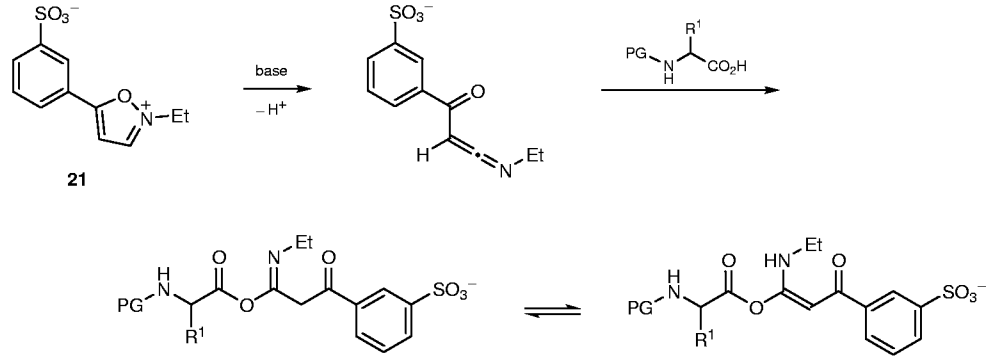
for references see p 531

3.5.3 Related Reagents

3.5.3.1 Woodward's Reagent K

2-Ethyl-5-phenylisoxazolium-3′-sulfonate (**21**, Woodwards Reagent K), a commercially available reagent, ring opens on treatment with base to a ketenimine (which can considered as a carbaanalogue of a carbodiimide) and consequently reacts with carboxylic acids to give iminoanhydrides, whose reactivity towards nucleophiles is similar to *O*-acylureas (Scheme 11). The reagent was introduced for peptide couplings,[99,100] however, **21** has not become a commonly used reagent. Comparative studies showed that **21** was sometimes superior to DCC/HOBt in DMF, when this polar solvent was required.[101,102] In the cyclization of a linear precursor of virginiamycin S_1, **21** was found to lead to better results compared to **2**, BOP-Cl, or BOP, since racemization during this reaction was significantly lower. However, yields have been quite poor (14%) with this reagent.[103]

Scheme 11 Woodward's Reagent K (**21**) as a Carbaanalogue of Carbodiimides[100]

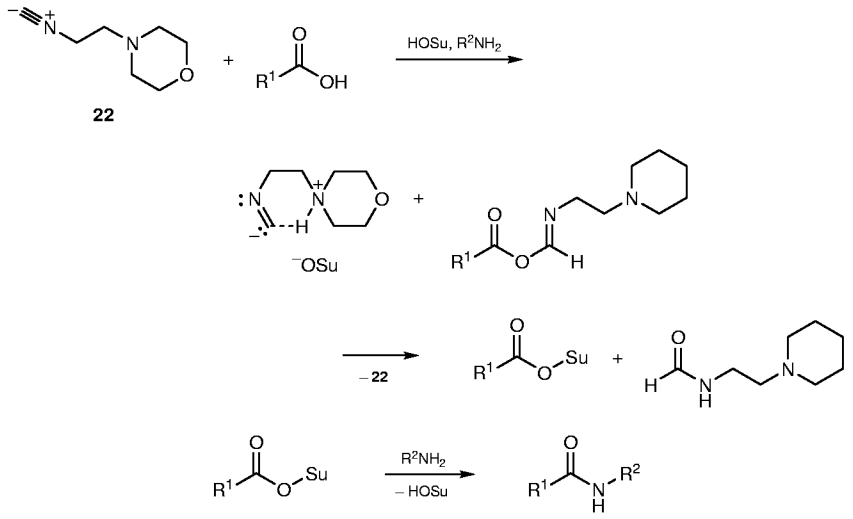


Other reagents with similar mechanistic modes of activation are diphenyl-*N*-tolylketenimine[104] and diphenylketene,[105] but these have been used even more rarely in peptide synthesis, as they are not able to be stored over a long period and thus are not commercially available.

3.5.3.2 *N*-(2-Isocyanoethyl)morpholine

Although attempts to use simple alkyl and aryl isocyanides as coupling reagents have not been satisfying in terms of yields of peptides,[109,110] the use of *N*-(2-isocyanoethyl)morpholine (**22**), particularly in presence of additives such as HOSu or HOBt, readily enables peptide bond formation in high yields.[111] A comparative analysis performed on a difficult peptide segment coupling clearly revealed that this reagent is as efficient as DCC.[71] Since preactivation of the carboxy component is required with this reagent, the coupling reaction proceeds via intermediate formation of the active ester (Scheme 12). The degree of epimerization was found to be similar, if not lower than that observed by DCC/HOSu segment condensation.

Scheme 12 Peptide Coupling with *N*-(2-Isocyanoethyl)morpholine[111]



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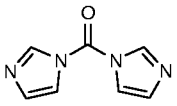
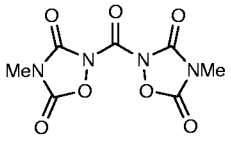
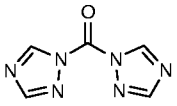
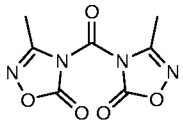
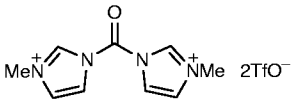
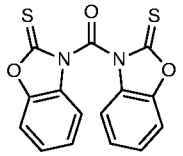
3.6 1,1′-Carbonyldiimidazole

J. Podlech

N-Acylimidazoles were recognized in the early 1950s as reactive intermediates suitable for acylation of amino compounds and therefore of potential usefulness in peptide chemistry.[1] A convenient method for the preparation of these acylimidazoles starting from carboxylic acids and 1,1′-carbonyldiimidazole (**1**, CDI) was introduced by Staab[2,3] and was then used for peptide synthesis by Anderson and Paul.[4,5] Although CDI represents an alternative to reagents such as carbodiimides, it is less frequently used, most probably due to cost and its sensitivity to moisture. Several variations of CDI are listed in Table 1. Though most of them are either less hygroscopic or more reactive, none of them are commercially available. The chemistry of CDI and related compounds has been comprehensively reviewed.[6–9]

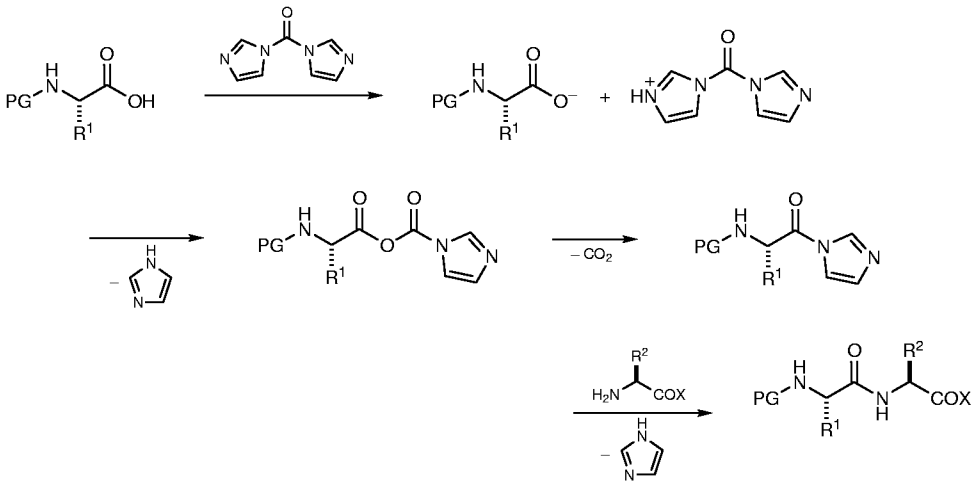
Apart from peptide synthesis, CDI is used for the preparation of ureas[2] and carbamates[10] from amines and alcohols, respectively. In these cases it efficiently replaces the highly toxic phosgene.

Table 1. Carbonyldiimidazoles and Related Compounds Suitable for Peptide Synthesis

Reagent	Ref	Reagent	Ref
 **1**	[5]	 **4**	[14]
 **2**	[11]	 **5**	[15]
 **3**	[12,13]	 **6**	[16]

The mechanism of CDI-mediated acylation of amines is well understood. The first step involves a partial protonation of the basic imidazole-nitrogen, protonated *N*-acetylimidazole has a pK_a of 3.6,[17] leading to an activated species which is then attacked by the carboxylate. The resulting mixed anhydride extrudes carbon dioxide giving rise to *N*-acylimidazole which on treatment with an amine compound leads to the desired amide (Scheme 1). An important advantage of this method over the carbodiimide method is that the byproducts carbon dioxide and imidazole are readily and quantitatively separated from the reaction product by simple washing procedures.

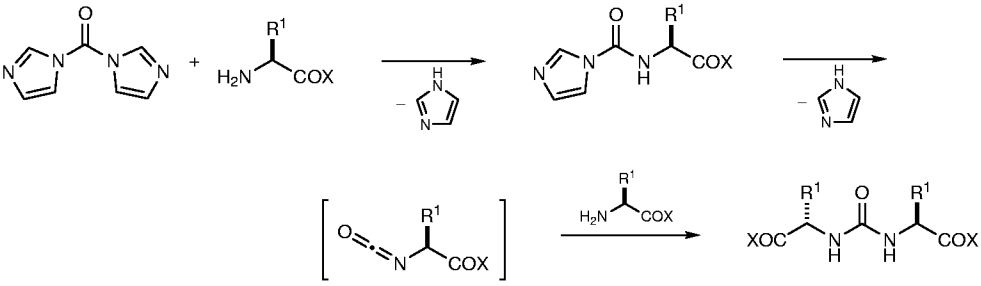
Scheme 1 Peptide Coupling Mediated by 1,1′-Carbonyldiimidazole



COX = ester, amide, peptide

Since CDI is very hygroscopic, it has to be stored and used with careful exclusion of moisture. Usually it is added in equimolar amounts to the carboxy component. The evolution of carbon dioxide can be monitored and the effervescence subsides within about 15 minutes indicating completion of the activation. After addition of the amine coupling is usually completed within 1–2 hours. An excess of CDI has to be avoided, since it reacts with the amine to form an isocyanate which is trapped by a further equivalent of amine to yield urea derivatives (Scheme 2).[18]

Scheme 2 Formation of Ureas by Nucleophilic Attack of Amines on 1,1′-Carbonyldiimidazole



COX = ester, amide, peptide

If water cannot be excluded, for example, due to the presence of water of crystallization in the carboxylic acid, or when the purity of the CDI is not precisely known, a small excess of CDI can be used. Nevertheless, unreacted CDI should be removed by a short washing of the mixture with water at −5 °C prior to the addition of the amino component. CDI is immediately hydrolyzed by water while the activated carboxylic acid is considerably more stable and survives this washing routine. Traces of water remaining in the solution usually do not reduce yields in the coupling step.[19]

Couplings with CDI can be carried out in THF, CH_2Cl_2, or $CHCl_3$, although best results are achieved with DMF as the solvent. The reaction can also be run in water or water-containing solvents, but poor yields have been reported in these cases.

The amount of racemization during activation is generally low, even when no racemization-suppressing agents are added. Anderson and Paul found in the sensitive coupling of

Z-Gly-Phe-OH with H-Gly-OEt 5% racemization using THF as solvent at −10 °C, but only <0.5% in DMF.[5] The rates of racemization increased at higher temperature (rt).[11]

Occasionally, O-acylation of hydroxy groups present in the carboxylic acid such as in the side chains of unprotected threonine, serine, or tyrosine residues has been observed.[20]

Several variations of CDI have been synthesized and tested for their applicability in the activation of amino acids and peptides (Table 1). 1-(1*H*-1,2,4-Triazol-1-ylcarbonyl)-1*H*-1,2,4-triazole (**2**) was reported to be even more reactive than CDI and thus, the shorter reaction times were expected to reduce racemization in peptide condensations with this reagent. Even when the activation was carried out at room temperature in DMF no racemization was detected.[11] 3-Methyl-1-[(3-methyl-1*H*-imidazol-3-ium-1-yl)carbonyl]-1*H*-imidazol-3-ium (**3**) was developed as a more reactive alternative to CDI, which does not lead to a basic by-product like imidazole.[12] Consequently, no racemization was observed with this reagent when carbamate-protected amino acids were activated. The considerable amount of epimerization during activation of peptide fragments could be effectively suppressed when one equivalent of $CuCl_2$ or $Cu(OTf)_2$ was added.[13] The higher reactivity of the activated species allowed even for acylation of sterically hindered amines. Problems might arise when acid-sensitive protecting groups are used in the activated fragment, since especially in older samples of 3-methyl-1-[(3-methyl-1*H*-imidazol-3-ium-1-yl)carbonyl]-1*H*-imidazol-3-ium traces of TfOH might be present, which can lead to a partial cleavage of such protecting groups.[12]

The 1,2,4-oxadiazolidine-3,5-dione derivative **4** is a nonhygroscopic compound and the resulting byproducts are readily removed by simple aqueous washings. Almost no racemization has been detected using this reagent in peptide condensations.[14] Conversely, the oxadiazolone derivative **5** which is also more reactive than CDI, leads to quantitative racemization when peptide fragments are activated. However, no racemization was observed in the acylation of amino components with urethane-protected amino acid derivatives.[15] The 1,3-benzoxazole-2(3*H*)-thione derivative **6** was shown to be nonhygroscopic, but detailed investigations with this reagent have not been reported.[16]

Boc-Pro-Phe-OMe:[19]
To a soln of Boc-Pro-OH (10.8 g, 50.0 mmol) in THF (50 mL) was added CDI (9.78 g, 50.0 mmol). After the effervescence subsided, the soln was stirred for 1 h and H-Phe-OMe (8.95 g, 50.0 mmol) was added. After stirring at rt overnight, the solvent was removed and the residue was taken up in Et_2O (300 mL) and washed with $NaHCO_3$, 1 M HCl, and H_2O. The organic layer was dried (Na_2SO_4) and the solvent removed under reduced pressure to give the product; yield: 18.3 g (98%); mp 72–75 °C. Recrystallization (iPr_2O) afforded analytically pure material; yield: 13.5 g (72%); mp 74–76 °C; $[\alpha]_D^{25}$ −53 (*c* 1, $CHCl_3$).

Z-Asp-Arg(HNO_3)-Val-Tyr-Ile-His-Pro-Phe-OMe·2H_2O:[19]
To Z-Asp-Arg(HNO_3)-Val-Tyr-OH·½H_2O (2.22 g, 3.00 mmol) in DMF (15 mL) was added CDI (818 mg, 4.50 mmol) at 10 °C and the mixture was stirred for 2.5 h at this temperature. Meanwhile, H-Ile-His-Pro-Phe-OMe·2HBr (3.58 g, 5 mmol) was dissolved in H_2O (20 mL) and washed with EtOAc (20 mL). The organic layer was discarded and the aqueous layer was made basic with sat. Na_2CO_3 (20 mL) and extracted with $CHCl_3$ (4 × 30 mL). The organic layers were extracted with sat. Na_2CO_3, sat. Na_2SO_4 (20 mL each), dried (Na_2SO_4), concentrated, dissolved in DMF (10 mL), and added to the reaction soln. The mixture was allowed to warm to rt and was stirred overnight. Removal of the solvent and trituration of the residue with H_2O gave on drying 4.68 g (theoretical yield 3.71 g) of material mp 160–172 °C. This product was taken up in hot MeOH and cooled to give a gel. This process was repeated. The product was boiled with $CHCl_3$, the insoluble part collected from the hot soln, washed with Et_2O, and dried to give the product; yield: 1.67 g (45%); mp 221–223.5 °C; $[\alpha]_D^{25}$ −37 (*c* 0.67, DMF).

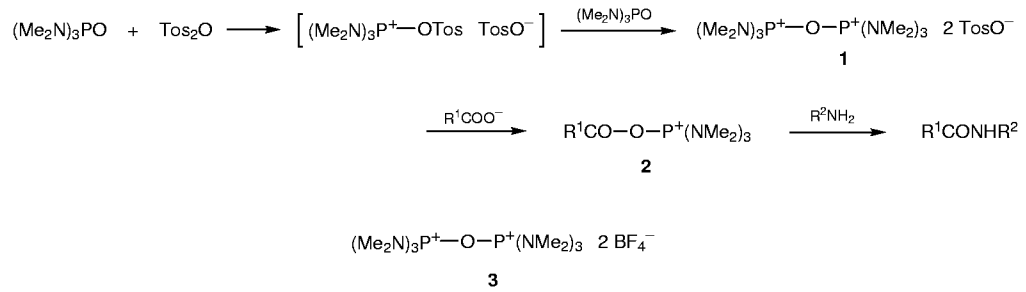
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3.7 Phosphonium Salts

J. COSTE

In 1969, Gawne et al.[1] considered that the acyloxyphosphonium salt **2**, derived from HMPA, would be an effective acylating agent and therefore proposed the use of the bisphosphonium salt **1** as a coupling reagent (Scheme 1). This reagent can be either formed in situ[1] or isolated as the tetrafluoroborate salt **3** (Bates' reagent).[2]

Scheme 1 The Formation of the Bisphosphonium Salt **1** and Its Use as a Coupling Reagent[1,2]

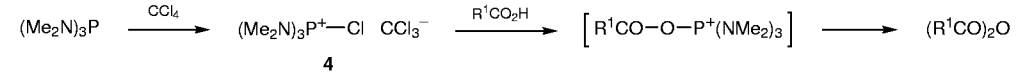


Subsequently, it was proposed by several authors[3–5] that peptide bonds could be formed by the activation of an N-protected amino acid using the phosphine/CX_4 system (Scheme 2). However, Castro et al.[6] proposed that in such a reaction the acyloxyphosphonium ion is not stable, but reacts with a carboxylate ion to give an anhydride (Scheme 3) which is subsequently aminolyzed. They showed that the chlorotris(dimethylamino)phosphonium **4** plays a role in the activation of carboxylic acids by the tris(dimethylamino)phosphine/CCl_4 system (Scheme 3) and also proposed the use of the corresponding perchlorate as a coupling reagent.[7]

Scheme 2 Peptide Bond Formation by Activation of the N-Protected Amino Acid Using Phosphine/CX_4[3–5]

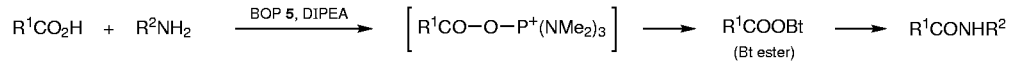


Scheme 3 Formation of Anhydrides from Acyloxyphosphonium Ions[6]



Finally, on the basis of the work of König and Geiger[8] concerning the importance of HOBt as an additive in peptide coupling, Castro et al. both suggested benzotriazol-1-yloxy-tris(dimethylamino)phosphonium hexafluorophosphate (BOP, **5**) as a suitable coupling reagent[9] and subsequently used it.[10–16] The coupling reagent BOP (**5**) contains both the phosphonium activation residue and the OBt residue, the phosphonium salt activates the carboxylic acid group by forming a benzotriazolyl active ester (Bt ester) (Scheme 4) which is the acylating species.[17] Appel et al.[18,19] have used chlorophosphonium salts, both isolated or prepared in situ by R_3P/C_2Cl_6, with or without additives such as HOBt, and Ramage and coworkers obtained the tetrafluoroborate homologue of BOP from HOBt and Bates' reagent **3**.[20]

Scheme 4 The Use of BOP as a Coupling Reagent[9–16]



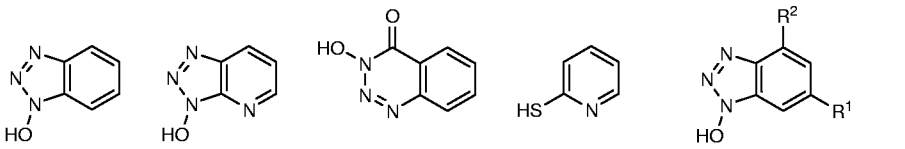
However, it is only since the efficiency of BOP as a coupling reagent was proven by several laboratories,[21–23] that BOP (**5**) has become widely used. Numerous variations of this phosphonium salt have been reported (Scheme 5), among which are the very efficient AOP (**8**) and PyAOP (**20**) reagents.[24] Phosphonium salts are powerful peptide coupling reagents which are easy to use and are now widely used, together with the uronium analogues, in all fields of peptide chemistry. Several of these phosphonium salts are commercially available.

Scheme 5 Phosphonium Salt Coupling Reagents



5 (BOP): W = OBt; Y = PF_6
6 (NOP): W = NO_2—OBt; Y = PF_6
7 (CF_3—BOP): W = CF_3—OBt; Y = PF_6
8 (AOP): W = OAt; Y = PF_6
9 (BroP): W = Br; Y = PF_6
10 W = Cl; Y = PF_6
11 W = Cl; Y = ClO_4
12 W = Y = Cl
13 W = N_3; Y = PF_6
14 W = ODhbt; Y = PF_6
15 W = SPy; Y = PF_6

16 (PyBOP): W = OBt; Y = PF_6
17 (PyNOP): W = NO_2—OBt; Y = PF_6
18 (PyFOP); (CF_3—PyBOP); W = CF_3—OBt; Y = PF_6
19 (CF_3—NO_2—PyBOP): W = CF_3—NO_2—OBt; Y = PF_6
20 (PyAOP): W = OAt; Y = PF_6
21 (PyBroP): W = Br; Y = PF_6
22 (PyCloP): W = Cl; Y = PF_6
23 W = N_3; Y = PF_6
24 W = ODhbt; Y = PF_6
25 W = SPy; Y = PF_6
26 W = OPfp; Y = PF_6



HOBt HOAt HODhbt HSPy

NO_2-HOBt: $R^1 = NO_2$; R^2 = H
CF_3-HOBt: $R^1 = CF_3$; R^2 = H
CF_3-NO_2-HOBt: $R^1 = CF_3$; $R^2 = NO_2$

3.7.1 Phosphonium Salts Used in Peptide Synthesis

Castro's reagent (BOP, **5**)[9] is a very effective coupling reagent in all fields of peptide synthesis. However, the synthesis of BOP (**5**) and its utilization both involve the handling of HMPA, the toxicity of which is well-known.[25] To avoid this problem, the use of PyBOP (**16**) (benzotriazol-1-yloxytripyrrolidinophosphonium hexafluorophosphate) has been proposed,[26,27] and it has been found to give comparable results to BOP and occasionally to give better results. In contrast, the tris(diethylamino), tripiperidino, or trimorpholino homologues[26] show poorer coupling properties.[27] Several homologs of BOP or PyBOP that contain substituted benzotriazolyl groups have also been proposed as suitable coupling reagents, these include NOP (**6**),[28] PyNOP (**17**),[28] CF_3-BOP (**7**),[29] PyFOP (CF_3-PyBOP) (**18**),[28,29] and CF_3-NO_2-PyBOP (**19**).[30] These reagents have been used for the synthesis of peptide thioamides (**6**, **17**, **18**)[28,31] and for Aib (**7**, **18**)[29] and *N*-methylamino acid coupling

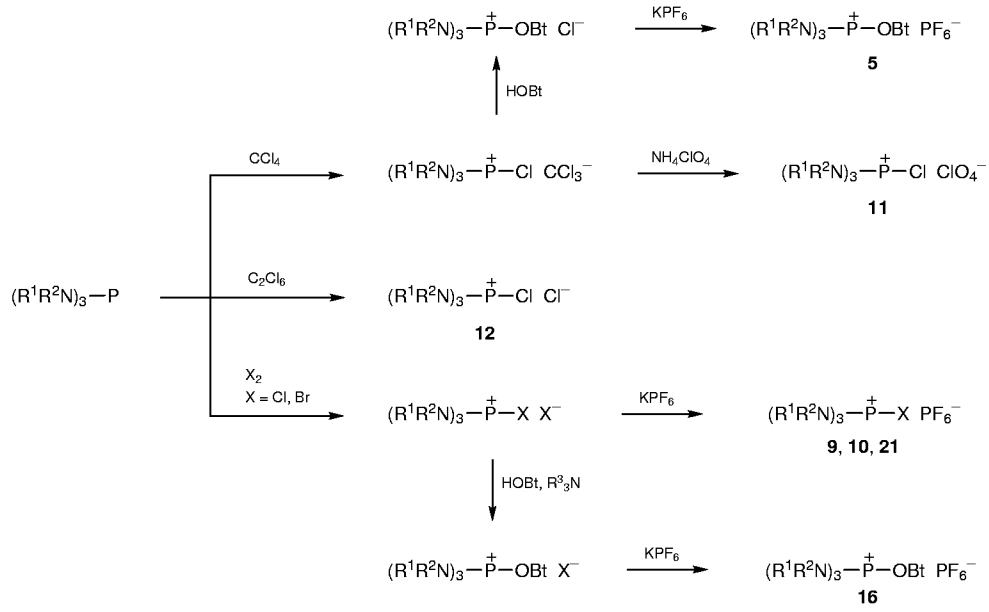
(**7**, **19**).[29,30] Carpino has shown that HOAt is a better additive than HOBt for peptide coupling[32] and as a consequence introduced[24] the related reagents AOP (**8**) and PyAOP (**20**) which are more effective than BOP or PyBOP,[24,33] notably in the case of hindered amino acids.[33]

The halophosphonium compounds BroP (**9**), PyBroP (**21**), or PyCloP (**22**) are efficient reagents when coupling *N*-methylamino acids[34–36] or α,α-disubstituted amino acids.[37] Conversely, the fluorophosphonium PyFloP (fluorotripyrrolidinophosphonium hexafluorophosphate), (Scheme 5, general formula **B**, W = F, Y = PF_6) lacks coupling properties, while its uronium analogue TFFH is a good coupling reagent.[38] Other phosphonium salts corresponding to the general formula **A** (Scheme 5) with W = CN, NCS, N_3, CCl_3, OPh, 4-$O_2NC_6H_4$ are more restricted in their use or are not effective coupling reagents.[39] Of these compounds, azidotris(dimethylamino)phosphonium hexafluorophosphate (**13**)[40] has potentially the same application as DPPA.[41] Azidophosphonium hexafluorophosphate **13** is a stable solid which is easily synthesized, but, nevertheless, it has not been used to a great extent. The pyrrolidino homologue of **13**, **23** has also been obtained.[42] Reagents **24**, **25**, and **26** have been tested for the activation of monothiocarboxylic acids in order to obtain a thioamide linkage containing pseudopeptides.[28] Reagents **14** and **24** have been tested for segment coupling,[42] they give optically purer peptides than BOP or PyBOP, however, a side reaction occurs, namely the formation of an azido compound, the formation of which has also been observed when using DCC/HODhbt.[43] This reaction is probably the main reaction when activating monothiocarboxylic acids with **24**.[28] The phosphonium salt **15** is a very poor peptide coupling reagent.[44]

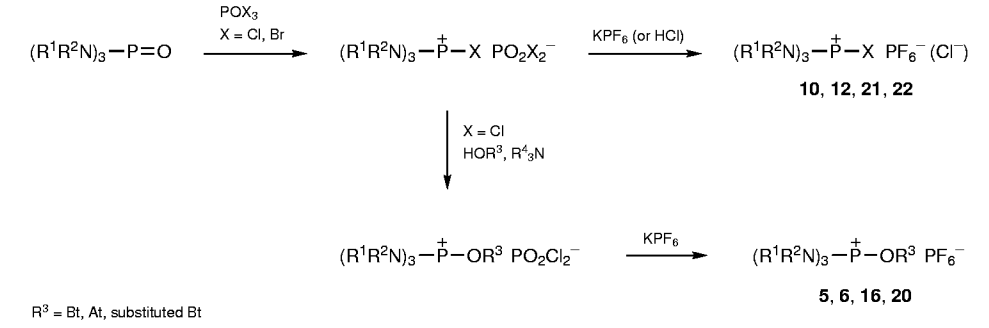
3.7.1.1 Synthesis of Phosphonium Salts

The most frequently used phosphonium salts **5**, **9**, **16**, and **20–22** are stable solids, which are easy to use and all are commercially available. These compounds and the others listed in Scheme 5 are readily prepared following Schemes 6–9, from aminophosphines (Scheme 6) or aminophosphine oxides (Schemes 7, 8). Only compounds **15**[44] and **25**[28] are obtained by a different pathway. Halophosphonium salts, which are intermediate compounds in these syntheses, are also coupling reagents and are generally isolated as the hexafluorophosphates (Schemes 6 and 7) as the chlorides are hygroscopic.[7] The halophosphonium hexafluorophosphates are also used as starting materials (Scheme 9) in the synthesis of other reagents. Examples of the most frequently used synthetic methods are described (*vide infra*). It is worth noting that the toxic[25] HMPA is used in some syntheses (Schemes 7 and 8).

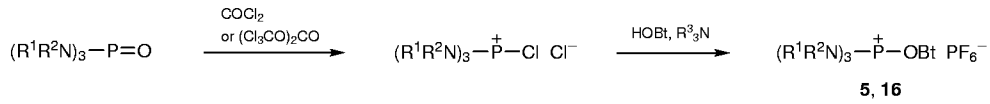
Scheme 6 The Synthesis of **5**,[9] **9**,[40,45] **10**,[42] **11**,[7] **12**,[19] **16**,[26] or **21**[45]



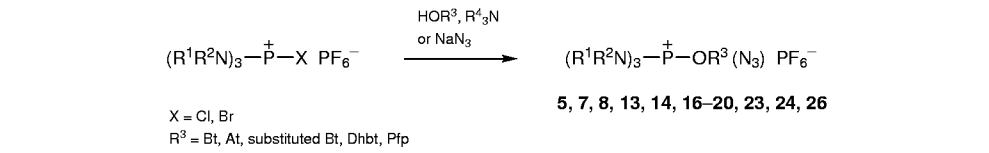
Scheme 7 The Synthesis of **5**,[46] **6**,[28] **10**,[42] **12**,[46] **16**,[26] **20**,[47] **21**,[36] or **22**[28,36,45]



Scheme 8 The Synthesis of **5**,[48,49] or **16**,[26,49] Using $COCl_2$[26,48] or $(Cl_3CO)_2CO$[49]



Scheme 9 The Synthesis of **5**,[42] **7**,[29] **8**,[47] **13**,[40] **14**,[42] **16**,[42] **17**,[28] **18**,[28,29] **19**,[30] **20**,[31,47] **23**,[42] **24**,[28,42] or **26**[28]



BroP 9; Halophosphonium Salts from Aminophosphine:[40,45]
To vigorously stirred $(Me_2N)_3P$ (36.6 mL, 0.2 mol) in Et_2O (600 mL), was added over a 30-min period at 0–10 °C, Br_2 (10.2 mL, 0.2 mol). The mixture was then stirred for 20 min and KPF_6 (36.8 g, 0.2 mol) in H_2O (800 mL) was added at 15 °C. After filtration, the solid was dissolved in CH_2Cl_2, dried (Na_2SO_4), and the solvent removed under reduced pressure. A soln of the crude product in acetone (500 mL) was

for references see p 551

added dropwise to stirred Et_2O (3 L). After filtration and removal of the solvent under reduced pressure, a white solid was obtained; yield: 53 g (68%); mp >250 °C; ^{31}P NMR (81 MHz, CD_2Cl_2, δ): 49.7 (s, 1P, P^+), −142.6 (hept, 1P, $J_{P,F}$ = 713 Hz, PF_6^-).

PyCloP 22; Halophosphonium Salts from Aminophosphine Oxide:[36]
To stirred $POCl_3$ (freshly distilled under N_2) (9.3 mL, 0.1 mol) in dry, EtOH-free CH_2Cl_2 (15 mL), was added dropwise at rt over a 20-min period tripyrrolidinophosphine oxide (25.7 g, 0.1 mol) in dry, EtOH-free CH_2Cl_2 (15 mL); the temperature rose to 30–35 °C. The mixture was then stirred for 30 min at rt. KPF_6 (18.5 g, 0.1 mol) in H_2O (150 mL) was rapidly added under stirring. The mixture was decanted and extracted with CH_2Cl_2. The organic phase was washed with H_2O, dried (Na_2SO_4); and the solvent removed under reduced pressure. After recrystallization (CH_2Cl_2/Et_2O), a white solid was obtained; yield: 34.4 g (82%); mp 150–151 °C; ^{31}P NMR (81 MHz, $CDCl_3$, δ): 36.5 (s, 1P, P^+), −143.7 (hept, 1P, $J_{P,F}$ = 713 Hz, PF_6^-).

PyBOP 16; Reagents from Aminophosphine Oxide:[26]
To stirred $POCl_3$ (freshly distilled under N_2) (2.3 mL, 0.025 mol) in dry, EtOH-free CH_2Cl_2 (3.5 mL), was added dropwise, at a temperature below 30 °C over a 20-min period, tripyrrolidinophosphine oxide (6.9 g, 0.027 mol) in EtOH-free CH_2Cl_2 (3.5 mL). The mixture was then stirred for 30 min at rt. Anhyd HOBt (3.38 g, 0.025 mol) and TEA (3.5 mL, 0.025 mol) in CH_2Cl_2 (15 mL) were added dropwise at a temperature below 20 °C over a 20-min period. The mixture was then stirred for 15 min at rt and the solvent removed under reduced pressure. The crude solid was dissolved in cold H_2O (30 mL) and acetone (20 mL) and the soln added dropwise under stirring to KPF_6 (4.6 g, 0.025 mol) in H_2O (80 mL). After filtration, the white solid was washed with H_2O until pH 7 and then dissolved in CH_2Cl_2. The soln was decanted, dried (Na_2SO_4), and the solvent removed under reduced pressure. After recrystallization (CH_2Cl_2/Et_2O), a white solid was obtained; yield: 10 g (77%); mp 156–157 °C; ^{31}P NMR (81 MHz, $CDCl_3$, δ): 31.8 (s, 1P, P^+), −143.7 (hept, 1P, $J_{P,F}$ = 713 Hz, PF_6^-).

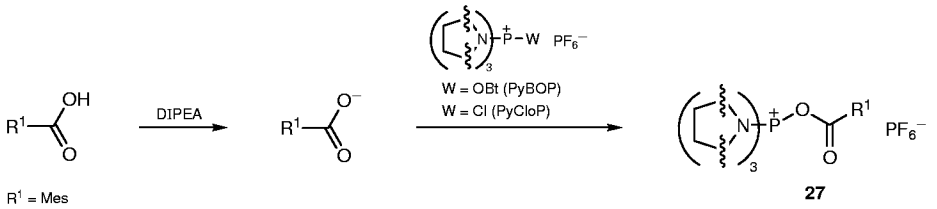
PyAOP 20; Reagents from Isolated Halogenophosphonium Hexafluorophosphates:[31]
To a stirred soln of PyBroP **21** (2.19 g, 4.70 mmol) and HOAt (0.639 g, 4.70 mmol) in anhyd CH_2Cl_2 (8 mL), was added TEA (655 µL, 4.70 mmol). The soln was stirred for 2 h and diluted with Et_2O (75 mL). The precipitate was collected on a filter, washed with H_2O (3 × 25 mL) and Et_2O (2 × 25 mL), and recrystallized (acetone/Et_2O); yield: 1.86 g (76%); mp 172–174 °C (dec); ^{31}P NMR (101 MHz, $CDCl_3$, δ): 32.5 (s, 1P, P^+), −143.2 (hept, 1P, PF_6^-).

3.7.1.2 Phosphonium Salt Mediated Peptide Bond Formation

Reaction Mechanism

Formation of a peptide bond involves two steps: (1) activation of the carboxylic acid, and (2) aminolysis of the activated species. The first step starts with the reaction of the carboxylate ion with the phosphorus atom of the coupling reagent. The resulting acyloxyphosphonium ion **27** (Scheme 10) has only been observed for PyBOP- or PyCloP-mediated activation by using low temperature ^{31}P NMR experiments and the sterically hindered 2,4,6-trimethylbenzoic acid.[36] With N-protected amino acids, the acyloxyphosphonium intermediate reacts to give more stable species, the structure of which depends on the type of the reagent involved (Scheme 11).[36]

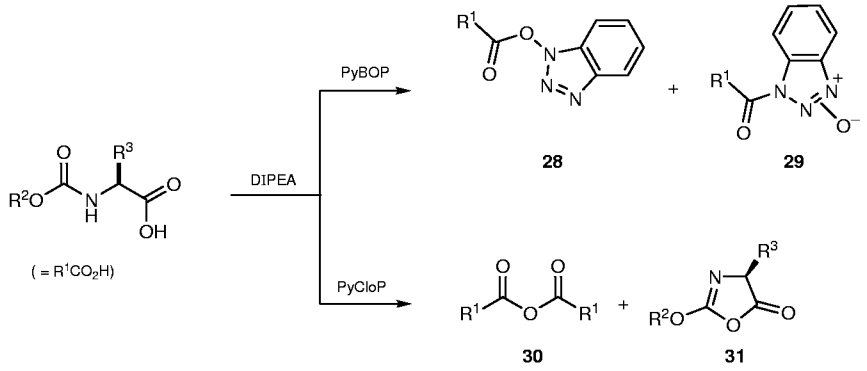
Scheme 10 Reaction of the Carboxylate Ion with the Phosphorus of the Coupling Reagent[36]



R^1 = Mes

27

Scheme 11 Structure of the Species Produced from the Reaction of Acyloxyphosphonium Intermediates[36]

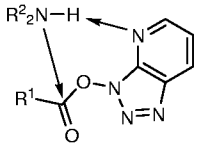


When N-protected amino acids are activated by BOP in the presence of a base, the acyloxyphosphonium ion is not observed.[36,50] A benzotriazolyl ester (Bt ester) is immediately formed,[50] which is,[51] as also demonstrated for PyBOP-mediated activation,[36] a mixture of O-acylated **28** and N-acylated **29** forms (Scheme 11). In the case of OAt-based reagents, the formation of At esters is also assumed to occur.[52–54]

Two cases must be considered for the peptide bond formation step. Phosphonium coupling reagents are often used in a true one-pot reaction (concomitant activation and coupling, Scheme 13, eq 2), however, some authors preactivate the carboxylic acid with the phosphonium coupling reagents and then perform the coupling reaction itself a few minutes later (Scheme 13, eq 3). When using the preactivation mode, the benzotriazolyl, substituted benzotriazolyl, or azabenzotriazolyl esters (Xt esters) are formed during the preactivation time, so that these esters are the aminolyzed species. Anhydride[23] or oxazolone[55] have been postulated as intermediates, but they have not been observed.[50] Under one-pot conditions, the aminolyzed species are most probably also the Xt esters;[36] it was claimed that acyloxyphosphonium was the alcoholized intermediate in BOP-mediated ester synthesis,[56] but this hypothesis seems unfounded.[51]

Reagents containing an OBt residue bearing an electron-withdrawing group (**6**, **7**, **17–19**) or an OAt residue (**8**, **20**) are more effective coupling reagents than those containing the OBt residue (**5**, **16**). This improved efficiency is thought to correspond to the leaving group properties of the OXt oxyanion (Xt = Bt, substituted Bt, At) and to the pK_a values of the corresponding HOXt compounds:[28,52] the pK_a values in water for HOBt, CF_3-HOBt and 7-HOAt are 4.60, 3.80, and 3.47, respectively.[57] However, in the case of the 7-At esters, Carpino et al.[32,52] suggested that the neighboring group effect depicted in Scheme 12 is an important factor.

Scheme 12 Neighboring Group Effects in 7-At Esters[32,52]



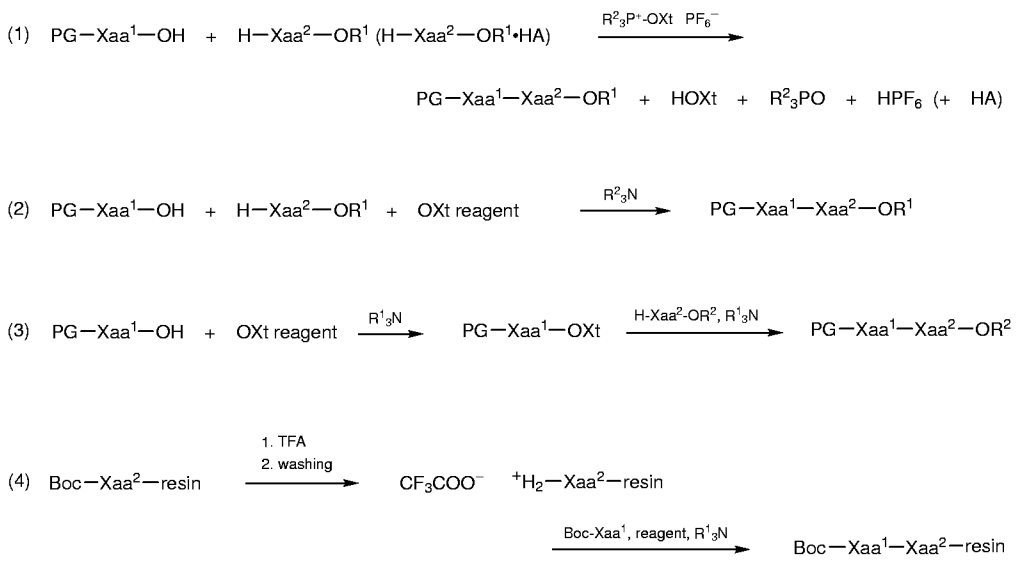
In the case of halophosphonium reagents, as demonstrated for PyCloP, the acyloxyphosphonium intermediate (Scheme 10) reacts to give a mixture of anhydride **30** and oxazolone **31** (Scheme 11) which are aminolyzed. In the case of the azido reagent **13**, an acyl azide is the aminolyzed compound.[40]

for references see p 551

Practical Consequences

The coupling reaction using OXt-based reagents is depicted in Scheme 13, eq 1. In the case of BOP, two (or three) equivalents of base must be used.[15] The first equivalent of base is used to generate the carboxylate of the N-protected amino acid and allow reaction with the reagent, the second equivalent of base neutralizes HOXt, the release of which results in a decrease in pH (the optimal pH value is between 8 and 9), and the third equivalent of base is used to regenerate the amine group when an ammonium salt of a C-protected amino acid is involved. To summarize, the HPF_6, HOXt, and, if it is formed, HA formed in the reaction presented in Scheme 13, eq 1 must be neutralized. The influence of the amount of base on the rate and the completeness of the coupling reaction has been studied: when coupling Fmoc-Ala-OH (2 equiv) and H-Phe-NH_2 (1 equiv) in DMF with BOP (2 equiv), without NMM the yield was 48%, using 1 equivalent of NMM it reached 82%, with 2, 4, or 8 equivalents the yield was >99% and the reaction proceeded more rapidly (>99% yield after reaction for 5, 2, and 1 min, respectively).[58] In these experiments, despite the use of only two equivalents of base, that is, one equivalent per equivalent of BOP, the reaction goes to completion. This result seems to contradict Castro's proposal[15] that 4 equivalents of base should be used; however, the situation is further complicated by the use of a twofold excess of Fmoc-Ala-OH and BOP over H-Phe-NH_2. Studies on the influence of the amount of base when using stoichiometric amounts of reactants and reagent have not yet been performed. In practice, authors use various amounts of base, both in solution and in solid-phase synthesis. In general, 1 to 3 equivalents per equivalent of BOP are used when coupling C-protected amino acids and 2 to 3 equivalents are used when using the salts of C-protected amino acids. When reactants and reagent are in stoichiometric amounts the use of two equivalents of base can be recommended for coupling C-protected amino acids and the use of three equivalents of base in the coupling of salts of C-protected amino acids. Finally, the pH of the reaction mixture can be monitored (moist pH paper) and readjusted to the optimal value of 8–9.[15] It should be noted that even though an excess of base accelerates the reaction, it can produce epimerization, at least in fragment couplings.

Scheme 13 Synthetic Protocols



HA = acid (A = CO_2CF_3, Cl, Br, OTos); Xt = Bt, substituted Bt, At

It was found that addition of HOBt improves the efficiency of the BOP-mediated coupling.[23] Therefore, HOBt is widely used as an additive (one equiv per equiv of reagent) in the case of BOP or PyBOP, as well as HOAt, when AOP or PyAOP are used. The cause of this effect is unknown, but it is worth noting that the addition of HOBt should result in a decrease in pH and thus in a decrease of the reaction rate. The usefulness of HOBt or HOAt as additives was recently questioned (see Table 2, entry 1 vs 2 and 4 vs 5).[24,52] It is, however, established that these additives generally reduce epimerization in fragment coupling steps (*vide infra* and Table 1).

Since BOP,[16] PyCloP,[36] PyBOP,[53] PyAOP,[53] and probably all other phosphonium reagents do not react with amines, the reaction is preferably performed as a true one-pot reaction by mixing all the reactants and reagents with the base added as the last component (Scheme 13, eq 2). Because in situ activation is often used to qualify experiments using the preactivation mode, the expression one-pot reaction is preferred to avoid errors. Preactivation (Scheme 13, eq 3), that is, the preliminary formation of the activated species, is also often used but the basis for its use is unfounded. Moreover, epimerization can arise during the preactivation period either when coupling fragments[17,59] or when using urethane-protected amino acids;[50,60] in these cases addition of HOBt is beneficial.[55] Whereas for solution[14] or manual solid-phase synthesis[15] the one-pot method is easy to perform, in some automatic synthesizers the preactivation mode is often dictated by the instrument, however, in other instruments the one-pot reaction has been introduced[61–63] or the program modified.[64]

In contrast to carbodiimides, the BOP family of reagents does not activate TFA or its anion. This permits the coupling of C-protected amino acid trifluoroacetates without the formation of trifluoroacetamides both in solution and in solid-phase syntheses. This is particularly useful in Boc-SPPS methodology where the in situ neutralization protocol was proposed by Le Nguyen et al.[14,15] and then adopted by numerous authors when using phosphonium[27,61,65–69] or uronium[70] reagents. This method, summarized in Scheme 13, eq 4, saves both time and solvent, avoids extensive formation of piperazine-2,5-dione (DKP) when coupling the third amino acid,[14,15,65,66] and probably increases (as demonstrated in the case of HBTU[70]) the efficiency of chain assembly especially for difficult sequences. After the Boc group has been removed and the peptidyl-resin salt washed to eliminate excess TFA, instead of neutralizing the trifluoroacetate salt in a separate step, the next Boc-protected amino acid and the coupling reagent are added with sufficient excess of base to neutralize the peptidyl-resin salt. In some automatic synthesizers, in situ neutralization is allowed.[61,62,70]

Almost all of these conclusions are applicable in the case of halophosphonium reagents, including the one-pot reaction, in situ neutralization and the influence of the amount of base on the reaction yield.[36] However, a case of BroP-mediated TFA activation has been reported (see footnote 65 in ref[36]).

Experimental Conditions

The reaction is generally conducted at room temperature and in the presence of a base. When coupling segments[59] or coupling *N*-methylamino acids with halophosphonium salts,[36] ice-bath cooling can be applied at the onset of the reaction to limit the degree of epimerization. The concentration of the reactants and reagents is preferably as high as possible, this is feasible due to the high degree of solubility of the phosphonium reagents. The solvents CH_2Cl_2 and DMF are most commonly used; other solvents such as THF, MeCN, and, in particular, NMP and DMSO are also suitable. In the case of fragment coupling, using a polar solvent, such as DMF, increases epimerization compared to using CH_2Cl_2[59,60] (see also Table 1). With halophosphonium reagents, CH_2Cl_2 is generally used; NMP and DMF are also

for references see p 551

effective, however, a reaction of DMF with PyBroP has been described.[71] With PyBroP, the use of DMSO is occasionally superior to CH_2Cl_2.[72]

Numerous bases can be used, particularly DIPEA or NMM. In the case of fragment coupling, other tertiary amines, such as collidine, are recommended to decrease the degree of epimerization.[54,60] Addition of DMAP improves the halophosphonium-mediated activation of α,α-dialkylamino acids.[37]

Coupling reactions are generally carried out by using 10% (solution synthesis) or two- to threefold (solid-phase synthesis) molar excesses of the N-protected amino acid and of the reagent which can be increased for difficult coupling reactions. In the case of BOP or PyBOP, the coupling of secondary amines is not easy and as a consequence the DCHA salts of N-protected amino acids, for example Boc-His(Boc)-OH, are reacted without separate regeneration of the carboxylic acid.[14,15,27] Either the free base or the salts (halide, trifluoroacetate, tosylate) of C-protected amino acids or peptide fragments can be used. Hydrates of amino acids [Boc-Leu-OH·H_2O; Boc-Ile-OH·0.5 H_2O] can be used without decreasing the reaction yield with BOP,[14,15] PyBOP,[27] or PyAOP.[33]

The reagent BOP and the more recent reagents PyBOP, AOP, and PyAOP are widely used in solution[9,12–14,27,37] and in solid-phase synthesis with manual,[10,15,21–23,27,67,69,73] automated,[33,55,61–64,74] tea bag,[68] or pin[63,75] methods. One-pot or preactivation conditions as well as in situ or separate neutralization procedures are used (*vide supra*). The preparation of stock solutions for automated synthesis is possible due to the stability of the reagents BOP, PyBOP,[33,76] AOP, or PyAOP.[33,53] However, when the coupling reagents are used in open vessels, some decomposition of the reagents occurs and this must be taken in account.[53] The halophosphonium salts are not widely used in SPPS. A preliminary study on the use of PyCloP gave disappointing results for the synthesis of ACP(65–74).[36,45] Nevertheless, some solid-phase syntheses of peptides containing *N*-alkyl- or α,α-dialkylamino acid residues have been achieved with the PyBroP reagent (*vide infra*).

The general procedure reported here for the synthesis in solution can be used for all the phosphonium coupling reagents except the azido reagent **13** for which a temperature below 0 °C, preactivation conditions (30–45 min), and a long coupling time (12–16 h) are required.[40] When using OXt-based reagents the released HOXt is eliminated during the workup. With reagents containing the tris(dimethylamino) residue, HMPA is produced; it is eliminated during the reaction workup so that the crude peptides are generally practically pure and usable without any purification. However, when the reaction is carried out in CH_2Cl_2, the elimination of HMPA is only possible if the solvent is evacuated before the workup. **CAUTION:** HMPA[25] is toxic and adequate precautions must be taken to avoid exposure to this compound. When using tripyrrolidino-containing reagents, tripyrrolidinophosphine oxide is produced which is still present in the crude product and can be removed by silica gel column chromatography. However, despite the presence of tripyrrolidinophosphine oxide in the crude product, continuous solution syntheses are possible.[77]

Phosphonium Salts in Synthesis in Solution; General Procedure:[37]

To a soln of the C-protected amino acid or peptide fragment (1 equiv), N-protected amino acid (1.1 equiv), and coupling reagent (1.1 equiv) in the solvent (CH_2Cl_2, DMF, etc.; preferably 1 mL·mmol^{-1}), was added a base (DIPEA, NMM, etc.; 2 equiv; 3 equiv when using the salts of C-protected amino acids) at rt under stirring. During the reaction, the pH of the mixture was monitored (moist pH paper) and readjusted to 8–9. After completion of the reaction (TLC monitoring), EtOAc was added to the mixture and the soln successively washed with 5% $KHSO_4$, 5% $NaHCO_3$, and brine, dried (Na_2SO_4), and concentrated. For usual couplings, the reaction time is <0.5 h; average yields of purified products: 90–95%.

Phosphonium Salts in Solid-Phase Synthesis with Boc Chemistry; General Procedure:[15]

In the following an example is given for manual synthesis using the in situ neutralization protocol. Starting from Boc-Xaa-Merrifield resin, the each coupling cycle consists of (1) CH_2Cl_2 (2 × 0.5 min); (2) TFA/CH_2Cl_2 (1:1) + 5% EDT (1 × 1 min + 1 × 30 min); (3) 3% EDT/iPrOH (0.5 min); (4) CH_2Cl_2

(2 × 0.5 min); (5) Boc amino acid (2 equiv) + BOP **5** (2 equiv) + CH_2Cl_2 + DIPEA (6 equiv), 10 min, pH checked (monitored with moist pH paper) and readjusted to pH 8–9 with DIPEA if required, coupling efficiency judged by the Kaiser test; (6) MeOH (2 × 0.5 min); (7) CH_2Cl_2 (2 × 0.5 min).

Phosphonium Salts in Solid-Phase Synthesis with Fmoc Chemistry; General Procedure:[78]
Using Dpa/MBHA resin, one cycle of the synthesis consists of the following operations: (1) CH_2Cl_2 (2 × 1 min); (2) DMF (2 × 1 min); (3) 20% piperidine in DMF (1 × 3 min + 1 × 15 min); (4) DMF (2 × 1 min); (5) CH_2Cl_2 (2 × 1 min); (6) MeOH (2 × 1 min); (7) CH_2Cl_2 (2 × 1 min); (8) DMF (2 × 1 min); (9) Fmoc amino acid (2 equiv) in DMF + BOP **5** (2 equiv) + DIPEA (4 equiv), 60 min, coupling efficiency judged by Kaiser test; (10) DMF (2 × 1 min); (11) CH_2Cl_2 (2 × 1 min).

3.7.1.3 Use of Phosphonium Salts as Coupling Reagents

The phosphonium salts AOP and PyAOP are, together with their uronium analogues, among the best coupling reagents, PyAOP being slightly better than AOP.[53] They are definitely superior to BOP or PyBOP, not only in the case of proteinogenic[24,33] but also of hindered amino acids.[33,53] In the latter case, PyBroP/HOAt is also very efficient (*vide infra*). The reagents BOP and PyBOP are of general use, but are less efficient for acylating *N*-methyl or α,α-dialkylamino acids whereas BroP, PyBroP, and PyCloP, which are suited for these latter couplings, are not generally used for proteinogenic amino acids.[36] Reagents **7** and **18** are promising in the case of Aib coupling[29] and **19** in that of *N*-methylamino acid condensation.[30] Reagents **17** and **18** are the best among the phosphonium reagents for the construction of the thioamide bond (CSNH) in pseudopeptides.[28,31] However, the general usefulness of reagents **7**, **17–19** has not been yet been determined.

No particular side reaction is encountered when using phosphonium reagents. In addition to the possible reaction of PyBroP with DMF[71] and the possible activation of TFA with BroP,[36] only the side reactions generally encountered in peptide synthesis are observed. This is an advantage over carbodiimides, which activate TFA, and over uronium salts, which can produce a chain-terminating side reaction at the amino terminus. Indeed, uronium salts, can react with the amine group of the C-protected moiety to give guanidino by-products;[33,53,79] this side reaction can take place during one-pot or cyclization[80] reactions or when hindered amino acids or fragments are involved.[53] Nevertheless with PyBOP, PyAOP, or PyBroP, pyrrolidine amides have occasionally been observed as byproducts in the case of slow coupling reactions. This is due to the presence of small amounts (e.g. 0.5% w/w) of pyrrolidine in the commerical reagents. The side reaction does not occur when the reagents are recrystallized before use.[133]

3.7.1.3.1 Occurrence and Minimization of Side Reactions

Using BOP or PyBOP, epimerization is not observed during the coupling of urethane-protected amino acids,[9,21,23,27,50,55] however as it is necessary to use a base with phosphonium (or uronium) reagents the possibility of epimerization should be considered.[81] Indeed, epimerization has been observed under preactivation-like conditions[50,60] and in this case the addition of HOBt is useful as it decreases the level of epimerization.[55] Histidine is known to racemize easily, indeed BOP-mediated coupling of Boc-His(Tos)-OH results in some epimerization (0.5 to 3% depending on the amount of base used),[67] but epimerization is not observed with either Boc-His(Boc)-OH[82] or Fmoc-His(Trt)-OH (in the latter case PyBOP was used).[55] Halophosphonium reagents seem to be more prone to producing epimerization during coupling, e.g. some epimerization occurs when coupling Z-Leu-OH with H-Aib-OAllyl using BroP (1.5%) or Z-Leu-OH with H-Aib-Pro-OtBu using PyBroP (1%), whereas this side reaction is not observed using PyBOP.[37] Conversely, in the case of the *N*-methylamino acids, when coupling Z-(Me)Val-OH with H-(Me)Val-OMe using PyBOP or HBPyU a high degree of epimerization (18 and 13%, respectively) is observed, whereas

for references see p 551

BroP, PyBroP, PyCloP, or HPyClU give almost optically pure peptides (see Table 3, entry 3).[34]

In fragment condensation steps, epimerization is generally observed with all the known coupling reagents.[81] The degree of epimerization depends on the reagent, the residue (Table 1, compare entries 1, 3 and 6),[11,60,83] the type[54,59,60,84] and the quantity[54,67,85] of base, the solvent (Table 1, entries 4 vs 5),[59,60,83] the reaction temperature,[36,59,85] and the use of HOBt (Table 1; entries 1, 3 and 6 vs 2, 4 and 7, respectively),[83] HOAt,[84] or $CuCl_2$[86] as additives. Using HOBt or HOAt as additives generally reduces epimerization,[83,84] but the opposite effect has also been observed.[59,134] It should be noted that the preactivation mode increases epimerization.[17,59] Halophosphonium reagents as well as DCC result in a high degree of epimerization (Table 1, entry 6), whereas BOP or PyBOP lead to better results and PyBOP in this context is superior to BOP (Table 1, entries 1 and 3–6). By using HOBt as additive, PyBOP (Table 1, entries 2, 4 and 5) and sometimes BOP (Table 1, entries 2 and 7) give results comparable to those obtained with TBTU (HBTU) or DCC/HOBt. However, DCC/HOBt is the best reagent in CH_2Cl_2.[83] In the Young test (Bzl-Leu-OH + H-Gly-OEt), **11**,[7] BOP,[17] and PyBOP[27] give 50, 30, and 15%, respectively, of DL-dipeptide, whereas only <3% is produced with the azido reagent **13**.[40] With OAt-based reagents the degree of epimerization is also decreased; e.g. H-Phe-Ser-Pro-NH_2 obtained by coupling Fmoc-Phe-Ser(tBu)-OH onto H-Pro-PAL-PEG-PS-resin contains 5 (AOP, HATU) or 7% (PyAOP) of the LDL-isomer vs 12% when using the OBt-based HBTU;[59] other results are displayed in Table 4 (see compound **33**).

Table 1 Epimerization in Fragment Coupling Steps[a]

Entry	Peptide	Conditions	Epimerization (%)[a]					Ref
			12	BOP **5**	PyBOP **16**	DCC	TBTU	
1	Z-Gly-Leu-Val-OBzl	CH_2Cl_2		14.6	4.9	25.9		[83]
2	Z-Gly-Leu-Val-OBzl	HOBt, CH_2Cl_2		<0.1	<0.1	<0.2		[83]
3	Z-Gly-Val-Val-OBzl	CH_2Cl_2		5.8	1.2	2.4		[83]
4	Z-Gly-Val-Val-OBzl	HOBt, CH_2Cl_2		3.7	<0.1	<0.2		[83]
5	Z-Gly-Val-Val-OBzl	HOBt,DMF		8.6	3.2	2.4	3.0[b]	[83]
6	Z-Gly-Phe-Val-OMe	CH_2Cl_2	35[c]	3.9[d]	3.3	36[c]	1.4[c]	[27]
7	Z-Gly-Phe-Val-OMe	HOBt	2.4	1.2		1.6	0.2	[87]

[a] % DL Isomer. For entries 1–6: Z-Gly-Xaa-OH + H-Val-OR · HCl. For BOP, PyBOP and HBTU, DIPEA (2 equiv for entries 1–5 and 2.75 equiv for entry 6) was added. For DCC, NMM (1 equiv) was added. When pertinent, HOBt (1 equiv) was added. One-pot reaction at 5 °C for entries 1–5 and at rt for entry 6. For entry 7 and some data (from ref[87]) of entry 6, conditions are not specified.
[b] Reagent was HBTU.
[c] From ref[87].
[d] 4.8% in ref[87].

The coupling of side-chain unprotected Asn with the chlorophosphonium reagent **11** produces intensive dehydration of the side-chain with formation of β-cyanoalanine.[7] This side reaction is not observed with BOP or PyBOP in the solution synthesis of dipeptides from Boc- or Z-Asn-OH[9] or it is reduced by adding HOBt in the case of Fmoc-SPPS[88] and in the case of cyclization in solution.[89] Protecting the Asn side-chain avoids the side reaction[88] and is essential with PyAOP.[33] Piperazine-2,5-dione (DKP) formation can arise when coupling the third amino acid of a peptide. Using in situ neutralization[14,15] (*vide supra*) reduces piperazine-2,5-dione formation in solution[14] as well as in Boc-SPPS methodology,[15,65] especially when the amount of DIPEA is reduced.[65] This method has been adapted in the case of Fmoc-SPPS by using *N*-trityl protection (instead of *N*-Fmoc) on the second amino acid and PyAOP as reagent.[66]

Esterification can be carried out with BOP.[51,56,90] Consequently, although the formation of the ester linkage is slower than that of the amide linkage,[91] the question of the protection

of the side-chain hydroxy group arises. Side-chain unprotected Ser and Thr have been coupled by solution synthesis with BOP.[9] The side-chains of Ser,[21] Thr,[92,93] or Tyr[21,92] can also be unprotected in SPPS, however, the number of coupling cycles after their incorporation must be minimal (<10).[21]

The BOP- or PyBOP-mediated SPPS of phosphino[94] or phosphono[95] peptides can be carried out without protecting the phosphinic or phosphonic acid group. Indeed, phosphinic acids are activated, but the P—N bond is not formed.[94] Phosphonic acids are also activated[91,96] and in fact an amide of methyl phenylphosphonic acid has been obtained,[91] however, in the case of protected α-amino phosphonic acids, phosphonamides are only obtained by using *N*-phthalyl protection.[96] Side-chain unprotected phosphorylated tyrosine [Tyr(PO_3H_2)] can be coupled by solid-phase synthesis using BOP (PyBOP),[97] but pyrophosphate formation is also observed[98] and some particular Tyr(P)-peptide syntheses are prone to total failure.[99]

Although the acylation of secondary amines is not easy to perform (*vide infra*), during the BOP-mediated synthesis of peptides containing ψ(CH_2-NH) amide replacement, using Z protection of the ψ(CH_2-NH) surrogate leads to more homogeneous products.[100]

3.7.1.3.2 Formation of Peptide Bonds with Proteinogenic Amino Acids

The reagents BOP and PyBOP have enabled the synthesis of numerous peptides or pseudopeptides in solution or solid-phase as shown, for example, by the large-scale solution synthesis of hGRF(1–44)-NH_2 (hGRF = human growth hormone releasing factor)[101] and the solid-phase syntheses of [Ala^{15}]-GRF(1–29)-NH_2,[21] various PACAP fragments (PACAP = pituitary adenylate cyclase activating polypeptide),[102] and ubiquitin (76 residues).[103]

The superiority of the OAt-containing phosphonium (AOP, PyAOP) and uronium (HATU, HAPyU) reagents versus their OBt analogues has been shown by Carpino et al.[24,33,52,104] For example, during the solid-phase synthesis of the model peptide H-Val-Gln-Ala-Ala-Ile-Asp-Tyr-Ile-Asn-Gly-NH_2 [ACP(65–74)] when the coupling time is intentionally decreased to 1.5 minutes and the excess of reagent reduced to 1.5 equivalents in order to exaggerate the differences among the reagents, better results are obtained with AOP and PyAOP or the corresponding uronium than with BOP or PyBOP (Table 2).

Table 2 Comparison of the Coupling Efficiencies of OBt- and OAt-Based Reagents[a]

Entry	Reagent	ACP(65–74)	des-Xaa-ACP(65–74)					Ref
			-2Ile	-Ile^{72}	-Ile^{69}	-Val	-Asn	
1	PyBOP	10	22	13	13	3	<1	[33]
2	PyBOP, HOBt	11	21	14	14	3	<1	[33]
3	AOP	49	3	6	13	1	18	[24]
4	PyAOP	46	3	5	5	2	18	[33]
5	PyAOP, HOAt	60	3	6	7	4	2	[33]
6	HATU, HOAt	50	5	9	12	2	8	[24]
7	HAPyU, HOAt	74	0	2	3	2	5	[104]

[a] Assembly of H-Val-Gln-Ala-Ala-Ile-Asp-Tyr-Ile-Asn-Gly-NH_2 [ACP(65–74)]: percent of products, including various deletion peptides [des-Xaa-ACP(65–74)]. Solid-phase synthesis was carried out in DMF in the presence of 1.5 equiv of Fmoc amino acid and coupling reagent and 3 equiv of DIPEA. Coupling time was 1.5 min.

3.7.1.3.3 Formation of Peptide Bonds with Non-Proteinogenic Amino Acids

Condensation of *N*-methyl or α,α-disubstituted amino acids is difficult.[105] In the coupling of *N*-methylamino acids problems can occur with reaction times and yields when the amine component or both the amine and the acid components are N-methylated.[36] In these cases OBt-based reagents such as BOP, PyBOP, HBPyU, or DCC/HOBt are often ineffective since

for references see p 551

the OBt esters are fairly unreactive[34,35] (see Table 3, entries 2, 3, 4, 6). In contrast, halophosphonium reagents such as BroP, PyBroP, and PyCloP give good results,[34–36] particularly when using an excess of reagent and base (Table 3, Method B). The yields can be poor for Boc-protected amino acids due to NCA formation[36] (Table 3, entry 6), however, this is not always the case (see Table 3 entries 5 and 7). The reagent CF_3-NO_2-PyBOP **19**[30] appears to be more effective than the halophosphonium salts, it gives good yields, in shorter reaction times, without using an excess of the reagent (Table 3). Moreover, a comparative study conducted under identical experimental conditions has shown the greater efficiency of **19** compared to PyBroP.[30] The reagents BroP, PyBroP, or PyCloP have been used to synthesize peptides containing N-alkylated residues both in solution[106–112] or in solid-phase,[76,113,114] including the cyclization step in the synthesis of aureobasidin A.[107] Finally, AOP and PyAOP have been used in the cyclization step of the (Me)Ile^{11}-CsA[108] and (Me)Leu^{1}-CsA,[33] respectively, and PyBroP/HOAt is as good or even better than HATU when coupling (solution or solid-phase) N-protected amino acids with N-alkylated (Me, Et, allyl) amino acid residues.[115]

Table 3 Comparison of the Coupling Efficiencies of Phosphonium Salts[a,b,c]

Entry	Peptide	Method, Time (h), Yield (%) [Epimer (%)]				Ref
		PyBOP **16**	PyBroP **21**	PyCloP **22**	**19**	
1	Z-(Me)Val-Val-OMe	A ,1 , 90 [0.2]	A, 1, 85 [0.3]	A, 1, 88 [0.15]		[36]
2	Z-Val-(Me)Val-OMe	A, 1, 11	A, 1, 70 [0]	A, 1, 85	C, 1, 76	[30,36]
3	Z-(Me)Val-(Me)Val-OMe	A, 72, 58 [18]	B, 3, 87 [≤0.3]	B, 3, 84 [≤0.3]	C, 1, 71	[30,36]
4	Fmoc-Val-(Me)Val-OMe	A, 24, 30	B, 3, 84		C, 1, 85	[30,36]
5	Boc-(Me)Val-Val-OMe	A, 1, 76	B, 3, 81			[36]
6	Boc-Val-(Me)Val-OMe	A, 24, 45	B, 3, 44	B, 3, 58	C, 1, 62	[30,36]
7	Boc-(Me)Leu-(Me)Val-OMe		B, 3, 88 [≤0.6]	B, 3, 97 [≤0.6]	C, 1, 76	[30,36]
8	Z-Aib-Aib-OMe	A, 16, 89	D, 1, 80		C, 1, 83[c]	[30,37]

[a] From ref[36] (entries 1–7) and ref[37] (entry 8) for PyBOP **16**, PyBroP **21**, and PyCloP **22**. From ref[30] for **19**.

[b] Method A: N-protected amino acid (1 equiv), C-protected amino acid hydrochloride (1.1 equiv), and coupling reagent (1 equiv) at 0 °C in CH_2Cl_2 (1 mL·mmol^{-1}), then DIPEA (3 equiv), 1 min at 0 °C, then rt. Method B: As above with N-protected amino acid (1.5 equiv), coupling reagent (1.5 equiv), and DIPEA (4 equiv). Method C: N-protected amino acid (1 equiv), C-protected amino acid hydrochloride (1 equiv), and coupling reagent (1 equiv) in CH_2Cl_2 (6 mL·mmol^{-1}), then DIPEA (3 equiv) at rt. Method D: N-protected amino acid (1 equiv), C-protected amino acid hydrochloride (1.1 equiv), and coupling reagent (1.2 equiv) in CH_2Cl_2 (1 mL·mmol^{-1}), then DIPEA (2.4 equiv) and DMAP (0.6 equiv) at rt.

[c] Results obtained with the reagents **7** or **18**.[29]

In the case of Aib, using BroP or PyBroP in the presence of DMAP improves reaction times (Table 3, entry 8) and efficiency versus BOP or PyBOP.[37] Indeed, PyBroP/DMAP is used to couple C- or N-protected α-methylcysteine or α-methylserine and also to couple two protected α-methylcysteine (or α-methylserine) together.[116,117] PyBroP is as efficient as UNCA when coupling Aib or (Me)Aib.[118] However, in the solid-phase assembly of Aib-containing peptides, acid fluorides are definitely superior.[119] When coupling two Aib, CF_3-BOP **7** or CF_3-PyBOP **18** give similar results to PyBroP without the use of DMAP (Table 3, entry 8).[29] Finally, PyAOP is very efficient in this condensation reaction.[33] For the coupling of Fmoc-Deg-OH with H-Phe-OFm, AOP or PyAOP are better than BOP or PyBOP and give identical results to HATU or HAPyU.[53]

3.7.1.3.4 Cyclization of Peptides with Phosphonium Salts

The reagent BOP is more effective than DCC/HOBt, DPPA, or BOP-Cl in solid-phase cyclizations.[22] It has been reported that reactions with BOP proceed more rapidly than with DIC/HOBt to give products of lower[120] or equivalent[121] purity. The reagent PyBOP is also

used[122,123] but PyAOP is more efficient (Table 4, see compound **32**).[33] For solution cyclization, BOP is either more[124,125] or less[126] effective than other reagents. The phosphonium salt PyBOP is also effective,[122,125,127] however OAt-based reagents give better results (Table 4, compound **33**).[128] The reagents BroP or PyBroP are also very efficient for lactamization.[107,125,129] As shown in Table 4 (see compound **33**), during cyclization reactions, epimerization generally occurs at the α-carbon of the residue bearing the carboxylic acid function. The level of epimerization decreases when using OAt-based reagents (see Table 4) or by reducing the amount of base when using BOP/HOBt.[130] Uronium coupling reagents can react with amines to give guanidino species[79] and as a result phosphonium salts often give better results in cyclization reactions.[80]

Table 4 Comparison of Various Coupling Reagents in Cyclization of Peptides

Compound[a]	Reagent	Cyclomonomer (%)	Purity (%)	Linear (%)	Ref
32	PyBOP, HOBt	69	51	5	[33]
32	PyAOP, HOAt	74	61	5	[33]
32	HATU, HOAt	70	50	5	[33]
Compound[a]	Reagent	Cyclomonomer (%)	Cyclodimer (%)	D-Tyr-Isomer (%)	Ref
33	BOP	38	13	20.2	[128]
33	PyBOP	52	17	13.0	[128]
33	PyAOP	56	20	10.9	[128]
33	HAPyU	55	25	8.8	[128]

[a] **32**: cyclo(Tyr-D-Trp-Val-D-Trp-D-Trp-Arg-Asp); solid-phase cyclization; reaction for 2 min at 25 °C in DMF; molar ratio reagent/HOXt/DIPEA (5:5:10).[33] **33**: cyclo(Arg(NO_2)-Lyz(Z)-Asp(OBzl)-Val-Tyr); solution cyclization; reaction for 1 h; 1.1 equiv of coupling reagent and 3 equiv of DIPEA.[128]

3.7.1.3.5 Fragment Condensations with Phosphonium Salts

In the coupling of peptide fragments with BOP (BOP/HOBt) in solution[101] or in the solid phase[85,131,132] the yields are often better than with DCC/HOBt and can be improved by using 15% DMSO in NMP as the solvent.[132] When the C-terminal residue is Gly, it is possible to use a greater amount of base (5 equiv DIPEA per equiv of BOP) and to avoid the addition of HOBt, to improve the yield.[132] Epimerization occurs when the C-terminal residue of the N-protected fragment can racemize (see above and Table 1). For example, the synthesis of human gastrin-I [pGlu-Gly-Pro-Trp-Leu-$(Glu)_5$-Ala-Tyr-Gly-Trp-Met-Asp-Phe-NH_2] was achieved [PAL-Val-MBHA-resin, Fmoc chemistry, BOP/HOBt/NMM (1:1:1) in DMF, 0 °C, 1 h and 25 °C, 23 h] by using two fragment couplings at Ala (>98% estimated coupling yield) and Leu, (89% estimated coupling yield) with epimerization levels of 4 and 11% at Ala and Leu respectively.[131] Low temperature (4 °C) and the minimization of base are critical factors in reducing epimerization.[85] As in the case of cyclization reactions, phosphonium salts can be more effective because uronium salts can react with the amine group of the C-protected moiety.[53]

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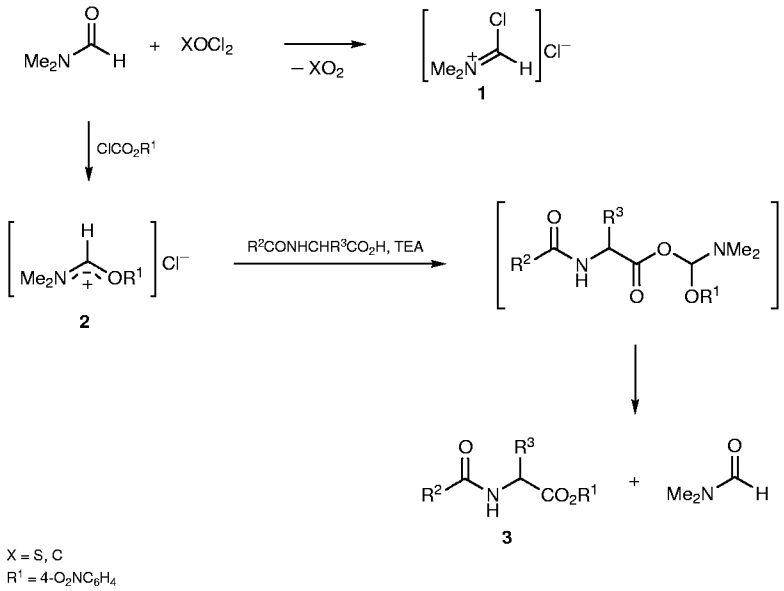
3.8 Uronium/Guanidinium Salts

M. Bienert, P. Henklein, M. Beyermann, and L. A. Carpino

Although numerous methods for peptide bond formation have been available for some time,[1] more recent developments have focused on the design of reagents that allow the rapid activation of a protected amino acid and its coupling to an amino acid or peptide derivative in a single step, accompanied by minimal racemization and side reactions. Benzotriazol-1-yloxytris(dimethylamino)phosphonium hexafluorophosphate (BOP)[2] was the first example of this new generation of coupling reagents. However, BOP-mediated activation results in the formation of hexamethylphosphoric triamide as a byproduct, a known carcinogen, thus, provoking the search for less-toxic compounds either by replacing the dimethylamino groups (see Section 3.7), or by shifting from a phosphonium to a uronium structure.

The development of uronium compounds is based on earlier findings that showed the successful application of iminium salts as activating reagents for carboxylic acids. Treatment of dimethylformamide with thionyl chloride or phosgene resulted in the formation of the crystalline complex *N,N*-dimethylchloromethyleneiminium chloride (**1**),[3] which was used in the formation of peptide bonds.[4] Similarly, reaction of dimethylformamide with 4-nitrophenyl chloroformate gave *N,N*-dimethyl(4-nitrophenoxy)iminium chloride (**2**); this reagent was found to be useful in the preparation of amino acid 4-nitrophenyl esters **3**[5] (Scheme 1). In an analogous manner, activation of a urea instead of dimethylformamide resulted in the generation of a formamidinium salt.[6]

Scheme 1 Iminium Salts in Peptide Activation

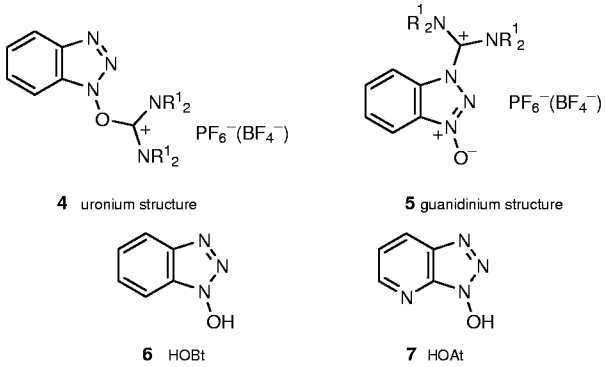


Thus, with the compound believed to have a uronium structure, 2-(1*H*-benzotriazol-1-yl)-1,1,3,3-tetramethyluronium hexafluorophosphate (**8**, HBTU)[7,8] and the corresponding tetrafluoroborate **9** (TBTU)[9] appropriate alternatives to BOP were found. More recently, similar so-called uronium coupling reagents of type **4**, based on the substitution of 3-hydroxy-3*H*-1,2,3-triazolo[4,5-*b*]pyridine (**7**, 7-azabenzo-1,2,3-triazol-1-ol, HOAt),[10–13] for benzo-1,2,3-triazol-1-ol (**6**, HOBt),[14] have been developed (Scheme 2).[12,13] The structures initially assigned to these compounds were subsequently shown to be incorrect and a thorough investigation by X-ray crystallography and solid-state and solution NMR techniques established their correct

for references see p 576

assignment as the corresponding guanidinium *N*-oxides **5**[15,16] (Scheme 2). Although the abbreviations HBTU, TBTU, etc., are maintained in the literature, the systematic names of these compounds must be reformulated as *N*-[(1*H*-benzotriazol-1-yl)(dimethylamino)-methylene]-*N*-methylmethanaminium hexafluorophosphate *N*-oxide for HBTU (**8**), and the corresponding tetrafluoroborate *N*-oxide for TBTU (**9**). Further study showed that certain specific derivatives of this type in fact exhibited the corresponding uronium structure, also established conclusively by X-ray analysis and NMR examination.[17] All such cases examined to date carry a substituent in the 4-position (4-Me or 4-Cl) of the HOBt or HOAt moiety suggesting that steric effects are decisive in fixing the uronium-type structure. On reflection it appears reasonable that the guanidinium structure is, indeed, expected to be the more stable in the absence of a 4-substituent,[15,16] a fact which may be explained by the concept of Y aromaticity.[18] In this section the uronium nomenclature is used for all systems for which the uronium structure is the only reasonable one, and arbitrarily and for the sake of simplicity for other systems for which X-ray or NMR spectroscopic data is not yet available. The authentic uronium isomers of HBTU and HATU (O-HBTU, O-HATU) have been synthesized and their reactivities compared with those of the guanidinium forms (N-HBTU, N-HATU).[217] The correct nomenclature according to *Chemical Abstracts* is N-HBTU, 1-[bis(dimethylamino)methylene]-1*H*-benzotriazolium hexafluorophosphate 3-oxide; N-HATU, 1-[bis(dimethylamino)methylene]-1*H*-1,2,3-triazolo-[4,5-*b*]pyridinium hexafluorophosphate 3-oxide; O-HBTU, *N*-[(1*H*-benzotriazol-1-yloxy)(dimethylamino)methylene]-*N*-methylmethanaminium hexafluorophosphate; O-HATU, *N*-[1*H*-1,2,3-triazolo[4,5-*b*]pyridin-1-yloxy)(dimethylamino)methylene]-*N*-methylmethanaminium hexafluorophosphate).

Scheme 2 Structures of Uronium and Guanidinium Salts from HOBt and HOAt



This section also includes the guanidinium salts formed with benzotriazole-based moieties such as HOBt and HOAt. In addition, haloformamidinium salts, which are the usual intermediates for the synthesis of uronium/guanidinium compounds and have occasionally themselves been employed in special coupling reactions, will be included in this section.

3.8.1 Uronium/Guanidinium Salts Used in Peptide Chemistry

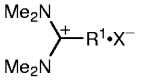
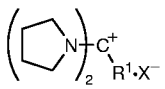
The development of uronium salts has followed strategies similar to those described in Section 3.7 for the design of phosphonium salt reagents. The reagent should possess sufficient solubility (≥0.3 M) and stability in organic solvents and, most importantly, the activating reagent should suppress epimerization/racemization during fragment condensations and the coupling of sensitive amino acid derivatives. Moreover, during peptide bond formation the coupling reagents should ideally generate only harmless byproducts.

Known uronium and guanidinium salts are compiled in Table 1. The design of the carbenium structures was mainly influenced by the availability of the corresponding urea

and the solubility of the final product. Although the urea-derived substituents in the carbenium moieties have only a marginal influence on the reactivity of these reagents, in contrast to the corresponding phosphonium compounds,[19] the structure of the carbon skeleton is of marked importance in connection with the stability of the uronium/guanidinium salt[20] (see Section 3.8.1.1).

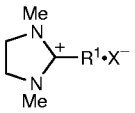
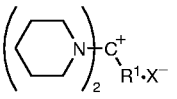
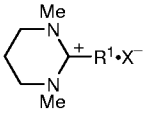
Although the chloroformamidinium salts **23**,[21] **24**,[7,22] and **31**[22–24,51] are highly reactive coupling reagents, they are often hygroscopic and their tendency to cause loss of stereochemical integrity in the activated amino acid moiety and other side reactions limits their broad application to peptide synthesis[24,26,27] and requires their conversion into more versatile derivatives. Most suitable are the products formed from reaction with *N*-hydroxy compounds in which one finds a strong influence on both the reaction rate and the racemization potential. Keeping in mind that the proposed mechanism of carboxy activation using uronium reagents involves the formation of active esters, it is not surprising that HOBt- and HOAt-derived compounds are appropriate reagents both in terms of reactivity and suppression of racemization, results that support previous studies on the positive effects of these *N*-hydroxy compounds when used as additives in carbodiimide-mediated couplings.[12,14,28]

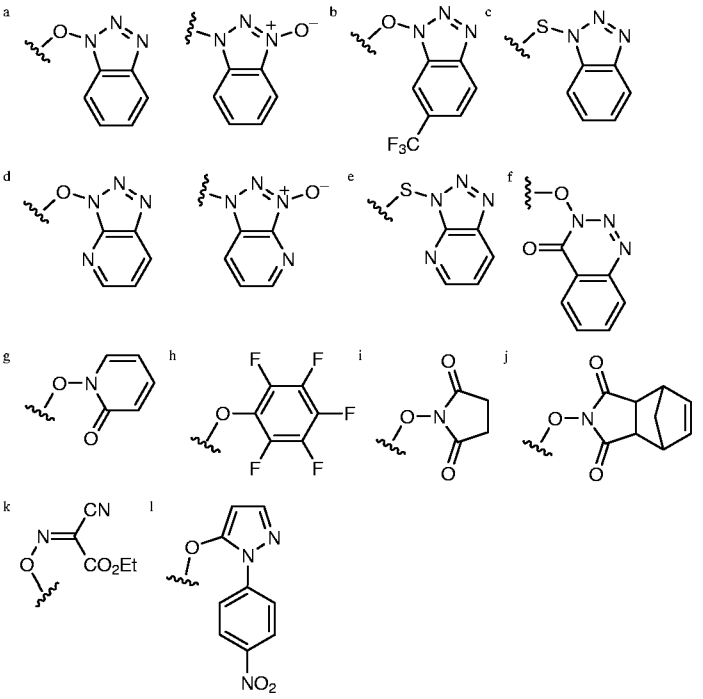
Table 1 Structures of Uronium, Guanidinium, and Amidinium Salts

Compound Number	Structure			Abbreviation	Ref
					
		R^1	X		
8		a	PF_6	HBTU	[7,8,38]
9		a	BF_4	TBTU	[9]
10		b	PF_6	CF_3-HBTU	[58]
11		c	PF_6	HSBTU	[60]
12		d	PF_6	HATU	[12,13,39]
13		e	PF_6	HATTU	[61]
14		f	PF_6	HDTU	[56]
15		f	BF_4	TDBTU	[9]
16		g	BF_4	TPTU	[9,54]
17		h	PF_6	PfTU	[22]
18		i	BF_4	TSTU	[9,52,54]
19		j	BF_4	TNTU	[9]
20		k	PF_6	HOTU	[57]
21		k	BF_4	TOTU	[64]
22		l	BF_4	HppTU	[65]
23		Cl	Cl	TCFC	[21]
24		Cl	PF_6	TCFH	[7,22]
25		F	PF_6	TFFH	[67,104]
					
		R^1	X		
26		a	PF_6	HBPyU	[23,51,52]
27		a	BF_4	TBPyU	[53,54]
28		b	PF_6	HAPyU	[12,55]
29		e	PF_6	HAPyTU	[61]
30		h	PF_6	PfPyU	[22]
		h	PF_6	HPyOPfp	[61]
31		Cl	PF_6	HPyClU	[22,24,51]
			BF_4		[24]
32		F	PF_6	BTFFH	[27,68]

for references see p 576

Table 1 (cont.)

Compound Number	Structure			Abbreviation	Ref
					
		R^1	X		
33		a	PF_6	BOI	[25]
		a	PF_6	HBMDU	[20,55]
34		d	PF_6	HAMDU	[20,55]
35		Cl	PF_6	CIP	[25]
		Cl	PF_6	DCIH	[27]
36		F	PF_6	DFIH	[27]
					
		R^1	X		
37		a	BF_4	TBPipU	[53,54]
38		d	PF_6	HAPipU	[20]
39		g	PF_6	HOPPipU	[53]
40		g	BF_4	TOPPipU	[53]
					
		R^1	X		
41		d	PF_6	HAMTU	[20]



In particular, reagents HBTU (**8**), TBTU (**9**), and HATU (**12**) have been widely used in both solution and solid-phase peptide synthesis (see refs[29–42] for reviews) as well as in the generation of peptide libraries and multiple peptide synthesis[34,43–49] and in the synthesis of PNAs.[50] These reagents possess good solubility in DMF (HBTU: 0.5 mol · L^{-1}, TBTU: 0.6 mol · L^{-1},[38] HATU: 0.6 mol · L^{-1}).[39]

There are indications that substitution of the dimethylamino groups by the pyrrolidine residue, resulting in the uronium/guanidinium compounds HBPyU (**26**)[51,52] or the corresponding tetrafluoroborate salts TBPyU (**27**)[53,54] and HAPyU (**28**),[12,55] gives slightly better results with regard to suppression of racemization.[51,55–57] For HBPyU and HAPyU it was shown that these compounds crystallize as the corresponding guanidinium salts.[15]

The introduction of an electron-withdrawing trifluoromethyl group into HOBt is expected to increase the reactivity of the corresponding uronium salt. Indeed it has been observed that compound CF_3-HBTU (**10**)[58] is useful in the coupling of C^{α}-alkyl amino acid residues, a process not easy to achieve with the simple HOBt analogues.[58]

Previous observations on the high reactivity of (benzoxazol-2-yl) thioesters toward various amines[59] prompted the development of uronium salts HSBTU (**11**)[60] as well as HATTU (**13**) and HAPyTU (**29**).[61] However, HATTU and HAPyTU were less reactive and provoked more epimerization in both cyclization studies and fragment condensations than the corresponding oxygen analogues HATU (**12**) and HAPyU (**28**).[61]

Uronium salts derived from 3-hydroxy-1,2,3-benzotriazin-4(3*H*)-one (HODhbt), e.g. HDTU (**14**)[56] and TDBTU (**15**),[9] are highly effective in reducing racemization in susceptible substrates, in accordance with the high efficiency of HODhbt in carbodiimide/additive couplings.[213] However, these reagents also result in the formation of an azide byproduct, which reacts readily with N^{α}-amino groups (Section 3.8.1.2). This side reaction reduces their usefulness and for that reason HODhbt-derived uronium salts have not achieved the popularity of HOAt- and HOBt-derived reagents, although special applications have been described.[62]

Uronium salts containing *N*-hydroxysuccinimide-derived residues, TSTU (**18**)[9,52,54] and TNTU (**19**)[9] are less efficient coupling reagents than those derived from HOBt and HOAt. However they generate active esters that are more stable in mixed organic/aqueous solvents than the OBt esters, thus, for example, allowing the application of TSTU in protein chemistry.[54]

The pentafluorophenyl-derived compounds PfTU (**17**)[22] and PfPyU[22,63] or HPyOPfp[61] (**30**) are somewhat inefficient coupling reagents, which exhibit a relatively high potential to racemize susceptible amino acid residues.[61] On the other hand, the compounds can be easily synthesized[22] and in the presence of HOAt as an additive, reagent **30** is as efficient as HAPyU (**28**).[61]

The use of the oxime 2-cyano-2-hydroximinoethyl acetate as the *N*-hydroxy component led to the development of TOTU (**21**)[64] and the corresponding hexafluorophosphate HOTU (**20**).[57] TOTU results in less epimerization in segment couplings than BOP or TBTU (**9**).[64]

Fmoc amino acid active esters derived from 5-hydroxy-1-(4′-nitrophenyl)pyrazole (Hpp) are of high intrinsic reactivity, they are self-indicating, and their properties render them highly suitable for solid-phase synthesis of complex sequences.[65] This prompted the synthesis of the uronium salt HppTU (**22**), which was found to be useful in the synthesis of active esters, and itself functions as an effective coupling reagent in its own right, allowing the synthesis of several peptides.[65]

In order to further improve the solubility of uronium salts and to eliminate any absorbance in the range 300–350 nm as this will improve Fmoc-assisted monitoring of peptide synthesis, uronium compounds derived from the oxo-pyridine system have been synthesized.[9,53,54] TPTU (**16**) has been recommended for segment condensations (low epimerization level),[9,66] and couples *N*-methyl amino acids more efficiently than TBTU.[54] Uronium salt TOPPipU (**40**), although causing substantial epimerization, is unique in possessing significantly enhanced solubility (1.5 mol · L^{-1} in DMF).[53]

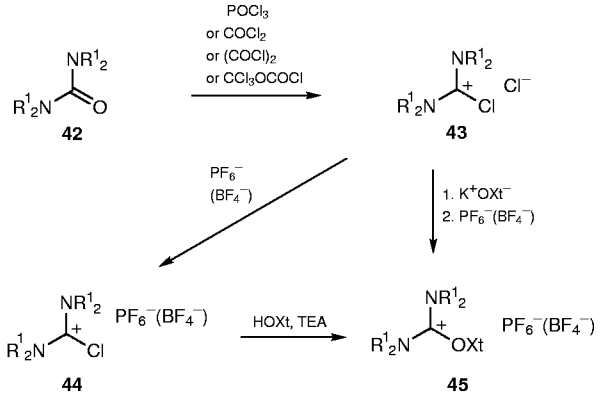
for references see p 576

Recently, the iminium-type coupling reagents represented by structures **1** and **2** have been updated in the form of the corresponding OBt and OAt analogues.[214,215] As expected, in view of the heightened electropositive character of the cationic center such compounds are significantly more reactive in peptide coupling than the analogous uronium/guanidinium reagents. Although initially assigned an O-structure, X-ray analysis has shown that, as in the well-known uronium/guanidinium systems, the stable, isolable species is N-substituted.[215]

3.8.1.1 Synthesis of Uronium/Guanidinium Salts

The synthesis of uronium/guanidinium salts **45** can be accomplished by preparation of the corresponding chloroformamidinium intermediates **43** and their reaction with *N*-hydroxy or other compounds bearing acidic hydroxy functions (Scheme 3). Phosgene,[7,12,22,51,53] oxalyl chloride,[9,21,39,52,53,60,67,68] trichloromethyl chloroformate,[25] and phosphoryl chloride[27,69] can all be used for the conversion of substituted ureas **42** into the chloroformamidinium compounds **43**. In most cases, the intermediate chloroformamidinium chloride **43** is converted into the less hygroscopic hexafluorophosphate or tetrafluoroborate **44** both of which possess higher solubility in nonpolar solvents, thus improving the separation of the triethylammonium hydrochloride liberated during reaction with the alcohol (Scheme 3). However, the amidinium chlorides **43** themselves may be successfully used when the potassium salts instead of the TEA salts of the alcohols are used in generating the uronium/guanidinium compounds **45**.[9] The synthesis of thiuronium salts has also been performed in this manner.[60] Furthermore, one-pot preparations of the hexafluorophosphates, without isolation of the corresponding chloroformamidinium chloride, have been described.[52]

Scheme 3 General Routes for the Synthesis of HOBt- (X = B) and HOAt- (X = A) Derived Uronium/Guanidinium Salts



Stability of Uronium/Guanidinium Salts

In order to determine the compatibility of uronium/guanidinium salts with peptide synthesis, the stability of DMF solutions (0.05 M) of some of these reagents has been investigated (Table 2). When stored in open vessels at 25 °C, the HOAt-derived compounds are less stable than the corresponding HOBt analogues, and the uronium/guanidinium derivatives are in general more stable than the corresponding phosphonium salts.[20] Furthermore, the stability is remarkably influenced by the structure of the uronium/guanidinium moiety, with the compounds BOI[25] and HAMDU[20] being very unstable, whereas the dimethylamino compounds HBTU and HATU are among the most stable. As expected, the coupling reagents are more stable when stored in closed vials or under a nitrogen atmosphere, conditions used in some automated synthesizers. Although the data in Table 2 indicates that the content of

the coupling reagent after 21 days is significantly decreased, the smooth synthesis of the model peptide ACP (65–74) demonstrates the feasibility of the use of HATU solutions after prolonged storage under nitrogen.[20] Similarly, the applicability of solutions of 0.45 M HBTU/HOBt in DMF was demonstrated for syntheses carried out after a storage period of 3 months.[36]

Table 2 Hydrolytic Stability of Selected Uronium/Guanidinium Salts[20]

Reagent	Stability (%)				
	Open Vial		Closed Vial	With DIPEA (1 equiv)	
	5 h	20 h	48 h	2 min	1 h
HATU	>99	95	>99	93	36
HBTU	100	98	100	95	62
HAPyU	96	79	93	59	9
HAPipU	98	95	97		
HAMDU	0		0		
HBMDU	0				
HAMTU	71	31	52		
HDTU	0				
AOP	91	23	78	96	76
BOP	96	53	98	96	82

As peptide bond formation mediated by uronium/guanidinium compounds is usually performed in the presence of an extra equivalent of base, the stability of these compounds has also been examined in the presence of DIPEA. The data in Table 2 demonstrates that these reagents degrade at least partially within one hour, with HBTU being the most stable of the compounds tested. This observation may have practical consequences if activation of a carboxylic acid is very slow. In such cases the coupling reagent may undergo degradation and no longer function properly in the reaction medium.[20]

Synthesis of Chloroformamidinium Salts:

***N*-[Chloro(dimethylamino)methylene]-*N*-methylmethanaminium Chloride (Tetramethylchloroformamidinium Chloride, TCFC, 23):**[9,70]

To a soln of tetramethylurea (120 mL, 1 mol) in CH_2Cl_2 (500 mL) was added slowly oxalyl chloride (84 mL, 0.95 mol) under an inert atmosphere with intermediate evolution of CO_2/CO. The mixture was refluxed for 2 h. On addition of anhyd Et_2O (1 L) the hygroscopic white crystals were filtered off and washed with anhyd Et_2O; yield: 145 g (85%).

***N*-[Chloro(dimethylamino)methylene]-*N*-methylmethanaminium Hexafluorophosphate (Tetramethylchloroformamidinium Hexafluorophosphate, TCFH, 24) Using Phosgene:**[7]

CAUTION: *Phosgene is a colorless, poisonous gas or volatile liquid. This reagent should be handled with the utmost care.*

To a soln of tetramethylurea (11.6 g, 0.1 mol) in toluene was added dropwise 20% $COCl_2$ in toluene (100 mL) under dry conditions in an efficient hood. When the CO_2 evolution had stopped, anhyd Et_2O (350 mL) was added with vigorous stirring. The precipitate was filtered and washed with anhyd Et_2O (3 × 50 mL). The highly hygroscopic solid was immediately dissolved in CH_2Cl_2 (500 mL) and to the soln was added with continuous stirring sat. NH_4PF_6 soln (50 mL). The organic phase was washed with H_2O (40 mL), dried ($MgSO_4$), and the solvent removed. The solid salt **24** was washed with Et_2O and dried (P_2O_5) to give a hygroscopic material, which was recrystallized (acetone/Et_2O) to give colorless needles; yield: 24 g (86%); mp 86–88 °C.

***N*-[Chloro(dimethylamino)methylene]-*N*-methylmethanaminium Hexafluorophosphate (Tetramethylchloroformamidinium Hexafluorophosphate, TCFH, 24) Using Phosphoryl Chloride:**[69]

A stirred mixture of tetramethylurea (100 g, 0.86 mol) and $POCl_3$ (90 mL, 0.95 mol) was heated to 100 °C for 1 h. After cooling to rt, CH_2Cl_2 (2.2 L) was added, followed by dropwise addition of 75% aq HPF_6 (100 mL) with good stirring. (**CAUTION:** *This reaction is highly exothermic!*) The mixture was washed

for references see p 576

with H_2O (400 mL), dried ($MgSO_4$), filtered, and concentrated under reduced pressure to give compound **24** as a white material; yield: 143.5 g (60%); mp 92–94 °C.

1-[Chloro(pyrrolidin-1-yl)methylene]pyrrolidinium Hexafluorophosphate [Bis(tetramethylene)chloroformamidinium Hexafluorophosphate, HPyClU, 31]:[22,51]

CAUTION: *Phosgene is a colorless, poisonous gas or volatile liquid. This reagent should be handled with the utmost care.*

To a soln of 1,1-carbonyldipyrrolidine (60.56 g, 0.36 mol) in toluene (120 mL) was added 20% $COCl_2$ in toluene (242.2 mL, 0.48 mol) with vigorous stirring. After 30 min, excess $COCl_2$ was removed under reduced pressure and the soln was concentrated to one third of its volume and under vigorous stirring poured into Et_2O (1 L). The precipitate was collected and again stirred with Et_2O (1 L). The precipitate was collected and dissolved in H_2O. To the aqueous soln was added slowly under vigorous stirring sat. aq KPF_6 (66.75 g, 0.36 mol). The precipitate was filtered off, dried, and dissolved in dry acetone; the soln was filtered, and to the filtrate an excess of Et_2O was added slowly. The precipitated colorless needles were filtered off and dried under reduced pressure; yield: 91.75 g (77%); mp 149.5–150.5 °C.

2-Chloro-4,5-dihydro-1,3-dimethyl-1*H*-imidazolium Chloride (35, X = Cl):[25]
Trichloromethyl chloroformate (15 g, 0.076 mol) was added dropwise to a soln of 1,3-dimethylimidazolidin-2-one (20.7 g, 0.18 mol) in CCl_4 (50 mL) and the mixture stirred for 5 h. The precipitate was collected and washed with CCl_4 and hexane; yield: 23.2 g (91%).

2-Chloro-4,5-dihydro-1,3-dimethyl-1*H*-imidazolium Hexafluorophosphate (35, X = PF_6):[25]
To a soln of 2-chloro-4,5-dihydro-1,3-dimethyl-1*H*-imidazolium chloride (**35**, X = Cl) (47.6 g, 0.282 mol) in CH_2Cl_2 (2 L) was added aq NH_4PF_6 (51.1 g , 0.313 mol) and the mixture was stirred for 10 min at 25 °C. The organic layer was washed with H_2O (200 mL), dried ($MgSO_4$), and the solvent removed; yield: 71.1 g (91%); mp 207 °C (dec).

Synthesis of Uronium/Guanidinium Reagents from Chloroformamidinium Salts:

***N*-[(1*H*-Benzotriazol-1-yl)(dimethylamino)methylene]-*N*-methylmethanaminium Hexafluorophosphate *N*-Oxide (HBTU, 8):**[7]
To chloroformamidinium salt **24** (20 g, 0.07 mol) in CH_2Cl_2 (300 mL) was added HOBt (9.45 g, 0.07 mol), followed by TEA (9.74 mL, 0.07 mol, 1 equiv). The precipitate was filtered, washed with CH_2Cl_2 (100 mL), and recrystallized (MeCN); yield: 26 g (100%); mp 250 °C.

Synthesis of the Tetramethylurea-Derived Uronium/Guanidinium Salts 9, 15, 16, 18, and 19 via Potassium Salts of *N*-Hydroxy Compounds:[9,70]
To a hot soln of the desired *N*-hydroxy compound (1 mol) in a minimum of EtOH was added slowly 2 M KOH in EtOH (500 mL, 1 mol). The potassium salt precipitated immediately after slight cooling of the mixture or upon addition of Et_2O. The salt was filtered off, washed with anhyd Et_2O, and dried. To a soln of compound **23** (128 g, 0.75 mol) in anhyd MeCN (1 L) was added NH_4PF_6 or $NaBF_4$ (0.75 mol). A suspension of the potassium salt of the desired *N*-hydroxy compound (0.75 mol) in MeCN (500 mL) was added with slight cooling and the mixture was stirred overnight at rt. Insoluble KCl was filtered off, and the filtrate concentrated to an oil. The uronium salt precipitated by addition of dry Et_2O (300 mL). The crude material was recrystallized (acetone/Et_2O), filtered off, and dried at rt; yield: 72–84%.

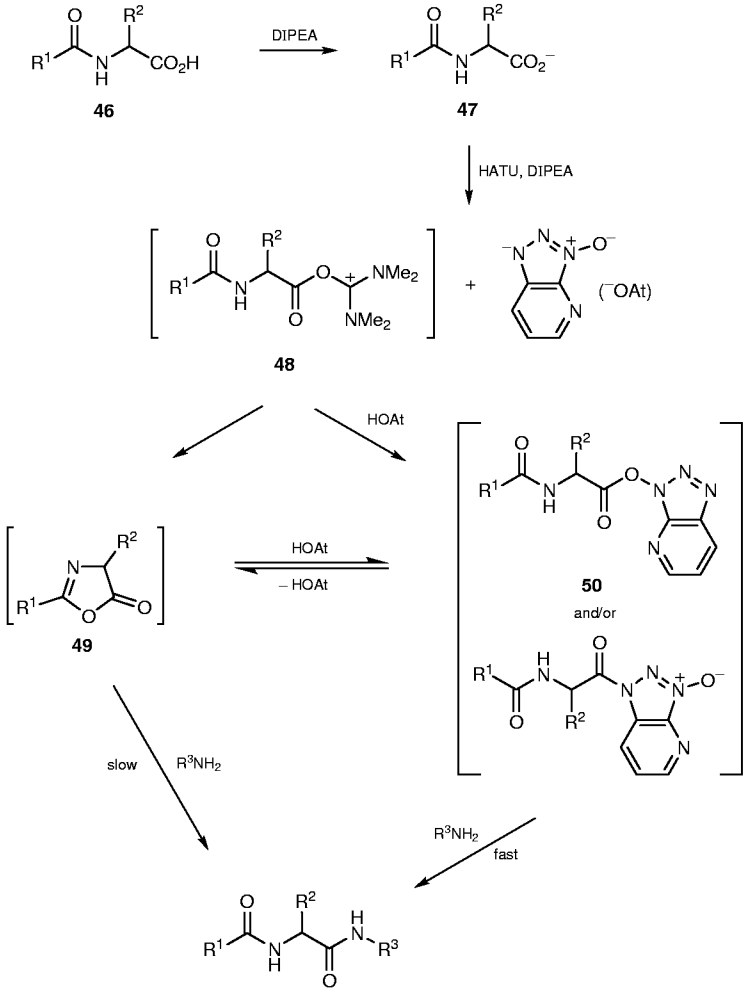
***N*-[(Dimethylamino)(1*H*-1,2,3-triazolo[4,5-*b*]pyridin-1-yl)methylene]-*N*-methylmethanaminium Hexafluorophosphate *N*-Oxide (HATU, 12):**[13]
To a suspension of tetramethylchloroformamidinium hexafluorophosphate (**24**) (0.88 g, 3.16 mmol) in CH_2Cl_2 (30 mL) was added HOAt (0.43 g, 3.16 mmol)[13,71] followed by TEA (0.44 mL, 3.16 mmol). A clear soln formed and then a granular solid precipitated. After stirring at rt for 30 min the solid was filtered, washed with CH_2Cl_2 (2 × 10 mL), and dried in air to give the crude guanidinium salt (0.79 g, 66%). Recrystallization [dissolution in MeCN (5 mL) at rt, filtration to remove some insoluble crystalline solid, and dilution of the filtrate with CH_2Cl_2 to a total volume of 25 mL] gave the pure guanidinium salt **12** as shiny white crystals; yield: 0.51 g (45%); mp 190–194 °C (dec).

2-(1*H*-Benzotriazol-1-yloxy)-1,3-dimethylimidazolidinium Hexafluorophosphate (BOI, 33):[25]
To a suspension of **35** (71.1 g, 0.255 mol) and HOBt (34.5 g, 0.256 mol) in CH_2Cl_2 (1.5 L) was added dropwise TEA (25.75 g, 0.255 mol) at 10 °C. The mixture was stirred for 3 h at 25 °C and then cooled in an ice bath. The resulting precipitate was collected and washed with CH_2Cl_2. The crude salt was reprecipitated (MeCN/CH_2Cl_2); yield: 43 g (46%); mp 173 °C (dec).

3.8.1.2 Peptide Bond Formation by Means of Uronium/Guanidinium Salts

Mechanistically, uronium salts are thought to function in a manner similar to that of the phosphonium analogues (Section 3.7). The carboxylate ion **47**, formed from the carboxy component **46** by reaction with a tertiary amine, attacks the uronium/guanidinium species, leading to the highly reactive acyluronium intermediate **48** (Scheme 4), which is converted into the corresponding active ester **50** that is thought to be the predominant species undergoing aminolysis.[7,29] Whether or not the acyluronium intermediate **48** itself is involved in direct aminolysis is not clear. In segment condensations, an alternate pathway for stabilization of the acyluronium cation involves formation of an oxazol-5(4*H*)-one **49**,[28] which may be in equilibrium with the corresponding active ester.[72,73] The formation of oxazolones is undesirable since facile deprotonation and re-protonation lead to epimerization, as outlined in Section 7.3. The existence of both active esters **50** and oxazolones **49** as intermediates in uronium/guanidinium mediated activation has been demonstrated in model reactions using HPLC analysis[74,75] and also by IR studies.[72,73] In a few cases, especially during the coupling of sterically hindered amino acid residues, the intermediate active ester partially resists aminolysis and can be isolated and characterized.[23,76]

Scheme 4 Uronium/Guanidinium-Mediated Formation of Peptide Bonds



for references see p 576

Although the rate of formation and the stereochemical integrity of the oxazolone depend critically upon the nature of the *N*-acyl residue of the activated amino acid, the results are also influenced by the nature of the oxygen moiety built into the uronium/guanidinium salt that eventually takes part in active ester formation. Of the various *N*-hydroxy derivatives studied to date the OAt residue is the most highly efficient in reducing racemization.[12,39,55,56]

In contrast to the case of uronium/guanidinium salts, haloformamidinium salts rapidly transform urethane-protected amino acid derivatives into a mixture of the corresponding anhydride and oxazolone[26] which are the species aminolyzed.

In normal couplings of amino acid residues, during both solution and solid-phase syntheses, the most commonly used uronium/guanidinium salts HBTU, TBTU, and HATU may be equally effective. However, for fragment condensations involving a chiral amino acid residue located at the C-terminus, or for the activation of racemization prone amino acids or couplings of sterically hindered amino acids, the selection of the appropriate coupling reagent as well as optimization of proper reaction conditions (base, solvent, additives, preactivation time) may be decisive for the success of the coupling.

General Experimental Conditions

Regarding the use of base, additives, preactivation time, and solvents, the requirements in uronium/guanidinium-mediated peptide synthesis are quite similar to those described for couplings using phosphonium salts (Section 3.7.1.2). According to Scheme 4, two equivalents of base must be used, one to generate the carboxylate anion of the N-protected amino acid and a second to neutralize the released HOXt to prevent a decrease of the pH of the system as this would tend to slow down the coupling process. A third equivalent of base is required when the amino component is used in its protonated form. Excess base may speed up the coupling reaction, but can also increase the level of racemization. Thus, for coupling reactions involving proteinogenic amino acids that are not likely to racemize the reactions are performed in the presence of 1–2 equivalents of DIPEA or NMM, when HBTU, TBTU, HATU, or HAPyU are used for activation.[7,38,39,51,55,56] For SPPS, the use of 1.5–2 equivalents of DIPEA permits the racemization-free coupling of those urethane-protected amino acid derivatives which are not sensitive to racemization.[33,36,74,77] In large-scale SPPS, the pH is adjusted to 7–7.5 by the addition of DIPEA.[32] For in situ neutralization protocols, originally developed for BOP-mediated SPPS-couplings[78,79] and later adapted for the rapid, high-yield assembly of difficult peptides[37,80–84] using HBTU or HATU and Boc amino acid derivatives, 1 equivalent of DIPEA for neutralization of the Boc amino acid plus 2 equivalents of the same base, relative to the protonated resin are recommended.[37]

For the coupling of racemization-prone amino acid derivatives and in fragment condensations involving protected peptides not having a C-terminal Gly or Pro unit, the use of 1–1.5 equivalents of DIPEA[35] or a weaker and more-hindered base, such as 2,4,6-trimethylpyridine (collidine) is recommended.[55,56,85–89] The favorable effect of collidine on the suppression of epimerization is demonstrated in Table 3 for couplings carried out in solution (see also the experimental procedures for peptides **57**, **60**, and **63**, Schemes 6 and 7).

Table 3 Effect of Base and Additives on Epimerization Occurring during the Uronium/Guanidinium Salt Mediated Coupling of Z-Phe-Val-OH and H-Pro-NH_2 in Dimethylformamide[56]

Coupling Reagent	Base (Equiv)	% LDL
HBTU	DIPEA (2)	27.4
HBTU	collidine[a] (2)	14.2
HATU	DIPEA (2)	13.9
HATU	collidine[a] (2)	5.3
HATU/HOAt	DIPEA (2)	10.9
HATU/HOAt	collidine[a] (2)	2.4
HATU/HOAt	collidine[a] (3)	4.5
HAPyU	DIPEA (2)	10.8
HAPyU	collidine[a] (2)	3.5
HAPyU/HOAt	collidine[a] (3)	2.3

[a] 2,4,6-trimethylpyridine.

For fragment condensations on solid phase, the effect of collidine on the level of uronium salt induced epimerization is less convincing than for solution couplings[41,90] and, moreover, the activation step often proceeds slowly.[91] Activation can be speeded up and the benefits of collidine maintained if a 1:1 mixture of collidine and DIPEA is used. Collidine has also been recommended as a base in uronium/guanidinium salt promoted couplings for the incorporation of the extremely racemization-prone derivatives Fmoc-Ser(X)-OH[92] and Fmoc-Cys(X)-OH.[93]

As described in the case of phosphonium salts (Section 3.7), uronium/guanidinium compounds are also often used in combination with additives. However, the effect of adding an equivalent of the corresponding *N*-hydroxy compound, HOAt or HOBt in the case of HATU and HAPyU or HBTU and TBTU mediated couplings, is poorly understood but is expected to depend both on the coupling reagent used and the coupling reaction under investigation. For fragment condensations carried out in homogeneous solution, additives clearly suppress epimerization (Table 3), however, the coupling yield may be diminished due to the reduced basicity of the system.[56]

In SPPS the advantages of using HOBt or HOAt as additives are less clear. For valine-valine couplings in automated SPPS[36] as well as coupling reactions involving Asn(Trt)[94] or C-terminal aspartylamide[95] derivatives a positive effect of the addition of 1 equivalent of HOBt on the reaction rate and suppression of the formation of cyano derivatives from C-terminal Asp(OX)-NH_2[95] has been described. On the other hand, HOAt addition showed no significant effects on HATU-mediated couplings of other amino acid derivatives. Curiously, the use of either HOBt or HOAt caused an increase in epimerization during HBTU- or HATU-mediated couplings of Fmoc-Phe-Ser(tBu)-OH onto resin-bound proline.[90] Nevertheless, the use of these additives has been recommended in SPPS[36,96–99] and also for segment condensations on solid phase.[100]

The HOAt additive has a remarkable influence on those uronium salts, which possess a relatively weak coupling efficiency and a relatively high tendency to epimerize C-terminal amino acid residues. Thus, by addition of one equivalent of HOAt to coupling reactions involving PfPyU both the reaction rate and the level of epimerization can be positively altered.[61] Similar effects have been demonstrated in coupling reactions mediated by the chloroamidinium compound CIP (Scheme 2). Although carboxy activation by CIP results in rapid formation of oxazolone which reacts slowly with the amino component, addition of HOAt or HOBt transforms the oxazolone rapidly into the corresponding highly reactive active ester.[26,96,101–103] Analogously, HOAt has been found to suppress racemization strongly

for references see p 576

when used in fluoroamidinium salt mediated couplings.[27,104] For uronium/guanidinium salts incorporating HOBt or HOAt, the activation step, resulting predominantly in the formation of the corresponding active ester, proceeds extremely rapidly. Although preactivation has been often recommended[33,35–37,66,98,99,105,106] the preactivation time should be kept to a minimum, especially for couplings of racemization-prone amino acid derivatives or peptide fragments, since on standing alone the activated species can give rise to several side reactions including epimerization.[30] Whether coupling is carried out in solution or on solid phase careful attention is required with regard to the preactivation time. Although in the case of some automated synthesizers the preactivation time is dictated by the instrument, in others and for manual syntheses it can be adjusted as desired. For example, a preactivation time of seven minutes for the HBTU/DIPEA-mediated reaction of Fmoc-Phe-Ser(tBu)-OH with resin-coupled proline leads to 40% of the LDL isomer, whereas the extent of epimerization is reduced to 12% if the reaction is carried out without preactivation.[90] The racemization of Fmoc-Cys(X)-OH derivatives[107,108] during HBTU- or HATU-mediated coupling is significantly reduced by avoiding the preactivation step.[93]

The solvent used during peptide synthesis influences both the reactivity of the coupling reagent and the level of racemization. Although DMF is the most commonly used solvent, coupling reactions involving uronium salts proceed in less polar solvents with a reduced risk of racemization.[55,57,90] Thus, for the synthesis of short peptides in solution the use of CH_2Cl_2 or $CHCl_3$ is recommended,[76,86,109,110] and CH_2Cl_2/DMF (1:1) mixtures are found to be useful for the reduction of epimerization during fragment couplings[88,90] or for the coupling of racemization-prone Fmoc-Cys(X)-OH.[93] However, the limited solubility of some of the uronium/guanidinium salts in less polar solvents allows only relatively low final coupling concentrations (e.g., 0.05 M for HBTU or HATU).[93] Mixtures of fluorinated alcohols (TFE, HFIP) with $CHCl_3$, successfully applied in solution-based fragment condensations,[97,111] have also been used in SPPS.[35,66,90,112] For HATU/DIPEA-mediated fragment condensations the level of epimerization was not influenced by the TFE/$CHCl_3$ solvent mixture[90] whereas the epimerization of a C-terminal Ala residue during TPTU-mediated segment condensation, amounting to 27% in NMP + 0.4 M LiCl, was drastically reduced (1%) when the reaction was performed in $CHCl_3$/TFE (9:1).[66] TFE is rather acidic and may consume additional base and coupling reagent, thus resulting in the formation of byproducts.[35,111,113] Although individual coupling reactions may proceed more rapidly in CH_2Cl_2 or $CHCl_3$ than in DMF,[55] the synthesis of larger peptides by the solid-phase technique is generally more efficient in DMF, presumably due to the greater structure-breaking effect of the more polar solvent. To overcome such aggregation-induced difficulties in the SPPS of difficult peptide sequences, solvents or solvent mixtures such as NMP,[97,98,114] DMSO,[115] DMF/DMSO,[35] DMA/NMP,[35] DMF/NMP,[116] and NMP/DMSO[114] have also been successfully applied in uronium/guanidinium salt mediated couplings of amino acid derivatives[114] or peptide fragments[35,116] (see also ref[112]).

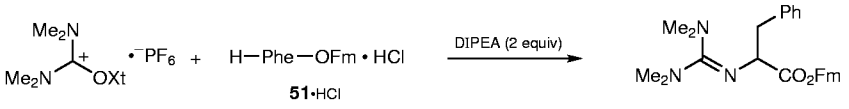
Uronium/Guanidinium Salt Induced Side Reactions

Unlike phosphonium salts, the uronium/guanidinium compounds may react with amino groups under the conditions commonly employed in peptide synthesis, forming guanidino-type side products (Scheme 5).

When HBTU is used in SPPS, a loss of reactive amino groups is noted when the reagent is added to the resin before the amino acid derivative which is to be activated.[117] In model experiments using TBTU and H-Phe-NH_2 the formation of a guanidino derivative by nucleophilic attack of the amino group on the positively charged carbon atom of the uronium salt has been demonstrated. The resulting tetramethylguanidinium derivative is stable to the reaction conditions of peptide synthesis and survives even the final treatment with TFA.[117]

In order to examine the extent of this side reaction, the relatively slow coupling of Fmoc-Deg-OH with H-Phe-OFm (**51**) in the presence of 2 equivalents of DIPEA and HBTU or HATU was analyzed. After 75 minutes, the desired dipeptide was formed in 85 and 94% yield, respectively, with <0.5% formation of the guanidino side products in either case.[20]

Scheme 5 Formation of Guanidino Derivatives Induced by HBTU (X = B) and HATU (X = A) in the Absence of a Carboxy Component[20]



Reagent	% Guanidino Formation	
	2 min	180 min
HBTU	<1	44
HATU	78	

To circumvent or minimize this side reaction, an excess of the reagent should also be avoided.[117] Further it is recommended that the amino acid to be activated is first added to the resin or alternatively, the amino acid and activator are first mixed prior to addition to the amino component. However, the increased risk of racemization caused by the preactivation process should also be considered.[90,93]

Although solution-based cyclization of linear peptides can be achieved with a slight excess of uronium salt without detectable N-terminal modification,[118] on-resin cyclizations may be accompanied by significant guanidino formation.[119–121] In such cases the use of phosphonium salts such as PyBOP or PyAOP (Section 3.7), which do not produce these side products, may be advantageous.

Epimerization occurs especially readily during fragment condensations and for couplings of racemization-prone amino acid residues such as Cys(X)[93,107,108] and Ser(X)[92] derivatives, *vide supra*. The use of HBTU has been shown to give higher levels of epimerization than DCC/HOBt in model couplings.[28,122] In analogy to phosphonium salt mediated couplings (Section 3.7), the use of HOAt-derived reagents (HATU, HAPyU) significantly reduces epimerization.[39,55–57,61,85,86,90,118,123]

Although 3-hydroxy-1,2,3-benzotriazin-4(3*H*)-one (HODhbt) is reported to be more effective than HOBt in reducing epimerization[213] allowing efficient segment condensations in solution,[62] the performance of HDTU or TDBTU (Table 1) is inferior to that of HATU or HAPyU, when used in SPPS,[56] probably because of side reactions involving the triazine moiety. Irreversible attack by the amino component with opening of the triazinone ring leads to chain termination.[216]

The use of less basic and sterically hindered amines [e.g., collidine[55,56] or 2,6-di-*tert*-butyl-4-(dimethylamino)pyridine[89]], the avoidance of preactivation, and the use of less polar solvent systems are additional appropriate strategies for the suppression of racemization. Except for fragment condensations in solution and couplings involving racemization-prone derivatives, the addition of HOXt to the reaction mixture has no pronounced effect on a uronium salt mediated racemization/epimerization, in contrast to the striking effects of HOAt on epimerization caused by PfPyU and haloamidinium salts such as CIP or TFFH (Section 3.8.1.2.1).

Other side reactions that are not specific for uronium/guanidinium-type reagents concern the modification of amino acid side chains. For example, the carboxamide group of asparagine and glutamine when left unprotected may be dehydrated by the coupling reagent to give the corresponding nitrile, as shown for carbodiimide/HOBt, BOP, and uronium salt (HBTU) couplings.[124] Dehydration of a C-terminal aspartylamide peptide during side-chain

for references see p 576

cyclization with HBTU has been described. This side reaction can be prevented by the addition of one equivalent of HOBt.[95] Chloroamidinium salts such as **31** lead to the formation of large amounts of Z-Ala(CN)-Gly-OEt when Z-Asn-OH and H-Gly-OEt are coupled in DMF/DIPEA for 3 hours. Addition of HOBt to the reaction mixture or use of the uronium salt TBPyU, reduce the formation of β-cyanoalanine drastically.[24]

The incorporation of unprotected side-chain derivatives of Asn and Gln derivatives has also been investigated using HATU. In the SPPS of ACP no cyano derivatives of Asn and Gln were detected, however, the incorporation of Asn was not complete, possibly due to the low solubility of Fmoc-Asn-OH in DMF.[91] Due to this low solubility of Fmoc-Asn-OH and the unfavorable effects of unprotected side chains on the aggregation behavior of the growing peptide chain, the use of side-chain protection is recommended (Section 2.3.3).[66] For reasons still not clear, even side-chain protection of Asn does not fully eliminate deficiencies in phosphonium- or uronium-mediated couplings. Thus, the incorporation of Fmoc-Asn(Trt)-OH is more satisfactory if one equivalent of HOAt is present during HATU-mediated couplings.[94]

In cases where loading of the first amino acid onto hydroxymethyl resins was not complete, HBTU-activated amino acid derivatives were shown to react also with the remaining hydroxy groups on the resin.[36] To avoid C-terminal deletion byproducts, a capping step is required.[36] Accordingly, in HBTU/HOBt-mediated peptide bond formation the hydroxy groups of serine and threonine should be protected to prevent O-acylation.[125] Although *tert*-butyl protection is normally a proper choice, peptides containing a high proportion of Ser and Thr can present problems in Fmoc/HBTU chemistry due to incomplete coupling reactions, probably caused by Ser(tBu)/Thr(tBu) induced secondary structure effects. Substitution of *tert*-butyl by trityl side-chain protection inhibits this effect.[126] However, the direct incorporation of side-chain unprotected serine using TBTU/HOBt/DIPEA has also been successfully accomplished,[127] and the resulting peptides were used as precursors for the corresponding phosphopeptides.[127,128] In solution, the successful HBTU-mediated synthesis of dipeptides using side-chain unprotected serine derivatives has been described[110] (see compound **54**, Scheme 6).

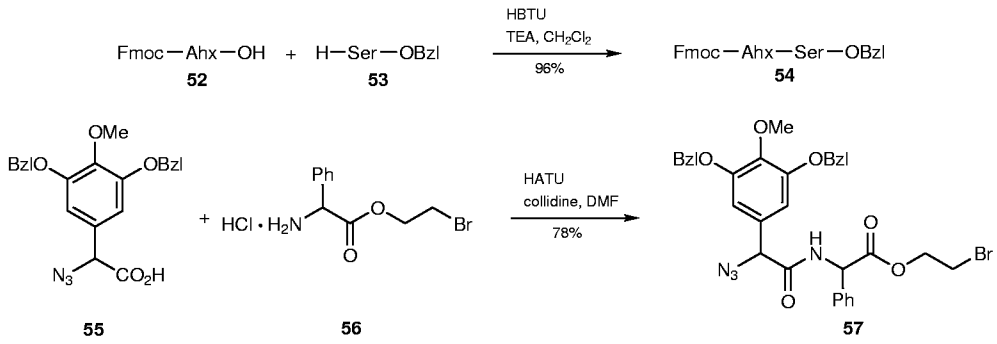
The use of unprotected α-hydroxy acids may also be compatible with the preparation of depsipeptides. However, attempts to couple L-α-hydroxyisovaleric acid and benzyl phenylalaninate using TBTU resulted in the formation of considerable amounts of the hydroxycarbamoylated compound,[129] in contrast to the smooth reaction mediated by PyBOP. Transfer of the dimethylcarbamoyl moiety of uronium/guanidinium salts appears to be restricted to α-hydroxy acids and has not been observed in coupling reactions between Fmoc-Ser-OH, Fmoc-Thr-OH, or L-3-hydroxybutyric acid and H-Phe-OBzl in the presence of TBTU/NMM in CH_2Cl_2.[129]

3.8.1.2.1 Uronium/Guanidinium Salt Mediated Peptide Synthesis in Solution

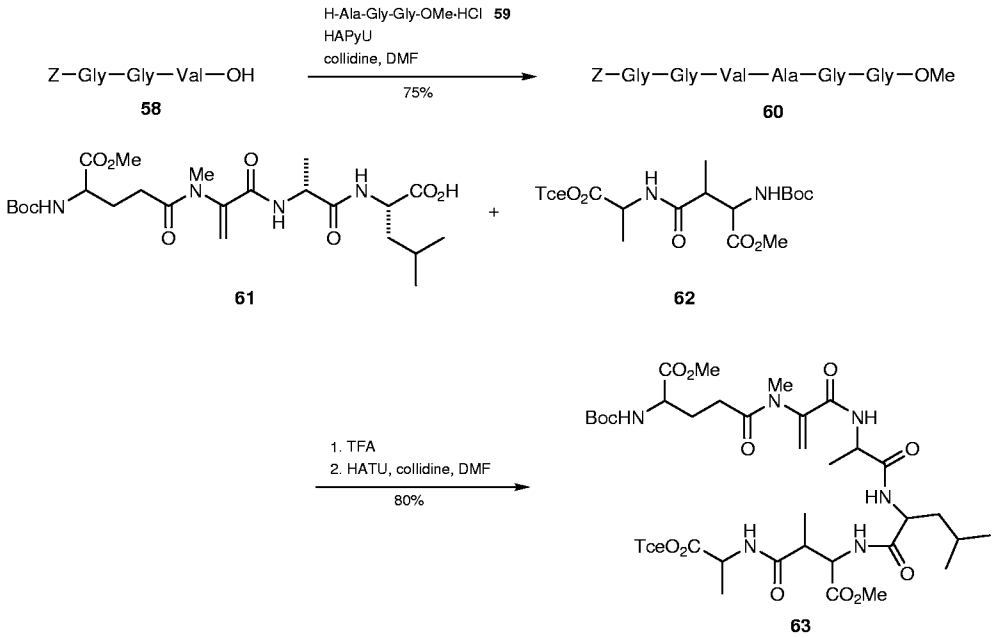
The most commonly used uronium/guanidinium salts HBTU, TBTU, and HATU may be equally effective in the stepwise synthesis of peptides in solution.

Although in uronium/guanidinium-mediated peptide bond formations the hydroxy groups of serine and threonine should be protected, the synthesis of short peptides can be performed in solution using side-chain unprotected serine derivatives, even in the presence of TEA, e.g. to give **54** (Scheme 6). In order to minimize epimerization, which may occur readily during couplings of racemization-prone amino acid residues (e.g., acid **55**, Scheme 6) and in segment condensations (Scheme 7), HOAt-derived coupling reagents in the presence of collidine can often be successfully applied, as also discussed in detail in Section 3.8.1.2.

Scheme 6 HBTU- and HATU-Mediated Coupling of Amino Acid Derivatives in Solution[85,110]



Scheme 7 Segment Condensation in Solution Mediated by HOAt-Derived Guanidinium Salts HAPyU and HATU[55,87]



Fmoc-Ahx-Ser-OBzl (54):[110]

To a soln of H-Ser-OBzl·HCl (**53**·HCl; 0.50 g, 2.16 mmol) in CH_2Cl_2 (20 mL), was added Fmoc-Ahx-OH (**52**; 0.84 g, 2.38 mmol), HBTU (1.20 g, 3.23 mmol), and TEA (1.2 mL, 8.64 mmol). The reaction was stirred at rt overnight, after which it was poured into 2 M HCl (3 mL). The organic layer was separated and the aqueous layer was extracted with CH_2Cl_2 (2 × 10 mL). The combined organic layers were washed with sat. aq $NaHCO_3$ (3 mL), brine (5 mL), dried, and concentrated. The crude product was purified by flash chromatography (hexane/EtOAc 1:3) to give **54**; yield: 1.10 g (96%); $[\alpha]_D^{20}$ +5.91 (*c* 1.10, $CHCl_3$).

***N*-[(*R*)-2-Azido-2-[3,5-bis(benzyloxy)-4-methoxyphenyl]acetyl]-D-phenylglycine 2-Bromoethyl Ester (57):**[85]

To a flask containing acid **55** (435 mg, 1.04 mmol), amine hydrochloride **56** (368 mg, 1.25 mmol, 1.2 equiv), and HATU (475 mg, 1.25 mmol, 1.2 equiv) at 0 °C was added DMF (10 mL) followed by collidine (412 µL). The mixture was stirred for 2.5 h at 0 °C and then for 2.5 h at rt, diluted with EtOAc (150 mL), and the organic layer was washed successively with 1 M HCl (3 × 50 mL), aq $NaHCO_3$ (2 × 50 mL), and brine (50 mL). It was dried (Na_2SO_4) and the solvent was removed under reduced pressure. Chromatography (silica gel, acetone/$CHCl_3$ 1:99) of the crude product gave the purified dipeptide **57**; yield: 530 mg (78%); mp 147–148 °C; $[\alpha]_D^{25}$ −110 (*c* 1.1, $CHCl_3$).

for references see p 576

Z-Gly-Gly-Val-Ala-Gly-Gly-OMe (60); Fragment Condensation with HAPyU/Collidine:[55]
The tripeptides Z-Gly-Gly-Val-OH (**58**; 147 mg, 0.37 mmol), H-Ala-Gly-Gly-OMe · HCl (**59**; 106 mg, 0.33 mmol) and HAPyU (159 mg, 0.35 mmol) were dissolved in DMF (1 mL). The mixture was cooled in an ice bath and collidine (133 μL, 1 mmol) was added. Stirring was continued in the ice bath for 2 h and than at rt for 3–4 h. The crude hexapeptide methyl ester **60** contains less than 0.1% of the corresponding DL isomer as determined by HPLC. The solvent was removed and the crude peptide purified by chromatography (silica gel, MeOH/$CHCl_3$/AcOH 3:7:0.1) and subsequent precipitation (MeOH/Et_2O); yield: 156 mg (75%); mp 199–204 °C (dec).

Boc-D-*iso*-Glu(OMe)-(Me)ΔAla-D-Ala-L-Leu-D-*erythro*-β-Me-*iso*-Asp(OMe)-L-Ala-OTce (63):[87]
N^{α}-Boc-protected dipeptide **62** (314 mg, 0.68 mmol) was treated with TFA (2 mL) for 30 min and concentrated, and the residue was partitioned between EtOAc and sat. $NaHCO_3$. The aqueous layer was extracted with two additional portions of EtOAc, and the combined organic extracts were washed with brine, dried (K_2CO_3), filtered, and concentrated. To the residue was added the tetrapeptide **61** (431 mg, 0.82 mmol) and toluene (4 mL). The mixture was then concentrated and the residue was dissolved in DMF (2 mL). To the soln was added collidine (247 mg, 2.04 mmol) and, after cooling to 0 °C, HATU (310 mg, 0.82 mmol). The soln was stirred for 1 h at 0 °C and for 19 h at rt. The mixture was partitioned between 50% sat. citric acid (20 mL) and Et_2O/EtOAc (1:1, 20 mL). The organic layer was washed with sat. $NaHCO_3$ (15 mL) and the aqueous layers were extracted with Et_2O/EtOAc (1:1, 2 × 15 mL). The combined organic layers were washed with brine, dried ($MgSO_4$), filtered, and concentrated to give crude product (729 mg). Chromatography (silica gel, iPrOH/CH_2Cl_2 5:95 to elute the nonpolar impurities, then iPrOH/CH_2Cl_2 10:90 to elute the product) gave **63**; yield: 476 mg (80%).

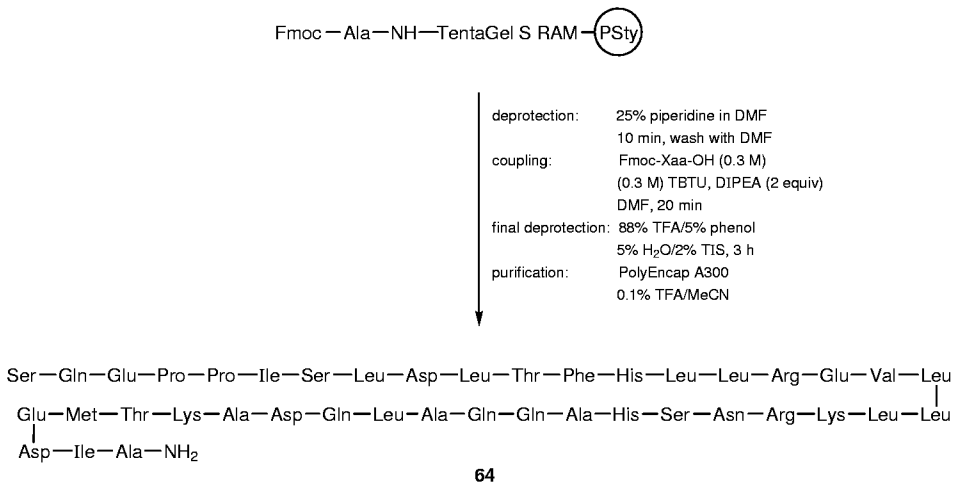
3.8.1.2.2 Solid-Phase Peptide Synthesis Using Uronium/Guanidinium Salts

By analogy to the phosphonium salts (Section 3.7), the related uronium/guanidinium salts have also gained acceptance in solid-phase synthesis since the first of these derivatives was introduced. Unlike the phosphonium analogues, however, the latter react with amino groups when used in excess, forming guanidino derivatives[117] (Section 3.8.1.2). Caution is, therefore, advised and the use of excess uronium salt should be avoided.[117,119]

The following experimental procedures describe the solid-phase synthesis of peptides by means of Fmoc chemistry and TBTU activation (Scheme 8), as well as by HBTU-mediated coupling of Boc amino acid derivatives and in situ neutralization (Scheme 9). Segment condensations on solid phase with C-terminal glycine or proline residues can be easily achieved, even on a multigram scale (Scheme 10). Segment condensations involving the activation of other amino acids require the careful optimization of reaction conditions in order to suppress extensive epimerization (see Scheme 11 and Section 3.8.1.2).

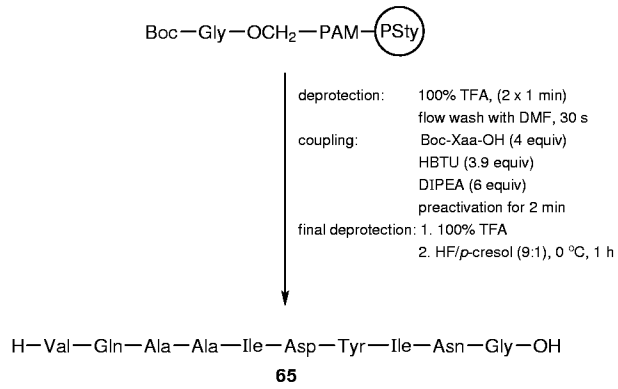
Although the following examples are limited to solid-phase synthesis of peptides, the use of HBTU, TBTU, and HATU also allows the successful application to small proteins by either stepwise or convergent methods.[37,79–83,97–100,116,130–133]

Scheme 8 TBTU-Mediated SPPS of Ovine Corticotropin Releasing Factor (oCRF)[134]



64

Scheme 9 SPPS of ACP (65–74) by the HBTU In Situ Neutralization Protocol[37]

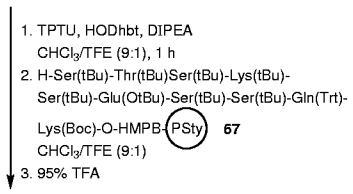


65

Scheme 10 TPTU-Mediated Segment Condensation of **66** with **67** To Give **68** on Solid Phase in $CHCl_3$/TFE[66]

Fmoc—Lys(Boc)—Arg(Pmc)—Arg(Pmc)—Arg(Pmc)—Leu—Ala—Ser(tBu)—Leu—Arg(Pmc)—Ala—OH

66

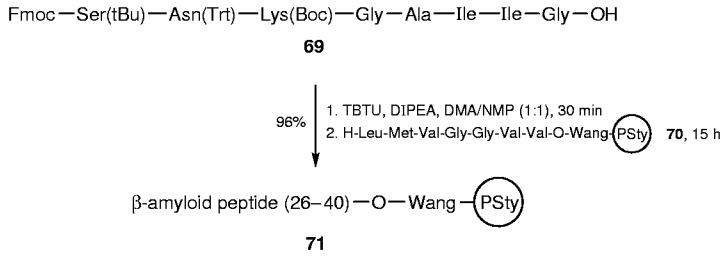


[Ala^{235}]-7OK S6 Kinase (230–249)

68

for references see p 576

Scheme 11 TBTU-Mediated Segment Coupling of β-Amyloid Peptides **69** and **70** on Solid Phase[35]



TBTU-Mediated SPPS of oCRF 64:[134]
CRF was synthesized automatically using Fmoc chemistry in the continuous flow mode. Synthesis was carried out on 4-[(2,4-dimethoxyphenyl)aminomethyl]phenoxyacetamido resin (0.5 g, 0.22 mmol · g^{-1}) using N^{α}-Fmoc-protected amino acid derivatives [0.3 M; protecting groups: Asn(Trt), Gln(Trt), His(Trt), Arg(Pmc), Asp(OtBu), Glu(OtBu), Lys(Boc), Thr(tBu), Ser(tBu)] and TBTU (0.3 M) as coupling reagent in the presence of DIPEA (2 equiv) in DMF. Single couplings were allowed to proceed for 20 min, N-terminal deblocking was carried out with 25% piperidine in DMF for 10 min, and all washes were made with DMF. Final cleavage from the resin and deprotection of side-chain functionalities was achieved by a mixture of 88% TFA/5% phenol/5% H_2O/2% TIS for 3 h. Purification of 100-mg samples was carried out by preparative HPLC (PolyEncap A300, 10 μm, 250 × 20 mm i.d.) using an MeCN/H_2O/0.1% TFA solvent system (linear gradient 25–45% MeCN in 70 min), to give **64** with a purity of >98% (HPLC); yield: 16.4 mg (13%). MALDI-MS *m/z*: 4671 $[M + H]^+$.

Manual SPPS of ACP (65–74) 65 **by the HBTU In Situ Protocol:**[37]
Synthesis was carried out on a 0.2 mmol scale. The N^{α}-Boc group was removed by treatment with 100% TFA (2 × 1 min) followed by a flow wash with DMF (30 s). Boc amino acids {0.8 mmol; side-chain protection: Asp(OBzl), Tyr[Z(Br)]; side-chain unprotected Boc-Asn-OH was coupled in the presence of 0.8 mmol of HOBt} were activated with HBTU (0.76 mmol) and DIPEA (1.2 mmol) for 2 min and added to the protonated resin. Coupling times were 10 min throughout without any double coupling. Coupling yields were typically >99.9%. The N^{α}-Boc group was removed from the peptide-resin by treatment with 100% TFA (2 × 1 min). The partially deprotected peptide-resin was dried under reduced pressure after washing with DMF and CH_2Cl_2 and treated with HF/*p*-cresol (9:1 v/v, 0 °C, 1 h). The HF was removed under reduced pressure at 0 °C and the crude peptide **65** precipitated and washed with Et_2O, then dissolved in dilute AcOH and lyophilized. HPLC analysis showed the purity of the crude target peptide **65** to be >90%. Des-Val-ACP (65–74) was not detectable.

β-Amyloid Peptide (26–40) 71:[35]
Peptide **69** (3.6 g, 2.62 mmol, 2 equiv) was dissolved in DMA/NMP (1:1, 60 mL). TBTU (0.84 g, 2.62 mmol, 2 equiv) and DIPEA (0.684 mL, 3.93 mmol, 3 equiv) were added, and the mixture was gently agitated for 30 min and then added to the peptide-resin **70** (synthesized on Fmoc-Val-O-Wang resin, loading 0.6 mmol · g^{-1}) previously washed with the coupling solvent mixture. The coupling was allowed to proceed overnight, with the yield as determined by HPLC analysis after TFA cleavage being over 96%.

3.8.1.2.3 Coupling of Sterically Hindered Amino Acid Derivatives

Both N^{α}-alkyl and $C^{\alpha,\alpha}$-dialkyl groups greatly hinder peptide bond formation under certain conditions. For *N*-alkyl amino acids, bond formation is generally not difficult when only the carboxy component is N-alkylated, and these reactions proceed well under standard coupling conditions. On the other hand, due to steric hindrance the reaction rate can be strikingly diminished when the amino component or both components are N-alkylated.[23,40,135] Moreover, racemization readily occurs during the activation of N-protected *N*-methyl amino acids via oxazolonium ions.[40,136] Due to the formation of oxazolonium intermediates, Boc protection is a poor choice for the acylation of secondary amines, because Boc-oxazolonium ions decompose to *N*-carboxyanhydride derivatives much more readily than the corresponding Z- or Fmoc-protected derivatives.[40,137]

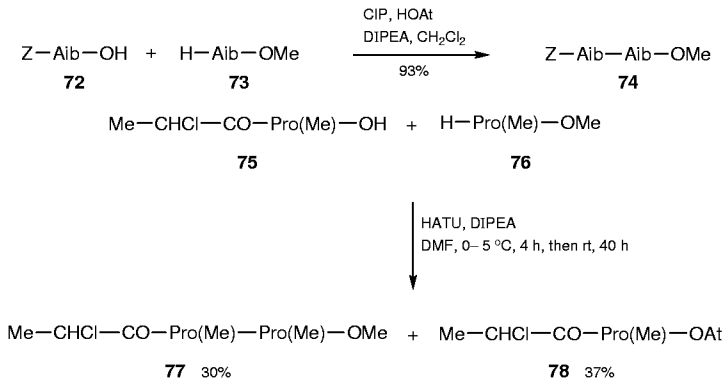
Numerous studies have demonstrated that coupling reagents which generate OBt esters as intermediates do not allow the formation in high yield of peptide bonds between amino acids which are both N-methylated,[23,40] and DCC-mediated couplings are more effective in the absence of HOBt.[23] Therefore, activation methods involving more reactive intermediates such as pivalic mixed anhydrides, BOP-Cl (Section 3.7), protected *N*-methyl amino acid chlorides (Section 3.3), BroP, and PyBroP (Section 3.7) as well as aminium chlorides such as compound **31**, have been applied more successfully to these difficult couplings. This is demonstrated, for example, by the synthesis of the dipeptide Z-(Me)Val-(Me)Val-OMe using HPyClU[23] or the corresponding tetrafluoroborate.[24] Both for solution and solid-phase syntheses, HOAt-derived coupling reagents exhibit good reactivity toward *N*-methyl amino acid residues,[87,88,94,138–144] in contrast to their HOBt counterparts and halogenated phosphonium and uronium salts. As a representative example, HATU or carbodiimide/HOAt gave nearly quantitative yields in the coupling of Fmoc-MeLeu-OH onto H-(Me)Val-Ala-resin and considerably better couplings for solid-phase syntheses of the cyclosporin segments H-Abu-Sar-(Me)Leu-Val-(Me)Leu-Ala-OH and H-D-Ala-(Me)Leu-(Me)Leu-(Me)Val-Phe-Val-OH (Scheme 13), than BOP, PyBroP, BOP-Cl, and HBTU.[94,140,144] However, hindered secondary amines often give incomplete reaction even with HATU under double coupling conditions.[139,140,145]

As is the case for *N*-methyl amino acids, the coupling of $C^{\alpha,\alpha}$-disubstituted amino acids such as Aib and C^{α}-ethylalanine is also complicated by steric hindrance. As expected, these couplings are especially difficult when both the carboxy and the amino component are $C^{\alpha,\alpha}$-disubstituted. During the activation of urethane-protected Aib, oxazolone formation is facilitated by the geminal dialkyl effect which, especially in the case of Boc-protection, may lead to *N*-carboxyanhydride formation (see ref[40] for review). Although a number of coupling reagents are able to generate Aib-Aib peptide bonds[40] in poor to moderate yields in solution, $C^{\alpha,\alpha}$-disubstituted amino acid fluorides are the reagents of choice for both solution and solid-phase syntheses[40,146–148] (Section 3.3). HOAt-derived guanidinium salts, such as HATU, also give excellent results for Aib-Aib couplings, as demonstrated by the solid-phase synthesis of H-Tyr-Aib-Aib-Phe-Leu-NH_2.[94] Although fluoride activation proved to be the most successful method for coupling Fmoc-Aib-OH to the resin-bound sequence H-Aib-Tyr-Cha-D-Glu, HATU was the most efficient among the various uronium/guanidinium and phosphonium salts tested.[147,149] Even coupling of the extremely hindered $C^{\alpha,\alpha}$-dicyclopropylglycine (Dcp) with itself using HATU (or carbodiimide/HOAt) in CH_2Cl_2 gave 40% of the dipeptide Z-Dcp-Dcp-OMe although 9 days were required,[150] and coupling of two C^{α}-methylproline units provided 39% of the target dipeptide.[76] Interestingly, in the latter case a considerable amount of the unreacted azabenzotriazolyl ester intermediate could be isolated from the reaction mixture after a coupling time of 44 hours[76] (Scheme 12).

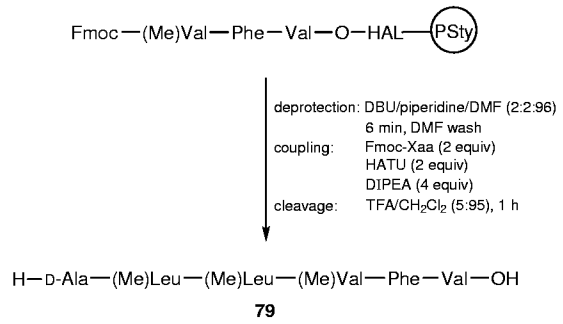
$C^{\alpha,\alpha}$-Dialkyl amino acids have also been successfully coupled by use of CF_3-HBTU[58] and of chloroimidazolidium reagents such as CIP when rapid oxazolone formation is prevented by addition of HOAt. Employing the CIP/HOAt system, both Fmoc-Aib-Aib-OMe[103], alamethicin[101] and C^{α}-methylcysteine-derived intermediates for the synthesis of mirabazole and dolastatin 15[96] have been obtained. Due to their superiority in forming peptide bonds involving sterically hindered amino acid derivatives, HATU and related HOAt-derived coupling reagents have become widely accepted in the synthesis of peptidomimetics,[76,115,139,145,151–156] for solid-phase syntheses of peptidyl *N*-alkyl amides,[157] *N*-alkyl hydroxamic acid derivatives,[158] or peptides containing metal-binding amino acids,[159] and in solid-phase combinatorial chemistry[160] in order to achieve difficult coupling reactions such as those involving $C^{\alpha,\alpha}$-disubstituted[161–164] and *N*-alkyl amino acids,[165–172] as well as substituted benzoic acids.[173–178]

for references see p 576

Scheme 12 Coupling of C^{α}-Alkyl Amino Acids with CIP/HOAt[101] and HATU[76]



Scheme 13 HATU-Mediated SPPS of a Cyclosporin-like Sequence[94]



Z-Aib-Aib-OMe (74):[101]
To a soln of Z-Aib-OH **72** (2.0 g, 8.4 mmol) in CH_2Cl_2 (35 mL) were added HOAt (0.57 g, 4.2 mmol), DIPEA (5.9 mL, 33.7 mmol), CIP (2.4 g, 8.4 mmol), and HCl · H-Aib-OMe (**73** · HCl) (1.4 g, 9.2 mmol). The mixture was stirred for 60 min at 25 °C. The product was dissolved in EtOAc and the extract was washed with 5% citric acid, 5% $NaHCO_3$, and brine. The organic layer was dried (Na_2SO_4) and the solvent was removed under reduced pressure. The residue was purified by recrystallization (EtOAc/hexane) to give the product as a white powder; yield: 2.7 g (93%); mp 107–108 °C.

Methyl (2*S*)-*N*-[(2*S*)-*N*-[(2*S*)-2-Chloropropanoyl]-2-methylprolyl]-2-methylprolinate (77):[76]
A soln of (2*S*)-*N*-[(2*S*)-2-chloropropanoyl]-2-methylproline (**75**) (68 mg, 0.31 mmol), methyl (2*S*)-2-methylprolinate hydrochloride (**76** · HCl) (37 mg, 0.21 mmol), and DIPEA (0.11 mL, 0.63 mmol) in dry DMF (1 mL) under N_2 at 0 °C was treated with HATU (120 mg, 0.32 mmol) and the mixture was stirred at 0 °C for 4 h and then at rt for 40 h. The soln was diluted to 30 mL with EtOAc and then washed with H_2O, 0.5 M aq $NaHCO_3$, and brine. The organic phase was dried ($MgSO_4$) and the solvent was removed under reduced pressure. The residual oil was purified by chromatography (silica gel, EtOAc/light petroleum 3:2) to give two products, the peptide **77** as a white solid (21 mg, 30%) and the 7-azabenzotriazolyl ester **78** as a colorless gum (38 mg, 37%). Analytical data for **77**: mp 124–126 °C; R_f 0.45 (EtOAc); $[\alpha]_D$ −42.5 (*c* 0.8, MeOH).

H-D-Ala-(Me)Leu-(Me)Leu-(Me)Val-Phe-Val-OH (79):[94]
The synthesis was performed automatically starting with Fmoc-Val-O-HAL-PEG-PS-resin and using the following protocol: Fmoc deprotection was carried out with DBU/piperidine/DMF (2:2:96) for 6 min, followed by DMF washing (7 min); couplings were performed for 2 h using Fmoc amino acid (2 equiv), HATU (2 equiv) and DIPEA (4 equiv), except in the case of Fmoc-Phe-OH and Fmoc-MeVal-OH which gave quantitative couplings within 10 min. After cleavage of the peptide from the resin via treatment with TFA/CH_2Cl_2 (5:95) for 1 h, the crude peptide **79** was obtained in a purity of 85% as shown by HPLC analysis.

3.8.1.2.4 Uronium/Guanidinium Salts in Peptide Cyclization

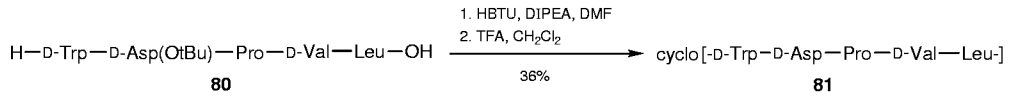
In the synthesis of homodetic cyclic peptides, the readiness of an open-chain precursor to cyclize depends on the size of the ring to be closed, and usually no difficulties arise for the cyclization of peptides containing seven or more amino acid residues (see Section 6.8). Although ring closure with hexa- and pentapeptides is more hampered, the ease of cyclization is often enhanced by the presence of glycine or of turn-structure-inducing amino acids such as *N*-methyl or D-amino acids. Such cyclization-prone sequences can be cyclized by means of a variety of coupling procedures (Section 6.8), including activation with phosphonium (Section 3.7) and uronium/guanidinium salts. Due to their high reactivity, both phosphonium and uronium/guanidinium salts tend to speed up the cyclization process,[179–194] although in some cases the less-reactive DPPA or pentafluorophenyl ester give more favorable results in terms of yield and purity.[195,196] As an example, HBTU was used in the preparation of the highly potent endothelin receptor antagonist (ET_A) **81**, cyclo[-D-Trp-D-Asp-Pro-D-Val-Leu-], by head-to-tail cyclization of the linear precursor **80**, H-D-Trp-D-Asp(OtBu)-Pro-D-Val-Leu-OH (Scheme 14).[197] Although the multigram cyclization was performed under relatively concentrated conditions (27 mM), cyclodimerization or oligomerization was not observed. Clearly, high dilution is not required when the linear precursor adopts a conformation that favors efficient ring closure. Other examples involve linear, modified sequences of gonadotropin-releasing hormone, such as H-Nal-D-Cpa-D-Pal-Gln-Tyr-D-Arg-Leu-Arg-Pro-Lys(Ac)-OH which give exclusively the cyclomonomer, even at extremely high peptide concentration (0.1 M).[198]

Especially for linear penta- and hexapeptides that do not contain turn-structure-inducing amino acid residues, the cyclization reaction may be an inherently improbable or slow process and the increased lifetime of the intermediate activated linear peptide provides an opportunity for increased epimerization at the C-terminal residue. In such cases[47,61,109,118,199–209] and in other cyclizations involving a high risk of epimerization, the epimerization level can be minimized by use of HOAt-derived coupling reagents such as HATU or HAPyU. The cyclization proceeds extremely rapidly, with about 90% of the desired product being formed within 1 minute after addition of the coupling reagent.[118] However, for certain systems even the most highly effective guanidinium salts are not able to shift the conformationally driven ratio of cyclomonomer to various cyclooligomers. Thus, for example, the sequences H-Arg-Pro-Ala-Val-Tyr-OH and H-Arg-Lys(Ac)-Ala-Pro-Tyr-OH gave exclusively cyclooligomers when cyclization was effected by means of HAPyU at 1 mM peptide concentration.[200,210]

Uronium/guanidinium salt mediated cyclizations may also be performed on solid supports.[47,119–121,179–182,193,211,212] However, in contrast to the case involving phosphonium salts (Section 3.7), uronium/guanidinium compounds tend to modify the amino group when used in excess[119–121] (Section 3.8.1.2). These limitations need to be considered when uronium/guanidinium salts are employed for peptide cyclization, especially when used under solid-phase conditions.[119–121] For a detailed discussion of the synthesis of cyclic peptides in general, see Section 6.8.

for references see p 576

Scheme 14 HBTU-Mediated Head-to-Tail Cyclization[197]



80 **81**

Cyclo[-D-Trp-D-Asp-Pro-D-Val-Leu-] (81); Cyclization with HBTU:[197]

A soln of H-D-Trp-D-Asp(OtBu)-Pro-D-Val-Leu-OH (**80**) (4.00 g, 7.0 mmol) in DMF (20 mL) was added dropwise to a soln of HBTU (3 equiv) and DIPEA (6 equiv) in DMF (200 mL) at rt over a period of 30 min. The cyclization was complete after an additional 30 min, as monitored by HPLC. The soln was concentrated and the crude protected cyclic peptide was treated with 50% TFA in CH_2Cl_2 for 30 min to yield the crude cyclopeptide **81** which was purified in 3×1.2 g batches on a Vydac C18 column (5.2×25.0 cm) by eluting with a linear gradient of A (0.1% TFA in H_2O) and B (0.1% TFA in MeCN), from 70% A to 40% A over 129 min (30.0 mL · min^{-1}). Fractions containing pure product were combined, concentrated and lyophilized; yield: 2.4 g (36%); white powder (purity >98%, HPLC), which was converted quantitatively into its monosodium salt.

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3.9 Miscellaneous Reactions

N. Schaschke, S. Alefelder, H.-J. Musiol, and L. Moroder

In addition to the most common coupling methods described in Sections 3.1–3.8, various alternative procedures have been proposed which either are used only for specific purposes or have not yet received due attention. For example, the oxazol-5(4*H*)-one coupling method is exclusively applied to the activation of C^{α}-tetrasubstituted amino acids and particularly of related *N*-peptidyl derivatives in segment coupling steps (see Vol. E 22c, Section 10.3.2). Conversely, 2-morpholinoethyl isocyanide,[1] although highly promising in terms of both peptide yields and epimerization in segment condensations,[2] has not found general application. More interest is focused on organophosphorus reagents, which are increasingly utilized both for synthesis in solution and on solid support in the case of sterically hindered amino acids (see Section 3.9.1). Recently, derivatives of triazines have been proposed for peptide synthesis (see Section 3.9.2), whereas di-*tert*-butyl dicarbonate (Boc_2O), in addition to representing the reagent of choice for Boc-protection of amine functionalities[3] (see Section 2.1.1 and 2.1.2), can be advantageously applied to the esterification and amidation of amino acid derivatives (see Section 3.9.3).

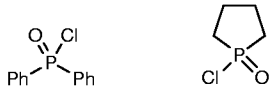
3.9.1 Organophosphorus Reagents

In addition to phosphonium salts, which are widely used for the activation of urethane-protected amino acids in peptide synthesis (see Section 3.7), other organophosphorus reagents derived from phosphinic, phosphonic, and phosphoric acids have proved to be efficient coupling reagents.

3.9.1.1 Phosphinic Acid Based Reagents

Mixed carboxylic–phosphoric anhydrides derived from *O,O*-dialkyl and *O,O*-diaryl phosphoric acids have been proposed as coupling reagents,[4,5] but have not found general application. Conversely, mixed carboxylic–phosphinic anhydrides[6–8] have been successfully applied as coupling reagents in solution[7] and solid-phase[9] synthesis. The anhydrides are prepared from phosphinic chlorides and show very low rates of disproportionation compared to aminolysis at low temperatures (−10 to 0 °C); thus acylation proceeds without formation of the racemization prone oxazolone intermediate.[7] Correspondingly, this type of mixed anhydride could also be applied in segment condensations.[10–12] Among the phosphinic chlorides used to prepare the mixed anhydrides, the readily available diphenylphosphinic chloride (Dpp-Cl, **1**)[13] and 1-chlorophospholan-1-one (Cpt-Cl, **2**)[14] (Scheme 1) are the most advantageous reagents;[7,14] the latter is preferred from a preparative point of view due to its better solubility properties. However, because of the tedious preparation of Cpt-Cl (**2**)[13] the method has not found general application, although the high reactivity of these mixed phosphinic anhydrides allows sterically hindered couplings in good yields as well documented by the synthesis of various cyclosporin analogues.[15,16]

Scheme 1 Structure of Diphenylphosphinic Chloride (Dpp-Cl, **1**)[13] and 1-Chlorophospholan-1-one (Cpt-Cl, **2**)[14]



1

2

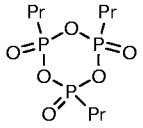
for references see p 587

Coupling in Solution with 1-Chlorophospholan-1-one (Cpt-Cl, 2); General Procedure:[12]
To an N^{α}-protected amino acid (1.2 equiv) in CH_2Cl_2 or DMF was added at −10 °C NMM (1.2 equiv) followed by Cpt-Cl (**2**; 1.2 equiv). After 10 min, the amino component was added (1 equiv) in the minimum volume of DMF. The mixture was kept for 1 h at −10 °C and then allowed to warm to rt. Upon completion of the reaction, the solvent was removed and the residue worked up according to standard procedures.

3.9.1.2 Phosphonic Acid Based Reagents

Alkylphosphonic anhydrides (Scheme 2) are commercially available reagents that are conveniently used in one-pot procedures for stepwise synthesis or segment condensations in solution.[17,18] The commercially available propane-1-phosphonic anhydride (T3P, **3**) is highly soluble in aprotic organic solvents, but couplings are also performed in biphasic systems such as ethyl acetate/water. In segment condensations under carefully optimized conditions racemization occurs at levels not superior to those observed by the DCC/additives method.[18] A comparative study of the effectiveness of T3P (**3**) and HOAt-derived coupling reagents in peptide cyclization reactions showed that the phosphonic anhydride is a superior reagent when sterically hindered amino acids are at the cyclization site.[19] Its application in a larger scale cyclization of an RGD-pentapeptide has been reported.[20] Low levels of racemization were also observed in esterification of Fmoc-protected amino acids to 4-alkoxybenzyl alcohol resin using T3P (**3**).[21]

Scheme 2 Structure of Propane-1-phosphonic Anhydride (T3P, **3**)[17]



3

Coupling with Propane-1-phosphonic Anhydride (T3P, 3); General Procedure:[17]
To a 10–20% soln or suspension of amino component (0.1 mol of suitably protected amino acid or peptide) and carboxy component (0.1 mol of suitably protected amino acid or peptide) in DMF at 0 °C was added 4-ethylmorpholine (60 mL, 0.47 mol) followed by a 50% T3P (**3**) in CH_2Cl_2 soln (88 g, 0.14 mol). After 1 h stirring at 0 °C and 24 h at rt, the mixture was concentrated and the product precipitated with sat. $NaHCO_3$. The solid was washed with citric acid soln and H_2O. Alternatively the crude product was partitioned between EtOAc and H_2O and then worked up in standard manner by basic, acid, and neutral washings.

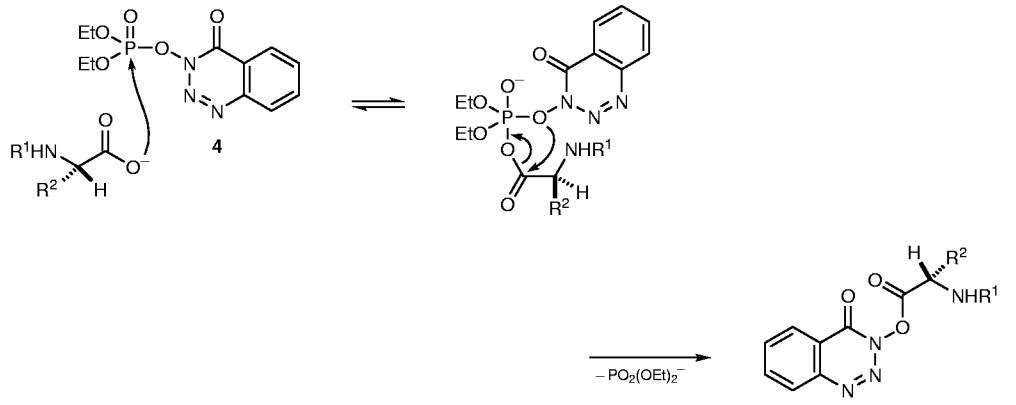
***c*[Asp(OtBu)-D-Phe-Lys(Boc)-Arg(Pbf)-Gly-]:**[20]
The solvent was removed from the soln of the crude linear peptide obtained by cleavage of H-Asp(OtBu)-D-Phe-Lys(Boc)-Arg(Pbf)-Gly-trityl-resin (loading: 17.60 mmol) with AcOH/TFE/CH_2Cl_2 (1:1:3; 250 mL) at rt for 1 h and the obtained acetate salt dissolved in CH_2Cl_2 (200 mL) was added slowly to a soln of 50% T3P (**3**) in EtOAc (54.24 mL), TEA (63.2 mL), and DMAP (200 mg) in CH_2Cl_2 (8 L). After 12 h, the mixture was concentrated and purified by flash chromatography (MeOH/EtOAc, 1:10); yield: 17.23 g (97%).

3.9.1.3 Phosphoric Acid Based Reagents

Of the derivatives of phosphoric acid, diphenyl phosphorazidate (diphenylphosphoryl azide, DPPA)[22] is largely used in azide coupling reactions (see Section 3.1). Other reagents like diethylphosphoryl cyanide,[23] 3-(diethoxyphosphoryl)benzoxazol-2(3*H*)-one,[24] 3-(2-oxo-1,3,2-dioxaphosphorinan-2-yl)benzoxazol-2(3*H*)-one,[24] 3-[(2-oxo-1,3,2-dioxaphosphorinan-2-yl)oxy]benzotriazin-4(3*H*)-one,[25] and benzotriazol-1-yl diethyl phosphate,[25] although useful reagents for synthesis in solution, have not found larger application. Conversely, 3-(diethoxyphosphoryloxy)-1,2,3-benzotriazin-4(3*H*)-one (DEPBT, **4**)[24] which activates the

carboxy component via intermediate formation of the HODhbt ester and release of diethyl phosphite (Scheme 3), was shown to be a highly efficient reagent in stepwise synthesis and fragment couplings, and superior to HBTU, HATU, BOP, and PyBroP in terms of racemization.[26] Ideal coupling conditions utilize 2 equivalents of DEPBT (**4**) and 2 equivalents of DIPEA in aprotic organic solvents.

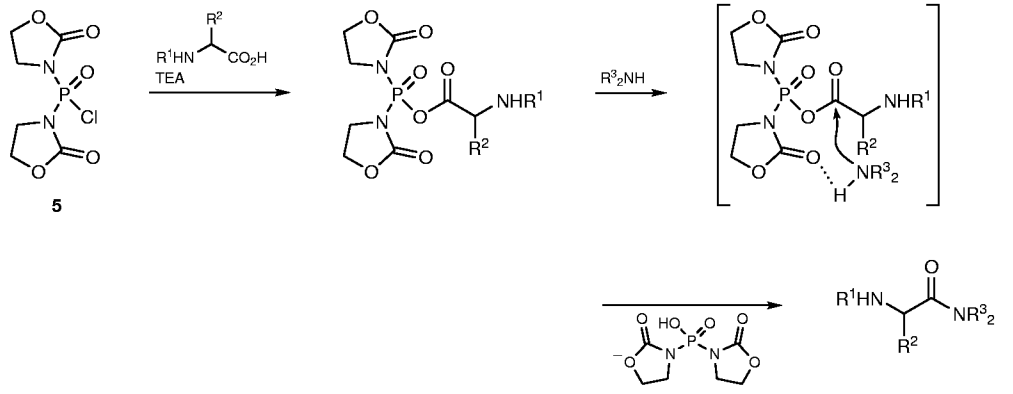
Scheme 3 Reaction of N-Protected Amino Acids with 3-(Diethoxyphosphoryloxy)benzotriazin-4(3*H*)-one (DEPBT, **4**) To Produce the Active Ester Intermediate[26]



More synthetic experience has been obtained with the commercially available bis(2-oxo-oxazolidin-3-yl)phosphinic chloride (BOP-Cl, **5**),[27,28] which has been extensively used for the acylation of sterically hindered amino components like *N*-methylamino acids (for a recent review see ref[29]). Generally improved yields and reduced racemization result as demonstrated in the synthesis of cyclosporin analogues.[30,31] The efficiency of BOP-Cl (**5**) in contrast to other phosphorus-based reagents, is attributed to a general intramolecular base catalysis by the oxazolidinone carbonyl of the mixed anhydride formed, as outlined in Scheme 4.[28] Acylations with BOP-Cl (**5**) are performed in a one-pot reaction when secondary amines are acylated, since BOP-Cl (**5**) reacts faster with carboxylate anions than with secondary amines, whilst primary amines can compete for the reagent.[31] For BOP-Cl (**5**) mediated acylations of primary amines good results are obtained using low-temperature preactivation of the carboxy component as the mixed anhydride, hence the addition of additives (HOBt or imidazole) has proved to be advantageous,[32] whereas in the case of N-methylated amino acids such additives are of no use.[33]

for references see p 587

Scheme 4 Activation of N-Protected Amino Acids with Bis(2-oxooxazolidin-3-yl)phosphinic Chloride (BOP-Cl, **5**) and Intramolecular Base-Catalysis in the Acylation



3-(Diethoxyphosphoryloxy)-1,2,3-benzotriazin-4(3*H*)-one (DEPBT, 4):[26]
To a soln of HODhbt (14.3 g, 0.088 mol) and TEA (8.92 g, 0.088 mol) in CH_2Cl_2 (100 mL) at 0 °C was added dropwise diethyl phosphorochloridate (17.3 g, 0.1 mol) in CH_2Cl_2 (50 mL). After 3 h, the precipitated TEA·HCl was removed by filtration and the solvent was removed. The residue was dissolved in EtOAc (150 mL) and washed successively with 0.1 M HCl, H_2O, and brine, and dried ($MgSO_4$). The crude product was recrystallized (EtOAc/petroleum ether) to give colorless crystals; yield: 21.5 g (82%); mp 72–74 °C; MS (EI).

Coupling with BOP-Cl (5); General Procedure:[33]
To a soln of N-protected acid component (1 mmol) in CH_2Cl_2, THF, or DMF (5 mL), was added at 0 °C BOP-Cl (279–305 mg, 1.1–1.2 mmol) (it is important to use purified BOP-Cl[28]) followed by HOBt·H_2O or imidazole (1–1.5 mmol) if required, and by TEA (or NMM) (1.2 mmol). After 5 min stirring, the amino component was added and TEA or NMM (1 mmol) dropwise over 30 min. Upon completion of the reaction, the mixture was taken to dryness and the residue partitioned between EtOAc and H_2O. The organic layer was washed with 1 M HCl, $NaHCO_3$, and brine, and dried (Na_2SO_4). The product was isolated by precipitation or crystallization.

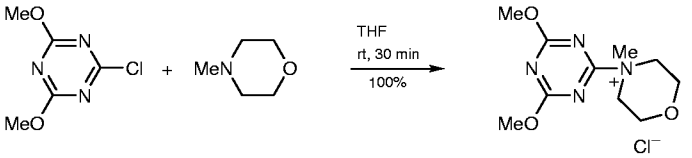
Boc-αAbu-Sar-OBzl:[31]
To a soln of Boc-αAbu-OH (3.225 g, 15.87 mmol) and H-Sar-OBzl (3.17 g, 17.76 mmol) in CH_2Cl_2 (100 mL) under an inert atmosphere, was added at 0 °C DIPEA (5.4 mL, 31.74 mmol) followed by BOP-Cl (**5**; 4.040 g, 15.87 mmol). After completion of the reaction, the mixture was poured into Et_2O (300 mL) and H_2O (200 mL). The organic layer was separated, washed with 10% aq $KHSO_4$, H_2O, 1 M $NaHCO_3$, 50% brine, and brine, and then dried ($MgSO_4$). The crude product was purified by column chromatography (silica gel, acetone/hexane, 1:4) and was obtained as pale yellow, viscous oil; yield: 4.84 g (84%); $[\alpha]_D$ −6.7 (*c* 1.0, $CHCl_3$).

3.9.2 4-(4,6-Dimethoxy-1,3,5-triazin-2-yl)-4-methylmorpholinium Chloride

Cyanuric chloride has been used for the preparation of acyl chlorides, amides, and peptides.[34] Conversion of cyanuric chloride into 2-chloro-4,6-dimethoxy-1,3,5-triazine (CDMT, **6**) leads to a reagent that upon reaction with carboxylic acids produces the highly reactive 2-acyloxy-4,6-dimethoxy-1,3,5-triazines.[35,36] The resulting active ester is a powerful acylating agent for alcohols and amines. The activation is performed in presence of a base, preferentially NMM, which leads to intermediate formation of 4-(4,6-dimethoxy-1,3,5-triazin-2-yl)-4-methylmorpholinium chloride (DMTMM, **7**)[37] (Scheme 5). This addition product is readily prepared from the commercially available CDMT (**6**) and NMM in THF and can be stored as solid compound in the cold.[38,39] It offers the advantage that it can be used in a one-

pot reaction for amide bond formation, whilst with CDMT (**6**) N-protected amino acids have to be preactivated prior to reaction with the amino component.[36] The coupling efficiency of Fmoc amino acids with DMTMM (**7**) on resin is comparable to that of PyBOP[40] whereby racemization was found to be below the detection limit.[41] Optimal coupling conditions on resin are four equivalents Fmoc amino acid, three equivalents DMTMM (**7**), and six equivalents DIPEA in NMP. This new reagent may well represent a convenient economical alternative to PyBOP or other similar coupling reagents.

Scheme 5 Reaction of 2-Chloro-4,6-dimethoxy-1,3,5-triazine (CDMT, **6**) with 4-Methylmorpholine To Form 4-(4,6-Dimethoxy-1,3,5-triazin-2-yl)-4-methylmorpholinium Chloride (DMTMM, **7**)



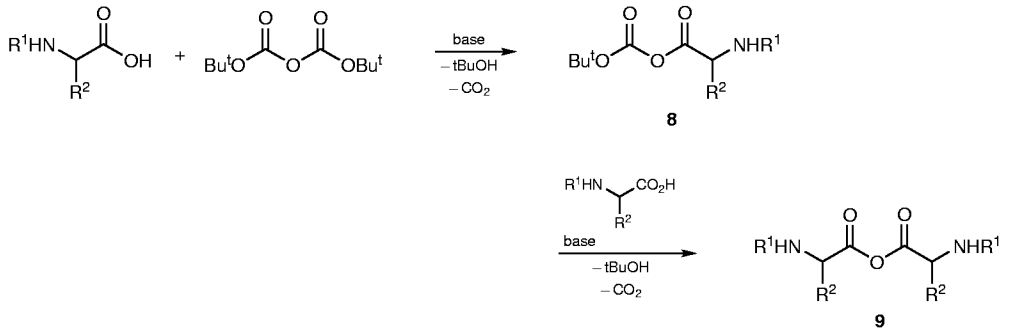
6 **7**

3.9.3 Di-*tert*-Butyl Dicarbonate

In amino acid and peptide chemistry, the di-*tert*-butyl dicarbonate (Boc_2O) is an extensively used reagent for the clean and rapid Boc-protection of amine functionalities.[3,42] It is also an efficient *tert*-butoxycarbonylating agent for alcohols, thiols, and carbon nucleophiles,[43–45] and it has been used for the conversion of amines into isocyanates, carbamates, and urea derivatives.[46–50] The reaction of amino acids with chloroformates to produce N-urethane-protected amino acids if not performed under optimal conditions is accompanied by the formation of N-protected oligomers; this has been well documented in the case of Z-[51,52] and Fmoc amino acids.[51,53–55] These side products result from activation of the carboxy group as a mixed anhydride (see Section 2.1.1). Similar side products are formed in the preparation of *N*-Boc amino acids with Boc_2O.[51] In fact, reaction of carboxylic acids with Boc_2O in the presence of 1-methylpiperidine[56] or pyridine[57–60] at room temperature leads to the mixed anhydrides **8** with liberation of carbon dioxide and *tert*-butyl alcohol (Scheme 6). However, the mixed anhydride **8** is not obtained as a single product; upon gradual accumulation in the reaction mixture, the mixed anhydrides **8** react with the starting carboxylic acid to form symmetrical anhydrides **9**. In fact, reaction of 2 equivalents of carboxylic acid with 1 equivalent Boc_2O leads to symmetrical anhydrides, a procedure which could represent a valid alternative to the DCC-mediated preparation of symmetrical anhydrides (see Section 3.4.2). The advantages of the Boc_2O/pyridine reagent is not limited to the synthesis of symmetrical anhydrides, as it was found that addition of nucleophilic agents to a Boc_2O/carboxylic acid 1:1 mixture in the presence of pyridine leads to acylation of the nucleophiles. It is, however, essential that the dicarbonate does not react at all or reacts more slowly with the nucleophile than with the carboxylic acid. Appropriate nucleophiles for this reaction are phenols and primary, secondary, and tertiary alcohols[57,60–63] as well as aromatic amines.[58] Since the dicarbonates with primary and secondary alkyl groups are more readily degraded in the presence of pyridine, the reactions in these cases are to be carried out with TEA, NMM, or 1-methylpiperidine as bases.

for references see p 587

Scheme 6 Di-*tert*-Butyl Dicarbonate Mediated Formation of the Mixed and Symmetrical Anhydride of a N-Protected Amino Acid

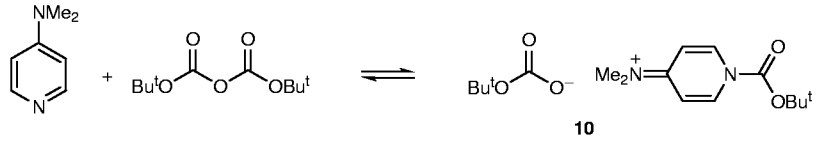


3.9.3.1 Synthesis of Amides

For the synthesis of N-protected amino acid *N*-arylamides a one-pot reaction of N-protected amino acids with arylamines by the Boc_2O/pyridine system in aprotic solvents or preformation of the mixed anhydride/symmetrical anhydride mixture followed by reaction with the amine component can be used. As possible side products urea derivatives are formed. By this procedure various *N*-arylamides were obtained in satisfactory yields (60–90%), including amides of 4-nitroaniline, 2-naphthylamine, 4-phenylazoaniline, 4-methyl-7-coumarylamine, or 6-quinolylamine.[58] Using ammonium hydrogen carbonate as the nucleophile even N-protected amino acid amides are obtained in high yields by this procedure.[64] As expected, amidation of peptides was accompanied by significant racemization which is, however, largely suppressed by addition of copper(II) chloride.[65]

Initial attempts to extend this procedure to the synthesis of peptides gave the products in good yields, at least at the level of model dipeptides when DMAP was added in catalytic amounts.[66] This may be attributed to the intermediate formation of the highly reactive 1-(*tert*-butoxycarbonyl)-4-(dimethylamino)pyridinium *tert*-butyl carbonate (**10**, Scheme 7) which facilitates the nucleophilic addition of the amino acid carboxylate anion at the *tert*-butoxycarbonyl group of the pyridinium system. Such an intermediate has been confirmed by Knölker and Braxmeier.[49] The addition of DMAP was not found to induce racemization. Whether this procedure is applicable in repetitive steps of peptide synthesis, is still questionable since *tert*-butoxycarbonylation of the amino component, at least to some extents, cannot be excluded.

Scheme 7 Reaction of Di-*tert*-Butyl Dicarbonate with 4-(Dimethylamino)pyridine[49,66]



The great advantages of this method compared to other more conventional methods, such as coupling with DCC, are the easy separation of the products from reagent-derived coproducts, the absence of racemization, low cost, and the stability of the reagent.

Z-Arg(Z)$_2$-βNA:[58]
A soln of Z-Arg(Z)$_2$-OH (2.9 g, 5 mmol), Py (0.4 mL, 5 mmol), and Boc_2O (1.2 g, 5.5 mmol) in dioxane (10 mL) was stirred at rt for 30 min. Then a soln of 2-naphthylamine (0.72 g, 5.0 mmol) in dioxane (5 mL) was added dropwise over 20–40 min. The mixture was stirred for an additional 4–5 h and the product was precipitated with EtOAc (5 mL) and Et_2O (10 mL); yield: 2.7 g (77%), mp 202–203 °C.

Boc-Pro-Phe-OMe:[66]
To a soln of Boc-Pro-OH (215 mg, 1 mmol), Py (80.5 μL, 1 mmol), and a catalytic amount of DMAP (20 mg) in anhyd THF (2 mL) was added a soln of Boc_2O (218 mg, 1 mmol) in anhyd THF (3 mL). The mixture was stirred for 30 min and then added to a soln of H-Phe-OMe (179.2 mg, 1.1 mmol) in anhyd THF (2 mL). After completion of the reaction, the solvent was removed, the residue dissolved in EtOAc (50 mL), washed with ice-cooled 10% aq HCl, sat. aq $NaHCO_3$, and brine, dried (Na_2SO_4), and purified by column chromatography (silica gel, hexane/EtOAc, 9:1 to 5:1); yield: 342 mg (91%); mp 65–66 C; $[\alpha]_D$ −50.4 (*c* 1.4, MeOH); HRMS.

3.9.3.2 Synthesis of Esters

Esterification of carboxylic acids by the Boc_2O/pyridine procedure with DMAP as catalyst in the case of secondary and tertiary alcohols is an easy to handle reaction, which proceeds in high yields (61–87%) without the need for acid catalysts;[60] thus it can be applied to acid-labile derivatives. In contrast to esterification of N-protected amino acids by symmetrical anhydride/DMAP where racemization is known to occur,[67] with the Boc_2O/pyridine procedure and DMAP as catalyst racemization was not detectable.[57] Esters of N-protected amino acids with primary, secondary, and tertiary alcohols including *tert*-butyl esters are obtained in straightforward manner. Therefore, this procedure has also been utilized for the ester-type linkage of Fmoc-protected amino acids to 4-alkoxybenzyl alcohol resin (Wang resin). Again goods yields were obtained (30–86%) and minimal racemization (0.1–4.7%).[61]

This procedure has also successfully been used for macrolactonization of ω-hydroxy acids at high dilution to produce a large number of macrolides.[62,63]

N-Protected Amino Acid *tert*-Butyl Esters; General Procedure:[57]
In a dry atmosphere Boc_2O (5.0 mmol) was added to a stirred soln of N-protected amino acid (5.5 mmol), DMAP (15–20 mg), Py (0.3–0.5 mL), and tBuOH (3–5 mL) in EtOAc or dioxane (10 mL). The mixture was stirred at rt for 5–16 h, EtOAc was added and after standard work-up the products were purified by crystallization or column chromatography in the case of oily products (hexane/EtOAc, 4:1); yields: 50–90%.

Esterification of Fmoc Amino Acids on Resin:[61]
To 4-alkoxybenzyl alcohol resin (100 mg, 0.45 mmol · g^{-1}, prewashed with CH_2Cl_2) in CH_2Cl_2 (2 mL) was added at 0 °C Fmoc-Val-OH (4 equiv), Boc_2O (4 equiv), Py (14 μL), and DMAP (2 mg). The mixture was gently stirred at 0 °C under anhydrous conditions for 16 h. Then the resin was washed with DMF (3 × 3 mL) and CH_2Cl_2 (3 × 3 mL) and the substitution determined by cleavage of the Fmoc group and absorbance measurement at 301 nm of the dibenzofulvene–piperidine adduct. Optical purity of the resin-bound amino acid was determined by the Marfey method.[68] The yield of the derivatization of the 4-alkoxybenzyl alcohol resin was 86% and the content of D-amino acid 0.3%.

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4 Synthesis of Peptides

A. M. Felix

It is essential for the peptide chemist to have a fundamental knowledge of the appropriate protecting groups (Section 2) and coupling methods (Section 3) before undertaking any synthesis. There are two general methods that are currently being used for the preparations of peptides; synthesis of peptides in solution (Section 4.1),[1] and synthesis of peptides on solid supports (Section 4.3).[2-5] Prior to undertaking a peptide synthesis each of these methodologies should be carefully considered since each method offers important advantages and opportunities for the peptide chemist. The peptide structure, including size, solubility characteristics, side-chain functionality, potential for racemization,[6] and quantity of target peptide should all be considered.

Peptide synthesis in solution may be carried out by stepwise chain elongation (Section 4.1.1) using urethane-protected amino acids to avoid racemization. The efficiency of this method usually decreases with chain length due to the insolubility of the growing protected peptide intermediate. Although this problem may be overcome with the use of solvents such as trifluoroethanol or hexafluoroisopropanol, larger peptides are more efficiently prepared by the segment condensation strategy (Section 4.1.3) even with minimal side-chain protection (Section 4.1.2). Although the azide procedure is preferable, DCC-mediated segment condensations have been used quite successfully. Both the stepwise and segment condensation strategies offer the option of purification at each synthetic stage. Final deprotection of the target peptide is usually carried out by catalytic hydrogenolysis, trifluoroacetic acid treatment or with hydrogen fluoride.[7]

The stepwise synthesis of peptides on solid supports (Section 4.3) offers important advantages over peptide synthesis in solution. Since the peptide derivative remains covalently attached to the insoluble support, coupling reactions can be driven to near completion through the use of excess protected amino acids and coupling reagents. Excess of reagents and byproducts are removed by simple washing procedures. A variety of solid supports, resins, and handles have been developed that have improved the solid-phase technology (Section 4.3.1). Following the introduction of the Boc/Bzl strategy (Section 4.3.2.1), other protection strategies have been developed including the Fmoc/tBu (Section 4.3.2.2) and allyl-based methods that offer an important element of orthogonality (Section 4.3.2.3). Different methods have been developed for the cleavage of the peptide from the resin in the final stage depending upon the solid-phase strategy (Section 4.3.5). These include hydrogen fluoride (both S_N2 and S_N1 mechanisms), trifluoroacetic acid (both concentrated and dilute solution), nucleophilic cleavage (e.g., ammonolysis), and others.

The ease, simplicity, and repetitive nature of solid-phase synthesis led to the development of automated synthesizers (Section 4.3.6). In addition, the advent of solid-phase peptide synthesis laid the groundwork for the development of combinatorial synthesis of peptide libraries[8] that are now being used for the discovery of lead analogues for drug discovery (Section 4.3.7)

Large peptides may be prepared by chemical ligation methods (Section 4.1.5) using thioester chemistry[9,10] to effect a fragment coupling between two fully unprotected peptides. Although the presence of a Cys residue is required at the site of ligation, hundreds of proteins have been prepared by this method. Larger proteins can be prepared by intein-mediated protein ligation using synthetic peptides and large proteins produced by molecular biological procedures. This combination of synthetic and biologically derived polypeptides are becoming the method of choice for generating large C-terminal thioester peptides that can be ligated to N-terminal Cys peptides and have great potential for the preparation of large protein domains.

for references see p 590

Procedures for the enzymatic synthesis of peptides[11] have been developed (Section 4.2) that are based on the stereo- and regiospecificity of enzymes for peptide-bond formation. These offer the advantage of peptide synthesis without any side-chain protection and without danger of racemization. The general methods of enzymatic synthesis involve either thermodynamic control in which the carboxy component has a free carboxy terminus or kinetic control in which the carboxy component is slightly activated (Section 4.2). Peptide-bond formation is limited to proteases that have reverse hydrolysis potential. Synthetic peptides ranging from dipeptide (e.g., aspartame) through ribonuclease A (124 residues) have been ligated by these methods on scales ranging from milligram through kilogram quantities.

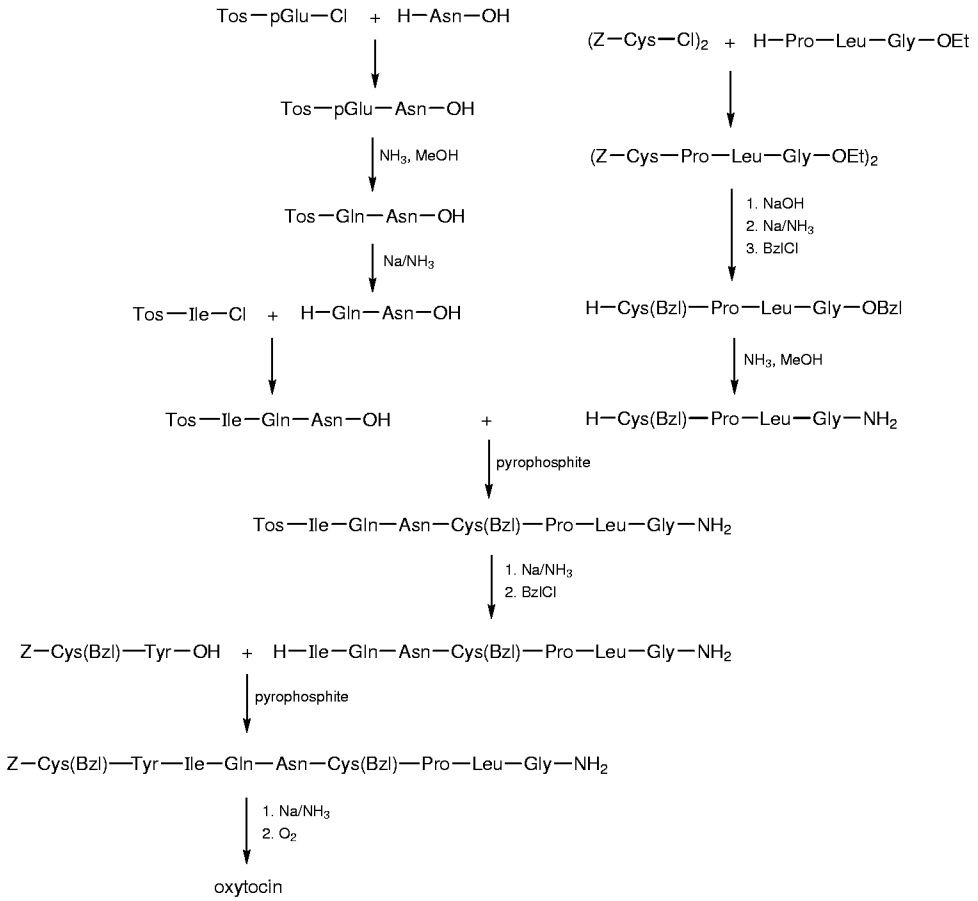
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4.1 Synthesis of Peptides in Solution

Y. Okada

The synthesis of benzoylglycylglycine by Curtius[1] has led to the development of various methodologies for peptide bond formation through the mixed anhydride method developed by Wieland.[2] Ethoxycarbonylglycylglycine ethyl ester was synthesized by E. Fischer.[3] The ethoxycarbonyl group then provided the basis for the development of the benzyloxycarbonyl group (Z) as a protecting group by Bergmann and Zervas[4] in 1932. Benzyloxycarbonyl as a protecting group is removed by catalytic hydrogenation at room temperature and atmospheric pressure, a process that leaves the peptide bond and the various side-chain groups unaffected and generates only relatively harmless byproducts: toluene and carbon dioxide. The Z group can also be removed by several alternative procedures, among them reduction by sodium in liquid ammonia[5] and acidolysis[6] being particularly noteworthy. In the historically important synthesis of oxytocin by duVigneaud et al.,[7] limited protecting groups and peptide bond formation methods had to be employed, resulting in the complicated and time-consuming synthesis of oxytocin shown in Scheme 1. The development of various protecting groups and peptide bond formation methods has now made it possible to select the most suitable tactics and strategies from the various types of methods available for peptide synthesis.

Scheme 1 Synthetic Route to Oxytocin[7]

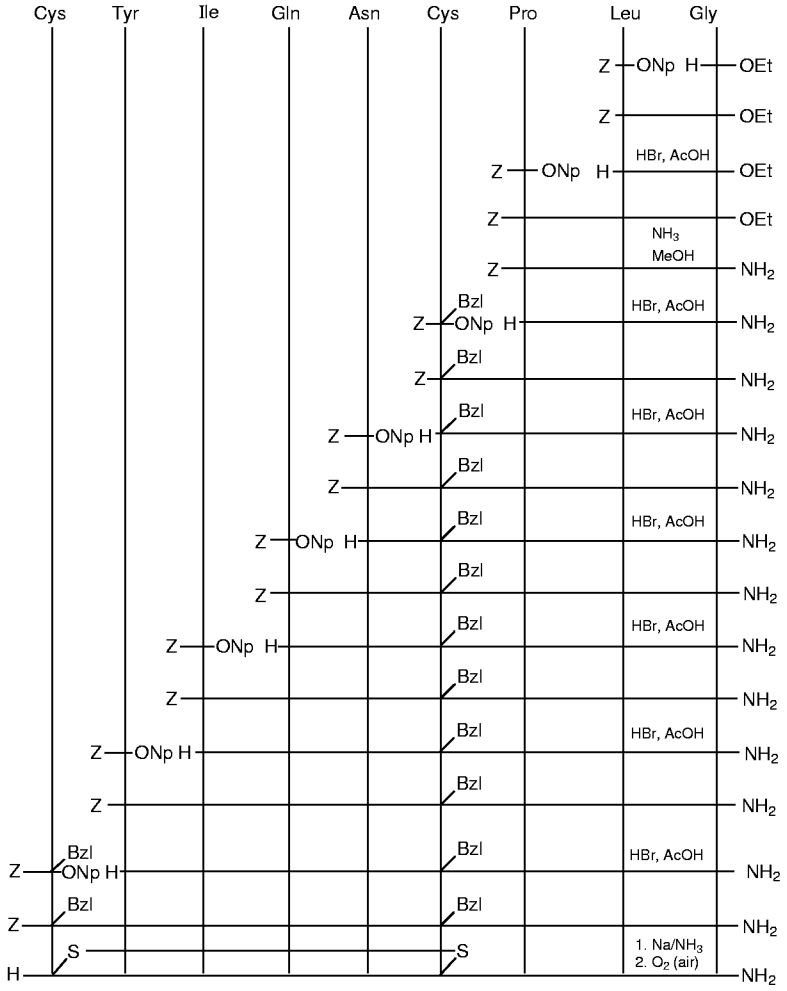


for references see p 623

4.1.1 Stepwise Synthesis

It is very important to avoid racemization during the synthesis of peptides. Stepwise elongation from the carboxy terminal by one amino acid at a time (Scheme 2)[8] using urethane-protected amino acids, such as the Z amino acid or the Boc amino acid, is advantageous for the avoidance of racemization during the peptide bond forming reaction.[8–10]

Scheme 2 Synthetic Route to Oxytocin by Stepwise Synthesis[8]



The stepwise elongation procedure is effective for the synthesis of small peptides and for preparing peptide segments for the construction of larger peptides and proteins. The acylation reactions are carried out in an appropriate solvent such as DMF, and excess acylating agent (activated urethane-protected amino acid) is used to ensure completion of the acylation. It has also been reported that the efficacy of the stepwise elongation procedure decreases with the growth of chain length[9,10] mainly due to the decreasing solubility of the peptide intermediates in organic solvents.

Racemization

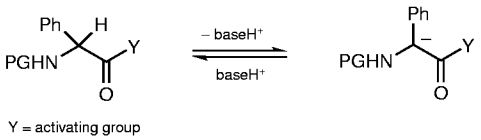
In peptide synthesis, only alanine and leucine out of the 20 amino acid constituents of proteins are free from characteristic side reactions, although not from racemization. In order to synthesize optically active peptides, it is important to understand the mechanism of racemization during peptide synthesis.[11]

Mechanism of Racemization

Racemization can occur by two mechanisms: Racemization by direct abstraction of the α-proton by strong base[12,13] and through oxazol-5(4*H*)-one formation.[14–16]

The proton-abstraction mechanism is the pathway only in very special cases such as the rapid racemization of derivatives of phenylglycine (Scheme 3). Racemization through the proton-abstraction mechanism can be prevented by employing suitable reaction conditions, in particular by controlling the use of tertiary amine.[17–22]

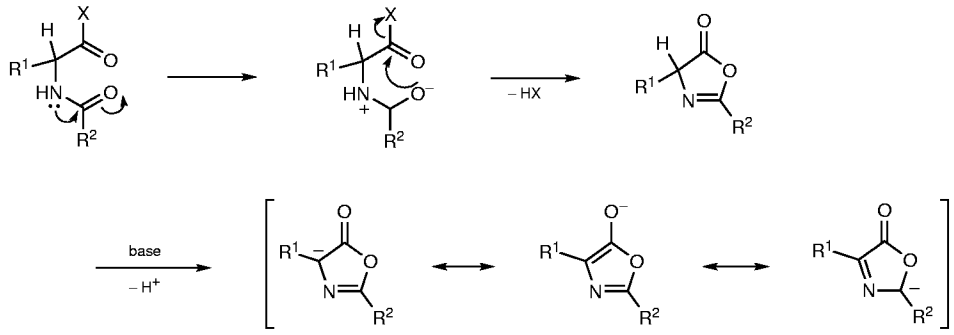
Scheme 3 Proton-Abstraction Mechanism



Y = activating group

The most important mechanism of racemization involves the formation of oxazol-5(4*H*)-ones.[14] In this mechanism it was shown that the properties of the amino protecting group might be associated with the tendency for racemization. The activation of the carboxy moiety of *N*-acyl amino acids (acyl = Ac, Bz, peptidyl, etc.) readily results in the formation of oxazol-5(4*H*)-ones, which give rise to chirally unstable intermediates via tautomerization of the oxazol-5(4*H*)-one (Scheme 4).

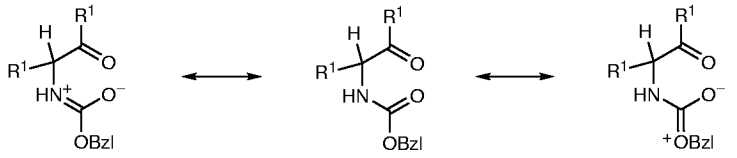
Scheme 4 Chirally Unstable Oxazol-5(4*H*)-one



In the case of urethane-protected amino acids, such as Z- or Boc-protected amino acids, it has been the accepted view that oxazol-5(4*H*)-ones are not produced from such amino acid derivatives, resulting in resistance to racemization during activation and coupling (Scheme 5).

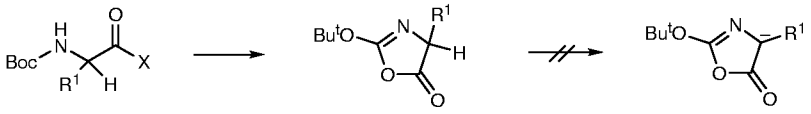
for references see p 623

Scheme 5 A Urethane-Protected Amino Acid Resisting Oxazol-5(4*H*)-one Formation



However, the isolation of optically pure oxazol-5(4*H*)-ones from Boc- or Z-protected amino acids by carbodiimide, mixed anhydride, or acyl chloride activation contradicts the above assumptions. It was suggested that these oxazol-5(4*H*)-ones are 2-alkoxyoxazol-5(4*H*)-ones and are chirally stable (Scheme 6).[23,24]

Scheme 6 Chirally Stable Oxazol-5(4*H*)-ones



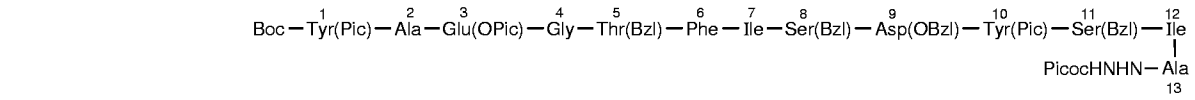
Solubility Problems

One of the most frustrating problems in the solution method of peptide synthesis is the insolubility of protected intermediates even in solvents as powerful as DMF, HMPA, and NMP. In the case of the synthesis of secretin,[9,10] it was preferable to start the reaction with the reactants at a concentration of no more than 0.1 M. The molecular weights and solubilities of the partially protected peptide intermediates are often such as to prevent the preparation of solutions of desirable concentration; however, the excess of the active ester can be raised to the point where, at least with respect to this component, the concentration reaches a desirable range. In this way, practical rates are maintained. The possibility of counterbalancing the difficulties that originated from the increasing molecular weight during the synthesis of long chains is a valuable feature of the stepwise approach.[25] In order to maintain the solubility of peptide intermediates suitable for coupling reactions, two main approaches have been attempted. One is the development of protecting groups that can lead to peptide intermediates that are soluble in organic solvents and another is the development of solvent systems that have high solubilizing potential.

Development of Protecting Groups for Increasing the Solubility of Peptides in Organic Solvents

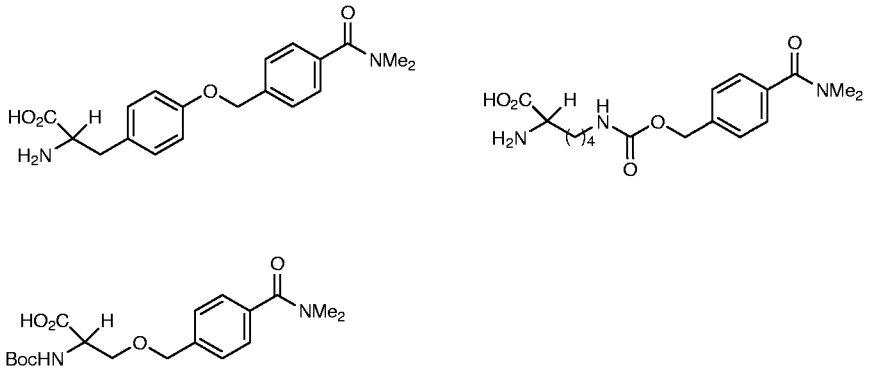
In the syntheses of peptides by the solution method, the difficulty in obtaining homogeneous products increases rapidly as the size of the target peptide increases. Low solubility in commonly used solvents and low molecular concentrations because of the high molecular weight of large segments results in slow reaction rates. If intermediate peptides are soluble in solvents, it is easy to purify them even at this stage. In 1973, a protected benzyloxycarbohydrazide of a tridecapeptide, corresponding to the sequence of residues 1–13 of porcine gastric inhibitory peptide and having benzyl side-chain protection, was described,[26] but the sparing solubility of the intermediates caused difficulty. With the objective of increasing solubility, the picolyl group instead of the benzyl group was employed to prepare protected tridecapeptide 4-picolyloxycarbohydrazide shown in Scheme 7.[27]

Scheme 7 Structure of the Protected Tridecapeptide 4-Picolyloxycarbohydrazide[27]



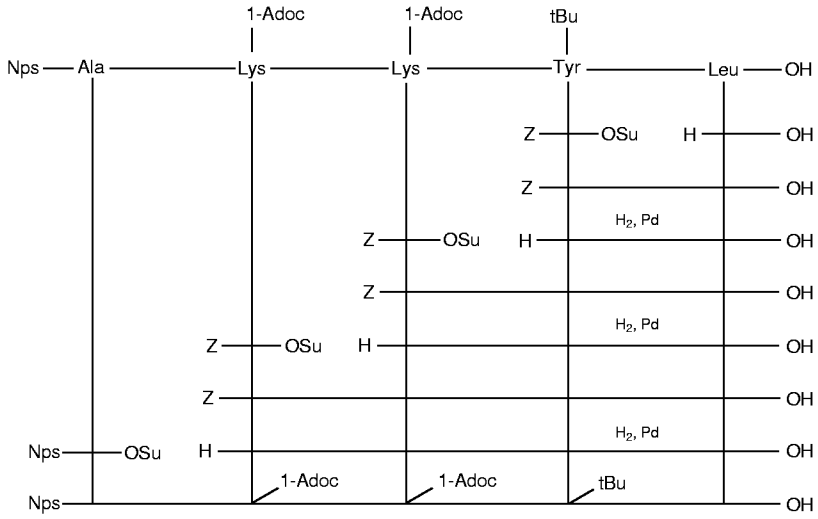
However, there was no significant improvement in the solubility of the higher protected peptide in DMF.[27] It was also attempted to increase the solubility of peptide fragments with benzyl-based protection by the introduction of the dimethylcarbamoyl group into the phenyl ring (Scheme 8), with the intention of improving the solubility in DMF.[28]

Scheme 8 The Structures of Benzyl-Based Protecting Groups Containing a Dimethylcarbamoyl Moiety[28]



However, in the cases examined so far, the molar solubility in DMF has not increased significantly. The 1-adamantyloxycarbonyl (1-Adoc) group could potentially improve the solubility of amino acids and peptides protected by 1-Adoc in organic solvents due to its strong hydrophobicity.[29,30] Lys(1-Adoc) was successfully employed in the synthesis of peptide histidine isoleucine (PHI) (Scheme 9)[31] and in the synthesis of a peptide fragment of a lysozyme.[32]

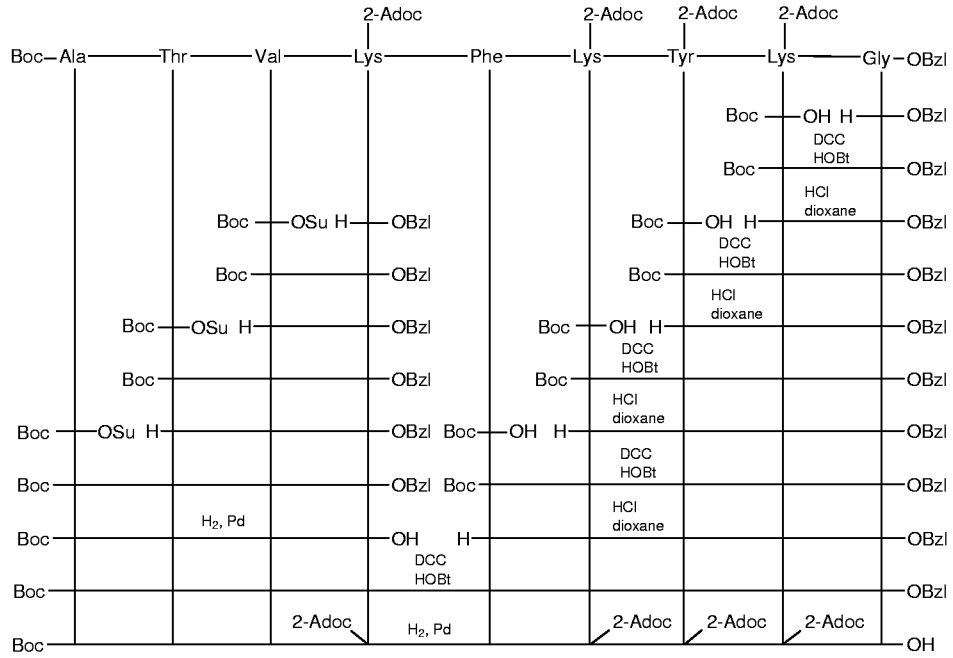
Scheme 9 Synthetic Route to a Protected Peptide Fragment of Sequence 19–23 of PHI[31]



for references see p 623

With a similar intention, 1-adamantyl and 2-adamantyl residue-based protecting groups were developed. β-1- and β-2-Adamantylaspartates,[33–35] γ-(2-adamantyl)glutamate,[36] N^{ε}-(2-adamantyloxycarbonyl)lysine,[37,38] N^{τ}-(2-adamantyloxycarbonyl)histidine,[39,40] N^{π}-(1-adamantyloxymethyl)histidine,[41,42] N^{π}-(2-adamantyloxymethyl)histidine,[43,44] and *O*-(2-adamantyloxycarbonyl)tyrosine[45] were developed and employed for peptide synthesis (Scheme 10).

Scheme 10 Synthesis of Protected Peptide Fragments Corresponding to the Sequence 1–9 of *S. solfataricus* RNase[45]



It was also reported that Fmoc-Cys(Acm)-Lys(2-Adoc)-Cys(Acm)-Thr(Bzl)-Ser(Bzl)-Cys-(Acm)-Lys(2-Adoc)-Lys(2-Adoc)-Ser(Bzl)-Cys(Acm)-Cys(Acm)-OH was more soluble in DMF than was the N^{ε}-Z homologue of the compound (>50 mg·mL^{-1} and <30 mg·mL^{-1}, respectively).[38,46] Thus, the solubility of the protected peptide fragments in organic solvents was actually improved.

Development of Effective Solvent Systems

Another approach to improving the limited solubility of peptide fragments is through the development of effective solvent systems.[47] The insolubility of the protected peptides is caused by β-sheet aggregation of the peptide chain resulting from inter- or intra-chain hydrogen bonds.[48,49] In order to dissolve such peptides, the β-sheet structure must be disrupted and transformed into a helical or random structure. TFE and HFIP have been recommended as solvents for disrupting β-sheet aggregation.[47,50] It has been shown that mixtures of halohydrocarbons and halo alcohols are extremely efficient at disrupting self-association and enhancing the solubility of peptides. The dissolving power follows the order: HFIP/$CHCl_3$>HFIP/CH_2Cl_2>TFE/$CHCl_3$>TFE/CH_2Cl_2.[51] The use of a $CHCl_3$/TFE (3:1) mixture was recommended as a suitable solvent for peptide synthesis from sparingly soluble, fully protected intermediates.[52] Sakakibara et al. attempted to couple Boc-(1–10)-OH with H-(11–28)-OBzl by the WSCI/HOBt method in a synthesis of protected rat atrial natriuretic peptide (1–28). Both carboxy and amino components are known to be soluble in a mixture of

NMP and DMSO, but the coupling reaction did not reach completion in 24 hours even though a 1.2 excess of both the carboxy component and reagents was used. After addition of excess carboxy component and coupling reagents, the entire mixture became a gel. However, in $CHCl_3$/TFE, the same coupling reaction proceeded smoothly by the DCC/HODhbt method without any detectable racemization. The reaction was completed within 5 hours and the mixture remained a clear solution. Based on these results, they applied this procedure to the synthesis of sparingly soluble protected peptides such as human amylin, porcine C-type naturiuretic peptide [pCNP (1–53)], and human parathyroid hormone [hPTH (1–84)].[51]

4.1.1.1 Application of Synthetic Fragments with or without C-Terminal Protection

The stepwise solution-synthesis procedure has been successfully employed for the synthesis of peptides. Its applicability was demonstrated in a synthesis of oxytocin.[8] A few larger peptides have been synthesized in solution entirely by this strategy, a notable example being the synthesis of secretin, a 27 amino acid residue peptide.[9,10] Another use of this strategy is for the preparation of peptide fragments as building blocks of larger peptides and proteins.

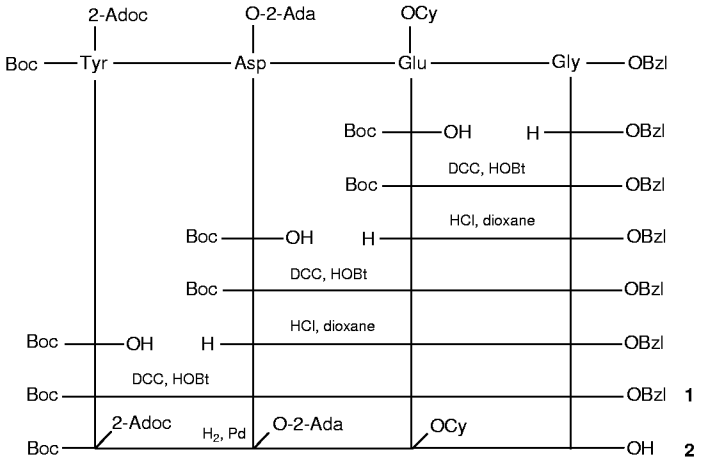
4.1.1.1.1 Synthesis of Peptide Fragments with C-Terminal Protection

The C-terminal carboxy group of the C-terminal peptide fragments should be protected throughout the chain-building process. However, C-terminal protected peptide fragments as carboxy components must be converted into the corresponding free C-terminal peptides or the corresponding hydrazides before coupling.

4.1.1.1.1.1 Synthesis of a Protected Tetrapeptide with a C-Terminal Benzyl Ester

As shown in Scheme 11, a protected tetrapeptide, a building block for the synthesis of *S. solfataricus* RNase, was prepared starting from H-Gly-OBzl in a stepwise manner. After construction of the fragment, the protected peptide was converted into the free C-terminal peptide for the next coupling.[45]

Scheme 11 Synthesis of the Protected Tetrapeptide Corresponding to the Sequence 33–36 of *S. solfataricus* RNase[45]



for references see p 623

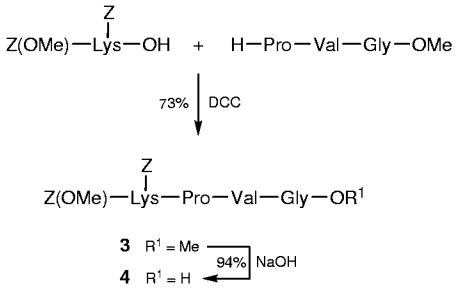
Boc-Tyr(2-Adoc)-Asp(O-2-Ada)-Glu(OCy)-Gly-OBzl (1):[45]
DCC (3.7 g, 18.1 mmol) was added to a soln of H-Asp(O-2-Ada)-Glu(OCy)-Gly-OBzl·HCl (10.0 g, 15.1 mmol), Boc-Tyr(2-Adoc)-OH (6.9 g, 15.1 mmol) and HOBt (2.0 g, 15.1 mmol) in DMF (100 mL) containing TEA (2.1 mL, 15.1 mmol) under cooling with an ice–salt bath and the mixture was stirred at 4 °C overnight. After removal of the urea and the solvent, the residue was extracted with EtOAc. The extract was washed with 10% citric acid, 5% Na_2CO_3, and H_2O, dried (Na_2SO_4), and concentrated. Petroleum ether was added to the residue to afford crystals which were collected by filtration and recrystallized (MeOH); yield: 12.5 g (77.5%); mp 107–108 °C; $[\alpha]_D^{27}$ −9.5 (*c* 1.0, DMF); Anal. Calcd for $C_{59}H_{78}N_4O_{14}$: C, 66.4; H, 7.37; N, 5.25. Found: C, 66.2; H, 7.54; N, 5.51.

Boc-Tyr(2-Adoc)-Asp(O-2-Ada)-Glu(OCy)-Gly-OH (2):[45]
Boc-Tyr(2-Adoc)-Asp(2-Ada)-Glu(OCy)-Gly-OBzl (**1**; 2.5 g, 2.3 mmol) in MeOH (100 mL) was hydrogenated over a Pd catalyst for 2 h. After removal of Pd and the solvent, Et_2O was added to the residue to afford an amorphous powder; yield: 2.0 g (89%); $[\alpha]_D^{27}$ −12.5 (*c* 1.0, DMF); Anal. Calcd for $C_{52}H_{72}N_4O_{14}$: C, 63.9; H, 7.43; N, 5.73. Found: C, 64.1; H, 7.28; N, 5.78.

4.1.1.1.1.2 Synthesis of a Protected Tetrapeptide with a C-Terminal Methyl Ester

The protected tetrapeptide corresponding to the sequence 11–14 of porcine adrenocorticotropic hormone was prepared by coupling Z(OMe)-Lys(Z)-OH with H-Pro-Val-Gly-OMe[53] in the presence of DCC. Saponification of Z(OMe)-Lys(Z)-Pro-Val-Gly-OMe (**3**) by alkali gave Z(OMe)-Lys(Z)-Pro-Val-Gly-OH (**4**) as shown in Scheme 12.[54]

Scheme 12 Synthetic Route to the Protected Tetrapeptide, Z(OMe)-Lys(Z)-Pro-Val-Gly-OH[54]



Z(OMe)-Lys(Z)-Pro-Val-Gly-OMe (3):[54]
Z-Pro-Val-Gly-OMe[53] (4.19 g, 10 mmol) in THF (50 mL) containing 1 M HCl (10 mL) was hydrogenated over a Pd catalyst in the usual manner until the evolution of CO_2 ceased. The soln was filtered and the filtrate was condensed to about one-third of the original volume in vacuo. The residue was dissolved in DMF (50 mL) and TEA (1.4 mL, 10 mmol) was added. This soln was combined with a soln of Z(OMe)-Lys(Z)-OH (4.44 g, 10 mmol) and DCC (2.3 g, 11.2 mmol) in THF (40 mL). The mixture was stirred at rt for 24 h and then the soln was filtered, the filtrate was concentrated in vacuo, and the residue was extracted with EtOAc. The extract was washed with 5% Na_2CO_3, 10% citric acid, and H_2O, dried (Na_2SO_4), and then concentrated. The residue solidified on treatment with Et_2O; yield: 5.20 g (73%); mp 114–117 °C; $[\alpha]_D^{27}$ −37.6 (*c* 0.8, DMF); Anal. Calcd for $C_{34}H_{47}N_5O_8$: C, 60.9; H, 7.35; N, 10.4. Found: C, 60.8; H, 7.15; N, 10.7.

Z(OMe)-Lys(Z)-Pro-Val-Gly-OH (4):[54]
Z(OMe)-Lys(Z)-Pro-Val-Gly-OMe (**3**; 5.40 g, 7.6 mmol) in MeOH (25 mL) was treated with 1 M NaOH (10 mL) at rt for 60 min. The solvent was removed in vacuo at 20 °C and the residue was dissolved in H_2O. The aqueous phase was washed with Et_2O and then acidified with citric acid. The resulting precipitate was extracted with EtOAc, which was washed with H_2O and concentrated. The residue was triturated with petroleum ether; yield: 4.96 g (94%); mp 77–82 °C; $[\alpha]_D^{27}$ −29.7 (*c* 0.8, DMF); Anal. Calcd for $C_{33}H_{45}N_5O_8 \cdot 1.5\ H_2O$: C, 59.5; H, 7.22; N, 10.5. Found: C, 59.4; H, 7.23; N, 10.0.

4.1.1.1.1.3 Removal of the Phenacyl Group from the Protected Peptide Phenacyl Ester

The phenacyl ester (Pac), first introduced into peptide synthesis by Hendrickson and Kendall in 1970,[55] can be removed by treatment with zinc powder in AcOH at 40–50 °C. If the segment is insoluble in AcOH or its mixture, another reduction condition must be sought. Hagiwara et al.[56] reported that Pac esters can be removed by zinc with acetylacetone in a mixture of pyridine, NMP, and DMSO at 35–50 °C. However, this procedure is only applicable when the segment is soluble in a mixture of NMP and DMSO, and requires a long time (4–6 h) for the reduction because of the presence of DMSO in the medium. Furthermore, it may lead to side reactions such as base-catalyzed succinimide formation or partial cleavage of the Z(Br) group from the Tyr residue because of the basicity of the medium. To overcome this obstacle, the DMSO-insoluble segments were dissolved in a small volume of CH_2Cl_2/TFE, then diluted with AcOH, and the solution was treated with zinc powder at around 35 °C under a nitrogen atmosphere; the Pac ester was removed within 30–60 min. The products thus obtained were homogeneous and the yields were almost quantitative. When $CHCl_3$ was used instead of CH_2Cl_2, partial Boc-group cleavage occurred due to the formation of HCl from $CHCl_3$ by zinc reduction. This result clearly demonstrates that zinc reduction in AcOH in a mixture of TFE and CH_2Cl_2 is much more efficient than the acetylacetone procedure for removing the Pac esters from sparingly soluble segments.

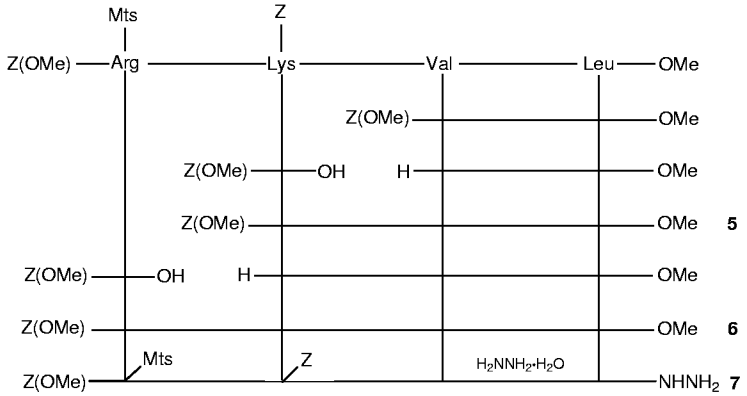
Z-pGlu-Phe-Thr(Bzl)-Asp(OCy)-Cys(Acm)-Ser(Bzl)-Val-Ser(Bzl)-OH; Removal of the Pac Ester from a Sparingly Soluble Protected Peptide:[51]
Z-pGlu-Phe-Thr(Bzl)-Asp(OCy)-Cys(Acm)-Ser(Bzl)-Val-Ser(Bzl)-OPac (6.64 g, 3.6 mmol) was dissolved in a mixture of CH_2Cl_2 (60 mL) and TFE (20 mL), and the soln was diluted to 300 mL with AcOH. Zn powder (10 g) was added to the soln under N_2, and the suspension was stirred for 40 min at 35 °C. After the Zn powder had been removed by filtration, the solvents were removed in vacuo. To the residue, was added 0.1 M HCl and the resulting precipitates were collected, washed with H_2O and then successively with MeOH, EtOAc, and hexane, and dried to give the Pac-deprotected product; yield: 5.90 g (95%); mp 258–262 °C; Anal. Calcd for $C_{89}H_{115}N_{11}O_{22}S \cdot 2\ H_2O$: C, 60.8; H, 6.82; N, 8.76. Found: C, 60.6; H, 7.01; N, 9.01.

4.1.1.1.1.4 Synthesis of a Protected Tetrapeptide Hydrazide from the Methyl Ester

The protected peptide hydrazides as building blocks for the synthesis of large peptides or proteins are usually prepared from the corresponding alkyl esters by treatment with hydrazine hydrate.[57–59] As an example, the synthesis of Z(OMe)-Arg(Mts)-Lys(Z)-Val-Leu-$NHNH_2$ (**7**) is reported. According to Scheme 13, starting from H-Val-Leu-OMe, Z(OMe)-Lys(Z)-OH and Z(OMe)-Arg(Mts)-OH were coupled successively by the mixed anhydride method to give Z(OMe)-Arg(Mts)-Lys(Z)-Val-Leu-OMe (**6**), which was converted into the corresponding hydrazide **7** by treatment with hydrazine hydrate.[57]

for references see p 623

Scheme 13 Synthetic Scheme for the Protected Tetrapeptide Hydrazide Corresponding to the Sequence 11–14 of Growth Hormone Releasing Factor[57]



Z(OMe)-Lys(Z)-Val-Leu-OMe (5):[57]
Z(OMe)-Val-Leu-OMe[60] (5.0 g, 12.2 mmol) was treated with a mixture of TFA (10.4 mL) and anisole (2.6 mL); the N^{α}-deprotected peptide was isolated as usual and dissolved in DMF (20 mL) containing TEA (1.7 mL, 12.2 mmol). A mixed anhydride [prepared from Z(OMe)-Lys(Z)-OH (5.40 g, 12.2 mmol)] in THF (70 mL) was added to the above ice-chilled soln and the mixture was stirred in an ice bath for 3 h. The solvent was removed and Et_2O was added to the residue to give a solid. The solid mass was washed with 5% citric acid, 5% $NaHCO_3$, and H_2O, and reprecipitated from DMF and MeOH; yield: 5.10 g (62%); mp 192–193 °C; $[\alpha]_D^{27}$ −4.8 (*c* 0.6, DMF); Anal. Calcd $C_{35}H_{50}N_4O_9$: C, 62.7; H, 7.51; N, 8.35. Found: C, 62.7; H, 7.50; N, 8.24.

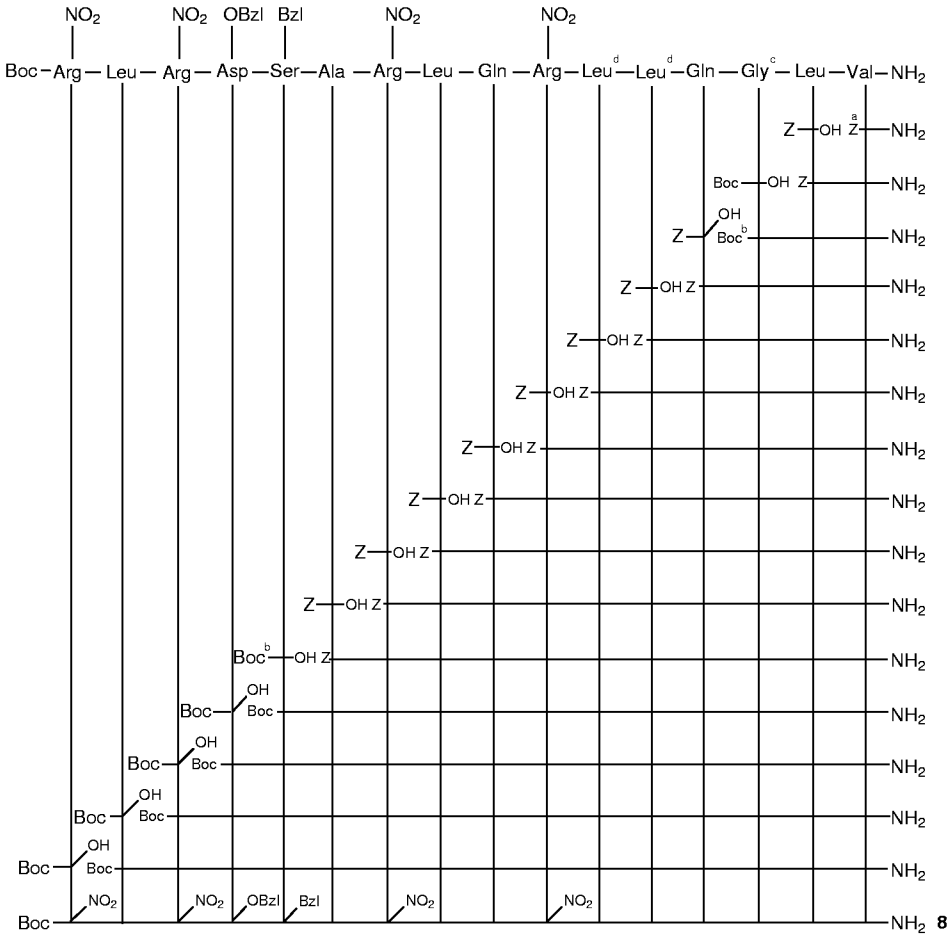
Z(OMe)-Arg(Mts)-Lys(Z)-Val-Leu-OMe (6):[57]
Z(OMe)-Lys(Z)-Val-Leu-OMe (**5**; 5.0 g, 7.5 mmol) was treated with a mixture of TFA (9.6 mL) and anisole (2.4 mL); the N^{α}-deprotected peptide, isolated as usual, was dissolved in DMF (40 mL) containing TEA (1.0 mL, 7.5 mmol). A mixed anhydride [prepared from Z(OMe)-Arg(Mts)-OH·CHA (5.08 g, 8.2 mmol)] in DMF (50 mL) was added to the above ice-chilled soln and the mixture was stirred in an ice bath for 3 h. The solvent was removed. To the residue was added Et_2O to give a powder which was collected, washed with 5% citric acid, 5% $NaHCO_3$, and H_2O, and reprecipitated from DMF and EtOAc; yield: 5.30 g (71%); mp 180–181 °C; $[\alpha]_D^{27}$ −8.7 (*c* 1.3, DMF); Anal. Calcd for $C_{50}H_{72}N_8O_{12}S$: C, 59.5; H, 7.19; N, 11.1. Found: C, 59.3; H, 7.08; N, 10.8.

Z(OMe)-Arg(Mts)-Lys(Z)-Val-Leu-NHNH$_2$ (7):[57]
Z(OMe)-Arg(Mts)-Lys(Z)-Val-Leu-OMe (**6**; 5.30 g, 5.3 mmol) was dissolved in DMF/MeOH (1:1, 50 mL) and treated with 80% $H_2NNH_2 \cdot H_2O$ (1.6 mL, 5 equiv). After standing for 14 h, additional $H_2NNH_2 \cdot H_2O$ (1.6 mL, 5 equiv) was added and the soln was kept at 30 °C for a further 14 h. After removal of the solvent, the residue was treated with H_2O and the resulting powder was reprecipitated from DMF with EtOH; yield: 4.50 g (85%); mp 204–205 °C; $[\alpha]_D^{27}$ −10.3 (*c* 1.4, DMSO); Anal. Calcd for $C_{49}H_{72}N_{10}O_{11}S$: C, 58.3; H, 7.19; N, 13.9. Found: C, 58.0; H, 7.19; N, 13.9.

4.1.1.1.1.5 Synthesis of a Protected C-Terminal Hexadecapeptide of Secretin by the Mixed Anhydride Method

A secretin[61] heptacosapeptide was successfully synthesized in a stepwise manner by the mixed anhydride method starting from H-Val-NH_2.[62,63] The synthesis used Boc or Z amino acids in which side-chain functional groups were blocked as mixed anhydrides, in excess amounts. A 1.4-fold excess of mixed anhydride was used to make the hexadecapeptide; after that a twofold excess was used. The synthetic route to the C-terminal hexadecapeptide **8** is shown in Scheme 14.[62] Beyermann called this method the repetitive excess mixed anhydride method (REMA).

Scheme 14 REMA Synthesis of the Protected C-Terminal Hexadecapeptide of Secretin (12–27)[62]



[a] Z groups were removed by acidolysis (2 M HBr/AcOH).
[b] Deprotection by TFA.
[c] Gly was also incorporated by means of Z-Gly-ONp.
[d] Leu was also coupled as the Boc derivative.

Synthesis of the C-Terminal Hexadecapeptide of Secretin (8); General Procedure:[62]
A 1.4-fold excess of mixed anhydride over the amino compound was obtained by adding isobutyl chloroformate (1.4 equiv) to a soln of the N-protected amino acid (1.5 equiv) and NMM (1.5 equiv) in THF or DMF at a temperature of −15 °C. The time for the formation of the mixed anhydride was 2 min, after which the amino component (1 equiv) was added at once. The reaction was performed with mechanical stirring at −15 °C. After a reaction time of 3 h the temperature was raised to <0 °C and 2 M $KHCO_3$ was added until a slightly alkaline reaction was obtained. The mixture was stirred vigorously for 30 min, after which the peptide was precipitated by addition of 75% sat. aq brine (generally four times the quantity of organic solvent), or in the case of larger peptides by addition of H_2O. The protected peptides, after washing with H_2O, were usually dried in vacuo. The N-protecting groups were removed by acidolysis. The Z group was deprotected by dissolving the peptide in 2 M HBr/AcOH and keeping the soln at rt for 1.5 h. The Boc group was removed on treatment of the peptide with TFA at rt for 30 min. Deprotected peptides were isolated either by precipitation with Et_2O or by removal of the solvent in vacuo. The synthesis was repeated as described.

for references see p 623

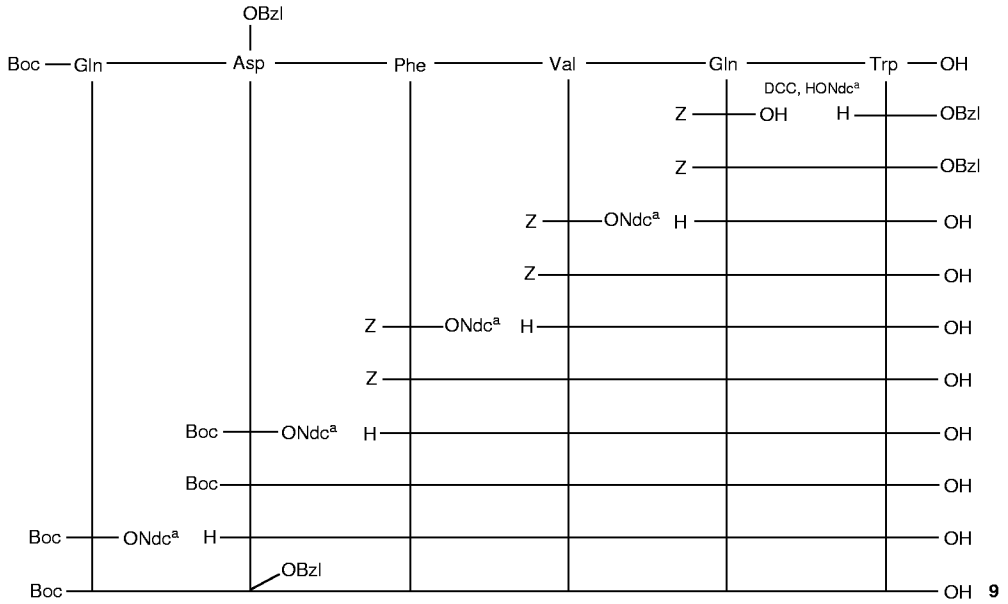
4.1.1.1.2 Synthesis of Peptide Fragments without C-Terminal Protection

In the case of the stepwise synthesis of peptide fragments starting from an amino acid without C-terminal protection, peptide bond forming methods are limited to the mixed anhydride method,[2,64,65] the active ester method,[8] and the azide method.[66,67] Although such peptide fragments can be directly used for the segment condensation reaction, the purification of the products after each coupling reaction is not so easy.

4.1.1.1.2.1 Synthesis of a Protected Peptide Fragment of Mammalian Glucagon (20–25) by the Active Ester Method

As one of the building blocks for the synthesis of mammalian glucagon, the protected hexapeptide **9** was prepared in a stepwise manner starting from C-terminal H-Gln-Trp-OH using N^{α}-protected amino acid active esters as shown in Scheme 15.[68]

Scheme 15 Preparation of Protected Hexapeptide Corresponding to the Sequence 20–25 of Mammalian Glucagon[68]

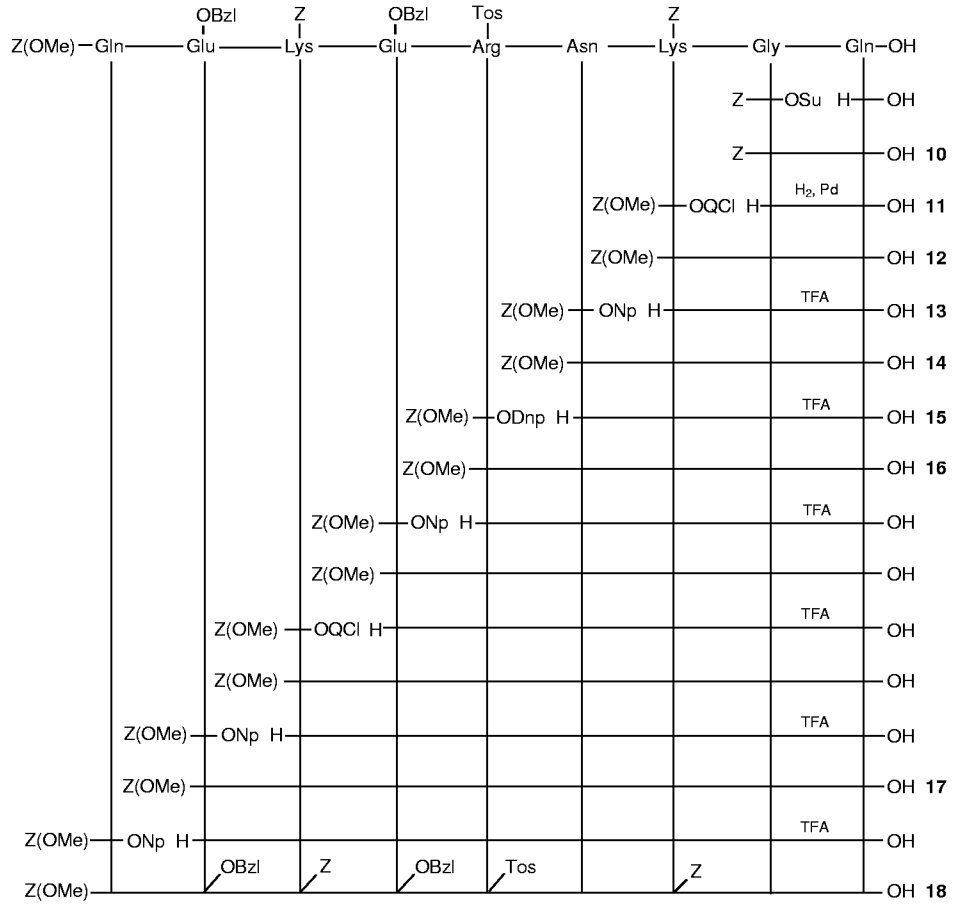


Boc-Gln-Asp(OBzl)-Phe-Val-Gln-Trp-OH (9):[68]
Boc-Asp(OBzl)-Phe-Val-Gln-Trp-OH (6.0 g, 6.79 mmol) was treated with TFA (50 mL) under N_2 and the TFA salt of the free base that was obtained as a fine powder was dissolved in DMF (100 mL) together with TEA (2.0 mL, 14.3 mmol). To the soln was added Boc-Gln-ONdc [prepared from Boc-Gln-OH (1.76 g, 7.13 mmol)] and the mixture was stirred at rt for 16 h, and then concentrated. The residue was triturated with aq AcOH to give a powder, which was purified by reprecipitation (aq MeCN); yield: 5.50 g (83%); mp 210–212 °C; $[\alpha]_D^{27}$ −10.1 (*c* 1.1, DMF); Anal. Calcd for $C_{51}H_{65}N_9O_{13}$: C, 60.5; H, 6.47; N, 12.5. Found: C, 60.2; H, 6.37; N, 12.2.

4.1.1.1.2.2 Synthesis of Protected Nonapeptide Z(OMe)-Motilin(14–22)-OH by the Active Ester Method

In order to construct porcine motilin,[69] a protected nonapeptide corresponding to the sequence 14–22 of motilin was prepared in a stepwise elongation manner by the active ester method starting from C-terminal glutamine, as shown in Scheme 16.[70]

Scheme 16 Preparation of a Protected Nonapeptide Corresponding to Sequence 14–22 of Motilin[70]



The triethylamine salt of H-Gln-OH was condensed with Z-Gly-OH by the *N*-hydroxysuccinimide ester procedure.[71] After the reaction, the solvent was removed and the residue was dissolved in water. The aqueous phase was washed with EtOAc to remove unreacted active ester and then acidified with citric acid. From the aqueous solution, Z-Gly-Gln-OH (**10**) was obtained in crystalline form. The Z group was removed by catalytic hydrogenation to give H-Gly-Gln-OH (**11**). To this free dipeptide, Z(OMe)-Lys(Z)-OH was introduced by the 5-chloro-8-quinolyl (QCl) ester procedure according to Jakubke.[72] The resulting Z(OMe)-Lys(Z)-Gly-Gln-OH (**12**), after treatment with TFA, was condensed with Z(OMe)-Asn-OH by the 4-nitrophenyl ester procedure[8] to give Z(OMe)-Asn-Lys(Z)-Gly-Gln-OH (**14**). Removal of the Z(OMe) group from this tetrapeptide was performed as usual by TFA treatment. To the resulting tetrapeptide **15**, Z(OMe)-Arg(Tos)-OH was introduced by the 2,4-dinitrophenyl ester method[73] without isolating the corresponding active ester. The desired pentapeptide, Z(OMe)-Arg(Tos)-Asn-Lys(Z)-Gly-Gln-OH (**16**), was no longer ex-

for references see p 623

tractable with EtOAc. Therefore, purification was carried out by batchwise washing with citric acid and water, followed by recrystallization from MeOH and Et_2O, or reprecipitation from DMF with EtOAc. The combination of the TFA treatment for removal of the Z(OMe) group and the active ester procedure was employed to elongate the above pentapeptide chain to the nonapeptide (**16** → **18**). Two residues of Glu(OBzl) and one of Gln were introduced by the 4-nitrophenyl ester procedure and Lys(Z) by the 5-chloro-8-quinolyl ester procedure. All intermediates and the protected nonapeptide were purified by the batchwise washing and reprecipitation methods.

Z-Gly-Gln-OH (10):[70]
Z-Gly-OSu (29.0 g, 0.095 mol) in THF (200 mL) was added to a soln of H-Gln-OH (16.1 g, 0.11 mol) in H_2O (100 mL) containing TEA (30.8 mL, 0.22 mol) and the mixture was stirred at rt for 48 h. The solvent was removed and the residue was dissolved in H_2O (200 mL), which after washing with EtOAc, was acidified with citric acid. The aq soln was saturated with NaCl and stood at rt for 4 d. The crystalline mass formed was collected by filtration, washed with a small amount of H_2O and then recrystallized (MeOH) twice; yield: 21.1 g (62%); mp 149–151 °C; $[\alpha]_D^{27}$ +5.4 (*c* 0.5, MeOH); Anal. Calcd for $C_{13}H_{19}N_3O_6 \cdot 0.5\ H_2O$: C, 52.0; H, 5.82; N, 12.1. Found: C, 51.9; H, 5.60; N, 11.9.

H-Gly-Gln-OH (11):[70]
Z-Gly-Gln-OH (**10**; 10.1 g, 29.9 mmol) in MeOH (100 mL) containing AcOH (1 mL) was hydrogenated in the usual manner over a Pd catalyst for 12 h. The soln was filtered, the filtrate was condensed in vacuo and the residue was treated with EtOAc. The resulting solid was recrystallized (H_2O/EtOH); yield: 5.79 g (95%); mp 198–200 °C; $[\alpha]_D^{27}$ −1.6 (*c* 1.0, H_2O); Anal. Calcd for $C_7H_{13}N_3O_4 \cdot 0.5\ H_2O$: C, 39.6; H, 6.65; N, 19.8. Found: C, 39.9; H, 6.74; N, 19.9.

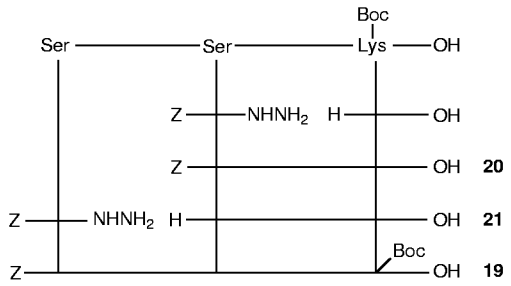
Z(OMe)-Lys(Z)-Gly-Gln-OH (12):[70]
Z(OMe)-Lys(Z)-OQCl (21.8 g, 36.0 mmol) in DMF (100 mL) was combined with a soln of H-Gly-Gln-OH (**11**; 6.1 g, 30.0 mmol) in H_2O (100 mL) containing TEA (8.4 mL, 60 mmol) and the mixture was stirred at rt for 48 h. The solvent was removed in vacuo and the residue was dissolved in 5% NH_4OH (100 mL), which after washing with EtOAc, was acidified with 5% citric acid. The resulting precipitate was extracted with EtOAc. The organic phase was washed with brine, dried (Na_2SO_4), and then concentrated. Treatment of the residue with Et_2O afforded the solid, which was recrystallized (EtOH); yield: 13.8 g (73%); mp 163–166 °C; $[\alpha]_D^{27}$ −6.6 (*c* 1.0, MeOH); Anal. Calcd for $C_{30}H_{39}N_5O_{10}$: C, 57.2; H, 6.24; N, 11.1. Found: C, 56.9; H, 6.25; N, 11.0.

Z(OMe)-Gln-Glu(OBzl)-Lys(Z)-Glu(OBzl)-Arg(Tos)-Asn-Lys(Z)-Gly-Gln-OH (18):[70]
Octapeptide **17** (3.0 g, 1.71 mmol) was treated with TFA (8 mL) in the presence of anisole (1.5 mL) as usual. The TFA salt formed by addition of dry Et_2O was dried (KOH pellets) and then dissolved in DMF (30 mL) containing TEA (0.48 mL, 3.43 mmol) and the soln, after addition of Z(OMe)-Gln-ONp (1.0 g, 2.32 mmol), was stirred at rt for 48 h. The solvent was removed and the residual solid formed by addition of 3% citric acid and EtOAc was washed batchwise with 5% citric acid and H_2O and then dissolved in $CHCl_3$/MeOH/H_2O (8:3:1). The soln was filtered and condensed. The residue was treated with EtOAc to form a fine powder; yield: 2.62 g (81%); mp 207–211 °C; $[\alpha]_D^{27}$ −6.2 (*c* 1.0, DMF). Amino acid ratios in an acid hydrolysate: Glu, 4.41; Lys, 1.87; Arg, 0.95; Asp, 1.00; Gly, 1.00 (average recovery 91%); Anal. Calcd for $C_{90}H_{115}N_{17}O_{26}S \cdot H_2O$: C, 56.9; H, 6.21; N, 12.5. Found: C, 56.9; H, 6.20; N, 12.7.

4.1.1.1.2.3 Synthesis of a Protected Tripeptide by the Azide Procedure

A building block, Z-Ser-Ser-Lys(Boc)-OH (**19**), for the synthesis of a triacontapeptide corresponding to the C-terminal sequence 116–145 of human chorionic gonadotropin (hCG)[74–76] was prepared without protection of the side chain of Ser by the route shown in Scheme 17.[77]

Scheme 17 Synthetic Route to a Protected Tripeptide by the Azide Procedure:[77]



Z-Ser-Lys(Boc)-OH (20):[77]
Z-Ser-N_3 [prepared from Z-Ser-$NHNH_2$ (2.53 g, 9.57 mmol) and *tert*-butyl nitrite (1.18 mL, 9.93 mmol) in the usual manner[78]] in DMF (15 mL) was mixed with a soln of H-Lys(Boc)-OH (1.97 g, 7.03 mmol) in 50% aq DMF (20 mL) containing TEA (1.1 mL, 7.86 mmol). The mixture was stirred in a cold room (4 °C) for 2 d, then the solvent was removed. The residue was dissolved in H_2O and the soln was washed with EtOAc. The H_2O layer was acidified with citric acid and the resulting precipitate was collected by filtration, washed with H_2O and recrystallized (MeOH); yield: 2.6 g (70%); mp 126–128 °C; $[\alpha]_D^{27}$ +6.7 (*c* 1.0, DMF); Anal. Calcd for $C_{22}H_{33}N_3O_8 \cdot 0.5\ H_2O$: C, 55.5; H, 7.19; N, 8.8. Found: C, 55.4; H, 7.19; N, 8.9.

H-Ser-Lys(Boc)-OH (21):[77]
Z-Ser-Lys(Boc)-OH (**20**; 1.2 g, 2.57 mmol) was hydrogenated over a Pd catalyst in 80% aq MeOH (70 mL). After removal of Pd and the solvent, the residue was washed with EtOAc and lyophilized from dioxane and H_2O; yield: 805 mg (94%); mp 219–221 °C (dec); $[\alpha]_D^{27}$ +6.0 (*c* 1.0, 50% AcOH); Anal. Calcd for $C_{14}H_{27}N_3O_6$: C, 50.4; H, 8.10; N, 12.6. Found: C, 50.3; H, 8.20; N, 12.8.

Z-Ser-Ser-Lys(Boc)-OH (19):[77]
A soln of Z-Ser-N_3 [prepared from Z-Ser-$NHNH_2$ (759 mg, 3.00 mmol)] in DMF (5 mL) was combined with a soln of **21** (500 mg, 1.50 mmol) in 50% aq DMF (3 mL) containing TEA (0.42 mL, 3.00 mmol). The mixture was stirred in a cold room (4 °C) for 20 h and the solvent was removed. The residue was dissolved in H_2O (10 mL), and the soln was washed with EtOAc. The H_2O layer was acidified with citric acid and the resulting precipitate was extracted with EtOAc. The extract was washed with H_2O and dried (Na_2SO_4). After removal of the solvent, the residue was precipitated from EtOAc and Et_2O; yield: 540 mg (77%); mp 72–75 °C (dec); $[\alpha]_D^{27}$ +6.6 (*c* 0.9, DMF); Anal. Calcd for $C_{25}H_{38}N_4O_{10} \cdot H_2O$: C, 53.3; H, 7.0; N, 9.9. Found: C, 53.2; H, 6.8; N, 9.8.

4.1.2 Segment Condensation with Minimum Protection

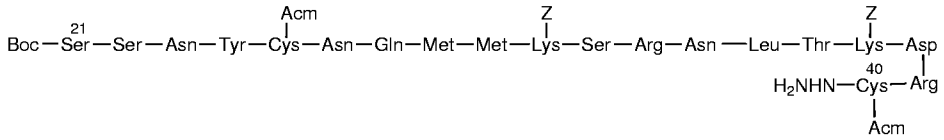
For the synthesis of larger peptides by the solution method, the condensation of peptide segments is an attractive strategy. The separation of the product from truncated or deleted sequences is considerably facilitated. The main drawback of this method lies in the facile racemization of the carboxylic segment of the C-terminal residue when the residue is neither glycine nor proline. This racemization proceeds primarily through an oxazol-5(4*H*)-one intermediate by the reaction of the penultimate residue on the terminal, activated carboxylate.[79] The other impediments confronting segment condensation are the decreasing coupling rates with increasing fragment size, the lower solubility even in solvents of high dissolving power, and difficulties in the removal of protecting groups from large peptides. The primary problem is the solubility, which may be influenced by changing the solvent or the mode of protection, that is, from the tactics of maximum protection to that of minimum protection. The maximum protection procedure provides optimum safety against side reactions and the freedom of choice for coupling methods. This approach has its limitations since the many protecting groups may result in larger peptides that are insoluble. As an alternative, minimum protection of side-chain functions has been employed by Hirschman et

for references see p 623

al.[80] and Hofmann et al.[81] in their synthesis of different ribonucleases. Some naturally occurring amino acids possess a third functional group.

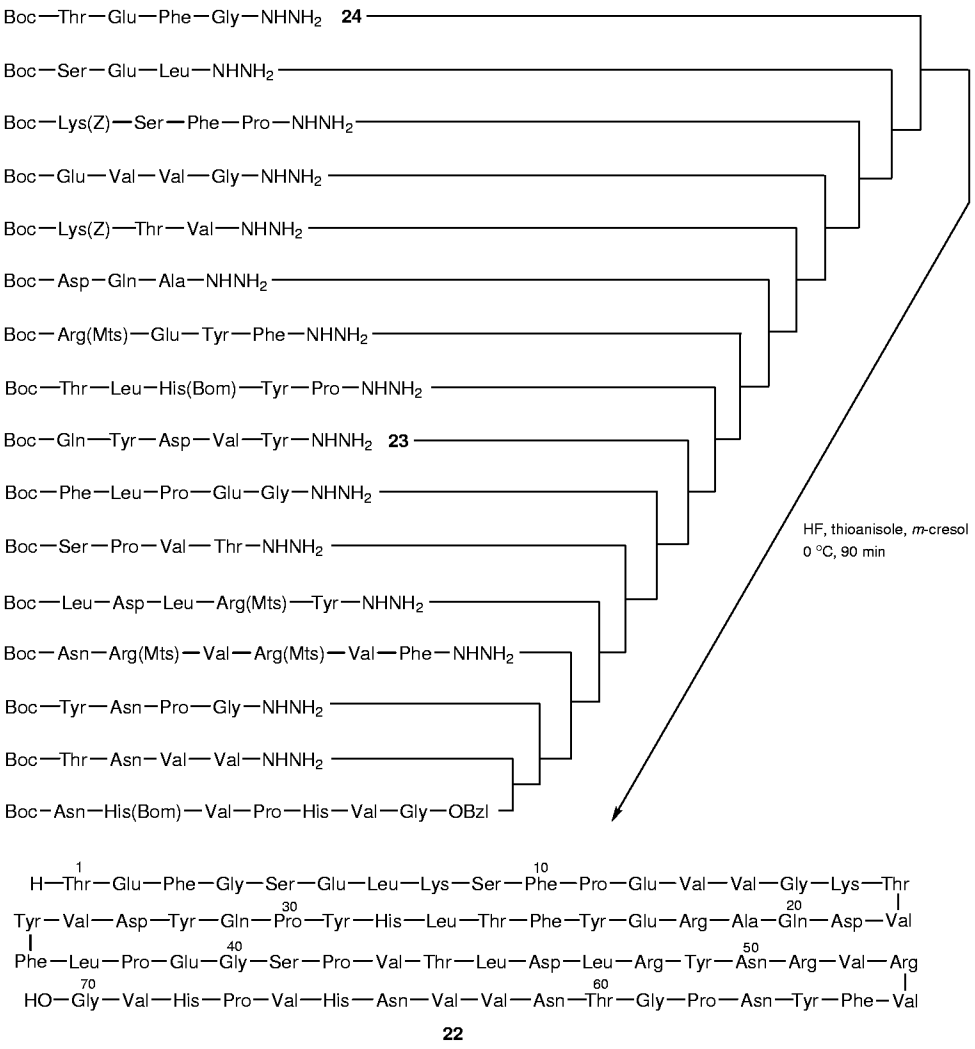
For the construction of peptides, the α-amino group of the carboxy component (amino acid or peptide), the ε-amino group of Lys (or δ-amino group of ornithine) in either the carboxy or amino component, and the β-thiol group of Cys in either component require protecting groups. In contrast, there is no general rule as to the requirement of protection for other side-chain functional groups such as the carboxy groups of Asp and Glu, the guanidino group of Arg, the phenolic hydroxy group of Tyr, aliphatic hydroxy groups of Ser and Thr, the imidazole of His, the primary carboxamide of Asn and Gln, the aliphatic thioether of Met, and the indole ring of Trp. Hirschman et al.[80] used the differences in the reactivity of functional groups and employed the tactic of minimum protection for the synthesis of the S-protein of RNase A. This peptide consists of 104 amino acids and contains all of the trifunctional amino acids with the exception of tryptophan. As indicated in Scheme 18 for the protected 21–40 fragment, only the ε-amino group of Lys and β-thiol group of Cys were protected. At the final coupling, two large segments, Boc-(21–64)-N_3 and H-(65–124)-OH, were employed. The azide of the protected 44-peptide was applied in about fourfold excess over the amino component of 60 residues. Nevertheless, the 104-peptide (S-protein) formed only in low yield and did not lend itself to isolation. The fact that the components carried only the mandatory protecting groups and were, therefore, reasonably soluble in the solvent, was in itself not sufficient for satisfactory results.

Scheme 18 Minimum Protection Tactic in the Synthesis of the S-Protein of RNase A (21–40) Fragment[80]



In a later, more successful synthesis of RNase A,[82] small segments were activated (in the form of azides) and incorporated into the growing peptide chain. The activated, low-molecular-weight intermediates were used in considerable excess and this excess increased with the increase in the molecular weight of the amino components. Okada et al.[83,84] synthesized eglin c (**22**), which consists of 70 amino acid residues, by the route shown in Scheme 19 by the minimum protection method using excess amounts of small-segment azides. Thus, they could solve the main problems such as the low solubility of protected large peptides even in solvents of high dissolving power, and the difficulties in the removal of protecting groups in the final stage. At each fragment-condensation step, the product could be purified using DMF/MeOH. This method of protection (or nonprotection), however, restricts the choice of coupling methods to only a few, such as the azide procedure, but enhances the solubility of the peptides.

Scheme 19 Synthetic Route to Eglin c (1–70)[83,84]



4.1.2.1 Synthesis of a Peptide Fragment Building Block by the Minimum Protection Method

In order to prepare Asp- or Glu-containing peptide fragments, in which side-chain carboxy groups are unblocked, one method is to introduce aspartic or glutamic acid into the peptide through its *N*-carboxyanhydride.[85] Another method is to introduce aspartic or glutamic acid with the side chain protected as the benzyl ester and then to remove the benzyl ester by H_2/Pd and finally convert the C-terminal methyl ester into the corresponding hydrazide.

Boc-Gln-Tyr-Asp-Val-Tyr-OMe [Boc-(Eglin c 31–35)-OMe]:[86]
The title compound was prepared from Boc-Gln-Tyr-Asp(OBzl)-Val-Tyr-OMe (500 mg, 0.57 mmol) by catalytic hydrogenation over a Pd catalyst in MeOH; yield: 420 mg (94%); mp 171–190 °C; $[\alpha]_D^{27}$ −18.9 (*c* 0.8, DMF); Anal. Calcd for $C_{39}H_{52}N_6O_{13} \cdot H_2O$: C, 55.7; H, 6.64; N, 10.3. Found: C, 55.7; H, 6.63; N, 10.3.

for references see p 623

Boc-Gln-Tyr-Asp-Val-Tyr-NHNH$_2$ [Boc-(Eglin c 31–35)-NHNH$_2$, 23]; General Procedure for Hydrazinolysis:[86]
$H_2NNH_2 \cdot H_2O$ (90%, 0.44 mL, 8.8 mmol) was added to a soln of Boc-Gln-Tyr-Asp-Val-Tyr-OMe (1.4 g, 1.8 mmol) in MeOH (15 mL). The mixture was stored at rt overnight. A precipitate was collected by filtration and washed with 10% AcOH, H_2O, and EtOH; yield: 850 mg (61%); mp 207–213 °C; $[\alpha]_D^{27}$ −32.8 (*c* 0.6, DMF); Anal. Calcd for $C_{37}H_{52}N_8O_{12} \cdot 0.5\ H_2O$: C, 54.9; H, 6.59; N, 13.8. Found: C, 54.5; H, 6.81; N, 14.2.

4.1.2.2 Synthesis of Protected Eglin c by the Minimum Protection Method

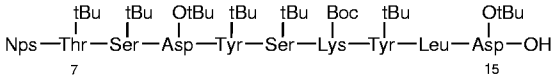
Of the fragment condensation procedures, the azide procedure[78] was employed for each fragment condensation in order to avoid racemization and side reactions (over reaction).[87]

Boc-Thr-Glu-Phe-Gly-Ser-Glu-Leu-Lys(Z)-Ser-Phe-Pro-Glu-Val-Val-Gly-Lys(Z)-Thr-Val-Asp-Gln-Ala-Arg(Mts)-Glu-Tyr-Phe-Thr-Leu-His(Bom)-Tyr-Pro-Gln-Tyr-Asp-Val-Tyr-Phe-Leu-Pro-Glu-Gly-Ser-Pro-Val-Thr-Leu-Asp-Leu-Arg(Mts)-Tyr-Asn-Arg(Mts)-Val-Arg(Mts)-Val-Phe-Tyr-Asn-Pro-Gly-Thr-Asn-Val-Val-Asn-His(Bom)-Val-Pro-His-Val-Gly-OBzl [Protected Eglin c, Protected-22]):[84]
Boc-Thr-Glu-Phe-Gly-N_3 [prepared from Boc-(1–4)-$NHNH_2$ (**24**; 58 mg, 0.10 mmol) and isopentyl nitrite (0.014 mL, 0.1 mmol)[78]] in DMF (10 mL) was added to a soln of H-(5–70)-OBzl · TFA [prepared as usual from Boc-(5–70)-OBzl (115 mg, 13 μmol), TFA (1.0 mL, 12 mmol), anisole (0.1 mL, 0.92 mmol) and *m*-cresol (0.1 mL, 0.96 mmol)] in DMF (50 mL). The mixture was stirred at 0 °C for 2 d. After removal of the solvent, MeOH was added to the residue to give a solid material, which was collected by filtration and washed with MeOH; yield: 104 mg (81%); mp 229–239 °C; $[\alpha]_D^{27}$ −42.7 (*c* 0.1, DMSO); Anal. Calcd for $C_{453}H_{632}N_{96}O_{117}S_4 \cdot 9\ H_2O$: C, 56.7; H, 6.85; N, 14.0. Found: C, 56.4; H, 6.50; N, 13.7.

4.1.3 Segment Condensation with Maximum Protection

In contrast to the minimum protection strategy, in the maximum protection strategy, the maximum number of functional groups are masked in order to avoid side reactions. Wuensch et al. followed this strategy in the synthesis of the peptide hormone glucagon[88–90] and protected side chains of all hydroxyamino acids, His, Lys, and Asp with acid-labile protecting groups. The resulting glucagon (7–15) intermediate (Scheme 20) indicates that a distinctly greater effort must be made to prepare such synthetic intermediates than is demanded by the minimum protection tactic. Sakakibara also recommended solution synthesis of peptides by the maximum protection procedure using the DCC coupling method and HF final deprotection method.[91] As described above, segments with full side-chain protection show less byproduct formation. It is difficult to construct highly soluble segments with full protection. However, if the principle of maximum protection is followed, the carboxy component can be activated by various methods and to almost any extent.

Scheme 20 Maximum Protection Tactic in the Synthesis of Glucagon (7–15) Fragment[88]



4.1.3.1 Fragments from Solution Synthesis

The most frequently used segment condensation methods are: (1) azide coupling, (2) the DCC method using acidic additives, (3) the active ester method, and (4) the mixed anhydride method. The application of the azide procedure is preferable when the segment to be activated does not contain C-terminal Gly or Pro residues, although racemization using the azide procedure was demonstrated by Kemp et al.[17] A further advantage is that the azides do not acylate hydroxy groups, thus the principle of minimum protection can be applied.

However, its disadvantage is the considerably decreased rate of coupling which favors the rearrangement reaction leading to carbamido-peptides that are difficult to separate. A modified azide procedure, the DPPA method, was also developed.[92] The most widely used acidic additives in the DCC-mediated segment condensations are *N*-hydroxysuccinimide (HOSu),[93] 1,2,3-benzotriazol-1-ol (HOBt),[94] and 3-hydroxy-1,2,3-benzotriazin-4-(3*H*)-one (HODhbt).[95] The DCC/HOSu method was employed successfully by Wuensch et al. for the preparation of glucagon.[88–90] However, it was shown that HOSu reacts with DCC (in molar proportion 3:1) forming dicyclohexylurea and succinimide oxycarbonyl-β-alanine hydroxysuccinimide ester through Lossen rearrangement.[96] The DCC/HOBt method was successfully employed for the synthesis of human ACTH.[97] Sakakibara et al. synthesized calciseptine, an L-type specific calcium channel blocker,[98] Na^+,K^+-ATPase inhibitor-1 (SPAI-1),[99] and O-palmitoylated 44-residue peptide amide (PLTXII) blocking presynaptic calcium channels in *Drosophila*[100] by using the EDC/HODhbt segment coupling method. In this section, the synthesis of protected peptide hydrazides and protected peptides with or without C-terminal protection as building blocks to construct large peptides is described.

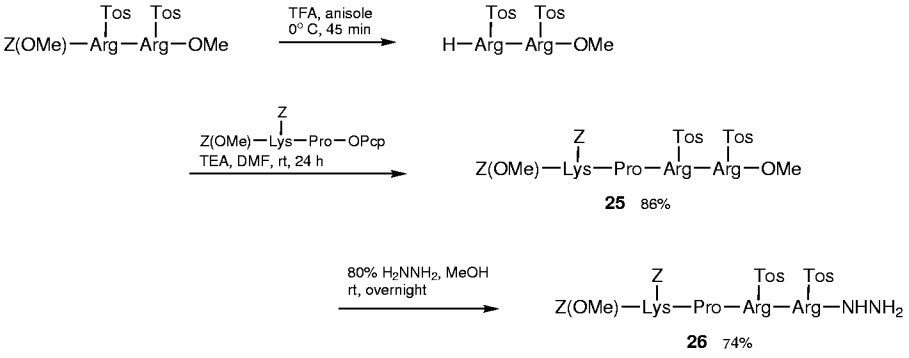
4.1.3.1.1 Synthesis of Protected Peptide Hydrazides

In the maximum protection procedure, the protected peptide hydrazide is used as a carboxy component for the construction of large peptides. As described in Sections 4.1.1.1.1.4 and 4.1.2, protected peptide hydrazides are derived from the corresponding alkyl esters by treatment with hydrazine hydrate. Another method for the synthesis of protected peptide hydrazides is by conversion from substituted hydrazides.

4.1.3.1.1.1 Synthesis of a Protected Peptide Hydrazide from the Alkyl Ester

As shown in Scheme 21, a protected tetrapeptide hydrazide corresponding to sequence 6–9 of bovine neurotensin was prepared.[101] Z(OMe)-Arg(Tos)-Arg(Tos)-OMe was prepared by the DCC procedure. After treatment of this dipeptide with TFA, the resulting amine was coupled with Z(OMe)-Lys(Z)-Pro-OPcp[102] to give Z(OMe)-Lys(Z)-Pro-Arg(Tos)-Arg-(Tos)-OMe (**25**). This methyl ester was treated with hydrazine hydrate to give the desired protected peptide hydrazide **26**.

Scheme 21 Synthesis of the Protected Tetrapeptide Hydrazide Corresponding to Sequence 6–9 of Bovine Neurotensin[101]



for references see p 623

Z(OMe)-Lys(Z)-Pro-Arg(Tos)-Arg(Tos)-OMe (25):[101]
Z(OMe)-Arg(Tos)-Arg(Tos)-OMe (8.17 g, 10 mmol) was treated with TFA (16 mL) in the presence of anisole (4.0 mL) in an ice bath for 45 min. The excess TFA was removed and the residue was treated with Et_2O. The resultant powder was collected by filtration, dried over KOH pellets in vacuo and then dissolved in DMF (100 mL). TEA (3.15 mL, 22.5 mmol) and Z(OMe)-Lys(Z)-Pro-OPcp (9.85 g, 12.4 mmol) were added and the mixture was stirred at rt for 24 h. The solvent was removed and the residue was extracted with EtOAc, which was washed with 5% citric acid, 5% Na_2CO_3, and H_2O, dried (Na_2SO_4), and then concentrated. The residue was dissolved in $CHCl_3$ and the soln was column chromatographed (silica gel, 2.8 × 14 cm, $CHCl_3$/MeOH 95:5). Fractions containing the desired product were combined and the solvent was removed. Treatment of the residue with Et_2O afforded a fine powder, which was washed with H_2O and then recrystallized (MeOH/Et_2O); yield: 10.1 g (86%); mp 95–101 °C; $[\alpha]_D^{27}$ −23.6 (*c* 0.8, MeOH); Anal. Calcd for $C_{55}H_{73}N_{11}O_{14}S_2 \cdot 2\ H_2O$: C, 54.5; H, 6.40; N, 12.7. Found: C, 54.5; H, 6.42; N, 12.4.

Z(OMe)-Lys(Z)-Pro-Arg(Tos)-Arg(Tos)-NHNH$_2$ (26):[101]
To a soln of **25** (10.2 g, 8.7 mmol) in MeOH (100 mL), 80% $H_2NNH_2 \cdot H_2O$ (5.0 mL, 80 mmol) was added and the soln was allowed to stand overnight. The solvent was removed and the residue was treated with Et_2O. The resulting powder was washed with Et_2O and H_2O and then recrystallized (MeOH/Et_2O); yield: 7.53 g (74%); mp 108–111 °C ; $[\alpha]_D^{27}$ −17.7 (*c* 0.6, MeOH); Anal. Calcd for $C_{54}H_{73}N_{13}O_{13}S_2$: C, 55.1; H, 6.25; N, 15.5. Found: C, 55.0; H, 6.36; N, 15.2.

4.1.3.1.1.2 Synthesis of a Protected Peptide Hydrazide from the Substituted Hydrazide

In the case of the protected peptide alkyl ester containing Arg(NO_2), Asp(OtBu), Asp(OBzl), or Glu(OBzl), treatment with hydrazine hydrate results in side reactions, such as hydrazinolysis of side-chain functional groups. In order to avoid such side reactions, it is preferable to synthesize the protected peptide hydrazide from the corresponding substituted hydrazide. The representative substituted hydrazides for $R^1C(O)NH\text{-}CHR^2\text{-}C(O)\text{-}NHNH\text{-}X$ are: X = Z,[103] C(O)OTrt,[104] Boc,[105] Troc,[106] $C(O)OCH_2$-4-pyridyl,[107] and $C(O)OCMe_2\text{-}CCl_3$.[108]

Synthesis of Boc-Glu(OBzl)-Ile-Gln-Leu-NHNH$_2$ from the Corresponding Troc-Hydrazide:[109]
Boc-Glu(OBzl)-Ile-Gln-Leu-NHNH-Troc (15.0 g, 20.7 mmol) in a mixture of DMF (40 mL) and AcOH (6.8 mL, 10 equiv) was treated with Zn dust (6.67 g, 10 equiv) at rt for 17 h. The soln was filtered, the filtrate was concentrated under reduced pressure at 35 °C and the residue was treated with a sat. soln of EDTA, then with $NaHCO_3$ to adjust the pH to neutral. The resulting powder was washed with H_2O and precipitated from DMF with MeOH; yield: 5.31 g (74%); mp 260 °C (dec); $[\alpha]_D^{27}$ −22.3 (*c* 0.6, DMSO); Anal. Calcd for $C_{34}H_{55}N_7O_9$: C, 57.9; H, 7.85; N, 13.9. Found: C, 57.8; H, 8.00; N, 13.6.

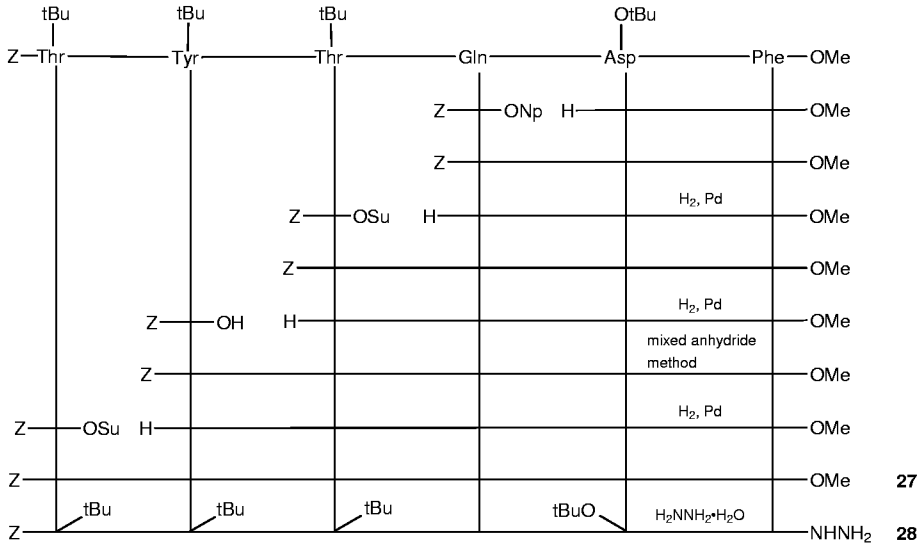
4.1.3.1.2 Synthesis of Protected Peptides Using Various Protecting Groups

Peptides and proteins have been synthesized by carefully selecting N^{α}-protecting groups and side-chain protecting groups. In order to avoid losing the side-chain protecting groups during the synthesis of peptides, orthogonal protecting strategies are preferable.

4.1.3.1.2.1 Synthesis of a Protected Peptide Using the Benzyloxycarbonyl/*tert*-Butyl Strategy

As previously described, the Z group can be removed by catalytic hydrogenation,[4] sodium in liquid ammonia,[5] or acidolysis.[6] Of these removal methods, the catalytic hydrogenolysis method is the most preferable in the Z/tBu strategy, as shown in Scheme 22.[110–113]

Scheme 22 Synthesis of the Protected Hexapeptide Corresponding to the Sequence 11–16 of Calcitonin M[112]



Each Z group was removed by hydrogenation over a Pd catalyst. Starting from H-Asp(OtBu)-Phe-OMe, Z-Gln-ONp, Z-Thr(tBu)-OSu, Z-Tyr(tBu)-OH, and Z-Thr-(tBu)-OSu were coupled successively to give Z-Thr(tBu)-Tyr(tBu)-Thr(tBu)-Gln-Asp-(OtBu)-Phe-OMe (**27**), which was converted into the corresponding hydrazide **28** by treatment with hydrazine hydrate.

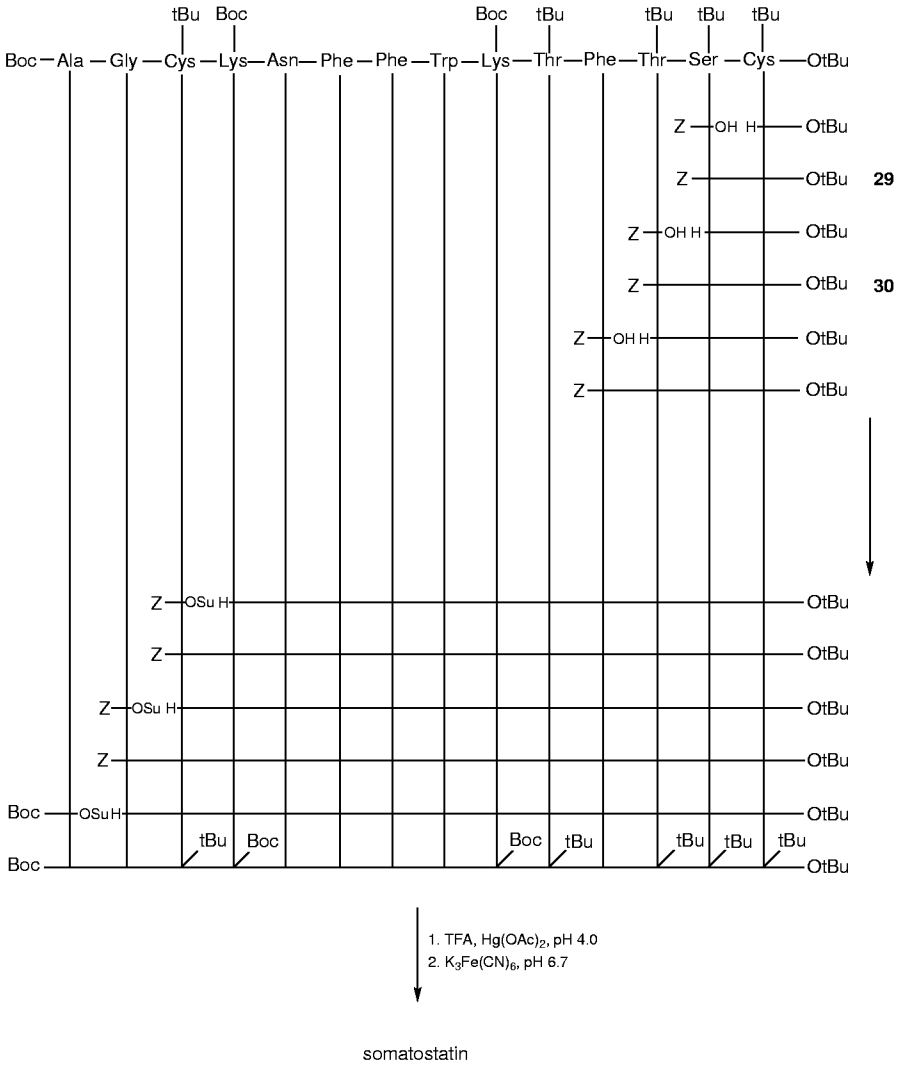
Z-Thr(tBu)-Tyr(tBu)-Thr(tBu)-Gln-Asp(OtBu)-Phe-OMe (27):[112]
Z-Tyr(tBu)-Thr(tBu)-Gln-Asp(OtBu)-Phe-OMe (2.36 g, 2.38 mmol) in MeOH (450 mL) was hydrogenated over a Pd catalyst. After removal of the solvent and Pd, Et_2O was added to the residue to afford crystals. This amine (6.31 g, 7.38 mmol) and Z-Thr(tBu)-OSu (4.33 g, 11.0 mmol) were dissolved in DMF (13 mL). The soln was stirred at 25 °C for 21 h. After removal of the solvent, the residue was extracted with EtOAc, which was washed successively with acidic and basic H_2O, dried (Na_2SO_4), and concentrated to afford solid product. The crude product was precipitated (MeOH/Et_2O) and recrystallized (MeOH); yield: 6.67 g (79%); mp 180–182 °C (dec); $[\alpha]_D^{27}$ +9 (*c* 2.1, DMF); Anal. Calcd for $C_{60}H_{87}N_7O_{15} \cdot H_2O$: C, 61.9; H, 7.70; N, 8.42. Found: C, 62.2; H, 7.55; N, 8.37.

4.1.3.1.2.2 Removal of the Benzyloxycarbonyl Group by Hydrogenation in Liquid Ammonia

The H_2/Pd hydrogenolysis system for the removal of the Z group from peptides is limited to sulfur-free peptides. In order to overcome this limitation, Meienhofer successfully conducted hydrogenolysis in liquid ammonia as a solvent to remove the Z group from Cys- or Met-containing peptides.[114,115] As shown in Scheme 23, protected somatostatin was prepared in a stepwise manner starting from H-Cys(tBu)-OtBu.

for references see p 623

Scheme 23 Synthesis of Somatostatin[115]

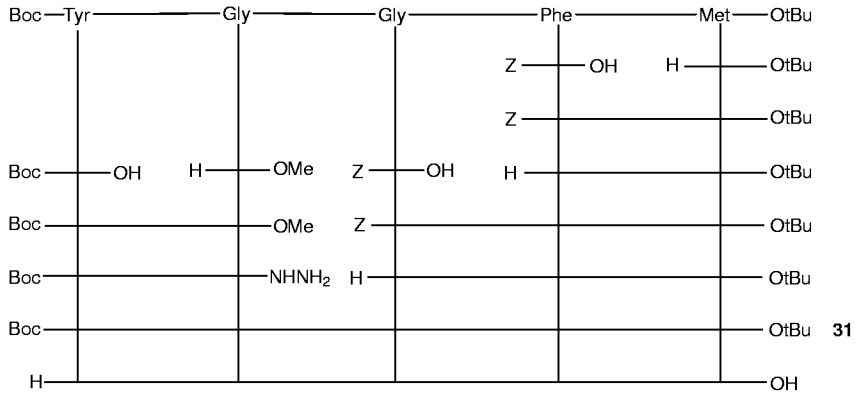


Z-Thr(tBu)-Ser(tBu)-Cys(tBu)-OtBu (30):[115]
Z-Ser(tBu)-Cys(tBu)-OtBu (**29**; 21.5 g, 4.2 mmol) in liq NH_3 (3 L) was hydrogenated in the presence of Pd for 6 h. Removal of NH_3 under N_2 was followed by addition of MeOH, filtration, and concentration in vacuo. The residue was crystallized (petroleum ether) to give a white crystalline product; yield: 10.7 g (68%); mp 71–73 °C; $[\alpha]_D^{27}$ −6.0 (*c* 1.0, MeOH). Condensation with Z-Thr(tBu)-OH by a mixed anhydride procedure afforded the crystalline tripeptide **30** from petroleum ether; yield: 92%; mp 73.5–76 °C; $[\alpha]_D^{27}$ +6.9 (*c* 0.7, MeOH).

4.1.3.1.2.3 Removal of the Benzyloxycarbonyl Group by Catalytic Transfer Hydrogenation

The above limitation of the H_2/Pd hydrogenolysis system for the removal of the Z group from peptides due to catalyst poisoning by Cys or Met residues was overcome by using catalytic transfer hydrogenation.[116–118] Ammonium formate,[119] cyclohexene,[116] and *cis*-Decalin[118] have been employed as hydrogen donors. As shown in Scheme 24, Met-enkephalin was prepared using cyclohexene as a hydrogen donor.[118]

Scheme 24 Synthesis of Met-Enkephalin Using Transfer Hydrogenation[118]



Boc-Tyr-Gly-Gly-Phe-Met-OtBu (31):[118]
A soln of Z-Gly-Phe-Met-OtBu (130 mg, 0.24 mmol) in dry EtOH (8 mL) and cyclohexene (4 mL) was refluxed over a Pd catalyst for 30 min. After removal of Pd, the solvent was removed to give the corresponding amine. Boc-Tyr-Gly-N_3 [prepared from Boc-Tyr-Gly-$NHNH_2$ (88 mg, 0.25 mmol) with isopentyl nitrite in the usual manner] in DMF (5 mL) was added to a cold soln of the tripeptide amine in DMF (5 mL). The mixture was stirred overnight at 4 °C and concentrated. The residue was extracted with EtOAc which was then washed with H_2O, dried (Na_2SO_4), and concentrated. The residue in $CHCl_3$ (5 mL) was column chromatographed (silica gel, 1.2 × 18 cm). The column was eluted with $CHCl_3$ (200 mL) followed by 1% MeOH in $CHCl_3$ (500 mL) and 3% MeOH in $CHCl_3$ (200 mL). The eluate containing 3% MeOH in $CHCl_3$ provided the desired pentapeptide **31**; yield: 80 mg (48%); mp 125–130 °C; $[\alpha]_D^{27}$ −11.0 (*c* 1.0, DMF); Anal. Calcd for $C_{36}H_{51}N_5O_9S$: C, 59.2; H, 7.04; N, 9.6. Found: C, 59.5; H, 7.06; N, 9.5.

4.1.3.1.2.4 Synthesis of a Protected Peptide Using the 2-Nitrophenylsulfenyl/*tert*-Butyl Strategy

The combination of N^{α}-Nps protection and side-chain tBu protection is a useful strategy to prepare peptides, because the Nps group can be removed by 2 equivalents of HCl[120] and other specific methods[121] which leave the tBu protecting group intact.

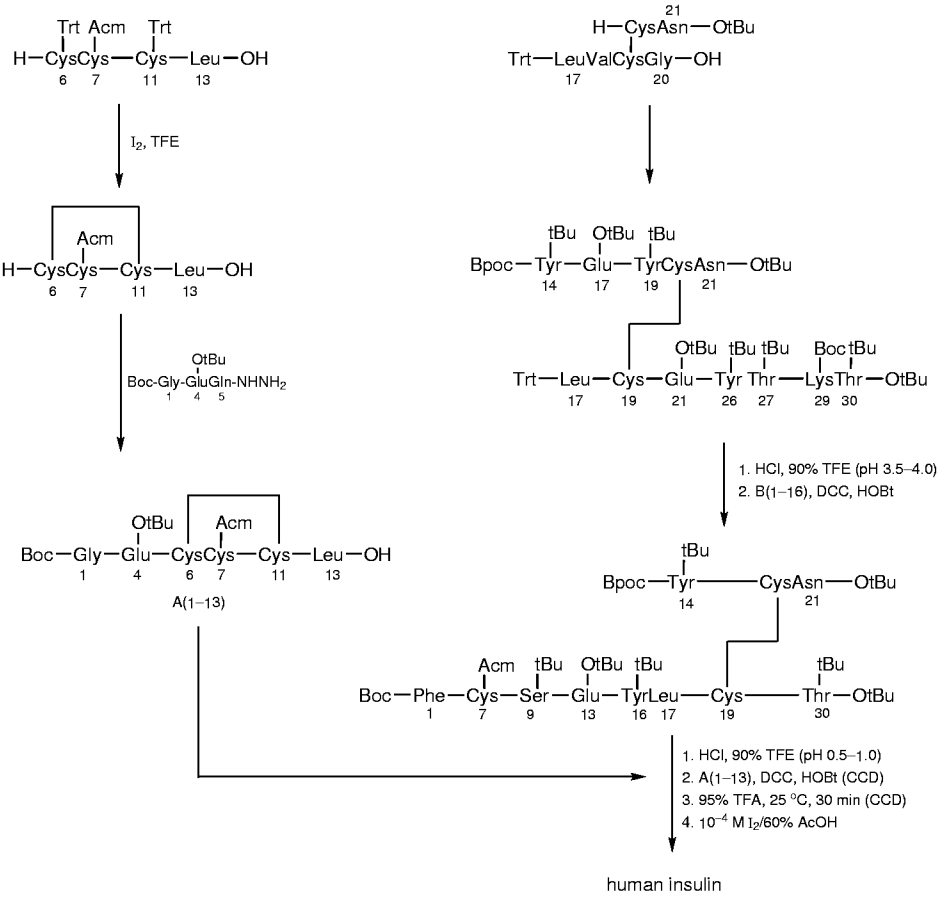
Nps-Ser(tBu)-Asp(OtBu)-Tyr(tBu)-Ser(tBu)-Lys(Boc)-Lys(Boc)-Tyr(tBu)-Leu-Asp(OtBu)-OH:[122]
A soln of H-Asp(OtBu)-Tyr(tBu)-Ser(tBu)-Lys(Boc)-Lys(Boc)-Tyr(tBu)-Leu-Asp(OtBu)-OH (2.57 g, 2.4 mmol), Nps-Ser(tBu)-OSu (0.98 g, 2.4 mmol), and TEA (0.28 mL) in DMF (25 mL) was stirred at rt for 15 h. The solvent was concentrated to half its volume. To the residue, 10% citric acid (500 mL) was added to afford a precipitate. This product was collected by filtration and washed with 10% citric acid and H_2O, and then reprecipitated (DMF/Et_2O); mp 227 °C (dec); $[\alpha]_D^{27}$ −9.7 (*c* 0.7, DMF); Anal. Calcd for $C_{79}H_{122}N_{10}O_{21}$: C, 60.1; H, 7.75; N, 8.88; S, 2.02. Found: C, 59.9; H, 7.91; N, 8.69; S, 1.79.

4.1.3.1.2.5 Synthesis of a Protected Peptide Using the Triphenylmethyl/*tert*-Butyl Strategy

The trityl[123] and Bpoc groups[124] can be removed under mild acidolysis. Sieber et al.[125] synthesized human insulin by using the Trt/tBu and Bpoc/tBu procedures by a route shown in Scheme 25. In the synthesis, the Trt and Bpoc groups were removed by HCl in 90% TFE (pH 3.5–4.0) and HCl in 90% TFE (pH 0.5–1.0), respectively. During the former treatment, the Bpoc group was stable. Although the synthesis of insulin proves that this methodology can be quite successful, it has not found widespread use in peptide chemistry due to the potential for losing protecting groups required later in the synthesis.

for references see p 623

Scheme 25 Synthesis of Human Insulin Using Triphenylmethyl/*tert*-Butyl and Biphenylpropyloxycarbonyl/*tert*-Butyl Strategies[125]

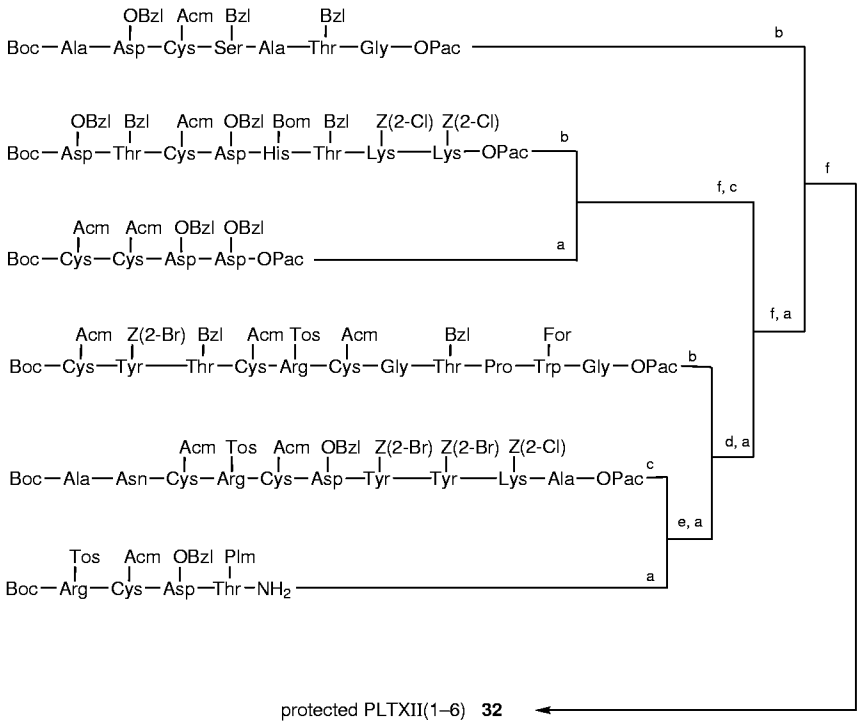


Trt-Leu-Val-Cys(Trt)-Gly-OMe; Sequence 17–20 of Human Insulin B Chain:[126]
Trt-Leu-Val-OH (2.30 g, 5 mmol), H-Cys(Trt)-Gly-OMe (2.17 g, 5 mmol) and HOBt (0.4 g, 3 mmol) were dissolved in $CHCl_3$ (25 mL), and DCC (1.13 g, 5.5 mmol) was added to the above soln at 0 °C. After 15 h at 20 °C, the solvent and DCU were removed. The residue was extracted with EtOAc and the extract was washed and concentrated in the usual manner to give crystals which were recrystallized ($CHCl_3$/MeOH); yield: 3.2 g; mp 252–254 °C; $[\alpha]_D^{27}$ –41.0 (*c* 1.2, $CHCl_3$); Anal. Calcd for $C_{55}H_{60}N_4O_5S$: C, 74.3; H, 6.80; N, 6.30. Found: C, 74.3; H, 6.75; N, 6.30.

4.1.3.1.2.6 Synthesis of a Protected Peptide Using the *tert*-Butoxycarbonyl/Benzyl Strategy

Sakakibara et al. prepared various peptides using fully protected peptide segments with Boc/Bzl protection.[98–100] As shown in Scheme 26, this strategy is exemplified by the synthesis of an O-palmitoylated 44-residue peptide amide (PLTXII).[100]

Scheme 26 Synthesis of Protected PLTXII[100]



(a) TFA; (b) Zn, AcOH; (c) Zn, HCO_2NH_4, AcOH, CH_2Cl_2/TFE (3:1); (d) EDC, HOBt; DMF
(e) EDC, HODhbt, DMF; (f) EDC, HODhbt, $CHCl_2$/TFE (3:1)

Synthesis of Protected Peptide PLTXII (32):[100]
Protected PLTXII was assembled in solution from six segments (in Scheme 26), each of which was elongated step-by-step from the C-terminus with Boc amino acids using the EDC/HOBt method. The C-terminus of each segment was temporarily protected by the phenacyl (Pac) ester, except for the C-terminal segment, which was amide. After removal of the Pac ester by Zn dust in AcOH or in a mixture of CH_2Cl_2 and TFE (3:1) and AcOH (60 equiv), the segments were coupled with amino components using EDC in the presence of HOBt or HODhbt in DMF, or EDC/HODhbt in a mixture of $CHCl_3$ and TFE (3:1).

4.1.3.2 Fragments from Solid-Phase Synthesis

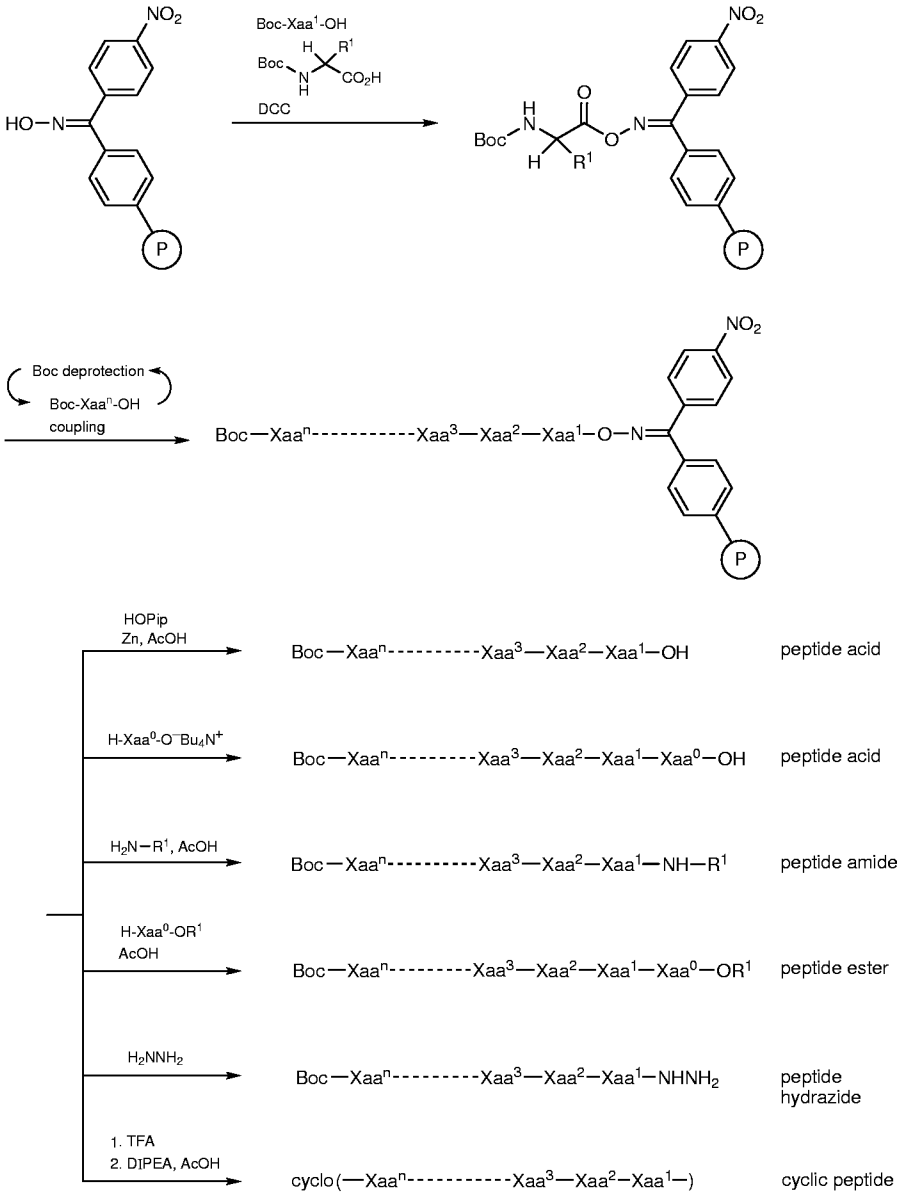
Stepwise solid-phase peptide synthesis suffers from the inherent problem that amino acid deletion impurities accumulate during the synthesis and are extremely difficult to remove from the desired product. The convergent solid-phase fragment-coupling approach combines the practicality of the solid-phase method with the advantages of a solution-phase fragment-coupling strategy. The deletion impurities are removed at an early stage by HPLC purification of the protected peptide fragments. Peptide fragments as building blocks can be prepared by the solid-phase method. The choice of resin and choice of cleavage procedure from the resin provides us with various kinds of peptide segments, such as C-terminal free peptide acid, peptide ester, peptide hydrazide, or peptide amide.

4.1.3.2.1 Fragments from Oxime Resin

As summarized in Scheme 27, various peptide segments can be prepared from the protected peptide oxime resin.

for references see p 623

Scheme 27 Synthesis of Protected Peptides with Oxime Resin[163]



Protected peptides are cleaved from the resin with *N*-hydroxypiperidine (HOPip), followed by treatment with zinc in AcOH to afford the free acid.[127–129] Resin-bound peptides can be cleaved from the oxime resin using the tetrabutylammonium salts of side-chain protected amino acids to directly provide the α-amino-protected (Boc), side-chain-protected peptide.[130–132] Peptide esters can be obtained from peptidyl oxime resin by treating with amino acid esters.[133,134] Peptide hydrazides are derived from peptidyl oxime resins on treatment with anhydrous hydrazine.[135] Protected peptide amides can be also derived from peptidyl oxime resins on treatment with ammonia.[133] Cyclic peptide acids as carboxy components were successfully prepared by this procedure through the Pac ester.[136,137]

4.1.3.2.1.1 Synthesis of a Protected Peptide Acid: Treatment of Peptidyl Oxime Resin with *N*-Hydroxypiperidine/Zinc/Acetic Acid

Peptidyl oxime resin was prepared in the usual manner starting from the resin-bound Boc amino acid. This peptidyl oxime resin was treated with HOPip to afford the corresponding protected peptide active ester. The active ester was converted into the protected peptide acid by reduction with zinc/AcOH.[132]

Boc-Val-Phe-Phe-Ala-Glu(OBzl)-Asp(OCy)-Val-Gly-OH:[132]
Boc-Val-Phe-Phe-Ala-Glu(OBzl)-Asp(OCy)-Val-Gly-oxime-resin was synthesized using the standard procedure. An acetylation step [Ac_2O (10 equiv), DIPEA (5 equiv), 30 min] was performed after each coupling. The peptide was cleaved with HOPip, reduced with Zn/AcOH to the free acid, and purified by HPLC [semiprep: 15 mL·min^{-1}, H_2O/MeCN 60:40, 0.1% TFA, 3 min isocratic followed by linear gradient to 30:70 (17 min)]; yield: 15% (based on Gly-oxime-resin); MS-FAB (*m/z*): $[M + H]^+$ calcd, 1155; found, 1155.

4.1.3.2.1.2 Synthesis of a Protected Peptide Acid: Treatment of Peptidyl Oxime Resin with the Tetrabutylammonium Salt of an Amino Acid

Peptidyl oxime resin reacted with the tetrabutylammonium salt of an amino acid to form the protected peptide acid.[132]

Boc-Asp(OtBu)-Ala-Glu(OBzl)-Phe-Arg(Mts)-His(Bom)-Asp(OCy)-Ser(Bzl)-Gly-OH:[132]
Boc-Asp(OtBu)-Ala-Glu(OBzl)-Phe-Arg(Mts)-His(Bom)-Asp(OCy)-Ser(Bzl)-oxime-resin was synthesized according to the modified standard procedure [Bop (5 equiv), amino acid (5 equiv), DIPEA (8.9 equiv) used for each coupling]. An acetylation step [Ac_2O (10 equiv), DIPEA (5 equiv)] was performed after each coupling. The peptide was cleaved with the tetrabutylammonium salt of glycine. The combined filtrate from the cleavage was concentrated to a solid residue and dissolved in DMF. The soln was purified by HPLC [semiprep: H_2O/MeCN 37:63, 0.1% TFA, R_V = 1040 mL (R_V = retention volume)]; yield: 45% [based on Ser(Bzl)-oxime-resin]; MS-FAB (*m/z*): $[M + H]^+$ calcd, 1752.8; found, 1753.1.

4.1.3.2.1.3 Synthesis of a Protected Peptide Ester: Treatment of Peptidyl Oxime Resin with Amino Acid Ester

Protected peptidyl oxime resin was treated with H-Leu-OtBu[133] and H-Ala-OBzl[134] to afford protected peptidyl-Leu-OtBu and peptidyl-Ala-OBzl, respectively.

Z-Leu-Leu-Glu(OBzl)-Ser(Bzl)-Leu-OtBu:[133]
Boc-Ser(Bzl)-oxime-resin (3.5 g, 3.0 mmol) was deprotected and coupled with the appropriate symmetrical anhydride by using an automated peptide synthesizer. Coupling times were 30 min. The peptide was cleaved from the resin by treatment with a 1.5-fold excess of H-Leu-OtBu acetate in CH_2Cl_2 for 12 h. The resin was filtered and washed with CH_2Cl_2 (3 ×). The filtrate was then washed with 5% aq citric acid (3 ×), and with H_2O (3 ×), and dried ($MgSO_4$). The CH_2Cl_2 was concentrated under reduced pressure to a final volume of 20 mL, and MeOH (50 mL) was added. The solvent was again concentrated to about 20 mL, and another portion of MeOH (50 mL) was added. This process was repeated again and the peptide crystallized upon standing at 5 °C overnight; yield: 1.68 g (60%); mp 208–219 °C; amino acid analysis: Leu, 3.00; Glu, 1.06; Ser, 0.78.

4.1.3.2.2 Fragments from Peptidyl (Nitrobenzoylamino)benzyl Resin by Photolysis

Photolabile peptidyl Nbb resin was successfully employed to prepare peptide segments by the solid-phase method.[138–140]

Photolysis of Boc-Asn-Ala-Cys(Acm)-Tyr(Cy)-Cys(Acm)-Tyr(Cy)-Lys(Z)-Leu-Pro-oxymethyl-Nbb-Resin:[138]
The photolysis was carried out in ca. 500-mg batches. The Boc-nonapeptidyl-OCH_2-Nbb-resin was suspended in a three-necked reaction vessel with 20% TFE/CH_2Cl_2 (100 mL) and the suspension was sonicated for 15 min at 0 °C. The reaction vessel was submerged in MeOH refrigerated by a cryostat. The

for references see p 623

suspension was cooled, purged with argon, and irradiated for 10 h, keeping the temperature inside the vessel below 20 °C. The resin was then filtered, washed with CH_2Cl_2, CH_2Cl_2/MeOH, and MeOH and the combined filtrate concentrated to dryness, yield: 295 mg (69%). The crude product was purified by chromatography (Ultrasphere ODS column, 1 × 25 cm); overall purification yield: 28%; mp 216–218 °C; $[\alpha]_D^{27}$ −56.1 (*c* 1.0, DMF); amino acid analysis: Pro, 1.00; Leu, 0.93; Lys, 0.98; Tyr, 1.91; Cys, 1.00; Ala, 1.09; Asp, 1.04.

4.1.3.2.3 Fragments from Peptidyl (4-Alkoxybenzyl Ester) Resin Using Trifluoroacetic Acid

Protected peptide fragments were prepared by treatment of peptidyl (4-alkoxybenzyl ester) resin prepared by solid-phase synthesis with TFA.[141,142] Aimoto et al. prepared peptide fragments based on the minimum protection strategy using the same method.[143]

4.1.3.2.3.1 Synthesis of a Protected Peptide from Peptidyl (4-Alkoxybenzyl Ester) Resin Using Trifluoroacetic Acid

As shown in Scheme 28,[143] peptide fragments as building blocks (based on the minimum protection strategy) for the synthesis of bovine pancreatic trypsin inhibitor (BPTI) were prepared by solid-phase methods.[143]

Scheme 28 Synthesis of Partially Protected Bovine Pancreatic Trypsin Inhibitor (1–13)[143]



Arg(Tos)-Pro-Asp(OCy)-Phe-Cys(Acm)-Leu-Glu(OCy)-Pro-Pro-Tyr-Thr-Gly-Pro-OH (33):[143]
Npys-Pro-resin (0.86 g, 0.5 mmol) was placed in a reaction vessel, and the synthesis was carried out manually according to the usual manner by using Npys derivatives of Gly, Thr(tBu), Tyr(tBu), Pro, Glu(OCy), Leu, Phe, and Asp(OCy). Cys(Acm) was introduced by using its Fmoc derivative. Arg(Tos) was introduced by using its Boc derivative, and washed with DMF, CH_2Cl_2, and iPrOH. The weight of the peptide resin, thus obtained was 1.59 g. The whole peptide-resin was treated with 25% TFA/CH_2Cl_2 (25 mL) for 30 min at rt. The resin was removed by filtration, and the filtrate was concentrated to dryness under reduced pressure, leaving a powder (720 mg). This crude product was purified by HPLC (with the MeCN concentration increasing from 30% to 80% in 0.05% aq TFA at a flow rate of 2 mL·min^{-1}) to give **33**; yield: 217 mg; amino acid analysis: Asp, 0.97; Thr, 1.01; Glu, 0.89; Gly, 1.00; Cys, 0.14; Leu, 0.92; Tyr, 0.93; Phe, 0.97; Arg, 0.27; Pro, 3.98; MS (*m/z*): $[M + 1]^+$ calcd for $C_{90}H_{130}N_{17}O_{23}S_2$, 1881.0; found, 1881.0.

Z-[Arg(Tos)^1, Asp(OCy)^3, Cys(Acm)^5, Glu(OCy)^7]-BPTI(1–13) (34):[143]
Peptide **33** (23.4 mg, 12.4 μmol) and Z-OSu (8 mg, 32.2 μmol) were dissolved in DMF (1 mL) containing NMM (10 μL), and the soln was stirred for 2 h. A product was precipitated by the addition of Et_2O to the soln, and washed successively with EtOAc and dioxane. The product was further purified by HPLC under the same conditions as compound **33** and gave compound **34**; yield: 14.0 mg; amino acid analysis: Asp, 0.98; Thr, 0.95; Glu, 0.97; Pro, 3.77; Gly, 1.00; Cys, 0.29; Leu, 0.99; Tyr, 1.00; Phe, 0.98; Arg, 0.36; MS (*m/z*): $[M + 1]^+$ calcd for $C_{98}H_{136}N_{17}O_{25}S_2$, 2015.1; found, 2014.9.

4.1.4 Final Deprotection of Protected Peptides

The peptide synthesis consists of stepwise-condensation and segment-condensation procedures. In both procedures, it is possible to purify the reaction product at each reaction step. Therefore, protected peptides with a high degree of purity can be obtained from a solution method. The choice of the reagents used in the final deprotection step determines the main strategy of peptide synthesis. The final deprotection procedures are mainly (1) catalytic hydrogenolysis in the presence of palladium,[105] (2) sodium treatment in liquid ammonia for 10 to 15 seconds, repetitively over 30 minutes,[144] (3) TFA treatment at room temperature for 1 hour,[145] (4) HF treatment at 0 °C for 1 hour,[146] or (5) hard-soft acid-base procedure.[147]

4.1.4.1 Final Deprotection by Catalytic Hydrogenation

In the synthesis of sulfur-free peptides, protecting groups removable by catalytic hydrogenation, such as Arg(NO_2), Asp(OBzl), or Glu(OBzl), can be used. Boissonnas et al. employed this procedure for the synthesis of bradykinin in 1960,[105] and Arakawa and Bumpus utilized this method in the synthesis of angiotensin II in 1961.[148]

Synthesis of Angiotensin II by Catalytic Hydrogenation:[148]
To a soln of Z-Asp(OBzl)-Arg(NO_2)-Val-Tyr-Ile-His-Pro-Phe-OBzl(4-NO_2) (2.176 g, 1.5 mmol) in MeOH (60 mL) and AcOH (20 mL) was added Pd (1.5 g) suspended in H_2O (15 mL). H_2 was bubbled through the soln for 30 h. After removal of Pd and the solvents, the residue was dried to remove AcOH completely. The crude material was purified by column chromatography (cellulose powder, 3.6 × 60 cm) and reprecipitated (EtOH); yield: 800 mg; mp 230–247 °C (dec); $[\alpha]_D^{27}$ −67.0 (*c* 0.4, 1 M HCl); Anal. Calcd for $C_{50}H_{71}N_{13}O_{12}\cdot C_2H_4O_2$: C, 56.5; H, 6.83; N, 16.5. Found: C, 56.0; H, 6.55; N, 16.7.

4.1.4.2 Final Deprotection of Protected Insulin A-Chain Using Sodium/Liquid Ammonia

This method was first employed by duVigneaud for the synthesis of oxytocin[7] and later for the synthesis of insulin.[149–151] In this procedure Cys(Bzl), Lys(Tos), and Arg(Tos) can be

for references see p 623

used for peptide synthesis. As a side reaction in this procedure, Hofmann and Yajima reported that the peptidyl proline bond was cleaved during the synthesis of ACTH.[152]

Synthesis of Sheep Insulin A-Chain:[144]
H-Gly-Ile-Val-Glu-Gln-Cys(Bzl)-Cys(Bzl)-Ala-Gly-Val-Cys(Bzl)-Ser-Leu-Tyr-Gln-Leu-Glu-Asn-Tyr-Cys(Bzl)-Asn-OBzl(4-NO_2) (200 mg) was dissolved in anhyd NH_3 (100 mL) in a 200-mL round-bottomed flask fitted with a magnetic stirrer. The reaction was carried out at the boiling point of the soln. Cleavage was accomplished by adding very small pieces of Na to the soln until a faint blue color appeared throughout. The blue color was allowed to persist for 1 min and was then discharged by the addition of a few crystals of NH_4Cl. The soln was removed at atmospheric pressure to about 10 mL and dried from the frozen state. This deprotected peptide was converted into the corresponding *S*-sulfonate.

4.1.4.3 Final Deprotection of Protected Glucagon Using Trifluoroacetic Acid

Schwyzer and Rittel synthesized Lys(Boc) and employed this derivative for the synthesis of porcine ACTH (1–19).[145] Wuencsh et al. synthesized glucagon by this procedure.[88,90]

Synthesis of Glucagon by Treating the Protected Glucagon with TFA:[88,90]
Protected nonacosapeptide (0.97 g, 0.2 mmol) was dissolved in TFA (20 mL) containing diethyl phosphite (2 mL) and EtSMe (2 mL). The mixture was stored under an argon stream and shielded from light for 2 h. After removal of TFA, Et_2O was added to the residue to afford a solid mass, which was collected by filtration and washed with Et_2O. This powder was dissolved in a H_2O suspension of Dowex 4 (OH^- form). After addition of a small amount of AcOH, the mixture was stirred for 1 h, filtered and lyophilized; yield: 750 mg.

4.1.4.4 Final Deprotection of Protected Eglin c Using Hydrogen Fluoride

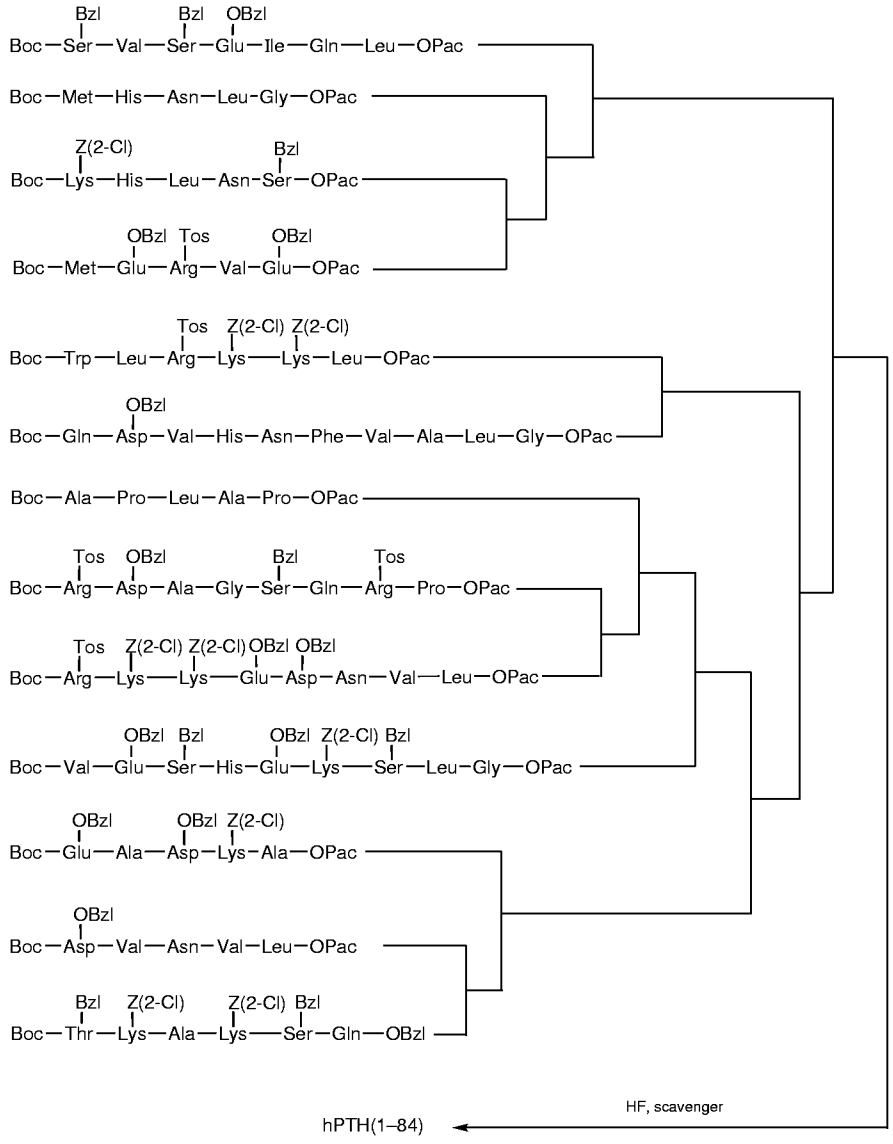
A protected eglin c was synthesized by the minimum protection strategy.[84] Of the several deprotection methods available, eglin c was obtained using the HF procedure.[146]

Deprotection of a Protected Eglin c:[84]
A protected eglin c (50 mg) was treated with HF (1.5 mL) containing thioanisole (0.7 mL, 6.7 mmol) and *m*-cresol (0.7 mL, 6.7 mmol) at 0 °C for 90 min. After removal of HF at reduced pressure, dry Et_2O was added to the residue. The resulting powder was collected by filtration and dried over KOH pellets in vacuo. This product was again treated with HF in the same way described above in order to complete the deprotection. The resulting powder was dissolved in H_2O (10 mL) and the soln was treated with Amberlite IRA-45 (acetate form) for 30 min. The pH of the filtrate was adjusted to 8 with 1 M NH_4OH. After 30 min, the pH of the soln was adjusted to 6.5 with 1 M AcOH and the solvent was removed by lyophilization to give a crude hygroscopic powder. This crude peptide was purified by gel filtration on Sephadex G-50, followed by RP-HPLC. Eglin c obtained exhibited a symmetrical single peak on analytical HPLC. Amino acid ratios from an acid hydrolysate were in good agreement with theoretically expected values.

4.1.4.5 Final Deprotection of a Maximally Protected Peptide Using Hydrogen Fluoride

Human parathyroid hormone, hPTH (1–84) was synthesized by the maximum protection method using Bzl-based protecting groups. The synthetic route is shown in Scheme 29.

Scheme 29 The Total Synthesis of Human Parathyroid Hormone [hPTH(1–84)][153]



Synthesis of Protected hPTH(1–84):[51]

Protected hPTH(23–84) (4.07 g, 0.4 mmol) was treated with TFA (30 mL) in an ice bath for 60 min. After removal of excess TFA under reduced pressure, 5.7 M HCl in dioxane (0.14 mL) was added to the residue, and the product was precipitated by adding Et_2O. The product was washed with Et_2O and dried over KOH pellets in vacuo. The product thus obtained and Boc-hPTH(1–22)-OH (1.68 g, 0.48 mmol) were dissolved in $CHCl_3$/TFE (3:1, 120 mL) and then HODhbt (78 mg, 0.48 mmol) and EDC (0.09 mL, 0.48 mmol) were added to the soln. After stirring for 5 h, the mixture was concentrated under reduced pressure. To the residue was added excess chilled 1% $NaHCO_3$ soln. The precipitate was collected by filtration, and washed with 1 M HCl and H_2O. The product was then dissolved in $CHCl_3$/TFE (3:1) and concentrated again. MeCN was added to the residue, and the precipitate was collected and washed successively with MeOH, EtOAc, and hexane; yield: 4.63 g (93%); amino acid analysis after hydrolysis in 6 M HCl containing phenol at 110 °C for 22 h: Asp (2), 1.98; Ser (5), 4.51; Glu (1), 1.00; Gly (5), 5.02; Ala (1), 1.03; Ile (2), 1.89; Leu (2), 2.03; Tyr (1), 1.00; Phe (2), 2.01; Arg (5), 5.04; Cys (2), 0.94.

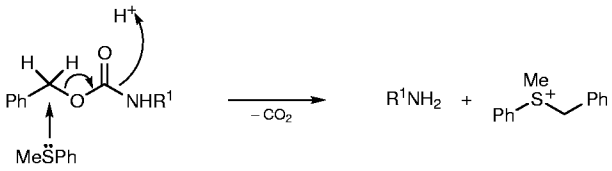
for references see p 623

Final Deprotection of Protected hPTH(1–84) Using the HF Procedure:[154]
The fully protected product (1 g) was treated with HF at 0 °C for 1 h in the presence of anisole, DMS, and ethanedithiol to obtain the free peptide. The product was purified on CM-cellulose, and the resultant major product was further purified on Sephadex G-50; yield of crude product at this stage was 60 mg, which was further purified by HPLC. The purified hPTH(1–84) exhibited a single peak on analytical HPLC and a single band on SDS gel electrophoresis at pH 6.8; yield: 10 mg; $[\alpha]_D^{25}$ −81.2 (*c* 0.3, 1% AcOH).

4.1.4.6 Final Deprotection of a Protected Peptide by the Hard-Soft Acid-Base Procedure

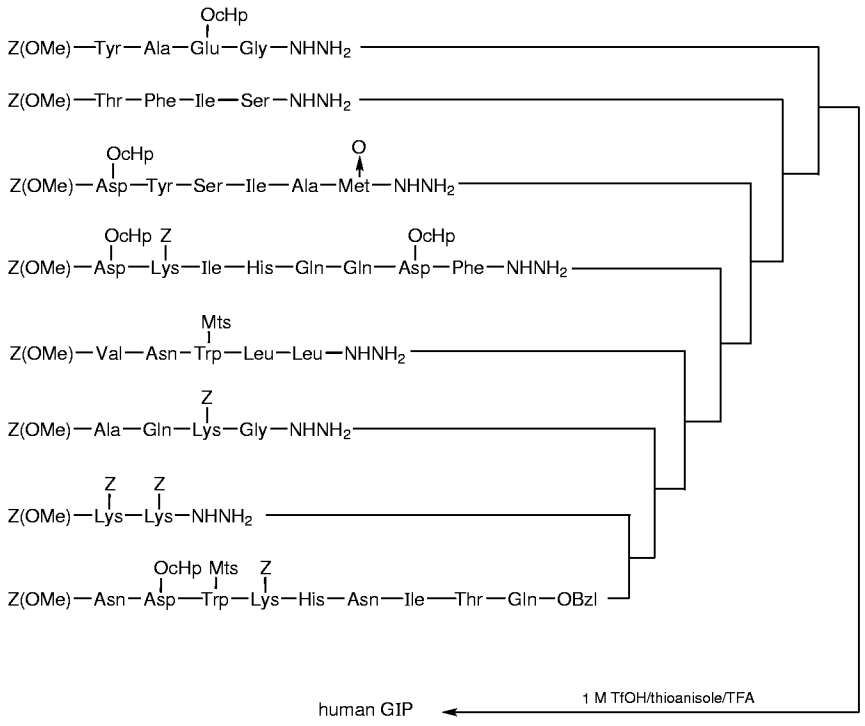
Yajima et al. introduced the organosulfonic acid deprotecting procedure in 1974.[147] MsOH or 1 M TfOH in TFA was found to cleave the Bzl ester and the Z group smoothly. Thioanisole,[155] used as a scavenger, was found to accelerate acidolytic deprotection,[156,157] the mechanism of which could be explained by the hard-soft concept as shown in Scheme 30.[158,159]

Scheme 30 Possible Mechanism of TfOH/Thioanisole/TFA Deprotection[158,159]



Human-glucose-dependent insulinotropic polypeptide (human GIP) was synthesized by the route shown in Scheme 31.[160] Trimethylsilyl trifluoromethanesulfonate (TMSOTf) has also been employed as a super hard acid[161] with thioanisole as a soft base[159] for the final deprotection of protected human GIP.[162]

Scheme 31 Synthetic Route to Human GIP[160]



Synthesis of Human GIP from Protected GIP by the Hard-Soft Acid-Base Procedure:[160]
The protected human GIP (50 mg, 73.9 μmol) was treated with 1 M TfOH/thioanisole in TFA (3 mL) in the presence of *m*-cresol (108 μL, 140 equiv) and EDT (43 μL, 70 equiv) in an ice bath for 180 min, then dry Et_2O was added. The resulting powder was collected by centrifugation, dried over KOH pellets in vacuo for 2 h, and dissolved in H_2O (5 mL). The pH of the ice-chilled soln was adjusted to 8.0 with 5% NH_4OH and after 30 min, to 5.0 with 1 M AcOH. β-ME (207 μL, 400 equiv) was added and the soln, after being incubated for 20 h, was applied to a column of Sephadex G-50 (2.3 × 142 cm, 1 M AcOH). The fractions (6 mL each, monitored at 280 nm) corresponding to the front main peak (tubes 41–64) were combined and the solvent was removed by lyophilization to give a powder; yield: 36 mg (98%). This crude product was further purified with CM-cellulose and RP-HPLC. The overall yield from the protected human GIP was 10%.

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4.1.5 Chemical Ligation

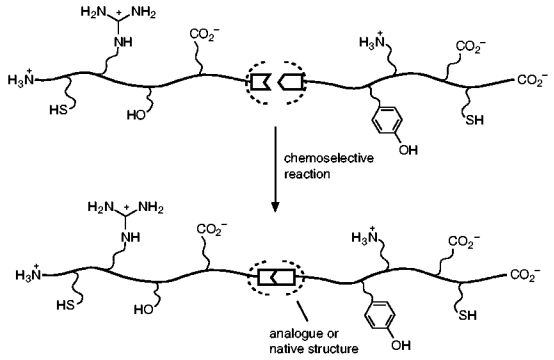
P. Dawson

The understanding of the molecular basis of protein function is a fundamental goal of the biochemical sciences. Consequently, a tremendous effort has been expended modifying the polypeptide chains that correspond to proteins using the techniques of molecular biology. These powerful techniques, such as site-directed mutagenesis, have dominated the field of protein chemistry in recent years and greatly added to our understanding of protein structure and function. However, the structural diversity attainable through ribosomal synthesis is largely limited to naturally occurring amino acids. Routine access to chemically synthesized proteins would allow the precise manipulation of covalent structure that has been developed for the study of peptides (peptidomimetics) to be applied to proteins.

The assembly of proteins by chemical synthesis has been an important goal of organic chemistry for many years.[1] A powerful demonstration of the effectiveness of chemical protein synthesis as an alternative for the production of native proteins was the original work on the enzymatic properties and X-ray crystal structures of the HIV protease.[2–5] This work was accomplished using a chemically synthesized enzyme before efficient biological expression of this protein was achieved. This 99 amino acid polypeptide was generated by the Kent laboratory using SPPS, and folded to make the dimeric enzyme in tens of milligram quantities.[2] In addition to wild-type proteins, the chemical synthesis of proteins allows researchers to create protein analogues that make use of unnatural amino acids. In the case of the HIV protease, β-turn mimics,[6] NMR isotopic labels,[7] and polypeptide backbone analogues[8,9] have been incorporated to gain insight into the molecular basis for its proteolytic activity. This ability to go beyond the genetic code promises to greatly expand our understanding of protein structure and function.

Chemical ligation approaches have proven to be one of the most effective methods for the synthesis of large polypeptides in homogeneous form.[10,11] These methods rely on the principle of chemoselective reaction, rather than the fully protected peptide strategies usually used for the assembly of peptide fragments. The general approach, shown in Scheme 1, is to synthesize two fully unprotected peptides with uniquely reactive groups on their respective termini.[12,13] These peptides are combined in denaturing aqueous solutions such as 6 M guanidinium chloride or 8 M urea to attain peptide concentrations of 1–10 mM. The peptides react in a rapid and specific manner to produce the full-length polypeptide product. The ligation reactions typically proceed to completion in less than 24 hours and are immediately purified by RP-HPLC. For many studies, a fully native structure at the site of ligation is not a requirement for the generation of fully functional protein analogues for protein engineering uses. It is important, however, for the ligation chemistry to produce a unique structure and not a mixture of isomers.

Scheme 1 Principle of Chemoselective Peptide Ligation



for references see p 640

The use of unprotected peptides takes advantage of the ability to synthesize peptides of 20–50 amino acids by SPPS in a straightforward manner. These peptides can be easily purified by HPLC and characterized by electrospray or MALDI mass spectrometry.[14–16] The high resolution and ease of use of electrospray mass spectrometry allows impurities such as the oxidized form of methionine to be easily identified for every peptide synthesized. Preparative HPLC fractions containing these impurities can be identified immediately following collection, allowing highly purified peptides to be obtained following a single HPLC separation. The straightforward handling and analysis of unprotected peptides allows for the rapid generation of high-quality segments for the assembly of the larger protein molecules. In this manner, chemical ligation avoids one of the greatest obstacles to the synthesis of large polypeptides: the difficult handling and characterization of fully protected peptides.

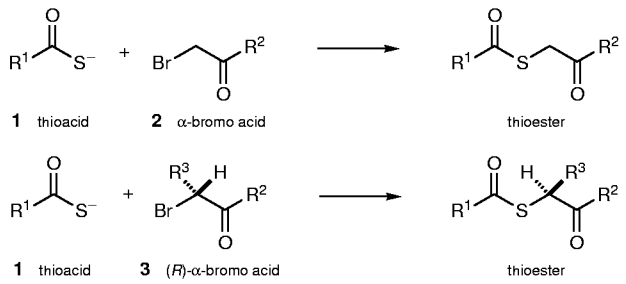
There are several essential elements required for a chemical ligation reaction to be applicable to the synthesis of proteins. The ligation chemistry must be highly chemoselective, and all natural (and any desired unnatural) functional groups must be stable to the ligation conditions. The requirement for any side-chain protection severely limits the utility of the ligation chemistry. In addition, the ligation chemistry must be compatible with denaturing solvents capable of solubilizing large unprotected peptides. The utility of solvents such as 6 M guanidinium chloride and organic/aqueous mixtures is a serious pitfall to enzyme-based approaches. It is also important that the unique functional groups for the ligation chemistry must be compatible with standard solid-phase peptide synthesis and cleavage conditions. Addition of these groups should not require manipulation of a fully protected peptide that has been cleaved from the solid support, which would frustrate the original intent of using unprotected peptides.

This section focuses on a set of ligation reactions that have proven to be of general utility for the synthesis of proteins and protein analogues. These ligations form either thioester or native bonds[17] at the site of ligation and utilize the specific and rapid reactions of thioacid/bromoacetyl groups[12] or thioester/N-terminal Cys (native chemical ligation).[17,18] Several other ligation reactions are used for peptide conjugation that form native[18] and nonnative[19] bonds at the site of ligation. These reactions have been described in detail elsewhere.[20] The highly chemoselective reaction between aldehydes and aminooxy/hydrazone nucleophiles is also of widespread use and is covered in detail in the context of protein semisynthesis (see Vol. E 22b, Section 5.4).

4.1.5.1 Thioester Forming Ligation

The first total synthesis of a protein by the chemical ligation approach utilized nonnative bonds at the site of ligation.[12] In the early 1990s the Kent laboratory pioneered a thioester forming ligation that utilized a highly chemoselective reaction between a peptide with a C-terminal carbothioate group **1** and a N-terminal bromoacetyl peptide **2** at pH <4[21] (Scheme 2).

Scheme 2 Thioester Forming Ligation



This ligation reaction was an important methodology for the synthesis of HIV protease analogues to study the molecular basis of HIV protease enzymatic activity.[9] In addition, several transcription factors,[22] a fibronectin 10F3 domain,[23] and a branched template-assembled synthetic protein[24] have been synthesized using this ligation chemistry.

Mechanistic Rationale

The chemoselectivity of the reaction originates from the strongly nucleophilic thioacid group that has a pK_a of ~3. At this pH, no other groups in a protein are nucleophilic; in particular, Cys side-chain thiols have a pK_a of ~9. Through the synthesis of a variety of protein targets, it has been demonstrated that the thioester forming ligation is chemoselective for all amino acid functional groups.[25] The peptide carbothioate group reacts with a N-terminal bromoacetyl peptide through an S_N2 mechanism. This reaction proceeds to completion in <24 hours to produce a thioester analogue of a Gly-Gly sequence at the ligation site. Reaction of thioacids with chiral α-halo carbonyl compounds results in inversion of the α-carbon (*vide infra*).

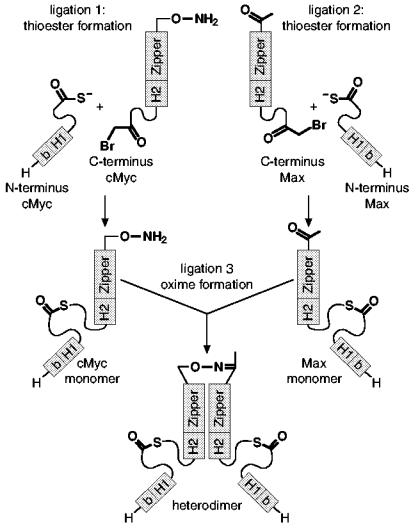
Scope and Limitations

The ligation product formed by the thioester ligation is not limited to mimicking Gly-Gly. Thioester ligated analogues, "Xaa-Gly", can be formed using a thioacid peptide in which the C-terminal residue is not Gly. For example, thioacid peptides containing C-terminal Ala, Phe, Lys, and Leu have been synthesized,[26] and a peptide-Leu-COSH was successfully ligated to a bromoacetylated peptide to yield a Leu-Gly ligated analogue.[22] Similarly, other α-bromo acids **3** can be utilized to add functionality to the C-terminal side of the ligation junction (Scheme 2). For example, a ligation using a N-terminal bromo(methyl)pentanoic acid was used to create a Gly-Ile analogue.[28] It is important to note that the S_N2 reaction will invert the chirality of the α-carbon. Consequently, the α-bromo acid must be of the opposite chirality to that in the desired product. In the Gly-Ile example, (2*R*,3*S*)-2-bromo-3-methylpentanoic acid was utilized. One limitation to adding functionality to the α-carbon is that the ligation can become sluggish. Thus, only 10% product was obtained in the Gly-Ile ligation example due to the highly hindered "Ile" side chain. However, this example is the most hindered of the natural side chains and non-β-branched amino acids would be expected to ligate more efficiently.

Peptide alkyl thioesters are stable at neutral pH in the absence of added thiols such as dithiothreitol or 2-sulfanylethanol, or strong nucleophiles such as hydroxylamine. Surprisingly, a sample of HIV protease (1–99) containing a thioester between residues 51 and 52 (Gly-Gly) has been stored at 4 °C in a pH 2 buffer for eight years with less than 20% thioester hydrolysis. The thioester forming ligation has been used to synthesize linear polypeptides up to 100 amino acids in length and has been combined with disulfide formation to produce polypeptides up to 200 amino acids.[25] This work demonstrated a thioester ligation in the presence of free thiols on both peptide chains. The compatibility of the thioester forming reaction with an oxime forming ligation was demonstrated by Canne et al. in the elegant synthesis of a tethered form of the transcription factor Max in which both ligations were performed simultaneously at pH 4.5 (Scheme 3).[22] This work exemplifies the principle of using one type of ligation for the synthesis of linear domains and a second ligation to link the domains together.

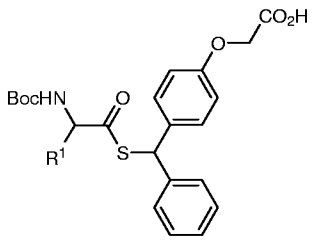
for references see p 640

Scheme 3 Convergent Synthesis of a Transcription Factor Using both Thioester and Oxime Forming Ligations[22][a]



[a] Reprinted from (Canne; Ferre-D'Amare; Burley; Kent, *Journal of the American Chemical Society*, Vol. 117), Copyright (1995), p 2998, with permission from The American Chemical Society.

Scheme 4 C-Terminal Thioacid Linker[28,29]



4

Thioacid Peptide; General Procedure:

Peptide carbothioic acids were synthesized using a thioacid linker **4** developed by Blake (Scheme 4).[28,29] A generalized synthesis of this Boc-Xaa-thioacid-linker has been described by Canne et al.[26] The thioacid-resin was synthesized by dissolving Boc-Xaa-thioacid-linker·DCHA salt (1.1 equiv) and HBTU (1.1 equiv) in a minimum volume of DMF and activated for 2 min. This mixture was added to aminomethyl-polystyrene-resin (1.0 equiv, ~1.0 mmol·g^{-1} loading of amine) and coupled for 1 h or until the reaction proceeded to >99.5% as monitored by the quantitative ninhydrin test.

Peptide assembly was by standard Boc chemistry protocols, in particular the in situ neutralization cycles for manual or machine-assisted synthesis developed by Kent.[30] Standard Boc/benzyl side-chain protection was used except for unprotected Boc-Trp-OH and Boc-His(Bom) due to the incompatibility of the thioacid linker with strong nucleophiles. Similarly, Fmoc amino protection is incompatible with this linker. The peptide-resin (400 mg) was cleaved in a standard HF apparatus with 5% *p*-cresol/anhyd HF (10 mL) at 0 °C for 1 h. The HF was rapidly removed under reduced pressure (<10 min) using a liq N_2 trap. The peptide was precipitated with cold Et_2O and filtered under reduced pressure with care to avoid pulling air through the filtrate. The crude peptide was dissolved in 30% MeCN/H_2O, 0.1% TFA, immediately purified by RP-HPLC and lyophilized. The thioester peptide was stable for several months as a lyophilized powder at −20 °C. The HF was slowly quenched into a CaO trap by allowing the liq N_2 trap to warm to rt over 1 h. Full safety procedures for HF cleavage have been described in detail.[31] Thiol scavengers such as *p*-thiocresol should be avoided.

N^{α}-Bromoacetyl Peptides; General Procedure:[32]
Following chain assembly of the desired peptide to produce 0.25 mmol of N^{α}-Boc-peptide-resin, the Boc group was removed by treatment with neat TFA (10 mL) for 2 × 1 min. The TFA was drained and the resin was flow-washed for 40 s with DMF. The resin was then neutralized with 10% DIPEA/DMF (10 mL) for 1 min, followed by a 10 s DMF flow wash. Bromoacetic acid (2.0 mmol) was dissolved in CH_2Cl_2 (2 mL) at 0 °C and DIC (1.0 mmol) was added and activated for 10 min in an ice bath. This mixture was added to the neutralized peptide-resin (swollen in DMF) and coupled for 20 min. The peptide was cleaved as described above.

Thioester Forming Ligation; General Procedure:[12,25]
Thioester ligation buffer contains 6 M guanidinium chloride and 100 mM NaOAc, pH 4.0. To ligation buffer, peptide-COSH (1 equiv) and $BrCH_2$-CO-peptide (1 equiv) was added to a concentration of 1 mM in a closed reaction vessel. The reaction was stirred at 4 °C for 18 h and continued at 25 °C if necessary. Ligation times of ~1 h at higher peptide concentrations (5–10 mM) were observed, depending on peptide size and solubility. Product formation was monitored by direct injection of ~1 µL of the ligation mixture on RP-HPLC and analysis of the fractions by electrospray mass spectrometry. The final polypeptide products were purified by HPLC and lyophilized. Yields were typically quantitative by HPLC and >50% yield of the purified protein product was recovered. It has also been observed that polar organic solvents such as DMF are excellent solvents for thioester forming ligations.[8]

4.1.5.2 Native Chemical Ligation

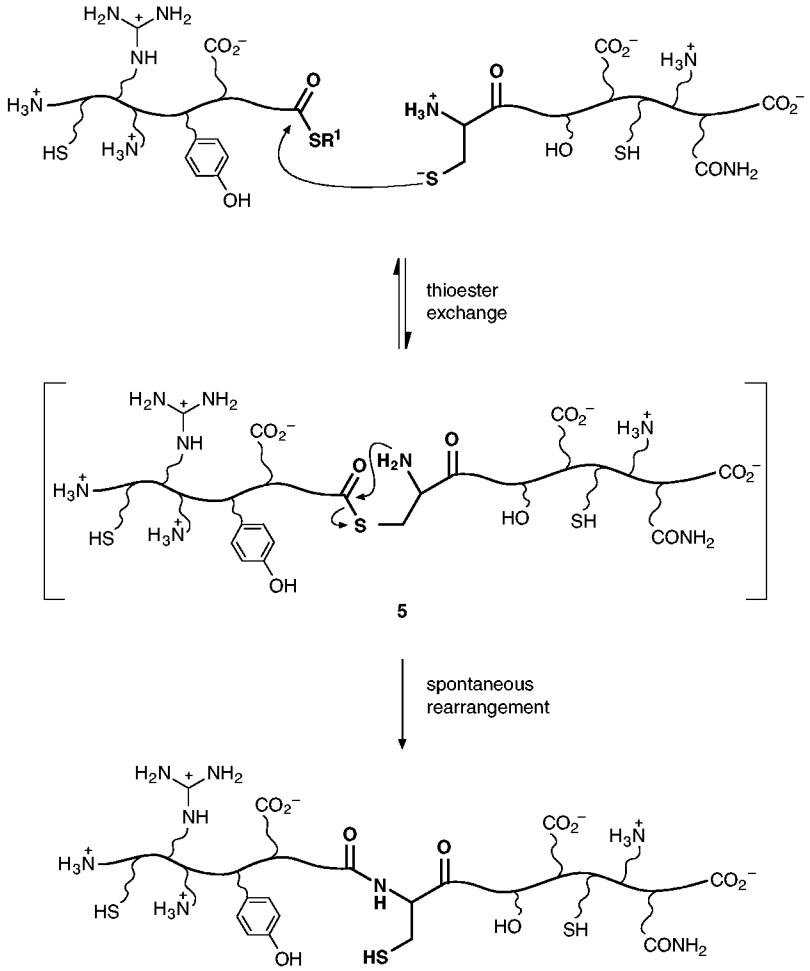
The development of chemoselective reactions to give a native peptide bond at the site of ligation allows the synthesis of proteins with little or no modification to the covalent structure. A native structure at the ligation site is often desirable in the middle of protein structural domains (amino acid 60–120). The challenge of this approach is to form an amide bond chemoselectively in the presence of free amine side chains (Lys) and carboxylate side chains (Glu/Asp). Ideally, no protecting groups should be used for any of the amino acid side chains as they limit peptide solubility and require additional deprotection steps that can severely reduce the yield and convenience of the synthesis.

Mechanistic Rationale

Native chemical ligation[17] allows two unprotected peptides to be joined through formation of a native peptide bond at the ligation site. The reaction is schematically illustrated in Scheme 5.[35] The ligation is initiated by a specific reaction between a cysteine thiol side chain of one peptide and the C-terminal thioester residue (Xaa) of a second peptide. Exchange of the peptide thioester with the thiol side chain gives a thioester-linked intermediate **5** as the initial covalent product. It is important to note that the initial step is under equilibrium conditions involving all cysteine side chains and free thiols in solution. When the thiol side chain is a N-terminal Cys residue, the thioester-linked intermediate undergoes a S-to-N acyl transfer[33,34] to generate a native peptide bond at a Xaa-Cys site in the polypeptide sequence. Native chemical ligation is highly chemoselective; it is compatible with all side-chain functional groups found in proteins and both denaturing aqueous solution and aqueous/organic solvent mixtures.

for references see p 640

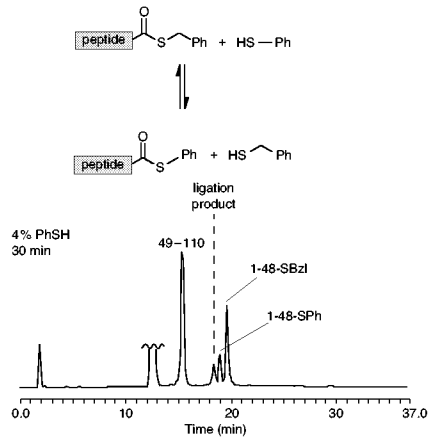
Scheme 5 Native Chemical Ligation



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In the first step of native chemical ligation, a rapid and reversible thiol-exchange reaction takes place to form a new thioester bond.[17,35] Peptides in a typical ligation reaction often contain several Cys residues, each of which has a free thiol capable of thioester exchange. Since only the N-terminal Cys residue can rearrange to form an amide bond, the reaction of an internal Cys thiol simply produces a new thioester group that is still competent for further thiol exchange. Typically, thiols such as phenylmethanethiol and benzenethiol are added to the mixture to facilitate this reversible exchange of thiols (Scheme 6).[35] In addition to promoting exchange, the benzenethiol additive acts as a nucleophilic catalyst by generating highly reactive phenyl thioester peptides while the phenylmethanethiol acts as a strong reducing agent to prevent disulfide bond formation during ligation.[36]

Scheme 6 Native Chemical Ligation of the Ribonuclease Barnase Using Thioester Exchange[35][a]



[a] Reprinted from (Dawson; Churchill; Ghadiri; Kent, *Journal of the American Chemical Society*, Vol 119), Copyright (1997), p 4325, with permission from The American Chemical Society.

Under these ligation conditions, essentially no hydrolysis of the thioester component is observed, even over prolonged (>48 h) reaction. Alternatively, phosphines such as tris(2-carboxyethyl)phosphine can be used as a reducing agent during or after ligation.[17,18,37] Thioester exchange allows peptides to be synthesized and stored with relatively unreactive thioester groups that can be converted into more reactive phenyl thioester peptides during the ligation reaction.

Scope and Limitations

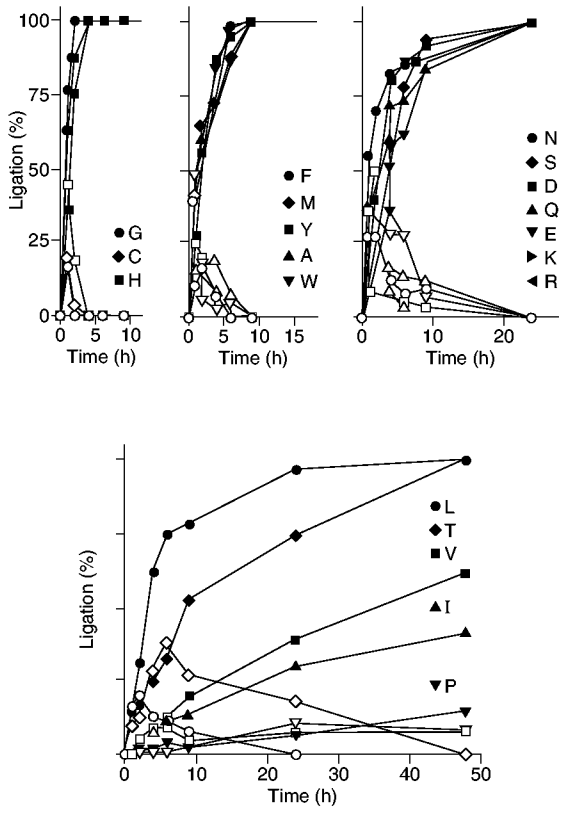
The native chemical ligation approach has proven to be a remarkably robust method for the synthesis of native proteins. Using this method, hundreds of proteins and protein analogues have been obtained in homogeneous form for protein structure–function studies.[38] From these studies, there does not appear to be any limitations on the size of the peptides or on their sequence. Using recently introduced protein-splicing techniques to generate C-terminal thioester peptides, ligation reactions on peptides over 50 kDa have been successfully performed.[39] The only apparent limitation to the methodology is the synthesis and handling of the individual thioester peptide fragments. In general, these difficulties are not due to the thioester component but to the general properties of the peptide sequence. Recent reports of the synthesis of membrane proteins by native chemical ligation illustrate the generality of the methodology to include peptides that have been demonstrated to be difficult to handle.[40,41]

One requirement for native chemical ligation is the incorporation of a Cys residue at every ligation junction. Many proteins contain natural Cys residues in an appropriate position for ligation, allowing for synthesis of the protein without modification. Other proteins, such as the ribonuclease, barnase, have no Cys residues and require substitution of Cys into the polypeptide sequence. In the example of barnase, a solvent-exposed Lys at position 49 in the polypeptide sequence was replaced by Cys, creating a Gly-Cys ligation site.[35] This modification had no effect on the catalytic activity of the final protein product. In cases in which free thiols are undesirable, specific alkylation of the Cys side-chain thiol has been established as a straightforward and high yielding procedure. An interesting variation of this approach involves ligation of a N-terminal homocysteine residue and subsequent methylation to form a Met residue.[42]

for references see p 640

It has been shown that there is great flexibility in the choice of the C-terminal residue of the thioester peptide.[43] As shown in Figure 1,[43] all peptide thioester groups can be synthesized and all but the β-branched amino acids and proline are competent for efficient ligation. Surprisingly, the presence of a functional group on the amino acid side chain is well tolerated, and can even increase the rate of ligation in the case of Cys and His. C-Terminal Asp can be prone to hydrolysis in ligation reactions that are slow due to low peptide concentrations. However, Asp has been used successfully in protein ligation reactions under standard conditions. In practice, 17 of the 20 amino acids appear to be highly compatible with the ligation chemistry, with Val, Ile, and Pro being less favorable selections. Syntheses using Val and Ile should be attainable using high peptide concentrations and prolonged ligation times. Native chemical ligation reactions progress without significant epimerization (<1%) of the thioester-activated amino acid.[43,44]

Figure 1 Relative Native Chemical Ligation Rates of C-Terminal Thioester Peptides Corresponding to L-Y-R-A-X (X is the designated amino acid); Percent Ligation Product (Filled Symbols) and Phenylmethanethiol–Thioester Exchange Intermediates (Unfilled Symbols) as a Function of Time[43]

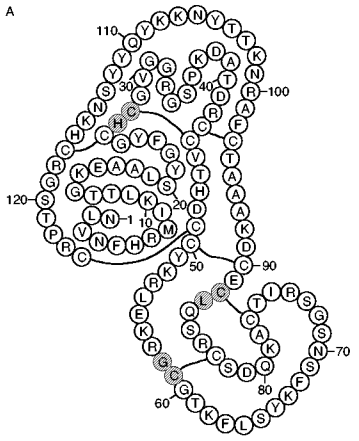


Selection of the Ligation Site

Two syntheses of human secretory phospholipase A2 (hsPLA2) provide an illustration of the choices that can be made in selecting ligation sites.[43,45] As shown in Scheme 7A, hsPLA2 is a 124 amino acid protein that has 14 cysteine residues that form seven disulfide bonds upon oxidative folding. Consequently, there are many sites available for ligation. The first synthesis of this protein utilized a Gly-Cys site near the center of the protein (Scheme 7B).[45] This site was selected for a two-segment approach in which one segment was synthesized on

a Boc-Gly-thioacid-resin and the other on a Boc-Arg(Tos)-PAM resin. The resulting peptides 1–58-COSR and 59–124 were ligated, purified by HPLC and folded. The advantage of this approach was that only a single ligation reaction was necessary. This approach was limited, however, by the challenging synthesis of two long polypeptides.

Scheme 7 Synthesis of hsPLA2: A: Amino Acid Sequence and Disulfide Pattern; B: Two-Component Synthetic Strategy; C: Four-Component Synthetic Strategy



B

1-NLVNFHRMIKLTTGKEAALSYGFYGCHCGVGGRGSPKDATDRCCVTHDCCYKRLEKRG-58

59-CGTKFLSYKFSNSGSRITCAKQDSCRSQLCECDKAAATCFARNKTTYNKKYQYYSNKHCRGSTPRC-124

C

1-NLVNFHRMIKLTTGKEAALSYGFYGCH-27

28-CGVGGRGSPKDATDRCCVTHDCCYKRLEKRG-58

59-CGTKFLSYKFSNSGSRITCAKQDSCRSQL-87

88-CECDKAAATCFARNKTTYNKKYQYYSNKHCRGSTPRC-124

A second approach utilized four segments and three ligations.[43] As shown in Scheme 7C, peptides corresponding to hsPLA2(1–27)-COSR, hsPLA2(28–58)-COSR, hsPLA2(59–87)-COSR and hsPLA2(88–124)-CO_2H. These peptides utilized ligation sites of His-Cys, Gly-Cys and Leu-Cys that demonstrated a range of chemical functionality and ligation rates. Alternative ligation sites could have been selected at 25–26 (Gly-Cys), 49–50 (Cys-Cys), or 89–90 (Glu-Cys). Although it would have made the C-terminal segment shorter, 96–97 (Thr-Cys) was not chosen due to the relatively slow reaction rate of C-terminal Thr thioester peptides. An advantage of using multiple peptide segments is that every peptide is intermediate in length (27–36 amino acids) and can be synthesized routinely and in high yield. Multiple-segment ligations require the use of a single protecting group on the N-terminal Cys of the middle segments (described in detail in Vol. E 22b, Section 5.3).

Synthesis of Thioester Peptides

Most work on native chemical ligation has utilized Boc/benzyl protocols to produce the required thioester peptides. Initial studies utilized a thioacid-yielding peptide linker[29] that was converted into a thioester by two methods. The first method is identical to the thioester forming ligation; the crude peptide is reacted with bromoacetic acid at pH 4.0 before purification to generate the thioester in quantitative yield.[17] This reaction is straightforward and can be done immediately following HF cleavage or following lyophilization of the crude peptide. The second approach involves the reaction of a peptide thioacid with an aryl di-

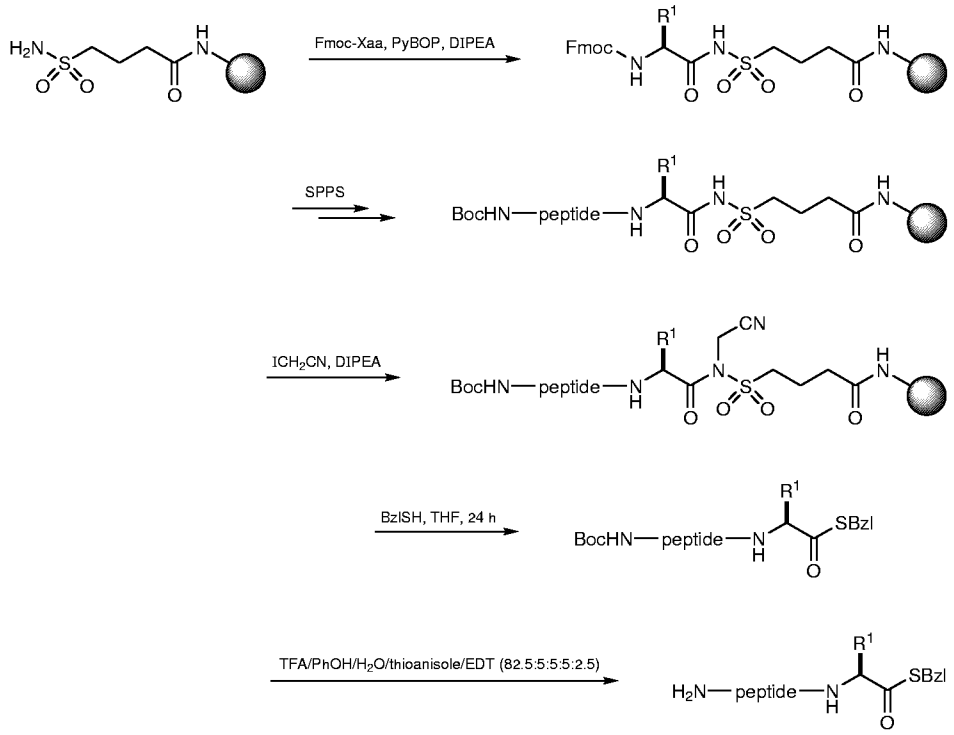
for references see p 640

sulfide such as 5,5′-disulfanediylbis(2-nitrobenzoic acid) (Ellman's Reagent).[17] This reaction produces a perthioester as the initial product that is immediately converted into a thioester in aqueous solution at pH >5. This product has been unambiguously characterized by mass spectrometry and with model studies. Performing this reaction at low pH can result in the stable generation of the highly reactive perthioester product that is prone to hydrolysis upon purification. The major limitation in the generation of thioester peptides from thioacid peptides is the synthesis of a Boc-Xaa-thioacid linker for every C-terminal amino acid.[26,29]

In order to facilitate the generation of peptide thioesters, several groups[43,46,47] have developed generalized versions of the thioester linker pioneered by Hojo and Aimoto.[48,49] A 3-sulfanylpropanoic acid residue is generated on an acid-labile linker such as Boc-Leu-PAM-resin or Boc-Leu-MBHA-resin. It is important to have a one-residue spacer between the MBHA linker and the thiol for optimal acid stability of the amide bond.[48] All twenty Boc-protected amino acids can be coupled to this thiol on the solid support to generate the thioester.[43] Despite the potential reactivity of the thioester to the amino terminus, the formation of a dioxopiperazine is not generally observed when using in situ neutralization protocols. However, when the sequence was Leu-Tyr-Arg-Ala-Pro, ~20% of a dipeptide deletion product, Leu-Tyr-Arg, was observed. It is likely that sequences such as C-terminal Pro-Gly would also be subject to this side reaction.[43]

The sensitivity of thioester groups to aminolysis by piperidine has prevented the use of thioester-based linkers for Fmoc SPPS. Two groups[27,61] have adapted a modified form of Kenner's acylsulfonamide safety-catch linker[50] for the synthesis of C-terminal thioester peptides using Fmoc protocols (Scheme 8). This approach is promising and future studies will more fully analyze the limitations of the method.

Scheme 8 Synthesis of C-Terminal Thioester Peptides Using an Fmoc-Compatible Linker[27,61]

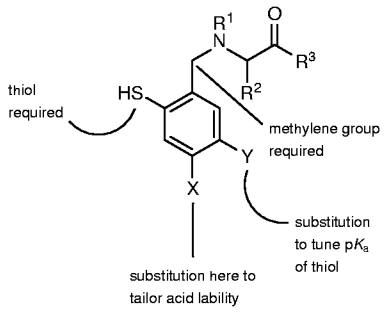


Future Directions

Native chemical ligation has already been utilized to synthesize hundreds of natural and unnatural proteins. The methodology has proven to be very robust. Typically, the most challenging aspect of a synthesis is the generation of the individual polypeptides by SPPS rather than their assembly through ligation. Recently, improvements have been made for the synthesis of long polypeptide chains from multiple peptide segments. Solid-phase chemical ligation,[51,52] described in Vol. E 22b, Section 5.3, will greatly facilitate the synthesis of proteins over 100 amino acids in length.

One important frontier for the chemical ligation of proteins is the development of efficient ligation chemistries that do not rely on Cys residues. One current limitation of native chemical ligation is the requirement for a Cys residue at the site of ligation. Although Cys residues can be easily modified through alkylation or disulfide formation, incorporation of Cys residues is not convenient for all proteins. One approach has been described by Canne et al. that utilizes a removable aminooxy thiol handle to facilitate thioester exchange and S-to-N acyl transfer.[53] Following ligation, this handle can be removed by treatment with zinc/acetic acid. However, this promising approach has not yet been utilized for protein synthesis. A second approach has been described that uses a N^{α}-2-sulfanylbenzyl auxiliary (Scheme 9).[54] This method is designed to facilitate thioester exchange followed by S-to-N acyl transfer. Although efficient ligation has been demonstrated, the sulfanylbenzyl auxiliary has not been successfully removed from the ligated product. Future development should produce a group that can ligate efficiently and be subsequently removed by acidolysis.

Scheme 9 N^{α}-2-Sulfanylbenzyl Auxiliary for Peptide Ligation[54][a]

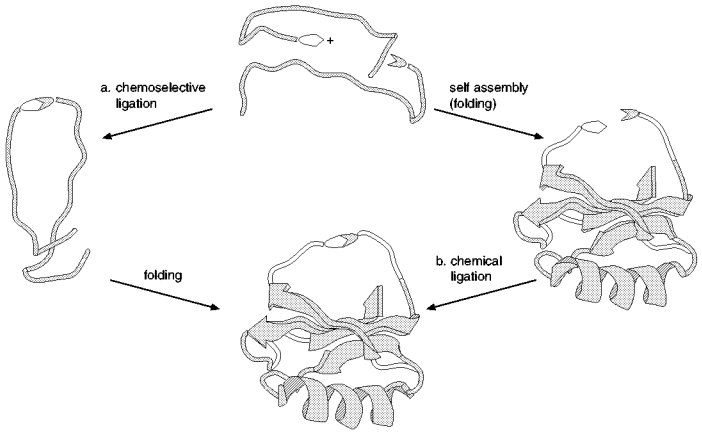


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Another approach for ligating peptides without Cys residues is to use the folding of the protein to align the N-terminus of one polypeptide with a weakly activated thioester group on the C-terminus of a second peptide (Scheme 10).[55] Many proteins fold into a nativelike structure even when the polypeptide chain has been cleaved into two or more polypeptide segments. This approach has been used for the generation of many semisynthetic analogues of cytochrome c using C-terminal homoserine lactone or weakly activated esters.[56,57] The conformationally assisted ligation approach using thioester peptides has been demonstrated with the synthesis of several analogues of chymotrypsin inhibitor 2 (CI2) and a semisynthesis of ribonuclease A. The ligation reaction is highly efficient and proceeds to completion in 2–12 hours.[55] Interestingly, ligation of N-terminal Cys peptides under folding conditions can greatly accelerate the reaction. In the example of CI2, even a relatively hindered C-terminal Thr residue formed a ligation product in a matter of minutes.[55]

for references see p 640

Scheme 10 Conformationally Assisted Ligation[55][a]



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The combination of synthetic and biologically derived polypeptides, often termed semi-synthesis, is a powerful approach for the synthesis of proteins. Native chemical ligation is becoming the method of choice for these studies with the advent of protein splicing techniques for the generation of C-terminal thioester peptides from biological sources.[58,59] It is important to note that Intein Mediated Ligation[59] and Expressed Protein Ligation[58] are not ligation methods but rather elegant methods for generating large C-terminal thioester peptides that can be subsequently ligated to N-terminal Cys peptides by native chemical ligation. The ability to mix and match expressed and synthesized polypeptide segments has great potential for the development of chemosensors, posttranslationally modified proteins and isotopic labeling of individual protein domains.[60]

Standard Native Chemical Ligation Protocol; General Procedure:
Native Chemical Ligation:[17,35,43]
The ligation of unprotected synthetic peptide segments was performed as follows: To the lyophilized peptides, 6 M guanidinium chloride, 0.1 M Na_3PO_4, pH 8.5, 2% (v/v) BzlSH and 2% (v/v) PhSH was added, resulting in a final peptide concentration of 1–3 mM at a pH $\approx$ 7 (lowered due to addition of thiols and TFA from the lyophilized peptide). No adjustment of this pH is necessary. The ligation reaction was vortexed for 5 s and placed in a heating block, stored in a ventilation hood, set at 25–37 °C. The reaction was vortexed periodically to equilibrate the thiol additives. The reaction was monitored by MALDI-MS and/or HPLC and ESI-MS until completion (5–30 h depending on the C-terminal thioester and concentration). Typically, the ligation reaction proceeded to >90% completion by HPLC and no hydrolysis of the thioester component was observed.
Over the course of the ligation reaction, a white precipitate of oxidized thiols accumulated. A brief centrifugation separated the ligation mixture from these thiols. The ligation mixture was filtered and purified directly on RP-HPLC using a C_4 derivatized column. Typically, linear gradients of 1% MeCN min^{-1} are sufficient to separate the ligation product. Occasionally, peaks corresponding to reduced thiols will overlap with the ligation product; this problem can be avoided by starting the HPLC run at 20% MeCN for 10 min to elute the thiols. The ligation product was recovered in 40–70% yield following HPLC purification.

Alternative Ligation Buffers:
Hydrophobic peptides corresponding to membrane proteins or proteins that tend to aggregate may have limited solubility even in 6 M guanidinium chloride. Several alternative ligation buffers have been used successfully for native chemical ligation reactions. For example, Kochendoerfer et al.[40] used 70% 0.1 M Na_3PO_4, 6 M guanidinium chloride, pH 7.5, 30% TFE in the synthesis of Influenza A Virus M2. The poorly soluble peptide was pretreated in this buffer at 60 °C prior to ligation. Bianchi et al.[41] utilized a detergent, β-octylglucoside, to solubilize a transmembrane peptide in a synthesis of the HCV protease cofactor protein NS4A. The buffer was composed of 25 mM Hepes, pH 7.0, 2% (v/v) β-octylglucoside, plus 1% (v/v) PhSH.

Synthesis of Thioester Peptides; General Procedure:
General Considerations:
As with thioacid peptides, Boc SPPS of thioester peptides is incompatible with strong nucleophiles. Consequently, side-chain protecting groups such as His(Dnp), Trp(For) and Lys(Fmoc) cannot be removed from the peptide before ligation. To avoid this problem, unprotected Trp is used and Lys(Aloc) can be used for orthogonal Lys protection. Although His(Bom) could be used in principle, inefficient removal of the formaldehyde released upon cleavage can interfere with ligation through thiazolidine formation with the N-terminal Cys peptide. Fortunately, it has been shown that His(Dnp) is efficiently deprotected during ligation, presumably by PhSH.

Boc-Xaa-Thioester-Resin:
TAMPAL Resin:[43] N^{α}-Boc-Leu (4.0 mmol) was activated with HBTU (3.6 mmol) in the presence of DIPEA (6 mmol) and coupled for 20 min to MBHA resin (2.0 mmol) (99.9% coupling yield). (Note: Boc-Leu-PAM-resin has also been used as the initial solid support). Next, 3-(tritylsulfanyl)propanoic acid (3 mmol) was activated with HBTU (2.7 mmol) in the presence of DIPEA (6 mmol) and coupled for 16 min to Leu-MBHA-resin (99.94% coupling). The resulting trityl-associated sulfanylpropanoic acid–leucine (TAMPAL) resin can be used as a starting resin for polypeptide chain assembly following removal of the trityl protecting group by treatment (2×1 min) with 2.5% TIS and 2.5% H_2O in TFA. The thioester bond can be formed with any desired amino acid using standard in situ neutralization coupling protocols for 1 h. Treatment of the final peptide with anhyd HF yields the C-terminal activated sulfanylpropanoic acid–leucine (MPAL)-thioester peptides that are ready for participation in native chemical ligation.

Disulfide formation on the resin following deprotection of the trityl-protected thiol could decrease the synthetic yield. This side reaction has not been a problem although care should be taken to avoid pulling air through the resin following trityl deprotection. Synthetic yields of human secretory phospholipase A2, all fragments synthesized on TAMPAL resin, were between 73 and 85% of the theoretical final peptide-resin weight. In comparison, the synthesis yield of fragment 88–124 (on Cys-PAM-resin) was 72%. Moreover, when the resin was treated with 2-sufanylethanol (5%) in DMF for 5 min after trityl removal, no increase in synthetic yield was observed.

Boc-Gly-[COS]-PEGA-Resin:[46] Amino-PEGA-resin (1 g, 0.3 mmol·g^{-1}) was swollen in DMF (10 min) and washed with 10% DIPEA/DMF and then DMF. A symmetrical anhydride was generated by adding DIC (315 µL, 2 mmol) to 3-bromopropanoic acid (611 mg, 4 mmol) in CH_2Cl_2 (3 mL) and coupled to the resin (2×30 min). The coupling was monitored by ninhydrin after the second coupling until it was >99% complete. The residual free amine was acetylated using Ac_2O/DIPEA/DMF (15:15:70) for 10 min. The thiol was introduced by treatment with AcSH/DIPEA/DMF (10:10:80, 10 mL, 2×20 min) followed by treatment with 2-sulfanylethanol/DIPEA/DMF (10:10:80, 10 mL, 2×20 min). The thiol-PEGA-resin was then acylated using Boc-Gly-OSu (545 mg, 2 mmol) in DMF (4 mL) for 2 h.

$HSCH_2CH_2CO$-MBHA-Resin:[47] To a suspension of MBHA resin (6 g, 3.2 mmol) in DMF (60 mL), 3-sulfanylpropanoic acid (1.2 mL, 13 mmol), HOBt (1.8 g, 13 mmol), and DIC (2 mL, 13 mmol) were sequentially added. The mixture was shaken at rt for 30 min. The resin was washed with DMF, CH_2Cl_2, MeOH, CH_2Cl_2, and DMF. The resulting resin was treated with a mixture of Cys(OMe)·HCl (0.55 g, 3.2 mmol), Ph_3P (0.85 g, 3.2 mmol), and DIPEA (0.68 mL, 3.9 mmol) in DMF/CH_2Cl_2 (3:1, 60 mL). After 2 h, the resin was thoroughly washed with DMF, CH_2Cl_2, MeOH, and CH_2Cl_2, and dried under reduced pressure; yield: 6.3 g.

Peptide Thioacid Alkylation:[17,35]
The crude peptide thioacid (100 mg) and bromoacetic acid (10 equiv) were added to 6 M guanidinium chloride and 100 mM NaOAc, pH 4.6 (10 mL). The peptide soln was mixed briefly and reacted at 25 °C for 1 h. The peptide thioester was immediately purified by HPLC and lyophilized.

Peptide Thioacid; Reaction with Ellman's Reagent:[11,17]
The crude peptide thioacid (100 mg) and 5,5′-disulfanediylbis(2-nitrobenzoic acid) (10 equiv) were added to 6 M guanidinium chloride and 100 mM Na_2HPO_4, pH 6.0 (10 mL) and mixed briefly. After 1 h, the peptide thioester was purified by HPLC and lyophilized.

Synthesis of C-Terminal Thioester by Fmoc/tBu Chemistry; General Procedure:
4-Sulfamylbutyryl-AM-Resin:[27,61]
The resin with the alkanesulfonamide "safety-catch" linker is commercially available, 4-sulfamylbutyryl AM-resin (NovaBiochem). It can also be prepared by the procedure of Ingenito et al.:[61] PEG-PS·HCl resin or Tentagel (NovaSyn TG, NovaBiochem) was washed with DIPEA (2×5 min) and DMF (5×2 min). Then a mixture of 3-carboxypropanesulfonamide, DIC, and HOBt (0.95 mmol) was added in the

for references see p 640

minimum amount of DMF required to keep the resin covered with liquid. After stirring on a rotary plate for 24 h, the resin was extensively washed with DMF before it was used to attach the first amino acid.

Attachment of the First Amino Acid and Peptide-Chain Assembly:
To the swollen 4-sulfamylbutyryl-AM-resin (110 mg, 0.10 mmol), DMF (2 mL), DIPEA (157 µL, 0.9 mmol), and the Fmoc amino acid (119 mg, 0.400 mmol) were added.[27] The mixture was stirred for 10 min followed by cooling to −20 °C. After 20 min, PyBOP (156 mg, 0.300 mmol) was added to the mixture as a solid. The mixture was stirred at −20 °C for 8 h and then warmed to rt overnight. The resin was filtered, washed with DMF and CH_2Cl_2, and subjected to the same coupling conditions.[62]
Pessi and co-workers employed a slightly different method.[61] To the swollen 3-carboxypropane-sulfonamide-resin (0.2 mmol) was added CH_2Cl_2 or $CHCl_3$ (5 mL), DIPEA (342 µL, 2 mmol), and the Fmoc amino acid (1 mmol). The mixture was stirred for 20 min, followed by cooling to −20 °C. Then PyBOP (520 mg, 1 mmol) was added as a solid, and the mixture was stirred for 8 h at −20 °C, filtered, and washed with CH_2Cl_2 or $CHCl_3$ (5 × 5 mL). Double coupling is usually necessary for high loading of the first residue. After the loading of the first amino acid, all the following amino acids are coupled using the standard Fmoc protocols, except that the last residue is incorporated as the *N*-Boc derivative.

Alkylation of the Peptide-Acylsulfonamide-Resin and Peptide Cleavage:
After the peptide-chain assembly was complete, the *N*-acylsulfonamide-resin (8 µmol) was washed with NMP, and then NMP (2.2 mL), DIPEA (82 µL, 0.48 mmol), and ICH_2CN (174 µL, 2.4 mmol), previously filtered through an alumina basic plug, were added to a final concentration of 1 M. After stirring on a rotary plate for 24 h, the resin was washed with NMP (5 × 3 mL) and DMF (5 × 3 mL). The peptide was cleaved from the resin using 1 M ethyl 3-sulfanylpropanoate in either DMF or CH_2Cl_2, with or without a catalytic amount of NaSPh.[61] After 24 h at rt, the resin was removed by filtration and the filtrate was concentrated in vacuo. Alternatively, BzlSH in THF was utilized for the cleavage by Bertozzi et al.[27]

Removal of N-Terminal and Side-Chain Protecting Groups:
The cleaved protected peptide was treated with a cocktail[61] of TFA/PhOH/TIS/H_2O (88:5:2:5) for 2 h at 21 °C or with a mixture[27] of TFA/PhOH/H_2O/thioanisole/EDT (82.5:5:5:5:2.5) for 4 h at rt. The unprotected peptide was then precipitated and washed with Et_2O. The precipitate was dissolved in aq MeCN and lyophilized.

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4.2 Enzymatic Synthesis

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Owing to the stereo- and regiospecificity of enzymes, their application for peptide bond formation offers several principal advantages over chemical procedures, e.g. mild reaction conditions, no need for side-chain protection, and simple scale up after optimization of the process. One of the most important attractions of using enzymes for chemical syntheses is the control of stereochemistry: the danger of racemization that attend chemical C–N ligation methods are completely avoided. However, in order to make full use of these advantages, a close to universally applicable peptide ligase is needed which principally would display high catalytic efficiency for all possible combinations of amino acids to be coupled.[1] Of course, the term "peptide ligase" here refers to a biocatalyst for peptide bond formation and is not in agreement with the IUPAC-IUB Enzyme Commission nomenclature of enzymes. In deviation from this classification, such a peptide ligase should be nonspecific, e.g. its catalytic potential should not be affected by the various side-chain functions of the amino acids involved in the peptide bond forming step. Protein biosynthesis requires a tRNA to convey the coded aminoacyl moiety to the growing peptide chain. Cells solve the problem of peptide bond formation with the help of one universal enzyme: the ribosomal peptidyl transferase. Unfortunately, this ideal biocatalyst seems to be a ribozyme[2] and also is not suitable for simple practical use as a catalyst for C—N ligation outside the ribosome. Since, in addition, multienzyme complexes for bacterial peptide synthesis are limited to very special purposes,[3] only the "reverse hydrolysis potential" of proteases can be used for peptide bond formation. In 1898, evoking the reversibility of chemical reactions, van't Hoff predicted that proteases could be used as catalysts for peptide bond formation, in contrast to their biological hydrolysis function to degrade proteins. In 1938, Bergmann and Fraenkel-Conrat[4] confirmed this idea experimentally in simple model reactions. Up to now the reverse hydrolysis potential of proteases has been developed as a new approach, complementary to that of chemistry, either in thermodynamically controlled synthesis reactions or demonstrated kinetically with proteases acting as transferases. The application of new innovative strategies of medium engineering, e.g. biotransformation in solid–liquid heterogeneous systems, synthesis in frozen aqueous systems, cryoenzymatic synthesis, and unexpected synthesis tools which allow practically irreversible peptide bond formation by zymogen catalysis, or new ligation strategies using substrate mimetics,[5–10] will be presented.

4.2.1 Approaches to Protease-Catalyzed Synthesis

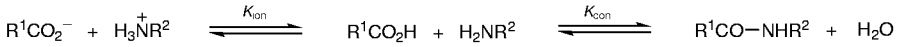
Generally, the methods of protease-catalyzed peptide bond formation can be classified into two basic strategies according to the type of carboxy component used. In the first approach, this component has a free carboxy terminus, and the formation of the peptide bond occurs with thermodynamic control as the reverse of peptide hydrolysis. In the second approach, the carboxy component is employed in a slightly activated form, mainly as an alkyl ester, and the synthesis occurs with kinetic control by competitive partitioning of a rapidly formed acyl-enzyme intermediate between the nucleophilic amino component and water. These two approaches are fundamentally different owing to the energy required for the conversion of the substrates into the peptide products. Independently of the mechanistic scheme discussed below, a successful way of reversing the equilibrium towards synthesis is the formation of insoluble products. Besides this solubility-controlled process of precipitation, in the biphasic approach, product extraction is a useful alternative where the product is much more soluble

in the organic phase and is thus continuously removed from the aqueous phase. In special cases the formed peptide product is removable from the equilibrium using molecular traps, e.g. by specific complex formation as demonstrated in the course of the clostripain-catalyzed segment condensation of the RNase fragments 1–10 with 11–15 using RNase S (21–124) as a suitable trap.[11]

4.2.1.1 Equilibrium-Controlled Synthesis

The equilibrium-controlled synthesis (thermodynamic approach) is the direct reversal of proteolysis. For this reason, independent of the individual mechanism, all proteases can be used. The thermodynamic barrier to peptide bond synthesis is predominantly due to the energy required for the transfer of a proton from the positively charged α-amino group to the negatively charged α-carboxy group. The equilibrium constant K_{con} for the conversion of the un-ionized reactants into the product is determined by the concentration of the un-ionized forms in K_{ion} (Scheme 1).

Scheme 1 Equilibrium-Controlled Peptide Bond Synthesis



Taking the water concentration into the equilibrium constant, the following results:

$$K_{syn} = K_{ion} \times K_{con} = [R^1CONHR^2]([R^1CO_2^-][H_3N^+R^2])^{-1}$$

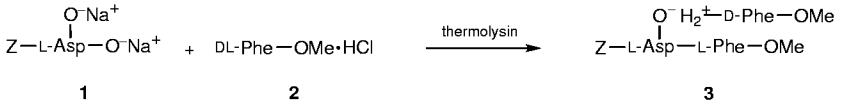
Owing to the unfavorable equilibrium position, relatively high concentrations of the reactants are required. In addition, the low reaction velocity and the high enzyme requirement are drawbacks of this approach. In order to shift the equilibrium in favor of peptide product formation, it is necessary to influence the ionization equilibrium. Synthesis will be promoted by increasing the equilibrium constant K_{ion} for proton transfer from H_3N^+ to CO_2^- end groups of the reactants. One method of choice is adding high concentrations of water-miscible organic solvents to the aqueous reaction mixture, thereby lowering the dielectric constant of the medium and resulting in reduced acidity of the carboxy group. The only function of the enzyme is to accelerate the attainment of the equilibrium constant K_{con}.[12] In addition, solvent systems consisting of an aqueous phase and an immiscible phase (nonpolar organic solvent), so-called biphasic systems,[13] are also usable. Furthermore, anhydrous media containing minimal water concentration and, as mentioned above, reaction conditions promoting product precipitation and product extraction, respectively, are often used.

4.2.1.1.1 Thermolysin-Catalyzed Synthesis of Aspartame

A number of synthetic peptides are significant commercial products, ranging from the sweet dipeptide aspartame (L-aspartyl-L-phenylalanine methyl ester) to clinically used hormones such as insulin and oxytocin. L-Aspartyl-L-phenylalanine methyl ester (**3**; Scheme 2) is the methyl ester of the C-terminal dipeptide of gastrin. It was found accidently during the synthesis of gastrin that this synthetic sweetener is about 200 times as sweet as sucrose.[14] This pleasant sweetness without a bitter aftertaste was the reason that L-aspartyl-L-phenylalanine methyl ester was approved in many countries as a food additive, receiving much attention as a low-calorie sweetener. L-Aspartyl-L-phenylalanine methyl ester can be prepared by various chemical routes and the first enzymatic procedure of commercial interest was described by Isowa et al.[15] In the industrial process,[16] L-Asp and DL-Phe were chosen as inexpensive raw materials. L-Asp is available very inexpensively, whereas L-Phe is more expensive than DL-Phe. Z-D-Asp acts as a competitive inhibitor, while D-Phe-OMe

does not. Owing to the regiospecificity of the preferred enzyme, thermolysin, only α-peptide linkages are formed without side-chain protection. However, chemical methods with an industrial potential provide 20–40% of the β-isomer, which has a bitter taste. In addition, the enzymatic coupling reaction can be done under very mild conditions in aqueous solution. In this reaction, peptide synthesis and optical resolution of the amino acid are carried out simultaneously.

Scheme 2 Route to Aspartame[16]



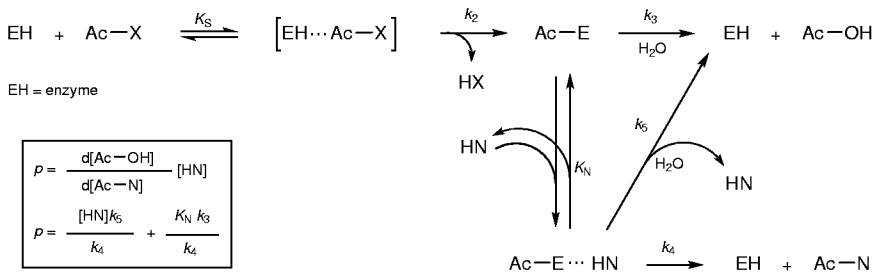
1 2 3

L-Aspartyl-L-phenylalanine Methyl Ester·D-Phenylalanine Methyl Ester (3):[16]
A soln of Z-L-Asp disodium salt (**1**; 53.4 g, 0.2 mol) in H_2O (139.6 g), a soln of DL-Phe-OMe·HCl (**2**; 89.5 g, 0.5 mol) in H_2O (209.4 g), and a 22 wt% aq NaOH soln (14.6 g) were added to a 2-L glass reactor equipped with a stirrer and thermostated at 40 °C. To this mixture an enzyme soln was added which had been prepared by dissolving NaCl (24 g), $CaCl_2 \cdot 2H_2O$ (0.75 g), and crude thermolysin (3 g) in distilled H_2O (220 g). The enzymatic coupling reaction was then stirred at 40 °C. During the coupling reaction the addition product **3** precipitated. After 11 h of reaction, the conversion of the reaction (calculated on Z-L-Asp) reached about 55%, as determined by HPLC. Toluene (17.6 g; 0.0145 part by weight relative to one part by weight of the sum of Z-L-Asp + DL-Phe-OMe) was added to the mixture. The mixture was stirred for a further 18 h, at which time no significant further conversion could be observed by HPLC; yield: 92.2%; the enzyme retention percentage was 81.5%. The N^{α}-unprotected aspartame (H-L-Asp-L-Phe-OMe) can be easily obtained by well-known deprotection procedures for the Z group, for example, according to the procedure described by Lindberg.[17]

4.2.1.2 Kinetically Controlled Synthesis

In contrast to the equilibrium-controlled approach which ends with a true equilibrium, in the protease-catalyzed kinetically controlled synthesis[18] the product appearing with the highest rate and disappearing with the lowest velocity would accumulate. This approach requires the use of acyl donor esters as carboxy components (Ac-X) and is limited to proteases which rapidly form an acyl-enzyme intermediate (Ac-E). Serine and cysteine proteases are known to catalyze acyl transfer from specific substrates to various nucleophilic amino components via an acyl-enzyme intermediate. In reactions of this type, the protease reacts rapidly with an amino acid or peptide ester, Ac-X, to form a covalent acyl-enzyme intermediate, Ac-E, that reacts, in competition with water, with the amino acid or peptide-derived nucleophile HN to form a new peptide bond (Scheme 3). The partitioning of the acyl-enzyme intermediate between water and the added nucleophile is the rate-limiting step. Under kinetic control, and if $k_4[HN] \gg k_3[H_2O]$, the peptide product Ac-N should accumulate. However, the soluble peptide product will be degraded if the reaction is not terminated after the acyl donor ester is consumed.

Scheme 3 Kinetic Scheme of Protease-Catalyzed Acyl-Transfer Reactions



Therefore the reaction conditions must be carefully considered in order to obtain the optimal product yield. Peptide bond formation by partitioning of acyl-enzyme intermediates may also be controlled by the specificity of the enzyme. Structural factors in the nucleophile which increase the extent to which it is bound productively at the S′ subsite of the enzyme promote peptide bond formation. The type and concentration of the nucleophilic amino component influence the partitioning of the acyl-enzyme intermediate. Since only the free-base form of the amino component reacts with the acyl-enzyme intermediate, and the pK of this type of reactant is about 8, the pH of the reaction mixture should preferably be higher than this value.

As shown in Scheme 3, by analogy to the definition of the Michaelis constant we defined the partition value p[19] corresponding to the concentration of the amino component [HN] at which hydrolysis and aminolysis of the acyl-enzyme intermediate proceeds with the same velocity. The value of p allows a rapid estimation of the peptide yield of any acyl-transfer reaction, and $[HN] \gg p$ is necessary for successful peptide synthesis in high yield. According to the shorthand notation of protease–substrate interactions of Schechter and Berger,[20] proteases have binding grooves (S_n···S_3-S_2-S_1-S_1'-S_2'-S_3'···S_n') on both sites of the scissile bond of the substrate (-P_1-P_1'-), whereas substrate binding (P_n···P_3-P_2-P_1-P_1'-P_2'-P_3'···P_n') in proteolysis is mostly reduced to the P_1 residue (primary specificity) beside additional P–S interactions. In contrast to proteolysis, it follows from Scheme 3 that enzymatic peptide bond formation is a two-step reaction. Therefore, the efficiency of the nucleophilic attack of the added nucleophilic amino component in peptide synthesis depends essentially on an optimal binding in the S′ subsite region within the active site. Since the S′–P′ interactions determine the outcome of the ligation reaction, a useful prediction of the efficiency of protease-catalyzed kinetically controlled synthesis can be established by S′ subsite mapping studies.[18] Owing to solubility limitations, di- and tripeptides can only be synthesized in a continuous large-scale procedure using solubilizing protecting groups.[21,22]

4.2.1.2.1 Large-Scale Synthesis of Kyotorphin

Kyotorphin (Tyr-Arg), with a Met-enkephalin releasing activity,[23] was chosen by Fischer et al.[24] as a model peptide for kinetically controlled dipeptide synthesis using highly solubilizing N^{α}-protecting groups. The maleyl (3-carboxyacryloyl) group strongly increases both the solubility of the amino acid substrate in purely aqueous medium as well as the α-chymotrypsin activity. This allows operation at extremely high Mal-Tyr-OEt concentrations, up to 1.5 mol·L^{-1}, and with equimolar H-Arg-OEt concentrations. The application of an organic cosolvent is no longer necessary.

Kyotorphin (Tyr-Arg):[24]
The pH of a suspension of H-Tyr-OEt·HCl (17.86 kg, 72.7 mol) in H_2O (300 L) was adjusted to 8.2 with 50% NaOH, and maleic anhydride (7.84 kg, 79.97 mol) was added. At the end of this operation the pH (8.0) of the homogeneous soln was adjusted to the optimum for the synthesis (9.5) with 50% NaOH. H-

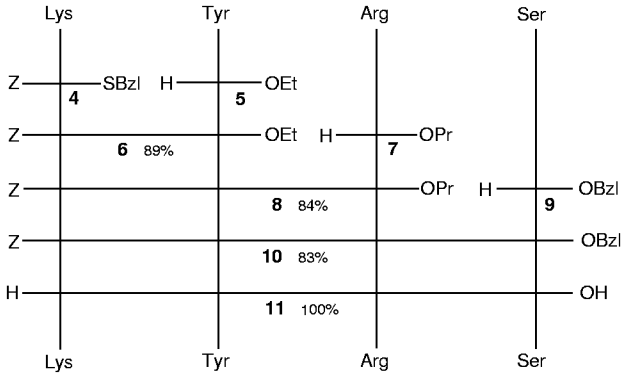
for references see p 663

Arg-OEt·2HCl (20 kg, 72.7 mol) and α-chymotrypsin (7.5 g) were added at constant pH of 9.5. After about 20 min, Mal-Tyr-Arg-OEt precipitated. After stirring for another 2 h, the product was filtered through a 150-L pressurized filter and washed with H_2O (4 × 5 L). For protecting group removal, the dipeptide derivative was suspended in H_2O (150 L), the pH adjusted to 0 with 50% H_2SO_4, and stirred for 72 h at 50 °C. Reaction progress was monitored by HPLC. The resulting soln was purified by cation-exchange chromatography, eluting with 10% NH_3. The eluant was removed using a thin-film evaporator and the soln concentrated to 20 L. After addition of a stoichiometric amount of AcOH and charcoal, the dipeptide diacetate was isolated by spray drying; yield: 12 kg (50.4%).

4.2.1.2.2 Synthesis of H-Lys-Tyr-Arg-Ser-OH from the N- to the C-Terminus

A step-by-step peptide synthesis from the N- to the C-terminus is not possible with chemical methods as it risks partial epimerization due to the repeated carboxy activation procedures.[25,26] In constrast, the stereo- and regiospecificity of serine and cysteine proteases ensures integrity of the stereogenic center and allows ecological reaction conditions without side-chain protection. Scheme 4 shows the synthesis scheme using clostripain and chymotrypsin as catalysts.[27] The second coupling reaction was carried out by enzyme catalysis in a frozen aqueous system (see Section 4.2.3.1).

Scheme 4 Synthesis of H-Lys-Tyr-Arg-Ser-OH[27]



H-Lys-Tyr-Arg-Ser-OH (11); Stepwise Synthesis Procedure:[27]

Step A: Z-Lys-SBzl (**4**; 250 mg, 0.59 mmol) and H-Tyr-OEt (**5**; 436 mg, 1.77 mmol) were dissolved in 0.05 M borate buffer (73.9 mL, pH 8.0, 0.2 M NaCl). The reaction was started by addition of clostripain (0.182 mg, 3.0 mL; stock soln: 1.1×10^{-6} M in 2.5 mM DTT and 1.0 mM $CaCl_2$). After 2.0 h, 8% TFA in H_2O (35 mL) was added to stop the enzymatic reaction. After concentration on a vacuum evaporator, **6** was separated by preparative RP-HPLC ($MeCN/H_2O$ gradient with 0.1% TFA). The solvent was removed under reduced pressure and the product was lyophilized from aqueous soln; yield: 300 mg (89% analyzed by analytical RP-HPLC; the yield was based both on the difference of the measured concentration of acid and the small amount of remaining substrate ester as well on the tyrosine content).

Step B: Z-Lys-Tyr-OEt·TFA (**6**; 281 mg, 0.48 mmol) and H-Arg-OPr·2HCl (**7**; 971.7 mg, 3.36 mmol) were dissolved in 0.025 M borate buffer (48 mL, pH 9.0, 0.2 M NaCl). The apparent pH was readjusted to 9.0 with 6 M NaOH (small portions, pH control). To ensure fast freezing of the solvent, the whole reaction volume was divided into 10 portions of 4.8 mL each using 10-mL plastic vials. α-Chymotrypsin (0.144 mg, 192 μL) in 1 mM HCl containing 10 mM $CaCl_2$ (stock soln: 3.0×10^{-5} M) was added consecutively to each reaction vial. After a short period of shaking the vials were placed immediately into liquid N_2 for approximately 2 min to achieve shock freezing. (In order to achieve rapid freezing in liquid N_2, all solns were precooled in a refrigerator. This was absolutely necessary to avoid ester hydrolysis after enzyme addition and mixing of the components before the soln was put into the liquid N_2. It was also necessary to keep this interval as short as possible and to work very quickly. HCl should not be used to stop the reaction. It will not be removed during vacuum concentration, resulting in product hydrolysis.) Subsequently, the vials were transferred into a cryostat at −25 °C for 48 h. The reactions were stopped by adding 8% TFA in H_2O (2.0 mL) to each vial. After thawing, the samples were combined

and concentrated on a vacuum evaporator. The crude peptide product was purified by preparative RP-HPLC (MeCN/H_2O 1:4 with 0.1% TFA). The solvent was removed under reduced pressure and the product **8** was lyophilized from aqueous soln; yield: 372 mg (84% by analytical RP-HPLC).

Step C: Z-Lys-Tyr-Arg-OPr · 3TFA (**8**; 354 mg, 0.36 mmol) and H-Ser-OBzl · HCl (**9**; 417.1 mg, 1.8 mmol) were dissolved in 0.05 M borate buffer (36 mL, pH 8.0, 0.2 M NaCl). The apparent pH was readjusted to 9.0 with 6 M NaOH (small portions, pH control). The reaction was initiated by addition of clostripain (0.09 mg, 1.5 mL, stock soln as in step A). After 1.0 h, 8% TFA in H_2O (20 mL) was added to stop the enzymatic reaction. The product **10** was obtained as above by preparative RP-HPLC (MeCN/H_2O gradient with 0.1% TFA) and lyophilization; yield: 306 mg (83% by analytical RP-HPLC).

Step D: The protecting groups of Z-Lys-Tyr-Arg-Ser-OBzl (**10**) were removed in a one-step reaction by catalytic hydrogenation using 10% Pd/C as catalyst, which occurred easily and completely (yield: 100%); overall yield of **11**: 62% (by analytical HPLC). Anal. calcd for H-Lys-Tyr-Arg-Ser-OH · 3TFA · $4H_2O$ ($C_{30}H_{51}F_9N_8O_{17}$; 966.77): C 37.27, H 5.32, N 11.59; found: C 37.14, H 5.11, N 11.21; MS: $m/z = 553$ ($M^+ + H$).

4.2.2 Manipulations to Suppress Competitive Reactions

The kinetic approach can be more efficiently manipulated than the thermodynamic approach, but serine and cysteine proteases are not perfect acyl transferases. Undesired reactions may take place parallel to acyl transfer, for instance hydrolysis of the acyl-enzyme, secondary hydrolysis of the formed peptide bond, and other undesired proteolytic cleavages of possible protease-labile bonds in reactants and product. The elimination or minimization of these disadvantages can be performed by various manipulations on the level of the reaction medium, the enzyme, and the substrate, as well as on mechanistic features of the process.

4.2.2.1 Medium Engineering

The solubility of the reactants in enzymatic synthesis dramatically influences the course of the reaction, but not only for ecological reasons should the solvent of choice for enzymatic reactions be water. However, owing to solubility limitations, di- and tripeptides can only be synthesized in a continuous large-scale procedure using solubilizing protecting groups (Section 4.2.1.2.1). In Table 1, various approaches to medium engineering are compiled. Water-miscible organic solvent systems increase the solubility of partially protected starting compounds. Despite some disadvantages, such media with small organic solvent concentrations have been used in various enzymochemical peptide syntheses, for example the chymotrypsin-catalyzed fragment coupling synthesis of [D-Phe^6]LHRH.[28] Biphasic systems[29] lead to preservation of enzyme activity and easy product separation but suffer from solubility limitations. Undesired hydrolytic side reactions can ultimately be prevented by syntheses in monophasic organic solvents. Independently of the enzyme preparation, trace amounts of water (ca. 0.3–1%) are necessary to maintain catalytic activity. The main drawbacks of this approach are enzyme deactivation and specificity changes of the enzymes. Frequently, enzymatic syntheses in monophasic organic solvents[30] are described as reactions with so-called solvent-modified enzymes. Nevertheless, enzyme modifications with poly(ethylene glycol) (PEG) have provided modified proteases which are soluble in organic solvents. PEG-modified chymotrypsin, papain, and thermolysin have been used for synthesis in monophasic organic solvents with complete elimination of undesired proteolytic side reactions.

for references see p 663

Table 1 Medium Engineering in Protease-Catalyzed Peptide Synthesis[28–30]

Medium	Advantages	Disadvantages	Alternatives	Ref
water	optimal ecological conditions	poor solubility of partially protected reactants	use of solubilizing protecting groups	[24]
water/water-immiscible organic solvents	preservation of enzyme activity; easy product extraction	high enzyme requirements; limited reactant solubility	–	[29]
water/water-miscible organic solvents	increased reactant solubility; promoting thermodynamic approach	difficult product isolation	use of immobilized enzymes or engineered proteases	[28]
monophasic organic solvents	prevention of hydrolysis; no solubility problems	reduced enzyme activity; change of specificity; higher enzyme requirements	use of chemically or genetically altered enzymes	[30]

4.2.2.1.1 α-Chymotrypsin-Catalyzed Segment Synthesis of [D-Phe^6]LHRH

The (3+7) fragment coupling is the preferred method for the synthesis of LHRH decapeptides. However, the application of chemical coupling procedures causes up to 3% racemization at the activated Trp residue, leading to a reduced stereochemical purity. In contrast, the enzymatic reaction gives high yields without racemization. The coupling of two fragments with α-chymotrypsin (enzyme/substrate = 1:800, by weight) was carried out in DMF/H_2O (48:52) at pH 8.0 and room temperature. There were no problems in scaling up, thus making the procedure convenient for the synthesis of kilogram quantities of the peptide.

Pyr-His-Trp-Ser-Tyr-D-Phe-Leu-Arg-Pro-Gly-NH_2:[28]
Pyr-His-Trp-OEt (21 g, 43.7 mmol) and H-Ser-Tyr-D-Phe-Leu-Arg-Pro-Gly-NH_2 (20 g, 23.9 mmol) were suspended in a mixture of DMF (66 mL) and H_2O (70.5 mL) the pH being adjusted to 8.0 by addition of 2 M KOH. The reaction was started by addition of α-chymotrypsin (50 mg). After 6.5 h, AcOH (10 mL) was added to stop the enzymatic reaction. After concentration, precipitation by addition of acetone gave [D-Phe^6]LHRH; yield: 24.4 g. The crude peptide was purified by RP chromatography to give the LHRH peptide which was characterized by HPLC (t_R = 15.5 min, 99%) and amino acid analysis; yield: 16.2 g (53%).

4.2.2.1.2 Trypsin-Catalyzed Semisynthesis of Human Insulin

Semisynthesis has become a suitable technique for chemical variations of large polypeptides or proteins from natural sources. Human insulin differs from porcine insulin only in position B30, where alanine is the C-terminal B-chain residue of porcine insulin but threonine is the C-terminal of the human insulin B-chain. Among the various ways to obtain human insulin from porcine insulin by protease-catalyzed semisynthesis, the one-step conversion by exchange of the C-terminal alanine B^{30} of porcine insulin with a threonine residue seems to be the method of choice. Several proteases, e.g. trypsin, achromobacter protease, and carboxypeptidase Y, are able to catalyze the conversion and various groups have independently described an essential improvement of the enzymatic semisynthesis of insulin.[7,8] The Hoechst procedure,[31] which was developed as an industrial process, is described below as an example of a large-scale conversion of porcine insulin to human insulin in kilogram amounts for therapeutic application. Based on this type of transamidation reaction, it is easy to prepare various B30-insulin analogues.

Di-*tert*-butylthreonine-B30-human Insulin:[31]
Porcine insulin (1 g, 0.173 mmol) was dissolved in H_2O (5 mL). Crystalline H-Thr(tBu)-OtBu · AcOH (6 g, 3.43 mmol) was added and the pH adjusted to 4.5 by addition of AcOH. An aq trypsin soln (80 mg, 0.5 mL) was stirred into the chilled mixture. The mixture was then kept overnight in the cold room.

Conversion rates were monitored by HPLC: the reaction was finished when about 90% of the porcine insulin had been converted into the product. The crude material was isolated by precipitation with MeOH/Et_2O, extensively washed with Et_2O, and dried under reduced pressure. HPLC of this material showed 80–85% (0.82–0.87 g) di-*tert*-butylthreonine-B30-human insulin. Separation of the human insulin ester could be achieved by various chromatographic methods, e.g. partition or ion-exchange chromatography.[32] Column chromatography (silica gel, $CHCl_3$/MeOH/H_2O/HCO_2H/TEA 325:270:65:8:2) resulted in an excellent separation of the less acidic and more hydrophobic di-*tert*-butylthreonine-B30-human insulin.
To obtain human insulin, cleavage of the tBu protecting groups was done in the usual way by treatment with TFA. The deblocked human insulin was isolated by precipitation with Et_2O and purified by column chromatography (silica gel) and then ion-exchange chromatography. The main fraction contained highly purified human insulin; overall yield: 0.55–0.60 g (55–60%); purity: 99%.

4.2.2.2 Enzyme Engineering

Enzyme engineering[33] is defined as a range of techniques from deliberate chemical modification to remodeling a wild-type enzyme by gene technology. The application of immobilized enzymes will probably be of technological and economical importance despite the increase in stability towards organic solvents.[34] Of special interest is the continuous synthesis of the aspartame precursor Z-Asp-Phe-OMe, with thermolysin immobilized on Amberlite XAD-7 in a plug flow-type reactor, and the conversion of porcine insulin into human insulin, catalyzed by *Achromobacter lyticus* protease I[35] immobilized on SiO_2-polyglutamic acid or by immobilized trypsin.[34] Specially chemically modified proteases are prepared in order to reduce the peptidase activity with some of the esterase activity remaining, thus preventing the proteolytic cleavage of peptide bonds.[33,36] Examples for these strategies are methyl-chymotrypsin obtained by N-methylation of His^{57} (practically without peptidase activity but drastically reduced esterase activity), thiosubtilisin, or selenosubtilisin. By multiple site-directed mutagenesis, subtilisin could be transformed into a mutant which catalyzes peptide synthesis in the presence of high concentration of DMF.[37] A mutant of subtilisin BPN′, called subtiligase, was used by Jackson et al.[38] for enzymatic segment condensation of fragments synthesized by the solid-phase technique, forming RNase A.

4.2.2.2.1 (4+3) Segment Synthesis Catalyzed by Subtilisin 8350

Subtilisin 8350, derived from subtilisin BPN′ via six site-specific mutations (Met50Phe, Gly169Ala, Asn76Asp, Gln206Cys, Tyr217Lys, and Asn218Ser), was found by Wong et al.[37] to be 100 times more stable than the wild-type enzyme in aqueous solution at room temperature and 50 times more stable in anhydrous DMF. The enzymatic peptide-coupling reaction was carried out under high concentrations of DMF (ca. 50%) in order to improve reactant solubility and to minimize unwanted enzymatic peptide cleavage.

Boc-Tyr-D-Ala-Phe-Gly-Tyr-Pro-Ser(Bzl)-NH_2[37]
Boc-Tyr-D-Ala-Phe-Gly-OEt (0.585 g, 1 mmol), H-Tyr-Pro-Ser(Bzl)-NH_2 (1.41 g, 3 mmol), and subtilisin 8350 (5 μmol) in H_2O/DMF (1:1, pH 8.5–9.0, adjusted with TEA) were incubated for 20 min. The product was then isolated by extraction with EtOAc or BuOH and purified by gel-filtration chromatography (Sephadex LH-20, MeOH); yield: 80%; mp 153–155 °C; $[\alpha]_D^{25}$ –26.6 (*c* 1, DMF); anal. calcd for $C_{52}H_{64}N_8O_{12}$: C 64.71, H 6.68, N 8.71; found: C 64.70, H 6.66, N 8.80.

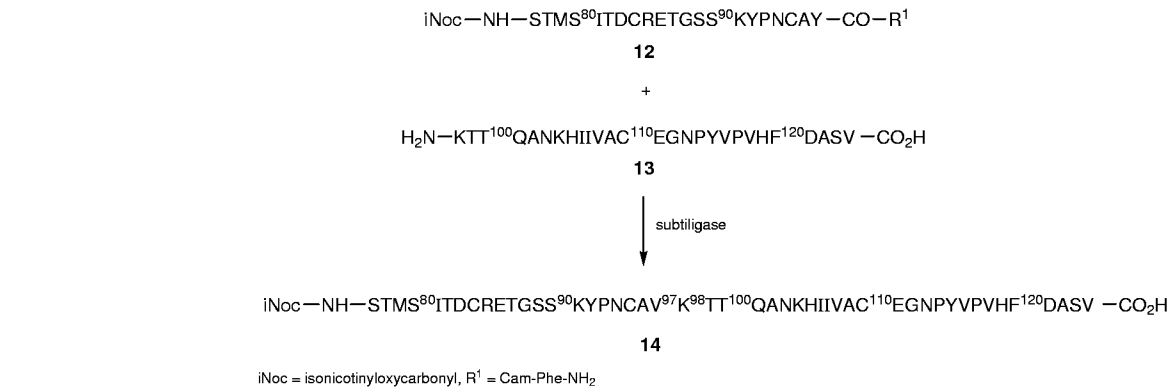
4.2.2.2.2 Ribonuclease A (77–97+98–124) Segment Synthesis Catalyzed by Subtiligase

A peptide ligase, termed subtiligase, an engineered variant of subtilisin BPN′ (Ser221Cys, Pro225Ala), was used by Jackson et al.[38] for the complete synthesis of ribonuclease A (RNase A) in milligram quantities by stepwise ligation of six esterified peptide fragments (each 12–30 residues). In the double mutant of subtilisin, BPN′ the catalytic Ser221 is

for references see p 663

converted to Cys. The new enzymatic ligation principle was demonstrated by the first ligation step of the RNase A synthesis, forming the CO_2H-terminal fragment (77–124) (Scheme 5). The strategy for RNase A synthesis with subtiligase starts with the solid-phase synthesis of fragment 98–124 followed by deprotection. The next-most H_2N-terminal fragment (77–97) is used as Phe-NH_2-modified carboxamido methyl ester, which even as an unmodified leaving group was found to be a very good acyl donor in other enzymatic syntheses,[39] and contains an isonicotinyloxycarbonyl protecting group on its N-terminus in order to prevent self-ligation. The isonicotinyloxycarbonyl group, incorporated at the last step of the solid-phase synthesis, is stable to anhydrous hydrofluoric acid and can be removed under mild reducing conditions (zinc/acetic acid). Since subtiligase is efficient with large and hydrophobic P_1 donor substrates and inefficient with negatively charged residues or Pro at P_1', the ligation was determined to be between Tyr and Lys residues.

Scheme 5 Route to RNase A (77–124)[38]



Ribonuclease A (77–97+98–124) (14); General Procedure:[38]
Ligations were performed at 25 °C in ligase buffer (100 mM tricine, pH 8, freshly prepared and degassed by vacuum filtration through a 5-μm filter) containing the amino component (**13**; 2–5 mM) and subtiligase (ca. 5 μM). The lyophilized acyl donor ester (**12**; 3–5 equiv) was dissolved in the soln, and the mixture was incubated for 1–2 h at 25 °C. The coupling reaction was monitored by analytical RP-HPLC (MeCN/H_2O gradient with 0.1% TFA). If hydrolysis of carboxy component **12** precluded complete disappearance of **13**, an additional 3 equiv of **12** was added. Unfortunately, the literature[38] does not contain exact data on the concentrations of **12** and **13** in this special ligation procedure. Yield of **14**: 63%; MS: *m/z* calcd 5370.7, found 5368.9. The deprotection of the isonicotinyloxycarbonyl group from **14** can be performed[40] by stirring HCl-activated Zn dust (50 mg) with **14** in AcOH (2 mL) for 2 h.

4.2.2.3 Substrate Engineering

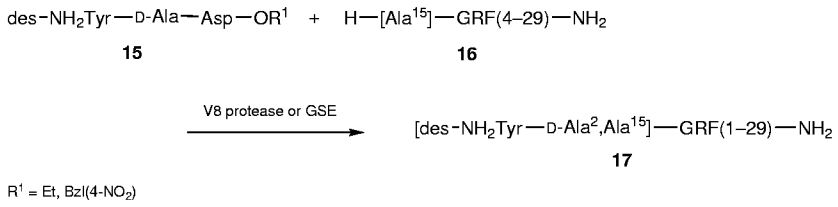
If undesired subsequent reactions are observed during kinetically controlled coupling reactions, then it is of minor importance which bond is cleaved by the enzyme. These side reactions merely show that the specificity of the enzyme for the acyl donor ester does not lie sufficiently above its specificity for the peptide product. Since structural changes in the amino acid building blocks of the reactants are ruled out, the leaving group of the acyl donor ester remains as the only variable to suppress competitive reactions. In order to circumvent this limitation, specific leaving group manipulations are carried out at the carboxy component in the kinetic approach.[41] Model experiments proved that by choosing a specific leaving group of the acyl donor, even protease-labile peptide products can be obtained in very good yields and high purity.

4.2.2.3.1 Enzymatic Semisynthesis of [des-NH_2Tyr^1,D-Ala^2,Ala^{15}]-Growth Hormone Releasing Factor (1–29)-NH_2

According to the strategy mentioned above, Bongers et al.[42,43] described a two-step enzymatic semisynthesis of the super potent analogue of growth human hormone releasing factor [des-NH_2Tyr^1,D-Ala^2,Ala^{15}]-GRF(1-29)-NH_2 (**17**) (GRF = growth hormone releasing factor) from the precursor H-[Ala^{15}]-GRF(4–29)-NH_2 (**16**) and the carboxy component des-NH_2Tyr-D-Ala-Asp-OR^1 [Bzl(4-NO_2)] **15**; R^1 = Et, catalyzed by V8 protease or Glu/Asp-specific endopeptidase (GSE) from *Bacillus licheniformis* (Scheme 6). GRF-(1–44)-NH_2, Y^{1}ADAIFTNSY^{10}RKVLGQLSAR^{20}KLLQDIMSRQ^{30}QGESNQERGA^{40}RARL-NH_2, from the hypothalamus stimulates the release of growth hormone from the pituitary gland.[44,45] The coupling of a small peptide segment containing noncoded amino acids to the N-terminus of an intermediate-length polypeptide, without side-chain protection, is described below. The product **17** possesses a four- to fivefold higher intrinsic potency than the parent hormone in vitro and has a significantly longer duration of action.

GSE is potentially less costly and more stable in comparison to V8 protease. The commercial detergent Alcalase is a rich source of GSE, from which it can be purified. GSE has the same P_1 specificity as the protease from the V8 strain of *Staphylococcus aurens* (V8 protease, endoproteinase Glu-C).[46,47]

Scheme 6 Route to The Super Potent Analogue of Growth Hormone Releasing Factor[44]



[des-NH_2Tyr^1,D-Ala^2,Ala^{15}]-GRF(1-29)-NH_2 (17):[42,43]
The tripeptide ester **15** (R^1 = OEt; 1.64 mg, 4.32 μmol) was dissolved in DMF (8.0 μL), followed by the addition of 1M triethanolamine (4.0 μL) and H_2O (28 μL). The amine component **16** · 6TFA (2.0 mg, 0.54 μmol) and CaO (75 μg, 1.3 μmol) were dissolved in DMF (4.0 μL), H_2O (1.0 μL), and 1M triethanolamine (4.0 μL), and combined with the soln of **15** (20 μL). The resulting mixture was then titrated to pH 8.2 with 1.0 μL increments of 0.5 M NaOH (5 μL total) and diluted to 37 μL. After equilibrating at 37 °C, the coupling reaction was initiated by adding aq enzyme (3.0 μL, 0.67–33.3 mg · mL^{-1}, i.e. substrate/enzyme = 20–1000 w/w) and maintained at 37 °C. The progress of the reaction was monitored by quenching 1 μL aliquots in 200 μL portions of 0.1% TFA and analyzing by RP-HPLC.

The preparative-scale GSE-catalyzed semisynthesis was performed by coupling as described above using **15** [R^1 = Bzl(4-NO_2); 15.8 mg, 32.4 μmol], **16** · 6TFA (30.0 mg, 8.1 μmol), and GSE (0.060 mg, substrate/enzyme = 500 w/w). The conversion reached 90% after 6 h, and the synthesis was stopped by adding glacial AcOH (600 μL); purification was by RP-HPLC. The product-containing fractions were pooled and lyophilized to give **17** · 5TFA; yield: 19.5 mg (61%); MS (FAB): *m/z* calcd for (M+H$^+$) 3357.9, found: 3357.1.

4.2.3 Nonconventional Approaches to Irreversible C—N Ligations

General limitations to the wide application of the reverse hydrolysis potential of proteases result from the restrictive specificity of proteases and the permanent danger of undesired proteolytic side reactions. Such hydrolytic reactions could be minimized by carrying out the synthesis in a medium of low water activity, e.g. in an organic solvent of controlled water

for references see p 663

content. Since enzyme deactivation and specificity changes are serious drawbacks, new strategies in reducing water concentration by freezing the aqueous reaction mixtures as well as by synthesis in an organic solvent-free medium containing a minimum water content have been developed. To avoid proteolytic side reactions completely, enzymatic syntheses have to be altered to a practically irreversible process.[48] Investigations along this line are oriented towards the use of proteolytically inactive zymogens of proteases as a new type of biocatalyst as well as in substrate mimetic-supported enzymatic synthesis.

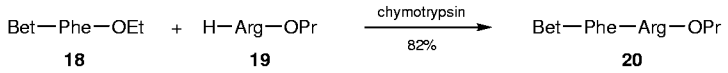
4.2.3.1 Synthesis in Frozen Aqueous Systems

The success of enzymatic synthesis is strongly affected by competitive hydrolysis of the acyl-enzyme and the newly formed peptide bond, as well as by other proteolytic side reactions in the reactants and the final peptide product. In order to suppress competitive reactions based on hydrolysis, frozen state protease catalysis opens up a new approach to enzymatic synthesis.[49] In addition, it could be established that amino acid derivatives, peptides, and even free amino acids, which are considered to be inefficient nucleophilic amino components at room temperature, provide significantly higher yields under frozen state conditions. Furthermore, N^{α}-unprotected dipeptide esters and acyl donors containing noncoded building blocks could be successfully coupled in frozen reaction mixtures.[50,51] The coupling of H-Asp-Phe-OMe with H-Ala-Leu-OH catalyzed by chymotrypsin in ice provided H-Asp-Phe-Ala-Leu-OH in a yield of 86% under semipreparative conditions. Starting from H-Tyr-OEt and H-Arg-OH, Meos et al.[52] synthesized kyotorphin in a yield of 80% in a single-step procedure in a frozen state catalyzed by chymotrypsin. This approach represents the simplest type of peptide synthesis which cannot be performed by chemical methods and does not require any protecting groups at all (cf. Section 4.2.1.2.1). The "freeze-concentration effect" was considered as a reasonable basis for explaining the yield-increasing effect of freezing.[53] However, specific properties of peptide synthesis in frozen systems suggest that there are probably other factors involved in peptide yield enhancement by freezing. Using the ^1H NMR relaxation time technique, the amount of unfrozen water has been determined in frozen samples at $-15\,^{\circ}C$ and compared to relaxation time measurements for the same system at room temperature.[54] An apparent concentration factor of 50 for the unfrozen water concentration was determined. However, simulation of these concentration relations at room temperature in papain-catalyzed peptide synthesis experiments could not confirm this effect. Therefore, freeze-concentration cannot be the only cause of the yield-increasing effect. Other physical and reaction parameters must be considered, e.g. increased proton mobility in ice, changes of dielectric behavior, reduced water activity, and imposition of a favorable orientation of substrate and enzyme.[49]

4.2.3.1.1 Chymotrypsin-Catalyzed Synthesis of Bet-Phe-Arg-OPr

In principle, the use of amino acid or peptide esters as nucleophilic components in protease-catalyzed synthesis is possible, but with a drastically decreased efficiency. However, acyl transfer to arginine or lysine alkyl esters in ice using α-chymotrypsin with regard to its strong preference for basic residues in the P_1' position enabled synthesis of a N-protected tripeptide ester in high yield (Scheme 7, see Section 4.2.1.2.2). Furthermore, it was found to be the method of choice in synthesizing new potential protease substrates (for proteases with a preference for basic residues in the P_1 position).[55] Neither enzymatic synthesis at room temperature nor synthesis in organic solvents has been shown to proceed in a successful manner.

Scheme 7 Route to Bet-Phe-Arg-OPr[55]



18 **19** **20**

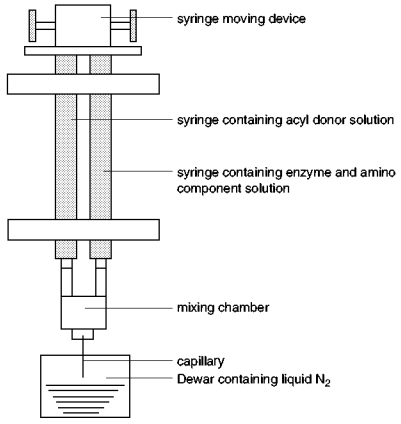
Bet-Phe-Arg-OPr (20):[55]
Bet-Phe-OEt (**18**) can be synthesized from chloracetyl-Phe-OEt by reaction with Me_3N in an autoclave. Solns of **18** (355 mg, 1.08 mmol) in H_2O (13.5 mL) and **19**·2HCl (781 mg, 2.7 mmol) in 0.4 M Na_2CO_3/$NaHCO_3$ buffer (12.2 mL, pH 10.0, containing 0.2 M NaCl) were stored in ice/NaCl. To avoid freezing, the solns were shaken from time to time. The pH value of soln **19** was adjusted in the buffer to pH 10 by deprotonating the hydrochloride with 2 M NaOH (ca. 1.3 mL, small portions, pH control). After combining the solns and thorough shaking, the mixture was divided into nine portions of 3 mL each using 5-mL polypropylene tubes. α-Chymotrypsin (30 μL) in 1 mM HCl containing 10 mM $CaCl_2$ (stock soln: 0.3 mM, 2.0 mg in 270 μL) was added consecutively to each reaction tube. After minimal shaking the tube was immediately placed into liquid N_2 for 2 min to achieve shock-freezing, and then transferred from liquid N_2 into the cryostat at −25 °C. In order to achieve rapid freezing in liquid N_2, all tubes and solns were precooled in a refrigerator (−20 °C). This was essential to avoid ester hydrolysis after enzyme addition and mixing the reaction components before the soln was put into the liquid N_2. It was necessary to keep this interval as short as possible and to work very quickly. HCl was not used to stop the reaction since it will not be removed during vacuum concentration, resulting in product hydrolysis. After a reaction time of 48 h the reaction was stopped by adding 8% TFA in H_2O (1.5 mL) to each tube. After thawing, the samples were combined and concentrated on a vacuum evaporator (temperature <60 °C). The product was purified portionwise by preparative RP-HPLC (H_2O/MeCN gradient with 0.1% TFA). The product-containing fractions were pooled and, after evaporation, lyophilized to give **20**; yield: 440 mg (82%).

4.2.3.1.2 α-Chymotrypsin-Catalyzed Synthesis of β-Phenyllactyl-L-arginine Amide Using Special Equipment

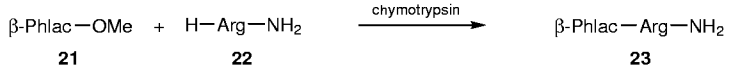
Shock freezing of the liquid reaction mixture is necessary to avoid acyl donor hydrolysis during the initiation phase of the synthesis. For this reason, most synthesis procedures are carried out on an analytical scale using reaction tubes of approximately 5 mL volume (see Sections 4.2.1.2.2 and 4.2.3.1.1). In order to circumvent this limitation, Haensler et al.[56] constructed a special apparatus which enables shock freezing of large reaction volumes. This equipment is shown schematically in Figure 1. A 10-mL and a 20-mL syringe, containing enzyme plus nucleophilic amino component and the acyl donor ester, respectively, were connected to a conical mixing chamber by hoses. At the outflow of the chamber a capillary (1 × 50 mm) was positioned, through which drops of the mixed reaction solution fell into a Dewar (2 L) containing liquid nitrogen, forming ice spheres with a diameter of approximately 3–6 mm. The spheres were collected in a plastic container adapted to the Dewar and incubated in a freezer at −18 °C. To guarantee a constant mixture ratio of both reaction solutions, injection was performed manually by constant pressure using a rack gear. In contrast to shock freezing of large reaction volumes in a vessel, the contact of enzyme and acyl donor ester is limited to the short distance from the mixing chamber to the cryogenic agent and the drops formed at the capillary outlet are frozen rapidly. For this reason, shock freezing of large volumes can be carried out without the danger of acyl donor ester hydrolysis during the starting period of the reaction. Several hundred milligrams of peptides, including those containing noncoded building blocks, were synthesized in yields of 43–95%. In principle, further scaling up is possible by simple repetition of the procedure. The procedure is demonstrated by enzymatic coupling of methyl β-phenyllactate (β-Phlac-OMe) with H-Arg-NH_2 (Scheme 8).[56]

for references see p 663

Figure 1 Schematic Diagram of Preparative Enzymatic Frozen State Synthesis Apparatus[56]



Scheme 8 Route to β-Phlac-Arg-NH_2[56]



21 **22** **23**

β-Phlac-Arg-NH_2 (23); General Procedure:[56]
The synthesis was carried out using the apparatus shown in Figure 1. Solns of α-chymotrypsin (7.5 μg mL^{-1}) and H-Arg-NH_2 (**22**; 50 mM) in H_2O were placed in the 10-mL syringe. The 20-mL syringe contained a soln of β-Phlac-OMe (**21**; 25 mM) in H_2O adjusted to pH 7.0. The pH of the soln of the enzyme/amino component was pH 9.2 after mixing with the acyl donor. Injection speed was approximately 5 mL · min^{-1}. To obtain single ice spheres, the outlet capillary was positioned 8–10 cm above the liquid N_2 surface. After 2 min of shock freezing in liquid N_2, the spheres were transferred into a freezer and incubated at −18 °C. The reactions were stopped by covering the ice spheres with 1% TFA. It was established in statistical studies that single spheres represent the composition of the whole frozen mixture. To monitor the course of the reaction, single spheres were stopped after definite reaction times and analyzed by RP-HPLC (running isocratic elution using MeCN/H_2O mixtures containing 0.1% TFA). Since acyl donor, hydrolysis, and aminolysis products contain the same chromophore, their molar extinction coefficients were assumed to be equal. After complete ester consumption, the whole frozen mixture was stopped and the product was isolated by preparative RP-HPLC (H_2O/MeCN 95:5 with 0.1% TFA). The oil obtained after removal of the eluant was dissolved in H_2O and lyophilized to give **23**; yield: 120.5 mg (75%). Anal. for $C_{15}H_{23}O_3N_5$ (321.4): calcd C 56.06, H 7.21, N 21.79; found C 55.56, H 7.35, N 20.98; MS: $m/z = 322$ ($M^+ + H$).

4.2.3.2 Synthesis in Solid–Liquid Heterogeneous Reaction Systems

Based on an extensive theoretical treatment of equilibrium positions,[57] an interesting approach of solid-to-solid conversion catalyzed by proteases has been developed. According to this, a favorable equilibrium shift toward the peptide product is given when the starting reactants are largely undissolved in the reaction medium and the product precipitates. The use of solid-phase substrate pools combines the equimolar or nearly equimolar supply of reactants with high obtainable yields, easy workup procedures and, in principle, compatibility with conventional chemical peptide synthesis standard procedures. Both the advantage of solid-phase substrate pools mainly in equilibrium-controlled synthesis[57–59] and the extension of this approach to protease-catalyzed acyl-transfer reactions[60] have been successfully demonstrated.

4.2.3.2.1 Thermolysin-Catalyzed Pilot-Scale Synthesis of Z-His-Phe-NH_2

The protected dipeptide Z-His-Phe-NH_2 is a precursor for cyclo[-His-Phe-], which is known as a catalyst for addition of cyanide to aldehydes, with high enantiomeric excess.[61] Such optically active cyanohydrins are valuable starting materials for the synthesis of a variety of chiral, bioactive compounds.[62] The easy scale-up (to mole scale) of the suspension-to-suspension conversion approach has been demonstrated by the thermolysin-catalyzed synthesis of Z-His-Phe-NH_2.[59]

Z-His-Phe-NH_2:[59]
Z-His-OH (553 g, 1.91 mol) and H-Phe-NH_2 · HCl (553 g, 1.91 mol) were placed in a 5-L pilot-scale mixer. Tris · HCl buffer (457 mL, 0.5 M, pH 7, 40 °C) and 10 M NaOH (181 mL) were added and the resulting suspension was stirred at 200 rpm. After heating to 40 °C the mixer speed was lowered to 100 rpm and the reaction was started by the addition of thermolysin (3.83 g). Samples were removed, dissolved in MeCN (0.3 mL) and 5% aq TFA (0.7 mL), and analyzed by RP-HPLC. After a reaction time of 6.5 h, the precipitate was washed with ice-cold H_2O (10 L) and dried under vacuum at 55 °C to give Z-His-Phe-NH_2; yield: 718.8 g (86%).

Cyclo[-His-Phe-]:[59]
A portion of Z-His-Phe-NH_2 (87.1 g, 0.2 mol) was deprotected with H_2 over Pd/C in MeOH (1.5 L) in the presence of AcOH (0.44 mol). After careful filtration, the solvent was removed under reduced pressure. A portion of the resultant crystalline H-His-Phe-NH_2 · 2AcOH (50 mmol) was dissolved in MeOH (250 mL), NEM (2 equiv) was added, and the mixture was refluxed. The precipitated cyclo[-His-Phe-] was washed several times with MeOH and dried under vacuum in a desiccator over P_4O_{10}.

4.2.3.3 Zymogen-Catalyzed Synthesis

In 1994, it was first established that zymogens, which are known as catalytically inactive precursors of proteases, can be used as biocatalysts for practical irreversible peptide bond formation.[63] In order to confirm true zymogen catalysis, it was essential to make it clear that the zymogen preparations were not contaminated with traces of active protease. Owing to the significantly different affinity of enzyme and zymogen to the basic pancreatic trypsin inhibitor it was possible to analyze the esterase activity of the zymogen which is an efficiency parameter in estimating their peptide bond forming potential. Since the differences in the specificity constant k_{cat}/K_M cover a range of about 5 orders of magnitude, for general use of zymogen catalysis it is essential to improve the acylation rate (k_2). The application of the carboxamido methyl ester led to an improvement of the acylation efficiency. Furthermore, guanidiated trypsinogen (which cannot be activated by trypsin) could be used for successful acyl-transfer experiments. Investigations have confirmed that zymogen catalysis can be regarded as an unexpected synthesis tool which allows practical irreversible peptide bond formation.[48] Starting with the first successful experiments, the improvement of this new strategy necessitates simultaneous development of mutants based on proteolytically inactive zymogens with higher efficiency combined with the application of acyl donor esters with an improved acylation potential.

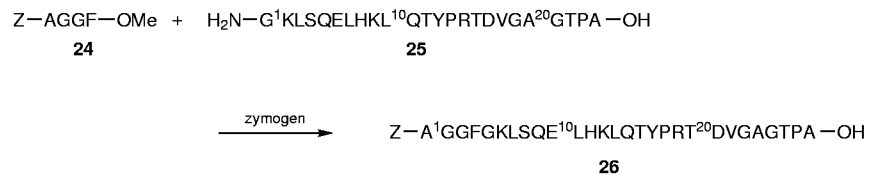
4.2.3.3.1 Zymogen-Catalyzed (4+24) Segment Synthesis

In order to evaluate the application of zymogens for irreversible fragment condensations, the coupling reaction of a synthetic tetrapeptide **24** with a recombinant 24-peptide **25** was studied under different reaction conditions using as biocatalyst both commercial chymotrypsinogen and trypsinogen (Scheme 9).[64] The coupling reaction was studied comparatively by dropping portionwise the carboxy component **24** into a solution of the amino component **25**, and also under conventional batch conditions. The first route was chosen to minimize the

for references see p 663

unwanted hydrolysis of **24** and, simultaneously, to manipulate a large excess of the amino component **25** in comparison to the temporary concentration of the acyl donor ester.

Scheme 9 Route to the 28-Peptide Z-AGGFGKLSQELHKLQTYPRTDVGAGTPA-OH[64]



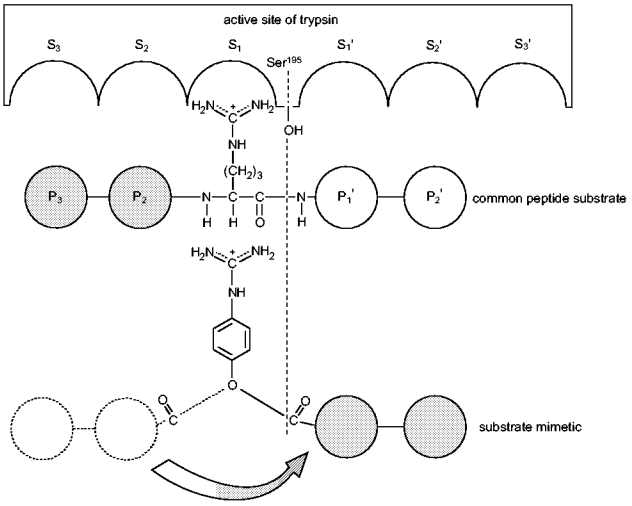
Z-AGGFGKLSQELHKLQTYPRTDVGAGTPA-OH (26):[64]
The 24-peptide **25** (5.4 mg, 0.002 mol) was dissolved in a mixture of 0.2 M sodium veronal buffer (20 μL, pH 9.5) and DMSO (16 μL), and then chymotrypsinogen (0.5 mg) in 1 mM HCl (4 μL) was added. Carboxy component **24** (4 mg, 0.008 mmol) was dissolved in DMSO (64 μL) to form a stock solution. At 10-min time intervals, the stock soln of **24** (1.6 μL) was added to the stirred soln of **25** and the biocatalyst, chymotrypsinogen, followed by sodium veronal buffer (2.4 μL, pH 9.5) at rt. The content of DMSO was kept constant at 40%. After 200 min, i.e. after addition of 2 equiv of the carboxy component, analytical HPLC analysis showed **26** (43%), **25** (52%), and 5% of a side product; the conversion of **24** was 100%. After 400 min (complete addition of 4 equiv of the carboxy component), RP-HPLC analysis showed 60% of the desired **26**, 31.5% of **25**, and 8.5% of a side product. The conversion of the acyl donor ester **24** was 100% as analyzed separately by RP-HPLC. The fragment condensation was stopped by addition of 1% TFA (0.5 mL). Using preparative RP-HPLC (H_2O/MeCN gradient with 0.1% TFA), **26** was isolated; yield: 2.5 mg (41%); which was homogeneous by analytical HPLC and high-voltage paper electrophoresis. The determined amino acid composition corresponded with the calculated values. MS (FAB): *m/z* calcd for $(M+H^+)$ 3035.4; found: 3035.1.
In the batch procedure, chymotrypsinogen (0.5 mg) in 1 mM HCl (4 μL) was added to the mixture of **25** (5.4 mg, 0.002 mmol) and **24** (4 mg, 0.008 mmol) in 0.2 M sodium veronal buffer (116 μL, pH 9.5) and DMSO (80 μL). The mixture was stirred for 90 min at rt. HPLC analysis showed 52% of the desired product **26** at 100% conversion of **24**.
Using trypsinogen as biocatalyst under similar conditions in the dropping experiment, after 400 min analytical HPLC showed 21% of **26** and 79% of **25**. The conversion of the acyl donor ester was only 80%. Preparative RP-HPLC provided **26**; yield: 1 mg (7%). The determined amino acid composition corresponded with the calculated values. In the appropriate batch experiment the analytical yield was 17% of **26** (95% conversion of **24**), whereas the application of trypsinogen inhibited by basic pancreatic trypsin inhibitor provided only 8% of **26** after 24 h at 48% conversion of the acyl donor ester **24**. Using trypsin instead of the zymogen, after approximately 20 min, more than 50% of **26** (after 1 min, 20% product formation at 100% conversion of **24**) and **25** were digested.

4.2.3.4 Substrate Mimetic-Mediated Synthesis

In 1991 the application of 4-guanidinophenyl (Gp) esters of the acyl amino acids of proline, D-alanine, or β-alanine in the specificity-determining P_1 position for irreversible trypsin-catalyzed peptide synthesis was first established.[65–67] This kind of substrate, originally developed as time-dependent irreversible inhibitors of trypsin,[68] interacts with the active site of this protease in a way that mimics the specific amino acid side chains of arginine or lysine in the P_1 position. By theoretical docking simulations, it was demonstrated that this leaving group moiety binds in place of the specific basic side-chain functionality of common acyl donors largely independent of its C-terminal amino acid.[69] The acylation of the enzyme by the nonspecific acyl residue results from the reverse recognition of these substrates, as demonstrated by Thormann et al.[70] in 1999, and illustrated in Scheme 10.[70] Consequently, if there are no arginine or lysine residues in the reactants to be coupled, peptide ligation occurs irreversibly and independent of the S_1 specificity of the enzyme. Because of this basic mechanism of catalysis for these special acyl donors, the term "substrate mimetics" was introduced[69] instead of the older description "inverse substrates" which referred to the enzymological inhibition studies mentioned above. Improvements in the synthesis of guani-

dino- and amidinophenyl esters and their application in trypsin-catalyzed synthesis[71,72] have initiated an extension of this approach to other arginine-specific proteases such as thrombin and the procaryontic cysteine protease clostripain.[69] Moreover, it was shown that special kinds of this new artificial enzyme substrate act also as acyl donors for proteases specific for hydrophobic amino acids such as α-chymotrypsin or subtilisin[73] and even Glu-specific enzymes such as V8 protease.[74] These results qualify the strategy of substrate mimetics as a powerful concept for programming the enzyme specificity in the direction of a more universal application of proteases in biocatalysis. Especially, the combination of the substrate mimetic approach with the solid-phase fragment synthesis allows irreversible C—N ligations since in classical protease-catalyzed segment condensations the irreversibility of the peptide bond forming step is normally not guaranteed.

Scheme 10 Schematic Binding Model of Substrate Mimetics to Trypsin[70]

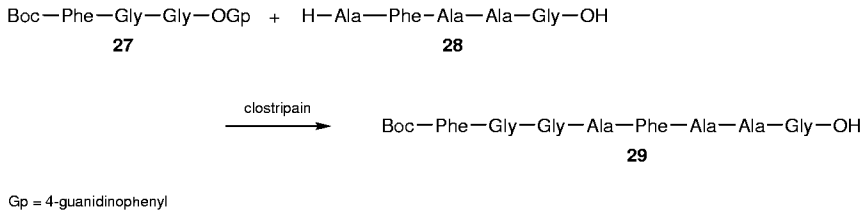


4.2.3.4.1 Clostripain-Catalyzed Synthesis of Boc-Phe-Gly-Gly-Ala-Phe-Ala-Ala-Gly-OH by (3+5) Segment Condensation

In order to evaluate the general function of 4-guanidinophenyl (Gp) esters to act as a substrate mimetic for trypsin-, thrombin-, and clostripain-catalyzed irreversible fragment condensations, the coupling of Boc-Phe-Gly-Gly-OGp (**27**) with H-Ala-Phe-Ala-Ala-Gly-OH (**28**) was studied (Scheme 11).[69] It was shown that all three enzymes were capable of catalyzing the formation of Boc-Phe-Gly-Gly-Ala-Phe-Ala-Ala-Gly-OH (**29**) in an irreversible manner, but with largely varying yields. The highest yield was obtained with the cysteine protease clostripain, which also provided yields of over 90% for other 4-guanidinophenyl esters of acylated amino acids (Boc-Xaa-OGp, Xaa = Gly, Ala, Leu, Phe, Pro). A decisive advantage of clostripain in substrate mimetic-mediated peptide syntheses results from its low S′ subsite specificity for the N-terminal amino acid residue of the amino component.[75] As a result, the product yields are largely unaffected by the individual sequence of the nucleophilic amino component, which demonstrates the excellent utility of clostripain for substrate mimetic-mediated peptide ligations. Universal application of clostripain is only limited when using arginine-containing reagents. In these cases the Arg residue should be placed in noncleavable positions (e.g., on the C-terminus of the amino component) to avoid competitive proteolytic side reactions in the starting material.

for references see p 663

Scheme 11 Route to Boc-Phe-Gly-Gly-Ala-Phe-Ala-Ala-Gly-OH[69]

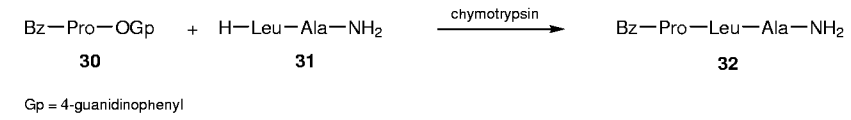


Boc-Phe-Gly-Gly-Ala-Phe-Ala-Ala-Gly-OH (29):[69]
Boc-Phe-Gly-Gly-OGp · Tos · H_2O (**27**; 122 mg, 0.174 mmol) and H-Ala-Phe-Ala-Ala-Gly-OH · TFA (**28**; 157 mg, 0.286 mmol) were dissolved in 0.05 M Hepes buffer (58 mL, pH 8.0, 0.1 M NaCl and 0.01 M $CaCl_2$). The apparent pH was readjusted to 8.0 with 6 M NaOH (small portions, pH control). The reaction was started by addition of clostripain (5.72 mg, 8 mL; stock soln: 1.3×10^{-5} M in 2.5 mM DTT and 1.0 mM $CaCl_2$). After approximately 30 min the enzymatic reaction was stopped by addition of aq 1% TFA (50 mL). This led to precipitation of the desired peptide product, which was subsequently washed (H_2O). After drying, the Boc-protected octapeptide **29** was obtained; yield: 177 mg (91%). Anal. calcd for Boc-Phe-Gly-Gly-Ala-Phe-Ala-Ala-Gly-OH · 2.5TFA · $2H_2O$ ($C_{43}H_{58.5}F_{7.5}N_8O_{18}$; 1117.98): C 46.20, H 5.27, N 10.02; found: C 45.22, H 5.80, N 10.81; MALDI-TOF: *m/z* calcd for ($M+Na^+$) 819.36; found: 819.29.

4.2.3.4.2 α-Chymotrypsin-Catalyzed Synthesis of Bz-Pro-Leu-Ala-NH_2

A closer analysis of the features of the 4-guanidinophenyl (Gp) ester moiety reveals that, in addition to the basic guanidino functionality, a partially hydrophobic character is imposed by the phenyl ring system. For that reason, this type of substrate mimetic was also investigated for its function to act as an artificial recognition site for proteases specific for hydrophobic amino acid residues, e.g. chymotrypsin.[73] Computer-simulated docking studies confirmed this assumption, showing a specific binding of the 4-guanidinophenyl ester moiety at the S_1 position of chymotrypsin stabilized both by hydrophobic contacts of the phenyl ring system and hydrogen bonds between the guanidino functionality and Ser189 and Ser190 at the bottom of this binding pocket.[73] Mediated by such binding, the distance between the carbonyl group of the scissile ester bond becomes very close to the active Ser195, indicating a possible nucleophilic attack leading to the hydrolysis of 4-guanidinophenyl esters. As predicted by these theoretical studies, both specific hydrolysis and peptide bond formation by chymotrypsin could be found experimentally.[73] Moreover, it was demonstrated that in addition to nonspecific coded amino acids in the P_1 position, also noncoded ones, like D-Ala, D-Phe, βAla, and even phenylbutyric acid, were coupled successfully by the enzyme. Investigations suggest that a universal application of this strategy for chymotrypsin-catalyzed reactions seems only to be limited when P_1-specific aromatic amino acids are present in the reactants. In these cases the results indicate that peptides containing tyrosine could not be coupled satisfactorily owing to undesired cleavages. In contrast, analogous Trp- or Phe-containing peptides were cleaved to a significantly lesser extent, resulting in distinctly higher yields. In order to demonstrate the utility of 4-guanindinophenyl esters for chymotrypsin-catalyzed reactions, an experimental procedure for a semipreparative synthesis of the model peptide Bz-Pro-Leu-Ala-NH_2 (**32**) is presented in Scheme 12.

Scheme 12 Route to Bz-Pro-Leu-Ala-NH_2[73]

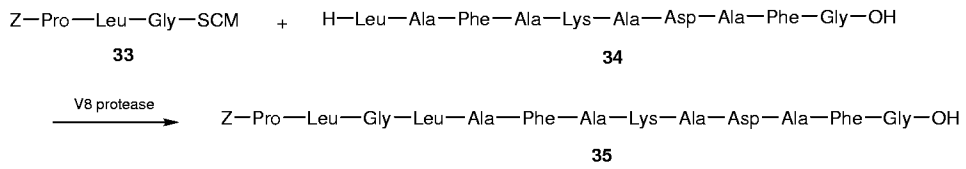


Bz-Pro-Leu-Ala-NH_2 (32):[73]
Bz-Pro-OGp · Tos · H_2O (**30**; 2.17 mg, 4.0 μmol) and H-Leu-Ala-NH_2 (**31**; 8.05 mg, 40 μmol) were dissolved in 0.2 M Hepes buffer (2 mL, pH 8.0, 0.1 M NaCl and 0.01 M $CaCl_2$) containing 10% MeOH. The apparent pH was readjusted to 8.0 with 6 M NaOH (small portions, pH control). The reaction was started by addition of α-chymotrypsin (0.83 mg, 200 μL; stock soln: 1.65×10^{-4} M in 1 mM HCl). After approximately 2 h the enzymatic reaction was stopped by addition of 8% TFA in MeOH/H_2O (1:1; 1 mL) and **32** isolated by RP-HPLC (H_2O/MeCN gradient with 0.1% TFA); yield: 1.1 mg (98% by analytical RP-HPLC); MS ($C_{21}H_{30}N_4O_4$): *m/z* calcd for ($M+H^+$) 403.23; found: 403.

4.2.3.4.3 V8 Protease-Catalyzed (3+10) Segment Condensation of Z-Pro-Leu-Gly-Leu-Ala-Phe-Ala-Lys-Ala-Asp-Ala-Phe-Gly-OH

As mentioned above, an important condition for the recognition of substrate mimetics by the protease is a high affinity of the leaving group for the appropriate S_1 subsite. Therefore, in the case of the high preference of V8 protease for Glu residues, the use of a new type of substrate mimetic bearing anionic specificity determinants is strongly required. Unfortunately, the unknown structure of this enzyme allows only the design of suitable mimetic moieties by empirical structure–function relationship studies based on the known native enzyme specificity. In this way the carboxymethyl (Cm) thioester functionality was selected as a potential suitable leaving group for imitating specific Glu residues.[74] The general utility of this ester type to serve as a new artificial recognition site for V8 protease was successfully demonstrated both by hydrolysis kinetic studies and enzymatic peptide syntheses when compared to a common ester substrate.[74] For this reason and because of its restrictive substrate specificity, V8 protease is a suitable biocatalyst for substrate mimetic-mediated condensations of segment fragments, demonstrated by the semipreparative (3+5) condensation reaction shown in Scheme 13.

Scheme 13 Route to Z-Pro-Leu-Gly-Leu-Ala-Phe-Ala-Lys-Ala-Asp-Ala-Phe-Gly-OH[74]



Z-Pro-Leu-Gly-Leu-Ala-Phe-Ala-Lys-Ala-Asp-Ala-Phe-Gly-OH (35):[74]
Z-Pro-Leu-Gly-SCm (**33**; 1.97 mg, 3.99 μmol) and H-Leu-Ala-Phe-Ala-Lys-Ala-Asp-Ala-Phe-Gly-OH · 2TFA · 2.5H_2O (**34**; 29.64 mg, 23.1 μmol) were dissolved in 0.2 M Hepes buffer (1 mL, pH 8.5) containing 5% DMSO. The apparent pH was readjusted to 8.5 with 6 M NaOH (small portions, pH control). Because of the temperature optimum of V8 protease it is important to carry out the reaction at 37 °C. After thermal equilibration the reaction was started by addition of V8 protease (0.011 mg, 33.3 μL; stock soln: 1.53×10^{-4} M in distilled H_2O). After approximately 2 h the enzymatic reaction was stopped by addition of 10% TFA in 50% MeOH/H_2O (0.5 mL) and the product **35** separated by RP-HPLC (H_2O/MeCN gradient with 0.1% TFA); yield: 2.8 mg (55% by analytical RP-HPLC, calculated at complete ester consumption based on the difference between the measured concentration of Z-Pro-Leu-Gly-OH for reactions with and without the decapeptide, respectively); MALDI-TOF ($C_{69}H_{98}N_{14}O_{18}$): *m/z* calcd for ($M+Na^+$) 1433.71; found: 1434.17.

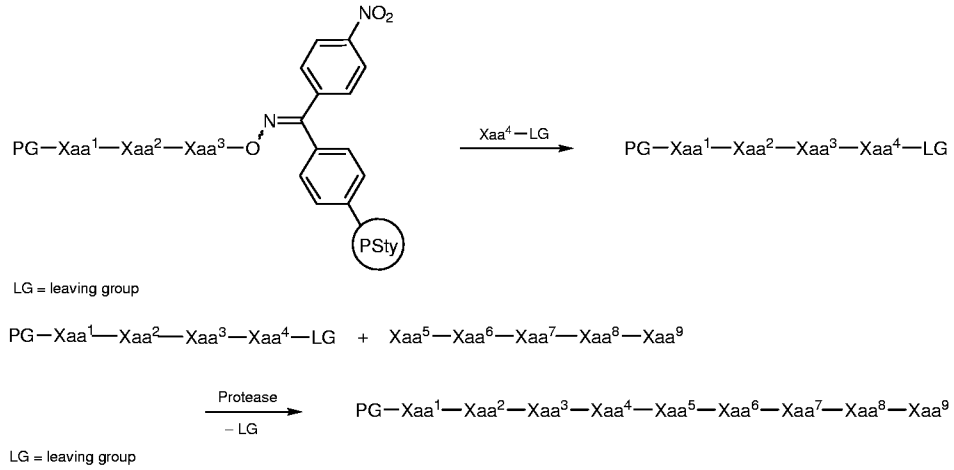
4.2.3.4.4 Synthesis of Boc-Leu-Asn-Lys-Ile-Val-Arg-Ala-Ala-Ala-Gly-OH by Chymotrpysin-Catalyzed (5+5) Fragment Condensation Using the Combined Solid-Phase Peptide Synthesis Substrate Mimetic Approach

In order to synthesize longer polypeptides and small proteins, the condensation of suitable starting fragments is an essential prerequisite. Since chemical fragment condensations suffer from the danger of enantiomerization, enzymatic ligations are promising alternatives (see

for references see p 663

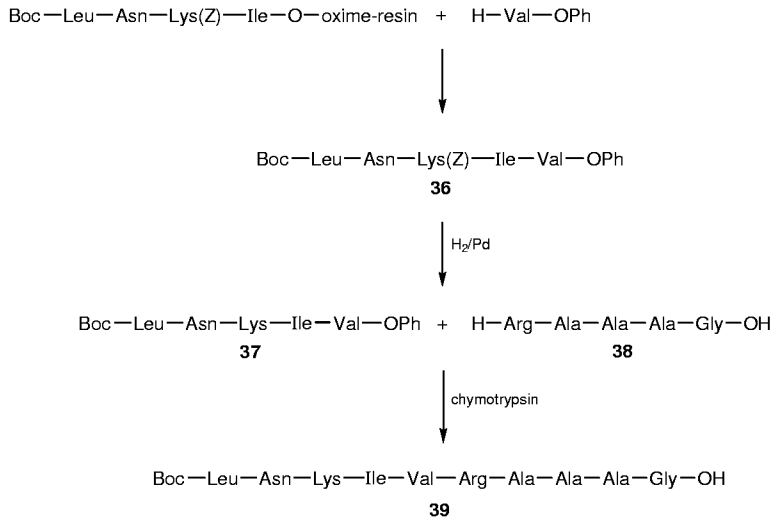
Section 4.2.2.2.2). However, an absolutely irreversible formation of the new peptide bond formed between the two fragments is presently only possible using the substrate mimetic approach. Consequently, the application of the substrate mimetic strategy for the protease-mediated condensation of peptide fragments indisputably needs to be combined with the solid-phase peptide synthesis (SPPS) approach. This enzymochemical substrate mimetic approach has been developed by Cerovsky and Bordusa,[76] and combines the solid-phase synthesis of peptide fragments according to Kaiser's oxime resin strategy[77,78] with the irreversible substrate mimetic-mediated fragment coupling (Scheme 14).

Scheme 14 General Course of the Combined Solid-Phase Peptide Synthesis Substrate Mimetic Fragment Condensation Approach: Generation of the Protected Peptide Fragment by Aminolysis Using an Amino Acid Substrate Mimetic, and Irreversible Protease-Catalyzed Fragment Condensation[76–78]



The synthesis protocol involves covalent attachment of the first N^{α}-Boc-protected amino acid to the oxime resin, blocking free hydroxy groups with acetic anhydride, and deprotection of the N^{α}-amino function of the attached amino acid, followed by successive chain elongation according to the well-known SPPS methodology.[77,78] The generation of the protected fragment in form of substrate mimetic **36** has been performed by aminolysis of the oxime ester linkage between the peptide fragment and the resin with an appropriate amino acid phenyl ester, as shown in Scheme 15. After deprotection of the blocked amino acid side-chain function by catalytic hydrogenation, the resulting N^{α}-protected peptide ester **37** is coupled with the amine component **38** under chymotrypsin catalysis.

Scheme 15 Route to Boc-Leu-Asn-Lys-Ile-Val-Arg-Ala-Ala-Ala-Gly-OH[76–78]



In principle, other substrate mimetics, e.g. trypsin- or V8 protease-catalyzed fragment condensations, could be generated according to the same principle and have been used for substrate mimetic-mediated fragment condensations.[76] There is no doubt that longer fragments can also be generated according to this general protocol, when suitable safety-catch linker/resin combinations are available followed by substrate mimetic-supported coupling. The principle strategic constraint is minimizing the risk of undesired proteolysis within the fragments being coupled by the highly specific protease used for fragment condensation. The number and distribution of amino acid residues that are related to the substrate specificity of the protease used are important determinants in the detailed design of a fragment condensation strategy. If there is a problematic constituent (e.g., Arg in the case of clostripain) in the target, it will be preferable to select only an amine fragment having the appropriate amino acid as the C-terminal amino acid. If it is not possible to place Arg at the C-terminus of the peptide fragment then other combinations of substrate mimetics and highly specific proteases that do not cleave Arg residues should be used as an alternative to this strategy. Generally, fragment condensations in frozen aqueous systems (see Section 4.2.3.1) avoid undesired proteolytic cleavages and, furthermore, suitable mutants of highly specific proteases with minimized amidase activity are useful biocatalysts, both for substrate mimetic-mediated fragment condensations and for selective introduction of marker and reporter groups at the N-terminus of peptides and proteins.

Boc-Leu-Asn-Lys-Ile-Val-Arg-Ala-Ala-Ala-Gly-OH (39):[76]
Tetrapeptide resin Boc-Leu-Asn-Lys(Z)-Ile-O-oxime-resin (1.2 g), obtained according to the standard SPPS protocol,[77,78] was shaken with H-Val-OPh (260 mg, 0.9 mmol), DIPEA (153 μL, 0.9 mmol) and AcOH (52 μL, 0.9 mmol) in THF (20 mL) for 50 h. The resin was filtered, washed with THF, and the combined filtrate was concentrated. The remaining residue was solidified by 5% citric acid, and washed extensively with 5% citric acid and H_2O. The crude dry **36** (210 mg) was recrystallized (EtOAc/petroleum ether); yield: 182 mg (55%). Anal. for $C_{46}H_{69}N_5O_{11} \cdot 0.5\ H_2O$ (905.1): calcd C 61.04, H 7.80, N 10.83; found: C 60.80, H 7.85, N 10.74; Amino acid analysis: Asp 1.05, Val 0.87, Ile 1.00, Leu 1.07, Lys 1.01; FAB-MS: $m/z = 897.2$ (MH^+).

The protected pentapeptide ester **36** (170 mg) was hydrogenated over Pd on charcoal in 80% aq THF (10 mL) containing HCl (1 equiv) for 8 h; the reaction was followed by HPLC. The hydrochloride of **37**, isolated by filtration and removal of THF, was recrystallized (Et_2O); yield: 110 mg (73%); FAB-MS: $m/z = 762.5$ (MH^+).

The chymotrypsin-catalyzed condensation of **37** (10 mM) with **38** (20 mM) was performed in 0.2 M sodium veronal buffer (pH 9) containing 40% DMSO (0.5 mL). The chymotrypsin concentration was 1.6×10^{-5} M. The reaction course was monitored by isocratic reverse-phase HPLC using a C-18 column

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(Merck, WP-300, 5 μm, 25 × 0.4 cm), and terminated by addition of 1% TFA. The condensation product **39** was isolated by gradient HPLC with 0.1% TFA/MeCN system at a flow rate of 3 mL · min^{-1} on a 25 × 1 cm Vydac C-18 column (detection was at 280 or 225 nm); yield: 56%. Amino acid analysis: Asp 1.06, Gly 0.97, Ala 2.95, Val 1.02, Ile 1.00, Leu 1.11, Lys 0.85, Arg 1.01; FAB-MS: $m/z = 1113$ (MH^+).

4.2.4 Conclusion and Outlook

It has been confirmed since the 1980s that enzymatic peptide synthesis, on a preparative scale, is a very useful synthetic approach and has been applied by development and production divisions both in pharmaceutical companies and by fine-chemical sections in industry. The synthetic biotransformations can be achieved with commercially available enzymes that are easy to handle. Due to their application in catalytic amounts, the costs of the proteases are, in most cases, insignificantly higher compared with highly sophisticated chemical coupling reagents. Owing to the stereo- and regiospecificity of proteases, their application for the formation of peptide bonds certainly offers several advantages over chemical synthesis procedures, e.g. mild reaction conditions, temporary protection of functional groups in side chains of the amino acid building blocks is not required, racemization-free course of the reaction guarantees chirally integral products, and simple scale-up after optimization of the process. Once the optimal synthesis conditions have been established, kilogram amounts of biologically active peptides can be obtained. The semisynthetic synthesis of human insulin (see Section 4.2.2.1.2) and the production of aspartame (see Section 4.2.1.1.1) on a ton scale are examples of the industrial importance of enzymatic peptide synthesis. However, due to the high degree of specificity of proteases and the protein character of these biocatalysts, enzymatic peptide synthesis does not have the versatility of chemical coupling methods and, therefore, suffers from some limitations.

Unfortunately, a universal C—N ligase with a high catalytic efficiency for all possible combinations of the proteinogenic amino acids, both as C- and N-terminal residues in fragments to be coupled, has not yet been developed. However, the ribosomal peptidyl transferase seems to be a ribozyme that lacks specificity for the amino acid side-chain functions, only catalyzing the acyl-transfer reaction of the selected aminoacyl-tRNAs fixed in the P and A sites of the ribosome. Even in this case, it would be possible to separate ribozyme activity from the ribosome, or to isolate an in vitro selected ribozyme that can catalyze the same type of peptide bond formation like the peptidyl transferase; such a biocatalyst would be of limited value for practical use. Therefore, the reverse hydrolysis potential of proteases is the only alternative for enzymatic peptide synthesis as shown in this section. The ingenious combination of chemical and enzymatic strategies has been demonstrated in the course of a new synthesis of RNase A using the subtilisin mutant subtiligase (see Section 4.2.2.2.1). Despite this impressive example of five subtiligase-catalyzed fragment condensations resulting in the synthesis of a small protein, this procedure could not solve the problem of reversibility in protease-catalyzed peptide synthesis. The new irreversible C—N ligation strategy based on the substrate mimetic concept (see Section 4.2.3.4) should promote progress in the practical synthesis of polypeptides and small proteins. Further systematic investigations along this line may be directed to enzymatic native ligations using protease mutants with minimized amidase activity, allowing the coupling of recombinant polypeptide thioesters as substrate mimetics with other chemically synthesized fragments or recombinant products.

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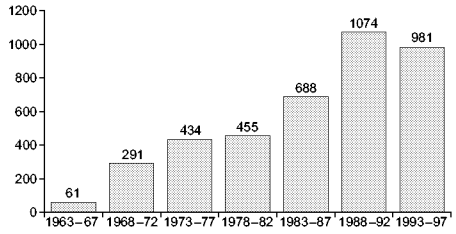
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4.3 Synthesis of Peptides on Solid Supports

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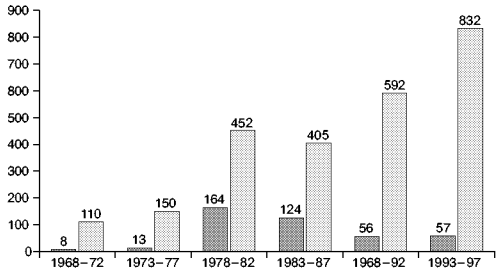
In July 1963, an article by R. B. Merrifield entitled "Solid Phase Peptide Synthesis. I. The Synthesis of a Tetrapeptide" appeared in *The Journal of the American Chemical Society.*[1] It is very difficult today to try to assess the enormous impact on the further development of chemistry and biology of this pioneering article, as well as related work from the same laboratory that soon started appearing regularly. To some extent the Nobel Prize awarded to Merrifield in 1984 was the first acknowledgement of this impact. The seminal 1963 Merrifield paper has been cited more than 4000 times in the scientific literature (Figure 1).

Figure 1 Scientific Citations of the Article "Solid Phase Peptide Peptide Synthesis. I. The Synthesis of a Tetrapeptide" Published by Merrifield in 1963[1]



Another indication of the scientific and technological impact of the discovery of peptide synthesis on solid supports comes from the fact that the term "solid-phase peptide synthesis" (SPPS) has been mentioned more than 2000 times in the scientific literature during the last 30 years. As can be seen in Figure 2, over this period in the primary literature there has been growth in the use of solid-phase methodologies and the impact on the patent literature has also been non-negligible. This latter aspect is the result of synthesis of peptides on solid supports being used not only at the laboratory scale but also, in several instances, for the industrial production of peptide molecules at the kilogram scale.

Figure 2 Frequency of the Phrase "Solid-Phase Peptide Synthesis" Appearing in *Chemical Abstracts.* Light Gray Bars Correspond to Articles in Scientific Journals and Dark Gray Bars Correspond to Patents



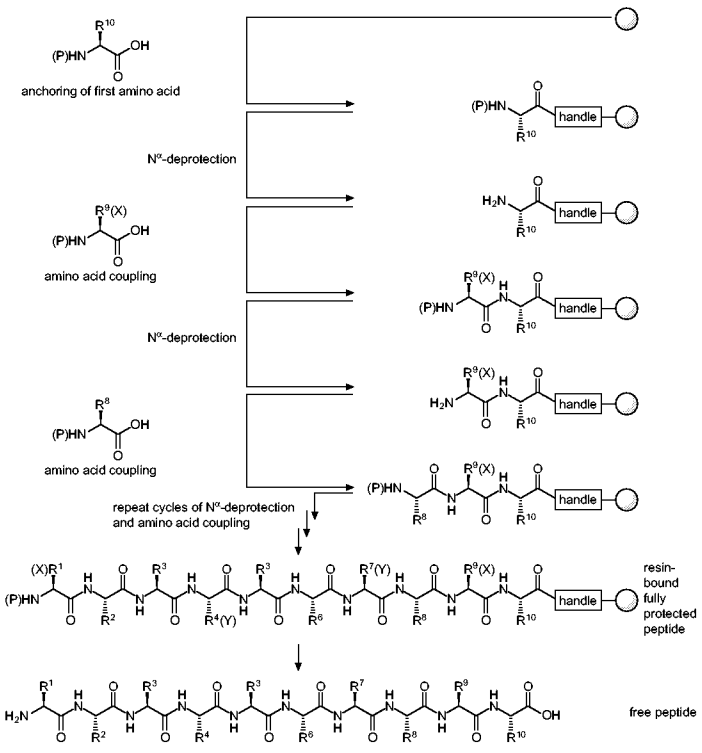
The development of the synthesis of peptides on solid supports has also had a strong impact on other areas of organic synthesis. From the very beginning, some research groups tried, with some success, to export the idea from peptide synthesis to, for instance, the monoalkylation and monoacylation of carboxylic acids,[2,3] the synthesis of aldehydes and ketones,[4] or to the use of resin-bound dienes or dienophiles to "trap" reactive intermediates.[5] This pioneering work has been fundamental to the much more recent explosion of combinatorial chemistry on solid supports.[6,7] Synthesis on solid supports has also found application in the synthesis of oligonucleotides,[8] oligosaccharides,[9] peptide-DNA hybrids,[10] and peptide nucleic acids (PNAs).[11]

for references see p 670

The general concept of the synthesis of peptides on solid supports is represented in Figure 3.[12] The first (C-terminus) amino acid of the peptide to be synthesized is protected at its N^{α}-terminus (and at its side chain if necessary) and it is then anchored covalently to the resin. Peptide synthesis is then continued by repetitive cycles of deprotection of the N^{α}-protecting group on the peptide resin and of coupling of the next protected amino acid. When the entire desired peptide chain has been assembled, the peptide is cleaved from the support and all the amino acid protecting groups are removed. The crude free peptide is subsequently purified and characterized. This method, which was a completely new concept, revolutionized some areas of organic synthesis and has several advantages but also some limitations.

The first advantage is that throughout the synthesis, since the peptide derivatives remain bound to the insoluble support, excess reagents and byproducts can be removed simply by washing the peptide resin with suitable solvents. Secondly, and this in our opinion is the key to the success of the method, this facile purification by filtration allows large excesses of the required reagents to be used for all synthetic operations. In particular, large excesses of the soluble N-protected amino acids can be used at each synthesis cycle to drive the coupling reactions to completion. Thus, very high coupling yields are achieved with little, if any, racemization. Finally, the simplicity and repetitive character of the mechanical laboratory operations used in the synthesis of peptides on solid supports has led to the development of automatic peptide synthesizers. The appearance of these instruments has contributed greatly to the popularization of the method. The good performance of modern automated synthesizers is the result of more than 30 years of effort in numerous laboratories around the world and has led to a rigorous optimization of all aspects of the chemistry, from the solid support itself to the protecting groups, coupling reagents, and cleavage conditions.

Figure 3 Synthesis of Peptides on Solid Supports Following a Linear Strategy.[12]



The major drawbacks of the synthesis of peptides on insoluble polymeric supports, when the traditional linear approach is applied, derive from the difficulty of purifying the various synthetic intermediates. First of all, this limitation has an impact on the synthesis of very large compounds.[13] It is easy to calculate that the maximum overall yield that can be achieved in the synthesis of a 20-residue peptide, if the mean yield per cycle is 90%, is only 12% ($0.90^{20} \times 100$). A mean yield per cycle of 99% affords a modest overall 37% yield in the synthesis of a peptide of 100 amino acids (see Table 1).

Table 1 Overall Synthetic Yield as a Function of the Peptide Length and the Mean Yield per Synthetic Cycle for Linear Solid-Phase Peptide Synthesis

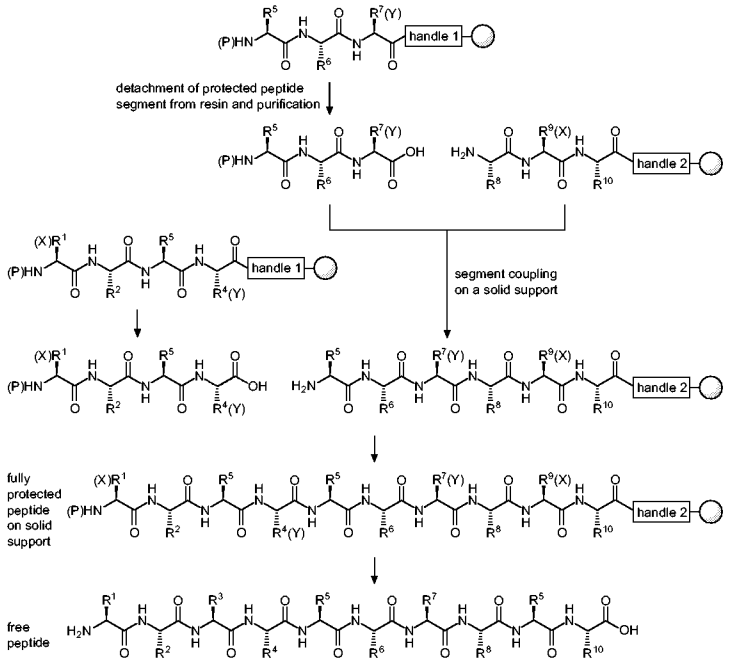
Peptide Length	Overall Synthetic Yield (%) for Selected Mean Yields per Synthetic Cycle				
	90%	95%	99%	99.5%	99.8%
20 residues	12	36	82	91	96
50 residues	<1	8	61	78	91
100 residues	<1	<1	37	61	82
200 residues	<1	<1	13	37	67
300 residues	<1	<1	5	22	55

Nonquantitative yields in amino acid coupling steps and in N^{α}-deprotection reactions lead to the formation of deletion peptides and truncated sequences, all of which may be present in the crude mixture after cleavage from the solid support. The major problem associated with linear solid-phase peptide synthesis is that most of these undesired peptides very closely resemble the target molecule. For some large peptides, their isolation in a pure state at the end of the synthesis can be unobtainable even when the more potent modern chromatographic methods are used. Capping the unreacted growing peptide chain with the appropriate chemical reagents, or tagging the target molecule with "purification aids", are partial solutions to this problem that will be discussed in detail in other parts of this volume.

A more ambitious approach to try to circumvent the major problems associated with linear peptide synthesis on solid supports consists of shifting from a linear to a convergent strategy relying on the union of peptide segments. Several successful syntheses of large peptides have been carried out by solid-phase fragment condensation of protected peptide segments that, themselves, have also been synthesized on appropriate solid supports and conveniently purified and characterized as synthetic intermediates. This strategy has been termed convergent solid-phase peptide synthesis[14,15] and is represented diagrammatically in Figure 4.

for references see p 670

Figure 4 Convergent Solid-Phase Peptide Synthesis.[14,15]



Shifting from a linear to a convergent solid-phase peptide synthesis offers several advantages. Firstly, if fragment coupling yields are reasonable, the overall yields of the synthesis can be higher than when linear approaches are used. Secondly, purification and characterization of the protected segments provides a series of checks on the nature of the material being synthesized. Finally, and most importantly, the undesired peptides that accompany the target molecule at the end of the synthesis are not so closely related to it as in linear approaches, making possible the isolation of complex and large molecules in a homogeneous state.[16–18]

Although many protected peptide segments can be coupled successfully on solid supports, problems are sometimes encountered in achieving high yields, even if excesses of a protected segment are used. This has led to the development of approaches in which protected peptide segments are synthesized on a solid support but are coupled in solution.[19–21]

Another source of problems in convergent solid-phase peptide synthesis is the often poor solubility of protected peptide segments. Ligation methods, either in solution[22,23] or on solid supports,[24] provide a good alternative for such situations and open up a new route for the synthesis of large or complex peptide compounds.

4.3.1 Solid Supports

Insolubility is, of course, the most fundamental property of a polymer support for it to be useful in peptide synthesis, but it is not the only relevant one. Particle size and shape as well as mechanical stability are important in order to permit easy manipulation and rapid filtration from liquids. Mechanical stability is especially critical when the solid support is to be used in a continuous-flow peptide synthesizer.

From a chemical point of view the support must be chemically inert to all the reagents and solvents used for peptide synthesis but, at the same time, must be reactive enough to allow the covalent attachment of the first amino acid of the sequence to be synthesized. However, the most important consideration for a successful solid-phase synthesis is, probably, the extent of solvation and swelling of the peptide resin in the reaction medium.[12,25] In general, maintenance of a highly swollen state of the peptide resin throughout the synthesis is a requirement for rapid and complete chemical reactions.

Cross-linked poly(styrene-divinylbenzene) is still the most popular solid support used in peptide synthesis. It is composed of beads, of ca. 50-μm diameter when dry, of a synthetic resin prepared by copolymerization of styrene with a small (typically 1%) amount of divinylbenzene. The polymeric support used by Merrifield in his pioneering studies[1] had a higher degree of cross-linking (2% divinylbenzene) in order to ensure good mechanical stability. However, lower degrees of cross-linking enhance the swelling properties, and 1% divinylbenzene represents a good compromise.

The influence of cross-linking on the self-diffusion of solvents and small molecules in cross-linked poly(styrene-divinylbenzene) was studied by Ford et al.[26] using pulsed-field-gradient spin-echo (PGSE) NMR, who observed lower self-diffusion constants with decreasing swelling ratios. Later, Wilson et al.[27] reported on the use of fluorescence quenching to study intra-resin diffusion in polystyrene cross-linked with various amounts of oligo(ethylene glycol).

As our laboratory reported in 1984, magnetic relaxation ^{13}C NMR measurements provide a clear indication of the high amount of flexibility of a growing peptide chain attached to a polymer support when the choice of the polymer, the degree of functionalization, and the solvent are appropriate.[28] Magnetic relaxation measurements are also very useful for comparing the mobility of a growing peptide chain attached to different polymers, as we showed using gel-phase ^{19}F NMR to study the behavior of Boc-5-fluoro-L-tryptophan attached to five different solid supports.[29]

The good swelling properties of a peptide resin not only determines its diffusion properties, and therefore the kinetics of a solid-phase reaction,[30,31] but also are a requirement for ensuring a uniform distribution of the growing peptide chains throughout the polymeric support. This uniform distribution can actually be achieved, at least in favorable cases, as was shown by an elegant experiment performed by Sarin et al.[25] where Boc-[^{3}H]valine was incorporated into a peptide synthesis and autoradiographs of cross sections of peptide resin beads showed a completely uniform pattern.

The swelling properties of the peptide resin are influenced by each of the components of the system: the peptide, the resin, and the solvent. Therefore, during the course of peptide synthesis, the solvation changes from being purely a property of the polymer that makes up the resin to being one strongly influenced by the protected peptide.[25,30] Dichloromethane, for instance, is a better solvating solvent for polystyrene than dimethylformamide, but dimethylformamide is normally better than dichloromethane for medium- and high-loaded peptide resins. To some extent, peptide resins behave as microscopic composites so that binary mixtures of solvents, such as dichloromethane plus trifluoroethanol, often provide optimum swelling properties.[30]

In 1971, Sheppard proposed that peptide synthesis would proceed more efficiently if the polymeric support was designed to have solvation properties similar to those of the peptide product.[31] This line of reasoning led to the development of polyamide resins.[32,33] The most successful of this type of support is based on cross-linked poly(dimethylacrylamide) (Pepsyn). Polyamide resins swell up to 10 times their dry volume in dimethylformamide and even more in water. On the other hand, they swell much less in dichloromethane. The few comparative studies that have been carried out between polystyrene and polyamide resins indicate that both give very similar results in routine solid-phase peptide synthesis.

for references see p 670

The usual cross-linked poly(styrene-divinylbenzene) resins are not suitable for continuous-flow peptide synthesizers in which the various solvents and reagents are pumped through a column of the functionalized resin. The pores between the polymer beads necessary for the flow are closed rapidly by the swollen resin, which leads to high pressures and the collapse of the gel phase.[6] A solution to the problem is to embed the resin in which peptide synthesis takes place in a rigid but highly permeable matrix. The first flow-stable synthesis resin was obtained by polymerization of a soft polyamide inside a solid matrix of supporting kieselguhr (Pepsyn K).[34] The second consisted of polyamide embedded within a highly cross-linked polystyrene[35] and is sold as Polyhipe. Other polyacrylamide resins developed for use in both batchwise and continous-flow syntheses include the polyacrylic support,[36] Expansin, and poly{*N*-[2-(4-hydroxyphenyl)ethyl]acrylamide}.[37]

The past few years have witnessed a great proliferation of work on the use of poly(ethylene glycol)-grafted polystyrene as a solid support in peptide synthesis. Such resins can be prepared either by grafting poly(ethylene glycol) (PEG) chains on to standard amino-functionalized polystyrene beads[38] or by anionic copolymerization of ethylene oxide and polystyrene.[39] PEG-grafted polystyrene resins have the mechanical stability necessary for synthesis under continuous-flow conditions. Nowadays there is no doubt that the performance of this new family of resins is at least equal and, in some cases, clearly better than that of the standard cross-linked poly(styrene-divinylbenzene). However, it is still too soon to say whether or not they will completely supersede them. PEG is also at the basis of other recently developed solid supports such as PEGA[40,41] and CLEAR.[42] PEGA is a copolymer of bis(acrylamido)poly(ethylene glycol), mono(acrylamido)poly(ethylene glycol), and *N,N*-dimethylacrylamide and has been introduced as a hydrophilic, biocompatible, and flexible support in peptide synthesis and is finding widespread application not only in standard peptide synthesis but also in combinatorial chemistry[41] and in solid-phase chemical ligation.[24].

The above list of solid supports is by no means exhaustive. Many others have been proposed and evaluated for use in peptide synthesis, including polypropylene membranes,[43] cotton,[44] controlled-pore glass beads,[45] polyethylene sheets,[46] and others.

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4.3.1.1 Resins

W. Rapp

Since the introduction of solid-phase peptide synthesis (SPPS) on lightly crosslinked polystyrene (PSty) by Merrifield in 1962,[1–4] this technique has been optimized in all its aspects and has subsequently found widespread application in many other fields of organic chemistry, e.g. for the synthesis of oligonucleotides,[5–7] for catalysis,[8–16] for the immobilization of enzymes in biotechnology,[17–28] and for the preparation of polymeric reagents.[29–37] A new but very rapidly growing field of application of this revolutionary method is the use of solid-phase techniques in combinatorial chemistry for the synthesis of libraries of compounds whose constituents differ widely in their chemical nature.[38–40]

The use of insoluble polymeric carriers has greatly simplified the synthesis of peptides: purification of the growing peptide chain in the repetitive steps is achieved by filtration procedures that simply remove all soluble reagents and byproducts from the reaction medium, whilst the covalently resin-linked macromolecule is retained on the insoluble polymeric support. In all solid-phase reactions of this type, the polymeric support represents the medium on or in which the chemical reaction takes place. Correspondingly, this medium is represented by the total amount of insoluble polymer present, which in the case of polymeric beads is divided into small, individual reaction compartments. Resin parameters such as the degree of crosslinking, the polarity of the resin, its swelling properties, mass-transport, phase transitions, bead size, and the particle size distribution therefore have to be taken into serious consideration.

4.3.1.1.1 Resin Types for Peptide Synthesis

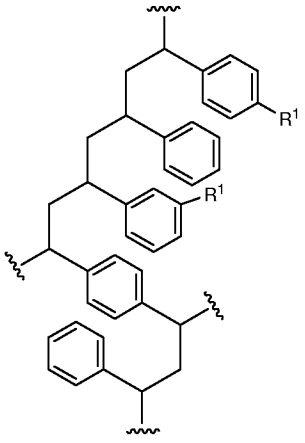
Besides the classical polymer introduced by Merrifield (1%-crosslinked chloromethylated polystyrene),[1–4] a broad variety of polymeric supports is available for SPPS and some of the most popular resins[1–4,41–71] are summarized in Table 1. The chemical structures of some selected resins are presented in Figure 1 and electron micrographs of several examples are displayed in Figure 2. In addition to the solid supports listed in Table 1, there are several other carriers used in peptide synthesis such as the gel-type and macroporous poly(methacrylates),[72–74] coated surfaces like polystyrene films on polyethylene (PEt) sheets,[75] polystyrene-coated polyethylene or polytetrafluoroethylene,[76] and modified glass surfaces.[77] (For recent reviews on polymeric carriers see refs[78–85].)

Table 1 Polymer Supports for Peptide Chemistry and Organic Synthesis

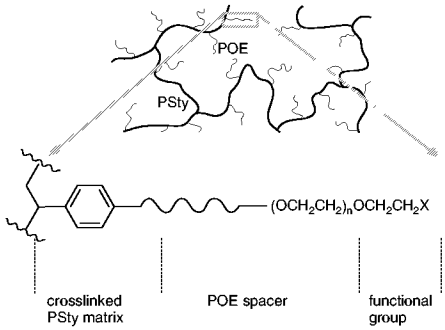
Entry	Resin	Composition	Resin Type	Ref
1	PSty	PSty 1–2% crosslinked by DVB	gel-type support, beaded shape (Figures 1a and 2a)	[1–4]
2	Pepsyn	polyacrylamide	gel-type support, beaded shape (Figure 1c)	[41–43]
3	TentaGel	PSty–PEG graft copolymer	gel-type support, beaded shape (Figures 1b and 2b)	[44–50]
4	NovaGel, Champion	PSty–PEG graft copolymer	gel-type support, beaded shape (Figures 1f and g)	[51]
5	CLEAR	crosslinked ethoxylate acrylate	gel-type support, beaded shape	[52,53]
6	PEGA	acrylamide–PEG copolymer	gel-type support, beaded shape (Figure 1d)	[54,55]
7	SPOC[a] resin	crosslinked PEG	gel-type support, beaded shape (Figure 1e)	[56]
8	ArgoGel	PSty–PEG graft copolymer	gel-type support, beaded shape (Figure 1h)	[57]
9	poly(4-acryloylmorpholine)		gel-type support, beaded shape	[58]
10	PEG–PS	PSty–PEG graft copolymer	gel-type support, beaded shape	[59,60]
11	Pepsyn K	polyacrylamide gel in kieselguhr	composite resin, irregular shape (Figures 2e and f)	[61]
12	PolyHipe	polyacrylamide gel in macroporous PSty	composite resin, beaded shape (Figures 2c and d)	[62,63]
13	PEt pins	grafted PEt	grafted surface, pin-like shape	[64,65]
14	paper	modified cellulose	hydrophilic sheets	[66–68]
15	cotton	cellulose-based natural fibers	sheets or strips	[69]
16	membranes	polypropylene, PTFE		[70,71]
17	ArgoPore	macroporous PSty		[57]

[a] SPOC = superpermeable organic combinatorial chemistry.

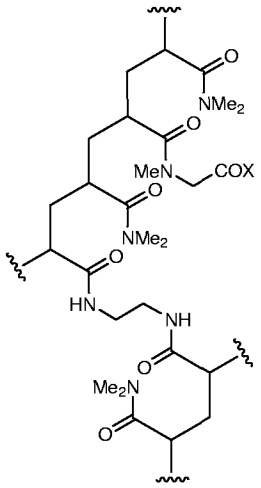
Figure 1 Chemical Structures of Selected Polymeric Supports



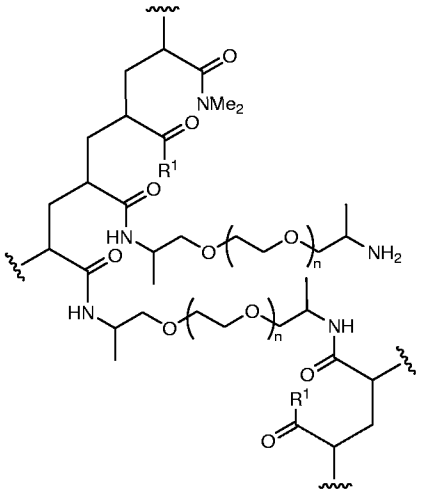
a: PSty crosslinked by DVB (1–2%)



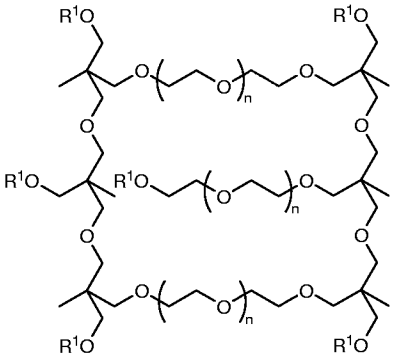
b: TentaGel (PS-POE; PSty-PEG graft copolymer)



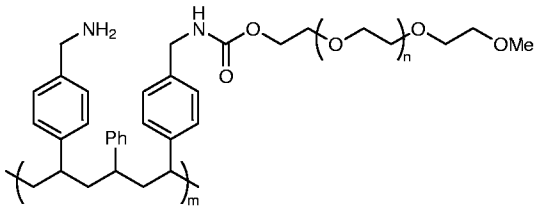
c: Pepsyn (polyacrylamide)



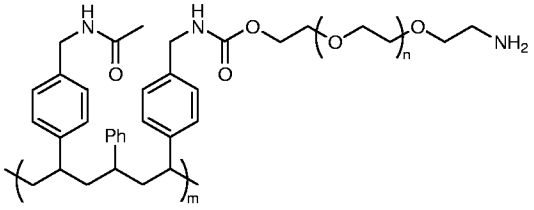
d: PEGA



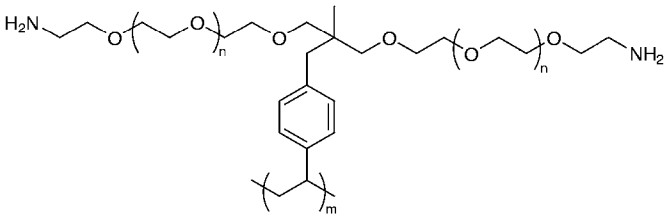
e: SPOC (crosslinked PEG)



f: NovaGel (PSty-PEG graft copolymer)

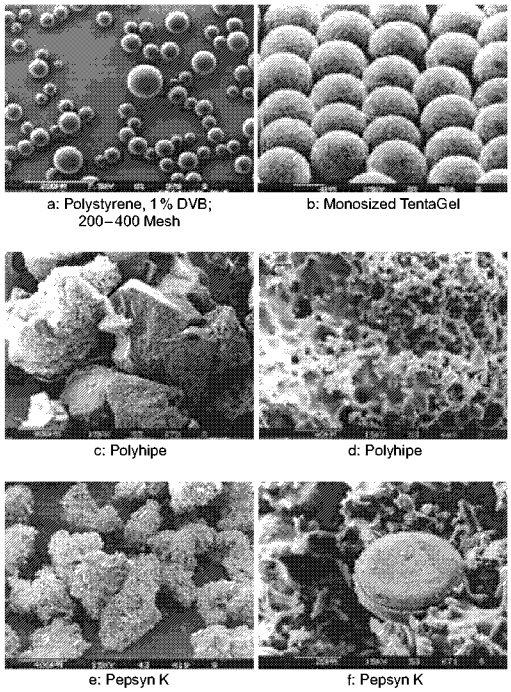


g: Champion (PSty-PEG graft copolymer)



h: ArgoGel (PSty-PEG graft copolymer)

Figure 2 Electron Micrographs of Selected Polymeric Supports



a: Polystyrene, 1% DVB; 200–400 Mesh

b: Monosized TentaGel

c: Polyhipe

d: Polyhipe

e: Pepsyn K

f: Pepsyn K

Most of the carriers used in organic synthesis are lightly crosslinked gel-type resins. In contrast to macroreticular (macroporous) resins which are characterized by a permanent porosity, the gelatinous resins have to swell in appropriate solvents in order to build up the polymeric network and to make the reactive sites located on the polymeric strands accessible to the reactants. Good swelling properties are therefore essential for these gel-type resins. The functional groups are introduced either by copolymerization with functionalized monomers or by *a posteriori* functionalization of the polymeric component. Thus the reactive sites are statistically distributed on the polymer chains and more than 99% are positioned inside the polymeric beads and not on the surface.

4.3.1.1.2 Effect of Chemical Composition on Resin Swelling Properties

Depending on the nature of the polymer, the resins exhibit distinct swelling properties in different solvents. The swelling behavior of various resins in solvents commonly used in peptide synthesis is summarized in Table 2. By comparing the single resins it is evident that 2%-crosslinked polystyrene has a significantly lower swelling capacity than the 1%-crosslinked resin. Similarly, the effect of resin modifications on swelling are significant; this is well

for references see p 682

illustrated by the aminomethylated resin (Table 2, entry 4) when modified with the Rink amide handle (Table 2, entry 5). This functionalization with a relatively small linker (M_r 511.6) has a remarkable impact on the swelling and capacity of the resin: the modified resin now consists of 38.5% of the attached linker and only 61.5% of polymer matrix. The capacity is changed not because of loss of functionalities but because of added mass. The capacity of the modified resin is calculated as follows:

$$\frac{\text{capacity of starting polymer (mmol} \cdot \text{g}^{-1})}{[\text{capacity of starting polymer (mmol} \cdot \text{g}^{-1}) \times M_r \text{ of attached molecule (g} \cdot \text{mmol}^{-1}) + 1]}$$

Table 2 Swelling Volumes of Gel-Type Resins in Various Solvents

Entry	Resin	Capacity[a] (mmol · g^{-1})	Swelling Volume[b] (mL · g^{-1})								
			H_2O	MeOH	EtOH	CH_2Cl_2	Toluene	DMF	MeCN	THF	Dioxane
1	PSty, 1% DVB			1.6	1.68	8.3	8.5	5.6	3.2	8.8	7.8
2	PSty, 2% DVB		1.7	1.6	1.7	6	5.2	3.2	1.9	5.6	5.4
3	PSty aminoethylated, 1% DVB	1.05		2.0	2.0	12.4	10.0	7.8	2.9	8.2	10.5
4	PSty aminomethylated, 1% DVB	1.24	1.8	2.0	2.0	9.2	8.4	5.9	2.0	8.0	8.0
5	PSty aminomethylated, 1% DVB, with Rink handle	0.75		1.9	1.9	7.8	5.1	6.4	2.2	6.6	7.4
6	TentaGel S resin	0.3	3.6	3.6	2.9	6.3	4.8	4.7	4.2	5.0	5.4
7	PEGA resin, amino form	0.4	14.25	12.62	9.90	14.73	7.86	10.69	7.92	8.14	12.65
8	CLEAR resin, amino form	0.35	6.0	6.2	5.0	8.0	4.4	7.2	5.2	6.4	7.0

[a] Defined as the amount of linking groups (mmol) per gram of dry polymer.
[b] Defined as the swollen volume of 1 g dry resin in a given solvent expressed as the total volume of the solvent associated with the swollen polymer gel.

In contrast to the polystyrene resins, all of the polar poly(ethylene glycol) modified resins swell in water and in alcohol. In comparison to the other resins only TentaGel shows an almost constant swelling which is independent of the solvent utilized. Due to the amphiphilic nature of poly(ethylene glycol) (PEG), the PEG-grafted copolymers swell in all solvents capable of dissolving PEG itself; conversely, their swelling is negligible in solvents which do not dissolve PEG, e.g. aliphatic hydrocarbons and diethyl ether. This broad range of good swelling solvents allows the use of the PEG-based resins in almost every solvent system. In fact, after reaction, e.g. in dichloromethane, a solvent gradient from dichloromethane to aqueous systems (via THF) can be used and the system can be reversed via a gradient from the hydrophilic environment to pure organic solvent again, e.g. from water (via THF or EtOH) to toluene.

Measurement of Resin Swelling Volume; General Procedure:
The resin was first washed carefully and consecutively with toluene, CH_2Cl_2, DMF, and MeOH (5 times each solvent) and dried, in order to remove soluble byproducts and impurities. To the dry resin (1 g) in a 10- to 15-mL graduated cylinder was added solvent (12–18 mL), and the resin was mixed thoroughly with the solvent. The cylinder was sealed, and after 2 h the resin was again stirred to remove all remaining air bubbles and to form a homogenous suspension of the resin in the solvent. The cylinder was sealed again and after 22 h of settlement the resin volume was measured (see Table 2).

4.3.1.1.3 Effect of Crosslinkers on Resin Swelling Properties

Not only does the nature of the polymer affect the swelling and thus the reaction volume (e.g., polystyrene is an example of a very nonpolar and hydrophobic resin whilst PEG resins are polar, hydrophilic materials), but also the degree of crosslinking and the nature of the crosslinker have similar effects. Crosslinkers are bifunctional monomers which crosslink the polymer chains during the polymerization process. Crosslinking of polymers is necessary to make the resins insoluble in the solvents generally used in peptide synthesis. Generally, polystyrene resins are crosslinked by divinylbenzene (DVB), as shown in Figure 1a. However, even homobifunctional polymers (macromers) can act as crosslinking agents, as illustrated in Figure 1d for acrylamide–PEG (PEGA) copolymer (Table 1, entry 6). Calculations of theoretically achievable volumes have been performed for polystyrene with varying degrees of crosslinking under the assumption that the polymer chains are fully stretched and the crosslinking sections are equally distributed throughout the polymer matrix (Table 3).[86] In practice, due to entropy effects, these theoretical swelling volumes are not achieved because even in a swollen state the polymer chains are still in a random-coil form.

Table 3 Theoretical Swelling Volumes for DVB-Crosslinked Polystyrenes

Degree of Crosslinking (%)	Distance Between Crosslinked Sections (Å)	Maximum Volume ($mL \cdot g^{-1}$)
2	65	49
1	130	196
0.5	260	784
0.25	520	3136

In the early stages of solid-phase peptide chemistry, 2%-crosslinked polystyrene supports were used almost exclusively. A lower degree of crosslinking such as 0.5% was found to render the resin too fragile and the swollen volume too high.[87] Presently, 1%-crosslinked polystyrene, which seems to be optimal in terms of mechanical stability and swelling behavior, is the most commonly used resin. Good swelling properties are required to facilitate access of the reactants to the functional groups, and thus for good reaction kinetics. Conversely, too high a degree of swelling is not desirable because of the resulting high reaction volume and thus low concentration of reactive sites; extremely high molar excesses of reagents are then required in order to achieve proper concentrations for the reaction to occur with suitable kinetics. In the latter case, even for the washing steps, a large amount of solvent is necessary to displace quantitatively the excess reagents from the resin. Although these types of resin are of restricted use in synthesis, they have an advantage if enzymes or proteins are involved since the latter form globular structures and have molecular masses of up to several hundred kilodaltons (kDa). In this case high swelling is both necessary and desirable in order to make the functionalities located within the polymer bead accessible to the biomolecules, and to achieve good diffusion kinetics for the biomolecules.[88,89] The degree of crosslinking is therefore a compromise between mechanical stability, swelling, and kinetic behavior.

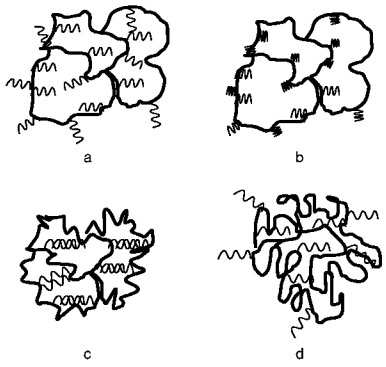
4.3.1.1.4 Kinetics of Diffusion

In the swollen state the gel-type supports can best be described as "polymeric solutions" and the reactions take place on solubilized polymer chains within each bead, which act as small, individual reaction vessels. Depending on the degree of crosslinking and the degree of swelling, the polymer chains are more or less flexible, a fact that affects the reaction rates. The mass transport of reactants to the reactive sites inside the polymer beads is driven by

for references see p 682

diffusion. Since the "polymeric gel phase" can be described as a second phase, the microviscosity of a polymer system influences decisively the mass transport as well as adsorption and partition coefficients between the surrounding liquid and the polymer phase. A partition coefficient >1 indicates a higher concentration of the reactants inside the beads than that of the surrounding liquid whereas a partition coefficient <1 indicates a lower concentration. The partition coefficient is influenced by the polarity of the resin and of the reactant and by the solubility of the reactant in the polymeric gel phase.[50,90]

Figure 3 Simplified Swelling Simulations for a Gel-Type Grafted Polymer



Four extreme situations of swelling that occur during chemical synthesis are illustrated in Figure 3 for a gel-type polymer. In the ideal case the resin is fully swollen, the polymer chains extended, and the growing peptide solvated (Figure 3a). Reactive sites at the N-terminus of the growing peptide chain are readily accessible to reactants. Figure 3b depicts a situation where the polymer is again fully swollen, but the peptide is not solvated. The reactive sites positioned at the termini of the peptide chains in such a case may be buried and thus not easily accessible to reactants. In Figure 3c the attached peptide molecules interact with each other to form an additional physical crosslinking of the resin. Such a phenomenon generates additional hindrance to the mass transport of reactants into the resin and the reactive sites are not freely accessible. Finally, in Figure 3d the resin is not optimally solvated which leads to a restricted mass transport.

4.3.1.1.5 Chemical Architectures of Gel-Type Resins

From their chemical architectures the gel-type resins can be classified as belonging to one of two different types of solid supports. For lightly crosslinked polystyrene and polyacrylamide resins the reactive sites are located along the polymer chains in a statistical manner. Reaction kinetics can be expected that are similar to those associated with a soluble polystyrene that was proposed by Shemiakin et al.[91] for peptide synthesis. However, reaction rates were found to depend on the location of the reactive sites on such linear noncrosslinked polystyrenes.[92] In fact, reactive sites located on a flexible polymer loop should exhibit a different kinetic behavior to those close to a more rigid, crosslinked section.

Polystyrene is a rather hydrophobic resin and therefore nonpolar solvents like dichloromethane or THF are required for a good swelling. Due to the high solubility of amino acid derivatives in DMF, this solvent is the favored one for peptide synthesis despite its average swelling properties. In contrast to the polystyrene backbones, the growing peptide chains are polar and therefore better solvated by polar solvents. Therefore in highly substituted resins the growing peptide can drastically affect the swelling properties of the resin.[86] Due to this observation, Sheppard et al.[41] and later Sparrow et al.[73] developed the

polyacrylamide gels which are more polar in nature and have "peptide-like" structures (due to the amide bonds). These features are more compatible with the properties of the growing peptide chain and thus allow the use of appropriate polar solvents.

The other gel-type supports are the PEG-grafted resins, where the reactive sites are separated from the crosslinked matrix by a PEG spacer-arm, and thus are located at the termini of the spacers. In this family of supports NovaGel (Figure 1f) constitutes an exception, because the reactive sites are still located on the polymer matrix, whilst the PEG components act as polymer modifiers. The reactive sites which are located at the termini of the PEG spacer-arms demonstrate solution kinetics. This was demonstrated by kinetic measurements where the coupling constants of the active ester, Boc-Gly-ONp, in both low molecular mass compounds and PEG-grafted copolymers, were of the same order of magnitude.[93] Due to the flexibility and good solvating properties of the PEG tentacles on the polymer matrix, chemical reactions on this solid support occur under quasi-homogeneous conditions. A comparison of the ^{13}C NMR relaxation times (T_1) of various polystyrene resins (Pepsyn K, Polyhipe, and TentaGel) in different solvents revealed the narrowest linewidth for TentaGel, confirming the high degree of flexibility of this type of spacer.[94] In contrast to other gel-type resins which are soft and not sufficiently stable toward pressure, TentaGel resins are pressure-stable due to their polymer architecture and can be used in continuous-flow-column processes.

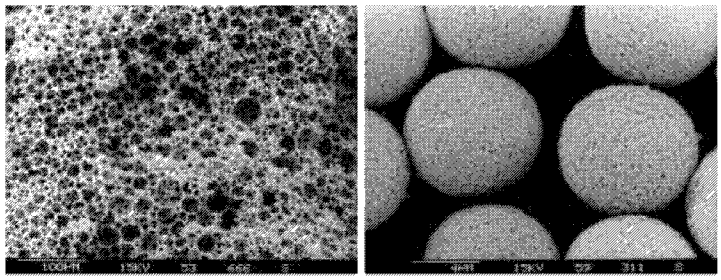
4.3.1.1.6 Chemical Architectures of Composite Resins

Composite resins are another type of solid support which were developed for use in column processes. Composite supports consist of macroporous, rigid skeletons which are either kieselguhr (Pepsyn K resin, Figures 2e and f) or polystyrene (Polyhipe, Figures 2c and d). Lightly crosslinked polyacrylamide is then polymerized within the pores. The rigid skeleton protects the soft, lightly crosslinked polyacrylamide gel from collapsing due to the applied pressure. Although this type of resin is stable toward pressure, mechanically it is very fragile and the use of these resins is strictly restricted to continuous-flow processes. Again the syntheses take place in the gelatinous polyacrylamide portions of the resin. Typical capacities are in the range of 0.1 to 0.5 $mmol \cdot g^{-1}$.

Macroporous polystyrene resins or macroporous methacrylates consist of highly crosslinked polymer phases and pores inside these polymeric solids (Figures 4a and b). In contrast to gel-type resins, highly crosslinked macroreticular resins do not swell in solvents and do not require swelling before use in synthesis. In the ideal case, the polymer/solvent system consists of a solid polymer host structure and pores through which the solvent is free to flow. The use of this type of resin is therefore (with some restrictions) more or less independent of the solvent employed. Nevertheless, this type of resin is not ideal. The performance of the resins is markedly dependent on the more or less broad pore size distribution within each bead resulting from the manufacturing processes. There are large pores (macropores) and very small micropores distributed within each bead. The liquid in the macropores is easily accessible whereas the liquid is occluded from the micropores. Moreover, in a manner similar to the lightly crosslinked polymer resins, the surface of the highly crosslinked polymer matrix can swell to a certain extent which depends on the solvent used. Although this swelling does not change the overall volume of the bead, it creates in microenvironments a highly crosslinked polymer-gel phase. Thus, the reactive sites are not only on the surfaces of such pores, but also in these highly crosslinked gel-phase regions. Correspondingly, the reactive sites located on the surfaces of the pores are easily accessible, whilst the diffusion of reactants to the reactive functions located in these partially swollen, highly crosslinked sections is hindered, leading to nonuniform kinetics.

for references see p 682

Figure 4 Electron Micrographs of Porous Structures of Composite Resins: a) Polyhipe Manufactured by Inverse Polymerization; b) 20%-Crosslinked Macroporous Polystyrene



a b

4.3.1.1.7 Size of Resin Beads and Size Homogeneity

Most of the resin beads employed in peptide synthesis have a spherical shape. In the 1980s the resins showed a very broad size distribution and almost no attention was paid to utilizing beads with a more narrow particle size distribution. All solid supports which are normally used in solid-phase chemistry show a more or less broad particle size distribution. Typical bead sizes are in the range of 75–150 μm (100–200 mesh) and 38–75 μm (200–400 mesh), with Pepsyn K and Polyhipe (a carrier with an irregular shape) being in the range of 500–1000 μm. Table 4 correlates the size of beads in μm (microns) to the ASTM (American Society for Testing and Materials) and British (BS) standards which are both reported in mesh. The size of the polymer particles and the particle size distribution are of major importance and have to be taken into account when performing organic synthesis on solid supports.

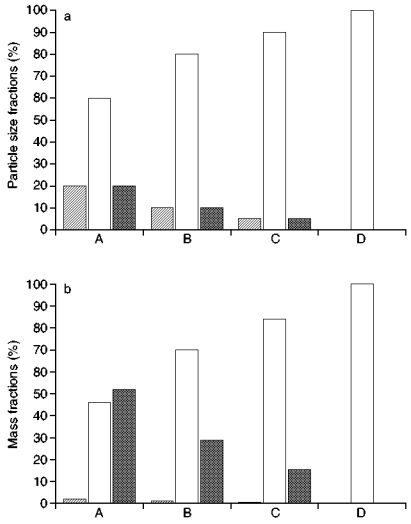
In all polymer-supported reactions there must be a mass transport of the reacting components from the surrounding solvent to the reactive sites which are located primarily inside the resin bead. The driving force for this mass transport is diffusion which in turn depends on the pathlength. In Figure 5 the particle size fraction (Figure 5a) is correlated to the corresponding mass fraction (Figure 5b). Assuming a theoretical size distribution in the range from 20 to 60 μm where 80% of the beads correspond to the main fraction, 10% belong to the smaller size fraction, and 10% to the larger one (case B in Figure 5a), then the 10% size fraction of the larger beads represents 29% of the total mass (case B in Figure 5b). Assuming that there is an equal distribution of the reactive groups on the mass fraction, this 10% size fraction of larger beads carries 29% of the reactive groups. Presupposing that the kinetic rate of reaction is much faster than the rate of diffusion which depends on the particle size, then the largest beads control the overall reaction time. Only monosized beads (case D) show a homogeneous distribution of reactive sites within all beads and therefore only with monosized beads can we be assured of identical reaction conditions in each bead at any given time.

Mass transport and reaction rates are individual parameters for each particle whereas the overall reaction time is controlled by the largest bead. The monosized nature of the beads divides the total reaction space (represented by all beads) into identically sized, small reaction compartments. Thus, independent of the type of resin employed, only polymer particles which are monosized in nature and have a uniform architecture allow significant optimization of the reaction conditions, because identical reaction conditions are generated for each individual bead.[95] Detailed investigations by HPLC and MS have confirmed the theory that a uniform reaction space within the beads yields purer products.[96]

Table 4 Conversion Table for Bead Sizes (μm to mesh)

μm	ASTM E 11–95 (mesh)	BS 410:86 (mesh)	μm	ASTM E 11–95 (mesh)	BS 410:86 (mesh)
1000	18	16	150	100	100
850	20	18	125	120	120
710	25	22	106	140	150
600	30	25	90	170	170
500	35	30	75	200	200
425	40	36	63	230	230
355	45	44	53	270	300
300	50	52	45	325	350
250	60	60	38	400	400
212	70	72	32	450	440
180	80	85	20	635	–

Figure 5 Correlation of Particle Size Fractions with the Corresponding Mass Fractions for Polydisperse (A–C) and Monosized (D) Beads Based on Theoretical Size Distribution



A broad variety of particle sizes are available, from microspheres with less than 5-μm particle size up to resins of 700-μm particle size. The large beads can be used as polymeric microreactors.[97] As the separation between resin and liquid is still performed by filtration, different glass filters have to be used; the porosity and nominal pore size of these filters are listed in Table 5. In those applications where single beads are used (split/mix, single-bead analysis, bead-based assays) it is essential to know the capacity of the single beads. For practical purposes a correlation between bead size and capacity per bead is reported in Table 6.

Table 5 Nominal Pore Size of Glass Filters

Porosity	00	0	1	2	3	4	5
ISO 4793	P 500	P 250	P 160	P 100	P 40	P 16	P 1.6
Nominal pore size (μm)	250–500	160–250	100–160	40–100	16–40	10–16	1.0–1.6

for references see p 682

Table 6 Correlation of Bead Size to Bead Number and Capacity of Single Beads

Size (μm)	Volume (mm^3)	Beads per g Dry Resin	1 mmol·g^{-1} Loading Results on PSty in Capacity per Bead [nmol]	0.28 mmol·g^{-1} Loading Results on TentaGel in Capacity per Bead [nmol]
750	0.220	4619	232	65
700	0.180	5013	196	55
650	0.144	6262	157	44
600	0.113	7962	125	35
550	0.870×10^{-1}	10336	96	27
500	0.654×10^{-1}	13757	71	20
450	0.476×10^{-1}	18872	53	15
400	0.335×10^{-1}	26871	36	10
350	0.224×10^{-1}	40110	25	7
300	0.141×10^{-1}	63694	14	4
250	0.817×10^{-2}	110063	9	2.5
200	0.418×10^{-2}	214968	4.6	1.3
180	0.305×10^{-2}	294880	3.4	0.94
150	0.176×10^{-2}	509554	2	0.55
130	0.115×10^{-2}	782769	1.3	0.35
100	0.523×10^{-3}	1.72×10^{6}	0.6	0.16
90	0.314×10^{-3}	2.86×10^{6}	0.4	0.10
35	0.148×10^{-4}	4.55×10^{7}	2.2×10^{-2}	6.2×10^{-3}
20	0.375×10^{-5}	2.40×10^{8}	3.6×10^{-3}	1×10^{-3}
10	0.462×10^{-6}	1.95×10^{9}	3.6×10^{-4}	1×10^{-4}

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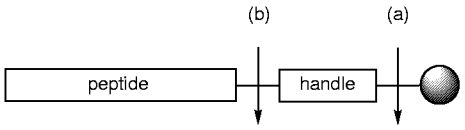
4.3.1.2 Handles and Supports

F. Albericio and E. Giralt

Solid-phase construction of peptides is traditionally carried out in the C → N direction; therefore the synthetic process begins by anchoring the C-terminal amino acid to the solid support.[1] Other alternatives that can be convenient for the preparation of C-terminal modified peptides involve the anchoring of the first amino acid through the side chain or through the backbone amide nitrogen between the first two amino acids. The inverse approach, i.e. the N → C direction,[2–5] has not been extensively used because, during the activation of the free α-carboxy group situated in the C-terminus of the protected peptide-resin, an oxazol-5(4*H*)-one can be formed with epimerization at the C-terminal residue.[6]

The classical Merrifield scheme begins with esterification of the protected C-terminal residue onto a chloromethyl-functionalized PSty-resin.[7] An alternative to functionalized resins is the use of bifunctional spacer molecules known as handles or linkers.[8] Such handles become attached (a) permanently to a functionalized resin at one end, often through a stable amide bond, and are linked (b) temporarily to a growing peptide chain at the other end (Scheme 1).

Scheme 1 Bifunctional Spacer Molecules[8]



Although many different mechanisms, e.g. photolysis, fluoridolysis, and base-catalyzed β-elimination, have been exploited for handle cleavage, the majority of handles rely on acidolysis for release of the final peptide from the support.[8]

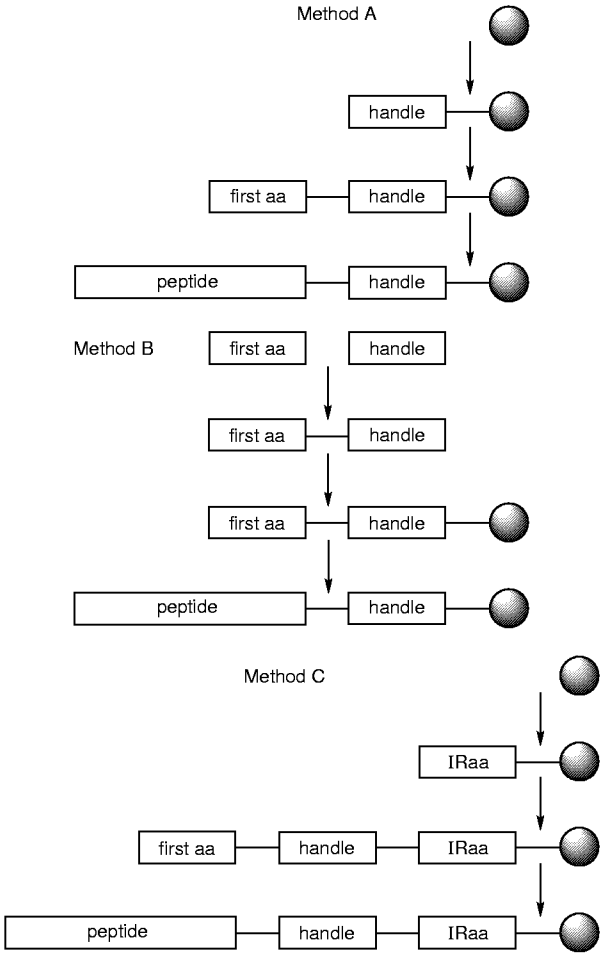
4.3.1.2.1 Handle versus Functionalized Solid Support; Internal Amino Acid Reference

Although a number of excellent functionalized solid supports are commercially available, the use of handles can be more convenient because they provide control and flexibility for the synthetic process. These advantages include: (1) any resin support or material that has been previously functionalized can be used as a parent system for synthesis; (2) substitution levels or loadings can be controlled better by using handles; (3) two discrete chemical steps, each of which can be optimized, are used to create the temporary and permanent links; (4) handle structures can be fine-tuned for optimal yields of both anchoring and cleavage steps; (5) handles can be used readily in conjunction with internal reference amino acids (IRaas),[9–11] which facilitates monitoring of the yields of the various steps by amino acid analysis of resin-bound peptides.

Both of the bonds involving the handle are very often formed on the solid support (Method A, Scheme 2). An alternative procedure, however, is to attach the first amino acid to the handle in solution, usually by an ester group (Method B, Scheme 2). This amino acid handle unit is now called a preformed handle and is incorporated into the resin, again usually by amide formation. Thus, one can achieve quantitative attachments to the support and therefore circumvent problems associated with free polymer-bound functional groups. The preformed handle can be purified to remove side products that arise during creation of the temporary linkage. Furthermore, racemization can be controlled and, more importantly, minimized. Finally, IRaas can be used with any of the strategies discussed previously (Method C, Scheme 2).

for references see p 706

Scheme 2 Attaching Amino Acids to a Solid Support[1]



The preparation in solution of the preformed handle requires the protection of its carboxy group prior to the formation of the ester bond. Phenacyl esters, which can be removed with zinc in acetic acid and are therefore compatible with the presence of both Boc- and Fmoc-protecting groups, have been used.[12] An alternative procedure, however, involves the use of 2,6-dichloro- or 2,4,5-trichlorophenyl esters,[13,14] which serves a dual purpose of first protecting and then activating the carboxy group of the handle.

Incorporation of both IRaa and handles with free carboxy groups can be performed using the methods commonly used for the formation of the peptides. If the handle contains a hydroxy group, for instance in the case of 4-hydroxymethylphenoxyacetic acid, rather mild coupling conditions, such as DIC/HOBt, have to be used in order to avoid double or triple incorporation of the handle.

In the design of a handle, the presence (once anchored to the solid support) of the carboxamido moiety has to be taken into account, because it can modify the conditions required for the cleavage step. This is especially important in those handles that are labile in the presence of acids and bases. Thus, the ester bond formed between the first amino acid and a 4-hydroxymethylphenoxyacetamidomethyl resin is more stable to TFA than when the first amino acid is anchored to the Wang resin, which also incorporates a 4-alkoxybenzyl alcohol.[14] In order to circumvent this effect, additional methylene groups should be added.

Thus, in the case of 4-hydroxymethylphenoxypropionamidomethyl resin, the second methylene unit separates the electron-donating oxygen *para* to the anchored amino acid from the electron-withdrawing alkylamide group linking the handle to the support, and this results in increased lability to acids of the anchoring linkage.[14]

The presence of an IRaa between the handle and the functional group of the resin has the additional benefit of increasing the stability of the bond between the solid support and the handle, which is sometimes labile to the acid conditions used to cleave the peptide from the support.[15]

IRaa-Resin; General Procedure:[16]
This protocol and those that follow were developed for use in conjunction with either polystyrene (PSty)- or poly(ethylene glycol)-polystyrene (PEG-PSty)-based supports. Aminomethyl resin was washed with DMF (all washings were carried out with 5 mL of solvent per g of resin, 5×1 min) and CH_2Cl_2 (5×1 min). DIPEA (8 equiv) was added to a soln of Boc- or Fmoc-Xaa (4 equiv), HATU (4 equiv), and HOAt (4 equiv) in DMF (4 mL · g^{-1} of resin) and, after preactivation for 1 min, the entire soln was added to the resin. Coupling was allowed to proceed for 2 h, after which time the resin was checked by the Kaiser ninhydrin test. If this test was negative or only slightly positive, the resultant Boc/Fmoc-IRaa-resin was washed with DMF (5×1 min) and CH_2Cl_2 (5×1 min), and then treated with Ac_2O/DMF (1:9, 20 min), washed with DMF (5×1 min), CH_2Cl_2 (5×1 min), and MeOH (2×1 min), and finally dried in vacuo. If the ninhydrin test was clearly positive, the coupling step was repeated as described above.

Handle-IRaa-Resin; General Procedure:[16]
The same protocol as described above was used, but 2 equiv of the corresponding handle, HATU, and HOAt, and 4 equiv of DIPEA were used.
For those handles containing hydroxy groups, the handle (2 equiv) and HOBt (2 equiv) were dissolved in DMF (4 mL · g^{-1} of resin), DIC (4 equiv) was added and, after preactivation for 5 min, the entire soln was added to the resin.

4.3.1.2.2 C-Terminal Acids

For the solid-phase preparation of peptide C-terminal acids, the first amino acid should form an ester bond with either the handle or the solid support. Regardless of the nature of the handle or the functionalized support used, a potential side reaction in the solid-phase preparation of peptide C-terminal acids is the formation of piperazine-2,5-diones (dioxopiperazines), which are the smallest cyclic peptides.[17–19] Piperazine-2,5-diones are formed by intramolecular aminolysis at the dipeptide level. Piperazine-2,5-dione formation has been established to be either acid[17] or base catalyzed[18] and is strongly dependent on the nature and the sequence of the amino acids. Thus, the reaction is favored by the presence, in either the first or second position, of amino acids (Gly, Pro, or *N*-alkyl) that can easily adopt a *cis* conformation in the resulting amide bond. Another favorable combination for this side reaction is to have one L- and one D-amino acid in the dipeptide, and this is due to the minimal steric interference between the two side chains. Furthermore, rates of piperazine-2,5-dione formation differ considerably depending on the *C*-carboxy ester linkage.[20,21] Thus, 2-nitrobenzyl handles are more prone to give piperazine-2,5-diones because the *o*-nitrobenzyl alcohol is a good leaving group. Formation of piperazine-2,5-diones not only leads to a lower overall yield, but also to the presence in the crude product of several deletion peptides lacking the first amino acid residues. These compounds are formed by acylation of the hydroxymethyl groups formed by the following protected amino acids and can start new peptide chains.[22,23]

The use of hindered handles or functionalized solid supports would minimize piperazine-2,5-dione formation. Thus, the tertiary alcohol handle 4-(1′,1′-dimethyl-1′-hydroxypropyl)phenoxyacetic acid (DHPP, **1**, Table 1)[24] and the 2-chlorotrityl chloride resin [Trt(2-Cl)-Cl resin, **2**][25,26] are both very convenient for this purpose.[23,27,28] An alternative procedure avoids the presence of the free amino group of the penultimate residue and the incorporation of the third residue with neutralization in situ. For the Boc/Bzl strategy this can

for references see p 706

be accomplished easily by removal of the Boc group with TFA/CH_2Cl_2 (3:7), and addition of the diisopropylethylammonium salt of the third residue in the presence of a phosphonium salt coupling reagent, such as PyBOP or PyAOP.[29] For the Fmoc/tBu strategy, the penultimate residue has to be incorporated as its N^{α}-trityl derivative, followed by detritylation with dilute TFA, and incorporation of the third residue as its N^{α}-Fmoc derivative under in situ neutralization–coupling conditions mediated by a phosphonium salt coupling reagent (see Section 4.3.1.2.5 for an example of this protocol).[30]

Table 1 Handles and Resins for Solid-Phase Synthesis of C-Terminal Peptide Acids

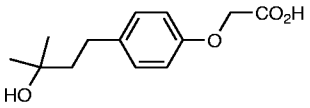
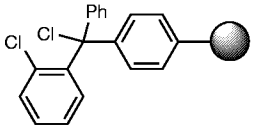
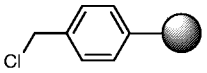
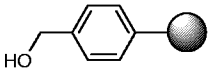
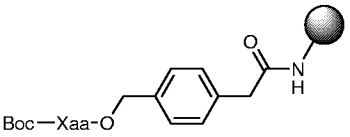
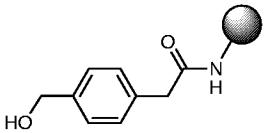
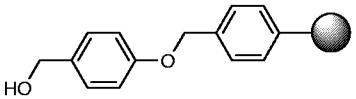
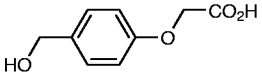
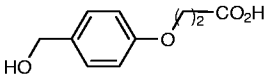
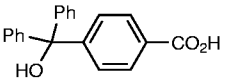
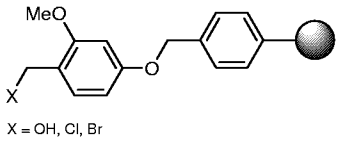
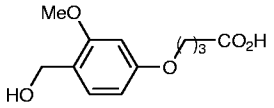
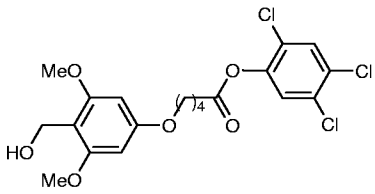
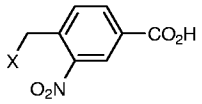
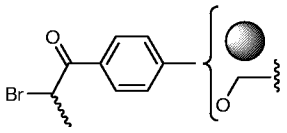
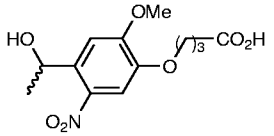
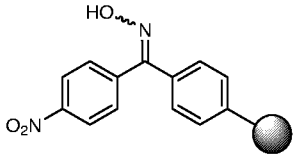
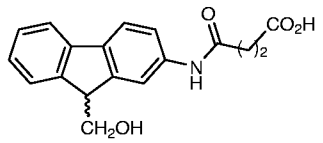
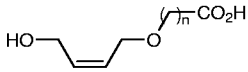
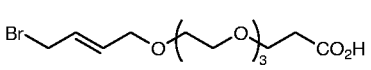
Handle/ Resin No	Name and/or Abbreviation(s)	Structure	Cleavage Conditions	Ref
1	4-(1′,1′-dimethyl-1′-hydroxypropyl)phenoxyacetic acid (DHPP)		TFA, scavengers	[24]
2	2-chlorotrityl chloride resin [Trt(2-Cl)-Cl resin]		dil TFA, AcOH/TFE, HFIP/CH_2Cl_2	[25,26]
3	chloromethyl resin (Merrifield resin)		HF, scavengers	[7]
4	hydroxymethyl resin		HF, scavengers	[7]
5	phenylacetamidomethyl resin (Phacm resin)		HF, scavengers	[12,31]
6	hydroxymethyl-PAM resin		HF, scavengers	[32]
7	4-alkoxybenzyl alcohol resin (Wang resin)		TFA, scavengers	[33]
8	4-(hydroxymethyl)-phenoxyacetic acid (HMPAA, PAC)		TFA, scavengers	[11,33,34]
9	3-[4-(hydroxymethyl)-phenoxy]propanoic acid (HMPPA)		TFA, scavengers	[14]
10	trityl handle		dil TFA	[40]

Table 1 (cont.)

Handle/ Resin No	Name and/or Abbreviation(s)	Structure	Cleavage Conditions	Ref
11	SASRIN resin (super acid sensitive resin)	 X = OH, Cl, Br	dil TFA	[42–44]
12	4-(4-hydroxymethyl-3-methoxyphenoxy)-butanoic acid (HMPB)		dil TFA	[45]
13	2,4,5-trichlorophenyl 5-(4-hydroxymethyl-3,5-dimethoxyphenoxy)valerate		dil TFA	[46]
14	2-nitrobenzyl handle (Nbz)	 X = OH, Br	*hν* (360 nm)	[49–51]
15	phenacyl resin/handle		*hν* (360 nm)	[54–57]
16	α-methyl-6-nitroveratryl handle		*hν* (360 nm)	[52,65]
17	oxime-based (Kaiser) resin		nucleophiles	[66–68]
18	*N*-[(9-hydroxymethyl)fluoren-2-yl]succinamic acid (HMFS)		morpholine	[72,73]
19	allyl handles	 n = 1, 5 	Pd^0, allyl acceptors	[76–80]

for references see p 706

4.3.1.2.2.1 Acid-Labile Handles and Functionalized Solid Supports

The three main classes of acid-labile handles are based on substituted benzyl, benzhydryl, and trityl derivatives. Introduction of alkoxy substituents at *para* and *ortho* positions increases the acid lability, and is used to fine-tune the properties of the handle. For example, *p*-alkylbenzyl ester handles require HF to release the peptide acid, whereas analogous *p*-alkoxy handles release the peptide upon treatment with TFA.

Originally in the Boc/Bzl strategy, the peptide was attached to the resin by the benzyl ester formed from the functionalized PSty matrix itself [chloromethyl resin **3** (Merrifield resin) or hydroxymethyl resin **4**]. However, these systems are not sufficiently stable to the repetitive acidolytic treatment necessary to remove the Boc group. Each acidolysis leads to a 1.1% loss of peptide chain from the support.[31] This loss, although unimportant in the synthesis of shorter peptides, becomes a serious problem in the synthesis of larger ones. This problem can be avoided by increasing the stability of the peptide-resin anchor to the acidic conditions used in the synthesis. The most successful approach for accomplishing this has been the use of phenylacetamidomethyl resin (Phacm resin, **5**).[12,31] The presence of the electron-withdrawing carboxamidomethyl group in the *para*-position of the peptide benzyl ester increases the acid stability of Phacm resin by approximately 100 times compared to that of standard polystyrene. This in turn reduces the loss of peptide per deprotection step to levels that permit the synthesis of large peptides; at the same time the peptide-resin bond of the Phacm resin may be cleaved by treatment with liquid HF at 0 °C in the presence of the corresponding scavengers. The best way to use Phacm resin is through the preformed handle approach because the acylation of hydroxymethyl-Phacm resin **6** occurs with very poor yields.[32]

For the Fmoc/tBu approach, several handles and resins are available for the synthesis of free peptide acids. Among the most commonly used are the Trt(2-Cl)-Cl resin **2**, the 4-alkoxybenzyl alcohol resin (Wang resin, **7**),[33] and the 4-(hydroxymethyl)phenoxyacetic acid (HMPAA, **8**)[11,13,34] and 3-[4-(hydroxymethyl)phenoxy]propanoic acid (HMPPA, **9**)[14] handles. The last forms an ester bond with the first amino acid that is slightly more labile in the presence of TFA solutions than that derived from HMPAA. In the case of 4-alkoxybenzyl alcohol handles, the incorporation of the protected amino acid directly onto the resin, as well as the use of the preformed handle, works well. The main advantages of using Trt(2-Cl)-Cl resin are that the esterification step is a nucleophilic displacement of the halogen group by the carboxylate of the protected amino acid derivative, and that racemization levels are significantly lower than when protected amino acids are incorporated in the presence of coupling reagents or in the form of active species (see Section 4.3.1.3). This is extremely important for the incorporation of Cys and His derivatives, which show a marked tendency to undergo racemization (see Section 4.3.1.3). In the case of the Fmoc/tBu synthesis of C-terminal Cys peptide acids, racemization can also be caused by piperidine during the Fmoc removal step.[35] This side reaction has been extensively studied and is particularly significant when 4-hydroxymethylphenoxy-type resins are used and when Acm is the protecting group of the Cys unit.[36] This racemization is almost completely avoided when Trt(2-Cl)-Cl resin is used, probably owing to the high degree of steric hindrance. Another side reaction associated with the Fmoc/tBu synthesis of C-terminal Cys peptide acids with 4-hydroxymethylphenoxy-type resins is the formation of a 3-(piperidin-1-yl)alanine residue at the C-terminus.[37] The mechanism of this side reaction, which is related to that of racemization, might be explained in terms of a base-catalyzed β-elimination of the protected sulfur of the side chain to give a dehydroalanine intermediate, which is trapped by a nucleophilic addition of the piperidine. The existence of the side reaction is diagnosed by mass spectrometry, i.e. the observation of a peak that is 51 Da higher in mass for peptides synthesized with Trt and 20 Da lower for those synthesized with Acm. The amount of undesired byproduct is greater with PSty resins compared with PEG-PSty supports, and with Acm protection compared

with Trt, although it has not been reported with Trt(2-Cl)-Cl resins. Both side reactions, racemization and formation of the 3-(piperidin-1-yl)alanine adduct, are negligible when C-terminal peptide amides are prepared.

Cleavage of the peptide from Trt(2-Cl)-Cl resin **2** can be accomplished in excellent yield by treatment with less than 1% TFA acid or a mixture of AcOH and TFE in CH_2Cl_2.[25] An alternative cleavage reagent is HFIP in CH_2Cl_2, which has the advantage of avoiding the contamination of the protected peptide segment with a carboxylic acid.[38] This allows the preparation of protected peptides, which after purification and/or characterization can be coupled with other protected peptides in a convergent solid-phase synthesis scheme.[39] Furthermore, the hindered Trt(2-Cl)-Cl resin is also very convenient for avoiding the formation of piperazine-2,5-dione.[23,28] Thus, Trt(2-Cl)-Cl resin is becoming the method of choice for the solid-phase preparation of C-terminal peptide acids. A convenient handle **10** based on the trityl structure has been developed and used in conjunction with PEG-PSty resins.[40] In this case, and after the incorporation of the handle onto the solid support, the resin must be converted into the chloride form by heating with acetyl chloride or thionyl chloride in toluene,[41] and the unstable trityl chloride resin should then be used immediately.

Other highly acid-labile functionalized solid supports and handles incorporate one or two additional electron-donating substituents, such as methoxy groups in 4-alkoxybenzyl alcohol-base-resins. Among these systems, SASRIN resin (super acid sensitive resin, **11**),[42–44] 4-(4-hydroxymethyl-3-methoxyphenoxy)butanoic acid (HMPB, **12**),[45] and 2,4,5-trichlorophenyl 5-(4-hydroxymethyl-3,5-dimethoxyphenoxy)valerate (**13**)[46] handles are the most widely used.

For experimental cleavage procedures carried out with strong acids (e.g., liq HF, CF_3SO_3H) and high concentrations of TFA, see Section 4.3.5.

Cleavage of Protected Peptides from Highly Acid-Labile Resins; General Procedure:[38,45,47]
The peptide-resin was pre-swollen with washes of CH_2Cl_2 (3 × 10 min, 10 mL · g^{-1} of resin). Cleavage was carried out at 25 °C by alternating washes of TFA/CH_2Cl_2 (1:99) and CH_2Cl_2 (5 mL · g^{-1} of resin, 20 s) and filtering into MeOH/pyridine (9:1) (4 × cleavage volume). This operation was repeated (3×). For peptides containing Trp and Cys, 5% EDT was added to the TFA soln. The combined cleavage portions were concentrated to one-third of the original volume and the peptide was precipitated with H_2O (usually about one-half of the organic volume). The precipitate was filtered off and dried under reduced pressure over P_2O_5. All common protecting groups, including His(Trt), were stable under these conditions.

Protected peptides were also quantitatively detached from Trt(2-Cl)-Cl resin using HFIP/CH_2Cl_2 (1:4) (5 mL · g^{-1} of resin, 3 min). Neutralization of the effluents with MeOH/pyridine (9:1) was carried out because several tBu-based protecting groups were not completely stable to the cleavage conditions.

4.3.1.2.2.2 Photolabile Handles

Photolysis of the peptide-resin bond is a mild method that is, in principle, compatible with both the Fmoc/tBu and the Boc/Bzl strategies, and therefore the use of these handles allows the preparation of both unprotected and protected peptides (see Section 4.3.2.4 for additional details on the use of photoreactive N-protecting groups).[48] The photolabile handles most widely used are those that contain the 2-nitrobenzyl (Nbz, **14**)[49–53] and the phenacyl moieties **15**.[54–57] This technique is compatible with all encoded amino acids except Met. It has been found that under the conditions of photolytic cleavage, oxidation of Met to the corresponding sulfoxide occurred to a significant extent.[58,59] An alternative method involves the use of protected Met(O) instead of unprotected Met. Boc-Met(O)-OH is completely stable under prolonged photolysis conditions, with no evidence of further oxidation to the corresponding sulfone. An additional advantage of protected peptides containing Met(O) is that they are often more soluble in common solvents, facilitating their manipulation.[60] Once the protein/peptide has been fully assembled, the final reduction of the sulfoxide to the free Met can easily be achieved. One drawback with the use of Met(O) is that it leads to the

for references see p 706

formation of diastereomeric peptides that, in our experience, usually (but not always) give a complex HPLC trace.[61]

The main limitation in the use of the Nbz handle **14** is that only relatively small amounts (up to 700 mg) of peptide-resin can be photolyzed at one time. This is thought to be a consequence of the formation of azo and azoxy compounds from the 2-nitrosobenzaldehyde produced in the photolysis reaction.[62] These compounds are a deep red color and act as an internal light filter. This effect is more pronounced with increasing amounts of resin. It has been found that optimum results are achieved by suspending up to 700 mg of peptide-resin in a mixture of TFE in CH_2Cl_2 or toluene (1:4), degassing, and irradiating at 360 nm with stirring for 7–15 h. This procedure gives reproducible cleavage yields of up to 85%.[63] The peptide-resin bond is not completely stable to treatment with piperidine. The release of the peptide from the resin depends on the steric hindrance of the first amino acid. Thus, for Gly it is approximately 3% per 20 min treatment with a piperidine/DMF (1:4) cycle, but for Val or Phe it appears to be negligible.[64] The synthesis of long peptides with nonhindered first amino acids is not advisable by this method. More recently, the use of the α-methyl-6-nitroveratryl handle **16** has been described. The incorporation of two additional alkoxy groups onto the benzene ring and a methyl group on the benzylic carbon allows the photolytic release of peptides in good yield in short periods of time (1 h).[52,65] The use of this handle for the preparation of Met-containing peptides with low (<4%) amounts of methionine sulfoxide peptides has been described. These handles have a marked tendency to provoke piperazine-2,5-dione formation and, therefore, the in situ neutralization protocol described above should be used.

The use of phenacyl handles **15** is only compatible with the Boc/Bzl strategy since the peptide-resin anchor is not stable to the nucleophilic conditions used in the Fmoc/tBu strategy. Besides the formation of piperazine-2,5-diones, another side reaction can take place on incorporation of the second residue. Attack by the free amino group of the first amino acid on the carbonyl group of the phenacyl resin leads to a cyclic Schiff base.[55] Both types of side reaction can be minimized using the in situ neutralization protocol. Photolysis is carried out as for nitrobenzyl resins.

Cleavage of Peptides from Photolabile Resins; General Procedure:[60]
A two-necked cylindrical reaction vessel was silylated before photolysis in order to prevent resin adhering to its walls. This procedure was performed by rinsing the vessel 3 or 4 times with TMSCl/toluene (1:9), followed by washing with abs EtOH and drying. The peptide-resin (0.5–0.7 g) was suspended in TFE/CH_2Cl_2 or TFE/toluene (1:4, 100 mL) in the reaction vessel. Prior to photolysis the peptide-resin suspension was degassed by evacuation at water-pump pressure and purging with argon (3×) in succession. The resin was photolyzed at 360 nm in a Rayonnet RPR-100 apparatus for 1–15 h, while maintaining vigorous magnetic stirring and keeping the temperature below 40 °C using the incorporated fan. The crude mixture was filtered and the resin washed with TFE/CH_2Cl_2 (1:4), CH_2Cl_2, DMF, and MeOH (3 × 5 mL in each case). The filtrates were combined and the solvent removed under reduced pressure.

4.3.1.2.2.3 Base- and Nucleophile-Labile Handles and Functionalized Solid Supports

These handles and functionalized solid supports can only be used for the Boc/Bzl strategy. The conditions necessary for Fmoc group removal would lead to a premature release of the peptide from the resin. The most widely used systems of this kind are oxime-based (Kaiser) resin (Section 4.3.1.2.2.3.1) and handles that incorporate the fluorenylmethyl moiety.

4.3.1.2.2.3.1 Oxime-Based (Kaiser) Resin

The 4-nitrobenzophenone oxime resin **17** has been used extensively for the preparation of Boc/Bzl protected peptide segments.[66–68] Various methods have been used to cleave pro-

tected peptides from the resin, including hydrazinolysis, ammonolysis, and aminolysis using a suitable amino acid ester. An alternative and more convenient procedure for the cleavage reaction is by transesterification of the peptide-resin with HOPip. This reaction initially leads to the formation of the OPip ester of the protected peptide, which after treatment with zinc and acetic acid affords the corresponding free carboxylic acid. For all couplings, an in situ neutralization protocol is recommended because a premature loss of peptide from the resin on neutralization with base (DIPEA) after acidolytic removal of the Boc can occur. This loss is much more important after the incorporation of the second residue owing to dioxopiperazine formation.

4.3.1.2.2.3.2 Fluorenylmethyl-Based Handles

Several handles incorporating the fluorenylmethyl moiety have been described.[69–71] Among them, the most widely used is *N*-[(9-hydroxymethyl)fluoren-2-yl]succinamic acid (HMFS, **18**).[72,73] HMFS circumvents some of the problems associated with other handles of the same family, such as the premature release of protected peptides upon treatment with the DIPEA used to neutralize the resin after Boc group removal with TFA. Although excellent yields (>95%) have been obtained after treatment of peptide-resins with both piperidine/DMF and morpholine/DMF (both 1:4), the morpholine-based cleavage gives products with cleaner HPLC profiles than those afforded by piperidine treatments. Furthermore, in order to minimize the risk of aspartimide formation during the cleavage of Asp-containing peptides, dry morpholine is preferred to piperidine[74] for cleavage, and cyclohexyl ester protection is preferred to benzyl ester protection for the β-carboxylic acid group of aspartic acid. Piperazine-2,5-dione formation has not been detected during the preparation of several peptides containing Pro and Gly at the C-terminal position.

Cleavage of Peptides from Oxime-Based (Kaiser) Resins; General Procedure:[75]
The protected peptide-oxime-resin **17** (1 g) was pre-swollen in a silylated screw-cap tube reaction vessel with a Teflon-lined cap, a sintered glass frit, and a stopcock, with washes of CH_2Cl_2 (10 mL, 3 × 10 min). The cleavage was carried out by treatment of the peptide-resin with HOPip (3 equiv, freshly recrystallized from hexane) for 3–16 h, at 25 °C, with mechanical shaking. The resin was filtered off and washed with CH_2Cl_2, DMF, and MeOH (3 × 5 mL in each case). The filtrates were combined and the solvent removed under reduced pressure. The residue was triturated with Et_2O or hexane to give the crude peptide OPip ester. This material was then dissolved in AcOH/H_2O (9:1, 15 mL) and Zn dust (30 equiv) was added. After vigorous stirring for 15–30 min at 25 °C, the Zn was removed by filtration and washed with AcOH/H_2O (9:1, 3 × 5 mL). The solvent was removed from the combined filtrates under reduced pressure. Gel-filtration chromatography using Sephadex LH-60 in DMF was used for the removal both of residual Zn and of byproducts originating from HOPip.

Cleavage of Peptides from Fluorenylmethyl-Based Resins; General Procedure:[72]
The peptide-resin (1 g) was pre-swollen, in a 20-mL polypropylene syringe fitted with a polyethylene disc and a stopcock, with washes of DMF (10 mL, 3 × 10 min). The cleavage was carried out by treatment of the peptide-resin with a freshly prepared soln of dry morpholine/DMF (1:4, 10 mL) for 2 h, at 25 °C, with occasional agitation. After filtration the resin was washed with DMF (3 × 5 mL), the cleavage portions combined, and the solvent removed under reduced pressure.

4.3.1.2.2.4 Allyl-Based Handles

Allyl-based handles **19** are very useful systems since cleavage of the protected peptide from the resin can be achieved under practically neutral conditions by palladium(0)-catalyzed allyl transfer to weakly basic nucleophiles.[76–80] These conditions are, in principle, compatible with both of the major SPPS strategies.

Reproducible, high-yielding cleavage of protected peptides from allyl-based resins requires strict control of the reaction conditions. Two general protocols have been developed.[81] The first involves treating a suspension of the peptide-resin in DMSO/THF/0.5 M HCl (2:2:1) with morpholine, as a nucleophile, in the presence of tetrakis(triphenyl-

for references see p 706

phosphine)palladium(0).[82] However, the use of a nucleophilic secondary amine as the allyl acceptor in the cleavage of Fmoc-protected peptides leads to deprotection of the Fmoc group. Furthermore, the presence of DMSO in the medium causes the oxidation of methionine to the sulfoxide. In these cases, morpholine can be replaced by *N*-methylaniline (NMA) and $CHCl_3$/AcOH (9:1) can be used as the solvent.[83] The use of other nucleophile scavengers, such as silanes and borane derivatives, is very useful for the removal of Aloc protecting groups but has not been applied to the release of peptides from allyl-based resins. Cleavage yields appear to depend upon the quality of the catalyst and can sometimes be improved by increasing the amount used. Alternatively, a hydrostannolytic allyl transfer method based on the use of dichlorobis(triphenylphosphine)palladium(II) in the presence of tributyltin(IV) hydride and using DMF/CH_2Cl_2 (1:1) as solvent can be applied. Since no base is required, this cleavage protocol is compatible with the Fmoc group. Palladium-catalyzed allyl transfer cleavages are best carried out under an argon atmosphere. For this type of handle, piperazine-2,5-dione formation, mainly in the Fmoc/tBu strategy, is a risk and must be avoided by using the techniques described earlier.

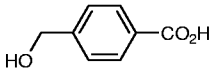
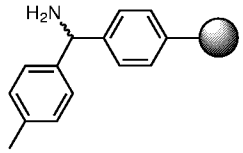
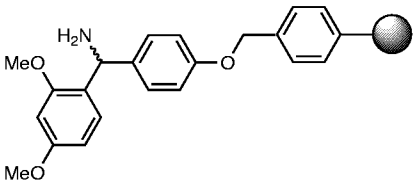
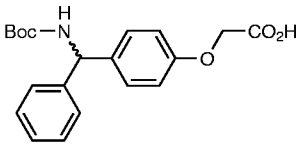
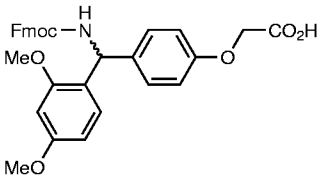
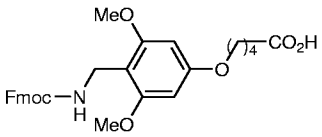
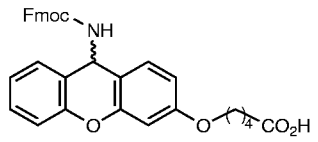
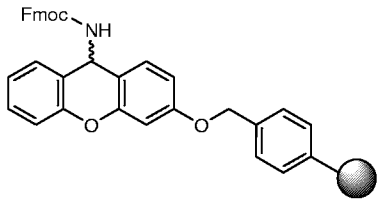
Cleavage of Peptides from Allyl-Based Resins; General Procedure:[81]
Method A: The peptide-resin (100 mg, ~0.075 mmol) and crystalline, yellow $Pd(PPh_3)_4$ (30 mg, 0.026 mmol) were suspended in a previously degassed mixture of DMSO/THF/0.5 M HCl (2:2:1, 5 mL) in a silylated two-necked round-bottomed flask and stirred vigorously under argon. The catalyst dissolved, giving the suspension a yellow color. NMA (385 μL, 3.5 mmol) was added and the mixture stirred under argon for 12 h at 25 °C. Filtration, followed by washing of the resin with DMF and $CHCl_3$ (3 × 1 mL in each case), and solvent removal under high vacuum gave the crude peptide. For Met-containing peptides, $CHCl_3$/AcOH (9:1) can be used as solvent.

Method B: The peptide-resin (100 mg, ~0.075 mmol) and $PdCl_2(PPh_3)_2$ (2 mg, 2.5 μmol) were suspended in a previously degassed mixture of DMF/CH_2Cl_2 (1:1, 5 mL) in a silylated two-necked round-bottomed flask and the mixture was stirred vigorously under argon. The catalyst dissolved, giving the suspension a yellow color. Bu_3SnH (75 mL, 0.25 μmol) in CH_2Cl_2 (1 mL) was added over 30 min and the mixture stirred for a further 10 min at 25 °C. After filtration, the resin was washed with DMF/CH_2Cl_2 (1:1, 3 × 1 mL), and the filtrate extracted repeatedly (4 × 5 mL) with pentane in order to remove Sn byproducts. Then 1 M HCl (5 mL) was added to convert the Sn carboxylate into the peptide free acid, followed by H_2O (5 mL) to precipitate the crude peptide, which was isolated by centrifugation and filtration. In this kind of cleavage, gel-filtration chromatography through Sephadex LH-60, using DMF as eluent, is advisable in order to completely remove any Sn byproducts.

4.3.1.2.3 C-Terminal Amides

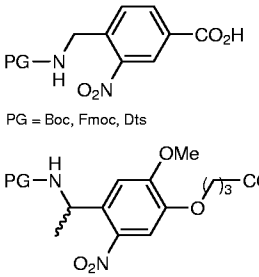
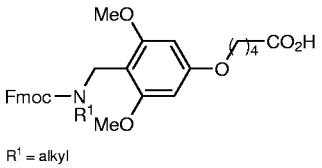
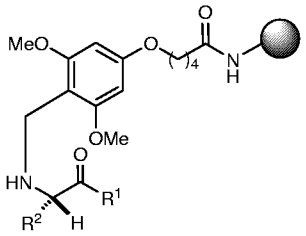
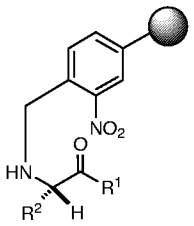
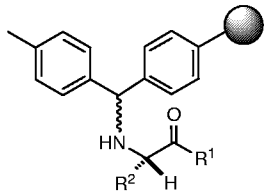
In SPPS, peptide amides could be obtained by ammonolysis in an alcoholic solvent of peptide-resins having ester anchorages to the solid support. The 4-hydroxymethylbenzoic acid handle **20** (Table 2) or its bromo derivative are the most suitable for this purpose.[84] This strategy is unsatisfactory, however, because of frequently low cleavage yields, especially for peptides having sterically hindered C-terminal residues such as Val or Ile. Furthermore, in the case of bulky residues the product can be contaminated with the corresponding peptide ester. Prolonged exposure to ammonia can also lead to epimerization at one or more stereogenic centers. Another limitation is that side-chain protecting groups should be removed independently either before cleavage of the peptide from the resin or once the protected peptide has been liberated from the support. In this case, undesired ammonolysis or rearrangement of sensitive side-chain functionalities, such as Asp/Glu(OBzl), may take place. The use of the photolabile 2-nitrobenzyl derivative,[64] which is also more labile to nucleophiles than the benzyl derivative, can be more convenient than the use of the oxime (Kaiser) resin.[66]

Table 2 Handles and Resins for Solid-Phase Synthesis of C-Terminal Peptide Amides and Backbone Attachment

Handle/ Resin No	Name and/or Abbreviation(s)	Structure	Cleavage Conditions	Ref
20	4-hydroxymethylbenzoic acid		NH_3	[84]
21	4-methylbenzhydryl-amine resin (pMeBHA resin)		HF, scavengers	[86,87]
22	4-(α-amino-2′,4′-dimethoxybenzyl) phenoxymethyl resin (Rink resin)		TFA, scavengers	[88]
23	*N*-Boc-(α-phenyl)amino-methylphenoxyacetic acid (BHA handle)		HF, scavengers	[89]
24	4-[(*R,S*)-α-[1-(9*H*)-fluoren-9-yl)methoxy-formamido]-2,4-dimethoxybenzyl]phenoxyacetic acid (Fmoc-Rink linker, linker AM)		TFA, scavengers	[90]
25	5-[4-({[(9*H*-fluoren-9-ylmethoxy)carbonyl]amino}methyl)-3,5-dimethoxyphenoxy]valeric acid (PAL handle)		TFA, scavengers	[91,92]
26	5-[(9-{[(9*H*-fluoren-9-ylmethoxy)carbonyl]amino}-9*H*-Xanthen-3-yl)-oxy]valeric acid (XAL handle)		dil TFA	[93]
27	XAL resin		dil TFA	[94,95]

for references see p 706

Table 2 (cont.)

Handle/ Resin No	Name and/or Abbreviation(s)	Structure	Cleavage Conditions	Ref
28	3-nitro-4-amino-methylbenzyl handles	 PG = Boc, Fmoc, Dts	*hν* (360 nm)	[96,97]
29	5-(4-{[[(9*H*-fluoren-9-ylmethoxy)carbonyl](alkyl)amino]methyl}-3,5-dimethoxyphenoxy)valeric acid [(R)PAL, handle]	 R^1 = alkyl	dil TFA	[98]
30	tris(alkoxy)benzylamide backbone amide linker (BAL)		TFA, scavengers	[16,121,122]
31	-		*hν* (360 nm)	[118]
32	-		HF, scavengers	[119]

A much more useful approach is the cleavage of peptide-resins having a C-terminal amide bond to the solid support. The most common solid supports for the synthesis of peptide amides are benzhydrylamine derivatives,[85] which liberate the peptide by acidolysis. Examples include the 4-methylbenzhydrylamine resin (pMeBHA resin, **21**)[86,87] for the Boc/Bzl strategy (release of the peptide is carried out with liquid HF at 0 °C in the presence of the corresponding scavengers) and the 4-(α-amino-2′,4′-dimethoxybenzyl)phenoxymethyl resin (Rink resin, **22**)[88] for the Fmoc/tBu strategy (release of the peptide is carried out with high concentrations of TFA in the presence of the corresponding scavengers). In both strategies, several handles have been developed that are compatible with a broad range of solid supports. Examples include *N*-Boc-(α-phenyl)aminomethylphenoxyacetic acid (BHA handle,

23)[89] for Boc/Bzl and Rink-resin-related handles for Fmoc/tBu, such as the 4-{(*R,S*)-α-[1-(9*H*-fluoren-9-yl)methoxyformamido]-2,4-dimethoxybenzyl}phenoxyacetic acid (Fmoc-Rink linker, linker AM, **24**).[90] For this last strategy, the 5-[4-({[(9*H*-fluoren-9-ylmethoxy)carbonyl]amino}methyl-3,5-dimethoxyphenoxy]valeric acid (PAL handle, **25**),[91,92] which is also the precursor of the backbone attachment strategy (see Section 4.3.1.2.5),[16] is perhaps one of the most convenient ones.

There are two possibilities for the preparation of protected peptide amides: the 5-[(9-{[(9*H*-fluoren-9-ylmethoxy)carbonyl]amino}-9*H*-xanthen-3-yl)oxy]valeric acid (XAL handle, **26**)[93] and its precursor XAL resin **27**[94,95] provides Fmoc/tBu-protected peptide amides by treatment with low concentrations of TFA (~1%), and the 3-nitro-4-aminomethylbenzyl handles **28**, which are photolabile handles and are compatible with both Boc/Bzl and Fmoc/tBu strategies.[65,96,97]

The best method for the preparation of *N*-alkylamide peptides employs 5-(4-{[[(9*H*-fluoren-9-ylmethoxy)carbonyl](alkyl)amino]methyl}-3,5-dimethoxyphenoxy)valeric acid [(R)-PAL handle, **29**],[98] which is derived from the PAL handle. The (R)PAL handle, when used in the same way as PAL, would transfer the alkylamine to the peptide. Similarly, the use of *N*-alkyl XAL[99] or *N*-alkyl Rink resins[100] also gives rise to *N*-alkylamide peptides. The same conditions as those used to release peptides from PAL, Rink, and XAL apply for the *N*-alkyl analogues. The preparation of *N,N*-dialkylamide peptides requires a backbone attachment strategy (see Section 4.3.1.2.5).[16]

For experimental cleavage procedures carried out with strong acids (e.g., liq HF, CF_3SO_3H) and high concentrations of TFA, see Section 4.3.5. The cleavage of peptides from photolabile-based resins is carried out as discussed above for acid handles, and from XAL and its *N*-alkyl-based resins as discussed for highly acid-labile resins.

Ammonolysis of Support-Bound 4-(Oxymethyl)benzamido Esters; General Procedure:[84]
The peptide-resin, after removal of all protecting groups, was dried in vacuo over P_2O_5 for 2 h, swollen in DMF (15 min), and washed with DMF (5 × 30 s), DIPEA/CH_2Cl_2 (1:19, 3 × 1 min), and DMF (5 × 30 s). The drained resin was transferred to a round-bottomed flask and a sat. soln of NH_3 in MeOH (120 mL · g^{-1} of resin) at 0 °C was added, and the flask sealed and shaken for 16 h. The suspension was filtered and the resin was washed with MeOH. The filtrate and MeOH washings were combined and the solvent removed. Well-sealed containers were required as a trace of moisture could result in the production of the acid product as a contaminant.[101] Furthermore, the time of each treatment was critical, since shorter times could lead to the presence of the methyl ester as a contaminant.[102]

A useful vapor-phase ammonolysis has been developed[103,104] for simultaneous multiple synthesis of peptide amides by both the resin and multipin[105] methods. Pins or peptide-resins are deprotected and washed with CH_2Cl_2, DMF, and CH_2Cl_2, and are then air-dried. A bottle of NH_3/THF (3:7 v/v, 50–100 mL) was cooled to −78 °C and placed in a well-greased desiccator together with the multipin pin holders (1–4) or the open vials containing resins (130 mg/vial). The desiccator was clamped shut, partially evacuated (10 s), and then left to stand at 20 °C for 24 h. After this time, the multipin pin holders and the resins were removed from the dessicator and allowed to stand for 30 min. Cleaved peptides were eluted from the pins by immersing them in MeCN/H_2O (4:6) for 3 h, or from the resins by washing with AcOH/MeCN/H_2O (3:4:3, 5 × 1 mL).

4.3.1.2.4 Side-Chain Attachment

An attractive alternative for the solid-phase synthesis of acid peptides containing Asp, Glu, Asn, or Gln as the C-terminal residue involves the side-chain anchoring of aspartate and glutamate (with both α-amino and α-carboxy groups protected) through their ω-carboxy groups onto resins (handles) for acid and amide peptides, respectively.[106,107] In the cases of Asn or Gln residues, this method overcomes the particularly low yields of incorporation of these derivatives, either with free or protected ω-carboxamide side chains, to 4-alkoxybenzyl resins.[108] If the α-carboxy groups of these residues are protected with an orthogonal group (e.g., allyl), selective removal of this protecting group will allow further solid-phase manipulation of the α-carboxy group, such as its activation and subsequent coupling with the

for references see p 706

free N-terminal amino group for the solid-phase preparation of a "head-to-tail" cyclic peptide[109] (see Vol. E 22b, Section 6.8 for details on the synthesis of cyclic peptides).

Side-chain anchoring of protected Asp or Glu to the Phacm resin **5** can take place with both low yields and substantial levels of epimerization at the α-carbon. The best route to overcome these problems is to use the corresponding cesium or zinc salts in conjunction with bromomethylbenzyl resins,[32,110] or through a Mitsunobu reaction using triphenylphosphine and DEAD.[111,112]

Side-chain attachment of Lys, Orn, Ser, and Tyr has been achieved using HMPPA **9**, Nbz **14**, and hydroxymethyl-PSty **4** resins.[112,113] The key step in this approach is the conversion of the hydroxymethyl group into an active carbonate-resin, which can react with the corresponding amino (Lys, Orn) or hydroxy (Ser, Tyr) group present in the side chain. In early work, hydroxymethylated polystyrene resins were reacted with phosgene to yield the corresponding chloroformate derivatives. However, this method has certain problems associated with it, especially that phosgene is a highly toxic gas. In addition, loss of functionalization in the resin has been reported and is probably a consequence of internal carbonate formation. Finally, phosgene is harsh and highly reactive and may be incompatible with certain acid-labile linkers.[2] 4-Nitrophenyl chloroformate and CDI did not give satisfactory results either. The best results (>95%) were obtained when either DSC[113] or bis(pentafluorophenyl) carbonate[112] were used in the presence of a base. These active carbonate-resins allow the incorporation of Boc- and Fmoc-protected Lys and Orn (>95%) through the ω-amino group. Boc derivatives of Ser and Tyr can also be incorporated with moderate yields (>40% and 65% for Ser and Tyr, respectively) to active carbonate Nbz **14** and hydroxymethyl-PSty **4** resins.[113] The carbonate linkage between Ser/Tyr and the handle is not totally stable to piperidine and therefore this strategy is not compatible with a Fmoc/tBu synthesis. Cleavage of peptides from these resins is achieved using the same conditions as those used for cleavage of acid peptides from the hydroxymethyl parent resins [liquid HF, TFA, and photolysis for hydroxymethylpolystyrene-, HMPPA-, and Nbz-based resins, respectively].[112,113]

The hydroxy group of a Boc-protected Ser can be acylated with succinic anhydride and then coupled to an pMeBHA resin **21** (Table 2).[112] The final peptide is released from the resin by ammonolysis. The hydroxy group of protected Tyr can be attached to the hydroxymethyl-PSty resin **4** using the Mitsunobu reaction.[112,114] In this case, treatment with liquid HF affords the corresponding peptide at the end of the synthetic process.

The thiol group of an Fmoc-protected Cys unit can be attached to a XAL handle **26**, which then becomes attached to amino-functionalized supports.[115] This method circumvents the racemization in the synthesis of C-terminal Cys peptide acids. The peptides are released by treatment with TFA.

Side chains of Fmoc-protected Cys, Lys, Orn, Asn, Ser, Thr, Tyr, Trp, Arg,[116] and His[117] can be attached to a Trt(2-Cl)-Cl resin **2**. The yields obtained range from 15% to 55%.

Preparation of SuO-CO_2-HMPPA-Ala-pMeBHA-Resin; Incorporation of the First Amino Acid; General Procedure:[113]

HMPPA-Ala-pMeBHA-resin was swollen with CH_2Cl_2 (5 × 0.5 min) and DMF (5 × 0.5 min). DSC (10 equiv) and DMAP (1 equiv) in DMF (4 mL · g^{-1} resin) were added to the resin under an argon atmosphere and the reaction was allowed to proceed at 25 °C for 30–120 min, with continuous mechanical shaking and occasional bubbling of argon through the suspension. The resin was filtered off, then washed with DMF (8 × 0.5 min) and CH_2Cl_2 (5 × 0.5 min). Similar protocols were used for Nbz-IRaa-pMeBHA and hydroxymethyl-PSty resins.

N^{α}-Fmoc,C^{α}-protected Lys/Orn bearing a free ω-amino group (10 equiv) and DIPEA (20 equiv) in DMF (4 mL · g^{-1} resin) were added to the SuO-CO_2-HMPPA-Ala-pMeBHA-resin under an argon atmosphere and the reaction was allowed to proceed at 25 °C for 5 h with continuous mechanical shaking and occasional bubbling of argon through the suspension. Finally, the resin was washed with DMF (5 × 0.5 min) and CH_2Cl_2 (5 × 0.5 min), and dried under reduced pressure.

Boc-Ser-OAl and Boc-Tyr-OAl (10 equiv) were incorporated in the presence of DMAP (0.2 equiv) in DMF (4 mL · g^{-1} resin) onto either the SuO-CO_2-Nbz-Ala-pMeBHA-resin or the SuO-CO_2-CH_2-PSty for 17 h.

Preparation of PfpO-CO_2-CH_2-PSty Resin; Incorporation of the First Amino Acid; General Procedure:[112]

$HOCH_2$-PSty resin **4** was swollen with CH_2Cl_2 (5 × 0.5 min) and DMF (5 × 0.5 min). Bis(pentafluorophenyl) carbonate (5 equiv) and pyridine (5 equiv) in DMF (8 mL · g^{-1} resin) were added to the resin and the reaction was allowed to proceed at 25 °C for 4 h, with continuous mechanical shaking. The resin was filtered off and washed with DMF (2 × 0.5 min) and CH_2Cl_2 (3 × 0.5 min).

N^{α}-Boc,C^{α}-protected Lys/Orn bearing a free ω-amino group (5 equiv), HOBt (5 equiv), and DIPEA (10 equiv) in CH_2Cl_2 (12 mL · g^{-1} resin) were added to the PfpO-CO-oxymethyl-PSty resin and the reaction was allowed to proceed at 25 °C for 20 h with constant stirring. Finally, the resin was washed with CH_2Cl_2 (3 × 0.5 min), DMF (3 × 0.5 min), CH_2Cl_2 (3 × 0.5 min), MeOH (3 × 0.5 min), Et_2O (3 × 0.5 min), and hexane (3 × 0.5 min), and dried under reduced pressure.

Succinimylation of the Hydroxy Group of Ser and Incorporation onto an Amino-Resin; General Procedure:[112]

Succinic anhydride (3 equiv) and pyridine (2.1 equiv) were added to a 3 M soln of N^{α}-Boc,C^{α}-protected Ser (bearing a free hydroxy group) in CH_2Cl_2 and the mixture was allowed to react for 18 h at 25 °C. Further quantities of succinic anhydride (3 equiv) and pyridine (2.1 equiv) were added and the reaction was continued for 4 d. The mixture was partially concentrated, dissolved in EtOAc, and extracted with H_2O (×2), 1 M HCl (×2), H_2O (×2), and 5% aq $NaHCO_3$. The last extract was collected, acidified, and re-extracted with EtOAc. The EtOAc extract was dried (Na_2SO_4), the solvent removed, and the residue crystallized (EtOAc/iPr_2O/hexane) to obtain the Boc-Ser($COCH_2CH_2CO_2H$)-O-protected product; yield: 78%.

Boc-Ser($COCH_2CH_2CO_2H$)-O-protected (2 equiv), BOP (2 equiv), HOBt (2 equiv), and DIPEA (6 equiv) in DMF (10 mL · g^{-1} of resin) were added to aminomethyl resin and the mixture was shaken for 90 min at 25 °C. The resin was then washed with DMF (3 × 0.5 min), CH_2Cl_2 (3 × 0.5 min), MeOH (3 × 0.5 min), Et_2O, and hexane, and dried.

Incorporation of Tyr through a Mitsunobu Reaction; General Procedure:[112]

DEAD (4 equiv) was added to a mixture of N^{α}-Boc,C^{α}-protected Tyr bearing a free phenol group (4 equiv), DIPEA (12 equiv), and $HOCH_2$-PSty resin **4** in THF (10 mL · g^{-1} resin) at 0 °C. The mixture was allowed to react for 90 min at 25 °C and the resin was then washed with with DMF (3 × 0.5 min), CH_2Cl_2 (3 × 0.5 min), MeOH (3 × 0.5 min), Et_2O, and hexane, and dried.

Incorporation of the First Amino Acid via the Side Chain to the Trt(2-Cl)-Cl Resin; General Procedure:[116]

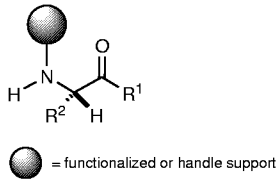
Protected Fmoc amino acids (4 equiv) bearing the free side chain were dissolved in dry CH_2Cl_2 (4 mL · g^{-1} resin) containing a small amount of DMF and the resulting soln was added to the Trt(2-Cl)-Cl resin, which had previously been swollen with CH_2Cl_2 (5 × 0.5 min). DIPEA (10 equiv for Orn, Lys, Cys, Ser, Thr, and Tyr) or TEA (20 equiv for Arg, His, and Trp), and DMAP (0.4 equiv for Ser, Thr, and Tyr) were added and the mixture shaken for 1–2 h. The reaction was terminated by addition of MeOH (1 μL · $μmol^{-1}$ of starting chloride). After 10 min stirring the resin was filtered off and subjected to the following further washings and treatments: DMF, CH_2Cl_2, DMF, piperidine/CH_2Cl_2/DMF (1:10:10, 10 min), DMF, iPrOH, DMF, iPrOH, and MeOH.

4.3.1.2.5 Backbone Attachment

Besides C-terminal and side-chain attachment of the first amino acid derivative onto the solid support, there is a third possibility, which involves attachment of a backbone amide nitrogen to an appropriate handle/support, as shown in Scheme 3.[16] This strategy represents a novel and general concept for the solid-phase preparation of peptides having a variety of C-terminal functionalities, including peptide alcohols, *N,N*-dialkylamides, esters, thioesters, aldehydes, and head-to-tail cyclic peptides.

for references see p 706

Scheme 3 Attachment of a Backbone Amide Nitrogen to a Handle/Support[16]

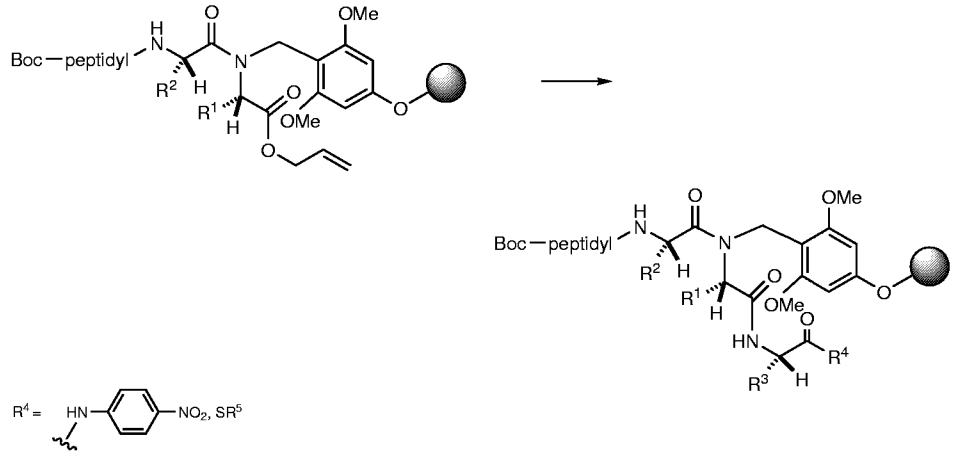


Although there are three different handles described in the literature for this purpose, the backbone amide linker (BAL, **30**) strategy, which is related to the PAL system[91,92] used for the preparation of C-terminal amides, is the most commonly used. The key linker is established by reductive amination with an amino acid residue (or an appropriately modified derivative) of the corresponding tris(alkoxy)benzylaldehyde system; after N-acylation by an appropriately protected second amino acid residue, a dipeptidyl unit is formed that is linked to the support through a backbone amide bond. Further chain growth is then expected to proceed normally. In a photosensitive strategy (see **31**),[118] the first amino acid derivative is attached to a 2-nitrobenzyl support by nucleophilic substitution of corresponding chloro moieties with an amino acid residue. A similar approach is used for the incorporation of the first residue in a BHA strategy (see **32**).[119]

In the BAL strategy, special care should be taken in two steps: the acylation of the hindered BAL-anchored amines by the penultimate amino acid residue, and the deprotection of that penultimate residue when the C-carboxy group contains a good leaving group, such as allyl or an unhindered alkyl alcohol, which can lead to the formation of a piperazine-2,5-dione. In the first case, the use of either the symmetrical anhydride or HATU[120] in the presence of DIPEA using CH_2Cl_2 containing the minimum amount of DMF as solvent is highly recommended. In order to avoid the formation of piperazine-2,5-diones, an acidolytically removable N^{α}-amino protecting group should be used. Thus, the amine end-group of the BAL-anchored dipeptide would remain protonated until the time of coupling, which should be carried out with in situ neutralization.[30]

An interesting extension of the BAL methodology has been described to accomodate C-terminal modifications that are labile to bases, e.g. piperidine (used in Fmoc chemistry), or to circumvent other synthetic challenges.[121,122] This aim is achieved by applying BAL anchoring to the penultimate residue of the target peptide. An orthogonal allyl ester makes it possible to selectively liberate, and then activate, a carboxy group for coupling with the C-terminal residue. During the latter step, a readily epimerizable oxazolonium ion may form by attack of the oxygen from the BAL-amide group on the activated carboxy group. Despite this risk, the use of HATU and DIPEA in CH_2Cl_2 allows the preparation of peptide 4-nitroanilides (Scheme 4) and thioesters with minimal racemization and in excellent yields and purities using Fmoc chemistry.

Scheme 4 Preparation of Peptide 4-Nitroanilides



PALdehyde-IRaa-Resin; General Procedure:[16]
Aminomethyl resin containing an IRaa was washed with DMF (all washings were carried out with 5 mL of solvent per g of resin, 5 × 1 min) and CH_2Cl_2 (5 × 1 min). DIPEA (4 equiv) was added to a soln of 5-(4-formyl-3,5-dimethoxyphenoxy)valeric acid (2 equiv) and HATU (2 equiv) in DMF (4 mL · g^{-1} of resin) and, after a 1-min preactivation period, the entire soln was added to the resin. Coupling was allowed to proceed for 2 h, after which time the resin was only slightly positive to the Kaiser ninhydrin test. The resultant PALdehyde-IRaa-resin was washed with DMF (5 × 1 min) and CH_2Cl_2 (5 × 1 min), and then treated with Ac_2O/DMF (1:9, 20 min), washed with DMF (5 × 1 min), CH_2Cl_2 (5 × 1 min), MeOH (2 × 1 min), and finally dried under reduced pressure.

H-(BAL-IRaa)Leu-OtBu-Resin and H-(BAL-IRaa)Leu-Ooctyl-Resin; General Procedure:[16]
A soln of H-Leu-OtBu · HCl or H-Leu-Ooctyl · HCl (10 equiv) and $NaBH_3CN$ (10 equiv) in DMF (5 mL · g^{-1} resin) was added to the PALdehyde-IRaa-resin. Reaction at 25 °C for 1 h gave H-(BAL-IRaa)Leu-OtBu-resin or H-(BAL-IRaa)Leu-Ooctyl-resin, which in each case was washed consecutively with DMF (5 × 0.5 min), CH_2Cl_2 (3 × 0.5 min), DMF (3 × 0.5 min), piperidine/DMF (1:4, 3 × 1 min), DMF (5 × 0.5 min), and CH_2Cl_2 (3 × 0.5 min), and dried under reduced pressure.

Fmoc-Gly-Phe-(BAL-IRaa)Leu-OtBu-Resin and Fmoc-Gly-Phe-(BAL-IRaa)Leu-Ooctyl-Resin; General Procedure:[16]
Fmoc-Phe-OH (10 equiv) was dissolved in CH_2Cl_2/DMF (9:1, 4 mL · g^{-1} resin) and DIPEA (20 equiv) was added. The resulting soln was added to the H-(BAL-IRaa)Leu-OtBu-resin and coupling was initiated 30 s later by addition of solid HATU (10 equiv) [double coupling, 2 h each, washings with DMF (5 × 0.5 min) and CH_2Cl_2 (5 × 0.5 min) between couplings]. Subsequent peptide chain elongation proceeded normally for the remaining residues using, for instance, Fmoc-Xaa-OH (5 equiv), DIC (5 equiv), and HOAt (15 mg, 5 equiv) in DMF (4 mL · g^{-1} resin) for 1 h per residue, to provide the completed peptide-resin.
Incorporation of Ddz-Phe-OH onto the H-(BAL-IRaa)Leu-Ooctyl-resin was performed as described above. The Ddz group was removed by treatment with TFA/H_2O/CH_2Cl_2 (3:1:96, 6 × 1 min), followed by washing with CH_2Cl_2 (5 × 0.5 min). Fmoc-Gly-OH (10 equiv) and PyAOP (10 equiv) were dissolved separately in DMF (4 mL total · g^{-1} resin) and the two solns combined and added to the resin. In situ neutralization-coupling was then initiated by the addition of DIPEA (20 equiv) and was carried out for 2 h. Peptide chain elongation and cleavage of the free peptide from the resin proceeded normally.

4.3.1.2.6 Other C-Terminal Functionalities

Peptides bearing functionalities other than acid or amide have significance owing to their biological importance,[8] as well as their use as intermediates for convergent strategies.[39] The solid-phase preparation of these peptides is not straightforward owing to the intrinsic characteristics of the C → N direction of the strategy that is mainly adopted for the synthesis. The method most generally used is based on nucleophilic cleavage of the appropriate resins

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(see Sections 4.3.1.2.2.3.1 and 4.3.1.2.3). Another possibility is the use of either side-chain or backbone attachment strategies where the carboxy group of the first amino acid is replaced by the functionality desired, or a precursor of it (see Sections 4.3.1.2.4 and 4.3.1.2.5). A third option is the use of handles or resins designed to anchor the C-terminus of the peptide through the functionality ultimately desired. Thus, the C-modified peptide is released without the need for further transformation or with only minor manipulation necessary. Examples of these strategies include the ones described previously for the side-chain anchoring of amino acids containing an alcohol group (see Section 4.3.1.2.4).

4.3.1.2.6.1 C-Terminal Peptide Alcohols

Besides the methods described above for the side-chain anchoring of Ser and Thr, the 3,4-dihydro-2*H*-pyran-2-functionalized-resin and handle **33**[123,124] (see Table 3) allows the preparation of peptide alcohols by treatment with TFA in the presence of scavengers. Furthermore, the alkylation of Fmoc amino alcohols with a diphenyldiazomethane polymeric resin **34** gives rise to a benzhydryl ether, which can be cleaved with TFA/CH_2Cl_2 (2:98) to give protected peptide alcohols.[125]

Table 3 Handles and Resins for Solid-Phase Synthesis of C-Terminal Modified Peptides

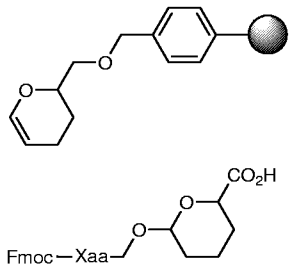
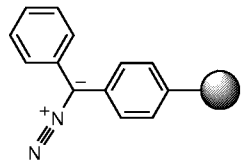
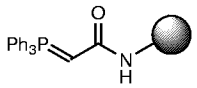
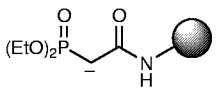
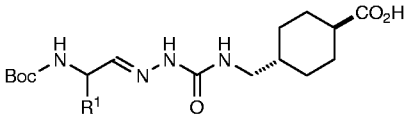
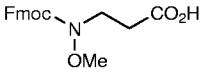
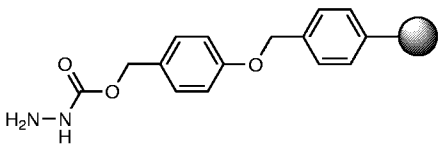
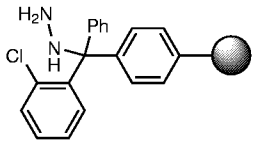
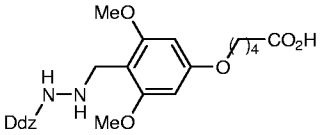
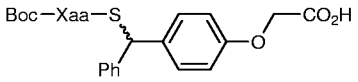
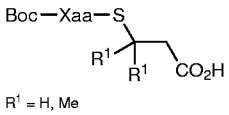
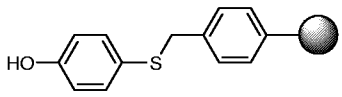
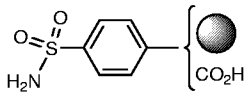
Handle/ Resin No	Name and/or Abbreviation(s)	Structure	C-Terminal Produced	Cleavage Conditions	Ref
33	hydropyran resin		alcohol	TFA, scavengers	[123,124]
34	diphenyldiazomethane polymeric resin		alcohol	dil TFA	[125]
35	Wittig resin		aldehyde	O_3, thiourea	[128]
36	Wittig–Horner resin		aldehyde	O_3, thiourea	[128]
37	semicarbazone handle		aldehyde	formic acid, dil aq acid	[132]
38	Weinreb handle		aldehyde	$LiAlH_4$	[133]

Table 3 (cont.)

Handle/ Resin No	Name and/or Abbreviation(s)	Structure	C-Terminal Produced	Cleavage Conditions	Ref
39	4-alkoxybenzyloxycarbonylhydrazide resin		hydrazide	TFA, scavengers	[33]
40	Trt(2-Cl)hydrazine resin		hydrazide	dil TFA	[41,139]
41	Ddz-hydrazine BAL resin		hydrazide, bis-hydrazide	TFA, scavengers	[140]
42	thioacid handle		thioacid	HF, scavengers	[143]
43	3-sulfanylpropanoic acid handles	 R¹ = H, Me	thioester	HF, scavengers	[146]
44	4-sulfanylphenol resin		-	(1) H_2O_2; (2) nucleophiles	[151]
45	sulfonamide resin/handle (Kenner resin)		-	(1) CH_2N_2 or ICH_2CN; (2) nucleophiles	[152–154]

4.3.1.2.6.2 C-Terminal Peptide Carbamates

C-Terminal peptide carbamates are prepared by reaction of the mixed succinimidyl carbonate of the N^{α}-protected amino alcohol (e.g., Fmoc-Phe-*o*-CO-OSu) obtained from the N^{α}-protected amino alcohol with DSC in the presence of DMAP, without isolation, with an aminomethyl resin (e.g., H-PAL-Nle-PEG-PSty).[113] Standard Fmoc/tBu assembly of the remaining sequence is subsequently performed. Cleavage of the resin with TFA/H_2O (19:1) affords the peptide carbamates with good purity. Use of isolated and purified 4-nitrophenyl carbonate monomers has also been described for the same purpose.[126]

C-Terminal Peptide Carbamates; General Procedure:[113]
A soln of DSC (121 mg, 18 equiv) in MeCN (1.5 mL) and a soln of DMAP (6.4 mg, 2 equiv) in CH_2Cl_2 (0.3 mL) were added successively to a soln of Fmoc-Phe-OH (196 mg, 20 equiv) in CH_2Cl_2 (2 mL) and the mixture was stirred at 25 °C for 4 h. The mixture and DMAP (0.6 mg, 0.2 equiv) were added to the resin, H-AM-Nle-PEG-PSty, and the suspension was continuously shaken under an argon atmosphere at 25 °C for 12 h.

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4.3.1.2.6.3 C-Terminal Peptide Aldehydes

The solid-phase synthesis of peptide aldehydes can be carried out from either protected amino aldehydes or common protected amino acids. While the first kind of method requires prior synthesis of a convenient aldehyde derivative, in the second type the final cleavage of the product is performed in the presence of a reducing reagent.

Peptide aldehydes can be synthesized by backbone attachment of the corresponding amino acetal to a BAL resin **30**.[16] A modification of this method involves the preparation of a peptide alcohol (with the free alcohol group) by side-chain attachment, and after elongation of the peptide chain the alcohol is converted into the aldehyde using sulfur trioxide and pyridine.[127]

Wittig **35** or Wittig–Horner **36** resins can also be used to incorporate N^{α}-protected amino aldehydes, which after removal of the protecting group are suitable for the elongation of the peptide chain using a Boc strategy. At the end of the synthesis the peptide aldehyde is liberated by ozonolysis.[128] Alternatively, the preformed handle containing the unsaturated protected amino derivative can be prepared, purified, and incorporated into an amino-resin.[129]

Peptide aldehydes can also be prepared by copper salt mediated neutral hydrolysis of the corresponding C-terminal thiazolidinyl peptides, which have been elongated on a solid support.[130] A related strategy has been used to obtain peptide aldehydes from an oxazolidine-resin formed by reaction of an N^{α}-protected amino aldehyde with a residue of Thr, with free amino and hydroxy groups, anchored to a solid support.[131] Cleavage is carried out with aqueous AcOH.

A handle based on a C-terminal semicarbazone **37** that is compatible with Boc/Bzl chemistry has also been developed. At the end of the chain elongation, peptide aldehydes are cleaved by treatment with dilute aqueous acid and formaldehyde.[132]

The best way to prepare peptide aldehydes from the corresponding N^{α}-protected amino acids is by using a handle **38**[133] based on the Weinreb amide.[134] This commercial handle allows classical solid-phase elongation of peptides using protected Boc or Fmoc amino acids and, at the end of the synthesis, the peptide aldehyde is formed by reduction and concomitant cleavage from the resin with lithium aluminum hydride. Although the 4-hydroxybenzoic acid handle also allows the preparation of peptide aldehydes by reduction of the resin-bound phenyl ester with lithium tri-*tert*-butoxyaluminum hydride, a mixture of the aldehyde and the alcohol is always formed.[133]

Preparation and Incorporation of Boc-Amino Aldehydes onto Wittig or Wittig-Horner Resins; General Procedure:[128]

Method A: Chloroacetic acid was incorporated onto a pMeBHA resin with isobutyl chloroformate as the activating reagent, followed by reaction of the modified resin with Ph_3P to form the phosphonium salt. BuLi (3 equiv) was then added to the resin at 0 °C in THF to form the phosphorane, and the resin was washed with CH_2Cl_2 and THF. The resin in THF was warmed to 65 °C, the Boc amino aldehyde was added, and the resin shaken for 20 h. Using these conditions, 60% yields, based on the substitution of the Wittig resin, and 7% epimerization were obtained. Avoiding the washing with CH_2Cl_2 and THF led to almost quantitative yields, but with 20% epimerization.

Method B: Diethyl phosphonoacetic acid was incorporated onto a pMeBHA resin with BOP as the activating reagent. The carbanion was generated with DBU/$MgBr_2$ (3 equiv) in DMF at 25 °C, then the Boc amino aldehyde was added. The resin was shaken for 20 h at 25 °C. On using these conditions, 96% yield and 9% epimerization were obtained.

Cleavage of Peptide Aldehydes from Wittig or Wittig–Horner Resins; General Procedure:[128,129]

The peptide-resin was suspended in CH_2Cl_2 at −80 °C and subjected to a stream of ozone. Once the reaction medium turned blue (usually after a few min), the ozone stream was continued for 5 min. Argon was then bubbled through the reaction for 10 min and thiourea (1.2 equiv) in MeOH was added over 10 min. The resin was filtered off, washed with CH_2Cl_2, and the organic soln washed with H_2O, brine, and dried (Na_2SO_4).

Peptide Aldehydes from Thr-Resins; General Procedure:[131]
The Fmoc amino aldehyde in DIPEA/MeOH (1:99) was added to the Thr-resin and the suspension shaken for 2 h at 60 °C. After removal of the Fmoc group, the rest of Fmoc amino acids were incorporated using the DIC/HOBt coupling method. The use of more effective methods should not be undertaken in order to avoid acylation of the secondary amine of the oxazolidine ring. At the end of the synthetic elongation, the peptide-resin was treated with TFA/H_2O (98:2) for 3 min, washed with H_2O, and then cleaved with AcOH/H_2O (1:19) for 30 min at 60 °C.

Cleavage of Peptide Aldehydes from Weinreb Amide-Based Resins; General Procedure:[133]
The peptide-resin was suspended in anhyd THF (5 mL · g^{-1} resin), placed in an ice bath, and $LiAlH_4$ (5 equiv) was added. The mixture was shaken for 30 min at 25 °C. Then 1 M $KHSO_4$ was added and the resin was filtered and washed with CH_2Cl_2. The combined washings were diluted with CH_2Cl_2 and washed with 1 M $KHSO_4$, sat. aq $NaHCO_3$, brine, and dried (Na_2SO_4).

4.3.1.2.6.4 C-Terminal Peptide Hydroxamic Acids

Peptide hydroxamic acids have been synthesized from TFA-labile aminooxy-resins.[135–138] The most convenient resin is probably that derived from the *N*-Fmoc-aminooxy-Trt(2-Cl)-resin, which is easily obtained by reaction of *N*-Fmoc-hydroxylamine with the Trt(2-Cl)-Cl resin **2**[137] (see above for incorporation of protected alcohols into this resin). Similarly, *N*-Fmoc-aminooxy-Rink resin can be obtained from Cl-Rink resin.[138] Furthermore, Wang resins can be first converted into the *N*-hydroxyphthalimide derivative using either Mitsunobu conditions or by displacement of the corresponding mesyl-resin. Removal of the phthalimido group with hydrazine leads to the aminooxy-Wang resin.[135,136] Incorporation of the first Fmoc amino acid should be carried out with an efficient coupling method, such as HATU/HOAt/DIPEA, owing to the poor nucleophilicity of the aminooxy group.[137] Furthermore, in order to prevent TFA-induced hydrolysis of acid-sensitive hydroxamates to the corresponding carboxylic acids, cleavage of the peptide from the resin should be carried out under anhydrous conditions.[136] Cleavage of the C-terminal peptide hydroxamino acids are carried out in a similar way to the parent handles and resins.

4.3.1.2.6.5 C-Terminal Peptide Hydrazides

Protected peptide hydrazides are convenient intermediates because they can be converted easily into the corresponding azides, which can later be coupled to prepare large peptides by a convergent approach.[39] These compounds can be synthesized with the 4-alkoxybenzyloxycarbonylhydrazide resin **39**,[33] Trt(2-Cl)-hydrazine resin **40**,[41,139] or Ddz-hydrazine BAL resin **41**.[140] The last system also allows the preparation of bis-peptide hydrazides. Thus, the first acylation of the hydrazine α-nitrogen is selective because the β-nitrogen is blocked. Following the construction of the first peptide chain, the Ddz group is removed and the acylation of the β-nitrogen will begin the elongation of the second peptide. At the end of the synthesis, cleavage with TFA renders the bis-peptide hydrazide.

Preparation of Ddz-NH-NH-BAL-Resin; General Procedure:[140]
A soln of Ddz-NH-NH_2 (26 mg, 10 equiv) in DMF/AcOH (99:1, 0.25 mL) was added to 2,4-PALdehyde-Ala-PEG-PSty (50 mg, 0.22 mmol · g^{-1}), and imine formation was allowed to proceed for 2 h. The resin was washed with DMF (2 × 2 min) and a soln of $NaBH_3CN$ (7 mg, 10 equiv) in the minimum amount of DMF/AcOH (99:1, 0.25 mL) was added. The reductive amination reaction was carried out for 2 h. The resin was then washed with DMF (2 × 2 min) and CH_2Cl_2 (2 × 2 min).

4.3.1.2.6.6 C-Terminal Peptide Thioacids and Thioesters

C-Terminal peptide thioacids and thioesters are the key intermediates for the preparation of proteins by chemical ligation of unprotected peptides in aqueous solution.[141] The synthesis of peptide thioacids is normally carried out from a benzhydrylsulfanyl resin[142] or its cor-

for references see p 706

responding handle **42**.[143] The preparation of the peptide thioesters is performed using the 3-sulfanylpropanoic acid handle[144,145] or its *tert*-butylsulfanyl derivative **43**[146] linked to pMeBHA-type resins **21**. In principle, these strategies are only compatible with Boc/Bzl chemistries owing to the instability of the thioester linkage to piperidine, and the peptides are released from the resin by HF in the form of the thioacid or propanamide thioester. Recently, a method has been described for the removal of Fmoc, based on 1-methylpyrrolidine/hexamethyleneimine/HOBt in NMP/DMSO (1:1), which allows direct preparation of peptide thioesters using the Fmoc solid-phase method.[147]

Peptide thioacids can be prepared using an oxime resin by cleavage with hexamethyldisilathiane/TBAF.[148] Furthermore, the 3-sulfanylpropanoic acid linker allows the detachment of unprotected peptides using nucleophiles to give the corresponding C-terminal modified peptides, such as esters, thioesters, thioacids, hydroxamic acids, hydrazides, alcohols,[149] and even aryl amides if the cleavage is performed using a silver ion–amine complex.[150]

4.3.1.2.6.7 "Safety Catch" Resins for the Preparation of C-Terminal Modified Peptides

The "safety catch" principle for the solid-phase preparation of C-terminal modified peptides requires the bond between the handle and the first residue to be stable to the normal conditions of SPPS. However, at the end of the synthesis, a chemical transformation of the linker substituent makes the key bond labile to nucleophiles. The 4-sulfanylphenol **44**[151] and the sulfonamide **45** (Kenner)[152–154] resins are two examples of "safety catch" resins. Thus, in the first case, which is only compatible with the Boc/Bzl strategy, once the peptide chain is elongated, treatment of the peptide-resin with hydrogen peroxide converts the sulfide into the corresponding sulfone, which makes the bond labile to nucleophiles. In the second case, N-methylation with diazomethane leads to an N-methylated peptidyl-sulfonamide-resin, from which peptides can also be cleaved by nucleophiles.

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4.3.1.3 Resin Attachment

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The majority of peptides are prepared in the form of C-terminal amides or acids. For the former class, the loading of the first protected amino acids onto the handle (see Section 4.3.1.2), or functionalized resin, is through an amide bond. Although this bond formation can be carried out using similar reagents or active species as used for the construction of the peptide bond (see Section 3), it is often convenient to use the most efficient methods such as onium salts derived from 7-aza-1,2,3-benzotriazol-1-ol (HOAt, **1**; see Table 1), mainly *N*-[(dimethylamino)(3-oxido-1*H*-1,2,3-triazolo[4,5-*b*]pyridin-1-yl)methylene]-*N*-methylmethanaminium hexafluorophosphate *N*-oxide (HATU, **2**) and 7-azabenzotriazol-1-yloxytripyrrolidinophosphonium hexafluorophosphate (PyAOP, **3**),[1] due to the poor nucleophilicity of the amino group present in the handle or functionalized resin (see Section 4.3.1.2.5). If, after the incorporation of the first protected amino acid, the Kaiser ninhydrin test[2] is negative or only slightly positive, the resultant handle-IRaa-resin is acetylated (IRaa = internal reference amino acid, see Section 4.3.1.2). If the ninhydrin test is clearly positive, the coupling step is repeated using the described method.

Table 1 Less-Common Coupling Reagents, Active Species, and Additives Used for the Incorporation of the First Protected Amino Acid onto a Solid Support

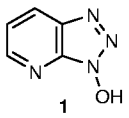
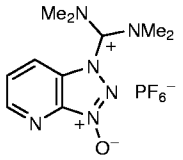
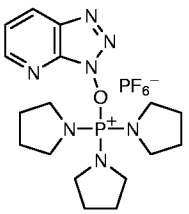
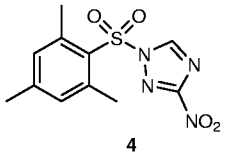
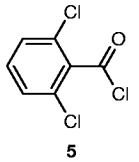
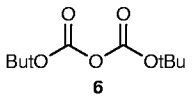
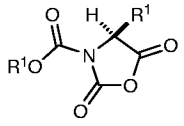
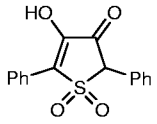
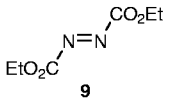
Name (Abbreviation)	Structure
7-aza-1,2,3-benzotriazol-1-ol (HOAt)	 **1**
N-[(dimethylamino)(3-oxido-1*H*-1,2,3-triazolo[4,5-*b*]pyridin-1-yl)methylene]-*N*-methylmethanaminium hexafluorophosphate *N*-oxide (HATU)	 **2**
7-azabenzotriazol-1-yloxytripyrrolidinophosphonium hexafluorophosphate (PyAOP)	 **3**
1-(mesitylsulfonyl)-3-nitro-1*H*-1,2,4-triazole (MSNT)	 **4**

Table 1 (cont.)

Name (Abbreviation)	Structure
2,6-dichlorobenzoyl chloride	 **5**
di-*tert*-butyl dicarbonate (Boc_2O)	 **6**
urethane-protected α-amino acid *N*-carboxyanhydrides (UNCAs)	 **7**
4-hydroxy-3-oxo-2,5-diphenyl-2,3-dihydrothiophene 1,1-dioxide (TDO)	 **8**
diethyl azodicarboxylate (DEAD)	 **9**

4.3.1.3.1 Solid-Phase Ester Formation

An ester bond is typically constructed either by the activation of a carboxylic acid followed by reaction with an alcohol group, or by the displacement of halides or activated alcohol moieties by the carboxylic/carboxylate group.

4.3.1.3.1.1 By Activation of Carboxylic Acids

The alcohol group is less nucleophilic than the amino group, and therefore the carboxylic acid must be activated more efficiently for ester formation than for the corresponding amide formation. A practical activation of optically active α-amino acids requires that the configurational integrity must be maintained. The balance between yield and absence of racemization is often difficult to achieve because the best methods usually involve conversion of the acid into a derivative bearing a good leaving group. Such leaving groups tend to increase the acidity of the α-proton, which favors both enolization and, if applicable, formation of an oxazol-5(4*H*)-one (oxazolonium ion in the case of *N*-alkyl amino acids), and these are the two main mechanisms of racemization (see Vol. E 22b, Section 7.4).[3]

Activation of the carboxylic acid can be carried out either in situ immediately prior to the reaction, with coupling reagents, or before the reaction with the formation of an activated species that is isolated, purified, and characterized. As discussed below, the precise experimental conditions, e.g. organic solvent, preactivation step (when applicable), presence of base, temperature, time, and concentration, are often critical and should be considered carefully.

After the coupling, unreacted hydroxy groups should be blocked. Although they are considered to be poor nucleophiles in comparison with the amino group, they can undergo

acylation on subsequent treatment with the protected amino acids and initiate new peptide chains.[4]

Blocking Unreacted Hydroxy Groups; General Procedures:[5,6]
Hydroxymethyl PSty resin was treated with Ac_2O/pyridine/CH_2Cl_2 (1:2:2, 10 mL · g^{-1} resin) for 30 min at 25 °C and then the resin was washed with CH_2Cl_2, DMF, and CH_2Cl_2.[5]

4-Hydroxymethylphenoxy base resin was treated with BzCl (0.3 M) and pyridine (0.4 M) in CH_2Cl_2 (10 mL · g^{-1} resin) for 30 min at 25 °C and then the resin was washed with CH_2Cl_2, DMF, and CH_2Cl_2.[6]

4.3.1.3.1.1.1 By Coupling Reagents

In recent years there has been an important evolution in the development of new coupling reagents and their application to solid-phase methodology; however, whereas the formerly predominant carbodiimide reagents are being replaced with onium (phosphonium and aminium) salts, carbodiimides, through a symmetrical anhydride species, are still the coupling reagents most commonly used for loading a protected amino acid onto the solid support by ester formation.[7–9]

4.3.1.3.1.1.1.1 With Carbodiimides

For SPPS using the Boc/Bzl strategy *N,N'*-dicyclohexylcarbodiimide (DCC) is the most widely used carbodiimide because the byproduct, *N,N'*-dicyclohexylurea (DCU), is easily removed from the reaction vessel owing to its solubility during treatment with trifluoroacetic acid (TFA), which is used to remove the Boc group. The carbodiimide of choice for syntheses conducted using the Fmoc/tBu strategy is *N,N'*-diisopropylcarbodiimide (DIC) due to the high solubility of the derived urea. It is important to note that DCU can very easily plug the frits of a reaction vessel or a column in the case of batch or continuous flow syntheses. Replacing DIC with DCC has no effect on the activation rate.[10] Carbodiimides, as well as other coupling reagents, should be treated with care; they are acute skin irritants for susceptible individuals. Thus, manipulation in a well-ventilated hood, using safety glasses, gloves and, if possible, a face mask is recommended. Due to its low melting point DCC can be handled as a liquid by gentle warming of the reagent container.[11] Even the most active intermediates derived from carbodiimides, the *O*-acylisourea and the symmetric anhydride, are not efficient enough for a practical loading. The use of 4-(dimethylamino)pyridine (DMAP) as a catalyst for the enhancement of reaction rates has been proposed;[12,13] however, in this case the potential for racemization is increased. If the amount of DMAP is reduced to 10% with respect to protected amino acids and carbodiimide, it is added to the resin when the active species has already formed, and the temperature is kept at 0 °C, the degree of racemization is reduced substantially and, in most cases, this is a convenient method for incorporating the first amino acid onto the resin.[13–15] Special care, as discussed below, should be taken for the incorporation of those protected amino acids, such as derivatives of Cys and His, which are prone to racemize. Another side reaction associated with the use of DMAP is the presence of Fmoc-dipeptides anchored to the resin when less-hindered Fmoc amino acids, mainly Gly, are incorporated using this method.[16] This can be explained on the basis of the instability of Fmoc amino acids in the presence of DMAP.[17]

Acylation of Hydroxymethyl Resins Using Fmoc Amino Acids via Symmetric Anhydrides in the Presence of DMAP; General Procedure:[13]
The Fmoc amino acid (8 equiv) was dissolved in CH_2Cl_2 (5 mL · g^{-1} resin), containing the minimum amount of DMF required for dissolution, at 4 °C, then DIC (4 equiv) was added and the mixture was stored for 5–15 min at 4 °C. The soln was added to the resin containing the free amino group and DMAP (0.1 equiv) was then added. After stirring for 30–60 min, the resin was filtered and washed with DMF. If

the temperature was maintained at 0 °C during the whole process, the racemization level was kept to a minimum, although the reaction time should be prolonged to 20 h.[15]

Acylation of Hydroxymethyl Resins Using Boc Amino Acids via Symmetric Anhydrides in the Presence of DMAP; General Procedure:[18]
The Boc amino acid (8 equiv) was dissolved in CH_2Cl_2 (5 mL · g^{-1} resin) at 4 °C, then DCC (4 equiv) was added and the mixture was stored for 5–15 min at 4 °C. The soln was filtered and added to the resin containing the free amino group. DMAP (0.1 equiv) was then added and, after stirring for 30–60 min, the resin was filtered and washed with DMF. If the temperature was maintained at 0 °C during the whole process, the racemization level was kept to a minimum, although the reaction time should be prolonged to 20 h.

4.3.1.3.1.1.1.2 With Onium Salts and Arenesulfonyl Derivatives

Although onium salt base-coupling reagents, mainly phosphonium and aminium/uronium salts, are being rapidly adapted for the construction of peptides both in solution and solid-phase modes, there is not a great deal of information regarding their performance in the esterification of the first protected amino acid onto a hydroxymethyl resin. The main reason for this is due to the fact that, when onium salts are used, the intermediate species formed are the corresponding benzotriazole or azabenzotriazole esters, and these are less-reactive and more-stable species. A method has been described for incorporation of the Fmoc/trityl (Trt) derivatives of Cys and His using the phosphonium salts derived from HOAt (**1**) and PyAOP (**3**).[19] The reaction is carried out at low temperatures with the aim of taking advantage of the possible presence of the highly reactive acyloxyphosphonium salt, as suggested by Kim and Patel.[20] The mechanism of this transformation is subject to debate since Coste and Campagne[21] have also suggested that this species is very unstable and even at low temperatures undergoes conversion into an active ester.

Arenesulfonyl-1,2,4-triazoles, which were originally used as coupling reagents in phosphotriester oligonucleotide synthesis,[22] have also been successfully applied to the incorporation of the first amino acid onto a hydroxymethyl resin.[23] Thus, the use of 1-(mesitylsulfonyl)-3-nitro-1*H*-1,2,4-triazole (MSNT, **4**; see Table 1) in the presence of *N*-methylimidazole (NMI) is a very useful method for acylating hydroxy groups in polystyrene resins and even for acylating the most unreactive hydroxy groups present in cellulose discs.

Mixed anhydrides produced from the commercially available 2,6-dichlorobenzoyl chloride (**5**, see Table 1) and Fmoc-protected amino acids have proven to be convenient reagents for acylating Wang-type resins.[24] Reaction takes place with good yields and minimal racemization, except for the case of Fmoc-Cys(Trt)-OH (4% D-isomer) and Fmoc-His(Trt)-OH (27% D-isomer). In the latter case, the use of Trt-His(Trt)-OH (<1.0% D-isomer) circumvents the problem.

Acylation of Hydroxymethyl Resins Using PyAOP; General Procedure:[19]
A soln of the Fmoc amino acid (5 equiv) and DIPEA (10 equiv) in the minimum amount of DMF was added to the resin at −20 °C (dry ice/CCl_4 bath). After 5 min, solid PyAOP (**3**; 5 equiv) was added, and the mixture was stirred for 2 h at −20 °C. The cooling bath was then removed and stirring was continued for an additional 2 h at 25 °C. The soln was filtered and the resin was washed thoroughly with DMF.

Acylation of Hydroxymethyl Resins Using MSNT in the Presence of NMI; General Procedure:[23]
The support material was carefully washed with DMF and anhyd CH_2Cl_2 (distilled from P_2O_5). The dry amino acid derivative (freeze-dried from dioxane) (2 equiv) and NMI (1.5 equiv), in a septum-stoppered flask, were dissolved in anhyd CH_2Cl_2, with a few drops of THF if required, to give a 0.1 M soln. This soln was transferred by syringe to another stoppered flask containing MSNT (**4**; 2 equiv) and the mixture was immediately added to the support. Upon completion of the reaction (30–60 min), the soln was filtered and the support was extensively washed with CH_2Cl_2, DMF, and CH_2Cl_2.

for references see p 724

Acylation of Hydroxymethyl Resins Using 2,6-Dichlorobenzoyl Chloride; General Procedure:[24]
The hydroxymethyl resin and the Fmoc-protected amino acid (2 equiv) were shaken in DMF (5 mL · g^{-1} resin) for 15 min at 25 °C. Pyridine (3.3 equiv) and 2,6-dichlorobenzoyl chloride (**5**; 2 equiv) were successively added and the suspension was shaken for 15–20 h. After washing, any remaining hydroxy groups were capped as described in Section 4.3.1.3.1.1.

4.3.1.3.1.1.3 With Di-*tert*-butyl Dicarbonate

Di-*tert*-butyl dicarbonate (Boc_2O, **6**) has mainly been used for the introduction of the Boc group into amino functions;[25] however, it has recently been used for the preparation of esters of N-protected amino acids from alcohols,[26] as well as for the esterification of Fmoc amino acids with Wang resins,[27] and protected Asp and Glu through the ω-carboxylic acid to HO-CH_2-PSty.[28]

Acylation of Hydroxymethyl Resins Using Di-*tert*-butyl Dicarbonate; General Procedure:[27]
The Fmoc amino acid (4 equiv), Boc_2O (**6**; 4 equiv), pyridine (4 equiv), and DMAP (0.3 equiv) in CH_2Cl_2 (5 mL · g^{-1} resin) were added at 0 °C to the hydroxymethyl resin. The mixture was gently stirred at the same temperature under anhydrous conditions for 16–20 h, after which the resin was washed with DMF and CH_2Cl_2.

4.3.1.3.1.1.2 Via Isolated Active Species

The major difficulty in the preparation of an ester bond via isolated intermediates is the prerequisite of first preparing and obtaining an activated derivative. In order to apply this technique, the full range of L-amino acids and nonproteinogenic amino acids prepared from the corresponding carboxylic acids is required. On the other hand, the starting materials implemented in a coupling-reagent-mediated synthesis are simpler since carboxylic acid groups are used. The use of activated species, however, has some important advantages: (1) high reactivity for some derivatives, (2) simplicity of the manipulation such that in some cases additives or base is not required, and (3) simplicity of the byproducts released (e.g., carbon dioxide and the fluoride ion). The most frequently used active species are active esters, urethane-protected α-amino acid *N*-carboxyanhydrides **7** (UNCAs, see Table 1), acid fluorides, and acid chlorides. Symmetrical anhydrides can also be prepared in an analytically pure form;[29,30] however, these reagents are preferably preformed in situ in CH_2Cl_2 immediately prior to incorporation into a peptide chain (*vide supra*).

4.3.1.3.1.1.2.1 Use of Active Esters

4-Hydroxy-3-oxo-2,5-diphenyl-2,3-dihydrothiophene 1,1-dioxide (TDO, **8**; see Table 1) esters[31] have been described which can incorporate the first amino acid onto hydroxymethyl resins with minimal racemization.[32] The use of other active esters, such as pentafluorophenyl esters, requires the use of DMAP, which carries the risk of racemization.

Acylation of Hydroxymethyl Resins Using TDO Esters; General Procedure:[32]
A suspension of the resin in the minimum volume of DMF was shaken gently with a TDO ester of the amino acid[33] (3 equiv with respect to the hydroxy group) and DIPEA (3 equiv) for 2 h at 25 °C. The red soln was filtered and the resin was washed thoroughly with DMF.

4.3.1.3.1.1.2.2 Use of UNCAs

The main advantage of UNCAs **7** (see Table 1)[34] relative to classical NCAs or Leuchs' anhydrides[35] is their greater stability and lack of sensitivity toward polymerization. Upon coupling, carbon dioxide is released, which is a distinct advantage compared to other

methods.[36] These derivatives are crystalline solids that are stable in the absence of water, for extended periods of time, at low temperatures. UNCAs are very soluble in most organic solvents and are most reactive in DMF. Hydroxy-containing resins can be efficiently esterified with high levels of substitution and low degrees of racemization.[37] Excellent incorporation of Fmoc-His(Trt)-OH, a residue prone to racemize under more commonly used conditions, has been observed.[38]

Acylation of Hydroxy-Containing Resins Using UNCAs; General Procedure:[38]
A soln of the UNCA **7** (4 equiv) in toluene and NMM (4 equiv) were added to the hydroxy-containing resin and allowed to react for 15 min for Wang-type resins and 90 min for a Rink resin.

4.3.1.3.1.1.2.3 Use of Protected Amino Acid Halides

Acid Chlorides

Although the first application of acid chlorides in peptide synthesis was reported in the 1930s, the tendency for urethane-protected amino acid halides to convert into oxazol-5(4*H*)-ones, which are less reactive intermediates, has sometimes negated the benefits from employing this kind of active species.[39,40]

The practical use of amino acid chlorides is restricted to amino acid derivatives that do not contain acid-labile protecting groups such as Boc or tBu.[41] Typically, only bifunctional Fmoc amino acids fulfill this requirement. The coupling of these derivatives requires the presence of a hydrogen chloride acceptor. In the presence of a tertiary amine, such as DIPEA or NMM, the corresponding oxazol-5(4*H*)-one is formed. If a hindered base such as 2,6-di-*tert*-butylpyridine is used as a hydrogen chloride scavenger, only a small amount of oxazol-5(4*H*)-one is formed.[40] Fmoc amino acid chlorides have been successfully applied to the incorporation of the first amino acid onto the highly hindered 4-(1′,1′-dimethyl-1′-hydroxypropyl)phenoxyacetyl (DHPP) resin, as well as onto Wang-type resins.[42,43]

Acid Fluorides

Advantages of fluorides relative to the chloro derivatives include their total compatibility with tBu-based protecting groups, as well as their greater stability, even in the presence of moisture, and lack of conversion into oxazol-5(4*H*)-one in the presence of tertiary amines.[41] These derivatives are very soluble in organic solvents (>1 M in DMF), thus facilitating acylation through a high concentration effect. Fmoc amino acid fluorides have been successfully applied to the incorporation of the first amino acid onto hydroxymethyl resins.[44,45]

Acylation of Hydroxymethyl Resins Using Protected Amino Acid Chlorides; General Procedure:[43]
The Fmoc amino acid chloride (5 equiv) was dissolved in CH_2Cl_2/pyridine (6:4, 5 mL · g^{-1} resin) and added to the resin. After 1 h, the soln was filtered and the resin was washed with CH_2Cl_2.

Acylation of Hydroxymethyl Resins Using Protected Amino Acid Fluorides; General Procedure:[44,45]
CH_2Cl_2/pyridine (6:4, 5 mL · g^{-1} resin) was added to a mixture of the Fmoc amino acid fluoride (3–5 equiv) and the resin. After 2 h, the soln was filtered and the resin was washed with CH_2Cl_2. Alternatively, a soln of the Fmoc amino acid fluoride (3 equiv) in either toluene or THF was added to the resin, followed by DMAP (2 equiv). The mixture was stirred for 30 min and the resin was filtered off and washed with toluene or THF and CH_2Cl_2.

4.3.1.3.1.2 Through Displacement of a Leaving Group by the Carboxylic/Carboxylate Function

In the original Merrifield procedure, the ester linkage is prepared from the chloromethyl-functionalized polystyrene resin and the protected amino acid in the presence of a tertiary

for references see p 724

base (TEA). Most protected amino acids can be introduced by this or modified procedures and there is no detectable racemization.[46] Care should be taken with Met because it can undergo S-alkylation by chloromethyl groups.[47] Use of Met(O) circumvents this side reaction. Drawbacks of this method are (1) unreacted chloromethyl sites can give rise to N-alkylation of a free α-amino function belonging to the peptide chain in subsequent steps of the synthetic assembly,[48] and (2) creation of quaternary ammonium sites, which can confer permanent ion-exchange characteristics to the resin. Thus, with the goal of moving away from the use of tertiary amines and of increasing the esterification yields, other counterions have been tested. Of these methods, the most widely used employs the cesium salt.[49] Furthermore, bromide derivatives are more reactive than chloride analogues, and therefore tend to give better results.

A clear example of the superiority of the displacement method compared to acylation with DCC/DMAP for Cys and His derivatives has been shown by Mergler et al.,[15] who studied the incorporation of these amino acids onto SASRIN-type resins (Table 2).

Table 2 Racemization during the Attachment of Selected Fmoc Amino Acids onto SASRIN Resins[15]

Fmoc-Amino Acid	D-Enantiomer Formation (%)	
	Acylation of HO-SASRIN-Resin[a]	Displacement of Cl-SASRIN-Resin[b]
Cys(Acm)	4.0	0.5
Cys(tBu)	4.7	0.3
Cys(Trt)	18.3	2.5
His(Trt)	26.0	0.4
Ile	1.1 (D-*allo*)	0.1 (D-*allo*)

[a] Conditions: Fmoc-Xaa-OH (1.5 equiv), DCC (1.7 equiv), DMAP (0.01 equiv) in CH_2Cl_2/DMF (3:1) at −20 to 0 °C for 20 h.
[b] Conditions: Fmoc-Xaa-O^- Cs^+ (1.5 equiv), NaI (1.0 equiv), DMA, 20 °C, 6 h.

As discussed in Section 4.3.1.2.1.1, incorporation of the first protected amino acid onto Cl-TrtCl-resin[50,51] takes place with negligible racemization, even when Cys and His derivatives are incorporated.[52] Furthermore, the use of Cl-TrtCl-resin precludes additional racemization of the Cys C-terminal as well as formation of the 3-(piperidin-1-yl)alanine adduct, during the piperidine step,[53] and formation of piperazine-2,5-diones during removal of the Fmoc group corresponding to the penultimate residue.[4,54] Special care should be taken during the initial loading (capacity) of the Cl-TrtCl-resin. Thus, if the loading is superior to 1 mmol·g^{-1}, as is the case for most of the commercially available resins of this kind, analysis by HPLC/ES-MS of the crude product shows additional peaks that correspond to terminated sequences. On the other hand, if the synthesis is started with a limited incorporation of the first residue (ca. 50% with a loading of ca. 0.50 mmol·g^{-1}), the quality of the final product is clearly superior without the presence of peaks corresponding to terminated peptides.[4] Remaining Cl groups are easily blocked by reaction with methanol.

These results confirm previous reports in the literature. Thus, loadings >1 mmol·g^{-1} have not been considered optimal for solid-phase peptide synthesis due to the fact that reactions commonly occur on mobile, well-solvated, and reagent-accessible polymer strands throughout the *interior* of the supports and relatively few of the sites occur on surface regions.[46,55] Therefore, high loading increases the risk that some of the reactive sites are not accessible to reagents. Furthermore, Merrifield et al.[56] have recommended that during the chain assembly, the weight of the protected peptide should not be greater than that of the starting polymer. If this situation occurs the properties of the peptide-polymer will be more similar to the protected peptide than to the polymer, and common solvents cannot properly solvate the reactive sites. Cl-TrtCl-resin with a limited incorporation of the first amino acid should be the method of choice for the preparation of protected Fmoc/tBu peptides. As a

general practice, amino acyl esters on the Cl-TrtCl-resin are best stored with the free amino group after Fmoc removal.

Another alternative for incorporation of the first protected amino acid onto hydroxymethyl base resins without activation of the carboxy group is by a Mitsunobu reaction.[57] The esterification of 4-hydroxymethylphenoxy-type resins by Fmoc-protected amino acids using this method was first reported by Sieber,[24] who observed the presence of an unidentified hydrophobic impurity in the amino acids cleaved from the resin; however, independent studies by the groups of Krchnák[58] and Spatola[59] have shown that the esterification of 4-hydroxymethylphenoxy-type PSty and PEG-PS resins using this procedure proceeds rapidly with acceptable to good yields, with practically no racemization and without the presence of additional contaminants in the final product. On the other hand, this method did not provide satisfactory results for cellulose supports.[58]

Incorporation of the First Protected Amino Acid through Displacement of a Leaving Group by the Carboxylic/Carboxylate Group; General Procedures:
Preparation of N^{α}-Protected Amino Acid Cesium Salts:[49]
A soln of the protected amino acid in MeOH/H_2O (9:1) was titrated to pH 7.0 with 20% aq Cs_2CO_3 then taken up in DMF and the solvent removed repeatedly under reduced pressure until dry. The white powdery salt was dried over P_2O_5 for 5 h and used without further purification.

Using Cl-SASRIN-Resin:[15]
The Fmoc amino acid cesium salt (1.5 equiv) and NaI (1.0 equiv) were dissolved in DMA (5 mL · g^{-1} resin) and added to the resin. The resulting suspension was shaken at 20 °C for 6 h. The resin was filtered then washed with DMA, DMA/H_2O (9:1), DMA, and MeOH.

Using Chloromethyl Resin:[49]
The Boc amino acid cesium salt (3 equiv) was dissolved in DMF (5 mL · g^{-1} resin) and added to the resin. The resulting suspension was shaken at 50 °C for 12 h. The resin was filtered then washed with DMF, DMF/H_2O (9:1), DMF, and MeOH.

Using Cl-TrtCl-Resin:[4]
Method A: The Cl-TrtCl-resin (typical nominal loading 1.35 mmol · g^{-1}) was washed with CH_2Cl_2 and a soln of the Fmoc-protected amino acid (0.5 equiv) and DIPEA (0.55 equiv) in the minimum volume of CH_2Cl_2 was added. The mixture was stirred for 5 min at 25 °C, then a further quantity of DIPEA (1.1 equiv) was added and stirring was continued for 2 h. The reaction was terminated by the addition of MeOH (1 μL · 1 μmol^{-1} starting chloride). After stirring for 10 min, the resin was filtered off and subjected to the following further washings and treatments: DMF, CH_2Cl_2, DMF, piperidine/CH_2Cl_2/DMF (1:10:10; 10 min), DMF, iPrOH, DMF, iPrOH, and MeOH.

Method B: For Cl-TrtCl-resin (typical loading 0.5 mmol · g^{-1}) the same procedure as in Method A was used but 2 equiv of the Fmoc-protected amino acid and 2.2 equiv of DIPEA were used.

Using the Mitsunobu Procedure:[58]
4-Hydroxymethylphenoxy base resin was washed with anhyd THF, and a soln of the Fmoc-protected amino acid (4 equiv) and Ph_3P (8 equiv) in THF (5 mL · g^{-1} resin) was added. DEAD (**9**; 8 equiv) in THF (0.6 mL · $mmol^{-1}$ DEAD) was slowly sucked into the reactor, which was then rotated on a tumbler for 1 h. When large batches of resin were esterified, a mixture of the resin, the Fmoc-protected amino acid and Ph_3P in THF was cooled to −15 °C prior to the addition of DEAD. Finally, the resin was washed with DMF.

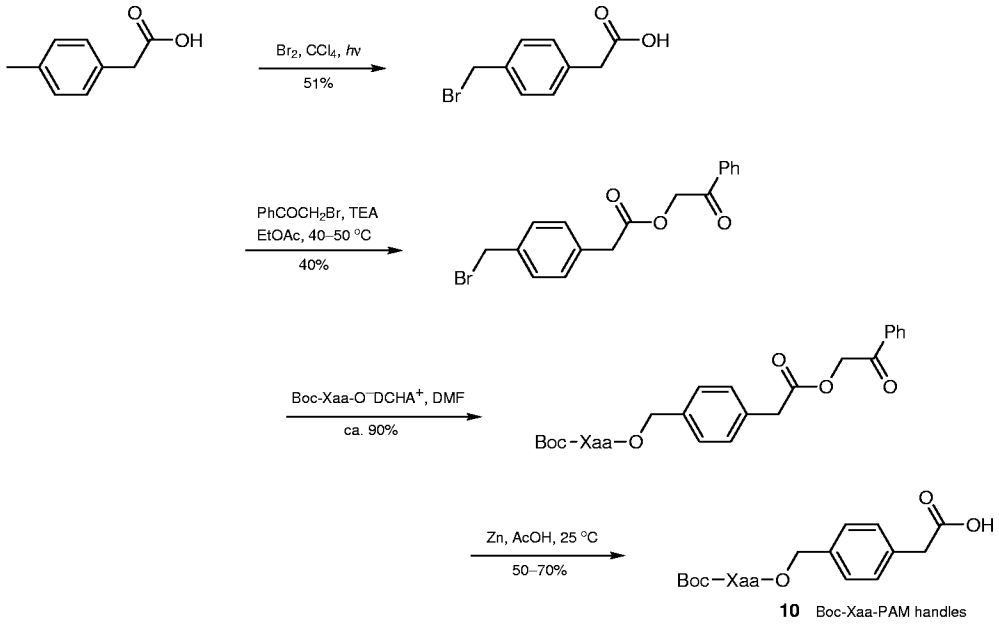
4.3.1.3.2 Preformed Handles

As discussed in Section 4.3.1.2, the use of preformed handles provides control and flexibility to the synthetic process. The two most important preformed handles are the Boc-Xaa-PAM handles **10** (Scheme 1)[60,61] and the Fmoc-Xaa-HMPAA handles **11** (HMPAA is also known as PAC) (Scheme 2),[62] which allow the preparation of acid peptides using the Boc/Bzl and Fmoc/tBu strategies, respectively. In both cases, the key ester bond is formed in solution by nucleophilic displacement of a bromomethyl derivative. In the first case, the temporary

for references see p 724

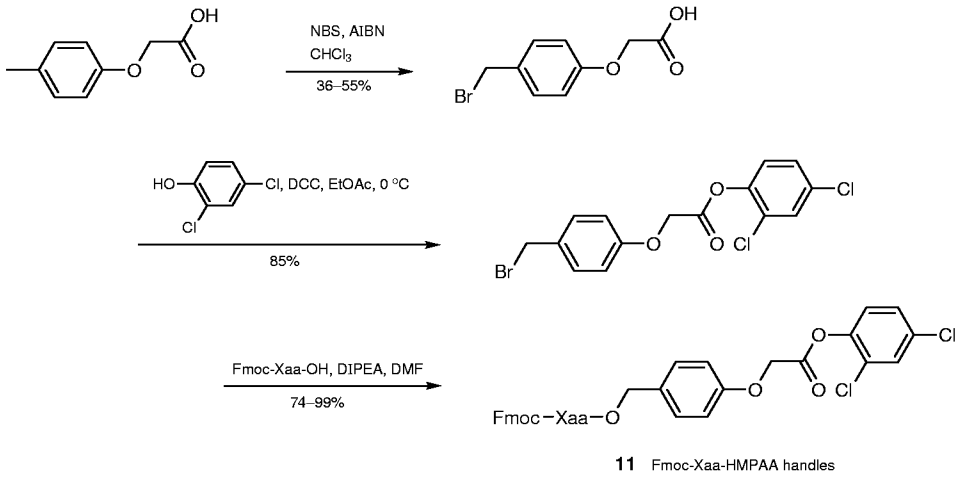
protection of the carboxylic acid is carried out in the form of a phenacyl ester. In the second case, 2,4-dichlorophenyl esters serve the dual purpose of protecting and activating the carboxy group of the handle.

Scheme 1 Synthesis of the Boc-Xaa-PAM Handles



10 Boc-Xaa-PAM handles

Scheme 2 Synthesis of the Fmoc-Xaa-HMPAA Handles



11 Fmoc-Xaa-HMPAA handles

Boc-Xaa-PAM Handles 10:[61]

4-Tolylacetic acid was brominated by the action of Br_2 in CCl_4 at reflux while the reaction was illuminated with a tungsten lamp. This procedure gives 4-(bromomethyl)phenylacetic acid as a white crystalline compound; yield: 51%; mp 177–178 °C. The bromo acid was converted into its phenacyl ester by reaction with bromoacetophenone in the presence of TEA in EtOAc at 40–50 °C. The ester was obtained by crystallization (CH_2Cl_2/petroleum ether); yield: 40%; mp 85–87 °C. Displacement of bromide by dicyclohexylammonium salts of Boc amino acids in DMF occurs cleanly in good yields (ca. 90%) and with high purity. The resulting products were used without further purification. The phenacyl group was removed with Zn in AcOH at 25 °C without cleavage of the Boc or benzyl ester group. Special care has

to be taken in removing traces of Boc amino acid from the penultimate reaction and AcOH from the last reaction. These handles, after recrystallization from EtOAc/petroleum ether (yield: 50–70%), were coupled to amino-resins using common coupling reagents. (In both displacement reactions the amine can be replaced by potassium fluoride dihydrate.[63])

Fmoc-Xaa-HMPAA Handles 11:[62]
4-Methylphenoxyacetic acid was brominated by the action of NBS and AIBN in $CHCl_3$ at reflux to give 4-(bromomethyl)phenoxyacetic acid as a white crystalline compound; yield: 36–55%; mp 121–123 °C. The bromo acid was converted into its 2,4-dichlorophenyl ester by reaction with DCC and the corresponding phenol in EtOAc at 0 °C. The active ester was obtained by crystallization (EtOAc/hexane); yield: 85%; mp 114–116 °C. Displacement of bromide from the active esters by Fmoc amino carboxylates, generated with DIPEA in DMF, occurs cleanly in good yields (74–99%) and with high purity without transesterification being observed. These handles (3 equiv) were easily coupled to amino resins in DMF in the presence of pyridine (3 equiv).

4.3.1.3.3 Racemization

Activation of a protected α-amino acid is always accompanied by the risk of loss of the chirality. This side reaction can be more important if the nucleophile to react with the activated carboxylic acid is rather weak, such as the hydroxy group. As discussed in Section 4.3.1.3.2, incorporation of the first amino acid through a preformed handle strategy, where the reaction conditions can be more easily optimized, or, even better, by a displacement reaction, is advisable. For cases where it is not possible to use any of the above strategies, activation of the protected amino acid should be carried out with caution. Section 4.3.1.3.1 contains a comprehensive list of different activation methods, but even given this diverse range of methods it is difficult to definitively state which is the best. The purpose of this section is to analyze this problem globally, taking into account amino acids, handles, supports, yields, degree of racemization, and experimental convenience of the method. Note that it is considered acceptable when <1% racemization and >75% incorporation yield is obtained.

Incorporation of the first amino acid onto PAM resin should be carried out through either preformed handles,[60,61] or by displacement of the bromo derivative with the cesium salts.[64] The reason for this is that the hydroxymethyl derivative is not very reactive. This lack of reactivity is more important when Asp or Glu derivatives are anchored through their ω-carboxylic acid. Thus, the incorporation of Boc-Asp-OFm with DCC in the presence of DMAP takes place in low yields and with racemization at the α-carbon.[64]

The optimal preparations of acid peptides by the Fmoc/tBu strategy, in regard to the absence of both racemization and formation of dioxopiperazines, are usually performed with Cl-TrtCl-resin.[4,52,54] If 4-alkoxybenzyl resins are used, a careful choice of the method for the incorporation of the first Fmoc amino acid should be made. Methods based on the displacement of halides are usually more convenient.[15,65] The acylation of hydroxymethyl base resins is always accompanied by some racemization. This can be extremely important for Cys and His derivatives. Carbodiimides with DMAP in CH_2Cl_2/DMF at low temperatures give acceptable results for all amino acids, apart from Cys and His.[15] Use of the dichlorobenzoyl chloride method at 25 °C gives good results for all derivatives except for Fmoc-His(Trt)-OH.[24] The incorporation of this residue has been achieved with low racemization through a Mitsunobu protocol[24] and the NCA derivative.[38] Fmoc amino acid fluorides also give good results.[44,45] The best method for incorporation of the first protected amino acid onto the highly hindered DHPP resin involves the use of the Fmoc amino acid chloride.[42] This method is very dependent on the solvent. Thus, it gives excellent results when used with CH_2Cl_2/pyridine (6:4)[43] but very poor results are obtained with DMA containing either pyridine or NMM.[16] Finally, MSNT with NMI is the method of choice for incorporation of the first amino acid onto cellulose supports[23] where other methods, such as the Mitsunobu reaction, fail.[58]

for references see p 724

4.3.1.3.4 Analytical Procedures

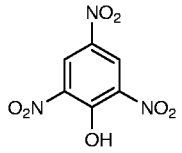
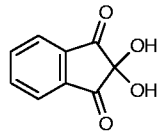
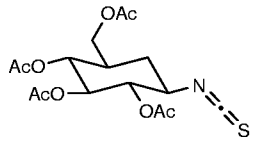
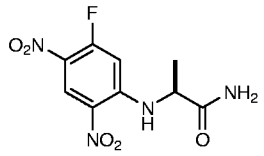
A precise knowledge of the loading is desirable in order to determine the amount of protected amino acid to be used in each coupling step, as well as for assessing the overall yield of the synthetic process.

After incorporation of the first protected amino acid, it is convenient to determine the degree of racemization, especially when residues that are prone to racemize, such as Cys and His, are used. This is advisable as it would be much more demanding to assess racemization at the end of the synthetic process, when other residues are present.

Although spectroscopic analysis of resins does not lead to a quantitative determination of the loading, spectroscopic techniques can be very useful for the characterization of resins, as well as for the identification of possible undesired functional groups arising from incomplete reactions.

Reagents used for the determination of loading and racemization are given in Table 3.

Table 3 Reagents Used for the Determination of Loading and Racemization

Name (Abbreviation)	Structure
picric acid	 **12**
ninhydrin	 **13**
2,3,4,6-tetra-*O*-acetyl-β-D-glucopyranosyl isothiocyanate (GITC)	 **14**
N^{α}-(5-fluoro-2,4-dinitrophenyl)-L-alaninamide (Marfey's reagent)	 **15**

4.3.1.3.4.1 Load Determination

Several methods for load determination have been described in the literature; however, it is important to point out that application of more than one method very often leads to different results.[66] For those resins where incorporation of an amino acid is easily achieved, amino acid analysis of the acid-hydrolyzed amino acid (peptide) resins should be the method of choice because it gives the real amino acid content. The amino acid to be incorporated into the resin should be stable to the acid conditions. Thus, Ser, Thr, Cys, and Tyr should be avoided. After incorporation of the amino acid, washing of the peptide-resin is advisable in order to remove the excess protected amino acid that is not covalently attached to the resin.

Removal of the protecting group, Boc or Fmoc, is also helpful for this purpose. Furthermore, Fmoc amino acyl resins can be partially resistant to the acid hydrolysis depending on the conditions.

Alternatively, if the incorporated amino acid is protected with an Fmoc group, the loading can be determined by quantitative spectrophotometric monitoring following piperidine deblocking. Other methods based on titration of the amino groups usually give higher values because these methods are not selective and all basic sites are titrated. For those resins that contain Cl or Br atoms, both elementary analysis and a modified Volhard titration are also convenient methods.

For determination of the loading when a substituent is present, it is necessary to correct for the weight change during the reaction. Thus, loading of resins are expressed in mmol per gram of resin according to the formula $f = a/(1 - a \times e)$,[67] where f is the loading in mmol per gram, a is the analytical concentration of substituent on the resin in mmol per gram, and e is the equivalent weight of the substituent added.

Acid Hydrolysis; General Procedure:
Amino acyl resins (4–8 mg) were hydrolyzed in propanoic acid/12 M HCl (1:1) for 12 h at 130 °C or 90 min at 150 °C in sealed tubes. For greater accuracy, the acid mixture may contain a known amount of an internal amino acid standard.

UV Determination of Fulvene-Piperidine Adduct; General Procedure:[68]
After incorporation of the Fmoc amino acid, extensive washing of the peptide-resin is advisable in order to remove any excess protected amino acid that is not covalently attached to the resin.

The Fmoc amino acyl resin (4–8 mg) in a 10-mL polypropylene syringe fitted with a polyethylene disc was stirred or shaken in piperidine/DMF (3:7, 1 mL) for 30 min. MeOH (5 mL) was then added and the soln was filtered and diluted with MeOH up to a total volume of 25 mL. The resultant fulvene-piperidine adduct has a characteristic UV absorption at 301 nm (ε = 7800 $M^{-1} \cdot cm^{-1}$). For reference, a piperidine/DMF/MeOH (0.3:0.7:24) soln was prepared and the result was compared to a free Fmoc amino acid soln of known concentration treated under identical conditions. The loading (mmol $\cdot g^{-1}$ or μmol $\cdot mg^{-1}$) = ($A_{301} \times 0.025$ L $\times 10^6$ μmol/mol/7800 $M^{-1} \cdot cm^{-1} \times 1$ cm $\times$ mg of resin).

Picric Acid Titration (Gisin Method); General Procedure:[69]
The amino acyl resin (4–8 mg) in a 10-mL polypropylene syringe fitted with a polyethylene disc was washed (1-mL portions) with CH_2Cl_2 (3 $\times$ 0.5 min), DIPEA/CH_2Cl_2 (1:19, 3 $\times$ 0.5 min), CH_2Cl_2 (3 $\times$ 0.5 min), 0.1 M picric acid (**12**) in CH_2Cl_2 (3 $\times$ 1 min), and CH_2Cl_2 (6 $\times$ 0.5 min). The resin was then washed with DIPEA/CH_2Cl_2 (1:19, 3 $\times$ 0.5 min) and CH_2Cl_2 (7 $\times$ 0.5 min) and the washings retained. The combined soln was diluted with EtOH up to a total volume of 50 mL. The absorbance was measured at 358 nm (ε = 14500 $M^{-1} \cdot cm^{-1}$). For reference, EtOH/CH_2Cl_2 (8:2) was prepared. The loading (mmol $\cdot g^{-1}$ or μmol $\cdot mg^{-1}$) = ($A_{358} \times 0.050$ L $\times 10^6$ μmol/mol/14500 $M^{-1} \cdot cm^{-1} \times 1$ cm $\times$ mg of resin).

Back Titration with Perchloric Acid in Acetic Acid; General Procedure:[66,70]
The amino resin (10–20 mg) was treated with 0.01 M $HClO_4$ in glacial AcOH (5 mL) and shaken for 100 h. Excess $HClO_4$ was back titrated with 7×10^{-3} M potassium hydrogenphthalate in glacial AcOH using crystal violet as indicator.

Quantitative Ninhydrin Test; General Procedure:
Amino groups can be quantified based on their reaction with ninhydrin (**13**, see Table 3) to produce Ruhemann's purple.[2,68,71] Testing for Pro and Cys using this procedure should be avoided because they give a different response with ninhydrin.

Three reagent solns are required; soln A: phenol/EtOH (7:3), soln B: 0.2 mM KCN in pyridine, and soln C: 0.28 M ninhydrin (**13**) in EtOH. The amino resin (4–8 mg) in a test tube was incubated with soln A (75 μL), soln B (100 μL), and soln C (75 μL) for 7 min at 100 °C. EtOH/H_2O (6:4, 5 mL) was added and the sample was vigorously mixed in a vortex shaker. The soln was filtered and diluted with EtOH/H_2O (6:4) to give a total volume of 50 mL. The absorbance was measured at 570 nm (ε = 15000 $M^{-1} \cdot cm^{-1}$). For reference, EtOH/H_2O (6:4) was prepared and the result was compared to a free amino acid soln of known concentration treated under identical conditions. The loading (mmol $\cdot g^{-1}$ or μmol $\cdot mg^{-1}$) = ($A_{570} \times 0.050$ L $\times 10^6$ μmol/mol/15000 $M^{-1} \cdot cm^{-1} \times 1$ cm $\times$ mg of resin).

for references see p 724

Halide Analysis by the Modified Volhard Method; General Procedure:[72,73]
The resin (250 mg) was quaternized by heating in pyridine (2.5 mL) for 1 h at 100 °C, and the suspension was then diluted with 50% aq AcOH (25 mL). The halide was displaced by the addition of HNO_3 (5 mL), and precipitated with a measured excess of standard $AgNO_3$ soln. The AgCl that was formed was coated with toluene and the excess $AgNO_3$ was back-titrated with a standard NH_4SCN soln, using ferric alum [$FeNH_4(SO_4)_2 \cdot 12H_2O$] as an indicator. A red color, due to the formation of $Fe(SCN)_3$, indicates that an excess of SCN^- is present and that the end point has been reached.

4.3.1.3.4.2 Racemization Determination

Usually, the most convenient method for the determination of the degree of racemization is to generate diastereomers and to analyze them by reversed-phase HPLC, which is the most common chromatography technique available in most laboratories. The use of ion-exchange chromatography in a modification of the Manning–Moore procedure[74,75] requires the use of an amino acid analyzer and also the search for special conditions for each case, which is always tedious in this kind of chromatography, and is therefore less appropriate. For the special case of Ile, the same conditions as used for amino acid analysis based on ion-exchange chromatography are also suitable for separation of L-Ile and D-*allo*-Ile and, therefore, the method can be easily used. Diastereomeric dipeptides can be prepared in the solid phase. For an Fmoc/tBu strategy, the dipeptide Z-Phe-Xaa-OH usually allows a clear HPLC separation of both diastereomers; however, a more attractive method is the HPLC analysis of diastereomers generated by reaction of 2,3,4,6-tetra-*O*-acetyl-β-D-glucopyranosyl isothiocyanate (GITC, **14**; see Table 3)[76] or N^{α}-(5-fluoro-2,4-dinitrophenyl)-L-alaninamide (Marfey's reagent, **15**; see Table 3)[77,78] with cleaved and deprotected amino acids. The high reactivity of Marfey's reagent and large extinction coefficient of the adduct, coupled with the speed and resolving power of HPLC, make this method the most sensitive and efficient.[77]

Derivatization of the amino and/or carboxy groups of cleaved and deprotected amino acids with volatilizing reagents provides enantiomers that are suitable for chiral GC analysis. This method requires equipment that may not be available in some laboratories.

GITC-Based HPLC Method; General Procedure:[76]
The Fmoc amino acyl resin (ca. 10 mg) in a 5-mL polypropylene syringe fitted with a polyethylene disc was stirred or shaken in piperidine/DMF (3:7, 1 mL) for 30 min, followed by filtration and washing with CH_2Cl_2. TFA (1 mL) was then added and the mixture was shaken for 1–2 h. The TFA soln was then filtered off and the solvent removed by repeated coevaporation with Et_2O. The residue was dissolved in 50% aq MeCN containing 0.4% (w/v) TEA (pH should be between 8 and 9) to give a final volume of 2 mL. To a 50-μL portion of this soln was added 0.2% (w/v) GITC in MeCN (50 μL). The resulting mixture was allowed to stand at 25 °C for 30 min, and 2 μL of the soln was then analyzed by HPLC [for most amino acids: Develosil ODS-5 C_{18} column with a MeOH/10 mM phosphate buffer (pH 2.8) (55:45) mobile phase, flow rate 0.9 mL/min, detection at 250 nm].

Marfey's Reagent Based HPLC Method; General Procedure:[78]
For a TFA-labile resin, the Fmoc amino acyl resin (2–5 mg) in a 5-mL polypropylene syringe fitted with a polyethylene disc was stirred or shaken in DBU/CH_2Cl_2 (1:99, 1 mL) for 30 min, followed by filtration and washing with CH_2Cl_2. TFA (1 mL) was then added and the mixture was shaken for 1–2 h. TFA was then removed by evaporation with N_2 bubbling or lyophilization, 0.1 M $NaHCO_3$ (400 μL) and freshly prepared 5 mM Marfey's reagent (**15**) in acetone (400 μL) were added, and the mixture was kept at 40 °C for 1 h with frequent shaking. 0.2 M HCl (200 μL) was added to the cooled suspension, and 5–100 μL of the filtered soln was then analyzed by HPLC (for most amino acids: Waters μBondapak C_{18} cartridge with a linear gradient over 50 min of 0.1% TFA in MeCN to 0.1% aq TFA from 1:9 to 6:4, flow rate 2.0 mL/min, detection at 340 nm).

GC Method; General Procedure:[23,79,80]
Unprotected amino acids (0.5 mg) cleaved from the resin were dissolved in 2 M HCl in iPrOH (1 mL), and heated in a sealed tube at 100 °C for 1 h. The solvent was removed from the mixture under reduced pressure, the residue was dissolved in CH_2Cl_2 (0.5 mL), and $(Tfa)_2O$ (0.1 mL) was added. After 1 h at 25 °C, the solvent was removed and the residue was dissolved in CH_2Cl_2, which provided a sample that was ready for GC. GC was carried out on a 25-m Chirasil-Val column (Alltech Associates).

4.3.1.3.4.3 Spectroscopic Analysis

Gel-phase ^{13}C NMR spectroscopy is a very convenient technique for the spectroscopic characterization of functionalized solid supports.[81] Optimal conditions for acquiring standard gel-phase ^{13}C NMR spectra from a variety of resins are well-established; optimal conditions, i.e. maximum swelling, result in reasonably narrow lines for the pendant groups. This method has the additional advantage of not requiring any special instrumentation besides a conventional FTNMR spectrometer and a conventional ^{13}C probe. In addition to ^{13}C NMR spectroscopy, ^{19}F NMR[82] and ^{31}P NMR[83,84] spectroscopy can also be used in those cases where fluorine or phosphorous is present in the handle or in the first amino acid.

Due to strong line-broadening, standard gel-phase NMR spectroscopy cannot be used for the analysis of protons. High resolution magic angle spinning (HRMAS) ^{1}H NMR spectroscopy has recently been developed to cover this gap.[85–87] This method has the limitation that specialized HRMAS probes are required; however, the method has been applied successfully to a variety of solid supports. Amazingly narrow lines are observed for the proton resonances and 2-D versions of HRMAS experiments can be very easily implemented.

IR spectroscopy of functionalized resins also has some value for monitoring the loading reaction, especially in those cases where the linkers have chemical functionalities, such as a carbonyl, with strong IR absorption properties. The standard method consists of measuring difference spectra of loaded resin, relative to the unloaded resin, using either conventional IR or FTIR spectrometers.[61,88,89] Samples of between 2 and 10 mg are used to prepare standard KBr pellets. More recently, FTIR microspectroscopic methods have been described that allow single-bead measurements, a technique that opens the way for analyzing solid-phase reactions during the course of the reaction.[90,91]

Gel-Phase ^{13}C NMR Method; General Procedure:[81–83]
Resins were dried to constant weight under vacuum at 40 °C or at rt. A sample of ca. 300 mg was weighed into a 10-mm NMR tube and the solvent was introduced with a pipet. The use of 5-mm tubes or microcells allows the experiment to be performed with a much smaller quantity of support (ca. 80 and 10 mg, respectively). Swelling of the polymer was crucial for the success of the experiment and the choice of the solvent must be carefully made in each case in order to optimize this property. Complete swelling of the polymer was usually achieved by allowing the resin to equilibrate with the solvent for 30 min after agitation with a vortex shaker; however, ultrasonic agitation was sometimes required to obtain homogeneous samples. The best results were obtained when the amount of solvent in the tube or in the microcell was the exact amount required to swell the resin, i.e. there was no excess free solvent outside the polymer beads. The ^{13}C NMR spectra can then be acquired using a relatively fast pulsing regime in order to take advantage of the fast relaxation properties of the resin. The mathematical weighting of the FIDs before Fourier transformation can be easily optimized in order to discriminate the pendant group signals from the much faster relaxing polymer-backbone ^{13}C atoms. Sensitivity can also be substantially increased using line-broadening routines (at least 1 Hz) during the weighting of the FID.

for references see p 724

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4.3.2 Protection Strategies

J. M. Stewart

The essential primary requirement for successful solid-phase peptide synthesis (SPPS) is achievement of quantitative deprotection and coupling at every step of the synthesis. This is no problem for simple sequences, and was achieved early in the development of solid-phase synthesis (SPS). However, when the peptide contains sterically hindered amino acid residues or an uninterrupted series of hydrophobic residues, either deprotection or coupling may fail dramatically to go to completion, causing extensive failure of chain propagation. Chain termination can arise from inadvertent acylation of the chain by a group not removable by the deprotection reagent being used or by cyclization of N-terminal glutamine residues. Fortunately, reagents and procedures have been developed that will now allow successful SPS of most peptides.

It is axiomatic that ready access of reagents to the reaction site is essential for efficient completion of reactions. In SPS the physical presence and chemical nature of the resin support is a major factor in the environment of the coupling reactions taking place on the resin. In the earliest years of SPPS it became apparent that adequate swelling of the resin was necessary for effective peptide chain growth. The original reagent used for deprotection of Boc groups was 1 M HCl in glacial acetic acid; this reagent swelled polystyrene only poorly, and peptide sequences were soon found that failed to grow in good yield. Replacement of the HCl/AcOH reagent by 4 M HCl in dioxane gave a great increase in swelling of the resin and concomitant increased effectiveness of coupling reactions, although preparation of the reagent was tedious.[1] Later, adoption of TFA in CH_2Cl_2 as a deprotection reagent greatly simplified preparation of reagents. Although use of neat (undiluted) TFA received some support for a time, use of this reagent was not acceptable generally, as it does not swell the resin well and may leave some chains undeprotected. The most commonly used reagent is 25% TFA in CH_2Cl_2, which swells the resin effectively. A 50% TFA concentration may be used, and allows somewhat shorter deprotection times, although this reagent does not swell the resin as well as the 25% reagent.

The degree of cross-linking of the resin is important. To achieve maximum swelling of the resin, the degree of cross-linking of the polymer should be the minimum that will give sufficient physical stability and allow effective filtration. Most synthesis is done on resins cross-linked with 1% divinylbenzene during polymerization. Reactions used for introduction of a functional group into polystyrene for attachment of the peptide chain can also cause additional cross-linking; these substitution reactions must be done under carefully controlled conditions. The user cannot be sure that the resin supplier has not inadvertently introduced additional cross-linking, so each new batch of resin should be tested. A 1% cross-linked polystyrene resin swollen in CH_2Cl_2 should occupy 1.0 mL per 100 mg resin; this can be determined conveniently in a 10-mL graduated cylinder.[2] Polystyrene having 0.4 mmol · g^{-1} of chloromethyl groups will occupy 0.6 mL · 100 mg^{-1}.

It was found that stretches of predominantly hydrophobic amino acids in sequences being synthesized caused failure of chain propagation, particularly when CH_2Cl_2 was used as solvent for coupling reactions on standard polystyrene resin. This caused failure of sequences as simple as decaalanine, and the critical region was the zone about 8–10 residues out from the resin.[3,27] It was then shown that, in such sequences, β-structure formed, causing aggregation of the peptide chains and leading to severe shrinkage of the resin. The aggregation was demonstrated by NMR linewidth broadening of residues in this region.[4] Different approaches for improving this condition have been applied, and the most widely adopted is use of powerful solvent mixtures that break up the β-structure and allow effective resin swelling in these "difficult" cases. A mixture of 10% HFIP in CH_2Cl_2 was used with good

results in Boc synthesis.[5] In a sequence where even acetylation to terminate chains failed, inclusion of HFIP made the acetylation possible. A "magic mixture" solvent has been used in Fmoc syntheses and was shown to be very effective for difficult sequences.[6] It is a 1:1:1 mixture by volume of DMF, NMP, and CH_2Cl_2 containing 1% Triton X-100 detergent. Most effective were the reactions carried out at 50 °C. The deprotection reagent used was 20% piperidine in DMF/NMP (1:1) containing 1% Triton. An interesting approach involves washing the resin and conducting the coupling in the presence of chaotropic salts such as sodium perchlorate and potassium thiocyanate, which effectively break up the β-structure and allow accessibility of coupling reagents to the growing peptide chain.[7] Another approach is incorporation at critical positions in the chain of N-substituted amino acids to break up incipient β-structure. The *N*-Hmb group has been used for this purpose; in Fmoc syntheses it can be removed during final cleavage of the peptide-resin.[8] Acylation of the 2-hydroxy function in this group gives more stable derivatives which can persist during purification of very insoluble peptides and can markedly increase peptide solubility; successful synthesis of the notoriously insoluble amyloid Aβ peptide is a good example. The Hmb group is also useful for inhibiting aspartimide formation in Asp-Gly and Asp-Ala sequences.[9]

After several residues of protected amino acids have been assembled on polystyrene, the swelling property of the resin changes; it may swell less in CH_2Cl_2. To overcome this problem, polyamide resins were introduced by Sheppard and modified by Meldal;[10] these polymers, being polyamides similar in chemical nature to the peptide, should swell in a similar manner. However, if appropriate powerful solvents are used, polystyrene can probably give equally good results.

A critically important problem in peptide synthesis is minimization of racemization during coupling reactions. Racemization of activated amino acid derivatives depends on the degree of activation (reactivity) of the acylating intermediate (a function of the activating reagent used), the structure of the amino acid being activated, and the rate of the coupling reaction. Once effective methods for measuring racemization were developed (see Vol. E 22b, Section 7.4), it became apparent that there is a direct relationship between the degree of activation for coupling and the degree of racemization. Reagents such as DCC, which produce a very high degree of activation, also are more likely to cause racemization, owing to the very low electron density at the reactive center. This low density causes drift of electrons away from the α-carbon, labilizing the C–H bond and facilitating removal of the α-hydrogen as a proton in the presence of base. When a proton returns to the α-carbon, it can approach from either side, yielding a partially or totally racemized amino acid residue. Active esters and symmetrical anhydrides were originally favored for minimization of racemization, but their lower reactivity caused problems in chain elongation. The present solution to this problem is use of phosphonium and uronium salts, which yield high reaction rates with minimal racemization (see Sections 3.7 and 3.8). Aromatic amino acids are particularly susceptible to racemization, as the electron-withdrawing action of the aromatic ring further lowers electron density at the α-carbon. Of all the natural amino acids, phenylalanine is most susceptible to racemization. The nature of the amine protecting group on the activated amino acid is very important. If the amino acid is derivatized by acylation with a normal acyl group, such as acetyl or peptidyl, racemization can be very rapid, again due to the low electron density at the α-carbon of the activated acylamino acid. In this case, intermediate formation of oxazol-5(4*H*)-ones is a cause of racemization. Urethane protecting groups (Boc, Fmoc, Aloc, etc.) are excellent for minimization of racemization; the oxygen atom in the urethane reduces the electron deficit at the α-carbon. The polarity of the solvent used in the coupling reaction is also a factor in racemization; activated amino acids racemize more rapidly in polar solvents, such as DMF, than they do in nonpolar solvents, such as CH_2Cl_2. However, the limited solubility of some amino acid derivatives in CH_2Cl_2 and the much faster rate of many couplings in polar solvents (DMF, NMP) limits use of nonpolar

solvents for general work. The degree of racemization occurring in a coupling reaction is also more or less proportional to the length of time the activated amino acid remains unreacted and subject to base-mediated removal of the α-proton. Originally, the philosophy of SPS was to allow slow coupling reactions extended times for completion, e.g. overnight. When it became apparent that this could cause increased racemization, the philosophy changed to that of allowing a relatively short reaction time, followed by recoupling with fresh, unracemized amino acids. In modern times, with the cost of amino acids and coupling agents being very low relative to the cost of purifying a poor product, use of higher excesses of activated amino acids and short coupling times has become common, and leads to a higher success rate in syntheses. Modern coupling agents such as HATU, HBTU, and HOAt, used in powerful, polar solvents, give rapid coupling with little racemization[11] (see Peptide Bond Formation, Section 3).

Use of DMAP to accelerate coupling reactions is another cause of racemization. DMAP has been most often used to accelerate coupling of the first protected amino acid to hydroxymethyl resins; without catalysis, this reaction is often unacceptably slow. Use of an equivalent amount of DMAP in this situation can cause extensive racemization, especially of phenylalanine; only a catalytic amount of the DMAP additive should be used.[12] As a precautionary measure, optical purity of purchased amino acid derivatives should be ascertained before use by measuring their optical rotation. Optical purity of naturally occurring L-amino acids should not be a problem, since they are generally prepared biologically. However, D-amino acids are prepared chemically by resolution of racemic mixtures, and there are examples of suppliers' mistakes in optical purity.

Practical SPS was introduced by Merrifield in 1964 with use of the Boc α-protecting group with benzyl-related groups for blocking side-chain functions and linking the peptide to the resin,[13] and Boc/Bzl chemistry has been used for countless successful syntheses since that time. Introduction in 1965 by Sakakibara of anhydrous HF for final cleavage of benzyl-related blocking groups and the peptide-resin link was an extremely important step for facilitating practical Boc-based SPPS.[14] Although there are potential hazards associated with use of anhydrous HF, use of proper equipment and exercise of reasonable precautions have made HF the reagent of choice for cleavage of benzyl-related groups, and it has been used in many laboratories for many years without any problems. Large-scale work with HF is awkward, tedious, and hazardous, but for research-scale work it is rapid, clean, and effective.

The application of the Fmoc α-protecting group to SPS in 1978[15,28] had two important features. First, α-deprotection and final cleavage were orthogonal processes, as the Fmoc group is removed by an organic base (usually a secondary amine), and final cleavage of tBu-based blocking groups and peptide-resin links was done with acid, typically TFA. Second, the entire process was carried out with milder reagents, and the limitations of anhydrous HF were no longer a problem. This latter feature has been the major factor in the widespread adoption of Fmoc/tBu chemistry, and has outweighed the higher cost of Fmoc derivatives and certain chemical problems associated with use of Fmoc chemistry.

Another problem associated with chain elongation is loss of the C-terminal dipeptide as a piperazine-2,5-dione (DKP) when coupling the third residue in the sequence.[16] Proline and other N-substituted amino acids especially tend to form the six-membered piperazine-2,5-dione ring, so when they occur in the C-terminal dipeptide, loss can be a serious problem. Since the cyclization is catalyzed by weak acid or base, the solution is to minimize the length of time the deprotected and neutralized dipeptide-resin is exposed to weak acid (Boc amino acid) or base. In Boc chemistry, this is accomplished by making the neutralization step very brief and adding the DCC coupling agent before the Boc amino acid. In certain very bad situations, the activated amino acid has been added to the peptide-resin before the neutralizing base is added, so that there is no exposure to base alone. Simultaneous neutralization–coupling can be used with certain uronium activating agents; see Section 4.3.2.1.3. The problem is much more severe in Fmoc chemistry, where secondary amines are used to

deprotect the Fmoc group and cyclization–cleavage proceeds during the deprotection step. In worst cases, the second and third residues can be added as a preformed dipeptide. Contemporary Fmoc-deprotection procedures use very short exposure to specialized bases, such as 1% DBU, which removes Fmoc groups rapidly and does not catalyze piperazine-2,5-dione formation.[17,29] Another approach for Fmoc synthesis of sequences prone to piperazine-2,5-dione formation is use of the chlorotrityl resin, in which steric hindrance at the site of the C-terminal amino acid attachment prevents piperazine-2,5-dione cyclization.[18]

Unwanted peptide chain termination can be a problem in some circumstances. In Boc/Bzl syntheses, if TFA is used for deprotection and is not totally removed from the resin, it can be activated by diimide coupling agents and form a stable trifluoroacetylated chain which can grow no further. Quaternary ammonium groups can be formed on the resin during attachment of the first amino acid to chloromethyl resins; these bind TFA strongly and may be a cause of trifluoroacetylation. N-Terminal deprotected glutamine chains can cyclize in the presence of weak acids to form a stable pyroglutamyl group, which stops chain growth. This reaction can be prevented by blocking the side-chain amide. Xanthyl groups have been used for this purpose in Boc/Bzl chemistry. The originally used procedure of equilibrating the protected amino acid with the neutralized peptide-resin for several minutes cannot be recommended; the coupling agent should be added immediately to help prevent this cyclization reaction. If anhydrous halogenated solvents are stored in clear glass bottles and exposed to bright light, some loss of HCl and formation of phosgene can occur. If such a contaminated solvent is used for synthesis, chain termination can occur.

In synthesis of peptides containing adjacent sterically hindered or N-substituted amino acids, complete chain propagation may not be obtainable. More accessible sites on the resin may continue to grow well, whereas more hindered sites may fail, but couple in a later step with a less hindered amino acid, e.g. glycine. In these cases, deliberate termination by acetylation can prevent contamination of the product by "omission sequences" which may be more difficult to remove during purification than a shorter acetylated peptide. Both acetylimidazole and acetic anhydride have been used successfully for such acetylation; acetylimidazole is the more reactive of these two reagents.

4.3.2.1 *tert*-Butoxycarbonyl/Benzyl Strategy

Many aspects of the Boc/Bzl SPPS strategy are covered in detail in other sections of this work, and will not be repeated here. For resins, linkers, and substitution methods, see Sections 4.3.1–4.3.8. For discussion of the chemistry of coupling reactions, see Section 4.3.3. Peptide-resin cleavage reactions are discussed in Section 4.3.5. Discussions of protecting groups for amino acid functions are given in Section 2.1, and coupling reactions in general are covered in Section 3. In this section, specific detailed protocols are given for standard varieties of SPPS chain-elongation procedures. Hudson[19] compared various coupling agents in a competition comparison and recommended Boc amino acid/BOP/HOBt/NMM (1:1:1:1.5) as the most effective coupling combination.

4.3.2.1.1 Instrumentation for Manual Solid-Phase Peptide Synthesis

Automatic instruments for SPPS and their use are discussed in Section 4.3.6. Apparatus for manual SPPS will be discussed here, along with notes on the use of the various items.

Probably the one greatest advantage of SPPS is that all of the synthetic operations are carried out in a single vessel after addition of the resin carrying the first (C-terminal) protected amino acid. Thus, the many manipulations of peptide synthesis in solution, with concomitant losses of material, are eliminated. Vessels for SPPS must:

for references see p 739

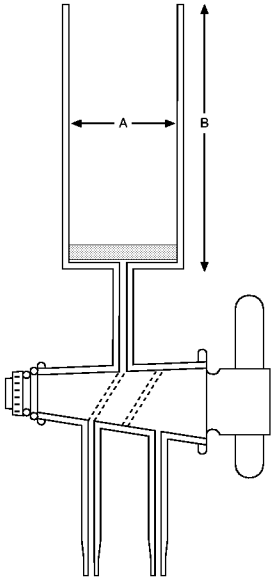
1. Hold the resin beads in a chemically resistant container (glass or inert plastic) and provide for filtration, usually through a fritted glass disk.
2. Provide for addition and removal of wash solvents and reagents.
3. Provide for mixing of the resin beads with solvents and reagents.

CAUTION: *The silane reagent is volatile and corrosive. Use in a good fume hood. Do not breathe the vapor.*

All vessels used in SPPS should be silanized to minimize the tendency of resin beads to stick to the vessel walls. Do not silanize with water-soluble silanizing agents. Treat dry glassware for 15 min with a 10% solution of trimethylsilyl chloride in an anhydrous solvent (toluene, CH_2Cl_2, THF); rinse with dry solvent.

One of the simplest possible vessels is a fritted glass funnel having a filter disk of coarse porosity and fitted with a three-way stopcock in the stem (see Figure 1). A tube supplying low-pressure nitrogen is attached to one arm of the stopcock, and suction to the other. The funnel is clamped in a vertical position. Solvents and reagents are added manually to the top of the funnel, the resin is agitated by a slow stream of nitrogen through the fritted disk, and liquids are removed by turning the stopcock to connect to a waste flask and the vacuum line. A stopper carrying a tube of drying agent is fitted to the top of the funnel to avoid condensing moisture in the resin, which becomes cold during filtration of a volatile solvent. An advantage of this method of mixing is that minimum volumes of liquid can be used during coupling reactions in order to have the highest possible concentration of activated amino acid, while allowing for use of much larger volumes of wash solvents. For 1.0 g of resin, the dimensions are A = 30 mm and B = 50 mm. For a 1.0-mmol run using 2.5 g of resin, A = 40 mm and B = 60 mm.

Figure 1 A Fritted Glass Funnel Fitted for SPPS

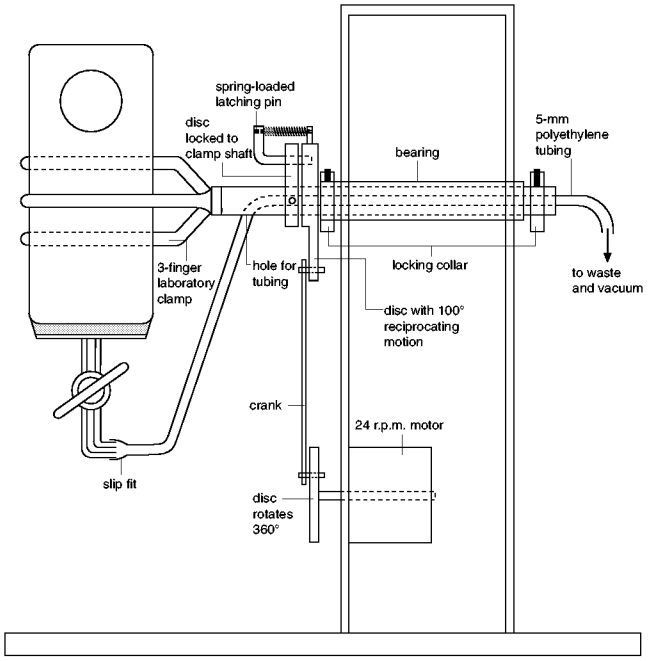


An even simpler device uses a polypropylene clinical syringe of a size to accommodate the largest solvent volume to be used in the synthesis run. A tight-fitting disc of porous polypropylene is cut and forced into the syringe, pushing the disc to the bottom of the barrel. The syringe is loaded with resin, and solvents and reagents are sucked into the syringe from a beaker or flask, followed by enough air to allow mixing. The syringe is clamped in a horizontal position on a slow laboratory mixer and rocked for the needed length of time. Liquid

is forced from the syringe by pushing the plunger. One caution with this device is that some solvents may attack the plastic sealing ring on the syringe plunger and cause possible contamination, which may or may not be significant.

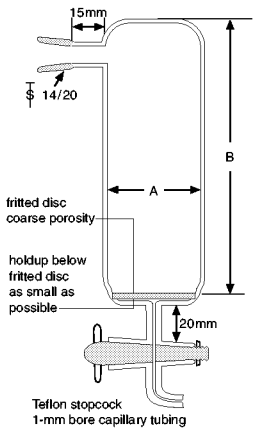
Manual SPPS has classically used gentle mechanical agitation of resin–solvent mixtures by rotation of the synthesis vessel through 100 or 180 degrees of arc.[20] This reciprocating motion, provided by a 24 rpm motor with an eccentric drive arm, allowed polyethylene tubing for liquid removal to be permanently attached to the reaction vessel. A device of this kind, fitted with a vessel for 100° rotation, is shown in Figure 2, and the vessel is shown in detail in Figure 3. In this apparatus, solvents are added from a polyethylene wash bottle or graduated cylinder, and are directed against the inner vessel walls to wash down adherent resin particles. The spring-loaded pin holds the vessel clamp to the driving mechanism and allows rapid disconnection for addition of solvents in a way that they wash all inside walls of the vessel. For filtration, the driving mechanism is stopped with the vessel in the vertical position. For a 0.4-mmol synthesis with 1.0 g of resin, dimension A = 40 mm and B = 75 mm. For a 1.0-mmol synthesis these dimensions can be 45 and 95 mm, respectively.

Figure 2 Simple Apparatus for SPPS (adapted from Stewart and Young,[1] with permission)



for references see p 739

Figure 3 A Synthesis Vessel Designed for 100° Rotation (adapted from Stewart and Young,[1] with permission)



The more complex vessel shown in Figure 4 is designed for 170–180° inversion. This vessel is fitted with permanent tubing connections to the top and bottom ports for fluid addition and removal and for application of nitrogen pressure for filtration, all selected by valves. Vessel dimensions are similar to those above. A possible valving system is shown in Figure 5. Solvents and reagents are stored in bottles under low pressure of nitrogen, and are forced into the reaction vessel through the fritted disc. Protected amino acids and coupling agents are added through the side-arm port, and more concentrated solutions in lower volume may be added.

Figure 4 A Synthesis Vessel for Mixing by 180° Rotation (adapted from Stewart and Young,[1] with permission)

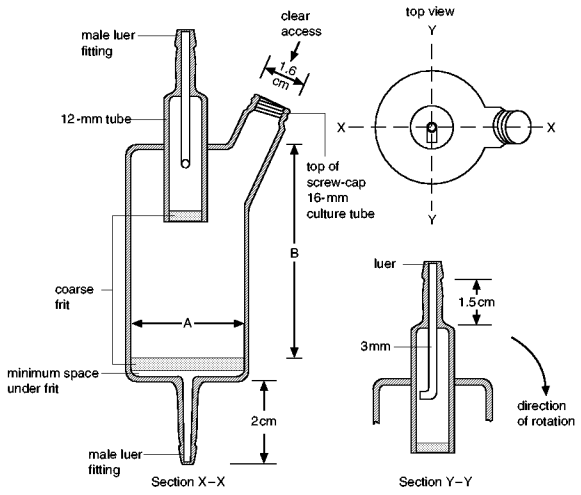
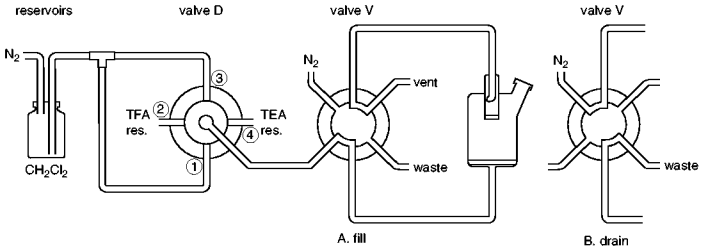


Figure 5 A Possible Valving System for the Vessel in Figure 4 (adapted from Stewart and Young,[1] with permission)



This vessel was used in the first automatic peptide synthesizer designed by Stewart and Merrifield.[21] The volume of solvent/reagent used should always fill the vessel over half full, so that all vessel walls are washed. An unfortunate consequence of this half-full requirement is that coupling reactions are carried out in more dilute conditions than are optimal.

4.3.2.1.2 Coupling Procedures

4.3.2.1.2.1 Standard Protocol for Adding One Residue to the Peptide Chain Mediated by Diimide or Other Coupling Methods

The schedule of operations for deprotection of the Boc group and addition of one residue to a peptide-resin by manual SPPS using TFA deprotection and DCC-mediated coupling is given in Table 1. Explanatory comments on each step are given below.

Table 1 Schedule of Operations for One Step of Chain Elongation by DCC Coupling[a]

Step	Reagent	Volume[b] (mL)	Time[c] (min)
1	CH_2Cl_2 wash (3×)	15	3 (approx)
2	TFA/CH_2Cl_2/indole prewash	15	1 (approx)
3	TFA/CH_2Cl_2/indole deprotection	15	30 (clock)
4	CH_2Cl_2 wash (6×)	15	5 (approx)
5	TEA/CH_2Cl_2 neutralization (2×)	15	2 (approx)
6	CH_2Cl_2 wash (6×)	15	5 (approx)
7	Boc amino acid in CH_2Cl_2[d]	7	0.5
7a	DCC (1.0 M in CH_2Cl_2)	1.0	120 or until complete
8	CH_2Cl_2 wash (3×)	15	3

[a] See text for explanatory description of operations.
[b] Volumes given are for a 0.4-mmol synthesis using 1.0 g of resin.
[c] Times given for washes (steps 1, 4, 6, and 8) are approximate. Add solvent, mix thoroughly with resin, and filter.
[d] Use DMF/CH_2Cl_2 if the Boc amino acid is not soluble in CH_2Cl_2.

The suggested scale of operations is for 1.0 g of resin having 0.4 mmol · g^{-1} substitution of the C-terminal Boc amino acid. At each step, 1.0 mmol of Boc amino acid and DCC are added, giving a 2.5-fold excess of activated amino acid for coupling. DCC is used as a 1.0 M soln in CH_2Cl_2 (206 mg DCC · mL^{-1} of CH_2Cl_2). Although the procedure is given here in terms of DCC-mediated couplings, the modified protocols given below allow application to other coupling methods.

In a real laboratory situation it can be very difficult to keep track of the operations of manual SPPS, especially if more than one synthesis is being done at once. A data sheet such as that shown in Figure 6 can be a great help to keep an accurate record of the synthesis as it is being done. The data sheet is shown filled in as it might be after a hypothetical synthesis of

for references see p 739

leucine enkephalin. For the timed operations, the actual time of day is entered at the start of the operation; in this way interruptions, conversations, etc., will not cause the operator to lose track of the duration of the reaction. For SPPS with automatic synthesizers that do not print a detailed record of the syntheses, a similar data sheet is valuable.

Figure 6 A Record Sheet for Manual SPPS, Illustrating Its Use for Synthesis of Leucine Enkephalin

Data Sheet for Manual Solid-Phase Synthesis

Synthesis of Leu-enkephalin: Tyr-Gly-Gly-Phe-Leu Run No. N-645 Date 3/15/88

Deprotection reagent TFA/CH_2Cl_2 (1:3) Boc-aminoacyl-resin: Lot number N-638, Weight 0.93 g Amino acid Boc-Leu, Total amino acid 0.4 mmol Substitution 0.43 mmol·g^{-1}, Resin type Merrifield

Coupling reagent DCC 1.0 M Amino acid calculation: 0.4 mmol Xaa-resin × 2.5 molar excess = Weight factor (mmol Boc-Xaa) 1.0

Weight of finished peptide-resin 1.08 g Weight factor × Mwt = wt of Boc-aa (mg)

					Reagent	CH_2Cl_2 wash 3 ×	TFA/CH_2Cl_2 wash	Deprot. TFA/CH_2Cl_2	CH_2Cl_2 wash 6 ×	TEA/CH_2Cl_2 (1:9)	CH_2Cl_2 wash 6 ×	Boc-Xaa in CH_2Cl_2	coupler	CH_2Cl_2 wash 3 ×	
					Volume	mL	mL	mL	mL	mL	mL	mL	mL	mL	
Resid. No.	Boc amino acid	Mwt	Wt of Boc-Xaa (mg)	Mfg. and Lot No.	Time → / Date ↓	3 × mix	mix	30 min clock	6 × mix	2 × mix	6 × mix	mix	2 h clock	3 × mix	Comments
2	Phe	265	265	B-698	3/15	III	I	9:20	IIIIII	II	IIIIII	I	10:10	III	K⁻ @ 10 min
3	Gly	175	175	B-1090	3/15	III	I	11:45	IIIIII	II	IIIIII	I	12:40	III	K⁻ @ 10 min
4	Gly	175	175	B-1090	3/15	III	I	2:40 P	IIIIII	II	IIIIII	I	3:30	III	K⁻ @ 10 min
5	Tyr(CIB)	440	440	B-1152	3/16	III	I	9:05	IIIIII	II	IIIIII	I	10:00	III	K⁻ @ 30 min
					3/16	III	I	1:05 P	III						
					EtOH	III	→	TRANSFER							
					CH_2Cl_2	III	→	DRY							

Step 1 (CH_2Cl_2 wash): Add CH_2Cl_2 (15 mL) to the vessel. Agitate to wet all resin with CH_2Cl_2 and remove solvent by suction or N_2 pressure. Repeat 2 times.

Step 2 [prewash with TFA/CH_2Cl_2 (15 mL)]: Add the reagent to the peptide-resin and agitate thoroughly; remove). The reagent is 25% (by volume) of TFA in reagent grade CH_2Cl_2. Add indole (1 mg·mL^{-1}) to the reagent. Prepare the reagent in advance and allow to stand overnight before use. The reagent will have a purple color, owing to condensation of contaminating aldehydes in the TFA with the scavenger indole. The purple compound does not cause any problems in synthesis. Scavenging aldehydes is critically important for sequences containing tryptophan (otherwise you get a purple peptide, or one totally lacking tryptophan!). Do not distil TFA to remove the aldehydes; some anhydride is usually formed during distillation, with serious consequences (chain termination). The prewash prevents excessive dilution of the reagent during deprotection. As discussed above, a 50% TFA/CH_2Cl_2 reagent may be used.

Step 3 [deprotection with TFA/CH_2Cl_2 (15 mL)]: Add a fresh batch of deprotection reagent and agitate for 30 min. CO_2 is formed from decomposition of the Boc group, and should be vented in closed systems. With 50% TFA/CH_2Cl_2, use 15 min deprotection time.

Step 4 [CH_2Cl_2 wash (5×15 mL)]: As in step 1.

Step 5 [neutralization with TEA/CH_2Cl_2 (2×15 mL)]: The neutralization reagent is prepared by adding 1 volume of TEA to 9 volumes of CH_2Cl_2. Use this reagent within 2 d, as it can decompose. Agitate the resin with the reagent to mix thoroughly; filter.

Step 6 [CH_2Cl_2 wash (6×15 mL)]: As in step 4.

Step 7 (coupling): For rapid, accurate, and efficient operation, use one tared vessel (beaker) for weighing all Boc amino acids. Prepare a list of Boc amino acid derivatives with a column for their molecular weights and a column for the combined weight of the weighing vessel

plus 1.0 mmol of each Boc amino acid. Note those derivatives that require DMF for solubility.

Add a soln of Boc amino acid (1.0 mmol, 2.5 equiv) in the minimum amount of CH_2Cl_2 needed to mix the resin well. Agitate briefly. Immediately add a 1.0 M soln of DCC in CH_2Cl_2 (1.0 mL, 2.5 equiv). Mix for 2 h. If the Boc amino acid is not completely soluble in CH_2Cl_2, dissolve the Boc amino acid first in a small amount of pure DMF and then add CH_2Cl_2 to the desired amount. DMF will generally be needed for Boc-Arg(Tos), Boc-Asn (Xan), Boc-Cys(Acm), Boc-Gln, Boc-His(Dnp), Boc-His(Tos), Boc-Hyp, and Boc-Trp, as well as pyroglutamic acid. Keeping the solvent volume low during coupling will increase the effective concentration of the activated amino acid and increase the rate of coupling. Be sure all resin beads are thoroughly equilibrated with the reaction mixture. Coupling can be accelerated by addition of an equivalent amount (1.0 mmol) of HOBt (153 mg), HOAt (155 mg), or HODhbt (163 mg); this is not recommended if His(Tos) is used, as the Tos group will be removed. Use of HODhbt has the advantage that it is intensely yellow in the presence of free amino groups, and disappearance of the yellow color indicates completeness of coupling.[22] If sterically hindered amino acids are not being used and if the sequence is not a "difficult" one, the coupling reaction will usually be complete in 10 min. The Kaiser test for completeness of coupling may be applied at that time (see Section 4.3.2.1.3.1). If the test is negative, proceed to the next step. If the Kaiser test is positive at 10-min coupling time, continue coupling for up to 2 h. If the coupling is still not complete, it is probably best to proceed to Step 8 and immediately recouple by the procedure in Section 4.3.2.1.2.4.

CAUTION: *DCC is very toxic and can produce severe allergic reactions on contact with skin or mucous membranes.*

When transferring DCC from the manufacturer's bottle, it may be preferable to warm the bottle slightly to melt the DCC (mp 35 °C) and use a pipet for transfer. Do NOT ever pipet by mouth! Immediately wash contaminated surfaces with 1 M AcOH.

DIC may be substituted for DCC in this procedure. An advantage is that it gives more soluble urea byproducts than DCC. However, it may be less reactive, and couplings may take longer to go to completion. DIC is a liquid and is much simpler to transfer (DIC $M_r = 126.2$; 0.157 mL = 1.0 mmol).

Step 8 [CH_2Cl_2 wash (3×15 mL)]: As in step 1. If DMF was used in the coupling reaction, the first two washes should contain a comparable amount of DMF. *N,N'*-Dicyclohexylurea is not very soluble in CH_2Cl_2, and can be seen as a white precipitate floating on top of the resin. It can be removed with DMF, or ignored, as it will be dissolved and removed in the next TFA deprotection step.

Some peptide chemists believe that a shrink–swell step at this point will "exercise" the resin and help efficiency of synthesis, particularly when coupling is incomplete and recoupling will be used. To apply shrinking, wash the resin with MeOH (3 × 15 mL) to shrink the resin, then with CH_2Cl_2 (3×) to swell the resin again before recoupling or the next coupling cycle.

When the chain assembly is complete, remove the Boc group from the N-terminus of the peptide by a standard deprotection, using steps 1–4 of the above procedure. After three CH_2Cl_2 washes, wash the resin with EtOH (3×), and use this solvent to transfer the resin to a fritted glass funnel. Then wash with CH_2Cl_2 (3×) to swell the resin in preparation for cleavage of the peptide.

4.3.2.1.2.2 Coupling with Preformed Active Esters

Use the procedure in Section 4.3.2.1.2.1, but replace steps 6–8 with the following:

for references see p 739

Step 6: Wash resin (3×15 mL) with the solvent to be used in the coupling reaction (DMF, NMP or 25% DMSO in DMF).

Step 7: For resin containing 0.4 mmol peptide, add the pentafluorophenyl or other active ester (1.0 mmol) dissolved in the minimum amount of the chosen solvent needed to suspend the resin. Add DIPEA (0.39 mL, 1.5 mmol). The coupling can be accelerated by addition of HOBt (153 mg, 1.0 mmol), HOAt (155 mg), or HODhbt (163 mg) (see Section 4.3.3 for a discussion of active esters and solvents). Mix until the Kaiser test is negative.

Step 8: Wash with the solvent used for coupling (2×), then with CH_2Cl_2 (2×).

4.3.2.1.2.3 Coupling with Active Esters Generated In Situ

CAUTION: *The side product HMPA, formed in BOP couplings, is a suspected carcinogen. Be careful to avoid any contact with the spent reaction solution.*

This procedure is like the previous one (Section 4.3.2.1.2.2) except for step 7 (coupling).

To the filtered resin after step 6, add a soln of Boc amino acid (2.0 mmol), BOP (884 mg, 2.0 mmol), and HOBt (306 mg, 2.0 mmol) in the minimum amount of solvent needed to mix the resin. Add DIPEA (0.39 mL, 2.0 mmol). Mix until the Kaiser test is negative. The procedure is similar for couplings mediated by HATU (1 mmol = 380.2 mg), but in this case the DIPEA should be increased to 4 mmol. Since BOP does not activate TFA for reaction, the entire procedure in Section 4.3.2.1.2.1 can be shortened even further by elimination of a separate neutralization step (step 5). In this case the well-washed resin after TFA deprotection is treated directly with the coupling mixture, and DIPEA is increased to neutralize all TFA salts on the resin. If the Kaiser test does not rapidly become negative, an additional equivalent of DIPEA should be added. The fact that TFA can bind nonstoichiometrically to peptide backbones means that additional DIPEA may be needed in certain cases when this modification is used. A large excess of base should be avoided, as it can promote racemization.

4.3.2.1.2.4 Recoupling When Coupling is Not Complete

This procedure for recoupling uses steps 5 to 8 of the procedure in Section 4.3.2.1.2.1. Use of neutralization before recoupling helps to wash out byproducts of the reaction (dicyclohexylurea, *N*-acylurea, etc.) and will give more effective recoupling. If standard DCC coupling was used initially, recommended is the use of a more vigorous reagent for recoupling, such as BOP/HOBt or HATU, using steps 5–8 of the procedures in Sections 4.3.2.1.2.2 or 4.3.2.1.2.3.

4.3.2.1.2.5 Using a Chaotropic Salt for "Difficult" Sequences

Salts such as $NaClO_4$ and KSCN are "chaotropic" because they break up peptide secondary structure, which is a major cause of failure of propagation of sequences containing many residues that promote β-structure of the peptide on the resin. Some difficult sequences have been synthesized successfully by using chaotropic salts.[7] $NaClO_4$ gives better results than KSCN. Chaotropic salt may also be used with symmetrical anhydride or BOP-mediated couplings. The following modification of steps 6–8 of the procedure in Section 4.3.2.1.2.1 should be used.

Step 6: After neutralization, wash resin with CH_2Cl_2 (3 ×15 mL), then with a 0.4 M soln of $NaClO_4$ (M_r = 122.46) in DMF/CH_2Cl_2 (1:1, 3 ×15 mL). Dissolve the salt first in the DMF and then add the CH_2Cl_2.

Step 7: Carry out the coupling reaction in the presence of 0.4 M $NaClO_4$ in the same solvent mixture.

Step 8: Wash with DMF/CH_2Cl_2 (1:1, 3 × 15 mL), then with CH_2Cl_2 (3×).

4.3.2.1.3 Tests for Completeness of Coupling

4.3.2.1.3.1 Kaiser Test for Non-Proline Residues

Of several possible methods to estimate the amount of free amino groups remaining on the peptide-resin during coupling, the Kaiser test has probably received widest application.[23] It can, however, give false negative results in "difficult" sequences where there is serious aggregation or steric hindrance. The Kaiser test is not applicable to N-terminal secondary amine groups (e.g., proline); use the chloranil test for these.

Equipment: glass test tubes (75 mm long); heating block for tubes held at 100 °C.

Kaiser reagents:

Reagent A: dissolve KCN (33 mg) in H_2O (50 mL). Dilute this soln (2 mL) with pyridine (98 mL).

Reagent B: dissolve ninhydrin (500 mg) in BuOH (10 mL).

Reagent C: dissolve phenol (80 g) in BuOH (20 mL). Alternate: 80% liq phenol (80 g) and BuOH (10 mL).

Reagent D: EtOH/AcOH (1:1, by vol).

Store these reagents in the refrigerator.

With a Pasteur pipet, remove a few resin beads from the reaction mixture, transfer to a 75-mm test tube, and wash with reagent D (3×), decanting carefully after each wash. Wash once with EtOH. Add 3 drops each of reagents A, B, and C. Heat at 100 °C for 5 min. Examine beads against a white background. The Kaiser test is negative if the beads are white and the soln is yellow. A very weak blue color of the beads and soln may indicate 99% complete coupling. Dark blue beads and soln indicate that coupling is no more than 75% complete. N-Terminal asparagine, aspartic acid, serine, and some sequences give a brown color. Proline does not react or may give yellowish colors.

4.3.2.1.3.2 Isatin Test for Proline Residues

When a Boc amino acid is being coupled to a N-terminal proline, hydroxyproline, or other secondary amino group, the Kaiser test is negative or gives an inconclusive brownish color. Such N-terminal residues give a good blue color with isatin.[24]

Stir isatin (4 g) with BzlOH (120 mL) for 2 h at rt; filter. Dissolve Boc-Phe (5 g) in the filtrate (100 mL). If stored in an amber bottle, the reagent is stable at rt. Wash solvents: EtOH/AcOH (1:1), EtOH, acetone.

Remove a small sample of peptide-resin beads and place in a 75-mm test tube. Wash as for the Kaiser test with EtOH/AcOH, then wash once with EtOH. Add the isatin reagent (5 drops). Heat at 100 °C for 5 min, then cool briefly. Wash the resin beads several times with acetone, decanting each time, until the soln is clear and colorless. Observe the color of the beads against a white background. If the beads have any color, from light green to blue, coupling is incomplete.

4.3.2.1.3.3 Chloranil Test

The chloranil monitoring test can be modified for either N-terminal proline or non-proline residues.[25] Its sensitivity is similar to that of the Kaiser test.

Reagents: chloranil (sat. soln in toluene), acetaldehyde, acetone.

Wash resin beads in a 75-mm test tube as for the Kaiser test. For N-terminal proline or hydroxyproline residues, add chloranil (2 drops) and acetone (2 drops). For all other N-terminal residues, add chloranil (2 drops) and acetaldehyde (2 drops). Mix and allow to stand at rt for 5 min. A blue color indicates N-terminal free amino groups.

for references see p 739

4.3.2.1.4 Alternate Deprotection Method

A stronger and more rapid deprotection reagent for Boc groups is made by adding a very low concentration of methanesulfonic acid (MsOH) to TFA.[26]

The reagent is 0.01 M MsOH (0.065 mL, 1 mmol) and 0.1 M TFA (0.074 mL, 1.0 mmol) in CH_2Cl_2. Use for steps 2 and 3 of the procedure in Section 4.3.2.1.2.1. The deprotection time in step 3 is 5 min. This reagent is more vigorous than TFA, and may cause excessive loss of the C-terminal residue from the resin if it is labile (e.g., Gly, Ala). PAM resin may be more appropriate for long sequences if this deprotection reagent is to be used.

4.3.2.1.5 Acetylation to Terminate Peptide Chains

This can be perfomed with either acetylimidazole or acetic anhydride.

Acetylation with Acetylimidazole: For 1.0 g resin containing 0.4 mmol peptide, suspend the peptide-resin in CH_2Cl_2 (10 mL) and add a soln of acetylimidazole (550 mg, 5 mmol) in CH_2Cl_2 (5 mL). Mix until the Kaiser test is negative. Wash with CH_2Cl_2 (3×).

Acetylation with Ac_2O: Suspend the resin in DMF. Add a soln of Ac_2O (0.47 mL, 5 mmol) and TEA (0.70 mL, 5 mmol) in DMF. Mix until Kaiser negative. Wash with DMF (3×).

4.3.2.1.6 Purification of Dimethylformamide

The DMF used in SPPS coupling reactions must be anhydrous and free of dimethylamine. Spectroscopic quality DMF may be anhydrous, but may contain excessive amounts of dimethylamine. Every new batch of DMF received should be tested with 2,4-dinitrofluorobenzene (DNFB) for the presence of dimethylamine as follows.

Reagent: a soln of DNFB ($1\ mg \cdot mL^{-1}$) in 95% EtOH.

Mix equal volumes of the reagent and the DMF to be tested. Let stand at rt for 30 min. Read absorbance at 381 nm. The blank ($0.5\ mg \cdot mL^{-1}$ of DNFB in 95% EtOH) has $A_{381} = 0.2$. The DMF test should not have A_{381} more than 0.1 or 0.15 higher than the blank.

If the dimethylamine content of the DMF is only slightly higher than recommended, store the DMF in a brown bottle over 4 Å molecular sieves for a few days, with occasional swirling, and retest. If the DMF then passes the test, bubble the solvent with N_2 and store in the cold and dark. Warm to rt before opening the bottle to avoid condensing atmospheric H_2O in the cold solvent.

If the level of amine is still too high, it can be lowered by distillation of the DMF. Add an equal volume of dry benzene (**CAUTION:** *carcinogen*) to the DMF and distil at atmospheric pressure until the benzene is removed. Distil the DMF under water-aspirator vacuum, using a drying tube in the line to prevent back-flow of water vapor from the aspirator. The amine content, as indicated by the DNFB test, should now be satisfactorily low.

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4.3.2.2 9-Fluorenylmethoxycarbonyl/*tert*-Butyl Strategy

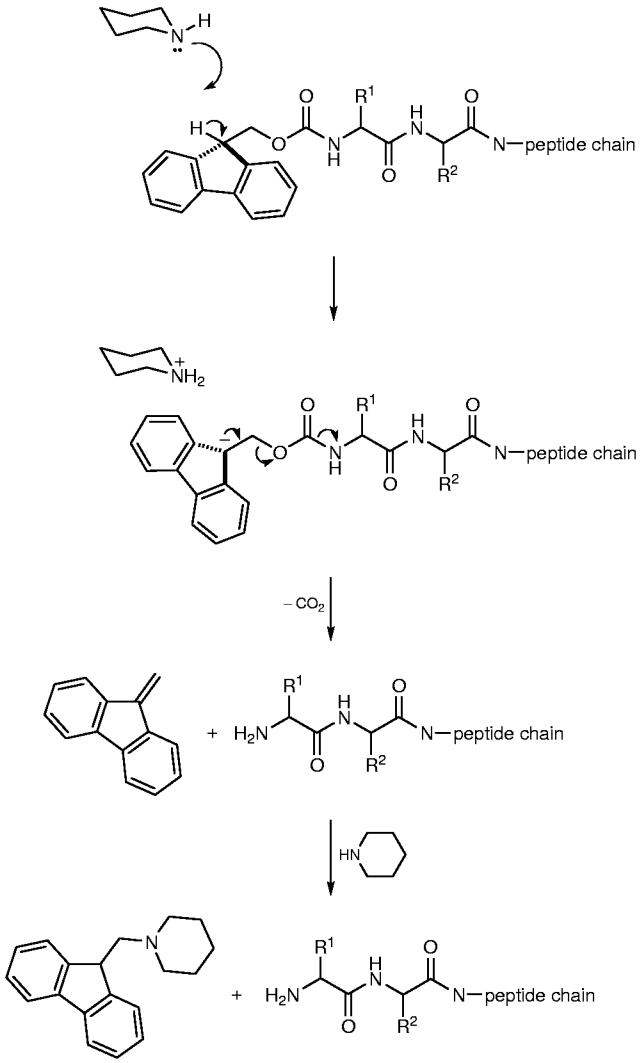
E. Atherton and D. A. Wellings

After the introduction of solid-phase peptide synthesis in 1963 by Merrifield[1] the method enjoyed a great deal of attention. By the early 1970s the rush of enthusiasm for the method had somewhat waned. This was primarily due to non-chemist users who had not achieved the results expected from a procedure still in its infancy. Also, experienced peptide chemists were discovering pitfalls in the procedures due to complications associated with driving reactions to completion in the heterogeneous conditions employed and side reactions associated with the support and the chemistry of assembly and cleavage.

This prompted the search for different supports and milder chemistries to further advance the brilliant Merrifield invention. Two groups with these objectives in mind, independently embarked on this course. Sheppard's group in Cambridge initially pursued ideas that were proposed at the 11th European Peptide Symposium in 1971.[2] Firstly looking at new solid supports for peptide synthesis[3] and then seeking alternative chemistries. Meienhofer's group, in the USA, also perceived the need for milder methods of solid-phase synthesis and were simultaneously seeking different techniques to the Boc/Bzl system. Both groups reached the conclusion that the orthogonal Fmoc/tBu combination provided a milder alternative and published almost simultaneously.[4,5]

The Fmoc group had been suggested earlier as an α-amino protecting group by Carpino in 1970,[6] however, the Fmoc group had not been adopted widely by peptide chemists since the recommended deprotection conditions were exposure to liquid ammonia. The discovery that the Fmoc group was extremely labile to secondary amines, by a base-catalyzed elimination process[7] (Scheme 1), under conditions that did not effect tBu protection resulted in adoption of the Fmoc/tBu combination for solid-phase synthesis.[8] The fact that Fmoc amino acids are crystalline, prepared in high yield, and are stable, very important commercial factors, added to the attraction of the procedure.

Scheme 1 Base-Catalyzed Elimination of the Fmoc Group[7]



The method is relatively mild, versatile, and extremely flexible and it has considerably widened the scope of the solid-phase concept. Earlier milder procedures had been sought through the investigation of α-amino protecting groups that were more acid labile than the Boc group. Thus increasing the differential acid lability between α-amino group protection and benzyl-based side-chain protection and linkage to the solid support. Notable amongst these were the Nps,[9] Ddz,[10] and the Bpoc[11] groups. However, the introduction of the base-labile Fmoc group for α-amino protection with side-chain protection based on the tBu group and equivalently acid-labile linkage agents to the solid support gave an orthogonal combination which has received worldwide prominence.

This combination together with protecting groups such as Aloc,[12] allyl,[13] and the Dde[14] groups, has widened the synthetic options considerably for solid-phase peptide synthesis. This is now the method of choice for making most side-chain-modified peptides, for example phospho- and glycopeptides by solid-phase techniques.

Early synthetic endeavors, notably the comparative synthesis of β-endorphin[15] and the first published solid-phase synthesis of gastrin,[16] illustrated the usefulness of the method.

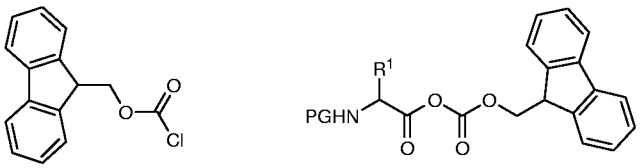
The Fmoc group lends itself particularly to the solid-phase procedure and is used almost exclusively in this mode. This is because in the deprotection reaction the initially formed dibenzofulvene moiety is efficiently scavenged by the secondary amine, usually piperidine, used for cleavage and is then washed away effectively from the solid support, whereas in solution phase the resulting moieties are not readily removed. However, the group still finds occasional utility in solution phase.[17]

The Fmoc/tBu method has been adopted universally and can be used in any operational mode, batch, or flow and can be utilized for any peptide or modified peptide objective. The methodology has been comprehensively described in books and reviews.[18–20,103]

4.3.2.2.1 Preparation of 9-Fluorenylmethoxycarbonyl Amino Acids

The fact that Fmoc amino acids are relatively simple to prepare, are obtained in high yield in a crystalline form, and are stable, has added to the attraction of the method. Several reagents have been proposed for the addition of the Fmoc group.[19] Initially Carpino[7] proposed the chloroformate **1** and this was used predominantly until it was discovered that there was a potential side reaction occurring to a small extent with amino acids such as Gly and Ala. This resulted in the formation of Fmoc dipeptides and in some cases tripeptides were encountered.[21–23] A plausible mechanism for the formation of these products is via the mixed anhydride **2** which can react at either activated carboxy to give the expected product or the unwanted dipeptide (Scheme 2).

Scheme 2 Fluorenylmethyl Chloroformate and the Fmoc/Amino Acid Mixed Anhydride



1 **2** PG = Fmoc, H
R^1 = amino acid side chain

This unwanted side reaction can be eliminated by the use of Fmoc-OSu, which was proposed by several groups[21,22,24] and is now the long preferred option for the introduction of the group. A typical reaction protocol for the synthesis of Fmoc-Trp-OH on a five-mole scale is given.

A similar procedure is employed for other amino acids. However, caution should be exercised when acidifying Fmoc amino acids with sensitive *tert*-butyl- or trityl-based side-chain protection. It is advisable in these cases to use dilute acid, e.g. HCl (1.5 M), aqueous $KHSO_4$, or an organic acid such as citric acid. Yields and physical properties of Fmoc amino acids have been tabulated previously.[19,20]

Fmoc-Trp-OH:[90]
To a mixture of 2 M aq Na_2CO_3 (4.5 L) and acetone (9 L) was added H-Trp-OH (1.02 kg, 5.0 mol) and Fmoc-OSu (1.65 kg, 4.9 mol) with rapid stirring. Further 2 M aq Na_2CO_3 (1 L) was added to this and the mixture stirred at rt overnight. EtOAc (9 L) was then added with continuous stirring and the mixture acidified with 6 M aq HCl (~5 L) to pH 2. Caution was exercised at this point due to the effervescent release of CO_2. To this mixture a further portion of EtOAc (6 L) was added with stirring. The stirrer was then stopped and the mixture allowed to settle into two layers. The organic top layer was isolated, washed with acidified H_2O and then brine. The organic layer was then transferred to a clean, dry glass vessel and dried ($MgSO_4$, 0.5 kg). The EtOAc soln was then filtered off and concentrated down to an oil at a temperature of 35 °C. EtOAc (4 L) was added to the oil, the oil redissolved, and the soln allowed to stand at rt. Petroleum ether (bp 60–80 °C) was added to this with stirring until the product started to crystallize, the mixture was left overnight, collected by filtration, and dried on trays in a vacuum oven set at 35 °C to give the crystalline product; yield: 1.96 kg (94%); mp 176–180 °C; $[\alpha]_D$ −26.8 (*c* 1, DMF).

4.3.2.2.2 Side-Chain Protection

Utility of the Fmoc group for α-amino protection in solid phase allows the side-chain reactive functions of trifunctional amino acids to be protected by orthogonal, base-stable, preferably mildly acid-labile protecting groups. Paramount amongst these are the groups based on a tertiary geometry, hence the Fmoc/tBu method. The Boc group is used for side-chain amino functions, tBu esters for acidic groups, and tBu ethers for alcoholic groups. Generally, the Boc group is eminently satisfactory for Lys and His, tBu esters for the side chains of Asp and Glu (although in some peptides Asp protected in this manner yields aspartimide derivatives), and tBu ethers for Tyr, Thr, and Ser side chains. These derivatives are freely available commercially from a number of suppliers. Many alternatives have been proposed, for instance in special circumstances Dde[25] protection of the ε-amino group of Lys allows one additional degree of orthogonality since it can be removed with a dilute organic solution of hydrazine. In addition, the mildly acid-labile groups Mtr[26] and Mmt[27] have been used. Also in circumstances where an increase in polarity is needed, to influence peptide chain assembly, the Tfa group has found application.[28]

Although Boc can be utilized for His imidazole side-chain protection many other proposals have been made. Trt[29] is an alternative to Boc that is widely used and the Mtt and Mmt[30] derivatives have been proposed. There has always been a problem regarding the tendency of His to undergo racemization and the π-nitrogen, which is closer to the α-carbon atom, has been implicated in the racemization mechanism. Unfortunately, the τ-nitrogen is the one that is routinely protected and consequently, depending on the mode of activation and the sequence being assembled, racemization can occur at His residues. An effort to overcome this was the introduction of the Bum group.[31] This was placed on the π-nitrogen by first protecting the distal τ-nitrogen. This suppresses racemization but unfortunately the formaldehyde released on deprotection can have an adverse effect on other residues if it is not effectively scavenged. This is especially so with peptides containing Cys when thiazolidine derivatives can be generated.[32]

Cys poses special problems in terms of side-chain protection and the group of choice depends on the utility of the peptide once it is cleaved from the solid support. The Acm, tBu, and *tert*-butylsulfanyl protecting groups were explored in early syntheses of oxytocin.[33] All gave acceptable results and it was concluded that the choice of sulfur protection for Fmoc-Cys-OH will depend upon the relative polarity of the side-chain protecting group in relation to the overall hydrophobic/hydrophilic nature of the peptide being assembled. These groups are orthogonal to the Fmoc/tBu/Bzl combination and consequently if cyclic disulfides are required, choices can be exercised as to when the S—S bridge is made. An example is the successful synthesis of conotoxin G1, which has two disulfides; a combination of Acm and *tert*-butylsulfanyl groups are employed and the sulfur bridges made in solution.[34] The Acm group has been employed in a synthesis of a peptide related to the HIVgp41 protein in combination with Boc/Bzl chemistry using an N-terminal Fmoc amino acid and on-resin cyclization following Acm group removal.[35] More recently mildly acid-labile protection of Cys has grown in prominence with the introduction of the Trt[36] and other groups such as Xan,[37] Tmb,[38] Mmt,[39] and tris(4-methoxyphenyl)methyl.[40]

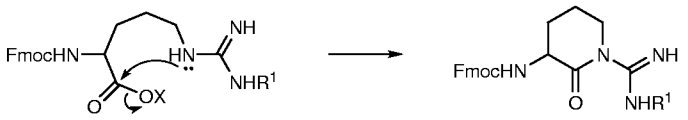
Arg, with its highly basic side-chain guanidino group, has received some special attention. Initially the Mtr[41] group was used in a synthesis of conotoxin,[34] but more recently the Pmc[42] and the Pbf[43] variations on this sulfonyl-based protecting group have received attention. The order of acid lability for this series is Pbf>Pmc>Mtr. However, caution must be exercised during the cleavage and deprotection process since electron-rich residues, such as the Trp indole nucleus are susceptible to back addition of the released group if it is not effectively scavenged.[44,45] This side reaction is suppressed but not completely eliminated by using the groups Sub, MeSab, or the Suben groups.[46] These groups are cleaved under mild TFA conditions, but they do not fully protect the guanidine group against unwanted

for references see p 752

acylation reactions during chain assembly. However, if deprotection is carried out at low temperature, back addition to Trp residues seems to be prevented.

A common side reaction with activated derivatives of Arg is the well-documented intramolecular δ-lactam formation (Scheme 3).[47] This consumes the Arg derivative, therefore lowering the concentration of the activated species in the acylation reaction. Consequently when Fmoc-Arg-OH derivatives are activated they must be brought into contact with the free amino group on the solid support as quickly as possibly so as to maximize the acylation concentration.[34]

Scheme 3 Intramolecular γ-Lactam Formation with Activated Arginine Derivatives[47]



R^1 = guanidine protecting group
X = active ester

Other more acid-labile groups have been tried for the side-chain protection of Arg and notable amongst these are the Boc,[48] Adoc,[49] and Trt groups.[50] However, they have not come into prominent utility probably due to incomplete protection leading to side reactions which can ultimately lead to Orn containing peptides[51] and/or side-chain instability. Also because of the highly basic nature of the guanidino group it has been suggested that protonation of the side chain can be effective.[52] In this instance Fmoc-Arg-OH free base or the hydrochloride can be used. The method of choice for activation uses carbodiimide in the presence of an excess of HOBt promoting in situ protonation to reduce the effects of intramolecular lactam formation. Before acylation with the next amino acid, reprotonation of the guanidine side chain is essential. This is done using a 10% HOBt solution in DMF following the wash cycle after piperidine treatment.[52]

Normally the side chains of Asn and Gln can be left unprotected during incorporation of these residues. However, with longer synthetic objectives, where cyclization to aspartimide derivatives[53] or dehydration of the amide side chain is a perceived problem,[54] these side-chain amide groups can be protected. Many acid-labile side chains have been suggested[20] but Trt[55] is currently in vogue. Insolubility and side-reaction problems with Fmoc-Asn-OH and Fmoc-Gln-OH are largely overcome when using the Trt group for protection of the amide functionality. Generally the Trt group can be removed very easily by mild acid treatment, but when Asn is at the N-terminus then it may be more sluggish to remove.[56] The Dmcp group has been suggested as an alternative in these cases.[57]

Protection of the Trp side chain in Fmoc synthesis is not always essential. However, if it is in close proximity to an Arg in the sequence where sulfonyl-based protection is employed then protection of the indole nitrogen with Boc is often recommended.[58] This eliminates the back addition of the sulfonyl-based protecting group since the indole is protected as a carbamic acid during the TFA treatment. Exposure to mildly acidic aqueous conditions releases the free indole nucleus.

4.3.2.2.3 Peptide Bond Formation

Most of the commonly applicable methods for activating the carboxy group of Fmoc amino acids to form amide bonds in solid-phase synthesis have been explored.[18,20,103] Generally speaking, most methods are suitable but, in practice, relatively few methods are routinely used especially when scale-up of Fmoc/tBu solid-phase procedures are of prime consideration. Methods employed can be grouped into four classes:

1. in situ activation,
2. preformed symmetrical anhydrides,
3. pre-activated derivatives, and
4. coupling agents

Initially, normal diimide-based[4] and symmetrical anhydride couplings[5] were employed successfully. The anhydride gives unambiguous acylation reactions and when freshly prepared, and employed at optimum concentration, they are clean and efficient coupling agents. The fact that two molecules of the activated amino acid are consumed for every amino group resin-bound make them a less economic choice. However, they can play a role in special circumstances, for example in the coupling of Cys to avoid racemization problems.

Methods employing pre-activated species have also been utilized, but they need to be highly activated, such as the active esters OPfp,[59] ODhbt,[60] and OTdo.[61] The latter two are useful for color-based monitoring of acylation reactions. Also included in this class are Fmoc amino acid *N*-carboxyanhydrides,[62] acid chlorides,[63] and acid fluorides.[64] The latter can also be formed in situ using TFFH.[65]

Coupling agents such as the phosphorus-based BOP[66] and PyBOP[67] and uronium salts including HBTU and TBTU[68] are the most widely used today. Other more specialized variations on these coupling agents exist, such as HATU[69] for more hindered reactions. However, the procedures used to carry out the couplings are generally the same.

Peptide Formation:[90]
The following procedures are based initially on coupling to 5 g of support with a loading of 1 mmol · g^{-1}. The wash volumes following each acylation reaction will depend upon the type of resin used and the state of the peptide-resin. However, for batchwise manual operation a rough guideline for the wash volume is to use an amount equivalent to 50% of the solvated bed volume of the peptide-resin.

In Situ Activation:
Fmoc amino acid (12.5 mmol) and HOBt (25 mmol) were added to the reactor followed by the minimum amount of DMF required to make the resin just mobile to agitation. DIC (16.7 mmol) was added and the coupling reaction was allowed to proceed with mild agitation and monitored to completion (monitoring of peptide bond formation is described in Section 4.3.2.2.5). When the coupling was complete the reaction soln was drawn off and the resin washed with DMF (10 × 1 min).

Preformed Symmetrical Anhydrides:
Fmoc amino acid (30 mmol) was transferred to freshly distilled CH_2Cl_2 (60 mL) and dissolved with stirring at ice-bath temperature (in some instances DMF was added dropwise to aid dissolution of the Fmoc amino acid). DCC (15 mmol) was added to the stirred soln and the mixture stirred for 15–20 min in an ice bath. The fine filtrate of dicyclohexylurea was removed by filtration and washed with CH_2Cl_2. The solvent was removed from the filtrate by rotary evaporation at 30–35 °C to leave an oil. The oil was dissolved in DMF (the minimum required for the type of resin used) and added immediately to the drained solid support following the previous washing procedure. The coupling reaction was allowed to proceed with mild agitation and monitored to completion (monitoring of peptide bond formation is described in Section 4.3.2.2.5). When the coupling was complete the reaction soln was drawn off and the resin washed with DMF (10 × 1 min).

Pre-activated Derivatives:
Fmoc amino acid OPfp ester (15 mmol) and HOBt (15 mmol) were added to the reactor followed by the minimum amount of DMF required to make the resin just mobile to agitation. The coupling reaction was allowed to proceed with mild agitation and monitored to completion (monitoring of peptide bond formation is described in Section 4.3.2.2.5). When the coupling was complete the reaction soln was drawn off and the resin washed with DMF (10 × 1 min).

Coupling Agents:
Fmoc amino acid (12.5 mmol) and TBTU (11.88 mmol) were placed in a beaker. DMF (minimum required to obtain soln) was added followed by DIPEA (25 mmol) and the mixture allowed to react for 3 min. This pre-activated amino acid soln was added to the swollen resin and further DMF added, if required, until the resin was just mobile to agitation. The coupling reaction was allowed to proceed with mild agitation and monitored to completion (monitoring of peptide bond formation is described in Section 4.3.2.2.5). When the coupling was complete the reaction soln was drawn off and the resin washed with DMF (10 × 1 min).

for references see p 752

4.3.2.2.4 9-Fluorenylmethoxycarbonyl Removal

The discovery that the Fmoc group was labile to secondary amines[7] stimulated its proposal for use in conjunction with acid-labile tBu-based side-chain protection for solid-phase chemistry.[8] Removal of the Fmoc group is thought to proceed via an E1cB mechanism initiated by proton abstraction from the 9 position of the fluorene ring (Scheme 1). The degree of instability to various bases is well-documented in the literature,[18,20,70,103] but the more routinely applied procedures using a solution of piperidine in DMF result in a reaction with a half life of less than 6 seconds. Use of the same base in less-polar solvents such as dichloromethane reduces the rate of reaction. Table 1 summarizes some of the conditions tested for removal of the Fmoc group with various bases, the table is ordered with the most reactive conditions at the top.[19,20]

Table 1 Comparison of the Conditions Tested for the Removal of the Fmoc Group from Fmoc-Val-OH with Various Bases[19,20]

Base	$t_{1/2}$ for Removal of Fmoc	Ref
20% piperidine/DMF	7 s	[19,20]
5% piperazine/DMF	20 s	[19]
50% morpholine/DMF	60 s	[19]
50% Cy_2NH/DMF	2100 s	[19]
50% DIPEA/DMF	10.1 h	[20]

The removal of the Fmoc group can often be incomplete in cases where the growing peptide chain becomes aggregated during assembly.[71] This leads to deletion sequences in the final product and many methods including addition of solvents such as DMSO[72] and the use of alternative base mixtures containing DBU and piperidine have been used to improve the situation.[70,73] However, caution should be exercised when using DBU because of its ability to promote side reactions, particularly aspartimide formation. Addition of chaotropes known to disrupt peptide aggregation has also shown a minor effect, but heating the reaction mixture to 45 °C produced a more clearly defined improvement.[71] TBAF[74] has also been used for Fmoc removal and is particularly useful for solution-phase peptide synthesis using Fmoc amino acids.

Standard Procedure for the Removal of Fmoc Groups:[18,103]
Piperidine/DMF (twice the swollen bed volume, 20%) was added to the resin and the mixture agitated for 3 min. The solvents were drawn off and this procedure repeated for a further 7 min. The piperidine/DMF soln was drawn off and the resin washed with DMF (10 × 1 min).

4.3.2.2.5 Monitoring

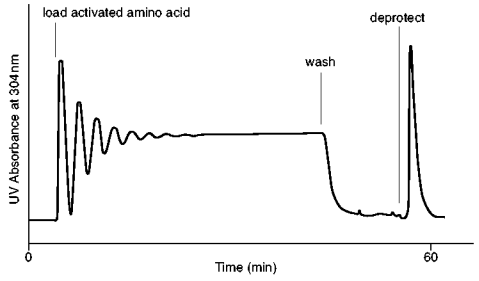
The monitoring of solid-phase synthesis in a real time quantitative manner has always been a goal since invention of the process. Many qualitative color tests exist, principal amongst these being the ninhydrin-based Kaiser test.[75] The Kaiser test is less sensitive when the N-terminal is a secondary amine and for this reason many other tests have been applied.[18,20] Although a quantitative Kaiser test has been proposed,[76] it has never achieved prominence as a routinely applicable method and the results are not usually available in time to influence progress of the reaction.

The introduction of the low-pressure continuous-flow mode of operation,[77] that is particularly suited to the mild Fmoc chemistry led to other noninvasive monitoring techniques. These include the usage of ODhbt[60,78] and OTdo active esters.[61] When ODhbt esters are employed the released HODhbt ionizes with the amino groups attached to the

support and forms a yellow anion which can be spectrometrically monitored at 440 nm directly on the resin. The disappearance of this color indicates completion of reaction leading to the option for truly automatic procedures.[79] The OTdo esters are particularly useful for monitoring ester formation when anchoring the C-terminal amino acid to the solid support.[61] Amongst many other noninvasive techniques is the use of bromophenol blue.[80]

The UV absorbing properties of the Fmoc group offers a unique advantage over the Boc group for monitoring both acylation and N-terminal deprotection when combined with the low-pressure continuous-flow mode of operation.[77] For the first time the prospect of quantitative monitoring became a possibility.[81] Passing the eluent from the synthesis column through a narrow pathlength flow cell of a UV detector allows a spectroscopic recording of the reaction cycle (Figure 1). Attempts to quantify the amount of Fmoc amino acid taken up during acylation and the amounts of dibenzofulvene adduct released on deprotection were made through refined mathematical models.[82] However, because of interfering factors including the presence of more than one UV absorbing species in reactions and fluorescent effects the accuracy required to determine 100% reaction at each stage could not be achieved. This technique is the closest that the technology has come to feedback control in real time.

Figure 1 UV Trace of a Complete Cycle in Fmoc Solid-Phase Flow Synthesis[90]



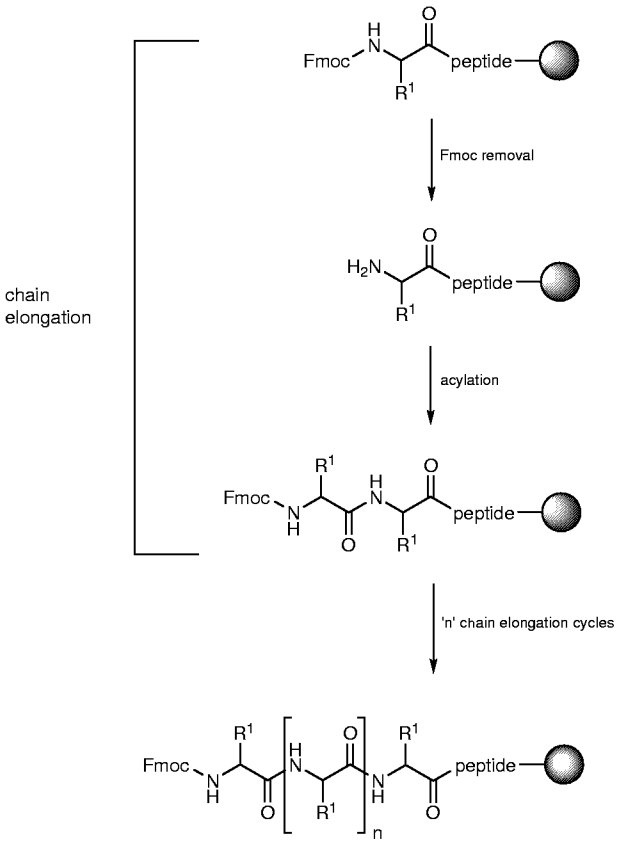
An additional benefit gained from monitoring the deprotection reaction arises from a visual interpretation of the peak shape produced. This reflects the rapidity of release of the dibenzofulvene adduct from the support. Rapid release is signified by a sharp peak whereas slow diffusion from the support, signified by a broad peak with extending tail is associated with the onset of aggregation problems within the matrix (*vide infra*).

4.3.2.2.6 Overall Reaction Protocol for Stepwise Addition of 9-Fluorenylmethoxycarbonyl Amino Acids

The overall reaction protocol for single residue addition is shown in Scheme 4. The acylation and deprotection reactions are separated by a series of DMF washes and the overall cycle comprises of four simple steps.

for references see p 752

Scheme 4 Typical Reaction Cycle for Fmoc/tBu-Based Solid-Phase Peptide Synthesis[18,103]



Complete Fmoc Solid-Phase Cycle:[18,103]
The cycle involved in single residue addition was carried out as summarized below,
1. Washed with DMF (10 × 1 min).
2. Acylated with activated Fmoc amino acid derivative (2.5 equiv).
3. Monitored by color test.
4. Washed with DMF (10 × 1 min).
5. N-Terminal deprotected with piperidine/DMF (20%, 1 × 3 min, 1 × 7 min).

4.3.2.2.7 Side Reactions in 9-Fluorenylmethoxycarbonyl/*tert*-Butyl Strategy

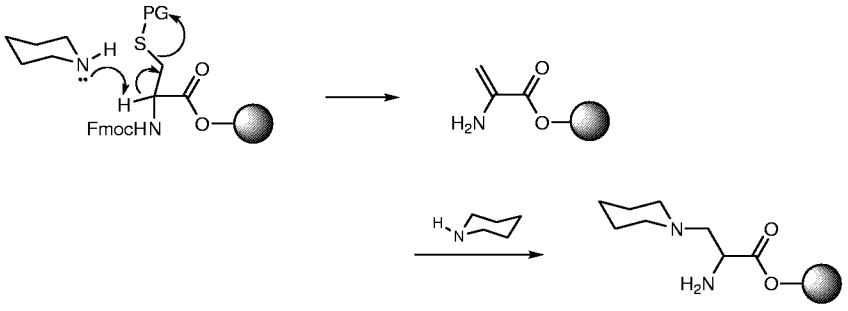
There are numerous side reactions that can occur during solid-phase peptide synthesis, some of which are specific to the chemistries employed using Fmoc-based methodology. This section describes some of the prevalent reactions that will lead to heterogeneity in the resultant peptide.

4.3.2.2.7.1 Side-Chain Related

Base-Catalyzed Elimination

A reaction specific to C-terminal esters of Cys results from the base-catalyzed elimination of the protected sulfhydryl side chain. The reaction, shown in Scheme 5, initially involves formation of dehydroalanine, which is followed by nucleophilic addition of piperidine to the alkene. The reaction has been known for many years, but a recent investigation[83] showed that the reaction can be virtually avoided if Trt side-chain protection is used in preference to Acm.

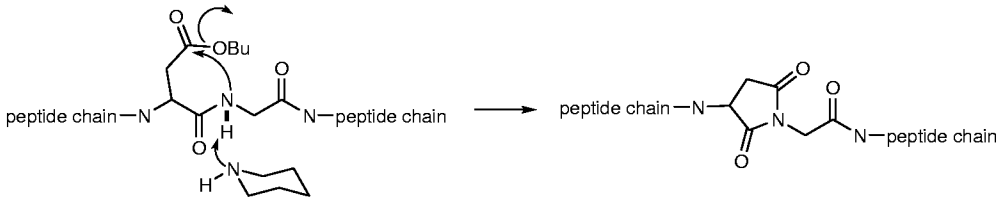
Scheme 5 Formation of 3-(1-Piperidinyl)alanine Adduct



Aspartimide Formation

Although aspartimide formation can occur during base-catalyzed acylation in Boc-based chemistry, the reaction is more pronounced during Fmoc removal with piperidine.[84] The reaction, shown in Scheme 6, is sequence dependent, but aspartimide formation can be up to 0.3% per N-terminal deprotection cycle when the required peptide contains -Asp-Gly-. When the resultant peptide is cleaved from the support the aspartimide ring may remain intact, but can also open to form the β-branched peptide.

Scheme 6 Sequence Dependent Aspartimide Formation



4.3.2.2.7.2 Piperazine-2,5-dione Formation

Piperazine-2,5-dione (DKP) formation, first discussed by Lunkenheimer and Zahn,[85] results from the base-catalyzed cleavage of the peptide from a peptide ester linkage to the solid support.[86] Peptides containing C-terminal Pro are particularly prone to this side reaction due to the steric features of the secondary amide. The reaction is by no means limited to Fmoc-based chemistry but the N-terminal salt formation following removal of Boc groups lends some temporary protection against this process. The reaction does not occur in the preparation of a C-terminal prolinamide using the Rink-type linker or in the preparation of free acid using trityl-based ester linkers.

Where convenient, the reaction can be avoided in Fmoc chemistry by using a suitable orthogonally stable ester and employing a Boc amino acid at the second residue. Coupling the third amino acid as the Fmoc derivative with in situ neutralization of the salt formed during Boc removal will avoid the reaction completely. Clearly this type of approach is not suitable if the second amino acid requires TFA-labile side-chain protection for the duration of the synthesis.

4.3.2.2.7.3 Aggregation

The formation of deletion sequences in solid-phase peptide synthesis either occurs as a result of incomplete coupling or incomplete N-terminal deprotection.[71] The former is less common nowadays due to the extremely efficient coupling agents used and this problem is now

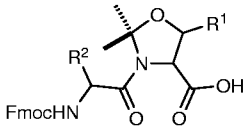
for references see p 752

generally limited to cases where the amine to be acylated is extremely hindered, such as N-substituted amino acids. In Fmoc/tBu-based methodology the main cause of deletion sequences is incomplete Fmoc removal. This is usually only a problem when the growing peptide chain becomes aggregated. This aggregation, generally attributed to β-sheet formation between the peptide's chains, brings about collapse of the peptide-resin matrix therefore hindering the diffusion of reagents to and from the reactive sites. The problem is noticeable when polystyrene is employed as the solid support and is less common when the more readily solvated polyacrylamide or polyethylene glycol based supports are used.

Aggregation is not completely overcome by changing the solid support however, since the root of the problem lies in the interactions between peptide chains. Changes in the secondary structure of the molecule such as the presence of secondary amino acids including Pro, can disrupt β-sheet formation. The recognition of this factor led to the development of the Hmb protecting group,[87] which when introduced as backbone amide protection at key positions in a molecule can result in disruption of interchain and intrachain interactions. The Hmb group is removed by TFA, but has the advantage of being completely stable to TFA when acetylated at the 2-hydroxy position. The 2-acetylated Hmb group has been shown to improve the solubility of the resultant peptides allowing the HPLC-based purification of otherwise insoluble molecules.[88] Subsequent removal of the acetyl group under mildly basic conditions followed by TFA treatment yields the required peptide. An added advantage when using the Hmb group protection at the carboxy terminal next to Asp and Asn residues is the reduction of aspartimide formation (Section 4.3.2.2.7.1).

However, it can be difficult to couple to the hindered Hmb-protected amino acids so in cases where the peptide chain contains Ser or Thr, oxazolidine dipeptide derivatives[89] **3** have been employed (Scheme 7). These pseudoproline derivatives induce a turn in the peptide chain disrupting β-sheet formation.

Scheme 7 Oxazolidine Dipeptide Derivatives[89]



3 R^1 = H, Me

4.3.2.2.7.4 Multiple Addition

Multiple addition of amino acids can occur when the acylation conditions are basic enough to lead to premature Fmoc removal during the coupling. This has been observed when using TBTU/DIPEA-based activation[90] and can be overcome by addition of excess HOBt or changing the conditions for peptide bond formation.

4.3.2.2.7.5 Racemization

Base-Catalyzed Racemization

Racemization of activated amino acids under mildly basic conditions has been studied in some detail for Boc-protected amino acids,[91] but recent reports confirm that the problem is not limited to Boc chemistry. Fmoc-Cys(Trt)-OH has been shown to racemize in TBTU-based couplings using excess DIPEA under certain conditions.[92] In this instance the problem was avoided by using the preformed symmetrical anhydride for peptide bond formation. In a more recent publication the racemization of Fmoc-Ser(tBu)-OH under similar conditions using NMM as base, was reduced by using a hindered collidine base.[93]

The racemization of His as discussed in Section 4.3.2.2.2 is promoted by its own internal base, the π-nitrogen of the imidazole ring. Histidine is usually protected at the τ-nitrogen, but is commercially available protected at the π-nitrogen as Fmoc-His(Bum)-OH. However, the less-expensive alternatives protected by Boc or Trt at the τ-position can be used effectively if the acidity of the acylating mixture is maintained by the addition of HOBt.

Racemization of Cysteine as a C-Terminal Ester

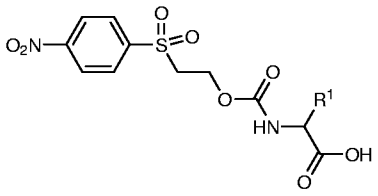
Racemization of C-terminal Cys bound to the solid support as an ester is promoted by repetitive Fmoc removal.[94] Cys is often introduced at the C-terminal of a peptide to provide an anchoring point, via the thiol, for subsequent conjugation so in some circumstances racemization at this point may not be important. Where it is essential to retain the chiral integrity, preparation of the peptide amide using a Rink-type linkage agent will overcome the problem.

4.3.2.2.8 2-[(4-Nitrophenyl)sulfonyl]ethoxycarbonyl and Other Base-Labile Protecting Groups

Recently the Nsc group **4** (Scheme 8) has been proposed as an alternative to the Fmoc group for solid-phase synthesis.[95–97] These types of sulfone-based urethane protecting groups were first proposed in the early 1960s when the 4-methyl equivalent of the Nsc group was utilized for amine protection.[98] The claimed advantages of the Nsc group include better coupling efficiencies, higher stability in neutral and slightly basic media, and a low tendency to polymerize.

Also the relatively polar nature and hence greater solubility of Nsc compared to the Fmoc group may offer benefits when aggregation becomes a problem in the peptide chain.

Scheme 8 Nsc-Protected Amino Acid **4**



4

Protocols for use of the Nsc amino acids are essentially the same as those described for Fmoc derivatives, but because of the increased stability to base, longer exposure to 20% piperidine/DMF is required (5 min and 10 min) to effect complete removal of the group. Otherwise the cycle is the same as above and nominally all of the side-chain protecting groups available in Fmoc chemistry can be applied with Nsc chemistry. A recent comparative study of the

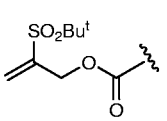
for references see p 752

synthesis of the pentapeptide Leu-enkephalin in two separate simultaneous assemblies in the author's laboratory using Fmoc and the Nsc amino acids produced similar crude HPLC profiles.[90]

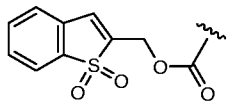
One major advantage for employment of the Nsc amino acids is that they are potentially less expensive to manufacture than Fmoc amino acids, a crucial factor when large-scale production is under consideration. However, the general utility of the Nsc group still has to be proven over a wide range of synthetic objectives before it can be widely accepted.

Three other base-labile groups have been proposed by Carpino. These are the 2-(*tert*-butylsulfonyl)prop-2-enyloxycarbonyl (Bspoc) (**5**),[99,100] Bsmoc (**6**)[99,101] and the [(*E*)-2-(methylsulfonyl)-3-phenylprop-2-enyl]oxycarbonyl (Mspoc)[102] group (**7**) (Scheme 9). Examples of the utility of the Bspoc and Bsmoc have been given both in solid- and solution-phase syntheses and particularly the Bsmoc and Mspoc may compete with the Fmoc group in the future, but as with Nsc, they need to be used more widely before general acceptance.

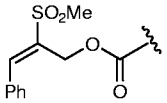
Scheme 9 Other Base-Labile Groups[99–102]



5 Bspoc



6 Bsmoc



7 Mspoc

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4.3.2.3 Orthogonal Allyl-Based Strategy

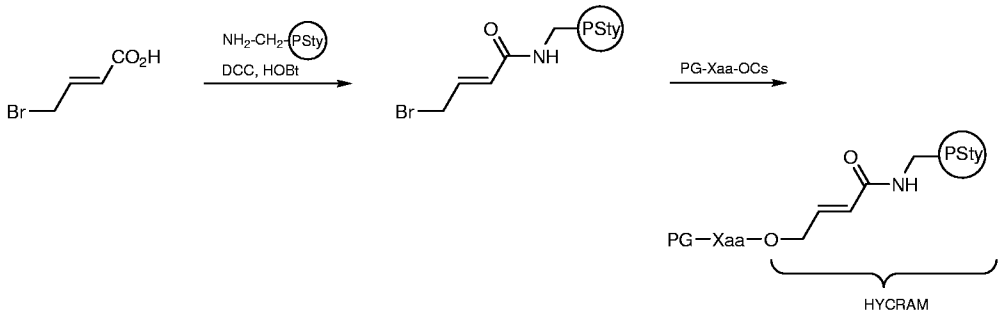
A. Loffet and F. Guibé

The need for new orthogonal protection/deprotection methods for the solid-phase peptide synthesis (SPPS) of proteins by a convergent strategy, as well as for the synthesis of branched and cyclic peptides, has stimulated intensive research in the field. The introduction of the allyl ester and allyloxycarbonyl groups for carboxy and amino groups, respectively,[1,2] enabled the establishment of a new orthogonal protection scheme, which is based on the highly selective and smooth deprotecting methods available for the allyl-based protection. These methods involve palladium-catalyzed transfer of the allyl group to a nucleophilic species.[1] The orthogonality of allyl groups in the *tert*-butyl- and fluorenylmethyl-based strategies has been well assessed.[1] The orthogonality of allyl groups toward trichloroethyl-based protecting groups[1,3] and Dde protection[4,5] has also been established.

4.3.2.3.1 Allylic Handles

The development of convergent SPPS,[6–8] which entails the synthesis of fully protected fragments for subsequent use in block condensation reactions, has stimulated the research into a wide variety of handles (linkers) in order to allow for specific attachment/detachment of the peptide chain on the solid support. A good handle must fulfill the criteria required for any protecting group, but must also allow for mild and specific cleavage of the protected peptide under conditions which ensure maximum compatibility with other protecting groups. In this context, allylic handles are obviously of high potential utility owing to their orthogonality with many other protecting groups. Allylic handles devised so far are structurally simple and readily accessible. They all contain at one end an allylic halide or hydroxy group for anchoring the peptide chain, while at the other terminus, attachment to the support is achieved through a carboxy group. The first allyl linker proposed by Kunz and Dombo[9] was from 4-bromocrotonic acid, which corresponds to the shortest possible structure of this kind. 4-Bromocrotonic acid is condensed onto an aminomethyl resin by the DCC/HOBt procedure. Attachment of a C-terminal amino acid onto the bromocrotonoyl resin is achieved by the cesium salt method.[10] This resin is commercially available under the name HYCRAM (Scheme 1).

Scheme 1 Formation of the First Allylic Handle from 4-Bromocrotonic Acid[9,10]

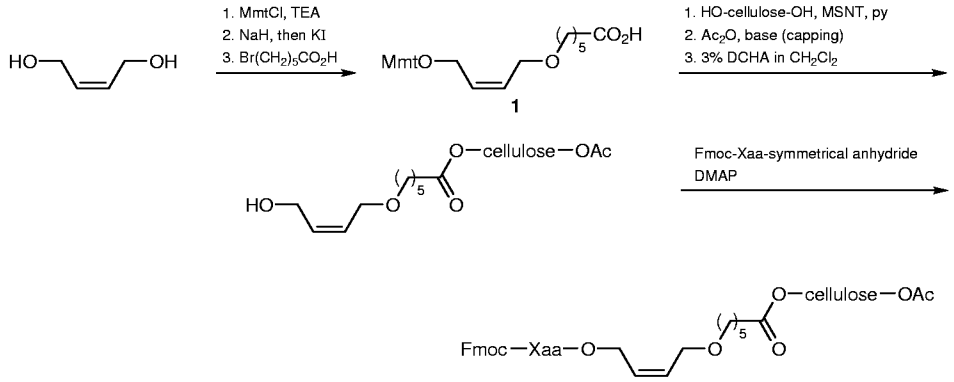


Peptide cleavage from the allyl-linker resin is achieved with morpholine in large excess in the presence of tetrakis(triphenylphosphine)palladium (10 mol% based on the allylic substitution of the resin). β-HYCRAM resin is a modification of HYCRAM containing a β-alanine residue as a spacer between the crotonoyl unit and the resin. It has been used for the solid-phase synthesis of glycopeptides by the Boc and Fmoc strategies.[11] Difficulties may be

for references see p 763

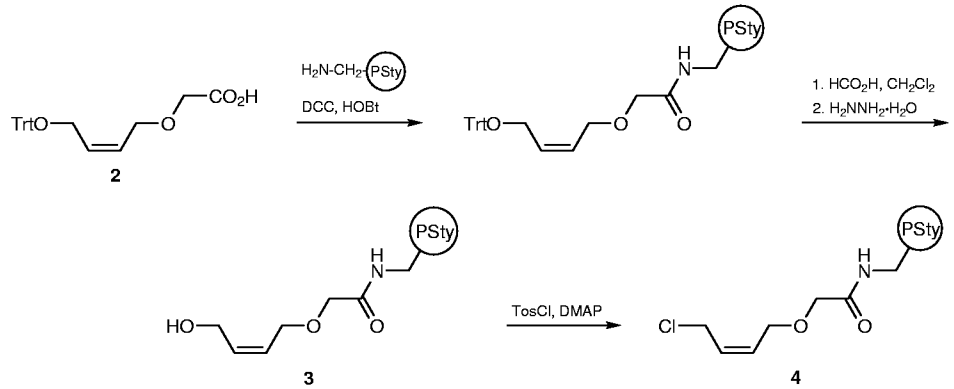
encountered in some cases with the latter linker[12] since in basic media the crotonoyl unit, as a Michael acceptor, may undergo a nucleophilic attack. Alternatively, the use of 6-{[(*Z*)-4-hydroxybut-2-enyl]oxy}hexanoic acid has been described,[13] which as the *O*-(4-methoxytrityl) derivative **1** is coupled to cellulose disks via esterification with 1-(mesitylsulfonyl)-3-nitro-1*H*-1,2,4-triazole in pyridine (Scheme 2). Cleavage from the resin is achieved by the use of HOBt as an allyl scavenger.

Scheme 2 Formation of an Allylic Handle from 6-({4-[(4-Methoxytrityl)oxy]-(*Z*)-but-2-enyl}oxy)hexanoic Acid[13]



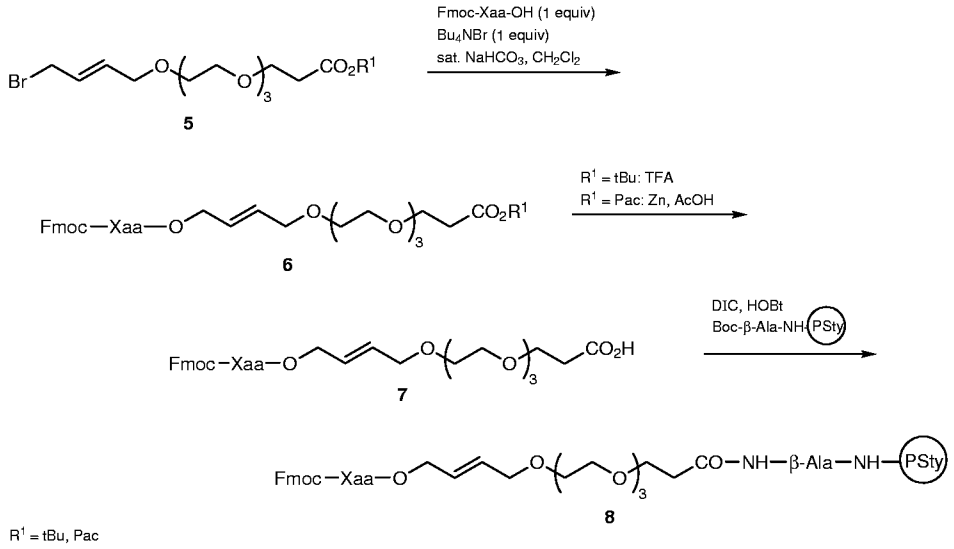
A third handle, very similar to the previous one, has been described by Guibé et al.[14,19] In this case, {[(*Z*)-4-hydroxybut-2-enyl]oxy}acetic acid is condensed as the *O*-trityl derivative **2** onto aminomethyl resin by the DCC/HOBt method; the trityl group is then removed and the free hydroxy group is used for esterification of the C-terminal amino acid using DCC/DMAP. Alternatively, the hydroxyallyl resin **3** is converted into the chloroallyl resin **4** (Scheme 3), which is then esterified by the Gisin method.[10] An additional option for minimizing the risk of racemization at the C-terminal residue involves prior attachment of the C-terminal amino acid to the free linker in solution followed by anchoring to the support. Cleavage of the peptide from the resin is effected by the hydrostannolytic procedure or by using *N*-methylaniline as the allyl group scavenger.[15]

Scheme 3 Formation of an Allylic Handle from {[(*Z*)-4-(Trityloxy)but-2-enyl]oxy}acetic Acid[14,19]



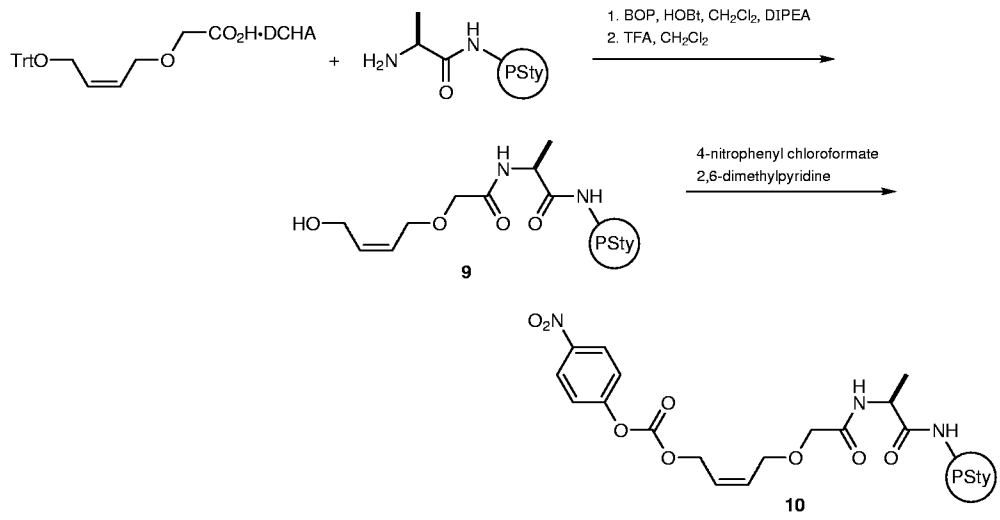
In the mid to late 1990s, Seitz and Kunz[12,16] described a new allylic linker called HYCRON in which a polyether chain is inserted between the allylic group and the carboxy group. The loading on the polymer is best achieved by the preformed handle approach, as outlined in Scheme 4.

Scheme 4 Synthesis of HYCRON-Anchor Conjugates and Resin Loading[16]



By converting the {[(*Z*)-4-hydroxybut-2-enyl]oxy}acetyl resin **9** into its 4-nitrophenyl carbonate derivative, an allyloxyacetyl resin **10** is obtained which has been used to prepare pseudoargiopinine III by SPPS (Scheme 5).[17]

Scheme 5 Conversion of {[(*Z*)-4-Hydroxybut-2-enyl]oxy}acetyl Resin into its 4-Nitrophenyl Carbonate Derivative[17]



An allylic handle with a terminal double bond, the *O*-9-(4,4′-dimethoxytrityl)undec-10-enyl linker, more specifically designed for the solid-phase synthesis of DNA and RNA fragments, has also been described.[18]

The isolation of peptides fully protected according to the Fmoc strategy is rather challenging, as the choice of allyl scavengers is limited to those whose basicity is sufficiently low to avoid competitive removal of the N-terminal Fmoc group (essentially HOBt, *N*-methylaniline, or tributyltin hydride). The problem is more acute than for the selective deprotection of simple allyl or Aloc groups because allylic linkers with their disubstituted

for references see p 763

double bond are cleaved more slowly by palladium. However, it may be concluded that allylic handles allow the SPPS of fully protected fragments, both by the Boc and Fmoc strategies. Final cleavage may sometimes cause problems, probably due to the poor access of the catalytic species to the reactive allylic site. To circumvent this problem and facilitate the formation of the π-allylpalladium complex, use of poly(ethylene glycol) spacers appears highly advisable.[1,16] It is also believed that the allyl scavengers phenylsilane[20] and borane–amine complexes[21] could be advantageous for allylic cleavage from the resin owing to their high reactivity and their nonbasic character. Finally, a possible drawback to the use of allylic linkers is their relatively pronounced tendency to lead to DKP formation during the third amino acid introduction step.[12,16,22]

Chloroallylic Aminomethyl Resin 4; Preparation and Attachment of the C-Terminal Amino Acid:[19]
To a soln of {[(*Z*)-4-(trityloxy)but-2-enyl]oxy}acetic acid (**2**; 2.5 g, 6.5 mmol) in CH_2Cl_2 (50 mL) under a dry atmosphere was added HOBt (1.8 g, 13 mmol), followed by dry DMF (few mL) until complete dissolution of the HOBt occurred. A soln of DCC (1.4 g, 6.5 mmol) in CH_2Cl_2 (2–3 mL) was then added while stirring. Within a few min, DCU precipitated out of the soln. The supernatant was transferred through a cannula and under dry atmosphere into a flask containing H_2N-CH_2-PSty (loading: 0.54 mequiv · g^{-1} of NH_2 groups; 4 g). After 16 h shaking at rt, the resin was filtered off, washed with CH_2Cl_2 (2 ×), MeOH (2 ×), again with CH_2Cl_2 (2 ×), and then dried in vacuo. The dry resin was suspended in a mixture of CH_2Cl_2 (20 mL) and HCO_2H (30 mL) and shaken for 3 h at rt. The resin was filtered and washed with CH_2Cl_2. The whole process of formolysis was then repeated. At this stage, IR analysis of the resin (KBr pellet) showed two C=O bands of equal intensity at 1680 cm^{-1} (amide band) and at 1725 cm^{-1} (formate ester band). Hydrazinolysis of the formate ester was achieved by reacting the resin in suspension in DMF (50 mL) with hydrazine hydrate (5.7 mL) for 3 h at rt. Treatment of the hydroxyallyl resin **3** included successive washings with CH_2Cl_2 (2 ×), MeOH (2 ×), and CH_2Cl_2 (2 ×), and drying in vacuo. Conversion of the hydroxyallyl resin into the chloroallyl resin was achieved in the following way with exclusion of moisture. To a soln of TosCl (1.8 g, 9.5 mmol) in dry DMF (50 mL) was added DMAP (1.1 g, 9 mmol) and the mixture was stirred at rt for a few min. The soln was transferred to a flask containing the hydroxyallyl resin (3.6 g, ca. 1.8 mmol of hydroxy groups) and the suspension was shaken for 16 h at rt. The resin was filtered, washed with CH_2Cl_2 (2 ×), MeOH (2 ×), and again with CH_2Cl_2 (2 ×), and then dried in vacuo. The IR spectrum of the resin revealed that the sulfonate ester band was absent, while Cl content determination by the Volhard method gave 80% of the theoretical value based on aminomethyl substitution of the starting resin. This value could not be improved by repeating the reaction. Capping of the remaining hydroxy groups was carried out in DMF (30 mL) with BzCl (1.8 mL, 15 mmol) and DIPEA (2.6 mL, 15 mmol) for 2 h at rt.

A soln of Boc-Tyr[Bzl(2,6-Cl_2)]-OCs (1.1 g, 2 mmol) in DMF (30 mL) was added to the chloroallyl resin **4** (3.6 g, ca. 1.4–1.5 mequiv of Cl) and the suspension was heated at 60 °C for 16 h. The resin was washed successively and several times with DMF, H_2O, MeOH, and CH_2Cl_2, and finally dried in vacuo. The substitution of Cl by Boc-Tyr[Bzl(2,6-Cl_2)]-OH residues was found to be complete.

HYCRON Linker 8; Synthesis of Anchor Conjugates and Resin Loading:[16]
A 0.25 M soln of Fmoc-Xaa-OH (1 equiv) and the anchor bromide **5** (1 equiv) in CH_2Cl_2 was added to a soln of Bu_4NBr (1 equiv) in an identical volume of sat. aq $NaHCO_3$. The two-phase system was stirred until TLC monitoring showed completion of the reaction (2–18 h). The organic layer was separated and the aqueous phase extracted with CH_2Cl_2 (2 ×). The combined organic layers were dried ($MgSO_4$) and concentrated. Further purification of **6** was accomplished by chromatography (silica gel).
The *tert*-butyl ester **6** (R^1 = tBu) was treated with TFA for 45 min. After removal of the TFA, the residue was dissolved in $CHCl_3$. The soln was washed with 1 M HCl and the combined organic layers were washed with brine and dried (Na_2SO_4) prior to removal of the solvent. The residue was purified by chromatography (silica gel, EtOAc/AcOH 100:1). Removal of trace amounts of AcOH from the eluent was achieved by coevaporation with toluene.

The phenacyl ester **6** [R^1 = Pac; Xaa = Thr(tBu)]; (2.62 g, 3.13 mmol) was dissolved in AcOH (40 mL) and freshly activated Zn (3.5 g, 54 mmol) was added. After 2 h, the Zn was removed by filtration over Hyflo and the solvent was removed under reduced pressure. The residue was dissolved in CH_2Cl_2 and washed with 0.5 M HCl. The aqueous layer was re-extracted with CH_2Cl_2 and the combined organic layers were dried ($MgSO_4$) and the solvent was removed under reduced pressure. The residue was purified by chromatography (silica gel, EtOAc/AcOH 100:1) and the solvent was coevaporated with toluene to give a clear and colorless, highly viscous oil; yield: 2.11 g (97%).

Boc-β-Ala-(aminomethyl)polystyrene (Boc-β-Ala-NH-PSty; 1 g, 1.0–1.2 mmol of β-Ala · g^{-1}) was shaken with CH_2Cl_2/TFA (1:1, 20 mL) for 45 min and then the resin was washed with CH_2Cl_2 (6 × 10 mL). Conjugates **7** were coupled to give **8** using carboxylic acid (2.0 equiv), DIC (3.3 equiv), and

HOBt (4.5 equiv) in the minimal volume of CH_2Cl_2 required for suspending the resin. After 10–15 h of shaking, the resin was washed with CH_2Cl_2 (6 × 10 mL) and treated with pyridine/Ac_2O (3:1, 20 mL) for 10 min in order to accomplish capping. The resin was washed with CH_2Cl_2 (6 × 10 mL) and dried.

Cleavage of Fmoc-Leu-Phe-Ala-OH from Fmoc-Leu-Phe-Ala-HYCRON-NH-β-Ala-NH-PSty with a Palladium Catalyst and *N*-Methylaniline:[16]
To a suspension of the peptide-resin (1.40 g, 0.40 mmol of Leu · g^{-1}, 0.33 mmol of Phe · g^{-1}, 0.40 mmol of Ala · g^{-1}, 0.78 mmol of β-Ala · g^{-1}) in CH_2Cl_2/DMSO (1:1, 30 mL), was added NMA (1.1 mL, 10.1 mmol). The reaction vessel was evacuated until bubbling of the suspension occurred and then subjected to a stream of argon. After five repeats, $Pd(PPh_3)_4$ (100 mg, 0.087 mmol) was added. The mixture was shaken under argon with exclusion of light for 15 h. The resin was washed with DMF (3 × 20 mL) and CH_2Cl_2 (4 × 20 mL) and dried; yield: 1.42 g of polymer (0.08 mmol of Leu · g^{-1}, 0.07 mmol of Phe · g^{-1}, 0.09 mmol of Ala · g^{-1}, 0.85 mmol of β-Ala · g^{-1}; cleavage yield: 81%). The solvent was removed under reduced pressure from the combined filtrates and the residue was dissolved in $CHCl_3$ (50 mL) and washed with 1 M HCl (30 mL). The aqueous soln was re-extracted with CH_2Cl_2 (4 ×) and the combined organic layers were dried (Na_2SO_4) and the solvent was removed under reduced pressure. After chromatography (silica gel, EtOAc/AcOH 100:1 and $CHCl_3$/MeOH), the product was reprecipitated from CH_2Cl_2 with heptane; yield: 290 mg (62%).

4.3.2.3.2 Temporary Allyl-Based Protection

The Aloc group can be used for protection of the α-amino group instead of Boc or Fmoc as it is orthogonal to the semipermanent benzyl- or *tert*-butyl-type side-chain and linker protections. This Aloc protection can be useful when some features of the peptide or pseudopeptide prevent the use of bases or acids in the deprotection step. In model studies, pseudometallic hydrides (tributyltin hydride and borane–dimethylamine complex), which are neutral reagents, have been used as allyl scavengers.

To evaluate the efficacy of N^{α}-Aloc protection in SPPS, the oligopeptide amides Tyr-Ala-Gly-Phe-Leu-NH_2 and Lys-Gly-Phe-Leu-Glu-Glu-Val-NH_2 have been synthesized on a PAL-PSty resin. The Boc group is used for side-chain protection of the Lys residue and the tBu group for side-chain protection of the Tyr and Glu residues. Coupling reactions are carried out either with free N^{α}-Aloc amino acids by the DIC/HOBt procedure in DMF or with their pentafluorophenyl esters in the presence of HOBt in DMF. Removal of the N^{α}-Aloc groups is accomplished under argon in dichloromethane at room temperature with 10 mol% (based on the substitution of the resin) of tetrakis(triphenylphosphine)palladium and 6 equivalents of borane–ammonia or –dimethylamine complex. The borane–amine complex is added as first reagent to the resin while stirring, followed a few minutes later by the catalyst. After 10 min, the palladium solution is drained, the resin washed with dichloromethane, and the deprotection procedure repeated once more. Substance P has been synthesized and shown to be equipotent to the peptide prepared by more classical methods.[23]

In order to explore the interesting features of the *N*-hydroxyamide group using automated solid-phase synthesis and combinatorial techniques, a method for the preparation of *N*-hydroxy-substituted peptide ligands for major histocompatibility complex class 1 (MHC-I) molecules has been designed.[25] After incorporation of the *O*-benzyl-*N*-hydroxyglycine derivative, the peptide chain is elongated using N^{α}-Aloc amino acids since the deprotection of Fmoc amino acids gives a complex mixture of products. Deprotection to remove the N^{α}-Aloc group is achieved using tetrakis(triphenylphosphine)palladium.[26]

Aloc Deprotection by Palladium-Catalyzed Hydrostannolysis (Synthesis of Substance P):[23]
Methylbenzhydrylamine-resin (1.85 g, 0.40 mequiv · g^{-1} of resin) was applied as the solid support and N^{α}-Aloc-protected amino acids were used. Side-chain protection utilized Lys(Z) and Arg(Tos) derivatives. All amino acid derivatives were coupled by the DCC/HOBt method in DMF/CH_2Cl_2 (1:9), except the N^{α}-Aloc-Gln-OH residues which were coupled as 4-nitrophenyl esters in DMF. Coupling efficiency was monitored by the Kaiser test.[24] The N^{α}-Aloc protection was removed by adding to the peptide-resin a soln of $PdCl_2(PPh_3)_2$ (0.04 equiv · $mmol^{-1}$ of amino group linked to the resin) and AcOH (3.5 equiv) in CH_2Cl_2 (3 mL · g^{-1} of resin). The deprotection was completed in less than 5 min after addition of Bu_3SnH

for references see p 763

(3 equiv). After washing the resin, cleavage with HF/anisole/Me_2S (10:1.5:0.25), and lyophilization, the crude product (625 mg) was found to contain 60% of Substance P and 13% of Substance P sulfoxide according to HPLC. The crude product (320 mg) was purified by CM-cellulose cation-exchange chromatography followed by partition chromatography on Sephadex G-25 F using BuOH/AcOH/H_2O (4:1:5) as eluent; yield: 110 mg of Substance P (H-Arg-Pro-Lys-Pro-Gln-Gln-Phe-Phe-Gly-Leu-Met-NH_2) and 17 mg of Substance P sulfoxide.

4.3.2.3.3 Semipermanent Allyl-Based Protection

The orthogonality of allyl-based protection toward Boc and Fmoc groups also allows its use for the side-chain protection of linear and branched peptides as well as for glycopeptides. It also enables specific deprotection at preselected side-chain groups for macrocyclization of peptides.

4.3.2.3.3.1 Synthesis of Cyclic Peptides by Allyl-Based Protection

In the synthesis of cyclic peptides, the cyclization step is best achieved on a resin-bound peptide by taking advantage of the pseudo-dilution effect offered by this technique. Such a strategy, however, requires the combination of multiple sets of orthogonal or semi-orthogonal protecting groups and linkers. Owing to its orthogonality with many acid-labile and base-labile protecting groups, allyl-based protection (OAl, Aloc) has been successfully applied on several occasions for the temporary protection of carboxy and/or amino groups preselected for the cyclization.[26–33] Head-to-tail, head-to-side-chain, side-chain-to-tail, and side-chain-to-side-chain cyclopeptides or pseudo-cyclopeptides have been synthesized in this manner. Standard Boc/Bzl methods using Merrifield resin together with orthogonal Fmoc/Fm and Aloc/Al protection of the side-chain functionalities taking part in the cyclization reactions have been used for the synthesis of a bicyclic peptide with a tripodal side-chain bridge.[34] In a later example, the protected sequence Fmoc-Ala-Lys(Boc)-Lys(Aloc)-Tyr(tBu)-Leu-Asn-Asp(O-Sasrin)-OBzl has been assembled stepwise on the resin.[35] Cleavage of the Aloc group with tributyltin hydride and dichlorobis(triphenylphosphine)palladium is performed on the Sasrin (super acid sensitive resin) solid support and the protected peptide is cleaved with dilute TFA and then cyclized in solution.

The synthesis of a monocarba analogue of deaminooxytocin has been described in which incorporation of the cysteine moiety as a *S*-3-propanoic acid allyl ester allows, after assembling of the peptide chain, the selective deprotection and cyclization of the peptide.[36]

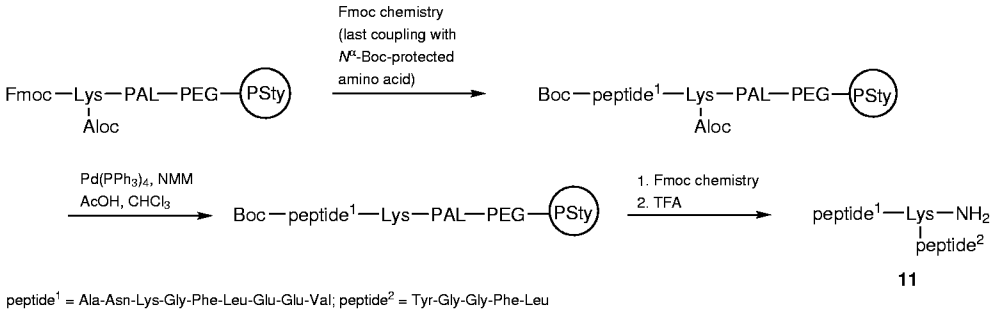
Partial Synthesis of Bacitracin A:[33]
The resin-bound linear decapeptide Fmoc-Leu-D-Glu(OtBu)-Ile-Lys(Aloc)-D-Orn(Boc)-Ile-D-Phe-His(Trt)-D-Asp(OtBu)-Asp(OAl)-PAL was assembled on a 0.25 mmol scale starting with PAL resin (0.39 g). Coupling reactions were performed with HBTU/DIPEA in NMP; single coupling cycles using a fourfold excess of Fmoc amino acid derivatives to resin-bound peptide were employed. Fmoc deprotection was effected by two treatments with 20% piperidine in NMP (5 and 15 min). The resin-bound linear decapeptide was dried and then swollen with $CHCl_3$ under an argon atmosphere. To this was added one-half of a mixture of $Pd(PPh_3)_4$ (1.73 g, 1.49 mmol), AcOH (0.8 mL), and NMM (1.6 mL) in $CHCl_3$ (30 mL). After stirring at rt for 4 h, the Pd soln was drained, and the resin was treated with the second portion of the Pd mixture and stirred at rt for additional 12 h. The ^1H NMR spectrum of an aliquot of peptide cleaved from the resin with TFA showed quantitative cleavage of the allyl groups. The partially deprotected resin-bound linear decapeptide was cyclized by washing with DMF and NMP and treatment with PyBOP (0.26 g, 0.5 mmol), HOBt (0.077 g, 0.5 mmol), and DIPEA (0.26 mL, 1.5 mmol) in NMP (10 mL) for 24 h.

Cleavage of Aloc from a Sasrin-Bound Protected Peptide before Cyclization in Solution:[35]
The resin-bound protected peptide Fmoc-Ala-Lys(Boc)-Lys(Aloc)-Tyr(tBu)-Leu-Asn-Asp(O-Sasrin)-OBzl (7.2 g, 2.8 mmol) was suspended in CH_2Cl_2 (130 mL) and selectively deprotected with a mixture of AcOH (0.33 mL, 5.75 mmol), $PdCl_2(PPh_3)_2$ (0.138 g, 0.196 mmol), and $SnBu_3H$ (3.17 mL, 11.9 mmol) in a stream of He for 1.5 h. After draining off the reactants, this process was repeated four more times and the partially deprotected resin was washed with CH_2Cl_2 (2 × 120 mL) and MeOH (4 × 120 mL).

4.3.2.3.3.2 Synthesis of Linear and Branched Peptides by Allyl-Based Protection

A branched peptide **11** [prothrombin(1–9)-Lys(Leu-enkephalin)-NH_2] has been synthesized on a PAL-PEG-PSty resin in a continuous flow peptide synthesizer[26] by combining N^{α}-Fmoc for temporary protection, N^{ε}-Aloc for semipermanent lysine protection, and Boc for permanent protection of the terminal amino group of prothrombin, as outlined in Scheme 6. Deprotection of Aloc in the continuous flow synthesizer is achieved by use of the $Pd(PPh_3)_4$/NMM/AcOH/$CHCl_3$ system.

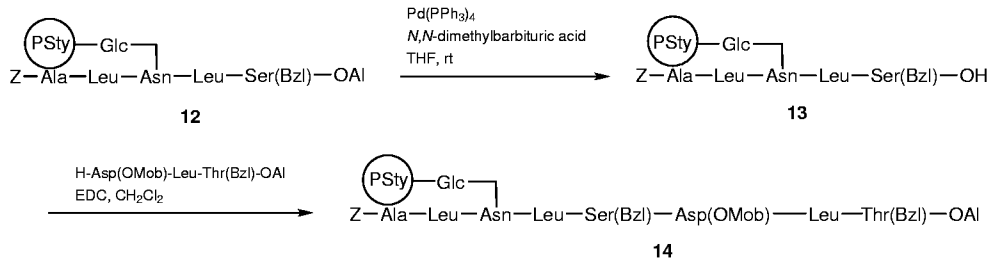
Scheme 6 Synthesis of a Branched Peptide by Allyl-Based Protection[26]



Another example is H-Thr(tBu)-Gln(Trt)-Ile-Asp(OtBu)-Ser(tBu)-Pro-Leu(N^{α}-Aloc)-Dap-MBHA-resin, synthesized by the Fmoc strategy.[37] Following cleavage of the Aloc group with phenylsilane and tetrakis(triphenylphosphine)palladium, two (aminooxy)acetyl moieties were linked to the α-amino groups of Thr and Dap by reacting the peptide-resin with [(*N*-Boc-amino)oxy]acetyl *N*-hydroxysuccinimide ester.

Allyl protection of terminal α-carboxy groups has been used in convergent syntheses of N-linked glycopeptides on a solid support.[3] For instance, the polymer-bound trisaccharide pentapeptide allyl ester **12** (Scheme 7) is built up by condensation of the polymer-linked trisaccharide with Z-Ala-Ile-Asp-Ile-Ser(Bzl)-OAl. Further conversion into the polymer-bound trisaccharide octapeptide derivative **14** is achieved by cleavage of the allyl ester with tetrakis(triphenylphosphine)palladium and *N,N*-dimethylbarbituric acid to give **13**, followed by condensation with H-Asp(OMob)-Leu-Thr(Bzl)-OAl. The presence of the allyl ester group at the C-terminus of the octapeptide provides the opportunity for further chain elongation.

Scheme 7 Convergent Synthesis of Linear Peptides by Allyl-Based Protection[3]



Deprotection of the Resin-Bound Glycopeptide Allyl Ester 12 **with a Palladium Catalyst and *N,N*-Dimethylbarbituric Acid:**[3]

The polymer-bound glycosylated pentapeptide **12** (125.4 mg, ca. 0.030 mmol) was swollen in THF (6 mL) for 1 h at rt. To this mixture, *N,N*-dimethylbarbituric acid (140 mg, 0.9 mmol) and $Pd(PPh_3)_4$ (17 mg, 0.015 mmol) were successively added, and the dark orange mixture was stirred at rt. After 14 h, the

for references see p 763

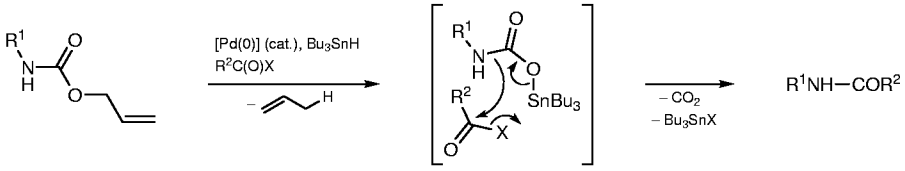
solvent was filtered off, and the resin was washed with CH_2Cl_2 (4 × 10 mL) and dried to give **13**; yield: 113.5 mg.

A similar experiment, performed in soln on the same, but resin-free, glycopeptide allyl ester using the Pd catalyst (0.4 equiv) and *N,N*-dimethylbarbituric acid (5.9 equiv), afforded the deprotected peptide in quantitative yield after chromatographic purification.

4.3.2.3.4 Tandem Allyloxycarbonyl Deprotection and Aminoacylation Reactions

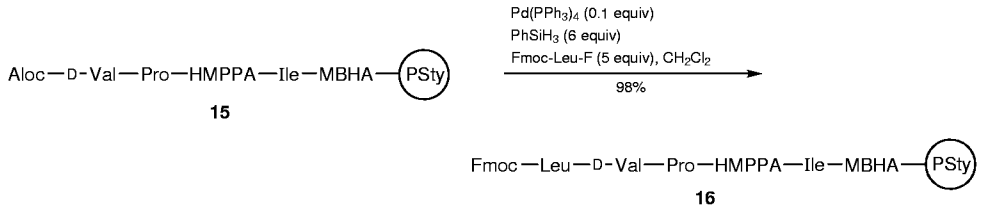
When deprotection reactions of the Aloc group with a palladium catalyst and pseudometallic hydrides, namely tributyltin hydride, sodium borohydride, or phenylsilane, are performed in the presence of an acylating agent, the deprotected amino group is directly obtained in its acylated form.[20,38–42] This tandem deprotection–acylation reaction is believed to involve a concerted decarboxylative condensation of the initially formed pseudometallic carbamate with the acylating agent, as represented in Scheme 8 in the case of hydrostannolysis.[42]

Scheme 8 Tandem Deprotection–Aminoacylation by Hydrostannolysis in the Presence of an Acylating Reagent[42]



This method can be used for transprotection,[20,38,41,42] e.g. Aloc to Boc by using Boc_2O as the acylating agent, or for peptide coupling by using such activated amino acid derivatives as hydroxysuccinimide or pentafluorophenyl esters,[39–42] in situ formed HOBt esters,[38,41,42] UNCAs,[39,40] or acyl fluorides.[39,40] Acyl chlorides[40] may also be used, but in this case the deprotection and the acylation steps must be carried out sequentially. In SPPS, the tandem deprotection–coupling methodology, which takes place under mild, near to neutral conditions, efficiently represses the formation of DKP at the stage of the introduction of the third amino acid residue, e.g. in the formation of **16** (Scheme 9).[43,44]

Scheme 9 Tandem Deprotection–Coupling To Give a Bound Tripeptide[43,44]



Coupling of Fmoc-Leu-F with 15 by the Tandem Deprotection–Coupling Procedure:[43,44]
N^{α}-Aloc-dipeptide resin **15** (50 mg, 0.57 mmol · g^{-1}) was washed under argon with CH_2Cl_2 (5 × 30 mL). To this resin were added successively Fmoc-Leu-F (51 mg, 5 equiv based on resin substitution) in CH_2Cl_2 (0.2 mL), $PhSiH_3$ (21 μL, 6 equiv) with a microsyringe, and $Pd(PPh_3)_4$ (3.3 mg, 0.1 equiv) in CH_2Cl_2 (0.2 mL). Argon was passed through the suspension and the reaction was mechanically stirred for 10 min. HPLC and amino acid analysis data of both the peptide-resin prior to cleavage and the corresponding peptide following release from the support confirmed that DKP formation did not occur.

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4.3.2.4 Photoreactive N-Protection/*tert*-Butyl Strategy

C. P. Holmes

The use of photoreactive N-protection for the assembly of peptides has received little attention in comparison to the more common Boc and Fmoc strategies optimized over the last four decades. Photodeprotection remains an elegant option for deprotection because light is clean, non toxic, easy to control, and is a mild reagent (*vide infra*), yet it suffers from limitations which makes it difficult to utilize in everyday situations and is essentially relegated to highly specialized applications and laboratories. In contrast to chemical deprotections which can be carried out by anyone with access to reagents and common glassware, there is a capital investment required for purchasing photolysis equipment which represents an additional hurdle to those wishing to explore photolabile-protecting groups for the first time. In addition, photolysis experiments are relatively fickle, and can require a great deal of fine-tuning before being rendered routine. Moreover, differences are often observed in comparing solution-phase deprotections with those performed on a solid phase. In solution, where there is no difficulty to expose the protected substrate to light, a significant concern is that the photolysis byproducts remain in close contact with the liberated product, and scavengers are generally included into the photolysis cocktail to suppress side reactions of the liberated molecules with the photolysis byproducts. In contrast, for deprotections performed on a solid phase, where the liberated products remain bound to the support while the photolysis byproducts are allowed to diffuse away, the greatest concern is the inherent challenge of efficient light delivery to the interior of the resin to achieve maximum yields. Thus, many factors come into play during removal of photoreactive protecting groups which render the approach critically dependent on experimental setup.

Although many chemical transformations on a variety of substrates can be triggered by exposure to UV light, natural amino acids and peptides are relatively stable to irradiation above 300 nm. The use of nonnatural amino acids, wavelengths lower than 300 nm, or high light intensities can severely degrade peptides. Literature photolysis studies, however, generally remain difficult to interpret due to the scarcity of experimental details reported. For example, the power level of the light at the sample is rarely reported even when the exact light source is described, presumably because light meters are not common in organic chemistry laboratories. In our hands, we have found widely varying results dependent on light intensity, even when allowing for different reaction times.[1] Hence it can be difficult to reproduce a desired photodeprotection by merely surveying the literature. Of the natural amino acids, tryptophan is the most sensitive to irradiation and its stability to light has been the subject of many studies. Detailed NMR examinations have been carried out on 200–600 nm irradiation of tryptophan-containing peptides and the local environment (i.e., flanking residues) was noted to partially govern its decomposition.[2] The effect of flanking residues was also noted to bias the degradation of tryptophan and tryptophan-containing peptides when exposed to gamma irradiation.[3] Aqueous photolysis of dipeptides under a variety of conditions with a medium-pressure mercury arc lamp was shown to afford many decomposition products with the principle pathway being thought to involve the <220 nm absorbance of the amide bonds.[4] Demonstration that the amide backbone can participate in extensive photoinduced electron transfers was elegantly shown later by the same authors, but again the irradiation was performed without filtering away the <300 nm light, and most likely occurs due to the 190–220 nm region where the amide bonds absorb.[5] Proline-rich sequences such as collagen are sensitive to irradiation with a mercury arc lamp when no precaution is taken to eliminate the lower wavelengths.[6] The effect of solar UV light on amino acids and peptides has also been studied and shown not to racemize the amino acids, and afforded only partial decomposition of Asp and Glu.[7] As one begins to employ nonnatural amino acids into peptides, one must remain cognizant that the novel side chain may

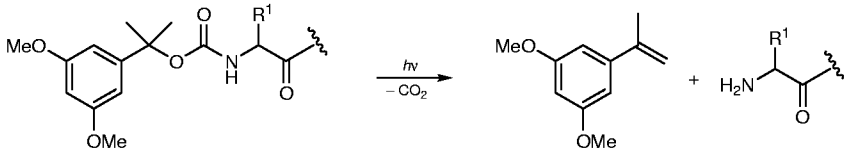
begin to act as a sensitizer and catalyze the decomposition of an otherwise stable peptide; such phenomena are well known with photoprotecting groups.[8,9]

Additional evidence demonstrating that peptides and amino acids can be readily photolyzed without decomposition is inferred by the numerous reports of photolabile linkers employed for the assembly and subsequent cleavage of peptides from resins. Here the combination of irradiation time, lamp intensity, photolysis cocktail, and wavelength of the UV light can all be balanced to afford high yields of liberated peptides.[10–12] Even proteins containing as many as 15 photoprotecting groups can survive photolytic deprotection provided that irradiation is performed at 365 nm.[13] Reactive side chains of amino acids such as cysteine and histidine can be problematic during photodeprotections if not suitably protected due to their nucleophilic properties. The removal of photoreactive protecting groups from Cys and His and the absolute dependence on scavengers to be present in the photolysis cocktail is a clear indication of the high reactivity of these side chains to standard photolysis byproducts.[14,15]

4.3.2.4.1 The Use of 2-(3,5-Dimethoxyphenyl)prop-2-yloxycarbonyl as N-Protecting Group

There have been several reports of using the α,α-dimethyl-3,5-dimethoxybenzyloxycarbonyl [2-(3,5-dimethoxyphenyl)prop-2-yloxycarbonyl, Ddz] group **1** for N-protection of peptides (Scheme 1).[16,17] The Ddz group was evolved from the parent 3,5-dimethoxybenzyl group which also affords good yields (42–86%) of amino acids and a dipeptide upon photolysis.[18] Both protecting groups exhibit absorptions around 280 nm and can be conveniently photolyzed with a high-pressure mercury arc lamp; the alkene byproduct **2** dimerizes during the deprotection rendering it unreactive. The Ddz group is also highly acid labile and can be cleaved with 5% TFA in dichloromethane. The sensitivity to TFA, making the traditional acid-labile side-chain protecting groups unusable, and the low wavelengths required for photocleavage have rendered the Ddz and related protecting groups[19] unpopular, particularly in comparison to the 2-nitrobenzyl groups described in Section 4.3.2.4.2.

Scheme 1 The 2-(3,5-Dimethoxyphenyl)prop-2-yloxycarbonyl Protecting Group[16]



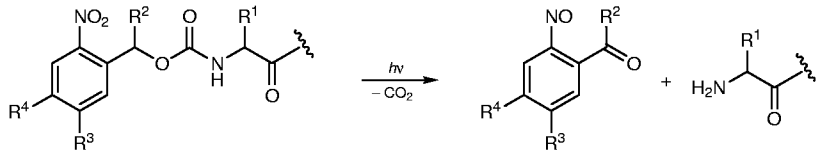
1 **2**

4.3.2.4.2 2-Nitrobenzyl N-Protecting Groups

The seminal paper describing the use of 2-nitrobenzyl photoprotecting groups for amino acids and dipeptides appeared in 1970[20] and the early literature was reviewed in 1980.[19] Patchornik employed various 2-nitrobenzyloxycarbonyl groups **3** and demonstrated quantitative yields of liberated amino acids and a dipeptide upon photolysis (Scheme 2). The initial photocleavage occurs readily upon exposure to >320 nm UV light and the chemistry of these early groups is dominated by the highly chromaphoric and reactive nitrosocarbonyl photolysis byproducts **4a** and **4b**, which possess a reactive aldehyde in addition to the reactive nitroso functionality. Critical to achieving quantitative yields was the inclusion of scavenging reagents (e.g., sulfuric acid, semicarbazide, hydrazine, or hydroxylamine) to either react with the nitrosocarbonyl compounds generated or to render the liberated amine unreactive (Scheme 3). The initially liberated nitrosoaldehyde **4b** has been shown to photochemically dimerize to produce diazodiacid **8**. Hydrazine can reduce the nitroso func-

tionality to an aniline such as **5** in addition to forming stable imines such as **6**. The yields of amino acids drop to roughly 40% if appropriate scavengers are not included into the solvent. The aldehyde group of **4a**, **4b**, **4c** appears to account for most of the reactivity of the liberated nitrosocarbonyls, since generation of the ketone **4d**, **4e**, **4f** is less dependent on having scavengers present. Thus, photolytic cleavage of the $Moc(2Np)_2$ group of $Moc(2Np)_2$-Phe-OH afforded nitrosocarbonyl **4d** and phenylalanine in 95% yield under neutral conditions.[20] Although high yields of products could be measured, the purification proved to be difficult due to highly colored byproducts in the medium; Patchornik revealed later that polymeric carbonyl scavengers could be employed during photolysis which greatly facilitated product isolation.[21]

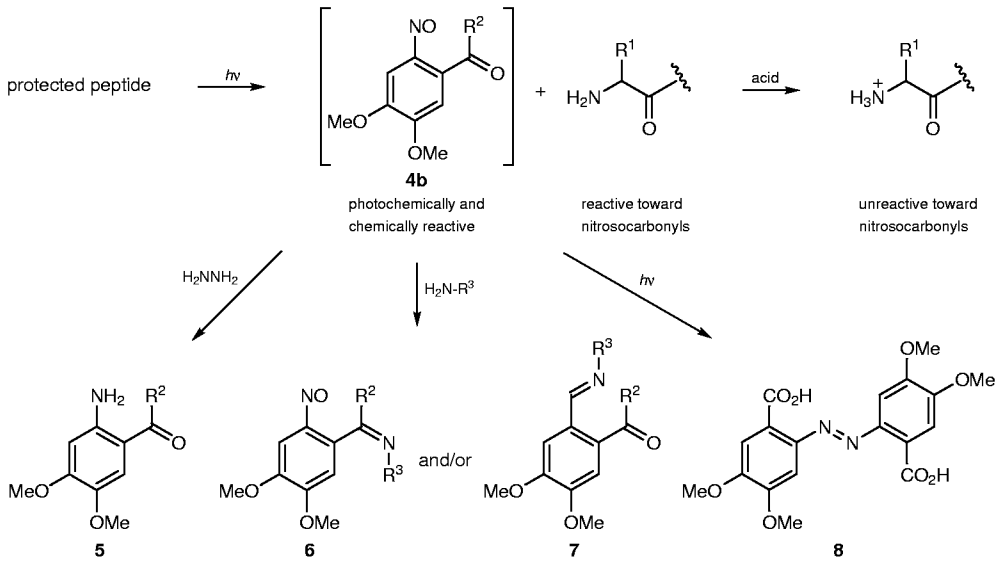
Scheme 2 2-Nitrobenzyl Protecting Groups[19,22]



3 4

Compound	Protecting Group on **3**	R^2	R^3	R^4
4a	$Z(2\text{-}NO_2)$	H	H	H
4b	Nvoc	H	OMe	OMe
4c	Npoc	H	OCH_2O	
4d	$Moc(2Np)_2$	$Z(2\text{-}NO_2)$	H	H
4e	Menvoc	Me	OMe	OMe
4f	Menpoc	Me	OCH_2O	

Scheme 3 Fate of Products and Byproducts upon Photolysis



We have built upon the foundation of Patchornik's work and have pursued novel ortho-nitrobenzyl groups **3e**and **3f** as an N-protection strategy.[22] Since our application involved the complete assembly of peptides with photoreactive N-protecting groups by combining photolithography and solid-phase peptide synthesis,[23,24] we were concerned with the rate of

photocleavage. Repetitive cycles of photodeprotection followed by coupling of activated amino acids would prove impractical if the rates of deprotections were unduly long. Thus, we prepared and examined the photolysis rates for the deprotection of several 2-nitrobenzyl protected amino acids (Scheme 4). Nvoc-Phe-OMe and Npoc-Phe-OMe were subjected to photolysis at 364 nm in a wide range of solvents and the rates of disappearance of starting material measured (Table 1). Ethereal solvents such as THF and dioxane were observed to afford the fastest rates, while protic solvents such as ethanol dramatically slowed the cleavage. Sacrificial amine scavengers such as *tert*-butylamine or tertiary amines also slowed the rate of cleavage. The 6-nitro-1,3-benzodioxol-5-yl based protecting group Npoc **3c** exhibited moderately faster cleavage kinetics than did the corresponding 4,5-dimethoxy-2-nitrobenzyl group **3b**.

Scheme 4 Preparation of Nvoc- and Npoc-Phe-OMe

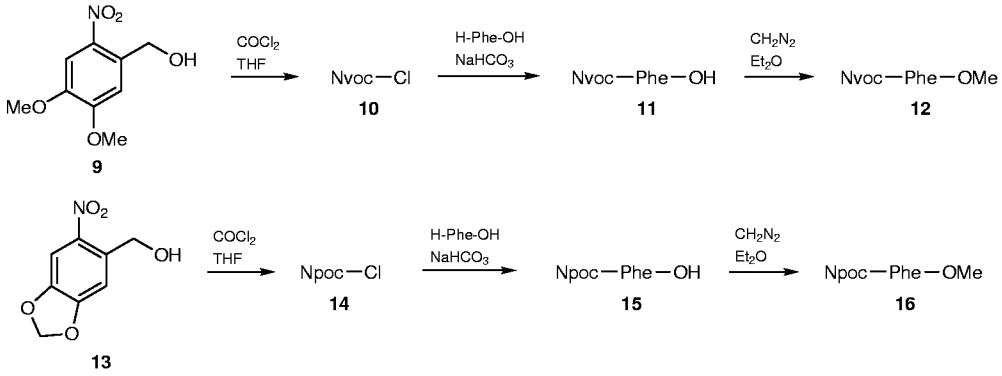


Table 1 Solvent Dependencies on Photolysis Rates of N-Photoreactive-Phe-OMe

Solvent	Nvoc-Phe-OMe[a]	Npoc-Phe-OMe[a]
THF (250 ppm BHT[b])	77	58
THF	62	65
dioxane	94	83
5 mM H_2SO_4/dioxane	105	96
50 mM H_2SO_4/dioxane	265	131
100 mM H_2SO_4/dioxane	213	80
500 mM H_2SO_4/dioxane	306	216
DMSO	196	185
MeCN	316	248
EtOH	336	249
5 mM DIPEA/dioxane	252	161
5 mM tBuNH$_2$/dioxane	584	434
5 mM $NH_2NHCONH_2 \cdot HCl$/10 mM DIPEA/dioxane	323	156

[a] Photolysis half lives ($t_{1/2}$) in seconds for disappearance of starting material via HPLC analysis. Substrate concentration 0.1 mM in indicated solvent, photolysis at 10 mW·cm^{-2} at 364 nm.

[b] BHT is 2,6-di-*tert*-butyl-4-methylphenol.

To extend the study beyond Phe, the natural amino acids were derivatized with Nvoc-Cl and those with TFA-labile side-chain protecting groups measured for stability to 5 mM H_2SO_4 in dioxane. All but two were found to be stable over 48 h. Nvoc-His(Trt)-OH slowly lost the trityl protecting group (14% over 2 h), which was deemed not detrimental during peptide assembly, since the trityl group would survive the initial coupling reaction. The Boc group of Nvoc-Lys(Boc)-OH cleaves in 5 mM H_2SO_4 in dioxane at a rate of 1–2% per hour which would severely compromise a synthesis, and was hence replaced by the more stable Nvoc-

for references see p 771

Lys(Mob)-OH. The photolysis half lives for the amino acids containing polar side chains (Arg, Trp, and His) were noted to cleave slower than the remaining amino acids, but nevertheless cleaved within a reasonable length of time (Table 2). It is clear that the local environment of molecules protected with 2-nitrobenzyl groups, which is governed by both the molecule itself and the photolysis solvent, can greatly influence the relative rates of cleavage.

Table 2 Photolysis Rates of Nvoc Amino Acids[a]

Amino Acid	Dioxane	H_2SO_4/ Dioxane	Amino Acid	Dioxane	H_2SO_4/ Dioxane
Ala	43	61	Ile	53	59
Arg(Pmc)	64	96	Leu	56	62
Asn	49	54	Lys(Boc)	54	64
Asn(Trt)	50	66	Lys(Mob)	33	38
Asp(OtBu)	64	62	Met	64	59
εAhx	56	46	Phe	59	61
Cys(Trt)	51	60	Pro	58	67
Gln	46	70	Ser(OtBu)	50	62
Gln(Trt)	68	67	Thr(OtBu)	46	62
Glu(OtBu)	57	66	Trp	122	80
Gly	54	62	Tyr(OtBu)	50	59
His(Trt)	48	90	Val	57	69

[a] Photolysis half lives ($t_{1/2}$) in seconds for disappearance of starting material via HPLC analysis. Substrate concentration 0.1 mM in indicated solvent, photolysis at 10 mW·cm^{-2} at 364 nm.

Nvoc-Phe-OMe (12):

CAUTION: *Phosgene is a colorless, poisonous gas or volatile liquid. This reagent should be handled with the utmost care.*

A soln of **9** (300 mg, 1.32 mmol) in THF (4 mL) was added to a soln of 1.93 M $COCl_2$ in toluene (3 mL, 5.79 mmol) and was stirred at rt for 14 h. The solvent was removed under reduced pressure and the resultant chloroformate taken up in dioxane (15 mL) and added to a soln of H-Phe-OH (245 mg, 1.48 mmol) and $NaHCO_3$ (278 mg, 3.31 mmol) in dioxane (15 mL). Additional dioxane (5 mL) was added to make a homogeneous soln. After stirring for 1.5 h the mixture was partitioned between $CHCl_3$ and 1 M H_2SO_4. The combined organic phase was dried ($MgSO_4$), filtered, and the solvent removed to afford **11** (0.72 g) as a pale yellow solid. The solid was taken up in Et_2O (50 mL) and treated with excess ethereal CH_2N_2 until the yellow color persisted. The solvent was removed under reduced pressure and the resultant solid was purified via flash chromatography (silica gel, 2% MeOH/CH_2Cl_2) to afford **12**; yield: 0.48 g (87% for 2 steps).

Npoc-Phe-OMe (16):

CAUTION: *Phosgene is a colorless, poisonous gas or volatile liquid. This reagent should be handled with the utmost care.*

A soln of **13** (300 mg, 1.52 mmol) in THF (4 mL) was added to a soln of 1.93 M $COCl_2$ in toluene (3 mL, 5.79 mmol) and was stirred at rt for 14 h. The solvent was removed under reduced pressure and the resultant chloroformate taken up in dioxane (15 mL) and added to a soln of H-Phe-OH (272 mg, 1.65 mmol) and $NaHCO_3$ (319 mg, 3.80 mmol) in dioxane (15 mL). Additional dioxane (5 mL) was added to make a homogeneous soln. After stirring for 1.5 h the mixture was partitioned between $CHCl_3$ and 1 M H_2SO_4. The combined organic phase was dried ($MgSO_4$), filtered, and the solvent removed to afford **15** as a pale yellow solid; yield: 0.37 g. The solid was taken up in Et_2O (50 mL) and treated with excess ethereal CH_2N_2 until the yellow color persisted. The solvent was removed under reduced pressure and the resultant solid was purified via flash chromatography (silica gel, 2% MeOH/CH_2Cl_2) to afford **16**; yield: 0.51 g (83% for 2 steps).

Photolyses; General Procedure:

Photolyses (Table 1) were performed in a darkened room with an argon laser with the 364-nm band selected. The beam was defocused with a lens and sample placed where the power level measured

10 mW·cm^{-2}. A 10 mM soln of protected amino acid ester in DMSO was diluted to 0.1 mM with the selected solvent and placed in a 0.2-cm path length quartz cuvette for various times ranging from 10 to 240 s. Remaining starting material was determined via HPLC analysis on a C_{18} reverse-phase column with detection at 350 nm. Photolyses (Table 2) were alternatively conducted with a 1000-W Hg(Xe) ARC lamp fitted with a water filter and a 350–450 dichroic reflector to select the 365-nm band. The output was passed through a collimating lens (Oriel Corp.) and the power level was adjusted to 10 mW·cm^{-2} as measured with a Optical Associates UV power meter with a 365-nm probe.

4.3.2.4.3 Solvent and Scavenger Effects on Deprotection Yields of a Generic Peptide

The photodeprotection of Nvoc-GGG-NBzl$_2$ as a generic peptide was examined in detail to determine yields under a variety of photolysis conditions (Table 3). The C-terminal amide was designed to facilitate analysis via reversed-phase HPLC with 220 nm detection. The photolytic removal of the Nvoc group was performed as described above employing the 365-nm light from a high-pressure Hg(Xe) ARC lamp at 10 mW·cm^{-2}. Substrate concentration was observed to greatly influence the yield of H-GGG-NBzl$_2$. The overall observation is that deprotections performed in solution at 10 mW·cm^{-2} are rapid and typical reaction times are 30 minutes or less. High concentrations of substrate, in addition to suppressing the overall rate of cleavage, also serves to greatly diminish the yields when cleaving under neutral conditions. The reaction between liberated nitrosoaldeyde **4b** with free tripeptide, being a bimolecular process, accounts for the observed concentration dependence. Under dilute (0.1 mM) conditions, less peptide is trapped by **4b** than under conditions of high concentration (10 mM). Examination of the HPLC traces reveals a variety of photolysis byproducts were formed under the various conditions in addition to the peptide, with diazodiacid **8** being one of the few identified. The effect of added scavenger is quite dramatic and the use of sulfuric acid as described by Patchornik[20] proved to be superior to the use of DTT employed by Walker.[25] The combination of both scavengers produces lower yields of free peptide compared to just the mineral acid as scavenger.

The effect of several acidic reagents included in the photolysis solvent for cleavage of Nvoc-GGG-NBzl$_2$ was examined (Table 4). In order to more thoroughly assess the effectiveness of the additive, a relatively high concentration of protected peptide (1 mM) was chosen and the photolysis time was fixed at 15 minutes. The strongly acidic mineral and methanesulfonic acids afforded the highest yield of H-GGG-NBzl$_2$, while the weakly acidic reagents such as tetrazole and HOBt gave poor yields, suggesting that the liberated amine must be fully protonated to avoid cross reacting with the nitrosoaldehyde produced upon cleavage. Moreover, the distribution of products from further reactions of the liberated nitrosoaldehyde **4b** depends on which acid scavenger is present, indicating that many reaction pathways are accessible to these highly reactive species.

Table 3 Yields of H-GGG-NBzl$_2$ from Deprotection of Nvoc-GGG-NBzl$_2$

Solvent	Substrate Concentration (mM)	Yield[a,b,c] (%) (% conversion)			
		10 min	15 min	20 min	30 min
dioxane	0.1	71	58	67	n.r.
	10	19 (34)	31 (34)	29 (41)	25 (57)
5 mM H_2SO_4/dioxane	0.1	92	94	97	n.r.
	10	97 (24)	98 (33)	99 (42)	96 (56)
5 mM DTT/dioxane	0.1	68	62	61	n.r.
	10	37 (23)	24 (40)	30 (40)	28 (55)
5 mM H_2SO_4/5 mM DTT/dioxane	0.1	86	86	88	n.r.
	10	88 (28)	87 (40)	64 (57)	78 (64)

[a] Yields determined via HPLC analysis of crude reaction mixtures with 220-nm detection.
[b] n.r. = not reported.
[c] Photolysis time was fixed at the values specified.

for references see p 771

Table 4 Acid Scavenger Study for Photolysis of 1mM Nvoc-GGG-$NBzl_2$

Acid	Concentration	Yield[a] (%)
sulfuric acid	10	96
trichloroacetic acid	10	56
dichloroacetic acid	10	32
chloroacetic acid	10	14
acetic acid	10	35
trifluoroacetic acid	10	74
sulfanylacetic acid	10	80
4-toluenesulfonic acid	10	66
methanesulfonic acid	10	95
hydrochloric acid	10	94
pyruvic acid	10	0
1-hydroxybenzotriazole	10	7
1*H*-tetrazole	10	35

[a] Yields determined via HPLC analysis of crude reaction mixtures with 220-nm detection.

4.3.2.4.4 Applications and Outlook

Despite not being commonplace, the use of photoreactive protecting groups for N-terminal protection for peptides affords high yields of liberated product. The caveats pointed out in this work, where critical attention must be paid to scavengers, irradiation wavelength, choice of solvent, and substrate concentration, should guide others to consider such routes. In transferring such methods to the solid phase, the nature of the polymer support also must be added to the list of caveats. The benefits of releasing the photobyproducts (e.g., reactive nitrosocarbonyls) from the support where they can diffuse away are obvious; what is not obvious, however, is that it can be difficult to photocleave protecting groups in high yields on a solid support. Large amounts of resin exhibit their own peculiarities with respect to photolysis due to light scattering and shielding, inherent opaqueness of conventional supports, and often lengthy irradiation times being employed to ensure adequate exposure of the substrate to UV light. Long photolysis times have been shown to promote oxidation of methionine to methionine sulfoxide,[26] whereas short photolysis times minimize oxidation.[12] While single deprotections can be performed under controlled conditions, the repetitive use of photoreactive N-protections on traditional solid supports (e.g., resins) may remain generally elusive in comparison to highly ideal systems such as glass surfaces.[23,24] Here the substrate may be considered to be a monolayer on a transparent support and therefore offers unhindered exposure to UV light while allowing photolysis byproducts to diffuse away from the surface. Many thousands of peptides have been made in a combinatorial fashion via this technique for a range of applications such as epitope mapping, protease substrate and inhibitor arrays, etc.[24] A laser printer has even been proposed as light source to perform the repetitive deprotections.[27] Other idealized systems for exploiting photoreactive N-protection strategies in solid-phase peptide synthesis lie in the area of soluble supports, where the concepts of bead interiors and exposed surfaces are meaningless.[28] Although there are numerous reports of employing photoreactive linkers on soluble PEG polymers to liberate peptides,[29,30] the use of photolabile N-protection to assemble peptides appears to remain unexplored, yet within reason. There will be improved photoprotecting groups with improved properties such as faster and cleaner cleavage, which means that the N-protection with photoreactive groups will continue to gain acceptance for the assembly of peptides.

[1] For example photolysis at a lower power level for a longer time (e.g., 2 h photolysis at 10 mW·cm^{-2} at 364 nm) often affords greater yields than higher intensities for shorter times (e.g., 0.2 h at 100 mW·cm^{-2} at 364 nm) despite the fact that the delivered amount of light is the same.
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4.3.3 Acylation Methods

M. Rolland, M. Amblard, J.-A. Fehrentz, and J. Martinez

Dedicated to the memory of Dr. Marc Rodriguez

All the methods of peptide bond formation described in Section 3 are useful for the synthesis of peptides on solid support [solid-phase peptide synthesis (SPPS)]. The ideal method would rapidly allow quantitative coupling reactions under mild conditions, avoiding side reactions, racemization, and generating side-products which could be easily removed by single washings from the peptide-resin. Various techniques have been explored by peptide chemists, but the focus of this section will be on the small number which have been widely used. These techniques have been applied to "stepwise synthesis", the addition of one residue at a time, and to "fragment condensation", which is the condensation of long sequences of amino acids.

The mechanism of peptide bond formation involves the attack of a nucleophilic α-amino group on the activated carboxylic group of a protected amino acid or peptide fragment. The method of activation of the carboxylic group will control the rate of the desired reaction, but can also influence the levels of side reactions. One of the most important side reactions that must be avoided is racemization. In the common practice of stepwise solid-phase synthesis, using a urethane group for α-amino group protection, racemization is not a serious concern. For fragment condensation, the rate of racemization will depend on the activated intermediates. A few techniques for fragment condensation with acceptable levels of racemization are available.

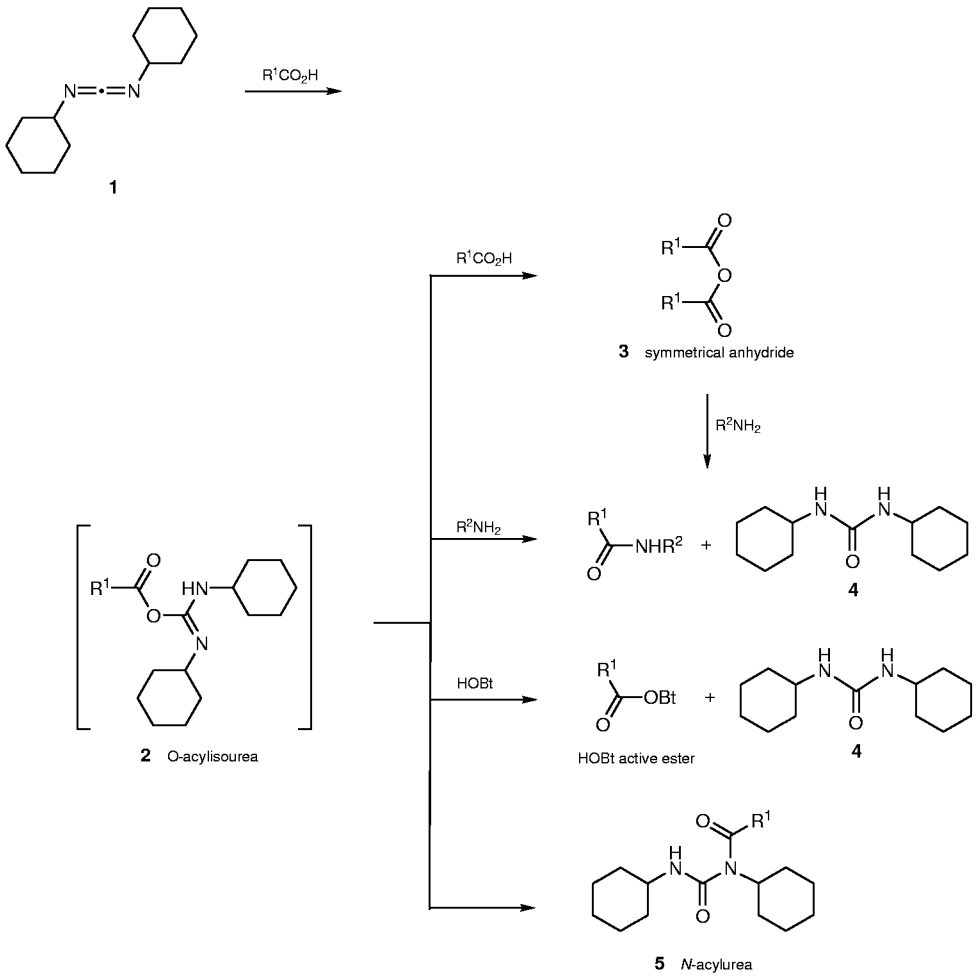
4.3.3.1 Stepwise Chain Elongation

In this strategy, the C-terminal amino acid of the desired peptide is anchored to the solid support generally through its α-carboxy group and monodirectional chain elongation proceeds by repetitive coupling/deprotection steps of successive protected α-amino acid units. This method was originally explored by Merrifield[110] and is still the most frequently used. The most important difference with peptide synthesis in solution is that in peptide synthesis on solid support, a large excess of activated N-protected amino acid can be used to allow the acylation reaction to go to completion. This excess of reagents can be easily removed from the peptidyl resin by simple washing.

4.3.3.1.1 *N,N'*-Dicyclohexylcarbodiimide

The carbodiimide method[1] was introduced into peptide synthesis by Sheehan and Hess[2] with the use of *N,N'*-dicyclohexylcarbodiimide (**1**, DCC). It has been the most important reagent for activating the carboxylic group and is used to generate activated carboxylic derivatives such as symmetrical anhydrides and active esters, or as a direct coupling reagent. In all cases the primary activating event is the formation of an *O*-acylisourea intermediate **2** which is a potent acylating agent. *O*-Acylisourea formation is rapid, leading to the peptide bond formation by direct aminolysis or via a symmetrical anhydride **3**, with formation of *N,N'*-dicyclohexylurea (**4**, DCU) (Scheme 1).

Scheme 1 Mechanism of Activation by *N,N'*-Dicyclohexylcarbodiimide[1–4]



Reactions are carried out at room temperature in dichloromethane which is an excellent swelling agent for polystyrene supports. Dimethylformamide can be added to increase the solubility of some amino acid derivatives e.g. Boc-Arg(NO_2), Boc-Arg(Tos), Boc-His(Tos), Boc-Trp, and Boc-Asn(Xan), but it increases the rate of undesirable rearrangement of *O*-acylisourea to *N*-acylurea which is not reactive.[3,4] The principal limitations in using carbodiimides are racemization, rearrangement of *O*-acylisourea to *N*-acylurea **5**, and dehydration of Asn and Gln side-chain carboxamide groups. Fortunately this dehydration problem is completely avoided by the use of additives such as *N*-hydroxysuccinimide (HOSu) and 1,2,3-benzotriazol-1-ol (HOBt). Additives also reduce racemization and *N*-acylurea formation. Other carbodiimides have also been used in SPPS such as DIC[5,6] or *N-tert*-butyl *N*-methylcarbodiimide[7,8] which form more soluble ureas in dichloromethane than **4**.

As indicated in Scheme 1, the use of two equivalents of a Boc amino acid with one equivalent of DCC leads to formation of the Boc amino acid symmetrical anhydride **3**. These preformed Boc amino acid symmetrical anhydrides have been found to give good results in many difficult sequences. Another advantage of anhydride coupling reactions is that peptide chains cannot be terminated by DCC activation of any residual TFA or AcOH remaining in the resin at the coupling step. Also there is evidence to indicate that the symmetrical anhydrides are possible intermediates in many SPPS coupling reactions, even when equimolar

amounts of Boc amino acid and DCC are used. Boc amino acid symmetrical anhydrides are usually prepared immediately before use, so their application to automatic synthesizers creates some inconvenience. In manual SPPS, the slight additional work due to their preparation is of little consequence. Symmetrical anhydrides should not be used for coupling Boc-Arg(Tos)-OH since undesired insertion reactions occur. The main disadvantage of amino acid symmetrical anhydrides is that only one equivalent of amino acid is consumed to form the peptide bond, so in SPPS where large excesses of activated amino acid are generally used, this excess is twice as large when symmetrical anhydrides are chosen.

Acylation of Peptide-Resin with Boc-Protected Amino Acid Using DCC; General Procedure:.[9]

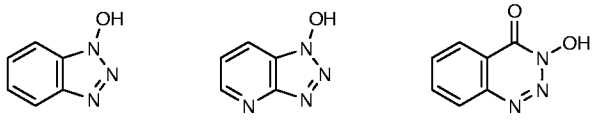
CAUTION: *DCC may cause severe allergic reactions to sensitive persons and the use of disposable gloves is recommended. Avoid all contact with the skin or mucosa. Wash immediately with acetone after any contact. In order to prevent DCC contamination of the balances, transfer DCC in the fumehood over disposable paper into a tared vial. Reweigh the vial with DCC and add dichloromethane to obtain a 1 M solution. Prepare enough DCC reagent for the entire synthesis. Do not pipette the DCC solution by mouth.*

This coupling procedure is the one recommended by Stewart and Young.[9] Before the acylation step, the N-terminal protecting group was removed and the classical washings were carried out. The number of moles of Boc amino acid is usually 2.5 times the number of moles of the first amino acid on the resin. Larger excesses of Boc amino acid and DCC can be used where steric hindrance has caused lower yields in the coupling reaction. The volume of solvent should be kept low during the coupling step to favor the coupling reaction over the rearrangement to the acyl urea byproduct. If the amino acid derivative is insoluble or only sparingly soluble in CH_2Cl_2, a mixture of purified DMF and CH_2Cl_2 is used in the diimide coupling reaction instead of CH_2Cl_2. Such derivatives are Boc-Arg(NO_2), Boc-Arg(Tos), Boc-His(Tos), Boc-Trp, and Boc-Asn(Xan). The amino acid is dissolved in the minimum amount of DMF, then CH_2Cl_2 is added. DMF should be used only when necessary and kept to a minimum since it can contain Me_2NH and the rearrangement of the reactive *O*-acylisourea to the unreactive *N*-acylurea is faster in DMF than in CH_2Cl_2. If the coupling reaction is performed with some DMF, it should be added as a washing solvent after the coupling step. Generally a time of 1.5 to 2 min is satisfactory to allow the equilibrium of Boc amino acid with the resin. Then an equimolar (to the Boc amino acid) quantity of DCC (1 M soln in CH_2Cl_2) is added. After 30 min most coupling reactions are complete, then three CH_2Cl_2 washes can be performed before the Kaiser ninhydrin test[109] or other monitoring reactions may be carried out on a small amount of peptidyl resin. If the test is positive, recoupling can be carried out. After complete reaction N-terminal deprotection can be performed and the cycle repeated.

4.3.3.1.2 *N,N'*-Dicyclohexylcarbodiimide with Additives

The inconveniences of DCC, *N*-acylurea formation, dehydration of Asn and Gln residues, and racemization, can be reduced by performing the acylation step in the presence of a suitable nucleophile which reacts very quickly with the *O*-acylisourea before side reactions can occur. An acylating agent of lower potency but still sufficiently reactive toward aminolysis and more resistant towards side reactions and racemization is formed. The additive is usually applied in equimolar amount with the two components to be coupled. As a result, there are two moles of nucleophiles, the additive and the amino component, present in the reaction mixture for each mole of carboxylic acid component or carbodiimide. Therefore the lifetime of highly reactive intermediates, such as *O*-acylisoureas and symmetrical anhydrides is considerably reduced. The concentration of the additive that acts as a second nucleophile increases with respect to the carboxylic component during the coupling reaction, because it is continuously regenerated. The most commonly used additives used in combination with DCC are shown in Scheme 2.

Scheme 2 The Most Commonly Used Additives in Combination with *N,N'*-Dicyclohexylcarbodiimide in Stepwise Solid-Phase Peptide Synthesis[12,16–18]



1,2,3-Benzotriazol-1-ol

Numerous additives which generate an active ester in situ have been investigated. The most common are *N*-hydroxysuccinimide (HOSu), 1,2,3-benzotriazol-1-ol (**6**, HOBt), 7-aza-1,2,3-benzotriazol-1-ol (**7**, HOAt) and 3-hydroxy-1,2,3-benzotriazin-4(3*H*)-one (**8**, HODhbt). *N*-Hydroxysuccinimide active esters[10,11] have not given sufficiently good results to be applied in SPPS. However, in situ generated HOBt esters are the most commonly used. HOBt esters of protected amino acids[12] are easily formed from DCC or DIC/HOBt in situ. Efficient couplings were obtained by preforming the active esters in a separate reaction vessel before addition to the peptide-resin. HOBt esters can be used for coupling Asn and Gln without side-chain protection. The DCC/HOBt combination is an extremely efficient method for stepwise solid-phase synthesis. It is not expensive, suppresses dehydration of Asn and Gln side-chain carboxamides, trifluoracetylation side reactions,[13] and racemization of Boc-im-benzyl-L-histidine.[14] These esters can be also prepared from BOP, PyBOP, TBTU, and HBTU coupling reagents and react rapidly in most cases. The use of HOBt with Boc-His(Tos)-OH should be avoided since the tosyl group is removed by HOBt.[15]

Acylation of Peptide-Resin Using DCC/HOBt; General Procedure:[12]

CAUTION: *DCC may cause severe allergic reactions to sensitive persons and the use of disposable gloves is recommended. Avoid all contact with the skin or mucosa. Wash immediately with acetone after any contact. In order to prevent DCC contamination of the balances, transfer DCC in the fume hood over disposable paper into a tared vial. Reweigh the vial with DCC and add dichloromethane to obtain a 1 M solution. Prepare enough DCC reagent for the entire synthesis. Do not pipet the DCC solution by mouth. Violent decomposition may occur when either HOAt or HOBt are dried at elevated temperatures.*

Before coupling, the carboxylic group of the N-protected amino acid must be activated. Activation by DCC/HOBt is illustrated in Scheme 1. The N-protected amino acid was dissolved in NMP then 1 M HOBt (2 mL) in NMP and 1 M DCC (2 mL) in NMP was added. After approximately 40–50 min the active ester was formed. The precipitated DCU was filtered off and the activated amino acid was transferred into the reaction vessel containing the peptide-resin. After 20–40 min DMSO, 10–15% of the total volume, was added to enhance solvation of the peptide-resin and to increase coupling in difficult sequences. After coupling, the peptide-resin was washed with NMP and CH_2Cl_2 and was then ready for the next step. These experimental conditions are similar for the use of HOBt (**6**), HOAt (**7**), and HODhbt (**8**).

Other Additives

Other auxiliary nucleophiles can be used in combination with DCC. The most important ones include 7-aza-1,2,3-benzotriazol-1-ol (**7**, HOAt)[16] and 3-hydroxy-1,2,3-benzotriazin-4(3*H*)-one (**8**, HODhbt) which will be described in the section concerning the fragment condensation (Section 4.3.2.2). The use of isolated 3-hydroxy-1,2,3-benzotriazin-4(3*H*)-ones (**8**, HODhbt)[19] will be discussed under active esters, (Section 4.3.3.1.5). The use of HOAt in peptide chemistry has been discussed by Carpino.[16] It decreases the time course of coupling reactions, reduces the loss of chiral integrity, and provides a visual indication, yellow to colorless, of the reaction endpoint in solution. HOAt was shown to be a superior coupling additive in both solution[16] and solid-phase syntheses[17] when compared with HOBt. HOAt can be used as a direct substitute for HOBt. Furthermore, HOAt like HOBt is compatible with a variety of activation methods. The performance of HOAt derivatives was compared

for references see p 786

with HOBt based analogues using Fmoc-PAL-PEG-PSty for the synthesis of the 65–74 fragment of the acyl carrier protein. The conclusions were that HOAt is superior to HOBt as an additive in DIC and pentafluorophenyl ester (Pfp) mediated couplings. Solid-phase synthesis of cyclosporin peptides containing *N*-methyl amino acid derivatives also proved the efficiency of HOAt active esters.[20] HOAt is compatible with Boc- and Fmoc-based methods, solution and solid-phase synthesis, batch and continuous-flow synthesizers.

HOAt is very stable, not hygroscopic, and can be stored indefinitely. HOAt solutions in DMF (0.3 M) can be stored in an inert atmosphere for weeks.

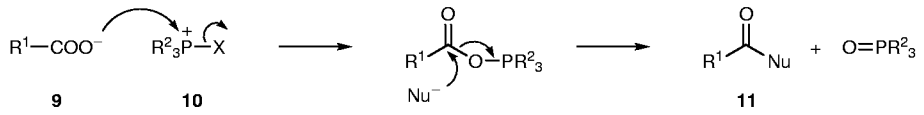
CAUTION: *When HOAt is dried at elevated temperatures violent decomposition may occur.*

4.3.3.1.3 Phosphonium Salts

Acyloxyphosphonium species, which can be generated by the attack of carboxylate anions **9** on suitable phosphonium cations **10**, react rapidly with nucleophiles at the acyl carbon to yield **11** (Scheme 3). They are intermediates in the so-called oxidation–reduction condensation,[21,22] and recently a great number of phosphonium cations have been developed as direct coupling reagents. The most common is Castro's reagent: benzotriazol-1-yloxy-tris(dimethylamino)phosphonium hexafluorophosphate (**12**, BOP) (Scheme 4).[23] This coupling agent, introduced in 1975, has been applied widely in peptide synthesis and was recognized as a powerful reagent early in the 1980s. Although it has been successful, the fact that hexamethylphosphoric triamide (HMPA), a carcinogenic material, is released during the reaction stimulated the development of a large number of phosphonium and uronium salts in peptide chemistry. Among them, two classes can be distinguished: phosphonium salts derived from HOBt or HOAt and halogenophosphonium salts (Scheme 4).[23,31,33]

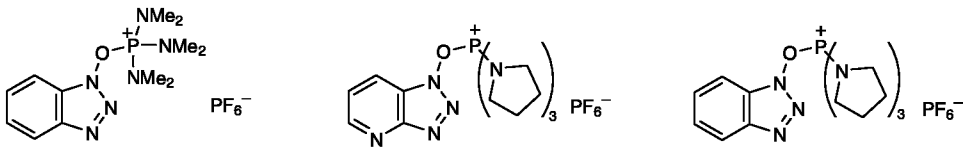
CAUTION: *HMPA is a substance suspected of carcinogenic potential in man. Appropriate safety precautions and procedures should be adopted when handling this reagent.*

Scheme 3 Generation of Acyloxyphosphonium Species[21,22]



The advantage of these reagents relies on their simplicity of use. An equivalent of the reagent is added to a 1:1 mixture of the carboxylic component and tertiary base in an inert solvent at room temperature. The mixing can be carried out in the presence of the amino component. These coupling reagents are compatible with Z, Boc, and Fmoc chemistry, solution and solid-phase synthesis, polystyrene, polyamide, and continuous flow resins.

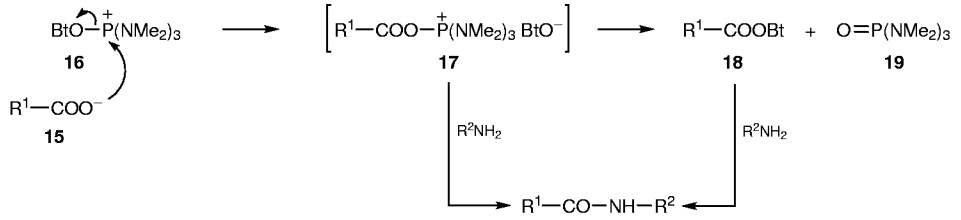
Scheme 4 Coupling Reagents Derived from 1,2,3-Benzotriazol-1-ol and 7-Aza-1,2,3-benzotriazol-1-ol[23,31,33]



Benzotriazol-1-yloxytris(dimethylamino)phosphonium Hexafluorophosphate

The mechanism of action of BOP has been studied.[24] In the first step, the carboxylate anion **15** of the N-protected amino acid displaces an oxybenzotriazolate anion from the phosphonium salt **16**, the resulting ion pair **17** forms highly reactive nucleophilic and electrophilic counter ions that collapse immediately to give the neutral active ester **18**, and phosphine oxide **19**. Either active ester **18** or acyloxyphosphonium **17** can be trapped by the amino component present in the solution to yield an amide (Scheme 5). The solvent can be THF, MeCN, dichloromethane, DMF, DMSO, HMPT, or NMP according to the reactant's solubility. Although all these inert solvents, which result in the swelling of resins, can be used, the lower the solvent polarity, the faster the reaction proceeds. No reaction occurs until triethylamine, *N*-methylmorpholine, or preferably DIPEA is added. The tertiary amine must be used in sufficient amount, three equivalents toward BOP and N-protected amino acids in Boc chemistry, and two equivalents in Fmoc chemistry.

Scheme 5 Proposed Coupling Pathway with Benzotriazol-1-yloxytris(dimethylamino)phosphonium Hexafluorophosphate[24]



Side reactions with most amino acid derivatives are avoided. In Boc/Bzl strategy dehydration of asparagine is usually not observed, since the final treatment of the protected peptide-resin with hydrofluoric acid reverses nitrile formation.[25] In Fmoc chemistry however, the final cleavage conditions are too mild to rehydrate the nitrile formed during synthesis, so the side reaction can be observed,[26] but it is reduced by addition of HOBt. The use of side-chain protected Asn or Gln gives the best results since formation of nitrile containing peptide does not occur. In Boc chemistry, trifluoroacetic acid is not activated with BOP and trifluoroacetylation is never observed as a side reaction, allowing the suppression of a separate neutralization step. Indeed, after TFA treatment and washing, it is convenient to introduce the next Boc amino acid and BOP as solids, in a minimal volume of solvent and then DIPEA. This neutralization step in the presence of the N-protected activated amino acid derivative suppresses or reduces to an acceptable minimum the piperazine-2,5-dione (DKP) formation. Interestingly and surprisingly, dicyclohexylamine (DCHA) salts have been used directly without neutralization in the BOP procedure, only small amounts of dicyclohexylamide were formed which were washed from the resin. This procedure is especially convenient in the case of Boc-His(Boc) and Boc-Nle which are commercially available as DCHA salts.[27] BOP reagent has been found to be very efficient in the syntheses of (22–29), (13–29), and (1–29) analogs of [Ala^{15}]-GRF-NH$_2$[28] (GRF = growth hormone releasing factor) and of cyclic GRF analogs.[29] It was also used with success in coupling the unprotected Boc(hydroxy) amino acids Boc-Thr-OH and Boc-Tyr-OH.[30]

Protocol for Use of BOP in Coupling Fmoc or Boc-Protected Amino Acids; General Procedure:[27]

CAUTION: *Hexamethylphosphoric triamide (HMPA) is formed as a side product. HMPA is a substance suspected of carcinogenic potenital in man. Appropriate safety precautions and procedures should be adopted when handling this reagent.*

for references see p 786

In a typical coupling reaction, BOP and the N-protected amino acid are added to the peptide-resin in the presence of a free (Fmoc strategy) or salted (Boc strategy) amino group. BOP and the N-protected amino acid can be weighed on the same disposable paper or in the same container as solids and then introduced as solids into the reaction vessel containing the peptide-resin. BOP and the N-protected amino acid are used in twofold or more excess compared to the amino component. A minimal volume of solvent is then added, usually CH_2Cl_2, DMF, or NMP, and the reaction components are mixed. A threefold excess of DIPEA or another tertiary amine, a twofold excess of BOP and carboxylic acid component are used compared to the amine component, and the pH is maintained above 8 (measured with moistened pH paper) by addition of DIPEA in order to improve the acylation reaction. A negative Kaiser test is generally observed after 15 to 30 min. Washing steps are very fast and easy since no insoluble byproduct such as DCU is formed. In Boc chemistry, a solid-phase manual procedure allows a cycle time, including acylation and deprotection, of less than 1 h. In Fmoc chemistry, this time is shortened due to the rapid deprotection step. Washings were performed twice with the coupling solvent, twice with methanol to shrink the resin, and then twice with CH_2Cl_2 (Boc chemistry) or DMF (Fmoc chemistry) to swell the resin and prepare it for the deprotection step.

SAFETY: BOP is a white, crystalline and non-hygroscopic solid. It can be kept indefinitely in the dark. The safety reservation regarding this reagent concerns the side product, hexamethylphosphoramide, which is toxic. This is one reason why PyBOP was introduced as a replacement.

Benzotriazol-1-yl-oxytripyrrolidinophosphonium Hexafluorophosphate

This reagent, benzotriazol-1-yl-oxytripyrrolidinophosphonium hexafluorophosphate (**14**, PyBOP)[31] (Scheme 4), was designed in order to avoid the formation of toxic HMPA during acylation. As with BOP, it is assumed that the first step is the carboxylic acid activation which involves formation of an acyloxyphosphonium salt.[32] This initial salt is then attacked by the benzotriazolyloxy anion to form the benzotriazolyl active ester which then reacts with the amino component. PyBOP can easily replace the BOP reagent and is especially suitable for solid-phase peptide synthesis. It is soluble in a wide range of solvents such as DMF, dichloromethane, THF, and NMP. PyBOP is more useful in peptide synthesis on solid support than in solution. The byproduct, tris(pyrrolidino)phosphine oxide is partially water-soluble and is easily removed by washing. PyBOP is used under the same experimental conditions as BOP. Note that PyBOP is a white, crystalline and non-hygroscopic solid. It can be kept as a solid, but solutions of PyBOP cannot be stored for more than 24 hours.

7-Azabenzotriazol-1-yl-oxytripyrrolidinophosphonium Hexafluorophosphate

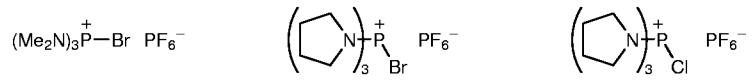
7-Azabenzotriazol-1-yl-oxytripyrrolidinophosphonium hexafluorophosphate (**13**, PyAOP)[33] (Scheme 4) as well as PyBOP **14**, is a pyrrolidine-derived phosphonium salt which is preferable to the corresponding dimethylamino derivative **12** due to the reduced toxicity of the phosphoramide side product. It is a phosphonium salt derivative of HOAt. As demonstrated by Carpino et al.,[33] it was shown to be superior to PyBOP in solid-phase synthesis of the 65–74 (H-Val-Gln-Ala-Ala-Ile-Asp-Tyr-Ile-Asn-Gly-NH_2) fragment of the acyl carrier protein by a Fmoc/tBu-protection scheme. Poly(ethylene glycol)-polystyrene (PEG-PS) resin bearing a peptide-amide linker derived from 5-[4-(9-fluorenylmethoxycarbonyl)-aminomethyl-3,5-dimethoxyphenoxy]valeric acid was used on a continuous-flow Millipore 9050Plus synthesizer. In this study coupling times were shortened to 1.5 minutes, excess of reagents were reduced to 1.5 equivalent of Fmoc amino acid and PyXOP (X = A, B) and 3 equivalents of DIPEA versus amino component, in order to emphasize the differences between coupling reagents. The superiority of PyAOP was also demonstrated in the synthesis of the model pentapeptide H-Tyr-Aib-Aib-Phe-Leu-NH_2[33] and in the synthesis of cyclic tachykinin antagonist cyclo(Tyr-D-Trp-Val-D-Trp-D-Trp-Arg-Asp).[34] Another advantage of PyAOP relative to the corresponding uronium salts is that excess of PyAOP does not participate in any side reactions at the amino terminus by formation of a guanidinium group which is stable and terminates the growing chain.[35–37] PyAOP is compatible with hydrated

amino acids such as Boc-Ile-OH hemihydrate. All the experimental conditions described with BOP are reproducible with PyAOP.

As a solid PyAOP is rather stable. At 55 °C it is completely stable for two days and less than 2% decomposition occurs over a period of one week. Solutions of PyAOP (0.5 M) in DMF in a nitrogen atmosphere are usable for one week with less than 10% decomposition. In these tests[34] PyBOP is more stable than PyAOP, suggesting that the latter is the more reactive of these two species.

The great interest in phosphonium salts prompted by the usefulness of BOP, resulted in the reinvestigation of phosphorous derivatives as coupling reagents. Among all the new phosphonium salts, three can be cited which were applied to the coupling of N-methylated amino acids, bromotris(dimethylamino)phosphonium hexafluorophosphate (**20**, BroP),[38] bromotripyrrolidinophosphonium hexafluorophosphate (**21**, PyBroP),[39] and chlorotripyrrolidinophosphonium hexafluorophosphate (**22**, PyCloP)[39] (Scheme 6). They allow efficient coupling between N-methylated amino esters and *Z*-N-methylated amino acids, in contrast to PyBOP or BOP.

Scheme 6 Phosphonium Salt Coupling Reagents[38,39]



20 BroP **21** PyBroP **22** PyCloP

The latter two reagents, **21** and **22**, have been used for the synthesis of natural products bearing N-methylated amino acids.[40,41] They can also be employed for the difficult coupling of α,α-disubstituted amino acids such as α-aminoisobutyric acid (Aib),[42] or α-methylcysteine,[43] and for coupling *N*-methyl-α,α-dialkyl amino acids.[44] However, a limitation in the use of these reagents with Boc amino acids was the formation of *N*-carboxyanhydrides in some cases.[45,46] The use of BroP, like BOP, results in the release of HMPA as a byproduct and it has not been widely used in SPPS. PyCloP and PyBroP gave identical results in difficult coupling reactions.[32] PyBroP was successfully used with the "tea bag" simultaneous multiple peptide synthesis method for coupling to secondary amines,[47,48] for cyclization on the support of Oscillamide Y and analogues,[49] and for elaboration of peptoids.[50]

These reagents allow coupling without preactivation and they do not react with primary amine groups. As discussed for BOP, the quantity of base is important and the pH should be maintained at 9–10. They are soluble in the common solvents (CH_2Cl_2, DMF, NMP) used in SPPS and are not water sensitive. They are solids, stable, and usable without preactivation.

Coupling of Boc-Protected Amino Acids using PyBroP; General Procedure:[48]
The coupling reaction was performed on a secondary amine (0.18 mmol) on a derivatized Wang cross-linked polystyrene resin which was suspended in CH_2Cl_2 (3 mL). The resin was treated with a soln of Boc amino acid (0.2 mmol) in CH_2Cl_2 (0.5 mL), and DMF if necessary to achieve complete soln, followed by addition of a soln of PyBroP (0.2 mmol) in CH_2Cl_2 (0.5 mL), and finally DIPEA (0.4 mmol). After 24 h, the resin was filtered, washed (DMF, H_2O, THF), dried and the coupling process repeated.

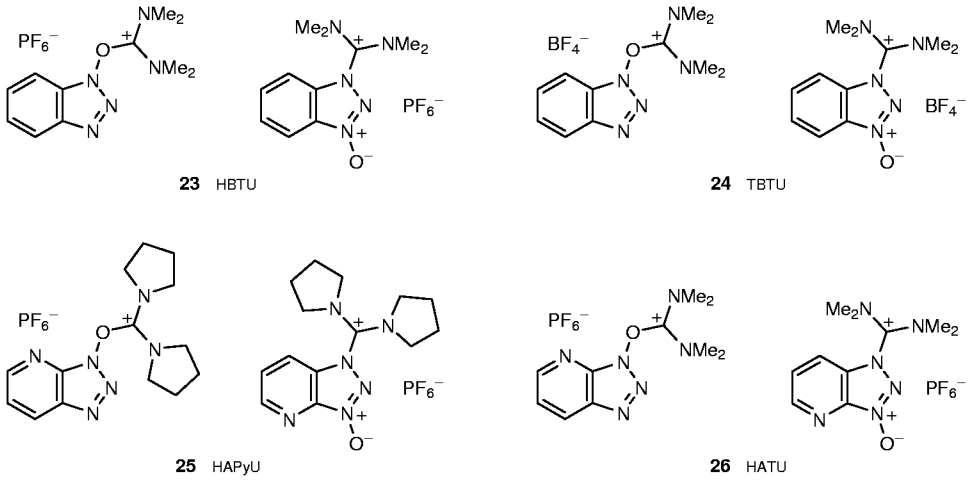
4.3.3.1.4 Uronium Salts

In parallel with BOP, 2-(1*H*-benzotriazol-1-yl)-1,1,3,3-tetramethyluronium hexafluorophosphate (**23**, HBTU) was introduced as a peptide bond coupling agent.[51] The mechanism of the peptide bond formation proceeds through benzotriazole active ester intermediates and through the tetramethyluronium ion.[52] The reaction time is very short, about 15 minutes, with yields higher than 87% and no detectable racemization.[51] The main advantage over BOP is that the byproduct is tetramethylurea, which is not toxic. This reagent, which is a good alternative to BOP for the in situ formation of hydroxybenzotriazolyl esters, was

for references see p 786

successfully applied to solid-phase synthesis in batch and continuous flow synthesizers. Couplings proceed smoothly without unwanted side reactions such as nitrile formation. Also free aliphatic hydroxy groups can be present.[53] Several other uronium reagents were introduced subsequently (Scheme 7).[53,54] One of the most useful is 2-(1*H*-benzotriazol-1-yl)-*N*,*N*,*N*′,*N*′-tetramethyluronium tetrafluoroborate (**24**, TBTU). Other new reagents based on HOAt are *O*-(7-azabenzotriazol-1-yl)-1,1,3,3-bis(tetramethylene)thiouronium hexafluorophosphate (**25**, HAPyU), and the aza analog of HBTU, *O*-(7-azabenzotriazol-1-yl)-1,1,3,3-tetramethyluronium hexafluorophosphate (**26**, HATU) (Scheme 7).[16]

Scheme 7 The Most Common Uronium Salts, also Represented as Aminium Salts[16,51–54]



These derivatives were described as uronium salts by analogy with phosphonium salts, but X-ray analysis demonstrated that HBTU, HATU, and HAPyU crystallize as aminium salts (guanidinium *N*-oxides) rather than uronium salts.[55,56] NMR studies of HAPyU in solution show the same structure as that observed in the crystalline form.[56] This discovery changes the nomenclature of these compounds, but their acronyms are unchanged, and they are described in the literature as uronium or aminium salts without discrimination. HBTU should be reformulated as *N*-[(1*H*-benzotriazol-1-yl)(dimethylamino)methylene]-*N*-methylmethanaminium hexafluorophosphate *N*-oxide, HATU as *N*-[(dimethylamino)(1*H*-1,2,3-triazolo[4,5-*b*]pyridin-1-yl)methylene]-*N*-methylmethanaminium hexafluorophosphate *N*-oxide, and HAPyU as *O*-(7-azabenzotriazol-1-yl)-*N*,*N*,*N*′,*N*′-bis(tetramethylene)uronium hexafluorophosphate.

Mechanistically, these salts are thought to act in a similar way to their phosphonium analogues, with HBTU for example, forming an OBt active ester. The superiority of the 7-aza-based coupling reagents HATU and HAPyU over OBt-based derivatives HBTU and TBTU was demonstrated by SPPS of the decapeptide model compound ACP(65–74), H-Val-Gln-Ala-Ala-Ile-Asp-Tyr-Ile-Asn-Gly-NH_2.[33] In this study, it was found that the addition of HOBt or HOAt to these coupling agents did not improve the coupling yields, with the exception of the coupling of Asn(Trt) to C-terminal Gly. The efficiency of HOAt-derived coupling reagents was also demonstrated for the incorporation of hindered amino acids such as *N*-methyl amino acids, Aib and $C^{\alpha,\alpha}$-diethylglycine (Deg).[33,57] They were found to be slightly inferior to acid fluorides which are the reagents of choice for coupling such hindered amino acids.[58]

A problem in solid-phase peptide synthesis arises when an ester linkage between the peptide and the solid support is used and results in the undesirable formation of piperazine-2,5-diones (DKP). This side reaction occurs during the deprotection of the N-temporary

protecting group of the second residue and the coupling of the third one. This intramolecular cyclization is favored by the presence of Pro and Gly residues in the first two positions of the peptide sequence. In Boc/Bzl chemistry, the piperazine-2,5-dione formation can be diminished using coupling reagents in which the neutralization step of the generated amine can occur in situ during coupling of the third amino acid, rather than neutralization in a separate step and then addition of the activated third residue. This alternative was well documented when BOP is used as the reagent.[27,59] The Boc group is removed using TFA/dichloromethane and without a neutralization step, the next amino acid is coupled with BOP, PyBOP, or PyAOP in the presence of DIPEA. In the Fmoc/tBu strategy, the situation is different since the deprotection of the Fmoc protecting group by piperidine/DMF or diethylamine/DMF leads to a free amino group. Alsina et al. described a general methodology available for classical alkoxybenzyl based linkages[60] where the second residue is introduced using the *N*-Trt (trityl) protecting groups. These authors demonstrated that this group can be removed by 0.2–2% TFA, or 1% H_2O, in dichloromethane in the presence of tBu protecting groups without subsequent cleavage of the ester linkage, 0% cleavage with Wang-type resin[61] and about 5% cleavage with HMPB resin.[62] Incorporation of the third residue was carried out without prior neutralization of the amine TFA salt, with the third amino acid *N*-Fmoc protected, the coupling reagent PyAOP and DIPEA in DMF were employed. This methodology suppresses DKP formation and allows protected peptides to be prepared with the HMPB handle. Another alternative is to use the 2-chlorotritylchloride resin,[63,64] the extreme steric hindrance of the resin suppressing the DKP formation.

Uronium salt based coupling agents are a powerful addition to the peptide chemist's arsenal, especially useful for the coupling of sterically hindered amino acids. Special care must be taken when they are used. In the presence of amino groups they form tetramethylguanidinium or dipyrrolidinylguanidinium derivatives,[26,36,37] terminating peptide chain elongation. The reaction of the uronium coupling reagent with the amine is fairly easy, 30% of the amine groups on the resin react with HBTU within 30 minutes.[35] Therefore, these authors recommended avoiding an excess of the uronium reagent toward the N-protected amino acid to be coupled and either adding this amino acid first and then the uronium salt or preactivating the amino acid prior to addition to the resin. All these reagents are similar and can be used in the experimental conditions described for HATU.[33] These reagents are all solid crystals and stable on storage in cold and dry conditions.

Standard Coupling Protocol using Fmoc-Protected Amino Acids and HATU; General Procedure:[33] Fmoc amino acid (2 equiv), HATU (2 equiv), and DIPEA (4 equiv) were stirred at r.t. for 10 min. For Asn and Gln couplings, HOAt (2 equiv) as an auxiliary nucleophile is added. Using this protocol ACP (65–74) has been obtained with a purity of 96%. For most complex peptides, extended coupling times and extra equivalents of reagents may be advisable.

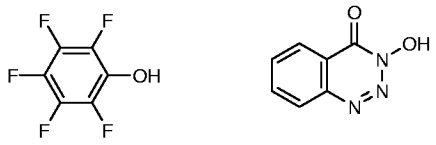
4.3.3.1.5 Active Esters

In this section, only isolable active esters will be described. The rate of peptide or amide bond formation proceeding by ester aminolysis depends, among other factors, on the rate-limiting expulsion of the leaving group. Schwyzer and his associates[65,66] investigated the aminolysis of cyanomethyl esters and Gordon, Miller, and Day[67,68] explored the reactivity of thiophenyl esters towards aminolysis. Bodanszky[69] introduced electron-withdrawing substituents to the aromatic ring of phenol to yield the efficient 4-nitrophenyl esters which allowed the synthesis of oxytocin[70,71] with a new strategy, the stepwise elongation of a peptide chain. The research on other withdrawing groups in the aromatic ring led to the pentachlorophenyl esters (OPcp), introduced by Kupryszewski,[72,73] with high reactivity but suffering from the steric effect of the bulky chlorine atoms. The logical replacement of the five chlorine atoms by fluorine[74–76] produced very potent active esters which are suitable for

for references see p 786

SPPS. In contrast to OPcp esters, these latter OPfp esters derived from **27** and those derived from HODhbt **8**[18] are the most commonly used active esters in SPPS and have been applied to automated continuous flow Fmoc/tBu synthesis. Their use notably simplifies the course of the solid-phase synthesis by avoiding individual preactivation procedures.

Scheme 8 Precursors of Pentafluorophenyloxy and 3-Hydroxy-1,2,3-benzotriazin-4(3*H*)-one Active Esters[18]



27 HOPfp **8** HODhbt

Pentafluorophenyloxy Active Esters

Pentafluorophenyloxy (OPfp) active esters are synthesized from the protected amino acids, the hydroxy compound **27** (Scheme 8) and DCC,[77] but can also be prepared from pentafluorophenyl trifluoroacetate.[78] Pentafluorophenyloxy esters react somewhat slowly, although the addition of HOBt[79–81] or HOAt[16] enhances their reactivity. Their use allows incorporation of Asn without protection of the side-chain group.[79,26] Fmoc derivatives of Tyr and Ser have also been incorporated as preformed active esters without side-chain protection.[82,83] Their efficiency and simplicity of use was demonstrated by the synthesis of the difficult ACP (65–74) sequence.[79]

Synthesis of the Acyl Carrier Protein Decapeptide (65–74) Sequence, H-Val-Gln-Ala-Ala-Ile-Asp-Tyr-Ile-Asn-Gly-OH; General Procedure:[79]
Fmoc amino acid pentafluorophenyloxy esters were prepared following the procedure of Kisfaludy and Schön.[77] The poly(dimethylacrylamide) resin was functionalized with an internal reference norleucine residue and with the acid-labile 4-alkoxybenzyl alcohol linkage agent. Esterification of the C-terminal Fmoc-Gly residue was performed using the preformed symmetrical anhydride with DMAP as catalyst. The preformed ester of the amino acid (2–5 equiv) and HOXt (HOBt or HOAt) (2–5 equiv) were dissolved in DMF, and the resulting soln was added to the resin bearing the free amino group. In the presence of HOXt, the acylation is generally complete within 30 min and washing steps can be carried out. The completed decapeptide was cleaved from the resin with 95% TFA and the yield determined to be 91% by quantitative amino acid analysis.

3-Hydroxy-1,2,3-benzotriazin-4(3*H*)-one Active Esters

The favorable acylating properties of 3-hydroxy-1,2,3-benzotriazin-4(3*H*)-one (HODhbt) esters were recognized by König and Geiger in 1970.[18] Atherton et al. found that Fmoc amino acid esters are very easily prepared and are generally stable crystalline solids, most of them may be stored at low temperature for long periods.[84] During the preparation of HODhbt esters it is possible to remove the triazine side product. As described by König and Geiger, the preparation of these esters with the aid of DCC may be accompanied by the formation of a byproduct, 3-[(2-azidobenzoyl)oxy]-1,2,3-benzotriazin-4(3*H*)-one, which can be easily detected by HPLC. All Fmoc amino acid derivatives should therefore be rigorously purified before use in solid-phase synthesis since this byproduct is an effective chain-terminating agent. In addition, these esters provide a sensitive color indicator of the presence of residual unreacted amino groups. In the presence of free amino groups, a bright yellow color is observed, this coloration disappears with complete acylation.[84,85] It should be noted that oxodihydrobenzotriazine esters are ineffective in ester-forming reactions, even in the presence of DMAP catalyst and cannot be used for esterification of resin hydroxy groups. Synthesis of the difficult ACP (65–74) sequence was achieved with high purity and yield with 2 equivalents of Dhbt esters, except for the final valine which was incorporated using 4

equivalents. The long-term stability of these active esters is not yet known. The authors recommend storage at low temperature (−20 °C), especially for asparagine, glutamine, and arginine derivatives.

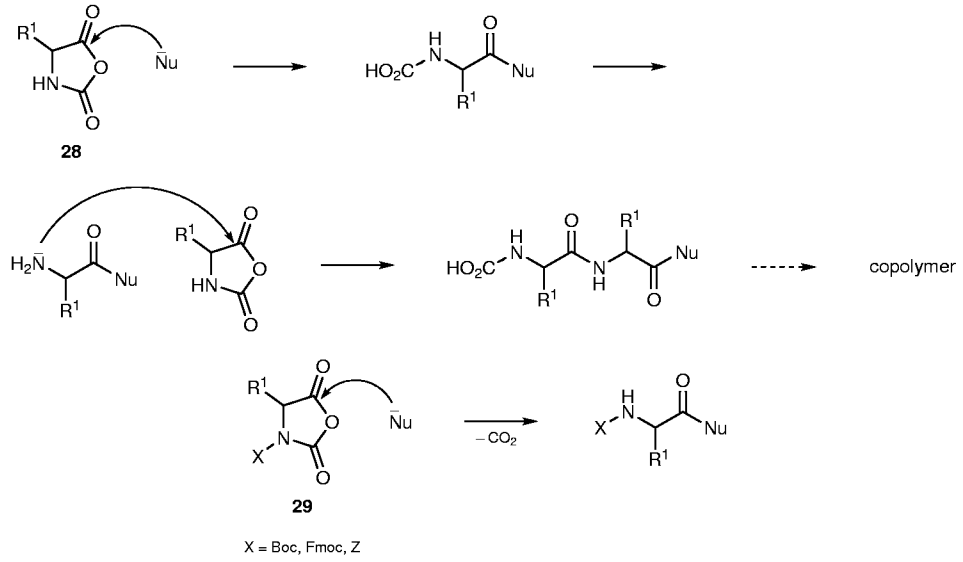
Synthesis of the Acyl Carrier Protein Decapeptide (65–74) Sequence: H-Val-Gln-Ala-Ala-Ile-Asp-Tyr-Ile-Asn-Gly-OH; General Procedure:[84]

The poly(dimethylacrylamide) resin was functionalized with an internal reference norleucine residue and with the acid-labile 4-alkoxybenzyl alcohol linkage agent. Esterification of the C-terminal Fmoc-Gly residue was performed using the pentafluorophenyl ester (5 equiv) with DMAP as catalyst (1 equiv) and was complete in 1–2 hours. All peptide bond-forming reactions used Fmoc amino acid ODhbt ester (4 equiv) in DMF but a later synthesis using only 2 equiv except for the final valine (4 equiv) also gave satisfactory results. The completed decapeptide was cleaved from the resin with 95% TFA and purified by HPLC.

4.3.3.1.6 Urethane-Protected α-Amino Acid *N*-Carboxyanhydrides

α-Amino acid *N*-carboxyanhydrides (**28**, NCAs) or Leuchs' anhydrides[86] were not widely used in peptide synthesis owing to their propensity to polymerize (Scheme 9). The reaction conditions must be carefully controlled and optimized for each coupling step in order to avoid this problem. Except for the solution synthesis of short peptides, NCAs are not used in solid-phase synthesis.

Scheme 9 Reaction of α-Amino Acid *N*-Carboxyanhydrides and Urethane-Protected α-Amino Acid *N*-Carboxyanhydrides with a Nucleophile[86]



However urethane-protected α-amino acid *N*-carboxyanhydrides, UNCAs **29**,[87] are more stable and are not sensitive to polymerization. UNCAs represent a new, broadly applicable class of stable protected and activated amino acid derivatives which can be used directly to form peptide bonds, having the advantage of generating carbon dioxide as the only by-product (Scheme 9). They can be used in solid-phase continuous-flow or batch reactors as well as in solution. Fuller et al. demonstrated their efficiency by synthesizing the acyl carrier decapeptide (65–74) via a Fmoc strategy[87] in a peptide-synthesis flow reactor and obtained the crude peptide in 73% yield with a high homogeneity. Xue and Naider[88] studied the coupling kinetics, the influence of solvents and base in the case of Boc-UNCAs and showed

for references see p 786

that the fastest coupling rate was found in DMF in the presence of one equivalent of DIPEA. The comparison of UNCAs with BOP and HBTU was studied in a standard procedure for the synthesis of tetra, penta, undeca, and pentadeca-peptides and it was found that UNCAs coupling were equally efficient to the other methods. Furthermore, tests of racemization showed that UNCAs do not lead to significant loss of chirality at the α-carbon of amino acid residues. These amino acid derivatives have been used in solid-phase synthesis and reviewed by Fuller et al.[89] One of their advantages is their use for esterification of a hydroxy-containing resin with less than 0.1% racemization. The esterification of Fmoc-His(N^{im}-Trt) to the Wang resin has been problematical, the best reported procedure using DCC/DMAP afforded 15.4% of the D-enantiomer.[90,91] The use of Fmoc-His(N^{im}-Trt)-UNCA afforded racemization free esterification of the Wang resin in toluene in presence of NMM at room temperature.[92] UNCAs are compatible with both Fmoc and Boc chemistry and are also used with allyl side-chain protection.[93] They are superior to classical coupling reagents for coupling of hindered peptides or amino acids when used over a prolonged reaction time or at elevated temperature in solution synthesis.[94]

UNCAs are able to dimerize in the presence of some bases to yield pyrrolidine-2,4-diones,[95,96] therefore the solvent and the base have to be chosen with care since, if the acylation step is slow, the risk of dimerization increases. This problem is not very important in solid-phase synthesis as the formed pyrrolidine-2,4-diones can be easily eliminated by the washing steps, but the concentration of reactive UNCAs can be drastically reduced by this phenomenon. No pyrrolidine-2,4-diones are formed when NMM is used as a base in dichloromethane, DMF, and THF as solvents, while DIPEA in dichloromethane and DMF can yield this side-product.

UNCAs should be stored at −15 °C in polyethylene or polypropylene bottles. For use they are routinely warmed to room temperature and the stock compound is returned to storage at −15 °C. Storage under nitrogen is not necessary, but exposure to water must be avoided. Under these conditions, they are stable for one to three years. They can be purified from decomposition compounds by dissolving in organic solvents such as ethyl acetate and then washed with a saturated aqueous solution of sodium bicarbonate and brine. The organic layer, after drying over sodium sulfate, is concentrated to give pure UNCA. In solution, most of the Fmoc and Boc-UNCAs have been evaluated after 6 h at 50 °C as 0.3 M solutions in THF or DMF. The majority of them are not degraded under these harsh conditions.[89] They are soluble in many common organic solvents including THF, DMF, and dichloromethane which are usually used in peptide synthesis.

Synthesis of GRF (1–29) by Fmoc Continuous-Flow Strategy on Fmoc-Arg(Pmc)-polyhipe Polystyrene Resin:[89]

The following stepwise protocol was adopted:

1. Deprotection of the Fmoc group by a soln of 20% piperidine in DMF (7 min, 30 mL · min^{-1}).
2. Washing with DMF (5 min, 30 mL · min^{-1}).
3. Coupling step with UNCA (3 equiv, 20 min, 30 mL · min^{-1}) except for incorporation of Fmoc-Arg(Pmc)-OH which was coupled with BOP/DIPEA (6 equiv) in 20 min.
4. Recoupling step with one equiv of UNCA and DIPEA from the Ile^{26} incorporation to the end of synthesis (20 min, 30mL · min^{-1}) except for incorporation of Fmoc-Arg(Pmc)-OH using BOP/DIPEA.
5. Washing with DMF (5 min, 30 mL · min^{-1}).
6. Repeat steps 1–5.

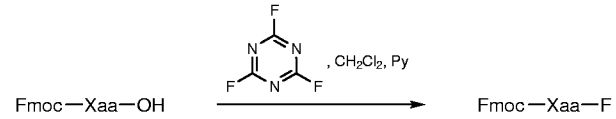
Starting from Fmoc-Arg(Pmc)-polyhipe polystyrene-resin (1.7 g, 0.493 mmol), the weight increase of the resin was 1.6 g. The crude peptide after cleavage; yield: 1.3 g (93%, based on starting resin), showed two main peaks on RP-HPLC which were the GRF(1–29) peptide and the corresponding oxidized methionyl compound, as determined by ^1H NMR.

4.3.3.1.7 Fluorides

The use of amino acid halides goes back to the very first days of peptide synthesis, when in 1903, Fischer reacted *N*-carbethoxyglycine chloride with amino acid esters to yield dipeptides. Further progress had to await the discovery of the benzyloxycarbonyl (Z) group as a temporary amino protecting group.[97] These applications are very limited because of the poor stability of these compounds. Boc analogues are less stable. The development of the new Fmoc N-protecting group[98] promoted the use of Fmoc amino acid chlorides, since the Fmoc group is very stable in acidic conditions. Such Fmoc amino acid chlorides can be prepared from the Fmoc amino acids and thionyl chloride and are stored in a dry atmosphere. Their practical utility is restricted to bifunctional amino acids or trifunctional amino acids whose side chains are benzyl-protected because of the incompatibility of Boc- and OtBu-based protecting groups to the conditions required to form the acid chlorides. The coupling of these derivatives requires the presence of a hydrogen chloride scavenger. In the presence of tertiary amines such as DIPEA or NMM, acid chlorides are converted into corresponding oxazolones which are less reactive than acyl chlorides.[99] The use of a hindered base such as 2,6-di-*tert*-butylpyridine almost avoids the formation of oxazol-5(4*H*)-one. The addition of one equivalent of HOBt in the coupling reaction involves the initial formation of the OBt ester. Fmoc amino acid chlorides have been successfully applied in solid-phase peptide syntheses.[100] All these reactions occur without significant racemization.

Owing to their compatibility with tBu, Boc, or Trt side-chain protecting groups, Fmoc amino acid fluorides[101] have much greater utility than the corresponding chlorides. They can be easily prepared from the Fmoc amino acids with 2,4,6-trichloro-1,3,5-triazine in the presence of pyridine according to the Olah procedure[102,103] (Scheme 10). More recently *N,N*-diethylaminosulfur trifluoride (DAST) was used for this conversion without base.[104] Other advantages of the acid fluorides relative to the chlorides are their greater stability towards water, and their relative lack of conversion to oxazol-5(4*H*)-one in the presence of tertiary amines. They are also very soluble in organic solvents (such as DMF) allowing the acylation step to be carried out at a high concentration of acylating agent. Acid fluorides are stable when dissolved in DMF for extended periods, 3 days or longer, and are suitable for use in automatic peptide synthesizers. Another difference from the corresponding chlorides is the ability of fluorides to carry out acylation in the absence of base, thereby reducing the danger of racemization.[105] Excellent solid-phase syntheses of ACP (65–74) acyl carrier protein, magainin-II-amide, and hCRH, three model peptides which are considered to be difficult sequences were achieved with Fmoc amino acid fluorides.[106] Furthermore, they have been tested in very highly hindered systems such as peptides incorporating two or more consecutive α-amino isobutyric acid (Aib) units. Elongation of hCRH (32–41) with four consecutive Aib units was successful resulting in a peptide of excellent quality.[106] Protected acid fluorides also enabled the first solid-phase syntheses of the peptaibols, naturally occurring peptides containing up to 60% Aib residues, to be accomplished.[107]

Scheme 10 Formation of Fmoc Amino Acid Fluoride Using 2,4,6-Trichloro-1,3,5-triazine[102,103]

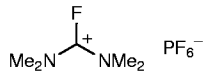


The conversion in situ of N-protected amino acids into the corresponding fluorides via the use of a new reagent, tetramethylfluoroformamidinium hexafluorophosphate (**30**, TFFH) has been reported (Scheme 11).[108] This discovery enabled the use of TFFH under conditions similar to those used for uronium or phosphonium reagents derived from HOBt or HOAt.

for references see p 786

The advantage of using TFFH for in situ generation of acid fluorides is the stability of TFFH, which is a non-hygroscopic salt under ordinary handling conditions, and the ability to couple His and Arg by this method. These amino acids could not be converted into shelf-stable amino acid fluorides for use in peptide synthesis, but generated as transient intermediates with TFFH, they can be routinely coupled. So TFFH appears to be an ideal coupling reagent for solid-phase syntheses, being readily available, inexpensive, and able to provide peptides of high quality.

Scheme 11 Tetramethylfluoroformamidinium Hexafluorophosphate Structure[108]



30

TFFH (30):[108]
To a soln of TFFH (5.6 g, 21.5 mmol) in dry MeCN (30 mL) was added oven-dried anhyd KF (1.16 g, 20 mmol) and the mixture was stirred at rt for 2–3 h (monitoring by ^{1}H NMR spectroscopy). Longer times were required for large-scale preparations. Following filtration of KCl, the filtrate was concentrated and the residue recrystallized (MeCN/Et_2O) to give the non-hygroscopic fluoroformamidinium salt as white crystals; yield: 4.3 g (92.3%); mp 111–112 °C; ^{1}H NMR ($CDCl_3$-DMSO-d_6) δ 3.175 (d, CH_3). Calcd for $C_5H_{12}F_7N_2P$: C, 22.72; H, 4.55; N, 10.61. Found: C, 22.73; H, 4.50; N, 10.63.

Coupling Procedure for the Synthesis of H-Tyr-Aib-Aib-Phe-Leu-NH_2 Using TFFH:[108]
A Biosearch 9050 instrument was used with DMF as solvent, and programmed for a 7 min preactivation time, 7 min deblocking, 30 min coupling times [5 fold excess of the acid and TFFH and 10 fold excess of the base (DIPEA)] were used for all amino acids. With Aib-Aib a 1 h double coupling was used. The peptide was obtained in approximately 88% yield with a purity of 92%.

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4.3.3.2 Acylation Methods in Fragment Condensation Approach

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Dedicated to the memory of Dr. Marc Rodriguez

Two main strategies can be applied to the synthesis of large peptides, stepwise synthesis and convergent synthesis. Stepwise synthesis consists of the peptide chain elongation from the C-terminus to the N-terminus by repetitive N^{α}-amino group deprotection and N-protected amino acid coupling steps. Convergent synthesis, also called fragments (or segments) condensation, involves the synthesis and coupling of peptide fragments. Both strategies have been applied in solution, on solid support or in combination.

Obtaining quantitative yields in amino acid coupling steps and in the repetitive N^{α}-amino group deprotection is the main problem in stepwise solid-phase peptide synthesis (SPPS). Since purification is only possible in the last step accumulation of deleted peptides and truncated sequences occurs. Therefore these undesired compounds are present in the crude mixture after final cleavage from the solid support are usually closely related to the target peptide in their physical properties, resulting in difficult purification steps.

Despite the success of stepwise SPPS in the assembly of large peptides[1–3] these limitations led research groups to develop a methodology called convergent solid-phase peptide synthesis (CSPPS) for the chemical synthesis of large peptides and proteins. Homogeneous large target molecules are probably best prepared by convergent strategies that rely on the coupling of peptide fragments. Each fragment can be purified and characterized before coupling and the purification of the desired compound at the end of the synthesis is easier. Incomplete couplings between peptide segments results in peptides that are significantly different from each other. The potential improvement in the purification step of the final mixture along with the decrease in the number of steps are the most important advantages of the fragment condensation strategy.

In the solid-phase convergent methodology, all reactive functional groups must be protected except those required for amide bond formation to obtain an unambiguous coupling. Alternative approaches that allow unprotected peptide segments to be coupled are generally applied to the synthesis in solution, such as the ligation approach (see Sections 4.1.5 and 4.3.3.2.4). Preparation of protected fragments by SPPS is possible because extensive investigations of many different types of anchors on solid supports have been carried out. Such modified resins, in contrast with classical resins yielding unprotected peptides, allow selective cleavage of the linker between the peptide and the resin and are stable under conditions necessary for the peptide synthesis. A variety of different chemical reactions can be used for the cleavage of protected peptide fragments from solid supports. If the peptide is synthesized by the Boc/Bzl strategy, acidolysis must be avoided. Photolysis, base-mediated cleavage, or allyl transfer methods are available to remove protected peptide fragments from solid supports. In Fmoc/tBu chemistry, cleavage of the protected peptide from the resin cannot be performed under basic conditions; mild acidolysis is the method of choice. Resins and anchors for these purposes are described in Section 4.3.1.2.

Apart from the choice of the resins and the associated chemistry, other factors should be considered such as the size of the different fragments and the residues at the C- and N-terminal positions of each protected fragment. The best choices for the C-terminal residues of the fragments to be coupled are Gly and Pro [see the problem of oxazol-5(4*H*)-one formation in Vol. E 22b, Section 7.4]. This choice is possible when these residues are present in the sequence. Alternatively, they must be correctly distributed in the sequence to allow the synthesis of fragments of reasonable length. When the sequence does not contain Gly or Pro, an alternative is to use coupling methods that reduce epimerization. Some of these methods are described in the following sections. When Gly or Pro are present in the sequence but not "ideally" distributed, the length of the fragments can be a limiting factor due to its solubility or purity. Some improvements in the solubility of protected fragments will be

for references see p 804

described. However, Pro and Gly residues at the C-terminus are favorable to the formation of piperazine-2,5-dione (diketopiperazine, DKP) during the deprotection step of the amino function of the dipeptide linked to a resin. This is the case for benzyl-, allyl- and oxime-type resins. However, DKP formation can be suppressed or reduced when the C-terminus has a poor leaving group, as in fluorenylmethyl-based resins, or a large steric hindrance, as in the trityl-based resins. In Boc chemistry, this formation can be diminished using coupling methods allowing in situ neutralization (see Section 4.3.3.1).

In CSPPS, the nature of the support and its degree of functionalization are important. Polystyrene, polyacrylamide, and poly(ethylene glycol) PEG-grafted polystyrene have given the best results. Substitution is generally lower than in the stepwise solid-phase synthesis, and is typically in the range of 0.04–0.3 mmol · g^{-1}. This low substitution can be obtained by the coupling of a minimal amount of the first amino acid on the resin, followed by an acetylation step to cap the unreacted functional groups.

In this section the most common methods of acylation used in CSPPS will be described. Two important factors in this strategy must be kept in mind: the risk of epimerization of the activated C-terminal residue already mentioned and the fact that, for economic reasons, large excesses of the protected fragment cannot be used to drive the coupling to completion.

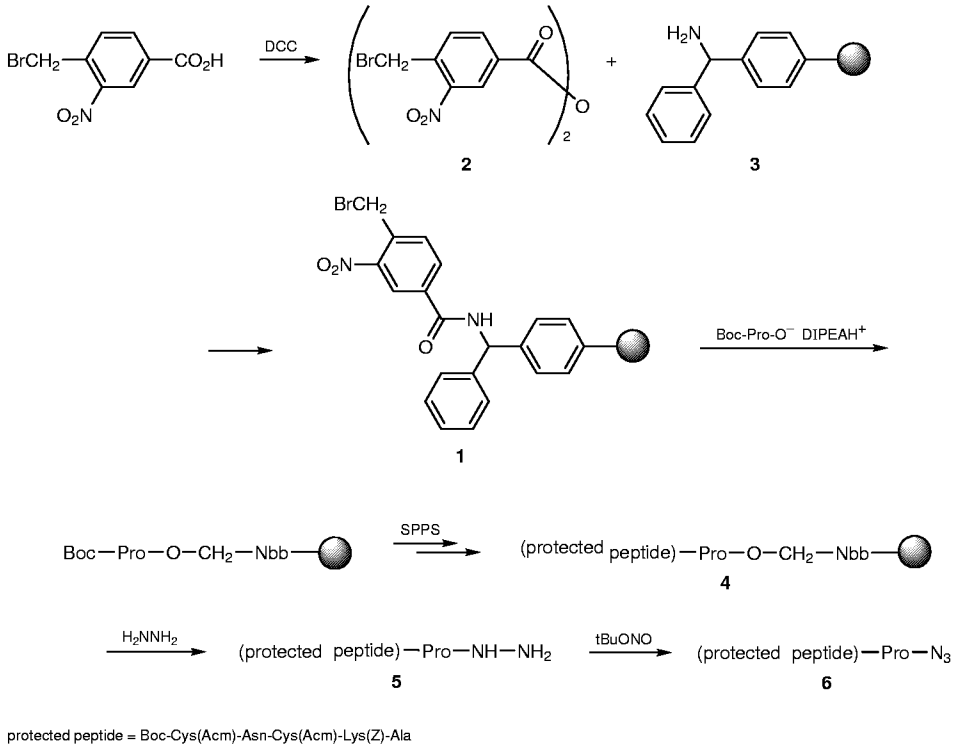
Few examples of solid-phase protected fragment condensations were reported in the early literature. Purification of protected fragments on an acceptable scale and a general methodology for rapid preparation of protected peptides by stepwise SPPS were the main problems which made the preparation of protected peptides in solution necessary. Furthermore, coupling of peptide fragments without racemization was initially known to be only possible by the way of azide couplings at low temperatures and for prolonged periods of time. The advent of additives in combination with the DCC method was an important step in the development of fragment couplings. In 1966 an important improvement in fragment couplings was afforded by Weygand's discovery of the racemization decreasing effect of HOSu when added to DCC and acyl amino acids.[4] A great variety of other coupling reagents and additives were tested for their utility in fragment condensation. Among them, HOBt and HODhbt in combination with DCC were the most popular. The reagent BOP was then successfully used to condense fragments on a solid support, followed by all the new phosphonium and uronium salts which are now the most frequently used.

4.3.3.2.1 Azides

Early peptide syntheses reported the use of azide for the coupling of protected segments. It was believed to be the only method free of racemization, making its use very popular and the most widely chosen in the coupling of protected peptide fragments in solution. However, it has been shown that even protected peptide acid azides can epimerize during the coupling step, although the amount of epimerization is usually very low when certain precautions are taken.[5] One theory explaining the low racemization levels observed during acid azide coupling proposed by Young[6] is that aminolysis of peptide acid azides is driven by an intramolecular general base catalysis without oxazol-5(4*H*)-one formation. The overall result favors aminolysis over racemization. The azide method is, nonetheless, not widely used in contemporary peptide chemistry. The activation level of the carboxylic group is weak, the coupling reactions are low yielding and need long reaction times; side reactions[7] can yield products that closely resemble the desired compound and are difficult to separate. Peptide chemists now prefer to use more rapid coupling techniques, which are described below.

The azide procedure[8] adopted for the synthesis of apamin by CSPPS included the preparation of 4-{[4-(bromomethyl)-3-nitrobenzoyl]amino}(phenyl)methylphenyl-copoly(styrene-1%-divinylbenzene) ($BrCH_2$-Nbb-resin) **1** which was used for the generation of the protected hydrazide peptide (Scheme 1). The Kaiser oxime resin can also be used for generation of hydrazide peptides.

Scheme 1 Synthesis of the Protected Azide from Boc-Pro-OCH_2-Nbb-Resin[8]



Synthesis of Protected 1–6-Apamin Azide Segment and Subsequent Coupling:[8]
$BrCH_2$-Nbb-resin **1** was synthesized by reaction of the anhydride of 4-bromomethyl-3-nitrobenzoic acid (**2**) with BHA resin **3** (Scheme 1); the C-terminal *N*-Boc-proline was incorporated into the resin by reaction of its diisopropylethyl ammonium salt with **1** in refluxing EtOAc for 48 h. After the deprotection–elongation repetitive steps, Boc-Cys(Acm)-Asn-Cys(Acm)-Lys(Z)-Ala-Pro-Nbb-resin (**4**) was obtained. To protected peptide-Pro-Nbb-resin **4** (527 mg, 243 μmol of peptide) suspended in DMF (5 mL) hydrazine hydrate (0.5 mL, 40 equiv) was added. After 71 h of magnetic stirring at 20 °C the resin was washed with DMF, DMF/MeOH (1:1), and MeOH. The protected peptide hydrazide was recovered and estimated by amino acid analysis; yield: 212 μmol (88%). Chromatographic purification on Sephadex LH-20 in MeOH followed by Biobeads S-X1 in CH_2Cl_2/MeOH (90:10) yielded the homogeneous protected peptide hydrazide Boc-Cys(Acm)-Asn-Cys(Acm)-Lys(Z)-Ala-Pro-NH-NH_2 (**5**); yield: 171 mg (69%); $[\alpha]_D^{25}$ −58.7 (*c* 0.89, MeOH). The hydrazide **5** (61 mg, 60 μmol) was dissolved in DMF (0.5 mL) at −20 °C to which HCl/THF (0.15 mL, 350 μmol) and tBuONO (9 mg, 87 μmol) in DMF (0.5 mL) was added. The soln was stirred at −20 °C for 30 min, cooled to −25 °C and DIPEA in DMF (0.5 mL, 360 μmol) was added to yield the 1–6-apamin azide **6**. To the soln was then added to the 7–18-C-terminal apamin-aminobenzyl-resin (143 mg, 50 μmol) followed by DIPEA in DMF (0.2 mL, 144 μmol). The mixture was left for 60 min at −20 °C, 48 h at 2 °C, and 72 h at rt. The resin was washed with DMF, CH_2Cl_2, and MeOH. Amino acid analysis of the peptide-resin showed that N-terminal amino acids were in a ratio of 65% compared to C-terminal amino acids. A second coupling under the same conditions did not improve the yield (67%).

4.3.3.2.2 Carbodiimides in the Presence of Additives

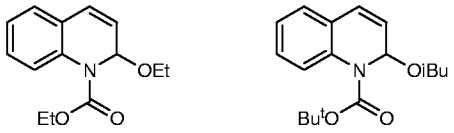
The coupling of protected fragments using carbodiimides proceeds with a high level of epimerization of the C-terminal residue of the activated fragment.[9] However, such epimerization can be reduced by the use of *N,N'*-dicyclohexylcarbodiimide (DCC), *N,N'*-diisopropylcarbodiimide (DIC), water-soluble carbodiimide (WSCI), e.g. *N*-[3-(dimethylamino)propyl]-*N'*-ethylcarbodiimide hydrochloride, in the presence of additives such as HOSu, HOBt, HODhbt or HOAt. These reagents react in situ with the highly reactive *O*-acyl-

for references see p 804

isourea or symmetrical anhydrides to yield the corresponding active esters, which are thought to form oxazol-5(4*H*)-ones less rapidly. The hydroxy group of these reagents also provides a proton source in the reaction mixture to neutralize any excess of base, avoiding further epimerization by abstraction of the α-proton.

Various studies with different coupling reagents and additives have been carried out to check their efficacy in CSPPS. DCC, DCC/HOSu have been compared to DCC/pentachlorophenol, triphenoxyphosphine/imidazole and ethyl 2-ethoxy-1,2-dihydroquinoline-1-carboxylate (**7**) (EEDQ) (Scheme 2).[10,11] The best results were obtained with EEDQ, which allowed the use of unprotected hydroxy amino acid groups. 2-Isobutoxy-1-isobutoxycarbonyl-1,2-dihydroquinoline (**8**) (IIDQ) (Scheme 2), introduced by Yajima et al.,[12] permitted almost quantitative coupling of a protected peptide fragment (in 4 fold excess) to a peptide-resin in 48 hours in DMF. Furthermore, less epimerization occurred using IIDQ compared to DCC (4.1% and 27.4%, respectively). On the other hand, Di Bello et al.[13] compared DCC/dichloromethane, DCC/HOSu/dichloromethane, DCC/HOBt/dichloromethane, DCC/HOBt/DMF and EEDQ/DMF in the condensation of Z-Pro-Arg-OH (5 equiv) with an amino decapeptide resin. In this case, the DCC/HOBt procedure gave 96% yield and virtually no racemization, whereas all other procedures failed.

Scheme 2 Structures of EEDQ and IIDQ Coupling Agents[12]



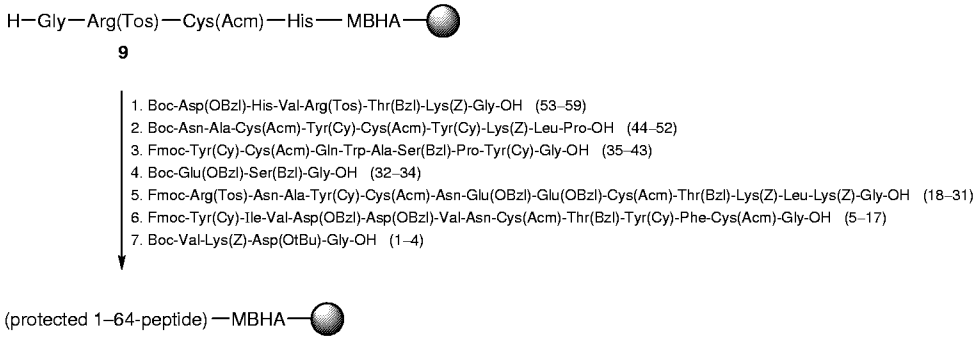
7 EEDQ **8** IIDQ

Even when a reasonable excess of the segment to be coupled was used and when solubility was not a problem, a segment-coupling reaction may require several hours if not days to go to completion. It is also important to treat each synthesis on a case-by-case basis. Different parameters such as solvents, reagents, reaction times, concentrations of the soluble components, temperature, and efficiency of the stirring of the resins can influence the yield of coupling. Among all the possibilities, HOBt and HOSu were the most popular as additives along with carbodiimides.

4.3.3.2.2.1 *N,N'*-Dicyclohexylcarbodiimide/1,2,3-Benzotriazol-1-ol Coupling Protocol

The DCC/HOBt procedure has been used extensively for the solid-phase coupling of protected fragments. This combination proved its efficiency in several examples of peptides having repetitive sequences. In the syntheses of poly-(L-prolyl-L-prolylglycyl) oligomers[14] of the cyclodepsipeptide valinomycin[15,16] as well as in the repeating peptides related to the glutelin-2 protein of maize,[17,18] the solid-phase coupling steps of the repeating units were all carried out using DCC/HOBt. The same method was used for the elaboration of the 64 amino acid peptide scorpion toxin II of *Androctomus australis*[19] which is illustrated in Scheme 3.

Scheme 3 Protected Peptide Fragments Used in the Assemblage of the Scorpion Toxin II of *Androctomus australis*[19]



Sato[20] has compared the preparation of gramicidin S by a stepwise SPPS approach with two separate segment coupling syntheses, one in which dipeptide fragments were coupled and the other in which tripeptide fragments were coupled. In each of the segment coupling strategies, the di- or tripeptide was coupled on the solid support using a four-fold excess of each peptide and of DCC/HOBt. Coupling yields were 92–100% after 24 hours shaking at room temperature. The overall yields based on amino acid analysis for each of the three strategies were 68%, 83%, and 86%, respectively, showing that the segment coupling strategies were superior to the linear approach. No epimerization was detected in the segment couplings. The presence of copper(II) salts in addition with DCC/HOBt has been reported earlier as a limiting factor of racemization,[21] and recently confirmed.[22]

Assemblage of the Scorpion Toxin II from *Androctomus australis* Using DCC/HOBt Coupling Protocol of Protected Fragments; Typical Procedure:[19]

The protected peptide fragments, prepared by SPPS on an UV-labile resin were purified and coupled onto the resin-bound 60–64 sequence, H-Gly-Arg(Tos)-Cys(Acm)-His-MBHA-resin **9** (the degree of resin functionalization was approximately 0.2 meq · g^{-1}). The fragments (1.3–2.8 equiv) were dissolved in the minimum required volume of DMF and then added to the peptide-resin together with an equimolar amount of HOBt. After two min stirring, DCC (1.3–2.8 equiv) was added and the mixture was stirred for 2 h at 0 °C and then at rt for the duration of the coupling (16–48 h). Portions of the peptide-resin were removed periodically and the extent of coupling was determined by amino acid analysis of the acid hydrolysates. When the yield appeared to be < 95%, the coupling was allowed to continue and if necessary was repeated. The 5–17 protected fragment could not be coupled to the 18–64 peptide-resin in acceptable yield. Shorter protected fragments corresponding to 11–16 and 5–10 sequences were synthesized and their couplings proceeded in high yields, demonstrating the significance of the segment size, solubility, and conformation.

4.3.3.2.2.2 *N,N'*-Dicyclohexylcarbodiimide/*N*-Hydroxysuccinimide Coupling Protocol

The DCC/HOSu coupling protocol has been frequently used, for example in the synthesis of BPTI,[23,24] in which the protected peptide fragments spanning the entire sequence of 58 amino acids were synthesized in solution and then coupled to H-Gly-Ala-O-Merrifield-resin. For the first segment a 1.5 fold excess was used for the coupling, but for other fragments the number of equivalents was increased to four in order to ensure high coupling yields. Coupling times were 4–6 days at room temperature and yields were 71–100%.

Kullmann and Gutte used the DCC/HOSu protocol for the synthesis of an open chain asymmetrical cystine peptide corresponding to the sequence A^{18-21}-B^{19-26} of bovine insulin[25] and of the asymmetrical insulin intermediate A^{1-21}-B^{18-26}.[26] Ragnarsson et al. also described the use of DCC/HOSu in the synthesis of bradykinin[27,28] and [Arg^8]-vasopressin[29] in high yields.

for references see p 804

4.3.3.2.2.3 *N,N'*-Dicyclohexylcarbodiimide/3-Hydroxy-1,2,3-benzotriazin-4(3*H*)-one Coupling Protocol

HODhbt was also used as an additive with DCC in the total synthesis of glucagon[30] as described in the early literature from the Shanghai Institute of Biochemistry. Four protected peptide segments, which were synthesized in solution, were coupled to the C-terminal tripeptide and linked to the Merrifield resin. Coupling reactions were carried out using a two to three fold excess of peptide fragment with DCC or EEDQ as coupling agent, in the presence of HODhbt as additive. Reaction times varied from 31 to 63 hours at 32 °C and yields were between 96 and 99%. Recently, HODhbt was used in solution as well as in solid-phase syntheses and in fragment condensations: Breipohl et al.[31] used HODhbt esters of Fmoc amino acids in stepwise synthesis of LHRH and secretin derivatives, Mostafavi et al.[32] compared the condensation of fragments in solution with EDC/HODhbt and with TBTU/HOBt in the synthesis of phospho-urodilatin, and Sakakibara et al.[33] synthesized human midkine, a 121 amino acid residue peptide, by condensation of two large fully protected intermediates, Boc-(1–59)-OH and H-(60–121)-OBzl, in a chloroform/TFE mixture using water-soluble carbodiimide in the presence of HODhbt. Furthermore, they proved that TFE/trichloromethane or TFE/dichloromethane solvent systems were extremely useful for coupling segments without danger of racemization and of trifluoro ester formation if WSCI was used as the coupling reagent in the presence of HODhbt.

4.3.3.2.3 Phosphonium and Uronium Salts

Coupling reagents based on phosphonium and uronium salts are now becoming more frequently used, both for the coupling of single amino acids in stepwise SPPS and for segment coupling in CSPPS.

Contemporary methods for solid-phase couplings of peptide fragments are based on the use of phosphonium or uronium salts. HOBt was introduced as an additive with BOP, and more recently HOAt was used as an alternative in carbodiimide couplings instead of HOBt and shown to be more effective. Phosphonium and uronium salts of HOAt, namely PyAOP, HATU, and HAPyU, have also been used with success in CSPPS.

Phosphonium Salts

For the synthesis of LHRH,[34] the protected peptide, pGlu-His-Trp-Ser(Bzl)-Tyr(Bzl)-Gly-OH was coupled onto the resin-bound peptide H-Leu-Arg(NO_2)-Pro-Gly-*O*-resin using BOP reagent and NMM in DMF for 36 hours at room temperature. After aminolysis in DMF, deprotection and purification, biologically active LHRH was obtained. In the synthesis of human gastrin I, Barany et al.[35,36] found activation with BOP/HOBt more efficient than with DCC/HOBt. However, some racemization occurred when C-terminal Ala and Leu fragments were activated. Efficient fragment coupling with BOP was again observed by Nokihara et al.[37] in the fragment synthesis of human pro-atrial peptide, but racemization occurred during activation with BOP/HOBt/DIPEA of a fragment containing a C-terminal Met residue compared to a similar stepwise synthesis. A general study by Benoiton et al.[38] on BOP couplings during segment condensation in solution stated that DIPEA was the base of choice, but that excess base promoted racemization. Another comparison which favored BOP for fragment condensation over DCC/HOBt was reported in the synthesis of β-amyloid protein (1–25) implicated in Alzheimer's disease.[39,40] BOP/HOBt/DIPEA was used with success for coupling four Boc protected fragments to a Kaiser resin. In this case racemization was suppressed at 0 °C. During the calcitonin synthesis[41] a more rapid activation of fragments by BOP/HOBt/DIPEA was observed, compared to DCC/HOBt, but leading to a less pure compound. The combination of BOP/HOBt/DIPEA has been used for

the coupling of protected peptide fragments by Ho et al.[42] in the synthesis of cyclic ($Asu^{1,7}$)-eel calcitonin.

Uronium Salts

Surovoy et al.[43] have used a combination of TBTU/HOBt/DIPEA to couple various protected peptide fragments in the syntheses of the nucleocapsid protein of the HIV virus and of a model peptide for the VP_1 protein of the foot and mouth disease virus. Coupling yields were generally up to 95% using a three-fold excess of peptide fragments and coupling times of about twelve hours.

Giralt et al.[44] have used HATU and HAPyU, the new reagents based on 7-aza-1,2,3-benzotriazol-1-ol, in the synthesis of the highly repetitive N-terminal domain of γ-zein protein. HATU/HOAt/DIPEA and HAPyU/HOAt/DIPEA were used for coupling the protected peptide Fmoc-Val-His(Trt)-Leu-Pro-Pro-Pro-OH with H-Val-His(Trt)-Leu-Pro-Pro-Pro-resin. Coupling yields of over 80% were obtained using an excess of four equivalents of the protected peptide and coupling reagents. The use of DIC/HOBt provided yields of 40% using an excess of five equivalents of the protected peptide and activating reagents. These results indicated that the triazolo[4,5-*b*]pyridine-based coupling reagents are more efficient for this coupling.

The results obtained with the triazolo[4,5-*b*]pyridine-based coupling reagents are in accordance with the studies of Carpino et al. who have shown that they are superior than those derived from HOBt.[45] Racemization studies using triazolo[4,5-*b*]pyridine-based coupling in SPPS concluded that it was better to minimize preactivation time.[45] These authors have recently shown that HATU in the presence of collidine (2,4,6-trimethylpyridine), a sterically hindered base causing less racemization than DIPEA, is able to induce fragment couplings in high yields and without racemization.[46–48] For example, the particularly epimerization-prone coupling of Z-Phe-Val-OH and H-Pro-OtBu was done by HATU/DIPEA/DMF in 80% yield with 6.5% epimerization, while HATU/collidine/DMF gave 86% yield and 1.7% epimerization. The reagent HAPyU yielded the best results for this segment coupling, providing only 0.1% epimerization with collidine and 6.3% with DIPEA. The most efficient results were obtained when one equivalent of HOAt was present as an additive during the coupling process.[49] The uronium salts show much promise in peptide couplings but their impact on CSPPS is not yet clearly defined, especially for their effect on racemization.

Uronium salts (or aminium salts) should be used with caution since they can react with the amino component to lead to guanidino species as described in Section 4.3.3.2.2.3. This side reaction is not so important during the coupling of single protected amino acids, such as in stepwise SPPS, since the activation is a fast step and the uronium (aminium) salt is rapidly consumed by the carboxylate to yield the activated species. However, the coupling of protected peptide fragments can require long reaction times and so the uronium (aminium) salt can react with the amino component to lead to guanidino species. It is therefore recommended to avoid excess of the uronium reagent.

A complete study on the solvents, bases, coupling reagents, and additives used for the condensation of fragments was performed in solution on test peptides by Carpino et al.[50] These authors demonstrated that HOAt and HODhbt offer significant improvement over HOBt. Examination of the data emphasizes that the goal of developing rapid, racemization-free fragment coupling processes applicable to peptide bond formation between any two amino acids still remains elusive. Furthermore in solid-phase chemistry, the question of preactivation time is of great significance and to date, fully appropriate techniques for safe coupling in solvents such as DMF are not available. A combined solid-phase/solution synthesis of a vasoactive intestinal peptide analogue has been published by Felix et al.[51] In this strategy, only equivalent amounts of each fragment were required to achieve almost quantitative coupling reactions using the HBTU/HOBt/DIPEA protocol. The protected

for references see p 804

segments were used without any purification in subsequent fragment condensations for the final assemblage of the VIP analogue. These workers combined the advantages of SPPS to synthesize the protected fragments and solution synthesis for the fragment condensations, which were monitored by HPLC.

Solid-Phase Coupling Using Phosphonium or Uronium Salts; General Procedure:[19,44]
Coupling of the fragments should first be tested on a small amount of peptide-resin (2–3 μmol) in a 1-mL polypropylene syringe fitted with a polyethylene disk and a stopcock before carrying out the large-scale experiment. The process of dissolving the protected peptide fragment may need to be started several hours before initiating the coupling reaction and may require special agitation such as ultrasound. The most concentrated protected peptide soln possible should be prepared. The excess of peptide fragment required depends on the coupling, but between 2 and 5 equivalents are usually used. Protected peptide-resins were preswollen in DMF (1 mL · 0.1 g^{-1} of resin, 3 × 10 min and 1 × 1 h). After removal of the N^{α}-amino protecting group with the appropriate reagent [TFA/CH_2Cl_2 (4:6) for Boc chemistry or piperidine/DMF (1:4) for Fmoc chemistry, with an extended time of reaction of about 1 h] and washing, a mixture containing the protected fragment, coupling reagent (uronium or phosphonium salt, 1 equiv versus protected fragment to be coupled), HOBt or HOAt (1 equiv), and base (1 equiv in the case of Fmoc chemistry and in the case of Boc chemistry if the trifluoroacetate salt was preneutralized), was added to the resin at 0 °C. Acetylation, when required, was performed with acetic anhydride/DIPEA (1:1, v/v, 30 equiv) in the minimum amount of DMF.

4.3.3.2.4 Other Convergent Peptide Synthesis Strategies

Although many syntheses succeeded by coupling of protected segments on solid supports, problems are sometimes encountered in achieving acceptable yields even in the presence of excesses of fragments to be coupled. This has led researchers to look to other approaches in which the fragments can be synthesized on solid phase but are coupled in solution. This goal was achieved by different strategies.

4.3.3.2.4.1 Lipophilic Segment-Coupling Strategy

The maximum protection of the desired segments, also called lipophilic segment-coupling strategy, was applied especially in Fmoc/tBu syntheses to make the fragments relatively soluble in organic solvents such as tetrahydrofuran, chloroform, or ethyl acetate. This approach was developed with success on highly acid-labile handle-resins by the Ciba-Geigy group[52,53] in the synthesis of several medium-size peptides including calcitonin and human neuropeptide Y.

4.3.3.2.4.2 Unprotected or Minimally Protected Fragment Strategy

Coupling of minimally protected peptides in aqueous solution has also been studied. The problems caused by the poor solubility of protected fragments stimulated the investigation of convergent strategies based upon the coupling of unprotected or minimally protected peptide fragments. These may be much more soluble in aqueous solvent systems allowing them to be purified to homogeneity using the standard reverse-phase chromatographic techniques for the purification of peptides.

Other convergent strategies for the synthesis of peptides are based on the solid-phase synthesis of unprotected or of minimally protected peptide fragments and their coupling in solution by chemical means[53–58] or by using enzymes.[59–61]

4.3.3.2.4.2.1 Enzymatic Acylation

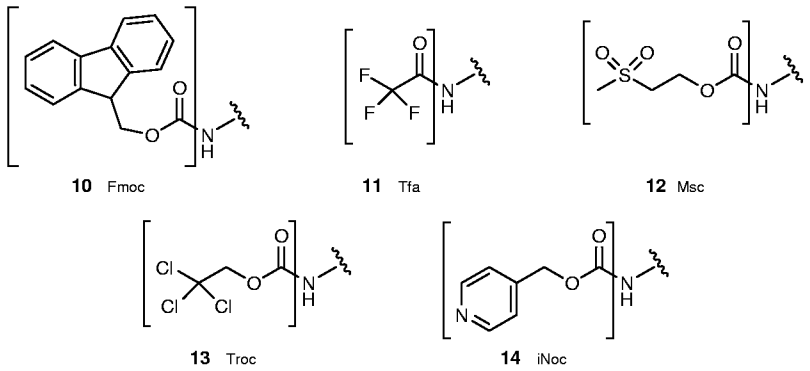
The benefit of using unprotected or partially protected fragments is the improved solubility in aqueous solutions, making the purification of the individual fragments easier. Enzyme-

catalyzed fragment condensation has been developed and applied to the total synthesis of RNase A,[62] semisynthesis[63] and peptide cyclizations.[64] It has been shown that enzymatic coupling can be used for fragment condensation with high yield without racemization, while chemical methods such as DIC/HOBt and BOP/HOBt/NMM give optically impure products with 10–20% epimerization.[65] These advantages have led to enzymes being used in solid-phase coupling of protected fragments.[66] Four different solid supports; polyacrylamide, PEG-PS, Kel-F-G-styrene, and controlled pore glass were tested. The controlled pore glass support was the most suitable for performing enzymatic reactions. The racemization-free assembly of partially protected peptides on this support was achieved in moderate yields of 31–54%.

4.3.3.2.4.2.2 Activation of C-Terminal Thiocarboxy or Thioester Groups

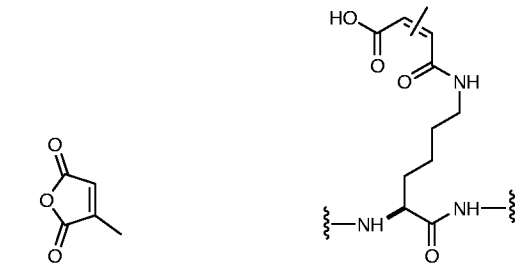
Since the fragments are minimally protected, coupling is a delicate operation when an unequivocal amide bond formation is required, which is not always the case, for example in immunochemistry. In the early 1980s the approach based on the coupling of minimally protected segments by specific activation of a C-terminal thiocarboxy group was developed.[67–74] The fragment to be activated incorporates a thiocarboxy amino acid as the C-terminus. Thioglycine is most commonly chosen as the C-terminus residue since racemization is not possible and the absence of steric hindrance tends to produce high coupling yields. Fragments can be synthesized using Boc/Bzl SPPS since thioester peptide-resin linkers are stable under the conditions that are used. Peptides can then be removed from the resin with strong acid and concomitantly their side chains can be deprotected, enhancing their solubility in aqueous solvent systems. Specific activation of the thiocarboxy group using silver salts or by acyl disulfide formation allows the two segments to be linked by an amide bond to give the desired target peptide. However, some precautions should be taken. The N-terminus of the thiocarboxy peptide should still be protected to avoid formation of dimers and polymers. N^{α}-Protecting groups that are not removed by strong acid under normal conditions include 9-fluorenylmethoxycarbonyl **10** (Fmoc), trifluoroacetyl **11** (Tfa), 2-(methylsulfonyl)ethoxycarbonyl **12** (Msc), 2,2,2-trichloroethoxycarbonyl **13** (Troc), and 4-pyridylmethoxycarbonyl **14** (iNoc) groups (Scheme 4). The Fmoc, Tfa and Msc groups are base labile while both the Troc and iNoc groups can be removed by treatment with zinc dust in acetic acid.

Scheme 4 N^{α}-Protecting Groups Stable in Acid Conditions[101]



for references see p 804

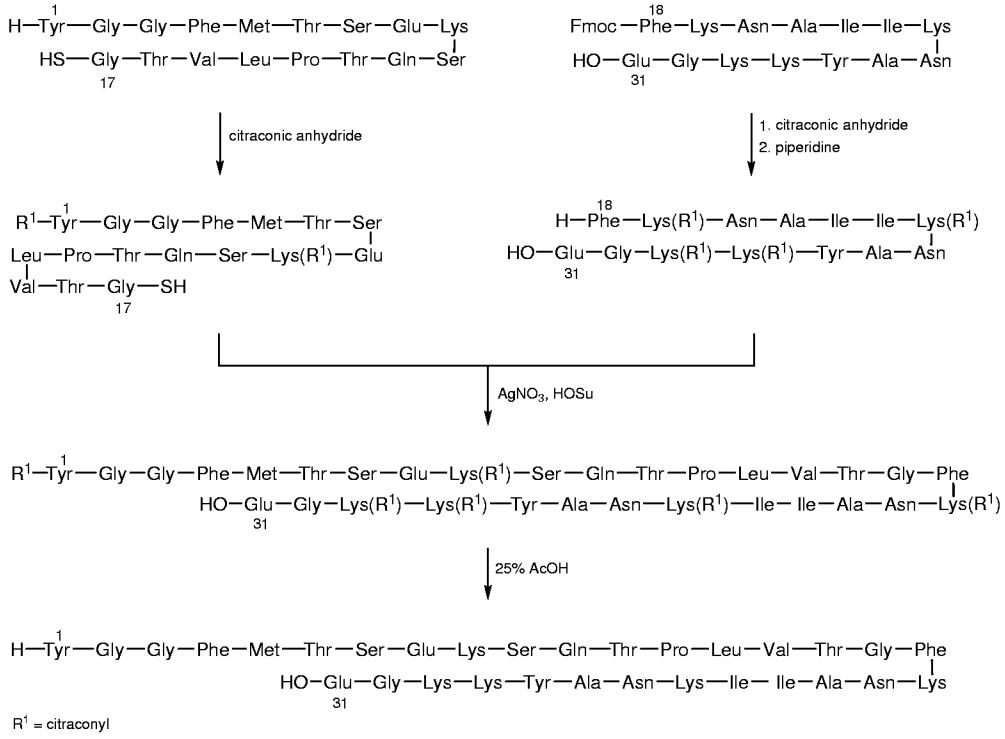
Scheme 5 Protection of Lysine Side Chains with the Citraconyl Group[68]



15 citraconic anhydride **16** citraconyl-protected lysine-containing peptide

The side chain of Lys residues contained in both segments must be protected as they are very nucleophilic amines able to undergo amide bond formation with the activated thiocarboxy C-terminus group. Blake et al.[68] introduced the use of citraconic anhydride (3-methylfuran-2,5-dione) (**15**) for the protection of Lys-containing peptides after cleavage from the solid support (Scheme 5), leading to peptides protected on the Lys side chain **16**. The citraconyl group can be removed under mild acidic conditions in 25% acetic acid.[75] Arg, Asn, Gln, His, or Trp side chains are not effective nucleophiles under normal conditions. Such an approach was applied to the synthesis of [Gly[17]]-β-endorphin[68] and is illustrated in Scheme 6.

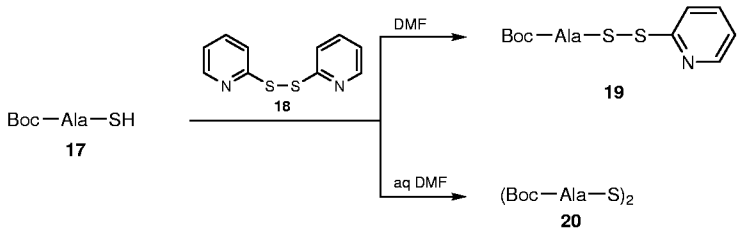
Scheme 6 Synthesis of [Gly[17]]-β-Endorphin[68]



This strategy was successfully applied with some modifications and refinements to the preparation of complex large proteins.[71,72] It has been very influential and recent syntheses of large peptides are based on the coupling or chemoselective "ligation" of unprotected or minimally protected peptide fragments activated as thiocarboxy[67,74] or thioester groups.[58]

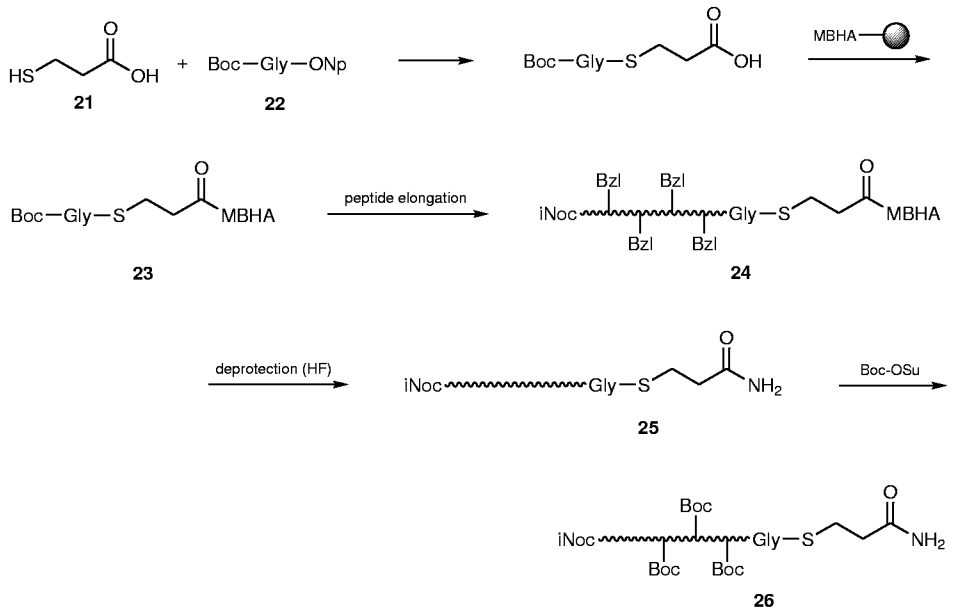
One alternative to the silver ion catalyzed coupling reaction of thiocarboxy C-terminus peptides uses acyl disulfides as the activated species.[73] The thiocarboxy group of **17** reacts with the diaryl disulfide **18** to give an unsymmetrical acyl disulfide **19** in DMF and a symmetrical diacyl disulfide **20** in aqueous DMF. Both acyl disulfides react with the α-amino group of another segment to form the desired peptide bond (Scheme 7).

Scheme 7 Reaction of the Thiocarboxy Group with Di(2-pyridyl)disulfide[73]



Di(2-pyridyl)disulfide **18** was found to be the most suitable disulfide for fragment coupling in dimethylsulfoxide (DMSO) in the presence of *N*-hydroxysuccinimide.[73] More recent variations[58,76–81] report on the use of specific activation of C-terminus thioesters. The *S*-alkyl thioester of glycine was synthesized by reaction of the thioacid **21** with Boc-Gly-ONp (**22**) and then anchored onto a 4-methylbenzhydrylamine resin to form **23**. Boc/Bzl peptide synthesis was then carried out. The last amino acid was incorporated with N^{α}-Troc or iNoc protection to yield **24**. After anhydrous HF treatment the peptide *S*-alkyl thioester **25** was allowed to react with *tert*-butyl succinimidyl carbonate (Boc-OSu) to protect side chains of Lys residues giving the minimally protected peptide **26**. This peptide was then activated at the C-terminus using silver salts and converted into the corresponding 4-nitrophenyl active ester which reacted with the N^{α}-amino group of the other minimally protected fragment (Scheme 8).

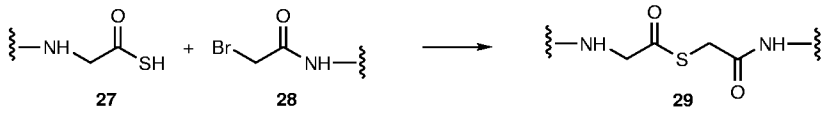
Scheme 8 C-Terminus Thioester Approach[76,77]



for references see p 804

Other developments using this approach were carried out, especially in chemical ligation. Chemical ligation also includes the union of peptides by bonds other than amides (Scheme 9). The method developed by Kent's group[82–84] is essentially thioester formation between two peptides, one possessing a thiocarboxy group at its C-terminus **27**, often chosen to be a thiocarboxy glycine residue in order to maintain chiral integrity at the ligation site, and the other a bromoacetyl group at its N-terminus **28**. Nucleophilic attack of the thiocarboxy group at the carbon atom bearing the bromine atom leads to thioester formation between the two peptides yielding **29**, as shown in Scheme 9.

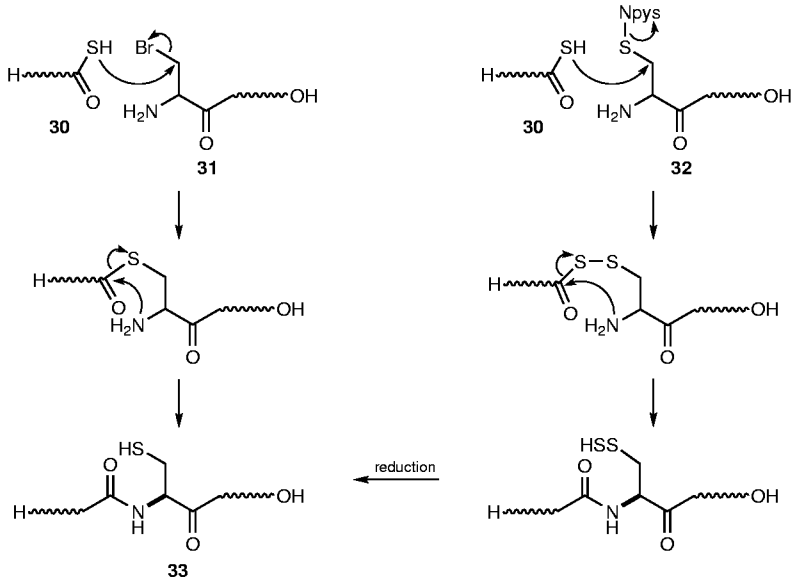
Scheme 9 Chemical Ligation by Thioester Formation[82–84]



27 28 29

Tam and his collaborators[85–87] have applied chemical ligation to the synthesis of complex peptides. The formation of chemical ligation products from peptide fragments is provided by nucleophilic attack of the thiocarboxy group of the N-terminal fragment **30** on the bromomethyl group of compound **31** or on the *S*-(3-nitro-2-pyridylsulfenyl) group of compound **32** of the C-terminal fragment (Scheme 10). In these cases the chemical ligation product **33** contains an amide bond at the ligation site followed by a cysteine residue.

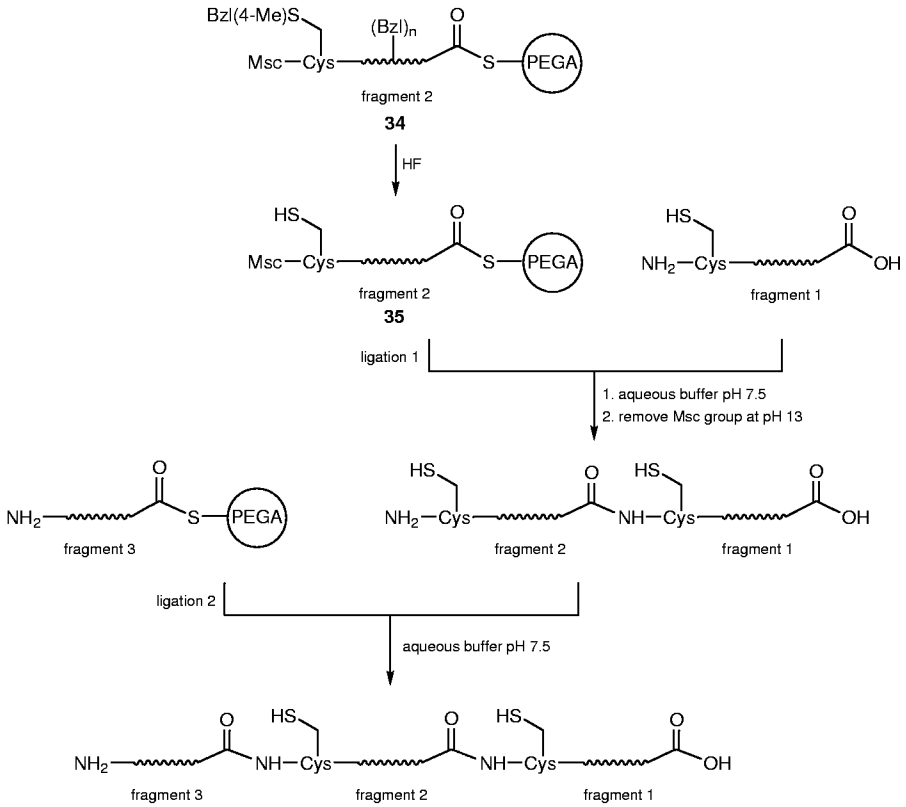
Scheme 10 Different Approaches of Native Ligation[85–87]



These approaches are described in more detail in Section 4.1.5. Following the success of the chemical ligation method in solution a recent study[88] describes a new convenient procedure to perform intramolecular and intermolecular chemical ligations of unprotected peptides directly on a solid support. This solid-phase ligation approach eliminates the manipulation of peptide thioacid and peptide thioester intermediates before the ligation step, resulting in an increase of the yield of the ligation reaction. This strategy involves the preparation of unprotected peptides linked to a solid support through an α-thioester linkage and was achieved using Boc solid-phase peptide synthesis on a 3-sulfanylpropionamido-PEGA sup-

port **34**. On completion of the synthesis of the fully protected peptide, the α-thioester resin was treated with anhydrous HF to yield the corresponding fully deprotected peptide α-thioester resin **35**. Using this procedure, several polypeptides ranging from 15 to 47 residues were successfully synthesized. These peptide-resins were then used to perform both intramolecular head-to-tail cyclizations, and intermolecular solid-phase ligations. The intramolecular solid-phase ligations proceeded much faster than their intermolecular counterparts, but in both cases the reactions were observed to be remarkably clean. The presence of aromatic thiol cofactors significantly accelerated the relatively slow intermolecular ligations. This novel methodology provides a general method for performing sequential intermolecular ligations, allowing easy access to much larger polypeptides and proteins. The principle of sequential solid-phase chemical ligation is represented in Scheme 11. The N-terminus residue within the C-terminal fragment containing the amino group to be acylated was replaced by cysteine to facilitate ligation.

Scheme 11 Intermolecular Solid-Phase Ligation Approach[88]

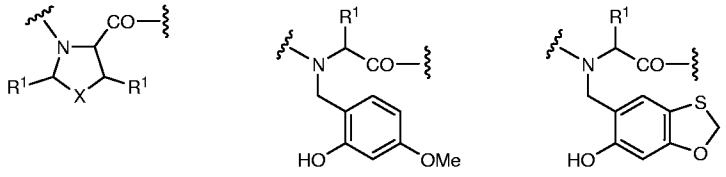


The preceding paragraphs demonstrate that several methods are available for the synthesis of large peptides. The two main problems remaining are solubility and racemization. Solubility problems seem to be solved on a case-by-case basis using solvent mixtures as described by Sakakibara.[33] An interesting method was reported recently which used crown ethers as noncovalent protecting groups of amine groups.[89] The ability of a crown ether molecule to complex charged amines and the subsequent gain of solubility in organic solvents of partially protected peptides was used to explore this noncovalent protection in peptide synthesis. Partially protected peptides corresponding to sequences 52–72 and 33–72 from *Rattus norvegicus* chaperonin 10 protein were synthesized and found to be soluble in water. These

for references see p 804

peptides were purified by RP-HPLC. Complexation of these peptides with 18-crown-6 gave two highly soluble products in organic solvents (CH_2Cl_2 and DMF) allowing their coupling to a tetrapeptide-resin using DCC/HOBt with, respectively, 68% and 50% yield. This approach is promising and merits further investigation. On the other hand, the use of unprotected or minimally protected fragments in ligation reactions or enzymatic syntheses seems to be practical. Racemization-free couplings are not yet available in all cases but new coupling reagents are under development to suppress racemization. As examples, 3-(dimethylphosphorothioyl)oxazol-2(3*H*)-one (MPTO)[90] and 3-(diethoxyphosphoryloxy)-1,2,3-benzotriazin-4(3*H*)-one (DEPBT)[91,92] were recently reported as new reagents for racemization-free couplings and they deserve further investigation. The addition of copper(II) chloride to the reaction has been shown to diminish racemization during segment condensation.[22,102–105] New copper(II) complexes associating 1,2,3-benzotriazol-1-ol, *N*-hydroxysuccinimide, and *N*-hydroxy-5-norbornene-2,3-dicarboximide and copper(II) have been described as racemization suppressors.[106,107] Bis(1,2,3-benzotriazol-1-yloxy)copper [$Cu(OBt)_2$] used in carbodiimide-mediated segment couplings proved its potential for the suppression or decreasing of racemization in two segment coupling model systems. Further studies are underway to understand the utility of $Cu(OBt)_2$ and to develop other copper(II) complexes. Another tactic for the prevention of racemization during segment condensation is to avoid oxazol-5(4*H*)-one formation. This can be achieved by reversible alkylation of the C-terminal amide bond of the protected peptide fragment to be activated. This strategy was used to overcome chain aggregation or disruption of secondary structure formation during solid-phase peptide synthesis.[93,94] In the first approach serine, threonine, and cysteine are incorporated as oxazolidine (or thiazolidine) protected "pseudoprolines" **36** which increases the solvation of the peptide chain and modulates the conformational and physico-chemical properties of the peptide. The second strategy uses 2-hydroxy-4-methoxybenzyl group **37** (Hmb) for amide protection (Scheme 12).[95,96] Both concepts of backbone amide protection were introduced for the assembly of difficult peptide sequences. Recently, taking into account that the Hmb protecting group prevents oxazol-5(4*H*)-one formation during activation, Offer et al. proposed this approach of reversible amide bond protection to suppress epimerization during segment condensation.[97] Hmb substitution at the C-terminal amide bond of a fully protected peptide segment significantly reduces the rate coupling. These authors have studied the influence of an electron-withdrawing group in the para position to the 2-hydroxy group of Hmb on coupling rate and epimerization. They developed a safety-catch amide bond protecting group, (6-hydroxy-1,3-benzoxathiol-5-yl)methyl (**38**) which couples with excellent yield without epimerization. Use of this protecting group may solve the epimerization problems encountered during fragment couplings.

Scheme 12 Backbone Amide Protecting Groups[93–97]



36 serine: X = O; R^1 = H
threonine: X = O; R^1 = Me
cysteine: X = S; R^1 = H

37

38

As we have seen in these chapters on acylation methods in stepwise and segment condensation peptide synthesis, considerable progress towards efficient procedures to couple amino acids and peptides by an amide bond have been made, particularly in the areas of coupling agents, enzyme-catalyzed coupling, and ligation methods. Problems still arise when

fragment condensation is performed without the Gly or Pro residue at the C-terminus. The identification of a protecting group which can be transformed into an activating group without loss of optical purity may be one of the solutions for these situations. Once again, one should note that each peptide synthesis has to be treated on a case-by-case basis. A remarkable synthesis including the combination of solid-phase/solution-phase synthesis was recently performed by Sakakibara's group for the preparation of a 238-residue protein, the green fluorescent protein,[98] from jellyfish *Aequorea victoria*. This synthesis demonstrates most of the current art of peptide science. Protecting groups of side-chain residues were selected with care, protected peptide segments were prepared by solid-phase synthesis using a HBTU/HOBt/DIPEA protocol on HMFS resin with a 4-{[9-(hydroxymethyl)-9*H*-fluoren-2-yl]amino}-4-oxobutanoyl linker, as developed by Albericio et al.[99,100] A fragment condensation method in solution was developed using a new solvent system, a mixture of phenol and chloroform, yielding after deprotection steps, purification and folding the 238 residue biologically active protein.

Synthesis of SH3 (Src Homology Type 3) Domain from the Protein Tyrosine Kinase, c-Abl, by Direct Ligation on Solid Phase LFVALYDFVASGDNTLSITKGCKLRVLGYNHNGEWBEAQTKNGQWVPS-NYITPVN-NH_2:[88]

H-CKLRVLGYNHNGEWBEAQTKNGQWVPSNYITPVN-NH_2 (B is an amino butyric residue replacing a cysteinyl residue) was manually synthesized on a MBHA resin on a 0.5 mmol scale according to the in situ neutralization/HBTU activation protocol for Boc SPPS with Dnp protecting group for His and the For protecting group for Trp. These groups were removed with 20% β-ME, 10% DIPEA in DMF (2×30 min), and 6% ethanolamine, 5% H_2O in DMF (2×30 min) respectively. The remaining side-chain protecting groups were removed and the peptide cleaved from the resin by treatment with anhydrous HF containing 4% *p*-cresol for 1 h at 0 °C. Soluble crude peptide was precipitated with Et_2O and washed with anhydrous Et_2O before being dissolved in aqueous MeCN (10–50%) and lyophilized to yield H-CKLRVLGYNHNGEWBEAQTKNGQWVPSNYITPVN-NH_2; fragment 1.

Amino-PEGA resin (1 g, 0.3 mmol · g^{-1}) was swollen in DMF for 10 min and washed with DMF, 10% DIPEA in DMF, and then DMF. Then 3-bromopropanoic acid (4 mmol, 611 mg) was coupled (2×30 min) as its symmetrical anhydride using DIC (315 µL, 2 mmol) in CH_2Cl_2 (3 mL). Acylation was monitored by a ninhydrin test showing that after the second coupling the reaction was complete. The residual free amino groups were acetylated using Ac_2O/DIPEA/DMF (15:15:70) for 10 min. The resulting brominated-PEGA resin was then washed with DMF, treated with 10% AcSH (2×20 min), 10% DIPEA in DMF (10 mL), washed with DMF, and finally treated with 10% β-ME and 10% DIPEA in DMF (10 mL, 2×20 min). The thiol-PEGA resin was then acylated using Boc-Gly-OSu (2 mmol, 545 mg) or Boc-Gly-OBt (2 mmol, formed in situ from HBTU) in DMF (4 mL) for 2 h. The final substitution of the Boc-Gly-[COS]-PEGA resin (0.2 mmol · g^{-1}) was calculated by a quantitative ninhydrin test after Boc deprotection with neat TFA (2×1 min) of a small aliquot of the corresponding resin. Peptide-resin was synthesized as described above, Boc-His(Bom)-OH and unprotected Boc-Trp-OH were used. The deprotection step was performed with liquid HF containing 4% *p*-cresol and 4% resorcinol for 1 h at 0 °C. Unprotected peptide-[COS]-PEGA resin was washed with anhydrous Et_2O and MeCN/phosphate buffer (1:1) at pH 7.5 and dried, to yield H-LFVALYDFVASGDNTLSITKG-[COS]-PEGA (COS is a 3-sulfanylpropionyl linker; fragment 2): To characterize the peptide-[COS]-resin, a small aliquot (1–2 mg) was cleaved using 50% MeCN in 0.5 M NaOH at 0 °C for 1 min and then analyzed by HPLC and ES-MS (observed mass: 2231.7±0.1 Da; calcd for $C_{103}H_{159}N_{23}O_{32}$ = 2230.0, monoisotopic, 2231.5, average isotope composition); amino acid analysis [Asx 3.0 (3), Ser 2.1 (2), Gly 2.0 (2), Thr 1.9 (2), Ala 2.1 (2), Tyr 1.1 (1), Val 2.0 (2), Ile 1.0 (1), Leu 3.0 (3), Phe 1.9 (2), Lys 0.8 (1)].

Intermolecular Ligation of Fragment 1 and Fragment 2 to yield LFVALYDFVASGDNT-LSITKGCKLRVLGYNHNGEWBEAQTKNGQWVPS-NYITPVN-NH_2.

The ligation reaction was initiated by combining fragment 2 (4.8 mg, 0.6 µmol) and fragment 1 (2.4 mg, 0.6 µmol) in freshly degassed 6 M Gn-Cl (100 µL), 0.1 M sodium phosphate, and pH 7.5 buffer containing 2% TCEP. The mixture was shaken at rt for 26 h. The supernatant was then removed from the resin by centrifugation, the resin was washed with fresh aliquots (6×50 µL) of the ligation buffer and washings and supernatant combined. The combined fractions were then purified by analytical HPLC to give the desired peptide of c-Abl SH3 domain; yield: 10% (as calculated by HPLC). The polypeptide was characterized by ES-MS [observed mass = 6173.2 ± 1.3 Da; calcd for $C_{279}H_{424}N_{75}O_{82}S$ = 6169.1, monoisotopic, 6173.0, average isotope composition] and amino acid analysis [Asx 8.1 (8), Glx 4.1 (4), Ser 3.1 (3), Gly 5.8 (6), His 1.0 (1), Arg 1.1 (1), Thr 4.1 (4), Ala 3.0 (3), Pro 1.9 (2), Tyr 3.1 (3), Val 5.3 (5), Ile 2.0 (2), Leu 4.9 (5), Phe 2.0 (2), Lys 2.8 (3), Cys, Trp and Abu not determined.

for references see p 804

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4.3.4 Solid-Phase Peptide Synthesis at Elevated Temperature

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As a result of past developments in nucleotide chemistry and instrumentation that insure rapid and almost quantitative assembly of oligonucleotides on solid supports, the question is why has there not been similar progress in the peptide/protein field as this uses the same principles of oligomer assembly? Optimistically, in the majority of cases, peptides up to 20 residues in length are readily and rapidly available in small quantities. However, more realistically, poor coupling yields and side reactions occur as a result of the diversity of natural amino acids, their relative complexity, and the multiple conformations that natural as well as unnatural amino acids (such as protected amino acids) may induce when incorporated in peptides. These induced secondary structures have been held responsible for increasing steric hindrance and concomitantly reducing coupling efficiency.[1] In addition, coupling reactions for amino acids must proceed without racemization, an added constraint that is not applicable to oligonucleotide synthesis. Numerous unpredictable side reactions in peptide chemistry can occur that may or may not be sequence dependent. While oligonucleotides are often used without further characterization, desired peptides must have their identities confirmed by mass spectrometry or other analytical methods. Finally and most importantly, the need for DNA fragments is in the order of micrograms while the need for peptides is 10^3 to 10^6 times that amount, thus creating purification challenges as compared to gel electrophoresis, which is used for one-step oligonucleotide purification and characterization. Although RP-HPLC systems have been described that are suitable for peptide purification, the rigorous process demands that the crude peptides be purified using more than one buffer or column system[2–4] (see Vol. E 22b, Section 7.2.2) and then analyzed in a third system to minimize and detect the possibility of contamination by closely related byproducts. Comparing analytical nucleotide preparations (μgs) with laboratory-scale (mgs) peptide syntheses is therefore unrealistic.

Matching coupling efficacy of building blocks in both oligonucleotide and peptide chemistries, however, is most likely possible. Improved coupling yields on solid phases have resulted from the use of more efficient and racemization-free coupling agents and protocols, optimized amino acid protecting groups, the use of resins with different swelling properties,[5] and/or the addition of chaotropic agents to the reaction mixture.[6] More recently, Schnölzer et al. and Kent et al.[7,8] have used in situ neutralization in Boc-chemistry SPPS and high concentrations of activated amino acids [1,2,3-benzotriazol-1-ol (HOBt) esters] to drive the coupling reaction to completion in a matter of minutes. In our experience, this is feasible as long as the substitution on the resins remains relatively low (<0.15 mequiv $\cdot g^{-1}$). An alternative approach would be to increase the reaction rates by increasing the temperature. Prior to our contributions discussed here,[9–11] protocols for the rapid assembly of peptides at elevated temperature were used by several groups with conflicting results.[12–16] It has also been reported that coupling at 30–80 °C using HOBt esters of amino acids did not induce racemization even in the presence of excess TEA.[17] In order to shorten the cycle time to below 30 min $\cdot$ cycle^{-1}, facilitate automation at constant temperature, and use resins with substitutions ≥ 0.4 mequiv $\cdot g^{-1}$ while keeping the excess of activated carboxy moieties below twofold for economical reasons (half of the concentration used by Schnölzer et al.[7]), we have investigated the possibility of performing all steps of SPPS (except for final cleavages and deprotection) at elevated temperature using model peptides known to give poor overall yields due in part to difficulties in coupling or susceptibility to side reactions (alkylation, oxidation, dehydration, and main-chain cleavage).

Comparative syntheses at room and elevated temperature of the following peptides (each presenting a different synthetic challenge) using either of two protocols (Boc-[18] or

Fmoc-based[19]) that were otherwise routinely followed in our laboratories[9,11,20] have been examined:

- Acyl carrier protein [ACP (65–74)].
- YLRPDWHQA-amide encompassing most trifunctional native amino acids.
- Human corticotropin releasing factor (hCRF, a 41-peptide amide).

From these experiments the following conclusions can be drawn: With few modifications to usual protocols (such as those presented here), carrying out SPPS at elevated temperature (preferably 60 °C) will increase efficiency by reducing cycle times to significantly less than 30 minutes under automated and non-automated conditions. Toluene/DMSO mixtures are also compatible with SPPS at room temperature and should solve some aspects (cost and environmental) of disposal associated with chlorinated solvents but have not been shown to have advantages over DMF or NMP.

The results suggest that the use of high temperature may or may not minimize problems associated with secondary structure formation that would interfere with coupling. Whether it is a case of appreciation or of sequence dependency will need further investigation. Regardless, couplings that were difficult at room temperature could be completed much faster at elevated temperatures, thereby reducing cycle times to less than 30 minutes. Whatever the advantages or disadvantages of different coupling reagents and their compatibility, it is likely that they will be carried over at elevated temperatures. In separate studies, the Merrifield resin and the Boc strategy for the synthesis of a number of peptides (including endothelin) with a free C-terminus carboxy group was also used. Weight gain was comparable to that experienced for syntheses carried out at room temperature, suggesting that loss of peptide from the resin during repetitive deblocking was not a major problem. Furthermore, the quality of the crude peptides compared favorably with that of peptides synthesized at room temperature. Other published studies using the Fmoc strategy and ACP (65–74) and unsulfated CCK-8 and -12 as model peptides indicated that the method can be successfully applied using a number of supports such as the TMBHA[21] and the commercially available Rink and SAMBHA resins.[9] Again, recovery and quality of crude peptides were comparable to those obtained using conventional SPPS. Interestingly, our conclusions regarding the problems associated with rapid and reliable SPPS and how to solve them are very similar to those reached by Schnölzer et al. and Kent et al.[7,8] These conclusions suggest that the protocols of Schnölzer et al. can be used at high temperature. Under such circumstances one could envision, using proper instrumentation, that the synthesis of relatively short peptides (<40 residues) could be achieved in parallel in a very short time. Using standard semi-preparative to preparative HPLC protocols with more than one mobile phase (TEAP at selected pHs followed by 0.1% TFA),[3] milligram to gram quantities of extremely pure peptides should become readily available.

In short, experimental results indicate that all phases (washing, deblocking, and use of common coupling reagents) of Boc-[18] or Fmoc-based[19] SPPS strategies, except for final cleavage from the resin and concomitant side-chain deprotection (HF or TFA treatments), can be carried out at temperatures $\leq$80 °C in different solvent systems. Whereas DMF and NMP are compatible solvents, DMSO (25%) in toluene promotes high, if not maximal, swelling of MBHA, TMBHA, Rink, and SAMBHA resins, dissolves all commonly used Boc- and Fmoc-protected amino acids, is compatible with all other reagents used in SPPS, and therefore favorably replaces polychlorinated solvents. One main advantage of high temperatures is accelerated speed of reaction and, under certain circumstances, better coupling yields resulting from disruption of deleterious/unwanted secondary structures known to interfere in conventional SPPS at room temperature. There is now consensus that 60 °C may be the optimal temperature at which to carry out SPPS, most particularly for difficult coupling steps.

4.3.4.1 Selection of Solvent Systems

Complementary studies have included optimization of solvent composition, as monitored by resin swelling in these solvents. Since poor swelling of resins may be a cause or an indication of poor couplings,[1,6,22–27] it was important at first to find a solvent system that was at least as good as any of the other solvents commonly used in SPPS. Both DMSO and toluene are identified as being environmentally friendly and obtainable at a reasonable cost. In fact, the choice of solvents is limited because of the low boiling points of both dichloromethane and chloroform, the instability of DMF at elevated temperatures and, to a lesser extent, the polarity (promoting racemization) of both DMF and NMP under certain conditions. On the other hand, toluene containing a minimum amount of DMSO (0–25%) dissolves all commonly used Boc and Fmoc amino acids. Toluene also dissolves, and is stable in TFA, piperidine, and TEA.[11,28] Since a swellographic method has been developed to study swelling behavior of gel-type reactive polymers,[29] resin swelling was determined quantitatively by measuring the volume variations in the absence (resins prewashed with methanol) and in the presence of the different solvents at different temperatures as presented earlier by Ludwick et al.[30] All of the measurements (triplicates for each solvent) shown in Table 1 for the MBHA, TMBHA,[21] Rink, and SAMBHA resins at 27 and 60 °C were performed in a graduated, water-jacketed glass vessel connected to a heated water bath circulator. Data shown in Table 1 demonstrate that the swelling properties of these resins in several solvents are nearly identical at both temperatures and that 25% DMSO/toluene and NMP promote high if not maximal resin swelling.

Table 1 Swelling of MBHA, TMBHA, Rink, and SAMBHA Resins in Different Solvents for a Constant Weight of Dry Resin

Resin	Solvent	Swelling Volume ΔV (mL)[a,b]		Ref
		27 °C	60 °C	
MBHA	25% DMSO/toluene	0.63 ± 0.06 (0.63)	0.73 ± 0.06 (0.73)	[9]
	CH_2Cl_2	0.33 ± 0.06 (0.33)	n.d.	[9]
	DMF	0.60 ± 0.00 (0.60)	0.63 ± 0.06 (0.63)	[9]
	NMP	0.73 ± 0.06 (0.73)	0.80 ± 0.00 (0.80)	[9]
	toluene	0.43 ± 0.06 (0.43)	0.50 ± 0.00 (0.50)	[9]
TMBHA	25% DMSO/toluene	1.07 ± 0.06 (2.70)	1.10 ± 0.04 (2.60)	[9]
	CH_2Cl_2	1.30 ± 0.00 (3.33)	n.d.	[9]
	DMF	0.90 ± 0.00 (2.30)	0.77 ± 0.03 (2.00)	[9]
	NMP	1.20 ± 0.00 (3.00)	1.10 ± 0.19 (2.60)	[9]
	toluene	1.10 ± 0.00 (2.80)	1.07 ± 0.03 (2.60)	[9]
Rink	25% DMSO/toluene	1.40 ± 0.26 (3.00)	1.43 ± 0.18 (2.90)	[9]
	CH_2Cl_2	1.53 ± 0.18 (3.30)	n.d.	[9]
	DMF	1.06 ± 0.26 (2.30)	1.00 ± 0.00 (2.00)	[9]
	NMP	1.53 ± 0.18 (3.30)	1.47 ± 0.18 (2.90)	[9]
	toluene	1.26 ± 0.26 (2.70)	1.20 ± 0.00 (2.40)	[9]
SAMBHA	25% DMSO/toluene	1.10 ± 0.19 (2.30)	0.97 ± 0.19 (2.00)	[9]
	CH_2Cl_2	1.10 ± 0.19 (2.30)	n.d.	[9]
	DMF	0.76 ± 0.19 (1.60)	0.69 ± 0.19 (1.40)	[9]
	NMP	0.98 ± 0.03 (2.10)	0.92 ± 0.03 (1.90)	[9]
	toluene	0.95 ± 0.04 (2.00)	1.02 ± 0.03 (2.10)	[9]

[a] n.d. = could not be determined.

[b] All the measurements were performed as described in the text and in Ludwick et al.[30] Amount of resin: 300 mg. The values shown in the parentheses refer to V/V_i, where V = final volume and V_i = initial volume.

4.3.4.2 General Synthetic Procedure

All syntheses were carried out manually as follows: dry resin was introduced into a 40–60-mL water-jacketed vessel with a coarse frit at the bottom and a 24/40 joint at the top, maintained at the reported temperature during the coupling, washing, and deblocking steps. Solvents and reagents were either poured or delivered from squeeze bottles onto the top. It should be noted that solvents kept at room temperature and used for washes did not have time to equilibrate to elevated temperatures. Two different kinds of washes corresponding to two modes of extraction were applied as appropriate. In the type I wash, solvent was delivered uniformly to the top of the column formed by the resin to achieve displacement of solvents and byproducts from the earlier step under continuous mild vacuum. In the type II wash, the resin was uniformly suspended by addition of the solvents from squeeze bottles under pressure followed by immediate drainage. Type I and II washing steps were carried out in seconds. Drainage by vacuum was controlled (pinching of the rubber hose leading to the vacuum source or closing of a valve) to minimize air passage through the resin to avoid condensation of water from the atmosphere. Amino acids were generally transferred as solids and the coupling reagent as a 1 M soln in the reported solvent (CH_2Cl_2, toluene, toluene/DMSO, DMF, NMP) to the drained resin in order to minimize solvent addition and obtain a thick (but flowing) slurry during coupling. The vessel was sealed under mild vacuum (generated by inversion of the capped flask and drawing of vacuum) and rocked during TFA or piperidine deblocking. HF cleavage was carried out in an all-KelF or in a Peptide Institute Inc. apparatus with vessel and traps cooled in liquid nitrogen or dry ice in acetone. Typically, the trap was refrigerated and high vacuum was drawn into the apparatus. The reaction vessel was then cooled and HF from the tank condensed under vacuum. Once enough HF had condensed, the cooled vessel was kept in ice water at which time ice formed around the vessel that had been cooled in liquid nitrogen. Within 30 minutes, an ice cake separated from the vessel indicating that the temperature in the vessel was around 0 °C. The reaction was left to proceed at this point for an extra hour at which time HF was removed under vacuum.

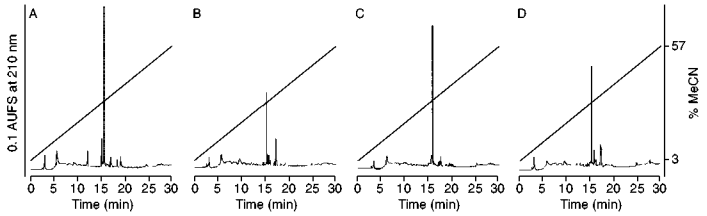
4.3.4.3 Acyl Carrier Protein [ACP (65–74)]

ACP (65–74) was built up manually on a Boc-Gly-PAM resin with a substitution of 0.69 mequiv $NH_2 \cdot g^{-1}$ (0.1 mmol). Amino acid derivatives were: Tyr[Z(2-Br)] and Ser(Bzl). Asn and Gln were not protected. Coupling reagents were DIC/HOBt (1:1) or TBTU. Deprotection and coupling steps were monitored by the Kaiser ninhydrin test.[31] The protocol for synthesis at 27 °C using mostly CH_2Cl_2 as solvent (conventional SPPS) was as follows: (1) initial washes: CH_2Cl_2, 10% TEA/CH_2Cl_2, MeOH, and CH_2Cl_2; (2) deblocking: 50% TFA/CH_2Cl_2 with 1% anisole; (3) deblocking washes: 1% anisole/iPrOH, MeOH, and CH_2Cl_2; (4) neutralization: 10% TEA/CH_2Cl_2, MeOH, and CH_2Cl_2; (5) coupling and recoupling in: (a) CH_2Cl_2 or CH_2Cl_2/NMP (1:1) or NMP (only when Boc-Asn or Boc-Gln were used) using Boc or Fmoc amino acid/DIC/HOBt (1:1:1, 2.5 equiv); (b) in NMP using BOP or TBTU (2.5 equiv) and DIPEA (pH 9–10); (6) coupling and recoupling washes: CH_2Cl_2 (or NMP when Boc-Asn or Boc-Gln were coupled), MeOH, 10% TEA/CH_2Cl_2, MeOH, and CH_2Cl_2. Each wash with each solvent was done twice. The TFA deblockings lasted 20 min. The couplings and the recouplings averaged 60 min. The protocol for syntheses at 45, 60, and 75 °C in 25% DMSO/toluene (SPPS at elevated temperatures) was as follows: (1) initial washes: 25% DMSO/toluene, MeOH, 10% TEA/toluene, MeOH, and toluene; (2) deblocking, 50% TFA/toluene with 1% anisole; (3) deblocking washes: 1% anisole/iPrOH, MeOH, and 25% DMSO/toluene; (4) neutralization: 10% TEA/toluene, MeOH, and toluene; (5) coupling and recoupling in: 25% DMSO/toluene using Boc or Fmoc amino acid/DIC/HOBt (1:1:1, 2.5 equiv) or in NMP using BOP or TBTU (2.5 equiv) and DIPEA (pH 9–10); (6) coupling and recoupling washes: 25% DMSO/toluene or NMP (when BOP or TBTU were used), MeOH,

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10% TEA/CH_2Cl_2, MeOH, and toluene. Each wash with each solvent was done twice. The TFA deblockings lasted 10 min. The couplings and the recouplings averaged 20 min. While DIC/HOBt (1:1) was used at all temperatures tested, TBTU was only used in conventional SPPS and SPPS at 60 °C. Peptide-resins were cleaved in HF (10 mL · g^{-1} peptide-resin) in the presence of 1% anisole. The HF was eliminated under vacuum, the resin/peptide was washed with Et_2O and the crude peptide extracted with dilute AcOH and lyophilized. Crude peptides were weighed to assess recovery and were analyzed by HPLC to assess purity (Figure 1). Peptides were purified by HPLC and were isolated as white powders >90% pure. The identity of purified ACP (65–74) was confirmed by MALDI-MS using a Brucker Reflex spectrometer and a thin film of α-cyano-4-hydroxycinnamic acid (dissolved in acetone) as matrix: MALDI-MS (*m/z*) Found: $[M + H]^+$, 1064.0. Calcd for $[M + H]^+$, 1063.5. Although it is difficult to carry out statistical analyses on synthetic procedures that inherently have multiple degrees of freedom, data presented in the original paper[9] and summarized here suggest that excellent peptide recovery and purity can be obtained at 60 °C but not necessarily at 75 °C.

Figure 1 RP-HPLC Profiles of the Syntheses of Acyl Carrier Protein (65–74) (10 μg Each Crude Peptide) by Conventional SPPS (27 °C) and SPPS at Elevated Temperatures (45, 60 and 75 °C).[9],a,b



[a] Column, Vydac C_{18} (300 Å, 5 μm, 0.46 × 25 cm); solvent A, 0.1% TFA in H_2O; solvent B, 0.09% TFA in 60% MeCN/H_2O; UV absorbance, 210 nm; flow rate, 1.0 mL · min^{-1}; gradient, 95% A/5% B to 5% A/95% B in 30 minutes.

[b] Adapted from (Varanda, L. M.; Miranda, M. T. M., *J. Peptide Res.*).

These results are in agreement with those reported by Lloyd et al.,[16] who also obtained the cleanest crude QFFYCNTTQLFNN at 60 °C and do not conflict with other results from our laboratory (synthesis of hCRF, pNPY and α-MSH[11]) at 75 °C described in part below. Significantly poorer results obtained at 45 °C may be due to the protocol used that called for 20-minute couplings; most residues had to be recoupled under these conditions thus lengthening the synthetic process and multiplying the number of steps, conditions that tend to have negative effects on the purity of the final products. Also encouraging was the observation that the swelling properties of both Boc-Gly- and ACP (65–74)-PAM resins in CH_2Cl_2 at 27 °C and in 25% DMSO/toluene at 60 °C were similar. Whereas TBTU couplings reported by Brandtner et al.[32] in DMF were more efficient than DIC/HOBt in DMF at 27 °C, the opposite was true at 37 °C. Our own results using ACP (65–74) on Boc-Gly-PAM in 25% DMSO/toluene at 60 °C suggest equivalence between the two coupling reagents based on peptide recoveries and purity.[9] The present conclusion is that DMF may not be ideal for standard SPPS steps at high temperature. While prudence suggests that TBTU with excess tertiary base DIPEA (threefold excess) be avoided at high temperatures, limited experimentation with that reagent in DMSO/toluene and elevated temperatures showed no untoward effects.[9,11] It is important to note that under the present experimental conditions, no difficulties caused by β-sheet formation[33] were encountered. This is in accordance with the hypothesis that the occurrence of aggregates can be avoided at high temperature[13] and in the presence of DMSO.[34,35]

4.3.4.4 YLRPDWHQA-Amide Encompassing Most Trifunctional Amino Acids

The sequence YLRPDWHQA-NH_2 (model peptide I) which encompasses most trifunctional amino acids, was synthesized to identify possible limitations (side reactions of individual amino acids) of SPPS and the Fmoc strategy at elevated temperatures. Model Peptide I was built up manually on a MBHA resin with a substitution $\geq$0.4 mequiv $NH_2 \cdot g^{-1} \leq 0.6$ mequiv $NH_2 \cdot g^{-1}$. Amino acid derivatives were: His(Trt)·DCHA, Asp(tBu), Arg(Mtr), Tyr(tBu). Trp and Gln were not protected. Coupling reagents were DIC/HOBt (1:1) or BOP. The synthesis procedure was as follows for 1 g of resin: (1) deblocking: 20% piperidine/toluene (20 mL, 5 min); (2) toluene type I wash (15 mL, 15 sec); (3) MeOH type II wash (15 mL, 15 sec); (4) repeat steps 2 and 3; (5) coupling in 25% DMSO/toluene (5 mL): Fmoc amino acid/HOBt (1:1, 2 equiv), DIC (2 equiv, 10 min) [exception: use BOP with Fmoc-His(Trt)·DCHA and Fmoc amino acid/HOBt (1:2) for Gln and Asn]; (6) 25% DMSO/toluene type I wash (15 mL, 15 sec); (7) MeOH type I wash (15 mL, 15 sec); (8) repeat steps 6 and 7, type II washes; (9) toluene type I wash (15 mL, 15 sec). In the synthesis of Model Peptide I, Pyr-Ala-amide resulting from ring closure of the deblocked QA-resin was also found. When this synthesis was repeated, this side reaction was prevented by deblocking the Boc-Gln-peptide-resin at room temperature and by adding the preformed Boc-Xaa-hydroxybenzotriazolyl esters during the following coupling step. Alternatively, this side reaction can be eliminated by protection of the γ-carboxamide moiety with a trityl or xanthydryl group. In these studies, couplings were monitored by the Kaiser ninhydrin test[31] and driven to completion by recoupling when necessary and not by extending the duration of the coupling step. This may be of some significance as it is recognized that activated amino acids (even at room temperature) may, over time, undergo undesired side reactions such as racemization of certain residues or dehydration of Gln and Asn (among others) that may contaminate the growing peptide chain. Acetylation (often used to limit the amount of deletion peptides) at high temperature was not performed systematically in the present studies (except for α-MSH: 3 min at 75 °C versus 15 min at rt; see ref[11]). In other studies[36] it was found that this protocol gave results similar to those obtained at room temperature.

4.3.4.5 Human Corticotropin Releasing Factor (hCRF, a 41-Peptide Amide)

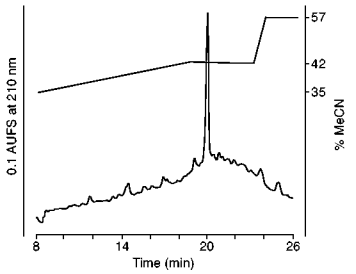
While the existence of corticotropin-releasing factor (CRF) was first proposed in 1955 it took 26 years to develop the isolation (RP-HPLC) and sequencing tools (microsequencing) that led to its characterization in 1981.[37] First isolated from ovine hypothalami,[37] CRF was subsequently characterized from rat hypothalami;[38] the identical structure was deduced for human CRF (hCRF) on the basis of the cDNA sequence of the hCRF precursor gene.[39] CRF plays a major role in the maintenance or restoration of homeostasis by regulating the activity of the hypothalamic-pituitary-adrenal (HPA) axis.[40] Its first synthesis stretched the limits of SPPS capabilities and of preparative HPLC.[37]

hCRF was built up manually on a MBHA resin with a substitution 0.3–0.5 mequiv $NH_2 \cdot g^{-1}$. Amino acid derivatives were: His(Tos), Tyr[Z(2-Br)], Ser(Bzl), Glu(Cy), Asp(Cy), Arg(Tos), Cys(Mob), Cys(Acm), Lys[Z(2-Cl)], and Thr(Bzl). Asn and Gln were not protected. Coupling reagents were DIC/HOBt (1:1) or TBTU. The protocol for synthesis at 75 °C was as follows with volumes per gram resin and reaction times shown in parenthesis: (1) deblocking: TFA/toluene/EDT (10:10:1, 15 mL, 3 min); (2) toluene type I wash (15 mL, 15 sec); (3) neutralization, 10% TEA/toluene (15 mL, 15 sec); (4) MeOH type II wash (15 mL, 15 sec); (5) repeat steps 3 and 4; (6) coupling in 25% DMSO/toluene (15 mL): Boc amino acid/HOBt (1:1, 2 equiv), DIC (2 equiv, 10 min) [exception: use BOP with Boc-His(Tos)·DCHA and Boc amino acid/HOBt (1:2) for Gln and Asn]; (7) 25% DMSO/toluene type I wash (15 mL, 15 sec); (8) MeOH type I wash (15 mL, 15 sec); (9) repeat steps 7 and 8, type II washes; (10) toluene type I wash (15 mL, 15 sec). hCRF resin was cleaved in HF/anisole/

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EMS (20:1:1, 15 mL · g^{-1}) as described above. The HF was eliminated under vacuum, the resin/peptide was washed with Et_2O and crude hCRF was extracted with dil AcOH and lyophilized. Crude hCRF was weighed to assess recovery and was analyzed by HPLC to assess purity (Figure 2). The crude peptide appeared to be as pure as any other preparation obtained using similar conditions (but different solvents and reaction times) at room temperature with equivalent yields. hCRF was purified by HPLC (TEAP 2.25, TEAP 6.4 and 0.1% TFA) and was isolated as a white powder shown to be >96% pure by HPLC and CZE. The identity of purified hCRF was confirmed by MALDI-MS: (*m/z*) Found: [M + H^+], 4755.7. Calcd for [M + H^+], 4755.51. Purified hCRF was tested for biological activity in the cultured rat pituitary cell assay of Vale et al.[37] and found to be equipotent to other preparations of hCRF.

Figure 2 RP-HPLC Profile of Human CRF (50 μg) Synthesized by Conventional SPPS at 75 °C.[10],a,b



[a] Column, Vydac C_{18} (300 Å, 5 μm, 0.46 × 25 cm); solvent A, 0.1% TFA in H_2O; solvent B, 0.09% TFA in 60% MeCN/H_2O; UV absorbance, 210 nm; flow rate, 2 mL · min^{-1}; gradient, 50% A/50% B to 30% A/70% B in 20 minutes followed by 1 minute gradient to 5% A/95% B.

[b] Adapted from (Rabinovich, Rivier, *Peptides: Chemistry, Structure and Biology*), Copyright (1994), p 71–73, with permission from ESCOM Science Publishers.

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4.3.5 Cleavage Reactions from Solid Supports

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An important feature of solid-phase peptide synthesis (SPPS) is that various kinds of cleavage reactions can be applied to the finished peptide-resin to yield a variety of products. This kind of versatility is available for peptide-resins prepared by either Boc/Bzl or Fmoc/tBu assembly strategies, although the reagents used to effect cleavage are necessarily different, due to the contrasting chemical labilities of the blocking groups and the peptide-resin link with the two strategies. Special resins and peptide-resin linkers have been developed to facilitate the production of the various types of products desired. The reader is referred to Section 4.3.1 for a discussion of these features. This section contains specific protocols for the basic kinds of Boc and Fmoc cleavage procedures. Additional details can be found in *Methods in Enzymology*, Vol. 289, in *Solid Phase Peptide Synthesis*, by Stewart and Young,[1] *Peptide Synthesis Protocols*,[2] the biennial volumes of proceedings of the American and European Peptide Symposia, and the annual volumes of the Japanese Peptide Society. A remarkable (and free) resource is the catalog of Novabiochem, which contains nearly 100 pages of SPPS information, including specific protocols and many references.

The major difference in resin attachments for Boc versus Fmoc chemistry is the acid lability of the peptide-resin link. For Boc chemistry, the resin link is a modified Bzl group, and requires strong ("hard") acids for cleavage to yield the peptide acid. Within this general category the original Merrifield resin offers moderate peptide-resin stability, and is suitable for most short peptides and for longer peptides where the C-terminal residue is sterically hindered. For longer peptides having labile C-terminal residues, the more stable PAM- or HMBA-resins are preferred in order to minimize loss of peptide from the resin. Both these types of peptide-resin are effectively cleaved by HF, TfOH, or TMSOTf. The peptide-resin link for Fmoc chemistry possesses similar acid lability to tBu groups, or is even more labile; it is cleaved by TFA, a much milder reagent than is required for Bzl groups. The reactivity toward nucleophilic cleavage (i.e., with amines or alcohols) is the inverse: resins used for Boc chemistry are relatively labile, whereas those for Fmoc chemistry are more stable. These properties affect the types of reactions that are practically available for cleavage of peptides. Again, Merrifield peptide-resins offer moderate lability toward nucleophilic cleavage, whereas those more stable toward acidolysis (PAM or HMBA) are more labile toward nucleophilic cleavage. Peptides containing sensitive residues (Cys, His, Met, Trp, Tyr) as well as residues bearing side-chain, Bzl-related (or tBu-related in the case of Fmoc synthesis) blocking groups are subject to irreversible alkylation of the sensitive residues by carbonium ions produced from the blocking groups. To minimize peptide damage, therefore, adequate quantities of reactive scavenger molecules must be present to trap the carbonium ions generated. For most standard HF cleavages, anisole is a satisfactory scavenger. If Trp, Cys, or other sensitive residues are present in the peptide, other scavengers may be added (see Section 4.3.5.1), or preferably the "low–high" cleavage technique[3] (see Section 4.3.5.1.5) may be used. The N-terminal Boc group should be removed by a standard deprotection procedure (see Section 4.3.2.1.2) to eliminate this source of tBu carbonium ions. For Fmoc-protected peptide-resins, anisole or TIS can be used as scavengers for simple peptides, but if sensitive residues are present, reagent K[4] should be employed. The Fmoc group is stable to acid, and may be left intact, if desired, to give an ultraviolet-absorbing handle to guide purification (λ_{max} 266 nm); it can be removed from the pure peptide by treatment with piperidine. If cleavage reactions are done in an overly concentrated solution, side reactions may be favored. For 0.5 g peptide-resin (0.2 mmol peptide), a volume of 10 mL HF with scavenger, or 10 mL TFA cleavage cocktail for labile (Fmoc) resins should be satisfactory.

Peptide amides can be synthesized by ammonolysis of Merrifield, PAM, or HMBA resins. This is a versatile approach, since peptides having more than one C-terminal function can be obtained from a single synthesis, but in many cases the presence of susceptible functional groups (such as Asp and Glu) in the peptide causes problems. The preferred approach is to use synthesis resins specifically designed to yield amides upon applying standard cleavage procedures, such as HF following Boc/Bzl synthesis or TFA-based reagents following Fmoc/tBu synthesis. For Boc/Bzl synthesis, the MBHA resin is satisfactory, and for Fmoc/tBu synthesis the Rink-amide or PAL resins are effective. A variant is to use Fmoc/tBu synthesis procedures on an HMBA resin. The peptide can be deblocked on the resin with TFA or Reagent K, and then cleaved from the resin by ammonolysis.

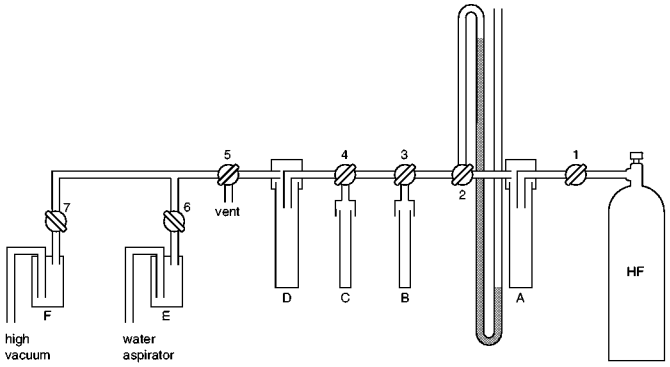
4.3.5.1 Cleavage after *tert*-Butoxycarbonyl/Benzyl Synthesis

To obtain free peptides having C-terminal carboxy groups from Boc/Bzl syntheses, the peptide-resin is cleaved with anhydrous strong ("hard") acid. Liquid anhydrous HF is the cleanest and most used reagent for this purpose.

CAUTION: *Due to the hazardous nature of HF [highly toxic vapor, highly corrosive liquid, a gas at room temperature (bp 19°C); dissolves glass vigorously], appropriate equipment MUST be used in a good fume hood.*

The necessary equipment is a fluorocarbon all-plastic vacuum line; such a commercial vacuum line is presented in Figure 1.

Figure 1 Diagram of a commercial plastic HF Line[1],[a]



[a] Reprinted from (Stewart, J. M.; Young, J. D., *Solid Phase Peptide Synthesis, 2nd ed.*), Copyright (1984), with permission from Pierce Chemical Co.

Alternate reagents for Boc/Bzl cleavage, if HF apparatus is not available, are TfOH and TMSOTf. These acids are nonvolatile, very corrosive, and more vigorous reagents than HF; they are also more potent Friedel–Crafts catalysts than HF, and thus they may cause more side reactions and peptide damage than HF.

Suitable scavengers must be present during the cleavage to prevent alkylation of sensitive residues by carbonium ions formed from blocking groups on side-chain groups. The best protection against such alkylation is to perform the HF cleavage in two stages, the so-called "low–high" procedure.[3] The peptide-resin is first cleaved by HF in the presence of a high concentration of DMS. The mechanism of cleavage in this case changes from the usual S_N1, where free carbonium ions are formed, to S_N2, where the concentration of DMS is so high that leaving groups are displaced by the sulfur nucleophile before a separate ion can form.

4.3.5.1.1 Preparation of the Peptide-Resin for Hydrogen Fluoride Cleavage

The last solvent used for washing the peptide-resin should be dichloromethane; this swells the resin and leaves it in a porous condition to facilitate penetration by the cleavage reagents. His(Dnp) is not cleaved by HF, and should be removed from the peptide-resin by thiolysis before cleavage with HF or other acids.

4.3.5.1.1.1 Thiolytic Removal of Dinitrophenyl Groups from Histidine

Suspend the peptide-resin in a round-bottomed flask in the minimum amount of DMF necessary to slurry the resin (ca. 5 mL per gram of resin) and add benzenethiol (20 equiv, $0.102\,mL \cdot mmol^{-1}$) per equivalent of His(Dnp) groups. Stir at rt for 1 h. Filter the resin and wash it thoroughly with DMF, H_2O, EtOH, and CH_2Cl_2. Dry thoroughly. It is not practical to remove all the intensely yellow Dnp-benzenethiol product from the resin, and this material will color the crude peptide. It is easily removed during purification of the peptide.[1]

4.3.5.1.2 Hydrogen Fluoride Cleavage: Precautions in the Use of Hydrogen Fluoride

All operations involving HF must be carried out in a fume hood with excellent air flow. Never bring condensed HF into the open laboratory. Never try to pour liquid HF; leave it in the vacuum line at all times. **CAUTION:** *Breathing HF can be fatal!* Contact with the liquid causes serious burns instantly. Make a saturated solution of calcium gluconate in glycerol, and have this available at all times when working with HF. If you get HF on the skin, wash immediately and thoroughly with water, soak in water, then apply the calcium gluconate solution. If careless operation of the vacuum line causes solids (CoF_3 or resin) to "bump" and completely occlude the line so the HF cannot be removed in the usual way, chill the vessel containing HF in dry ice/EtOH, carefully remove it from the vacuum line and stand it at the back of the fume hood. Carefully direct a stream of water from a wash bottle at the inner wall of the vessel. The HF will react vigorously and exothermically with the water. When all the HF has reacted, dilute the residue further with water and discard properly.[1]

HF is supplied in steel compressed-gas cylinders. Because the boiling point of HF is 19 °C, most of the HF in the cylinder is liquid, and the pressure is low (about 300 Torr). When HF cylinders are stored for a long time, pressure in the cylinders can build up due to reaction of HF with iron to form hydrogen. Newly purchased cylinders have been received with high pressure; open such cylinders very cautiously the first time. If there is high pressure in the cylinder, mercury in the vacuum-line manometer can be blown out of the top and scattered about the hood.[1]

Some people have constructed makeshift HF lines from low-density polyethylene bottles and tubing. Do not use this kind of apparatus: it is hazardous, and may also cause damage to the peptides processed. When polyethylene is chilled in the dry ice traps it becomes extremely brittle and may shatter, with disastrous consequences. Since such systems cannot be evacuated, the HF is evaporated at its boiling point of 19 °C; this prolonged high-temperature exposure of peptides to HF can cause side reactions.[1]

Although HF is a dangerous substance, any danger can be avoided through the use of proper equipment and by taking reasonable precautions. HF cleavage has been performed in the author's laboratory for more than 30 years, without any accidents.

4.3.5.1.3 Hydrogen Fluoride Cleavage: Use of the All-Plastic Vacuum Line

The vacuum line shown in Figure 1 is constructed of rigid fluoropolymer (similar to Kel-F) and is commercially available (Protein Research Foundation, Osaka, Japan; Peptides In-

ternational, Louisville, KY). It allows two cleavage reactions to be run concurrently, if the operator pays close attention to the timing of operations. The apparatus should be assembled in a good fume hood. Trap A, which should be calibrated and marked at 20- and 40-mL levels, contains a magnetic stirring bar and a few grams of CoF_3 to dry HF from the cylinder. Trap E is a safety trap between the line and the water aspirator. Trap F is filled with KOH to protect the vacuum pump. In localities where discharge of fluoride into the drain is illegal, a CaO trap is added between trap E and the water aspirator. The mercury manometer is essential to indicate the pressure status of the interior of the vacuum line. The mercury should be covered with a layer of fluorocarbon oil (infrared sample oil is satisfactory) on the arm adjacent to the vacuum. The joints between all parts of the line are sealed with ethylene–propylene O-rings (Parker Seals), and will need to be replaced occasionally. Once the line is set up and vacuum integrity has been established, do not move these joints, or they may leak. The elastomer O-rings take on a "set" and will probably need to be replaced if the joints are moved. To assure a vacuum-tight assembly, close stopcocks 1 and 7, turn 3-way stopcock 5 to open the line to the aspirator, and have stopcocks 3 and 4 turned to connect the cleavage vessels B and C, which should be calibrated and marked indelibly on the outside at 5- and 10-mL levels, to the line. Evacuate the system with the water aspirator; the manometer should read about 360 mm on the scale at sea level (the actual vacuum is double this number). Close stopcock 6 and wait to see that the vacuum is maintained.

To collect HF in drying reservoir A, open stopcock 6 and turn stopcock 2 to connect the manometer and the reservoir. Place a dry ice/EtOH bath under vessel A and wait 5–10 min. Do not use acetone for the bath, as it is too flammable. Fluorocarbons are poor conductors of heat, and with the heavy vessel walls, the contents of these vessels cool and warm only very slowly. With stopcock 1 closed, open the cylinder valve, then very carefully open stopcock 1, watching the pressure in the manometer. If the pressure is too high, with the danger of mercury blowing out of the manometer, close stopcock 1 and reevacuate the line. High pressure indicates the presence of hydrogen in the HF cylinder and it must be bled out first. Normally, if the HF is condensing at a good rate then the manometer will show a vacuum of ca. 20 Torr. Occasionally remove the dry ice bath from vessel A to check the amount of HF condensing. Collect enough HF for three or four cleavage reactions (30–40 mL). When an adequate volume of HF has been collected, close the cylinder valve and stopcock 1. Turn stopcock 2 to the position shown in Figure 1 to close off the HF reservoir. Stir the CoF_3 in the reservoir for a few min to dry the HF.

4.3.5.1.4 Hydrogen Fluoride Cleavage: Standard "High" Conditions

Reservoir A containing HF is at rt. Place a magnetic stirring bar, peptide-resin (0.5 g), and anisole (1.0 mL) in vessel B. A baffle must be in the top of the vessel to help prevent bumping of resin into the stopcock and line. Quickly turn stopcock 2 to connect vessel B and the line to the manometer; close stopcocks 4, 5, and 7. Place a dry ice/EtOH bath under vessel B and wait 10 min. Evacuate the system with the water aspirator and check for vacuum integrity. Do not pump for too long, otherwise the anisole will be lost. Turn stopcock 3 as shown in Figure 1 to close off vessel B. Slowly turn stopcock 2 to connect the manometer to the HF reservoir; the manometer will show a moderate, positive pressure (ca. 300 Torr in total). Start the magnetic stirrer under reservoir A and very carefully open stopcock 3 to connect vessel B to the manometer and the HF supply. Constantly watch the HF in the reservoir while opening the stopcock to be sure that the HF does not boil up too far and get CoF_3 particles in the line. Particles of CoF_3 will wreck the plastic stopcocks and may totally occlude the orifices in the line. Be very careful. Be sure that the reservoir stirrer is operating properly to prevent "bumping" of the HF. Once steady distillation has been established, the positive pressure in the manometer should become negative (ca. 20 Torr). Check vessel B to

for references see p 824

see that HF is condensing; continue distillation until HF is at the 10-mL mark on vessel B. Once distillation has become steady, it can be accelerated by placing a beaker of H_2O at rt under reservoir A. When the desired amount of HF has been collected, turn stopcock 2 to close off reservoir A and connect vessel B to the manometer for the duration of the cleavage. Replace the dry ice bath under vessel B with an ice–water bath, start the magnetic stirrer, and stir the reaction mixture at 0 °C for 45–60 min. This period of time includes 15 min to bring the vessel contents to the bath temperature. Once the anisole has melted, ensure that the stirrer is operating correctly in order to keep the resin mixing continuously with the HF. In the presence of anisole, the mixture has a brick-red color; without anisole, the resin is brilliant-blue in HF.[1]

At the end of the desired reaction time, close stopcock 5 and evacuate the line with the water pump. Very carefully turn stopcock 3 to open the cleavage vessel B to the vacuum. Constantly watch to see that the stirrer in vessel B is operating and that the HF does not boil too high or "bump". Continue concentration until the resin appears dry and the color has changed from red to buff, indicating that the HF has been removed. Turn 3-way stopcock 5 to vent the aspirator, but not the vacuum line. Turn off the water pump and close stopcock 6. Place a dry ice/EtOH bath under reservoir D; allow 15 min to chill, and start the high-vacuum pump. Turn stopcock 5 to close the vent and open stopcock 7. Pump the resin for ca. 1 h. Vent the line and remove vessel B, now containing a mixture of peptide and resin.[1]

Transfer the resin to a fritted glass funnel with EtOAc, using at least three washes to remove scavengers. If the peptide is very hydrophobic and might be soluble in EtOAc, use Et_2O for transfer and washing. Save the filtrate until you are sure that the peptide has not been lost. Allow the resin to air-dry. Extract the peptide with an appropriate solvent (3×) and allow the solvent to soak into the resin beads at each wash. For water-soluble peptides, 10% AcOH is often used. Many chemists prefer to extract the peptide with glacial AcOH, which can be lyophilized directly.[1]

Extremely hydrophobic peptides may often be extracted with TFA, DMF, or DMSO. Since TFA has a low boiling point (72 °C), it is removed readily. DMF and DMSO are to be avoided if possible, as they are high boiling and are not easily removed. Mixtures of H_2O/AcOH that are more than 90% or less than 10% AcOH will generally lyophilize satisfactorily; more equal mixtures will have depressed freezing points and will not stay frozen on the lyophilizer.[1] To reverse any N → O shifts that have taken place in the HF, dissolve the peptide in H_2O, neutralize with NH_4OH, and relyophilize. If Trp(For) was used in the synthesis, treat the crude formylated peptide with 0.03 M $HONH_2$ {use 50% aq $HONH_2$ [$HONH_2$ (0.2 mL) in H_2O (100 mL)]}. Adjust the pH to 9.0 with NH_4OH and allow to stand at rt for 2 h.[5]

4.3.5.1.5 Hydrogen Fluoride Cleavage: "Low–High" Cleavage

The best protection against damaging sensitive residues is provided by the "low–high" HF cleavage technique. The procedure is tedious, requiring more than twice the time of the standard HF cleavage; allow a full day to carry out the process. Another advantage of the "low–high" cleavage condition is that it removes the formyl group from the indole ring if Trp(For) is used in the synthesis. The essential DMS reagent has a powerful stench; have a beaker of commercial hypochlorite bleach available in the hood to decontaminate any glassware that has been in contact with it.

Following the general procedures given in the previous section for use of the HF vacuum line, place the peptide-resin (0.5 g, 0.2 mmol), *m*-cresol (1.0 mL), 4-methylbenzenethiol (0.25 mL), and DMS (6.5 mL) in the cleavage vessel. Immediately put all contaminated glassware in the bleach container to deodorize. Attach the vessel to the HF line, chill in dry ice, evacuate, and distill in the HF (2.5 mL). Do not prolong the evacuation or the

volatile DMS will be lost. Stir at 0 °C for 2 h. Distill off the HF and DMS using the water aspirator; the cresol and 4-methylbenzenethiol will remain in the vessel to serve as scavengers for the following standard "high" HF cleavage. Distill HF (10 mL) into the vessel and do a standard HF cleavage, as described above in Section 4.3.5.1.4.[3]

4.3.5.1.6 Hydrogen Fluoride Cleavage: Use of Hydrogen Fluoride/Pyridine

A solution of 70% HF in pyridine (Olah's reagent) is commercially available and can be used for peptide cleavage without the need for expensive apparatus. Place 0.5 g peptide-resin (0.2 mmol peptide) and a magnetic stirring bar in a thick-walled fluorocarbon or polypropylene vessel that can be tightly closed and evacuated. Add appropriate scavengers (see Section 4.3.5.1.5) and HF/Py (10mL). Close the vessel and stir at rt for 1 h. Transfer to a high-vacuum pump fitted with a CaO trap, to collect the HF, and KOH and cold traps to protect the vacuum pump. Remove the reagent, evacuating very carefully at first, and pump until the resin is dry and free of HF. Filter on a fritted funnel and work up as in Section 4.3.5.1.4.[6]

4.3.5.1.7 Standard "High" Cleavage with Trifluoromethanesulfonic Acid in Trifluoroacetic Acid

The peptide-resin should have been dried after washing with CH_2Cl_2. Boc and Dnp groups should have been removed (see Sections 4.3.2.1.2 and 4.3.5.1.1.1, respectively). Place the peptide-resin (0.5 g, containing 0.2 mmol peptide) in a 50-mL round-bottomed flask and add the scavenger mixture of thioanisole/EDT (2:1, 1.5 mL). After the scavenger has soaked into the resin, add TFA (10 mL) and chill to 0 °C in an ice bath. While stirring, slowly add the TfOH (1.0 mL). Allow the reaction to warm to rt and stir for 30–90 min, using the minimum time sufficient to cleave and deprotect the peptide. Longer reaction times are needed for hindered C-terminal residues and resistant blocking groups. Arg(Tos) and Cys(pMeBzl) groups and peptide links to BHA resins are not cleaved. Use Arg(Mts), Arg(Pmc), or Cys(pMeOBzl) blocking groups and MBHA resin for peptide amides.[7]

At the end of the reaction time, filter the soln through a fritted funnel directly into cold, stirred Et_2O or MTBE (200 mL). More complete precipitation of the peptide may be obtained if the Et_2O is chilled in dry ice/EtOH. Wash the resin with three small portions of cold TFA. Add dry Py (1.0 mL) dropwise to the cold, stirred Et_2O suspension to complete peptide precipitation. Collect the peptide on a fine-pore fritted glass funnel, wash with cold Et_2O (3 ×), and air-dry. Purify the crude peptide by an acceptable procedure.[7]

4.3.5.1.8 "Low" Cleavage with Trifluoromethanesulfonic Acid

A TfOH cleavage procedure analogous to the "low–high" HF cleavage technique can be achieved as follows. For 0.5 g peptide-resin, mix TFA (5.0 mL), DMS (3.0 mL), and *m*-cresol (1.0 mL). Chill in an ice bath and add TfOH (1.0 mL). If the peptide-resin contains Trp(For), use a reagent composed of TfOH/TFA/DMS/*m*-cresol/EDT (1:5:3:0.8:0.2, 5.5 mL). Add the cold reagent to the peptide-resin and stir at 0 °C for 4 h; work up as for "high" TfOH cleavage (Section 4.3.5.1.7). Follow with the "high" TfOH procedure to cleave the peptide from the resin.[8]

4.3.5.1.9 "High" Cleavage with Trimethylsilyl Trifluoromethanesulfonate

For 0.5 g peptide-resin (0.2 mmol peptide) use an ice-cold reagent composed of TFA (20 mL), thioanisole (2.4 mL), *m*-cresol (0.5 mL), and TMSOTf (4.1 mL). If the peptide contains Trp, add EDT (0.5 mL). Stir at 0 °C for 2 h. Removal of Arg(Tos) will usually require a longer time. Work up as for the "high" TfOH procedure (Section 4.3.5.1.7).[9]

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4.3.5.1.10 "Low" Cleavage with Trimethylsilyl Bromide in Trifluoroacetic Acid

The TMSBr cleavage is analogous to the "low" TfOH procedure (Section 4.3.5.1.8) for removal of blocking groups under conditions less likely to damage sensitive residues in the peptide. It can also be used for cleavage of peptides from resins used for Fmoc-based syntheses; TMSBr effectively cleaves Arg(Mtr) groups. For 0.5 g peptide-resin (0.2 mmol peptide) use an ice-cold reagent composed of TFA (20 mL), thioanisole (2.4 mL), *m*-cresol (0.5 mL), and TMSBr (2.65 mL). If the peptide contains Trp, add EDT (0.5 mL). Stir at 0 °C for 1 h. Filter the resin on a fritted funnel and wash with TFA (3 ×). Return the resin to the flask for a "high" cleavage to cleave the peptide from the resin.[10]

This reagent silylates hydroxy groups, but the resulting ethers are generally hydrolyzed readily when the peptide is dissolved in H_2O. If a TMS ether is stable, MS of the product will show a contaminant 70 amu higher than the theoretical peptide mass. Such ethers can be cleaved by treatment with a solution of ammonium formate in MeOH/H_2O (1:1).[10]

4.3.5.1.11 Cleavage with Boron Tribromide in Trifluoroacetic Acid: Boron Tris(trifluoroacetate)

Boron tris(trifluoroacetate) (BTFA) can be purchased as a 1 M soln in TFA. It can also be prepared from BBr_3 and TFA by adding TFA (3 equiv) to cold (0 °C) 1 M BBr_3 soln (1 equiv) and concentrating to dryness to remove the HBr byproduct.

Dissolve the solid BTFA in TFA to reconstitute a 1 M soln and chill to 0 °C. To a flask containing the peptide-resin and scavengers add 8 equiv of the BTFA soln per equiv of cleavable functional groups in the peptide (the peptide-resin link plus all Bzl-type blocking groups). For 0.2 mmol peptide-resin use thioanisole [1.5 mL; also add EDT (0.5 mL) if Trp is present]. Do not use anisole as a scavenger with BTFA. Stir at 0 °C for 1 h, filter through a fritted filter, and wash the resin with TFA (3 ×). Concentrate the solvent under reduced pressure and apply the residue to a suitable gel column to remove boron and scavengers. Alternatively, boron may be removed by repeatedly dissolving the residue in MeOH and concentrating under reduced pressure. This protocol cleaves Merrifield resin links, Arg(Tos), Cys(Mob), Bzl, and Z groups.[11]

4.3.5.1.12 Cleavage of Merrifield or (Hydroxymethoxy)benzyl Resins by Ammonolysis

The recommended procedure for the synthesis of peptide amides is to use a resin (e.g., MBHA) that yields amides directly upon cleavage with HF. Ammonolysis can be used to give amides of peptides on Merrifield or HMBA resins. Ammonolysis can also probably be used with PAM resins.

CAUTION: *All Bzl-blocked Asp and Glu residues will be converted into amides (Asn and Gln).*

If the C-terminal residue is sterically hindered, ammonolysis may fail or be very slow. Yields may be improved by adding peroxide-free dioxane to the alcohol [up to 50% (v/v)] to improve resin swelling. If ammonolysis fails, the peptide can be converted into the methyl ester (see Section 4.3.5.1.14) and then ammonolyzed in solution. Ethers are not cleaved. If Ser(Bzl), Thr(Bzl), or Tyr(ether) are present, these must be subsequently cleaved by a suitable procedure.

Working in a good fume hood, suspend the peptide-resin in anhyd iPrOH (20 mL · g^{-1}) in a heavy-wall pressure vessel. For small runs, a round-bottomed flask can be used if it is in perfect condition, free from nicks and scratches. Chill the suspension to −20 °C in an ice/$CaCl_2$ bath and saturate with anhyd NH_3 from a cylinder, using a KOH trap on the exit tube to prevent entry of moisture. Use a safety trap between the cylinder and the flask and watch constantly to be sure that solvent is not sucked back into the cylinder. Wire a tight rubber

stopper in place and stir the suspension at rt for 2–4 d. Chill the bottle, remove the stopper, and allow the contents to come to rt. Filter the resin on a fritted funnel and wash with iPrOH (3 ×). Save the resin for additional treatment if peptide recovery is poor. Concentrate the combined filtrate and washings on a rotary evaporator.[1]

For cleavages involving iPrOH/dioxane (up to 1:1), the latter should first be tested for the presence of peroxides by mixing it with an equal volume of 4% KI soln. If any brown color is formed after 1 min, remove the peroxides by passing the dioxane through a column of activated alumina (Al_2O_3). Not all commercially available activated alumina is effective; Baker No. 0537 is satisfactory. Do not use acid-washed alumina. The same procedure is used to purify THF.[1]

4.3.5.1.13 Cleavage of Merrifield or (Hydroxymethoxy)benzyl Resins by Hydrazinolysis

Hydrazinolysis can yield protected peptides for assembly into large peptides or proteins by the azide coupling method. A limitation is that Bzl esters on Asp and Glu will also be converted; tBu esters are stable. Boc and For groups on amines are stable, but His(Tos) is cleaved.

CAUTION: *Hydrazine is a skin and eye irritant; use care in handling and work in a good fume hood. Hydrazine can explode during distillation if oxygen is present.*

Suspend the peptide-resin in purified DMF (5 mL per gram of resin) and add anhyd H_2NNH_2 (30 equiv) per equiv of peptide. Stir at rt for 2 d. Filter the resin and wash with DMF (3 ×). Concentrate the liquid under a suitable high vacuum (H_2NNH_2, bp 113 °C; DMF, bp 153 °C) and purify the peptide.[12]

4.3.5.1.14 Cleavage of Merrifield or (Hydroxymethoxy)benzyl Resins by Transesterification

Simple esters (Me, Et, Bzl) can be prepared from Merrifield peptide-resins if the C-terminal residue of the peptide is not sterically hindered. During methanolysis, Bzl esters of Asp and Glu will be converted into methyl esters. Methyl esters can be prepared by reaction at rt; ethyl and Bzl esters should be carried out at 50 °C to accelerate the transesterification. A unique kind of transesterification is that with dimethylaminoethanol (DMAE).[13] The intermediate DMAE ester formed is extremely labile to aqueous hydrolysis, and gives the free carboxy peptide readily. Bzl esters of Asp and Glu will be similarly converted, yielding the free carboxy groups. If the peptide-resin does not have Asp or Glu residues, this procedure will yield protected peptides for further coupling reactions in segment synthesis ("convergent" synthesis). C-terminal Pro peptide-resins transesterify only very slowly; the reaction can be catalyzed in high yield preferably by thallous 2-(dimethylamino)ethoxide or sodium 2-(dimethylamino)ethoxide (yield may be lower).

CAUTION: *Thallium is extremely toxic. Avoid all contact of solutions with skin.*

Suspend the peptide-resin in anhyd MeOH (40 mL per gram of resin) and add TEA (50 equiv per equiv of peptide). Stir at rt for 24 h. Filter off the resin (save in case retreatment is needed), wash with MeOH (3 ×), and concentrate the solvent under reduced pressure. Purify the peptide ester by a suitable method. If the yield is not satisfactory, repeat the reaction in DMF/MeOH/TEA (2:2:1; 40 mL per gram of resin) at 50 °C for 24 h.[1]

For transesterification with DMAE, mix Merrifield or HMBA peptide-resin (0.5 g) with DMAE/DMF (1:1, 10 mL) and stir at rt for 70 h. Remove the solvents under high vacuum (DMAE, bp 139 °C; DMF, bp 153 °C). For C-terminal Pro peptides, mix the peptide-resin (1.0 g, 0.4 mmol) with DMAE/DMF (1:2) containing thallous 2-(dimethylamino)ethoxide (0.1 equiv, see below). Stir at rt for 24 h. Filter and pass the filtrate through a column of

for references see p 824

carboxy ion-exchange resin (such as IRC-50; poured in DMF) to remove the thallium. Concentrate the solvent under high vacuum and purify the peptide as desired.[1]

CAUTION: *Thallium is extremely toxic. Avoid all contact of solutions with skin.*

To avoid using thallium, sodium 2-(dimethylamino)ethoxide may be used, although yields may be lower. The catalyst, thallous 2-(dimethylamino)ethoxide, is prepared by mixing thallous ethoxide (0.75 g, 3.0 mmol) with DMF (42 mL), DMAE (21 mL), and benzene (8 mL), and distilling at 50 °C under water aspirator vacuum until 10 mL distillate is obtained. This operation removes EtOH, which will yield stable ethyl esters if it is allowed to remain. Use an appropriate aliquot of this soln to catalyze the transesterification. To avoid using toxic thallium, prepare the sodium salt in a similar fashion from sodium ethoxide (204 mg, 3.0 mmol).[1]

Hydrolyze DMAE esters by treating the ester with a mixture of DMF/H_2O (1:1) containing a trace amount of imidazole for 24 h at rt. Concentrate the solvents in vacuum and purify the peptide by an appropriate method.[1]

4.3.5.1.15 Cleavage of Merrifield Resins by Catalytic Transfer Hydrogenolysis

Add the peptide-resin (1.0 g) to a soln of $Pd(OAc)_2$ (1.0 g) in DMF (15 mL) and stir for 2 h. Add a soln of ammonium formate (1.5 g) in H_2O (1.5 mL) and stir for an additional 2 h. Filter the resin off and wash with AcOH (2 ×) and with H_2O (2 ×). Concentrate the solvent under vacuum and purify the peptide as usual.[14,15] This procedure may not cleave Ser(Bzl) and Thr(Bzl) completely. If these are present in the peptide, after the initial 2-h hydrogenolysis add AcOH (20 mL) and Pd/C (0.5 g), and stir for 2 h. Work up as above. Hydrogenolysis fails for peptides containing Cys or Met, since sulfur poisons the catalyst.[14,15]

4.3.5.1.16 Reductive Cleavage of Merrifield or (Hydroxymethoxy)benzyl Resins To Yield Peptide Alcohols

Stir the peptide-resin (1.0 g, 0.4 mmol) in anhyd, peroxide-free THF (30 mL), ensuring the exclusion of moisture. (Check the THF for peroxides and purify as described in Section 4.3.5.1.12). Add $LiBH_4$ (58 mg, 2 mmol) slowly in portions and stir at rt for 1 h. Slowly add glacial AcOH (1.0 mL) to destroy excess hydride, filter on a fritted funnel, wash the resin with small portions of AcOH, and lyophilize the soln directly. Purify the peptide alcohol by a suitable procedure. The whole procedure converts Asp and Glu esters into alcohols and deblocks Tyr[Z(Br)] groups.[16]

4.3.5.2 Cleavage after 9-Fluorenylmethoxycarbonyl/*tert*-Butyl Synthesis

For best results and highest yields of Fmoc-synthesized peptides, the selection of synthesis resin and side-chain blocking groups is critical. See Sections 2.2 and 4.3.2.2 for a full discussion of these points.[2,17]

If the peptide does not contain blocked side-chain functional groups that will produce carbonium ions or sensitive residues that may be attacked by them (Cys, Met, Trp, Tyr), peptide-resins may be cleaved with TFA containing 2.5% H_2O and 2.5% TIS (v/v) using 15 mL reagent per gram of peptide-resin. Most peptides containing any of the above groups should be cleaved with Reagent K. Arg(Mtr) is too stable for rapid cleavage by TFA or Reagent K; if several residues are present, up to 24 h may be required, in which case Arg(Pmc) or Arg(Pbf) are recommended. For more vigorous cleavage conditions, for example to facilitate deblocking of Arg(Mtr) when several residues are present, TMSBr in TFA (Section 4.3.5.1.10) can be used, or HBF_4 in TFA.

4.3.5.2.1 Reagents for Cleavage of 9-Fluorenylmethoxycarbonyl Peptide-Resins

Peroxide-free Ethers

Cleavage reactions following Fmoc syntheses are typically worked up by precipitating the peptide in Et_2O or MTBE. The latter is preferred because of its higher boiling point and lower risk of causing a fire or explosion. Ethers readily form peroxides which can oxidize Met or Cys residues in peptides. For this reason, test ethers for the presence of peroxides with KI soln before use, and purify as described in Section 4.3.5.1.12.[1]

Reagent K

CAUTION: *Phenol is caustic. Do not allow it to come in contact with skin or mucosa.*

To a mixture of TFA (85 mL), H_2O (5 mL), thioanisole (5 mL), and EDT (2.5 mL, 30 mmol) add phenol (5 g, 53 mmol). Some peptide chemists recommend the addition of TIS (1.0 mL, 4.9 mmol).[4]

Tetrafluoroboric Acid in Trifluoroacetic Acid

CAUTION: *HBF_4 is extremely corrosive and toxic. Use only in a good fume hood; wear protective gloves.*

Add $HBF_4 \cdot OEt_2$ complex (1.38 mL, 10 mmol), *m*-cresol (0.5 mL, 4 mmol), EDT (0.4 mL, 3.5 mmol), and thioanisole (1.22 mL, 10 mmol) to TFA (10 mL). Chill to 0 °C and use cold.[18]

4.3.5.2.2 Cleavage of Simple Peptides with Trifluoroacetic Acid

Place the peptide-resin (1.0 g, 0.4 mmol) in a flask containing a stirring bar. Add TFA (15 mL) containing 2.5% H_2O and 2.5% TIS (v/v), and stir for 1 h at rt. Filter through a fritted glass funnel into a round-bottomed flask, washing the resin with three small portions of TFA. Save the resin for further treatment if full cleavage is not obtained after the first run. Concentrate the combined filtrate and washings to a low volume under vacuum in a rotary evaporator. Add the residue dropwise to stirred, ice-cold, peroxide-free MTBE or Et_2O (200 mL). Filter the precipitated peptide on a fine-porosity fritted funnel and wash with cold Et_2O (3 ×). Dry the peptide, then dissolve it in an appropriate solvent and purify by a suitable procedure. Cleavage of Arg(Pmc) will require a longer reaction time, but Arg(Mtr) is not cleaved.

4.3.5.2.3 Cleavage of Peptide-Resins with Reagent K

Add Reagent K (15 mL) to the peptide-resin (1.0 g, 0.4 mmol) in a tightly stoppered flask and mix gently for 1 h, or the time found necessary to achieve full cleavage. At the end of the reaction time, pour the mixture slowly through a fritted glass filter into stirred, ice-cold Et_2O or MTBE (200 mL) to precipitate the peptide. Wash the resin with small volumes of cold TFA (3 ×), adding it to the ether. Filter the peptide on a fine-porosity fritted glass funnel, wash with three small portions of cold Et_2O, and dry the peptide. Dissolve the peptide in an appropriate solvent (H_2O, 10% AcOH, or glacial AcOH can be lyophilized directly), and purify as is appropriate.[4]

4.3.5.2.4 Cleavage of Peptide-Resins with Tetrafluoroboric Acid/Trifluoroacetic Acid

Add the ice-cold reagent (15 mL, prepared as in Section 4.3.5.2.1) to the peptide-resin (1.0 g) and mix at 0 °C for 1 h. All blocking groups used in Fmoc synthesis should be removed during this time, including Arg(Mtr). Work up the cleavage mixture as in the previous section (Section 4.3.5.2.3).[18]

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4.3.5.2.5 Cleavage of Labile Peptide-Resins with Dilute Trifluoroacetic Acid for Blocked Peptides

Protected peptides for further "convergent" synthesis can be synthesized on hyperlabile Trt(2-Cl), HMPB, Rink, SASRIN, Sieber, or TGT resins (see Sections 4.3.1, 4.3.1.1, 4.3.2.2, and 4.3.3.2) and cleaved with dilute TFA to yield the fully blocked peptide acid.

Be sure the peptide-resin contains no DMF, which is basic and will interfere with the cleavage. Place the resin (1.0 g) in a fritted funnel that can be stoppered or in a polypropylene syringe fitted with a sintered polypropylene filter disk (see Section 4.3.2.1.1). In a filter flask, place Py/MeOH (1:9, 2 mL). Add a 1% soln of TFA in CH_2Cl_2 (10 mL) to the resin and agitate for 2 min; filter into the Py/MeOH soln and mix the filtrate to neutralize the acid. Repeat the acid treatment up to 10 times to cleave the peptide, which may not be soluble in CH_2Cl_2 and may remain in the resin. Check the filtrates to see if the peptide has eluted. If not, extract the resin with MeOH, $CHCl_3$, or the less volatile DMF or DMSO (3×25 mL). $CHCl_3$/TFE (3:1) or $CHCl_3$/HFIP (3:1) have been reported to be the most powerful solvents available for protected peptides.[19] Some peptides, such as hydrophobic leader sequences, remain insoluble. Some investigators[17] recommend inclusion of 3% anisole and 1% EDT in the reagent for use with peptides containing Cys, Met, or Trp.

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4.3.6 Automated Synthesis

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The first automated peptide synthesizer was described by Merrifield and Stewart in 1965.[1] Since the invention of this instrument, which is presently located in the Smithsonian Institute in Washington, DC, more than seventy commercial synthesizers have been produced by over twenty companies. As the field of solid-phase synthesis evolved, manufacturers have designed instruments based on the synergy between chemistry and engineering. The chemistry advances have been in new protecting groups, activation methods, and resins. The engineering improvements have been in areas that include valve design and microprocessors. The chemistry and technological improvements in conjunction have resulted in instrumentation with enhanced performance and simplified design.

All commercial peptide synthesizers are designed to utilize the *tert*-butoxycarbonyl (Boc)/benzyl (Bzl) and/or the 9-fluorenylmethoxycarbonyl (Fmoc)/*tert*-butyl (tBu) protection group strategies. Synthesizers originally planned to automate the Boc approach may be readily adapted to perform Fmoc chemistry because the latter incorporates milder reagents. However, this is not the case for instruments suitable for Fmoc-based methods. Valves and tubing on these instruments typically are not compatible with trifluoroacetic acid (TFA), a corrosive reagent that with extended use may destroy proper mechanical operation.

There are two general classes of peptide synthesizers: batch and continuous-flow. In a batch synthesizer, the resin is contained in a vessel, the solvents and reagents are delivered, the resin is mixed, and the solvents and reagents are removed by filtration. Batch synthesizers can be used for both Boc- and Fmoc-based methods. In continuous-flow instruments, the resin is packed in a column and the solvents and reagents are pumped through the column. One requirement for continuous-flow synthesizers is that the resin cannot increase its swollen volume during chain elongation. To satisfy this requirement, special resins have been developed. Only Fmoc syntheses are performed on continuous-flow synthesizers since this chemistry is compatible with the recirculating pump.

Conventional batch and continuous-flow peptide synthesizers typically produce one to three peptides at a time in quantities from 0.025 to 2 mmol. To satisfy the demand for a greater number of sequences, robotic instruments have been adapted for peptide assembly which are able to construct 8 to 144 peptides simultaneously. Production scale synthesizers have also been designed that are able to prepare up to 5 moles of a peptide.

For sequences that are difficult to construct, it is very beneficial to have methods of monitoring the coupling and/or the deprotection efficiency. The data generated from monitoring may assist in the analysis of the assembly steps. Additionally, if the instrument contains feedback control, the quality of the synthesis may be improved.

Valve design is a critical feature in all automated synthesizers. While some instrument manufacturers use commercially available rotary valves, other companies have engineered their own custom valves that have zero dead volume. Regardless of the type of valve, the plumbing and cycles must be developed to eliminate cross contamination.

All materials that come in contact with the solvents and reagents must be chemically resistant. Some of the high-performance materials used in peptide synthesizers are listed in Table 1.

Except for the early commercial automated instruments, which used punched tape or drum controllers, all automated synthesizers are microprocessor controlled. The microprocessor controls the driver, which in turn, operates the valves, motors, heaters and sensors.

The use and study of peptides are relevant to scientists in many different disciplines. As a result, automated peptide synthesizers are sometimes operated by researchers not skilled with peptide synthesis or sophisticated engineered equipment. To address this market,

for references see p 841

manufacturers have designed synthesizers that are easy to use, have well-defined cycles, and are maintained with strong training, customer support, and service departments. Alternatively, peptide chemists experienced with automated equipment require instruments with an open architecture that allows greater flexibility. These two general classes of users, as well as the requirements for different numbers and amounts of peptides, partially explain the variety in the types of commercially available peptide synthesizers.

Table 1 High Performance Materials Used in Synthesizers

Name	Description	Trademark
Teflon PTFE	polytetrafluoroethylene	Teflon is a registered trademark of DuPont
Teflon FEP	fluorinated ethylene polypropylene	Teflon is a registered trademark of DuPont
Kel-F	chlorotrifluoroethylene	Kel-F is a registered trademark of 3M
PPS	polyphenylene sulfide	no trademark
Kalrez	perfluoroelastomer	Kalrez is a registered trademark of DuPont
Viton	fluoroelastomer	Viton is a registered trademark of DuPont
Zitex	microporous PTFE	Zitex is a registered trademark of Norton Performance Plastics

4.3.6.1 Batch Synthesis

In batch SPPS (solid-phase peptide synthesis), the resin is contained in a vessel, usually equipped for bottom filtration. After a washing solvent or a reagent is added, the resin is mixed and the solvent or reagent is removed by filtration. Batch synthesizers use a variety of methods to mix the resin: the wrist-action shaker, vortex mixing, nitrogen bubbling, or overhead stirring.

Most of the resins used in batch synthesis are derivatives of 1% crosslinked copoly-(styrene-divinylbenzene) resin in the form of beads 50 to 100 μ in diameter. These beads swell four to six times their dry volume in the solvents used in peptide assembly and during the chain elongation process the resin again may double its swollen volume. In batch synthesis, provided the proper size vessel is chosen, the increase in swollen volume is not a major concern.

Table 2 lists most of the commercial automated batch synthesizers. The name given for the manufacturer is the name of the company at the time the instrument was first introduced. The amount of resin the synthesizer can handle is given in grams or mmol, depending on the information found in the literature for that product.

4.3.6.1.1 Beckman 990

The Beckman Model 990 Synthesizer, introduced in 1972, was modeled after the synthesizers built by Merrifield, Stewart, and Jernberg.[2–4] The standard 200-mL, cone-shaped, Teflon reaction vessel could accommodate up to 10 g of resin. An optional 500-mL vessel was also available. Mixing was accomplished by a variable-speed overhead stirrer and there were twelve amino acid reservoirs.

The instrument was operated by a punched paper tape (a 42-channel mylar-backed tape). The punched holes specified a particular operation, such as the delivery of a solvent or a reagent into a metered vessel, transfer of that liquid to the reaction vessel, mixing, filtration, etc. The operator could program their own cycles, since an instrument to punch the tape was also provided. In 1984, Beckman introduced the System 990 Synthesizer, which was similar to the Model 990, except it was controlled by a microprocessor.

Table 2 Batch Synthesizers

Date	Company	Instrument	Boc/Fmoc	Mixing	Resin[a]	Other
1971	Vega Engineering Company	Series 70	Boc	180° shaker or ultrasonic agitation	2–20 g	8 Xaa reservoirs, drum controller
1972	Beckman	Model 990	Boc	overhead stirring	0.1–20 g	12 Xaa reservoirs, tape controller
1975	Vega Biochemicals	Model 96	Boc	180° shaker	0.5–50 g	20 Xaa reservoirs, drum controller
1976	Vega Biochemicals	Model 50	Boc	free-standing shaker	0.5–5.0 g	semiautomatic
1983	Vega Biotechnologies	Coupler 250	Boc/Fmoc	180° shaker	0.5–25 g	20 Xaa reservoirs, microprocessor
	Vega Biotechnologies	Coupler 1000	Boc/Fmoc	180° shaker	0.5–7.0 g	semiautomatic
	Biosearch	SAM TWO	Boc/Fmoc	gas bubbling	1–10 g	12 Xaa reservoirs
	LABORTEC	SP640 F	Fmoc	60° shaker	0.5–10 g	semiautomatic
1984	Beckman	System 990	Boc/Fmoc	overhead stirring	0.1–20 g	microprocessor
	Applied Biosystems	ABI 430A	Boc/Fmoc[b]	vortex	0.10 and 0.50 mmol	PSA in DMF, Xaa cartridges
	Vega Biotechnologies	Coupler 2000	Boc/Fmoc	180° shaker	1–12 g	Xaa cartridges
	Peptides International	Syn-Thor2000	Boc/Fmoc	wrist action shaker	0.5–50 g	6 Xaa reservoirs
	Peptides International	Syn-Thor2000 AT	Boc/Fmoc	wrist action shaker or gas bubbling	0.5–50 g	12 Xaa reservoirs
	Applied Protein Technologies	PSS-80	Boc	shaker	0.1–20 g	15 refrigerated Xaa reservoirs
1985	Biosearch/New Brunswick	9500	Boc/Fmoc	gas bubbling	0.1–2.0 g	12 Xaa reservoirs, 2–20 g optional
	LABORTEC	SP640 B	Boc/Fmoc	60° shaker	0.5–10 g	semiautomatic
1986	Biosearch/New Brunswick	9600	Boc/Fmoc	gas bubbling	0.1–1.0 g	20 Xaa reservoirs
	Advanced ChemTech	Model 100	Boc/Fmoc	oscillation, gas bubbling, or both	0.5–40 g	no preactivation, 10 Xaa reservoirs
	Advanced ChemTech	Model 200	Boc/Fmoc	oscillation, gas bubbling, or both	0.5–40 g	preactivation, 10 Xaa reservoirs
	DuPont	Coupler 2100	Boc/Fmoc	180° shaker	0.50 and 2.0 mmol	Xaa cartridges
1987	Biotech Instruments	BT 7500	Boc/Fmoc	180° shaker and gas bubbling	2–10 g	semiautomatic
1988	Applied Biosystems	ABI 431A	Boc/Fmoc	vortex	0.10–0.50 mmol	preactivation
	Neosystems	NPS 4000	Boc/Fmoc	gas bubbling	0.50–1.0 mmol	semiautomatic
	Neosystems	NPS 4000A	Boc/Fmoc	gas bubbling	0.025–0.25 mmol	automatic
	Kem-En-Tec	MKIII	Fmoc	stirring	up to 1 g	conductivity monitoring
1989	Biotronik	SYNOSTAT P	Boc/Fmoc	adjustable vortex	0.05–0.50 mmol	46 Xaa positions
	Biotech Instruments	BT 7600	Boc/Fmoc	180° shaker and gas bubbling	0.02–1.0 mmol	8 Xaa positions

for references see p 841

Table 2 (cont.)

Date	Company	Instrument	Boc/Fmoc	Mixing	Resin[a]	Other
1990	Advanced ChemTech	Model 90	Boc/Fmoc	oscillation, gas bubbling, or both	0.10–12 mmol	2 reactors, 4 Xaa reservoirs
1991	Protein Technologies	PS3	Fmoc optional Boc	gas bubbling	0.10–0.50 mmol	3 sequential reactions
1992	Eppendorf/Biotronik	ECOSYN P	Boc/Fmoc	adjustable vortex	0.1–10 mmol	46 Xaa positions
	C S Bio	CS036	Boc/Fmoc	180° shaker	0.2–5 g	4 Xaa reservoirs
	C S Bio	CS136	Boc/Fmoc	180° shaker	0.2–5 g	12 Xaa reservoirs
	C S Bio	CS536	Boc/Fmoc	180° shaker	0.2–50 g	12 Xaa reservoirs
1993	PE Applied Biosystems	ABI 433A	Boc/Fmoc	vortex	0.1–1.0 mmol	feedback monitoring
	LABORTEC	SP4000-LAB	Boc/Fmoc	60° shaker	0.5–10 g	semiautomatic
1995	Rainin/Protein Technologies	SONATA/Pilot	Boc/Fmoc	gas bubbling, recirculation, and oscillation	up to 100 g	20 Xaa reservoirs
1999	C S Bio	CS736 System 3	Boc/Fmoc	gas bubbling and shaking	0.05–0.50 mmol	3 reactions concurrently

[a] The amount of starting resin is given in either mmol or g, depending on the information provided by the manufacturer. [b] Fmoc protocol released in 1987.

The users of the Model 990 were experienced peptide chemists, who desired to automate their synthetic methods. Most of these techniques involved the Boc/Bzl group strategies and some of the activation/coupling methods employed were in situ *N,N'*-dicyclohexylcarbodiimide (DCC) coupling in CH_2Cl_2, preformed symmetrical anhydride (PSA) coupling in CH_2Cl_2 or DMF/CH_2Cl_2 and preformed 1-hydroxybenzotriazole (HOBt) active ester coupling in DMF.[4]

Boc Synthesis, DCC Activation:[5]
Boc amino acid resin (1.00 g, 0.43 mmol) was treated by the following schedule for the introduction of each residue: (1) CH_2Cl_2 (2 × 35 mL); (2) 50% TFA/CH_2Cl_2 (35 mL) for 1 min; (3) 50% TFA/CH_2Cl_2 (35 mL) for 10 min; (4) CH_2Cl_2 (3 × 35 mL); (5) 50% EtOH/$CHCl_3$ (3 × 35 mL); (6) $CHCl_3$ (3 × 35 mL); (7) 3% DIPEA in $CHCl_3$ (35 mL) for 5 min; (8) $CHCl_3$ (3 × 35 mL); (9) CH_2Cl_2 (3 × 35 mL); (10) the appropriate Boc amino acid (1.72 mmol) in CH_2Cl_2 (15 mL) for 10 min; (11) DCC (0.352 g, 1.71 mmol) in CH_2Cl_2 (4.10 mL) for 2 h; (12) CH_2Cl_2 (2 × 35 mL); (13) EtOH (2 × 35 mL); (14) CH_2Cl_2 (2 × 35 mL).

4.3.6.1.2 Applied Biosystems 430A, 431A, and 433A

The Model 430A was designed by Applied Biosystems and their consultants to automate highly optimized Boc cycles.[6–8] These improved cycles use DMF as the solvent during the neutralization and coupling steps in order to minimize sequence-dependent coupling difficulties.[6,9] The coupling is with preformed symmetrical anhydrides in DMF. Preloaded phenylacetamidomethyl (Phacm) resins are used as the solid support, since they are more stable to the conditions of chain assembly than the original Merrifield resins.[10,11] Additionally, the solid support is prepared from N^{α}-Boc amino acid 4-(carboxymethyl)benzyl ester and aminomethyl resin[12] to give a resin which will minimize or eliminate chronic side reactions, such as trifluoroacetylation[13] or Schiff base formation.[14]

The standard 40-mL reaction vessel for the 430A is constructed of Kel-F, with top and bottom Zitex filters attached with caps made of Teflon. Mixing of the resin is accomplished by rigorous vortexing at 1500 to 2000 rpm in a minimal amount of solvent. The amino acids are stored as solids in 10-mL, serum-capped polypropylene cartridges and arranged on a linear guideway according to the peptide sequence. In addition to the reaction vessel, the 430A has an activation vessel for forming the Boc amino acid symmetrical anhydride (SA) in CH_2Cl_2 and a concentration vessel to evaporate the CH_2Cl_2 followed by replacement with DMF. The 430A peptide synthesizer has either one or three reaction vessels. The three reaction vessels allow an automated start to a new synthesis after the previous synthesis is completed. The synthesizer has six zero-dead-volume valve blocks based on the Wittmann–Liebold design.[15] The solvents and reagents are delivered under 414 – 517 Torr of nitrogen pressure.

The 430A was designed as a Boc synthesizer and was engineered to accommodate TFA. All the tubing in the 430A is FEP Teflon except for the tubing to the TFA bottle. This tubing is polyethylene since FEP Teflon tubing is porous to TFA vapors. Also, the TFA bottle is only under pressure during the TFA delivery. After the delivery is completed, the bottle is vented and the TFA in the polyethylene tube is transferred back into the designated reagent bottle in order to keep the contact of the corrosive TFA with the valve block to a minimum.

In 1984, when the 430A was introduced, the only scale was 0.50 mmol which used 2 mmol Boc amino acid cartridges. In 1987, a 7-mL reaction vessel was created for a 0.10 mmol synthesis using 1 mmol of Boc amino acid. These cycles were called the small-scale rapid cycles[16] since the time was 20 minutes compared to 45 minutes for the standard cycles. Also in 1987, Fmoc cycles using HOBt active esters were released.[17] In 1992, Boc cycles with 2-(1*H*-benzotriazol-1-yl)-1,1,3,3-tetramethyluronium hexafluorophosphate (HBTU) activation for the 430A which did not have a neutralization step after the TFA deprotection, was reported.[18,19] Neutralization of the TFA-salt of the resin-bound amino

for references see p 841

group was accomplished with the excess DIPEA employed in the HBTU activation in a process referred to as in situ neutralization.

In 1988, Applied Biosystems introduced the 431A peptide synthesizer. The 431A incorporated many of the same components as the 430A but is a simpler instrument with half the number of valve blocks and no concentrator vessel. The 431A performs Boc and Fmoc cycles with HOBt/DCC activation using 1-methylpyrrolidin-2-one (NMP) as the solvent.[20,21] NMP is an excellent solvent for swelling the peptide-resin and, in the case of HOBt activation, the ester can be formed with DCC without the addition of CH_2Cl_2. In 1991, Fmoc cycles with HBTU activation, called FastMoc, for 0.10, 0.25, and, 1.00 mmol scale syntheses were released.[22] Additionally, cycles with NMP as the solvent were developed for the 430A.

In 1993, PE Applied Biosystems released the 433A, which is essentially a 431A with an external computer for storage of cycles, peptide sequences, and completed synthesis data. The 433A also has the capability of measuring the Fmoc deprotections by conductivity monitoring, which is discussed later in this section.

Boc Synthesis, PSA Activation, DMF Solvent:[23]
Boc amino acid Phacm resin (0.77 g, 0.50 mmol) was treated by the following schedule for the introduction of each residue: (1) 65% TFA/CH_2Cl_2 (6 mL) for 1 min; (2) 65% TFA/CH_2Cl_2 (6 mL) for 13 min; (3) CH_2Cl_2 (3 × 6 mL); (4) 10% DIPEA in DMF (2 × 10 mL) each for 1 min; (5) DMF (5 × 10 mL); (6) the appropriate Boc amino acid PSA in DMF for 16 min; (7) CH_2Cl_2 (5 × 9 mL).
The Boc amino acid PSA in DMF was prepared on the instrument for all amino acids except Arg, Asn, and Gln, which were activated as HOBt esters. The Boc amino acid (2 mmol) was dissolved in CH_2Cl_2 (3.0 mL) and, if necessary, DMF (0.3 mL), transferred from the amino acid cartridge to the activator where 0.5 M DCC in CH_2Cl_2 (2 mL, 1 mmol) was added. After 5 min, the activated amino acid was transferred to the concentrator vessel where CH_2Cl_2 was evaporated by purging with a continuous stream of N_2. The CH_2Cl_2 was replaced by DMF (6 mL) at various stages during the purging.

Boc Synthesis, HOBt/DCC Activation, NMP Solvent:[24]
Boc amino acid Phacm resin (0.77 g, 0.50 mmol) was treated by the following schedule for the introduction of each residue: (1) 25% TFA/CH_2Cl_2 (11 mL) for 3 min; (2) 50% TFA/CH_2Cl_2 (8 mL) for 16 min; (3) CH_2Cl_2 (5 × 6 mL); (4) 5% DIPEA/35% NMP/60% CH_2Cl_2 (13.6 mL) for 1 min; (5) 5% DIPEA/NMP for 1 min; (6) NMP (5 × 6.5 mL); (7) the appropriate preactivated Boc amino acid HOBt ester for 39 min; (8) DMSO (3.3 mL) added to the coupling soln for 16 min; (9) DIPEA (0.4 mL) added to the coupling soln for 5 min; (10) capped with 10% Ac_2O/5% DIPEA/NMP (13 mL) for 9 min; (11) CH_2Cl_2 (6 × 6 mL).
The preformed Boc amino acid HOBt ester in NMP was prepared on the instrument by dissolving the Boc amino acid (2 mmol) in NMP (3.4 mL) and 1 M HOBt in NMP (2.0 mL, 2 mmol), this soln was transferred from the amino acid cartridge to the activator vessel where 1 M DCC in NMP (2.0 mL, 2 mmol) was added. After 50 min, the soln was transferred to the reaction vessel.

Fmoc Synthesis, HOBt/DCC Activation, NMP Solvent:[25]
Fmoc amino acid resin (0.38 g, 0.25 mmol) was treated by the following schedule for the introduction of each residue: (1) NMP (1 × 7.5 mL); (2) 20% piperidine/NMP (9 mL) for 3 min; (3) 20% piperidine/NMP (10 mL) for 15 min; (4) NMP (6 × 6.5 mL); (5) the appropriate preactivated Fmoc amino acid HOBt ester for 71 min; (8) NMP (7 × 7 mL).
The preformed Fmoc amino acid HOBt ester in NMP was prepared on the instrument by dissolving the Fmoc amino acid (1 mmol) in NMP (2.1 mL) and 1 M HOBt in NMP (1.0 mL, 1 mmol), this soln was transferred from the amino acid cartridge to the activator vessel where 1 M DCC in NMP (1.0 mL, 1 mmol) was added. After 58 min, the soln was transferred to the reaction vessel.

Fmoc Synthesis, HBTU Activation, NMP Solvent:[26]
Fmoc amino acid resin (0.38 g, 0.25 mmol) was treated by the following schedule for the introduction of each residue: (1) NMP (1 × 6.5 mL); (2) 20% piperidine/NMP (9 mL) for 3 min; (3) 20% piperidine/NMP (10 mL) for 8 min; (4) NMP (5 × 6.5 mL); (5) the appropriate Fmoc amino acid (1 mmol) in NMP (2.0 mL), 0.45 M HBTU/0.45 M HOBt in DMF (2.1 mL, 0.95 mmol), and 2.0 M DIPEA in NMP (1.0 mL, 2.0 mmol) for 21 min; (6) capped with 0.5 M Ac_2O/0.125 M DIPEA/0.015 M HOBt in NMP (12 mL) for 5 min; (7) NMP (3 × 6 mL).

4.3.6.1.3 Advanced ChemTech Model 90

The ACT Model 90 is a dual reactor instrument capable of synthesizing two different peptides concurrently at scales from 0.1 to 3 mmol or one peptide up to 12 mmol in the single reactor configuration. The synthesizer is capable of either Boc or Fmoc syntheses using a variety of activation methods, including DCC, *N,N'*-diisopropylcarbodiimide (DIC), 2-(1*H*-benzotriazol-1-yl)-1,1,3,3-tetramethyluronium tetrafluoroborate (TBTU), benzotriazol-1-yloxytripyrrolidinophosphonium hexafluorophosphate (PyBOP), HBTU, pentafluorophenyl (Pfp) ester, and PSA or HOBt esters. There are four 90-mL amino acid vessels, three 4-L reagent bottles, four 1-L bottles, and an isolated TFA system. The standard reactors are 25 mL and optional reactors are available in sizes of 50, 100, 200, 300, and 500 mL. The reactors above 100 mL are used in the single reactor configuration. Mixing is accomplished by wrist-action shaking and/or nitrogen bubbling.

4.3.6.1.4 Rainin/Protein Technologies SONATA/Pilot

The SONATA/Pilot Synthesizer from Rainin/Protein Technologies is a batch instrument capable of synthesizing from 0.10 to 50 mmol of peptide by either the Fmoc or Boc method. Three reaction vessels are available in sizes of 75, 270, and 540 mL. The largest vessel can accommodate 100 g of resin. Mixing is performed by adjustable gas bubbling, recirculation, and variable speed oscillation. The primary solvent is delivered by a pump at 400 mL · min^{-1} from a container that can be as large as 200 L. There are six reagents in 0.5- to 20-L containers which are transferred by nitrogen or argon pressure. The twenty amino acid solutions are contained in 0.25-L bottles and these solutions are also delivered by gas pressure. Two measurement vessels are available for the transfer of precise amounts of reagents. The waste is collected in two separate 20-L containers.

Several options are available for nonroutine synthesis, such as heating and cooling the reaction vessel as well as feedback control using additional monitoring equipment. It is also possible to order custom sized reaction vessels, an optional mechanical stirrer, and a 200-L waste container.

4.3.6.2 Continuous-Flow Synthesis

The technique of continuous-flow peptide synthesis evolved from the Fmoc-polyamide approach to peptide synthesis championed by Sheppard and Atherton at the Medical Research Council Laboratory of Molecular Biology, Cambridge.[27] The Fmoc-polyamide method incorporates a polar polydimethylacrylamide support, a polar aprotic solvent, typically DMF, and the base-labile Fmoc amino acids. This technique became the continuous-flow Fmoc-polyamide method when the support was packed in a column, the washing was performed with a continuous flow of DMF, and the mixing during the coupling was obtained by recirculating the activated Fmoc amino acid solution (for Fmoc/tBu strategy, see Section 4.3.2.2). An additional feature of this method is the ability to continuously monitor the acylating and deprotection reaction by ultraviolet absorption, due to the UV absorption of the Fmoc group on the amino acid and the cleaved fulvene-piperidine adduct, respectively.[28]

The original resin applied in continuous-flow Fmoc synthesis was the polydimethylacrylamide–kieselguhr support, commercially known as Pepsyn K.[28–30] Another popular resin for continuous-flow synthesis is polystyrene resin grafted with poly(ethylene glycol) (PEG) which is commercially available as TentaGel[31] and as PEG-PS.[32] Other resins that have been demonstrated in continuous-flow synthesis include Polyhipe, a rigid and macroporous polystyrene support which has incorporated polyamide gel,[33] PEGA, a poly-(ethylene glycol)-polyacrylamide resin,[34] and CLEAR, a cross-linked ethoxylate acrylate resin.[35]

for references see p 841

The first commercially available continuous-flow synthesizers were manual instruments: the PEPSYNthesiser I from Cambridge Research Biochemicals (CRB) and the Model 4175 from LKB Biochrom. In these instruments, one manual valve switches between the flow and recirculate modes and another manual valve selects the solvent, reagent or the amino acid injection port. CRB later released the semiautomatic PEPSYNthesiser II before selling the rights to the instrument to MilliGen, who ultimately released the fully automated 9050 PepSynthesizer. Pharmacia-LKB Biochrom also developed an automated continuous-flow synthesizer, the Biolynx Model 4170. Several years later, the rights to this instrument were obtained by Novabiochem. Most of the automated, semiautomated, and manual continuous-flow peptide synthesizers are listed in Table 3.

Table 3 Continuous-Flow Synthesizers

Date	Company	Instrument	Boc/ Fmoc	Scale (mmol)	Comments
1983	Cambridge Research Biochemicals	PEPSYNthesiser I	Fmoc	up to 0.5	manual, 2 columns
	LKB Biochrom	Model 4175	Fmoc	up to 0.5	manual, 2 columns
1985	Cambridge Research Biochemicals	PEPSYNthesiser II	Fmoc	up to 0.5	semiautomatic, 3 columns
1987	MilliGen	9020	Fmoc	up to 0.5	semiautomatic, 3 columns
	MilliGen	9050 PepSynthesizer	Fmoc	0.05–5.0	automatic, 3 columns
	Pharmacia-LKB Biochrom	Biolynx Model 4170	Fmoc	0.25	automatic, 3 columns
1988	MilliGen/Biosearch	EXCELL	Fmoc	0.10	automatic, 1 column
1991	Novabiochem	NovaSyn Gem	Fmoc	0.05–2.0	semiautomatic, 2 columns
	Novabiochem	NovaSyn Crystal	Fmoc	0.05–0.70	automatic, 3 columns
1992	MilliGen/Biosearch	9050 Plus PepSynthesizer	Fmoc	0.05–5.0	automatic, 3 columns
	Applied Biosystems	Synergy (ABI 432)	Fmoc	0.025	automatic, 1 column
1994	ABIMED Analysen-Technik	EPS 221	Fmoc	0.025, 0.050, and 0.100	automatic, 1–3 peptides sequentially
1996	PerSeptive Biosystems	Pioneer	Fmoc	0.020–2.0	automatic, 2 columns

4.3.6.2.1 MilliGen 9050 PepSynthesizer

The Milligen 9050 PepSynthesizer was the first commercially available completely automatic continuous-flow instrument. The open architecture design provided the flexibility to integrate new chemistries in the synthesis of more complex peptides. Following improvements in software, the 9050 Plus PepSynthesizer allowed automated synthesis of cyclic and branched chain peptides as well as multiple antigenic peptides implementing only Fmoc-based methods. The instrument can assemble three peptides simultaneously or sequentially. FastPep protocols allow 16-minute cycle times. The most significant feature to the 9050 Plus PepSynthesizer instrument is the ability to perform multiple activation chemistry within a single synthesis. Thus, sequences can be produced both economically and in high purity.

The synthesis cabinet contains the electronics, pneumatics, and valves for performing automation. Main components are fluidic valves for controlling solvent or reagent delivery, a recirculating pump for solvent delivery, a UV/visible detector for monitoring pre- and post-column, and three syringe pumps for delivering reagents to the amino acid module and delivering activated amino acid to the three columns.

The system incorporates seven solvent/reagent reservoirs including bottles for main wash (DMF), deblock [typically 20% piperidine in DMF but other solutions such as 2% 1,8-

diazabicyclo[5.4.0]undec-7-ene (DBU)/2% piperidine in DMF can be applied], and additional wash (CH_2Cl_2 or MeOH), an auxiliary bottle (capping, solvent exchange, etc.), and three bottles for the syringes.

The amino acid module performs the dissolution, activation, and delivery of amino acids to the synthesis cabinet. This is accomplished via a robotic arm with a probe attachment. Liquids can be dispensed or withdrawn from amino acid vials through the probe. In addition, nitrogen can be bubbled to aid in the dissolution of the amino acid. After each amino acid coupling the probe is thoroughly washed, both internally and externally, to eliminate cross contamination. The amino acid module can hold up to 72 amino acids in three 24-position racks. A rack can be assigned to each column or all 72 can be used for single column synthesis.

Fmoc Synthesis, PyBOP Activation:[36]
The flow rate of the pump was set at 5.0 mL·min^{-1} and the following synthetic protocol was used with Fmoc PAL-PEG-PS (0.18 mmol·g^{-1}) for the introduction of each residue: (1) 20% piperidine in DMF for 7 min; (2) DMF flow wash for 12 min; (3) the appropriate Fmoc amino acid (4 equiv), HOBt (4 equiv), and PyBOP (4 equiv) were dissolved to a final concentration of 0.3 M with a soln of 0.6 M DIPEA (8 equiv) in DMF and recirculated for 30 min; (4) DMF flow wash for 8 min.

4.3.6.2.2 PerSeptive Biosystems Pioneer

The Pioneer Peptide Synthesis System is a two-column, Fmoc continuous-flow instrument capable of synthesizing two peptides simultaneously and independently at scales from 0.02 to 2.0 mmol.[37] Fluid flow is with a positive displacement pump capable of delivering up to 50 mL·min^{-1}. The dimensions of the standard column are 16 mm × 100 mm with a volume of 14 mL. A larger column is available with dimensions of 27 mm × 150 mm (70 mL). An adjustable end piece enables the volume of these columns to be changed for different amounts of resin. The Fmoc amino acids are stored as dry powders in vials positioned according to the peptide sequence. A probe delivers the activator solution to the vial followed by bubbling with nitrogen to dissolve the Fmoc amino acid. The pump transfers the amino acid solution through the column for the recirculation/coupling step. There are 101 amino acid positions in the transport system. The solvents and reagents are under 310 Torr of nitrogen pressure.

The standard activation method on the Pioneer is with *N*-[(dimethylamino)(3-oxido-1*H*-1,2,3-triazolo[4,5-*b*]pyridin-1-yl)methylene]-*N*-methylmethanaminium hexafluorophosphate (HATU). The concentration of activated amino acid depends on the scale of synthesis, ranging from 0.06 M (0.02-mmol scale) to 0.13 M (0.1-mmol scale) to 0.25 M (>0.2-mmol scale). The recommended resin for the Pioneer is PEG-PS. The volume of the column depends on the void volume of the solid support (the void volume for PEG-PS is ca. 4.4 mL·g^{-1}).

Fmoc Synthesis, HATU Coupling:[38]
The Fmoc amino acid PEG-PS resin (1.11 g, 0.20 mmol) was treated by the following schedule for the introduction of each residue: (1) 20% piperidine/DMF (49 mL) for a total of 5 min; (2) DMF flow wash (49 mL); (3) the appropriate Fmoc amino acid (0.80 mmol), 0.50 M HATU in DMF (1.6 mL, 0.80 mmol) and 1.0 M DIPEA in DMF (1.6 mL, 1.6 mmol) for 30 min; (4) DMF flow wash (39 mL). Steps 1 and 3 were extended if UV monitoring showed a difficult deprotection. The flow rate during the DMF washes and during the coupling was 30 mL·min^{-1}.

4.3.6.3 Multiple Synthesis

In the mid 1980s, several methods were reported to simultaneously assemble a large number of peptides, such as T-bags,[39] pins,[40] and RaMPS cartridges.[41] Unfortunately, these novel techniques were not yet developed for automation. In 1989, Schnorrenberg at Boehringer Ingelheim created the first fully automated multiple peptide synthesizer using a Tecan robotic processor.[42,43] Later that year, this simultaneous multiple peptide synthesizer (SMPS)

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was commercially available as the SMPS 350 by Zinsser Analytic[44] and as the Model 350 by Advanced ChemTech.[45] In 1990, researchers at the European Molecular Biology Laboratory developed an automated multiple synthesizer based on a Gilson autosampler.[46] This instrument was later commercialized by ABIMED Analysen-Technik and Gilson as the AMS 422 Multiple Peptide Synthesizer. In 1991, Shimadzu released a compact multiple peptide synthesizer, the PSSM-8.[47] In 1992, Protein Technologies and Rainin introduced the SYMPHONY/Multiplex, a nonrobotic multiple peptide synthesizer. In less than four years, multiple peptide synthesizers had radically changed the market for peptide instrumentation by enabling the researchers to inexpensively produce small quantities and large numbers of sequences by automated methods. A list of these synthesizers appear in Table 4.

Table 4 Multiple Peptide Synthesizers

Date	Company	Instrument	Boc/Fmoc	Number of peptides	Scale (μmol)
1989	Advanced ChemTech	Model 350 MPS	Fmoc	96	5–50
	Zinsser Analytic	SMPS 350	Fmoc	90/144	5–25
1990	ABIMED Analysen-Technik	AMS 422	Fmoc	48	5–50
1991	Gilson	AMS 422	Fmoc	48	5–50
	Shimadzu	PSSM-8	Fmoc	8	5–400
1992	Advanced ChemTech	Model 396 MBS	Boc/Fmoc	8/16/40/96	5–1000
	Rainin/Protein Technologies	SYMPHONY/Multiplex	Boc/Fmoc	12	5–400
1993	MultiSynTech	SyRo I	Fmoc	48	5–100
	ABIMED Analysen-Technik	ASP 222 (Auto-Spot Robot)	Fmoc	10000 on membrane	nmoles
	Advanced ChemTech	Model 357 FBS	Boc/Fmoc	36 with split-pool capabilities	5–250
1994	Advanced ChemTech	Model 348 MPS	Fmoc	8/16/40/96	5–1000
	Tyler Research	SAMPS (semiautomatic)	Fmoc	20/100	10–200
1997	PerSeptive Biosystems	Pioneer with MPS	Fmoc	16/32	25–50
1998	Zinsser Analytic	PepSy	Fmoc	32/96	5–50
1999	C S Bio	CS736 System 10	Boc/Fmoc	10	50–500

4.3.6.3.1 Zinsser Analytic SMPS 350 and PepSy

The SMPS 350 Synthesizer by Zinsser Analytic implements the Fmoc strategy to construct 144 and 90 sequences at the 5 and 25 μmol scale, respectively. The instrument is built with a Tecan RSP 5052 robotic sample processor with two independent robotic arms and 1-mL syringe plungers. One robotic arm performs the washing steps and the other arm adds the reagents and amino acid solutions. The arm for washing contains a double tip to differentiate aspirating and washing. The resin is filtered by aspirating the liquid through the washing tip, creating a vacuum with the syringe plunger. A stainless steel net on the aspirating tip prevents the loss of resin and the washing tip is positioned to wash any polymeric support. The double tip may also be cleaned in a wash station. There are up to 32 amino acid reservoirs with a capacity of 45 mL and five reagent bottles with a capacity of 400 mL to contain the coupling and deprotection reagents. There are two sizes of small polypropylene reaction vessels to accommodate 10 to 15 mg and 15 to 80 mg of resin. These reactors are open to the atmosphere and restrained by racks in a 3×48 configuration and a 3×30 configuration, respectively. Symmetrical anhydride, DIC, DIC/HOBt, and TBTU activation methods have been used on the synthesizer.

Zinsser later released two additional versions of the SMPS 350. The 350V is supplied with a vacuum block that holds 48 disposable plastic reactors. These reactors can hold up to 150 mg of resin (75 μmol). The 350X is an economic single arm robot which uses the 48

reactor vacuum block. Additionally, Zinsser released an upgrade kit for fully automated Pepscan synthesis on polyethylene pins.

In 1998, Zinsser introduced the PepSy, their next generation multiple peptide synthesizer, which is capable of synthesizing 32 or 96 peptides in reaction vessels of 3.5 and 1.8 mL, respectively. The reaction block can be vortexed for mixing of the resin and the reagents and solvents are distributed with four specially designed probes. The filtration of the resin is from the top.

Fmoc Synthesis, HOBt/DIC Activation:[43]
The Fmoc amino acid resin (10 mg, 5 μmol) was treated by the following schedule for the introduction of each residue: (1) DMF (300 μL); (2) 50% piperidine/DMF (300 μL) for 1 min; (3) 50% piperidine/DMF (300 μL) for 15 min; (4) DMF (10 × 300 μL); (5) the appropriate 0.25 M Fmoc amino acid/0.25 M HOBt in DMF (200 μL, 50 μmol); (6) 0.55 M DIC in DMF (100 μL, 55 μmol) for 40 min; (7) DMF (300 μL); (8) repeat steps 5 and 6; (9) DMF (10 × 300 μL).

4.3.6.3.2 Advanced ChemTech Models 350, 396, 357, and 348

In 1989, Advanced ChemTech (ACT) introduced the Model 350 Simultaneous Multiple Peptide Synthesizer capable of assembling up to 96 peptides at scales from 5 to 50 μmol using the Fmoc method. Two different sized reaction vessels are available: 600 μL and 1.5 mL. The first Model 350 was originally designed similar to the SMPS 350 by Zinsser, however during the next several years, ACT added additional features, such as an orbital shaker for resin agitation to differentiate the product.[48]

In 1992, ACT replaced the Model 350 with the Model 396 Multiple Biomolecular Synthesizer. This instrument possesses two robotic arms, a variable speed orbital mixer, nitrogen assisted bottom filtration, a built-in ventilation system, protocols for both Fmoc and Boc synthesis and fully automated on-board cleavage for the Fmoc syntheses. It has reactor blocks of 8, 16, 40, and 96 wells for the synthesis of peptides from 5 μmol to 1 mmol. The Model 396 MBS also has a heater/cooler option for applications in solid-phase organic synthesis.

ACT introduced the 357 Flexible Biomolecular Synthesizer in 1993.[49] The Model 357 FBS is also a two arm robotic system, with a reaction block that incorporates thirty six 9.0-mL reaction wells and a single 600-mL well. The larger vessel permits the split and mix technique for the generation of peptide combinatorial libraries.[50,51] The scale range is 5 to 250 μmol and 9 mmol for the 9.0-mL and 600-mL reactor, respectively. The Model 357 FBS also has a variable speed orbital mixer, nitrogen assisted bottom filtration and a built-in ventilation system.

ACT produces the Model 348 Multiple Peptide Synthesizer for researchers who do not require a sophisticated instrument for multiple peptide synthesis. This synthesizer is similar to the Model 396 MBS except it possesses a single robotic arm.

4.3.6.3.3 ABIMED AMS 422 and Gilson AMS 422

The AMS 422 from ABIMED and Gilson is an Fmoc-based robotic system for the multiple synthesis of up to 48 peptides at a scale of 5 to 50 μmol on standard resins or polypropylene pins.[46,52] The instrument is composed of a Gilson Model 222 Autosampler, a Gilson Model 401 micro-dispenser, a multi-needle assembly for washing six vessels at a time and a single dispense needle for adding the amino acids and reagents. The resin is contained in individual 2-mL open reactors with fritted filters at the bottom. These removable reaction columns are fastened in a reaction module in a 6 × 8 array. The reaction module has a valve manifold system that controls the filtration of reagents. To accomplish filtering, vacuum aspiration is applied to the space below the supporting frit filters. The solvent bottles are pressurized with nitrogen and the solvent and piperidine/DMF solution are dispensed to six columns at a time

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in a flow-through mode. There is no mixing on the AMS 422, except upon addition of CH_2Cl_2, which occurs at the conclusion of the coupling time and causes the beads to rise and distribute evenly in the reaction mixture. The AMS 422 employs in situ activation of Fmoc amino acids by PyBOP with *N*-methylmorpholine (NMM) as the base and DMF as the solvent.[53] ABIMED also developed a separate system, the APC 522, for the automated cleavage of the peptidyl resins and isolation of the peptides.[54]

Fmoc Synthesis, BOP Activation:[52]
Fmoc amino acid resin (50 mg, 25 μmol) was treated by the following schedule for the introduction of each residue: (1) 20% piperidine/DMF (2 × 1 mL) for 15 min; (2) DMF (6 × 1 mL); (4) the appropriate 0.67 M Fmoc amino acid in DMF (150 μL, 100 μmol), 0.67 M BOP in DMF (150 μL, 100 μmol) and 4.0 M NMM in DMF (50 μL, 200 μmol) for 30 min; (5) CH_2Cl_2 (300 μL); (6) DMF (6 × 1 mL).

4.3.6.3.4 Shimadzu PSSM-8

The PSSM-8 by Shimadzu may construct eight sequences by the Fmoc strategy at scales from 5 to 400 μmol.[47,55,56] The reaction vessels are 2.5 mL disposable polypropylene syringes with a polypropylene filter at the bottom. There are 160 disposable amino acid vials to contain up to 2 mL of predissolved amino acids. Each amino acid vial is used only one time during a synthesis. All amino acid solutions, solvents, and reagents are delivered by eight independent syringes to the eight vessels. Each reaction vessel has its own activation and capping reagents, allowing a different protocol for each sequence. The deprotection solution and wash solvent is in an external bottle. Agitation is by nitrogen bubbling. Some of the coupling methods that have been used with the PSSM-8 are: DIC, DIC/HOBt, TBTU, HBTU, OPfp esters, and PyBOP/HOBt.

4.3.6.3.5 Rainin/Protein Technologies SYMPHONY/Multiplex

The SYMPHONY/Multiplex from Rainin/Protein Technologies is a multiple batch peptide synthesizer capable of assembling twelve peptides at scales from 5 to 400 μmol. The SYMPHONY has a multiplexing matrix valve, constructed of Techtron PPS, which permits various solutions to be transferred to different reaction vessels simultaneously.[57] All solutions are added in aliquots of 1.25 mL. There are 9-, 16-, and 34-mL glass reaction vessels and a 26-mL disposable plastic reaction vessel. The 34-mL vessel can contain up to 650 mg of resin. Mixing is by nitrogen bubbling and there are twenty 100-mL amino acid reservoirs. The SYMPHONY/Multiplex is not a robotic system, and all reaction vessels, reagents and solvents are maintained in an inert atmosphere.

The SYMPHONY is designed for Fmoc synthesis and is compatible with HBTU, TBTU, PyBOP, or BOP activating agents, although alternative activation strategies are possible. The standard protocol uses a 5-fold excess of amino acids, NMM as the base, and a cycle time of approximately 45 minutes. Using the TURBO coupling program, a 10-fold excess of amino acids is applied and the cycle time is reduced to 18 minutes. The SYMPHONY can also automatically cleave the peptidyl resin. Following cleavage, the filtered TFA/peptide cocktail is transferred to centrifuge tubes in a vented compartment for later workup.

Fmoc Synthesis, HBTU Activation:[58]
The Fmoc amino acid resin (50 mg, 25 μmol) was treated by the following schedule for the introduction of each residue: (1) DMF (3 × 1.25 mL); (2) 20% piperidine/DMF (1.25 mL) for 10 min; (3) DMF (6 × 1.25 mL); (4) the appropriate 0.1 M Fmoc amino acid in DMF (1.25 mL, 125 μmol) and 0.1 M HBTU/0.4 M NMM/DMF (1.25 mL, 125 μmol HBTU/500 μmol NMM) for 15 min; (5) DMF (3 × 1.25 mL).

4.3.6.3.6 PerSeptive Biosystems Pioneer with MPS

The Pioneer Peptide Synthesis System has an option of a multiple peptide synthesis (MPS) accessory with a 16-column capacity. A maximum of two MPS units may be added to each Pioneer extending the capabilities from 2 to 32 columns. The MPS unit possesses scales at 25 and 50 μmol compatible with 1.0-mL and 1.6-mL columns. The Fmoc amino acids are predissolved and an 8-fold excess is used. While the DMF washings and 20% piperidine/DMF deprotections utilize continuous-flow conditions, HATU/DIPEA-mediated couplings incorporate static flow conditions.

Fmoc Synthesis, HATU Activation:[59]
The Fmoc amino acid PEG-PS resin (280 mg, 50 μmol) was treated by the following schedule for the introduction of each residue: (1) 20% piperidine/DMF (3 × 5 mL) for a total of 5 min; (2) DMF flow wash (15 mL); (3) the appropriate 0.50 M Fmoc amino acid in DMF (0.80 mL, 400 μmol), 0.5 M HATU in DMF (0.80 mL, 400 μmol) and 1.0 M DIPEA in DMF (0.80 mL, 800 μmol) with 1 min in-line preactivation, for 30 min; (4) DMF flow wash (9.3 mL).

4.3.6.4 Production-Scale Synthesis

The synthesizers described in this section and listed in Table 5 are capable of containing 100 g or more of resin.[60] The approach in a large-scale chain elongation is different from research scale methods. For example, a 0.25 mmol synthesis may use 4 to 5 equivalents of amino acid and couple for 30 minutes. By contrast, a 250 mmol synthesis may implement only 1.5 to 2.5 equivalents of amino acid and couple for 90 minutes or longer. Resin samples are usually removed after every coupling and the ninhydrin test is performed to verify reaction completion prior to starting the next cycle (for ninhydrin monitoring see Section 4.3.6.5.1).

In large-scale syntheses, the minimum amount of solvent is added at each step and the amount is adjusted at each cycle as the swollen volume of the peptidyl resin increases. These amounts are previously determined on a smaller scale during the process development phase of the project. There is also the possibility that during the process development phase, it becomes apparent that the most economical method of synthesizing large amounts of the peptide is by a combination of solution and solid-phase methods[61] or completely by solution chemistry.[62]

Table 5 Production-Scale Synthesizers

Date	Company	Instrument	Mixing	Starting Resin	Reaction Vessels
1974	Vega Biochemicals	Model 105	overhead stirrer	up to 3 kg	6–25 L
1975	Vega Biochemicals	Model 96-P	overhead stirrer	20–250 g	0.25–4 L
1983	Vega Biotechnologies	Coupler 296	overhead stirrer	100–300 g	5 L
1986	DuPont	Coupler 296	overhead stirrer	up to 400 g	5 L, optional 3 L
1989	Advanced ChemTech	Model 400 Mini Pilot Plant	gas bubbling, variable speed oscillation, or both	up to 5 kg	2–50 L
1991	Advanced ChemTech	Model 400S	gas bubbling, variable speed oscillation, or both	100 g–1 kg	1–5 L
1993	C S Bio	CS936	overhead stirrer or inversion	10 g–2 kg	1–20 L
1994	C S Bio	CS936S	overhead stirrer or inversion	5–500 g	0.5–5 L
1995	LABORTEC	SP4000-PRO	overhead stirrer	10–250 g	1–10 L
1996	C S Bio	CS936X	overhead stirrer or inversion	15 g–10 kg	5–200 L
1997	TechniKrom	Dan-Process	flow-through	up to 20 kg	up to 400 L
	LABORTEC	SP4000-PPS	overhead stirrer	250 g–2.5 kg	30–70 L

for references see p 841

4.3.6.4.1 Vega and DuPont Coupler 296

The Coupler 296 was designed for pilot and production-scale operations and was originally commercialized by Vega Biotechnologies and later by DuPont. It is a batch instrument equipped with a 5-L reaction vessel that can accommodate 200–400 g of resin. An optional 3-L reaction vessel for 100–200 g of resin is also available. Mixing is accomplished with an air-driven, variable-speed overhead stirrer and the reaction vessel has a baffled interior which also increases reagent mixing. Solvents are added from the top through a showerhead that flushes the vessel and filtration is through the bottom of the vessel. A combination of nitrogen pressure at 362 Torr and a vacuum delivers solvents at $2\,L \cdot min^{-1}$. There are 4×20-L solvent bottles, 3×5-L reagent bottles, 1×5-L TFA bottle, and 8×1-L amino acid reservoirs. There are metering vessels from 500 mL to 4 L. The waste is collected in two individually selected containers. Both Boc and Fmoc syntheses may be performed utilizing DCC or active ester coupling procedures.

4.3.6.4.2 C S Bio 936

C S Bio produces three large-scale synthesizers in the 936 series. The standard CS936 instrument has reaction vessels from 1 to 20 L that can hold 10–2 000 g of resin. Mixing is accomplished by 180° inversion or overhead stirring. The reaction vessels have removable bottom filters composed of either sintered glass or Zitex. All solvents, amino acid solutions, and reagents are first delivered to measuring vessels with adjustable light sensors for setting the desired volume and then the liquids are transferred to the reaction vessel. The solvents are delivered from a showerhead, which washes the resin down the side of the vessel. There are four amino acid reservoirs and the instrument is compatible with Boc and Fmoc cycles.

C S Bio also produces the CS936S and the CS936X. The CS936S has 500-mL to 5-L reaction vessels that can hold 5–500 g of resin. The CS936X has 5- to 200-L reaction vessels that can contain 15–10 000 g of resin.

Boc Synthesis, DCC Activation, Large Scale:[63]
Boc amino acid resin (500 g, 250 mmol) was treated by the following schedule for the introduction of each residue: (1) CH_2Cl_2 (3×2 L); (2) 50% TFA/CH_2Cl_2 (2 L) for 2 min; (3) 50% TFA/CH_2Cl_2 (2 L) for 30 min; (4) CH_2Cl_2 (2×2 L); (5) iPrOH (2×2 L); (6) CH_2Cl_2 (2×2 L); (7) 10% DIPEA in CH_2Cl_2 (2×2 L) each for 5 min; (8) CH_2Cl_2 (2×2 L); (9) the appropriate Boc amino acid (500 mmol) in CH_2Cl_2 (2 L) and 1.0 M DCC in CH_2Cl_2 (0.50 L, 500 mmol) for 90 min, after which the coupling efficiency was checked by the ninhydrin test; (10) CH_2Cl_2 (2×2 L). The washing volume was adjusted at each cycle as the peptidyl resin gained weight during the synthesis.

Fmoc Synthesis, DIC Activation, Large Scale:[63]
Fmoc amino acid resin (500 g, 250 mmol) was treated by the following schedule for the introduction of each residue: (1) DMF (3×2 L); (2) 20% piperidine/DMF (2 L) for 2 min; (3) 20% piperidine/DMF (2 L) for 10 min; (4) DMF (2×2 L); (5) iPrOH (2×2 L); (6) DMF (3×2 L); (7) the appropriate Fmoc amino acid (500 mmol) in DMF (2 L) and DIC (63 g, 500 mmol) for 90 min, after which the coupling efficiency was checked by the ninhydrin test; (8) DMF (2×2 L). The washing volume was adjusted at each cycle as the peptidyl resin gained weight during the synthesis.

4.3.6.5 Monitoring

Monitoring the progress of a solid-phase peptide synthesis is important for both quality control requirements and detecting difficult deprotections and coupling reactions. The ninhydrin reaction is the most common method used to measure the coupling efficiency.[64,65] Monitoring the Fmoc deprotection is possible by spectrophotometric measurements due to the UV absorption of the dibenzofulvene and fulvene-piperidine adduct[28,66–68] or by conductivity measurements due to the detectable conductive signal of the carbamic acid salt.[69,70] The spectophotometric and conductive measurements have also been used to follow

the progress of the acylation. The UV monitoring of the Fmoc amino acids during the coupling is adequate for a qualitative assessment of the acylation by interpretation of the profile in continuous-flow synthesis. However, there are limitations due to the incorporation of a large excess of Fmoc derivatives. Improved spectrophotometric methods have been developed for monitoring the coupling reaction, such as the visible monitoring of the yellow color during coupling reactions with the self-indicating activated ester, 3,4-dihydro-4-oxo-1,2,3-benzotriazin-3-yl ester[71] or the monitoring of the blue color with the addition of bromophenol blue.[72–75] The conductivity monitoring of the acylation is typically limited to couplings using pentafluorophenyl esters or 3,4-dihydro-4-oxo-1,2,3-benzotriazin-3-yl esters.[70,71,76–78] Other monitoring techniques adapted to automation are photometric methods involving the addition of dimethoxytrityl chloride,[79] acylation monitoring by equilibrium distribution of colored ions (referred to as counterion distribution monitoring),[80] and monitoring methods that measure the swelling of the resin.[81,82]

4.3.6.5.1 Ninhydrin Monitoring

The ninhydrin reaction is an extremely useful method for measuring the coupling efficiency during peptide synthesis. Originally developed for amino acid analysis,[83] the ninhydrin reaction was adapted to solid-phase peptide synthesis in 1970 by Kaiser[64] for a qualitative measurement of the unreacted amino groups remaining after the coupling reaction is completed. In the Kaiser test, following the coupling and washing steps, a few milligrams of the peptidyl resin are placed in a test tube and treated with the ninhydrin reagents (see experimental section for a detailed procedure). The tube is heated at 100 °C for 5 minutes and the color examined against a white background. If the beads are slightly blue and the solution has only a trace of blue color, then the coupling is greater than 99% complete. If the beads and the solution are dark blue, then the coupling is less than 80%. Estimating the coupling to proline is not possible since the imino acid proline does not release a blue chromophore into the solution.

In 1981, a modification to the original ninhydrin procedure was reported to quantify the measurement.[65] In this quantitative ninhydrin test, the resin is accurately weighed and following heating with the ninhydrin reagents, the solution is diluted to a known volume with 60% ethanol/water. The beads are allowed to settle, and the absorbance of the solution is measured at 570 nm. For routine monitoring, an extinction coefficient of $1.5 \times 10^4\ M^{-1} \cdot cm^{-1}$ is used to estimate the level of free amine.[83] For a more accurate result, the exact extinction coefficient for the particular peptide must be used. For example, the extinction coefficient at 570 nm is $1.01 \times 10^4\ M^{-1} \cdot cm^{-1}$ and $1.25 \times 10^4\ M^{-1} \cdot cm^{-1}$ for H-Gly-resin and H-Glu(OBzl)-resin, respectively. One important criteria with the quantitative ninhydrin test is to insure that all the blue chromophore is removed from the resin when diluting with 60% ethanol/water. Boc syntheses with PAM resins satisfy this requirement, but Boc chain elongation with resins derived from chloromethyl resins are not compatible, presumably due to side reactions leading to quaternary amine sites, which will retain some of the color.

Additionally, the quantitative ninhydrin test is not completely compatible for monitoring couplings of Fmoc amino acids. The pyridine in the ninhydrin reagents partially deprotects the Fmoc-peptidyl resin at 100 °C, which produces unprotected amines that can react with ninhydrin. Heating for 5 instead of 10 minutes minimizes this unwanted deprotection. To our knowledge, a careful examination of this phenomenon has not been reported in the literature.

The quantitative ninhydrin test has not been automated. The Applied Biosystems Models 430, 431, and 433 Peptide Synthesizers, however, have the capability to automatically remove 3–10 mg resin samples after each coupling for a manual ninhydrin test.

for references see p 841

Quantitative Ninhydrin Test:[65,84]
Dried peptidyl resin (2–10 mg) was placed in a test tube and the weight determined to the nearest 0.1 mg. To this tube and a blank tube was added 76% phenol/EtOH (w/w, 4 drops), 0.0002 M KCN/pyridine (8 drops), and 0.28 M ninhydrin/EtOH (4 drops). The test tubes were heated at 100 °C for 7 min and then diluted to 5 mL with 60% EtOH. The tubes were mixed thoroughly and the resin allowed to settle. The absorbance of the solutions was obtained at 570 nm. The $\mu mol \cdot g^{-1}$ of amine was calculated using the following equation:

$$\mu mol \cdot g^{-1} = \frac{[\text{abs. of sample} - \text{abs. of blank}] \times [\text{dilution (mL)}] \times 10^6}{[\text{extinction coefficient}] \times [\text{sample weight (mg)}]}$$

The percent coupling was determined by the equation:

$$\% \text{ coupling} = \left[1 - \frac{\text{amine } (\mu mol/g)}{10^3 \times \text{substitution (mmol/g)}}\right] \times 100$$

4.3.6.5.2 UV Monitoring

The chromophore of Fmoc amino acids, dibenzofulvene and the fulvene-piperidine adduct has allowed the monitoring of the coupling and deprotection reactions, respectively, during Fmoc solid-phase peptide synthesis by UV methods. Dryland and Sheppard published the first UV-trace of an Fmoc peptide synthesis, illustrating the continuous spectrophotometric monitoring of the coupling, washing, and deprotection during chain elongation.[28] Although only qualitative information on the coupling and deprotection is provided, this information has proven to be extremely valuable in detecting the sections in difficult sequences that are challenging to assemble.

The continuous-flow peptide synthesizers equipped for UV monitoring are the manual CRB PEPSYNthesiser I and LKB Biochrom 4175, the semiautomatic CRB PEPSYNthesiser II, MilliGen 9020, and Novabiochem NovaSyn Gem, and the automatic Pharmacia-LKB Biochrom 4170, MilliGen 9050 PepSynthesizer, Novabiochem NovaSyn Crystal, MilliGen/Biosearch 9050 Plus PepSynthesizer, and the PerSeptive Biosystems Pioneer. Many of the automatic continuous-flow synthesizers are also able to automatically optimize the deprotection time to ensure more complete Fmoc deprotection during difficult regions. The ABI 433 batch synthesizer, with an optional UV monitor, also has the ability to detect difficult Fmoc deprotections and extend the number of deprotections as well as the time for the next coupling.

4.3.6.5.3 Conductivity Monitoring

Three monitoring methods have been developed that measure the changes in conductivity during the deprotection, washing, and coupling steps of an Fmoc solid-phase synthesis.[69,70,76–78] These methods vary in the placement of the pair of electrodes and the addition of a weak base during the coupling reaction. All three methods give similar results and are optimal in conjunction with pentafluorophenyl (Pfp) esters. During the acylation reaction, the free acid (Pfp-OH) is released which causes a change in the conductivity of the solution. This signal is enhanced by the addition of a base, either DIPEA or NMM. Monitoring of the acylation cannot be achieved with uronium or phosphonium salt based activation methods.

Conductivity monitoring is also possible during Fmoc deprotections, since a piperidinyl carbamate salt is formed producing a detectable signal. The piperidinyl carbamate, however, has a low conductance and atmospheric carbon dioxide in the piperidine or DMF often interferes with the signal. Uronium and phosphonium salts also interfere with the deprotection conductivity signal in a difficult region due to inefficient resin washing.

Peptide synthesizers with conductivity monitoring include the Applied Biosystems Synergy and Model 433A, Biotech BT 7600, and Kem-En-Tec MKIII.

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4.3.7 Combinatorial Synthesis

J. Eichler and R. A. Houghten

Synthetic combinatorial library techniques emerged during the 1990s as powerful research tools in response to the rapidly growing demand for synthetic peptides in numerous fields of biomedical research and drug discovery. This issue had initially been addressed by various multiple synthesis techniques, which were developed during the 1980s (reviewed by Beck-Sickinger and Jung[1]) and which enabled the multiple parallel synthesis of up to hundreds of individual peptides in a fraction of the time previously required. The introduction of synthetic combinatorial library technologies was a logical continuation of this development with the aim to dramatically increase the speed and throughput of the synthesis and biological evaluation of peptides. The synthetic peptide combinatorial library concept involves the generation of all possible sequence permutations for a peptide of a given length (e.g., $20^6 = 64$ million hexapeptides if only the proteinogenic amino acids are used), in combination with a screening and selection process that enables the rapid identification of unique, highly active peptides in the presence of millions of less-active or inactive peptides. The individual synthesis of such large numbers of peptides was unrealistic using the methods available before the development of combinatorial technologies. Several approaches for the generation of highly complex peptide mixtures were established in the early 1990s and have been used successfully for the identification of numerous highly active and selective peptides, such as receptor ligands, antigenic or antimicrobial peptides, enzyme substrates, or inhibitors.

Using recombinant DNA techniques, large numbers of peptides can be expressed randomly in a fusion phage vector system. This method, however, remains restricted to the use of the 20 proteinogenic amino acids as building blocks. Chemical approaches to the generation of peptide libraries,[2–5] on the other hand, allow for the use of D-[6,7] and other nonproteinogenic amino acids,[8] or carboxylic acids[9] as building blocks, as well as the postsynthetic chemical modification of peptide libraries as a means of generating peptidomimetic libraries.[10]

The principal differences between the various methods of peptide library synthesis are (i) the solid support used, (ii) the means of incorporating amino acid mixtures, (iii) the state in which the libraries are screened (support-bound vs in solution), and (iv) the manner in which individual active peptides are identified within the library.

This section provides an account of synthetic approaches and experimental procedures for the generation of peptide and peptidomimetic libraries with special emphasis on solid-phase methods, along with approaches for library screening and deconvolution. Other library techniques, such as phage display libraries[11] and the generation of nonpeptide small-molecule libraries,[12,13] are not covered here.

4.3.7.1 Library Synthesis Methods

The principal difference between the synthesis of individual peptides and peptide libraries is that mixtures of amino acids, rather than individual amino acids, are incorporated into selected or all positions of the sequence of peptide libraries. However, all current peptide chemistry strategies can be used for the synthesis of peptide libraries. In general, library synthesis requires greater emphasis on simplicity and reproducibility of the synthesis process. Although soluble supports have also been used for peptide library synthesis,[14] the majority of methods used to synthesize peptide combinatorial libraries utilize Merrifield's concept of solid-phase synthesis,[15] which is based on the sequential assembly of peptides after covalent attachment of the C-terminal amino acid to a polymeric solid support. This enables the excess of reagents to be removed by simple wash and filtration processes, and avoids the

laborious isolation and purification procedures associated with the conventional synthesis in solution, thus, greatly facilitating and accelerating the synthesis process. Furthermore, the highly repetitive character of solid-phase synthesis (i.e., repetition of the cycle: couple–wash–deprotect–wash) makes it readily amenable to automation.

The Solid Support

Most synthetic methods for the generation of peptide libraries have been derived from various multiple parallel peptide synthesis techniques developed since 1984.[16–19] Consequently, the solid supports used for library synthesis are essentially the same as those used for multiple peptide synthesis. Standard divinylbenzene cross-linked polystyrene resins are typically used for libraries that are cleaved from the resin and screened in solution.[3,20,21] Polyoxyethylene-grafted polystyrene resins,[22,23] or acrylamide–polyoxyethylene copolymers,[24] on the other hand, are the solid supports of choice for the synthesis of resin-bound peptide libraries screened in solid-phase binding assays.[4] Such resins are compatible with both organic solvents used for peptide synthesis, as well as aqueous buffers used in the bioassays. Various segmental supports previously employed for multiple peptide syntheses have also been utilized for the synthesis of peptide libraries, including polypropylene pins,[2] cotton,[25] cellulose membrane,[26,27] and glass slides.[28]

4.3.7.1.1 Divide–Couple–Recombine (Split-and-Mix Synthesis)

One important issue in library synthesis is to obtain, as close to equimolar representation as possible, all individual peptides within the library for ease of deconvolution. Therefore, the various amino acids have to be incorporated into each of the library positions in a ratio as close to equimolar as possible. When using resin beads as the solid support for library synthesis, this can be achieved using a process that is known as "divide–couple–recombine" (DCR),[3] "split-and-mix synthesis",[4] or "portioning mixing".[5] This process involves coupling of each protected amino acid to be used for the library to separate portions of the resin, followed by combining and mixing all resin portions, before dividing the resin again for the next coupling step. By repeating this process for a total of five couplings and using 20 amino acids as building blocks, a library of 3200000 (20^5) pentapeptides can be readily prepared. Due to the mode of incorporating amino acids, the DCR process yields libraries containing an individual, unique peptide sequence on each resin bead. Care should be taken to use a sufficient amount of resin for the synthesis, i.e. the number of beads used should be at least 10 times the theoretical number of peptides in the library in order to ensure that all library components are present with statistical probability. After assembling the library on the resin, it can either be cleaved from the resin for bioassays in solution, or left on the resin for solid-phase assays.

DCR Synthesis:[29]
PSty-POE resin (TentaGel, 130-μm beads, 50 g) was divided into 19 equal aliquots and distributed into 19 polypropylene syringes equipped at the bottom with a plastic frit. Nineteen Fmoc-protected amino acids (all of the proteinogenic amino acids except Cys) (3 equiv of each) were added separately to the resin aliquots using a minimal amount of DMF. Coupling was initiated by adding DIC (3 equiv) and HOBt (3 equiv), and a trace amount of bromophenol blue was added to the mixture. The syringes were tightly sealed and rocked gently for approximately 30 min at rt, or until all the beads turned from blue to colorless (indicating completeness of coupling). If the coupling was incomplete, the beads were allowed to settle and the supernatant was gently removed. More of the freshly prepared Fmoc-protected amino acid was added and the reaction allowed to proceed for a further 60 min. All resin aliquots were then mixed in a siliconized cylindrical glass vessel fitted at the bottom with a frit. Dried N_2 was bubbled through in order to mix the resin, which was then washed with DMF (8 ×).

4.3.7.1.2 Coupling of Amino Acid Mixtures

An alternative means for the introduction of mixture positions is through the coupling of mixtures of protected amino acids. Due to the differences in coupling rates of the various amino acids, coupling of an equimolar amino acid mixture in a molar excess over resin-bound amino groups, as typically used in solid-phase peptide synthesis, leads to non-equimolar incorporation of amino acids into the mixture positions. This results in a highly nonequimolar distribution of individual peptides within the library. Two different strategies are used to overcome this problem. In the first strategy, the ratio of the protected amino acids within the coupling mixture is adjusted according to their different coupling rates, i.e. the higher the coupling rate of a particular amino acid, the lower the concentration of that amino acid in the coupling mixture.[2,20,25,30] Such ratios are established by adjusting the relative concentration of each amino acid according to its incorporation ratio after coupling of an equimolar amino acid mixture, as determined by amino acid analysis or HPLC. Since these amino acid mixtures are coupled in a large (i.e., 5- to 10-fold) molar excess over resin-bound amino groups, the coupling can be considered a pseudo-first-order reaction, and the ratio of coupling rates of the amino acids within the mixture should, therefore, be independent of the amino acid it is being coupled to. The relative molar ratio obtained for 18 of the 20 proteinogenic N^{α}-Boc-protected amino acids, determined by amino acid analysis, is shown in Table 1.

Table 1 Relative Molar Ratio of Amino Acids in the Coupling Mixture When Using a 10-Fold Molar Excess[30]

Boc-Protected Amino Acid	Relative Molar Ratio (Mol%)
Ala	3.58
Asp(OBzl)	3.70
Glu(OBzl)	3.84
Phe	2.66
Gly	3.04
His(Dnp)	3.76
Ile	18.3
Lys[Z(2-Cl)]	6.56
Leu · $0.5H_2O$	5.23
MetO	2.42
Asn	5.64
Pro	4.56
Gln	5.61
Arg(Tos)	6.87
Ser(Bzl)	2.93
Thr(Bzl)	5.04
Val	11.9
Tyr[Z(2-Br)]	4.36

The second strategy uses double couplings of equimolar amino acid mixtures in an equimolar or slightly less than equimolar ratio to the resin-bound amino groups,[21,31] thus avoiding the coupling of some amino acids at a higher rate than others. A potential problem associated with this strategy is the fact that the coupling is a second-order reaction, whose rate is influenced by both partners. Consequently, when an amino acid mixture is coupled to a resin-bound amino acid mixture using this method, some combinations are likely to be formed at a higher rate than others, although the overall incorporation of amino acids is close to equimolar at both positions.

Coupling of Protected Amino Acid Mixtures:[30]
A mixture of 18 Boc-protected amino acids (the 20 proteinogenic amino acids excluding Cys and Trp, in a molar ratio of amino acids derived from previous competition experiments[30]) and DIC in DMF

(equimolar to the amino acid mixture) was coupled to MBHA resin contained in polypropylene mesh packets[17] by means of DIC/HOBt (10 equiv) in DMF over 1 h. In order to obtain an as close to equimolar as possible coupling of amino acids to the resin, the ratio of protected amino acids in the coupling mixture was adjusted iteratively based on repetitive amino acid analyses of resin-bound amino acid mixtures (see Section 4.3.7.4.1).

4.3.7.2 Peptide Libraries with Specific Structural Elements

4.3.7.2.1 Cyclic Peptide Libraries

The reduced flexibility of cyclic peptides is generally thought to increase their potential for high-affinity binding. Cyclic peptides have also been shown to be less prone to proteolytic degradation than linear peptides. Therefore, cyclic peptide libraries are promising tools for the discovery of novel bioactive compounds.

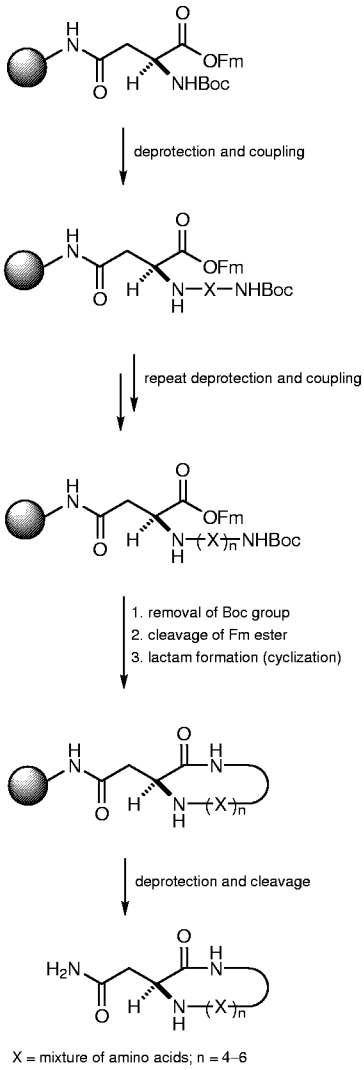
The two most common reactions for the generation of cyclic peptides are (i) disulfide formation between cysteine or penicillamine residues through oxidation, and (ii) lactam formation between the amino and carboxy terminal ends (head-to-tail cyclization), the amino terminus and the side chain of an aspartic or glutamic acid residue, or between the side chains of a lysine (or another diamino acid) and aspartic or glutamic acid residues. A special mode of the second type is backbone cyclization between an amino group (N-terminal or side chain) and a carboxyalkylated backbone nitrogen.[103] Both reactions have been used to generate cyclic peptide libraries.

Cyclic disulfide libraries have primarily been generated through solid-phase oxidation of support-bound peptide libraries, either on resin beads[32] or on a cellulose membrane support.[27] A study comparing several methods for solid-phase and solution cyclization[33] found oxidation with hydrogen peroxide (1.5 equiv at a peptide concentration of $0.5\,g \cdot L^{-1}$) to be best suited for the generation of cyclic disulfide peptide libraries in solution.

Like disulfides, cyclic lactam peptides can be generated while bound to the solid support, or in solution after cleavage. Both approaches have been applied to the generation of cyclic peptide libraries. Resin-bound head-to-tail cyclization is possible when the C-terminal amino acid is an aspartic or glutamic acid bound to the resin via its β- or γ-carboxy group, respectively, and the α-carboxy group is orthogonally protected, allowing for its selective deprotection after the linear sequence has been assembled. This method has been used for the preparation of head-to-tail cyclic peptide libraries following Boc/Bzl chemistry (see Section 4.3.2.1) with the α-fluoren-9-ylmethyl ester of Boc-Asp as the C-terminal amino acid attached to the resin (Scheme 1).[34,35] Alternatively, the linear library precursors can be cleaved from the resin with the side chains still protected, and cyclized in solution.[36] This method requires the use of linkers that are selectively cleaved under very mild conditions [e.g., the Trt(2-Cl) linker] without concurrent removal of the side-chain protecting groups.

for references see p 864

Scheme 1 Formation of Cyclic Lactam Peptides on a Solid Support[35]

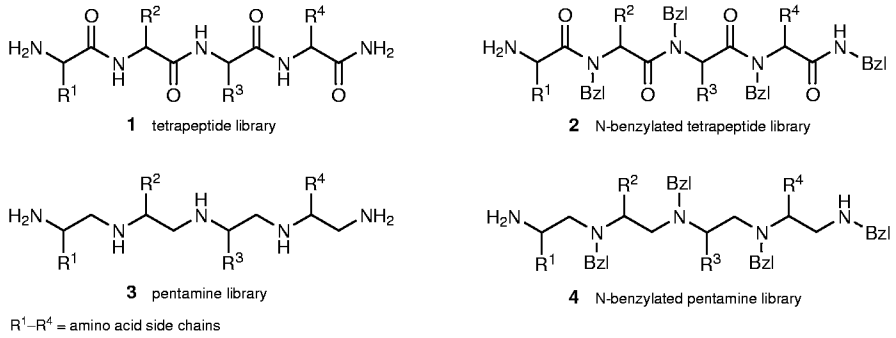


Synthesis of Cyclic Lactam Peptide Libraries:[35]
Fourteen cyclic penta-, hexa-, and heptapeptide libraries of the general formula *cyclo*-[X_nAsn] (n = 4–6) were synthesized simultaneously on MBHA resin (0.9 mequiv · g^{-1}) following the Boc/Bzl strategy (see Section 4.3.2.1) and using the method of simultaneous multiple peptide synthesis.[17] The resin for each library (100 mg) was contained in polypropylene mesh packets. Boc-Asp-OFm was coupled as the first amino acid to the resin via its β-carboxy group.[37] Mixture positions (X) were then introduced by coupling mixtures of 19 of the 20 proteinogenic amino acids (Cys excluded), or of the respective D-amino acids, in a predetermined molar ratio[20] (see Section 4.3.7.1.2). After the final amino acid mixture coupling and removal of the N-terminal Boc group, the Fm ester was cleaved with 20% piperidine/DMF over 1 h, followed by PyBOP/HOBt/DIPEA-mediated amide formation between the N-terminal amino and the α-carboxy groups of the C-terminal Asp (head-to-tail cyclization). The libraries were deprotected and cleaved from the resin using the low/high HF method[38] in combination with a multivessel cleavage apparatus.[39] Upon cleavage from the MBHA resin, which releases the peptides as carboxamides, the C-terminal Asp residue was released as Asn.

4.3.7.2.2 Peptidomimetic Libraries through Amide Alkylation and/or Reduction

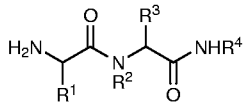
Existing peptide libraries can be chemically modified through N-alkylation and/or reduction of the peptide bonds,[10,40] thus, dramatically modifying the physicochemical properties of the peptides, and greatly extending the range and repertoire of chemical diversity presented. Due to the lack of the peptide bond, the components of such libraries, which have been termed "libraries from libraries",[10] are stable toward proteolytic degradation. This concept can be illustrated by the example shown in Scheme 2.[40] A tetrapeptide library is synthesized and divided into four aliquots. The first aliquot **1** is left untreated, the second aliquot **2** is N-benzylated, the amide bonds of the third aliquot are reduced yielding a pentamine library **3**, and the fourth aliquot is perbenzylated and reduced to an N-benzylated pentamine library **4**. The screening of these four libraries yields very different results, indicating the significance of the physicochemical character of the backbone for the biological activity.

Scheme 2 Modification of a Tetrapeptide Library through N-Benzylation and/or Reduction[40]



A dipeptidomimetic library **5** has been synthesized through successive amide alkylation using 46 different amino acids and five different alkylating agents (Scheme 3).[41] By alternating the amino acid coupling and alkylation steps, different alkyl groups can be introduced at different positions of the sequence.

Scheme 3 A Dipeptidomimetic Library through Successive Amide Alkylation[41]



Successive Amide Alkylation of Resin-Bound Amino Acids or Peptides:[41]
The cycle of alternating amino acid coupling and amide alkylation consisted of the following steps, which are further described below: (i) coupling of an Fmoc-protected amino acid, (ii) Fmoc removal followed by tritylation of the N-terminal amino group, (iii) amide alkylation, and (iv) removal of the trityl group.
(i) The MBHA resin solid support was contained in five polypropylene mesh packets (250 mg per packet), and an N^{α}-Fmoc-protected amino acid was coupled to each resin packet.
(ii) Following Fmoc deprotection, the amino group was tritylated using TrtCl (5 equiv) in a mixture of CH_2Cl_2/DMF (9:1) containing 7% DIPEA (10 mL per resin packet).
(iii) The nitrogen of the amide bond between the amino acid and the resin was alkylated as follows (all manipulations were performed under a N_2 atmosphere and strictly anhydrous conditions): The resin packets were dried overnight at 50 mTorr. Each resin packet was placed in one of five round-bottomed flasks. A soln of 1 M tBuOLi in THF (4 mL) and THF (4 mL) were added to each flask, which were then shaken at rt for 15 min. Excess base soln was removed by cannulation. Following addition of DMSO

for references see p 864

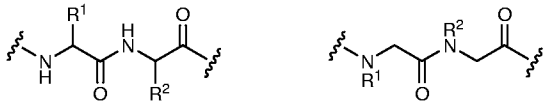
(8 mL), one of the following five alkylating agents was added to each vessel: MeI (0.77 mL), EtI (0.98 mL), allyl bromide (1.06 mL), BzlBr (1.46 mL), or 2-(bromomethyl)naphthalene (2.72 g, 12.3 mmol) dissolved in DMSO (8 mL). The mixtures were vigorously shaken for 2 h at rt and the alkylating solns removed by cannulation. The resin packets were washed with DMF (3 ×), iPrOH (2 ×), CH_2Cl_2 (3 ×), and MeOH (1 ×). After completely drying the resin packets overnight at 50 mTorr, the entire alkylation process was repeated three times for the allylation, benzylation, and naphthylmethylation, the first two repetitions for a period of 2 h and the third for 5 h.

(iv) The N-terminal trityl group was removed by shaking the resin packets in 2% TFA/CH_2Cl_2 for 2 min and then with fresh soln for 30 min, followed by washing with CH_2Cl_2, iPrOH, CH_2Cl_2 again, and MeOH, rendering the resin ready for the next amino acid coupling.

4.3.7.2.3 Peptoid Libraries

Peptoids are oligomers of N-substituted glycines. Hence, the principal difference between peptides and peptoids is the location of the amino acid side chains with respect to the backbone. While in peptides **6** the side chains are attached to the α-carbon of the amino acids, in peptoids **7** they are linked to the amide nitrogens of the peptide backbone (Scheme 4).[42,43] Similar to amide-alkylated or reduced peptides (see Section 4.3.7.2.2), peptoids lack the peptide bond CO–NH, and are, therefore, less susceptible to proteolytic degradation. Consequently, peptoids are attractive molecules for the generation of combinatorial libraries.

Scheme 4 The Location of the Amino Acid Side Chains on the Peptide and Peptoid Backbones[42,43]



6 peptide backbone **7** peptoid backbone

$R^1 = R^2$ = amino acid side chains

One potential problem associated with the generation of peptoid libraries is the supply of sufficiently diverse protected monomers. Initially, these monomers were individually prepared via reductive amination of glyoxylic acid.[42] The assembly of peptoids using these monomers follows the standard protocols of solid-phase peptide synthesis. Later, a submonomer synthesis method was devised[43] which involves alternating couplings of bromoacetic acid, followed by reaction with a primary amine containing the amino acid side chain or other functionalities, thus transforming it into a secondary amine, which subsequently is again acylated with bromoacetic acid. The principal advantage of this method is that it eliminates the need to prepare large sets of protected N-substituted glycine monomers as building blocks, and, in conjunction with the DCR synthesis described in Section 4.3.7.1.1, it is readily amenable to the generation of peptoid libraries.

Synthesis of a Protected N-Substituted Glycine Monomer via Reductive Amination:[42]

β-Alanine *tert*-butyl ester hydrochloride (10.0 g, 55.1 mmol) and glyoxylic acid hydrate (5.47 g, 59.5 mmol) were dissolved in H_2O (100 mL), and the pH was adjusted to 6.0 with 1 M NaOH. 10% Pd/C (400 mg) was added and the mixture placed in a hydrogenator under H_2 for 24 h, after which no further reduction in pressure was observed. The catalyst was removed by filtration and washed, and the pH of the filtrate adjusted to 9.5 with 1 M NaOH. After addition of acetone (100 mL), Fmoc-OSu (18.6 g, 55.1 mmol) was slowly added. The mixture was maintained at pH 9.5 with periodic additions of 1 M NaOH. After stirring overnight, the mixture was concentrated under reduced pressure to remove the acetone and the remaining aqueous soln extracted three times with Et_2O/hexane (1:1). The aqueous layer was acidified with aq citric acid to pH 2.5 and extracted with EtOAc. The organic layer was dried over Na_2SO_4 and the solvent removed under reduced pressure. The residue was recrystallized from EtOAc/hexane to give Fmoc-CEGly(OtBu) [where CEGly is *N*-carboxyethylglycine]; yield: 14.6 g (62%).

4.3.7.2.4 Conformationally Defined Peptide Libraries

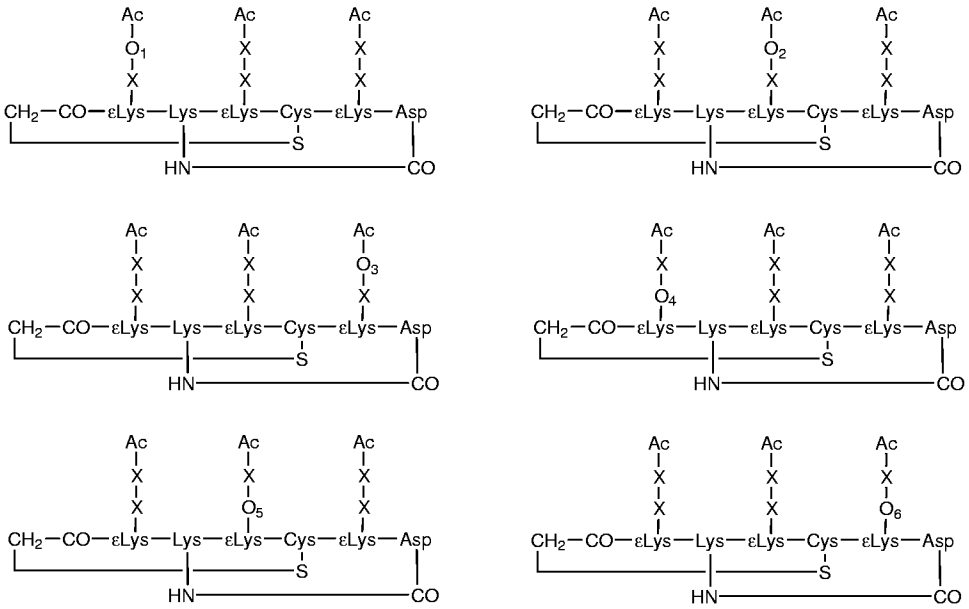
The combinatorial library concept is not limited to short (i.e., <10 amino acid residues) peptides, but can be equally well applied to longer sequences, which are able to adopt defined secondary structures. This thesis is the basis for the concept of conformationally defined peptide libraries for the de novo design of synthetic protein-like structures with specific structures and/or biological activities.[44–46] Positional-scanning libraries have been built by randomizing selected positions of an 18 amino acid long peptide known to adopt an α-helical conformation (YKLLKKLLKKLKKLLKKL-NH_2).[44] The randomized library positions were located either at the hydrophilic or hydrophobic face of the α-helix, or in one complete helix turn. Peptides capable of catalyzing the decarboxylation of oxalacetate have been identified through the screening of the library built on the hydrophilic face. The catalytic activity of these peptides was found to correlate well with their propensity to fold into an α-helical conformation. A similar approach has been taken by Bianchi et al.[46] by building a library through randomization of five positions of a 26 amino acid long zinc-finger motif.

4.3.7.2.5 Scaffolded Peptide Libraries

The regions of proteins responsible for binding to their ligands are often not contained in short, continuous stretches of the primary amino acid sequence, but are located in regions with a distinct secondary structure and/or in nonconsecutive amino acid residues ("contact sites"). These sites are isolated in the amino acid sequence, but brought into spatial proximity by protein folding. Therefore, the structures of libraries for the design of molecules capable of mimicking protein-binding sites ("synthetic receptors") should have a molecular architecture different from linear oligomers. This concept has been the basis for the design of scaffolded peptide libraries, which are not assembled in a continuous, linear fashion, but built on molecular scaffolds, such as steroids[47,48] or cyclic peptides.[9,49,50] Molecules generated in this fashion can be expected to be more likely than continuous short peptide sequences to mimic discontinuous regions of proteins. Boyce et al.[47] generated a "receptor library" on a rigid steroidal scaffold, to which two dipeptide units were attached at different sites of the scaffold. Another library based on this concept has the proportions of a hexapeptide, which is divided into three dipeptide segments attached to three different sites (α-amino groups of lysine) of a bicyclic peptide template (see Figure 1).[49,50] Each of the six sublibraries is composed of 23 separate mixtures, in which one of the six library positions (O) is individually defined with one of 23 amino acids, and the remaining five positions (X) are mixtures of the same 23 amino acids.

for references see p 864

Figure 1 A Scaffolded Positional-Scanning Peptide Library Built on a Bicyclic Peptide Template[49,50]



Synthesis of a Scaffolded Positional-Scanning Peptide Library Built on a Bicyclic Peptide Template:[49] The following describes the synthesis of the first sublibrary of a scaffolded positional-scanning peptide library (Figure 1), in which the first library position (O_1) was individually defined with one of 23 amino acids (the 20 proteinogenic amino acids, except Cys, plus Hyp, Cha, Tic, and Ahx) and the remaining five library positions (X) were mixtures of the same 23 amino acids. The library was synthesized following the Fmoc/tBu strategy [Fmoc-Xaa/DIC/HOBt (5 equiv); see also Section 4.3.2.2] in conjunction with simultaneous multiple peptide synthesis[17] using TentaGel resin as the solid support, which was contained in polypropylene mesh packets (100 mg per packet).

The bicyclic template was assembled simultaneously on all 23 resin packets by coupling Fmoc-Asp(OAl), followed by Boc-Lys(Fmoc), Fmoc-Cys(Mmt), Boc-Lys(Fmoc), Fmoc-Lys(Aloc), and Dde-Lys(Fmoc). After removal of the Fmoc groups, the N-terminus was acylated with $BrCH_2CO_2H$/DIC (10 equiv) in DMF, and the Mmt group removed from Cys by three consecutive treatments with fresh 1% TFA and 5% Et_3SiH in CH_2Cl_2, each over 10 min. The first ring, a sulfide between the α-carbon of the bromoacetyl group and the sulfanyl group of Cys, was formed by shaking the packets overnight in 5% NMM/DMF. The allyl ester and Aloc group were then removed under argon using catalytic amounts of $Pd(PPh_3)_4$ in a mixture of DMF/AcOH/NMM (5:1:0.5) for 3 h. The second ring, a lactam between the side chains of Asp and Lys, was formed by two consecutive treatments with HATU (2 equiv) and DIPEA (4 equiv) in DMF, each over 2 h with fresh soln for each treatment. The Boc groups were removed from the α-amino groups of two εLys residues and four mixture positions X were introduced by two consecutive couplings, each over a period of 2 h, of an equimolar mixture of the 23 Fmoc-protected amino acids (1.2 equiv). This was followed by acetylation of the N-termini of the two dipeptide mixtures formed in this manner. The Dde group was removed from the α-amino group of the third εLys residue by three consecutive treatments with 2% H_2NNH_2 in DMF over 5 min periods, using fresh soln for each treatment, and the fifth mixture position was then coupled. Then followed the introduction of the defined position through coupling of individual amino acids to each of the 23 resin packets.

The second through sixth sublibraries were generated accordingly by modifying the location of the N^{α}-Dde-protected εLys residue within the template, as well as the order of the coupling mixture and defined positions after Dde removal. After the final side-chain deprotection using TFA/CH_2Cl_2/iBu_3SiH/H_2O (70:20:5:5) over 3 h, the resin-bound mixtures were removed from the polypropylene mesh packets, suspended in 0.1 mM HCl, and stored at 4°C.

4.3.7.3 Library Formats

The term "library format" refers to the arrangement and organization of peptides within the library which determine the type of deconvolution (i.e., the manner in which individual active peptides are identified within the library) that can be used.

4.3.7.3.1 One-Bead-One-Peptide Libraries

The majority of support-bound peptide libraries are synthesized as one-bead-one-peptide libraries.[4] As indicated by its descriptive name, the characteristic feature of this library format is the fact that one, and only one, individual peptide sequence is generated on each bead of the solid support. This type of library can be readily generated using the DCR synthesis (see Section 4.3.7.1.1), which inherently produces one-bead-one-peptide libraries due to the fact that only individual amino acids, as opposed to amino acid mixtures, are coupled in each synthesis cycle. One-bead-one-peptide libraries are typically completely randomized, i.e. each position of the sequence is diversified.

A hybrid of the one-bead-one-peptide and positional-scanning library formats (see Section 4.3.7.3.2.2) is presented by an approach termed "library of libraries",[51] which is directed toward the identification of pharmacophore motifs, i.e. structural motifs necessary for the bioactivity of interest, rather than the complete structures of individual active compounds. This library format enables the identification of specific, i.e. nonreplaceable, positions of a peptide with a bioactivity of interest versus unspecific positions, which can be replaced by a variety of different amino acids without loss in activity. In the example presented,[51] a hexapeptide library is generated as 160000 ($20^3 \times 20$) sublibraries (beads), which represent all possible combinations of three defined and three mixture positions within the hexapeptide framework, i.e. 8000 (20^3) different combinations of three amino acids at the defined positions within a given arrangement of the hexapeptide multiplied by 20 possible different arrangements of three defined and three mixture positions. While the defined positions are coupled using the DCR method,[4] the mixture positions are introduced by the coupling of amino acid mixtures.

Support-Bound One-Bead-One-Peptide Pentapeptide Library:[29]
The library was assembled using DCR synthesis with 19 Fmoc-protected amino acids (all of the proteinogenic amino acids except Cys), as described in Section 4.3.7.1.1. After each coupling, the resin was combined and washed, and 20% piperidine in DMF was added. After 10 min of bubbling with N_2, the piperidine was removed and the resin washed with DMF (10 ×). The amount of released DBF–piperidine adduct was determined by measuring the absorbance at 302 nm. A stable level of substitution determined in this manner throughout the library synthesis served as one of the quality control measures. Then, the resin was again divided into 19 aliquots for the next coupling. The process of dividing the resin, coupling individual amino acids to the separate resin portions, combining the resin, and removing the Fmoc-protecting group was repeated for a total of five couplings. This was followed by removal of the side-chain protecting groups by treatment with a mixture of TFA/PhSMe/PhOH/EDT/H_2O (32:2:2:1:1) for 5 min and then a second treatment with fresh soln for 120 min. The resin was washed thoroughly with TFA, CH_2Cl_2, DMF, DMF/H_2O (1:1), and 0.01% HCl in H_2O, and stored in DMF at 4 °C.

4.3.7.3.2 Mixture-Based Libraries

Unlike one-bead-one-peptide libraries, which are typically completely randomized, libraries composed of separate peptide mixtures have one or more positions in the sequence individually defined with one or very few amino acids and the remaining positions are occupied by amino acid mixtures. Such libraries can be prepared using either the DCR method (see Section 4.3.7.1.1), or through the coupling of amino acid mixtures (see Section 4.3.7.1.2).

for references see p 864

4.3.7.3.2.1 Iterative Libraries

Iterative peptide libraries owe their name to the algorithm used for their deconvolution (i.e., the identification of individual active peptides), which follows an iterative synthesis and screening scheme (see Section 4.3.7.5.2.1). The initial iterative hexapeptide library,[3] with two defined positions at the N-terminal and the remaining four mixture positions as amino acid mixtures, has the general formula Ac-O_1O_2XXXX-NH_2, in which O represents positions individually defined with one of the 20 proteinogenic amino acids, and X represents positions occupied by mixtures of 19 of the 20 amino acids (cysteine excluded). This library is organized into 400 separate peptide mixtures (i.e., Ac-AAXXXX-NH_2, Ac-ACXXXX-NH_2, through Ac-YWXXXX-NH_2, and Ac-YYXXXX-NH_2), each containing 130321 (19^4) individual peptides; therefore, the entire library represents 52128400 (400×130321) individual peptides.

Synthesis of a Dual-Defined Iterative Hexapeptide Library (Ac-O_1O_2XXXX-NH_2):[52]
MBHA–PSty resin (20 g) was divided into 19 aliquots sealed in porous polypropylene mesh packets.[17] Each of 19 of the 20 proteinogenic amino acids (Cys excluded) was coupled to one resin packet following Boc/Bzl chemistry (Section 4.3.2.1) in conjunction with DIC activation. Completion of couplings was checked using the ninhydrin[53] or bromophenol blue[54] tests. After washing and air-drying the resin packets, the resin from all packets was combined and thoroughly mixed, before dividing it again into 19 aliquots. At this stage, the resin was referred to as X-resin. After Boc removal from all 19 packets using 55% TFA/CH_2Cl_2 for 30 min, and washing with CH_2Cl_2, iPrOH, 5% DIPEA/CH_2Cl_2, and CH_2Cl_2 again, coupling of one of the 19 individual amino acids to each resin packet, followed by Boc removal, was repeated three more times to generate an XXXX-resin. This resin was divided into 400 aliquots and the two defined positions (O_2 and O_1) were introduced, using all 20 proteinogenic amino acids, by coupling to the 400 resin packets in a manner which generated all possible combinations (20×20) at the two N-terminal positions. The 400 peptide mixtures were deprotected and cleaved using the high/low HF method[38] in conjunction with a multivessel cleavage apparatus.[39] Peptide mixtures were extracted from the resin with AcOH/H_2O, and lyophilized twice. The lyophilized mixtures were dissolved in H_2O (1–5 mg · mL^{-1}). Solubilization of peptide mixtures with hydrophobic amino acids at the defined position was facilitated by sonication. The library was stored at 4 °C for up to four weeks, and frozen for prolonged storage.

4.3.7.3.2.2 Positional-Scanning Libraries

A positional-scanning library with n diverse (i.e., variable) positions is composed of n sublibraries, in which one position is individually defined with one of a preselected set of building blocks and the remaining positions are mixtures of building blocks.[20,55] For example, a positional-scanning hexapeptide library (Figure 2) consists of six sublibraries. If 20 different amino acids are used as building blocks for each position of this library, each of the sublibraries is composed of 20 separate peptide mixtures, every one of them containing 32000000 (20^5) individual peptides. Each sublibrary addresses a different position of the sequence, i.e. a different position is defined in each sublibrary, but all of the six sublibraries represent the same collection of 64000000 (20^6) individual peptides. This library format has been used for the generation of numerous peptide and peptidomimetic libraries with varying diversity and complexity including tetra-,[56] hexa-,[20,57] deca-,[58] and undecapeptide libraries,[59] a multimeric pentapeptide library,[60] N-alkylated and/or reduced tetrapeptide libraries,[40] as well as conformationally defined[44,45] and scaffolded peptide libraries.[9,49]

Figure 2 A Positional-Scanning Hexapeptide Library[20]

1st sublibrary	2nd sublibrary	3rd sublibrary	4th sublibrary	5th sublibrary	6th sublibrary
AXXXXX	X**A**XXXX	XX**A**XXX	XXX**A**XX	XXXX**A**X	XXXXX**A**
CXXXXX	X**C**XXXX	XX**C**XXX	XXX**C**XX	XXXX**C**X	XXXXX**C**
DXXXXX	X**D**XXXX	XX**D**XXX	XXX**D**XX	XXXX**D**X	XXXXX**D**
⋮	⋮	⋮	⋮	⋮	⋮
VXXXXX	X**V**XXXX	XX**V**XXX	XXX**V**XX	XXXX**V**X	XXXXX**V**
WXXXXX	X**W**XXXX	XX**W**XXX	XXX**W**XX	XXXX**W**X	XXXXX**W**
YXXXXX	X**Y**XXXX	XX**Y**XXX	XXX**Y**XX	XXXX**Y**X	XXXXX**Y**

X = mixture of amino acids

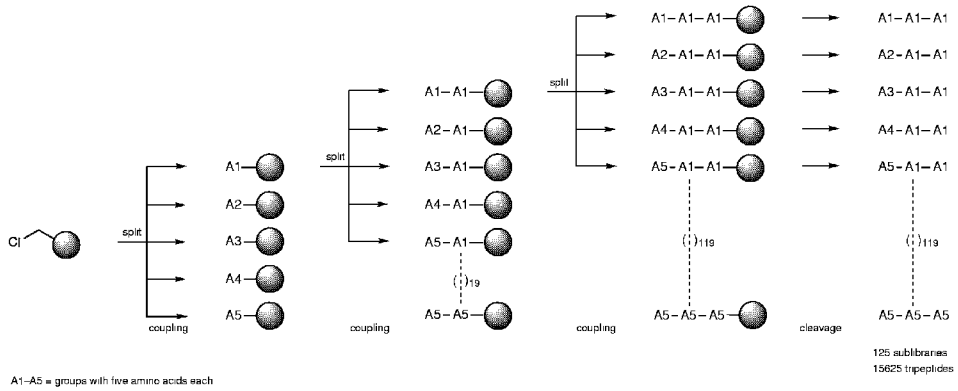
Synthesis of a Positional-Scanning Hexapeptide Library:[61]
MBHA–PSty resin (45.6 g) was divided into 114 aliquots (400 mg each) and placed into 114 polypropylene mesh packets, which were then sealed and labeled. A mixture of 19 Boc-protected amino acids (all proteinogenic amino acids except Cys) in a predetermined molar ratio in CH_2Cl_2 was activated with DIC, and coupled to packets 1 through 95 (6 equiv) over a period of 120 min. The same 19 amino acids were coupled individually to separate resin aliquots, i.e. packets 96 through 114, to give a defined amino acid at position six. After Boc-removal using 55% TFA/CH_2Cl_2 for 30 min, and washing with CH_2Cl_2, iPrOH, and 5% DIPEA/CH_2Cl_2, the 19 individual amino acids were coupled to packets 77 through 95, to give a defined amino acid at position five, and the amino acid mixture was coupled to the remaining packets. This procedure was repeated until all six positions were coupled with the correct arrangement of individual amino acids and amino acid mixtures at their respective positions. The peptide mixtures were deprotected, cleaved, extracted, and lyophilized as described for the dual-defined iterative hexapeptide library (Section 4.3.7.3.2.1), and then dissolved in H_2O (10 mg · mL^{-1}). This higher concentration, as compared to the dual-defined iterative library, takes into account the fact that each peptide mixture of the positional-scanning library contains 19 times more peptides than the dual-defined library (19^5 vs 19^4). This results in a 19-fold lower relative concentration of individual peptides within the mixtures of the positional-scanning library.

4.3.7.3.2.3 Orthogonal Libraries

As first described by Déprez et al.,[62] a two-dimensional orthogonal library is represented by two libraries, A and B, which have the exact same set of peptides. Both libraries are composed of separate peptide mixtures, i.e. sublibraries, each generated by coupling groups of five amino acids. For example, when using five groups each containing five amino acids, the two libraries making up a two-dimensional orthogonal tripeptide library are each composed of 125 (5^3) sublibraries, with each sublibrary containing 125 (5^3) individual peptides to give a total of 15 625 peptides in the library; this is illustrated in Figure 3 for library A. The amino acids used for this procedure are grouped differently for each of the libraries, i.e. each amino acid group used for the synthesis of library A has one, and only one, amino acid in common with each group used to synthesize library B (Figure 4). Consequently, each peptide of the orthogonal library is present in one sublibrary of A and one sublibrary of B, and that particular peptide is the only one these two sublibraries have in common.

for references see p 864

Figure 3 An Example of the Generation of a Two-Dimensional Orthogonal Tripeptide Library[62]a



a Reprinted from [Déprez, B.; Williard, X.; Bourel, L.; Coste, H.; Hyafil, F.; Tartar, A., *J. Am. Chem. Soc.*, (1995) **117**]. Copyright (1995). p 5405, with permission from American Chemical Society.

Figure 4 Amino Acid Groups Used for the Synthesis of a Two-Dimensional Orthogonal Tripeptide Library[62]a

		library A			
	group	A1	A2	A3	A4	A5
library B	B1	D-Leu	D-Pro	D-Ser	D-Phe(NO_2)	D-hGlu
	B2	D-Arg	D-Ile	Gly	D-Thr	D-hGln
	B3	D-Gln	D-Tyr	D-Val	D-C^{ε}-nicotinoyllysine	D-His
	B4	D-Trp	D-Asn	D-Glu	D-MetO	D-Ala
	B5	hexahydroiso-nicotinic acid	D-Lys	D-Tic	D-Asp	D-Phe

a Reprinted from [Déprez, B.; Williard, X.; Bourel, L.; Coste, H.; Hyafil, F.; Tartar, A., *J. Am. Chem. Soc.*, (1995) **117**]. Copyright (1995). p 5405, with permission from American Chemical Society.

Alternatively, orthogonal libraries can be generated by coupling individual amino acids in conjunction with mixing the intermediate peptide resins at certain stages of the synthesis. This is preferably done in the 96-well plate format by alternate mixing of the resins of both the rows and columns of the plate; thus, facilitating the generation of higher-dimensional orthogonal libraries.[63]

Synthesis of a Two-Dimensional Orthogonal Tripeptide Library:[62]
The library was synthesized on chloromethyl–polystyrene resin following Boc/Bzl strategy (Section 4.3.2.1). The resin was divided into two sets of 125 aliquots each (libraries A and B). The 25 protected amino acids listed in Figure 4 were grouped in two different ways by mixing the five amino acids of each group in equimolar amounts, generating groups A1 through A5 and B1 through B5, respectively. The amino acid groups were double-coupled in an equimolar ratio to resin-bound amino groups in a combinatorial manner, resulting in the generation of all possible combinations of the five groups within a trimer framework, i.e. 125 sublibraries with 125 peptides in each. The 250 sublibraries were cleaved from the resin by treatment with anhyd HF at 0 °C for 90 min in the presence of *p*-cresol and *p*-thiocresol. The cleaved mixtures were precipitated in Et_2O/pentane, dissolved in TFA, reprecipitated, and then lyophilized from AcOH.

4.3.7.4 Analysis of Combinatorial Libraries

As for all synthetic products to be tested in biological systems, a careful analytical characterization of peptide libraries is crucial in order to confirm their identity and establish their quality. Compared to individual peptides, however, the analysis of peptide libraries is complicated due to the fact that the peptides are either bound to a solid support or arranged in highly complex mixtures. This poses certain restrictions on which analytical methods can be used to characterize combinatorial libraries. For example, analytical methods that are based on the separation of product components, such as high performance liquid chromatography (HPLC) and capillary electrophoresis (CE), are only of limited use for the analysis of peptide libraries, in particular of those made up of complex mixtures (>100 peptides per mixture). The analytical methods beneficially applicable to peptide libraries include amino acid analysis, mass spectrometry, and sequencing.

4.3.7.4.1 Amino Acid Analysis

Amino acid analysis is often used to confirm the overall "equimolar" representation of amino acids within complex peptide mixtures[36] and is very useful for the analysis of amino acid incorporation after only one mixture coupling.[30] Amino acid analysis does not, however, provide information on the equimolar representation of peptides within complex mixtures containing more than one mixture position in the sequence, since the amino acid composition is measured after hydrolysis of the peptides. For example, the amino acid analysis results for a tripeptide mixture composed of the 20 peptides AAA, CCC, DDD, to VVV, WWW, and YYY, would be the same as the results obtained with an equimolar combinatorial tripeptide mixture composed of 8000 (20^3) peptides.

The amino acid analysis results of peptide mixtures should always be corrected for losses of amino acids through decomposition during hydrolysis, preferably by parallel analysis of a control mixture of known composition.

Amino Acid Analysis of Resin-Bound Peptide Mixtures:[30]
Resin-bound peptide mixtures (each approx. 5 mg, approx. 10 000 beads) containing one mixture position, introduced by coupling a mixture of 18 amino acids in a predetermined molar ratio (see Section 4.3.7.1.2), were hydrolyzed in 50% propionic acid/HCl at 130 °C for 2 h. Each hydrolysate was then submitted to amino acid analysis using a derivatizer–analyzer system (Applied Biosystems) in conjunction with PTH derivatization. Two samples of each peptide-resin were hydrolyzed and analyzed in parallel.

for references see p 864

The results were corrected as follows: Ten aliquots of a control resin mixture, prepared by physically mixing the respective 18 individual peptide resins, were hydrolyzed and analyzed, and an average value for each amino acid was determined. Deviations from the theoretical values for some amino acids obtained for this mixture reflected the experimental error inherent to amino acid analysis. The amino acid analysis data for the peptide mixtures prepared by coupling of amino acid mixtures were corrected by multiplying the values obtained for each amino acid by a correction factor. This factor was calculated by dividing the theoretical value for each amino acid (5.56 mol%) by the value obtained analyzing the physically prepared control peptide-resin mixture.

4.3.7.4.2 Mass Spectrometry

Using mass spectrometry techniques, the mass distribution of peptide mixtures can be experimentally determined and compared to the theoretical mass distribution; this distribution has the shape of a bell curve, and can be calculated using various computer programs.[64,65] These data provide important information on the composition of the analyzed peptide mixture. While it is not possible to verify the correct representation of each individual peptide within the mixture, the mass distribution curve can be used to confirm the correct overall composition of the mixture. It is also often possible to confirm the correct amino acid at a defined position of a peptide mixture. For example, the average mass (i.e., the "peak" of the mass distribution bell curve) of the mixture Ac-AXXXXX-NH_2 (716) is sufficiently different from that of the mixture Ac-YXXXXX-NH_2 (808). Furthermore, an incorrect mass distribution of the peptide mixture can indicate problems that occurred during the synthesis, such as incomplete side-chain deprotection of a particular amino acid.

The preferred mass spectrometry method for the analysis of peptide libraries is electrospray ionization (ESI),[66] one of the softest ionization techniques, enabling the formation of protonated ions, even from highly polar and/or thermolabile analytes. Fragmentation during ESI is generally low. The sample is introduced in dissolved form, thus, greatly facilitating sample preparation and injection, which can be automated using an autosampler. Thus, at a typical run time of one sample per minute, hundreds of samples can be analyzed overnight.

Electrospray Mass Spectrometry of Peptide Mixtures:[66]
Synthetic peptide mixtures (1–2 mg) were dissolved in a mixture of 1% aq HCO_2H/MeCN (1:1). Positive-ion ESI mass spectra were recorded using a triple-quadrupole mass spectrometer API III equipped with a nebulizer-assisted electrospray ("ion spray") source (Sciex, Thornhill, Ontario, Canada). The spectra were obtained as the sum of 5–10 scans at a step size of 0.2 μ and a dwell time of 5 ms.

4.3.7.4.3 Pool Sequencing

The term "pool sequencing" refers to the multiple sequence analysis through Edman degradation of peptide libraries composed of highly complex mixtures.[67] This analytical technique is valuable for the characterization of peptide libraries in that it enables the qualitative and quantitative determination of each amino acid at each position of the sequence, thus, not only verifying equimolar representation of amino acids at each position, but also reconfirming completeness of coupling at each step.

Sequence Analysis of Peptide Mixtures:[67]
Sequence analysis was carried out using a 477A protein sequencer (Applied Biosystems) equipped with an on-line PTH analyzer 120A, and the data analysis system 900A. A glass fiber filter disc activated with TFA was coated with BioBrene Plus (1.5 mg) prior to sequencing. Filters were not precycled, but a blank was run before each sequence run, followed by the standard program BEGIN-1 with a PTH amino acid (25 pmol) standard run. All other cycles were run using the program NORMAL-1. Calculations based on peak-height quantitation with respect to the standard run, as well as the results, were obtained as "tabulation reports", which list the amounts of PTH amino acids detected in each cycle.

4.3.7.4.4 Solid-State NMR and Fourier Transform IR Spectroscopy

Solid-state NMR and Fourier-transform IR (FTIR) spectroscopy are particularly useful for the characterization of compounds bound to a solid support and are mostly used to monitor reactions in solid-phase organic synthesis for the generation of nonpeptide libraries.

Recent developments in the areas of solid-state and gel-phase NMR spectroscopy have greatly facilitated the monitoring of solid-phase reactions. Gel-phase NMR[68,69] represents a combination of classical solution NMR and solid-state NMR spectroscopy in that it enables the measurement of a solid support swollen in a suitable solvent under the conditions of solution NMR spectroscopy. Although one-dimensional ^{1}H NMR spectra obtained in this manner cannot be interpreted due to strong line broadening, high-resolution ^{1}H NMR spectra can be obtained by combining magic-angle spinning,[70] a solid-state NMR method, with the above described gel-phase NMR spectroscopy, resulting in a greatly improved signal-to-noise ratio.[71]

When FTIR spectroscopy is carried out under defined conditions, it provides both qualitative and quantitative information on the performed solid-phase reaction.[72] Using difference measurements of the solid support before and after the reaction, absorption bands can be detected which otherwise would be seen only as weak shoulders. The absorption band of H–C–Cl stretching vibrations (1250 cm^{-1}), for example, can serve as the basis for the quantification of the coupling of carboxylic acids to a chloromethylated PSty–DVB resin.[73]

4.3.7.5 Library Screening and Deconvolution

The ultimate goal of all combinatorial library technologies is the discovery of novel compounds with strong and selective bioactivities. Therefore, library screening and deconvolution are an integral part of the library technology, where the term deconvolution refers to the algorithm and method of identifying individual active compounds within the library. This method is closely related to the library format, i.e. the library format determines the mode of deconvolution for a particular library.

4.3.7.5.1 Of One-Bead-One-Peptide Libraries

Most one-bead-one-peptide libraries (see Section 4.3.7.3.1) are resin-bound and screened in direct solid-phase binding assays. In a typical binding assay, the library beads are incubated with the target protein labeled with a reporter molecule, such as an enzyme that catalyzes a reaction; generating a colored compound, which in turn stains the beads to which the peptides that bind to the target protein are tethered.[4] The positive (colored) beads are identified among the vast majority of noncolored beads, isolated, and submitted to structure determination of the peptides bound to them.

Edman Degradation

Edman degradation is the method of choice to determine the sequence of peptides without N-terminal modification directly from the beads.[74] For the sequencing of peptides containing nonproteinogenic amino acids, a modified gradient for the chromatographic separation of the respective PTH amino acids is used.[75] During the sequencing of cyclic peptides with free N-termini (e.g., cyclic disulfide peptides and side-chain-to-side-chain cyclized lactam peptides), the first (N-terminal) amino acid involved in the ring formation is not detected at all, since it remains linked to the second amino acid involved in the ring formation, which is later detected as a modified amino acid.[63]

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Tagging of Library Compounds

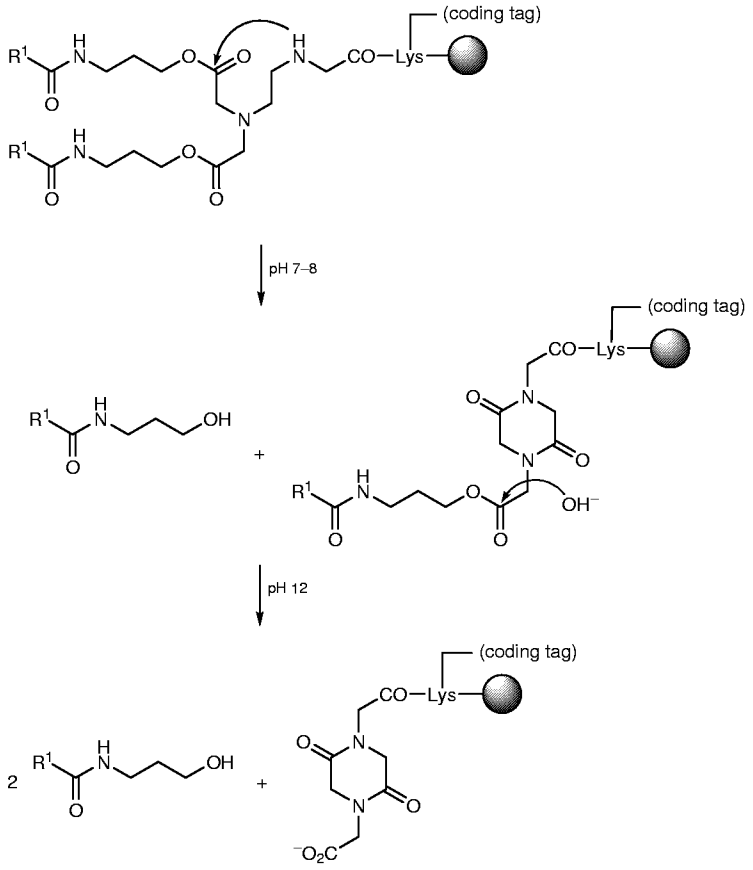
Sequencing through Edman degradation is not feasible for peptides containing anything other than α-amino acids and/or other nonsequenceable building blocks. For the deconvolution of libraries containing such "unnatural" peptides, several strategies for the coding of library structures have been developed. The initial coding method was based on tagging the resin beads with oligonucleotides, in which each nucleotide triplet encoded one building block used for the library sequence.[76,77] The structure of the compound on an "active" bead can then be decoded by sequencing the tag. The feasibility of this method, however, is limited due to the incompatibility of the rather sensitive oligonucleotide chemistry with many reaction conditions in peptide and organic synthesis. Other coding tags for nonsequenceable library compounds include peptides composed solely of proteinogenic amino acids,[78,79] as well as silylated halohydrocarbons, which can be separated by gas chromatography.[80,81]

The on-resin screening of one-bead-one-peptide libraries for enzyme inhibitors is possible when a quenched fluorogenic substrate, converted into a fluorescent molecule by the enzyme, is present on each bead along with the library peptide.[82] After incubation of the library beads with the enzyme, most beads fluoresce, since the substrate is cleaved and the fluorescent part of the molecule remains attached to the bead. Only beads containing peptides that inhibit the enzyme do not fluoresce, since the substrates bound to these beads are not converted.

Two-Step Peptide Release for Testing in Solution

Many bioassays, for example, those involving membrane-bound receptors or cells, require the assayed compounds be in solution. Since the release and testing of individual peptides from one-bead-one-peptide libraries composed of millions of peptide beads is unrealistic, several multiple-cleavable linkers have been developed for the stepwise release of peptides from the beads.[32,83–85] Using these linkers, resin beads can be pooled for the first release from one linker arm (e.g., 500 beads per pool), resulting in peptide mixtures in solution. After identifying the most-active mixture(s) in the bioassay of interest, the beads from which these active mixtures have been released are distributed individually, followed by releasing and testing individual peptides from the second linker arm in order to identify the most-active individual peptide(s). A double-cleavable linker based on iminodiacetic acid is shown in Scheme 5.[32] Two copies of the peptide are synthesized on the two chemically equivalent linker arms, since iminodiacetic acid is a symmetrical molecule. The first copy is released through piperazine-2,5-dione (DKP) formation at pH 7–8, and the second copy through alkaline ester hydrolysis. A third copy of the peptide, or a coding tag for nonpeptide library compounds, remains attached to the bead and, if the bead is identified as active, is sequenced later in order to identify the structure of the peptide (or other library compound).

Scheme 5 Two-Step Release of Library Compounds from a Double-Cleavable Linker Based on Iminodiacetic Acid.[32]



Two-Stage Release from a One-Bead-One-Peptide Library Screened in Solution:[29]
The peptide library was synthesized on the double-cleavable iminodiacetic acid linker (Scheme 5) using DCR synthesis (see Section 4.3.7.1.1). The library beads were suspended in a pH 4.5 buffer containing 1% carboxymethyl cellulose in order to slow down bead sedimentation. The resin suspension was then pipetted into 96-well filter plates, so that each well contained approximately 500 beads. The buffer was removed by vacuum filtration, and the peptides released via piperazine-2,5-dione formation from the first linker arm by incubating the beads overnight in a buffer or cell culture medium at pH 7–8, resulting in mixture solns of 500 peptides per well. These mixture solns were vacuum-filtered into standard 96-well plates, where they were tested for the bioactivity of interest in order to identify the most-active mixture(s) in that particular bioassay. The beads from which the identified active mixtures originated were then distributed one bead per well in a new filter plate using microforceps and a low-power stereomicroscope. The individual resin beads were incubated overnight with 0.2% NaOH to release the peptides from the second linker arm. The pH was adjusted to neutral, the peptide solns filtered into an assay plate, and tested. Alternatively, the beads were incubated overnight in NH_3 vapor, during which the peptides were cleaved but remained physically attached to the beads. The peptides were extracted by shaking the beads gently in the screening buffer before filtering the peptide solns into the assay plates. The tags on the beads from which the most-active individual peptide(s) originated where then sequenced in order to determine the structure of the active peptides (or other library compounds).

4.3.7.5.2 Of Mixture-Based Libraries

Combinatorial libraries composed of separate peptide mixtures with one or more defined positions in the sequence are typically in solution (as opposed to one-bead-one-peptide libraries) and can, therefore, be screened in a variety of bioassays, including those involving

for references see p 864

membrane-bound receptors and whole cells. Furthermore, the systematic array (i.e., the use of defined and mixture positions) eliminates the need for any kind of coding for building blocks other than α-amino acids, since the identification of the structure of active compounds within the library is inherent to the library screening and deconvolution process (*vide infra*).

Mixture-based libraries have been used to identify a variety of novel, highly active compounds, including receptor ligands,[6,56,86–88] antimicrobial compounds,[45,89–91] enzyme inhibitors,[9,25,35,92,93] antigenic peptides,[58,59,94–96] and catalytic peptides,[44] as well as to study other protein–ligand interactions, such as enzyme substrate specificity[97,98] and antigen recognition by T-cells.[21,99–101]

It should be noted, however, that mixture-based libraries are not limited to the solution state, but have also been screened while bound to a solid support, such as plastic pins,[2] cellulose membranes,[27,98] or resin beads.[49,50] This assay format is favorable for direct-binding assays involving soluble acceptors.

4.3.7.5.2.1 Iterative Deconvolution

The term "iterative library deconvolution" describes a process consisting of repetitive synthesis, screening, and selection steps, during which all mixture positions of the library (see also Section 4.3.7.3.2.1) are successively defined, resulting in the identification of individual peptides with the desired bioactivity.[3] At each step of this iterative process, the number of peptides per mixture is reduced by a factor equal to the number of amino acids in the mixture position that was defined at that step. At the final step of the process, individual peptides are synthesized and tested. For a hexapeptide library with two defined and four mixture positions, the iterative deconvolution consists of four cycles of synthesis, screening, and selection (see Table 2).[52]

Table 2 Screening and Iterative Deconvolution of a Dual-Defined Hexapeptide Library against mAb 17/9[52]

Step	Peptide Mixture[a]	Number of Mixtures× Peptides per Mixture	Total Number of Peptides	IC_{50} (nM)
screening	Ac-OOXXXX-NH_2	400×130321	52128400	
selection	Ac-DVXXXX-NH_2	1×130321	130321	58000
screening	Ac-DVOXXX-NH_2	20×6859	137180	
selection	Ac-DVPXXX-NH_2	1×6859	6859	15000
screening	Ac-DVPOXX-NH_2	20×361	7220	
selection	Ac-DVPDXX-NH_2	1×361	361	1100
screening	Ac-DVPDOX-NH_2	20×19	380	
selection	Ac-DVPDYX-NH_2	1×19	19	40
screening	Ac-DVPDYO-NH_2	20×1	20	
selection	Ac-DVPDYA-NH_2	1×1	1	2

[a] O = one of the 20 proteinogenic amino acids; X = a mixture of 19 proteinogenic amino acids excluding Cys (see Section 4.3.7.3.2.1).

Iterative Deconvolution of a Dual-Defined Hexapeptide Library by ELISA:[52]
An N-terminal acetylated and C-terminal amidated hexapeptide library (Ac-OOXXXX-NH_2), composed of 400 separate peptide mixtures,[3] was screened against the monoclonal antibody mAb 17/9, which recognizes the antigenic determinant DVPDYA.[102] The 400 peptide mixtures making up the library were tested by competitive ELISA for binding to mAb 17/9, and their relative affinities expressed as IC_{50} values (i.e., the concentration of peptide mixture necessary for 50% inhibition of binding of the known antigen to the antibody). Among those 400 mixtures, Ac-DVXXXX-NH_2 was found to have the strongest affinity (IC_{50} = 58000 nM). The iterative deconvolution process for this interaction is illustrated in Table 2.

Twenty new peptide mixtures (Ac-DVOXXX-NH_2) with three defined and three mixture positions were synthesized and tested, of which Ac-DVPXXX-NH_2 (IC_{50} = 15000 nM) was the most-active mix-

ture. The next two steps of the iterative process involved the synthesis and testing of 20 new peptide mixtures at each step (Ac-DVPOXX-NH_2 and Ac-DVPDOX-NH_2, respectively); the mixtures Ac-DVPDXX-NH_2 (IC_{50} = 1100 nM) and Ac-DVPDYX-NH_2 (IC_{50} = 40 nM) had the strongest affinities to mAb 17/9. The increase in affinity, expressed as decreasing IC_{50} values, of the most-active mixtures during the iterative process reflected the decreasing complexity of the peptide mixtures as more positions were defined, which lead to an increase of the relative concentration of individual peptides within the mixtures. At the final step of the iterative process, 20 individual peptides, Ac-DVPDYO-NH_2, were synthesized and tested. Ac-DVPDYA-NH_2 (IC_{50} = 2 nM) was identified as the peptide with the strongest affinity to mAb 17/9. This sequence exactly matched the antigenic determinant found in earlier studies to be recognized by mAb 17/9.

4.3.7.5.2.2 One-Step Deconvolution of Positional-Scanning Libraries

The principal advantage of the positional-scanning library format (see Section 4.3.7.3.2.2) is the fact that it enables the identification of active individual compounds following a single assay; thus, avoiding the iterative synthesis and screening process associated with the iterative library format. Screening of the 120 mixtures making up the six sublibraries (20 mixtures per sublibrary) of a positional-scanning hexapeptide library (see Figure 2) in a bioassay provides information on which amino acids are preferred at each of the six positions of the sequence for that particular bioactivity, as well as about the relative specificity of each position (i.e., the fewer amino acids found to be active at a particular position, the more specific the position). Based on the screening data, individual peptides representing all possible combinations of the amino acids found to be most effective at each of the positions are synthesized and tested to identify the most-active peptides, as well as to establish their individual bioactivities.[20,57] Alternatively, each of the six sublibraries can serve separately as the starting point for an iterative synthesis and screening process, as described in Section 4.3.7.5.2.1.

Identification of Antigenic Peptides from a Positional-Scanning Hexapeptide Library:[20]
The 120 mixtures making up the positional-scanning hexapeptide library (Figure 2) were tested by competitive ELISA against monoclonal antibody mAb 125–10F3. Among the 20 mixtures of the first sublibrary (Ac-O_1XXXXX-NH_2), Ac-PXXXXX-NH_2 was found to have the strongest affinity. At the second and third positions, Tyr and Pro, respectively, were the most effective amino acids, since the mixtures Ac-XYXXXX-NH_2 and Ac-XXPXXX-NH_2 were the most-active mixtures within their respective sublibraries. Two amino acids (Asn and Pro) were found to be effective at the fourth position, three amino acids (Ile, Leu, and Asn) at the fifth, and two amino acids (Leu and Arg) at the sixth. Based on that screening data, a set of 12 (1×1×1×2×3×2) individual peptides representing all possible sequence combinations of the amino acids found to be effective at their respective positions was synthesized and tested, i.e. Ac-PYPNIL-NH_2, Ac-PYPNIR-NH_2, Ac-PYPNLL-NH_2, Ac-PYPNLR-NH_2, Ac-PYPNNL-NH_2, Ac-PYPNNR-NH_2, Ac-PYPPIL-NH_2, Ac-PYPPIR-NH_2, Ac-PYPPLL-NH_2, Ac-PYPPLR-NH_2, Ac-PYPPNL-NH_2, and Ac-PYPPNR-NH_2. Of these peptides, Ac-PYPNLR-NH_2 and Ac-PYPNIL-NH_2 were found to have the strongest affinity to mAb 125–10F3 (IC_{50} = 13 and 18 nM, respectively). The first five and four residues of these two sequences, PYPNL and PYPN, respectively, correspond to the sequence of the antigenic determinant known to be recognized by mAb 125–10F3 (PYPNLS).

4.3.7.5.2.3 Deconvolution of Orthogonal Libraries

The deconvolution of orthogonal libraries is based on the fact that each peptide of the library is present in one sublibrary of A and one sublibrary of B, and that particular peptide is the only one these two sublibraries have in common (see Section 4.3.7.3.2.3). Consequently, after having determined the most-active sublibraries in both libraries A and B in a given bioassay, individual active peptides are identified by deciphering the peptide that the active sublibraries of A and B have in common. If more than one sublibrary is active in each library, many individual peptides representing the common peptides of the different sublibrary pairs from both libraries have to be synthesized and tested in order to identify the

for references see p 864

most-active peptide(s).[62] For example, if two sublibraries are found to be equally active in both the A and B libraries, four individual peptides representing the peptides common to all four A/B sublibrary combinations must be synthesized and tested.

Deconvolution of a Two-Dimensional Orthogonal Tripeptide Library:[62]
The orthogonal library consisted of two libraries, A and B, each composed of 125 tripeptide-containing sublibraries, which represented all possible combinations of five groups of amino acids within a tripeptide framework (see Section 4.3.7.3.2.3 and Figures 3 and 4). The 250 peptide sublibraries making up the entire library were tested in a vasopressin (V2) receptor binding assay. In library A, the sublibrary with the sequence A3-A2-A4, in which A3, A2, and A4 represent the groups of amino acids shown in Figure 3, was found to be the most active, and B5-B3-B1 was the most active in library B. The two most-effective amino acid mixtures at the first position, A3 and B5, had D-Tic in common. At the second position, the most-active mixtures (A2 and B3) had D-Tyr in common, and at the third position, the most-active mixtures (A4 and B1) had D-Phe(NO_2) in common. This indicated that the tripeptide D-Tic-D-Tyr-D-Phe(NO_2) was the most-active individual tripeptide in the library. This peptide was subsequently synthesized and found to be a high-affinity ligand to the V2 receptor (IC_{50} = 63 nM).

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4.3.8 Parallel Synthesis

V. ANTONENKO

Advances in high-throughput screening have dramatically increased the ability to screen large numbers of chemical entities. Encoded synthetic libraries have been reduced in practice, producing hundreds of thousands of compounds for primary screening; however, this approach has limitations. The encoding–decoding process is often expensive, time-consuming, and not always compatible with all chemistries. Usually, a single bead assay format is required. The reliability of such assays depends strongly on bead-to-bead variations in the quantities of synthesized materials. Several library equivalents need to be screened to ensure coverage of all members of the library. In contrast, parallel synthesis does not require encoding, it produces discrete compounds in measurable quantities, and allows for the collection of a full set of SAR data. It is becoming increasingly popular among other combinatorial methods.

4.3.8.1 Spatially Addressable Synthesis on Laminar Solid Supports

Many biological assays can be done when the compounds to be analyzed are immobilized on a solid support; usually, only very small amounts of compounds are required for such assays. The cost of biological reagents is a very important factor in high-throughput screening. These constraints call for miniaturization of the systems designed for the simultaneous synthesis and subsequent screening of peptides in large numbers. Compartmentalization of individual synthesis and screening sites in conventional three-dimensional systems imposes strict limitations on the density of peptide arrays. These limitations have been successfully eliminated by synthesizing very dense peptide arrays on laminar solid-phase supports.

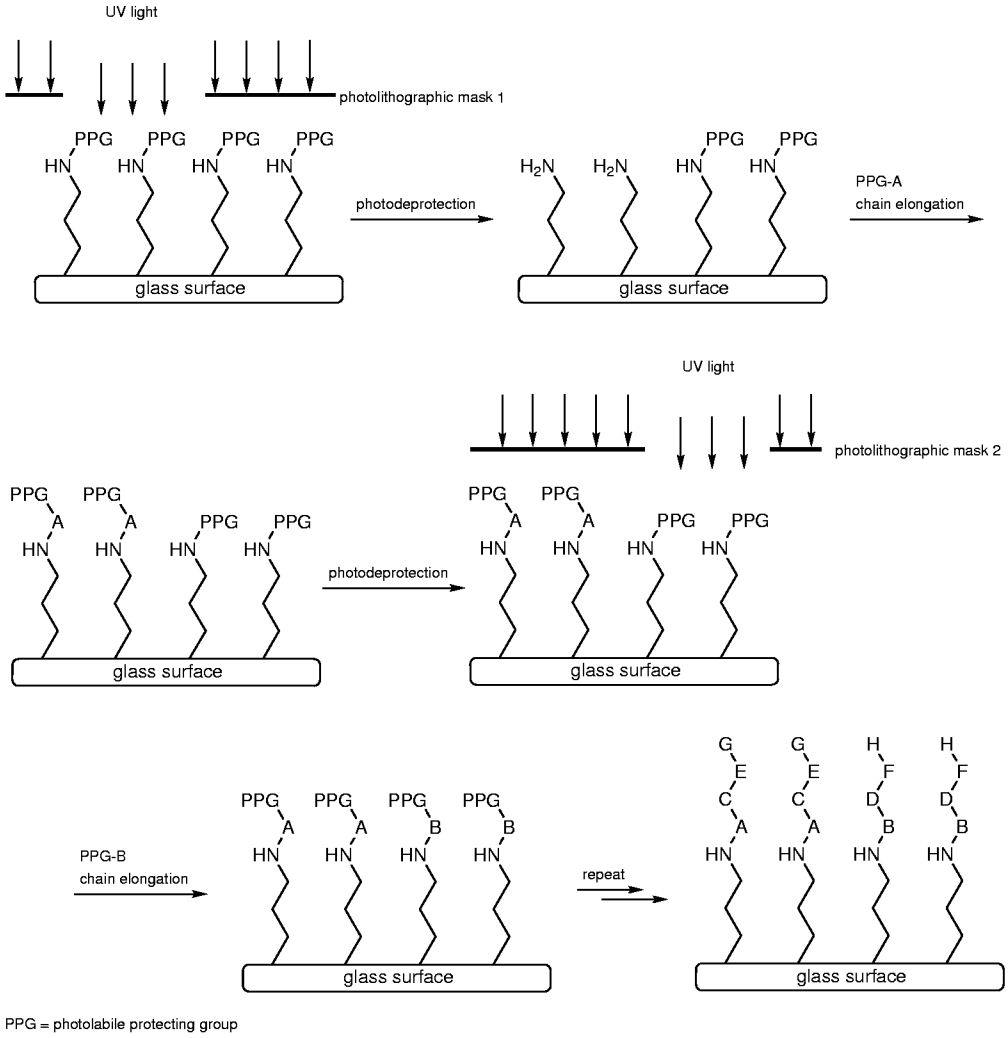
4.3.8.1.1 Light-Directed Synthesis

In 1991, scientists from the Affymax Research Institute combined photolithography, photochemistry, and solid-phase synthesis in a new technology[1–3] called light-directed, spatially addressable parallel synthesis; this technology has been reviewed.[4] The principal points of this technology are illustrated in Scheme 1.

The synthesis occurs on a flat glass surface, modified with an appropriate linker [e.g., (3-aminopropyl)triethoxysilane] to allow for the covalent attachment of protected amino acids. The entire synthesis area of the slide is derivatized with a photolabile protecting group (PPG). In the first step, selected sites of the synthesis area (typically 3 squares per slide, 1.28×1.28 cm^2 each) are exposed to UV light through photolithographic mask 1. The variety of patterns available for photolithography is essentially unlimited. Exposure to light then causes removal of the photolabile groups from the selected sites to give the free amino groups. In the next step, the synthesis area is treated with reagents necessary for the elongation of the peptide chain. Only sites previously exposed to UV light participate in the coupling reaction. The rest of the synthesis area remains protected. The synthesis is continued by illuminating a different part of the surface through photolithographic mask 2, followed by another peptide chain elongation reaction. By repeating photodeprotection and coupling steps, highly dense arrays, consisting of thousands of peptides, can be synthesized.

for references see p 876

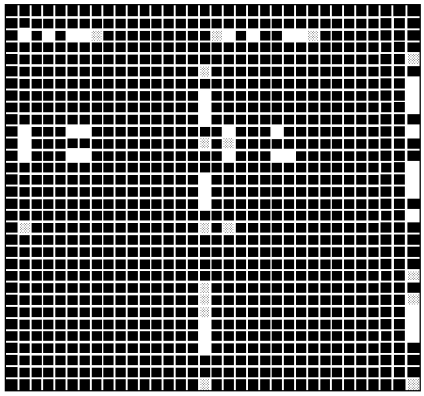
Scheme 1 The Concept of Light-Directed, Spatially Addressable Parallel Synthesis[1–3]



PPG = photolabile protecting group

Importantly, the primary structure of each peptide in the array is sufficiently defined by the sequence of coupling and photolysis steps, and by the photolithography mask patterns. Therefore, the structure can be easily deduced from the (x,y) coordinates of the peptide on the slide. This eliminates the need for encoding–decoding procedures required by some other combinatorial technologies. After synthesis is complete, the synthesis area is exposed to reagents necessary for the elimination of the side-chain protecting groups. To assess the binding properties of all the peptides synthesized, the entire array is incubated with a fluorescently labeled target molecule and scanned using a stage-scanning confocal fluorescence microscope.[1] The sites containing peptides that bind to the target become fluorescent. Affinity data on all peptides in the entire array are obtained in one step. An example of a fluorescence scan of 1024 undecapeptides synthesized on 1.28×1.28 cm^2 squares is presented in Figure 1.

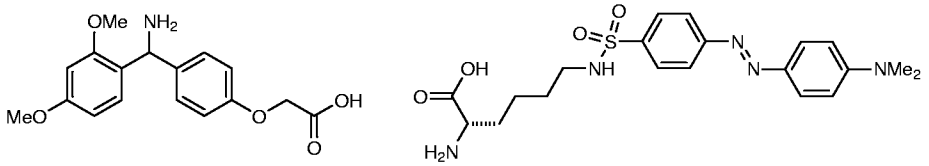
Figure 1 Fluorescence Scan of an Array of 1024 Undecapeptides



The consumption of the chemical reagents required for the syntheses of thousands of peptides comprising an array, or of biological reagents necessary for a bioassay, is very small because the capacity of the flat glass surface is only $5 \div 20$ pmol · cm^{-2}. Biological reagents used in this technology are recoverable and can be reused. Moreover, after performing an assay with one target molecule, the bound target can be easily dissociated from the array, for example, by treating with 6 M guanidine hydrochloride, making the array available for the subsequent screening with other targets. These arrays are reusable for at least six months.

Due to the very low capacity of the glass surface, traditional analytical tools cannot be used for adequate assessment of the fidelity of the light-directed synthesis. A highly sensitive analytical method to characterize the quality of peptides synthesized by this technology has been developed.[5] It involves derivatization of the surface with an acid-cleavable Knorr linker **1** followed by the attachment of a C-terminal chromophoric tag, Lys(DABS) (**2**) (Scheme 2). After completion of the synthesis, the resulting peptide containing C-terminal Lys(DABS)-NH_2 can be cleaved from the surface and characterized by HPLC. Its chromatographic behavior can then be compared with the retention time of the same compound synthesized using conventional methods and whose structure has been proven. This process proved to be crucial for optimizing all the methodological aspects of light-directed synthesis.

Scheme 2 Structures Used To Enable Characterization of Peptides Formed via Light-Directed Synthesis[5]

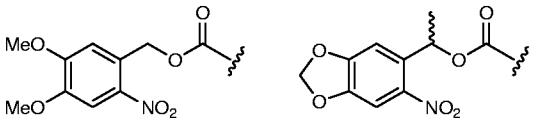


Originally, the photoremovable 4,5-dimethyl-2-nitrobenzyloxycarbonyl group (Nvoc, **3**) introduced by Patchornik (Scheme 3)[6] was utilized for temporary protection of the α-amino group of the growing peptide chain. However, photolysis under optimized conditions (12 min in 5 mM sulfuric acid/dioxane) sometimes causes side reactions and does not always proceed quantitatively.

Investigation of different α-amino photoprotective groups and conditions for their removal[5] resulted in selecting the α-methyl-substituted (6-nitro-1,3-benzodioxol-5-yl)methyloxycarbonyl (α-methyl-substituted Npoc) group[7] **4** (Scheme 3). It can be quantitatively removed by photolysis under neutral conditions in 3–4 min.

for references see p 876

Scheme 3 Structures of Photoremovable Groups Nvoc and α-Methyl-Substituted Npoc[6,7]



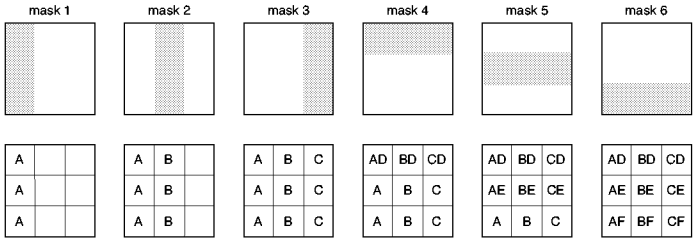
3 4

The flat glass surface behaves in a noticeably different way from conventional solid phase peptide synthesis supports.[5] Usually, higher concentrations of activated species (0.1–0.25 M) are necessary to achieve acceptable coupling efficiency. A number of coupling reagents have been examined and HATU[8] provided the most satisfactory results with short (5–15 min) coupling times.[5]

With the same number of chemical steps, the diversity and number of peptides synthesized using light-directed, spatially addressable chemical synthesis depends, among other factors, on the patterns of the photolithographic masks used for each photodeprotection. Different masking strategies can be used.

An example of an orthogonal stripe masking strategy for a six-step synthesis is shown in Figure 2. The reactants are the ordered set {A, B, C, D, E, F}, which represents amino acids protected with an N^{α}-photoremovable group. Synthesis begins by exposing the left third of the entire synthesis area to UV light through photolithographic mask 1. Building block A is then added in the presence of a coupling reagent and reacts only at the previously illuminated region. UV light exposure of the middle third of the synthesis area through mask 2 activates it allowing for covalent attachment of building block B. Similarly, the right third of the synthesis region is derivatized with building block C. Building block D is added after illuminating the upper third of the surface through mask 4, thus creating in one coupling step three different chemical entities AD, BD, and CD. The process is continued using masks 5 and 6 and building blocks E and F, respectively, to give a set of nine dipeptides, AD, AE, AF, BD, BE, BF, CD, CE, and CF, from six starting amino acids.

Figure 2 Orthogonal Stripe Masking Strategy for a Six-Step Synthesis

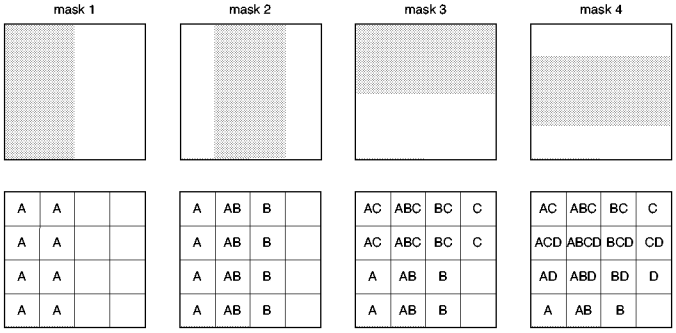


In general, orthogonal masking strategy allows generation of n^2 compounds from 2n starting materials in 2n chemical steps. For example, a set of 400 dipeptides can be formed from 20 L- and 20 D-amino acids using only 40 coupling reactions. This orthogonal stripe masking strategy can be continued to make longer peptides by subdividing previous stripes into a number of synthesis regions until the limit of photolithographic resolution is achieved (10–20 μm).[9] With this resolution, 250 000–1 000 000 compounds can be synthesized in 1 cm^2. Routinely, 50 μm resolution is practiced[1] and allows the production of 40 000 compounds in the same area.

Another masking strategy, binary synthesis, utilizing four photoprotected amino acids A, B, C, and D, is presented in Figure 3. In this approach, each building block is added after one half of the synthesis area is photolyzed. The left half is illuminated through mask 1 and is then derivatized with building block A, while the other half bears only the protected amino

group. Coupling of building block B after photolyzing the middle 50% of the synthesis region through mask 2 yields three compounds, A, AB, and B. During the third step of the process, the upper half of the area is exposed to UV light through mask 3, followed by coupling of building block C. This produces compounds A, AB, AC, ABC, B, BC, and C. Derivatization with building block D, following photodeprotection of the middle 50% of the synthesis area in a horizontal direction through mask 4, leads to the production of 16 unique regions of the glass support each, except one, bearing different chemical entities.

Figure 3 Binary Masking Strategy for a Four-Step Synthesis



Binary masking strategy generates 2^n-1 number of compounds in n chemical steps. A 16-step peptide synthesis would generate 2^{16} (65536) different areas, of which 65535 would represent different peptides or amino acids. This strategy also creates a distribution of chain lengths, ranging from 1 to n, with the maximum number of peptides being n/2 in length.[1] All possible permutations of amino acids used in the synthesis are generated while preserving their order of introduction. Interestingly, all possible deletion sequences as well as all possible truncation sequences are obtained. This strategy has been used in epitope mapping experiments with monoclonal antibody D32.39, which binds to the C-terminal region of dynorphin B. A binary synthesis of the decapeptide FLRRQFKVVT was carried out, generating 1023 different peptides. The results of this experiment identified heptapeptide RQFKVVT as the minimal binding sequence.[10]

Light-directed, spatially addressable synthesis is a powerful technology for generating chemical diversity not only of peptides, but also of oligonucleotides[1,11] and of other linear polymers.[12]

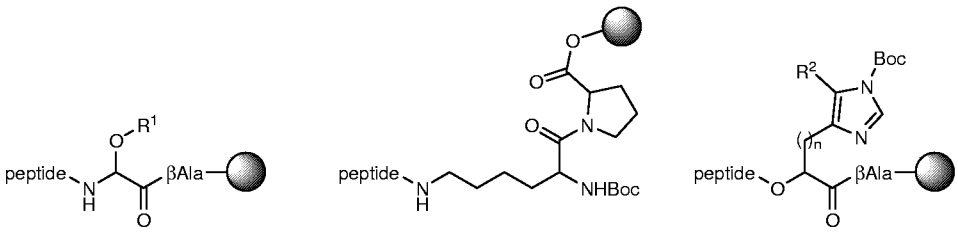
4.3.8.1.2 Spot Synthesis on Membrane Sheets

Frank reported[13–16] a method for the parallel synthesis of peptides on cellulose membranes derivatized with an appropriate linker. Reagents are delivered in drops to different sites on a flat surface. The volume of each drop, along with the other properties of the solid support, determines the diameter of the spot formed on the surface by the delivered reagents. This also determines the distance between the different spots and, therefore, the density of the array. Solvents of low volatility are used to prevent evaporation before a chemical step is complete. Reagents can be added to different sites on the array both manually and automatically, using liquid-handling equipment. In the course of the synthesis, common synthetic steps, such as deprotection of temporary protective groups and washing procedures, are carried out by treating the entire membrane with necessary reagents and solvents. At the end of the synthesis, all side-chain protecting groups are removed. The peptides comprising the array can be screened on the solid support. Alternatively, if a cleavable linker is used [e.g.,

for references see p 876

structures **5**,[17] **6**,[18] and **7**[19](Scheme 4)], individual synthesis sites can be physically separated and their peptides released in solution by an appropriate reagent.

Scheme 4 Cleavable Linkers for Spot Synthesis and Membrane Sheets[17–19]



5 R^1 = Me, iPr, tBu, TBDMS, SEM, Mom, Mem

6

7 n = 1; R^2 = H, Me
n = 2; R^2 = H

The amount of each peptide in the array is easy to control by the size of each spot. For example, synthesis on Whatman 540 paper (capacity 0.4–0.6 μmol · cm^{-2}) with 0.5–0.7 μL of reagents used for the coupling step yields 25 nmol of a peptide on a spot 7 mm in diameter.[17] These parameters are used in a standard format, an 8 × 12 array with 9-mm distance between the centers of the synthesis sites. Larger arrays of 40 × 50 can be prepared on Whatman 50 paper with the help of automation. Drops 0.03–0.05 μL in volume form spots that are 1 mm in diameter, producing an array containing 2000 peptides, 1.0 nmol each.[17]

The spot synthesis technique is simple and easily accessible for practical use. There are some limitations to the scope of chemistries that can be used with cellulose as a solid support. To eliminate some of these shortcomings, Gao and Esnouf[20,21] proposed using a different membrane, Immobilon AV-1, for spot synthesis. Before the synthesis of peptide arrays can be carried out, this polyvinylidene fluoride based material is derivatized with ethane-1,2-diamine, followed by coupling of Fmoc-βAla as a spacer. In most other aspects, this peptide array synthesis is very similar to the spot synthesis proposed by Frank.[13–16]

Spot Synthesis on Membrane Sheets:[17]

Derivatization of the Membranes:
A sheet and two small 1-cm^2 pieces of acid-hardened paper, such as Whatman 540 or Whatman 50, were dried overnight in a desiccator at 100 Pa. The sheet and the pieces were then soaked in 0.2 M Fmoc-βAla or Fmoc-Pro (2 mL), 0.24 M DIC, and 0.3 M of NMI in dry NMP and kept in a closed container for 1–4 h. One small piece was taken out, placed in a small beaker, washed with DMF and 20% piperidine in DMF for 5 min, and stained with 1% bromophenol blue in DMF until the supernatant remained yellow. After washing with EtOH or MeOH, the piece was dried with a fan and placed in a glass vessel containing 20% piperidine in DMF (5 mL). The optical density of the dark blue solution was measured and the loading of free amino groups was calculated (bromophenol blue ε_{605} = 950000). Loading should be within the range 0.2–0.4 μM · cm^{-2}. The sheet was washed twice in a flat trough with 2% Ac_2O in DMF (20 mL), left overnight in the same solution (20 mL), and washed with DMF (3 × 20 mL). Then, the sheet was treated with 20% piperidine/DMF (4 × 20 mL) for 15 min, washed with DMF (20 mL) for 2 min and EtOH or MeOH (3 × 20 mL) for 2 min, and dried in a desiccator at 100 Pa overnight. The sheet was stored at − 20 °C.

Generation of the Spot Array:
A list of peptides to be synthesized was prepared and the number of peptides in the required array, the spot size, and the scale were defined. The spot positions were marked on the membrane with pencil dots for manual synthesis. The membrane sheet was placed in the reaction trough. An NMP solution, which was 0.3 M with respect to the second anchor compound [e.g., Fmoc-βAla or Boc-Lys(Fmoc)], 0.45 M with respect to HOBt, and 0.35 M with respect to DIC, was prepared and allowed to stand for 30 min. Aliquots of this solution were then spotted onto all positions according to the chosen array configuration. After 15 min, the spotting was repeated and reactions were allowed to proceed for 30 min. The sheet was washed with 2% Ac_2O/DMF (20 mL) for 30 s, once again for 2 min, and finally left in the solution overnight. The array was washed with DMF (3 × 20 mL), incubated for 10 min with 20% pi-

peridine in DMF (20 mL), and washed again with DMF (4 × 20 mL). The effectiveness of the washing step was tested by incubation with 1% bromophenol blue in DMF (20 mL). The solution should be yellow and the spots light blue. Washes with DMF were repeated if the solution turned dark blue due to traces of any remaining piperidine. The array was washed with EtOH or MeOH (2 × 20 mL). After drying between a folder of Whatman 3MM paper using cold air from a hairdryer, the sheet was sealed in a plastic bag and stored at − 20 °C.

Assembly of Peptides on the Spots:
The spot positions on the membrane were numbered with a pencil, and the membrane was placed in a reaction trough. Solutions of all the Fmoc-protected amino acids (1.0 mmol) necessary for the first coupling in 0.25 M anhyd HOBt (5 mL), were prepared and activated with DIC (4 μL) for 30 min. Aliquots of these solutions were spotted onto their corresponding position in the array. After 15 min, spotting was repeated and reactions were allowed to proceed for another 30 min. The array was washed with 2% Ac_2O/DMF (20 mL) for 30 s, once again for 2 min, and then incubated for 10 min until all the remaining blue color disappeared. The sheet was washed with DMF (3 × 20 mL), followed by incubation with 20% piperidine/DMF (20 mL) for 5 min. The effectiveness of the washing step was tested by incubation with 1% bromophenol blue in DMF (20 mL). The solution should be yellow and the spots light blue. If necessary, the washing step was repeated. The sheet was washed twice with EtOH or MeOH (20 mL). After drying between a folder of Whatman 3MM paper using cold air from a hairdryer, the array was ready for the next peptide Whatman elongation reaction.

Side-Chain Deprotection:
The dried membrane was placed in the reaction trough and a freshly prepared solution containing TFA/CH_2Cl_2/H_2O/iBu_3SiH (50:45:2:3, 20 mL) was added. The trough was tightly closed and deprotection was allowed to proceed for 1 h. The deprotection solution was replaced with fresh solution (20 mL) and then drained after 1 h. The array was washed with CH_2Cl_2 (4 × 20 mL) for 10 min each time, DMF (3 × 20 mL) for 10 min each time, 1 M AcOH (3 × 20 mL) for 10 min each time, and finally EtOH or MeOH (3 × 20 mL) for 2 min each time. The sheet was then dried with cold air and stored at − 20 °C, or directly used for bioassay.

Peptide Elution:
This procedure is applicable for safety-catch linkers of types **5**, **6**,[18] and **7**.[19]

The dried membrane was placed in the reaction trough and a freshly prepared solution containing TFA/CH_2Cl_2/H_2O/iBu_3SiH (50:45:2:3, 20 mL) was added. The trough was tightly closed and deprotection was allowed to proceed for 1 h. The deprotection solution was replaced with fresh solution (20 mL) and drained after 1 h. The array was washed with CH_2Cl_2 (4 × 20 mL) for 10 min each time, 0.1% HCl in MeOH/H_2O (1:1) (3 × 20 mL) for 15 min each time, and 1 M AcOH (3 × 20 mL) for 10 min each time, and dried overnight in a vacuum desiccator. The membrane was cut into pieces containing individual spots and the pieces were placed into separate plastic tubes. A buffer of pH 7–8 and sufficient capacity was added (to dissolve more hydrophobic peptides, DMSO or EtOH may be added). The tubes were closed and shaken overnight at 30–37 °C so that the membrane pieces were gently agitated. Obtained solutions were kept at − 20 °C.

4.3.8.2 Manual Methods for Resin-Based Synthesis

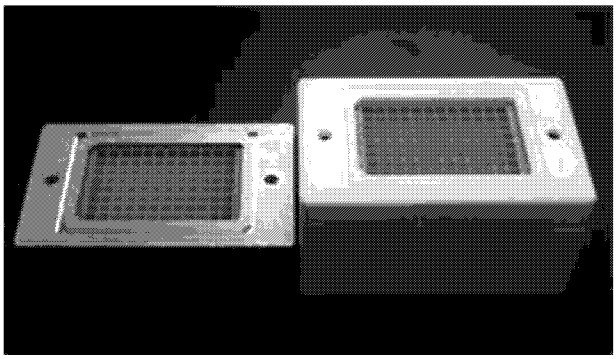
The 96-well microtiter format has been a common platform for high-throughput screening for years. It also offers many advantages as a platform for parallel synthesis. There are multiple-liquid handling devices, e.g. multichannel pipets, robotic systems, etc., that significantly simplify and accelerate the process of the reagent delivery to the wells on the plate. Moreover, all 96 compounds can be handled as one synthesis entity. Their structures are spatially encoded by their location on the plate. Contemporary analytical instrumentation, such as mass spectrometers and HPLC systems, can also easily handle compounds in the 96-well microtiter format. Combined with the commonly used 96-well based systems for screening, it eliminates the need for time-consuming redistribution and relabeling of a large number of synthesized compounds for analytical and biological characterization.

for references see p 876

High-Throughput Organic Parallel Synthesis System

The Affymax Research Institute has developed an original approach for the synthesis of organic compounds in 96 deep well plates. The high-throughput organic parallel synthesis (HiTOPS) system[22] (Figure 4) utilizes a variety of 96 deep well filtration microtiter plates available from Whatman Polyfiltronics.

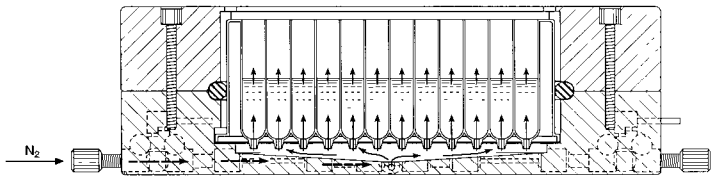
Figure 4 The High-Throughput Organic Parallel Synthesis System with a Synthesis Device Shown on the Left and the Cleavage Device on the Right[22]



The plates are made of polypropylene and other polymers and are available with a selection of different filters. The volume of each well is 2 mL and allows for the use of up to 50 mg of resin. For larger-scale syntheses, several wells or even the entire row or column can be used for the preparation of the same peptide. Reactants are retained in wells by the positive pressure of an inert gas (Figure 5).

At the beginning of the synthesis, a filtration plate is placed into a synthesis device. The bottom portion of the device has an opening in the center connected to an internal channel, which leads to two valves through a T-connector. One valve regulates the delivery of an inert gas to the bottom of the plate. The other valve links the bottom with a waste container connected to a vacuum. When the inert gas valve is open, positive pressure is created at the bottom. Resins and reagents can be delivered to the plate and kept in place by the positive pressure for the duration of the synthesis. The pressure can be increased to allow the inert gas to bubble through the wells, providing mixing and creating an inert atmosphere (Figure 5). Mixing can also be achieved by either placing the device on an orbital shaker, or by using magnetic stir bars in each well. At the end of a chemical step, liquid reagents or washing solvents can be removed into the waste container by opening the vacuum valve. At all times the plate remains in the synthesis device. There is no need to disassemble the device during the course of the synthesis, making the process very ergonomic. One person can also easily operate four devices simultaneously, synthesizing 384 compounds.

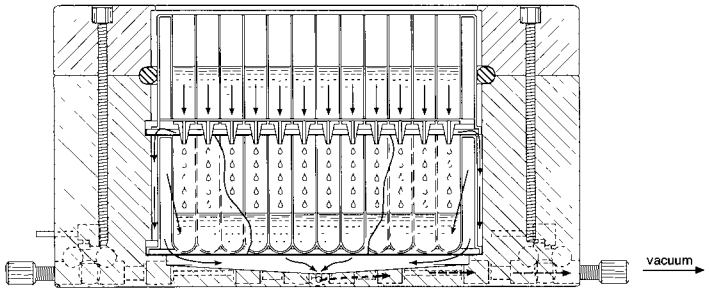
Figure 5 Synthesis Device Showing the Retention of Reactants by the Positive Pressure of Nitrogen[22]



For the final cleavage, the plate is transferred to the cleavage device (Figure 6), which is made of Teflon. It is deeper than the synthesis device, in order to accommodate a collection plate, but otherwise has the same design.

After creating positive pressure at the bottom, a cleavage reagent is delivered to the top plate, which contains all the resins with the synthesized peptides. Upon completion of the cleavage, a vacuum is applied and the filtrates are collected in the 96 deep well collection plate at the bottom. The collection plate is then placed in a vacuum centrifuge to eliminate the cleavage mixture.

Figure 6 Cleavage Device in Which the Products are Cleaved and Collected[22]



One of the advantages of this method is the commercial availability of the above devices. The HiTOPS device has been used in the synthesis of combinatorial libraries of piperazinediones[23] for the development of highly selective inhibitors of collagenase-1.

4.3.8.3 Synthesis on Macroscopic Polymer Supports

Traditionally, solid-phase synthesis of peptides is carried out on polymeric supports comprised of small beads of a polymer. In parallel synthesis approaches that are not spatially addressable, there is a need for the easy manipulation of all members of the array during the course of the synthesis. This is often easier when each member of the array is synthesized on a single macroscopic piece of solid support, or on a small batch of resin enclosed in an appropriate shell. Such macroscopic supports are also easy to label.

Synthesis on Multipins

In 1984, Geysen[24] introduced an alternative to synthesis on polymeric beads. As a solid support he proposed to use "pins", reusable polyethylene rods with 4-mm diameter and 40 mm in length, grafted with polyacrylic acid. The pins are attached to a supporting block arranged in an 8×12 array to fit into a 96-well microtiter plate. The 96-well plate is used as 96 separate reaction vessels; therefore, each well can be used for the attachment of a different building block. Washing and deprotection steps can be carried out in a separate common reaction vessel. After the synthesis is complete and side-chain protecting groups have been removed, peptides can be assayed while attached to the pins.[25,26] If a cleavable linker is used, individual peptides are separated from the solid support in a 96-well microtiter plate containing appropriate reagents,[27,28] or by exposing the array to vapor-phase ammonolysis.[29,30] Originally, the capacity of each pin was 10–100 nmol. Later, a much greater variety of multipins of different capacities with a wide selection of linkers became commercially available from Chiron Technologies Pty Ltd.

for references see p 876

Parallel Peptide Synthesis in "Tea Bags"

In 1985, Houghten suggested synthesizing peptides on a resin sealed in porous polypropylene packets.[31] The polypropylene mesh is porous enough (74 μm) to allow for the free access of chemicals to and from the contained resin. Each packet or "tea bag" can be individually labeled to identify the peptide synthesized on the entrapped polymer. Many "tea bags" can be combined in the same reaction to carry out common synthetic steps, such as deprotection and washing steps. The packets are sorted into separate reaction vessels according to the specific amino acid that is to be coupled next. Cleavage, depending on the amount of used resin, can be carried out in separate vessels or in a 96-well microtiter plate with 2-mL well volume. In the original paper,[31] 248 different tridecapeptides were synthesized in 10–20 mg quantities and characterized in less than four weeks.

The "tea bag" method is very practical. It does not require any special tools, except a sealing device. A number of mesh materials of different porosity are available from Spectrum Medical Industries, Inc. Almost all commercially available resins can be used as long as the size of the beads is larger than the mesh size. The scale of the synthesis is easy to control by the size of the packet chosen to contain the desired amount of a polymer support. The multiple synthesis can be carried out manually or on a synthesizer.

Synthesis on Paper, Cotton, and Other Laminar Supports

Similarly to the "tea bag" approach, pieces of any solid-phase support suitable for solid-phase synthesis can be used for parallel peptide preparation. One of the first materials used in this manner was paper. Frank used paper disks[32,33] (Whatman 3MM, 1.5-cm diameter) packed into the columns of a multicolumn continuous-flow synthesizer. Prior to the synthesis, the paper was derivatized with 4-alkoxybenzyl (Pab) linkage to allow for the cleavage with trifluoroacetic acid. The disks can be easily labeled with a pencil, sorted, and combined depending on the common amino acid to be coupled at the next step.

Cotton has mechanical properties superior to paper and has been used in a similar way by Lebl.[34,35] Polystyrene-grafted polyethylene film has also been used by researchers from Rockefeller University.[36,37] Researchers from Pfizer used material prepared from resin sandwiched in between two sheets of polypropylene mesh.[38] The "sandwich" was fused together with a polymer with a low melting point.

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4.4 Multicomponent Reactions

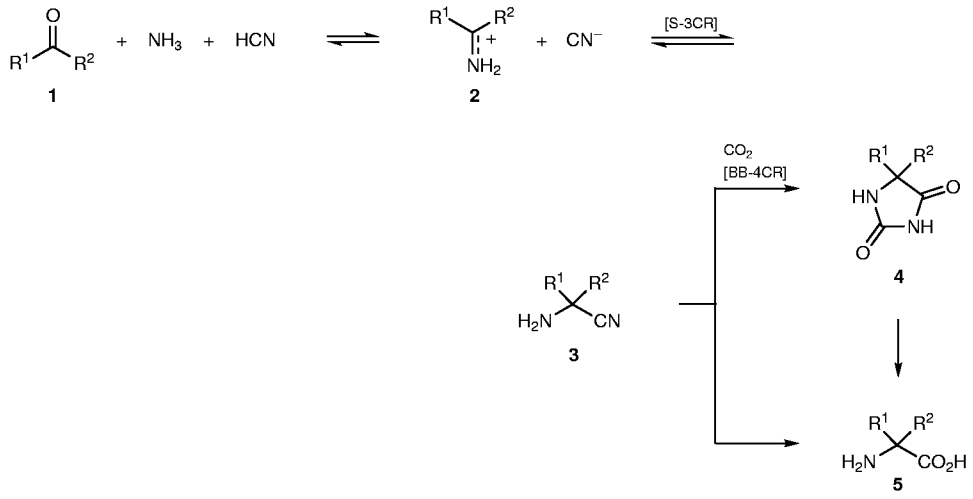
I. Ugi, A. Dömling, and B. Werner

Many chemical processes take place between one or two reactants, and in principle all such reactions will equilibrate. In practice, however, the actual preparative procedures proceed irreversibly and very high yields of pure products can result if no competing events participate.[1–4] A usual multistep synthesis where the product is formed from three or more educts consists of a number of successive reaction steps. The intermediate product of each step is isolated, purified, and reacted with the next educt in a further reaction step, until the final product is formed. In many syntheses, particularly in those of peptide derivatives and related compounds, some of the participating functional groups are activated, while others are protected and later deprotected.[5] The more preparative steps that are carried out, the more work there is to be accomplished toward obtaining the final desired product, and the overall yields of the final products decrease with each additional step.

In contrast to multistep syntheses with each step involving the reaction of at least two reactive groups, the starting materials of multicomponent reactions (MCRs) form their final products simply by mixing the components. These one-pot reactions are not necessarily direct conversions of the reactants into their products, but are also sequences of subreactions between (usually) two participating components. MCRs have their own automatic programs of subreactions that are determined by the structures of the participating reactants.

Three basic types of MCRs can take place.[4,6] MCRs of type I are equilibria that form mixtures of reactants, intermediate products, and final products, whereas the reactants and intermediate products of MCRs of type II equilibrate, but their final products are in practice formed preferentially and irreversibly. MCRs of type III are sequences of irreversible subreactions from the reactants toward their final products. Usually, only rather low yields of products are formed by MCRs of type I. In 1850, Strecker[7,8] introduced the formation of α-amino-substituted alkyl cyanides **3** from ammonia, hydrogen cyanide, and the corresponding carbonyl compound **1**, via the cationic intermediate **2** (Scheme 1). The introduction of the Strecker three-component reaction (S-3CR), an MCR of type I, heralded the beginning of MCR chemistry.

Scheme 1 The Strecker and Bucherer–Bergs Multicomponent Reactions[7–10]

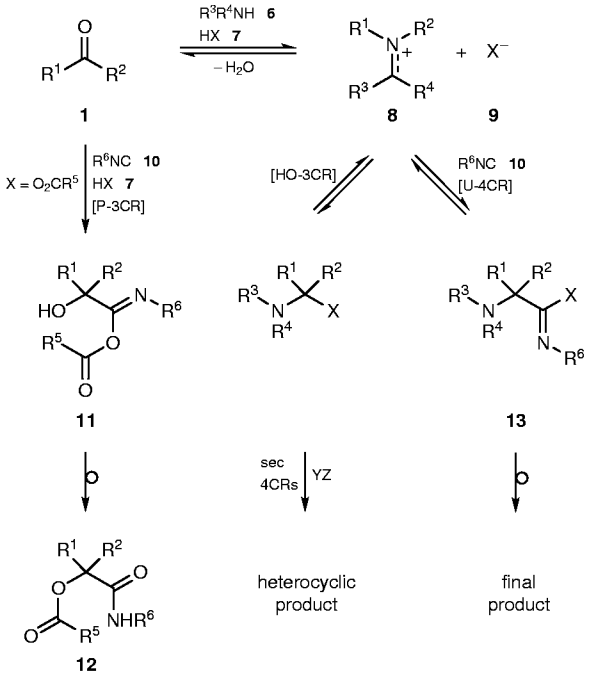


Until the 1920s, α-amino acids **5** were synthesized by the hydrolysis of the α-amino-substituted alkyl cyanides **3**,[8] but subsequently the α-amino acids were produced industrially by hydrolyzing the hydantoin derivatives **4** (Scheme 1). The latter can be prepared by the Bucherer–Bergs four-component reaction (BB-4CR),[9,10] an MCR of type II that forms **4** from carbon dioxide and the three components of the S-3CR. The α-amino acids thus prepared are obtained in much higher yields than by the S-3CR.

The Mannich reaction[11] and the related α-aminoalkylations of nucleophilic anions of weak acids are also examples of MCRs of type I,[8] whereas ring closures to heterocycles, e.g. the Hantzsch[12] and Radziszewki[13] reactions, correspond to MCRs of type II. The Asinger reaction[14,15] sometimes belongs to type I and in some other cases to type II MCRs.

In 1960, Hellmann and Opitz[8] realized that the majority of the previously discussed "named reactions" and other related processes are all α-aminoalkylations of nucleophilic compounds of sufficiently weak acids, corresponding to MCRs of type I, or they have secondary reactions of type II that are followed by the formation of heterocycles [the Hellmann–Opitz three-component reaction (HO-3CR),[8] Scheme 2]. In 1921, Passerini[16,17] introduced his three-component reaction (P-3CR), which involves the reaction of a carboxylic acid **7**, a carbonyl compound **1**, and an isocyanide **10**, to form (via rearrangement of the first-formed product **11**) an α-carboxy-substituted alkanamide **12** (Scheme 2). This was the first example of a MCR of type II that involved an isocyanide.

Scheme 2 The Passerini, Hellmann–Opitz, and Ugi Multicomponent Reactions[2,4,8,16,17]



There are only a few MCRs of type III known in preparative organic synthesis;[18] in living cells, however, the majority of bioorganic chemical compounds are formed by MCRs of type III. In these cases suitable enzymes selectively accelerate the required sequences of subreactions.

In 1958, the isocyanides **10** became readily available[1] and shortly afterwards the four-component reaction (4CR) was introduced, in which an amine **6** (e.g., ammonia, hydrazine, hydroxylamine, or a primary or secondary amine), a carbonyl compound **1**, and an acid **7**

(HX = H_2O, H_2Se, $H_2S_2O_3$, HN_3, H_2NCN, HOCN, HSCN, RCO_2H, RCOSH, $ROCO_2H$, etc.) form the ions **8** and **9**, which subsequently undergo an α-addition onto the unique divalent carbon of the isocyanide **10** (Scheme 2). The α-substituted adduct **13** then rearranges into its final product.[1,19] Since 1962 this reaction has been called the Ugi four-component reaction (U-4CR).[2,4] The U-4CR can formally be considered as the union of the HO-3CR and the P-3CR. Both of these three-component reactions have in common that they use carbonyl compounds and acids as reactants.

Later, when it was recognized that stereoselective syntheses could also be accomplished, syntheses of stereoisomeric α-amino acid and peptide derivatives by suitable U-4CRs were attempted.[1,19] As such reactions require the use of chiral components, the products of U-4CRs must contain selectively removable chiral auxiliary groups. Consequently, the search for such groups has become one of the key areas of contemporary U-4CR chemistry. Whilst not all problems have optimally been solved, it is nevertheless presently possible to synthesize many types of α-amino acid derivatives by the U-4CR.

Since 1993 it has been possible to form an even greater variety of products using isocyanide-containing MCRs, with five or more reactants being involved. It was also demonstrated that even seven-component reactions can lead to defined products,[20] and it was then realized that in principle even higher numbers of differing coreactants may generate their products by one-pot reactions. These MCRs involving more than four components are accomplished by combining the U-4CR with other reactions.[3]

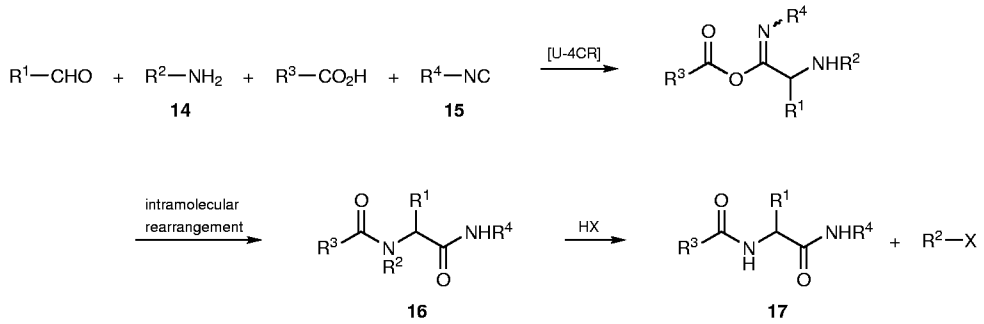
In the last few years the chemical industry has recognized that any synthesis utilizing higher MCRs can form a large variety of products and that MCRs have also great preparative advantages over conventional multistep procedures. The search for new desirable products from libraries of MCR products is now one of the preferred methodologies. The products have the advantage that they can be prepared by minimal work and in very high yields.

The libraries[1,2] of U-4CR products already mentioned in 1961,[21] were not prepared until 1995. Since then, this approach has found widespread application in industrial chemical research. It is expected that this one-pot, multicomponent approach will play an essential role in synthetic organic strategies in the future.[3,6]

4.4.1 Synthesis of α-Amino Acid and Peptide Derivatives by the Ugi Four-Component Reaction of Cleavable Amine Components

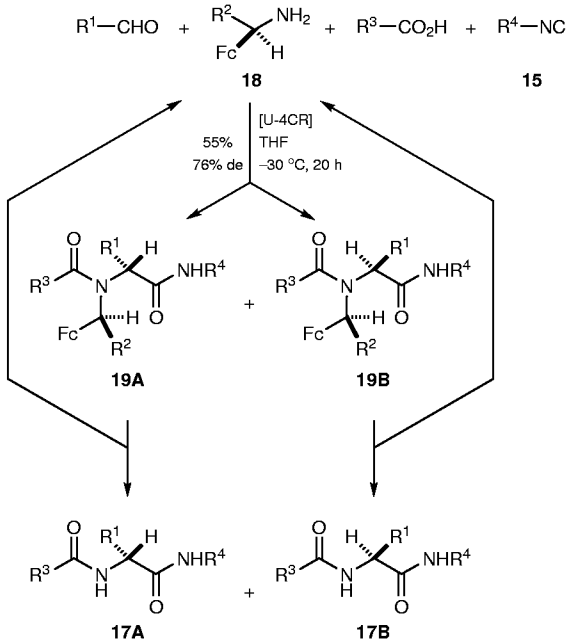
The first one-pot synthesis of a variety of α-amino acid and peptide derivatives **17** by the U-4CR of a chiral primary amine **14**, a carboxylic acid, a carbonyl compound (usually an aldehyde), and an isocyanide **15**, was reported in 1961 (Scheme 3).[1] The overall reaction involves condensation of the four components followed by an intramolecular rearrangement of the intermediate to give the final U-4CR product **16**. Many U-4CRs have been carried out since then, with many being directly related to peptide syntheses. The U-4CRs of bifunctional α-amino acids, with the involvement of five functional groups, form 1,1′-iminodicarboxylic acid derivatives,[1,22] whereas the U-4CRs of β-amino acids form β-lactam derivatives.[23–25] Amino acid derivatives with amino and carboxy groups that are separated by a larger spacer form larger ring derivatives.[26]

Scheme 3 U-4CR and the Formation of α-Amino Acid and Peptide Derivatives[1,27]



For the stereoselective formation of diastereomeric products, it was recognized at an early stage that the U-4CR can proceed stereoselectively if chiral primary amines and aldehydes are employed as the reactive components. The peptide derivatives **17** can only be formed if the amino component **14** contains an alkyl group R^2 that can be replaced by a proton in **16** producing **17** (Scheme 3).[27] The essential problems were first solved separately by a variety of experimental investigations.[28] It was found that chiral, α-ferrocenyl-substituted alkylamines **18** are able to fulfill all the requirements of peptide synthesis by stereoselective U-4CRs (Scheme 4).[1] New, efficient methods were developed for the preparation of alkylamines **18**,[28,29] in order to investigate their role in the synthesis of peptide derivatives by the U-4CR procedure. Recently, this chemistry has become useful in the development of chiral catalysts.[29] It was found that peptide derivatives can be produced in a stereoselective manner by the U-4CRs of suitable α-ferrocenyl-substituted alkylamines **18**, and that the resulting products **19A** and **19B** can be cleaved to yield **17A** and **17B**, respectively, in a procedure that ultimately regenerates the chiral α-ferrocenyl-substituted alkylamine **18**.[28]

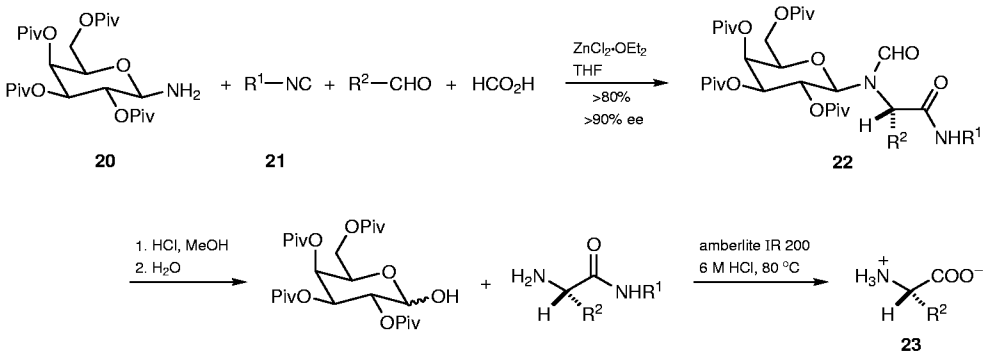
Scheme 4 Formation of Peptides Using the U-4CR and Chiral, α-Ferrocenyl-Substituted Alkylamines[28]



for references see p 888

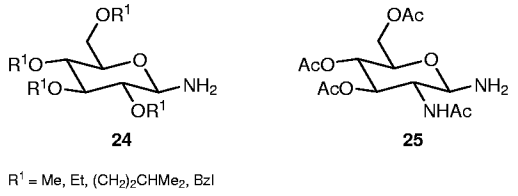
The disadvantages of the method outlined in Scheme 4 for the synthesis of peptides are that the U-4CRs are not sufficiently stereoselective, and that the preparative steps lead to unsatisfactory yields.[28] A new, highly stereoselective approach for the preparation of α-amino acid derivatives by the U-4CR was then developed by Kunz and Pfrengle:[30] the use of 1-amino-substituted carbohydrates such as **20** leads to highly stereoselective (>90% ee) formation of α-amino acid derivatives **22** in excellent yields (typically >80%). 2,3,4,6-Tetrakis-*O*-pivaloyl-β-D-galactopyranosylamine (**20**) proved to be particularly useful when applied in U-4CRs in the presence of $ZnCl_2 \cdot OEt_2$ (Scheme 5).[30] In these U-4CRs, only formic acid was used as the acid component, because more sterically hindered acids would probably not undergo the U-4CR sufficiently well. Furthermore, the products **22** have to be hydrolyzed with hot 6 M HCl to produce the free α-amino acids **23**. Thus, even this type of U-4CR does not fulfill all the suitable requirements for the synthesis of peptide derivatives, although it is a promising route for the synthesis of selected α-amino acids.

Scheme 5 Stereoselective Formation of α-Amino Acid Derivatives Based on Amino-Substituted Carbohydrates by the U-4CR[30]



As a result of this promising new approach, other 1-amino-substituted carbohydrate derivatives **24** were used to form any desirable α-amino acid and peptide derivative by the U-4CR, but which could not be cleaved under sufficiently mild conditions (Scheme 6).[31] The α-amino acid derivatives that are formed by the use of these new amines **24** and any usual acid component are somewhat easier to cleave than **22**, but still not under sufficiently mild conditions.

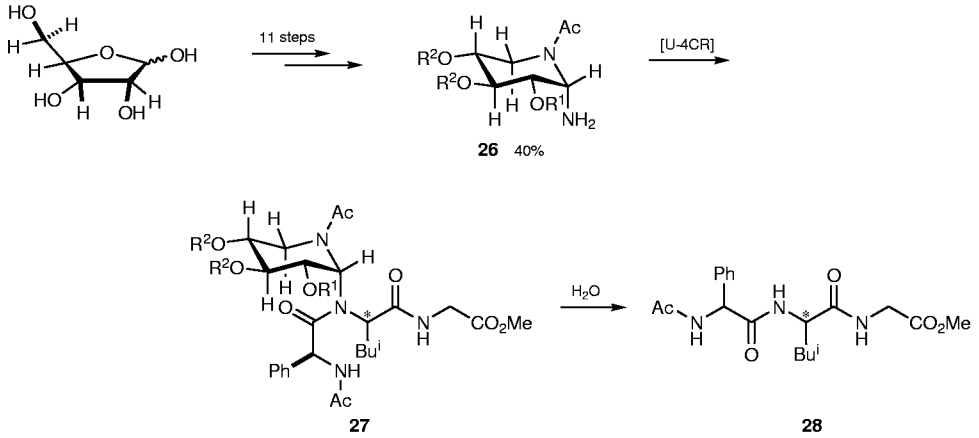
Scheme 6 Glucopyranosylamine Derivatives for the U-4CR[31]



Similarly, the product resulting from the U-4CR of amine **25** (Scheme 6), also requires relatively harsh cleavage conditions with the use of methanol and hydrochloric acid.[31] Nevertheless, the reaction of **25**, isobutyraldehyde, formic acid, and *tert*-butyl isocyanide gives the U-4CR product in a 90% yield.[31] A large number of U-4CR products with **25** were also found to contain >99% of one stereoisomer and even the yields were in many cases >80%.[31] Finally, U-4CRs with 1-amino-5-acetamino-5-deoxy-α-D-xylopiperidose **26** produce α-amino acid derivatives such as **27** in high stereoselectivity (>99%) and which are

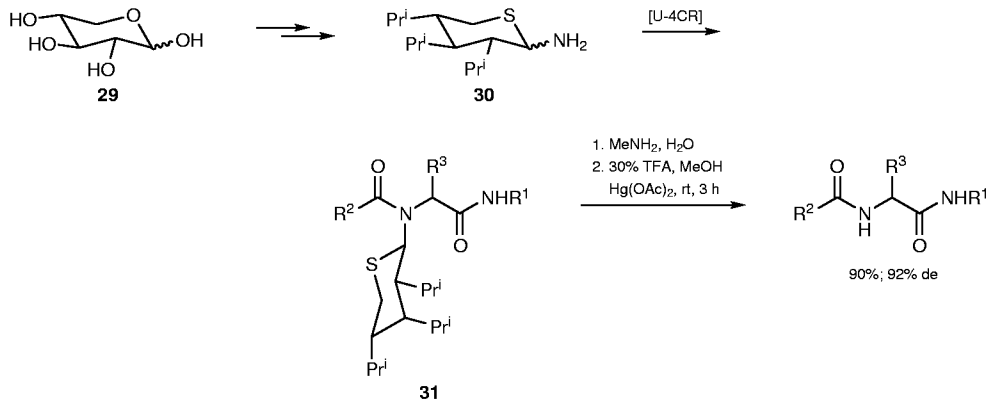
readily cleaved by water (Scheme 7).[32] The only disadvantage of this almost ideal amine component **26** is its availability: a minimum of 11 preparative steps are currently required for its synthesis with an upper limit on the total yield of only 40%.

Scheme 7 U-4CR Involving 1-Amino-5-acetamino-5-deoxy-α-D-xylopiperidose[32]



In fewer synthetic steps, the oxygen in the 5-position of xylose (**29**) can be replaced by sulfur, and the 1-amino group subsequently introduced (Scheme 8).[33,51,52,57,58] According to preliminary investigations, the use of the resulting 5-thio-D-xylopyranosylamine (**30**) in U-4CRs proceeds well, and the products **31** are cleaved under mild conditions. Due to the easier accessibility of **30** than **26**, and the stability of its U-4CRs, it represents a promising candidate for the stereoselective synthesis of peptides by the U-4CR procedure.

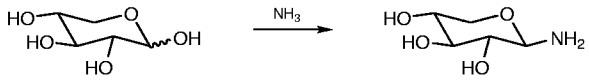
Scheme 8 U-4CR Involving 5-Thio-D-xylopyranosylamine[33,51,52,57,58]



Following the procedure of Likhosherstov et al.[34,35] for the preparation of 1-amino-substituted carbohydrates like **32** by reacting xylose (**29**) or glucose with ammonia or ammonium carbonates (Scheme 9), 1-amino derivatives of xylose, glucose, and 2-acetylamino-glucose[36,53] have been obtained and used in stereoselective U-4CRs. A systematic search for the best cleavage conditions for the resulting peptide derivatives showed that the best results were obtained using very dilute HCl in methanol at 20 °C. Correspondingly, this procedure may become one of the preferred methods for synthesis of peptide derivatives containing unnatural amino acids which usually require complex, multistep syntheses.

for references see p 888

Scheme 9 Synthesis of D-Xylopyranosylamine for Use in Stereoselective U-4CRs[34-36,53]



29 32

***N,N*-Phthaloylglycyl-*N*-benzylvaline *tert*-Butylamide (16, R^1 = iPr; R^2 = Bzl; R^3 = CH_2NPhth; R^4 = tBu):**[37]

CAUTION: *Benzene is carcinogenic!*

N,N-Phthalylglycine (3.08 g, 15 mmol), benzylamine (1.61 g, 15 mmol), isobutyraldehyde (1.08 g, 15 mmol), and *tert*-butyl isocyanide (1.25 g, 15 mmol) were dissolved in ice-cold MeOH, and the mixture was stirred for 2 h at 20 °C. The crude product was then recrystallized (benzene/Et_2O 1:1); yield: 6.4 g (92%); mp 165–167 °C.

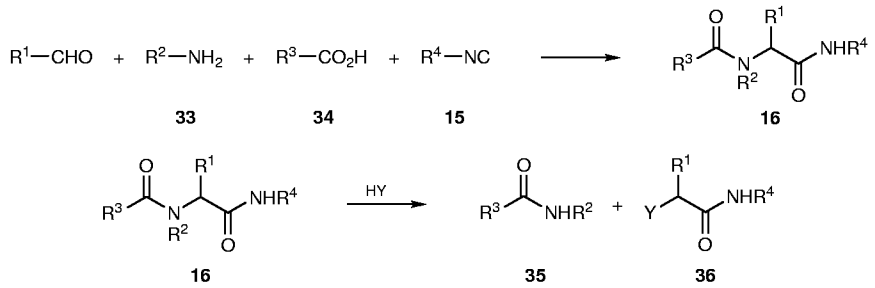
Stereoselective Synthesis of α-Amino Acid and Peptide Derivatives by the U-4CR of 25; General Procedure:[31]

2-Acetamido-3,4,6-tri-*O*-acetyl-2-deoxy-ß-D-glucopyranosylamine (**25**; 10 mmol), an aldehyde (10 mmol), 4 Å molecular sieves (2.5 g), and $ZnCl_2$·OEt_2 (11 mmol) were dissolved under an argon atmosphere at 20 °C in a solvent (MeOH or THF, 90 mL). After a few hours reaction time, an isocyanide (10.4 mmol) and a carboxylic acid (10.4 mmol) were added at a suitable temperature (20, −40, or −75 °C). Upon consumption of the amine, as monitored by TLC ($CHCl_3$/EtOH), the mixture was filtered through a layer of Celite (1 cm), diluted with CH_2Cl_2 (20 mL), and evaporated. The residue was dissolved in CH_2Cl_2 (100 mL) and stirred with 7.5% aq $NaHCO_3$ (15 mL). Isocyanides, if still present, were removed by treatment with AcOH (20 mL) for 16 h. (If *tert*-butyl isocyanide was utilized, it was removed by 3.5% aq tartaric acid.) The soln was finally washed with 7.5% aq $NaHCO_3$ (250 mL), dried (Na_2SO_4), and evaporated to dryness to yield the desired amino acid derivative.

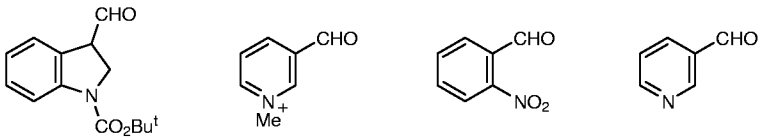
4.4.2 Synthesis of α-Amino Acid and Peptide Derivatives by the Ugi Four-Component Reaction of Cleavable Aldehydes

The protected α-amino acid derivatives **33** and **34** (Scheme 10; R^2 = C-protected amino acid residue, R^3 = N-protected amino acid residue) cannot only be condensed to peptides by the usual methods of peptide synthesis, but also by the U-4CR, if suitably cleavable aldehyde components are used.[28] Cleavage of the initially formed protected dipeptide **16** at the aldehyde position with acid, typically trifluoroacetic acid or by other methods (*vide infra*) can yield the products **35** and **36** (cf. amine cleavage, Scheme 3). Of the large number of different aldehydes analyzed for this purpose, only a few fulfill the requirements of efficient and mild cleavage (Scheme 11).[38]

Scheme 10 Synthesis of Dipeptides Using the U-4CR and Cleavable Aldehydes[28]



Scheme 11 Cleavable Aldehydes Suitable for the U-4CR[38–45]

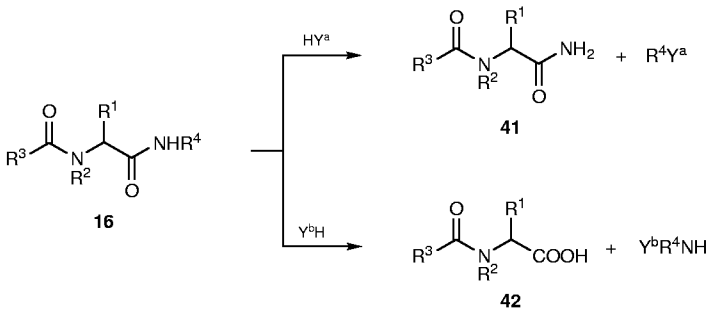


A variety of peptide derivatives have been prepared and cleaved using the U-4CR incorporating aldehyde **37** (Scheme 11);[39,40] racemization was found to be negligible.[41] The 3-formyl-1-methylpyridinium cation (**38**) forms U-4CR products that may be cleaved by reductive methods and by subsequent treatment with trifluoroacetic acid.[42] The use of 2-nitrobenzaldehyde (**39**), on the other hand, leads to U-4CR products that may be cleaved photolytically.[43] It was also found that the products obtained with nicotinaldehyde (**40**) may be cleaved electrochemically or by reduction with zinc, and even by photolysis in methanol in the presence of HCl. Cleavage is also accomplished by autoxidation in the presence of a Ni(II)–phthalocyanine complex.[28,44,45]

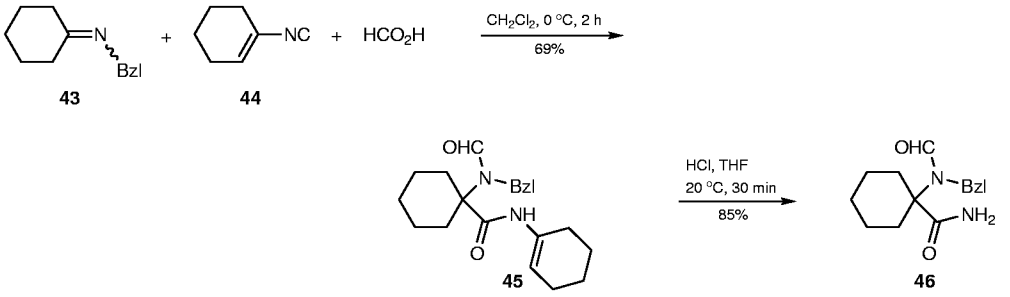
4.4.3 Synthesis of α-Amino Acid and Peptide Derivatives by the Ugi Four-Component Reaction of Cleavable Isocyanides

In some cases it is necessary to cleave the alkyl or aryl group R^4, or the NHR^4 substituent of the U-4CR products **16**, giving rise to the amide **41** and the acid **42**, respectively (Scheme 12). This requires the choice of isocyanides with suitable substituents R^4. For example, reaction of cyclohex-1-enyl isocyanide (**44**) with *N*-benzyl-*N*-cyclohexylideneamine (**43**) and formic acid yields the related U-4CR product **45** (Scheme 13). Treatment with acid converts the cyclohexenyl-substituted amide **45** into the corresponding free amide **46** in good overall yield.[46]

Scheme 12 Conversion of Isocyanide-Derived Substituted Amides into Free Amides or Carboxylic Acids



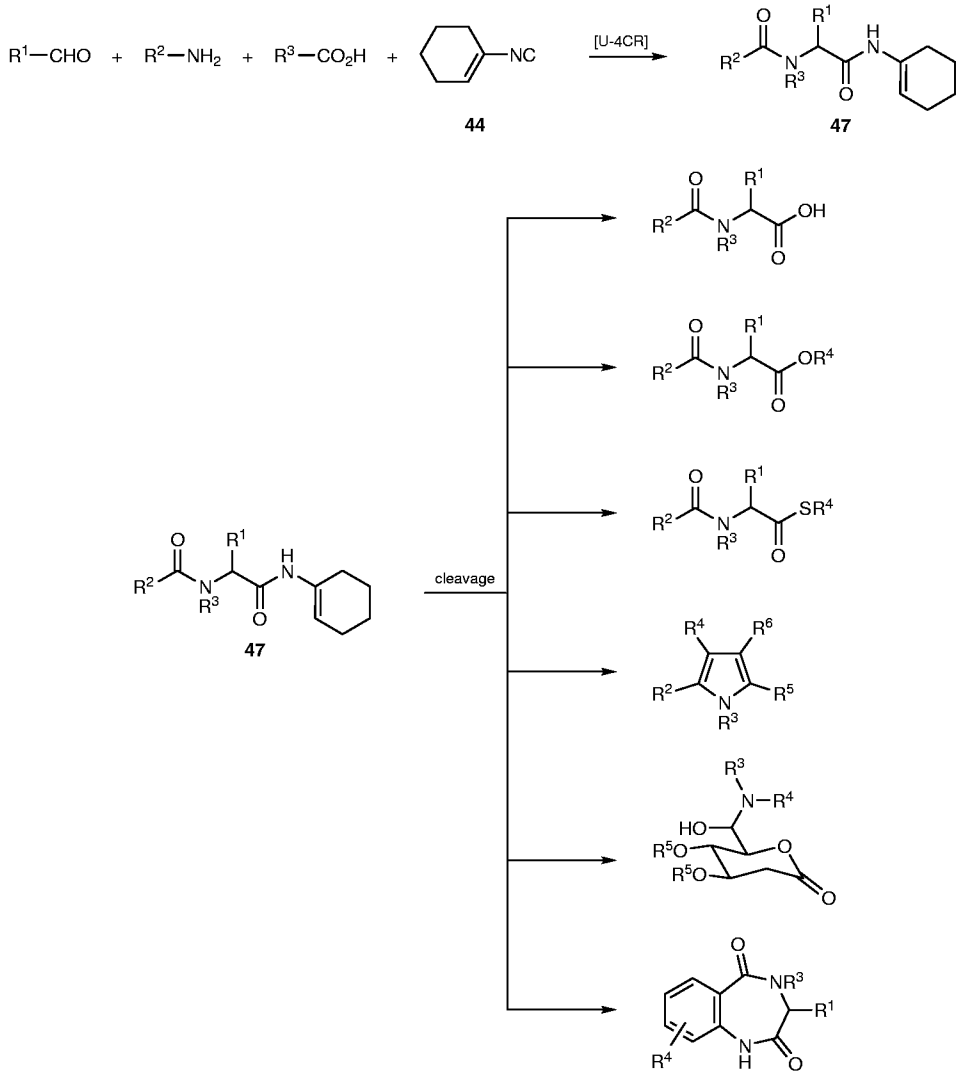
Scheme 13 U-4CR with Cyclohex-1-enyl Isocyanide[46]



for references see p 888

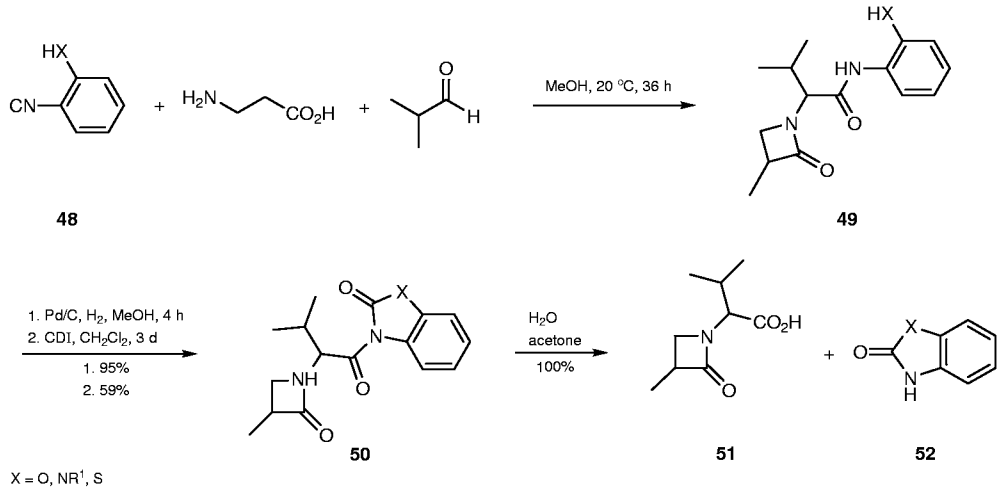
The isocyanide **44** has also been used to prepare the α-amino acid derivatives **47** illustrated in Scheme 14.[47,48] Cleavage of either the *N*-cyclohex-1-enyl or the *N*-cyclohex-1-enylamino group of **47** produces the compounds shown in Scheme 14.

Scheme 14 Cleavage of the *N*-Cyclohex-1-enyl or *N*-Cyclohex-1-enylamino Groups[47,48]



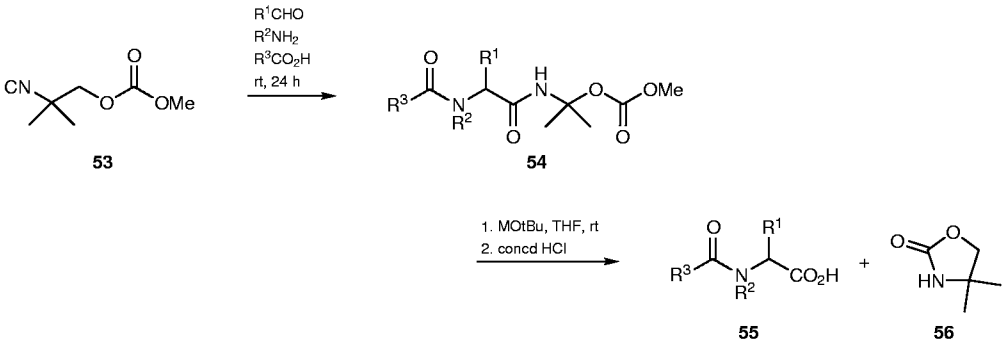
The U-4CR products **49** of the isocyanides **48** with β-alanine and isobutyraldehyde, as shown in Scheme 15, can be converted into their cyclized derivatives **50**, which are hydrolyzed by neutral water to yield the compounds **51** and **52**.[49] These β-lactam derivatives clearly show that even highly sensitive U-4CR products are not destroyed by this chemistry.[49]

Scheme 15 U-4CRs Involving β-Lactam Derivatives[49]



Recently, the isocyanide **53** was converted into the U-4CR products **54** (Scheme 16); these can be cleaved under mild conditions to give the products **55** and **56**.[50] In an analogous manner, many other ester derivatives of **53** could be employed. The chemistry of **53** and its related derivatives provides new preparative possibilities in peptide chemistry.

Scheme 16 The U-4CR with 2-Isocyano-2-methylpropyl Methyl Carbonate[50]



1-[Benzyl(formyl)amino]-*N*-cyclohex-1-en-1-ylcyclohexanecarboxamide (45):[46]
Cyclohex-1-enyl isocyanide (**44**; 2.14 g, 29.0 mmol) was added at 0 °C to a mixture of *N*-benzyl-*N*-cyclohexylideneamine (**43**; 4.67 g, 25.0 mmol) and formic acid (1.15 g, 25.0 mmol) in CH_2Cl_2 (20 mL). After 2 h, the solvent was removed, and the residue was washed with MeOH (10 mL) at 0 °C and recrystallized (MeOH) to give **45**; yield: 4.70 g (69%); mp 149–155 °C.

1-[Benzyl(formyl)amino]cyclohexanecarboxamide (46):[46]

CAUTION: *Benzene is carcinogenic!*

To *N*-cyclohex-1-en-1-ylcyclohexanecarboxamide (**45**; 1.50 g, 4.41 mmol) in THF (10 mL) was added at 20 °C concd HCl (0.5 mL). After 30 min, the mixture was neutralized with aq $NaHCO_3$ and the solvent was removed. The crude product **46** was recrystallized (MeOH/benzene); yield: 0.98 g (85%); mp 157–159 °C.

2-Isocyano-2-methylpropyl Methyl Carbonate (53):[50]
To a soln of 4,4-dimethyl-4,5-dihydrooxazole (19.83 g, 200 mmol) in dry THF (200 mL) was added slowly at −78 °C BuLi (12.81 g, 200 mmol), followed by dropwise addition of methyl chloroformate (12.9 g, 15.45 mL, 200 mmol) under stirring. After 30 min, the temperature was gradually raised to 25 °C and the

for references see p 888

soln was washed with brine and dried (Na_2SO_4). Evaporation of the solvent gave the crude product **53**; yield: 80%. The crude product can be used directly in U-4CRs; if necessary, **53** can be distilled prior to use; mp 31 °C; bp 52–53 °C/0.1 Torr.

U-4CR with 2-Isocyano-2-methylpropyl Methyl Carbonate (53); General Procedure:[50]
An aldehyde (8 mmol) and a primary amine (8 mmol) were stirred in MeOH (16 mL) at 25 °C for 2 h. 2-Isocyano-2-methylpropyl methyl carbonate (**53**; 8 mmol) and a carboxylic acid (8 mmol) were then added and the mixture was stirred for 24 h. The solvent was removed, and the product **54** was isolated by flash chromatography (silica gel, hexane/EtOAc 3:1) as a mixture of rotamers.

(*R,S*)-[Alkanoyl(alkyl)amino](alkyl)acetic Acids 55; General Procedure:[50]
The substituted methyl carbonate **54** (5mmol) and *tert*-butoxide (6 mmol) were stirred in dry THF (15 mL) at 25 °C. Upon completion of the reaction, as monitored by TLC (hexane/EtOAc 3:1), concd HCl (6 mmol) was added. After removal of the solvent, the product **55** was isolated by flash chromatography (silica gel, hexane/EtOAc 3:1) as a rotameric mixture.

4.4.4 Libraries of α-Amino Acid Derivatives Formed from Mulitcomponent Reactions

Since the 1980s, collections of different organic compounds have frequently been termed "chemical libraries" and the methods used for their preparation are often referred to as combinatorial chemistry. In principle, however, chemical libraries have existed since the beginning of organic chemistry eons ago, but were not exploited systematically until the late 20th century.

Merrifield introduced his solid-phase peptide synthesis (SPPS) methodology in 1963;[54] this has since become the preferred method of peptide synthesis. Two decades later, Furka[55,56] extended Merrifield's techniques to the synthesis of libraries of peptides used for the screening of new, desirable pharmaceutical products. Because peptides do not necessarily represent the ideal candidates for therapeutics, more recent attention has focused on other libraries of compounds that can be prepared by solid-phase multistep procedures. Among the approaches used for such purposes, the HO-3CR and the U-4CR, and their combination with other reactions, are finding increasing applications.

It was pointed out by Bodanszky and Ondetti[5] as early as 1966, that the U-4CR might represent a potentially important synthetic method that could be adopted in peptide chemistry, particularly if the stereoselectivity of the U-4CRs was improved. This is because it allows for the one-pot synthesis of unnatural α-amino acid derivatives[57,58] and of related libraries as an alternative approach to other known but time-consuming stereoselective syntheses of such derivatives. However, the anticipated preparative advantages are only now being generally recognized.[6]

[1] Ugi, I., *Isonitrile Chemistry*, Academic: New York, (1971).
[2] Ugi, I.; Lohberger, S.; Karl, R., In *Comprehensive Organic Synthesis: Selectivity for Synthetic Efficiency*, Trost, B. M.; Heathcock, C. H., Eds.; Pergamon: Oxford, UK, (1991); Vol. 2, pp 1083, 1090.
[3] Ugi, I.; Dömling, A.; Hörl, W., *Endeavour*, (1994) **18**, 115.
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